



JRC SCIENCE FOR POLICY REPORT

Preparatory study on textiles for product policy instruments

Ecodesign

EU Green Public Procurement

EU Ecolabel

3rd milestone

Delre, A., Gonzalez Torres, M., Kouloumpis, V.,
Senatore, V., Perez Arribas, Z., Bernad Beltran,
D., Perez Camacho, M., N., Syrus, A., Magrini, C.,
Albizzati, P., F., Cappucci, G., Ardente, F.,
Moldovan, S., Costamagna, M., Molina Vercher,
C., Mollá, K., Perucca, M., Gallego, E., Balaguer,
A., Garcia John, E.

This is a draft document that has not been adopted or endorsed by the European Commission. Any views expressed are the preliminary views of the Commission's services and may not in any circumstances be regarded as stating an official position of the Commission.

This publication is a Science for Policy report by the Joint Research Centre (JRC), the European Commission's science and knowledge service. It aims to provide evidence-based scientific support to the European policymaking process. The contents of this publication do not necessarily reflect the position or opinion of the European Commission. Neither the European Commission nor any person acting on behalf of the Commission is responsible for the use that might be made of this publication. For information on the methodology and quality underlying the data used in this publication for which the source is neither Eurostat nor other Commission services, users should contact the referenced source. The designations employed and the presentation of material on the maps do not imply the expression of any opinion whatsoever on the part of the European Union concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries.

Contact information

Name Joint Research Centre, Directorate B – Fair and Sustainable Economy, Unit B5 – Circular Economy and Sustainable Industry
Email: JRC-B5-TEXTILES@ec.europa.eu

EU Science Hub

<https://joint-research-centre.ec.europa.eu>

JRCXXXXXX

EUR XXXX XX

PDF ISBN XXX-XX-XX-XXXX-X ISSN XXXX-XXXX doi:XX.XXXX/XXXXXX XX-XX-XX-XXX-XX-C

Seville: European Commission, 2024

© European Union, 2024



The reuse policy of the European Commission documents is implemented by the Commission Decision 2011/833/EU of 12 December 2011 on the reuse of Commission documents (OJ L 330, 14.12.2011, p. 39). Unless otherwise noted, the reuse of this document is authorised under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence (<https://creativecommons.org/licenses/by/4.0/>). This means that reuse is allowed provided appropriate credit is given and any changes are indicated.

For any use or reproduction of photos or other material that is not owned by the European Union permission must be sought directly from the copyright holders. The European Union does not own the copyright in relation to icons from www.flaticon.com included in some Figures. All Figures with these icons report information in their source.

Disclaimer

This is a draft document, containing work in progress. Any reference made to this document should clearly state its draft character. The views expressed in this document are purely those of the authors and may not in any circumstances be regarded as stating an official position of the European Commission. Neither the European Commission nor any person acting on behalf of the Commission is responsible for the use that might be made of the information in this document.

DRAFT – work in progress

Contents

1	Introduction.....	10
2	Definitions.....	14
3	Scope.....	18
3.1	Selection of the scope.....	18
3.1.1	Potential improvement considering product aspects in Article 5(1) of the ESPR.....	20
3.1.2	Amount of sales and trade in the EU.....	22
3.1.3	Environmental impacts and waste generation across the value chain.....	22
3.2	Proposed scope.....	23
3.2.1	Products included in the scope.....	24
3.2.2	Products excluded from the scope.....	25
3.3	Composition, life-cycle stages and main negative environmental impacts of products included in the scope.....	26
3.3.1	Composition and life-cycle stages.....	26
3.3.2	Main factors influencing the negative environmental impacts.....	29
4	Legislation, strategies and voluntary environmental labels relevant for the textile sector.....	33
4.1	EU legislation.....	33
4.2	Legislation and initiatives in EU Member States and non-EU countries.....	39
4.3	Strategies of the United Nations.....	40
4.4	Strategies of the Organisation for Economic Co-operation and Development.....	40
4.5	Tests and standards.....	41
4.6	Voluntary environmental labels.....	42
5	Market analysis.....	45
5.1	The global market.....	45
5.2	The EU market.....	52
5.3	Role of the EU Member States.....	55
5.4	Main elements affecting the production of textile apparel.....	56
5.4.1	Fibres.....	56
5.4.2	Chemicals.....	63
5.4.3	Energy.....	65
5.4.4	Water.....	66
5.5	Market structure and business models.....	67
5.6	Characteristics of the value chain.....	70
5.7	Competitiveness and environmental compliance costs.....	71
5.8	Lifespan of textile apparel.....	73
5.9	Market penetration of environmental labels.....	75
6	User behaviour.....	76
6.1	Introduction.....	76

6.2	Pre-purchase aspects.....	78
6.2.1	Reasons for purchasing apparel.....	78
6.2.2	Criteria used when buying apparel.....	78
6.2.3	User quality assessment of apparel: key insights.....	80
6.2.4	Consumer behaviour towards labels on apparel.....	82
6.2.5	Attitudes towards second-hand apparel purchase.....	82
6.2.6	Attitudes towards the purchase of apparel made without harmful chemicals.....	83
6.2.7	Attitudes towards the purchase of apparel made with recycled materials.....	84
6.3	Post-purchase aspects.....	84
6.3.1	User behaviour during use: laundering practices.....	84
6.3.2	Following apparel care label instructions.....	85
6.3.3	Reparability.....	86
6.3.4	Storage of apparel.....	86
6.4	User behaviour related to the disposal of apparel.....	87
6.4.1	Reasons for the disposal of apparel.....	87
6.4.2	Disposal channels.....	89
6.4.3	Person-product attachment.....	90
6.4.4	Returns of apparel.....	90
7	Current EU Ecolabel criteria for textile products.....	92
7.1	Facts and figures.....	92
7.2	Suggestions for the revision of EU Ecolabel criteria.....	92
7.3	Revision of EU Ecolabel criteria within the ESPR framework.....	93
7.4	Looking for synergies with other Ecolabels used in the EU.....	94
8	Public procurement and current EU voluntary Green Public Procurement criteria.....	96
8.1	Public procurement in the EU.....	96
8.2	Current voluntary EU Green Public Procurement criteria.....	97
9	Technologies.....	99
9.1	Relevant product aspects.....	99
9.1.1	Methodology.....	100
9.1.2	Exclusion of non-relevant product aspects.....	100
9.1.3	Grouping of relevant product aspects.....	103
9.2	Analysis of technologies.....	109
9.2.1	Physical durability.....	110
9.2.2	Maintenance.....	122
9.2.3	Repairability.....	127
9.2.4	Waste generation.....	130
9.2.5	Recyclability and recycled content.....	134
9.2.6	Environmental impacts.....	144
9.2.7	Presence of substances of concern.....	148

9.3 Mutual influence of product aspects and product categorization.....	154
10 Environmental and economic analysis.....	156
10.1 Impacts via Life Cycle Thinking approach	156
10.1.1 Description of the model.....	156
10.1.2 Contribution analysis.....	160
10.1.3 Perturbation analysis.....	171
10.1.4 Scenario analysis.....	171
10.1.5 Propagation of uncertainties.....	175
10.2 Fragmented fibre release into the environment	176
10.2.1 The products of textile apparel fragmentation	176
10.2.2 Impacts on the environment and human health	178
10.2.3 Sources and pathways due to textile apparel consumption.....	179
10.2.4 Analysis of pollution hotspots.....	181
11 Assessment of the design options	182
11.1 Definition of design options.....	182
11.1.1 D01 - Product with an increased robustness	184
11.1.2 D02 - Product with a higher share of weight going to recycling.....	190
11.1.3 D03 - Product with higher recycled content.....	194
11.1.4 D04 - Product with a decreased environmental or carbon footprint.....	200
11.2 Combination of design options: paths.....	206
11.2.1 Estimates for the assessment.....	207
11.2.2 Results.....	208
11.2.3 Propagation of uncertainties.....	214
11.3 Main conclusions of the assessment of the design options	217
12 Chemical substances in textile apparel.....	218
12.1 ESPR and chemical substances.....	218
12.2 Methodology: substances and substances of concern	221
12.3 Substances in textile apparel.....	222
12.3.1 Inventory.....	228
12.3.2 Identification of information requirements on Substances of Concern	229
12.3.3 Substances of Concern type (a)	229
12.3.4 Substances of Concern type (b)	232
12.3.5 Substances of Concern type (c).....	235
12.3.6 Substances of Concern type (d)	238
12.3.7 Technical constraints	239
12.3.8 The case of elastane.....	239
12.3.9 Other identified substances.....	241
12.3.10 Regulatory and customer-driven limitations.....	241
12.3.11 Conclusion - Substances of Concern type (d)	246

12.3.12	Thresholds for information requirements for substances under Article 2(27)	246
12.3.13	Exemptions	249
12.3.14	Entry into application	251
12.3.15	Identification of performance requirements on substances	253
12.3.16	Product aspect (a) 'durability' and (p) 'expected generation of waste'	254
12.3.17	Product aspect (h) 'energy use and energy efficiency', (i) 'water use and water efficiency', (p) 'expected generation of waste' and (j) 'resource use and resource efficiency'	254
12.3.18	Product aspect (o) 'environmental impacts, including carbon footprint and environmental footprint'	255
12.3.19	Chemical substances in the life cycle assessment	255
12.3.20	Conclusion – performance requirements	258
	References	259
	List of abbreviations	293
	List of figures	294
	List of tables	298
13	Annexes	305
13.1	Consultation with stakeholders	305
13.1.1	Initial questionnaire	305
13.1.2	1 st milestone	305
13.1.3	2 nd milestone	306
13.2	PRODCOM and market analysis	307
13.2.1	General description	307
13.2.2	Features of PRODCOM data influencing market analysis	308
13.2.3	Market analysis for the selection of the scope	309
13.2.4	PRODCOM codes included in the scope	319
13.2.5	Supplementary information about the EU market	332
13.3	Life-cycle stages of textile apparel	350
13.3.1	Raw material and fibre manufacturing	350
13.3.2	Yarn manufacturing	352
13.3.3	Fabric manufacturing	353
13.3.4	Finishing processes	354
13.3.5	Confectioning	356
13.3.6	Retailing	357
13.3.7	Use	358
13.3.8	Waste management	359
13.4	Supporting information about tests and standards in the textile industry	360
13.5	Supporting information about environmental labels	378
13.5.1	Types of labels in general	378
13.5.2	Voluntary environmental labels used in EU	379
13.6	Supporting information on market analysis	382

13.6.1	Recycling plants.....	382
13.6.2	Comparison among BREFs.....	383
13.6.3	Service lifespan.....	384
13.7	Supporting information about user behaviour.....	387
13.7.1	Pre-purchase aspects.....	387
13.7.2	Post-purchase aspects: the user behaviour during use.....	394
13.7.3	User behaviour regarding the disposal of apparel.....	411
13.8	Supporting information about environmental labels and current EU Ecolabel criteria.....	420
13.8.1	Figures of EU Ecolabel for textile products.....	420
13.8.2	Contributions to the initial questionnaire.....	421
13.8.3	Comparison among Ecolabels.....	425
13.9	Supporting information about public procurement.....	463
13.9.1	The Common Procurement Vocabulary (CPV) codes for products in the scope.....	463
13.9.2	Number of Contract Award procuring apparel in EU.....	465
13.9.3	Contributions to the initial questionnaire.....	468
13.10	 Supporting information on relevant aspects	468
13.10.1	Qualitative assessment based on technical, socioeconomic and environmental dimensions.....	468
13.10.2	Supporting information on parameters affecting the physical durability.....	484
13.10.3	Supporting information on test methods to describe the physical durability.....	496
13.10.4	Supporting information on maintenance.....	497
13.10.5	Supporting information on repairability.....	499
13.11	... Supporting information on the environmental and economic assessment – the Life Cycle Thinking model.....	501
13.11.1	Intended application of the results.....	501
13.11.2	Limitations due to methodological choices.....	502
13.11.3	Limitations due to LCI information availability.....	504
13.11.4	Target audience.....	505
13.11.5	Comparative study to be disclosed to the public.....	505
13.11.6	Commissioner of the study and influential actors.....	505
13.11.7	Definitions.....	505
13.11.8	Goal and scope.....	506
13.11.9	Life Cycle Inventory analysis.....	510
13.11.10	 Assessment of Life Cycle Environmental Impacts and Costs	537
13.11.11	 Sensitivity and uncertainty Analysis	572
13.12	 Supporting information on the assessment of the availability of secondary materials	579
13.12.1	Detailed assessment of recycled cotton availability.....	579
13.12.2	Detailed assessment of recycled nylon availability.....	580

13.12.3 Detailed assessment of recycled polyester availability.....	580
13.12.4 Detailed assessment of recycled wool availability.....	581
13.13 Supporting information for the identification of substances and substances of concern	
582	

DRAFT – work in progress

Foreword

This document is the 3rd milestone of the preparatory study on textile products, which aims to provide a basis on which the European Commission can consider the development of ecodesign requirements, green public procurement criteria and revised EU Ecolabel criteria for textile products. In particular, the 3rd milestone addresses scope, market, user behaviour, current EU Ecolabel criteria, current EU Green Public Procurement criteria, the product technologies, the environmental and economic analysis of the products currently placed on the EU market and design options. The preparatory study is developed within the legal context of the Regulation (EU) 2024/1781 establishing a framework for the setting of ecodesign requirements for sustainable products and the Regulation (EC) No 66/2010 on EU Ecolabel. The project is led by the Joint Research Centre of the European Commission, addressing the request from the Directorate-General for Environment (DG ENV) and the Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs (DG GROW).

DRAFT – work in progress

Acknowledgements

The authors thank the colleague Loredana Napolano (JRC.D3) for her contribution in the initial questionnaire sent to the stakeholders, Gillian Foster (JRC.B5) for her inputs on the economic analysis. The authors thank all the colleagues in DG ENV and DG GROW for the constructive feedback provided during the development of the report. In particular, the authors thank Carsten Wentink (ENV.B4), Silvia Ferratini (ENV.B1) and Antonio De Sousa Maia (GROW.F.3) for the continuous exchanges. Finally, the authors thank Oliver Wolf and all colleagues supporting the development of this document.

Authors

DELRE Antonio ^(a), GONZALEZ TORRES Maria ^(a), Kouloumpis Viktor ^(b), Senatore Vincenzo ^(a), PEREZ ARRIBAS Zahara ^(a), BERNAD BELTRAN David ^(a), PEREZ CAMACHO Maria Natividad ^(a), SYRUS Anandu ^(b), MAGRINI Chiara ^(a), ALBIZZATI Paola Federica ^(a), CAPPUCCI Grazia ^(b), Ardente Fulvio ^(b), MOLDOVAN Simona ^(c), COSTAMAGNA Mattia ^(d), MOLINA VERCHER Cristina ^(c), MOLLÁ Korina ^(c), PERUCCA Massimo ^(d), GALLEGO Eder ^(c), BALAGUER Ana ^(c), GARCIA JOHN Enrique ^(a)

^(a) European Commission, Joint Research Centre, Directorate B – Fair and Sustainable Economy, Unit B5 – Circular Economy and Sustainable Industry

^(b) European Commission, Joint Research Centre, Directorate D – Sustainable Resources, Unit D3 – Land Resources and Supply Chain Assessments

^(c) Research and Innovation Centre – AITEX®

^(d) Project Hub-360

1 Introduction

2 Context

On 13 June 2024, the Regulation on Ecodesign for Sustainable Products (ESPR) ⁽¹⁾ was published on the Official Journal of the European Union. The ESPR delivers on the commitments made in both the European Green Deal ⁽²⁾ and the Circular Economy Action Plan ⁽³⁾ to make the European Union (EU) regulatory framework fit for a sustainable future and to ensure that products placed on the EU market become increasingly sustainable.

The ESPR aims to reduce the environmental impacts of products across their life cycle and to improve the functioning of the EU's internal market. It proposes to do this by building on the successful approach pioneered under the Ecodesign Directive ⁽⁴⁾, which applies to energy-related products only. The ESPR proposes to extend the Ecodesign Directive to cover a very broad range of physical products and to strengthen its provisions. This would enable the ESPR to set a range of far-reaching ecodesign requirements for specific product groups, to improve product circularity, energy performance and other environmental sustainability aspects. Ecodesign requirements can be related to the performance of the product or information that must accompany the product. A Digital Product Passport (DPP) will be required to hold and convey the information in question, with traceability features.

The ESPR provides a general framework for these rules, with specific product requirements to be set at a later stage via delegated acts dedicated to a particular product or to groups of similar products. These delegated acts will also serve as the reference for the adoption of implementing acts, which should establish minimum mandatory Green Public Procurement (GPP) requirements for public contracts.

Article 41(4) of the ESPR establishes synergies between the mandatory ecodesign legislative framework and the EU Ecolabel, which is the EU's official voluntary label for environmental excellence awarded to best-in-class products and services. The ESPR and EU Ecolabel requirements must be coherent and synergic to guarantee that products awarded the EU Ecolabel comply with the ESPR requirements set in the relevant delegated act. The EU Ecolabel remains regulated by the EU Ecolabel Regulation ⁽⁵⁾.

On 30 March 2022 the EC presented the EU Strategy for Sustainable and Circular Textiles ⁽⁶⁾ which aims, *inter alia*, 'to tackle fast fashion and textile waste and to make textiles more durable, repairable, reusable and recyclable'. The Textile Strategy lays out a forward-looking set of actions, which includes setting ecodesign requirements for textiles under the framework of the ESPR proposal ⁽⁷⁾. The Textile Strategy also announces a revision of the EU Ecolabel criteria for textile products to support its uptake among producers and offer consumers an easily recognisable and reliable way to choose eco-friendly textile products.

Since 2019, the EC has supported the project 'Product Environmental Footprint Category Rules (PEFCR): apparel and footwear', hereinafter named PEFCR A&F ⁽⁸⁾. Part of the textile industry leads this project, which aims to establish rules for the calculation of the Product Environmental Footprint. The PEFCR A&F can be used by the industry to voluntarily quantify the environmental impacts of specific textile products. The EC follows the development of the PEFCR A&F project as an observer.

¹ ESPR. Regulation (EU) 2024/1781 of the European Parliament and of the Council of 13 June 2024 establishing a framework for the setting of ecodesign requirements for sustainable products. Available at [this link](#).

² The Green Deal. Communication from the Commission to the European Parliament, the European Council, the Council, the European Economic and Social Committee and the Committee of the Regions. The European Green Deal. COM(2019) 640 final. Available at [this link](#).

³ The Circular Economy Action Plan. Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions. A new Circular Economy Action Plan For a cleaner and more competitive Europe. COM(2020) 98 final. Available at [this link](#).

⁴ Ecodesign Directive. Directive 2009/125/EC of the European Parliament and of the Council establishing a framework for the setting of ecodesign requirements for energy-related products. Available at [this link](#).

⁵ EU Ecolabel Regulation. Regulation (EC) No 66/2010 of the European Parliament and of the Council of 25 November 2009 on the EU Ecolabel. Available at [this link](#).

⁶ Textile Strategy. Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions. EU Strategy for Sustainable and Circular Textiles. COM(2022) 141 final. Available at [this link](#).

⁷ Proposal of ESPR by European Commission. Proposal for a Regulation of the European Parliament and of the Council establishing a framework for setting ecodesign requirements for sustainable products and repealing Directive 2009/125/EC. COM(2022) 142 final. Available at [this link](#).

⁸ 'Product Environmental Footprint Category Rules (PEFCR): apparel and footwear', available at [this link](#). Last accessed on 22 August 2025. It will set rules on the calculation and reporting of life cycle environmental impacts of apparel.

Aim

In this context, this report describes the preparatory study on textile products, which is hereinafter referred to as the PS. The PS aims to provide the scientific and technical basis for:

- the future development of the ecodesign requirements for textile products to be defined in a delegated act within the framework of the ESPR;
- the future development of the mandatory EU GPP requirements for textile products within the framework of the ESPR;
- the future revision of the EU Ecolabel criteria for textile products ⁽⁹⁾, within the framework of the ESPR and the EU Ecolabel Regulation.

Methodology

The PS will follow the structure prescribed by the Methodology for Ecodesign of Energy-related Products (MEErP) (MEErP, 2011). This methodology has over many years proved to be comprehensive and effective for preparing product-related legislation, and is therefore also deemed fit for non-energy-related products. The MEErP consists of seven tasks:

- **Task 1 on scope** defines the boundaries of the system under analysis. It identifies the textile products included in the scope of the PS. Additionally, it analyses legislative and voluntary frameworks at global, EU and Member State level.
- **Task 2 on market** investigates the market structure, trends and other market characteristics of the products in the scope.
- **Task 3 on user behaviour**. It describes how users relate to the products in the scope. It investigates:
 - (a) aspects influencing the design of the products, identifying potential barriers and restrictions to possible ecodesign measures, due to social, cultural or infrastructural factors;
 - (b) aspects influencing the modelling of environmental impacts and costs of the products in the life-cycle perspective, identifying behaviour not directly quantifiable with conventional standard tests on the products.
- **Task 4 on technologies** entails a general technical analysis of the products in the scope in order to describe the average products on the market, as well as the best available technologies (BAT) and the best not yet available technologies (BNAT).
- **Task 5 on environmental and economic analysis** of the average products on the EU market. It defines *base case* products, which are a conscious abstraction of the reality for practical reasons, and it represents a specific product category. The description of the base case is the synthesis of tasks 1 to 4 and the starting point for the following tasks. The base case is analysed via models based on Life Cycle Assessment (LCA) and Life Cycle Costing (LCC).
- **Task 6 on design options** investigates the monetary consequences of the design options in terms of cost within the life cycle for consumers and society. Environmental costs and benefits are also investigated using the Least Life Cycle Costs (LLCC) and the BAT. The BNAT indicate long-term possibilities.
- **Task 7 on possible policy scenarios** gathers the results of all previous tasks and investigates suitable policy means to achieve the potential improvements in the environmental impacts of the assessed products, as well as estimating economic impacts on consumers and the industry. Finally, it assesses the robustness of the outcomes via a sensitivity analysis.

The methodology used will be adapted to the specific features of the textile product group and the requirements set by the ESPR.

⁹ EU Ecolabel criteria for textile products. Commission Decision of 5 June 2014 establishing the ecological criteria for the award of the EU Ecolabel for textile products. Commission Decision (2014/350/EU). Available at [this link](#).

In addition to following the tasks set by the MEErP, the PS will include all elements of the Preliminary Report, to be used for the revision of the EU Ecolabel criteria for textile products ⁽⁹⁾, as set by the EU Ecolabel Regulation ⁽⁵⁾.

Due to their aims, future mandatory EU GPP requirements and future revised EU Ecolabel criteria of products included in the scope of the PS will be built up from the ecodesign requirements.

Involve ment of stakeholders

The development of the PS will include the direct involvement of stakeholders following the principles of the Sevilla process ⁽¹⁰⁾, which will enable the JRC to verify with stakeholders the work under development and collect additional evidence on the investigated topics.

Table 1 reports the stages of the project that require the participation of registered stakeholders. Any organisation, institution and citizen can register as a stakeholder by filling in a form on a dedicated webpage ⁽¹¹⁾. Stakeholders will be able to register at any stage of the development of the PS.

Table 1. Development of the preparatory study

Milestone	Topic addressed	Date
Initial questionnaire ⁽¹²⁾	Definitions, scope, market analysis, user behaviour, product aspects, EU Ecolabel, EU Green Public Procurement	30 March 2023 – 8 May 2023
1 st milestone	Scope, market, user behaviour, current EU Ecolabel criteria, current EU Green Public Procurement criteria	23 February to 22 April 2024
2 nd milestone	Technologies, framework and data gaps of environmental and economic model, and a questionnaire about substances and substances of concern	14 November 2024 to 17 March 2025
3 rd milestone	Analysis of base case and design options	16 December 2025 to 23 March 2026 (*)
4 th milestone	Policy scenarios, and elements to be included in the Digital Product Passport	To be communicated

The number and topics of the milestones could be modified to improve the involvement of the stakeholders during the development of the preparatory study.

(*) The 3rd milestone includes the following steps: (a) since 16 December 2025, registered stakeholders may read the working document of the 3rd milestone and its summary; (b) on 14 and 15 January 2026, registered stakeholders and JRC will attend the online consultation meeting; (c) from 16 January to 23 March 2026, registered stakeholders will be able to provide comments in writing to the working document via a dedicated questionnaire.

For each milestone, the stakeholder consultation will include the following steps:

- The JRC will communicate by email to all registered stakeholders the time and steps of the specific consultation, as well as the document(s) related to the milestone. All documents will be uploaded on the project's website ⁽¹³⁾.
- Registered stakeholders will read the document(s) before the online consultation meeting.
- JRC and registered stakeholders will attend the online consultation meeting. In this meeting, registered stakeholders will be able to have open discussions with the JRC and among themselves.
- Registered stakeholders will provide comments on the working document(s).
- The JRC will publish on the project's website ⁽¹³⁾ a document containing anonymised written comments submitted by the registered stakeholders.

All entities involved in the stakeholder consultations will have enough time to actively participate in the process.

¹⁰ With the 'Sevilla process', the Joint Research Centre of the European Commission works together with scientists, industry, non-governmental organisations (NGOs), national authorities and international institutions to take decisions that are underpinned by solid scientific and technical data, and that are based on consensus and transparency. The process is clearly structured within the framework of the Industrial Emissions Directive (2010/75/EU), and it is legally defined in Commission Implementing Decision 2012/119/EU available at [this link](#). More information is available at [this link](#). The process in the preparatory study on textiles is not legally defined, but it will follow the principles of the "Sevilla process".

¹¹ Stakeholders can participate in the development of the preparatory study on textile products by registering via this [web form](#).

¹² Initial questionnaire. Preparatory Study on textiles for product policy instruments – the initial questionnaire. Available at [this link](#).

¹³ The textile project website is available at [this link](#). Documents will be uploaded in the dedicated section available at [this link](#).

111
112 This version of the PS reports the 3rd milestone and it includes some of the information collected by the initial
113 questionnaire ⁽¹²⁾. Section 13.1 provides details about the previous consultations.

114

115 **Structure of the preparatory study – 2nd milestone**

116 After this introduction, Section 2 provides the definitions of words and expressions used in the document.

117 Section 3 defines the scope and describes the products included in it, providing also a first insight into their life-
118 cycle stages and main environmental burdens. This brief description provides the reader with a basic knowledge
119 useful to better understand the topics investigated in the following sections.

120 Section 4 analyses legislation and strategies at different levels: EU, Member State, third country, and global,
121 the latter describing initiatives of the United Nations. Section 4 is completed by the analysis of the most relevant
122 test methods, standards and environmental labels used internationally by the textile industry.

123 Section 5 analyses the market of the textile products belonging to the scope. The analysis is performed at
124 global, EU and Member State levels. Additionally, Section 5 investigates the market of the main elements
125 affecting the manufacturing of the products included in the scope. The market analysis includes some insights
126 into the market structure, most common business models, and it reports the main characteristics of the value
127 chain. A specific subsection is dedicated to market competitiveness, which is affected by the different costs
128 that companies face around the world due to the different requirements related to environmental protection.
129 Section 5 concludes with a subsection analysing the lifespan of products, because it largely affects the market
130 and the demand for new products.

131 Section 6 analyses user behaviour, which will be relevant for: (1) modelling the products in the economic and
132 environmental assessment, and (2) the analysis of possible ecodesign requirements within the product aspects
133 listed by Article 5 of the ESPR. The section addresses the behaviour before and after the purchase, as well as
134 behaviour related to the disposal of the textile product.

135 Section 7 analyses the current EU Ecolabel criteria for textile products in light of their revision. Facts and figures
136 of the current criteria are reported as well as the suggestions received via the initial questionnaire. The section
137 describes the relationship between the new mandatory ecodesign framework (ESPR) and the revision of the EU
138 Ecolabel criteria for textile products. To explore potential synergies with other officially recognised Ecolabels,
139 Section 7 provides a detailed comparison of the criteria for textile products set by the EU Ecolabel, Blue Angel
140 and Nordic Swan.

141 Section 8 collects information for the future development of mandatory EU GPP criteria. To this aim, this section
142 assesses the current public procurement of apparel in the EU, and the current status of the voluntary EU GPP
143 criteria.

144 Section 9 provides a general technical analysis of the products in the scope from the perspective of the relevant
145 product aspects listed by Article 5 of the ESPR.

146 Section 10 reports the environmental and economic assessment of products in the scope via a Life Cycle
147 Thinking approach and a specific assessment on pollution caused by fragmentation of textiles.

148 Based on the knowledge gathered in all previous sections, Section 11 analyses potential design options, which
149 could be implemented as requirements to the single products placed on the EU market.

150 Section 12 reports the analysis on chemical substances, identifying information and performance requirements.

2 Definitions

This section lists the definitions of terms and expressions used in the document. It will be complemented with more definitions as the project goes through its milestones (see Section 1).

Textile fibre

‘Textile fibre’ means either of the following ⁽¹⁴⁾:

- a unit of matter characterised by its flexibility, fineness and high ratio of length to maximum transverse dimension, which render it suitable for textile applications;
- a flexible strip or tube, of which the apparent width does not exceed 5 mm, including strips cut from wider strips or films, produced from the substances used for the manufacture of the fibres listed in Table 2 of Annex I to Regulation 1007/2011 and suitable for textile applications.

Textile product

‘Textile product’ means any raw, semi-worked, worked, semi-manufactured, manufactured, semi-made-up or made-up product which is exclusively composed of textile fibres, regardless of the mixing or assembly process employed, as well as a product containing at least 80% textile fibres by weight ⁽¹⁵⁾.

The definition of textile product is aligned with the definition provided by the Textile Labelling Regulation (TLR) ⁽¹⁴⁾.

Textile apparel

‘Textile apparel’ are textile products that aim to cover the body with everything except footwear. In addition, it also means a textile product worn as clothing or a clothing accessory by a person to clothe and/or adorn, and/or shield from, and/or feel comfortable with the outer environment and/or to express their personal and professional identity and/or belonging to a specific social group, with symbolic meanings and aesthetic values.

Technical textile

Technical textiles are textile products meeting technical rather than aesthetic criteria, even if, for certain markets like workwear or sports equipment, both types of criteria could be met. Technical textiles bring a functional answer to a wide range of specific requirements: lightness, resistance, reinforcement, filtration, fire retardancy, conductivity, insulation, flexibility, absorption and so on. The definition does not depend on the raw material, the fibre or the technology used, but on the expected inherent functionality of the product itself. Technical textiles can be used by professionals or not.

Textile apparel meeting the definition above is defined as *technical textile apparel*.

The definition of technical textiles is aligned with the definition provided by the European Economic and Social Committee ⁽¹⁶⁾.

Home/interior textiles

Textile products used indoor or outdoor to protect the object they are used on/with, decorate the home/interior environment, or dry the body or parts of the body after they have become wet.

Footwear

¹⁴ Textile Labelling Regulation. Regulation (EU) No 1007/2011 of the European Parliament and of the Council of 27 September 2011 on textile fibre names and related labelling and marking of the fibre composition of textile products and repealing Council Directive 73/44/EEC and Directives 96/73/EC and 2008/121/EC of the European Parliament and of the Council. Available at [this link](#).

¹⁵ In the preparatory study, weight and mass will be used interchangeably as it is done in the EU legislation, e.g. the Textile Labelling Regulation (Regulation (EU) No 1007/2011) and REACH (Regulation (EC) No 1907/2006). Nevertheless, the authors flag that from the technical perspective only the reference to mass should be made. The mass is the measure of the quantity of matter expressed in kilograms (kg). Differently, the weight is the force that the mass applies due to the acceleration of gravity. The weight is expressed in Newton (N). In the whole preparatory study, mass and weight are expressed in kg and refer to the quantity of matter.

¹⁶ CCMI/105, Technical textiles, Brussels, 17 April 2013, OPINION of the European Economic and Social Committee on Growth Driver Technical Textiles. (own initiative opinion). Rapporteur: Ms Butaud-Stubbs. Corapporteur: Ms Niestroy. Available at [this link](#). Last accessed on 30 March 2023.

'Footwear' means all articles with applied soles designed to protect or cover the foot, including parts marketed separately ⁽¹⁷⁾.

Textile waste

Textile waste is a textile product which the holder discards or intends or is required to discard ⁽¹⁸⁾.

Post-industrial textile waste

Post-industrial textile waste is textile waste generated during the manufacturing of textile products and their precursors (manufacturing of fibre, yarn and fabric, and during confectioning) (Huygens et al., 2023). This includes waste generated as a result of technical features of the manufacturing processes (e.g. cutting, roll ends), inaccurate resource planning (e.g. excess fabric) or production errors (e.g. defective or damaged fabric and products) (Aus et al., 2021). Moreover, it includes discarded finished products that the manufacturers do not send to their customers due to order change or cancellation.

Pre-consumer textile waste

Pre-consumer textile waste is textile waste generated at the retailer stage, as a result of discarding unsold textile products. This can be due to:

- overstock (products coming directly from the production line);
- obsolescence (products for which there is no longer a demand);
- damages (products that have been damaged in transit or storage);
- defects (products taken off the market due to quality issues).

Thus, this includes discarded products that have not been sold due to surplus stock, excess inventory and deadstock and products returned by a consumer on the basis of their right of withdrawal in accordance with Article 9 of Directive 2011/83/EU ⁽¹⁹⁾ or, where applicable, during any longer withdrawal period provided by the trader.

Post-consumer textile waste

Post-consumer textile waste is a textile product that have been discarded after consumption and use by the citizen or end-users of commercial and industrial activities (hotel, care, automotive, etc.). For this reason it is commonly referred to as household and commercial post-consumer textile waste, respectively (Huygens et al., 2023).

Final product

A final product is understood to be a product that is already suitable for users ⁽²⁰⁾.

Intermediate product

'Intermediate product' means a product that requires further manufacturing or transformation such as mixing, coating or assembling to make it suitable for end-users ⁽²¹⁾.

Table 2 reports the definitions of the 16 product aspects reported in article 5 of the ESPR. The hierarchy applied for selecting the most appropriate source for each definition was the following: the current legislation, standards and scientific literature. In some cases, the authors have adapted the definition from those sources.

¹⁷ Directive 94/11/EC of European Parliament and Council of 23 March 1994 on the approximation of the laws, regulations and administrative provisions of the Member States relating to labelling of the materials used in the main components of footwear for sale to the consumer. Available at [this link](#).

¹⁸ The definition of textile waste is inspired by the definition of waste reported by the Waste Framework Directive (WFD). Directive 2008/98/EC of the European Parliament and of the Council of 19 November 2008 on waste and repealing certain Directives (Text with EEA relevance). Available at [this link](#).

¹⁹ Directive 2011/83/EU on consumer rights. Available at [this link](#).

²⁰ The definition of final product is implied from the ESPR.

²¹ The definition of intermediate product is reported in Article 2(3) of the ESPR.

Product aspect	Definition	Source
Durability	The ability of a product to maintain over time its function and performance under specified conditions of use, maintenance and repair.	ESPR
Reliability	The probability that a product functions as required under given conditions for a given duration without an occurrence which results in a primary or secondary function of the product no longer being performed	ESPR
Reusability	Ability of a product or component that is not waste to be used again for the same purpose for which it was conceived.	JRC (adapted based on Directive 2008/98/EC definition for 'Re-use')
Upgradability	Ability of a product to be accessible for implementing actions to enhance its functionality, performance, capacity, safety or aesthetics.	JRC (adapted based on ESPR definition for 'Upgrading')
Repairability	Ability of a defective product or waste to return to a condition where it fulfils its intended purpose through one or more actions.	JRC (adapted based on ESPR definition for 'Repair')
Possibility of maintenance	Ability of a product to be kept in a condition where it is able to fulfil its intended purpose through one or more actions.	JRC (adapted based on ESPR definition for 'maintenance')
Possibility of refurbishment	Ability of a product or a discarded product to be prepared, cleaned, tested, serviced and, where necessary repaired to restore its performance or functionality within the intended use and range of performance originally conceived at the design stage at the time of the placing of the product on the market.	JRC (adapted based on ESPR definition for 'refurbishment')
Presence of substances of concern	The presence of substances that: - meet the criteria laid down in Article 57 of Regulation (EC) No 1907/2006 and are identified in accordance with Article 59(1) of that Regulation - are classified in Part 3 of Annex VI to Regulation (EC) No 1272/2008 in one of the hazard classes or hazard categories listed under Article 2(27)(b) of ESPR - are regulated under Regulation (EU) 2019/1021; or - negatively affect the reuse and recycling of materials in the product in which they are present	ESPR
Energy use and energy efficiency	Energy use: Use of energy in all lifecycle stages of a product. Energy efficiency: the ratio of output of performance, service, goods or energy to input of energy.	JRC Directive 2012/27/EU ⁽²²⁾
Water use and water efficiency	<u>Water use</u> : Use of water in all lifecycle stages of a product. <u>Water efficiency</u> : The creation of economic value with less use of water. In other words, minimization of the amount of water used in any of the lifecycle activities of a product.	JRC JRC
Resource use and resource efficiency	<u>Resource use</u> : Use of raw materials, mainly abiotic (minerals, metals, fossil fuels), in all lifecycle stages of a product. It can also include other biotic resources such as land, air, ecosystems. The use of natural resources can be accounted for as the volumes of resources consumed (materials) or used (land, air, ecosystems), or the impacts derived from the use of resources. Water and energy are resources addressed in specific product aspects. <u>Resource efficiency</u> : The ratio of output of performance, service, goods to input of resources, raw materials, air, land, soil and ecosystem services.	JRC and (BIO Intelligence Service, DG for Enterprise and Industry (EC), 2013)
Recycled content	Proportion, by mass, of recycled fibres, from post-industrial, pre- and post-consumer waste, in a textile product.	JRC (adapted based on ISO 14021)

²² Directive 2012/27/EU of the European Parliament and of the Council of 25 October 2012 on energy efficiency. Available at [this link](#).

Product aspect	Definition	Source
Possibility of remanufacturing	Possibility of producing a new product from objects that are waste, products or components and through which at least one change is made that may affect the performance, purpose or type of the product.	JRC (adapted based on ESPR definition for 'Remanufacturing')
Recyclability	Ability of product's waste material to be reprocessed into products, materials or substances whether for the original or other purposes. It includes the reprocessing of organic material but does not include energy recovery and the reprocessing into materials that are to be used as fuels or for backfilling operations.	JRC (adapted based on Directive 2008/98/EC definition for 'recycling')
Possibility of recovery of materials	Ability of the product waste generated to be recovered through any recovery operation, other than energy recovery and the reprocessing into materials that are to be used as fuels or other means to generate energy. It includes, inter alia, preparing for re-use, recycling and backfilling.	JRC (adapted based on Directive 2008/98/EC definition for 'material recovery')
Environmental impacts	Any change to the environment, whether adverse or beneficial, wholly or partially resulting from a product during its life cycle	ESPR
Expected generation of waste	Generation of any non-hazardous or hazardous substance, mixture, material or object along the whole life cycle of a product that the holder discards or intends or is required to discard.	JRC (adapted based on Directive 2008/98/EC definition for 'waste')

Source: own elaboration

Robustness

'Robustness' is the capability of a product to resist, i.e. maintain its physical structure and appearance, after undergoing external stresses, which could be of chemical or physical nature. The concept of robustness mainly differs from the physical durability because there is no reference to the product aging.

Product technologies

product technologies are products with defined characteristics related to a specific product aspect reported in Article 5(1) of the ESPR, such as durability, repairability, etc. in particular:

- The Base Case of technologies (BC) is the average product on the market.
- The Best Available Technologies (BAT) have the most ambitious performances available on the market. BAT are implemented at scale.
- The Best Not yet Available Technologies (BNAT) have the most ambitious performances, but they are not implemented at scale, therefore they are considered not available on the market.

3 Scope

This section selects and describes the products included in the scope of the PS. The first subsection (Section 3.1) reports the thought process followed to select the scope in accordance with the framework set by the ESPR. The second subsection (Section 3.2) lists the products included in the scope and specifies those that are excluded. The final subsection (Section 3.3) describes two important aspects of textile products that apply to the products in the scope, i.e. the life-cycle stages and their main factors influencing the negative environmental impacts. This last section is the essential basis for fully understanding a number of topics addressed in subsequent sections about environmental labels (Section 4.6), the market analysis (Section 5), and the analysis of product technologies (Section 9). Nevertheless, the topics mentioned in Section 3.3 will be detailed in the following milestones, when the PS addresses Task 5 of the MEeRP.

3.1 Selection of the scope

Within the framework of the European Industrial Strategy⁽²³⁾, the Annual Single Market Report 2021⁽²⁴⁾ identifies the products included in the **industrial ecosystem of textiles**: *‘transformation of natural (e.g. cotton, flax, wool), man-made and artificial (synthetic polyester and viscose) fibres into yarns and fabrics, production of yarns, home textiles, industrial filters, technical textiles, carpets and clothing. The ecosystem also includes production of footwear and leather.’*

In 2025, the EC communicated the first ESPR Working Plan⁽²⁵⁾, taking into account among other things a report⁽²⁶⁾ entitled ‘Ecodesign for Sustainable Products Regulation: Study on new product priorities’, hereinafter referred to as the **Product Priorities Study**. It served as the basis for an open public consultation⁽²⁷⁾. Among the many product groups in the scope of the ESPR, the Product Priorities Study assesses a product group named **‘textiles and footwear’**, which includes products belonging to the industrial ecosystem of textiles. Respondents to the open public consultation agreed to prioritise this product group (58% of respondents), and they considered it of highest priority (68% of respondents). 38% of respondents considered that apparel should be prioritised, followed by footwear (25%) and home/interior textiles (18%). The ESPR Working Plan reports “Textiles/Apparel” as product group to be addressed by the ESPR.

The scope of the PS was defined following the Product Priorities Study, and the selection criteria reported in Article 5(4), Article 5(7) and Article 18 of the ESPR.

Within the product group textiles and footwear, the Product Priorities Study identifies four main subgroups: (1) textile apparel, (2) home/interior textiles (e.g. bed linen, towels, tablecloths, curtains), (3) footwear, and (4) technical textiles not included in previous subgroups, usually or also meant for consumers, such as truck covers (tarpaulins) and cleaning products, or specifically meant for industry (automotive, construction, medical, agriculture, etc.). Although it is not specified in the Product Priorities Study, the subgroups textile apparel, home/interior textiles and technical textiles are understood to be products containing at least 80% by weight of textile fibres, as defined by the TLR⁽¹⁴⁾. This interpretation of subgroups is also adopted in the PS.

The product group textiles and footwear is considered to be too heterogeneous for the setting of common ecodesign requirements. This heterogeneity is driven by the specific functions and end uses of each subgroup, as reported in **Table 3**, as well as their material and chemical compositions. The specific function of a product requires specific tests to be performed on the product to verify its performance.

²³ European Industrial Strategy. Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions updating the 2020 New Industrial Strategy: Building a stronger Single Market for Europe’s recovery. COM(2021) 350 final. Available at [this link](#).

²⁴ Annual Single Market Report 2021. Accompanying the Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions Updating the 2020 New Industrial Strategy: Building a stronger Single Market for Europe’s recovery. SWD(2021) 351 final. Available at [this link](#).

²⁵ COMMUNICATION FROM THE COMMISSION Ecodesign for Sustainable Products and Energy Labelling Working Plan 2025-2030. COM/2025/187 final. Available at [this link](#).

²⁶ Ecodesign for Sustainable Products Regulation: Study on new product priorities. Available at [this link](#). Last accessed on 17 December 2024.

²⁷ New product priorities for Ecodesign for Sustainable Products. Available at [this link](#). Last accessed on 5 February 2024.

275 **Table 3.** Function and intended use of subgroups in the textiles and footwear product group

Subgroup	Function and intended use
Textile apparel	Textile apparel have the following functions and intended uses: (1) protect the human body from the outer environment, such as preventing heat loss, exposure to the sun, penetration of water, etc.); (2) comply with legal obligations related to public body exposure, because in EU countries nudity in public is forbidden; (3) self-representation, driven by expression of personal identity and/or expression of belonging to a specific social group, with symbolic meanings and aesthetic values; (4) social acceptance when gathering with other people, for example at parties, work places, religious ceremonies, etc. N.B. <i>Technical textile apparel</i> solves additional specific functions, which depend on the final application of the product (<i>Section 2</i>)
Home/interior textiles ^(a)	Most home/interior textiles have at least two functions: (1) protect the object they are used on/with, e.g. a cushion case protects the enveloped cushion, a tablecloth protects the table underneath, a curtain protects the indoor environment by filtering the light coming from outside, etc.; (2) decorate the home/interior environment. Products like towels have the function of drying the body or parts of the body after they have become wet (e.g. after washing, swimming, etc.).
Footwear ^(b)	Footwear has similar functions to apparel: (1) protect the feet from the outer environment; (2) self-representation, driven by expression of personal identity and/or expression of belonging to a specific social group, with symbolic meanings and aesthetic values.
Technical textiles	Technical textiles, which do not belong to the previous subgroups, have specific technical (non-aesthetical) functions according to their final application. Two examples are reported. Textile cleaning products have the function of removing dust and dirt from surfaces, interacting with detergents and other cleaning chemicals. Reusable textile absorbent hygiene products have the function of handling fluids, delivering protective properties like absorption, protecting the environment from unintentionally released body waste. In particular, reusable baby diapers must be flexible, so that they are comfortable and adaptable to changing body size.

^(a) Construction products, as defined in Regulation (EU) No 305/2011, are not considered home/interior textiles.

^(b) According to Directive 94/11/EC ⁽²⁸⁾, 'footwear' means all articles with applied soles designed to protect or cover the foot, including parts marketed separately. Due to the significant relative weight of soles in the overall weight of a footwear article and to the fact that soles are seldom composed of textile fibres, footwear products are seldom textile products.

The identified subgroups were further analysed following criteria reported in Article 5(4), Article 5(7) and Article 18 of the ESPR. Article 18 of the ESPR states the following:

When prioritising products to be covered by ecodesign requirements, the Commission shall analyse the potential contribution of those products to achieving Union climate, environmental and energy efficiency objectives, taking into account the following criteria:

- a) the potential for improving the product aspects without entailing disproportionate costs, [...];*
- b) the volume of sales and trade of the product within the Union;*
- c) the distribution across the value chain of the climate and environmental impacts, energy use, resource use and waste generation concerning those products;*
- d) the need to regularly review and adapt delegated acts adopted pursuant to Article 4 in light of technological and market developments.*

Article 5(4) of the ESPR states the following:

²⁸ Directive 94/11/EC of European Parliament and Council of 23 March 1994 on the approximation of the laws, regulations and administrative provisions of the Member States relating to labelling of the materials used in the main components of footwear for sale to the consumer. Available at [this link](#).

Ecodesign requirements shall be set for a specific product group⁽²⁹⁾. They may be differentiated for any specific product that belongs to that specific product group.

Article 5(7) of the ESPR states the following:

Where two or more product groups display one or more similarities allowing a product aspect to be effectively improved based on common information requirements or performance requirements⁽³⁰⁾, horizontal ecodesign requirements may be set for those product groups ('horizontal ecodesign requirements'). When considering whether to set horizontal ecodesign requirements, the Commission shall also take into account the positive effects of those requirements towards reaching the objectives of this Regulation, in particular the ability to cover a wide range of product groups in the same delegated act. The Commission may supplement the horizontal ecodesign requirements through the setting of ecodesign requirements for a specific product group.

For all subgroups of the textiles and footwear product group reported in the Product Priorities Study, the following subsections report information related to the three main scope selection criteria pointed out by Article 18: (a) potential for improvement (Section 3.1.1), (b) volume of sales and trade in the EU (Section 3.1.2), and (c) the distribution of the climate and environmental impacts, energy use, resource use and waste generation across the value chain (Section 3.1.3). Point (d) of Article 18 was not considered because there is no delegated act on textiles and footwear products to be reviewed.

Information related to the textiles and footwear sectors is reported with a granularity that changes according to the aims of the specific studies. The subsequent sections (from 3.1.2 to 3.1.3) try to interpret the available data in the most holistic way to meet the requirements set by Article 18 of the ESPR.

3.1.1 Potential improvement considering product aspects in Article 5(1) of the ESPR

The analysis reported in this section is only a first indicative investigation of potential improvements based on a literature review. Once the scope is defined, specific improvement potential will be assessed via environmental and economic assessments in the following stages of the PS. In particular, this will be performed in the following milestones addressing tasks 5 to 7 of the MEeRP (see Section 1).

Article 5(1) of the ESPR lists the following product aspects:

- durability;
- reliability;
- reusability;
- upgradability;
- repairability;
- the possibility of maintenance and refurbishment;
- the presence of substances of concern;
- energy use and energy efficiency;
- water use and water efficiency;
- resource use and resource efficiency;
- recycled content;
- the possibility of remanufacturing;
- recyclability;

²⁹ Article 2 of the ESPR establishes the following definition: 'product group' means a set of products that serve similar purposes and are similar in terms of use, or have similar functional properties, and are similar in terms of consumer perception.

³⁰ Article 2 of the ESPR establishes the following definitions:
'ecodesign requirement' means a performance requirement or an information requirement aimed at making a product, including processes taking place throughout the product's value chain, more environmentally sustainable;
'performance requirement' means a quantitative or non-quantitative requirement for or in relation to a product to achieve a certain performance level in relation to a product parameter referred to in Annex I;
'information requirement' means an obligation for a product to be accompanied by information as specified in Article 7(2).

- the possibility of the recovery of materials;
 - environmental impacts, including carbon footprint and environmental footprint;
 - expected generation of waste.
- Many studies underline the great potential for improvement regarding product aspects of textile apparel. In particular, the literature suggests six most important product aspects:
- Increasing all features of product **durability** (Ellen MacArthur Foundation, 2017; Bauer, Watson, Gylling, Remmen, Lysemose, Chatarina Hohenthal, et al., 2018; Sandin, Roos, Spak, et al., 2019; Claxton and Kent, 2020a; Goworek, L. Oxborrow, et al., 2020; Niinimäki et al., 2020; Cooper and Claxton, 2022a; Nordic Ecolabelling, 2022; OVAM, 2022; Horn et al., 2023; TAUW, 2023; Dodd and Gama, 2017; Aakko and Niinimäki, 2021).
 - Possibility of **maintenance** and how best to use the textile product during its lifetime (Ellen MacArthur Foundation, 2017; Bauer, Watson, Gylling, Remmen, Lysemose, Chatarina Hohenthal, et al., 2018; Nordic Ecolabelling, 2022; OVAM, 2022; Horn et al., 2023).
 - **Repairability** (Bauer, Watson, Gylling, Remmen, Lysemose, Chatarina Hohenthal, et al., 2018; Niinimäki et al., 2020; Botta, 2021; EEA, 2022a; OVAM, 2022; TAUW, 2023).
 - Regulating the **presence of substances of concern**, from the perspective of eliminating or reducing the use of these substances (Ellen MacArthur Foundation, 2017; Bauer, Watson, Gylling, Remmen, Lysemose, Chatarina Hohenthal, et al., 2018; Niinimäki et al., 2020; Mohapatra and Gaonkar, 2021; EEA, 2022b; Nordic Ecolabelling, 2022; OVAM, 2022; TAUW, 2023; Dodd and Gama, 2017).
 - **Recycled content and possibility of recovery of materials** (Ellen MacArthur Foundation, 2017; Bauer, Watson, Gylling, Remmen, Lysemose, Chatarina Hohenthal, et al., 2018; Niinimäki et al., 2020; ECOS et al., 2021; EEA, 2022b; Nordic Ecolabelling, 2022; OVAM, 2022; Horn et al., 2023; Dodd and Gama, 2017).
 - **Possibility of recycling** (Niinimäki et al., 2020; ECOS et al., 2021; EEA, 2022b; Nordic Ecolabelling, 2022; OVAM, 2022; Horn et al., 2023; TAUW, 2023; Dodd and Gama, 2017).
- Most of the studies listed above do not specify the kind of requirements, but they suggest focussing on the specific product aspect to best decrease the environmental impacts of the apparel:
- Significant improvement potential is associated with increasing the product lifespan, which could be achieved by increasing the physical and emotional durability⁽³¹⁾ of the product, improving product care, and facilitating/supporting reparability. Increasing the durability of products is considered an effective strategy to decrease the environmental impacts of this product group. It was estimated that, over the last 20 years, the use time of apparel decreased by 36%, with each product used only seven or eight times on average (Ellen MacArthur Foundation, 2017). This results in a significant amount of waste originating from used apparel. Extending the lifespan of apparel by an extra 9 months of active use would reduce carbon, water and waste footprints by around 4-10% each (WRAP, 2017a).
 - The literature reports a relevant concern about the hazards of substances used and contained in textile apparel. The use of alternative substances with a reduced hazard profile is considered a crucial aspect to pursue.
 - In line with the Circular Economy Action Plan, the literature considers product circularity aspects to show good improvement potential, such as increasing the recycled content and promoting recyclable products.
- Only Bauer et al. (2019), OVAM (2022) and TAUW (2023) further suggest specific criteria that would decrease the environmental footprint of textile apparel. Differently, (Sandin, Roos, Spak, et al., 2019) reports the possibility to decrease the environmental impacts of these products using a less impacting source of energy, such as solar energy.

³¹ Emotional durability is the product's ability to stay relevant and desirable to the consumer (Anthesis, 2015).

Most of the product aspects listed above could also be valid for home/interior textiles and footwear, which are usually included in studies focusing on generic textile products. This is the case of studies like Ellen Macarthur Foundation (2017), ECOS (2021), and Mohapatra and Gaonkar (2021). However, any ecodesign requirement under a specific product aspect should always consider the specific function that the textile product performs. Therefore, the potential adoption of the same product aspects for textile apparel and home/interior textiles should consider that the two subgroups, and their categories, perform different functions. This distinction is one of the numerous factors driving the grouping of textile products that are included in the scope of the PS.

The literature review did not reveal specific studies about potential improvements to generic technical textiles. This could be explained by the numerous specific applications of the technical textiles that are adopted in many sectors.

The literature review showed that textile apparel, but also to some extent home/interior textiles and footwear, have the potential for improvement under various product aspects. In the following milestones, the PS will assess all the product aspects set out in Article 5(1) of the ESPR.

3.1.2 Amount of sales and trade in the EU

Information reported in the PRODCOM database⁽³²⁾ was processed and analysed to quantify the amount of sales and trade in the EU in the textiles and footwear subgroups. The classification of products into subgroups was based on the European Statistical System, which is described in detail in Section 13.1.

The analysis targeted the market indicators production, import, export and apparent consumption⁽³³⁾ of the EU-27⁽³⁴⁾ in 2019. This year was chosen because it was the latest year free from market disruptions, such as the COVID-19 pandemic (2020) and the increase of EU energy prices (2022). **Table 4** reports the results of the analysis, which show that the textile apparel subgroup has the largest share of the market for all market indicators expressed in value and the largest share of the import and apparent consumption expressed in mass.

Table 4. Economic indicators of textiles and footwear in the EU-27 in 2019

Subgroup	Production		Import		Export		Apparent consumption	
	Mass (bn kg)	Value (bn EUR)	Mass (bn kg)	Value (bn EUR)	Mass (bn kg)	Value (bn EUR)	Mass (bn kg)	Value (bn EUR)
Textile apparel	0.8	33	4.67	80.84	0.62	36.02	4.85	77.82
Home/interior textiles	1.83	14.87	2.04	12.09	0.59	4.9	3.28	22.07
Footwear	0.57	19.83	1.72	21.27	0.28	14.46	2.01	26.64
Technical textiles	3.65	18.07	0.69	4.03	0.79	6.24	3.55	15.86

N.B. Colour shades from red to white rank the subgroups from the largest to the smallest share of the economic indicator.

Figures related to economic indicators expressed in mass could be underestimated because some PRODCOM codes did not contain information. This is caused by the data collection system that allows Member States to avoid, in specific cases, reporting information expressed as quantity of goods. More information is available in Section 13.1.

Source: own analysis using PRODCOM dataset DS-056120 (data extracted on 15 November 2023)

3.1.3 Environmental impacts and waste generation across the value chain

The literature shows that environmental impacts caused by textile products were studied using numerous levels of granularity. Most of the studies focused on 'textiles', 'textile sector', 'textile industry', without clearly specifying which subgroup was included. In this screening of the literature, the term 'textiles' is assumed to include textile apparel, home/interior textiles and technical textiles.

The literature seems to treat the terms 'apparel', 'clothing' and 'garments' as synonyms, which raises doubts on whether clothing accessories are always included; whereas it addresses footwear separately. The same literature refers to 'fashion', which forms a major component of the product group 'apparel and footwear', and

³² PRODCOM database (Sold production, exports and imports – DS-056120). Available at [this link](#).

³³ The apparent consumption is defined as the sum of production and import, minus the export. Production, import and export refer to the entire EU. Therefore, the economic indicators used from the PRODCOM database were: PRODQNT and PRODVAL, which describe production with mass-related units and in euro, respectively; IMPQNT and IMPVAL, which describe import with mass-related units and in euro, respectively; EXPQNT and EXPVAL, which describe export with mass-related units and in euro, respectively.

³⁴ The 27 Member States are: Austria, Belgium, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Poland, Portugal, Romania, Slovakia, Slovenia, Spain and Sweden.

it can be understood as including the subgroups textile apparel and footwear. The authors understand that fashion includes leather and fur apparel, which are not part of this PS in view of its focus on textile products according to the definition under the TLR.

Table 5 reports the result of a literature review focusing on specific subgroups of the textile and footwear product group, or textiles in general. For the home/interior textiles, footwear and technical textiles subgroups only a few articles were found, showing the very limited attention paid by the scientific community to these subgroups. Most of the literature focuses on textile apparel and textiles in general.

Table 5. Literature studies investigating environmental impacts of textile products

Granularity of textile products in the literature	Reports and scientific papers	Number of studies
Textile apparel	Cai and Choi (2020) reviewed 108 studies; Munasinghe et al. (2021) reviewed 57 studies.	108
Home/interior textiles	Kalliala and Nousiainen (1999).	1
Footwear	Gottfridsson and Zhang (2015); Quantis (2018); Jadhav and Jadhav (2020); Muthu (2020); Van Rensburg et al. (2020).	5
Technical textiles	Shishoo (1994); Sohail and Sun (2019); Aldalbahi et al. (2021).	3
Textiles in general	Amicarelli et al. (2022) reviewed 54 studies; Luo et al. (2023a) reviewed 107 studies.	107

Different literature reviews can refer to the same studies to analyse them from specific perspectives.

Source: own analysis

In 2022, the European Environment Agency published a study analysing the environmental impacts of textile apparel, home/interior textiles and footwear (EEA, 2022b). **Table 6** reports an overview of the environmental impacts related to the upstream supply chain of textile apparel, home/interior textiles and footwear. Most of the impacts generated by the subgroups investigated occur outside the EU and are due to textile apparel production. The contributions of footwear and home/interior textiles are very similar to each other.

Table 6. Environmental impacts related to the supply chain of textile products

Environmental aspect	Amount in 2020	Attribution of impacts (%)		Attribution to specific subgroup (%)		
		Inside EU	Outside EU	Apparel textiles	Home/interior textiles	Footwear
Use of primary raw materials	175 m t 391 kg/person	20	80	40	30	30
Water use	5 000 m m ³ 9 m ³ /person	Contribution not relevant	Most of the contribution	40	30	30
Land Use	180 000 km ² 400 m ² /person	8	92	43	23	35
Greenhouse gas emissions	121 m t CO ₂ eq 270 kg CO ₂ eq/person	25	75	50	30	20

Source: EEA (2022)

Huygens et al. (2023) performed an analysis of waste generation along the value chain of textile products including textile apparel, home/interior textiles and technical textiles. The analysis does not show which product subgroup generates more waste, but it distinguishes between the generation of post-industrial, pre-consumer and post-consumer waste. In 2019, the EU-27 generated 12.6 Mt of textile waste; 11% was post-industrial waste, 3% was pre-consumer waste, and 86% was post-consumer waste (Huygens et al., 2023).

The review of the available literature on the environmental impacts caused by textile products showed that textile apparel is the most impactful and the most investigated subgroup in the textiles and footwear product group.

3.2 Proposed scope

All products in the scope of the PS should be sufficiently homogeneous in terms of function, material composition, chemical composition and technologies used. This homogeneity should allow the development of common ecodesign requirements within the aspects listed in Article 5 of the ESPR, tailoring them to product subgroups as appropriate.

3.2.1 Products included in the scope

The analysis reported in Section 3.1 revealed that, within the product group textiles and footwear, textile apparel is the most suitable subgroup to be addressed by the PS because it:

- has potential improvements already investigated by the literature (Section 3.1.1);
- has the largest share in the EU market (Section 3.1.2),
- produces the largest share of the environmental impacts, based on the available literature (Section 3.1.3).

Table 7 reports the initial textile apparel categories and their description. **Table 115** (Section 13.2.4) reports all PRODCOM codes of products included in the scope of the PS. The identified product categories are aligned with the PEFCR A&F ⁽⁸⁾. This alignment allows a description of the products included in the scope. The PS includes the work performed by the project PEFCR A&F whenever it is appropriate within the frameworks of the ESPR and the EU Ecolabel.

Table 7. Initial product categories of textile apparel included in the scope of the preparatory study

ID	Category	Description
01	T-shirts	Garment to cover the upper body to the elbow (e.g. singlets, vests, t-shirts, polo shirts, other short-sleeved shirts)
02	Shirts and blouses	Garment to cover the upper body including the entire arm (e.g. long-sleeved shirts, blouses, base layers)
03	Sweaters and mid-layers	Garment to keep the upper body warm and covered (e.g. pullovers, cardigans, hoodies, jerseys, sweatshirts, sweaters)
04	Jackets and coats	Garments to put on top of a shirt or sweater or to protect from the natural elements (e.g. blazers, suit jackets, overcoats, other light jackets, rain jackets, outdoor winter jackets, parkas, outdoor vests, anoraks)
05	Pants and shorts	Garment to cover the lower body, may protect from the elements (e.g. casual pants, outdoor pants, dress pants, jeans, sports pants, capri pants, shorts)
06	Dresses, skirts and jumpsuits	One-piece garment that covers both the upper and lower body, or the lower body only, other than pants and shorts (e.g. short- and long-sleeved, strapless, wrap, long and short, one-piece suits)
07	Leggings, stockings, tights and socks	Tight garment to cover the legs and/or feet. (e.g. opaque and sheer tights, pantyhose, fishnets, ankle socks, knee socks, low-cut socks)
08	Underwear	Garment worn under clothes, often next to the skin of the upper or lower body (e.g. boxers, briefs, panties, bras, body-shaping suits)
09	Swimwear	Garment worn for water-based or sun-based activities (e.g. bikinis, bathing suits, racing-style swimwear, board shorts)
10	Textile apparel accessories	Hats – Garment to cover the head for warmth or as a fashion item (e.g. caps, flat caps, woollen hats/beanies, fedoras, panamas, bowlers, newsboys, berets); Scarves and ties – Garment worn around the neck for warmth or as a fashion item (e.g. warm and light scarves, buffs, neckerchiefs, headscarves, shawls, bowties); Belts – Flexible band or strap worn around the waist or over the shoulders used to secure or to hold up clothing such as pants (e.g. dress belts, casual belts, buckle belts, tie-up belts, suspenders); Gloves and mittens – Articles of clothing that protect hands and wrists from the elements or as a fashion item. Used in pairs (e.g. fingerless gloves, fashion gloves, outdoor sports gloves, mittens).

Source: own production based on the ongoing work performed within the development of PEFCR A&F.

Technical textile apparel, such as workwear and sportswear are included in the scope of the PS, as long as (1) it is textile apparel (containing at least 80% by weight of textile fibres, as defined above) and (2) they do not belong to the list of excluded products defined in section 3.2.2.

Sportswear could be considered technical textiles due to its high performance in terms of thermoregulatory properties ⁽³⁵⁾. Besides these thermoregulatory properties, sportswear and leisurewear have the same, or very

³⁵ The textile industry certifies or reports the performance of sportswear following the guidelines contained in the technical report PD CEN/TR 16422:2012, available at [this link](#). Last accessed on 5 February 2024. These guidelines establish an evaluation system with three levels of performance related to thermoregulatory properties: (a) thermal insulation, (b) water vapour transmission (breathability), (c) air permeability, (d) water penetration resistance and repellence, and (e) liquid sweat management.

similar, features that allow these two types of products to be addressed in the same PS with the same ecodesign requirements, which fall in the domain of the aspects listed by Article 5 of the ESPR.

Workwear textile apparel not excluded from scope (as per section 3.2.2) ⁽³⁶⁾ and leisurewear also have same, or very similar, features that allow these two types of products to be addressed in the same PS with the same ecodesign requirements, which fall in the domain of the aspects listed by article 5 of the ESPR.

Nevertheless, this approach on workwear and sportswear will be reassessed when the PS addresses task 6 of the MEErP on ecodesign options.

3.2.2 Products excluded from the scope

The following textile apparel are excluded from the scope because they are very different to products included in the scope in terms of their function and physical characteristics:

- **smart textiles** ⁽³⁷⁾, which are textiles able to sense and react to environmental conditions and external stimuli (e.g. mechanical, thermal, and chemical stimuli) thanks to a number of sensors incorporated in the textiles;
- **electronic textiles** or **e-textiles** ⁽³⁷⁾, which are textile-based systems that exhibit an intended and exploitable response as a reaction either to changes in their surroundings/environment or to an external signal/input ⁽³⁸⁾;
- **textile apparel identified as personal protective equipment (PPE)** in accordance with Regulation (EU) 2016/425 ⁽³⁹⁾;
- **textile apparel identified as medical devices or as an accessory for a medical device** in accordance with Regulation (EU) 2017/745 ⁽⁴⁰⁾;
- **textile apparel identified as toys** in accordance with the Directive 2009/48/EC ⁽⁴¹⁾.

E-textiles are excluded from the scope of the PS, however all products included in the scope will be able to use information carriers based on technologies such as Ultra High Frequency (UHF), Radio-frequency identification (RFID) and Near-Field Communication (NFC).

Additionally, all **intermediate products**, such as fabrics, yarns, fibres, etc., are excluded from the scope of the PS, because their characteristics should be strictly related to the function of the specific textile apparel they are part of. For example, a cotton fabric should have different characteristics when used as a t-shirt component to when used as a table cloth. Exclusion of intermediate textile products from the scope of this PS does not exclude the possibility to set specific requirements and parameters on the intermediate textile product once incorporated in the selected products in the scope. This means that all intermediate textile products, which are part of textile apparel in the scope, are indirectly addressed by the PS.

Aspects related to customised textile apparel and upcycled textile apparel ⁽⁴²⁾ will be addressed in the impact assessment that will follow the PS in the policy-making process.

In accordance with Article 5(2) of the ESPR, the initial questionnaire ⁽¹²⁾ investigated the possibility to include in the scope further textile products, such as bed linen, kitchen textiles, towels and bathrobes, textile cleaning

³⁶ Examples of workwear textile apparel included in the scope are uniforms used in hospitals, hotels and restaurants.

³⁷ Definition of smart textiles is inspired by ISO/TR 23383:2020. Textiles and textile products — Smart (Intelligent) textiles — Definitions, categorisation, applications and standardization needs. Available at [this link](#). Last accessed on 5 February 2024.

³⁸ Textile apparel containing batteries to produce lights and/or sounds are excluded from the scope of this preparatory study.

³⁹ Personal protective equipment (PPE) Regulation. REGULATION (EU) 2016/425 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 9 March 2016 on personal protective equipment and repealing Council Directive 89/686/EEC. Available at [this link](#).

⁴⁰ Medical devices Regulation. Regulation (EU) 2017/745 of the European Parliament and of the Council of 5 April 2017 on medical devices, amending Directive 2001/83/EC, Regulation (EC) No 178/2002 and Regulation (EC) No 1223/2009 and repealing Council Directives 90/385/EEC and 93/42/EEC. Available at [this link](#).

⁴¹ Toy Safety Directive (TSD). Directive 2009/48/EC of the European Parliament and of the Council of 18 June 2009 on the safety of toys. Available at [this link](#).

⁴² Upcycled textiles meant here as textile products manufactured by making use of used and/or waste components of other textile products.

products, reusable textile hygiene products. The very different functions of these textile products⁽⁴³⁾ compared to textile apparel does not allow the extension of the scope of the PS, because they cannot be considered similar to apparel. A product category with a different function to textile apparel requires a specific study which includes the investigation of the user behaviour, the testing methods to check the performance of the products, and consequently a different approach when establishing ecodesign requirements. Additionally, a different function of the textile product affects the fate of its end-of-life. For example, during their use phase, textile cleaning products interact with many chemicals that are absorbed and could hinder the recycling processes. These aspects deserve to be addressed by a specific study different to this PS. In particular, due to their functions and technical characteristics, textile cleaning products and reusable textile absorbent hygiene products are considered technical textiles.

3.3 Composition, life-cycle stages and main negative environmental impacts of products included in the scope

This section provides a literature review concerning the main characteristics of the textile industry in terms of composition and life-cycle stages (Section 3.3.1), and the main factors influencing the negative environmental impacts (Section 3.3.2). Implicitly, this section refers to textile apparel, which represent a major part of the textile industry production, import, export and apparent consumption (**Table 4**). An overview of the life-cycle stages of textile apparel and the main factors affecting the environmental impacts are essential bases to fully understand numerous topics addressed in the following sections. The main environmental impacts are crucial to understand the environmental areas addressed by environmental labels described in Section 4.6. Meanwhile, the description of the life-cycle stages is very important to put in context specific concepts addressed in the market analysis (Section 5), notably (1) manufacturing stages spread over different countries, (2) the main elements affecting apparel manufacture, (3) the characteristics of the value chain, and (4) global competitiveness related to costs of environmental compliancy.

The description reported in this section is further detailed in Section 9 and the following milestone, when the PS addresses Task 5 of the MEerP.

3.3.1 Composition and life-cycle stages

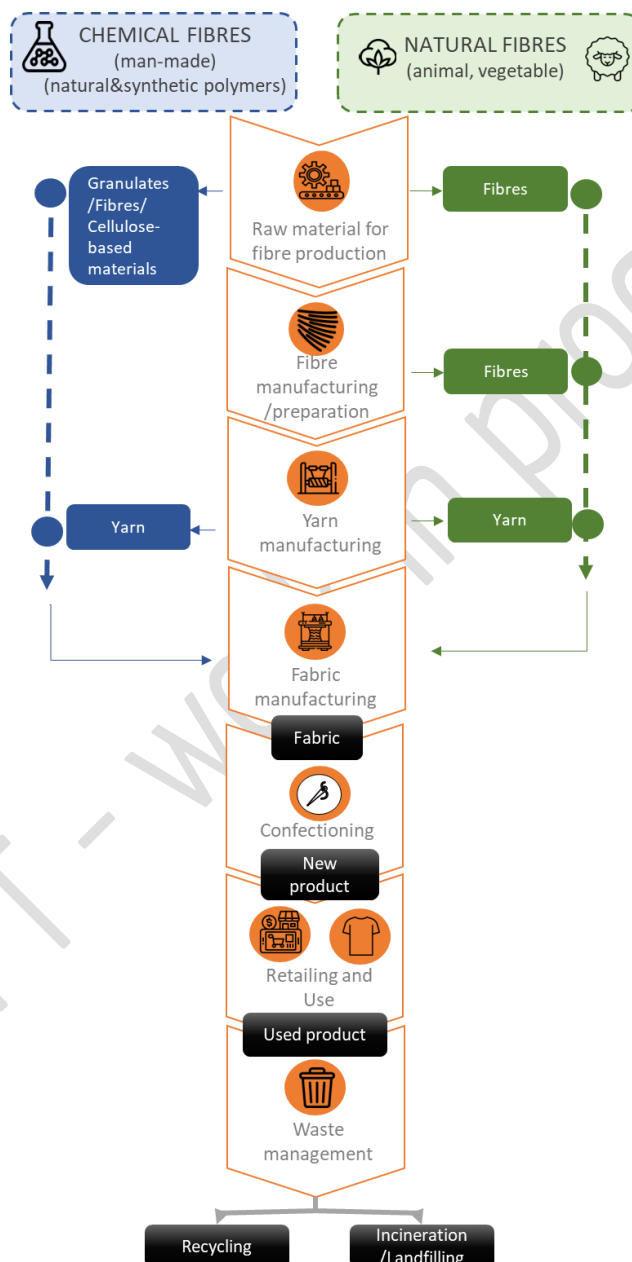
The textile products included in the scope of the PS are mostly made of textile fibres – at least 80% by weight. The Textile Labelling Regulation 1007/2011 lists 50 different types of fibres, but not all of them are commonly used for manufacturing textile apparel (e.g. carbon, ceramic, glass, and metal fibres). Textile fibres most commonly used by the textile apparel industry were classified in the Best Available Techniques (BAT) Reference Document (BREF) on textiles (Roth et al., 2023) as follows:

- Natural origin fibres:
 - animal origin, including wool, silk and hair;
 - vegetable origin, including cotton, flax and jute.
- Chemical fibres (man-made):
 - natural polymer fibres / man-made cellulosic fibres (MMCF), including viscose, cupro, lyocell, acetate, triacetate;
 - synthetic polymer fibres, specifically organic polymers, including polyester (PES), polyamide (PA), acrylic (PAC), polypropylene (PP), elastane (EL).

⁴³ *Bed linens* (bed sheets, pillowcases, duvet/blanket cover) have the function to cover the human body during sleep, and they have a decorative function for the bed.
Kitchen textiles (tablecloths, kitchen tea towels, napkins, aprons) have the function of protecting the surfaces they cover and have a decorative function for an interior environment.
Towels and bathrobes have the function of drying the body or parts of the body after they have become wet (e.g. after washing, swimming, etc.).
Textile cleaning products (floor cloths, dishcloths, dusters and similar cleaning cloths) have the function to remove dust and dirt from surfaces, interacting with detergents and other cleaning chemicals.
Reusable textile absorbent hygiene products have the function of handling fluids, delivering protective properties like absorption, protecting the environment from unintentionally released body waste. In particular, reusable baby diapers must be flexible, so that they are comfortable and adaptable to changing body size.

Despite the fact that textile apparel is produced with specific fibres and fibre combinations, technologies and resources, the life cycle of textile apparel can be described by eight distinct stages: (1) raw material for fibre production, (2) fibre manufacturing/preparation, (3) yarn manufacturing, (4) fabric manufacturing, (5) confectioning (often called 'Cut, Make-up and Trim' (CMT)), (6) retailing, (7) use, and (8) waste management. **Figure 1** shows these stages in a linear model that does not mark recycling routes, and it depicts some peculiarities of chemical and natural fibres. **Table 8** reports processes, technologies and resources used at each life-cycle stage of textile apparel. Section 13.3 provides more details.

Figure 1. Main textile life-cycle stages



Source: own production adapted from McKinsey & Company (2022), icons from www.flaticon.com

Table 8. Life-cycle stages of textile apparel – processes, techniques and resources

Stage	Input	Process and techniques	Main resources	Output
(1) Raw material for fibre production	Fibres from crops is used for vegetable origin fibres	Cultivation with subsequent processing.	Land Water Agrochemicals	Staple fibre

Stage	Input	Process and techniques	Main resources	Output
	Animals and insects are used for animal-based fibres	Animal farming with shearing and sericulture with silk extraction. Pre-treatments are needed.	Energy Chemicals Land Fossil fuels	
	Cellulose mainly from wood is used for man-made chemical fibres using natural polymers	Dissolving pulp from various sources is dissolved with chemicals and further processed.		Staple fibre, Filament fibres
	Petroleum-based material is used for man-made chemical fibres using synthetic polymers	Specific chemicals and processes are used to produce specific polymers.		Filament fibres, Granulates
(2) Fibre preparation / manufacturing	Natural staple fibre	Preparation stage usually includes scouring and/or cleaning processes before carding. The processes vary according to the natural fibre.	Energy Water Chemicals	Carded natural fibres
	Staple fibre bale of MMCF	Opening, carding, drawing and additional specific processes according to the spinning technology used.	Energy	Carded MMCF
	Chemical granulates	No process required.	None	Granulates
(3) Yarn manufacturing	Carded natural fibres	Ring spinning and open-end spinning .	Energy Water Chemicals	Yarns
	Carded MMCF	Ring spinning, air-jet spinning, and open-end spinning.		
	Polymers in solution or as granulates (for chemical fibres)	Wet or dry spinning from a solution and melt spinning from granulates		
(4) Fabric manufacturing	Yarn (for weaving and knitting) Fibres and filaments (for non-woven)	A two-dimensional structure is created by interlacing yarns. Weaving is the predominant fabric manufacturing technique. It involves interlacing two sets of yarns at right angles on a loom. Knitting is the second most used fabric manufacturing method. Yarns are interlooped using needles to form fabric on knitting machines. Non-woven technology produces textile structures by bonding fibres or filaments together, either mechanically, thermally, or chemically.	Energy	Fabrics
Finishing processes	Yarn and/or Fabric	Numerous wet treatments for sizing, desizing, pre-treatment, dyeing, printing, finishing, coating, laminating, and many more. These treatments are applied based on the specific requirements of the final products, and are not applied sequentially. Each treatment utilises specific technologies and chemicals.	Chemicals Energy Water (dyeing)	Finished yarn, Finished fabric
(5) Confectioning	Fabric Non textile components	Product design, fabric spreading and cutting, product assembly, sewing and ironing.	Energy Manual labour	Textile apparel
(6) Retailing	Textile apparel	Transportation and sale via numerous channels, such as shops, on the internet.	Fuel (transport) Energy Manual labour	Textile apparel owned by the user
(7) Use	Textile apparel owned by the user	Washing, cleaning, drying, ironing, as well as wear and tear.	Water Energy Chemicals	Used textile apparel

Stage	Input	Process and techniques	Main resources	Output
(8) Waste management	Used textile apparel	Used textile apparel are usually incinerated and landfilled. However, circular solutions are also possible, such as: textile apparel reuse, fabric recycling, fibre recycling (mechanical process), raw material recycling (chemical process).	Specific to the fate of the used product	Specific to the fate of the used product

Source: own elaboration

3.3.2 Main factors influencing the negative environmental impacts

This section aims to provide a general overview about the factors that negatively affect the environment due to the consumption of textile apparel, or textiles in general. This is done via a literature review to provide some background useful for the understanding of the environmental areas addressed by environmental labels described in Section 4.6. A complete environmental assessment will be performed in Task 5 and reported in the following milestone.

The textile production processes are characterised by a large use of resources and numerous emissions into water bodies and into the atmosphere. The European Environment Agency estimated that the textile industry is the fifth industrial sector for primary use of materials and greenhouse gas emissions, and the third industrial sector for water and land use (EEA, 2022b).

Use of land

The European Environment Agency estimated that in 2020 the purchase of apparel by European households used about 77 400 km² of land (EEA, 2022b). About 92% of this land is located outside Europe, and it is mainly used for cultivation of cotton in China and India (Manshoven et al., 2019). The grazing of sheep and cashmere goats for the production of animal fibres such as wool require also use of land. In this case the attribution of environmental impacts to the fibre should take into account that the same animals are used to produce meat, milk and leather. The United Nation Environment Program estimates that the use of land in cellulose supply for the production of MMCF and the production of synthetic fibres is smaller than what is requested by natural fibres (UNEP, 2020).

In general, use of land and land use change are relevant because they influence biodiversity and the emission of greenhouse gases (Newbold et al., 2016; IPCC, 2023).

Use and discharge into water

The textile industry's water consumption is estimated to account for 4% of global freshwater extraction (Ellen MacArthur Foundation, 2017). The European Environment Agency reported that the production of clothing, footwear and household textiles purchased in the EU in 2020 was equal to 4 000 million m³ of blue water⁽⁴⁴⁾, meaning about 9 m³ per person (EEA, 2022b). Water is used in numerous stages of the entire life cycle of textile apparel (**Table 8**). The BREF for the EU textile industry sets ambitious thresholds for emissions into water due to their highly negative effects. Numerous parameters are considered including carbon-load-related parameters (chemical oxygen demand (COD), biological oxygen demand (BOD), total organic carbon (TOC), suspended solids, nitrogen and phosphorus compounds, metals, adsorbable organic halides (AOX), pesticide, flame retardants and many more. Use of dyes generates highly polluted wastewater, which needs to be properly treated before its release into the environment. Also, other processes like production of polyester fibres and the treatment of knitted fabrics generate wastewater with a high load of heavy metals, such as antimony, and with a high hydrocarbon oil index (HOI), respectively (Roth et al., 2023). Additionally, when treating natural fibres, a series of substances, such as biocides, are released into the wastewater even before pre-treating the fibres (Zhang et al., 2022; Roth et al., 2023).

Use of chemicals

Numerous life-cycle stages of textile apparel involve the use of chemical substances and mixtures (hereafter called 'chemicals') (**Table 8**). Some of them are pesticides, solvents, surfactants, dyes and pigments, water and

⁽⁴⁴⁾ 'Blue' water is reported as 'surface water or groundwater that is consumed or evaporated during irrigation, industry processes or household use' in (EEA, 2022b).

stain repellents, flame retardants, biocides and many more (Ellen MacArthur Foundation, 2017). Some of these chemicals are directly in contact with soil and water bodies, like during the cultivation of natural fibres using fertilisers and pesticides. Other chemicals are usually dissolved and released in the wastewater produced in several processes (Zhang et al., 2022; Roth et al., 2023).

Humans are also directly exposed to chemicals used in textile apparel. Farmers and factory workers face exposure to these chemicals, and people wearing textile products are potentially at risk due to chemicals that remain bound to the products placed on the market (Ellen MacArthur Foundation, 2017).

Use of energy

Almost all life-cycle stages of the textile apparel have a significant energy consumption (**Table 8**). In particular, some processes are highly energy-demanding, like the extraction of raw materials, specifically in the cases of fibres like silk or synthetic fibres, and production processes such as spinning, knitting, and weaving. Furthermore, thermal treatments are crucial during production, as well as in the use phase for washing, ironing and drying (Niinimäki et al., 2020; Munasinghe et al., 2021).

Generation of waste

In 2019, the EU generated about 12.6 Mt of textile waste, including post-industrial, pre-consumer and post-consumer waste, representing 11%, 3% and 86% of the total, respectively. Nevertheless, regarding waste generation, it is not possible to distinguish the impact of textile apparel from other textile products (Huygens et al., 2023).

Huygens et al. (2023) estimated that, in 2019, about 90% of post-industrial and pre-consumer waste was incinerated and landfilled, and about 10% was recycled. Additionally, in the same year, about 8.5 Mt of post-consumer waste was not separately collected and was sent to incinerators or landfilled. The majority of post-consumer waste that is separately collected is sent outside the EU as 'used product', producing negative environmental and social impacts (Lingås et al., 2023; Huygens et al., 2023).

The landfilling and incineration of waste generates pollution, especially if it is not well controlled (Christensen, 2010). The Ellen MacArthur Foundation estimates that about 2 000 tonnes of hazardous substances annually are released in the EU due to degradation of textiles in landfills (Ellen MacArthur Foundation, 2017).

Emissions into the atmosphere

The apparel industry is responsible for about 6.5% of global greenhouse gas emissions. This mainly depends on the use of energy and the sources employed for its production (Niinimäki et al., 2020). The European Environment Agency estimated that textile consumption in the EU in 2020 emitted 121 million tonnes of greenhouse gases. Around 75% of the emissions occurred outside Europe, specifically in Asian countries (EEA, 2022b).

The BREF for the EU textile industry sets ambitious thresholds for emissions into the atmosphere due to their highly negative effects. Numerous parameters are considered including volatile organic compounds, formaldehyde, oil mist, dust, ammonia, carbon monoxide, sulphur oxides and many more (Roth et al., 2023). These compounds are generated in processes associated with thermal treatments, wet processes (coating, laminating and other finishing), and the use of agents and carriers in the production phase (Roth et al., 2023).

Pollution originated from textile fragmentation

Due to the publication of recent studies highlighting the adverse environmental and health impacts of microplastics, the release of synthetic fragmented fibres from textile products is under the spotlight (Amobonye, Bhagwat, Raveendran, et al., 2021; Leslie et al., 2022). One of the leading sources of microplastics pollution is the fragmentation of synthetic textiles (Boucher and Friot, 2017). Fibres are released throughout the textile value chain, from the production phase to the end-of-use phase. During the manufacturing stages, microplastics are released into the atmosphere and into the wastewater produced. Household laundering during the use phase is another leading source of microplastics release into wastewater. During the end-of-use phase, landfilling, incineration and recycling are believed to be an important source of airborne and terrestrial microplastics (UNEP, 2020). Estimations of microplastics releases exhibit considerable uncertainty, with annual figures for the EU oscillating between 1 649 tonnes and 61 078 tonnes (DG ENV, 2023). This uncertainty is primarily attributable to the scarce data available regarding the production and use phases of the microplastics life cycle, and a complete lack of information for the disposal stage. The imprecision comes from the challenge of quantifying microplastics emissions, rather than any issues with the underlying assumptions of the baseline projection (DG

ENV, 2023). Current patterns indicate that emissions of microplastics from textiles are projected to rise by approximately 22% by the year 2030 (DG ENV, 2023).

Once released into the environment, microplastics can be ingested by organisms, leading to problems due to physical and chemical properties of the microplastics. Their small size and persistence make them difficult to remove from the environment, and they can act as vectors for other pollutants, including persistent organic pollutants (POPs) that can adhere to their surfaces (Xiang et al., 2022). The ubiquity of microplastics has prompted global concern, necessitating research into their environmental distribution, fate, and long-term implications.

Despite the biodegradability of natural fragmented fibres, their potential risk is still under evaluation, because their release could be associated with harmful substances (UNEP, 2020). The release of natural fragmented fibres is an emerging environmental concern that parallels the issues posed by synthetic fragmented fibres (microplastics). Although fragmented natural fibres have a degree of biodegradability, their extensive release into aquatic systems may still result in ecological disruption (Athey and Erdle, 2021; Zambrano et al., 2021). The concern is that, similar to synthetic fragmented fibres, they can transport hazardous substances, introduce invasive species via attachment, and affect the feeding behaviour within food webs. Additionally, during the manufacturing of textile apparel, natural fibres and chemical fibres undergo a series of chemical treatments that could modify their biodegradability.

Understanding the full environmental impact of the fragmented fibres, including their degradation rates and interactions with aquatic, soil and atmospheric life, remains a critical area to be explored.

Table 9 provides an overview of important negative environmental impacts in the value chain of textile apparel based on available literature:

- Climate impacts are mainly generated during the processes of bleaching / dyeing and finishing.
- Fresh water is mainly affected during the use phase, the processes of bleaching / dyeing and finishing, and in raw material production.
- The impacts on land use are mainly generated during the raw material production.
- Impacts on both ecosystem quality and human health are mainly generated in raw material production, the processes of bleaching / dyeing and finishing, and during the use phase.

666 **Table 9** Negative environmental impacts across the global apparel value chain

Phase	Process	Climate (% in the value chain)	Water resources (% in the value chain)		Land use (% in the value chain)	Ecosystem quality (main factors)	Human health (main factors)
			Freshwater use	Water scarcity footprint			
Fibre production	Raw material production	12	21	33	56	Habitat loss when using land, Water use, Soil degradation, Agrochemicals use	Emission of harmful substances
	Material processing & sourcing	NA	NA	NA	NA	NA	NA
	Fibre preparation	NA	NA	NA	NA	NA	NA
Yarn and fabric production	Yarn preparation (spinning)	12	7	21	6	NA	NA
	Weaving, knitting, bonding	10	7	16	4	NA	NA
Textile production	Bleaching / dyeing and finishing	36	24	10	10	Chemicals released in fresh waters, High use of energy based on fossil fuels	Emission of harmful substances and due to the extraction and burning of fossil fuels
	Assembly	5	5	2	12	NA	NA
Consumption	Distribution and retail	1-11	< 1	< 1	< 1	NA	NA
	Use	14-24	35	18	13	High electricity consumption	Hazardous chemicals retained in the textile apparel
End-of-life	Collection and sorting	< 1	< 1	NA	< 1	NA	NA
	Landfilling / waste to energy	< 1	NA	NA	NA	NA	NA

667 NA: Not Available

668 Breakdown of phases is aligned with the analysis reported by [UNEP \(2020\)](#).

669 Source: own elaboration based on (Ellen MacArthur Foundation, 2017; Niinimäki et al, 2020; UNEP, 2020).

670

4 Legislation, strategies and voluntary environmental labels relevant for the textile sector

The sound development of the PS includes a good understanding of the EU legislation (existing and in preparation) and legislation from outside EU. This will allow the proposed ecodesign requirements to establish synergies with other legislation and strategies at EU and global level. For this reason, this section analyses the textile legislative context at different levels: EU, Member State, and third country (Sections 4.1 and 4.2). Textile strategies promoted by the United Nations are addressed in Section 4.3.

The textile industry currently uses numerous standards to measure and identify numerous technical aspects/parameters of a textile product. Section 4.4 analyses and classifies many standards used by the textile industry because they could potentially be used to propose and/or verify ecodesign requirements, when the PS addresses tasks 6 and 7 of the MEErP.

Finally, Section 4.6 provides an overview of the voluntary environmental labels used in the textile sector. This analysis focuses on the type and the environmental aspects mostly addressed by these labels. This investigation is mainly important for the following points:

- the background for the revision process of the EU Ecolabel criteria for textile products (see Section 1);
- the identification of the most recognised negative environmental impacts of the textile industry;
- the identification of requirement areas, which could inspire future requirements;
- the analysis of possible information on environmental labels to potentially be reported in the Digital Product Passport.

4.1 EU legislation

Currently the EU has no specific legislation addressing the mandatory sustainability of textiles. As mentioned in Section 1, there are the current EU Ecolabel criteria for textile products⁽⁴⁵⁾, which belong to a voluntary scheme for companies willing to show the good environmental performance of their products. Additionally, a number of EU legal acts have a direct impact on the placing on the market of textile products. Furthermore, profound changes in the EU acquis in the areas of consumer protection, consumer rights and product policy were, at the time of drafting this study, at different stages of the ordinary legislative procedure. In addition, a number of important policy documents, adopted by the Commission following the European Green Deal Communication, and mentioned in Section 1, address the relevance of textiles as a key product group. This section outlines the most relevant acts, existing and in preparation, related to textile products.

Textile Labelling Regulation (TLR)⁽¹⁴⁾

The TLR applies to textile products and products with textile components made up of at least 80% by weight of textile fibres. It defines rules on:

- the labelling and marking of the fibre composition of textile products;
- the labelling or marking of textile products containing non-textile parts of animals;
- the determination of the fibre composition of textile products, including of textile fibre mixtures.

The Regulation, which is currently under revision, aims to improve the functioning of the internal market and provide accurate information to consumers. It sets out a framework on how the textile composition is to be declared and how fibre composition should be determined. The introduction of rules on labelling domains such

⁴⁵ EU Ecolabel criteria for textile products. Commission Decision of 5 June 2014 establishing the ecological criteria for the award of the EU Ecolabel for textile products. Commission Decision (2014/350/EU). Available at [this link](#).

as sustainability and circularity, care, origin, size and presence of allergenic substances is currently under consideration, with a view to proposing a fundamental revision of the Regulation soon.

Industrial and Livestock Rearing Emission Directive

The Industrial and Livestock Rearing Emissions Directive (IED 2.0) ⁽⁴⁶⁾ as amended by Directive (EU) 2024/1785 ⁽⁴⁷⁾ is the main EU instrument regulating pollutant emissions from industrial installations and intensive livestock farms (pig and poultry). Industrial activities listed in Annex I to the Directive are required to operate in accordance with a permit based on the principles and provisions of the Directive. The IED 2.0 is based on several pillars, in particular an integrated approach, the use of best available techniques and public participation.

The integrated approach means that permits must take the whole environmental performance of the plant into account. This covers emissions to air, water and land, generation of waste, use of raw materials, energy efficiency, noise, prevention of accidents, and restoration of the site upon closure.

The permit conditions including emission limit values must be based on the Best Available Techniques (BAT). In order to define BAT and the BAT-associated environmental performance at EU level, the Commission organises an exchange of information with experts from Member States, industry and environmental organisations. This process results in BAT Reference Documents (BREFs) ⁽⁴⁸⁾; the BAT conclusions contained in the BREFs are adopted by the Commission as Implementing Decisions. The IED 2.0 requires that these BAT conclusions are the reference for setting permit conditions.

The IED 2.0 ensures that the public has a right to participate in the decision-making process, and to be informed of its consequences, by having access to permit applications, permits and the results of the monitoring of releases. Through the European Industrial Emissions Portal (EIEP), established by Regulation (EU) 2024/1244 ⁽⁴⁹⁾, environmental data reported by Member States, and some other countries ⁽⁵⁰⁾, are made accessible in a public register ⁽⁵¹⁾.

The IED 2.0 and the EIEP Regulation focus on emissions into air, water and land, prevention of waste generation, energy, water and material efficiency and reuse, in addition to promoting the use of safer, less toxic or non-toxic chemicals in industrial processes. They:

- ensure full and consistent implementation of the IED 2.0 across Member States, with tighter permit controls on air and water emissions;
- increase investment in new, cleaner technologies taking into account energy use, resource efficiency and water reuse whilst avoiding lock-in to obsolete technologies;
- support more sustainable growth of sectors that are key to building a clean, low-carbon and circular economy;
- cover intensive farming and industrial activities, ensuring that sectors with significant potential for high resource use or pollution also curb environmental damage at source by applying Best Available Techniques;
- establish an Innovation Centre for Industrial Transformation and Emissions (INCITE) ⁽⁵²⁾;
- integrate the previously separate requirements for depollution and decarbonisation so that future pollution control investments take better account of greenhouse gas emissions, resource efficiency and water reuse;

⁴⁶ Industrial Emissions Directive 2.0. Directive 2010/75/EU on industrial and livestock rearing emissions (integrated pollution prevention and control). Available at [this link](#).

⁴⁷ Directive (EU) 2024/1785 of the European Parliament and of the Council of 24 April 2024 amending Directive 2010/75/EU of the European Parliament and of the Council on industrial emissions (integrated pollution prevention and control) and Council Directive 1999/31/EC on the landfill of waste. Available at [this link](#).

⁴⁸ Best Available Techniques Reference Documents (BREFs), which are available at [this link](#). Last visited on 12 September 2024.

⁴⁹ EIEP Regulation (EC) No 2024/1244 on reporting of environmental data from industrial installations, establishing an Industrial Emissions Portal. Available at [this link](#).

⁵⁰ Iceland, Liechtenstein, Norway, Serbia, Switzerland and the United Kingdom.

⁵¹ European Industrial Emissions Portal. Available at [this link](#). Last visited on 12 September 2024.

⁵² The European Innovation Centre for Industrial Transformation and Emissions (INCITE). Available at [this link](#). Last visited on 12 September 2024.

- enhance data transparency and public access to environmental information by making permit summaries available online and providing more opportunities for public participation in the setting and review of permits.

The currently valid BREF for the textile industry dates back to 2023 (Roth et al., 2023). The associated Best Available Techniques (BAT) conclusions were adopted as Commission Implementing Decision (EU) 2022/2508⁽⁵³⁾. However, the industry belonging to the value chain of textile apparel could find relevant other BAT conclusions, such as Commission Implementing Decision (EU) 2019/2010⁽⁵⁴⁾ and Commission Implementing Decision (EU) 2018/1147⁽⁵⁵⁾, which are based on the corresponding BREFs dealing with waste incineration and waste treatments (Pinasseau et al., 2018a; Neuwahl et al., 2019).

REACH Regulation⁽⁵⁶⁾

The Regulation on Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) has the main purpose of ensuring the safe use of chemicals in the EU. More specifically, REACH aims to ensure a high level of protection of human health and the environment from risks resulting from the intrinsic properties of chemical substances, as well as the free circulation of substances on the internal market, while enhancing competitiveness and innovation. REACH regulates, amongst many other aspects, the restriction of the placing on the market and use of certain substances listed in its Annex XVII, including in some cases, the incorporation of substances into articles. Consequently, restrictions also cover recycled substances and the presence of restricted substances in recovered materials. The authorisation title of REACH applies to the placing on the market and use of substances of very high concern (SVHC), aiming at their progressive substitution by less hazardous substances or technologies and by subjecting their use to specific conditions.

A number of restrictions in Annex XVII to REACH specifically mention textiles in their scope. These include entries: 4 - Tris (2,3 dibromopropyl) phosphate; 7 - Tris(aziridinyl)phosphin oxide; 8 - Polybrominated biphenyls; 18 - Mercury compounds; 20 - Organostannic compounds; 43 - Azocolourants and Azodyes; 46/46a - Nonylphenol and Nonylphenol ethoxylates; 47 - Chromium VI compounds (relevant to leather articles); 68 - C9-C14 PFCAs and 72 - CMRs in textiles and footwear. Other restrictions, of general application to articles placed on the market for supply to the general public, or covering all articles placed on the market, may also apply to textile articles, for instance entries 50 - (certain) polycyclic aromatic hydrocarbons; 51 and 52 (certain phthalates); 61 - Dimethylfumarate; and 63 - Lead and its compounds.

In addition, Article 33 of REACH sets up a supply chain communication duty requiring suppliers of articles containing SVHC above 0.1% to communicate certain information to the recipients of those articles.

Regulation on the classification, labelling and packaging of substances and mixtures (CLP)⁽⁵⁷⁾

Regulation (EC) No. 1272/2008 (CLP) focuses on the identification and classification of the intrinsic hazards of chemicals, i.e. the hazardous effects of chemicals on human health or the environment, and on communicating them to users of chemicals and decision makers (consumers, industry and authorities). Identifying the intrinsic hazardous properties of substances to derive a hazard classification is the first step in assessing chemical risks.

⁵³ BREF for the textile industry. Commission Implementing Decision (EU) 2022/2508 of 9 December 2022 establishing the best available techniques (BAT) conclusions, under Directive 2010/75/EU of the European Parliament and of the Council on industrial emissions, for the textiles industry. Available at [this link](#).

⁵⁴ BREF for waste incineration. Commission Implementing Decision (EU) 2019/2010 of 12 November 2019 establishing the best available techniques (BAT) conclusions, under Directive 2010/75/EU of the European Parliament and of the Council, for waste incineration. Available at [this link](#).

⁵⁵ BREF for waste treatment. Commission Implementing Decision (EU) 2018/1147 of 10 August 2018 establishing best available techniques (BAT) conclusions for waste treatment, under Directive 2010/75/EU of the European Parliament and of the Council. Available at [this link](#).

⁵⁶ REACH Regulation. Regulation (EC) No 1907/2006 of the European Parliament and of the Council of 18 December 2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH), establishing a European Chemicals Agency, amending Directive 1999/45/EC and repealing Council Regulation (EEC) No 793/93 and Commission Regulation (EC) No 1488/94 as well as Council Directive 76/769/EEC and Commission Directives 91/155/EEC, 93/67/EEC, 93/105/EC and 2000/21/EC. Available at [this link](#).

⁵⁷ CLP Regulation. Regulation (EC) No 1272/2008 of the European Parliament and of the Council of 16 December 2008 on classification, labelling and packaging of substances and mixtures, amending and repealing Directives 67/548/EEC and 1999/45/EC, and amending Regulation (EC) No 1907/2006. Available at [this link](#).

CLP requires manufacturers, importers and downstream users to classify hazardous substances and mixtures and provides rules on how to classify them, applicable throughout the EU. For a substance that has a harmonised classification (an entry in Annex VI to CLP), that classification is legally binding for the hazard classes and differentiations covered in the entry. The substances not covered in the entry, as well as mixtures, must be evaluated and self-classified. The hazard classification determines the appropriate labelling and packaging of the chemicals in the supply chain, in particular to protect workers, consumers and the environment.

Hazard classifications under CLP, including the new hazard classes defined in the recently adopted Delegated Commission Regulation (EU) 2023/707, are very relevant in determining obligations imposed by many relevant EU acts, including on products. Article 2(28) of ESPR refers to specific hazard classes under Annex VI of CLP to define “substances of concern”, whereas under Regulation (EC) No. 66/2010 on the EU Ecolabel, its Article 6(6) specifies that an EU Ecolabel cannot be awarded to goods containing substances or mixtures meeting the criteria for certain hazard classifications in accordance with CLP.

POPs Regulation ⁽⁵⁸⁾

The POPs Regulation focuses on persistent organic pollutants, and it implements the international obligations of the EU as a party to the Stockholm Convention. The objective of the Regulation is to protect human health and the environment from POPs by prohibiting, phasing out as soon as possible, or restricting the manufacturing, placing on the market and use of substances subject to the Stockholm Convention. Substances listed in Annexes I and II to the Regulation are, respectively, prohibited or limited, in terms of their manufacture, placing on the market and use, with some specific exemptions. Some of these restrictions are relevant to textiles, for instance those associated with certain brominated flame retardants (e.g. certain PBDEs), surface-active substances such as PFOS or PFOA, or substances with biocidal properties such as pentachlorophenol.

Waste Framework Directive (WFD) ⁽⁵⁹⁾

The WFD sets the basic concepts and definitions related to waste management, including definitions of waste, recycling and recovery. It lays down waste management principles, which contribute to the reduction of the adverse impact of waste management on human health or the environment, with an emphasis on waste prevention. The WFD contains provisions on waste prevention, encouraging the reuse of products and the setting up of systems promoting repair and reuse activities, including in particular for textiles (Article 9), and requires Member States to set up separate collection for textiles by 1 January 2025 (Article 11) in order to promote high-quality recycling. Article 9 of the Directive also promotes the reduction of the content of hazardous substances in materials and products by defining a reporting obligation that applies to suppliers of articles (as defined under REACH), requiring them to provide information regarding the presence of SVHC in articles, including textile articles, pursuant to Article 33 of REACH, to the European Chemicals Agency (ECHA), as of 5 January 2021. This information is collected in a database, operated by ECHA, and access is provided to waste treatment operators and consumers.

Waste Shipment Regulation (WSR) ⁽⁶⁰⁾

This Regulation aims to (1) ensure that the EU does not export its waste challenges to third countries and contributes to environmentally sound management of waste; (2) strengthen enforcement to prevent illegal shipments of waste occurring within the EU, as well as from the EU to third countries; (3) increase traceability of waste shipments within the EU and facilitate recycling and reuse. Although the WSR entered into force on 20 May 2024, most provisions will apply from 21 May 2026 and most export rules will apply from 21 May 2027.

⁵⁸ POPs Regulation. Regulation (EU) 2019/1021 of the European Parliament and of the Council of 20 June 2019 on persistent organic pollutants (recast) (Text with EEA relevance) Text with EEA relevance. Available at [this link](#).

⁵⁹ Waste Framework Directive (WFD). Directive 2008/98/EC of the European Parliament and of the Council of 19 November 2008 on waste and repealing certain Directives (Text with EEA relevance). Available at [this link](#).

⁶⁰ Waste Shipment Regulation (WSR). Regulation (EU) 2024/1157 of the European Parliament and of the Council of 11 April 2024 on shipments of waste, amending Regulations (EU) No 1257/2013 and (EU) 2020/1056 and repealing Regulation (EC) No 1013/2006. Available at [this link](#).

833 **Corporate Sustainability Due Diligence Directive (CSDDD)** ⁽⁶¹⁾

834 The primary objective of the CSDDD is to promote sustainable and responsible corporate practices throughout
835 companies' operations and global value chains. The new rules aim to ensure that in-scope companies identify
836 and mitigate adverse human rights and environmental impacts arising from their activities, both within and
837 outside the EU.

838 This Directive establishes a mandatory corporate due diligence obligation, comprising the following key
839 elements:

- 840 — Companies must identify and address potential and actual human rights and environmental
841 impacts within their own operations, subsidiaries, and value chains, including those of their
842 business partners.
- 843 — Large companies are required to adopt and implement, to the best of their abilities, a transition
844 plan for climate change mitigation, aligned with the 2050 climate neutrality objective of the Paris
845 Agreement and intermediate targets under the European Climate Law.

846 The CSDDD applies to: (1) large EU limited liability companies and partnerships with more than 1,000 employees
847 and a global turnover exceeding EUR 450 million, and (2) large non-EU companies with a turnover of more than
848 EUR 450 million in the EU. Micro companies and small and medium-sized enterprises (SMEs) are exempt from
849 the proposed rules. However, the Directive provides supportive and protective measures for SMEs that may be
850 indirectly affected as business partners in value chains

851

852 **The Corporate Sustainability Reporting Directive (CSRD)** ⁽⁶²⁾

853 The CSRD modernised and strengthened the rules governing the social and environmental information that
854 companies must report. A broad range of large companies, including listed small and SMEs, are required to
855 report on sustainability. Non-EU companies generating over EUR 150 million on the EU market are also subject
856 to reporting requirements.

857 The CSRD aims to provide investors and other stakeholders with access to the necessary information to assess
858 the impact of companies on people and the environment, as well as to evaluate financial risks and opportunities
859 arising from climate change and other sustainability issues. Furthermore, the harmonisation of reporting
860 requirements is expected to reduce reporting costs for companies in the medium to long term.

861 Companies subject to the CSRD will be required to report according to the European Sustainability Reporting
862 Standards (ESRS), drafted by an independent body representing various stakeholders. The first set of ESRS was
863 published in December 2023, in the form of a Delegated Regulation ⁽⁶³⁾. These standards apply to all companies
864 within the scope of the CSRD, regardless of their sector of operation. The ESRS are aligned with EU policies,
865 while also building on and contributing to international standardisation initiatives.

866 The CSRD also requires assurance on the sustainability information that companies report and will provide for
867 the digital taxonomy of sustainability information.

868 The first companies will be required to apply the new rules for the 2024 financial year, with reports to be
869 published in 2025.

870

871 **Unfair Commercial Practices Directive (UCPD)** ⁽⁶⁴⁾

⁶¹ Corporate Sustainability Due Diligence Directive (CSDDD). Directive (EU) 2024/1760 of the European Parliament and of the Council of 13 June 2024 on corporate sustainability due diligence and amending Directive (EU) 2019/1937 and Regulation (EU) 2023/2859. Available at [this link](#).

⁶² Corporate Sustainability Reporting Directive (CSRD). Directive (EU) 2022/2464 of the European Parliament and of the Council of 14 December 2022 amending Regulation (EU) No 537/2014, Directive 2004/109/EC, Directive 2006/43/EC and Directive 2013/34/EU, as regards corporate sustainability reporting. Available at [this link](#).

⁶³ Commission Delegated Regulation (EU) 2023/2772 of 31 July 2023 supplementing Directive 2013/34/EU of the European Parliament and of the Council as regards sustainability reporting standards. Available at [this link](#).

⁶⁴ Unfair Commercial Practices Directive (UCPD). Directive 2005/29/EC of the European Parliament and of the Council of 11 May 2005 concerning unfair business-to-consumer commercial practices in the internal market and amending Council Directive 84/450/EEC,

The UCPD concerns unfair business-to-consumer commercial practices, and it aims to boost consumer confidence and make it easier for businesses, especially small and medium-sized enterprises, to trade across borders. It regulates unfair commercial practices that occur before, during and after a business-to-consumer transaction has taken place.

The Directive does not provide specific rules on environmental claims or specifically for textiles. However, it provides a legal basis to ensure that traders do not present environmental claims in ways that are unfair to consumers. It does not prohibit the use of 'green claims' as long as they are not unfair. On the contrary, the UCPD can help traders invest in the environmental performance of their products by enabling them to communicate these efforts to consumers transparently and by preventing competitors from presenting misleading environmental claims. The Directive states that claims should not be misleading and that all claims which lead a consumer to choose one product over another must be trustworthy. The Directive's Annex I lists which practices are considered unfair and misleading in all circumstances.

Directive on empowering consumers for the green transition ⁽⁶⁵⁾

The Directive on empowering consumers for the green transition amends the Unfair Commercial Practices Directive (2005/29/EC) and the Consumer Rights Directive (2011/83/EU). It addresses problems identified with consumer information at the point of sale, in particular the fact that consumers lack reliable information for choosing more environmentally sustainable products, and it aims to protect consumers against certain unfair commercial practices.

This is to be achieved through the improved participation of consumers in the circular economy by providing better information on the durability and reparability of certain products to consumers and stepping up the protection of consumers against unfair commercial practices by preventing: greenwashing, early obsolescence practices, use of unreliable and non-transparent sustainability labels and information tools. The co-legislator adopted this Directive on 20 February 2024.

Directive on common rules promoting the repair of goods ⁽⁶⁶⁾

The Directive introduces a new 'right to repair' for consumers, both within and beyond the legal guarantee. It aims to provide consumers savings and support the objectives of the European Green Deal by reducing waste. The Directive makes it easier and more cost-effective for consumers to repair as opposed to replace goods. Additionally, it aims to promote the repair sector, incentivising sustainable business models.

Amendment of the Waste Framework Directive ⁽⁶⁷⁾

The overall objective of the WFD revision is to reduce environmental and climate impacts, increase environment quality and improve public health associated with textiles waste management in line with the waste hierarchy. The Commission proposal aims to make producers responsible for the full life cycle of textile products and to support the sustainable management of textile waste across the EU. This initiative also aims to accelerate the development of the separate collection, sorting, reuse and recycling sector for textiles in the EU, in line with the EU Strategy for Sustainable and Circular Textiles.

The Commission proposal intends to introduce mandatory and harmonised Extended Producer Responsibility (EPR) schemes for textiles in all EU Member States. According to this Commission proposal, producers will cover the costs of management of textile waste, which will also provide incentives to reduce waste and increase the circularity of textile products, resulting in a better design of products from the start. The proposal envisages

⁶⁵ Directives 97/7/EC, 98/27/EC and 2002/65/EC of the European Parliament and of the Council and Regulation (EC) No 2006/2004 of the European Parliament and of the Council ('Unfair Commercial Practices Directive') (Text with EEA relevance). Available at [this link](#). Directive (EU) 2024/825 of the European Parliament and of the Council of 28 February 2024 amending Directives 2005/29/EC and 2011/83/EU as regards empowering consumers for the green transition through better protection against unfair practices and through better information. Available at [this link](#).

⁶⁶ Directive (EU) 2024/1799 of the European Parliament and of the Council of 13 June 2024 on common rules promoting the repair of goods and amending Regulation (EU) 2017/2394 and Directives (EU) 2019/771 and (EU) 2020/1828. Available at [this link](#).

⁶⁷ Waste Framework Directive (WFD). Directive (EU) 2025/1892 of the European Parliament and of the Council of 10 September 2025 amending Directive 2008/98/EC on waste (Text with EEA relevance). Available [at this link](#).

that the contribution to EPR schemes will be adjusted based on the environmental performance of textiles (eco-modulation) based on parameters aligned with ecodesign requirements under the ESPR.

Proposal for a Directive on substantiation and communication of explicit environmental claims (Green Claims Directive) ⁽⁶⁸⁾

The Commission Green Claims Directive proposal will address greenwashing by tackling false environmental claims made to consumers, and stopping the proliferation of public and private environmental labels. Together with the proposal for a Directive on empowering consumers for the green transition, the proposal establishes a clear regime for environmental claims and labels. Their purpose is to ensure that consumers receive trustworthy information about the environmental credentials of the products they buy.

The proposal targets 'green claims' made by businesses that state or imply a positive environmental impact, lesser negative impact, no impact, or improvement over time for their products, services, or organisation. It requires that green claims are substantiated and this substantiation be verified *ex-ante*. This only concerns claims that are not covered by other EU rules, notably the ESPR and the TLR. The proposal also addresses environmental labelling schemes, stopping the proliferation of public and private labels and ensuring the transparency and robustness of labelling schemes.

4.2 Legislation and initiatives in EU Member States and non-EU countries

The publication of Directive (EU) 2018/851 ⁽⁶⁹⁾ (the 2018 revision of the WFD) promoted the establishment of **Extended Producer Responsibility (EPR)** schemes for textile products in several Member States. For textile, France ⁽⁷⁰⁾, the Netherlands ⁽⁷¹⁾, Hungary (Decree 80/2023) and Latvia ⁽⁷²⁾ were pioneers in establishing systems where economic operators placing textile products on the market contribute to the collection, sorting, reuse, preparing for reuse and recycling infrastructure for the same products.

Other Member States establishing EPR schemes are Greece (Law 4819/2021), and Spain (Law 7/2022), while Italy ⁽⁷³⁾ is not far behind.

Within the national Climate Law, France is establishing the **Eco-Score** ⁽⁷⁴⁾, which is an online tool that provides the consumer with an idea about the environmental effects ⁽⁷⁵⁾ of apparel, with specific characteristics (e.g. weight, fibre composition) selected directly by the consumer from a defined list. In the future, it is planned that economic operators will be able to report the environmental performance of their products via the Eco-Score, so that the Eco-Score will work as an environmental labelling tool. Additionally, it is planned to connect the Eco-Score with the EPR scheme to deploy eco-modulation.

The Dutch Government established the **Denim Deal** ⁽⁷⁶⁾, which aimed to bolster the use of post-consumer recycled cotton in denim products marketed in the Netherlands. In particular, the signatories of the initiative aimed to use 20% post-consumer recycled cotton fibres in 3 million pairs of jeans produced until the end of 2023 ⁽⁷⁷⁾.

The Luxembourg Ministry of the Economy developed the **Product Circularity Data Sheet (PCDS)** ⁽⁷⁸⁾, which aims to provide basic product information about the several steps of the supply chain. Information is gathered

⁶⁸ Green Claim Directive. Proposal for a Directive of the European Parliament and of the Council on substantiation and communication of explicit environmental claims (Green Claims Directive). COM/2023/166 final. Available at [this link](#).

⁶⁹ Directive (EU) 2018/851 of the European Parliament and of the Council of 30 May 2018 amending Directive 2008/98/EC on waste (Text with EEA relevance). Available at [this link](#).

⁷⁰ French Anti-Waste law for a circular economy. AGECL - Law n°2020-105, 10 February 2020. Available at [this link](#). Code de l'environnement [L.541-10-27](#). Code de l'environnement [R.543-214 to R.541-219](#). Last accessed on 17 December 2023.

⁷¹ Dutch Decree on extended producer responsibility for textiles. Available at [this link](#). Last accessed on 17 December 2023.

⁷² Latvia Natural Resources Tax Law, July 2024. Available at [this link](#). Last accessed on 19 April 2025.

⁷³ Announcement of the Italian EPR scheme. Available at [this link](#). Last accessed on 17 December 2023.

⁷⁴ Eco-Score. Available at [this link](#). Last accessed on 17 December 2023.

⁷⁵ The French Eco Score uses the term "environmental costs" to include aspects besides the environmental impacts. In the main text the wording "environmental effects" was preferred to avoid any confusion with the assessment of environmental costs reported in Section 10 and Section 11.

⁷⁶ The Denim Deal. Available at [this link](#). Last accessed on 17 December 2023.

⁷⁷ Results and Lessons Learnt from the Denim Deal. Available at [this link](#). Last accessed on 19 April 2025.

⁷⁸ Product Circularity Datasheet. Available at [this link](#). Last accessed on 17 December 2023.

in five areas: general information, composition, design for better use, design for disassembly, and design for reuse.

While the EU is leading the way in legislating Extended Producer Responsibility for textiles, other countries and regions are gradually beginning to follow suit. This includes Australia, Chile, the United Kingdom, and the United States, though in the latter, EPR initiatives are being developed at the state level rather than federally, with examples including California, New York, and Washington ([Global Fashion Agenda, 2025](#)). In fact, in September 2024, the State of **California** (USA) has signed the **Responsible Textile Recovery Act** ⁽⁷⁹⁾ with Senate Bill 707 (SB-707), which aims to establish an Extended Producer Responsibility scheme for textile products. The bill introduces the concept of a programme operator to oversee the implementation of a stewardship programme. Additionally, the bill also establishes a Textile Stewardship Recovery fund which would receive fees paid by programme operators. This fund will cover the costs associated with programme implementation and enforcement. Products covered include apparel, workwear, footwear, accessories (such as handbags and backpacks), and home textiles ([Global Fashion Agenda, 2025](#)).

In contrast, countries such as Brazil, Japan, India, and South Korea have established EPR schemes for other waste streams, but have not yet extended these to textiles. At this stage, it remains uncertain whether they will consider doing so in the future ([Global Fashion Agenda, 2025](#)).

4.3 Strategies of the United Nations

The United Nations Environment Programme (UNEP) offers strategic guidance and promotes collaboration across sectors to foster a fair shift towards a sustainable and circular textile value chain. To this end, UNEP promotes numerous activities and publishes studies on its website ⁽⁸⁰⁾.

In particular, UNEP proposes a roadmap with collective actions focusing on nine points (UNEP, 2023a):

1. Adopt globally sustainable and circular business models.
1. Contrast overconsumption and overproduction.
2. Design sustainable and circular textile products.
3. Improve product care and durability.
4. Reduce the emissions into the environment during the production stages.
5. Address social issues along the value chain.
6. Use of sustainable or recycled materials.
7. Improve the shared infrastructure of the value chain.
8. Develop a suitable textile waste management system that avoids landfilling and incineration of textile products.

UNEP promotes the engagement of all stakeholders in the value chain to focus on the nine points of the roadmap (UNEP, 2023a).

4.4 Strategies of the Organisation for Economic Co-operation and Development

The Organisation for Economic Co-operation and Development (OECD) provides evidence-based policy guidance, promotes international cooperation, and supports best practices in sustainable development, including in the textile and apparel sectors. As a global policy forum and knowledge hub, the OECD supports governments and industry stakeholders through standards, analytical tools, and consensus-driven recommendations.

In the context of textiles, the OECD has developed specific instruments to encourage responsible business conduct and improve sustainability across global supply chains. Notably, the OECD Guidelines for Multinational Enterprises on Responsible Business Conduct ⁽⁸¹⁾ serve as a reference framework for aligning business practices

⁷⁹ SB-707 Responsible Textile Recovery Act of 2023. Available at [this link](#). Last accessed on 21 April 2025.

⁸⁰ United Nations environment programme. Building Sustainability and Circularity in the Textile Value Chain. Available at [this link](#). Last accessed on 17 December 2023.

⁸¹ OECD Guidelines for Multinational Enterprises. Responsible Supply Chains in the Textile and Garment Sector. Available at [this link](#). Last accessed on 19 April 2025.

with environmental and social sustainability goals. The dedicated guidance for the textile and garment sector outlines due diligence steps companies should undertake to identify, prevent, and mitigate adverse impacts in their supply chains (OECD, 2018). This includes actions related to labour rights, environmental degradation, and responsible purchasing practices.

4.5 Tests and standards

Textile standards in Europe are coordinated by CEN-CENELEC⁽⁸²⁾. Several working groups are related to the textile industry, mainly CEN/TC248 TEXTILES AND TEXTILE PRODUCTS. **Table 122** in Section 13.4 reports details of technical working groups and scientific committees. On the other hand, ISO/TC 38 TEXTILES encompasses several working groups that deal with international standards for textile properties, testing methods, and quality control. **Table 123** in Section 13.4 includes the technical working groups in ISO/TC 38 that focus on aspects of the textile industry.

The knowledge of the available standards in the textile sector is crucial in the development of the PS, because it provides an overview of the technical aspects/parameters of a textile product that can currently be measured via a commonly recognised test/method/tool. Specific standards could potentially be used to propose and/or verify ecodesign requirements, when the PS addresses tasks 6 and 7 of the MEErP.

Numerous standards used in the textile industry were classified and some of them could be related to the product aspects reported in Article 5 of the ESPR (see Section 13.4):

- Standards directly related to the intrinsic durability of the textile product, which could address abrasion, pilling, colourfastness, dimensional stability, seam slippage, tear strength, etc. (**Table 124**). Specific textile frameworks already use some of these standards to address durability aspects of textile products. **Table 125** compares how durability parameters for textile products are tested by PEFCR A&F, EU Ecolabel criteria, Blue Angel criteria, and Nordic Swan Ecolabel criteria (see Section 4.6 for information about ecolabels and other environmental labels).
- Standards related to the functionality of the textile products, such as antifungal activity, antiviral activity, oil stain repellency, resistance to chlorinated water, resistance to insects, resistance to surface wetting, stain repellency, water repellency, water resistance after aging, wicking (**Table 126**).
- Standards used for textile characterisation, such as identification of dyestuff and fibres, thickness, mass per unit area and composition (**Table 127**), which could be a reference for characterising and referring to specific textile products.
- Standards for the identification of specific substances, such as alkylphenol ethoxylates (APEO), formaldehyde, and other chemicals (**Table 128**), which are of environmental concern as reported in Section 3.1.3.
- Standards related to the loss of fragmented fibres (**Table 129**), which are of environmental concern as reported in Section 3.1.3.
- Standards potentially related to circularity and environmental aspects. Almost all of these standards are still under development (**Table 130**).
- Standards related to potential information on the care of textile products, because they describe and report labelling symbols (**Table 131**). Apparel properly maintained is more likely to have a longer lifespan (see Section 6).

Some of the standards available for the textile industry measure a specific parameter using different methods (e.g. Determination of the abrasion resistance on coated fabrics. Part 1 using taber abraser and Part 2 using martindale abraser). The selection of the method depends mainly on the laboratory equipment used for testing.

Only one standard was found addressing non-woven technology to measure the tensile strength and elongation; all the other standards focused on woven and knitted technologies.

⁸² CenCenelec website. Available at [this link](#). Last accessed on 17 December.

The analysis did not reveal any standard capable of recognising whether a fibre is of virgin or recycled origin, or of identifying the type of fibres being mechanically recycled. Specific methods to verify this type of information will be investigated in the following milestone, when the PS addresses task 6.

In addition to the investigated standards, the textile industry widely deploys a multitude of other standards that are tailored to specific industrial sectors or regional contexts. Notably, several standards are of relevance, including those promulgated by organisations such as the AATCC (American Association of Textile Chemists and Colorists), ASTM (American Society for Testing and Materials), British Standards Institution, JIS (Japanese Industrial Standards), and the GB standards (National Standards of the People's Republic of China), and many more. By adhering to such comprehensive and recognised frameworks, the textile industry can foster consistency, reliability, and quality assurance throughout its global operations.

4.6 Voluntary environmental labels

In general, there are many types of labels, addressing single or multiple environmental issues and covering different sectors and regions. See Section 13.5.1 for more information. Three ISO standards classify voluntary environmental labels as follows:

- **ISO Type I environmental labels** (ISO 14024:2018)⁽⁸³⁾, known as **Ecolabels**, are defined as ‘voluntary, multi-criteria-based and third party-verified labels that indicate an overall environmental preference in a life cycle perspective of a product or service within a specific product category’.
- **ISO Type II environmental labels** (ISO 14021:2016)⁽⁸⁴⁾, known as **Self-declared Environmental Claims**, are neither third-party verified nor based on a Life Cycle Thinking approach. Many self-declared environmental claims on the EU market do not necessarily follow ISO 14021:2016.
- **ISO Type III environmental labels** (ISO 14025:2016)⁽⁸⁵⁾, known as **Environmental Declarations**, are labels presenting ‘quantified environmental information on the life cycle of a product to enable comparisons between products fulfilling the same function’. The establishment of Product Category Rules ensures that the life cycle assessment is performed with specific rules aiming to foster transparency and facilitate comparisons between different Environmental Declarations.

Global environmental labels used in the textile industry were recently analysed (Ranasinghe and Jayasooriya, 2021) via an investigation including bibliographic research, the Ecolabel Index⁽⁸⁶⁾, and the Global Ecolabelling Network⁽⁸⁷⁾. Ranasinghe and Jayasooriya (2021) report that on 20 March 2021 there were 199 environmental labels for textiles in the world: 55 used in Europe, 54 used in the USA, 41 used in Asia, 18 used in Latin America, 19 used in Oceania, and 12 used in Africa. When looking for the same information on 7 February 2024, the ITC Standard Map⁽⁸⁸⁾ provided 297 environmental labels for textiles used globally: 58 used in Europe, 50 used in USA, 54 used in Asia, 42 used in Latin America, 48 used in Oceania, and 45 used in Africa.

Table 10 reports the topics that were addressed the most by environmental labels for textiles globally in 2021. The majority of the concerns focused on the use of harmful chemicals and toxic substances, as well as natural resources. Additionally, specific global regions used environmental labels addressing specific topics: such as Europe with ‘Pesticides/herbicides/fungicides’, and Latin America with ‘Carbon/GHG offsets’.

Table 10. Top five topics addressed by environmental labels for textiles globally in 2021

Topic	Europe	USA	Asia	Latin America	Oceania	Africa
Toxics	X	X	X	X	X	X

⁸³ ISO 14024:2018. Environmental labels and declarations. Type I environmental labelling. Principles and procedures. Available at [this link](#). Last accessed on 12 January 2024.

⁸⁴ ISO 14021:2016. Environmental labels and declarations. Self-declared environmental claims (Type II environmental labelling). Available at [this link](#). Last accessed on 12 January 2024.

⁸⁵ ISO 14025:2006. Environmental labels and declarations. Type III environmental declarations. Principles and procedures. Available at [this link](#). Last accessed on 12 January 2024.

⁸⁶ Ecolabel Index. Website available at [this link](#). Last accessed on 12 January 2024.

⁸⁷ Global Ecolabelling Network. Website available at [this link](#). Last accessed on 12 January 2024.

⁸⁸ ITC Standards Map App, selecting sectors of clothing and textiles. Available at [this link](#). Last accessed on 7 February 2024.

Topic	Europe	USA	Asia	Latin America	Oceania	Africa
Harmful chemicals	X	X		X	X	X
Natural resources	X	X	X		X	X
Pesticides/herbicides/fungicides	X					
Material use	X	X	X			
Waste		X	X	X	X	X
Energy use/efficiency			X	X	X	X
Carbon/GHG offsets				X		

N.B. The topic 'Toxics' addresses harmful substances that are already prohibited or regulated; whereas the topic 'Harmful chemicals' addresses substances that are known to be harmful to health, but are not officially banned. The authors of the referenced study use the term 'Chemicals' when referring to 'Harmful chemicals'.

The authors of the referenced study do not specify if the topic 'Waste' addresses any kind of waste generated along the value chain, or waste generated at a specific stage. No further explanation/specification is provided for other topics.

Source: Ranasinghe and Jayasooriya (2021)

Figure 2 shows the topics addressed by environmental labels used for textiles in 2021 in Europe and their number. These environmental labels addressed most of the environmental impacts described in Section 3.3.2.

The analysis showed that the textile industry largely relies on environmental labels, but most of the time actors do not state which ISO type standard they follow – if they follow any. This situation causes confusion for the consumers, who cannot identify governance, reliability, environmental scope and level of environmental ambition of all environmental labels. To fill this gap, some voluntary initiatives try to score, compare and describe some environmental labels⁽⁸⁹⁾. The lack of transparency of the numerous environmental labels could be further analysed in the 3rd milestone, when the PS addresses the content of the Digital Product Passport.

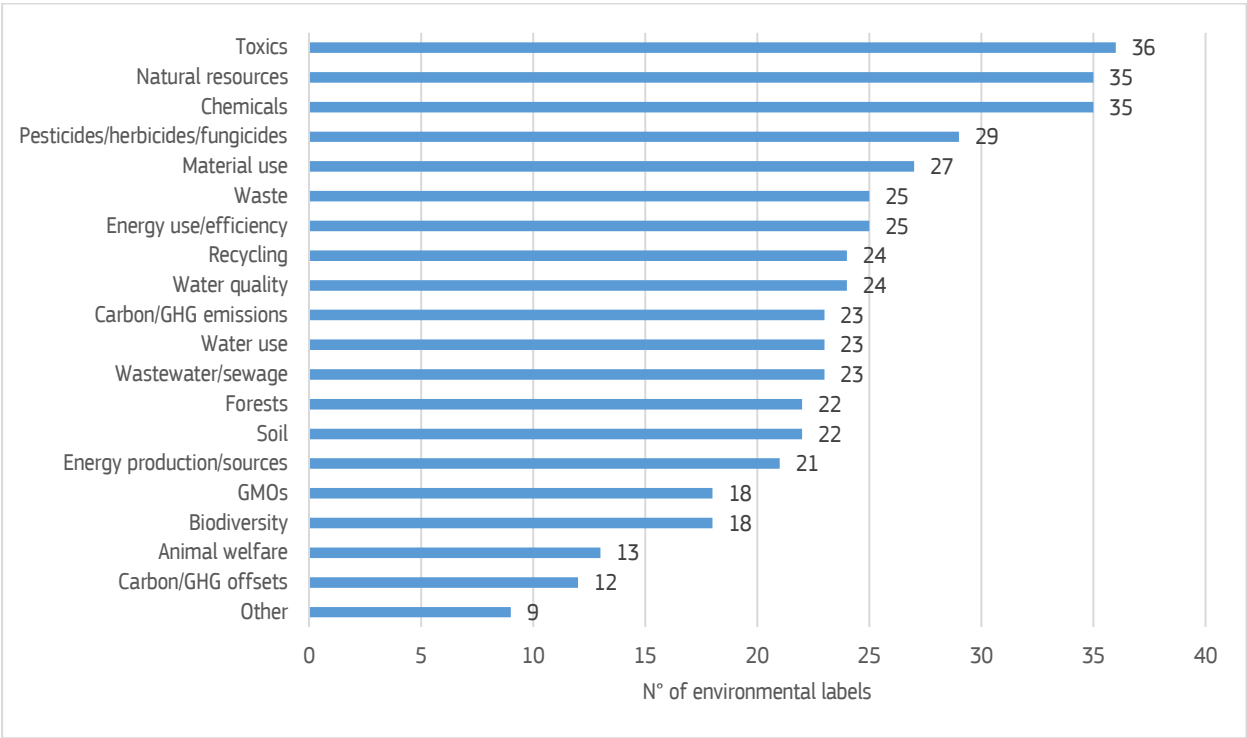
Section 7 assesses in detail the EU Ecolabel criteria for textile products in light of their revision process, as described in Section 1. As established by Article 6(3.f) of the EU Ecolabel Regulation, the analysis performed considers other ecolabels used in the EU (Blue Angel⁽⁹⁰⁾ and Nordic Swan Ecolabel⁽⁹¹⁾) to enhance synergies.

Further information about voluntary environmental labels used in Europe is reported in **Table 132** in Section 13.5.2.

⁸⁹ ITC Standards Map App. Available at [this link](#). Last accessed on 7 February 2024.
Label Directory – Siegelklarheit. Available at [this link](#). Last accessed on 7 February 2024.
Labels Environnementaux. Available at [this link](#). Last accessed on 7 February 2024.
⁹⁰ Blue Angel – The German Ecolabel. Available at [this link](#). Last accessed on 7 February 2024.
⁹¹ Nordic Swan Ecolabel. Available at [this link](#). Last accessed on 7 February 2024.

1095

Figure 2. Main topics addressed by environmental labels used in 2021 in Europe



1096

1097

1098

1099

1100

1101

N.B. The topic 'Toxics' addresses harmful substances that are already prohibited or regulated, whereas the topic 'Chemicals' addresses substances that are known to be harmful to health, but are not officially banned. The authors of the referenced study do not specify if the topic 'Waste' addresses any kind of waste generated along the value chain, or waste generated at a specific stage. No further explanation/specification is provided for other topics.
Source: (Ranasinghe and Jayasooriya, 2021)

5 Market analysis

This section analyses the market of the products included in the scope of the PS, addressing many aspects. The information collected at this stage will feed into the following tasks of the MEErP, in particular task 4 (Section 9) to task 7.

The first subsections analyse the market at different scales: global, EU and Member State (Sections 5.1, 5.2 and 5.3). Section 5.4 presents available market information of elements affecting the production and recycling of textile apparel. The market structure and the value chain are analysed in Section 5.5 and Section 5.6, respectively. Section 5.7 analyses the competitiveness at global level, focusing on different environmental compliance costs that companies placed in different countries must face. Section 5.8 analyses the lifespan of the products included in the scope. This information is particularly crucial to build a suitable model for the environmental and economic assessment, as well as the stock analysis (tasks 5 to 7 of the MEErP). The section concludes with a brief comment about the penetration in the market of the environmental labels used in the textile sector (Section 5.9).

5.1 The global market

Historical overview

In 1947, the Grant Agreement on Tariffs and Trade (GATT) was signed by 23 countries to promote free movement of goods among signing countries. The GATT set the bases for the future institution of the World Trade Organization (WTO) in 1994 ⁽⁹²⁾.

From 1974 to 1994, under a special regime outside the GATT framework, the Multifibre Arrangement (MFA) set rules for international trade of textile products. The MFA was a framework for bilateral agreements or unilateral actions that established quotas ⁽⁹³⁾ limiting imports into countries whose domestic industries could be damaged by a quick rise of imports ⁽⁹⁴⁾. The MFA applied only to trade between developed and developing countries, but not to trade among developed countries. Although the MFA did not comply with the principle of non-discrimination established in the GATT, developed countries used it to protect their textile industry (Stephen MacDonald, 2006).

In 1995, within the establishment of the WTO and the revised GATT, the MFA was substituted by the Agreement on Textiles and Clothing (ATC), which represented the 10-year-long transition towards an international textile market without quotas ⁽⁹⁵⁾. The ATC applied to all the WTO members ⁽⁹⁶⁾. Between 1995 and the end of 2004, quotas were gradually removed ⁽⁹⁷⁾.

Since 1 January 2005, the trade of textile products among WTO members has been free of quotas.

In June 2005, the European Commission signed a Memorandum of Understanding with China to keep the imports of specific textile products ⁽⁹⁸⁾ below some thresholds until the end of 2007. This action was intended to prevent distortion of the European textile sector ⁽⁹⁹⁾.

Since 2005, the European textile market has been completely open to imports of any product coming from any country belonging to the WTO.

In 2005, the removal of quotas from the global textile market caused an increase in production and employment in Asian countries, such as China, India, Türkiye, Hong Kong, Bangladesh, and Indonesia (Hildegunn Kyvik Nordås, 2004). Already in the 1970s, textile production had started moving from Europe and North America mainly to

⁹² The General Agreement on Tariffs and Trade (GATT 1947) – the legal text available on [this World Trade Organization webpage](#), last accessed on 29 September 2023.

⁹³ A quota is a government-imposed trade restriction that limits the number or monetary value of goods that a country can import or export during a particular period (Adam Barone, 2022)

⁹⁴ Textiles: back in the mainstream – [World Trade Organization webpage](#), last visited on 29 September 2023.

⁹⁵ Textiles Monitoring Body (TMB) The Agreement on Textiles and Clothing – [World Trade Organization website](#), last accessed on 29 September 2023. Marrakesh Agreement Establishing the World Trade Organization – [World Trade Organization website](#), last accessed on 29 September 2023.

⁹⁶ The list of WTO members and dates of membership are available here: https://www.wto.org/english/res_e/booksp_e/sli_e/4wtomembers.pdf

⁹⁷ Textiles: back in the mainstream – [World Trade organization webpage](#), last visited on 29 September 2023.

⁹⁸ Ten types of products were included in the Memorandum of Understanding between EU and China: pullovers, men's trousers, blouses, t-shirts, dresses, bras, flax yarn, cotton fabrics, bed linen, table and kitchen linen.

⁹⁹ EU – China textile agreement 10 June 2005. Available at [this link](#). Last accessed on 29 September 2023.

Asia and other developing regions of the world⁽¹⁰⁰⁾. In 1996, Asia was the predominant global exporter of apparel products, contributing to more than 32% of the world's apparel exports (ILO News, 1996). However, the general lack of transparency of the textile supply chain until the 2010s does not allow a description over time of EU textile outsourcing⁽¹⁰¹⁾.

In September 2008, the financial crisis that originated in the USA rapidly affected major economies due to their interconnections. The EU experienced the Great Recession between 2008 and 2009. Following a brief period of recovery, many Member States subsequently became vulnerable to the sovereign debt crisis (Szczepanski, 2019). This global crisis significantly impacted the European textile industry which had already been hit by offshoring and the increased competitiveness of the Chinese industry after 2005 (Maya Forstater, 2010).

In 2020, the outbreak of the COVID-19 pandemic led to international economic and social disruptions, including the most significant global recession since the time of the Great Depression in the 1930s (Gita Gopinath, 2020). The main effects of the pandemic on the European textile industry were: (a) a decrease in production, (b) the conversion of traditional production, for some companies, into the manufacture of sanitary products or face masks, and (c) an extraordinary growth in sales via online channels (Vet et al., 2021).

In 2022, two main factors negatively affected the competitiveness of the European textile industry. First, the increase of the energy price in Europe which was over six times that in the USA, China, and other Asian countries. In this context, numerous textiles and apparel companies either operated with a net loss or ceased their production activities (EURATEX, 2022b). Second, countries strongly supported their domestic textile industries, despite being minimally affected by the energy crisis (EURATEX, 2022b).

Table 11 sums up the historical events and international agreements affecting the European apparel sector.

Table 11. Main historical events and international agreements affecting the European apparel sector

Year	Event
1974 to 1994	The Multifibre Arrangement (MFA), with the establishment of import quotas, protected the domestic market of developed countries from products produced in developing countries.
1995 to 2004	The Agreement on Textiles and Clothing (ATC) established a progressive removal of import quotas set with the MFA.
2005 onwards	Among members of the World Trade Organization (WTO), any trade of textile products were free of barriers – all import quotas were removed.
2005 to 2007	The Memorandum of Understanding between the EU and China allowed the monitoring of specific types of textile products imported from China. It aimed to prevent market distortions in the European textile sector.
2008	The Great Recession impacted the European textile sector.
2020	The COVID-19 pandemic led to global recession.
2022	The price of energy strongly increased in EU.

Source: own elaboration based on World Trade Organization website

Global figures

Globally, the apparel sector includes two thirds of the textile industry (EURATEX, 2020; EURATEX, 2022a). It is part of one of the largest industries in the world, if teamed up with the footwear sector in the fashion industry (McKinsey & Company and BOF, 2016). **Table 12** reports the effects of the COVID-19 pandemic on the size of the fashion market from 2019 to 2021. The fashion market shrank about 4% and 21% at global and European scale, respectively. **Table 13** lists the largest global exporters and importers of apparel in 2019 in terms of value as reported by the World Trade Organization. A more detailed description of producing countries at specific phases of the apparel value chain is provided by a study published by the United Nations Environmental Programme (**Figure 3** and **Figure 4**)⁽¹⁰²⁾.

- The apparel export market is dominated by China and the EU, followed by Bangladesh, Vietnam, India and Türkiye.

¹⁰⁰ Globalization Changes the Face of Textile, Clothing and Footwear Industries. Available at [this link](#). Last accessed on 29 September 2023.

¹⁰¹ Follow the Thread - The Need for Supply Chain Transparency in the Garment and Footwear Industry. Available at [this link](#). Last accessed on 29 September 2023.

¹⁰² It is not clear to the authors which parameter, e.g. value, mass or any other, was used to develop the percentages reported in **Figure 4**, which is Figure 5 on page 16 of UNEP (2020).

- 1173 — The apparel import market is dominated by the EU and the USA, followed by Japan, the UK, Hong
1174 Kong, Canada and the Republic of Korea.
- 1175 — The role of China is prominent in all production phases of the value chain.
- 1176 — The role of the EU is prominent both in terms of exports and imports.

1177 **Table 12.** Market data of the apparel and apparel & footwear industries at global and European scale

Industry	Market	Economic parameter	Year	Reported value	Value (bn EUR) ^(a)	Source
Apparel and footwear	Global	Retail Sale Price	2019	USD 1 773 bn	1 644.3	(b)
			2021	USD 1 717 bn	1 592.3	
Apparel	European	Turnover	2019	EUR 72.8 bn	72.8 bn	(c)
			2021	EUR 65.3 bn	65.3 bn	(d)

1178 N.B. The retail sale price is the price declared by the producer or importer of the goods after deducting any tax included in that price.
1179 The turnover refers to the totals invoiced by the unit, and it corresponds to market sales of goods.

1180 ^(a) 1 USD = 0.92739 EUR Feb 08, 2024 10:12 UTC.

1181 Source: (b) Fashion United, based on Euromonitor ⁽¹⁰³⁾, (c) EURATEX (2020), (d) EURATEX (2022)

1182 **Table 13.** Top global exporters and importers of apparel in 2019

Rank	Exports			Imports		
	Country	Value (bn EUR)	Share in world exports (%)	Country	Value (bn EUR)	Share in world imports (%)
1	China ^(a)	141.0	30.8	EU	88.1	18.1
2	EU	39.9	8.8	USA	88.1	18.2
3	Bangladesh ^(b)	31.5	6.8	Japan	27.8	5.7
4	Vietnam ^(b)	28.7	6.2	United Kingdom	24.1	5.0
5	India	15.8	3.5	Hong Kong, China	10.2	2.1
6	Türkiye	14.8	3.2	Canada ^(c)	10.2	2.1
7	Hong Kong, China	11.1	NA	Republic of Korea	10.2	2.1
8	United Kingdom	8.3	1.8	China ^(a)	8.3	1.7
9	Indonesia	8.3	1.7	Russian Federation ^(c)	7.4	1.5
10	Cambodia ^(b)	8.3	1.7	Switzerland	7.4	1.5

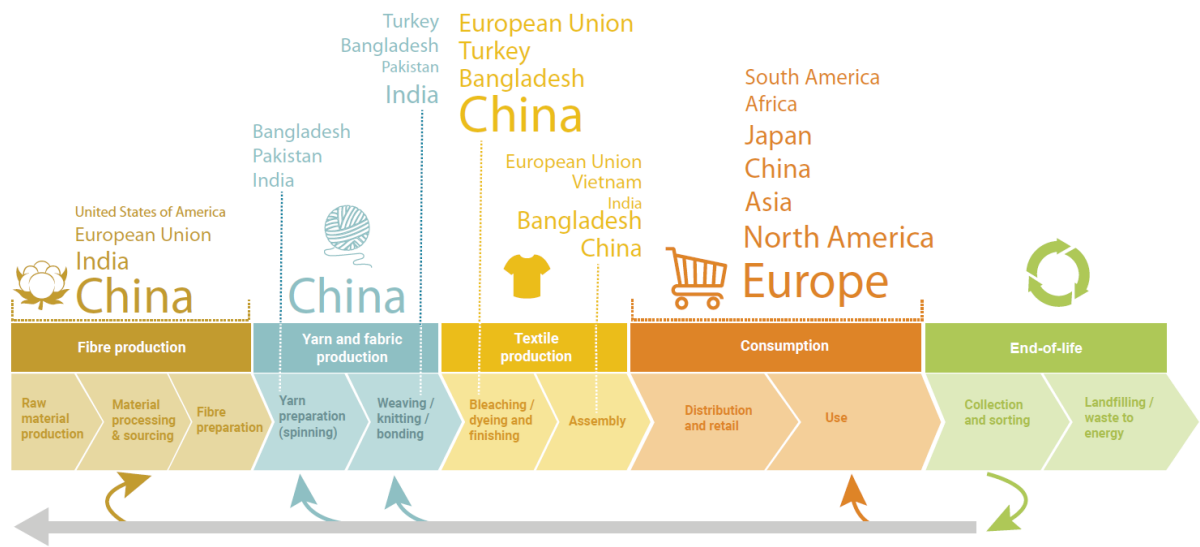
1183 ^(a) Includes significant shipments through processing zones. ^(b) Estimates of WTO. ^(c) Imports are valued free on board.

1184 1 USD = 0.92739 EUR Feb 08, 2024 10:12 UTC.

1185 Source: (WTO, 2020)

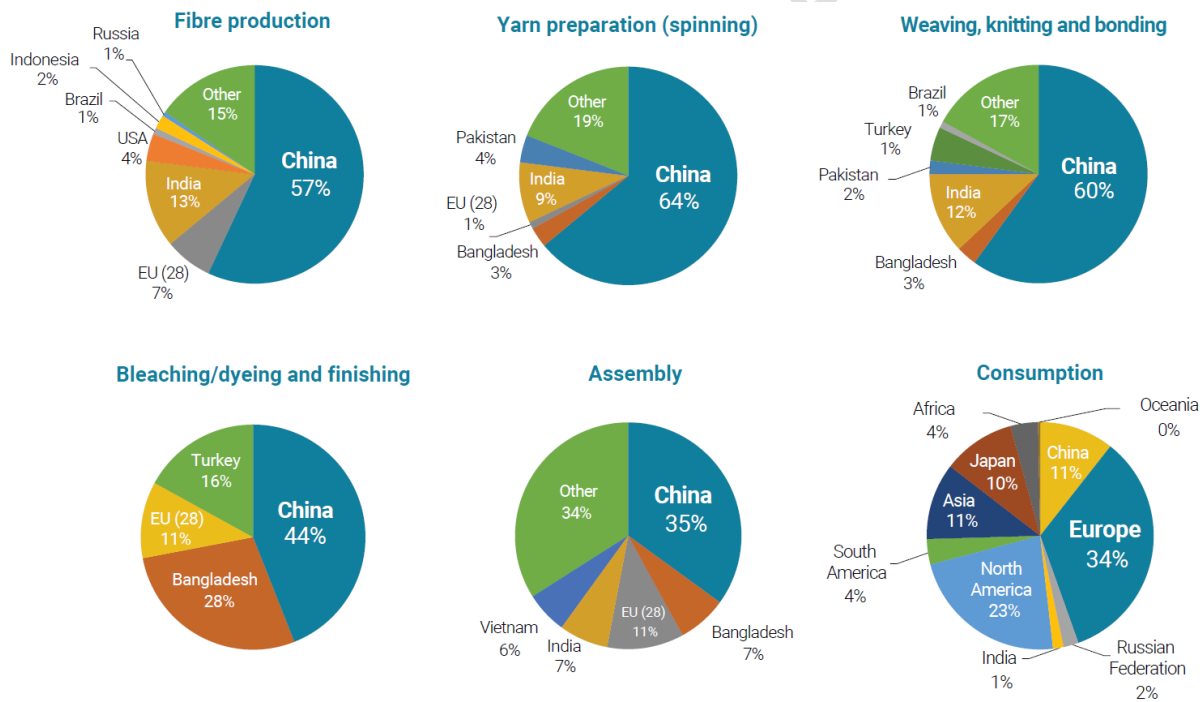
¹⁰³ Information collected from the website Fashion United, Global Fashion Industry Statistics, available [here](#), Euromonitor International, a market research provider, last accessed on 24 October 2023.

1186 **Figure 3.** Geographical breakdown of global apparel production and consumption – representation A



1187
1188 N.B. The European Union is different to Europe.
1189 Source: (UNEP, 2020)

1190 **Figure 4.** Geographical breakdown of global apparel production and consumption – representation B

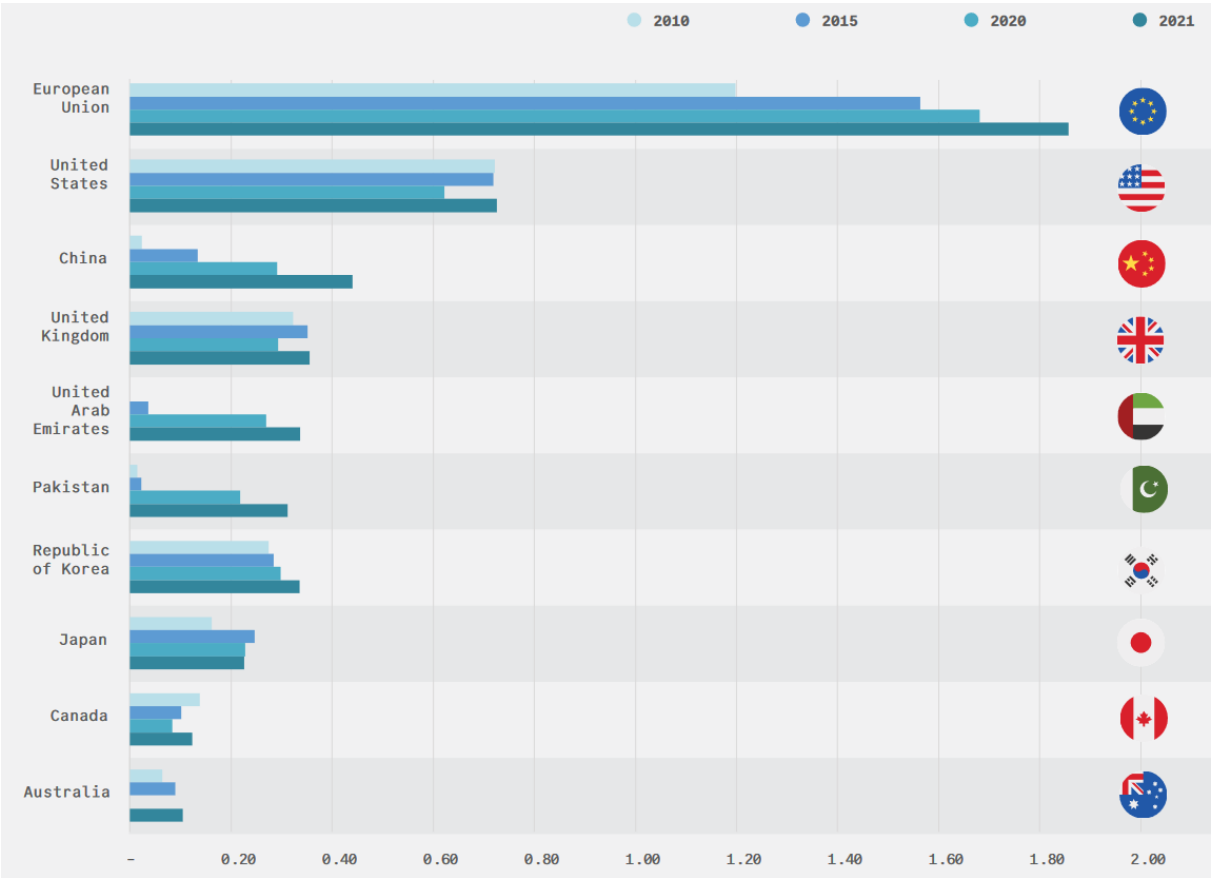


1191
1192 N.B. The EU figures include the United Kingdom. The EU is different to Europe.
1193 Source: (UNEP, 2020)

1194 A global mapping of the textile waste management is currently not available. However, the United Nations and
1195 the European Environment Agency investigated the fate of used textile products (Lingås et al., 2023; UNECE
1196 and ECLAC, 2024). **Figure 5** and **Figure 6** show the most exporting and importing countries in the world of
1197 second-hand apparel in terms of mass. The European Union is the largest exporter, while Pakistan is the largest
1198 importer. **Figure 6** shows that the European Union is the second largest importer, most probably because there

1199 are five countries (¹⁰⁴) in EU that export to countries outside EU what is collected locally and what it is imported
1200 by other EU countries (Lingås et al., 2023).

1201 **Figure 5.** Top ten exporting countries of second-hand apparel by mass (million tonnes)



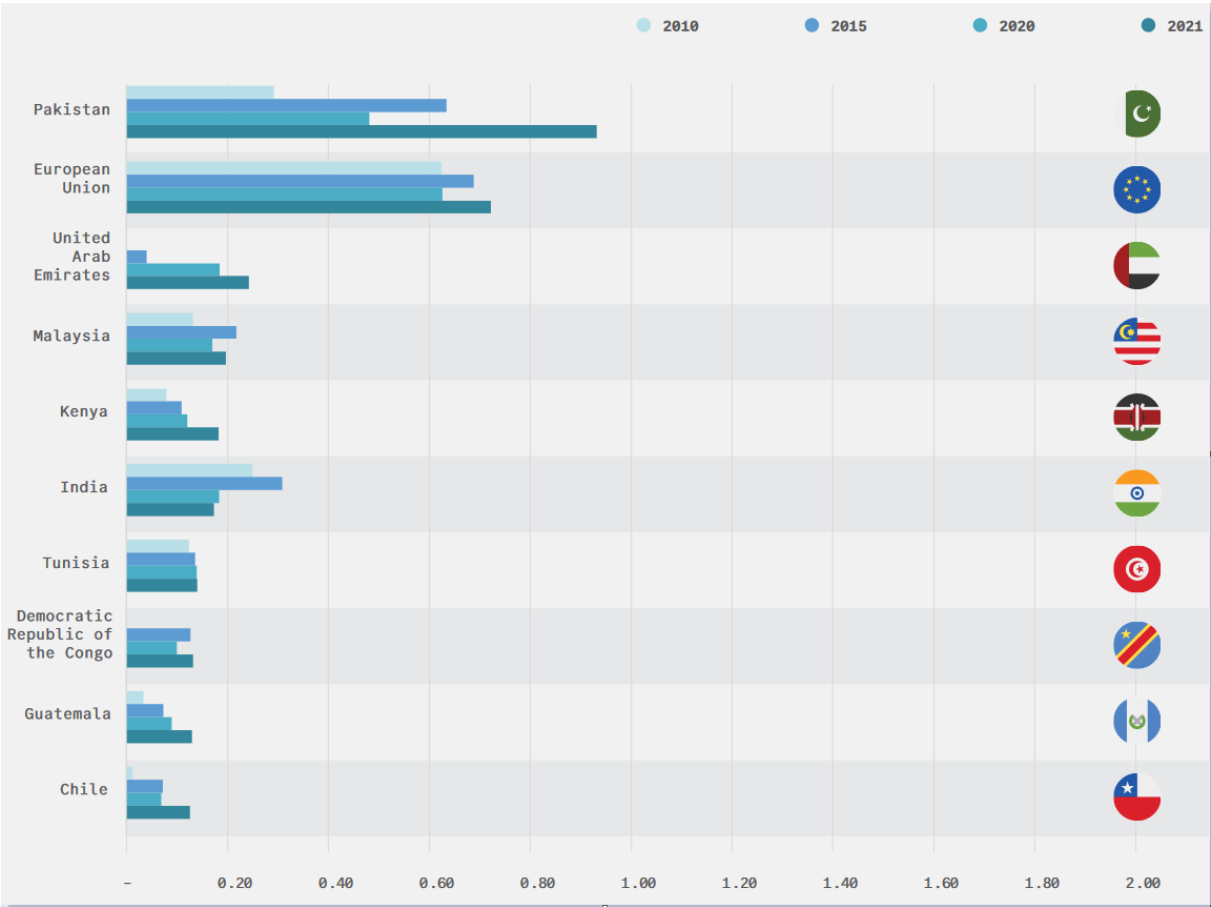
1202 Source: (UNECE and ECLAC, 2024) based on UN Comtrade - HS Code 6309: Worn Clothing and Other Worn Textile Articles
1203

DRAFT

¹⁰⁴ Overall Belgium, Germany, Italy, the Netherlands, and Poland, are the hubs responsible for most of the exports (Lingås et al., 2023)

1204

Figure 6. Top ten importing countries of second-hand apparel by mass (million tonnes)



1205

1206

Source: (UNECE and ECLAC, 2024) based on UN Comtrade – HS Code 6309: Worn Clothing and Other Worn Textile Articles

1207

Over the last 20 years, the EU exported an increasing amount of used textile waste:

1208

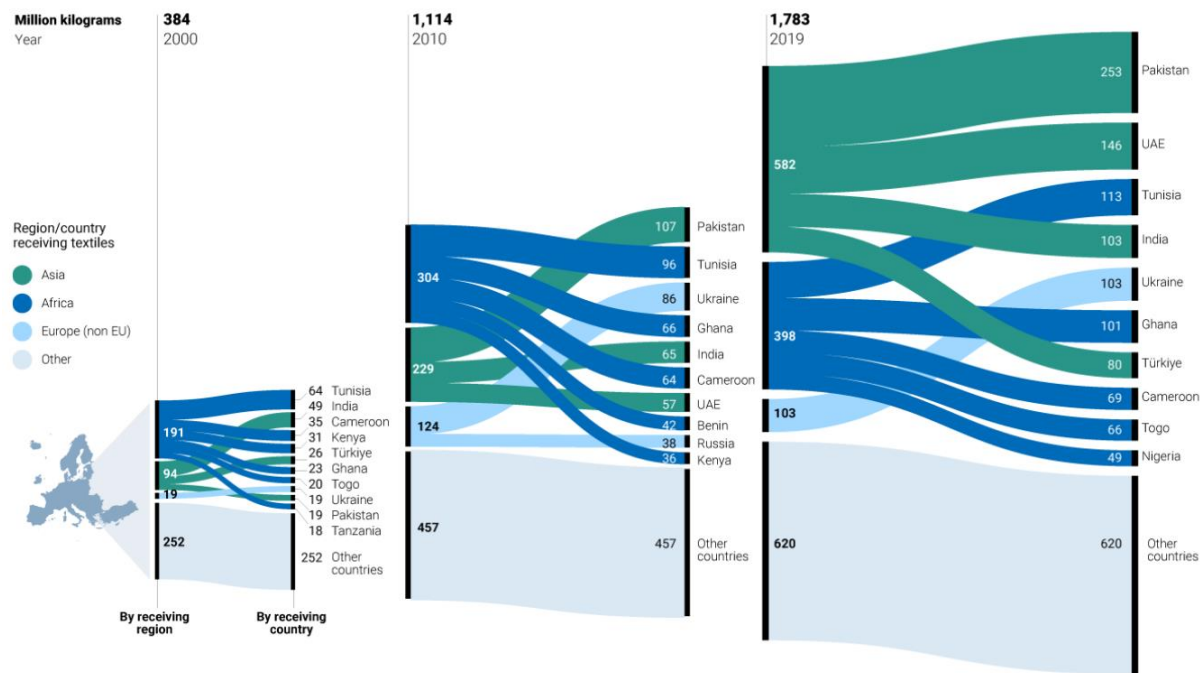
1209

1210

The export of used textiles from the EU has steadily risen over the last 20 years, from a little more than 384 000 tonnes in 2000, to almost 1.7 million tonnes in 2019 (**Figure 7**). The value of these exports has, however steadily decreased, from EUR 0.76 in 2000, to EUR 0.57 per kilogram in 2019.

1211

Figure 7. EU-27 exports of used textiles



UAE is an acronym for United Arab Emirates

Source: (Lingås et al., 2023) based on UN Comtrade

1212

1213

1214

1215

1216 EU global partners

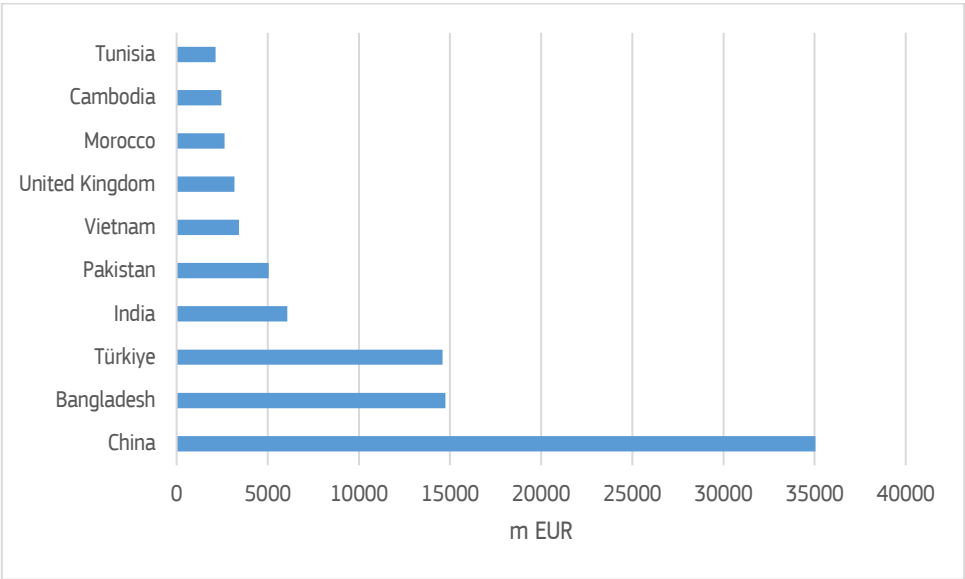
1217 Only a few key partners dominate the EU's trade interactions. In 2021, the top 10 EU suppliers of textiles and
1218 apparel made up 84% of all EU imports from non-EU countries. In the same year, the top 10 export partners of
1219 the EU purchased 68% of all exports to third countries. (EURATEX, 2022a) ⁽¹⁰⁵⁾: **Figure 8** and **Figure 9** show
1220 the main EU suppliers and customers of textiles and apparel in 2021.

1221 The pivotal role of the EU goes beyond economics due to Europe's consolidated history and reputation in fashion
1222 and design.

¹⁰⁵ EURATEX (2022) reports data as textile and clothing, which includes the following types of textile products: clothing and accessories, industrial and technical textiles, fabrics, home textiles, knitwear, man-made fibres, yarns, underwear, and workwear.

1223

Figure 8. The EU's main suppliers of textiles and apparel in 2021



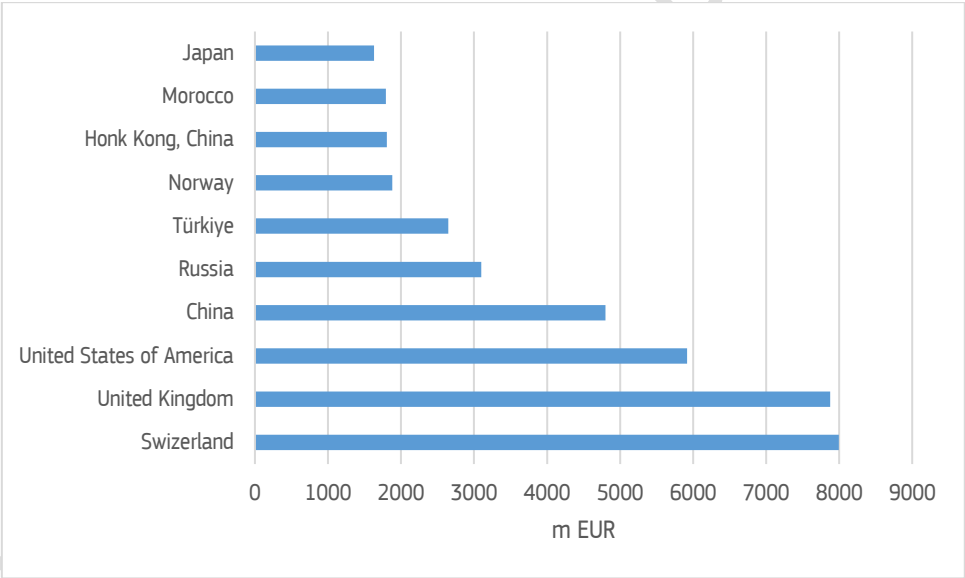
Source: EURATEX (2022a)

1224

1225

1226

Figure 9. The EU's main customers of textiles and apparel in 2021



Source: EURATEX (2022a)

1227

1228

Innovation

The European textile industry is an innovation leader worldwide. In 2019, more than 200 000 industrial designs were released by the EU-27, compared to about 50 000 and 20 000 from USA and China, respectively. Additionally, from 2015 to 2019, about 6 600 patents were filed by the EU-27, compared to about 6 000, 5 800, and 3 900, by the Republic of Korea, the USA and China, respectively (EURATEX, 2022a).

5.2 The EU market

The PRODCOM dataset ⁽³²⁾ collects information about the textile apparel market. This section reports an analysis focusing on the evolution of market indicators of the EU-27 from 1995 to 2022. Details about data used are available in Section 13.1. In particular, **Table 115** describes the codes used for the analysis and their allocation to the specific product categories. Additionally, **Table 116** describes how codes merged or were introduced over the years and specifies potential missing data.

1240 The available data allow an analysis of big trends over time, rather than interpretation of small changes, or
1241 differences, in a specific year.

1242 The textile apparel market in the EU-27 was affected by all the historical events reported in Section 5.1. **Figure**
1243 **10** shows its evolution via four market indicators: production, import, export, and apparent consumption ⁽³³⁾.
1244 These indicators were analysed from the perspective of the mass, the value and the value-to-mass ratio of the
1245 textile apparel. The analysis led to the following observations.

1246 *Apparent consumption*

1247 — Apparent consumption followed the evolution of imports in all analysed perspectives: mass, value
1248 and value-to-mass ratio. This means that the market is largely affected by imported products.

1249 — From 2004 to 2005, the apparent consumption increased by 90% if expressed as mass, and 50%
1250 if expressed as value. This corresponds to the year when import quotas were removed (**Table 11**).

1251 *Production*

1252 — From 2003 to 2009, production decreased by about 53% if expressed as mass and about 35% as
1253 value, most probably due to the expected import quota removal in 2005 and the subsequent
1254 increase in imports. Sector experts revealed that in those years many EU-27 producers relocated
1255 their production to third countries.

1256 — From 2009 to 2019, production evolved relatively constantly, fluctuating between 0.7-0.9 billion
1257 kg, and EUR 23.9-27.9 billion.

1258 *Export*

1259 — With the exception of a disruption around 2003-2004, export followed a relatively constant trend
1260 in terms of mass, and a slightly increasing trend in terms of value.

1261 — Between 2009 and 2019, while production evolved relatively constantly in terms of mass and
1262 value, exports increased by 63% and 110%, in terms of mass and value, respectively. These very
1263 different increase rates led export to overtake production in terms of value from 2015 to 2022.
1264 These figures are possible because, as reported in Section 13.2.2, PRODCOM accounts for every
1265 time a product passes through EU customs. This means that a product could be imported and
1266 subsequently exported without undergoing any mass modification, but with an increased value.
1267 Most probably, after 2015, export has higher values than production because many EU companies
1268 import products that afterwards are exported with a higher value.

1269 *Import*

1270 — The removal of the import quota established the largest change in the EU market of textile apparel.
1271 **Table 14** reports the change that occurred in the apparent consumption when comparing two time
1272 intervals: 1995-2004 and 2005-2019, before and after the EU removed the import quota. The
1273 apparent consumption of textile apparel in general increased by 86% in mass and 27% in value.
1274 This increase was mainly driven by product categories like t-shirts (236% in mass and 133% in
1275 value), pants and shorts (194% in mass and 58% in value), jackets and coats (129% in mass and
1276 14% in value), and shirts and blouses (110% in mass and 38% in value) (**Table 14**). For the
1277 majority of the product categories, the increase in mass is very much larger than the increase in
1278 value. This could flag the purchase every year of more products at lower prices. More details are
1279 reported in **Table 117** in Section 13.2.5.

1280 *Additional observations*

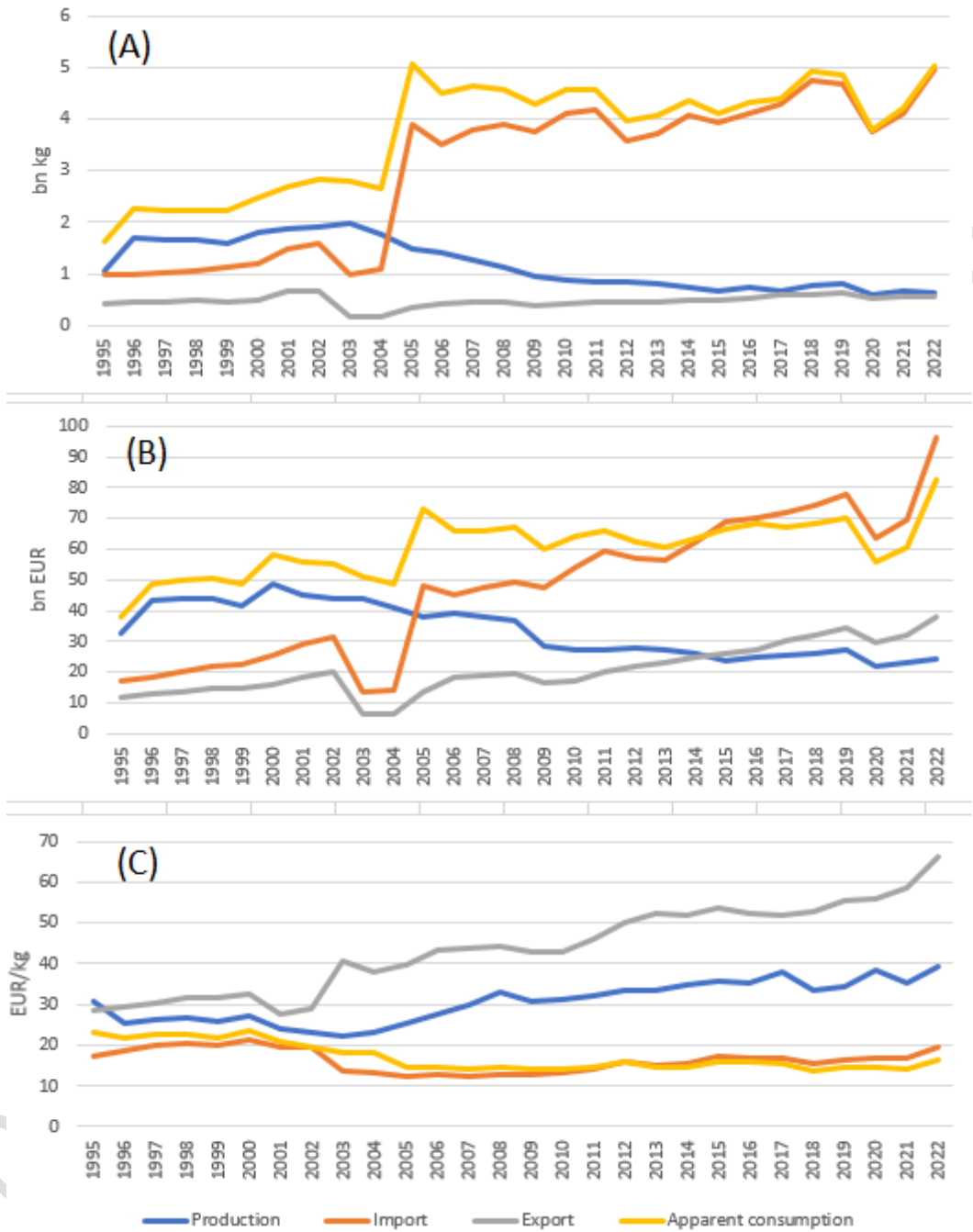
1281 — From 1996 to 2022, the value-to-mass ratio of export, production and import always had the
1282 highest, middle and lowest value each year, respectively. Before 2000, the gap among these
1283 market indicators was limited, but after 2002 the gap progressively increased.

1284 — In 2020, the COVID-19 pandemic strongly impacted the market: production, import and export
1285 decreased by 27%, 20% and 16%, respectively.

1286 The same market indicators were normalised by number of EU-27 citizens over the years to investigate the
1287 possible impact of the change in the EU population. Nevertheless, the analysis provided the same trends over
1288 the years (**Figure 64**).

All textile apparel categories were further investigated. The results of this analysis are reported in Section 13.2.5.

Figure 10. Market indicators for textile apparel in the EU-27



(A) Total mass; (B) Total value; (C) Value-to-mass ratio

Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

Table 14. Change of apparent consumption between the time intervals 1995-2004 and 2005-2019

Product category or subgroup	Change in mass (%)	Change in value (%)
1.T-shirts	236	133
2. Shirts and blouses	110	38
3. Sweaters and mid-layers	86	12
4. Jackets and coats	129	14
5. Pants and shorts	194	58
6. Dresses, Skirts and jumpsuits	33	20
7. Leggings, Stockings, Tights and socks	-33	-13
8. Underwear	15	-13
9. Swimwear	100	13
10. Accessories	11	20
Textile apparel	86	27

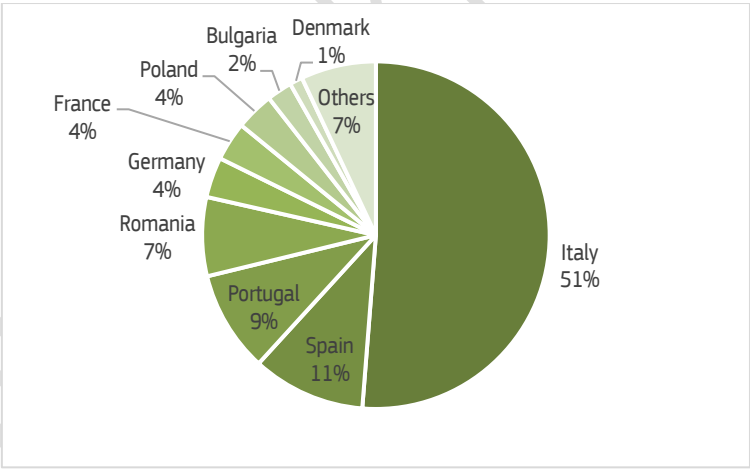
N.B. The change is calculated by comparing the average mass and the average value in the two time intervals. More details are reported in **Table 117** in Section 13.2.5.

Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

5.3 Role of the EU Member States

The PRODCOM database was investigated to understand which Member States play a more relevant role in the textile apparel market. **Figure 11** and **Figure 12** show the largest producers and exporters, respectively. The analysis focuses on 2019, because this is the last year without market disruptions. Italy accounted for half of the EU-27 production, followed by Spain, Portugal and Romania, which covered 11%, 9%, and 7% of the total, respectively. The most relevant exporters in 2019 were Italy, Germany, Spain and the Netherlands, with 19%, 18%, 12%, and 10% of the total, respectively.

Figure 11. Member States producing textile apparel in 2019

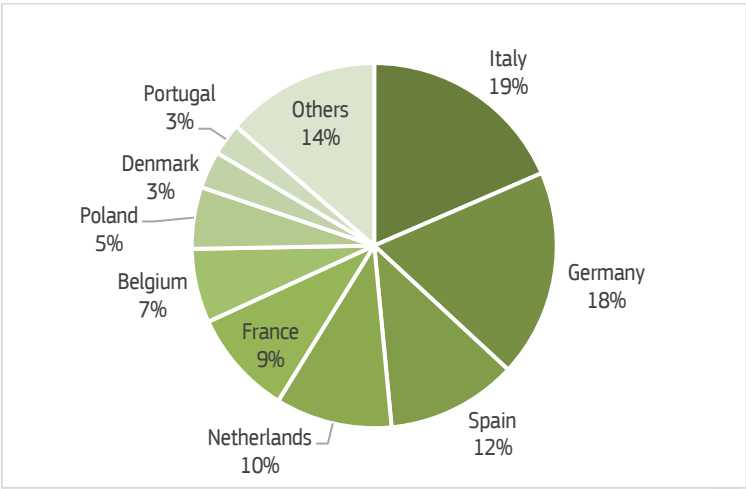


Composition based on value

Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

1311

Figure 12. Member States exporting textile apparel in 2019



Composition based on value

1312

1313

1314

Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

1315

1316

1317

1318

1319

1320

1321

The relevance of these countries is confirmed over the years. A study published in 2021 by the EC recognises Italy, Germany, France and Spain as the most prominent Member States in the textile ecosystem (DG GROW, 2021a). These countries have the largest number of companies and generate the highest value in terms of production and turnover. Additionally, most of the EU companies operating in the textile ecosystem have their headquarters in Italy, Germany, France and Spain. Central and eastern European Member States focus on more labour-intensive activities and generate a smaller share of turnover. Nonetheless, the main EU companies often locate certain production facilities in central and eastern Europe.

1322

1323

1324

1325

The European Apparel and Textile Confederation (EURATEX) confirmed in their last report on key facts and figures that Italy is recognised as the largest contributor to the textile and clothing industry in the EU. Other important countries are Germany, France, Spain, the Netherlands and Portugal. These are also the countries that invest the most in innovation (EURATEX, 2022a).

1326

5.4 Main elements affecting the production of textile apparel

1327

1328

1329

Section 3.3 describes the life-cycle stages of textile apparel, reporting all the main elements that are crucial for production. This section analyses the market characteristics of these elements, which are fibres, chemicals, energy and water.

1330

5.4.1 Fibres

1331

1332

1333

1334

1335

1336

1337

In recent years, the estimates of textile fibre global production followed an increasing trend going from 111 million tonnes in 2019 to 124 million tonnes in 2023 (Textile Exchange, 2020; Textile Exchange, 2024a). In 2023, about 57% of these fibres are polyester, about 20% cotton, about 6% man-made cellulosic fibres and 5% polyamide. In 2023, the production of viscose was equal to 6.3 million tonnes. **Table 15** reports the estimates of textile fibre production for 4 recent years, showing that about 70% are chemical fibres, while about 30% have a natural origin.

1338

1339 **Table 15.** Estimates of global production of textile fibres and group of fibres

Rank	2019 ^(a)			2021 ^(b)			2022 ^(c)			2023 ^(d)		
	Fibre	Mass (m tonnes)	Share of the total (%)	Fibre	Mass (m tonnes)	Share of the total (%)	Fibre	Mass (m tonnes)	Share of the total (%)	Fibre	Mass (m tonnes)	Share of the total (%)
1	Polyester	57.70	52	Polyester	60.50	51	Polyester	63.30	56	Polyester	71.10	57
2	Cotton	25.70	23	Cotton	24.40	20	Cotton	25.50	23	Cotton	24.70	20
3	MMCFs	7.10	6	MMCFs	7.20	6	MMCFs	7.30	6	MMCFs	7.90	6
4	Other plant based	6.50	6	Other plant based	6.70	6	Polyamide	6.20	5	Polyamide	6.70	5
5	Polyamide	5.60	5	Polyamide	5.90	5	Other plant based	6.03	5	Viscose	6.30	5
6	Wool-sheep	1.00	<1	Viscose	5.80	5	Viscose	5.80	5	Other plant based	6.10	5
7	Down	0.27	<1	Polypropylene	3.00		Polypropylene	3.10	3	Polypropylene	3.10	3
8	Other animal-based	0.05	<1	Acrylics	1.7		Acrylics	1.60	1	Acrylics	1.60	1
9	Silk	0.16	<1	Elastane	1.20	1	Elastane	1.20	1	Elastane	1.40	1
10	Acetate	NA	NA	Wool-sheep	1.00	<1	Wool-sheep	1.10	1	Wool-sheep	1.10	<1
11	Acrylics	NA	NA	Acetate	0.90	<1	Acetate	0.90	<1	Acetate	1.00	<1
12	Cupro	NA	NA	Down	0.57	<1	Down	0.61	<1	Flax	0.40	<1
13	Elastane	NA	NA	Lyocell	0.30	<1	Flax	0.38	<1	Lyocell	0.40	<1
14	Flax	NA	NA	Modal	0.20	<1	Hemp	0.30	<1	Hemp	0.20	<1
15	Hemp	NA	NA	Silk	0.17	<1	Lyocell	0.30	<1	Modal	0.20	<1
16	Lyocell	NA	NA	Other animal based	0.05	<1	Modal	0.20	<1	Silk	0.09	<1
17	Modal	NA	NA	Cupro	0.02	<1	Silk	0.09	<1	Other animal based	0.05	<1
18	Polypropylene	NA	NA	Flax	NA	NA	Other animal based	0.05	<1	Cupro	0.01	<1
19	Viscose	NA	NA	Hemp	NA	NA	Cupro	0.01	<1	Down	NA	NA

1340 N.B. Estimates of total production were between: 104 m and 111 m tonnes in 2019; 113 m and 115 m tonnes in 2021; 116 m and 124 m tonnes in 2022.

1341 MMCF: man-made cellulosic fibres; NA: Not available.

1342 Source: adapted from ^(a) Textile Exchange (2020), ^(b) Textile Exchange (2022), ^(c) Textile Exchange (2023)

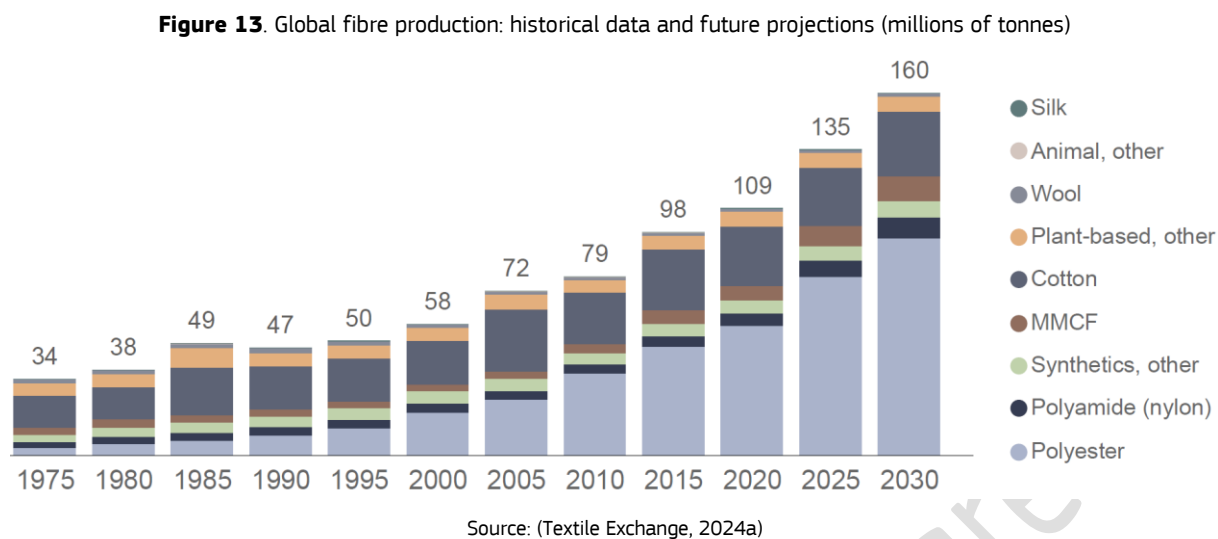


Figure 13 shows the evolution of the production of specific fibres from 1975 to 2020, and it foresees quantities for 2025 and 2030. In the first 25 years (from 1975 to 2000), the total production increased by 71% (from 34 million to 58 million tonnes), whereas in the second 20 years (from 2000 to 2020), the total production increased by 88% (from 58 million to 109 million tonnes). Future projections foresee a further increase up to 160 million tonnes in 2030. The evolution of fibre production is mainly driven by the increase in production of polyester. The data shown in **Figure 13** refer to fibres used for any textile products: textile apparel included in the scope of the PS and textile products not addressed by the PS.

Most of the fibres produced globally are of unknown origin, due to the difficulties of tracing information (see Section 5.6). Nevertheless, the available information shows the global dimension of fibre production, with China and India producing most of the fibres, and many more countries from all continents producing specific fibres (from **Table 16** to **Table 20**).

Table 16. Location of production for specific textile fibres

Polyester (PES)	Cotton	MMCF	Polyamide (PA)	Wool	Other animal fibres
Unknown 57.7% EU 17.6% China 10.7% Others 8.2% Türkiye 5.8%	Unknown 33.8% China 22.7% India 20.7% USA 15.1% Others 6.2% EU 1.4%	Unknown 58% China 18% Others 16% India 5% Indonesia 3%	Unknown 81% China 15% Taiwan 3% USA 1%	Unknown 44.2% Australia 22% Others 17.3% New Zealand 9.4% South Africa 2.9% EU 4.1%	China 88% Unknown 9% Hungary 2% Poland 1%

N.B. MMCF: man-made cellulosic fibres.

Source: adapted from (IWTO, 2022; Textile Exchange, 2022a; Textile Exchange, 2023), Europe polyester fibre market, available at [this link](#). Last visited on 26 May 2025.

Table 17. Production of cotton lint in the season 2021-22

Rank	Country	Mass (1 000 tonnes)	Share of the total (%)
1	India	5 900	22.6
2	China	5 730	21.9
3	USA	3 963	15.2
4	Brazil	2 678	10.2
5	Pakistan	981	3.8
6	Uzbekistan	940	3.6
7	Türkiye	833	3.2

World production was estimated equal to 26 134 000 tonnes

Source: International Cotton Advisory Committee website, available at [this link](#). Last visited on 13 December 2024

Table 18. Import of cotton lint in the season 2021-22

Rank	Country	Mass (1 000 tonnes)	Share of the total (%)
1	China	2 520	24.8
2	Bangladesh	1 688	16.6
3	Vietnam	1 576	15.5
4	Pakistan	1 200	11.8
5	Türkiye	1 170	11.5
6	Indonesia	539	5.3
7	Mexico	178	1.8

World import was estimated equal to 10 153 000 tonnes.

Source: International Cotton Advisory Committee website, available at [this link](#). Last visited on 13 December 2024

Table 19. Export of cotton lint in the season 2021-22

Rank	Country	Mass (1 000 tonnes)	Share of the total (%)
1	USA	3 375	33.2
2	Brazil	2 064	20.3
3	India	816	8.0
4	Australia	749	7.4
5	Benin	321	3.2
6	Greece	299	2.9
7	Mali	283	2.8

World export was estimated equal to 10 153 000 tonnes.

Source: International Cotton Advisory Committee website, available at [this link](#). Last visited on 13 December 2024

Table 20. Consumption of cotton lint in the season 2021-22

Rank	Country	Mass (1 000 tonnes)	Share of the total (%)
1	China	8 200	31.4
2	India	5 698	21.8
3	Pakistan	2 152	8.2
4	Bangladesh	1 660	6.4
5	Türkiye	1 617	6.2
6	Vietnam	1 541	5.9
7	Uzbekistan	836	3.2

World production was estimated equal to 25 629 000 tonnes

Source: International Cotton Advisory Committee website, available at [this link](#). Last visited on 13 December 2024

As mentioned in Section 3.1.1, the ESPR establishes two product aspects that address recycled material: recycled content and possibility of recycling. Within this framework, information about current recycled material and recycling plants is crucial for the potential future development of requirements within these two product aspects.

The current global availability of recycled fibres is very limited (**Table 21**). The highest share of recycled material is available for polyester (15%) and wool (6%) fibres. However, almost all recycled polyester fibres come from recycling of plastic bottles, which are made of a specific type of polyester that is called polyethylene terephthalate (PET) (Textile Exchange, 2022a). From 2020 to 2023 the estimates of recycled fibres were relatively constant (**Table 21**).

Table 21. Estimated percentages (%) of recycled fibres in recent years

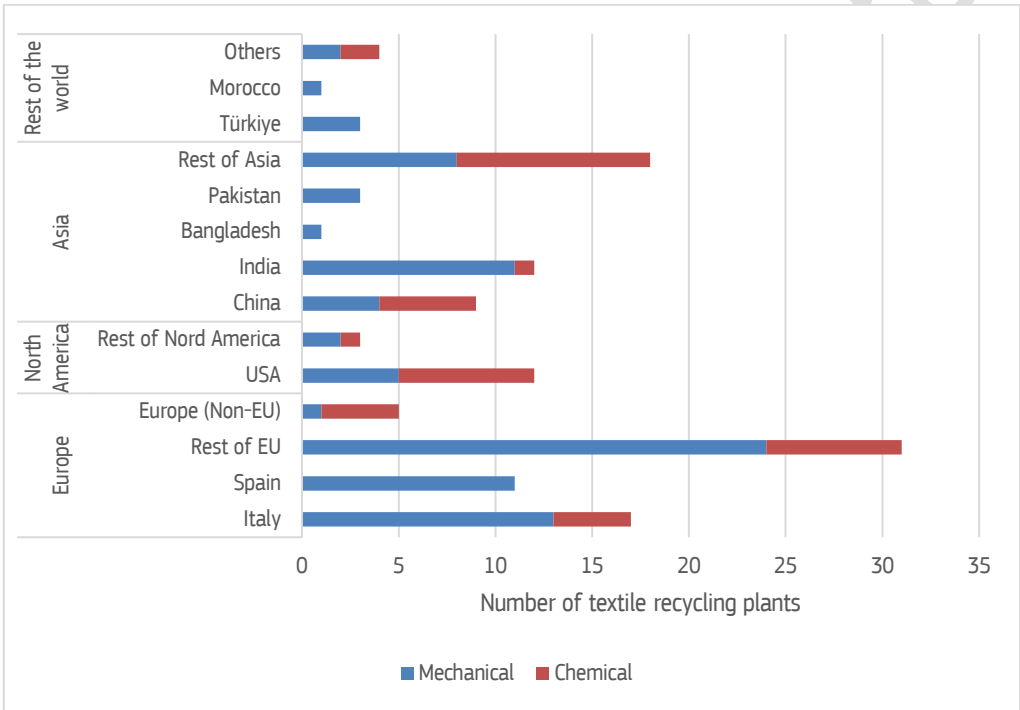
Fibre	2019	2020	2021	2022
Polyester (PES)	14	15	15	14
Wool	NA	6	6	7
Polyamide (PA)	NA	2	2	2
Cotton	NA	0.96	1	1
MMCF	NA	0.4	0.5	0.5

N.B. Approximately 99% of the polyester recycled fibres come from plastic bottles made of polyethylene terephthalate (PET).

Source: Own elaboration based on Textile Exchange (2020), DG GROW (2021b), Textile Exchange (2021), Textile Exchange (2022), Textile Exchange (2023)

The current availability of textile recycling plants was investigated in terms of location, technologies, possible input fibres and scale of the plants (pilot or full scale). **Figure 14** shows the location of these 130 recycling plants, highlighting consuming and producing countries (see Sections 5.1 and 5.3). Europe and North America, which are the largest consumers, host about 50% and 11% of the global textile recycling plants, respectively. Europe hosts mainly mechanical recycling plants, whereas North America mostly has chemical recycling plants. Italy and Spain are the largest EU producers and the countries with the largest number of textile recycling plants. Asia, which is the continent producing the most global apparel, has 33% of the textile recycling plants. In particular, China and India, the world's largest producers, host only 7% and 9% of the global textile recycling plants. **Figure 14** shows that there are currently more textile recycling plants in countries that consume the most apparel.

Figure 14. Current number of textile recycling plants classified by location



Source: own elaboration based on Airtable - Sorting for Circularity - Recyclers Database ⁽¹⁰⁶⁾, (Jørgensen et al., 2022;)

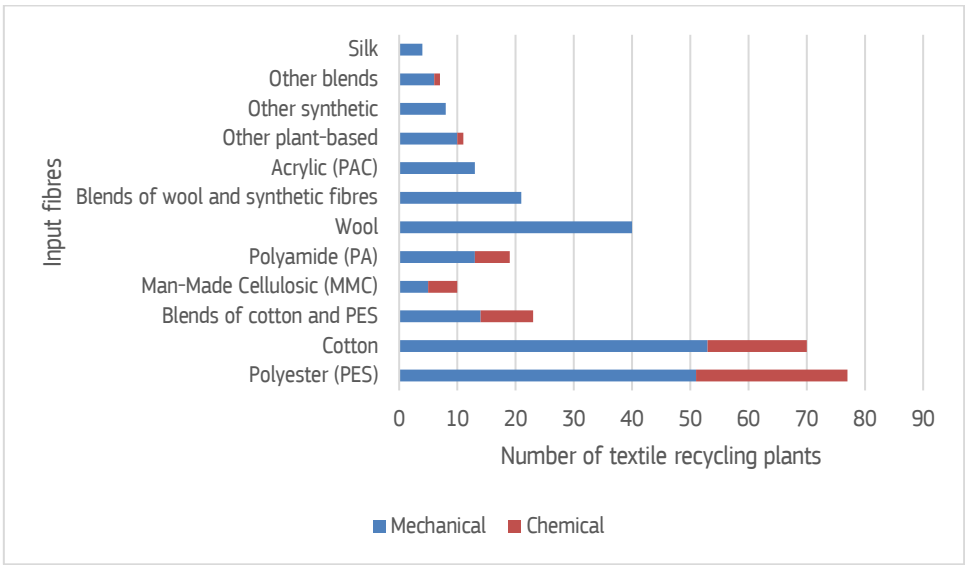
Figure 15 shows that some recycling plants are already capable of processing textile products made of many textile fibres. **Figure 15** shows that most of the investigated recycling plants are capable of processing polyester and cotton fibres, as well as wool fibres. Additionally, **Figure 16** shows that most of the recycling plants are full-scale facilities. This is in line with the availability of recycled fibres reported in **Table 21**.

More detailed data about current textile recycling plants is available in Section 13.6.1.

¹⁰⁶ Airtable - Sorting for Circularity - Recyclers Database. Available at [this link](#). Last accessed on 31 January 2024.

1404

Figure 15. Current number of textile recycling plants classified by input fibre



1405

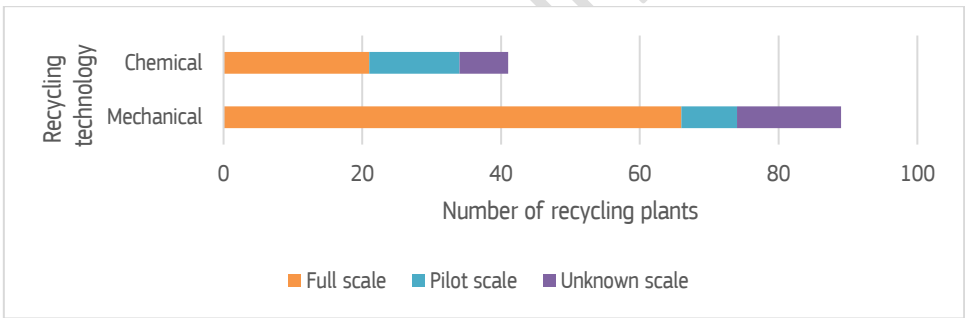
1406

1407

Source: own elaboration based on Airtable - Sorting for Circularity - Recyclers Database ⁽¹⁰⁷⁾, (Jørgensen et al., 2022; Textile Exchange, 2022a)

1408

Figure 16. Scale of current textile recycling plants



1409

1410

1411

Source: own elaboration based on Airtable - Sorting for Circularity - Recyclers Database ⁽¹⁰⁸⁾, (Jørgensen et al., 2022; Textile Exchange, 2022a)

1412

1413

1414

1415

1416

A study of the Joint Research Centre of the European Commission estimated that in the EU approximately 30% of the used and sorted textiles (0.55-0.60 Mt yr⁻¹) are sent for recycling. The resulting recycled fibres are mainly used for cleaning wipes, non-woven material and insulation material, due to the fact that the most abundant resulting recycled material is represented by non-spinnable fibres. In the EU, recycled material comes from post-industrial waste, which represents only 11% of all textile waste (Huygens et al., 2023).

1417

1418

1419

1420

1421

Table 22 reports the current fibre composition of EU textile waste, which is mainly made up of cotton and polyester. **Figure 17** shows the composition of single textile products, which are mainly made of blends of two or more fibres. **Figure 18** shows the composition of textile apparel sampled in the waste stream in three EU countries, and it shows that 53% of the items were made of single-fibre products, 45% were made of two-fibre products, whereas 2% were made of more than two fibres (Bakowska et al., 2025).

1422

1423

1424

1425

1426

The current EU recycling capacity is estimated to be equal to 0.7-0.85 Mt yr⁻¹; whereas the future projection for 2030-2035 is for it to reach 1.2 - 2.7 Mt yr⁻¹, based on the assumption that the textile industry is expected to generate more waste despite the future regulations in place. Based on the future projections for 2030-2035, a 30% increase in textile waste is expected, along with more sorting capacities, increased energy recovery and a reduction in landfilling. The majority of the future textile recycling capacity it is estimated to be based on

¹⁰⁷ Airtable - Sorting for Circularity - Recyclers Database. Available at [this link](#). Last accessed on 31 January 2024.

¹⁰⁸ Airtable - Sorting for Circularity - Recyclers Database. Available at [this link](#). Last accessed on 31 January 2024.

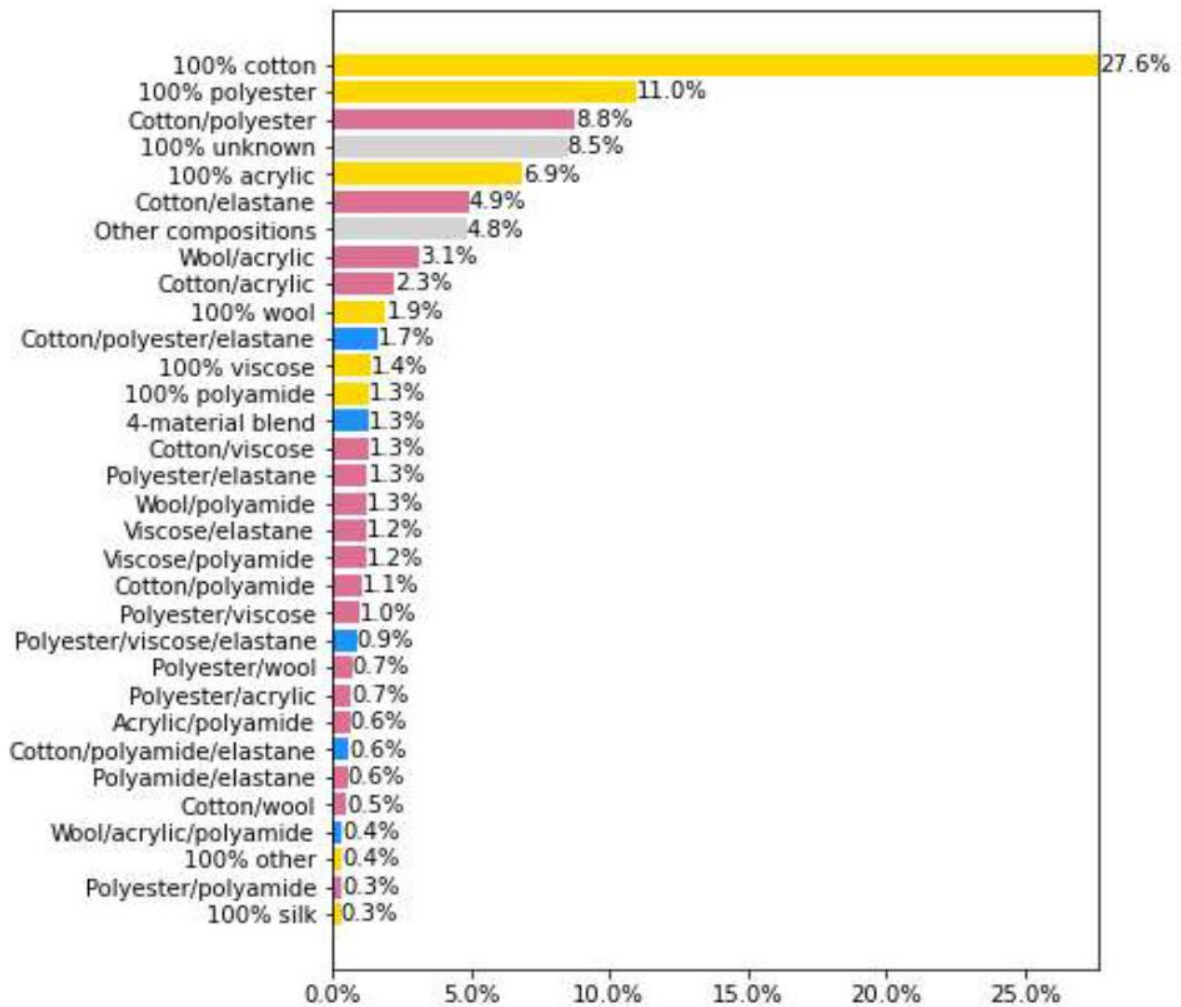
1427 mechanical recycling, while lower volumes are claimed by operators using chemical recycling (Huygens et al.,
1428 2023).

1429 **Table 22.** Material composition of EU post-consumer textile waste

Material	Percentage in the waste fraction (%)
Cotton	34
Polyester	29
Polyamide	7
Wool, polypropylene, acrylic	30
Non-textile	11

1430 *Source: Huygens et al. (2023)*

1431 **Figure 17.** Composition of textile products

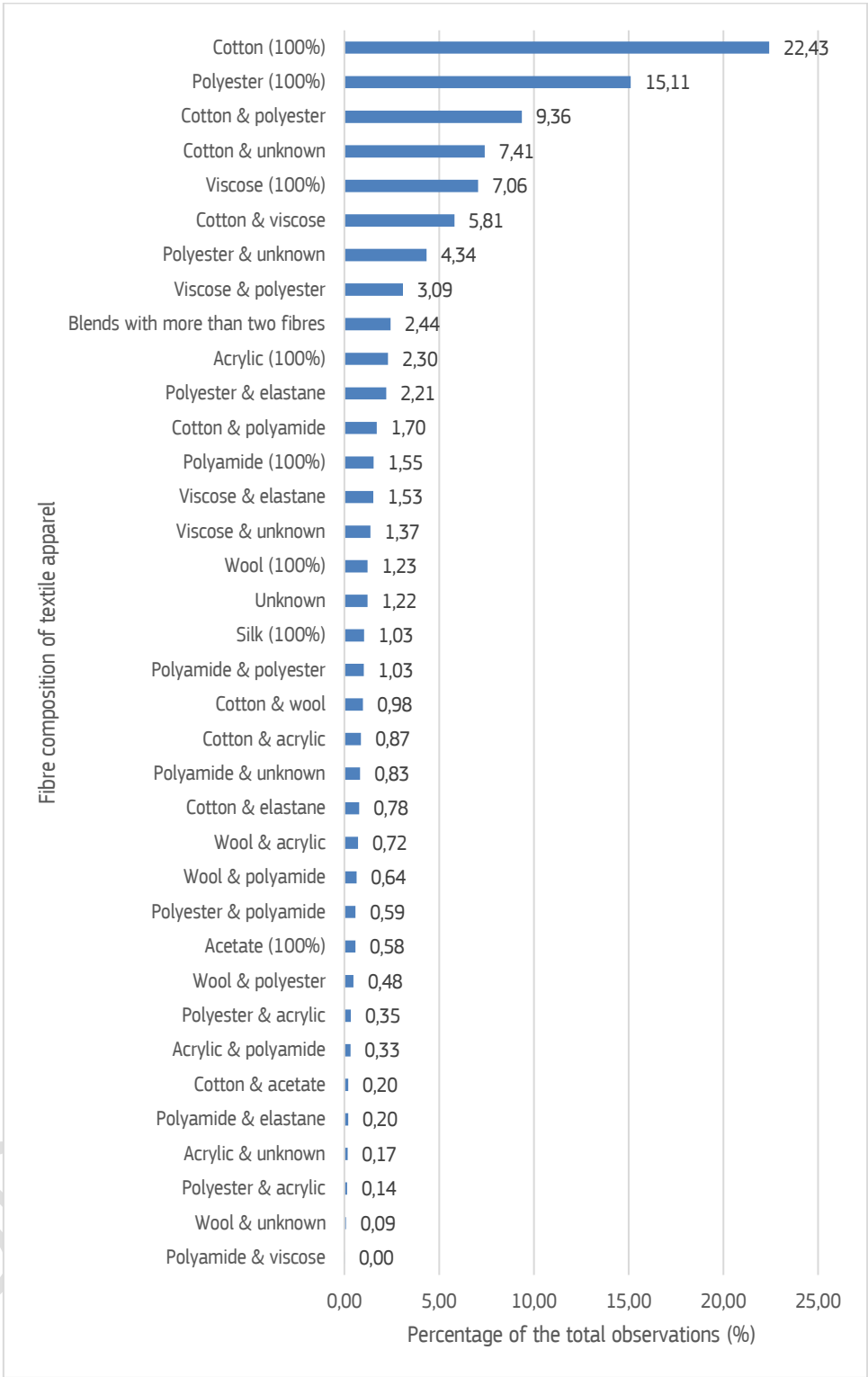


1432
1433 Analysis of textile products incoming to sorting facilities. The sample was composed of about 73% of apparel, 9% of home/interior
1434 textiles, 9% of footwear, and 9% of unclassified items. The colour code marks with yellow mono-fibre products, pink the blends of
1435 two fibres, and blue the blends with more than two fibres.

1436 *Source: (Refashion, 2023a)*

1437

Figure 18. Composition of textile apparel



1438

1439

The sample included 40 212 pieces of textile apparel sampled from the waste stream in Italy, Czechia, and Romania.

1440

Source: (Bakowska et al., 2025)

1441

5.4.2 Chemicals

1442

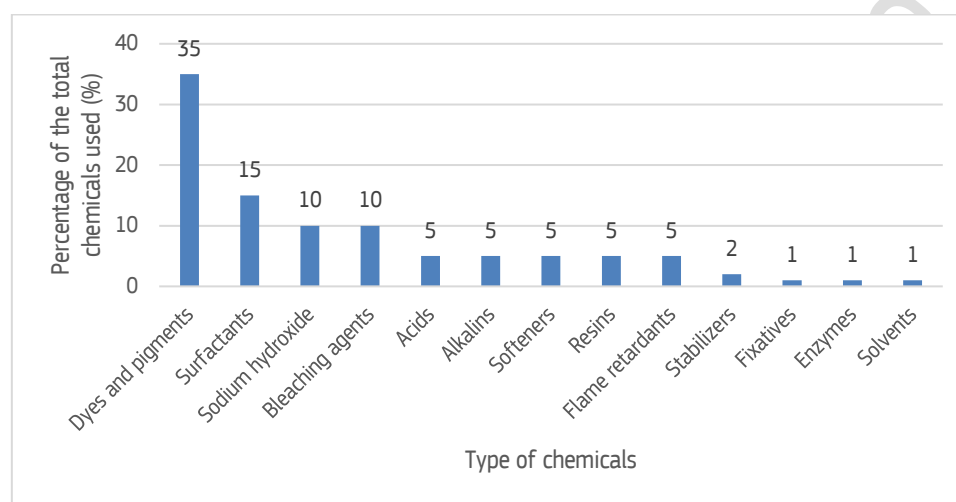
The chemical demand of the textile industry is estimated to use about 25% of the global chemical production, which in 2021 was equal to 9.3 million tonnes (Prasannamedha and Senthilkumar, 2021; Raj et al., 2022).

1443

The Ellen MacArthur Foundation estimated that the textile industry uses approximately 43 million tonnes of chemicals per year. In particular, the production of 1 kg of cotton requires 0.35–1.5 kg of chemicals, and the production of 1 kg of synthetic fibre requires 0.11–0.82 kg of chemicals, besides the polymers making the fibre itself (Ellen MacArthur Foundation, 2017). In general, the manufacturing of 1 kg of apparel usually requires 1–4 kg of chemicals. This range is relatively large because it depends on the type of apparel, the efficiency of the supply chain, and the processes used for production (Muthu, 2020).

Figure 19 shows the chemicals most commonly used by the textile industry. Dyes and pigments, surfactants, caustic soda, and bleaching agents represent 35%, 15%, 10% and 10% of the total consumption, respectively. These four types of chemicals represent about 70% of the total chemicals used (Rahman et al., 2023).

Figure 19. Share of the most commonly used chemicals in the textile industry



N.B. Sodium hydroxide is the caustic soda. Resins are formaldehyde-based.

Source: Own elaboration based on (Rahman et al., 2023)

According to the UN COMTRADE database⁽¹⁰⁹⁾ and the Observatory of Economic Complexity (OEC) database⁽¹¹⁰⁾, the largest global exporters of dyes, identified with HS 32⁽¹¹¹⁾ of the Harmonized System, are Germany, China and the USA, representing about 15%, 10% and 9% of the global exports (**Table 23**).

Table 23. Largest exporters of dyes in 2019

Reporter country	Trade value (m USD)	Share of the global market (%)
Germany	13 036.6	15.6
China	7 719.9	10.2
USA	7 548.3	9.22
Japan	4 624.0	5.92
Netherlands	4 474.6	4.91
India	3 504.3	4.52
United Kingdom	3 450.8	4.14
Italy	3 430.6	4.08
Spain	2 995.8	3.71
Belgium	2 631.8	3.38
Republic of Korea	2 352.4	2.88
Other Asia	1 409.9	1.72

Dyes are identified with the harmonised Standard code HS 32.

¹⁰⁹ UN COMTRADE database. Available at [this link](#). Last accessed on 12 January 2024.

¹¹⁰ Observatory of Economic Complexity (OEC) database. Available at [this link](#). Last accessed on 12 January 2024.

¹¹¹ HS 32: 'tanning or dyeing extracts; tannins and their derivatives; dyes, pigments and other colouring matter; paints, varnishes; putty, other mastics; inks'

Source: elaboration based on UN COMTRADE database ⁽¹⁰⁹⁾ and OEC database ⁽¹¹⁰⁾

5.4.3 Energy

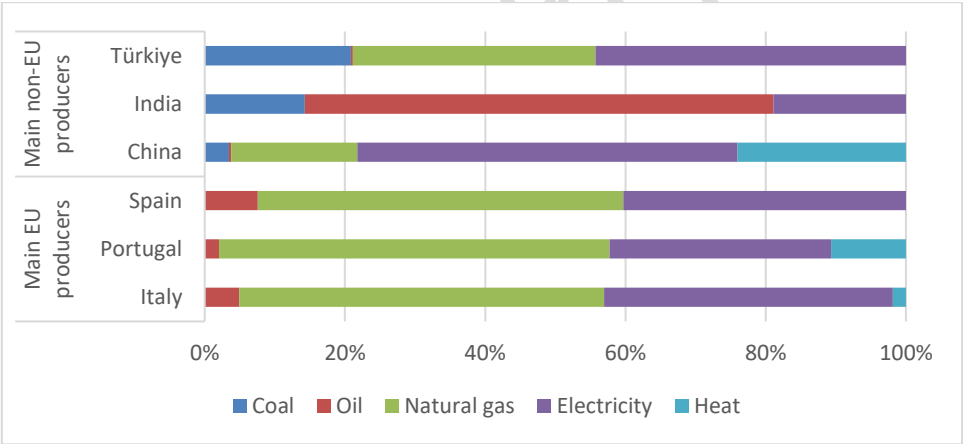
In 2004, the energy consumption in the global textile industry was estimated to be equal to 2% of the global energy consumption. Additionally, the production of 1 kg of generic textile product was estimated to require about 126 MJ (about 35 kWh) of energy (Muthu, 2015).

Table 8 in Section 3.3.1 reports that almost all the stages of the textile value chain require the use of energy. Nevertheless, the manufacturing stages cover about 70-80% of the total life-cycle energy consumption (Sandin, Roos, Spak, et al., 2019; Quantis, 2021).

The energy consumption largely varies according to the country where the textile production process occurs, and its impacts depend on the energy source used (Hasanbeigi and Price, 2012; Muthu, 2020). Figure 20 reports the energy mix of the textile and leather industries of the main producers inside and outside the EU. Meanwhile, **Figure 21** shows the electricity generation by source of the same countries. Both figures show that the energy used in the EU has a very different composition to the energy used outside the EU. The energy used in China and India is mainly generated with coal and oil, which are known to be more polluting than natural gas and other sources of energy.

Specific stages of textile production consume different amounts of energy (Muthu, 2015; Muthu, 2020; Roth et al., 2023). These aspects will be detailed in the following milestone, when the PS addresses tasks 4 (Section 9) and task 5 of the MEErP.

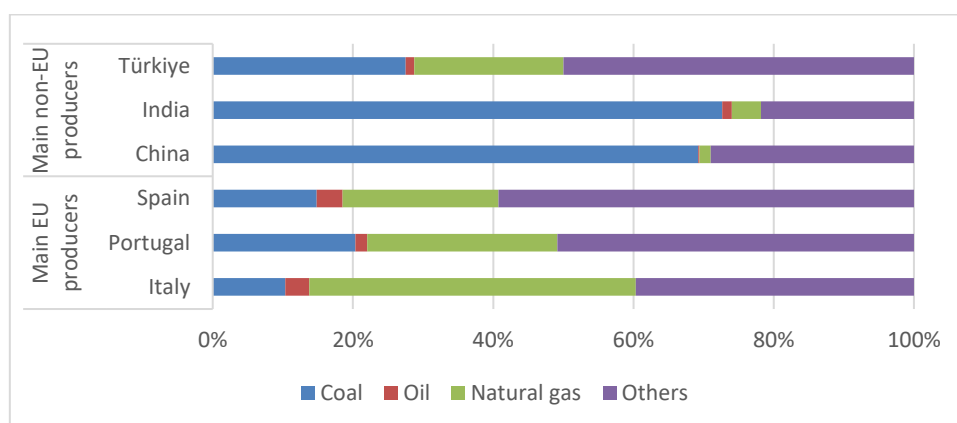
Figure 20. Energy mix of the textile and leather industries in 2021



Source: own elaboration based on IEA World Energy Statistics and Balances ⁽¹¹²⁾

¹¹² IEA World Energy Statistics and Balances (database). Available at [this link](#). Last accessed on 10 February 2024.

1483

Figure 21. Electricity generation by source

N.B. The group 'Others' includes nuclear power, hydropower and renewable sources.

Source: own elaboration based on IEA (2019)

1484

1485

1486

1487 In the manufacture of apparel, the cost of energy also plays an important role. As already mentioned in Section
 1488 5.1, the increase of the cost of energy in the EU in 2022 negatively affected the Union's textile production.
 1489 **Table 24** reports the price of electricity for the countries where most of the production and consumption occurs.
 1490 The data gathered report that the electricity price at industrial scale is more expensive in the EU than in China
 1491 and India.

Table 24. Electricity prices for selected countries for the base year 2019

Country	Residential - Electricity price (USD/MWh)		Industrial - Electricity price (USD/MWh)	
	2012	2022	2010	2022
China	NA	82.32	NA	88*
India	69.66	NA	105.51	NA
Türkiye	184.14	89.52	150.92	204.51
EU-27*	227.1	NA	160.38	NA
USA	115.77	150.89	67.89	83.69
Japan	243.76	NA	162.11	NA
Republic of Korea	101.73	106.81	75.65	104.81

N.B. NA: Not available.

Source: UN COMTRADE database ⁽¹¹³⁾ and * Global petrol prices ⁽¹¹⁴⁾

1493

1494

1495 5.4.4 Water

1496 **Table 8** in Section 3.3.1 reports that many stages of the textile value chain require the use of water.

1497 The water demand of the textile industry was estimated to be equal to about 79 billion m³ per year. Most of
 1498 this water is attributed to cotton production. In general, the manufacture of 1 tonne of textiles is associated
 1499 with the consumption of 200 tonnes of fresh water (Niinimäki et al., 2020).

1500 Cotton is the most water-demanding among the textile fibres; whereas polyester and polypropylene need the
 1501 least water for their production (**Table 25**). Among the manufacturing processes, the finishing processes are
 1502 the most demanding, requiring between 40 l kg⁻¹ and 80 l kg⁻¹ of fabric (Roth et al., 2023). More details will be
 1503 provided in the following milestone, when the PS addresses tasks 4 (Section 9) and task 5 of the MEeP.

¹¹³ UN COMTRADE database available at [this link](#). Last accessed on 15 December 2023.

¹¹⁴ Global Petrol Prices, available at [this link](#). Last accessed on 15 December 2023.

Table 25 Water consumption of fibres

Fibre	Water consumption (l/kg of fibre)
Cotton	1 559
Wool	530
Man-made cellulosic fibres	92
Hemp	89
Polyester	51 – 71
Nylon (known as polyamide)	185 – 633
Polyethylene and polypropylene (PP)	47 – 32
Acrylic	210
Viscose rayon	640

Source: Muthu (2015), Muthu (2020), Niinimäki et al. (2020)

Due to the large water demand of the finishing processes, the cost of water at industrial scale is a factor affecting its manufacture.

Table 26 reports the tap water price in several apparel-producing and -consuming countries. Assuming that the price difference between industrial and tap water has the same ratio worldwide, **Table 26** reports that the manufacturing of apparel in China and India can be performed with a lower water price than in Europe.

Table 26. Tap water price for selected countries

Country (city)	Tap water price (EUR/m ³)
China (Shanghai and Beijing)	0.36-0.68
India (Bangalore)	0.24
Turkey (Istanbul)	0.85
Europe (cities from 24 countries)	2.80
USA (30 cities)	2.23
Japan (Tokyo)	1.58
Republic of Korea (Seoul)	0.77

N.B. Reference year 2019, with the latest updates in 2021.

Source: Tap water price index ⁽¹¹⁵⁾ and Eur Eau (2020)

5.5 Market structure and business models

Composition

EU companies in the textile value chain are mostly microenterprises, covering all manufacturing stages (**Table 27**). However, medium and large enterprises generate most of the total turnover (**Table 28**).

After the manufacturing stages, textile apparel goes through wholesalers and subsequently retailers ⁽¹¹⁶⁾, enabling their placing on the market and their making available on the market for the first time. In the textile and apparel sector, wholesalers are understood as companies that sell products to other companies in large quantities and at low prices, whereas retailers are companies that sell in small quantities and generally with higher prices than wholesalers. In the EU, retailers in the apparel sector are constituted by a bigger number of enterprises, generate higher turnover, higher production value, and employ more people than wholesalers of apparel and footwear together (**Table 29**). Among the top 120 players of the EU textile ecosystem, 50 are retailers of fashion products and about 38 are wholesalers and agents for fashion products (DG GROW, 2021a).

¹¹⁵ Water Price Index. Available at [this link](#). Last accessed on 11 February 2024. Water Price Index examines and interprets the prices of tap water and bottled water in 120 cities globally. It presents the actual average expenditure for usage and the percent difference of the price from the median value of the dataset in each respective location. The factor 'Tap Water Price (EUR/m³)' refers to the monthly cost for 1 cubic metre of tap water, as indicated by the most recent applicable rate from the database of the International Benchmarking Network for Water and Sanitation Utilities (IBNET), which is available at [this link](#) (last accessed on 11 February 2024).

¹¹⁶ According to the Cambridge dictionary: (1) a wholesaler is a buyer and seller of goods in large amounts to shops and businesses; (2) retailer is a person, shop, or business that sells goods to the public.

Table 27. Share of the number of EU-27 enterprises per NACE group in 2021

Size of enterprises	Economic activity			
	Preparation and spinning of textile fibres [C131] (%)	Weaving of textiles [C132] & finishing of textiles [C133] (%)	Manufacture of other textiles [C139] (%)	Manufacture of wearing apparel, except fur apparel [C141] & Manufacture of knitted and crocheted apparel [C143] (%)
Microenterprises (from 0 to 9 employees)	80 (p)	85 (p)	92 (p)	95 (p)
Small enterprises (from 10 to 49 employees)	13 (ep)	9 (p)	6 (p)	3 (p)
Medium enterprises (from 50 to 249 employees)	6 (p)	6 (p)	2 (p)	2 (p)
Large enterprises (more than 250 employees)	: (c)	2 (pu)	0.3 (p)	0.2 (pu)

N.B. Data from relevant countries like Italy is missing in all the indicators.

p: provisional; ep: estimated, provisional; c: confidential; pu: provisional, unreliable or uncertain data with estimation error from $\pm 1\%$ to $\pm 5\%$; ":" not available.

Source: own elaboration based on EUROSTAT SBS_SC_OVW ⁽¹¹⁷⁾

Table 28. Share of the turnover of EU-27 enterprises per NACE group in 2021

Size of enterprises	Economic activity			
	Preparation and spinning of textile fibres [C131] (%)	Weaving of textiles [C132] & finishing of textiles [C133] (%)	Manufacture of other textiles [C139] (%)	Manufacture of wearing apparel, except fur apparel [C141] & Manufacture of knitted and crocheted apparel [C143] (%)
Microenterprises (from 0 to 9 employees)	10 (p)	7 (p)	13 (p)	19 (p)
Small enterprises (from 10 to 49 employees)	15 (p)	8 (p)	10 (p)	15 (p)
Medium enterprises (from 50 to 249 employees)	48 (p)	41 (p)	44 (p)	28 (p)
Large enterprises (more than 250 employees)	27 (p)	45 (p)	33 (p)	38 (p)

N.B. Data from relevant countries like Italy is missing in all the indicators. Italy is the EU country with the highest turnover and employment share compared with other MS in 2021 (EURATEX, 2022a). In this context, the data presented are indicative but not precise data.

p: provisional.

Source: own elaboration based on EUROSTAT SBS_SC_OVW ⁽¹¹⁷⁾

Table 29. Description of wholesalers and retailers in 2020, according to NACE classes

Economic parameter	Economic activity	
	Wholesale of clothing and footwear [G4642]	Retail sale of clothing in specialised stores [G4771]
Number of enterprises	52 212	262 403
Turnover (m EUR)	120 758.3	139 012.2
Production value (m EUR)	46 047.5	65 744.7
Number of employees	294 255	1 155 453

Source: own elaboration based on data from EUROSTAT SBS_NA_DT_R2 ⁽¹¹⁸⁾.

Besides numerous SMEs, the fashion industry includes large companies, which are usually active in many subsectors of the textile ecosystem, as a consequence of acquisitions and mergers, as well as subcontracting and outsourcing. The largest groups in the fashion industry include several brands, which supply diverse end

¹¹⁷ Enterprise statistics by size class and NACE Rev. 2 activity (from 2021 onwards). Available at [this link](#).

¹¹⁸ Annual detailed enterprise statistics for trade (NACE Rev. 2 G). Available at [this link](#).

1541 markets or propose different product lines that go beyond apparel and footwear. Other companies function
1542 under a single brand, but still with a variety of products (DG GROW, 2021a).

1543 Some large companies are vertically integrated, meaning made up of large groups that design, manufacture
1544 and sell a variety of products. Nevertheless, acquisitions, mergers, subcontracting and outsourcing strongly
1545 define the textile apparel market in a global value chain. The majority of the most important EU companies are
1546 also world players that manufacture, innovate and sell across the globe and acquire (and are acquired by) other
1547 world players (DG GROW, 2021a).

1548 Drawing the boundaries of the EU textile apparel market is not possible because it fully operates in the global
1549 market and value chains. Some important EU companies play a role in distribution, but they are not
1550 manufacturers in the EU. Many of them own hundreds (and in a few cases thousands) of subsidiaries outside
1551 the EU, which are responsible for production and/or retail. At the same time, subsidiaries of non-European
1552 companies are also frequently considered important players in the EU (DG GROW, 2021a).

1553 A recent analysis of the top 120 main players in the EU textile ecosystem identified four broad classes of
1554 companies (DG GROW, 2021a):

- 1555 — high-end luxury brands;
- 1556 — manufacturers and retailers producing mid- and low-end products;
- 1557 — manufacturers of intermediate textile products;
- 1558 — companies specialised in the manufacturing of specific goods.

1559 **General business models**

1560 Companies in the fashion industry follow different models in relation to the intangible value of the product and
1561 the management of its supply chain. As described in Section 3.1.1, textile apparel are products with physical
1562 and symbolic functionalities. Besides the tangible quality of the manufacturing process, textile apparel must
1563 meet symbolic and aesthetic values, known as intangibles. Since these intangibles change rapidly, companies
1564 working in the textile apparel industry try to gain value via the supply chain, reacting efficiently to the
1565 unpredictable changes in consumer tastes and demands. Within this context, textile apparel companies
1566 inevitably rely on outsourcing to compete in the rapidly changing market (DG GROW, 2021a). The reasons of
1567 the rapid change of consumer taste and demand are further investigated in Section 9, where business models
1568 and user behaviour are analysed in the context of all relevant product aspects reported in Article 5 of the ESPR.

1569 Regarding the integration of manufacturing and intangibles, two main models can be identified in the textile
1570 apparel industry (DG GROW, 2021a):

- 1571 — **Consumer-led operation model**, where the requests of the consumer are the centre of the
1572 business model. There is a huge effort to collect customer feedback, and produce what is desired
1573 by the market. In this model, consumers dictate the terms of production and affect the whole
1574 supply chain.
- 1575 — **Brand-led operation model**, where the brand dictates the design and manufacture. There is a
1576 huge effort in strategies and programmes to promote the interest in the brand.

1577 Two other main models can be identified regarding the approaches to the supply chain (DG GROW, 2021a):

- 1578 — **Integrated approach**, where the production is entrusted to internal suppliers and the logistics
1579 aims to quickly react to customer's demands.
- 1580 — **Centralised approach**, where the production is mostly outsourced, and supported by audit and
1581 quality control programmes, which could eventually change contractors.

1582 Companies also apply hybrid forms of these models.

1583 **Distribution, retailing and e-commerce**

According to McKinsey & Company (¹¹⁹), the distribution and retailing channels in the fashion industry are mainly controlled by fashion brands and retailers. **Table 30** describes the main distribution and retail models of companies selling textile apparel.

Table 30. Distribution and retail models of apparel companies

Brand		Retailer	
Only direct to consumer: Fashion brands that cover various stages within the fashion value chain, beyond being solely an apparel brand. This includes managing their own retail operations and potentially establishing their own e-commerce presence, all while avoiding wholesale distribution. (Vertically integrated apparel player)	Mix wholesale/direct to consumer: Apparel brand that sells both directly to consumers (in physical stores and/or e-commerce) and to other retailers. (Hybrid apparel player)	On- and offline: multi-brand retailer with its own brands, typically with a physical store and an online store. (Multibrand retailer)	Only online: Online retailer offering a variety of brands, including its own. (Multibrand pure e-commerce retailer)

Source: Reported from McKinsey & Company (¹¹⁹).

In the last few years, e-commerce has been rapidly increasing, especially for online market places and multi-brand retailers (DG GROW, 2021a). In particular, the EEA used the EUROSTAT database to report a steady increase in the percentage of individuals purchasing online apparel and footwear between 2020 and 2022 (Duhoux et al., 2024). In 2009, the percentage of textile and apparel turnover generated by e-sales was equal to 5% of the total. In the following years, it gradually increased up to 11% in 2020 (EURATEX, 2022a).

Second-hand and rental markets

Second-hand and rental of textile apparel are experiencing fast growth, but are still at the very first stages of development. Their future success requires the adoption of business models based on collaborative consumption practices, e.g. utility-based non-ownership, and redistributed ownership of textile apparel. The lack of consumer awareness is seen as a barrier to the development of these two markets. For the second-hand market, other barriers are the limited durability of clothes and the need for traceability and guarantee of authenticity (DG GROW, 2021a).

Figures about the second-hand market are limited. According to RREUSE (¹²⁰), in 2022, about 17 of their members were involved in the collection, sorting, and reuse of used and waste textiles. Members of RREUSE collect 360 000 tonnes of textiles annually, of which about 15% is reused locally. RREUSE estimates that a social enterprise employs between 20 and 35 people per 1 000 tonnes of textile waste collected. The same organization counts about 2 400 second-hand stores across the EU and beyond.

5.6 Characteristics of the value chain

The value chain of textile apparel is defined as global, long, complex, fragmented and opaque (Cai and Choi, 2020; UNECE and UNTRADE, 2020; Brondino, 2022).

The nodes of the value chain could be divided into two parts:

- a first part including the life-cycle stages from raw material for fibre production until retailing, when the product is placed on the market and it is made available on the market for the first time,
- a second part including use phase and waste management.

The first part of the value chain is composed of at least 15 nodes, i.e. the production of an item of textile apparel involves at least 15 economic operators via the production of raw materials, manufacturing and retailing operations. Companies placing products on the market do not know the full story behind their products – they do not know which economic operators were involved over the whole supply chain and the processes and material/chemicals used. Most of the companies in the value chain can get information from their immediate suppliers, but usually information is lost about suppliers further upstream (UNECE and UNTRADE, 2020). This

¹¹⁹ McKinsey & Company webpage on Retail. – Measuring the fashion world. Achim Berg, Miriam Lobis, Elizabeth Hunter, Felix Rölkens, Patrick Simon, and Hannah Yankelevich. Available at [this link](#). Last accessed on 10 December 2023.

¹²⁰ RREUSE website. Available at [this link](#). Last accessed on 13 December 2025. Reported figures were shared during consultation with stakeholders.

part of the value chain is highly interconnected with nodes spread worldwide at all stages (UNEP, 2020) ⁽¹²¹⁾. Although there are specific countries producing specific fibres, most fibres can be supplied from many parts of the world and mixed together (Textile Exchange, 2022a). Fibre processing as well as yarn spinning and fabric manufacturing occur in several countries and use resources, e.g. chemicals, which are produced in third countries. This was also shown in **Figure 3** and **Figure 4** in Section 5.1.

This description of the first part of the value chain shows that the manufacturing origin of textile apparel should not refer to just one country, but it should refer to the many countries where the main production stages occurred.

Usually, the textile apparel market mainly includes multinational companies based in developed countries that act as retailers or as producers and retailers. These multinational companies generally outsource production in developing countries due to the lower production costs (Brondino, 2022). For the first four stages of the product's life cycle (see **Figure 1** and **Table 8**), developing countries offer the possibility to produce textile apparel facing lower environmental compliance costs than in developed countries (see Section 5.7). Similar conditions occur for the confectioning stage, which is very labour-intensive: developing countries offer cheaper labour costs than developed countries (UNECE and UNTRADE, 2020).

Some studies envisage the possibility of reshoring and/or nearshoring production to developed countries, brought about by increased automation. However, this process is expected to happen only in the long term and subject to the sector investing strongly in this direction (Brondino, 2022).

The textile apparel industry has always operated according to seasons, which are associated with the release of new collections. Over the last decades, the number of seasons has been drastically increasing from two per year to almost one per week ⁽¹²²⁾. This seasonality largely affects the supply chain, from procurement to manufacturing capacity, planning and inventory management.

Within this framework, the first part of the value chain evolves among global dynamics, consumer demands and the strategies of big retailers (Ellen MacArthur Foundation, 2017; UNECE and UNTRADE, 2020; Brondino, 2022)

In order to increase transparency in the textile apparel value chain, in 2019 the United Nations Economic Commission for Europe (UNECE) launched a project for an international framework initiative to enhance transparency and traceability for sustainable value chains in the garment and footwear industry ⁽¹²³⁾.

The second part of the value chain is mainly constituted by the post-consumer textile waste treatment market and by the second-hand market (reuse). The European Environment Agency and the United Nations provide figures about used textiles, mainly textile apparel, exported outside the EU (Lingås et al., 2023; UNECE and ECLAC, 2024). Used textile apparel is part of a specialised and traded global value chain. Over the last two decades, the export of used textiles from the EU has tripled from about 550 000 tonnes in 2000 to almost 1 700 000 tonnes in 2019. These products can have several fates. Usually, what is exported to Africa is first screened in local markets and subsequently most is dumped via informal waste streams. Used textiles reaching Asia are usually downcycled into industrial rags or filling, or re-exported for recycling in other Asian countries or are further sent to Africa for reuse. Products that cannot be recycled or re-exported are likely to end up in Asian landfills (Lingås et al., 2023; UNECE and ECLAC, 2024).

5.7 Competitiveness and environmental compliance costs

The global value chain of textile apparel produces most of its environmental impacts in the production stages (**Table 9**). Usually, the production stages occur in Asian countries (**Figure 4**). This implies that most of the negative emissions to the environment occur outside the EU (**Table 6**), in particular in countries that allow production at lower costs due to poor labour conditions and less stringent measures about environmental protection (UNECE and UN TRADE, 2020).

¹²¹ An exchange with a large global economic actor revealed that a retailer/producer could have more than 1 700 suppliers, which may have more than 8 200 factories located in 50 supply markets.

¹²² Numerous fashion websites report this information. Some of them are: (1) [Digitally Empowering Fashion](#), (2) [InStyle](#), and (3) [techfashionista](#). All websites were last visited on 7 December 2023. The exchange with a large global retailer revealed that this company updates their collections every week to satisfy the requests of their customers.

¹²³ UNECE - Traceability for Sustainable Garment and Footwear. Available at [this link](#). Last accessed on 7 December 2023.

A recent analysis performed by the OECD investigated the requirements set by several frameworks establishing Best Available Techniques (BAT) for Preventing and Controlling Industrial Pollution (OECD, 2022). This publication aimed to gather information on existing BAT reference documents (BREFs) that could stimulate more countries to implement specific requirements in their territories. **Table 31** reports the comparison of environmental aspects that are addressed by several BREFs implemented around the world. The analysis showed the following:

- Comparison is made difficult by the different approaches used by each BREF, in terms of key environmental indicators and stages of production.
- The EU has the most ambitious mandatory system, covering almost all the environmental aspects (7 out of 8).
- Among the largest global producers, the BREFs applied in China and India were analysed ⁽¹²⁴⁾. Both China and India set limits for fewer environmental aspects than the original EU BREF. In particular, India addresses only emissions to water, with less stringent thresholds than the EU BREF. The Chinese BREF addresses more environmental aspects than India (4 out of 8), but it sets less stringent values for emissions to water compared to the EU BREF (**Table 135** in Section 13.6.1). Comparison with values related to emissions to air was not possible due to the different practices and key environmental indicators used by the several schemes.

In this context, it is evident that companies producing textile apparel, or their intermediate products, in the EU must face higher costs than companies producing in China and India due to prevention/reduction of emissions into the environment set by the EU BREF, within the framework of the EU Industrial Emissions Directive ⁽¹²⁵⁾.

Table 31. Environmental aspects covered by the Best Available Techniques (BAT) reference documents (BREFs) for preventing and controlling industrial pollution around the world

Country/organisation Environmental aspects	European Union (EU BREF)	China	India (MINAS)	South Korea	United States (US EPA)	World Bank (EHS Guideline)
Emissions to air	Yes	Yes	No	Yes	Yes ^(a)	No
Emissions to water	Yes	Yes	Yes	Yes	Yes	Yes
Consumption of energy	Yes	No	No	No	No	Yes
Water usage	Yes	No	No	Yes	No	No
Waste generation	Yes	Yes	No	No	No	Yes
Usage and management of chemicals	Yes	No	No	No	No	No
Energy efficiency	Yes	No	No	Yes	No	No
Noise emission	No	Yes	No	Yes	No	No

N.B. MINAS: Minimal National Standard; US EPA: United States Environmental protection Agency; EHS Guideline: World Bank Group Environmental, Health, and Safety Guidelines.

^(a) Covered by the Clean Air Act, National Emission Standards for Hazardous Air Pollutants (NESHAP) (OECD, 2022).

Source: (Ministry of Ecology and Environment, China, 2021; OECD, 2022).

China implemented its scheme on available pollution prevention and control techniques for the textile industry ⁽¹²⁶⁾ in 2021, but it was not included in the analysis of OECD (2022).

Besides the above-mentioned aspects, the current competitiveness in the global textile value chain is also influenced by the cost of energy, water and chemicals, as assessed in Section 5.4.3 and Section 5.4.4. Besides the Industrial Emission Directive, EU industries must comply with requirements established by REACH (Section 9.2.7).

¹²⁴ Information about the Minimum National Standard (MINAS) from India was included in the analysis provided by OECD (2022). Meanwhile, the analysis of the Chinese BREF was performed thanks to a machine translation of the document found on the internet.

¹²⁵ Industrial Emission Directive. Directive 2010/75/EU of the European Parliament and of the Council of 24 November 2010 on industrial emissions (integrated pollution prevention and control) (Recast) (Text with EEA relevance). Available at [this link](#).

¹²⁶ Chinese guidelines on available pollution prevention and control techniques for the textile industry. Available at [this link](#). Last accessed on 11 December 2023.

5.8 Lifespan of textile apparel

The apparent consumption in the EU-27 of textile apparel (**Figure 10** in Section 5.2) focuses on yearly market data, which provide a good understanding of the amount of textile apparel traded over the years. Future steps of the PS will focus on the economic and environmental assessment of textile apparel in the Union. This assessment will take into account the use phase and therefore the lifespan of the products, which affects their demand.

The lifespan of a product can be measured according to different descriptors (Murakami et al., 2010). Table 32 lists the different lifespan descriptors used for textile apparel.

Table 32. Types of lifespans

Type	Definition
Total lifespan	The period during which an item of textile apparel retains its original form, irrespective of its functional condition.
Service lifespan	The time an item of textile apparel remains functional and usable, considering its use by both the initial and subsequent owners. This timeframe initiates upon the product's acquisition by the first owner and concludes when the last owner disposes of it.
Possession span	The period of time in which an item of textile apparel is held by a specific owner. This timeframe does not discriminate whether the item of textile apparel is used subsequently by another person or it is disposed of.
Duration in use	The period of time a single owner utilises the item of textile apparel, considering only the use time (i.e. the time it is worn).
Physical lifespan	The period of time an item of textile apparel can be worn before it exhibits a level of wear beyond what is deemed acceptable.

Source: adapted from Murakami et al. (2010)

The lifespan of textile apparel is affected by many factors related to the specific type of textile apparel, the intrinsic physical properties of the product, and the behaviour of the users. Textile apparel is worn and cleaned with different frequencies according to their specific use, material composition and user choices. This is because specific textile apparel has seasonal use, or are related to specific activities, like sports. Additionally, many textile apparel remain stored and not in use in wardrobes. More investigation about factors related to user behaviour are addressed in Section 6.

The PS will focus on the service lifespan, which considers the time period from the purchase of the new item to its disposal, disregarding the number of users. Currently, there is no established system that allows the direct measurement of the service lifespan of textile apparel. The information available in the literature focuses on the possession span, which refers to the time a single user keeps the item, and the duration in use, which refers to how much time the owner uses the item (Table 32). This information is collected via surveys where users are asked to provide their estimates.

The possession span of textile apparel is often expressed as number of years. The duration in use is expressed as 'days of wear'. **Table 33** and **Table 34** report data gathered from the literature, using both metrics for most of the textile apparel categories. **Table 137** in Section 13.6.3 reports a brief description of the investigated studies.

The figures obtained from the analysis show a relatively large range of values, a consequence of the many factors affecting this parameter. The analysis addressed only specific products belonging to specific product categories. Section 13.6.3 provides additional, more granular information on the collected data.

Users and experts perceive that over the last 20 years the service lifespan of textile apparel has decreased. Some experts estimated that it decreased by 36% (Ellen MacArthur Foundation, 2017). The lack of a direct measurement method does not allow the provision of a better understanding of this important parameter. Nevertheless, educated assumptions will be made to be used in the stock model that will be produced in the following milestone, when the PS addresses task 7 of the MEeRP.

1730

1731 **Table 33.** Possession span of textile apparel expressed in years

Textile apparel category	Gray et al. (2022)	Laitala and Klepp (2020)	Laitala et al. (2018)	WRAP (2017a)	Drycleaning Institute (2015)
1. T-shirts	4	4.6	3.3-6.8	4.5	NA
2. Shirts and blouses	4.1	4.8	3.3-7.2	5 (Shirts)	2-3
3. Sweaters and mid layers	4.4 (Sweatshirts and hoodies)	6	3.7-10.8	NA	3-4
4. Jackets and coats	6.2 (Non-padded) 5.4 (Padded)	7 (Coats) 6.8 (Jackets)	4-11.6 (Coats) 4-11.5 (Jackets)	NA	2-4
5. Pants and shorts	4.8 (Shorts) 4.3 (Trousers) 4.1 (Jeans)	4.7 (Pants) 3.5 (Jeans)	2.5-6.2 (Pants) 2.5-4.3 (Jeans)	4 (Jeans)	2-4
6. Dresses, skirts and jumpsuits	4.9 (Skirts) 4.6 (Dresses)	7.1 (Dresses) 6.9 (Skirts)	4.1-15.2 (Dresses and skirts)	NA	1-3 (Dresses) 2-4 (Skirts)
7. Leggings, stockings, tights and socks	3.8 (Leggings) 2.9 (Socks and hosiery)	2.6 (Socks)	1.8-3.6 (Socks)	2.5 (Socks)	1 (Socks)
8. Underwear	2.7	3.5 (Bras) 3.1 (Underpants)	2.4-4.4	NA	1-2
9. Swimwear	NA	NA	NA	NA	2
10. Accessories	NA	NA	NA	NA	1-2

N.B. NA: Not available.

Beton et al. (2014) reports that apparel has a lifespan between 1 and 3 years, based on expert opinions.

Knitwear products were reported to have a possession span of 4.8 years (Gray et al., 2022) and 5 years (WRAP, 2017a).

All investigated studies address specific products belonging to specific textile apparel categories.

Source: own production

Table 34. Duration in use of textile apparel expressed in days of wear

Textile apparel category	Roos et al. (2015)	WRAP (2017a)	Klepp et al. (2020)	PEFCR (2022)
1. T-shirts	22	112.5	90.5	45
2. Shirts and blouses	NA	80 (Shirts)	90.5	40
3. Sweaters and mid layers	NA	NA	90.5	85
4. Jackets and coats	100 (Jackets)	NA	90.5	100
5. Pants and shorts	200 (Jeans)	300 (Jeans)	90.5	70 (Jeans)
6. Dresses, skirts and jumpsuits	10 (Dresses)	NA	90.5	70
7. Leggings, stockings, tights and socks	NA	125 (Socks)	90.5	50 (Socks) 70 (Leggings/tights) 50 (Hosiery)
8. Underwear	NA	NA	90.5	60
9. Swimwear	NA	NA	NA	30
10. Accessories	NA	NA	NA	100

N.B. NA: Not available.

Days of wear of knitwear products were reported to be equal to 150 (WRAP, 2017a) and 90.5 Klepp et al. (2020).

All investigated studies address specific products belonging to specific textile apparel categories.

Source: own production

5.9 Market penetration of environmental labels

Although numerous environmental labels are used in the textile sector, currently there is no direct method to quantify their market penetration. Therefore, it is not possible to understand the real success of these labels and the specific reasons behind their success.

Such information could be helpful for the revision of the EU Ecolabel criteria for textile products. The following milestone of the PS will address the potential inclusion of information requirements for products awarded with one or more environmental labels. The required information could also include a description of the specific environmental label in the Digital Product Passport.

6 User behaviour

6.1 Introduction

The aim of this section is to present and analyse information available in the literature regarding behavioural trends among users with regards to apparel.

The methodology used in the user behaviour analysis is twofold. First, a comprehensive screening and literature review of scientific papers and other studies on user behaviour related to apparel was conducted. The main goal of the systematic literature review was to identify the topics usually covered by studies on user behaviour regarding apparel. As a second step, several questions related to the most common aspects explored by user behaviour studies were prepared. These questions were included in the first stakeholder consultation through an online questionnaire ⁽¹²⁷⁾. Stakeholder references to additional scientific papers, consumer surveys and other relevant documents were analysed in order to complement the literature review carried out in the first step.

The information was analysed considering its potential use when modelling the user phase in the environmental and economic analysis taking place at a later stage of the preparatory study (task 5 of the MEErP), and also its connection with potential **ecodesign requirements** to be developed within the aspects listed in Article 5(1) of the ESPR as well as with potential **EU Ecolabel** and **GPP criteria**.

The user behaviour chapter covers aspects such as the way users choose to buy apparel, their habits during the use phase, and why they decide to dispose of. It distinguishes user behaviour aspects at pre-purchase (section 6.2.), post-purchase (section 6.3.) and disposal stages (section 6.4.).

The table below indicates the main aspects related to user behaviour that will inform the modelling phase and the development of potential ecodesign requirements. Knowledge of user behaviour can help to identify barriers and restrictions to possible ecodesign measures resulting from social or cultural factors. Having information about how frequently consumers buy apparel, the temperature of washing or whether softeners are used or not, is important for the modelling. Equally, knowing whether consumers can assess the quality of apparel, what are their priorities when purchasing, whether users follow care labels, reasons for disposal and if consumers are willing to repair the apparel or not, are relevant considerations for the development of ecodesign requirements. The table below indicates the main user behaviour aspects that will serve as relevant input in the modelling of the environmental impacts and the life cycle costs of apparel, and for the development of potential ecodesign requirements.

¹²⁷ Initial questionnaire. Preparatory Study on textiles for product policy instruments – the initial questionnaire. Available at [this link](#).

1779 **Table 35.** Potential application of user behaviour aspects in the next steps of the preparatory study

ID	Aspect	Information used for	Comments
1	Reasons for purchasing	Potential ecodesign requirements	Reasons for purchasing showcase the triggers for new apparel purchases and can shape the development of ecodesign requirements.
2	Criteria used when purchasing	Potential ecodesign requirements	The criteria used when purchasing could give an indication of which aspects of the apparel are more relevant to users, e.g. durability.
3	Quality assessment of apparel	Potential ecodesign requirements	Quality and longer apparel lifespans are important to consumers. This supports the focus on durability ecodesign requirements.
4	Attitude towards second-hand purchases	Potential ecodesign requirements	Supports the development of potential durability requirements.
5	Attitudes towards chemicals in apparel	Potential ecodesign requirements	Supports the development of requirements on substances of concern. Provides relevant input for the EU Ecolabel potential criteria on chemicals.
6	Attitudes towards the purchase of apparel made with recycled materials	Potential ecodesign requirements	Supports the development of requirements on recycled content. Provides relevant input for the EU Ecolabel and/or EU GPP potential criteria on recycled content.
7	Laundering practices ^(a)	Modelling and potential ecodesign requirements	Different washing temperatures, washing frequencies and sorting practices among users can be modelled to show the way they affect the durability and quality of apparel. Potentially relevant input for the assessment of the environmental impact and the life cycle costs.
8	Care labels	Potential ecodesign requirements	The user behaviour informs how care labels are used by users and what aspects may be missing from them ⁽¹²⁸⁾ , supporting the drafting of information requirements and the development of the Digital Product Passport.
9	Reparability	Potential ecodesign requirements	Potential reparability and durability ecodesign requirements can be shaped by user behaviour data on reparability e.g. the link between the apparel item and accessories such as buttons that tend to fail sooner.
10	Long-term apparel storage	Potential ecodesign requirements	The user storage of apparel could affect its lifespan, thus it may be considered in the LCA modelling phase.
11	Reasons for disposal ⁽¹²⁹⁾ and product-person attachment	Potential ecodesign requirements	The early disposal of apparel by users could trigger the need for the development of durability ecodesign requirements. Reasons for apparel disposal include loss of functionality and quality. These are aspects that create product-person attachment which translates into apparel being kept in the system for longer. This supports the focus on design for durability requirements.
12	Reasons for returning apparel	Potential ecodesign requirements	The lack of quality of ordered apparel is among the main reasons for returning products ordered online. This supports potential inclusion of durability ecodesign requirements.

^(a) It includes sorting before washing, washing temperature and frequency, choice of detergent and softener, drying, ironing, and storing after washing and drying

Source: own production

¹²⁸ The ongoing review of the Textile Labelling Regulation is also assessing this same issue.

¹²⁹ In this study the term 'disposal' used to refer to the generic action of getting rid of a product, regardless of whether the product becomes waste or it simply changes ownership as a product. The general use of the term 'disposal' in this Preparatory Study should not be understood to imply 'disposal operations' described in Annex I of the Directive 2008/98/EC on waste and repealing certain Directives Available at [this link](#).

Note: The following sections provide a summary of the main learnings and observations derived from an extensive literature search carried out on user behaviour with regards to apparel. **A more detailed review of literature corresponding to pre-purchase, post-purchase and disposal aspects is provided in the annex, Section 13.7.**

6.2 Pre-purchase aspects

6.2.1 Reasons for purchasing apparel

Consumer behaviour in the apparel industry is influenced by various factors, including individual perspectives on purchasing and the prevalence of spontaneous buying behaviour. Understanding the motivations behind consumer purchases, whether driven by functionality or self-representation, shows what qualities in apparel users consider to be important for them.

— **Diverse consumer perspectives and reasons for purchasing:** There are two primary consumer approaches towards apparel – those viewing it as purely functional and those seeing it as self-representational. These perspectives significantly influence consumer priorities during the purchase process (McNeill and Moore 2015). Research suggests that the primary purchase motivations include looking good, replacing old items (Ribeiro et al., 2023), buying essential wear, staying on-trend, and preparing for special occasions (D&B, 2020).

— **Spontaneous buying behaviour:** Apparel purchases often occur spontaneously, with a considerable proportion in the 40-70 age group engaging in frequent spontaneous buying. This trend indicates that a majority of consumers may not thoroughly inform themselves before making apparel purchases and act on impulse (Kleinhückelkotten et al., 2018a). Discount offers, including global phenomena like New Year sales, influence impulsive purchases, leading to more frequent buying and increased monthly spending on apparel (D&B, 2020a; Djafarova and Bowes, 2021; Heiny and Schneide, 2021; Amasawa and Kimita, 2023).

— **Digital vs. physical shopping dynamics:** While visual aids on social media attempt to replicate physical store experiences (Djafarova and Bowes, 2021), impulse purchases are more closely associated with physical stores, particularly affecting the younger population (Cook and Yurchisin, 2017). However, malicious interface design strategies, such as dark patterns, are prevalent in online shopping sites (Yada et al., 2022) where they mislead users into making purchases that may not align with their best interests, including unnecessary spending (Schäfer et al., 2023). An example of a dark pattern frequently observed in apparel purchases is the 'low-stock message,' which falsely claims that a product is nearly sold out to induce a sense of urgency and prompt quicker buying decisions (Schäfer et al., 2023).

The level of information provided by surveys on reasons for purchasing apparel is sufficient to get an idea of the different consumer preferences. It should be noted that the literature does not provide an order of importance or a priority list for the reasons for purchase mentioned by consumers.

The reasons users have for purchasing apparel are diverse and highly dependent on personal preferences. However, some seem to be linked to the need to replace apparel that is old, worn out or broken to a certain extent. In these cases, information requirements on reparability and disposal could guide the user concerning ways to repair the product to ensure optimum durability.

In the cases in which impulse purchases are involved, information requirements on the product itself regarding the life-cycle environmental impacts of the apparel item in question, would allow consumers to be more aware of the impact of their purchase and make evidence-based decisions.

6.2.2 Criteria used when buying apparel

When it comes to purchasing apparel, consumers navigate a complex landscape of decision-making criteria that encompass various factors influencing their choices. These criteria often include considerations such as product quality, price affordability, brand significance, durability, ease of care, and alignment with personal values and preferences. Additionally, age-related preferences and sustainability concerns further shape the decision-making process, highlighting the diverse range of factors at play in the apparel market.

— **Main apparel purchase decision attributes:**

- Quality and price are paramount factors influencing consumers' apparel purchasing decisions, with 97% emphasising the importance of product quality and 94% expressing a similar sentiment

- towards price in a survey run among 26 635 European citizens (European Commission. Directorate General for Environment, 2023). Similarly, price is for 68% of 11 483 consumers from ten European countries the most important factor when purchasing clothes, followed by quality (61%) and fit (56%) (YouGov, 2021).
- The brand's significance has less importance when purchasing apparel compared to quality and price (YouGov, 2021; European Commission. Directorate General for Environment., 2023).
 - Other studies indicate that the perceived value of an apparel item by a specific user is also a significant driver of purchasing behaviour, with quality and functionality following closely behind (AK Wienn and Greenpeace, 2023a; Mishra et al., 2023; Thredup, 2023).
 - Durability and ease of care seem not to be explicitly considered by users during apparel purchases but are associated with perceived quality according to De Klerk and Lubbe (2008) and Wakes et al. (2020). However, other studies highlight durability in the top three main aspects to influence the decision to purchase apparel items such as coats or jackets (Consumers, Health, Agriculture and Food Executive Agency et al., 2018) and is also the fourth most important aspect when purchasing apparel for 30% of 11 483 consumers from ten European countries (YouGov, 2021).
 - According to a Cotton Incorporated (2014) survey, the top pre-purchase drivers for sportswear among consumers are comfort (77%), fit (69%), washes clean (65%), quality (64%), and durability (62%).
- **Age-related purchase preferences:**
- Fashion trends hold greater significance for younger consumers, as they see them as a sign of success (AK Wienn and Greenpeace, 2023a; Spaepen et al., 2021).
 - Older generations prioritize new items, comfort, country of origin, textile material, and brand knowledge (Spaepen et al., 2021).
- **Sustainability considerations:**
- Some sources highlight that across all Member States, a majority of respondents reply that the environmental impact of a product is 'very' or 'rather important' in their purchasing decisions (European Commission. Directorate General for Environment, 2023). In fact, in an EU-wide survey involving 27 498 respondents, three-quarters express the view that apparel should be crafted from materials that are recyclable (European Commission, 2019). Moreover, around 54% of 2 500 Spanish consumers would appreciate having more information on the recyclability of the apparel item they purchase (Asociación para la Gestión del Residuo Textil y el Calzado, 2024).
 - Other sources indicate that sustainability and social standards rank lower in importance compared to price or quality considerations when purchasing (Consumers, Health, Agriculture and Food Executive Agency. et al., 2018; AK Wienn and Greenpeace, 2023a). In fact, approximately 32% of the 27 498 EU survey respondents express agreement with the notion that they are not concerned about the environmentally-friendliness of their apparel while 65% disagree with this (European Commission, 2019). 11 483 consumers from ten European countries prioritise price, quality and fit when they buy new apparel (YouGov, 2021). 30% consider product longevity while only 15% consider important the environmental impact of the apparel item when purchasing (YouGov, 2021). Moreover, approximately half of the 27 498 respondents indicate that apparel should be offered at the lowest possible price, irrespective of the environmental impact or working conditions during its production, with a similar proportion of respondents in disagreement with this statement (European Commission, 2019).
 - Approximately 82% of 27 498 survey respondents believe that there is insufficient information available regarding environmental aspects and working conditions associated with apparel (European Commission, 2019).
 - The willingness of having more information on the sustainability of products in the category of 'textiles, clothing/footwear' seems relatively high. In particular, 71% out of 26 635 survey respondents in the EU-27 Member States express a desire to find more of the above-mentioned products carrying the EU Ecolabel. This sentiment is prevalent in all countries, to varying degrees (from 54% in Czechia to 83% in Portugal and Romania) (European Commission. Directorate General for Environment, 2023).
 - Research by consumer organizations indicates that people are becoming increasingly aware of the environmental issues related to the textile sector and are willing to adjust their behaviours as a result (BEUC, 2023). For example, national-level surveys, particularly in Spain (OCU, 2018), Germany (VZBV, 2022), and Austria (AK Wienn and Greenpeace, 2023a), have highlighted that a significant majority of consumers prefer long-lasting clothing and are inclined to avoid purchasing new items to support climate protection. However, a recent survey showed that consumers report

1892 more frequent second-hand purchases, apparel repairs, and label checks, but often overlook
1893 brands' environmental and labour practices when buying (Brun, 2025).

1894 In summary, purchase decisions are very much influenced by the perception of quality and price, but there are
1895 other factors (trends, sustainability, comfort, perceived value), which are of varying importance to different
1896 population segments (based on their age, education level). Given the analysed information provided by surveys,
1897 there is no doubt that quality in apparel is very important for consumers. As apparel quality is closely linked to
1898 its durability, developing ecodesign requirements for this product aspect deserves further consideration.

1899 Moreover, the fact that a high number of consumers believes that environmental information about apparel is
1900 not sufficient is relevant for consideration of possible information requirements on environmental impacts of
1901 apparel.

1902 Equally important is the certainty that consumers would like to find more apparel holding the EU Ecolabel, which
1903 supports the revision of requirements set by this label.

1904 Criteria applied by consumers when buying apparel can support the inclusion and prioritization of ecodesign
1905 requirements on certain product sustainability aspects (e.g. durability) and the setting of classes of
1906 performance, as appropriate.

1907 **6.2.3 User quality assessment of apparel: key insights**

1908 Understanding how users assess the quality of apparel involves examining various phases and factors that
1909 influence their perceptions. This assessment is influenced by intrinsic attributes like material and fit, as well as
1910 extrinsic factors such as brand reputation and manufacturing location. Additionally, experienced features and
1911 personal values play a significant role in shaping perceptions of quality. Consumers also tend to associate
1912 durability and ease of care with higher quality apparel, with a strong expectation for long-lasting products.
1913 Despite the role of price in influencing perceptions of quality and longevity expectations, studies suggest that
1914 cheaper apparel is not necessarily synonymous with lower quality, indicating a complex relationship between
1915 price and perceived value.

1916 — **Phases of quality assessment by users:**

- 1917 • Consumer evaluates the quality in apparel in three phases: at the point of purchase, during use,
1918 and upon disposal (Piippo et al., 2022a).

1919 — **Factors influencing the perception of quality:**

- 1920 • Intrinsic attributes (material, fit), extrinsic factors (brand, manufacturing location), experienced
1921 features, and values influence the perception of quality in apparel (Niinimäki, 2011; Koszewska,
1922 2016; Henninger et al., 2017).

1923 — **Interlinkages between durability and quality in apparel:**

- 1924 • Durability and ease of care are associated with perceived quality (De Klerk and Lubbe, 2008;
1925 Wakes et al., 2020a).
- 1926 • Consumers expect high-quality apparel to be durable (Yuille, 2015). In fact, 43% of survey
1927 respondents declared that long-lasting products are generally better quality (Consumers, Health,
1928 Agriculture and Food Executive Agency et al., 2018).
- 1929 • In 2019, nine in ten respondents (88 % out of 27 498 EU-27 citizens) indicated that apparel should
1930 be made to last longer (European Commission, 2019). Among the reasons, over 45% of
1931 respondents of another survey indicated that they will save money if apparel items such as coats
1932 or jackets last longer (Consumers, Health, Agriculture and Food Executive Agency et al., 2018).

1933 — **Role of price in apparel quality perception and longevity expectations:**

- 1934 • Consumer expectations of apparel longevity are linked to the price, particularly for younger
1935 generations (Monitor, Cotton Incorporated Lifestyle, 2018; Wakes et al., 2020a).
- 1936 • Price may not always accurately reflect quality and durability (Ghaani Farashahi et al., 2018a;
1937 Wakes et al., 2020a). In fact, some studies indicate that cheaper apparel is not always synonymous
1938 with lower quality (Wakes et al., 2020a; Badgett, 2017).
- 1939 • 43% of Austrian survey participants express a willingness to buy long-lasting apparel despite the
1940 price, with 30% already practising this behaviour (AK Wienn and Greenpeace, 2023a) while 55%
1941 out of 1 000 German consumers would be willing to pay more for their apparel if being certain
1942 about increased durability (VZBV, 2022).
- 1943 • For coats or jackets, over 60% of 1 001 respondents' observations indicated that it is hard to tell
1944 how long a product will last (Consumers, Health, Agriculture and Food Executive Agency et al.,
1945 2018).

1946 Overall, the references analysed provide consistent views on the fact that the assessment of quality by users
1947 is multifactorial and somewhat subjective because, inter alia, it is largely based on tactile experience with the
1948 apparel.

1949 In the previous section, apparel quality has been highlighted as a key aspect users deem to be important in the
1950 apparel they purchase. However, how consumers assess quality is not always clear as this can be very
1951 subjective. For this reason, information on the performance of the product could provide consumers with
1952 additional, objective quality parameters for the apparel. Information requirements on product aspects linked to
1953 durability, and even reliability, could enable consumers to better understand and assess the quality of apparel.

1954 In order to facilitate the user's assessment of the quality of apparel, performance requirements based on
1955 product durability, reliability and reusability aspects could, as appropriate, include minimum levels. Another
1956 possibility could be to define requirements connected to the functional performance of apparel. Information
1957 requirements related to the above-mentioned product parameters could serve as metrics to guide users'
1958 assessment of quality in apparel.

1959

6.2.4 Consumer behaviour towards labels on apparel

Consumer behaviour towards labels on apparel plays a pivotal role in shaping purchasing decisions and influencing perceptions of product quality and sustainability. Expectations regarding label information vary, with consumers expressing preferences for receiving details about durability and/or reparability either via official EU labels, product descriptions, or retailers. Offering such information on labels has the potential to impact the relative importance of price in purchasing decisions, highlighting the growing interest in transparency and sustainability among consumers. This interest extends to EU Ecolabel products, with a substantial proportion of survey respondents expressing a desire for more environmentally friendly options in apparel.

— Consumer attention to labels in general:

- There is limited information regarding the level of consumer attention to apparel labels generally speaking, except for care labels (Section 6.3.2).

— Expectations regarding label information:

- Around 18% of over 4 880 user responses indicate that consumers expect to receive information regarding the durability of coats or jackets via an EU official label while the majority prefers to receive such information via product descriptions (39.9%) or via the retailer (36%) (Consumers, Health, Agriculture and Food Executive Agency. et al., 2018).
- Offering details about durability on labels could alter the relative importance of price when making purchasing decisions (Consumers, Health, Agriculture and Food Executive Agency. et al., 2018).
- A significant 74% of 1 000 Italian consumers expressed a strong need for clearer and more transparent information about the sustainability of apparel and the production processes involved (Altroconsumo and IPSOS, 2024).
- In the case of reparability information, 14% of survey respondents expect to have such information via an EU official label while the majority would opt to have it via the retailer (35.6%) and via product descriptions (33.6%) (Consumers, Health, Agriculture and Food Executive Agency. et al., 2018).

— Growing interest in EU Ecolabel products:

- There is increasing interest in EU Ecolabel textile products. A total of 71% of 26 635 EU survey respondents express a desire for more EU Ecolabel textile products. There is a consistent level of interest across countries, ranging from 54% in Czechia to 83% in Portugal and Romania (European Commission. Directorate General for Environment, 2023).

Understanding the way and extent to which consumers respond to information contained on labels can support the development of information requirements, for instance related to the environmental impacts of a product and its performance on specific product sustainability aspects.

6.2.5 Attitudes towards second-hand apparel purchase

There is a growing interest in second-hand apparel reflected in consumer attitudes. Evidence suggests a significant portion of consumers are open to purchasing second-hand items. While concerns about hygiene and a preference for new items persist among some, the appeal of financial savings and eco-friendly practices drives many to incorporate second-hand pieces into their wardrobe. Additionally, emotional attachment to clothing items further boosts positive attitudes towards second-hand shopping, highlighting a nuanced approach to fashion consumption.

— Consumer inclination towards second-hand apparel:

- There is a certain interest among consumers towards second-hand apparel, as indicated by 34% of the 26 595 respondents showing a readiness to buy second-hand apparel (European Commission. Directorate General for Environment, 2014).
- Over 70% of 27 498 respondents in the EU agree that the promotion of second-hand apparel should be increased (European Commission, 2019).
- About 37% of the population actively engages in buying second-hand apparel, while concerns about hygiene, perceived effort, and scepticism about the quality and availability of suitable items remain key barriers to participation (D&B, 2020a). These barriers are often reinforced by subconscious factors such as low self-efficacy, where individuals doubt their ability to find desirable items without significant effort, and concerns about personal image, including fears of appearing unkempt (D&B, 2020a).
- Those incorporating second-hand apparel tend to purchase fewer new items, with financial savings, sustainability, and a preference for unique pieces being primary drivers (D&B, 2020a).

- The emotional value users have for apparel items significantly boosts positive attitudes towards second-hand clothing (Rulikova, 2020; Amini et al., 2021; Koay et al., 2022).
 - Additionally, beyond financial and emotional motivations, moral considerations also play a role in second-hand apparel practices. Both donation and resale of apparel are perceived by consumers as ways to reduce waste and act responsibly, helping to alleviate feelings of guilt associated with discarding usable garments (Pera and Ferruli, 2023).
- **Marketing strategies in second-hand apparel:**
- Preliminary observations from Turunen and Gossen (2024) suggest that current second-hand business models often employ consumption-promoting marketing strategies and include a rise in unworn or lightly used items on second-hand platforms. Additionally, subscription models may lead to shorter apparel lifespans, and self-service flea markets, with their low prices, can attract shoppers more interested in the experience than in making responsible purchases (Turunen and Gossen, 2024).
- **The concept of replacement rates and second-hand apparel:**
- The 'replacement rate' refers to the extent to which purchasing second-hand apparel substitutes the need for new items (Nørup et al., 2019; Trzepacz et al., 2023). Some studies show that second-hand purchases could reduce the need for new apparel, potentially extending the lifespan of existing clothing (Nørup et al., 2019; Trzepacz et al., 2023). For example, a survey indicated that 39% of second-hand buyers would have otherwise bought a new product (Vinted, 2021). However, replacement rates vary, and factors such as the quality of second-hand items can influence their environmental benefits (Farrant et al., 2010; Sandin and Peters, 2018; Trzepacz et al., 2023).
- **Gender and age dynamics:**
- Women show a higher inclination to purchase second-hand apparel, with approximately 40% embracing this trend compared to just over 25% of men (European Commission. Directorate General for Environment, 2014; D&B, 2020a).
 - The younger population, especially those aged 18-25, lead in the adoption of second-hand apparel, while participation decreases in the 26-40 age group (D&B, 2020a).

The willingness of users to purchase previously used apparel (second-hand) can be relevant in view of the potential definition of ecodesign requirements on reusability.

6.2.6 Attitudes towards the purchase of apparel made without harmful chemicals

Consumer attitudes towards chemicals in apparel encompass a spectrum of perceptions shaped by factors such as fabric type, geographical location and evolving consumer preferences. Insights from surveys and analyses offer valuable perspectives on the perceived risks associated with chemicals in apparel fabrics, highlighting varying degrees of concern and priorities among consumers.

- **Chemical perception in apparel:**
- In a survey spanning 27 EU Member States, 60% of 26 718 respondents perceive chemicals in apparel fabrics as minimally risky, with most not seeing them as a threat to people (European Commission, 2009). However, more recent evidence shows that consumers are increasingly concerned about the use of hazardous chemicals in apparel (Evaluation report of the Textile Labelling Regulation, under development). In a survey of over 2 000 Spanish consumers, around 61% identified restricting hazardous chemicals in clothing as the second most important action for policymakers (CECU, 2023).
 - Synthetics are perceived as posing the highest risk among fabric types, with only 22% categorising them as significantly risky (European Commission, 2009).
- **Country-level variances in risk considerations:**
- A country-level analysis reveals that environmental and health risks are deemed very important by about two-thirds of respondents in Germany and Slovakia. In contrast, more than six in ten respondents in Denmark and the United Kingdom consider this aspect unimportant (European Commission, 2009).
 - Data from the Fashion Revolution (2020) survey, covering 5 000 consumers across the United Kingdom, France, Germany, Italy, and Spain, indicates that an average of 37% considers it important to buy apparel produced without harmful chemicals. Similarly, 36% of 11 483 consumers from ten European countries consider important that there are less or no chemicals in the apparel they are buying (YouGov, 2021). The surveys do not provide detailed information about the types of chemicals.

The user perception towards chemicals in apparel, especially as regards substances that pose a risk to human health or to the environment, could support the establishment of information requirements on the presence of substances of concern. This in turn can inform possible actions that could be addressed in specific legislation on the safe use of chemicals such as REACH (e.g. via prioritisation in the REACH restrictions “roadmap” which is subject to periodic review by the Commission).

6.2.7 Attitudes towards the purchase of apparel made with recycled materials

Knowledge of consumer attitudes towards apparel containing recycled material (recycled content) is relevant for the analysis of technologies reported in Section 9.

— Consumer preference for recycled materials:

- According to the Fashion Revolution (2020) survey of 5 000 consumers in the United Kingdom, France, Germany, Italy, and Spain, an average of 11% emphasize the importance of their apparel containing recycled materials. Nearly 60% of the 2 500 Spanish consumers surveyed expressed a desire to purchase textiles and/or footwear made from recycled materials, while around 25% reported that they have already made such purchases (Asociación para la Gestión del Residuo Textil y el Calzado, 2024).

— Age group differences:

- Younger respondents (12-24 years old) give it greater importance, with 14% indicating a preference for apparel containing recycled materials, compared to only 7% in the 55-75 age group (Fashion Revolution, 2020).

6.3 Post-purchase aspects

6.3.1 User behaviour during use: laundering practices

User laundering practices include sorting before washing, use of a certain washing temperature, washing frequency, choice of softeners and detergents, drying methods, ironing, and storage. Consumer behaviour related to apparel laundering practices influences the environmental impact during the life of the product and constitutes relevant input for the assessment of the environmental impact and the life cycle costs of an apparel product.

Moreover, research indicates that consumer behaviour during the use and care of apparel may significantly impact the lifespan of apparel too. It could therefore be a good idea for apparel items to be accompanied by information for consumers and other end users on how to maintain the product to ensure a longer lifespan.

— Sorting before washing:

- Sorting practices vary based on factors like washing temperature, colour, fibre type, and care labelling (Laitala et al., 2012).
- Sorting based on washing temperature is more common among consumers compared to washing everything together (Laitala et al., 2012).

— Washing temperature and washing frequency:

- The average European washing temperature is 42.4 °C (A.I.S.E, 2020).
- Differences in washing habits (i.e. washing temperature used, choice of detergent and softeners, etc.) are observed across countries. Factors like age, location and societal norms contribute.
- Sportswear often features distinct fibres and fabrics that can quickly absorb sweat and may lead to unpleasant odours, necessitating more frequent laundering compared to everyday casual wear (Wei et al., 2020; Chang and Wang, 2023) typically after a single use (Brice and Thorpe, 2021).
- User beliefs about hygiene and convenience may impact the apparel washing frequency (D&B, 2020a).

— Choice of softeners and detergents:

- Users often dose detergent arbitrarily; overdosing is common (A.I.S.E, 2020).
- Fabric softeners are used in approximately 55% of washing cycles in Europe (Klepp and Laitala, 2023a).
- Economic factors play a substantial role in users' choices of laundry products.

— Consumer awareness on microplastics release during washing cycles:

- A survey of 411 Belgian citizens revealed that while 68% are aware of plastic pollution, only 37% are aware of microfibre pollution and its connection to the issue. Awareness of synthetic fibre

- pollution is lowest among individuals under 25 (29%) and highest among those aged 41 to 60 and over 60, though still less than half in these groups are informed (43%) (Herweyers et al., 2020).
- In a 2020 survey by the Swedish Environmental Protection Agency (2020) (~3 000 respondents), 54% considered how to wash apparel to reduce microplastic release; 17% of them lacked knowledge on how to do so effectively.
- **Drying and ironing:**
- Analysing domestic apparel ironing and drying phases is challenging due to the limited availability of data (Munasinghe et al., 2021).
 - Diverse drying methods include air-drying, electric dryers and dedicated drying spaces. Survey data provide insights into specific drying methods where line-dried outdoors (natural drying) is one of the most commonly used drying methods for apparel (Laitala et al., 2020; GINETEX, 2017a).
 - Ironing practices have decreased over the years; habits vary across countries and demographics. The frequency of ironing is influenced by gender and age; men and younger respondents tend to iron less (Klepp and Laitala, 2023b).
- **Storage after washing and drying**
- The folding and storage of apparel after washing and drying may influence the lifespan of apparel. However, no studies have been found on post-washing, drying and storage behaviours.

Sufficient information is available on commonly used washing temperatures among users as well as washing frequency. When it comes to sorting before washing, drying and ironing practices, the literature is more limited. Additionally, the surveys analysed on choice of softeners and detergents provide valuable information for the LCA modelling phase given that softeners could be a source of substances of concern to be taken into consideration.

6.3.2 Following apparel care label instructions

Apparel care labels provide instructions on washing temperature, cycles, detergents, etc. Surveys and studies shed light on consumer attitudes and behaviours regarding care label instructions, revealing insights into adherence rates, preferences for accessing instructions, and associations with care label symbols.

— **Attitudes towards care label instructions:**

- Surveys indicate that a significant percentage of users follow care instructions on apparel labels (GINETEX, 2017a; GINETEX, 2019).
- Users' adherence to care labels diminishes after the initial wash (McLaren et al., 2015).
- Some users cut out care labels, affecting the resale potential of apparel (GINETEX 2017).
- Some users prefer new ways to access the apparel care instructions, such as QR codes on smartphones (COFREET, 2023; Ribeiro et al., 2023).

— **Association with care label symbols:**

- Surveys show consistent understanding of symbols like ironing and washing (AB-REOC and BV-OECO, 2019; GINETEX, 2017a).
- Users' poorer understanding of symbols for bleaching, drying, and professional cleaning may be due to them being less intuitive (AB-REOC and BV-OECO, 2019; GINETEX, 2017a).

Relevant information can be extracted from the available surveys on user interaction with care labels, especially when it comes to the extent to which users follow care labels in Europe. For other user behaviour patterns in relation to care labels, information is available only from one survey, and so not allowing the comparison of information from different sources. Nonetheless, all the surveys related to user behaviour and apparel care labels have an ample respondent base, making the results valuable for the preparatory study at hand.

Additionally, the user behaviour analysis provides insights on apparel maintenance aspects that may either be missing or need further clarification on care labels. This is valuable data for the development of information requirements for consumers on how to use and maintain the product in order to minimise its impact on the environment and to allow for optimal duration.

Finally, consumer opinions about care label instructions and their level of adherence to them can guide the manner in which the information could be provided (e.g. in the Digital Product Passport or directly on a label as referred to in Article 14 of the ESPR final compromise text).

6.3.3 Reparability

The consumer relationship with apparel often exhibits a transient nature, characterised by the disposal of items before they reach the end of their lifecycle. This disposability stems from factors such as overconsumption and the convenience of replacing rather than repairing apparel. Moreover, diminishing sewing skills and limited access to repair services further impact consumers' decisions regarding garment maintenance and repair.

— Transitory relationship with apparel:

- Consumers have a transient relationship with apparel, often disposing of items before they are worn out or broken (Harris et al., 2016).
- Reasons for this behaviour include overconsumption and the convenience of replacing unwanted items rather than repairing them. A lack of emotional attachment to the product is observed (Terzioğlu, 2021; EEA, 2022a).
- Most survey respondents rated reparability as unimportant when purchasing apparel items such as coats or jackets while only about 11% considered reparability to be important (Consumers, Health, Agriculture and Food Executive Agency. et al., 2018).

— User decisions and repair capabilities:

- Sewing skills for apparel repair are diminishing, attributed to a lack of dedicated teaching in schools and at home and a scarcity of time and repair equipment (Finnish Ministry of the Environment, 2023a; Pera and Ferruli, 2023).
- Trust in one's own skills is a significant motivator for engaging in sewing and repairs (Finnish Ministry of the Environment, 2023a; Pera and Ferruli, 2023).
- Users decide to mend apparel based on sewing skills; some can sew a button, but fewer can replace zippers (Laitala and Boks, 2012).

— The apparel repair sector in the EU:

- Repair in Europe is common, but the total cost is influenced by high labour costs, equipment, and materials (EEA, 2022a).
- On average, the share of survey respondents rating the availability of repair services as having a lot of influence on their purchasing decision was relatively low (Consumers, Health, Agriculture and Food Executive Agency et al., 2018).
- Difficulties accessing repair services notably diminished the appeal of repair (Consumers, Health, Agriculture and Food Executive Agency et al., 2018).
- Lack of information about, trust in, and satisfaction with repair services hinder consumer engagement (EEA, 2022a).
- Research by Laitala et al. (2023) emphasizes the need for clear guidelines on what constitutes unacceptable wear versus normal use.

The user behaviour analysis identifies a series of common apparel repairs that users generally carry out. These apparel repairs are generally limited to simple ones, such as substituting a button, and only a limited fraction of the population is skilled enough to engage in more demanding repairs such as changing a zipper. Performance requirements for reparability could be considered to ensure the availability of replacement components that, according to users, tend to break or wear out most frequently. Moreover, information on the period of availability of spare parts (e.g. buttons and other accessories) and the possibilities of repairing may also contribute to further user engagement in repair activities whether privately or by visiting repair shops.

It could also be envisaged that information requirements would provide instructions or 'use cues' in the product to guide the correct repair or replacement of the above-mentioned apparel components. This could facilitate the ease of self-repair by the user but also guide tailors when repairing apparel, guaranteeing that a certain level of quality in the apparel repair is achieved. Some of this information could be included in the Digital Product Passport.

6.3.4 Storage of apparel

User behaviour regarding storage of apparel is linked to aspects described in Section 6.2 and Section 6.4.3. The storage of apparel items, whether temporary or permanent, appears to be strongly connected to the emotional attachment individuals have to their apparel. Increasing the emotional durability could potentially make users keep their items for longer, avoiding disposal and potentially increasing the number of uses. However, such storage does not prevent users from continuing to buy more apparel.

— Types of apparel storage:

2227 • Apparel storage can be classified into two categories: active and inactive. Active apparel storage
 2228 involves placing regularly worn items inside wardrobes or easily accessible spaces. Inactive apparel
 2229 storage refers to keeping items at home without use for an extended period. Inactive storage
 2230 constitutes over 30% of apparel in European closets (European Parliamentary Research Service,
 2231 2019).

2232 — **Reasons for inactive apparel storage:**

2233 • Inactive apparel may include items that no longer fit, stored with the hope they will fit again, and
 2234 as a means of monitoring weight (Bye and McKinney, 2007).

2235 — **Temporary storage practices:**

2236 • Temporary storage occurs when users are contemplating disposal methods, waiting for apparel to
 2237 come back into fashion, repairing, or determining if there are still ways to utilise the apparel
 2238 (Cluver, 2008a).

2239 — **Storage of basic apparel:**

2240 • Essential but outdated or worn-out basic apparel is stored until substitutes are purchased (Cluver,
 2241 2008a).

2242 Information requirements related to the end-of-use options (e.g. reusability) for apparel that is no longer wanted
 2243 could encourage users to opt to give a second life to their apparel, for instance via donation. This would avoid
 2244 the storage of apparel users no longer wish to wear or to keep in their closets.

2245 **6.4 User behaviour related to the disposal of apparel**

2246 **6.4.1 Reasons for the disposal of apparel**

2247 Disposal ⁽¹²⁹⁾ of apparel happens when a user transfers its ownership to another person or entity (Cluver,
 2248 2008a).

2249 The main reasons for apparel disposal seem to be due to certain product characteristics changing over time,
 2250 damage or simply for reasons related to consumer preferences.

2251 — **Factors influencing disposal decisions:**

2252 • Individual characteristics, habits, demographics, product traits, and quality influence the decision
 2253 to dispose of an apparel products (Cluver, 2008; Goworek et al., 2012, as cited in Harris et al.,
 2254 2016; Sandin et al., 2019).

2255 • Apparel disposal may also be associated with major life transitions such as moving house or the
 2256 death of a family member, being these moments also key triggers for disposal behaviour (Sigaard,
 2257 2023)

2258 • Perceived quality is crucial; low quality may lead to early disposal, and owning new apparel
 2259 frequently shortens usage time (Aakko and Niinimäki, 2022).

2260 • Loss of symbolic perceived value also contributes to early disposal (Gwozdz et al., 2017a).

2261 — **Reasons for apparel disposal:**

2262 • A review by Laitala and Klepp, (2022) of 17 consumer studies involving around 20 000
 2263 participants identified the most common reasons for apparel disposal as intrinsic quality (34%),
 2264 perceived value (31.4%), and fit issues (25.8%). These findings are consistent with the results
 2265 summarized in **Table 36**, which also highlight these factors as the primary reasons for apparel
 2266 disposal. While quality-related concerns are the leading cause, perceived value and fit issues are
 2267 almost equally significant.

2268 • Price influences disposal frequency, with consumers valuing higher-priced items and disposing of
 2269 cheaper apparel more frequently (Morgan and Birtwistle, 2009; Joy et al., 2012).

2270 — **Intrinsic durability and user perceptions:**

2271 • Intrinsic durability, defined during product design, influences the ability of apparel to withstand
 2272 wear without compromising functionality and aesthetics (Alliance of Commerce and Deloitte,
 2273 2022).

2274 • Physical issues like abrasion, colour changes, and broken zippers contribute to the perceived loss
 2275 of intrinsic durability (Laitala and Boks, 2012).

2276 — **Disposed apparel problems:**

2277 • Predominant problems of disposed apparel: colour fading and fabric-related issues.

2278 • Other issues: pilling, fabric breakdown, accidental damage, loss of dimensional stability, logo
 2279 failure, discoloration, holes in seams, and trim failure (Cooper and Claxton, 2022b).

2280 The combined dataset in **Table 36** from various studies (detailed in Section **13.7.3.1**) provides a comprehensive
2281 view of global apparel disposal trends. Survey participants across countries commonly cite intrinsic quality and
2282 fit as reasons for disposal of apparel.

2283 Perceived value, taste-related factors, situational reasons, and fashion trends also play a role in apparel
2284 disposal. Overall, there is a complex interplay of individual factors contributing to disposal decisions.

2285

DRAFT - work in progress

Table 36. Main reasons for apparel disposal

Publication	Country	Main reasons for the disposal (expressed in % of respondents)				
		Intrinsic quality (e.g. worn-out items)	Fit issues	Perceived value (e.g. taste-related unsuitability)	Fashion changes	Other (e.g. Situational reasons, functional shortcomings)
Greenpeace. (2015)	Germany	92	72	64	40	NA
Ungerth and Carlsson. (2011)	Sweden	60	9	21	NA	4
Laitala and Boks. (2012))	Norway	49	19	11	NA	19
Laitala and Klepp (2020)	China, Germany, Japan, United Kingdom, and USA	44	13	35	NA	9
WRAP (2017b)	United Kingdom	18	42	33	NA	NA
Lang, Armstrong, and Brannon (2013)	USA	30	31	39	NA	NA
Morell-Delgado et al., (2024)	Spain	65	44	21.5	NA	2.5
YouGov. (2019)	Italy	60	47	16	12	2
Accenture. (2022)	Poland	38	21	7	6	21

N.B. NA: Not Available

Source: Own elaboration based on data provided in the indicated publications.

Changes in the product's characteristics may relate to apparel quality issues that can shed light on which types of performance and information requirements could be prioritised extend the duration of use of apparel products. For instance, in order to enhance technical durability for an extended usage period, it is crucial to understand the reasons behind consumer apparel disposal. Gaining such insights should aid in assessing possible ecodesign requirements that foster the durability of the apparel, while also addressing practices associated with premature disposal of apparel which have an overall negative impact on the environment.

Moreover, the reasons for disposal can focus on certain aspects of the product's performance.

6.4.2 Disposal channels

Users may dispose of apparel directly in the residual waste, donate it for reuse or pass on to family and friends.

— Disposal trends:

- The person-product type of attachment plays a key role in how apparel may be disposed of. Sharing within social circles is consistent behaviour when there are positive associations with the user's apparel. Negative associations with the apparel may result in donation, swapping, or disposal in the waste bin (Joung and Park-Poaps, 2013; Lewis, 2015). At the same time, throwing away usable apparel is generally perceived negatively and as socially reproachable behaviour.
- The overarching observation made by the European Commission. Directorate General Joint Research Centre (2021) study regarding the quality of apparel found in residual waste is that there exists a correlation between the proportion of apparel collected separately and the average quality and value of apparel discarded in residual waste. According to the above-mentioned study, this relationship is partly attributable to households' discerning decisions regarding the perceived monetary value of apparel, determining which items merit donation or resale for reuse, and which have minimal reuse potential. Supporting this notion, a 2018 study conducted in Denmark (Watson et al., 2018) lends credence to this theory. It suggests that the 42 000 tonnes of apparel disposed of in Danish residual waste for incineration in 2017 held an estimated value of 12-15 million euros prior to disposal. In contrast, the 36 000 tonnes of apparel separately collected were sold

2314 on reuse markets for an estimated 65 million euros, signifying a value per tonne that is four to
2315 five times higher.

2316 The way users dispose of their apparel provides valuable data that can help shape information requirements
2317 related to end-of-life behaviour. For instance, information requirements may guide consumers and public
2318 authorities towards more sustainable choices when disposing of apparel, which could also contribute to
2319 enhanced separate apparel collection rates as prescribed by the Waste Framework Directive by Member States
2320 given the significant reuse potential of apparel.

2321 **6.4.3 Person-product attachment**

2322 The emotional attachment formed between the apparel and the consumer is generally referred to as 'person-
2323 product attachment'. This type of connection between a user and an apparel item has an influence on how long
2324 consumers own certain apparel and how often they make use of it. This person-product attachment is usually
2325 embedded in the term 'emotional durability', which goes beyond just functionality. The Ellen MacArthur
2326 Foundation report (Ellen MacArthur Foundation, 2021) defines emotional durability as: 'the product's relevance
2327 and desirability to a user, or multiple users, over time'.

2328
2329 — **Ownership categories:**

- 2330 • Ownership categories include 'active' (daily use), 'seldom' (several times a year), and 'inactive/in
2331 storage' (rarely or never used) (Niinimäki and Armstrong, 2013).
- 2332 • A survey by Niinimäki and Armstrong (2013) identified elements fostering person-product
2333 attachment:
 - 2334 ○ Functionality: Comfort, good fit, multi-function, ease of matching, easy to put on.
 - 2335 ○ Memory: Memories, received from a special person, family ties.
 - 2336 ○ Emotional satisfaction: Looking/feeling good, receiving compliments, love for the brand.
 - 2337 ○ Design and style: Good design, in style.
 - 2338 ○ Fabric and material: Nice colour, pleasant touch, aesthetic, flexible.
 - 2339 ○ Personal values, quality, effort invested, and financial value: Uniqueness, feeling relaxed,
2340 durability, high quality, hand-made, reward for self, price (Niinimäki and Armstrong, 2013).

2341 — **Value of new apparel and barriers to reuse:**

- 2342 • Newly purchased apparel holds the highest emotional attachment due to its perceived intrinsic
2343 value (Forbrugerrådet Tænk, 2022).
- 2344 • New apparel is used more frequently than older apparel; second-hand apparel is used 30% less
2345 (Forbrugerrådet Tænk, 2022).
- 2346 • Reuse may extend the possession span but does not necessarily increase apparel usage (Laitala
2347 and Klepp, 2021).

2348 — **Emotional durability as a forward-looking issue:**

- 2349 • Emotional durability, linked to consumer perceptions, is challenging to measure, relying on social
2350 science concepts (Alliance of Commerce and Deloitte, 2022).

2351 — **Quality as a key factor in prolonged use:**

- 2352 • Quality is a significant factor in incentivising prolonged apparel use. In fact, consumers express
2353 willingness to wear apparel longer if it is of better quality, maintains shape and colour, and has a
2354 bonding effect (Kleinhüeckelkotten et al., 2019; Laitala et al., 2021a).

2355 — **Role of designers in increasing emotional attachment:**

- 2356 • Strategies to promote the product-person attachment include designing for long-term needs,
2357 incorporating added value, and creating adaptable, modular, or timeless products (Niinimäki and
2358 Armstrong, 2013; Alliance of Commerce and Deloitte, 2022).

2359 **6.4.4 Returns of apparel**

2360 The dynamics of online shopping and returns encompass various facets that influence consumer behaviour and
2361 preferences. Convenience emerges as a primary driver for the proliferation of online purchases, while return
2362 policies and processes significantly shape the shopping experience. This overview delves into return rates,
2363 underlying reasons for returns, and the factors influencing return behaviour, shedding light on the complexities
2364 of the retail returns landscape.

2365 — **Online shopping and returns:**

- 2366 • Convenience is a key reason for online shopping (AK Wienn and Greenpeace, 2023a).

2367 • Long return periods encourage more apparel orders, while inconvenient return processes lead to
2368 items being stored rather than returned (Forbrugerrådet Tænk, 2022).

2369 — **Return rates and demographics:**

2370 • Casual dresses, jackets and jeans have high return rates, with expensive products more likely to
2371 be returned (EEA, 2024).

2372 • Women tend to return more online purchases than men (AK Wienn and Greenpeace, 2023a; British
2373 Fashion Council's Institute of Positive and Roland Berger, 2023).

2374 • Young consumers, more active in online shopping, tend to return apparel more frequently, ordering
2375 multiple sizes (AK Wienn and Greenpeace, 2023a).

2376 — **Reasons for returns:**

2377 • **Fit issues:** Fit issues is the main reason for returns, followed by taste-related unsuitability
2378 (including shape, material, colour, or pattern dislike), quality and faulty items (Foresight Factory,
2379 2021; Zimmermann et al., 2021).

2380 • **Mis-buying:** Common issue with online shopping, driven by fitting problems and consumer dislikes
2381 (Forbrugerrådet Tænk, 2022).

2382 • **Quality concerns:** Low quality, lack of durability, and buyer's remorse also contribute to returns
2383 (Bernon et al., 2011).

2384 — **Factors influencing returns:**

2385 • **Sizing issues:** Difficulty interpreting sizing scales and inconsistent sizing contribute to fitting
2386 problems (Vladimirova et al., 2022).

2387 • **Return policies:** Consumers often check return policies before purchasing, and their return
2388 experience influences repeat purchases (Asdecker and Sucky, 2019).

2389 • **Assumptions about returns:** Many consumers assume returned items are always resold,
2390 impacting their return behaviour (Makov, 2023).

2391 Similarly to the reasons for apparel disposal, the causes for apparel returns may shed light on the product
2392 aspects that are important for users, e.g. quality. This could support the potential inclusion of durability
2393 ecodesign requirements.

7 Current EU Ecolabel criteria for textile products

This section analyses the current EU Ecolabel criteria for textile products⁽¹³⁰⁾ in light of their revision. Section 7.1 reports the facts and figures of the criteria, whereas Section 7.2 provides the main suggestions for the criteria revision received via the initial questionnaire (see Table 1 in Section 1). Section 7.3 describes the relationship between the new mandatory ecodesign framework (ESPR) and the revision of the EU Ecolabel criteria for textile products. Finally, Section 7.4 provides a detailed comparison of the criteria for textile products set by the EU Ecolabel, Blue Angel and Nordic Swan. This analysis set the bases to address Article 6(f) of the EU Ecolabel Regulation which promotes synergy with other officially recognised ISO Type I environmental labels (Ecolabels).

7.1 Facts and figures

In recent years, the number of licences and products awarded the EU Ecolabel for textile products has continuously increased (**Figure 93** in Section 13.8.1). In September 2023, the EU Ecolabel for textile products counted 86 licences and 9 250 products. The most licences were awarded by Denmark, Italy and Norway, representing 28%, 21% and 13% of the total, respectively. The most products were awarded by Portugal, Italy and Denmark, representing 42%, 36% and 14% of the total, respectively. More details are available in **Table 156** in Section 13.8.1.

Table 37 reports the figures published in September 2023 according to the type of products. Most licences and products awarded were textile apparel: 76% and 32% of the total products and licences, respectively. Home/interior textiles were 8% and 18% of the total products and licences, respectively. Cleaning products were 11% and 16% of the total products and licences, respectively. Intermediate products represent a significant percentage in terms of licences but less in products.

Table 37. Figures of types of products awarded the EU Ecolabel for textile products in September 2023

Type of product	Licences		Products	
	Number	Percentage of the total (%)	Number	Percentage of the total (%)
Textile apparel	27	32	6 947	76
Home/interior textiles	15	18	688	8
Textile cleaning products	14	16	1 012	11
Intermediate products, such as textile fibres, yarns, fabrics and knitted panels	27	32	512	6
Intermediate products, such as non-fibre elements	2	2	4	<1

N.B. The number of licences and products is affected by the reporting methods used by competent bodies. This results in small discrepancies compared to the total statistics.

Source: own elaboration based on data provided by EU Ecolabel Helpdesk

7.2 Suggestions for the revision of EU Ecolabel criteria

When voting on the final draft of the current EU Ecolabel criteria, in November 2013, the Commission and the Member States identified some aspects to potentially be assessed during the subsequent revision process of the EU Ecolabel criteria. The proposal suggested investigating the extension of the scope to silk, bamboo fibres, man-made fibres, as well as the use of additional recycled materials and potential alternatives to the use of fluorinated membranes.

The initial questionnaire (see **Table 1** in Section 1) allowed the collection of respondents' opinions on the current EU Ecolabel criteria and on the potential topics to investigate during the revision process. Out of 34 respondents, 10 belonged to the manufacturing industry and 6 to governmental institutions (4 competent bodies of the EU Ecolabel), 29% and 18% of the total, respectively. More details about the types of respondents are provided in **Table 157** in Section 13.8.2.

¹³⁰ EU Ecolabel criteria for textile products. Commission Decision of 5 June 2014 establishing the ecological criteria for the award of the EU Ecolabel for textile products. Commission Decision (2014/350/EU). Available at [this link](#).

2430 In general, current EU Ecolabel criteria are perceived to be too complex, and the application process cumbersome
2431 and bureaucratic, with most of the costs and difficulties being related to the involvement of suppliers, tests and
2432 certifications.

2433 In particular, two competent bodies declared that applicants usually need clarifications/guidance on which
2434 products are included in the scope, and which are the corresponding criteria that these specific products must
2435 meet. A couple of competent bodies reported that some applicants gave up with the application, because they
2436 lacked data and information on their supply chain. This is further confirmation of the features of the supply
2437 chain discussed in Section 5.6.

2438 Respondents had different general perceptions about the interest in the EU Ecolabel criteria: 32% think that
2439 consumers look for textile products with the EU Ecolabel, 52% think that consumers do not look for products
2440 with the EU Ecolabel, and 16% have no opinion.

2441 Respondents to the initial questionnaire gave specific suggestions on the requirements of the current EU
2442 Ecolabel criteria for textile products, reported in Section 13.8.2, **Table 158** to **Table 162**.

2443 General suggestions on how to improve the EU Ecolabel criteria included:

- 2444 • simplification of the application process;
- 2445 • harmonisation of the requirements with the ones of other Type I ecolabels;
- 2446 • use of more third-party certifications to prove compliance with the criteria (e.g. on chemicals)
2447 to streamline the verification process;
- 2448 • facilitating the retrieval of information from the supply chain actors outside the EU;
- 2449 • inclusion of criteria addressing product recyclability and packaging;
- 2450 • alignment with ecodesign requirements developed in the framework of the ESPR.

2451 **7.3 Revision of EU Ecolabel criteria within the ESPR framework**

2452 The establishment of the ESPR brings a change in the revision of the EU Ecolabel criteria for products addressed
2453 by a delegated act of the ESPR. This is the case of the EU Ecolabel criteria for textile products. As mentioned in
2454 Section 1, Article 34(3) of the ESPR establishes synergies between the mandatory ecodesign legislative
2455 framework and the EU Ecolabel, as the EU Ecolabel could be used as proof of compliance to Ecodesign
2456 requirements when ESPR and EU Ecolabel cover the same product groups. The ESPR and EU Ecolabel
2457 requirements must therefore be coherent and synergic to guarantee that products awarded the EU Ecolabel
2458 comply with the ESPR requirements set in the relevant delegated act. Therefore, the revision of the EU Ecolabel
2459 criteria for textile products remains regulated by the EU Ecolabel Regulation, but it should also follow the
2460 framework of the ESPR.

2461 In particular, this means that the EU Ecolabel criteria should complement ecodesign requirements, increasing
2462 the ambition level set by the ESPR, complying with the specificities of the EU Ecolabel Regulation. For instance,
2463 the EU Ecolabel criteria could restrict hazardous chemicals and consider ethical and social aspects where
2464 appropriate, as well as they could forbid or restrict the use of specific chemicals and substances.

2465 The EU Ecolabel criteria for textile products set the product environmental performance at fibre, fabric and
2466 product level:

- 2467 • specific fibres must meet requirements on sourcing, and emissions at their production stage;
- 2468 • fabrics must demonstrate that they satisfy specific physical durability parameters;
- 2469 • the product, as well as its components, must meet specific requirements regarding chemical
2470 content.

2471 These EU Ecolabel criteria disregard the final function of the product, e.g. there is no distinction as to whether
2472 a fabric is used as component of a T-shirt, a winter jacket, or a sofa. As explained in Section 3.2, the same
2473 approach cannot be used in the PS because ecodesign requirements must include aspects like durability and
2474 recycled content, which are closely related to the function and use of the textile product. A preliminary exchange
2475 with the textile industry and associations provided important information on this subject. First, a fabric must be
2476 designed considering its final use because it undergoes different wear and tear according to the function and
2477 use of the textile product. Second, the fact that mechanically recycled natural fibres are shorter than virgin

natural fibres significantly affects the physical characteristics of the yarn and consequently of the fabric. The longer the fibre, the thinner the manufactured yarn can be. Specific yarns are used to produce specific fabrics. In general, a T-shirt, which requires thin yarns, can contain less mechanically recycled natural fibre than a coat, which can be manufactured with thicker yarns. This means that the possibility to use a specific quantity of recycled natural fibres in a textile product depends on the function performed by the textile product including the intended lifespan. Additionally, the content of recycled fibres affects the characteristics of the yarn and influences thereby its aesthetic, and the comfort provided to the user.

7.4 Looking for synergies with other Ecolabels used in the EU

Article 6(f) of the EU Ecolabel Regulation promotes synergy with other officially recognised ISO Type I environmental labels (**Ecolabels**). In the EU, besides the EU Ecolabel for textile products, there are the following:

- Blue Angel – The German Ecolabel: criteria are set by DE-UZ 154 Basic Award Criteria. Edition January 2023, version 2 ⁽¹³¹⁾.
- Nordic Swan Ecolabel: criteria are set by ‘Textiles, hides/skins, and leather’. Version 5.4 ⁽¹³²⁾. Nordic Swan Ecolabel has a separate criteria set for cleaning textiles called “Supplies for microfibre based cleaning”, which includes cloths, mops, pads, and other cleaning products containing microfibres. Version 2.9 ⁽¹³³⁾.

Table 38 reports the scope of these Ecolabels according to the classification adopted in this PS. Besides some differences, the three Ecolabels include in their scope apparel, home/interior textiles, technical textiles like cleaning textiles, and intermediate textile products. Nordic Swan is the only Ecolabel addressing hide and leather products.

Section 13.8.3 (from **Table 163** to **Table 169**) describes in parallel the topics addressed by the criteria of the three Ecolabels, which all have the same approach. They set the product environmental performance at fibre, fabric and product level, as described for the EU Ecolabel criteria in Section 7.3.

The three Ecolabels address the same main topics, even if there are differences in terms of ambition level, specific substances and test methods used as verification.

¹³¹ Blue Angel – The German Ecolabel: DE-UZ 154 Basic Award Criteria. Edition January 2023, version 2. Available at [this link](#). Last accessed on 12 January 2024.

¹³² Nordic Swan Ecolabel: Textiles, hides/skins, and leather. Version 5.4. Available at [this link](#). Last accessed on 12 January 2024.

¹³³ Nordic Swan Ecolabel: Supplies for microfibre based cleaning. Available at [this link](#). Last accessed on 22 April 2025.

2504 **Table 38.** Scope of the textile Ecolabels used the most in the EU

Type of product	EU Ecolabel ^(a)	Blue Angel ^(b)	Nordic Swan ^(c)
Apparel	Textile clothing and accessories: clothing and accessories consisting of at least 80% by weight of textile fibres in a woven, non-woven or knitted form.	- Textile clothing and textile accessories consisting of at least 90% textile fibres by mass. - Functional clothing in which more than 90% by mass of the material is textile fibres or textile substances that have undergone finishing processes (impregnation, sealing, etc.).	Apparel and accessories, for example trousers, shirts, jackets, workwear, uniforms, underwear, handkerchiefs, scarves.
Home/interior textiles	Interior textiles: textile products for interior use consisting of at least 80% by weight of textile fibres in a woven, non-woven or knitted form.	- Textile products for use inside buildings (house and home textiles incl. uncoated carpets) consisting of at least 90% textile fibres by mass. - Bedding consisting of at least 90% textile fibres by mass.	Furnishing fabrics (for both private and professional use), such as towels, bedding, curtains, tablecloths, pillows, duvets, plus textiles for use in the furnishing of cars/trains/aircraft/boats.
Technical textiles	Cleaning products: woven or non-woven products made from textile fibres and intended for the wet or dry cleaning of surfaces and the drying of kitchenware	- Cleaning textiles: woven or non-woven textiles consisting of at least 90% textile fibres by mass that are designed for the wet or dry cleaning of surfaces or for drying household articles. - Textile products designed for food contact (e.g. waxed cloths).	NA
Intermediate products	Textile fibres, yarn, fabric and knitted panels: intermediate products intended for use in textile clothing and accessories and interior textiles, including upholstery fabric and mattress ticking prior to the application of backings and treatments associated with the final product. Non-fibre elements: intermediate products that are incorporated into textile clothing and accessories and interior textiles, including zips, buttons and other accessories, as well as membranes, coatings and laminates.	- Fibres, yarn, fabric, knitted and crocheted items, non-wovens (including textile composites). - Fibres made of stainless steel and mineral fibres are limited to a maximum of 10% by mass.	- Fibres*, yarn, fabric. * Only the following fibre types can be certified with the Nordic Swan Ecolabel as a certified fibre and only if the relevant fibre requirements of the criteria are met: Organic cotton fibres, wool, and other creatine fibres (either sheep, camel, alpaca, or goat), regenerated cellulose produced by closed loop process, flax (linen), silk, bamboo, sisal and other bast fibres. - Durable non-woven textiles that are to be used for apparel and accessories or in interior furnishings.
Other products	Accessories such as bags, handbags, purses, and wallets are eligible for the EU Ecolabel, provided they contain at least 80% textile fibre and that non-fibre elements comply with the requirements set by Commission Decision 2014/350/EU.	Handbags, bicycle bags, backpacks and school bags consisting of at least 70% textile fibres by mass.	- Purses, wallets, and bags. - Hide and leather products, such as jackets, trousers or bags, and hides/skins and leather as raw materials for clothing or home furnishings (including for cars/trains/aircraft/boats), from the following species of animal: sheep, goat, cow, horse, pig, elk, deer, and reindeer.

^(a) EU Ecolabel criteria for textile products. Commission Decision (2014/350/EU).

^(b) Blue Angel – The German Ecolabel: DE-UZ 154 Basic Award Criteria. Edition January 2023, version 2.

^(c) Nordic Swan Ecolabel: Textiles, hides/skins, and leather. Version 5.4.

N.B. The classification of the type of products follows the classification adopted in this PS, which does not necessarily correspond to the classification given by the specific Ecolabel. NA: Not available.

Source: own production

8 Public procurement and current EU voluntary Green Public Procurement criteria

As mentioned in Section 1, the PS will provide the scientific and technical basis for the future development of the possible mandatory EU GPP requirements for textile products within the framework of the ESPR. To this aim, it is important to assess the current public procurement of apparel in the EU (Section 8.1), and the current status of the voluntary EU GPP criteria, which could provide important learnings for the development of future mandatory criteria (Section 8.2).

8.1 Public procurement in the EU

Public procurement in the EU is regulated by Directive 2014/24/EU⁽¹³⁴⁾, Directive 2014/23/EU⁽¹³⁵⁾ and Directive 2014/25/EU⁽¹³⁶⁾. Monitoring public procurement is currently a challenging task. The first reporting and monitoring exercise (COM(2021) 245 final)⁽¹³⁷⁾ submitted by Member States showed that all reports contained more qualitative information than quantitative data. Additionally, the incomplete available data limited the analysis at EU level (COM(2021) 245 final). The monitoring exercise showed that Member States follow different methodologies for data collection, and that in several cases there were discrepancies between figures collected at country level and figures reported on Tenders Electronic Daily (TED)⁽¹³⁸⁾.

TED is the online version of the 'Supplement to the Official Journal' of the EU, which is dedicated to European public procurement. TED gathers information covering public procurement for the European Economic Area, Switzerland, and the Republic of North Macedonia from 1 January 2006 to 31 December 2021. These data include the most important information of the contract notice and contract award notice standard forms, such as who bought what from whom, for how much, and which procedure and award criteria were used. Therefore, TED is the only platform that can currently be used to analyse public procurement in the EU.

Generally, the data reported in TED consist of tenders above the procurement threshold of EUR 139 000. Nevertheless, many public authorities register on TED data of their tenders below this mandatory threshold.

Goods, works, and services that are being procured are classified with Common Procurement Vocabulary (CPV) codes. CPV codes help procurement personnel to classify their contract notices and to help suppliers find the notices which are of interest.

Each CPV code nine digits: the first two digits identify the divisions (XX000000-Y); the first three digits identify the groups (XXX00000-Y); the first four digits identify the classes (XXXX0000-Y); the first five digits identify the categories (XXXXX000-Y). Each of the last three digits gives a greater degree of precision within each category. A ninth digit serves to verify the previous digits.

Table 170 in Section 13.9.1 reports the CPV codes for products in the scope of the PS.

CPV codes related to apparel were investigated in the TED dataset⁽¹³⁹⁾ for 5 years before the pandemic (2015-2019). This time interval was chosen to investigate the latest evolution of public procurements without considering the market disruption caused by the COVID-19 pandemic in 2020.

The description of CPV codes reported in the Contract Award allowed the analysis of apparel public procurement as reported in **Table 39**. The number of Contract Awards procuring apparel in the EU gradually increased from 556 in 2015 to 1 261 in 2019. Products included in the CPV group of 'Occupational clothing, special workwear and accessories' were the most purchased by public authorities in the investigated years, representing between 42% and 46% of the total procured apparel reported in TED.

¹³⁴ Public Procurement Directive. Directive 2014/24/EU of the European Parliament and of the Council of 26 February 2014 on public procurement and repealing Directive 2004/18/EC (Text with EEA relevance)Text with EEA relevance. Available at [this link](#).

¹³⁵ Award of Concession Contracts Directive. Directive 2014/23/EU of the European Parliament and of the Council of 26 February 2014 on the award of concession contracts (Text with EEA relevance) Text with EEA relevance. Available at [this link](#).

¹³⁶ Directive 2014/25/EU of the European Parliament and of the Council of 26 February 2014 on procurement by entities operating in the water, energy, transport and postal services sectors and repealing Directive 2004/17/EC (Text with EEA relevance)Text with EEA relevance. Available at [this link](#).

¹³⁷ Report from the Commission - Implementation and best practices of national procurement policies in the Internal Market. COM/2021/245 final. Available at [this link](#).

¹³⁸ Tenders Electronic Daily (TED). The online version of the 'Supplement to the Official Journal' of the EU, dedicated to European public procurement. Available at [this link](#). Last visited on 12 January 2024.

¹³⁹ Tenders Electronic Daily (TED) (csv subset) – public procurement notices. Available at [this link](#). Last accessed on 12 January 2024.

2550 The analysis showed that the Member States with the highest number of Contract Awards reported in TED were
 2551 France, Germany, Poland and Czechia. More details about the analysis are available in Section 13.9.2.

2552 **Table 39.** Number of Contract Awards procuring apparel in the EU

CPV code	2015		2016		2017		2018		2019	
	N	%	N	%	N	%	N	%	N	%
181XXXXX-X Occupational clothing, special workwear and accessories	241	43	285	46	360	43	431	44	533	42
182XXXXX-X Outerwear	61	11	68	11	93	11	93	10	124	10
183XXXXX-X Garments	57	10	60	10	88	11	99	10	138	11
184XXXXX-X Special clothing and accessories	110	20	47	8	97	12	111	11	142	11
351134XX-X Protective and safety clothing	14	3	37	6	50	6	69	7	103	8
3741XXXX-X Sport goods and equipment	7	1	11	2	8	1	10	1	15	1
3581XXXX-X Individual and support equipment	66	12	108	18	133	16	158	16	206	16
Total	556	100	616	100	829	100	971	100	1 261	100

2553 N.B. N: number of Contract Awards. %: Percentage of Contract Awards compared to the total number of contracts related to textile products.

2554 Source: own elaboration based on Tenders Electronic Daily (TED) (csv subset) – public procurement notices ⁽¹³⁹⁾.

2555 **8.2 Current voluntary EU Green Public Procurement criteria**

2556 As mentioned in Section 1, the PS will provide the scientific and technical basis for the future development of
 2557 the mandatory EU GPP requirements for textile products within the framework of the ESPR. To this aim, the
 2558 analysis of current voluntary EU GPP criteria ⁽¹⁴⁰⁾ could provide some important learnings.

2559 The current voluntary EU GPP criteria are based on the current EU Ecolabel criteria for textile products. In
 2560 addition to EU Ecolabel criteria, the voluntary EU GPP criteria suggest some requirements about textile services
 2561 related to laundry operations and take-back systems.

2562 The initial questionnaire (see **Table 1** in Section 1) allowed the collection of respondents' opinions on the status
 2563 of the current voluntary EU GPP criteria and on the lessons learnt so far. Out of 34 respondents, 12 belonged
 2564 to the governmental institutions and 6 to the manufacturing industry, 35% and 18% of the total, respectively.
 2565 More details about the types of respondents are provided in **Table 176** in Section 13.9.3.

2566 The questionnaire showed that although many European countries ⁽¹⁴¹⁾ have GPP schemes on textile products,
 2567 it is not possible to understand the uptake of the EU GPP criteria. This is mainly due to their voluntary nature,
 2568 and the lack of a framework to collect this kind of data.

2569 In light of possible future mandatory EU GPP criteria, respondents provided the following suggestions:

- 2570 — Provide a clear and fixed set of requirements, which will allow manufacturers to produce goods
 2571 that meet the demand.
- 2572 — Assess administrative obstacles related to the premature disposal of textile products caused by
 2573 contracting or budget period reasons.
- 2574 — Facilitate the verification of product characteristics to procurers, who are usually not sustainability
 2575 experts.
- 2576 — Establish a framework based on Life Cycle Costing of durable textile products.

¹⁴⁰ EU GPP criteria for textile products and services. Commission Staff Working Document on EU green public procurement criteria for textiles products and services. SWD(2017) 231 final. Available at [this link](#).

¹⁴¹ These countries are: Austria, Belgium, Czechia, Denmark, Finland, France, Germany, Hungary, Ireland, Italy, Malta, the Netherlands, Portugal, Sweden and Norway.

- 2577 — Establish a framework which considers environmental and social aspects in the whole value chain,
2578 possibly promoting the purchase of second-hand products.
- 2579 — Support public authorities to develop a common strategy in purchasing and disposing of textile
2580 products.

2581 The positions expressed by the respondents to the initial questionnaire are complemented by COM(2021) 245
2582 final ⁽¹³⁷⁾. This Communication lists the main aspects hindering the adoption of current voluntary EU GPP criteria:

- 2583 • the lack of legal obligation;
- 2584 • the lack of legal certainty on the correct interpretation of the requirement for a 'link to the
2585 subject matter of the contract' and the general fear of litigation;
- 2586 • the lack of data proving the economic benefits of applying GPP criteria;
- 2587 • the difficulty to monitor their application;
- 2588 • the lack of specific knowledge and skills of the public servants engaged in tendering
2589 procedures;
- 2590 • the fact that GPP may be perceived as an obstacle to competition, specifically restricting SME
2591 participation in public tendering.
- 2592

9 Technologies

This section addresses the Task 4 of the MEErP providing a **general technical analysis of the products in the scope**. This analysis aims to describe the following types of product technologies:

- The Base Case of technologies (BC), which is the average product on the market.
- The Best Available Technologies (BAT), which have the most ambitious performances available on the market. BAT are implemented at scale.
- The Best Not yet Available Technologies (BNAT), which have the most ambitious performances, but they are not implemented at scale, therefore they are considered not available on the market.

In this context, product technologies are products with defined characteristics related to a specific product aspect reported in Article 5(1) of the ESPR, such as durability, reparability, etc.

Section 9.1 identifies the relevant product aspects for the products in scope, and groups those that can be potentially addressed with the same product parameters via future requirements. This approach aims to streamline the multi-criteria analysis that will be performed in the following Tasks of the PS.

Section 9.2 describes the product technologies for each relevant product aspect, analysing the complexity of their ecosystems.

Section 9.3 reports on the influence that product aspects have upon each other and proposes product categories based on the analysis performed in section 9.2. Information reported in Tasks 1 to 4 feed the environmental and economic models in Tasks 5 (section 10) and Task 6 (section 11), which describe the base cases and design options, respectively. Each product category reported in section 9.3 will be described with a representative product, which will be used as reference for all products belonging to the specific category.

As for product categories, previous sections of the preparatory study used the product categories suggested by the PEFCR A&F to describe the scope of the study (section 3) and characteristics of the market (section 5). The new categorization reported in section 9.3 serves the aims of the preparatory study based on the analysis of product technologies.

The analysis performed in Section 9 goes beyond the directions reported in the MEErP. It aims to address Task 4 of the MEErP in the context of the ESPR.

9.1 Relevant product aspects

Article 5(1) of ESPR establishes that, in order to address environmental impacts of products, the ecodesign requirements in the delegated acts shall improve the relevant aspects from a specific list of product aspects. This list includes 16 product aspects:

- (a) durability
- (b) reliability
- (c) reusability
- (d) upgradability
- (e) reparability,
- (f) the possibility of maintenance and refurbishment
- (g) the presence of substances of concern
- (h) energy use and energy efficiency
- (i) water use and water efficiency
- (j) resource use and resource efficiency
- (k) recycled content
- (l) the possibility of remanufacturing
- (m) recyclability
- (n) the possibility of recovery of materials

2637 (o) environmental impacts, including carbon footprint and environmental footprint

2638 (p) expected generation of waste

2639 Article 5 of ESPR also establishes that ecodesign requirements in the delegated acts shall be based on the
2640 above product aspects, when they are relevant to the product group concerned. Therefore, this section aims to
2641 evaluate the relevance of the 16 product aspects in the context of textile apparel, that deserve to be further
2642 analysed in the subsequent sections. To this aim, two main actions were performed:

2643 — Discarding product aspects that are not relevant in the context of textile apparel;

2644 — Grouping product aspects that can be addressed together in order to streamline the following steps
2645 of the study.

2646 Section 9.1.1 provides a general description of the methodology followed, whereas Section 9.1.2 describes the
2647 specific steps involved. Finally, section 9.1.3 describes the proposed grouping of product aspects. This approach
2648 aims to streamline any subsequent analysis, rather than deciding at this stage which product aspects could be
2649 regulated with ecodesign requirement.

2650 **9.1.1 Methodology**

2651 The analysis undertaken started from the list of 16 product aspects in Article 5 of ESPR, and aimed to: (a)
2652 exclude product aspects which were not relevant for textile apparel, and (b) group product aspects that could
2653 be potentially addressed with the same product parameters via future requirements. This approach streamlines
2654 the multi-criteria analysis that will be performed in the following tasks of the PS.

2655 Non-relevant product aspects were excluded following three steps:

- 2656 • Generic description of textile apparel,
- 2657 • Guiding questions, and
- 2658 • Qualitative assessment.

2659 First, textile apparel were described considering the following characteristics: final vs intermediate product;
2660 complex vs non-complex product; and durable vs consumable product. This description sets the understanding
2661 of the product characteristics that are crucial for steps 2 and 3. Second, key guiding questions identified product
2662 aspects that evidently were not relevant for textile apparel. Third, a qualitative assessment investigated
2663 technical, socioeconomic and environmental dimensions. This qualitative assessment was based on information
2664 found in the literature and did not aim to produce a comprehensive analysis given this will be performed in the
2665 following tasks of the PS. The purpose of the qualitative assessment is to further refine the justification to
2666 exclude product aspects that are evidently not relevant for textile apparel, and to serve as a basis for the
2667 grouping of product aspects.

2668 Finally, only the relevant product aspects were grouped taking into account the characteristics of each product
2669 aspect and the product parameters reported in Annex I to ESPR. This grouping streamlined the following steps
2670 of the PS because it allowed the use of the same parameters to address more than one product aspect.
2671 Therefore, this grouping decreased the number of relevant product aspects to be investigated in the analysis
2672 of technologies (see section 9.2) and Tasks 5 and 6 of the PS.

2673 **9.1.2 Exclusion of non-relevant product aspects**

2674 This section followed the methodology described in section 9.1.1 and identifies product aspects that are
2675 considered not relevant for textile apparel.

2676 **Generic description of textile apparel**

2677 Textile apparel were screened following three sets of characteristics:

- 2678 — Intermediate product or final product
- 2679 — Complex product or non-complex product
- 2680 — Durable product or consumable product

2681 According to Article 2(3) of the ESPR, an intermediate product is a product that requires further manufacturing
2682 or transformation such as mixing, coating or assembling to make it suitable for customers. Conversely, a final

2683 product is understood to be a product that is already suitable for users. One of the main differences between
2684 intermediate and final products is the lifecycle stages that can be considered. While all lifecycle stages could
2685 be evaluated in the assessment of final products, in the case of intermediate products a cradle-to-gate
2686 approach is followed, where the use stage is disregarded and the end-of-life stage is only partially assessed.
2687 Product aspects that serve to reduce the impacts of use and end-of-life stages would be considered as not
2688 relevant for intermediate products. By definition, textile apparel belongs to **final** products, because it is suitable
2689 for consumers.

2690 According to Council Regulation (EC) 6/2002 ⁽¹⁴²⁾, a complex product is a product which is composed of multiple
2691 components which can be replaced permitting disassembly and re-assembly of the product. In contrast, non-
2692 complex products do not include components or priority parts, thus they do not have components that can be
2693 replaced. A product's priority part is functionally important and is likely to fail or to be upgraded. The part will
2694 have high priority if it is necessary to deliver either primary (necessary to fulfil the intended use) or secondary
2695 functions (necessary to enable, supplement or enhance the primary function) (Cordella, Alfieri, et al., 2019).

2696 Textile apparel belongs to the category of **complex** products, because often it is composed of multiple
2697 components which can be replaced permitting disassembly and re-assembly of the product. Additionally, the
2698 complexity of textile apparel relies on the numerous fibre types used in blends and processed in several ways
2699 along spinning, fabric manufacturing and finishing processes ⁽¹⁴³⁾. Therefore, even textile apparel that is not
2700 made of components is considered to be a complex product.

2701 Finally, textile apparel belongs to **durable** products because it is conceived and designed to last. Conversely,
2702 once used, consumable products are expected to be consumed or discarded.

2703 Therefore, textile apparel encompasses final, complex and durable products whose relevant product aspects
2704 should consider all life-cycle stages, the technological product complexity and its durable function.

2705 **Guiding questions**

2706 Key guiding questions were used to identify product aspects that evidently are not relevant for textile apparel.
2707 These questions addressed only the circularity aspects and did not focus on substances of concern, generic
2708 environmental impacts or use and efficiency of water, energy and resources, because these product aspects
2709 were considered already relevant based on the analysis performed in Section 3.1.3 and 3.3.

2710 Durability

2711 *Is the functionality of the textile apparel expected to be provided for a long lifetime and/or for a high amount*
2712 *of multiple uses?*

2713 Yes, it is. Therefore, durability can be considered a relevant product aspect.

2714 Reliability

2715 *Can the use of the textile apparel or a part/component thereof cause a limiting event in the product?*

2716 Yes, fatigue and or ageing of the product (or of a specific part/component) due to its use can cause a limiting
2717 event that prevents its use. Therefore, reliability can be considered a relevant product aspect.

2718 Reusability

2719 *Could the reuse of textile apparel potentially entail a health risk?*

2720 In general, reuse of textile apparel do not entail health risks for consumers. Therefore, reusability can be
2721 considered a relevant product aspect.

2722 Upgradability

2723 *Would it be technically feasible to upgrade a priority part that could potentially stop functioning, that is already*
2724 *not allowing textile apparel to perform at its fullest and/or there is potential to redesign the product in a way*
2725 *that can be upgraded?*

¹⁴² Council Regulation (EC) No 6/2002 of 12 December 2001 on Community designs. Available at [this link](#).

¹⁴³ A description of this complexity is reported in Section 9.2.1.1.

2726 Yes, it is technically feasible to upgrade a priority part in an item of textile apparel. For instance, reassembling
2727 a zipper of better quality. It is also possible to redesign textile apparel in a way that they can be upgraded. For
2728 instance, improving their ability for disassembly. Therefore, upgradability can be considered a relevant aspect.

2729 Repairability

2730 *Would it be technically feasible to repair/replace the no longer functioning priority part and/or there is potential*
2731 *to redesign the product in a way that can be repairable?*

2732 Yes, it is. Therefore, repairability can be considered a relevant aspect.

2733 Possibility of maintenance

2734 *Will the use and/or storage of textile apparel in specific conditions maintain its expected lifetime or extend it?*
2735 *Would it be possible to postpone a limiting event by performing maintenance activities to any of the*
2736 *parts/components of textile apparel?*

2737 Yes, appropriate use and storage of textile apparel can extend their expected lifetime. Moreover, maintenance
2738 activities (proper washing, drying and ironing) can postpone limiting events. Therefore, possibility of
2739 maintenance can be considered a relevant aspect.

2740 Possibility of refurbishment

2741 *Is it technically feasible to refurbish textile apparel by the manufacturer and/or third party operators?*

2742 Yes, it is. Therefore, possibility of refurbishment can be considered a relevant aspect.

2743 Recycled content

2744 *Is there a waste stream from which material can be recycled and reintroduced in the manufacturing process of*
2745 *textile apparel?*

2746 *Is the waste stream available for using it in the manufacturing of textile apparel?*

2747 *Is the waste stream generated within an acceptable distance from the recycling and manufacturing sites?*

2748 *Can the recycled material be used for the manufacturing of textile apparel with the same or an acceptable*
2749 *minimum quality to keep the properties and function of the product?*

2750 Yes, there is a waste stream of textile apparel from which material can be recycled. This material is available
2751 for using in the manufacturing of textile apparel. Generally, the waste stream is not generated close to the
2752 most common sites for recycling and manufacturing of products. Only post-industrial textile waste is generated
2753 at manufacturing facilities. The possibility to incorporate recycled content largely depends on the type of fibre
2754 concerned: chemical recycling of cotton waste cannot generate new cotton fibres, but it can generate chemically
2755 modified (regenerated) fibres, such as man-made cellulosic fibres (MMCFs). Therefore, recycled content can be
2756 considered a relevant aspect.

2757 Possibility of remanufacturing

2758 *Would it be technically possible to disassemble textile apparel without damaging those components or parts*
2759 *that could have potential to be used in a new product and/or would be possible to design the product in a way*
2760 *that allows it?*

2761 Yes, depending on the specific product and manufacturing process it may be technically possible to disassemble
2762 textile apparel so that some parts are used in a new product. It is also possible to redesign textile apparel in a
2763 way that they can be remanufactured. Therefore, possibility of remanufacturing can be considered a relevant
2764 aspect.

2765 Recyclability

2766 *Is textile apparel made of components/parts or materials that can or have potential to be separated and*
2767 *recycled?*

2768 *Is there available waste derived from the product from which materials can be recovered and recycled?*

2769 *Is the technology to recycle textile apparel's material available? Is there an existing or potential demand for the*
2770 *recycled material?*

2771 Yes, textile apparel is made of materials that can be separated and recycled. There is available waste derived
2772 from textile apparel from which materials can be recovered and recycled. Some recycling technologies are

2773 already available. There is certain demand for material recycled from textile apparel, although it is limited due
2774 to non-competitive prices of recycled fibres. Therefore, recyclability can be considered a relevant aspect.

2775 Possibility of recovery of materials

2776 *Besides preparation for re-use and recycling, is there another way to recover materials from the products in the*
2777 *scope?*

2778 No, the only way to recover materials from textile apparel is via preparation for re-use and recycling. Therefore,
2779 possibility of recovery of materials via any other route can be considered a non-relevant aspect.

2780

2781 The outcome of the qualitative analysis based on key guiding questions is that only the product aspect
2782 'possibility of recovery of materials' can be considered as non-relevant at this point of the analysis.

2783 **Qualitative assessment**

2784 The qualitative assessment investigated technical, socioeconomic and environmental dimensions of the 15
2785 relevant product aspects screened in in the guiding questions. This qualitative assessment used information
2786 gathered through literature review and did not include a comprehensive analysis because this is the aim of the
2787 following steps of the PS, mostly the analysis of technologies (section 9.2), Task 5 and Task 6.

2788 The qualitative assessment aimed to further refine the exclusion of product aspects that are evidently not
2789 relevant for textile apparel, and to serve as a basis for the grouping of product aspects. In this way, the following
2790 steps of the PS can exclude their assessment for a streamlined process.

2791 The qualitative assessment included:

- 2792 — The technical dimension, which addressed product characteristics and/or improvement potential
- 2793 from a technical perspective.
- 2794 — The socioeconomic dimension, which addressed the economic feasibility, impacts on job
- 2795 loss/creation and user-related aspects.
- 2796 — The environmental dimension, which addressed environmental impacts caused by the consumption
- 2797 of the product and potential improvements.

2798 The results of the qualitative assessment showed that all screened product aspects had at least one dimension
2799 that resulted relevant for textile apparel. Section 13.10.1 in the Annex provides details of the qualitative
2800 assessment.

2801 Finally, **Table 40** reports the outcome of the analysis aiming to exclude non-relevant product aspects.

2802 **Table 40.** Relevance of product aspects for textile apparel

Relevant product aspects	Non-relevant product aspects
Durability Reliability Reusability Upgradability Repairability Possibility of maintenance and refurbishment Presence of substances of concern Energy use and energy efficiency Water use and water efficiency Resource use and resource efficiency Recycled content Possibility of remanufacturing Recyclability Environmental impacts Expected generation of waste	Possibility of recovery of materials

2803 *Source: own production*

2804 **9.1.3 Grouping of relevant product aspects**

2805 Grouping of relevant product aspects aimed to streamline the multi-criteria analysis that will be performed in
2806 the following tasks of the PS. Each product aspect represents a criterion of the analysis. If the number of these
2807 criteria decreases, the analysis is more efficient.

2808 To this aim, **Table 41** reports some of the characteristics typical for specific product aspects, using the results
 2809 of the qualitative assessment performed in section 9.1.2. Additionally, **Table 42** reports the direct interaction
 2810 between product aspects and product parameters reported in Annex I to the ESPR. The product aspects that
 2811 have common characteristics were grouped because they can be addressed via the same product parameters.

2812 **Table 41.** Product aspects and required characteristics of products

Product aspect	Required characteristics of products (Annex I of ESPR)
Durability	A highly durable item of textile apparel should have, among others, the following characteristics over time: high resistance to abrasion, tearing, pilling, colour-fastness, dimensional changes and seam slippage. It should have easily accessible information for repair and maintenance. Moreover, a durable product should be desirable to the user for a long time. If textile apparel is modular, its priority parts should have easy physical access. Additionally, the process, tools and fasteners required for disassembly should be simple.
Reliability	A highly reliable item of textile apparel should have, among others, the following characteristics: high resistance to abrasion, tearing, pilling, colour-fastness, soiling, dimensional changes and seam slippage. Additionally, a reliable product should be desirable to the user for long time.
Reusability	A highly reusable item of textile apparel should have, among others, the following characteristics: high resistance to abrasion, tearing, pilling, colour-fastness, soiling, dimensional changes and seam slippage. It should have easily accessible information for repair and maintenance. Moreover, a reusable product should be desirable to the user for a long time. If textile apparel is modular, its priority parts should have easy physical access. Additionally, the process, tools and fasteners required for disassembly should be simple.
Upgradability	A highly upgradable item of textile apparel should have, among others, the following characteristics: it should be highly modular in design, with easy physical access to priority parts. The process, tools and fasteners required for disassembly should be simple.
Repairability	A highly repairable item of textile apparel should have, among others, the following characteristics: it should be highly modular in design, with easy physical access to priority parts. The process, tools and fasteners required for disassembly should be simple. The user of the product should also have access to repair services.
Possibility of maintenance	An item of textile apparel with possibilities for maintenance should have, among others, the following characteristics: it should have easily accessible information for care in terms of cleaning, drying, ironing and storing the product.
Possibility of refurbishment	An item of textile apparel with possibilities for refurbishing should have, among others, the following characteristics: it should be highly modular in design, with easy physical access to priority parts. The process, tools and fasteners required for disassembly should be simple.
Presence of substances of concern	An item of textile apparel with good information on presence of substances of concern is accompanied with a comprehensive list of all the substances of concern that it contains (above specified thresholds and subject to exemptions, as appropriate). Substances are used in order to give specific characteristics to the product, facilitate the manufacturing process or to help during the treatment of the product when it becomes waste. Consequently, substances of concern could affect durability, recyclability and environmental impacts.
Energy use and energy efficiency	An item of textile apparel with low energy use or high energy efficiency should (1) be manufactured with low energy consumption, (2) use materials which are not energy intensive in their manufacturing stage, (3) allow to reduce the energy consumption during the use phase in laundering, drying and ironing activities, and (4) be treated at its end of life with non-energy intensive techniques.
Water use and water efficiency	An item of textile apparel with low water use or high water efficiency should (1) be manufactured with low water consumption, (2) use materials which are not water intensive in their manufacturing stage, (3) allow to reduce the water consumption during the use phase in laundering activities, and (4) be treated at its end-of-life with non-water intensive techniques.
Resource use and resource efficiency	An item of textile apparel with low resource use or high resource efficiency should, among other things, use materials that throughout its life cycle stages (1) consume raw materials produced in sustainable way, (2) indirectly use land assuring its future use with the same activity, (3) use ecosystems without damaging their biodiversity and general balance.
Recycled content	An item of textile apparel with recycled content should contain recycled materials, in substitution of virgin materials.
Possibility of remanufacturing	An item of textile apparel with possibilities for remanufacturing should have, among others, the following characteristics: it should be highly modular in design, with easy physical access to priority parts. The process, tools and fasteners required for disassembly should be simple. It should have easily accessible information for repair.

Product aspect	Required characteristics of products (Annex I of ESPR)
Recyclability	In order to be recyclable, an item of textile apparel should meet all the following five characteristics when it becomes waste: (1) it can be effectively collected; (2) it can be sorted, i.e. segregated from other waste and sent to the subsequent recycling pathways; (3) it can be prepared for recycling, or can be sent directly recycling without specific preparation; (4) its fibre content can be used as feedstock for one or more recycling techniques; (5) it has no elements or substances that disrupt the collection, sorting, preparation for recycling and recycling, or that prevent the use of the recycled fibre.
Environmental impacts	An item of textile apparel with low environmental impact should have, among others, the following characteristics: in all life cycle stages it should (1) use a limited quantity of energy and water, (2) release directly and indirectly a limited quantity of pollutants (e.g. SO _x , NO _x , COD, microplastics) into the environment, use in the product and emit into the environment minimum possible amounts of substances of concern.
Expected generation of waste	An item of textile apparel with low expected generation of waste should have, among others, the following characteristics: (1) in all life cycle stages, it should generate minimal amounts of waste, (2) it should be designed and manufactured to prevent the generation of post-industrial waste, (3) ideally it should be designed to increase emotional attachment to the user to limit the demand for new products, (4) it should be durable to postpone the demand for new products.

2813 **Table 42.** Interaction between product aspects and product parameters reported in Annex I to ESPR

Product parameters	Durability	Reliability	Reusability	Upgradability	Repairability	Possibility of maintenance	Possibility of refurbishment	Presence of substances of concern	Energy use and energy efficiency	Water use and water efficiency	Resource use and resource efficiency	Recycled content	Possibility of remanufacturing	Recyclability	Environmental impacts	Expected generation of waste
Product's guaranteed lifetime																
Technical lifetime																
Mean time between failures																
Indication of real use information on the product																
Resistance to stressor ageing mechanisms																
Characteristics, availability, delivery time and affordability of spare parts																
Modularity																
Compatibility with commonly available tools and spare parts																
Availability of repair instructions																
Availability maintenance instructions																
Number of materials and components used																
Use of standard components	Not applicable to textile apparel															

Product parameters	Durability	Reliability	Reusability	Upgradability	Repairability	Possibility of maintenance	Possibility of refurbishment	Presence of substances of concern	Energy use and energy efficiency	Water use and water efficiency	Resource use and resource efficiency	Recycled content	Possibility of remanufacturing	Recyclability	Environmental impacts	Expected generation of waste
Use of component and material coding standards for the identification of components and materials																
Number and complexity of processes and whether specialised tools are needed																
Ease of non-destructive disassembly and reassembly																
Conditions for access to product data	Not applicable to textile apparel															
Conditions for access to or use of hardware and software needed	Not applicable to textile apparel															
Conditions of access to test protocols or not commonly available testing equipment	Not applicable to textile apparel															
Availability of guarantees specific to remanufactured or refurbished products																
Conditions for access to or use of technologies protected by intellectual property rights																
Use of easily recyclable materials																
Safe, easy and non-destructive access to recyclable components and materials																
Material composition and homogeneity																
Possibility for high purity sorting																
Avoidance of technical solutions detrimental to reuse																
Avoidance of technical solutions detrimental to upgrading																
Avoidance of technical solutions detrimental to repair																

Product parameters	Durability	Reliability	Reusability	Upgradability	Repairability	Possibility of maintenance	Possibility of refurbishment	Presence of substances of concern	Energy use and energy efficiency	Water use and water efficiency	Resource use and resource efficiency	Recycled content	Possibility of remanufacturing	Recyclability	Environmental impacts	Expected generation of waste
Avoidance of technical solutions detrimental to maintenance																
Avoidance of technical solutions detrimental to refurbishment																
Avoidance of technical solutions detrimental to remanufacturing																
Avoidance of technical solutions detrimental to recycling																
Use of substances, in particular the use of substances of concern																
Use or consumption of energy, in one or more life cycle stages of the product																
Use or consumption of water, in one or more life cycle stages of the product																
Use or consumption of other resources, in one or more life cycle stages of the product																
Use or content of recycled materials and recovery of materials																
Use or content of sustainable renewable materials																
Weight and volume of the product and its packaging																
Incorporation of used components																
Quantity, characteristics and availability of consumables needed for proper use																
Environmental footprint of product																
Carbon footprint of product																
Material footprint of product																
Microplastic and nanoplastic release																
Emissions to air																

Product parameters	Durability	Reliability	Reusability	Upgradability	Repairability	Possibility of maintenance	Possibility of refurbishment	Presence of substances of concern	Energy use and energy efficiency	Water use and water efficiency	Resource use and resource efficiency	Recycled content	Possibility of remanufacturing	Recyclability	Environmental impacts	Expected generation of waste
Emissions to water																
Emissions to soil																
Noise																
Amounts of waste generated																
Functional performance																
Reduction of material consumption																
Load and stress optimisation of structures	Not applicable to textile apparel															
Integration of functions within the material or into a single product component	Not applicable to textile apparel															
Use of lower density or high strength materials and hybrid materials																
Waste reduction																

Coloured cells highlight relevant product parameters for specific product aspects.

Source: JRC own elaboration.

In the context of textile apparel, the product aspects of durability, reliability and reusability were grouped because they have common characteristics and have overlapping aims which can be reached with the same actions. A durable product is likely to be reliable and reusable. **Physical durability** is taken as the leading aspect of this group because it is expected to relate to intrinsic measurable properties of the product which are reported in Annex I to ESPR and allow to indirectly address reliability and reusability. Conversely, emotional durability refers to the emotional attachment that the user has to the product and it does not fall into the definition of "durability" reported in the ESPR (**Table 2**). Although emotional durability is not a product aspect, its relevance to the life-cycle environmental impacts of the textile apparel was largely taken into account in the following sections of the PS.

Although the product aspect of **maintenance** is strictly connected to the physical durability, it was addressed separately because it is mostly related to information to be provided to the user, rather than connected to the physical performance of the product.

The third group of product aspects is led by **repairability** which has a definition and characteristics that closely relate to upgradability, possibility of refurbishment, and possibility of remanufacturing. When addressing repairability with product modularity, use of standard components, and the other relevant parameters, the product aspects of upgradability, refurbishment and remanufacturing is indirectly addressed.

The expected **generation of waste** is considered as a product aspect to be addressed individually, providing the feedstock for the recycling system.

The fifth group gathers **recyclability and recycled content**, because recycled material should come from recyclable textile products. From this perspective, these product aspects share the same ecosystem and are

affected by the same process techniques, business models, legislation, and industrial practices. More details about this will be provided in the analysis of technologies in section 9.2.5.

The product aspect addressing **environmental impacts** includes the assessment of use (and efficiency) of water, energy and resources, which are the fundamental elements affecting the environment (see section 3.3). However, the use of recycled material and the durability of the product affect its environmental impacts. The following steps of the PS will develop the environmental and economic model that will take into account all resources used in the entire life-cycle of the textile apparel. In particular, the part of the model related to raw material production will gather available data coming from the most commonly used practices. This approach will support the identification of resource use that less negatively impacts the environment in this specific stage. Additionally, Task 5 will report a specific analysis on microplastics release in the whole lifecycle of textile products.

Although the **presence of substances of concern** strongly affects other product aspects, it is suggested to be addressed separately because it mainly refers to information requirements to be reported by the economic operator placing/making available the product on the market. Substances are used to give specific characteristics to the product, facilitate the manufacturing stage, or affects the treatment of the product when it becomes waste. Consequently, substances of concern could directly affect the physical durability, the recyclability, the recycled content and the environmental impacts.

Therefore, the next stages of the PS will address the following groups of relevant product aspects:

- Physical durability, which includes physical durability, reliability and reusability;
- Maintenance;
- Repairability, which includes repairability, upgradability, possibility of refurbishment, and possibility of remanufacturing;
- Generation of waste;
- Recyclability and recycled content;
- Environmental impacts, which include environmental impacts, energy use and energy efficiency, water use and water efficiency, resource use and resource efficiency; and
- Presence of substances of concern.

9.2 Analysis of technologies

For each relevant product aspect, this section describes the product technologies following a three-step analysis:

- Step 1: analysis of the ecosystem related to the specific product aspect;
- Step 2: identification of a methodology to describe the product technologies;
- Step 3: description of the product technologies based on previous steps. This description could be supported by a categorization to best describe the products in the scope.

In step 1, the ecosystem related to the specific product aspect was studied considering four elements:

- the **process techniques**, which are the instruments and practices used along the stages of the product's lifecycle to manufacture or treat product technologies,
- the **business models** of economic operators in the ecosystem,
- the **user behaviour**,
- the **legislative framework** and **industrial best practices**.

In step 2, the methodology used was specific to the product aspect and was based on the analysis of the specific ecosystem. In step 3, the categorization aimed to gather all products that can be subject to the same future requirements.

Sections 9.2.1, 9.2.2 and 9.2.3 address the extension of product lifespan in term of physical durability, maintenance and repairability, respectively. Section 9.2.4 analyses the waste generation, while section 9.2.5 addresses the recirculation of materials in terms of recyclability and recycled content. Section 9.2.6 analyses

the environmental impacts, including the use and efficiency of water and energy. Finally, section 9.2.7 addresses the presence of substances of concern.

9.2.1 Physical durability

This section addresses the physical durability of textile apparel, gathering information from currently available physical durability frameworks and from consultation with all stakeholders supporting the development of the PS. In particular, this section includes a description of how the topic is addressed by the PEFCR A&F to shed light on the characteristics of the physical durability frameworks belonging to different legislative contexts.

9.2.1.1 The ecosystem of physical durability

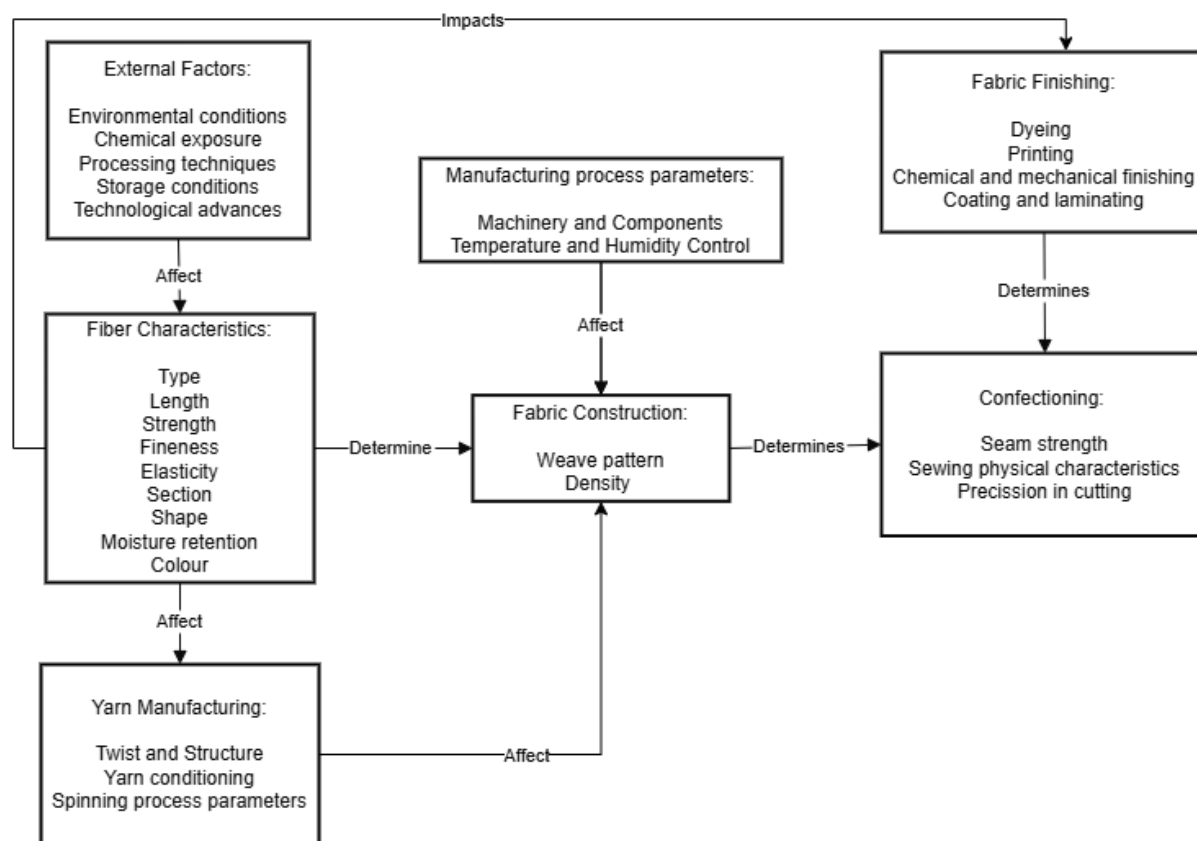
Manufacturing factors and process techniques

The characteristics of the product technologies related to physical durability are strictly related to all process techniques involved in the manufacturing stage, as well as the choice of raw materials and the choice of function of the product made at the design stage.

Figure 22 shows a simplification of the complex relations among the numerous factors influencing the physical durability of textile apparel. These factors are grouped according to the specific process during the manufacturing stage. The fibre characteristics directly affect yarn and fabric manufacturing and finishing processes. Fibre characteristics include, type, length, strength, fineness, elasticity, fibre section shape, moisture retention, and colour. At the same time, there are external factors such as environmental and storage conditions, and chemical exposure that directly affect the characteristics of the fibres. During the manufacturing of the yarn, the chosen spinning method provides specific levels of evenness, softness and strength. Moreover, spinning parameters such as the twist level, spinning tension, temperature and humidity control play a crucial role on the characteristics of the yarn. The degree of twist, chosen by yarn manufacturers, enhances tenacity and resistance to pilling but diminishes the softness and elasticity of the yarns. Additionally, it can increase spirality. The fabric manufacture is strictly related to the final application: weaving is generally used for relatively more rigid and stronger fabrics (e.g. used in pants and shirts), whereas knitting is usually used for relatively more elastic fabrics (e.g. used in T-shirts and dresses). Fabric manufacturing is affected by the specific process techniques used and the ability to control temperature and humidity during the process. During the finishing processes, dyeing and printing give colour-related characteristics, whereas chemical and mechanical finishing provide functional and physical properties, respectively. Fabric manufacturing and finishing directly affect the confectioning process where the physical characteristics of sewing, the precision of cuts and seam strength play an important role in the final appearance and physical durability of the textile apparel. Section 13.10.2 provides more details about the complexity simplified in **Figure 22**.

2912

Figure 22. Overview of the main factors influencing the physical durability of textile apparel



2913

2914

Source: Own production based on consultation with stakeholders and (Rahman et al., 2023)

2915 User behaviour

2916 Users influence the physical durability of textile apparel mainly at the moment of the purchase when choosing
 2917 a product with specific characteristics⁽¹⁴⁴⁾. Purchasing decisions are an important factor in shaping market
 2918 trends, as consumer preferences can drive brands to offer more physically durable products. This is particularly
 2919 true whenever the product is placed on the market by a company following the consumer-led operation model
 2920 (see Section 5.5). The analysis of user behaviour in Section 6 revealed that product quality is an important
 2921 parameter taken into consideration when users buy an item of textile apparel. Physical durability is considered
 2922 one of the main quality aspects. However, when choosing a textile apparel in physical shops, users are driven
 2923 by their subjective judgement based on the look and touch. When buying online, users still rely on the possibility
 2924 to check the look and touch of the textile apparel if there is the possibility to return it.

2925 Price is another important aspect that users take into account when buying textile apparel, but they generally
 2926 do not consider it an indicator of quality, and therefore they also do not consider it to be an indicator of physical
 2927 durability (Section 6). At the same time, the literature revealed that relatively cheap products are disposed of
 2928 more frequently than higher-priced ones (Morgan and Birtwistle, 2009a; Joy et al., 2012a). The authors found
 2929 only one study investigating the potential relation between price of products and their properties that could be
 2930 related to the physical durability. Guo et al. (2025) performed a series of standardized tests on 17 jeans and 33
 2931 T-shirts ranging from EUR 2.89 to EUR 173.27⁽¹⁴⁵⁾. The analysis of visual assessment, colour fastness, seam
 2932 strength, pilling, and spirality showed that the performances of these parameters were not correlated to the
 2933 price of the product. The sample investigated by Guo et al. (2025) and Morris et al. (2025) showed that
 2934 expensive products not necessarily have higher performances in terms of products robustness.

¹⁴⁴ Users also affect the actual lifespan of the products with their maintenance practices, but this topic is addressed in section 9.2.2, where the product aspect of maintenance is analysed.

¹⁴⁵ The conversion rate used was 1 GBP = 1.16 EUR.

2935 All in all, although physical durability refers to intrinsic properties of textile apparel, currently users have no way
2936 to access this information.

2937 *Legislative frameworks and industrial practices*

2938 Currently, the industry uses numerous standards to measure specific parameters related to the physical
2939 durability of textile apparel (Section 4.4). In France, economic operators placing products generating waste on
2940 the French market are requested by the Law n° 2020-105 (¹⁴⁶) to report information about extrinsic
2941 durability (¹⁴⁷) of the product.

2942 *Business models*

2943 Companies in the textile apparel industry adopt numerous business models (Sections 5.5. and 5.6).

2944 Some of them promote physically durable products and tend to communicate the intrinsic characteristics of the
2945 textile apparel. This approach is most common among economic operators dealing with sportswear and
2946 workwear as technical performance is prioritised. Other companies place on the market textile apparel with high
2947 frequency, either to satisfy the demand of customers for new items, or to promote consumption of new
2948 collections. Usually, economic operators using this business model do not pay attention to physical durability of
2949 the textile apparel because the item would be changed or disposed of by the user relatively soon after purchase
2950 (Piippo et al., 2022b).

2951 **9.2.1.2** Natural vs synthetic fibres – duality or complexity?

2952 The physical durability of textile apparel is often perceived to be higher for products made with synthetic fibres,
2953 compared to those made with natural fibres. This perception may be attributed to the fact that synthetic fibres
2954 are man-made fibres often designed to be stronger than natural fibres (**Table 177** in section 13.10.2 reports
2955 comparison on characteristics). However, the reality is more complex and the factors influencing the physical
2956 durability of textile apparel are multifaceted and interconnected (Section 9.2.1.1 and Section 13.10.2). In
2957 practice, a textile apparel with a dense cotton fabric and well-constructed seams can outlast a polyester-based
2958 product with loosely woven fabric and inadequately constructed seams. Additionally, each fibre and fibre blend
2959 has specific properties used for particular applications.

2960 To meet consumer needs, textile apparel must satisfy three key requirements: (1) perform a specific function,
2961 (2) meet consumer taste and comfort expectations, and (3) be reasonably priced. Achieving these requirements
2962 involves a complex engineering process that takes into account the various factors influencing physical
2963 durability. The diverse range of fibres, each with unique characteristics, plays a crucial role in providing specific
2964 properties to textile apparel.

2965 As a result, about 47% of textile apparel on the EU market is made from blends of natural and chemical fibres.
2966 Single-fibre products account for about 53%, with 22% made of cotton and 15% made of polyester (Bakowska
2967 et al., 2025).

2968 **9.2.1.3** How to assess physical durability

2969 The definition of physical durability (see Section 2) suggests the need to establish a methodology capable of
2970 assessing the performance of textile apparel over time, so to determine if it can still provide the same function.
2971 In theory, the methodology could include three steps:

- 2972 1. The **identification of key parameters**, which describe specific characteristics of the product.
2973 These parameters should consider:
- 2974 (a) The interaction among the manufacturing factors and process techniques (Section 9.2.1.1);
- 2975 (b) The most common failure modes, in order to address the main problems triggering the disposal of
2976 the products;
- 2977 (c) The parameters currently tested by the textile apparel industry;

¹⁴⁶ LOI n° 2020-105 du 10 février 2020 relative à la lutte contre le gaspillage et à l'économie circulaire. Available at [this link](#). Last accessed on 10 October 2024.

¹⁴⁷ The extrinsic durability is related to the emotional attachment of the user to the product. It is not related to physical properties of the product.

2978 (d) The principle of economy, to avoid redundant analysis and limit testing costs.

2979 2. The **assessment of the key parameters over time**, which should be done simulating the aging of

2980 the product via wearing during the actual use, cleaning, drying, potential ironing and storage.

2981 3. The **grouping of products** that can be described with the same key parameters and performance

2982 levels.

2983 These steps assume that the physical durability is driven by intrinsic properties of the product, when it is

2984 correctly maintained. The possibility to repair the product could extend its service lifespan. This aspect is

2985 separately addressed in section 9.2.3.

2986 The durability framework is capable to quantify the relation between:

2987 — initial performance level of the key parameters, and

2988 — performance level of the key parameters after a defined number of uses, which express the

2989 product aging.

2990 Therefore, the durability framework allows the quantification of the extension of the service lifespan due to a

2991 known improvement of the performance level of the key parameters.

2992 The approach of this methodology, which requires assessing the evolution of key parameters over time, does

2993 not seem implementable with the standardised tests currently available to the textile industry. The authors

2994 were unable to identify standardised tests that effectively simulate the ageing of textile apparel. Until now, the

2995 industry has developed numerous standards to assess specific parameters relevant to their products (see

2996 Section 13.4). The consultation with stakeholders revealed that nowadays each company adopts **its own**

2997 **acceptance level** for specific parameters measured with test methods that could vary from company to

2998 company. Currently, **there is no standardized test method that aims to assess the aging of textile**

2999 **apparel**.

3000 There are standardized tests applying physical and chemical stresses, which remind to those occurring during

3001 washing and drying (e.g. ISO 6330:2021) and dry-cleaning (e.g. ISO 3175-2:2017). Usually, companies ask

3002 certified laboratories to assess the performance of their products after about three to five cleaning cycles.

3003 There are also standardised tests capable to investigate the resistance of the fabric to a specific physical stress

3004 (e.g. ISO 12945:2020) (**Table 124**). Usually, companies ask certified laboratories to assess the resistance of

3005 their products to the specific level of stress they establish.

3006 The aim of the abovementioned tests currently used by the industry was never to assess the effect of the aging

3007 on the use of the product. The consultation with stakeholders revealed that the stresses applied to the textile

3008 product via these tests do not simulate the stresses occurring during the use phase, meaning during wearing,

3009 washing, drying, potential ironing, and storage. Therefore, the currently available tests do not simulate the aging

3010 process of the textile product.

3011 The Durhabi project ⁽¹⁴⁸⁾ was the first project investigating the performances of 308 items of textile apparel ⁽¹⁴⁹⁾

3012 as new item and over 1, 5, 15, 30 and 60 washing cycles. Although these 308 items are hardly representative

3013 of all products placed on the EU market, the results of the Durhabi project are the only source of information

3014 currently available. In theory, the collection of a representative sample should consider the numerous product

3015 parameters affecting the durability performances of the product, as reported in section 9.2.1.1 and section

3016 13.10.2. Most probably, the products analysed by Durhabi are relatively good-performing products because

3017 they were voluntarily provided by companies participating to the project. However, this statement cannot be

3018 verified because nowadays nobody knows which performance levels the products placed on the EU market

3019 have. This happens because, as reported above, each company adopts its own acceptance level for specific

3020 parameters measured with test methods that could vary from company to company.

3021 The analysis performed by the Durhabi project involved numerous parameters related to the most frequent

3022 failure modes determined via consumer surveys. The results showed that the performance in terms of the

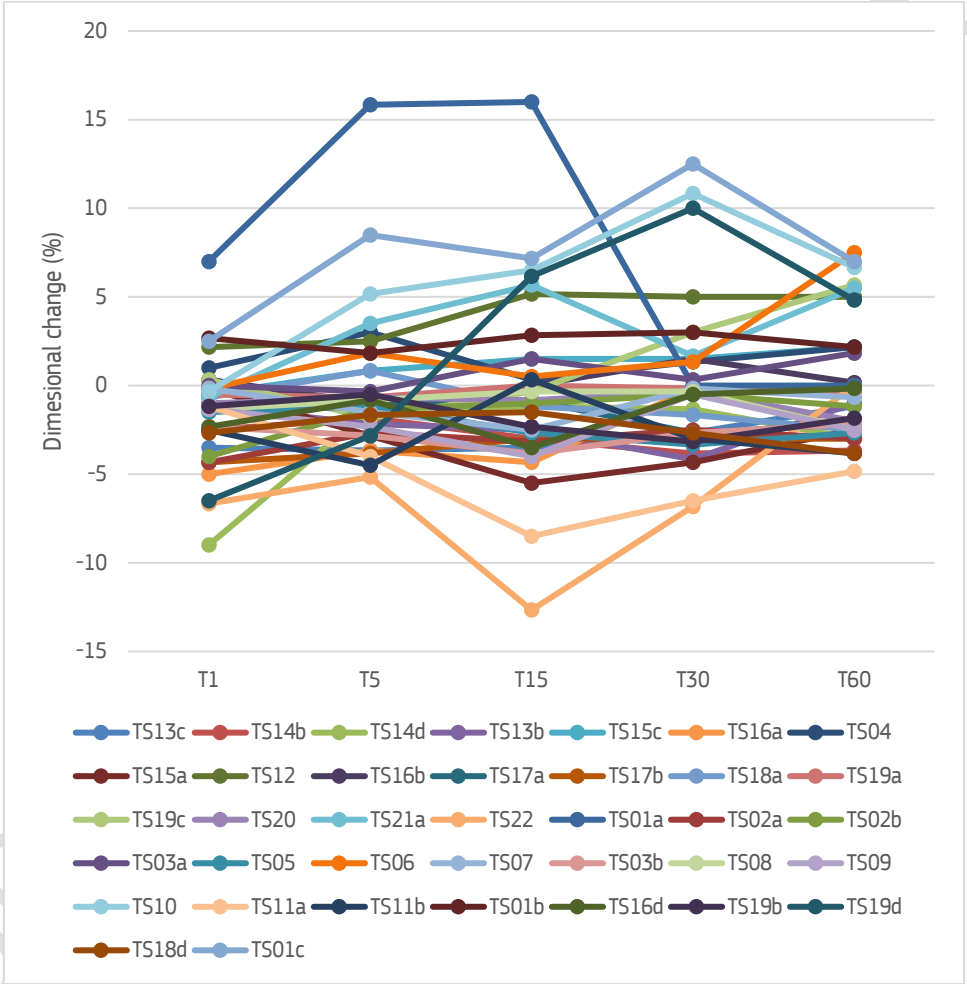
3023 selected key parameters does not necessary decrease over the 60 cleaning cycles, but sometimes even

¹⁴⁸ Livre blanc DURHABI — Mesurer la durabilité textile pour transformer la filière / juin 2025. Available at [this link](#). Last accessed on 13 August 2025.

¹⁴⁹ In particular, 37 t-shirts and bodies, 30 trousers and shorts, 28 sweaters and midlayers, 30 dresses and skirts, 41 items of underwear, 30 stockings, 30 shirts, 30 items of swimwear, 30 jackets and coats, 22 accessories.

improved over the washing cycles. In particular, the increase of stress over the washing cycles showed a worsening of the key parameters investigated via visual inspection following ISO 15487:2018⁽¹⁵⁰⁾. In this case, the parameters were colour change, fabric smoothness, pilling, fuzzing and matting, seams smoothness and slippage, and degradation of non-textile parts. On the contrary, the increased stress over the washing cycles showed sometimes improvements of dimensional change⁽¹⁵¹⁾ measured following ISO 5077:2007⁽¹⁵²⁾, and improvement of spirality⁽¹⁵³⁾ measured following ISO 16322-3:2021⁽¹⁵⁴⁾. **Figure 23** and **Figure 24** show examples of results obtained.

Figure 23. Dimensional change test run in the Durhabi project on 37 T-shirts over washing cycles



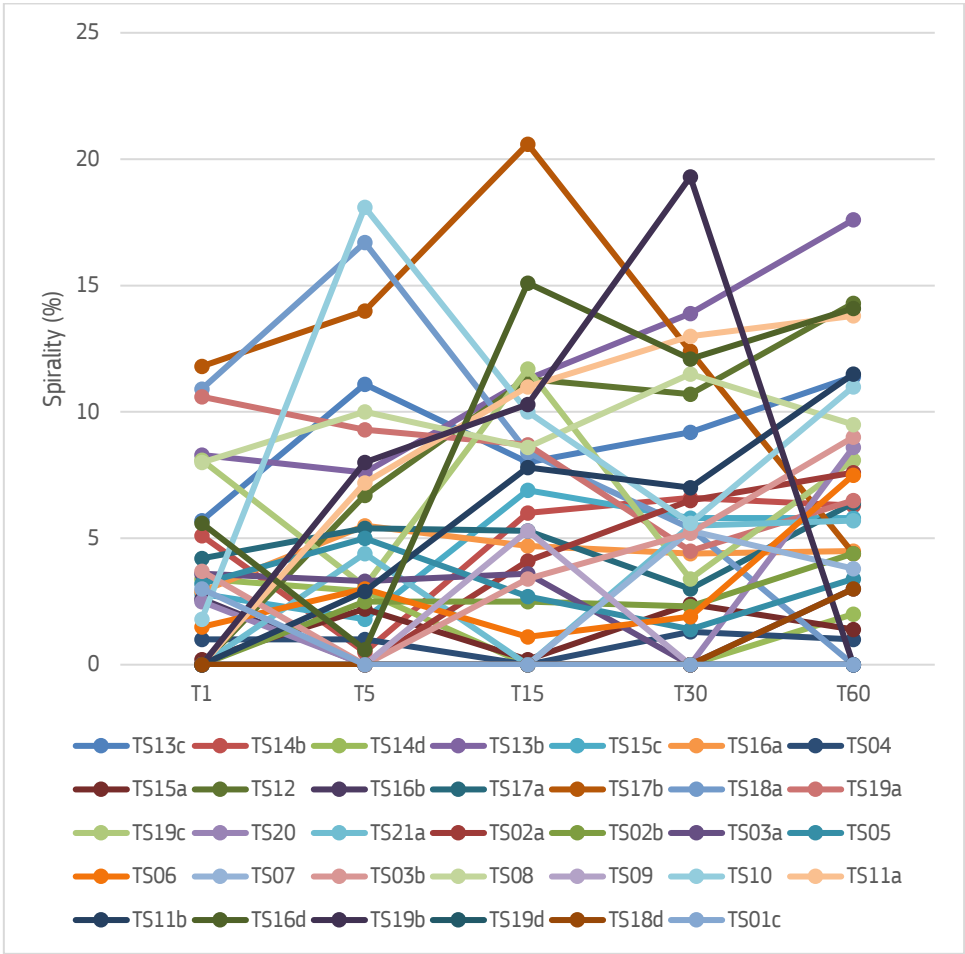
T1, T5, T15, T30 and T60 stand for the number of washing cycles. The Dimensional change was measured following ISO 5077:2007.

Source: Durhabi project. Available at [this link](#). Last accessed on 13 August 2025.

¹⁵⁰ ISO 15487:2018. Textiles — Method for assessing appearance of apparel and other textile end products after domestic washing and drying. Available at [this link](#). Last accessed on 13 November 2025.
¹⁵¹ As improvement of dimensional change here is meant that sometimes the value of the dimensional change decreases in absolute value over washing cycles.
¹⁵² ISO 5077:2007 Textiles — Determination of dimensional change in washing and drying. Available at [this link](#). Last accessed on 13 November 2025.
¹⁵³ As improvement of spirality here is meant that sometimes the value of the spirality decreases over washing cycles.
¹⁵⁴ ISO 16322-3:2021 Textiles — Determination of spirality after launderingPart 3: Woven and knitted garments. Available at [this link](#). Last accessed on 13 November 2025.

3035

Figure 24. Spirality test run in the Durhabi project on 34 T-shirts over washing cycles



3036

3037

3038

T1, T5, T15, T30 and T60 stand for the number of washing cycles. The spirality was measured following ISO 16322-3:2021.

Source: Durhabi project. Available at [this link](#). Last accessed on 13 August 2025.

3039

3040

Due to the lack of suitable method to simulate the aging process of textile apparel, an alternative approach could be to use **robustness as proxy for physical durability**. In this report, robustness is defined as:

3041

3042

“the capability of a product to resist, i.e. maintain its physical structure and appearance, after undergoing external stresses, which could be of chemical or physical nature”.

3043

3044

Based on the proposed definition of robustness, a methodology capable to assess the performance of textile apparel in terms of its robustness could include two steps:

3045

— Step 1: **Identification of key parameters**, which should consider:

3046

3047

3048

3049

3050

- The most common failure modes, in order to address the main problems triggering the disposal of the products;
- The parameters currently tested by the textile apparel industry;
- The interaction among the manufacturing factors and process techniques (Section 9.2.1.1);
- The principle of economy, to avoid redundant analysis and limit testing costs.

3051

3052

— Step 2: **Grouping of products** that can be described with the same key parameters and performance levels.

3053

3054

3055

The concept of robustness differs from that of physical durability because the time component is not addressed and therefore does not cover product aging. This makes the robustness framework not capable to quantify the relation between:

- 3056 — initial performance level of the key parameters, and
- 3057 — performance level of the key parameters after a defined number of uses, which express the
- 3058 product aging.

3059 This means that in the robustness framework, a known increase of performance level of key parameters

3060 corresponds to an unknown increase of physical durability. Therefore, it is not possible to quantify the

3061 consequent increase of the service lifespan related to physical properties of the product. Experts of the textile

3062 industry can only propose values based on their judgement, which will not be verifiable. Textile experts cannot

3063 base their judgement on direct experience, because nobody performs such measurements due to lack of aging

3064 tests.

3065 The literature reports two studies investigating the main failure modes of the products in scope of the PS.

3066 Cooper and Claxton (2022) assessed 1 476 donated products, which were considered not suitable for resale.

3067 Overall, 69% of assessed products had either a colour-related or logo problem (including fading or

3068 discolouration) and 75% had fabric-related problems (including pilling, fabric breakdown, loss of dimensional

3069 stability, holes in seams, trim failure). Therefore, many products had multiple problems (**Table 43**). The Durhabi

3070 project analysed 1 200 products returned to retailers and manufacturers, and performed consumer surveys to

3071 determine their reasons to stop using textile apparel. Most of the causes were related to deformation, holes,

3072 pilling, seam and trim faults (**Figure 25**).

3073 **Table 43.** Most common textile apparel failure modes investigated by Cooper and Claxton (2022)

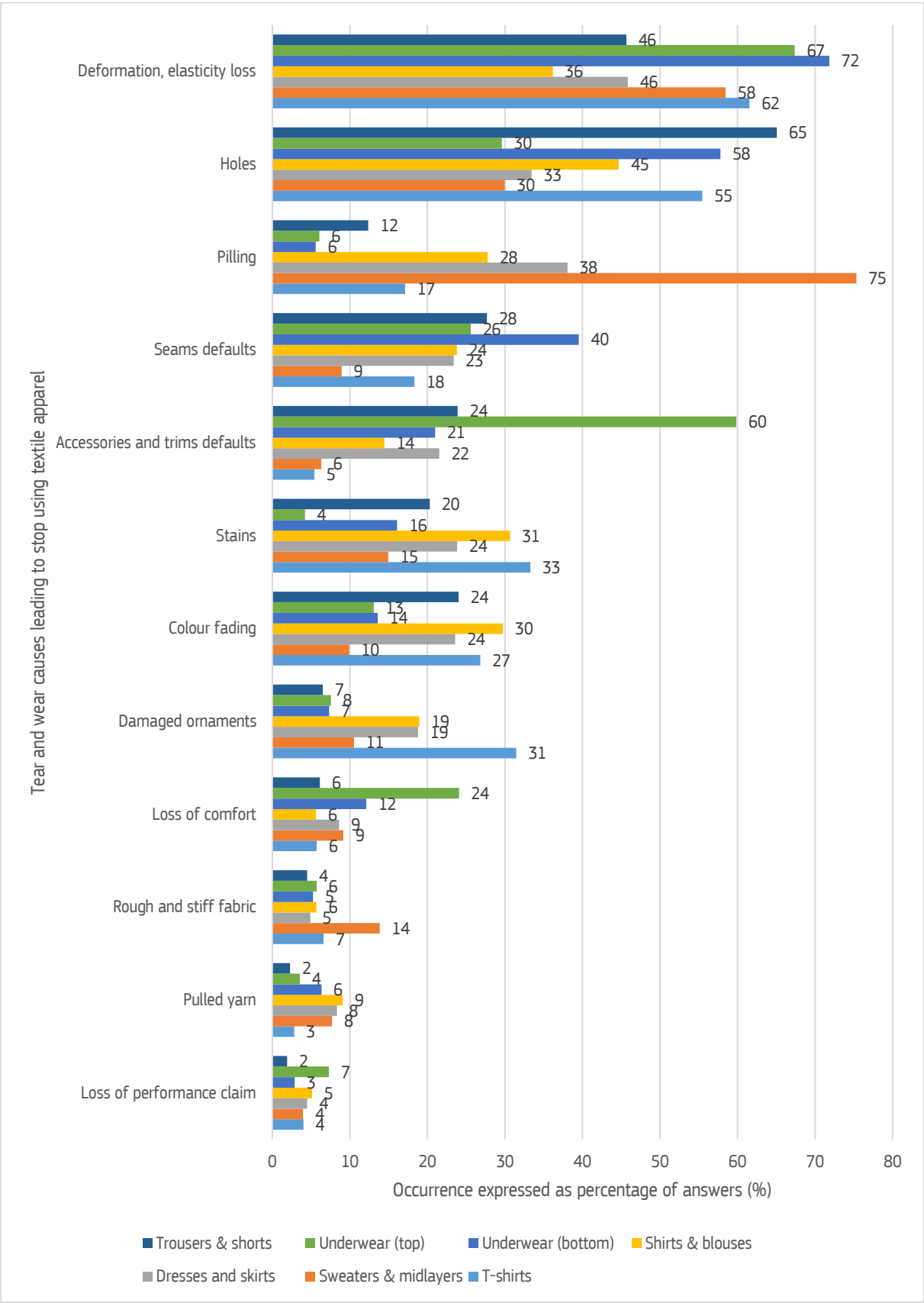
Failure mode group	Type of failure	Occurrence (*) (%)
Fabric related	Pilling	55
Colour related	Colour fading	53
Fabric related	Fabric breakdown	29
Fabric related	Accidental damage	29
Fabric related	Loss of dimensional stability	20
Logo failure	Logo failure	16
Colour related	Discolouration	15
Fabric related	Hole(s) in seams	14
Fabric related	Trim failure	8

3074 (*) Occurrence of failure in the sample analysed

3075 Source: Cooper and Claxton (2022)

3076

Figure 25. Most common textile apparel failure modes collected by the Durhabi project based on consumer surveys



3077

3078

3079

The analysis is based on 36 921 contributions provided by consumers that could identify three main failure modes.

Source: Durhabi project. Available at [this link](#). Last accessed on 13 August 2025.

9.2.1.4 Description of products based on robustness

Based on steps 1 and 2 of the robustness framework (section 9.2.1.3), **Table 44** reports a description of product technologies in the context of robustness. The identified key parameters were:

— **Visual inspection**, which includes the analysis of sub-parameters related to:

- Colour: colour change, staining, printing pattern
- Fabric: smoothness, pilling, fuzzing, matting
- Seams: smoothness, slippage
- Non textile parts: buttons, press fasteners, slide fasteners, touch and close fasteners, labels, embroidery, and others. This is an assessment of appearance and proper functioning.

— **Spirality** ¹⁵⁵, and

— **Dimensional change**

Visual inspection is considered as one parameter whose value is given by the worst performing sub-parameter assessed. This choice was made because a failure in a sub-parameter directly affects the whole visual inspection.

All these parameters are proposed to be analysed after 5 cleaning cycles, via washing machine or dry-cleaning according to the care label. Based on the available data and the consultation with stakeholders, this number of cleaning cycles is considered sufficient to bring about the stress required to induce most of the failures that occur in the products in scope of the PS.

This approach to describing robustness would lead to the identification of two product categories, chosen considering the fabric construction: **woven and knitted products**. Performance levels to distinguish the technologies have been defined based on expert judgement obtained via stakeholder consultation.

Products that by design cannot be cleaned, cannot be tested with this framework. The consultation with stakeholders revealed that these products represent a niche market – their number is negligible compared to the number of products included in the scope of the PS.

Table 44. Description of product technologies in the context of robustness

Parameter	ISO standard	Fabric construction	BC	BAT	BNAT
Visual inspection	15487	All	3. Moderate change in appearance	4. Negligible change in appearance	5. No change in appearance
Spirality	16322-3	Woven	5% < spirality < 6%	4% ≤ spirality ≤ 5%	spirality < 4%
		Knitted	5% < spirality < 7%	4% ≤ spirality ≤ 5%	spirality < 5%
Dimensional change	3759	Woven	±3% < change < ±4%	±2% ≤ change ≤ ±3%	change < ±2%
		Knitted	±5% < change < ±6%	±3% ≤ change ≤ ±5%	change < ±3%

BC: Base Case, BAT: Best Available Technologies, BNAT: Best Not yet Available Technologies.

All tests must be performed on the product ready to be bought by the user.

All tests must be performed after 5 cleaning cycles according to directions reported on the care label. ISO 6330 should be followed for washing machine, considering a type A machine, and flat dry. In-between washing cycle, the product should be flat dried for a maximum of 12 hours. After the last washing cycle, the product must be completely dry because the tests must be performed on a completely dry item. Detergent 7 will be used for products mainly made of wool and silk, and for products that need delicate or hand washing. For the rest of the products detergent 3 will be used. The detergent should never include bleaching. ISO 3175-2 should be followed for dry cleaning.

The visual inspection test should not include the qualitative assessment of the spirality because this is quantitatively assessed separately with ISO 16322-3.

When performing the visual inspection, many sub-parameters are assessed. In this case, the performance level of the visual inspection is given by the worst performing sub-parameter assessed.

Products which are designed to change their colour over time should not be assessed for colour change in the visual inspection. This applies only if the economic operator clearly communicates to the user this design choice. An example of this would be a product made of denim fabric dyed using indigo techniques to provide a vintage appearance to the product.

¹⁵⁵ According to the ISO 16322-3, spirality, or torque, in garments is the rotation, usually lateral, between different panels of a garment resulting from the release of latent stresses during laundering of the woven or knitted fabric forming the garment. N.B. The phenomenon is sometimes referred to as twist, for example, denim jean leg twist.

Source: own elaboration and consultation with stakeholders

The framework proposed in **Table 44** could be affected by the ongoing revision of the ISO 15487, which is due in 2026. Additionally, any potential ecodesign requirements tested in Task 6 of the PS should consider that products mainly made of viscose (fibre content higher than 80% by mass) should be ironed after washing to be included in the description of technologies reported in **Table 44**. This happens because textile apparel mainly made of viscose tends to shrink after washing. This shrinkage can be recovered simply via ironing the product.

The performance limits defining each of the product technologies (i.e. BC, BAT and BNAT in **Table 44**) have to be established considering the performance of the least robust (or weakest) products belonging to each of the categories in scope. i.e. woven and knitted products. For example, the minimum spirality level of BC woven products are given by those products showing spirality equal to 5% (**Table 44**).

A more exhaustive analysis of the main failure modes would lead to include in the proposed framework tests related to abrasion, tear, and seam slip. However, products included in the scope of the PS exhibit very different performances in terms of these parameters according to their specific application and the function that they are meant to fulfil. This would lead to the need to define a disproportionate number of categories to describe the numerous types of functionally distinct products. For example, following this reasoning, t-shirts should be classified according to their leisure and sport application. Among the leisure t-shirts a distinction should further be made between those with thin and very transpirable fabric used in hot summer days and thick ones used in winter, worn under a jacket or a midlayer. Among sport t-shirts, a distinction should also be made for the numerous sports requiring specific characteristics according to the stresses caused by the specific sport. This line of thinking would lead to many more categories than the 34 segments identified in the PEFCR A&F (see section 9.2.1.5), resulting in an obviously unmanageable level of complexity from the regulatory standpoint.

The proposed simplified robustness framework described above is based on what currently the authors consider feasible and on the current state of the art. This robustness framework has two **notable drawbacks**:

1. Since it does not consider product aging, it is not capable to describe the product performances over a defined number of uses, which are a common unit to express service lifespan of a product (see section 5.8). In other words, the robustness framework is not capable to describe how many times a person can use an item before the probable appearance of a physical failure due to its normal use⁽¹⁵⁶⁾. Therefore, the robustness framework is **not capable to describe the foreseen service lifetime related to intrinsic product properties**. The robustness framework is capable to describe only the resistance to external stresses, which do not correspond to the stresses occurring during product aging.
2. A known increase of performance level of key parameters corresponds to an unknown increase of physical durability. This implies that the consequent **increase of the service lifespan can only be based on expert judgements, which will not be verifiable**. Textile experts cannot base their judgement on direct experience, because nobody performs such measurements due to lack of aging tests.

Table 45 reports the main characteristics of the two potential frameworks: physical durability and robustness.

Table 45. Description of the two potential frameworks

Characteristic	Framework	Physical durability	Robustness
Compliance with the ESPR framework		Fully compliant. It is capable to: describe the foreseen service lifetime related to intrinsic product properties; objectively quantify the increased service lifespan due to improvement of the key parameters.	Used as a proxy for physical durability despite two notable drawbacks: is not capable to describe the foreseen service lifetime related to intrinsic product properties; the potentially increased service lifespan can only be based on expert judgements, which will not be verifiable.

¹⁵⁶ This concept is strictly related to the reliability of the product as reported in the ESPR (see **Table 2**). It is still suitable to report it together with the concept of durability because these two product aspects were gathered in the same group as reported in section 9.1.3.

Characteristic	Framework	Physical durability	Robustness
Availability of standardised tests		Standards simulating the aging process applicable to textile apparel are not available. The aging process should assess the evolution, or degradation, of the physical properties of the item over time, throughout several number of uses. Standards describing the resistance of the product to specific stresses are available for a list of physical parameters.	Standards describing the resistance of the product to specific stresses are available for a list of physical parameters.
Description of heterogeneous products in the scope		Since this framework focuses on the evolution, or degradation, of the physical properties over several uses, products with different initial characteristics could be assessed with the same framework.	Since this framework focuses on the initial physical properties of the item, products in the scope should be grouped in categories that have similar characteristics.
Identification of the performance level of selected parameters for the investigated product technologies.		Since it is a new framework, any performance level of parameters related to physical properties can be only based on expert judgement - nobody knows the performance level of products placed on the EU market. The ambition level related to the degradation of performances over several uses can be based on a trade-off between policy ambition in terms of improvement of performance and technical and economic feasibility. A known increase of performance level of key parameters corresponds to a quantifiable known increase of the service lifespan related to physical properties of the product.	Since it is a new framework, any performance level of parameters related to physical properties can be only based on expert judgement - nobody knows the performance level of products placed on the EU market. This framework implies that the performance level of products in a category is defined by the characteristics of the least robust products belonging to the category. A known increase of performance level of key parameters corresponds to an unknown increase of the service lifespan related to physical properties of the product. The increase of the service lifespan can only be based on unverifiable expert judgment.
Type of information provided in the perspective of potential ecodesign requirement		This framework describes how many times the item can be used before its properties will worsen of a certain degree.	This framework expresses the robustness of a new product. The physical characteristics analysed could be understood to resemble those checked visually and manually ("feel and touch") by users, who currently use their personal judgement to define the perceived quality of the product.

Source: own production

9.2.1.5 Durability in the context of the PEFCR A&F

As reported in the references, the analysis performed in sections 9.2.1.1 and 9.2.1.3 takes full advantage of the knowledge generated in the development of the PEFCR A&F and the additional knowledge obtained via stakeholder consultation in the development of the preparatory study. This section gathers the main features of the approach used in the PEFCR A&F when addressing physical durability and it tries to shed light on the differences between the PEFCR and ESPR frameworks.

The development of the PEFCR A&F started in 2019 and was finalised in May 2025⁽¹⁵⁷⁾. This project belongs to the voluntary framework of the environmental footprint methods⁽¹⁵⁸⁾ promoted by the EC and defined in

¹⁵⁷ PEFCR A&F publication. Available at [this link](#). Last accessed on 13 August 2025.

¹⁵⁸ Environmental Footprint Methods. Available at [this link](#). Last accessed on 13 August 2025.

Commission Recommendation (EU) 2021/2279 ⁽¹⁵⁹⁾. Therefore, this process is independent from the framework of the ESPR published on 13 June 2024.

The PEFCR A&F do not provide an explicit definition of durability, but it distinguishes intrinsic durability from the extrinsic durability. The former is related to intrinsic properties of the product, whereas the latter is related to the emotional attachment of the user to the product. When referring to intrinsic durability, the PEFCR A&F refers to “*product-specific attributes that contribute to its potential lifetime (e.g. physical toughness and design features)*”. In this context, the intrinsic durability is used to estimate the *Specific Duration of Service (DoS)* of a product, which the authors of the PS understand to be equivalent to the service lifespan described in section 5.8. The *Specific DoS* is described by Equation 1.

Equation 1. Specific Duration of Service

$$\text{Specific DoS} = \text{Default DoS} * \text{IDM} * \text{RM}$$

Where: *Default DoS* is a category-specific factor based on the literature; *IDM* is the Intrinsic Durability Multiplier, which is a factor based on a scoring system using the information related to failure modes collected in the project; *RM* is the Repairability Multiplier, which is a factor considering several parameters affecting the repairability. These factors try to express a correlation between robustness and extension of the duration of service.

The authors of the PS understand that the PEFCR A&F interprets intrinsic durability as the robustness of the product that is assumed to impact its service lifespan via Equation 1. This is an approach similar to the robustness framework described in section 9.2.1.3, which relies on the available standardized test methods. The development of the PEFCR A&F might had challenges adopting the legal definition reported in the ESPR, because its preliminary proposal (available since March 2022) could have been changed until its adoption (on 13 June 2024). The PEFCR A&F project was in its last stages when the ESPR was adopted.

Determining and reporting intrinsic durability in the PEFCR A&F is not mandatory. The method allows the Specific Duration of Service to be calculated using a default Intrinsic Durability Multiplier equal to 0.67, when no intrinsic durability tests are conducted.

The PEFCR A&F calculates the Intrinsic Durability Multiplier for 34 segments of textile apparel, excluding accessories. The first nine product categories described in Table 7 (see Section 3.2.1) were divided according to three main factors: the fabric construction (woven and knitted), the product’s generic application (sport and non-sport), and, only for underwear and swimwear, the part of the body covered (tops and bottoms). In the context of the ESPR, section 9.2.1.4 already reports the challenges given by a framework with many product categories.

Finally, the PEFCR A&F establishes three levels of intrinsic durability performance: basic, moderate and aspirational. These levels of performance do not correspond to the BC, BAT and BNAT, but they were based on the levels of performance determined analysing the 308 products of the Durhabi project. In particular, the aspirational performance level (the top level) is very much populated because under the PEFCR A&F products that demonstrate high level of robustness are rewarded. On the contrary, the BNAT represent products which are not available on the market.

The full adoption of the PEFCR A&F framework was not possible in the PS because:

1. The PEFCR A&F framework does not comply with the durability definition reported in the ESPR.
2. As reported in section 9.2.1.4, the product categories and segments proposed by the PEFCR A&F do not meet the specific needs of the development of the PS.
3. PEFCR A&F establishes robustness performance level disregarding the three product technologies defined in the MEERp, i.e. BC, BAT and BNAT.
4. The study performed in this PS has to take into account additional knowledge provided via stakeholder consultation for the mandatory framework of the ESPR.

¹⁵⁹ Commission Recommendation (EU) 2021/2279 of 15 December 2021 on the use of the Environmental Footprint methods to measure and communicate the life cycle environmental performance of products and organisations. Available at [this link](#).

9.2.2 Maintenance

Maintenance is the ability of a product to be kept in a condition where it is able to fulfil its intended purpose through one or more actions (**Table 2**). In the context of textile apparel, it comprises the set of activities that a user carries out, fundamentally during the use phase, to maintain the product in a condition that satisfies their needs. It includes activities such as cleaning, drying, ironing, storing and wearing, in specific environments and conditions. It does not include any operations addressing the repair of the product, which is addressed under a separate specific product aspect (see section 9.2.3).

Maintenance operations are specific to the item they refer to. In particular, the fibre composition, the fabric construction type, the presence of non-fibre-based parts and the finishing treatments of the item are crucial factors determining the suitable maintenance strategies to make the product last longer. This section mainly focuses on the information that the manufacturer provides to the user to properly maintain textile apparel.

9.2.2.1 The ecosystem affecting maintenance

Process techniques and design choices

Maintenance instructions are commonly communicated through specific symbols on care labels. These symbols are designed to be universally recognisable providing clear guidance on textile apparel care.

Despite 82 % of European consumers considering care instructions important and 75 % finding written explanations on labels very useful (GINETEX, 2025) the meaning of some care symbols remains unclear to many users, often leading to misinterpretation of recommended care practices. To improve clarity and support correct product care, researchers advocate for combining standardised symbols with explanatory text, an approach strongly supported by current consumer preferences (Yan et al., 2008; Nayak and Ratnapandian, 2018) (Section 13.7.2.7).

Figure 26. Example of existing care symbols under ISO 3758:2023



Source: GINETEX¹⁶⁰

Traditionally, care labels have been physically sewn during the confectioning phase onto textile apparel, providing essential instructions on washing, drying, and ironing. However, with the rise of digital technology, many brands now complement the physical labels with online care guides and mobile apps. These digital platforms offer more detailed explanations and interactive features that help consumers better understand and follow textile apparel care instructions (Nayak and Ratnapandian, 2018)

User behaviour

Consumer behaviour strongly affects textile apparel maintenance, because the choices made by the users may adhere or not to the best practices reported on the care label. When care information is reported on the item of textile apparel, users could disregard it, have difficulties interpreting the instructions or they could even remove the label because it results uncomfortable when wearing the item (see section 6.3.2 and 13.7.2.7). The removal of physical labels from textile apparel is indirectly promoted by labels displaying a cutting line.

Since price is one of the main factors consumers consider when purchasing textile apparel, consumers may be more attentive to care instructions when handling items they perceive as valuable or difficult to replace; however, there is no consistent evidence linking product price to increased care label compliance (Wakes et al., 2020b).

¹⁶⁰ GINETEX website. Available at this [link](#). Last accessed 18 July 2025.

3249 To optimise textile apparel maintenance, several best practices are recommended from a user behaviour
 3250 perspective in Sections 6.3.1 and 13.7.2, and specific literature (Nayak and Ratnapandian, 2018) (Gwozdz et al.,
 3251 2017b; Cotton et al., 2020; Balevičiūtė, 2021):

- 3252 • Following information provided in the care label,
- 3253 • Running laundry with sorted items according to their colour, fabric type, and washing
 3254 temperature,
- 3255 • Using the suitable washing temperature,
- 3256 • Using the right quantity and type of detergents and softeners,
- 3257 • Preferring air-drying to tumble drying and avoiding direct sunlight when not recommended by
 3258 the care label.
- 3259 • Minimising wash frequency and ironing,
- 3260 • Properly folding or hanging to prevent fabric deformation

3261 Store garments in dry, ventilated spaces, away from sunlight, folded or hung to prevent damage during long-
 3262 term non-use.

3263 **Legislative framework and industrial best practices**

3264 Although care instructions remain largely voluntary in the EU, manufacturers frequently include them, via text
 3265 or ISO 3758 ⁽¹⁶¹⁾/GINETEX symbols, to ensure correct product use, reduce liability ⁽¹⁶²⁾, and meet market
 3266 expectations ⁽¹⁶³⁾ and export requirements. Many countries voluntarily follow the GINETEX Standard ⁽¹⁶⁴⁾, which
 3267 establishes care labelling system for textiles based on symbols.

3268 The European Commission is currently considering the introduction of harmonised and partially mandatory
 3269 textile labelling rules on textile care, in the context of the review of the Textile Labelling Regulation (EU)
 3270 1007/2001 (TLR) ⁽¹⁶⁵⁾. In this context, also inspired by the best practices listed below, the Commission is even
 3271 considering requiring an explanation of symbols (including on required care content) used in physical labels,
 3272 either through accompanying documentation or (preferably) through a (voluntary) digital label. Additional rules
 3273 on legibility and durability of (notably physical) labels are also being considered, inspired the best practices
 3274 listed below.

3275 Twelve Member States already require mandatory care labels ⁽¹⁶⁶⁾ (GINETEX, 2017b).

3276 **Table 185** in section 13.10.4 reports the description of standards used in some regions of the world. They
 3277 mainly differ as regards: (a) temperature units, (b) requirement of captions alongside symbols, (c) adopted
 3278 symbols, and (4) legal requirements (mandatory vs voluntary).

3279 **Figure 27** shows legal requirements and the adoption of standards in specific regions of the world.

¹⁶¹ ISO 3758:2023 Textiles — Care labelling code using symbols. Available at [this link](#). Last accessed on 14 August 2025.

¹⁶² Ginetex website. Available at this link. Last accessed 12 November 2024

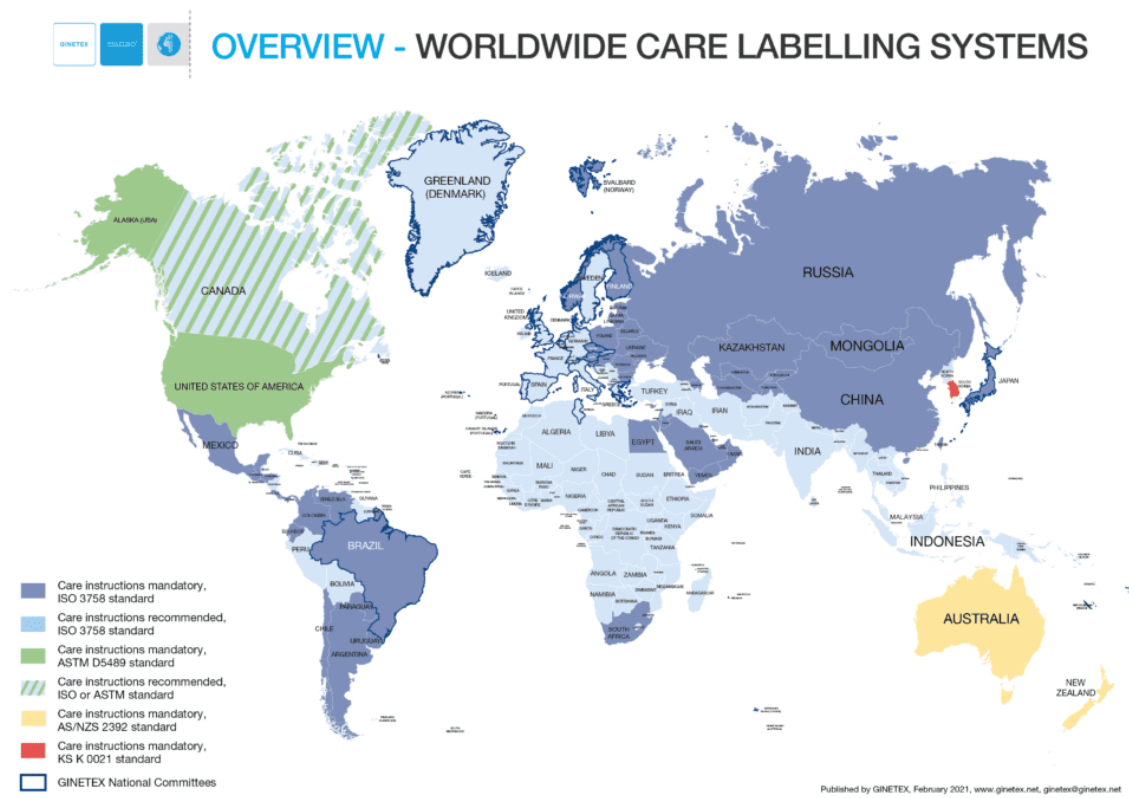
¹⁶³ Ginetex website on market standardisation. Available at this [link](#). Last accessed 17 July 2025.

¹⁶⁴ Ginetex website. Available at this link. Last accessed 12 November 2024

¹⁶⁵ Revision of the Textile Labelling Regulation (EU) 1007/2011. Available at this link. Last accessed on 12 November 2024.

¹⁶⁶ Austria, Bulgaria, Estonia, Finland, Hungary, Lithuania, Romania, Slovakia, Czech Republic, Poland, Denmark, Letonia and, as well as, in the EEA, Norway.

Figure 27. Care labelling systems in the world



3281

3282

3283

Source: UK Fashion and Textile Association. Available at [this link](#) . Last accessed 12 November 2024.
Previously published by GINETEX in February 2021

3284

Both the ESPR and the EU Regulation on general product safety (Regulation (EU) 2023/988) ⁽¹⁶⁷⁾ establishes a general obligation for products placed on the EU market to bear appropriate warnings and safety information, including written accompanying documentation or labelling where needed for safe use, which may include maintenance-related aspects in cases where the condition or upkeep of a product influences its safety.

3286

3287

3288

Table 46 reports the best practices that manufacturers and retailers try to implement.

3289

Table 46. Best practices on care labelling implemented by producers and retailers

Factor	Action
Legibility	Symbols and letters on labels should remain legible throughout the textile apparel's useful life. The label should use medium-width lettering, where no individual letter should be less than 1.5 mm high (also proposed by ISO 3578:2023).
Material-based indication	Care instructions should be based on the most sensitive or delicate material present in the product. This ensures that the guidance protects all components (e.g., linings, trims, decorations) under a unified instruction, avoiding user confusion or damage from under-specified labels.
Consistent label placement	Labels should be positioned in standardised positions for all similar items of textile apparel to improve visibility and reduce user confusion or removal.
Durability of labels	Care labels should be made of materials that are resistant to the care treatments indicated on the label, at least to the same extent as the textile apparel itself.
Enhanced understanding	To improve user comprehension, labels should combine standardised symbols with short explanatory text or send the user to care guides available online. Labels should use easily understandable symbols alongside (such as standardized by ISO 3758:2023), and where appropriate, with written instructions.

¹⁶⁷ Regulation (EU) 2023/988 on general product safety, Available at this [link](#).

Factor	Action
Care instructions verification	The care instructions should be verified by manufacturers before being placed on the market to ensure their correctness.

Source: Own production based on (Nayak and Ratnapandian, 2018) and (GINETEX, 2023)

Business models

As outlined earlier, care labels are typically included for a combination of strategic and regulatory reasons. While it is sometimes suggested that the level of detail in care instructions may reflect a company's product strategy, **the literature does not provide conclusive evidence of systematic differences between business models** in this regard.

9.2.2.2 How to assess maintenance

Based on the information gathered in sections 9.2.2.1 and 9.2.2.2, the assessment of maintenance should be based on three main aspects:

1. Type of information provided, addressing all steps of the use-phase:

(a) cleaning and laundering

(b) drying,

(c) ironing,

(d) bleaching,

(e) additional suggestions,

(f) storing,

2. Carrier; and

3. Communication strategy.

Information about **cleaning and laundering** should address all types of care operations the product can/should undergo. This includes:

- Domestic laundering, with clear guidance on washing temperature, spin speed, compatibility with similar colours and fibre types, and, where appropriate, the use and dosing of detergents and softeners ⁽¹⁶⁸⁾;
- Professional cleaning, including dry and wet cleaning, with indications of suitability for such treatments and any restrictions on solvents.

Information about **drying** should include directions on the optimal type of drying, with specific information regarding drying temperature in the tumble dryer, and sunlight exposure when line drying (air-drying) is used.

Information about **ironing** should include ironing temperature, and where relevant best practices regarding moisture level of the item during ironing.

Information about **bleaching**, should specify whether and under what conditions it is permitted.

Information about **additional suggestions** should include any further information that could support the suitable maintenance of the product.

Information on **storage** may be provided to support proper textile apparel care and help preserve product condition. Where relevant, this may include suggestions on whether folding or hanging is preferable, and general advice on environmental conditions to avoid (e.g. excessive humidity or direct sunlight). For certain items, different recommendations may apply for storage after wearing versus after cleaning. The inclusion of storage guidance could be provided by the manufacture/designer for groups of products in their portfolio, because they could be common to many items.

¹⁶⁸ In general, specific chemical formulations can weaken certain fibre types increasing the risk of damage (Cooper and Claxton, 2022c).

3328 The **carrier** of maintenance-related information can be either a physical label, where information is provided
3329 directly on the product, and/or a digital tool. The two options present advantages and limitations:

3330 — online information allow extensive and flexible descriptions with an availability over time assured
3331 by the economic operator selling the product;

3332 — the physical label can carry a limited amount of information with the risk to compromise wearer
3333 comfort, and increased the risk of label removal.

3334 Therefore, maintenance information should balance clarity, accessibility, and relevance. Essential maintenance-
3335 related information should remain physically available on the product in a clear and durable format. Meanwhile,
3336 digital tools, can be used to complement this information with contextual, product-specific, or less time-critical
3337 guidance.

3338 Overreliance on digital-only formats could limit access to information only to people willing to use the online
3339 platform. Finally, regardless of the carrier used, the information should be provided with a **standardised**
3340 **strategy** (or structure), so that users can easily navigate information provided by any economic actor. Initiatives
3341 in this sense are, for example, “My care label app” ⁽¹⁶⁹⁾ developed by GINETEX, or “Clevercare” ⁽¹⁷⁰⁾ supported
3342 by textile apparel brands.

3343 This framework is suitable to describe all products included in the scope of the PS without any further grouping
3344 or categorization.

3345 Although user behaviour plays a crucial role in the effects generated by product maintenance, the assessment
3346 of maintenance in the framework of the ESPR focuses only on the instructions that the manufacturer provides
3347 to the user. The behaviour of the user will be taken into account in the environmental and economic models to
3348 be built in the following tasks of the PS.

3349 9.2.2.3 Description of products based on maintenance

3350 **Table 47** reports the description of product technologies in the context of maintenance of textile apparel.

3351 **Table 47.** Description of product technologies in the context of maintenance of textile apparel applicable to the EU

Aspect	BC	BAT	BNAT
Type of information provided, addressing all steps of the use-phase	Using standardized symbols generally reported in ISO 3758:2023.	It reports standardized symbols reported in ISO 3758:2023, and where needed, complementary care information is made available through digital tools.	It reports standardized symbols reported in ISO 3758:2023 (minimal) and full digital guidance. It reports all types of care operations the product can/should undergo dry cleaning by professionals or home washing. In the case of professional cleaning, directions should be provided on compatible products for dry cleaning. In the case of laundering, directions should be provided on washing temperature, type and dosing of detergents and softeners, spin speed, type of textile products that can be washed with (e.g. similar colours and fibre types).
Additional suggestions	Not available.	Reduction of washing frequency and preference for line drying.	It includes any further information that could support the suitable maintenance of the product.
Carrier	Physical label.	Physical label and digital tools.	Physical label is used for basic most important information or warnings, while other information, is delivered through full digital guidance, is reachable with a support of a device (e.g. Radio Frequency Identification (RFID) system) incorporated in the product, so that it does not create discomfort and it is difficult to remove.
Communication strategy	Using standardized symbols generally reported in ISO 3758:2023 on	It reports standardized symbols generally reported in ISO 3758:2023 mainly on physical label with digitally supported	Users can easily navigate digital, smart system-based information provided by any economic actor because it is provided following a standardised format.

¹⁶⁹ GINETEX website, specifically section My care label app. Available [here](#). Last accessed 23 July 2025.

¹⁷⁰ Clevercare website. Available [here](#). Last accessed 23 July 2025.

Aspect	BC	BAT	BNAT
	physical label, with no user adaptation.	explanations for user adaptation.	

Physical labels are usually made of polyester and satin due to their softness and resistance properties (Nayak and Padhye, 2015). Care symbols use ISO 3758:2023 ⁽¹⁷¹⁾.

Source: own production

9.2.3 Repairability

According to the ESPR, the repairability is defined as the ability of a defective product or waste to return to a condition where it fulfils its intended use (**Table 2**). In the context of textile apparel, this return to the intended use includes the acceptance by the user, who should still be satisfied with the potential new aesthetics and the restored functionality of the product after repair operations. This could be the case for example in the use of patches, mending operations on fabrics, or the use of buttons different from the original ones.

9.2.3.1 The ecosystem affecting the repairability

Design principles for repairable items - choices and process techniques

Textile apparel is often composed by many pieces of fabrics and by non-textile parts such as buttons, zippers and many more. Fabrics could be organised on one or more layers, which could have the same or different fibre composition and properties. Additionally, specific parts of textile apparel could have exclusively an aesthetic function and be made of numerous materials such as leather, plastics, metal and fibre-based. All these parts composing textile apparel are mostly joined with seams, but other ways such as heat-bonding and laser-based techniques are also in use. Furthermore, an item of textile apparel can also be seamless and be made of only one piece of fabric.

The products included in the scope of the PS are very heterogeneous in terms of number, properties and functions of their parts. This heterogeneity goes beyond the intended function of the product, because the same product could be made in numerous ways. For example, a t-shirt can be made of a knitted fabric with no seams, or it can be made by numerous panels of fabric joined with seams, buttons and sequins, resulting in a product made of numerous parts, each of them with a specific function.

Repairability is strictly connected to the failure modes of these products. **Table 43** and **Figure 25** in section 9.2.1.3 report that the main discard causes are generally combined and are colour and fabric-related (e.g. pilling, breakdown, stains etc). Some of these failure modes could be tackled by a specific design, e.g. making a more robust fabric, but others are largely dependent on the way the product is maintained and used, e.g. the presence of stains.

From the general design perspective of energy-related products, modularity, use of standardised parts and availability of spare parts are considered to be important for product repairability (Cordella, Sanfelix, et al., 2019). **Modular products** are easily disassembled without damaging the product, so that the damaged part can be repaired. In the context of textile apparel, modularity leads to trade-offs with comfort, because seams or components joining different parts of the textile apparel create a discontinuity in the fabric in contact with the skin which is usually perceived in terms of discomfort. Additionally, modularity leads to a trade-off with the physical durability because seams are weak parts of textile apparel. In this context, also the combination of fabric construction and connector types plays an important role because it affects the disassembly complexity of a product. To the authors' knowledge, there is no literature reference proposing an objective framework evaluating the disassembly complexity of textile apparel items.

Although the **use of standard parts** in the textile industry is limited due to its intrinsically creative nature, the consultation with stakeholders revealed that standard parts are commonly used for "basic" products and sometimes for entire collections. The European Environment Agency identified fasteners as the main textile apparel part that could be subjected to standardization without affecting the creative nature of this industry (EEA et al., 2022). The size, material and attachment method of zippers, buttons, snaps and other fasteners could be standardised as well as the thickness of the yarns. However, it is difficult to envisage standardization involving the fabric, which is the main part of textile apparel and which is also the part that is most prone to

¹⁷¹ ISO 3758:2023 Textiles — Care labelling code using symbols. Available at [this link](#). Last accessed on 13 November 2024.

3397 being damaged The standardisation of fabric would potentially limit the creative nature of the textile apparel
3398 industry as well as the technical characteristics of products.

3399 The **availability of spare parts** in the textile apparel industry should take into account the high number of
3400 different collections placed on the market and the real use of these spare parts. Section 5.6 indicated that some
3401 retailers place on the market up to one collection per week. Making spare parts available for all these different
3402 items would require the manufacture of product parts that most probably would never be used due to the fast
3403 changing preferences and attitudes of the users. Additionally, the use of spare pieces of fabric is strictly
3404 connected to the fashion trends and user acceptance to wear a product composed by pieces of fabric in a
3405 different status (new and worn), which would resemble patches. The reparability of specific failures in textile
3406 apparel is constrained by the lack of suitable machinery for delicate or technical fabrics, availability of skilled
3407 professionals, inherent design that can compromise garment structure, performance, or aesthetics after repair.
3408 Some biological or chemical contamination of the product could also make repair technically unfeasible,
3409 economically inefficient or even unsafe.

3410 **Business models**

3411 The European Environment Agency estimated that companies working in the textile repair sector decreased in
3412 the last decade, reporting that they are small in size and have a turnover equal to about 0.25% of the textile
3413 apparel sales in Europe (EEA et al., 2022). A more detailed analysis of the apparel repair sector is hindered by
3414 the way in which collection of this data is performed by Eurostat, which aggregates under the NACE code 95.29
3415 repair of many personal and household goods, including bicycles, books, musical instruments, apparel, etc. ⁽¹⁷²⁾.

3416 In recent years, some fashion brands, in light of the increasing landscape of repair provider start-ups, have
3417 begun offering repair services to their customers and/or manuals and directions to address some simple repairs
3418 of their products, such as replacing a zipper or sewing a button.

3419 **Fashion trends** result from the action of many players such as stylists, influencers and fashion brands. They
3420 could potentially promote repaired products with for example visible mending and patches, but in reality, such
3421 products are not socially accepted in the majority of social gatherings, such as workplaces, parties, etc (SY Choi
3422 et al., 2022; Hong et al., 2024).

3423 **User behaviour and economic aspects**

3424 The decision of repairing an item of textile apparel is highly subjective and depends on individual values, fashion
3425 trends, price, skills and time available to users.

3426 The **emotional attachment** to the product largely influences the decision to repair an item or to replace it
3427 with a new one. Textile apparel is more likely to be repaired when it evokes meaningful memories, it is an
3428 expression of user identity or provides particular comfort (Page, 2014; Gwilt, 2021; Damhorst, 2019).

3429 The **price of repair** compared to the price of the new item plays a crucial role when the user decides to repair
3430 or replace an item of textile apparel. Expensive items tend to be repaired more often than cheap ones due to
3431 the cost-effectiveness of the choice (Gwilt, 2021). In particular, if the price of repair exceeds about 40% of the
3432 price of the new item, users tend to replace the product rather than repairing it (Cordella, Sanfelix, et al., 2019;
3433 Ribeiro et al., 2023). A simple exercise reported in section 13.10.5 shows that in the EU, professional repair
3434 operations can be much more expensive than the purchase of a new item. In this context, the user prefers to
3435 buy a new item rather than repairing the damaged one.

3436 When a user decides to repair an item, self-repair is the most common practice followed by unpaid repair via
3437 associations such as repair cafes ⁽¹⁷³⁾ and lastly via paid support using professional repairer. Besides economic
3438 reasons reported above, it seems that this ranking among different options is largely affected by user mistrust
3439 of the capabilities of professional repairers and the cost in terms of time that is associated to repairing
3440 operations including logistics (Laitala et al., 2021b; EEA et al., 2022; McQueen et al., 2023).

3441 However, self-repair is a viable option only if the user has suitable skills and available time. Generally, users do
3442 not have the right skills for self-repair. Although in general women are more likely to repair their defective
3443 apparel than men, younger generations, regardless of gender, tend to lack knowledge about the necessary tools,

¹⁷² Eurostat: NACE Rev. 2 – Statistical classification of economic activities in the European Community, available at [this link](#). Last accessed 10 October 2024.

¹⁷³ Repair cafes are local no-profit workshops where volunteers repair or help repairing numerous goods, including textile apparel. Available at [this link](#). Last accessed on 11 November 2024.

3444 materials and practices (EEA et al., 2022; McQueen et al., 2023; Hernandez et al., 2024). Disregarding user skills,
3445 the value given to the time spent in self-repair plays a crucial role when a user decides what to do with their
3446 damaged item (Jain, 2021).

3447 While official Refashion publications ⁽¹⁷⁴⁾ ⁽¹⁷⁵⁾ and the EEA circular economy report on the profile of France
3448 (European Environment Agency, 2024) confirm that the availability of a repair bonus lowers consumer repair
3449 costs, none provide conclusive evidence that this reduction has directly caused an increase in repair demand.

3450 **Legislative framework and best practices**

3451 Currently, there is no legal framework addressing the repair of textile apparel in EU. The promotion of
3452 reparability is left to the individual fashion brands' voluntary initiatives. However, on 30 July 2024, Directive
3453 2024/1799 on repair of goods ⁽¹⁷⁶⁾ entered into force, with the aim of promoting repair of goods both within
3454 and outside the legal guarantee. Member States have to transpose the Directive into national legislation and
3455 apply it from 31 July 2026.

3456 Under Directive 2024/1799, manufacturers of products that are subject to reparability requirements under EU
3457 law will have to repair those products within a reasonable time and for a reasonable price. For such an obligation
3458 to apply, reparability requirements would have to be set in product-specific legislation, potentially for instance
3459 in the future Delegated Act on textile products under the ESPR.

3460 Furthermore, manufacturers are required to make information available about their repair services to
3461 consumers in an easily accessible manner and consumers will be able to find repairers more easily through a
3462 new online European Repair Platform.

3463 Directive 2024/1799 has also amended the existing Sale of goods Directive (EU) 2019/771 ⁽¹⁷⁷⁾, adding an
3464 additional year to the legal guarantee if the choice is made to repair a product instead of replacing it under the
3465 legal guarantee.

3466 **9.2.3.2 How to assess reparability**

3467 Section 9.2.3.1 reported that reparability for textile apparel cannot be addressed in the same way as done for
3468 energy-related products, such as smartphones. Emotional attachment, fashion trends and repair price strongly
3469 affect the reparability of textile apparel. In theory, any item of textile apparel can be repaired, but its degree
3470 of complexity cannot be described with an objective framework because the factors influencing the reparability
3471 in the design phase are at odds with the creative nature of the textile industry. In this context, the establishment
3472 of such an objective framework is hindered by subjective consumer decisions, the entity of the product failure,
3473 and feasibility of repair.

3474 Acknowledging the lack of a reparability framework, the authors understand that, from the perspective of the
3475 manufacturer, making information available about repair services could facilitate the reparability of textile
3476 apparel.

3477 **9.2.3.3 Description of products based on reparability characteristics**

3478 Following the rationale provided in section 9.2.3.2, **Table 48** reports a qualitative description of product
3479 technologies in the context of reparability. However, the necessary condition for a repairable product is that
3480 the price of the repair should be acceptable to the user when compared to the price of a similar new item. This
3481 finally depends on the user's judgement.

¹⁷⁴ Refashion activity report 2024, available at this [link](#). Last accessed 27 August 2025.

¹⁷⁵ Refashion publication on the Decree on Mandatory Display of the Repair Fund, available at this [link](#).

¹⁷⁶ Directive (EU) 2024/1799 of the European Parliament and of the Council of 13 June 2024 on common rules promoting the repair of goods and amending Regulation (EU) 2017/2394 and Directives (EU) 2019/771 and (EU) 2020/1828 (Text with EEA relevance). Available at [this link](#).

¹⁷⁷ Directive (EU) 2019/771 of the European Parliament and of the Council of 20 May 2019 on certain aspects concerning contracts for the sale of goods. Available at [this link](#).

Table 48. Description of product technologies from the perspective of repair services

BC	BAT	BNAT
Professional repair services available, but characterized by long response time, reduced transparency and consumer mistrust. Repair support is very limited, with vague instructions and brand service limited to portions of the region where the product is sold.	Professional repair services available and transparent. The product has detailed, repair guidance about the failure of most common damaged product parts. Brand repair services are available in the whole region where the product is sold.	Interactive repair support is available real-time via augmented reality, artificial intelligence or other technologies. Brand repair services are available in the whole region where the product is sold.

BC: Base Case; BAT: Best Available Technologies; BNAT: Best Not yet Available Technologies.

Source: JRC own production.

9.2.4 Waste generation

In this PS, the following definitions are used:

- Textile waste is a textile product which the holder discards or intends or is required to discard ⁽¹⁷⁸⁾.
- Post-industrial textile waste is textile waste generated during the manufacturing of textile products and their precursors (manufacturing of fibre, yarn and fabric, and during confectioning) (Huygens et al., 2023).

This includes waste generated as a result of technical features of the manufacturing processes (e.g. cutting, roll ends), inaccurate resource planning (e.g. excess fabric) or production errors (e.g. defective or damaged fabric and products) (Aus et al., 2021). Moreover, it includes discarded finished products that the manufacturers do not send to their customers due to order change or cancellation.

- Pre-consumer textile waste is textile waste generated at the retailer stage, as a result of discarding unsold textile products. This can be due to:
 - overstock (products coming directly from the production line);
 - obsolescence (products for which there is no longer a demand);
 - damages (products that have been damaged in transit or storage);
 - defects (products taken off the market due to quality issues).

Thus, this includes discarded products that have not been sold due to surplus stock, excess inventory and deadstock and products returned by a consumer on the basis of their right of withdrawal in accordance with Article 9 of Directive 2011/83/EU or, where applicable, during any longer withdrawal period provided by the trader.

- Post-consumer textile waste is a textile product that has been discarded after consumption and use by the citizen or end-users of commercial and industrial activities (tourist accommodation services, automotive, etc.). For this reason, it is commonly referred to as household and commercial post-consumer textile waste, respectively (Huygens et al., 2023).

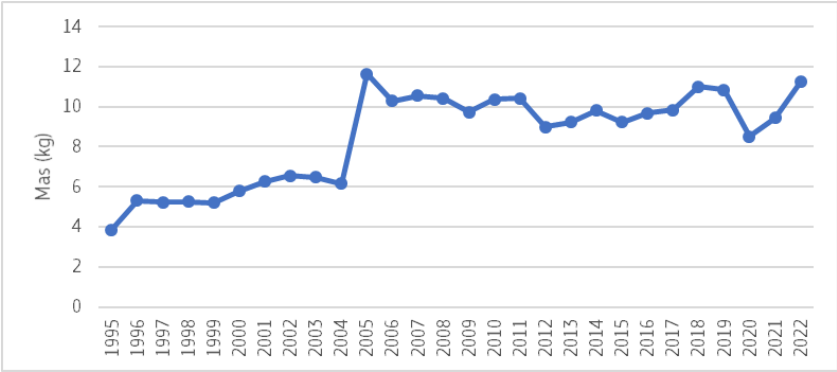
9.2.4.1 Textile waste in numbers

The consumption of textile apparel drives the generation of its waste. **Figure 28** shows that the yearly apparent consumption of an EU-27 citizen increased from about 6 kg before 2005 up to about 10 kg between 2005 and 2022. In particular, in 2019 the EU generated about 6 Mt of textile apparel waste (**Table 49**). Post-industrial waste was the dominant fraction, representing 87% of textile apparel waste generated. Despite post-industrial waste ranging from 25% to 45% of all fabric used in the production (Aus et al., 2021), it accounted only for 11% of the textile apparel waste generated in the EU because of the high volumes produced in third countries (so out of the scope of these statistics). Pre-consumer waste was indicated to contribute by 3%. These quantities

¹⁷⁸ The definition of textile waste is inspired by the definition of waste given in the Waste Framework Directive (WFD). Directive 2008/98/EC of the European Parliament and of the Council of 19 November 2008 on waste and repealing certain Directives (Text with EEA relevance). Available at [this link](#).

are currently considered underestimated mainly due to lack of transparent reporting (Aus et al., 2021; Duhoux et al., 2024). Overall, a substantial degree of uncertainty exists on these values because Member States and companies have different definitions of textile waste and standards for reporting waste generation, and reporting standards to official databases (e.g. Eurostat) (Huygens et al., 2023).

Figure 28. Apparent consumption of textile apparel per capita in EU-27



Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

Table 49. Amount of textile apparel waste generated in EU in 2019

Type of waste	Mass (Mt/y)	Contribution to the total (%)
Post-consumer	5.2	87
Pre-consumer	0.16	3
Post-industrial	0.64	11
Total	6	100

Reported figures do not include the textile waste of the production occurring outside EU.

Source: Huygens et al. (2023)

In terms of fate of textile waste, it is estimated that about 10% of post-industrial and pre-consumer textile waste is recycled, while the rest is incinerated, landfilled or exported to third countries with unknown final destiny. Only about 5% of the post-consumer textile waste is recycled in the EU, about 17% is exported to third countries and the rest is either incinerated or landfilled (Huygens et al., 2023).

Besides textile waste, the textile industry generates waste from packaging, hangers, which are usually made of plastics, and spent chemicals used in the manufacturing stage. Wastewater is also generated, but it is not included in this analysis because it is addressed in section 9.2.6 on environmental impacts as emissions into water (Roth et al., 2023).

9.2.4.2 Analysis of the ecosystem

Although the types of textile waste have some peculiarities, their generation has some common drivers when focussing on business models and user behaviours.

On one hand, **business models** of many fashion designers, fashion brands and retailers incentivise the continuous consumption of new products:

- placing on the market up to 47 new collections per year (see Section 5.6),
- offering multiple seasonal discounts over the year to incentivise the replacement of products that are considered obsolete, and
- generating a sense of urgency and exclusivity by placing on the market numerous limited editions and using the dark pattern in online sales (see Section 6).

In particular, the brand-led operation model promotes new trends via numerous Fashion Weeks and trade shows all around the world, as well as by advertising the proposals of fashion editors, stylists, celebrities and influencers. Additionally, companies using the consumer-led operation model aim to predict the demand of users via trend forecasting agencies or in-house forecasting models, which lead to overproduction to meet the fast changing requests of as many people as possible in different parts of the world (Roberts et al., 2023; McKinsey & Company, 2024; Brondino, 2022).

3552 On the other hand, in addition to functionality and perceived quality, **users** choose products based on (1) what
3553 is available on the market, (2) what is economically convenient, and (3) what makes them feel good and
3554 accepted by others. Therefore, the change of user demand depends on the new trends, prices and the needs of
3555 individuals to self-represent and feel accepted when meeting others at work, in their free time and in special
3556 occasions (see Section 6).

3557 The following three subsections deepen into aspects of the ecosystem that are specific to each type of textile
3558 waste.

3559 9.2.4.2.1 Post-industrial waste

3560 When focussing on the manufacturing stages, in addition to what results from technical features such as
3561 machine inefficiencies, cutting losses or production failures, business models and process techniques largely
3562 affect the generation of post-industrial textile waste.

3563 As described in section 5.6, the **business model** of European fashion companies mainly relies on the
3564 manufacture by companies located in third countries, which operate as:

- 3565 — Primary suppliers: main suppliers that provide the products directly to the fashion company.
- 3566 — Sub-contractors: secondary suppliers that primary suppliers might use to fulfil part of their job.
- 3567 — Licensed suppliers: companies producing goods under a licensing agreement, which allow them to
3568 use in third countries the patents and trademarks of the fashion companies.

3569 Within this structure, suppliers are asked to rapidly produce and adapt to the volatile requests of the market,
3570 which could lead to changes or even cancellation of orders. Lead time from the concept of the textile apparel
3571 to the potential customer purchase could be as short as 15-21 days. This business model challenges a careful
3572 design, resource planning, and quality control during the manufacturing processes. As consequence, post-
3573 industrial textile waste increases because (Aus et al., 2021; Zhao and Kim, 2021; Fernandez-Stark et al., 2022;
3574 Duhoux et al., 2024; McKinsey & Company, 2024):

- 3575 • The supply chain is characterised by the bullwhip effect ⁽¹⁷⁹⁾, where suppliers enlarge their
3576 inventories to tackle demand fluctuations;
- 3577 • Suppliers waste part of their inventories when stock location storage becomes limited;
- 3578 • Suppliers overproduce to benefit from economies of scope;
- 3579 • Material belonging to cancelled orders is directly wasted rather than remanufactured;
- 3580 • Lack of manufacturing quality control leads to low-performing products rejected by fashion
3581 brands.

3582 **Process techniques** adopted in the manufacture of textile apparel are crucial elements conditioning the
3583 generation of post-industrial textile waste. In particular, key factors include poor planning, inefficient
3584 manufacturing systems and cutting methods, as well as suboptimal fabric utilisation (Aus et al., 2021; Khairul
3585 Akter et al., 2022).

3586 At the confectioning stage, cutting operations are the largest contributors of this type of waste, followed by
3587 mistakes during sewing and damages to fabrics (Aus et al., 2021). On one hand, design software like CAD
3588 (Computer-Aided Design) and the use of artificial intelligence can optimise the use of fabric and decrease the
3589 generation of post-industrial waste by 27% when compared to manual cutting (Krsteva and Demboski, 2011;
3590 Palacios-Mateo et al., 2021). On the other hand, sewing operations are semi-manual activities affected by the
3591 performance of the machinery and the pressure put on the operators to meet the tight deadlines of rapid
3592 production (Aus et al., 2021).

3593 In general, the manufacturing process could be optimised adopting methodologies such as Lean Manufacturing,
3594 Total Quality Management, Kaizen, Just-In-Time (JIT) and Lean Six Sigma. The adoption of these methodologies

¹⁷⁹ The bullwhip effect refers to the phenomenon where order variability increases as the orders move upstream in the supply chain. The bullwhip effect is sometimes referred to as 'demand amplification', 'variance amplification' or the 'Forrester effect'. This effect becomes significant when the cost from fluctuations in production/ordering outweighs the cost of holding inventory (Wang and Disney, 2016).

3595 optimise processes, improves the alignment of demand and production and reduces the incidence of defective
3596 goods (Saleeshya et al., 2012; Harpa et al., 2024).

3597 However, despite technological developments, there is a lack of standardised industrial best practices that
3598 enable the reduction of post-industrial textile waste. The authors are not aware of any specific **legislation** in
3599 producing countries specifically addressing this type of textile waste.

3600 9.2.4.2.2 Pre-consumer waste

3601 When focussing on the retailer stage, all elements of the waste generation ecosystem play an important role.

3602 **Process techniques** related to the efficiency of the reverse logistics could decrease the number of destroyed
3603 returned products avoiding that more items get lost, damaged or with time become obsolete. The literature
3604 reports different improvement potential when adopting the Best Management Practices: pre-consumer waste
3605 due to returns could be decreased from 44% to 22% or from 25% to 13%. Moreover, detailed product
3606 descriptions, especially on size and fitting, could support informed purchase and therefore reduce the number
3607 of returns (Ahlström et al., 2020; Duhoux et al., 2024; Roichman et al., 2024).

3608 The **business model** concerning return policies of the retailer and the marketplace plays a relevant role. To
3609 overcome reverse logistics issues, some fashion brands offer an integrated shopping experience that combines
3610 e-commerce and physical stores. In this case, customers can purchase online and potentially return in physical
3611 stores where they can get customised recommendations (McKinsey & Company, 2024).

3612 The bullwhip effect characterising the manufacturing processes affects also retailers, who would try to enlarge
3613 their inventories to meet the requests of the changing market (Fernandez-Stark et al., 2022; Duhoux et al.,
3614 2024; McKinsey & Company, 2024). As a consequence, as soon as fashion trends change, new collections
3615 generate obsolete goods that remain unsold. Therefore, items less affected by changing trends and with high
3616 demand are less likely to remain unsold (Duhoux et al., 2024).

3617 Section 6.4.4 reported that **users** are not aware of the consequences of returning items of textile apparel after
3618 their online purchase – most of them think that the items are always re-sold. The investigation showed that
3619 customers find convenient purchasing online when returns are free of charge. In these cases, customers have
3620 the opportunity to try and touch the product directly at home without going to the physical store. This
3621 opportunity sometimes evolves to the extreme behaviours of bracketing and 'wardrobing'. The former describes
3622 users purchasing multiple sizes of the same item and returning those that do not fit. The latter describes
3623 consumers purchasing expensive items, wearing them, and then returning them (see Section 13.7.3).

3624 From a **legislative perspective**, the destruction of pre-consumer textile waste is often preferred by companies
3625 because there are taxation advantages related to VAT payment (Duhoux et al., 2024). With the entrance into
3626 force of the ESPR, from 19 July 2026, the destruction of unsold textile products is prohibited, thus preventing
3627 this type of waste, unless derogations apply in cases where destruction is necessary, for instance due to health
3628 and safety reasons. Only micro and small enterprises will be exempted from this prohibition. However, the ESPR
3629 addresses only the unsold products that are placed on the EU market.

3630 9.2.4.2.3 Post-consumer waste

3631 When focussing on the end-of-life of textile apparel, the analysis of reasons for textile apparel disposal sheds
3632 light on what mostly affects the generation of post-consumer waste. The investigation reported in Section 6.2.1
3633 highlighted three main reasons:

- 3634 1. The loss over time of product intrinsic performance,
- 3635 2. The change in perceived value, and
- 3636 3. The change over time of the body shape of the user resulting in unfitted textile apparel.

3637 The first reason is strictly connected to the physical durability of the product over aging processes. The second
3638 reason is related to the desire or need to change the textile apparel due to changing fashion trends, personal
3639 tastes, or social life. The third reason is connected to physiological change of the human body over time.

3640 In addition to these three factors, a study revealed that users are more prone to dispose of cheaper products
3641 than of higher-priced items because they usually develop higher emotional attachment to the latter than to the
3642 former (Forbrugerrådet Tænk, 2022).

From the **legislative perspective**, the Waste Framework Directive (2008/98/EC) and the recently published revision of the Waste Shipment Regulation (2024/1157) (WSR) ⁽¹⁸⁰⁾ regulate the management of waste in EU and its potential shipment within the EU and from the EU to third countries, respectively. In particular, the WFD in establishing Extended Producer Responsibility (EPR) schemes for textiles in all EU Member States should ensure that producers will cover the costs of textile waste management and research and development on e.g. recycling technologies (see Section 4.1). Additionally, the WSR allows the shipment of textile waste produced in EU only to non-OECD countries that have expressed their willingness to receive the waste and demonstrate their ability to treat it in an environmentally sound manner ⁽¹⁸¹⁾.

9.2.4.3 How to assess and describe product technologies in the context of waste generation

The generation of waste is not an intrinsic property of a single product technology, but rather depends on many elements of its ecosystem, and is driven by the production and consumption of textile apparel. Currently, it is not possible to understand if a product was manufactured using specific process techniques and under particular business models. Moreover, it is unknown how many companies use advanced technologies to minimise their post-industrial and pre-consumer waste. Nevertheless, the study of the literature and the consultation with stakeholders reveals that business models exist that incentivise overconsumption and overproduction as described in section 9.2.4.2. However, other stakeholders also related intentional overproduction with inefficient production and claimed it should not be considered as the most common model in the industry. Due to lack of evidence that allow the numerical estimation of the effects generated by different business models, the authors can only report a qualitative analysis.

Thus, the BC, BAT and BNAT related to the waste generation cannot be defined at product-level. The description of waste generation will be therefore modelled taking into account the variability and uncertainties of all factors playing a role: the influence of the business models, user behaviours and the general performances of available process techniques.

9.2.5 Recyclability and recycled content

This section analyses the recyclability and recycled content of textile apparel starting from their definitions reported in other regulatory frameworks and standards:

- Based on the Waste Framework Directive, **recyclability** is the ability of products after becoming waste to be reprocessed into products, materials or substances whether for the original or other purposes. It includes the reprocessing of organic material but does not include energy recovery and the reprocessing into materials that are to be used as fuels or for backfilling operations.
- According to the ISO 14021, the **recycled content** is the proportion, by mass, of recycled material, from pre- and post-consumer waste, in a product or packaging.

These definitions will be revised in the end of this section based on the analysis of this product aspect specific to the textile apparel product group.

9.2.5.1 The ecosystem of recyclability and recycled content

Process techniques and design choices

Figure 29 shows the description of the recycling system of textile waste aiming to produce recycled fibres.

In the first place, textile waste is collected separately from other waste fractions and is subsequently sorted following criteria that depend on the techniques used in the following stages of the recycling system. Then the waste is treated to prepare the material for the following process and finally it undergoes the actual recycling process that results in output recycled fibres. The recycling process includes recycling techniques and potential

¹⁸⁰ Regulation (EU) 2024/1157 of the European Parliament and of the Council of 11 April 2024 on shipments of waste, amending Regulations (EU) No 1257/2013 and (EU) 2020/1056 and repealing Regulation (EC) No 1013/2006. Available at [this link](#).

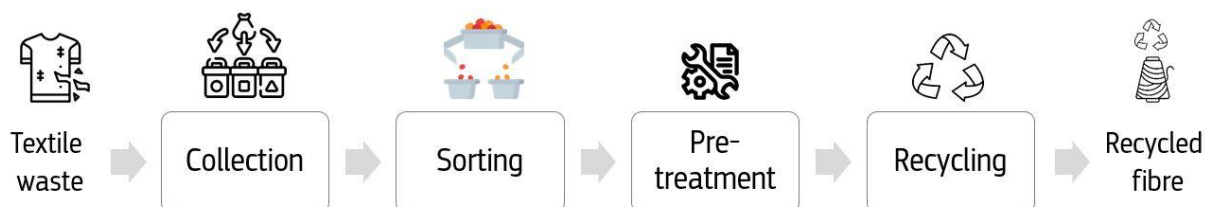
¹⁸¹ Decision of the Council on the Control of Transboundary Movements of Wastes Destined for Recovery Operations. Available at [this link](#). Last accessed on 18 October 2024.

The Waste Shipment Regulation (2024/1157) prohibits the exports of textile waste (Basel Convention entry B3030) destined for recovery in countries to which the OECD Decision does not apply. Specific countries can be included via a Delegated Act (see Article 41).

further treatments to obtain the recycled fibre ⁽¹⁸²⁾. **Figure 29** does not include the transportation of the waste potentially occurring between different processes because it is not relevant to the analysis performed.

In this section, only the waste undergoing recycling will be described but the reader should keep in mind that the waste management of textile waste includes all processes described in Section 13.3.8.

Figure 29. Recycling system of textile waste



Note that post-treatment operations may be required between Recycling and Fibre spinning.

Source: own elaboration using icons from www.flaticon.com

The separate **collection** of textile waste guarantees that this waste fraction can enter the recycling system.

Since post-industrial textile waste is managed by manufacturers, the waste is already separated from other waste fractions and it is directly delivered to other actors of the value chain. When the waste is generated at fibre stage, the spinnable part is usually reintegrated, either directly or with previous treatment, in the manufacturing process, whereas the non-spinnable part is usually given to other actors of the value chain for further applications, such as nonwovens filters, insulation, and filling (Boschmeier et al., 2024). When the waste is generated during spinning, broken ends of slivers and laps are usually reintegrated into the manufacturing process, whereas waste from blow-room machines and carding is usually given to other actors of the value chain for further treatments or applications (Bedez Ute et al., 2019). Differently, when the waste is generated during the manufacturing of fabric or confectioning, manufacturers may send the waste to specialised companies for further treatments or disposal.

The pre-consumer textile waste is managed by retailers. Also in this case, the waste is already separated from other waste fractions and is directly delivered to other actors of the value chain.

The separate collection of post-consumer textile waste includes pick-up and drop-off schemes. Pick-up schemes involve scheduled collection routes targeting specific waste types, such as bulky or frequently disposed waste. In pick-up schemes, consumers are usually provided with containers, especially in door-to-door or kerbside collection. Alternatively, drop-off schemes require individuals to deliver their waste either to designated containers or through take-back systems offered by retailers. In Europe, while the collection of commercial and professional users is often organised through take-back schemes, return logistics or business models in which the supplier remains owner or responsible for caring and end-of-use, the post-consumer textiles waste is mainly collected via drop-off containers (Huygens et al., 2023).

In the recycling system, **sorting** is relevant mainly for post-consumer waste. In the presence of appropriate waste management practices, post-industrial waste and pre-consumer waste could be already sorted at the generation site (manufacturers and retailers), where the waste would be already segregated according to similar characteristics, such as cut-offs of the same fabric, or the unsold items with similar characteristics as belonging to the same collection. However, returned items could be an exception – according to the retailer logistics, this waste could need to be sorted in facilities in the same way as post-consumer waste.

At sorting facilities, items are sorted according to the waste hierarchy assessing their reusability, and if they are not suitable for reuse, according to their composition, fabric construction or colour, depending on the requirements of the targeted recycling route (mechanical, chemical or thermomechanical).

Thus, the sorting criteria are set by second-hand traders and recyclers looking for items with specific characteristics. For example, textiles directed to mechanical recycling are sorted based on product type (e.g. jeans), on the construction of the product (mono, multilayered) and the fabric (woven, knitted) and on the

¹⁸² This happens because the output of recycling techniques can be an intermediate product that needs to be further treated before becoming a recycled fibre. **Table 50** provides more details.

3725 composition and colour. In contrast, textiles which sent to chemical or thermo-mechanical recycling are sorted
 3726 by fibre composition regardless of their product and fabric construction. Moreover, in the case of
 3727 thermomechanical recycling, a sorting by colour is required to produce a range of dyeable materials, while this
 3728 is not needed in the chemical recycling processes, which can remove colour.

3729 There are mainly three types of sorting techniques (EuRIC, 2021):

- 3730 — Manual sorting relies on human inspection of the waste. It is labour-intensive, but currently
 3731 remains the main practice.
- 3732 — Automated sorting currently uses machines equipped with near-infrared (NIR) spectroscopy
 3733 sensors to identify the colour and the surface composition of the items.
- 3734 — Hybrid sorting combines human inspection with automation to balance accuracy and efficiency
 3735 and to complement automated techniques with manual pre-sorting.

3736 When comparing these practices, human inspection is particularly accurate when sorting reusable items.
 3737 Automated solutions cannot detect whether the garment is reusable or not except for some cases such as
 3738 denim jeans, which are easily recognisable with artificial intelligence. In any case, this would require subsequent
 3739 quality control to ensure the items are intact and the trimmings, such as buttons and zippers, still work.

3740 Regarding the fibre composition, the use of near-infrared spectroscopy is normally more accurate than manual
 3741 sorting when identifying the material surface composition, obtaining purities of even 95-98% for pure fractions
 3742 and determining mixed compositions with +-5% accuracy. However, on the one hand, the NIR technology has
 3743 difficulties when used on multi-layered items, items with layered fabrics whose outer fibres are different from
 3744 the inner ones, blends if the content of the minor material is lower than 5%, and dark colours. Furthermore, it
 3745 may have problems to distinguish similar kinds of raw materials, such as viscose and cotton. Moreover, on the
 3746 other hand, the reading of the fibre composition reported on labels is time consuming if done with human
 3747 inspection. Information is not necessarily accurate or it could be even not accessible due to absence of the label
 3748 or removal of the printed elements due to ageing processes (Cura et al., 2021). Consequently, the identification
 3749 of the fibre composition sometimes challenges the sorting process. With growing discussions around Digital
 3750 Product Passports, and digital labels ID-based sorting using Radio-Frequency Identification (RFID) and Near
 3751 Field Communication (NFC) technology is also being explored to develop alternatives and improve automated
 3752 solutions.

3753 After sorting, the waste undergoes **pre-treatment**, which optimises the material processing in the specific
 3754 recycling technique to be used. Usually, during pre-treatment, non-textile parts are separated from the parts
 3755 containing fibres. This is commonly done manually, with loss of some part of the fabric, while new process
 3756 techniques use mechanical separation together with shredding of the waste. Alternatively, other new process
 3757 techniques allow disassembly of waste in cases where specific stitches have been used for the confectioning.
 3758 These stitches can be loosened or dissolved under specific electromagnetic or heat treatments, respectively.
 3759 During pre-treatment, fabrics are usually shredded to improve the efficiency of recycling technology adopted
 3760 in the following step of the recycling system.

3761 **Table 50** reports the status of the **recycling techniques** applicable to textiles waste. Available techniques
 3762 were described via their feedstock, the main recycling output, the possibility to deal with disruptors and their
 3763 maturity. Each technique can process feedstock with specific characteristics and can provide recycled fibres
 3764 with specificities that affect their application in textile products. Additionally, disruptors like non-textile parts,
 3765 dyes, coatings, and undesired fibres could be an obstacle for some techniques (e.g. elastane content above 5%
 3766 when mechanically recycled), but not for all of them.

3767 Existing recycling technologies have the potential to drive 80% circularity in the textile sector if fully scaled. The
 3768 challenge is providing investment and conditions for scaling, which include deployment of a wide collection
 3769 network (a legal requirement for Member States since January 2025 under the Waste Framework Directive)
 3770 and of sorting infrastructure. This requires confidence in supply of usable feedstock, and demand for recycled
 3771 output. The level of maturity of the techniques shown in **Table 50** refers to the global picture in 2024, updated
 3772 based on stakeholder feedback as the general landscape rapidly changes given the fast technical evolution of
 3773 the sector.

3774 **Table 50.** Status of recycling techniques for textile waste in 2024

Recycling technique	Feedstock	Main output	Disruptors	Maturity of the technique and comments
Mechanical recycling	Textile waste constituted of cotton, wool, or synthetic fibres. Blends can be also processed.	Spinnable and non-spinnable fibres. of the same type of the input material	Non-textile materials have to be removed. Fibre contaminants and colours of feedstock will be transferred to recycled fibres. This usually requires colour sorting before recycling.	It is the most commonly used technology at scale. In Europe, recycling companies handle wool and cotton feedstock from post-consumer waste at TRL = 7-8 (TRL = 9 for post-industrial textiles waste). Recycled fibres are shorter than virgin fibres and have lower mechanical properties. The mechanical characteristics of the recycled fibre limit its application. This usually forces the use of recycled fibres with virgin fibres. Moreover, some short fibres cannot be used in fine products. Instead, they are often used in low value applications as cleaning wipes or filling and non-woven material to produce insulation material or technical nonwovens for the automotive industry. A portion of recycled fibre is usually not spinnable. The physical and mechanical characteristics of the recycled fibres highly depend on the status of fibres used as feedstock. For polycotton blends (ideally >60% cotton), the mechanical treated textiles can undergo a thermochemical process to remove the contained polyester, elastane, etc. and recover the clean cotton fibres. In this case, the mechanical step is a pre-treatment at low strength which allows that the recycled fibres are not shortened and have the same characteristics of the virgin ones.
Thermo-mechanical recycling	Textile waste constituted of fibres based on thermoplastic polymers. Input textile waste should consist only of one polymer type (e.g. acrylic, nylon, polyester) or of compatible polymer types.	Polymers in the form of granulate or fibres	Non-textile materials have to be removed. Pigments and dyes remain in the output material.	It is at about TRL=9. During the process the polymers/fibres are deteriorated. Thus, recycled fibres should be blended with virgin fibres.
Chemical recycling for cellulose	Textile waste with cellulosic material. Presence of impurities and non-targeted fibres decreases the efficiency of the process. A specific process step removes impurities and non-targeted fibres.	Wet or semi-dried cellulosic pulp.	Non-textile materials are removed before or during the process. A process stage removes dyes and finishes. It has a bleaching step similar to the traditional wood pulp production.	Most technologies have reached high readiness (TRL=7-9). This is particularly true for feedstock with pure cotton. In Europe, pulp derived from cellulosic materials in post-consumer waste (used textiles, but also newspapers or wood) is recycled at TRL = 4. The potential degradation in length and strength of the cellulosic fibres influences the performance of the resulting pulp.

Recycling technique	Feedstock	Main output	Disruptors	Maturity of the technique and comments
Chemical recycling for synthetic fibres (mainly polyester (PES) and nylon (PA6))	Textile waste constituted of at least 80-90% PES or PA6.	PES or PA6 monomers/oligomers.	Non-textile materials are removed before or during the process. Contaminants like dyes can be handled to achieve homogeneous colours of recycled monomers/oligomers.	Depolymerisation techniques processing PA6 are operational at scale. Recycling facilities in Europe handle PA6 in pre- and post-consumer waste at TRL 9. Techniques processing PES have TRL=4-7. In Europe, facilities recycling polyester from post-consumer waste are at TRL 5-6.
(Thermo-) Chemical recycling for polycotton fabrics	Textile waste made of polycotton. A degree of contamination is accepted by specific techniques.	Depending on the process, the output is: - pulp; - PES fibres, polymers, or monomers.	Non-textile materials are removed before or during the process. Colours can be removed also via bleaching.	The solvent-based dissolution and filtration technique used for polycotton is currently at TRL=5-6. The hydrothermal techniques used for polycotton are at TRL=6-7. The enzymatic technique used for polycotton (also considered a biochemical or biological recycling process) is estimated to be at TRL=5.
Thermo-chemical recycling via pyrolysis and gasification	Textile waste with any fibre composition.	Syngas or pyrolysis oil, which could serve as a basis to produce methanol and then transform the methanol in different monomers for later polymer production.	Non-textile materials are removed before or during the process.	Pyrolysis has already been implemented as industrial scale (TRL 9), but applications for textile waste treatment are unknown. Syngas and pyrolysis oil are usually used as fuels, in that case not being considered as a recycling technique but as energy recovery.

TRL: Technology Readiness Level, which is a 9-grade scale. TRL=1: Basic principles observed; TRL=2: Technology concept formulated; TRL=3: Experimental proof of concept; TRL=4: Technology validated in lab; TRL=5: Technology validated in relevant environment (industrially relevant environment in the case of key enabling technics); TRL=6: Technology demonstrated in relevant environment (industrially relevant environment in the case of key enabling techniques); TRL=7: System prototype demonstration in operational environment; TRL=8: System complete and qualified; TRL=9: Actual system proven in operational environment (competitive manufacturing in the case of key enabling technologies; or in space).

PES: polyester; PA6: polyamide/nylon 6; Polycotton: blend made of cotton and polyester

Source: own elaboration based on (Duhoux et al., 2021; Lu et al., 2023; Dahlbom and Martvall, 2025; Huygens et al., 2023) and inputs from stakeholders.

Overall, the most used recycling technique is mechanical recycling. It is operational at scale, mainly handling post-industrial textiles waste, while recycling companies in Europe also handle wool and cotton feedstock from post-consumer waste at TRL = 7-8. As compared to other techniques, it is less energy intensive and does not require the removal of impurities in the feedstock, such as dyes, which makes the chemical recycling more complicated and expensive. It processes mono-fibre textiles, preferably cotton and wool. Blends can be also processed with mechanical recycling, while in the further processing, mainly dyeing gives some issues. Moreover, the output would be a mix of fibres with different properties that have more difficult application in textile products, if the feedstock is not homogeneous, since the characteristics of the feedstock are directly transferred to the recycled fibres.

Mechanically recycled fibres are shorter than virgin fibres and thus have lower mechanical properties than virgin fibres, which limits their applications. In order to improve their characteristics, recycled fibres need to be spun with virgin high-performing fibres. The non-spinnable recycled fibres that can also result from the mechanical recycling are often used outside the textiles sector for other low value applications, for instance as filling and non-woven material to produce insulation material or technical nonwovens for the automotive industry.

Chemical recycling of cellulose from pure cotton and of nylon is also implemented at operational scale, even though process capacities and the technological readiness level in the EU are still low. Regarding the recycling of cellulose, the potential degradation in length and strength of the cellulosic fibres influences the performance of the resulting pulp. Thus, mixing the recycled material with virgin material helps improving the performance of the yarn. In chemical recycling of synthetic fibres, a different output is obtained processing polyamide-rich textiles to obtain nylon 6 (PA6 – polyamide), where the monomers can be further processed to build a synthetic fibre with characteristics comparable to the virgin material.

Table 50 shows that blends are still difficult to recycle – apart from the mechanical recycling with the described limitations, only depolymerisation techniques processing PA6 are implemented at scale. This represents a big technological challenge that should be addressed because the majority of textile apparel placed on the EU market is made with blends of natural and chemical fibres (48-60%). Single-fibre products account for smaller shares: 18-28% made of cotton, and 11-17% made of polyester (Refashion, 2023a; Bakowska et al., 2025).

This analysis allows to better understand information reported in **Table 21**, which shows that recycled fibres represent a very low share of the market and that the most recycled ones are polyester (14%), wool (7%), polyamide (PA) (2%), cotton (1%), and MMCF (0.5%). From this list, it is important to note that polyester fibres of recycled origin are currently mainly manufactured from PET derived from packaging (separately collected plastic beverage bottles).

In 2023, the industry produced 8.9 million tons of recycled polyester globally, with 98% coming from bottles (Textile Exchange, 2024b). While fibre-to-fibre mechanical recycling of polyester is already operational at scale, the availability of materials remains insufficient for widespread adoption. Moreover, PET bottles currently provide a higher-quality source for recycled material compared to fibre-to-fibre, that is mechanically recycled with the limitations previously described. However, the rising demand for recycled PET for packaging due to the legally binding recycled content targets first in the Single Use Plastics Directive and then in the PPWR (30 % for contact-sensitive packaging made from PET as the major component, as of 2030) could constrain the availability of this material for other uses. Notably, once PET is recycled into apparel, it cannot be reconverted into bottles due to non-compliance with food contact material standards, representing a lower value application. Nonetheless, the availability of recycled plastic (primarily PET) is expected to increase due to the existing collection rate target for beverage bottles. Consequently, the European Commission Directorate General for Environment and Eunomia. (2023) estimate that the 30% recycled content target across the packaging market would mean a 56% of the material available in 2030. Moreover, recycled polyester can also be made from other post-consumer plastics such as ocean waste or other packaging waste.

In terms of the different types of waste, **Table 51** reports a description of post-industrial, pre-consumer and post-consumer textile waste in the context of recyclability and recycled content. Currently, the type of textile waste processed for the production of recycled fibres is mainly post-industrial. Also, pre-consumer waste is used in higher quantity than post-consumer. This happens because these types of waste:

- Are available at manufacturing or retailing sites, thus they can be already segregated from other waste fractions;
- Need less sorting, because they can already be when generated;

- 3836 — Are generally constituted by undamaged and clean fibres;
- 3837 — Have known composition.

3838 In particular, post-industrial waste is often free of non-textile parts, coatings and other disruptors for the most
 3839 used recycling techniques. This means that it can often be further processed without prior pre-treatment.

3840 Conversely, post-consumer waste has largely not been used as feedstock in fibre-to-fibre recycling processes
 3841 because its treatment is more challenging and expensive than the other waste types. The reasons for this are
 3842 that: (1) post-consumer textile waste needs to be segregated from other waste fractions and sorted; (2) fibres
 3843 can be damaged, which would result in less performing mechanically recycled fibres; (3) requires fibre and
 3844 chemical composition to be analysed – see sorting techniques. Instead, it has been used as filling and non-
 3845 woven material in other industries, such as automotive or construction.

3846 Still, although post-industrial and pre-consumer textile waste are currently the most common option, the
 3847 resulting recycled fibres are still more expensive than virgin fibres. Consequently, in 2019, about 90% of post-
 3848 industrial and pre-consumer waste was still incinerated and landfilled, and only about 10% recycled (Huygens
 3849 et al. 2023).

3850 **Table 51.** Description of the textile apparel waste in the context of the recyclability and the recycled content

Characteristic	Post-industrial	Pre-consumer	Post-consumer
Status of the material	Normally undamaged and clean fibres.	Normally undamaged and clean fibres.	Worn material, which may contain damaged fibres. It may be soiled, with various degrees of staining and moisture, generating mould and hygienic problems.
Non-textile parts	Absent	Present	Present
Knowledge of fibre composition	Known and detailed by the manufacturer.	Based on what is reported on the label in accordance with the TLR. The reading of one label allows the identification of fibre composition of the entire collection.	Based on the analysis of the item's surface via infrared spectroscopy or on what is reported on the label in accordance with the TLR. The latter may not be possible due to absence of the label or removal of the writing due to ageing processes.
Degree of heterogeneity	Very low – homogeneous waste	Low	Very high
Location of generation	Manufacturing stage (available in producing countries, which are mainly countries in Asia - see Sections 5.1 and 5.2)	Retail stage (available in EU)	User stage (available in EU)
Sorting	Not always needed, because it can be already sorted at the moment of generation.	Not always needed. - Products belonging to collections that do not reach the customer can be already sorted at the retailer site. - Returned products could need some sorting depending on the retailer logistic.	Needed
Pre-treatment	It is normally ready for recycling because it is cleaner, more standardised and includes mainly cuts of fabrics, so it does not require disassembly and washing. In those cases, only shredding is potentially needed. Pre-treatment is needed in case of finished products.	Needed	Needed

Characteristic	Post-industrial	Pre-consumer	Post-consumer
Regulating legislation	Most of the production occurs in Asian countries, where the authors did not find any legislation addressing the generation of this type of waste. Very little production occurs in the EU. The generation of post-industrial waste in the EU is not specifically regulated. The WFD sets only generic objectives to prevent waste generation. The shipment of post-industrial waste generated in EU is subjected to the WSR. The material described here as post-industrial waste could in some Member States be also identified as a by-product, which would put it out of the scope of the WSR.	In addition to the generic objectives in the WFD, the prevention of this type of waste is specifically addressed by the ESPR, which bans destruction of unsold products, with several derogations and exemption for micro and small enterprises. The shipment of pre-consumer waste generated in EU is subjected to the WSR. However, unsold products that did not become waste are outside the scope of the WSR.	This type of waste is fully generated in EU and is addressed by the WFD (EPR and EoW), and the WSR.

TLR: Textile Labelling Regulation (1007/2011)

ESPR: Ecodesign for sustainable Product Regulation (2024/1781)

WFD: Waste Framework Directive (2025/1892)

EPR: Expanded Producer Responsibility, under the WFD

EoW: End-of-Waste criteria, under the WFD

WSR: Waste Shipment Regulation (2024/1157)

The analysis does not include transportation of the waste because it depends on the location of the treatment facilities.

Source: own production

The study of the literature and the exchange with stakeholders revealed that designers and recyclers have apparently opposite needs. The former uses fibre blends, dyes, coating and other current recycling disruptors to improve the performances of the textile product and simultaneously meet the taste of the users, who would like to buy products at low price (see section 9.2.1.1). The latter, due to the current status of recycling techniques (see **Table 50**), would need to process textile products carrying a minimum amount of disruptors and, for certain technologies, made from one fibre only.

Business models and user behaviour

A recent analysis performed by the JRC reported that recycling is not adequately in place and that the lack of a strong business case for recycling has negative effects on the business case of other economic actors in the recycling system, such as collectors and sorters (Huygens et al., 2023). Conversely, the lacking infrastructure for collection and sorting also adversely affects the recyclers.

One of the main barriers to the development of a market for recycled fibres is the cost. Textile products containing recycled fibres are more expensive than the same products made only of virgin fibres. This is especially relevant when talking about low-priced products, in which a change in the price of the materials would greatly affect the final price. The research conducted by the JRC found that the insufficient internalisation of externalities in the global textile supply chain produces economic market barriers to recycling. Therefore, the economy of scale for the establishment of a profitable recycling system is challenged due to (Huygens et al., 2023):

- Technical limitations in recycling techniques (see **Table 50**);
- The design of non-recyclable textile products (technological externalities);
- Risk aversion to adopting recycled fibres by the next value chain user (consumption externalities);
- The cost associated to the identification of feedstock characteristics suitable for the specific recycling technology.

Despite the problem highlighted above, there are some fashion brands promoting the use of textile apparel with recycled content. Additionally, the preliminary exchange with some stakeholders revealed that a few large

fashion brands are also promoting the collaboration between designers and operators in the recycling system to design recyclable textile apparel.

With a growing interest in textile products containing recycled fibres, there should be also a growing demand for certification schemes based on chain of custody systems capable of tracking the recycled fibres used.

Users contribute to increasing the demand for textile products containing recycled fibres. The analysis performed in Section 6.2.7 revealed that it is mostly young and educated individuals who usually look for these products. The investigation performed by Pranta et al. (2024) revealed that individuals with higher income have a higher likelihood to purchase textile products with recycled content compared to individuals having a lower-income. This confirms the difficulty of recycled fibres to compete with virgin fibres.

Legislative framework

The legislative framework addressing the recycling system could be described focussing on the types of textile waste.

All types of waste are subject to the Waste Framework Directive (2025/1892/EU) as it establishes generic objectives to prevent waste generation. Other than that, post-industrial textile waste is not subject to any specific legislation in the EU nor in producing third countries.

The prevention of pre-consumer textile waste is addressed by the ESPR, which prohibits the destruction of unsold textile products in the EU. Only micro and small enterprises will be exempted from this prohibition. However, the ESPR only addresses unsold products in the EU, while it does not address pre-consumer textile waste in third countries. The authors are not aware of any legislation in third producing countries addressing pre-consumer textile waste.

Post-consumer textile waste is addressed by the Waste Framework Directive (2025/1892/EU) and the newly published Waste Shipment Regulation (2024/1157) (WSR)¹⁸³, regulating its separate collection in EU and its potential shipment within the EU and from the EU to third countries, respectively. In particular, the recently required Extended Producer Responsibility (EPR) schemes for textiles in all EU Member States, as per the latest amendment to the WFD, should ensure that producers will cover the costs of textile waste management and research and development on e.g. recycling technologies (see Section 4.1).

9.2.5.2 How to assess recyclability

The analysis of the ecosystem reported in section 9.2.5.1 suggests to define recyclability with an integral and dynamic approach assessing the evolution over time of all elements of the recycling system, from the feedstock to all process techniques involved. To this aim, in the framework of this PS, a recyclable textile apparel must have the following five characteristics, which apply as soon as it becomes waste:

1. It can be effectively collected;
2. It can be sorted, i.e. segregated from other textile waste and sent to the subsequent suitable recycling pathways;
3. It can be pre-treated before recycling, or can be sent directly to recycling without specific pre-treatment;
4. Its fibre content can be used as feedstock for one or more recycling techniques at scale, either for the original or other purposes;
5. It has no elements or substances in amounts that disrupt the collection, sorting, preparation for recycling and recycling, or that prevent the use of the recycled fibre.

Therefore, textile apparel that meets all these five characteristics is considered to be recyclable, otherwise it is not recyclable. Due to the qualitative nature of the approach, any potential ecodesign requirement to address this product aspect will need to be complemented with the description of a suitable standardised verification system.

¹⁸³ Regulation (EU) 2024/1157 of the European Parliament and of the Council of 11 April 2024 on shipments of waste, amending Regulations (EU) No 1257/2013 and (EU) 2020/1056 and repealing Regulation (EC) No 1013/2006. Available at [this link](#).

This integrated approach was chosen to assess the recyclability of textile apparel because all elements in the recycling system are important. This integral approach based on these five characteristics is a similar approach used by the French Law n° 2020-105 ⁽¹⁸⁴⁾ to define recyclable textile products.

The approach proposed in this PS does not include any geographical limitation of the processes involved provided it complies with the WSR, where it is required that the requirements applied in the recycling system of the country of destination ensure a similar level of protection of human health and the environment than the requirements stemming from Union legislation. Moreover, it does not refer to specific characteristics of the feedstock or particular process techniques, because it aims to promote the technological evolution from the perspectives of the product design, sorting and recycling techniques.

Despite its integral and dynamic nature, this approach is in line with the generic definition of recyclability reported in section 9.1, which is inspired by the WFD. Moreover, this approach is applicable in the same way to all products in the scope of the PS, making categorization unnecessary in the context of recyclability.

Upon the entry into force of a mandatory EPR for textiles under the WFD (or of an implementing act further developing the matter), it will support this integrated approach because economic operators (i.e. fashion brands and retailers) placing products on the EU market would pay eco-modulated fees to support the waste management of textile apparel. It is envisaged that eco-modulation of EPR fees for textiles will be based on future ecodesign criteria, which could include recyclability requirements. In this framework, designers and economic actors of the recycling system will be invited to work side-by-side to place on the EU market recyclable products.

9.2.5.3 Description of the product technologies in the context of recyclability

In the framework of recyclability, the product technologies could be described as follows.

- BC are mechanically recyclable, which is the most common technique at scale even if the recycling rates remain low at around 10%;
- BAT are products that currently can be processed by other techniques reported in **Table 50**, which are implemented at scale, i.e. (1) pure cotton textile apparel recycled via chemical recycling for the production of MMCs, (2) PA6-rich textile apparel recycled via chemical recycling for synthetic fibres, and (3) acrylics and polyester rich blends recycled via thermomechanical recycling.
- BNAT are products that can be processed with techniques at intermediate maturity level reported in **Table 50**.

9.2.5.4 How to assess recycled content

The analysis of the ecosystem suggests the refinement of the definition of recycled content for the purpose of this PS as *the proportion, by mass, of recycled fibres, from post-industrial, pre- and post-consumer waste, in a textile product*.

In comparison with the ISO 14021, the proposed definition further disaggregates the types of textiles waste considered, differentiating between post-industrial and pre-consumer which are clustered within the same category in the former. In the framework of the ESPR, this disaggregation is maintained in order to acknowledge the specificities of each type of waste (as described in **Table 51**) and to align with other EU textiles-related legislation. Specifically, the harmonisation in the definitions of the types of waste in the End-of-Waste criteria for textiles and the ESPR is crucial to create a consistent and viable circular economy framework.

Moreover, it is important to note that, while the proposed definition also considers the use of secondary material coming from sources other than fibre-to-fibre recycling, a closed-loop in the textiles sector is a desirable goal not to conflict with the targets in other sectors. Thus, a potential evolution of the definition could restrict the recycled content to recycled fibres coming from textile waste in the future.

At present, textile products including recycled fibres report their recycled content either via a manufacturer declaration, or via a third-party verification system. Since there is no laboratory test at scale capable of

¹⁸⁴ LOI n° 2020-105 du 10 février 2020 relative à la lutte contre le gaspillage et à l'économie circulaire. Available at [this link](#). Last accessed on 22 October 2024.

determining the recycled or virgin origin of fibres, the only possible verification tool has to rely on chain of custody systems.

The recycled content of a new textile apparel is fibre-specific, and depends on the availability of the secondary material and on the characteristics of the garment (i.e., the mechanically recycled fibres are shorter than virgin fibres and have lower mechanical properties, which could limit their uptake in certain items). A categorization of the products in scope based on all types of fibres and all items requiring a different recycled content uptake would result in a disproportionate number of categories, which would not meet the aims of the PS. For this reason, a more generic approach is followed to ensure that all products included in scope are described by the framework reported in this section, while the specificities of the recycled fibres will be taken into account when proposing and assessing potential ecodesign requirements in the following steps of the PS.

9.2.5.5 Description of the product technologies in the context of recycled content

In the framework of the recycled content, the base case of the product technologies would have no recycled content (see **Table 21**). Differently, the identification of BAT and BNAT should distinguish fibres and product functions. In addition, the BAT would consider recycled content coming from sources other than textiles waste, while the BNAT would be identified with a closed-loop recycling in the sector.

As for the categorisation regarding this product aspect, in general, when constructing fabrics, knitted products are generally better suited for incorporating mechanically recycled fibres due to their more flexible structure compared to woven fabrics (Boschmeier et al., 2024). However, a notable exception is represented by denim trousers, which can incorporate up to 20% post-consumer recycled material, or alternatively 30% post-industrial recycled material (Kuppen, 2024). This is a characteristic common to all items of textile apparel made of denim fabric, regardless their grammage and their stretchability/elasticity.

9.2.6 Environmental impacts

The product aspect on 'environmental impacts' is defined as any change to the environment, whether adverse or beneficial, wholly or partially resulting from a product during its life cycle.

This section analyses the elements affecting the environmental impacts generated by the consumption of textile apparel in EU to best describe the product technologies. Therefore, the analysis reported in this section does not aim to give a comprehensive overview of the environmental impacts caused. Task 5 of the PS will include a comprehensive environmental assessment of the base case of products included in the scope.

9.2.6.1 The ecosystem affecting the environmental impacts

Process techniques

For each stage of the life cycle of the product, there are numerous process techniques that contribute to the product development. Each of those techniques have different environmental performances. Therefore, the choice of process techniques plays a key role in the environmental impacts of textile apparel.

Different levels of environmental impacts, related to specific process techniques, can be seen in production stages such as yarn manufacturing. A range of spinning techniques is available. Ring-spinning is the most commonly used in the textile industry due to its versatility in terms of the range of fibres that can be processed. Its main advantage is that it produces finer and stronger yarns than other spinning techniques due to good fibre control, orientation and alignment during spinning. However, it has high rates of power consumption, caused by the necessity of rotating the bobbin at a rate of one turn for each twist inserted. In contrast, open-end spinning was developed in an attempt to overcome the speed limitations of ring spinning. Another advantage is a reduction in energy consumption (Elhawary, 2015a). In the case of wool spinning processes, significant differences can be seen between fabric derived from carded yarn (woollen process) and fabric manufactured using combed yarn (worsted process) in terms of the quantity of lubricants applied, with 5% in the former and 2% in the latter (Roth et al., 2023).

Knitting is a mechanical process and involves knotting yarn together with a series of needles. The main knitting production methods are straight knitting, fully fashioned knitting, integral knitting, and complete textile apparel knitting. Straight knitting most commonly relies on the weft technique, which consists of using one continuous yarn which is fed to and looped in rows by one or more needles at a time. Fully fashioned knitting machines can produce custom-shaped sheets of fabric by adding/removing the stitches, so that there is little or no need for cutting panels, reducing the amount of discarded material. Integral knitting is an advancement of the fully fashioned knitting technique. An integral knitting machine is able to add additional trimmings as an integrated

part of the fabric panel, reducing fabric losses from cutting, as well as sewing requirements. Finally, complete textile apparel knitting machines are able to knit complete textile apparel, eliminating the cutting and sewing steps altogether, decreases raw material consumption and producing higher-quality textile apparel (Roth et al., 2023).

Another example of these differences can be seen in the dyeing process, in this case in terms of water use. In batch dyeing, a certain amount of textile material is loaded into a dyeing machine and brought to equilibrium with a solution containing the dye and the auxiliaries. An important parameter in dyeing is the liquor ratio of the equipment: the weight ratio between the total dry material and the total liquor (a liquor ratio of 1:10 means 10 litres of water for 1 kg of textile material). This parameter not only influences the amount of water and energy consumed in the dyeing process, but also plays an important role in the level of exhaustion of the dye and in the consumption of chemicals and auxiliaries. Dyeing machines vary greatly in their liquor ratios, depending also on the type of substrate to be dyed and its hydrophilicity. For instance, in woven and knitted fabrics, liquor ratios can vary from 1:2 in airflow equipment, to 1:40 in winch beck equipment (Roth et al., 2023). These differences will significantly influence the environmental impacts of final products. Dyeing can also be carried out in batch or in continuous/semi-continuous mode. Batch dyeing processes generally require higher water and energy consumption than continuous processes. Continuous and semi-continuous dyeing processes consume less water, but this also means a higher dyestuff concentration in the dye liquor.

The term finishing covers all those treatments that serve to impart to the textile the desired end-use properties. These can include properties relating to visual effect, handle and special characteristics such as water and fire proofing. Finishing may involve mechanical/physical and chemical treatments. Among textile finishing processes, chemical ones are the most significant from the point of view of the emissions generated. As in dyeing, the emissions are relatively different between continuous and discontinuous processes. Continuous finishing processes do not require washing operations after curing.

In essence, the wide variability of process techniques available to manufacture textile apparel has a fundamental relevance on the environmental impact of final products.

In terms of waste management techniques, a number of options are available, from landfilling and incineration, to preparing for reuse and recycling. In this case, as reported in Solis et al. (2024), preparing for reuse is the most beneficial management pathway from both an environmental and a socio-economic perspective. Overall, recycling is the second-least impacting option for most environmental impact categories, while technology and processing costs will have to decrease to be economically competitive against incineration.

Business models

Companies have multiple choices in the definition of their strategies and business models. These choices imply different levels of environmental impact of the final product.

As described in Section 5.5, two main models can be identified in the textile apparel industry (DG GROW, 2021a). A consumer-led model, where the requests of the consumer are the centre; and a brand-led operation model, where the brand dictates the design and the manufacture. Other classification of business models can also be made. For instance, in terms of supply chain approach. On one hand, there is the integrated approach, where the production is entrusted to internal suppliers and the logistics aims to quickly react to customers' demands. On the other hand, there is the centralised approach, where the production is mostly outsourced and supported by audit and quality control programmes. Hybrid versions of those business models can also be found. Each business model has different characteristics, each of them with different implications on the environmental impact of products. In this section, a brief description of these characteristics is made.

The trend turnover is a fundamental factor that defines a business model. Some companies opt to implement business models characterised by continuous novelty and disposable trends in constant change (Centobelli et al., 2022), designing their products for rapid trend turnovers through obsolescence and early disposal. Other companies, in contrast, implement business models with less frequent trend turnovers, focusing on product durability and reverse logistics. The number of seasons or trends that are placed on the market largely affects the environmental impacts. Rapid trend turnovers tend to drive consumers (directly or indirectly) to replace clothes, even when it is still not necessary from a technical perspective.

Related with trend turnover is production time, which refers to the time between the design of the product and the availability of such product for the consumer. Production time largely affects several aspects of the supply chain, from procurement to manufacturing capacity, planning and inventory management. Shorter production times could increase the probability of manufacturing errors. This also could increase the generation of waste, due to the required destruction of products that cannot be sold (see section 9.2.4 and (Allwood et al., 2016)).

4078 Shorter production times also requires the use of air transport, rather than cargo ships, increasing the
4079 environmental impact of the distribution stage.

4080 The location of different stages of the supply chain also affects the environmental impact of products. Many
4081 companies outsource the transformation process of raw materials into completed textile apparel to third
4082 countries, often to allow access to low-cost labour and less stringent environmental regulations, such as water
4083 or air emission levels which are less ambitious than in the country of origin (Centobelli et al., 2022). If the
4084 numerous stages of the supply chain are scattered over the globe, more transport is required, increasing the
4085 environmental impact of the distribution stage.

4086 The location of different stages of the supply chain also affects the energy source used to produce textile
4087 apparel. The manufacturing of textile apparel is energy intensive. Each country has a different energy mix ⁽¹⁸⁵⁾
4088 that has a significant influence on the impact of the final product.

4089 Different environmental impacts can also be expected, related with the ownership of the apparel. There are
4090 companies that opt for renting products rather than selling them to individuals. When a product is rented, it
4091 has the possibility to be used by more consumers. This is particularly true for products that are not used very
4092 frequently, such as gala dresses, or for products that quickly become unusable, such as clothes for children. In
4093 principle, products sold under this business model (sold as a service rather than a product) produce lower
4094 environmental impacts, due to the higher intensity of use among different users across their lifetime.

4095 In essence, decisions made by companies in terms of the business model implemented to place textile apparel
4096 on the market have a significant relevance on the environmental impact of the final product.

4097 **User behaviour**

4098 Consumers influence the environmental impact of clothing when they make purchase decisions. Apparel is
4099 often bought spontaneously, simply because leisure and fun are associated with the act of buying clothes. Only
4100 a minority of consumers inform themselves correctly before buying apparel (Kleinhüchelkotten et al., 2018b).
4101 When a consumer chooses to acquire a product that has been placed on the market under a business model
4102 based on fast trend turnovers and short production times, they are contributing –possibly unknowingly– to the
4103 environmental impacts associated with that business model. The amount of products that are purchased also
4104 influences overall environmental impact of clothing.

4105 Consumers also have an influence on the environmental impact of textile apparel, depending on the way they
4106 use and maintain these products. Using clothes in the right environment contributes to increase their lifespan.
4107 Storing and folding apparel may also have an influence on the lifespan of apparel, since it may affect their
4108 intrinsic quality.

4109 Related with product use are the unwritten social rules that govern specific gatherings, such as workplaces,
4110 celebrations such as weddings or regional festivities that last several days. In these events, it is expected that
4111 attendees will not wear the same apparel on different days. The expectation is that different apparel will be
4112 worn on different days, even when it is not necessary from technical point of view. This contributes to an
4113 increased consumption and production of textile apparel, and therefore to their environmental impact.

4114 Maintenance activities can also contribute to increase lifespan of products and avoid (or delay) the acquisition
4115 of new ones, minimising the overall environmental impact. Clothes must be sorted appropriately before
4116 washing to avoid premature deterioration. On occasions, consumers use quantities of detergent and softeners
4117 which are merely based on their perception, rather on manufacturers' recommendations (A.I.S.E, 2017a),
4118 potentially increasing the impact of the product use stage. In terms of washing, water temperature and washing
4119 frequency are not always optimised, contributing to an excessive consumption of energy. Drying can be
4120 performed either naturally or using dryers, a choice which is highly related with user choice, availability of
4121 space in the household or geographic location. The choice of clothes that do not need ironing –or that require
4122 less frequent ironing– can also have an influence on the environmental impact of the final product.

4123 Consumers also have an effect on the environmental impact of textile apparel when they make decisions
4124 related with its end of life. As pointed out in section 13.7.3.1, this is influenced by the individual characteristics
4125 of the consumers, their habits, demographic context, product traits and perceived quality (Cluver, 2008b;
4126 Goworek, Lynn Oxborrow, et al., 2020). Perceived quality of the product is a decisive factor to discard textile
4127 apparel (Aakko and Niinimäki, 2021). Low perceived quality also triggers early disposal (Piippo et al., 2022b).

¹⁸⁵ A different combination of energy sources to produce electricity

This is also magnified by the loss of symbolic value⁽¹⁸⁶⁾ given by the consumer (Gwozdz et al., 2017b). Discarding products when they are still technically functional increases their environmental impact, due to a low intensity of use.

In essence, user behaviour of textile apparel can have a significant relevance in the environmental impact of final products.

Legislative framework and industrial best practices

The legislative framework has an influence as well on the environmental impact of final products.

In the EU, the Industrial Emissions Directive aims to reduce industrial pollution by requiring industries to adopt preventive measures and use the best available techniques (BATq). It mandates integrated permits covering all environmental impacts and sets strict emission limits for key pollutants. This Directive applies to the manufacturing of textile apparel. The BAT for the textiles industry are described in the Best Available Techniques Reference Document (Roth et al., 2023). The end of life is addressed in the Best Available Techniques Reference document for Waste Treatment (Pinasseau et al., 2018b).

As pointed out in Section 5.7, countries outside of the EU tend to have less stringent measures regarding environmental protection, in particular on emission limits for key pollutants. The EU has the most ambitious legislative framework worldwide, covering aspects such as emissions to air and water, energy consumption and energy efficiency, water use, waste generation, and usage and management of chemicals. Comparing this framework with that of China, the top global exporter of apparel, it can be seen that the latter only addresses emissions to air and water as well as waste generation. In contrast with the EU BREF, the Chinese framework also covers noise emissions. With regards to India (the fifth global exporter of apparel), only emissions to water are addressed. The full comparison of aspects covered in different Best Available Techniques reference documents around the world can be seen in **Table 31**.

A comparison in terms of specific substances is made in **Table 52**. The emission levels to water of three substances/parameters are compared: chemical oxygen demand (COD), chromium and zinc (the comparison of values on emissions to air was not possible due to different practices and key environmental indicators used in the different frameworks). In **Table 52** it can be seen that there are significant differences in the level of ambition between the EU, China and India.

Table 52. Environmental performance levels for emission into water in specific regions

Key Environmental Indicator	EU	India	China
COD (mg/l)	40-100	100	7-30 000
Chromium (mg/l)	0.01-0.1	2	n/a
Zinc (mg/l)	0.04-0.5	n/a	n/a

n/a: Not Addressed

Source: own production based on Section 5.7 and 13.6.2

The EU BREF sets the emission limits on COD between 40-100 mg/l for all activities and processes, whereas in India and China those limits are set at 100 mg/l in the former and between 7-30 000 mg/l in the latter. In the case of chromium, China does not establish a limit. The EU limit is established between 0.01-0.1 mg/l, more ambitious than the 2 mg/l set in India. In terms of zinc, only the EU establishes a limit value, between 0.04-0.5 mg/l.

These differences are particularly relevant for textile apparel, since most of the activities related with raw material production and manufacturing occur out of the EU (**Figure 8**). Deciding to manufacture products outside of the EU means that the emission levels of manufacturing plants will be subject to less stringent emission requirements. Therefore, considering the relevance of the manufacturing stage in textile apparel, producing textile apparel outside of the EU could increase the environmental impact of the final product. Moreover, companies producing textile apparel in the EU must face higher costs than companies producing outside of the EU, due to the prevention and reduction of emission activities that they need to implement.

¹⁸⁶ Symbolic value refers to the intangible meaning and significance that an object holds beyond its practical or material function. It represents the cultural, social, or emotional associations that people attach to something, shaping its perceived worth and importance.

9.2.6.2 How to assess the environmental impacts and description of product technologies

The analysis of the ecosystem affecting the environmental impact of textile apparel shows the difficulty to identify in a rigid framework the characteristics of specific product technologies. However, the information gathered in the previous sections allow to build a reasonable picture.

The BC takes into account that China, India, and Bangladesh manufacture the largest part of textile apparel consumed in EU. This means that the BC is described by process techniques adopted in these countries, where the legislation allows higher emission into the environment compared to what happens in EU. The business model that characterises the BC promotes overproduction and overconsumption, supported by users that tend to change frequently their wardrobe. The end of life of the BC is mainly described by landfilling and incineration of textile waste in the EU as well as in third countries, and some 10% going to recycling.

The BAT takes into account EU manufacture and the currently available less-impacting business models, user behaviours and waste management options. This means that the BAT is described by process techniques adopted in EU and described in the EU-BREF. The business model that characterises the BAT promotes a production rate similar to that before 2004, when the apparent consumption of EU was about half of the current one (**Figure 10** in Section 5.2). The end of life of the BAT is described by energy recovery and recycling in the EU of the textile waste.

Since the environmental impacts are affected by very numerous aspects, the description of BNAT will be simply more ambitious than BAT and will take into account all the BNAT reported for other product aspects.

This description of the product technologies is suitable for all products in the scope of the PS.

9.2.7 Presence of substances of concern

Substances of concern are defined in Article 2(27) of the Regulation and largely encompass substances that are of concern due to their negative effects upon the health of humans and to that of receptors in the environment as a consequence of their adverse long-term effects. These substances are specifically identified via being listed as (a) Substances of Very High Concern (SVHCs) in the REACH “Candidate List”, (b) by having a harmonised classification under specific hazard classes defined in the CLP Regulation or (c) by being identified as Persistent Organic Pollutants (POPs) by the POPs Regulation. In addition, a specific class of SoCs are defined in ESPR, which (d) hinder reuse and recycling (of materials in the product in which SoCs are present) and which should be identified in the product-specific delegated acts to be developed under ESPR. It is worth noting that a single substance could potentially be identified simultaneously under several SoC classes (a, b, c and d).

9.2.7.1 Ecosystem of Substances of Concern

Manufacturing factors and process techniques

A very large variety of chemical substances are used by the textile sector in the different manufacturing stages, from fibre production, to spinning, weaving or dyeing and finishing of fabrics, to name just a few of the many processes involved. Chemicals are used to make fabrics more durable, softer or to provide colour and colour-fastness or a water or stain-repellent finish, among many other uses. Some chemicals are specific to the different fibres used to make textiles (cotton, polyester, wool, etc.).

It is not a simple task to determine the number of chemicals used by the textile sector, given the large variety of processes in which these are used, the multitude of functions they provide and the global distribution of chemicals supply chains and textile manufacturers. Roos et al. (2019) claim that the number of chemicals in use in the textile industry exceeds 15 000, with over 10 000 dyes and pigments and about 5 000 auxiliary chemicals, quoting figures from the Colour Index⁽¹⁸⁷⁾ and TEGEWA⁽¹⁸⁸⁾, respectively. A recent report¹⁸⁹ on the use of PFASs in the textile sector indicates that more than 8 000 chemical substances are used by this sector. Regardless of which the precise figure may be, the sheer numbers involved provide an indication of the magnitude of the challenges that the sector faces in implementing supply-chain transparency and traceability of chemicals.

¹⁸⁷ <https://colour-index.com/>

¹⁸⁸ Textiles Auxiliaries e-market. <https://www.thk-online.net/>

¹⁸⁹ A Review of PFAS as a Chemical Class in the Textile Sector. Natural Resources Defense Council, 2021. <https://saicmknowledge.org/library/review-pfas-chemical-class-textiles-sector-policy-brief>

Although it is acknowledged that chemicals play an essential role¹⁹⁰ in achieving a competitive textile apparel sector that to the best of our knowledge largely meets the demands of consumers, reports such as that by the Swedish Chemicals Agency, that investigated 2 450 substances used by the textile sector and found 750 classified as hazardous for human health and 440 as hazardous for the environment, are a cause of concern (KEMI, 2014), although it should be noted that the likelihood of exposure differs from one substance to the another. As for other material streams, it is also acknowledged that legacy chemicals may be reintroduced back into the economy via recycled fibres, potentially posing risks to human and environmental health and interfering with the recyclability and market acceptance of the recycled fibre.

Similarly, a recent report by the European Environment Agency¹⁹¹ on per- and polyfluoroalkyl substances (PFAS) highlights the specific concerns that this group of substances pose both due to their health and environmental impacts as well as because of their consequences on circularity. According to this study, it is estimated that textiles account for approximately 35% of the total global fluoropolymer demand and that a third of all PFAS in the EU (41 000 – 143 000 tonnes) are used in the textile sector, which is thereby estimated to be the biggest contributing use sector to PFAS pollution in Europe. Furthermore, in its report on textiles, published in the year 2017, the Ellen MacArthur Foundation stated:

“The use of substances of concern in textile production has negative effects on farmers, factory workers, and the surrounding environment. While there is little data on the volume of substances of concern used across the industry, it is recognised that textile production discharges high volumes of water containing hazardous chemicals into the environment. As an example, 20% of industrial water pollution globally is attributable to the dyeing and treatment of textiles” (Ellen MacArthur Foundation, 2017).

In this report a call for action is made in moving towards a new textiles economy, requiring, as regards substances of concern, to develop *“a robust evidence base on the usage of chemicals, including the amount used, as well as identification of substances of concern and the impacts of these”*.

User behaviour

Section 6 reports information available on behavioural trends among users with regards to textile apparel. It covers aspects that include the way users choose to buy apparel, their habits during the use phase, and why they decide to dispose of them. It also distinguishes user behaviour aspects at pre-purchase, post-purchase and disposal stages.

In a survey spanning 27 EU Member States, 60% of 26 718 respondents perceive chemicals in apparel fabrics as minimally risky, with most not seeing them as a threat to people whereas, looking at the country level, health risks are deemed very important by about two-thirds of respondents in Germany and Slovakia whereas respondent in Denmark and the UK attributed a lower importance to this aspect (European Commission, 2009). Similar conclusions can be derived from a survey by Fashion Revolution (2020) which indicates that only about 37% of respondents in United Kingdom, France, Germany, Italy, and Spain considers it important to buy apparel produced without harmful chemicals. When looking at the broader picture, not specific to apparel, the Special Eurobarometer survey¹⁹² published in March 2020 revealed that 86% of respondents are worried about the impact on health of chemicals present in everyday products.

Despite the uneven level of concern by users regarding chemicals in apparel, especially as regards substances that pose a risk to human health or to the environment, the establishment of information requirements under ESPR on the presence of substances, in particular of substances of concern in apparel will contribute to the generation of information that can further provide transparency to enable an informed public opinion. This in turn can inform possible actions that could be addressed in specific legislation on the safe use of chemicals such as REACH.

Business models

¹⁹⁰ As indicated in the Chemical Strategy for Sustainability Communication (COM(2020) 667) “Chemicals are everywhere in our daily life and play a fundamental role in most of our activities, as they form part of virtually every device we use to ensure our well-being, protect our health and security, and meet new challenges through innovation”.

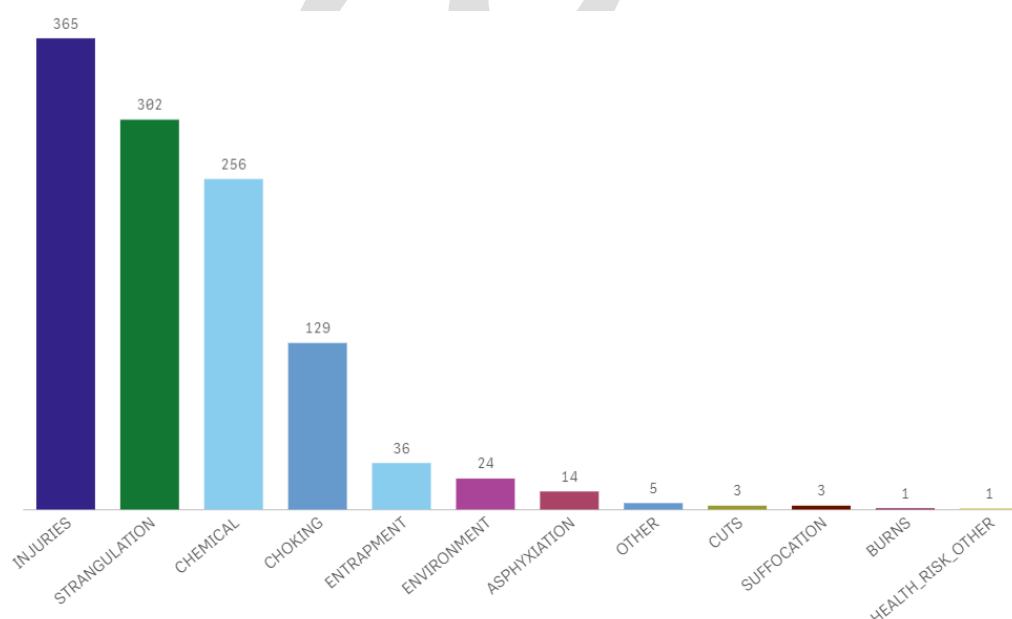
¹⁹¹ EEA 2024. Briefing no. 11/2024. PFAS in textiles in Europe’s circular economy. <https://www.eea.europa.eu/en/analysis/publications/pfas-in-textiles-in-europes-circular-economy>

¹⁹² Special Eurobarometer 501. Attitudes of Europeans Towards the Environment. European Union 2020. <https://europa.eu/eurobarometer/api/deliverable/download/file?deliverableId=72378>

The high throughput business models, already mentioned in previous sections, result in very short textile apparel manufacturing times, with the appearance of dozens of collections every year and where manufacturing occurs largely out of the EU. This model based on production happening largely in third countries also has consequences related to the use of chemicals, given regulations on the use and management of chemicals differ, as do levels of enforcement, control and monitoring. According to Niinimäki et al. (2020), the majority of the chemicals use connected to producing textiles for the EU occurs outside the EU. This highlights that although supply chain transparency as regards chemicals is challenging, for production that takes place both within the EU and in third countries, the latter face additional limitations, for instance related to hazard classes recognised under GHS and the uptake of GHS safety data sheets and their differences compared to those developed to be REACH compliant.

Although it is not possible to pin-point impacts of the business model on specific chemicals, it seems clear that the large increase in production, taking place in third countries where often legislation and environmental controls are less stringent than in the EU, results in increased emissions of chemicals, notably via waste water during dyeing and washing processes, and can result in the use of substances that are regulated in the EU and other constituencies, which subsequently risk entering these markets, incorporated within textile products. The latter problem is further exacerbated by the growing importance of purchases made via on-line platforms and the difficulties associated to effective enforcement. Ogugbue and Sawidis (2011) state that some 0.7 million tons of synthetic dyes are produced annually worldwide and that the textile industry releases up to 200 000 tons per year of these dyes via effluents resulting from inefficient dyeing and finishing processes. Most of these dyes escape conventional wastewater treatment processes and persist in the environment. Evidence of the use of regulated hazardous substances in textiles, and of their presence in products imported into the EU can be found consulting alerts in Safety Gate ⁽¹⁹³⁾, the EU rapid alert system for dangerous non-food products, where reports of textile apparel and footwear items can be found containing Chromium VI, nickel, different phthalates, cadmium and other regulated substances. An analysis of the total alerts registered in Safety Gate in the period 2019 – 2023 for the product category “Clothing_textiles” indicates that chemical risks are the third most frequent risk reported (22.5%).

Figure 30. Risks associated to textile apparel



Source: EU Safety Gate statistical tool. Available at [this link](https://ec.europa.eu/safety-gate-alerts/screen/search?resetSearch=true). Last accessed on 13 November 2024

Legislative frameworks and industrial practices

¹⁹³ <https://ec.europa.eu/safety-gate-alerts/screen/search?resetSearch=true>

As indicated in Section 4.1, currently there is no specific EU legislation addressing the mandatory sustainability of textiles, but EU Ecolabel criteria for textile products ⁽¹⁹⁴⁾ exist, this being a voluntary scheme for companies willing to show the good environmental performance of their products. Article 6(6) and 6(7) of the Ecolabel Regulation (EC) No 66/2010 specify that the Ecolabel may not be awarded to products containing substances classified under CLP as toxic, hazardous to the environment, carcinogenic, mutagenic or toxic for reproduction or containing substances meeting the SVHC criteria defined in Article 57 of REACH. Derogations from this prohibition are possible except for substances identified as SVHCs and thereby listed in the so-called “Candidate List” established under REACH. The specific implementation of these general provisions into chemical-specific criteria applicable to textiles are developed as criteria 13 and 14 of the EU Ecolabel criteria for textile products enacted in Commission Decision 2014/350/EU and in the restricted substance list defined in its Appendix 1 and the list of restricted dyes specified in its Appendix 2.

More specifically as regards limitations on chemicals, the REACH regulation has the main purpose of ensuring the safe use of chemicals in the EU. REACH regulates, amongst many other aspects, the restriction of the placing on the market and use of certain substances listed in its Annex XVII, including in some cases, the incorporation of substances into articles. The authorisation title of REACH applies to the placing on the market and use of substances of very high concern (SVHC), aiming at their progressive substitution by less hazardous substances or technologies and by subjecting their use to specific conditions.

A number of restrictions in Annex XVII to REACH specifically mention textiles in their scope. These include entries: 4 - Tris (2,3 dibromopropyl) phosphate; 7 - Tris(aziridinyl)phosphin oxide; 8 - Polybrominated biphenyls; 18 - Mercury compounds; 20 - Organostannic compounds; 43 - Azocolourants and Azodyes; 46/46a - Nonylphenol and Nonylphenol ethoxylates; 47 - Chromium VI compounds (relevant to leather articles); 68 - C9-C14 PFCAs and 72 - CMRs in textiles and footwear.

Other restrictions, of general application to articles placed on the market for supply to the general public, or covering all articles placed on the market, may also apply to textile articles, for instance entries 50 – (certain) polycyclic aromatic hydrocarbons; 51 and 52 (certain phthalates); 61 - Dimethylfumarate; and 63 – Lead and its compounds.

Article 33 of REACH sets up a supply chain communication duty requiring suppliers of articles containing SVHC above 0.1% to communicate certain information to the recipients of those articles. Furthermore, Article 9 of Directive 2008/98/EC, the Waste Framework Directive, promotes the reduction of the content of hazardous substances in materials and products by defining a reporting obligation that applies to suppliers of articles (as defined under REACH), requiring them to provide information regarding the presence of SVHC in articles, including textile articles, pursuant to the referred Article 33, to the European Chemicals Agency (ECHA). This information is collected in the SCIP database ⁽¹⁹⁵⁾, operated by ECHA, and access is provided to waste treatment operators and consumers.

Similarly to REACH, the POPs Regulation regulates persistent organic pollutants with the objective of protecting human health and the environment by prohibiting, phasing out as soon as possible, or restricting the manufacturing, placing on the market and use of substances subject to the Stockholm Convention. Some of these restrictions are relevant to textiles, for instance those associated with certain brominated flame retardants (e.g. certain PBDEs), surface-active substances such as PFOS or PFOA, or substances with biocidal properties such as pentachlorophenol.

Whereas the EU and many developed countries have legislation regulating chemicals, including their use by the textile sector, such legislation does not exist or is less stringent in other countries where textile production takes place. A recent report by Toxics Link ⁽¹⁹⁶⁾ highlights that “A review of the global regulatory requirements highlights that there are stringent regulations concerning most of these chemicals only in the developed world, especially in the EU; whereas, regulations on some of the chemicals either do not exist or are less stringent in the developing or emerging economies”.

¹⁹⁴ EU Ecolabel criteria for textile products. Commission Decision of 5 June 2014 establishing the ecological criteria for the award of the EU Ecolabel for textile products. Commission Decision (2014/350/EU). Available at [this link](#).

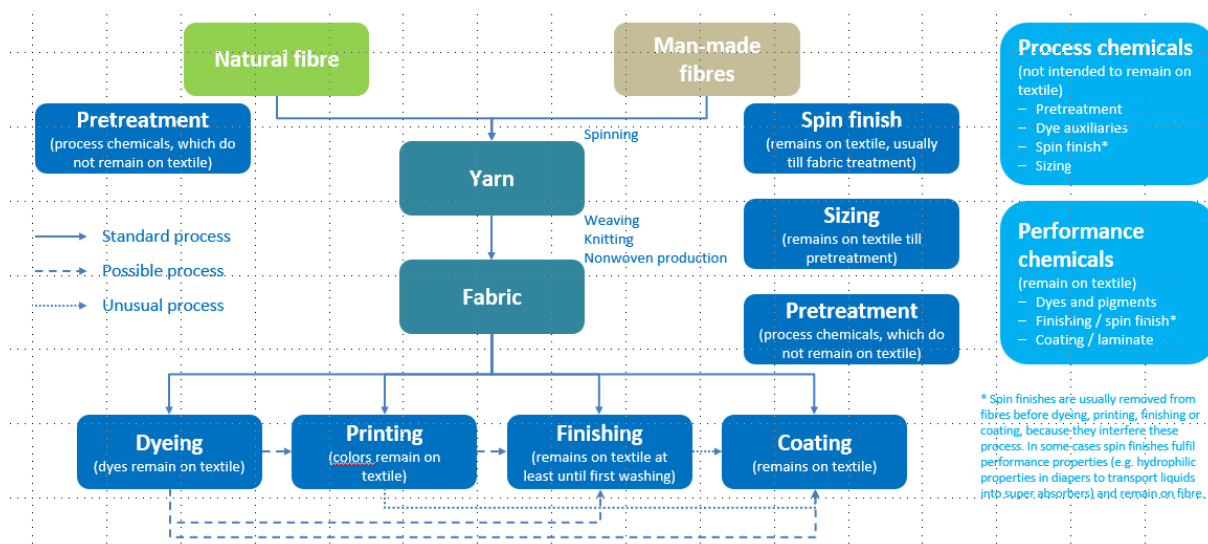
¹⁹⁵ SCIP database from the European Chemical Agency. Available at [this link](#). Last accessed on 13 November 2024.

¹⁹⁶ An overview of toxic chemicals in textiles. Toxics Link 2021. <https://toxicslink.org/publications/reports/toxics-chemical-in-textile-report>

9.2.7.2 Methodology to assess Substances of Concern

As explained in previous sections, retailers and brands in the textile apparel sector are facing increasing regulatory and reputational challenges to control chemicals present in their products. The sheer number of chemicals involved, the complexity of supply chains and the fact that the majority of processes involving the use of chemicals takes place in third countries, make tracking of chemicals in textiles a complex endeavour. **Figure 31** provides an overview of processes using process and auxiliary chemicals associated to the textile components of apparel. The chemical constituents of non-textile elements are also relevant, but are not covered by the diagram below.

Figure 31. Processes using process and auxiliary chemicals



Source: Adapted from EUCTL - European Chemistry for Textile and Leather, the European Association representing the companies that operate in Europe, producing and putting on the market chemicals used in the textiles and leather value chains.

KEMI (2014) describes the different types of chemicals used in textile production in terms of:

Functional chemicals, which are intended to remain in the final article and are expected to be present at certain concentrations in order to achieve the desirable function. Some examples include:

- Dyestuffs and pigments
- Crease resistant agents
- Anti-shrinking agents
- Oil, soil and water repellents
- Plasticisers
- Flame retardants
- Biocides for defined functionalities in articles, e.g. antibacterial agents
- Stabilisers
- Stiffening agents
- Reactive resins for various finishing treatments

Auxiliary (Process) chemicals, which are involved in some step of textile processing but are not intended to remain in the product. These include:

- Organic solvents
- Surfactants
- Softeners

- 4365 • Salts
- 4366 • Acids and bases
- 4367 • Biocides as preservatives in the process or during storage and transport

4368 Unintended chemical substances – contaminants and degradation products

4369 These are chemicals not intended to remain in the finished article, such as contaminants and degradation
4370 products, have no function in neither the textile production process nor in the finished textiles.

- 4371 ○ Formaldehyde released from certain reactive resins
- 4372 ○ Polyaromatic hydrocarbons (PAH) impurities in pyrolysed products e.g Carbon Black
- 4373 ○ Arylamines derived from certain azo dyestuffs and pigments
- 4374 ○ Toxic metals (e.g heavy metals) due to impurities from the raw material

4375 Many retailers and other operators placing textile apparel on the market have implemented chemical
4376 management strategies to enable compliance and to meet their due-diligence obligations as regards chemicals
4377 in their products. Effectively there are two types of approaches to obtain information about substances of
4378 concern in products.

4379 The **first** relies on **supply chain information** and the implementation of different **supply chain**
4380 **transparency measures**. In this sense retailers and other operators in the supply chain rely on declarations
4381 by suppliers, binding contractual terms and third-party certification to provide information about substances in
4382 products, in particular about those which should not be present in them (negative lists). International and
4383 European sectorial initiatives have created relevant Restricted Substance Lists (RSLs) and Restricted
4384 Manufacturing Substance Lists (RMSLs) to capture and harmonise reporting of compliance regarding process
4385 and auxiliary substances regulated under different legislations and in different countries. Examples of these
4386 are the AFIRM Restricted Substances List ⁽¹⁹⁷⁾ and the ZDHC Manufacturing Restricted Substances List ⁽¹⁹⁸⁾ used
4387 by the textiles and footwear sectors. Other restricted substance-based approaches rely on certification of
4388 compliance with the requirements in different ecolabels. The EU Ecolabel for textiles, which has a strong focus
4389 on chemicals, as well as the privately managed Oeko-tex 100 standard for the textile sector ⁽¹⁹⁹⁾ are examples
4390 of this approach. It is acknowledged that restricted substance lists are not as such a supply chain transparency
4391 measure, but help operators to support compliance claims related to the absence of regulated / controlled
4392 substances.

4393 Broader disclosure approaches for chemicals along textile supply chains, which require a high level of take-up
4394 and implementation and the necessary IT infrastructure, in a way not dissimilar to the digital product passport
4395 concept introduced under ESPR, have been the subject of numerous studies, pilots and initiatives. In this sense,
4396 UN/CTFACT has recently provided a recommendation providing industry actors with a set of internationally
4397 agreed practices for the harmonized collection and transmission of data for tracking and tracing materials,
4398 products, and processes across an entire value chain (UNECE, 2022). Further detailed information on related
4399 blockchain based pilot projects, on the developed business requirements specification for traceability and
4400 transparency and on the associated data model can be download from the dedicated website ⁽²⁰⁰⁾. A number
4401 of privately developed traceability information platforms ⁽²⁰¹⁾ focusing on the textile sector are already on the
4402 market. The EU funded Interreg project ECHT ⁽²⁰²⁾, which has the purpose to Enable Digital Product Passports
4403 with Chemicals Traceability for a Circular Economy is also relevant.

4404 The **second approach** is **analytical** and relies on direct testing of textile apparel articles to verify the presence
4405 and concentration of specific substances of concern. This approach requires the existence of suitable analytical
4406 methods for the target chemicals, involves representative sampling of products and is costly. Consequently
4407 only targeted analyses have to date been applied to check compliance as regards regulated substances, usually
4408 following a risk based approach. AFIRM, ZDHC or the Oeko-tex standard provide information on available

¹⁹⁷ https://afirm-group.com/wp-content/uploads/2024/04/2024_AFIRM_RSL_2024_0404_EN.pdf

¹⁹⁸ <https://mrsl.roadmaptozero.com/>

¹⁹⁹ https://www.oeko-tex.com/importedmedia/downloadfiles/OEKO-TEX_STANDARD_100_Standard_EN_DE.pdf

²⁰⁰ <https://unece.org/trade/traceability-sustainable-textile-apparel-and-footwear>

²⁰¹ Such as: TextileGenesis <https://textilegenesis.com/> ; Global Textile Scheme <https://www.globaltextilescheme.org/>

²⁰² <https://echt.nweurope.eu/>

analytical methods for specific substances. Other sources of test method information, for instance as regards substances restricted under REACH, can be found in the Compendium of Analytical Methods recommended by the ECHA Forum to check compliance of REACH Annex XVII restrictions⁽²⁰³⁾. Such targeted assessments, checking for compliance of textile products have been carried out under the REACH4Textiles⁽²⁰⁴⁾ project and, more recently, by IKEA and H&M in a project⁽²⁰⁵⁾ focusing on collected textile apparel targeted for fibre-to-fibre recycling.

In general it can be stated that a combination of the two approaches is necessary. Both still have limitations in terms in accessibility of data, costs and capacity, especially for small and medium sized enterprises.

9.2.7.3 Description of product technologies

The use of chemicals in textile apparel manufacture is often related to the specific fibre type as well as to specific finishings (e.g. water-proofing) which are generally not product category specific. Consequently, at this point in the development of the project it is difficult to envisage the possibility of potential information or performance requirements that would be category specific.

Building a base case (BC) and the definition of products representing the best available technology (BAT) is particularly challenging for chemicals, given the large number of substances used by the sector and the lack of quantitative information on the distribution of substances. For certain specific substances and substance families, BAT and BNAT products could potentially be defined in terms of products having switched to non-toxic or less toxic (or in general more sustainable) alternatives – e.g. alternatives to the use of PFAS. Given the current paucity of detailed information on SoCs in textile apparel, the setting of information requirements, as prescribed under ESPR, with justified exemptions and thresholds for declaration of substances, seems clearly warranted.

9.3 Mutual influence of product aspects and product categorization

The analysis reported in section 9.2 described the product technologies as result of the complex interaction among process techniques, business models, user behaviour and legislative frameworks in the perspective of each relevant product aspects. It was crucial to analyse product technologies in silos to better understand the complexity of the ecosystem belonging to each relevant ecodesign aspect. However, these product aspects are strictly connected to each other and influence each other. The analysis of product technologies already showed that:

- Waste generation is strictly connected to recyclability and recycled content especially due to the same process techniques and legislation, as reported in sections 9.2.4 and 9.2.5;
- The recycled content depends on the recyclability and directly affects the physical durability whenever the recycled fibres have lower performances than virgin fibres and are fed into the recycling process;
- Physical durability is affected by maintenance and affects the repairability, due to its definition given by the ESPR, where these product aspects are related to each other (**Table 2**);
- The use of specific chemicals and substances affects the physical durability, recycling and recycled content;
- The environmental impacts are influenced by all the other relevant product aspects.

The analysis of technologies must now find a synthesis in Task 5 (section 10) and Task 6 (section 11), where the environmental and economic model describes the BC and the Design Options for each product category. In this way, the interactions among the relevant product aspects will be quantified to best find potential trade-offs to be expressed in the Design Options.

As for product categories, previous sections of the preparatory study used the categorization suggested by the PEFCR A&F to describe the scope of the study (section 3) and characteristics of the market (section 5). The

²⁰³ ECHA 2021. https://www.echa.europa.eu/documents/10162/13577/compendium_of_analytical_methods_en.pdf/4c730fb9-1b48-2e14-6ee3-7a36391b7322

²⁰⁴ <https://www.centexbel.be/en/toxic-substances-textiles>

²⁰⁵ <https://hmgroupp.com/wp-content/uploads/2021/10/Press-release-HM-Group-and-IKEA-study.pdf>

4453 product categories reported in this section serve the aims of the preparatory study based on the analysis of
4454 product technologies, which was described in section 9.2.

4455 The first step to obtain a synthesis of the analysis of technologies is the adoption of product categories valid
4456 for all relevant product aspects. In the case of textile apparel, this exercise is very simple because the only
4457 product aspects described via categories were the physical durability, considering the framework of robustness,
4458 and the recycled content. This allows the adoption of three categories as the product categorization to be used
4459 in the following steps of the development of the PS.

4460 Therefore, Task 5 will analyse three representative products, one for each product category: (1) knitted products,
4461 (2) products made of denim fabric, and (3) woven products made without denim fabric.

10 Environmental and economic analysis

This section reports the analysis of the base cases of the product categories defined in section 9.3. Each product category is described with a representative product. The analysis uses the Life Cycle Assessment method based on international standard as in ISO 14040⁽²⁰⁶⁾ and ISO 14044⁽²⁰⁷⁾, and using to the extent possible the Product Environmental Footprint (PEF) framework as defined in the Commission Recommendation (EU) 2021/2279⁽²⁰⁸⁾ (Section 10.1). This analysis was complemented by an assessment of fragmented fibre release in the life cycle of the textile apparel (Section 10.2).

The main aim of this section is to describe the consumption of the products in the scope from the environmental and economic perspectives.

10.1 Impacts via Life Cycle Thinking approach

The environmental and economic analysis reported in this section is detailed in Section 13.11. The current section reports the main model characteristics and results via Life Cycle Assessment (LCA), Life Cycle Cost (LCC) and societal Life Cycle Cost (sLCC). Focussing on the same functional unit, the LCA expresses its environmental impacts, the LCC expresses its internalised costs, and the sLCC expresses the total costs borne by the society. The sLCC accounts for the internal costs of the LCC and the external costs, which are the monetarization of environmental impacts evaluated via the LCA.

The reported analysis includes the following subsections:

- A brief description of the model used (section 10.1.1).
- A contribution analysis that allows the identification of the most important elements defining the environmental and economic impacts (section 10.1.2).
- A perturbation analysis that investigated the most sensitive parameters in the model, i.e. those for which a small change would result in a large variation of the results (section 10.1.3).
- A scenario analysis investigating:
 - The influence of user behaviour when choosing the time of disposal of the product (section 10.1.4.1);
 - The effect of manufacturing the product entirely in EU or entirely in third countries (non-EU) (section 10.1.4.2).
- A propagation of input uncertainties that will support the interpretation of the results, especially when comparing the base cases with the design options presented in section 11 (section 10.1.5).

10.1.1 Description of the model

The environmental and economic model was built to achieve two goals:

1. Investigating the environmental and economic performance of textile apparel included in the scope of this preparatory study.
2. Assessing the environmental and economic effects of design options that could be considered to develop future potential ecodesign requirements on the same products.

The goal 1 supports the analysis performed in Task 5 of the MEErP, and reported in the current section, whereas the goal 2 supports the analysis performed in Task 6 of the MEErP and described in section 11.

²⁰⁶ ISO 14040:2006 Environmental management — Life cycle assessment — Principles and framework. Available at [this link](#). Last accessed on 4 September 2025.

²⁰⁷ ISO 14044:2006 Environmental management — Life cycle assessment — Requirements and guidelines. Available at [this link](#). Last accessed on 4 September 2025.

²⁰⁸ Commission Recommendation (EU) 2021/2279 of 15 December 2021 on the use of the Environmental Footprint methods to measure and communicate the life cycle environmental performance of products and organisations. Available at [this link](#).

4499 The analysis included all life cycle stages of the products consumed in EU and therefore included in the scope
4500 of the preparatory study (section 3.2). A representative product was modelled for each product category defined
4501 in section 9.3:

- 4502 — knitted products (Knitted),
- 4503 — products made of denim fabric (Denim), and
- 4504 — woven products made without denim fabric (Other woven).

4505 The functional unit was defined as the coverage of the body using a specified item that fulfils the consumer's
4506 needs over 6 years in EU. This timeframe was chosen because it represents the longest service lifespan of the
4507 products included in the scope (²⁰⁹). More information on the functional unit are available in section 13.11.8.

4508 Based on the goal and the functional unit, the inventory analysis collected information on the physical flows
4509 as input of materials, energy and resources in general, and the output of emissions, waste and processed
4510 products. The outcome of the inventory analysis was the life cycle inventory, which is a list of quantified
4511 physical elementary flows for the product system.

4512 The model described the following Life Cycle Stages (LCS):

- 4513 — LCS 1: raw material and fibre manufacturing
- 4514 — LCS 2: production process:
 - 4515 • LCS 2a: fibre pre-treatment
 - 4516 • LCS 2b: yarn production
 - 4517 • LCS 2c: fabric production
 - 4518 • LCS 2d: fabric finishing processes
 - 4519 • LCS 2e: confectioning
 - 4520 • LCS 2f: waste management (post-industrial)
 - 4521 • Transport LCS 1 & LCS 2
- 4522 — LCS 3: retailing and transport
- 4523 — LCS 4: use
- 4524 — LCS 5: end of life (waste management of post-consumer waste).

4525 The assessment carried out has some limitations, which are due to the methodological choices and the
4526 availability of data, whose details are reported in section 13.11.2 and section 13.11.3, respectively.

4527 The assessment of the environmental impacts consisted of five steps based on Sala et al. (2017):

- 4528 1. **Selection of impact categories.** A representative indicator was chosen for each impact category
4529 together with an environmental model that was used to quantify the impacts on the indicator.
- 4530 2. **Classification of elementary flows** from the inventory by assigning them to impact categories
4531 according to their ability to contribute by impacting the chosen indicator.
- 4532 3. **Characterisation** using environmental models. The ability of each elementary flow to impact the
4533 indicator of the category was calculated. The collection of aggregated indicator scores for the different
4534 impact categories (each expressed in its own metric) made the characterised impact profile of the
4535 product system.

²⁰⁹ For example, the main function of the representative product (RP) knitted products is designed to provide adequate body coverage for a consumer in the European Union over a 6-year period. The study assumes that a single RP knitted product has a service lifespan of approximately 3.83 years, accommodating around 72 uses and 13.66 care cycles. A consumer would require a total of 1.57 RP knitted product over the 6-year timeframe. By considering the entire service lifespan of these 1.57 RP T-shirts, the analysis can effectively assess the environmental impacts associated with exploitation of the consumer needs in 6 years, considering their production, distribution, usage, and end-of-life phases, offering valuable insights into resource use, waste generation, cost and potential sustainability enhancements within the specified geographic and temporal context.

4. **Normalisation** was used to express a relative magnitude of the characterised scores by expressing them relative to a common set of reference impacts. The result of the normalisation was the normalised impact profile of the product system in which all category indicator scores were expressed in the same metric.
5. **Weighting** supported comparison across the impact categories by weighting them using weighting factors that for each impact category gave a quantitative expression of how severe it is relative to the other impact categories. Quantitative weighting allowed aggregation of all the weighted impact scores into one holistic environmental impact score, which is named single score and it is expressed as environmental points.

The weighting step is particularly useful in the context of the PS because it gives a holistic understanding of the environmental impacts and it solves the multicriteria analysis. Additionally, when assessing alternative options, like the analysis of the design options (section 11), it avoids the potential overlook of burden shifting because all possible impact categories are accounted for and included in the single score. As described in (Sala et al., 2017), the adopted weighting, as currently in the Environmental Footprint (EF) method, took into account EU political priorities.

The environmental assessment investigated the 16 impact categories described in **Table 53**, and it used normalisation and weighting factors reported in **Table 207**.

Table 53. Description of investigated impact categories

Impact category	Description
Acidification	Acidification can be caused by emissions to the air and deposition of emissions in water and soil. The most significant sources are combustion processes in electricity, heat production, and transport. The more sulphur the fuels contain the greater their contribution to acidification. The potential impact of substances contributing to acidification is converted to the equivalent of moles of hydron (general name for a cationic form of atomic hydrogen, mol H ⁺ eq).
Climate change	This indicator refers to the increase in the average global temperatures as result of greenhouse gas (GHG) emissions. The greatest contributor is generally the combustion of fossil fuels such as coal, oil, and natural gas. The global warming potential of all GHG emissions is measured in kilogram of carbon dioxide equivalent (kg CO ₂ eq), namely all GHG are compared to the amount of the global warming potential of 1 kg of CO ₂ .
Ecotoxicity, freshwater	This indicator refers to potential toxic impacts on an ecosystem, which may damage individual species as well as the functioning of the ecosystem. Some substances tend to accumulate in living organisms. The unit of measurement is Comparative Toxic Unit for ecosystems (CTUe). This is based on a model called USEtox.
Eutrophication, freshwater	Eutrophication impacts ecosystems due to substances containing nitrogen (N) or phosphorus (P), which promotes growth of algae or specific plants. If algae grow too rapidly, it can leave water without enough oxygen for fish to survive. Nitrogen emissions into the aquatic environment are caused by fertilisers used in agriculture, but also by combustion processes. The most significant sources of phosphorus emissions are sewage treatment plants for urban and industrial effluents and leaching from agricultural land. The potential impact of substances contributing to freshwater eutrophication is converted to the equivalent of kilograms of phosphorus (kg P eq).
Eutrophication, marine	Eutrophication in ecosystems happens when substances containing nitrogen (N) or phosphorus (P) are released to the ecosystem. As a rule, the availability of one of these nutrients will be a limiting factor for growth in the ecosystem, and if this nutrient is added, the growth of algae or specific plants will increase. For the marine environment this will be mainly due to an increase of nitrogen (N). Nitrogen emissions are caused largely by the agricultural use of fertilisers, but also by combustion processes. The potential impact of substances contributing to marine eutrophication is converted to the equivalent of kilograms of nitrogen (kg N eq).
Eutrophication, terrestrial	Eutrophication arises when substances containing nitrogen (N) or phosphorus (P) are released to ecosystems. These nutrients cause a growth of algae or specific plants and thus limit growth in the original ecosystem. The potential impact of substances contributing to terrestrial eutrophication is converted to the equivalent of moles of nitrogen (mol N eq).
Human toxicity, cancer	This indicator refers to potential impacts, via the environment, on human health caused by absorbing substances from the air, water and soil. Direct effects of products on human health are currently not measured. The unit of measurement is Comparative Toxic Unit for humans (CTUh). This is based on a model called USEtox.

Impact category	Description
Human toxicity, non-cancer	This indicator refers to potential impacts, via the environment, on human health caused by absorbing substances from the air, water, and soil. Direct effects of products on human health are currently not measured. The unit of measurement is Comparative Toxic Unit for humans (CTUh). This is based on a model called USEtox.
Ionising radiation, human health	The exposure to ionising radiation (radioactivity) can have impacts on human health. The Environmental Footprint only considers emissions under normal operating conditions (no accidents in nuclear plants are considered). The potential impact on human health of different ionising radiations is converted to the equivalent of kilobecquerels of Uranium 235 (kg U235 eq).
Land use	Use and transformation of land for agriculture, roads, housing, mining or other purposes. The impacts can vary and include loss of species, of the organic matter content of soil, or loss of the soil itself (erosion). This is a composite indicator measuring impacts on four soil properties (biotic production, erosion resistance, groundwater regeneration and mechanical filtration), expressed in points (Pts).
Ozone depletion	The stratospheric ozone (O3) layer protects us from hazardous ultraviolet radiation (UV-B). Its depletion increases skin cancer cases in humans and damage to plants. The potential impacts of all relevant substances for ozone depletion are converted to their equivalent of kilograms of trichlorofluoromethane (also called Freon-11 and R-11), hence the unit of measurement is in kilogram of CFC-11 equivalent (kg CFC-11 eq).
Particulate matter	This indicator measures the adverse impacts on human health caused by emissions of Particulate Matter (PM) and its precursors (e.g. NOx, SO2). Usually, the smaller the particles, the more dangerous they are, as they can go deeper into the lungs. The potential impact of is measured as the change in mortality due to PM emissions, expressed as disease incidence per kg of PM2.5 emitted.
Photochemical ozone formation, human health	Ozone (O3) on the ground (in the troposphere) is harmful: it attacks organic compounds in animals and plants, it increases the frequency of respiratory problems when photochemical smog ("summer smog") is present in cities. The potential impact of substances contributing to photochemical ozone formation is converted into the equivalent of kilograms of Non-Methane Volatile Organic Compounds (e.g. alcohols, aromatics, etc.; kg NMVOC eq).
Resource use, fossils	The earth contains a finite amount of non-renewable resources, such as fossil fuels like coal, oil and gas. The basic idea behind this impact category is that extracting resources today will force future generations to extract less or different resources. For example, the depletion of fossil fuels may lead to the non-availability of fossil fuels for future generations. The amount of materials contributing to resource use, fossils, are converted into MJ.
Resource use, minerals and metals	This impact category has the same underlying basic idea as the impact category resource use, fossils (namely, extracting a high concentration of resources today will force future generations to extract lower concentration or lower value resources). The amount of materials contributing to resource depletion are converted into equivalents of kilograms of antimony (kg Sb eq).
Water use	The abstraction of water from lakes, rivers or groundwater can contribute to the 'depletion' of available water. The impact category considers the availability or scarcity of water in the regions where the activity takes place, if this information is known. The potential impact is expressed in cubic metres (m ³) of water use related to the local scarcity of water.

4554 Description based on the update version for the Environmental Footprint method - version 3.1.

4555 Source: (Sala et al., 2017)

4556 The economic performance of the representative products throughout their lifecycle were assessed via Life
4557 Cycle Costing (LCC). This methodology allows the estimation of the total cost of a product by evaluating all
4558 direct and indirect cost sources. LCC was conducted with reference to the LCA standards, to be able to combine
4559 the economic and environmental analysis. The performed LCC shares the same goal and scope, functional unit
4560 and system boundaries of the LCA, with small exceptions reported in the sections 13.11.8 and 13.11.9.

4561 This report adopts three types of economic indicators: the costs borne by the producer, the costs borne by the
4562 consumer and the costs borne by the society.

4563 **The costs borne by the producer** were based on conventional LCC, including the costs sustained by the
4564 producer until the moment in which the product is possessed by the consumer. This is named a cradle to gate
4565 approach, where VAT and tariffs (for the import of semi-finished and finished products into the EU) are included.

4566 **The costs borne by the consumer** were based on conventional LCC. These costs are related to what the
4567 consumer must bear to own, use and dispose of the product. The costs borne by the consumer included the

costs borne by the producer, with the addition of the margin obtained from the sale, plus the costs of the use phase of the product and the cost of product disposal (quantified as direct process costs and taxes).

The costs borne by the society were based on societal LCC. It quantifies the costs that originate from each unit process and for each stage of the product's lifecycle (from cradle to grave). Additionally, the societal LCC includes the costs of externalities deriving from environmental impacts, using the monetization factors reported in **Table 208**. Neither tariffs nor taxes were included in this calculation.

The holistic approach provided by the societal LCC allows to take into account of those costs that usually are not internalised into the price that the user pays at the moment of the purchase, but that the entire society must face directly or indirectly at some point in time. The use of this economic indicator is particularly crucial in the context of the PS, which aims to support policy decisions.

10.1.2 Contribution analysis

10.1.2.1 Composition of the total environmental impacts – contribution of environmental impact categories

Table 54 reports the lifecycle environmental impacts for all investigated impact categories. Whereas **Table 55** and **Figure 32** report the contribution of all investigated impact categories to the total environmental impacts expressed as environmental points. The most contributing impact categories were:

- Water use, contributing 27-42%
- Climate change, contributing 15-18%
- Resource use, fossils, contributing about 9%
- Particulate matter, contributing 4-7%
- Resource use, minerals and metals, contributing 4-7%
- Acidification, contributing 4-6%
- Eutrophication freshwater, contributing about 5%

Table 54. Environmental impacts reported for all investigated impact categories

Impact category	Unit	Knitted	Denim	Other woven
Acidification	mol H ⁺ eq.	1.11E-01	1.82E-01	1.41E-01
Climate change	kg CO ₂ eq.	1.41E+01	2.67E+01	1.75E+01
Ecotoxicity, freshwater	CTUe	2.03E+02	4.05E+02	2.24E+02
Eutrophication, freshwater	kg P eq.	6.95E-03	1.54E-02	7.80E-03
Eutrophication, marine	kg N eq.	5.95E-02	1.33E-01	5.79E-02
Eutrophication, terrestrial	mol N eq.	3.75E-01	5.47E-01	4.78E-01
Human toxicity, cancer	CTUh	4.51E-08	7.17E-08	6.01E-08
Human toxicity, non-cancer	CTUh	4.30E-07	5.37E-07	5.37E-07
Ionising radiation, human health	kBq U-235 eq.	1.34E+00	4.35E+00	1.24E+00
Land use	Pt	3.25E+02	4.48E+02	3.96E+02
Ozone depletion	kg CFC-11 eq.	3.32E-05	4.07E-05	3.88E-05
Particulate matter	disease incidences	9.56E-07	1.52E-06	1.37E-06
Photochemical ozone formation, human health	kg NMVOC eq.	5.18E-02	9.41E-02	6.65E-02
Resource use, fossils	MJ	1.67E+02	3.49E+02	1.99E+02
Resource use, minerals and metals	kg Sb eq.	1.17E-04	1.94E-04	1.58E-04
Water use	m ³ water eq. of deprived water	1.18E+02	2.89E+02	1.03E+02

CTUe: Comparative Toxic Unit for ecosystems; pt: points; NMVOC: Non-Methane Volatile Organic Carbon.

Source: own elaboration

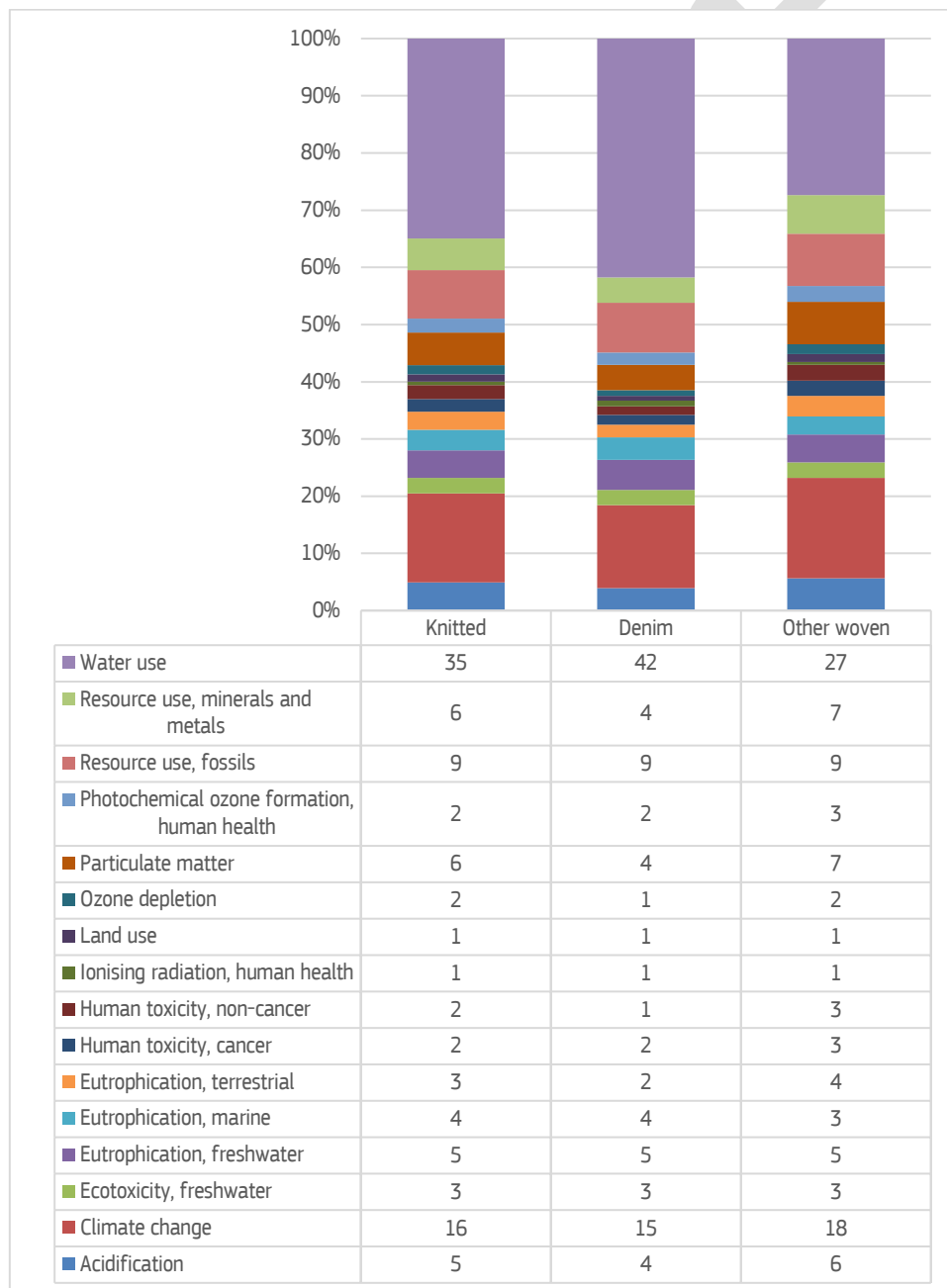
Table 55. Environmental impacts expressed as environmental points – Values of all investigated impact categories

Impact category	Knitted	Denim	Other woven
Acidification	1.23E-04	2.03E-04	1.57E-04
Climate change	3.92E-04	7.46E-04	4.89E-04
Ecotoxicity, freshwater	6.88E-05	1.37E-04	7.59E-05
Eutrophication, freshwater	1.21E-04	2.69E-04	1.36E-04
Eutrophication, marine	9.01E-05	2.01E-04	8.77E-05

Impact category	Knitted	Denim	Other woven
Eutrophication, terrestrial	7.87E-05	1.15E-04	1.00E-04
Human toxicity, cancer	5.56E-05	8.86E-05	7.42E-05
Human toxicity, non-cancer	6.15E-05	7.67E-05	7.68E-05
Ionising radiation, human health	1.59E-05	5.16E-05	1.47E-05
Land use	3.15E-05	4.34E-05	3.84E-05
Ozone depletion	4.00E-05	4.91E-05	4.68E-05
Particulate matter	1.44E-04	2.29E-04	2.07E-04
Photochemical ozone formation, human health	6.06E-05	1.10E-04	7.78E-05
Resource use, fossils	2.14E-04	4.47E-04	2.55E-04
Resource use, minerals and metals	1.39E-04	2.30E-04	1.87E-04
Water use	8.78E-04	2.15E-03	7.63E-04
Total impacts	2.51E-03	5.14E-03	2.79E-03

Source: own elaboration

Figure 32. Contribution to the single score of the impact categories to the lifecycle environmental impacts



Source: own elaboration

10.1.2.2 Most contributing life cycle stages

Table 56 and **Figure 33** report that the largest environmental impacts are generated by the production of raw materials (LCS1) (60–63%) and the manufacturing processes (LCS2) (21–29%). Environmental impacts generated by the production of raw materials largely depend on the specific fibres used. Each fibre has specific characteristics (**Table 177**), which are deployed for specific applications to meet users demand in terms of performances and price.

From the economic perspective, **Table 57** and **Figure 34** show that the largest societal costs are generated by externalities (external costs) (34–36%) and internal costs due to manufacturing processes (29–37%). The internal costs due to production of raw materials and the use stage give a smaller contribution (14–18% and 6–21%, respectively).

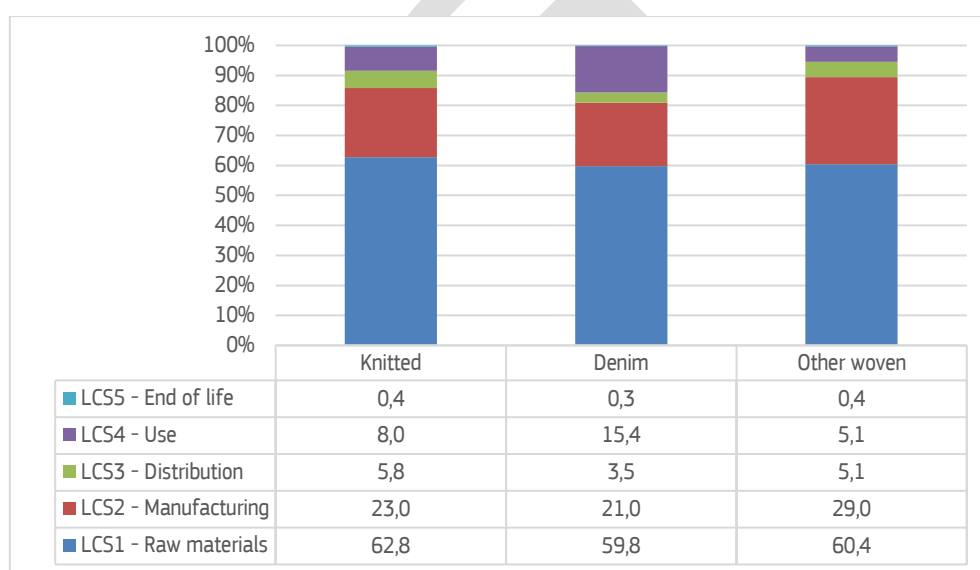
Table 56. Environmental impacts expressed as environmental points for the three product categories – contribution of lifecycle stages

Life Cycle stage	Knitted	Denim	Other woven
LCS1 - Raw materials	1.58E-03	3.07E-03	1.68E-03
LCS2 - Manufacturing	5.78E-04	1.08E-03	8.08E-04
LCS3 - Distribution	1.45E-04	1.78E-04	1.43E-04
LCS4 - Use	2.02E-04	7.92E-04	1.43E-04
LCS5 - End of life	1.00E-05	1.41E-05	1.01E-05
All LCSs (Total)	2.51E-03	5.14E-03	2.79E-03

LCS: Life Cycle Stage; Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.

Source: own elaboration

Figure 33. Share of environmental impacts generated in the lifecycle of the three product categories – contribution of lifecycle stages



LCS: Life Cycle Stage; Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric..

Source: Own production

Table 57. Societal costs expressed as EUR for the three product categories – contribution of internal costs in all lifecycle stages and external costs

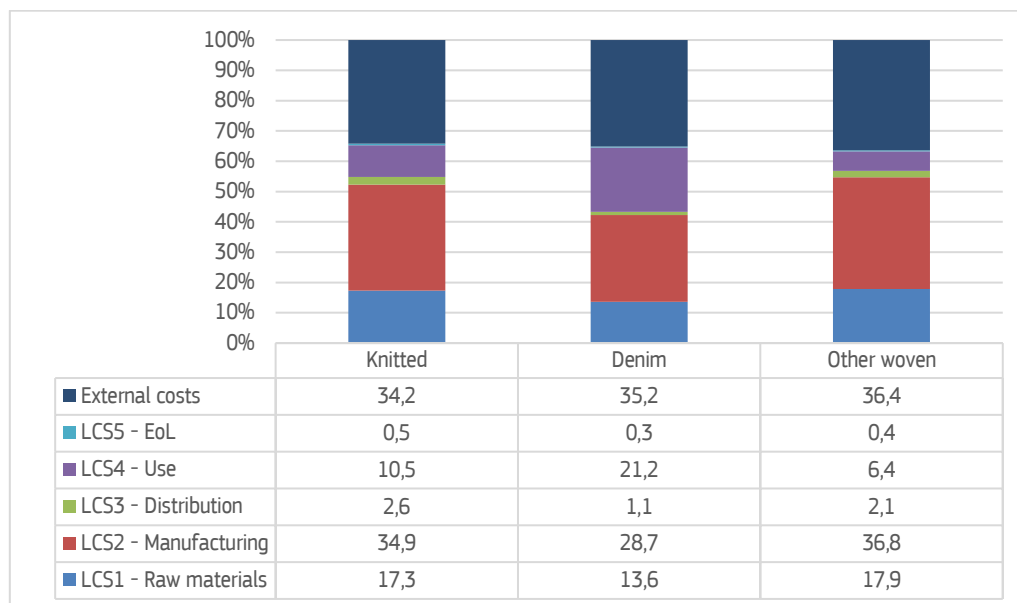
Life Cycle stage	Knitted	Denim	Other woven
LCS1 - Raw materials	2.02	2.97	2.36
LCS2 - Manufacturing	4.07	6.29	4.86
LCS3 - Distribution	0.30	0.23	0.27
LCS4 - Use	1.22	4.65	0.85
LCS5 - EoL	0.05	0.06	0.05

Life Cycle stage	Knitted	Denim	Other woven
Total internal costs	7.67	14.20	8.39
External costs	3.99	7.70	4.80
Societal costs	11.66	21.90	13.20

LCS: Life Cycle Stage; Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.

Source: own elaboration

Figure 34. Share of societal costs for the three product categories – contribution of internal costs in all life cycle stages and external costs



LCS: Life Cycle Stage; Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.

Source: Own production

10.1.2.3 Analysis of life cycle stage 1 – Raw materials

When investigating the raw material production stage, **Table 58** and **Figure 35** show that 35-43% of the environmental impacts are generated by the use of water, 18-31% are generated by the use of energy, 12-29% are generated by the use of chemicals, and 11-16% are generated by direct emissions into water, soil and atmosphere. Due to lack of alternative source of information, the life cycle inventory used some aggregated datasets accounting for 3-26% of the environmental impacts in this lifecycle stage.

The analysis of the societal LCC showed that internal costs are 46-49% of the total societal costs generated in the LCS1. This analysis could include only the final costs of the fibres used and the external costs due to environmental impacts. The contribution of specific components, such as energy, water and chemicals used, were not included due to lack of available data.

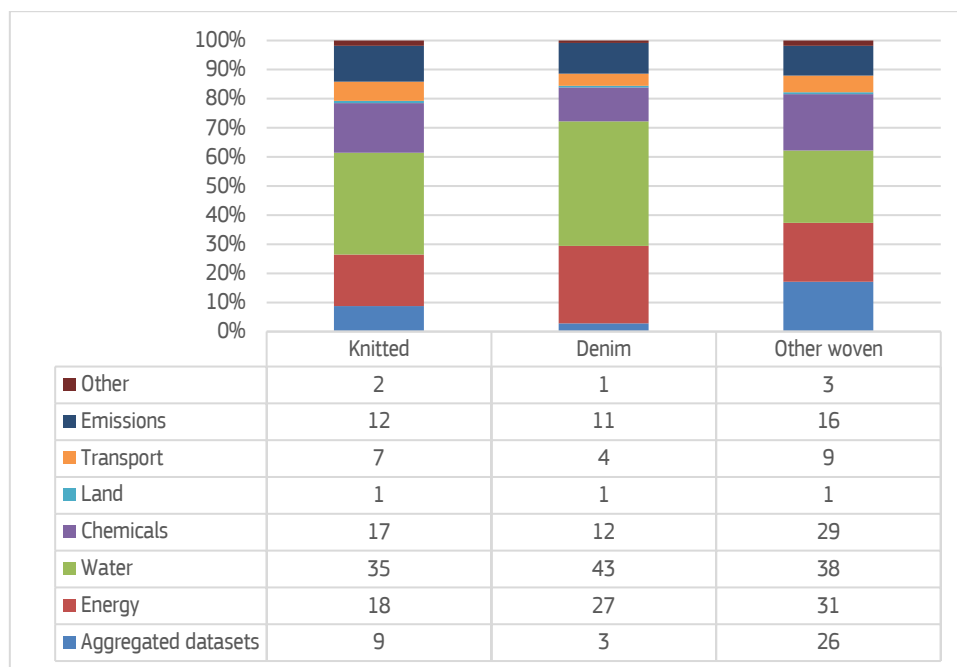
Table 58. Environmental impacts expressed as environmental points for the three product categories – contributors in LCS1 – Raw materials

Contributor	Knitted	Denim	Other woven
Aggregated datasets	2.19E-04	1.45E-04	4.78E-04
Energy	4.47E-04	1.37E-03	5.61E-04
Water	8.79E-04	2.20E-03	6.94E-04
Chemicals	4.30E-04	5.99E-04	5.39E-04
Land	1.88E-05	2.58E-05	1.80E-05
Transport	1.64E-04	2.15E-04	1.60E-04
Emissions	3.12E-04	5.48E-04	2.85E-04
Other	4.42E-05	3.95E-05	4.97E-05
Total - Raw Materials	2.51E-03	5.14E-03	1.83E-03

LCS: Life Cycle Stage; Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.

Source: own elaboration

Figure 35. Share of environmental impacts generated by specific contributors in raw material production (LCS1) of the three product categories



LCS: Life Cycle Stage; Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.

Source: Own production

10.1.2.4 Analysis of life cycle stage 2 - Manufacturing

When investigating the manufacturing stage, **Table 59** and **Figure 36** show that the fabric finishing processes are the largest contributors of environmental impacts (41-63%) followed by yarn production (15-23%) and fabric production (5-27%). From the economic perspective, **Table 60** and **Figure 37** report that the largest societal costs are generated by internal costs due to fabric finishing processes (23-30%), externalities (23-29%), and internal costs due to confectioning (17-20%). The internal costs due to yarn production and transport give a smaller contribution (12-14% and 9-11%, respectively).

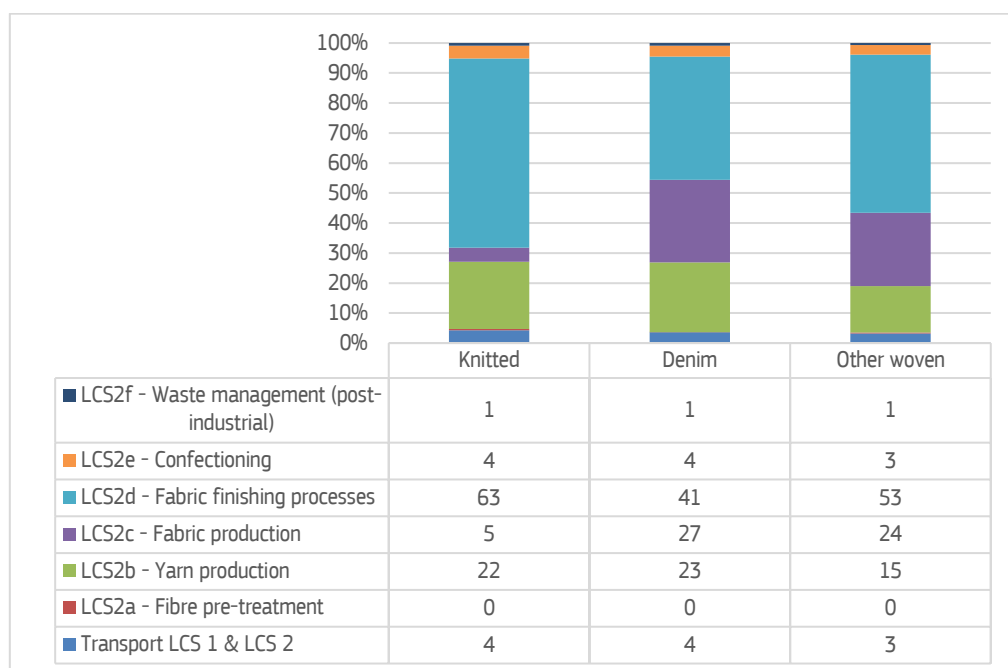
Table 59. Environmental impacts expressed as environmental points for the three product categories – contribution of manufacturing processes in LCS2

Life Cycle process	Knitted	Denim	Other woven
Transport LCS 1 & LCS 2	2.43E-05	3.91E-05	2.55E-05
LCS2a - Fibre pre-treatment	2.35E-06	3.53E-07	2.72E-06
LCS2b - Yarn production	1.30E-04	2.52E-04	1.25E-04
LCS2c - Fabric production	2.72E-05	2.97E-04	1.97E-04
LCS2d - Fabric finishing processes	3.64E-04	4.45E-04	4.26E-04
LCS2e - Confectioning	2.45E-05	3.83E-05	2.55E-05
LCS2f - Waste management (post-industrial)	5.24E-06	1.01E-05	5.60E-06
Total - Manufacturing	5.78E-04	1.08E-03	8.08E-04

LCS: Life Cycle Stage; Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.

Source: own elaboration

Figure 36. Share of environmental impacts generated in the manufacturing processes (LCS2) of the three product categories



LCS: Life Cycle Stage; Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.

Since the contribution of LCS2a – Fibre pre-treatment was less than 0.5%, it was rounded to 0%.

Source: Own production

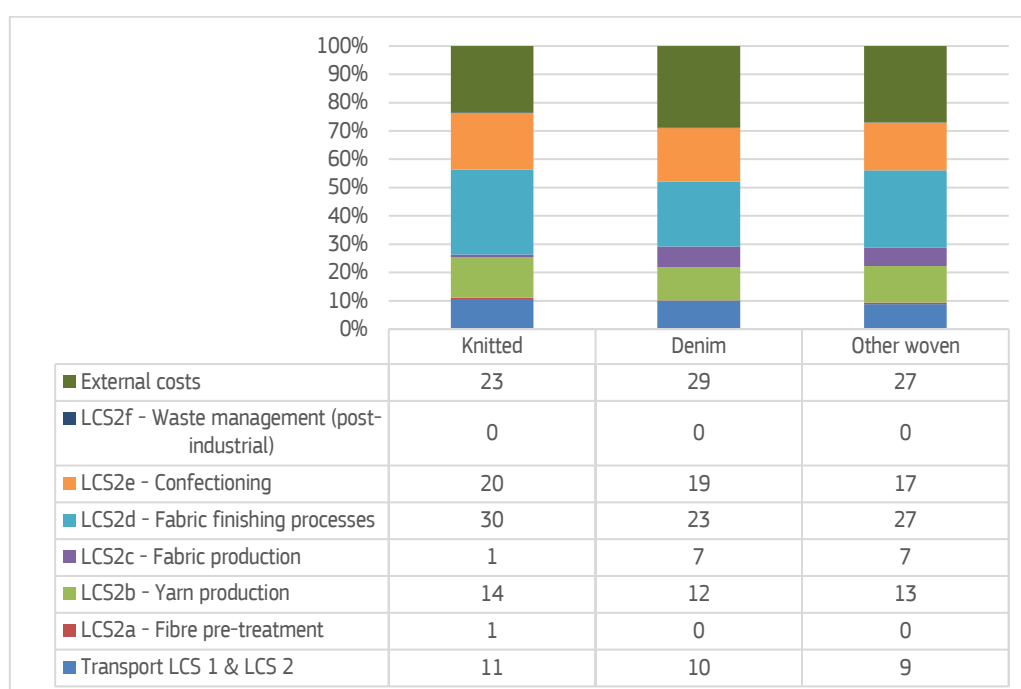
Table 60. Societal costs expressed as EUR for the three product categories – contribution of external costs due to the manufacturing stage (LCS2) and internal costs due to all manufacturing processes

Life Cycle stage	Knitted	Denim	Other woven
Transport LCS 1 & LCS 2	0.56	0.90	0.59
LCS2a - Fibre pre-treatment	0.03	0.00	0.03
LCS2b - Yarn production	0.75	1.03	0.87
LCS2c - Fabric production	0.05	0.66	0.43
LCS2d - Fabric finishing processes	1.59	2.03	1.83
LCS2e - Confectioning	1.07	1.68	1.12
LCS2f - Waste management (post-industrial)	0.01	0.02	0.01
Total internal costs	4.07	6.32	4.88
External costs	1.25	2.55	1.80
Total Societal LCC	5.32	8.87	6.68

LCS: Life Cycle Stage; Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.

Source: own elaboration

Figure 37. Share of societal costs for the three product categories – contribution of external costs due to the manufacturing stage (LCS2) and internal costs due to all manufacturing processes



LCS: Life Cycle Stage; Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.

Values lower than 0.5% were rounded to 0%.

Source: Own production

Table 61 and **Figure 38** show that the largest contributors in the manufacturing stage are the use of energy (44-63%) and the use of chemicals (25-41%), which correspond to 10-15% and 5-11% respectively of the environmental impacts of the entire life cycle.

The analysis of sub-stages showed that:

- In LCS2b - Yarn production, 92-97% of the impacts are generated by the use of energy,
- In LCS2c - Fabric production, 95-99% of the impacts are generated by the use of energy
- In LCS2d - Fabric finishing processes:
 - The use of chemicals contributed for 57-66%, and
 - The use of energy contributed for 20-25%.

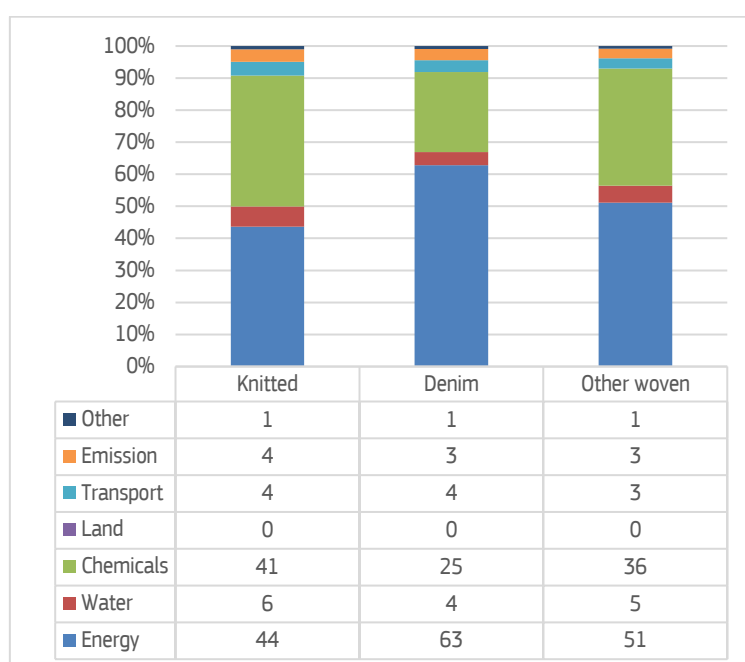
Table 61. Environmental impacts expressed as environmental points for the three product categories – contributors in LCS2 - Manufacturing

Contributor	Knitted	Denim	Other woven
Energy	2.52E-04	6.79E-04	4.13E-04
Water	3.63E-05	4.52E-05	4.30E-05
Chemicals	2.36E-04	2.70E-04	2.95E-04
Land	0.00E+00	0.00E+00	0.00E+00
Transport	2.50E-05	4.03E-05	2.62E-05
Emission	2.20E-05	3.74E-05	2.39E-05
Other	6.17E-06	1.01E-05	6.98E-06
Total - Single Score	5.78E-04	1.08E-03	8.08E-04

LCS: Life Cycle Stage; Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.

Source: own elaboration

Figure 38. Share of environmental impacts generated by specific contributors in the manufacturing life cycle stage (LCS2) of the three product categories



LCS: Life Cycle Stage; Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.

Contribution of land is equal to 0% because no land is used in this LCS.

Source: Own production

From the economic perspective, **Table 62** and **Figure 39** show that about half of the internal costs during manufacturing are due to human resources and capital expenditure (47-49%). The use of energy and chemicals gives a lower contribution: 15-21% and 14-20%, respectively.

The analysis of sub-stages showed that:

- In LCS2d – Fabric finishing processes:
 - the use of chemicals contributed for 40-52%,
 - human resources and capital expenditure contributed for 22-29%, and
 - the use of energy contributed for 20-23% of the LCS2d.
- In LCS2e – Confectioning:
 - human resources and capital expenditure contributed for 96%, and
 - the use of energy contributed for 4%.

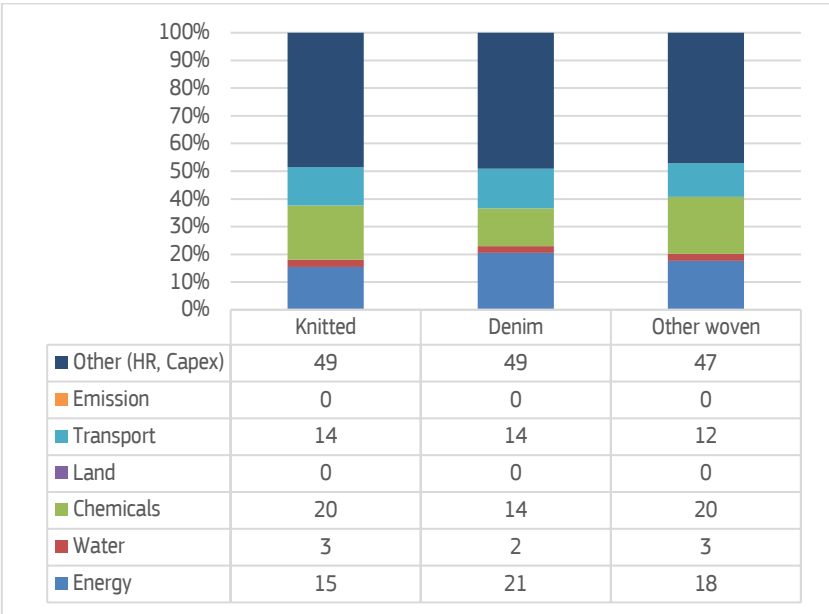
Table 62. Internal costs expressed as EUR for the three product categories – contributors in LCS2 - Manufacturing

Contributor	Knitted	Denim	Other woven
Energy	0.63	1.29	0.86
Water	0.11	0.15	0.12
Chemicals	0.79	0.86	1.00
Land	0.00	0.00	0.00
Transport	0.56	0.90	0.59
Emission	0.00	0.00	0.00
Other (HR, Capex)	1.98	3.09	2.29
Total internal costs	4.07	6.29	4.86

LCS: Life Cycle Stage; Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric; HR: Human Resources; CAPEX: CAPITAL EXpenditure.

Source: own elaboration

Figure 39. Share of internal costs for the three product categories – contributors in LCS2 - Manufacturing



LCS: Life Cycle Stage; Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric; HR: Human Resources; CAPEX: CAPital EXpenditure
Contribution of emissions and land is equal to 0% because no internal cost is paid due to emissions and land is not used in this LCS.

Source: Own production

10.1.2.5 Analysis of the waste management

Figures reported in sections 10.1.2.1 and 10.1.2.3 account for the impacts of the waste treatment and disposal in the stage where the waste is generated. This section focuses on the waste management of the three types of textile waste, which corresponds to 0.6-0.8% of the environmental impacts generated in the whole life cycle, so not only at the end of life.

Table 63 and **Figure 40** show that post-consumer waste generates half of the environmental impacts in the waste management, while post-industrial and pre-consumer waste generate 27-35% and 16-22%, respectively.

From an economic perspective, **Table 64** and **Figure 41** show that about half of the societal costs are due to externalities (45-53%), while the second largest contribution is given by internal costs of post-consumer waste.

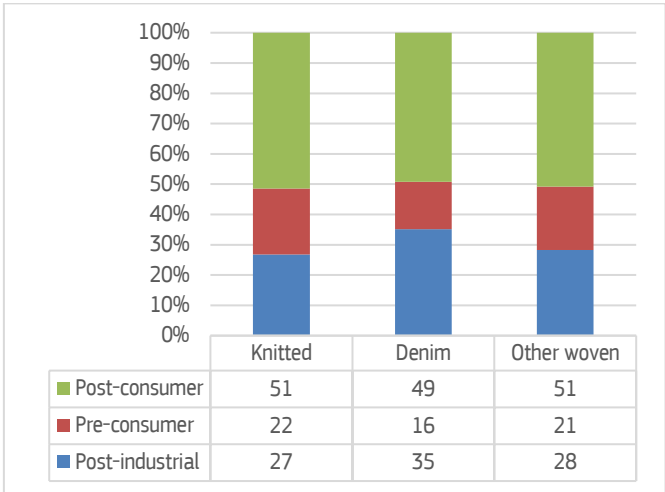
Table 63. Environmental impacts expressed as environmental points for the three product categories – contribution of the waste management

Type of textile waste	Knitted	Denim	Other woven
Post-industrial	5.24E-06	1.01E-05	5.60E-06
Pre-consumer	4.25E-06	4.47E-06	4.17E-06
Post-consumer	1.00E-05	1.41E-05	1.01E-05
All types of textile waste	1.95E-05	2.87E-05	1.98E-05

Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.
The environmental impacts generated by waste management for the three product categories are equal to 0.6-0.7% of the environmental impacts generated in the whole life cycle.

Source: own elaboration

Figure 40. Share of environmental impacts generated by the waste management



Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.

Source: Own production

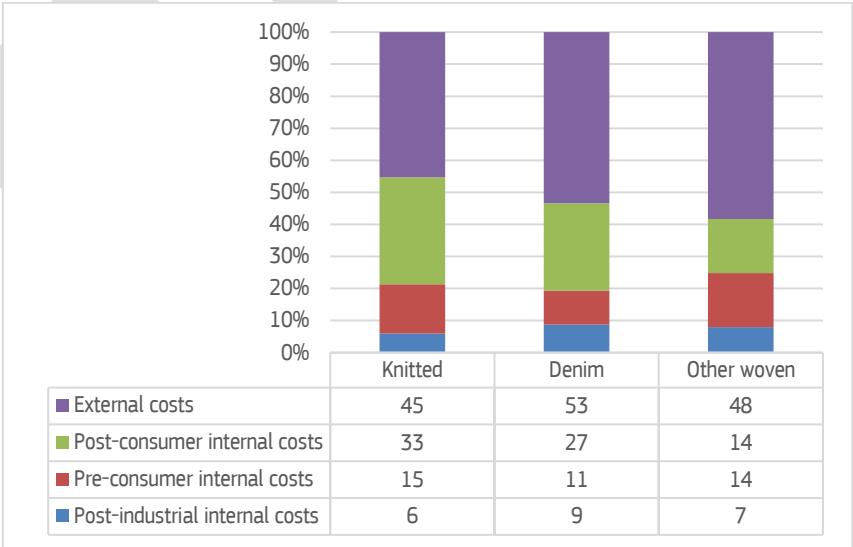
Table 64. Societal costs expressed as EUR for the three product categories – contribution of the waste management

Waste management	Knitted	Denim	Other woven
Post-industrial internal costs	0.01	0.02	0.01
Post-industrial external costs	0.02	0.04	0.02
Pre-consumer internal costs	0.02	0.02	0.02
Pre-consumer external costs	0.01	0.01	0.01
Post-consumer internal costs	0.05	0.06	0.05
Post-consumer external costs	0.04	0.06	0.04
Internal costs	0.09	0.10	0.08
External costs	0.07	0.11	0.07
Societal costs	0.16	0.21	0.16

Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.

Source: own elaboration

Figure 41. Share of societal costs for the three product categories – contribution of the waste management



Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.

Source: Own production

10.1.2.6 Costs borne by the producer and the consumer

Table 65 and **Figure 42** show that 71-72% of the costs addressed by the producer are due to the manufacturing process, while **Table 66** and **Figure 43** report that the largest part of the costs addressed by the consumer is due to the purchase including the first three lifecycles of product.

Section 5.6 reported that the supply chain of an economic operator placing a product on the EU market can be composed by many actors. Due to lack of information, the costs reported in this section include only margins of the economic operator placing the product on the EU market.

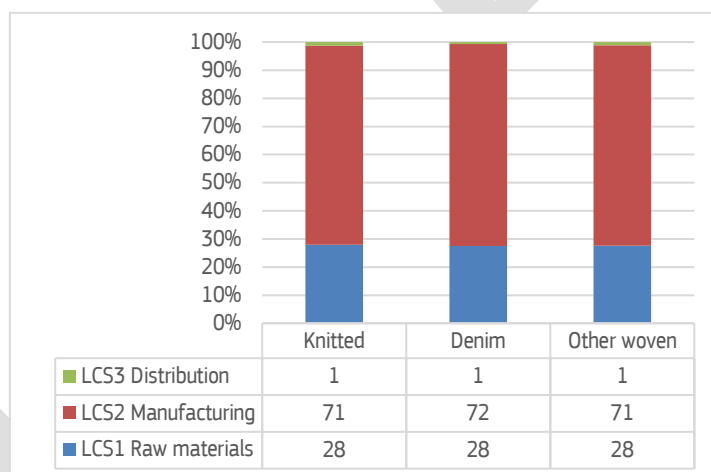
Table 65. Producer costs expressed as EUR for the three product categories

Life Cycle stage	Knitted	Denim	Other woven
LCS1 Raw materials	2.02	2.97	2.36
LCS2 Manufacturing	5.14	7.75	6.10
LCS3 Distribution	0.09	0.07	0.10
Producer costs	7.25	10.80	8.56

LCS: Life Cycle Stage; Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.

Source: own elaboration

Figure 42. Share of costs addressed by the producer



LCS: Life Cycle Stage; Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.

Source: Own production

Table 66. Consumer costs expressed as EUR for the three product categories

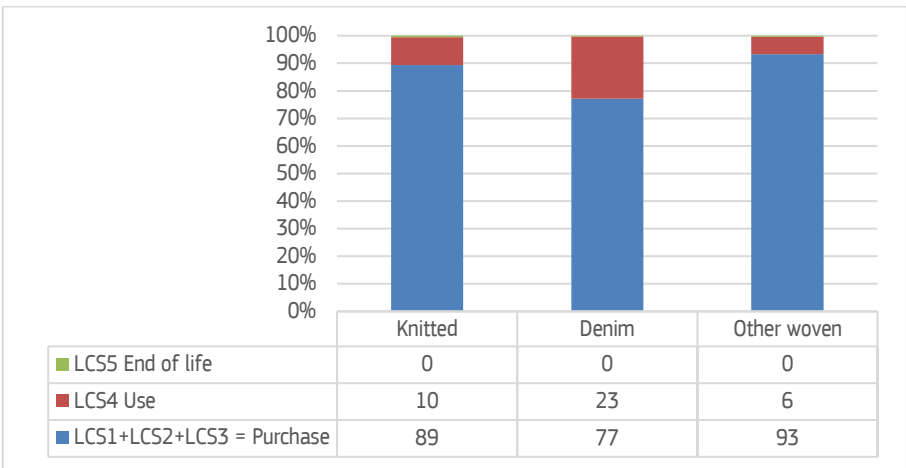
Life Cycle stage	Knitted	Denim	Other woven
LCS1+LCS2+LCS3 = Purchase	10.80	15.94	12.69
LCS4 Use	1.22	4.65	0.85
LCS5 EoL	0.06	0.06	0.05
Producer costs	12.09	20.65	13.59

LCS: Life Cycle Stage; Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.

Source: own elaboration

4773

Figure 43. Share of costs addressed by the consumer



4774

4775

4776

4777

4778

LCS: Life Cycle Stage; Knitted: knitted products; Denim: products made with denim fabric; Other woven: woven products made without denim fabric.

Since costs related to LCS5 End of life were lower than 0.5%, they were rounded to 0%.

Source: Own production

4779

10.1.3 Perturbation analysis

4780

4781

4782

For each product category, sensitivity ratios (SRs) were calculated for 98 parameters (see section 13.11.11.2) by considering a parameter increment of 10%. SR was defined as follows:

Equation 2. Sensitivity Ratio

4783

$$SR = (\text{variation of the results} / \text{parameter variation}) * (\text{initial parameter} / \text{initial result})$$

4784

4785

4786

The higher the SR is, the more sensitive is the model to a change of the specific parameter considered. This means that if SR is high, a 10% change in the parameter results in a large change in the carbon footprint. For example, if a parameter has an SR of 3, an increase of 10% of its value results in a result that is 30% higher.

4787

4788

The perturbation analysis was tested for all environmental and economic indicators, and it showed that the model includes parameters whose sensitivity does not largely affect the final results.

4789

Table 220 reports the detailed results.

4790

10.1.4 Scenario analysis

4791

10.1.4.1 Influence of the choice made by the consumer

4792

4793

4794

4795

4796

Worsening and improving scenarios were run to test the influence of the consumer choice when disposing a product regardless its physical performances. This investigation was driven by the results of the user behaviour (section 6.4) reporting that users dispose their textile apparel even if they are still fit for use. Additionally, the analysis of textile waste reported that about 62% of textile waste are still fit to be used and therefore enter the reuse market (Bakowska et al., 2025).

4797

The worsening scenarios included:

4798

4799

— “Worst (1 use)” simulates a disposable product, i.e. a product that is used only once and is subsequently disposed of.

4800

4801

— “BC weighted MIN uses” simulates a representative product influenced by the minimum values of “number of uses” reported in the literature.

4802

— “1 year” simulates that the consumer changes their textile apparel every year.

4803

The improving scenarios included:

4804

4805

— “BC plus 10% uses” simulates an increase of number of uses before disposal equal to 10%.

— “BC plus 30% uses” simulates an increase of number of uses before disposal equal to 30%.

- “BC plus 50% uses” simulates an increase of number of uses before disposal equal to 50%.
- “BC weighted MAX uses” simulates a representative product influenced by the maximum values of “number of uses” reported in the literature.

Table 67 and **Table 68** show the large effect that the choice of the users has on the environmental and economic impacts. For example, if the user decides to change their textile apparel every year, the environmental impacts and the societal costs increase about four times compared to the base case. In contrast, if the user decides to use the products more, the environmental impacts (and similarly the societal costs) decrease by 28% for knitted, 16% for other woven and 11% for denim products, when choosing the maximum number of uses reported by the literature.

Table 67. Number of times that the worsening scenario produces higher environmental impacts and societal costs compared to the base case

Analysis	Worsening scenario	Knitted	Other woven	High denim
Environmental impacts	Worst (1 use)	66.3	74.6	200.6
	BC weighted MIN uses	1.4	1.2	1.3
	1 year	3.6	4.2	4.4
Societal costs	Worst (1 use)	62.4	72.1	172.9
	BC weighted MIN uses	1.4	1.1	1.2
	1 year	3.4	4.1	4.0

LCS: Life Cycle Stage; Knitted: knitted products; High denim: products with high-grammage denim fabric; Other woven: woven products without high-grammage denim fabric.

Source: own elaboration

Table 68. Decrease of environmental impacts and societal costs of the improving scenario compared to the base case expressed as percentage (%)

Analysis	Improving scenario	Knitted	Other woven	High denim
Environmental impacts	BC plus 10% uses	8	9	8
	BC plus 30% uses	21	22	20
	BC plus 50% uses	31	32	28
	BC weighted MAX uses	28	16	11
Societal costs	BC plus 10% uses	8	8	7
	BC plus 30% uses	20	21	17
	BC plus 50% uses	29	31	25
	BC weighted MAX uses	27	15	10

LCS: Life Cycle Stage; Knitted: knitted products; High denim: products with high-grammage denim fabric; Other woven: woven products without high-grammage denim fabric.

The decrease is calculated as follows: $decrease = \frac{improving\ scenario - base\ case}{base\ case} * 100\ (\%)$

Source: own elaboration

More details about this scenario are reported in section 13.11.10.4.1.

This analysis shows that the choice of the user on when disposing the textile product largely affects the environmental impacts and the societal costs.

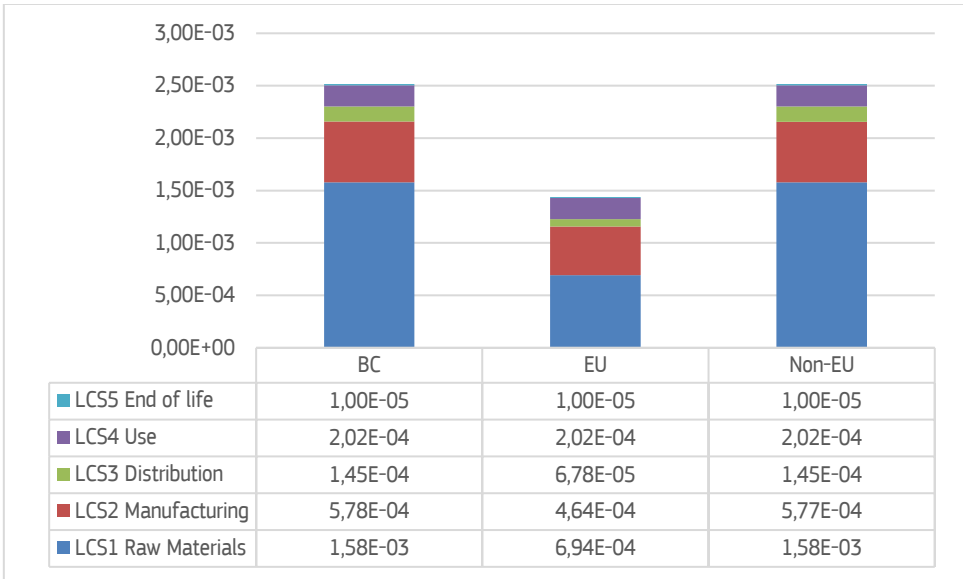
10.1.4.2 Location of raw material production and manufacturing processes

This analysis investigates the influence of the location of the raw material production and manufacturing processes occurring in EU or in third countries. Since the analysis for the three product categories shows the same insights, this section reports only the results related to knitted products. More details are reported in section 13.11.10.4.2.

Figure 44 shows that the production in third countries generates environmental impacts 75% higher than those generated when production occurs in EU. This difference is driven by the use of water in fibre production (LCS1) and the use of energy in LCS2 (**Figure 45**). The use of water in third countries generates more environmental impacts because it is withdrawn from ecosystems that are under more water stress compared those in EU (Chapagain et al., 2006; Boulay and Lenoir, 2020). The life cycle impact assessment method for the use of water based on the PEF uses the characterisation model of ‘Available Water Remaining’ (AWARE) (Boulay et al., 2018). Additionally, the energy mix in third countries is more impacting than the one in EU (section 5.4.3).

Figure 46 shows that the production in third countries generates societal costs 35% lower than those generated when production occurs in EU. **Figure 47** shows that this difference is mainly driven by the higher costs in EU related to HR and CAPEX. Due to dataset limitations, the externalities are underestimated. In fact, many of the datasets used to conduct the LCA analysis, the results of which form the basis for quantifying externalities, are not specific to the countries where textile production processes take place. To model chemical inputs (the second leading cause of environmental impact after energy, **Figure 45**), datasets based on global averages were used; therefore, this approach does not allow for an exhaustive representation of the production reality in the countries considered in the model (e.g., India, China, etc.).

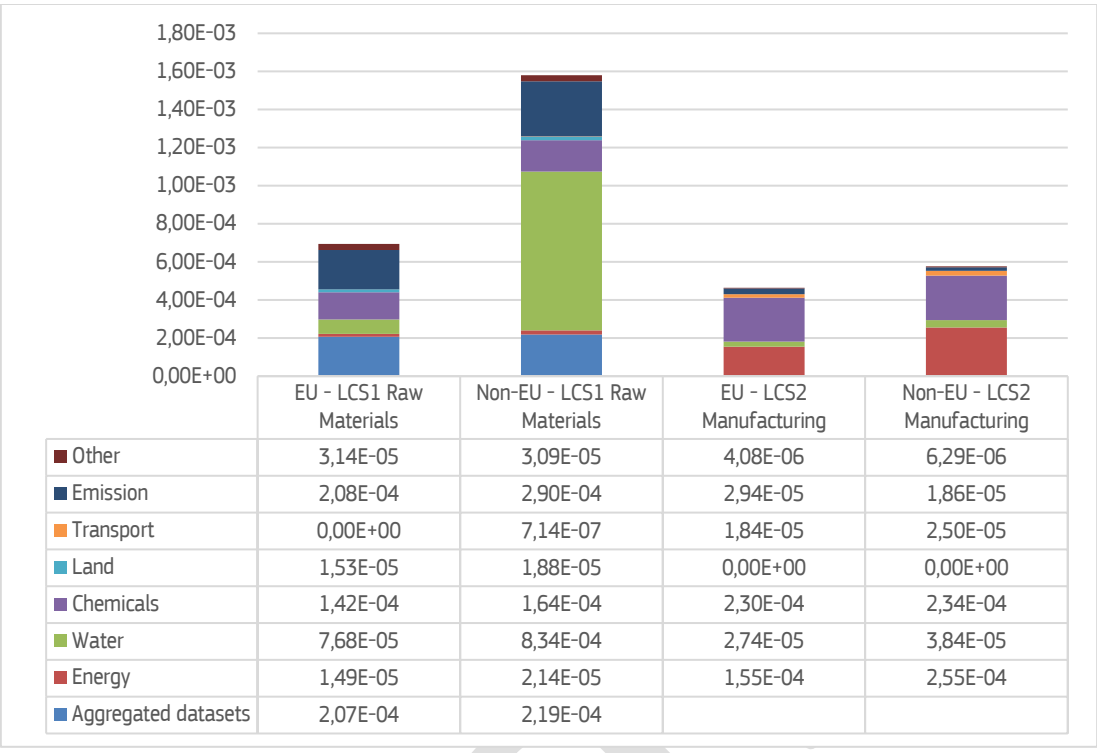
Figure 44. Environmental impacts expressed as environmental points for knitted products in scenarios investigating the influence of the location of the raw material production and manufacturing processes – all life cycle stages



BC: Base Case; LCS: Life Cycle Stage

Source: Own production

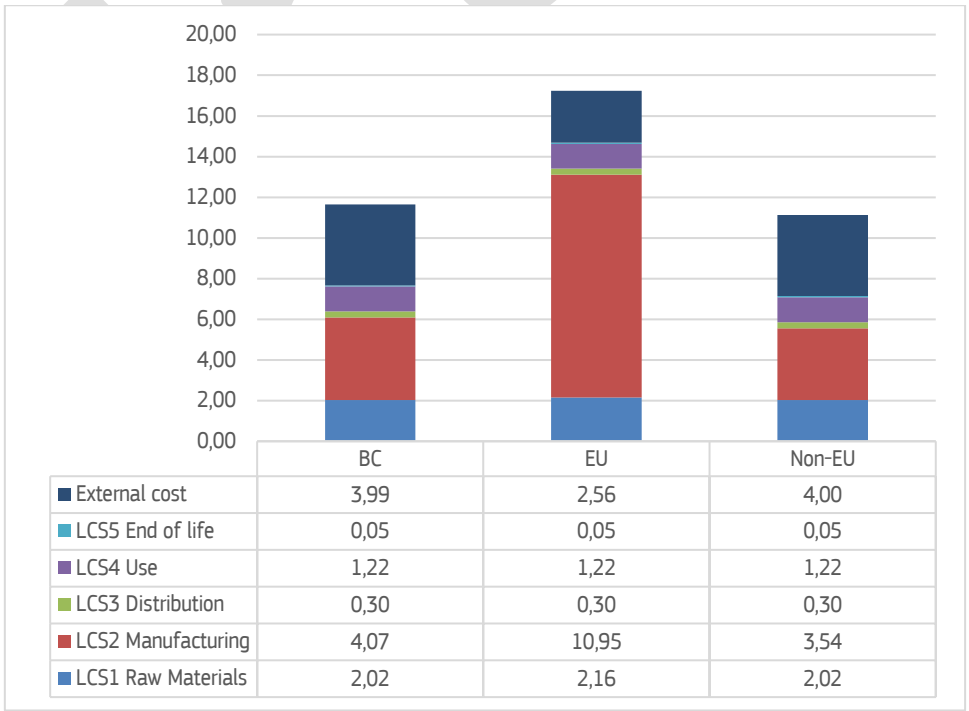
Figure 45. Environmental impacts expressed as environmental points for knitted products in scenarios investigating the influence of the location of the raw material production and manufacturing processes – contributors of the first two life cycles



LCS: Life Cycle Stage

Source: Own production

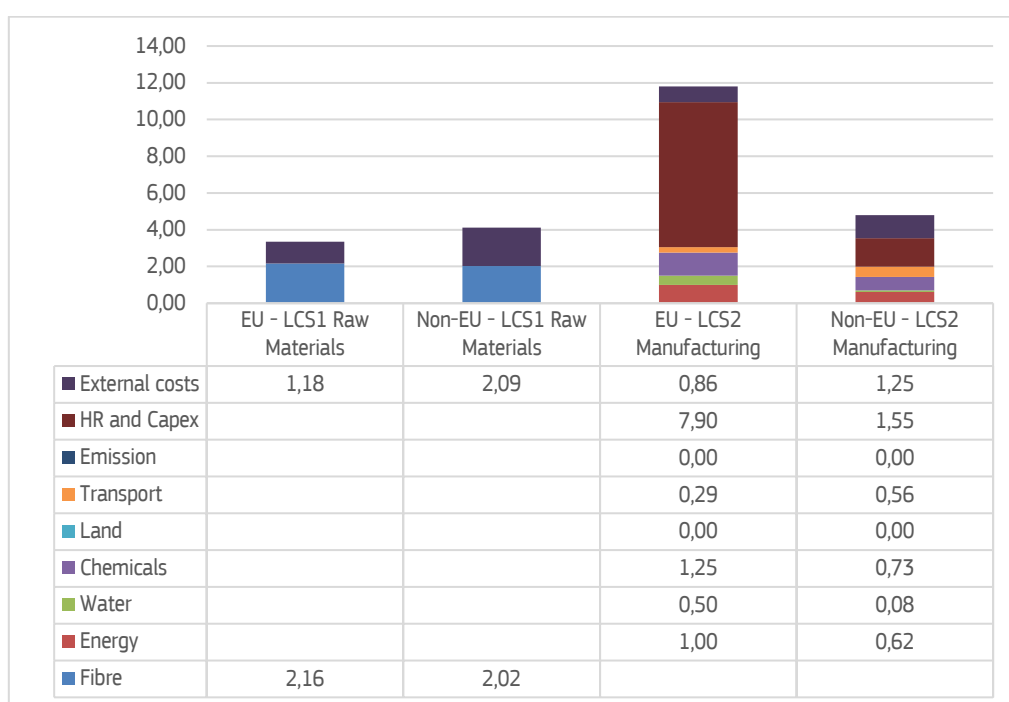
Figure 46. Societal costs expressed as EUR for knitted products in scenarios investigating the influence of location of raw material production and manufacturing processes – all life cycle stages



BC: Base Case; LCS: Life Cycle Stage

Source: Own production

Figure 47. Societal costs expressed as EUR for knitted products in scenarios investigating the influence of location of raw material production and manufacturing processes – contributors of the first two life cycles



LCS: Life Cycle Stage; HR: Human Resources; CAPEX: CAPital EXpenditure

Source: Own production

10.1.5 Propagation of uncertainties

The model developed was based on input values with some intrinsic uncertainty, whose propagation was accounted for in the results via a Monte Carlo analysis based on 500 random iterations as detailed in section 13.11.11, and in particular in section 13.11.11.3.

Table 69 reports the results of the environmental and economic assessment reported as central value and corresponding uncertainty expressed as 95% confidence interval. Compared to the central value, the uncertainty is equal to:

- 0.4-0.8% for the environmental impacts,
- 0.5-0.6% for the costs borne by the producer,
- 0.4-0.6% for the costs borne by the consumer,
- 0.5-0.7% for the costs borne by the society (societal cost).

Table 69. Results of the environmental and economic assessment reported as central value and corresponding uncertainty expressed as 95% confidence interval

Product category	Environmental impacts (environmental points)	Cost borne by the producer (EUR)	Cost borne by the consumer (EUR)	Societal cost (EUR)
Knitted	0.00251±0.00002	7.25±0.04	12.09±0.06	11.66±0.06
Denim	0.00279±0.00002	8.56±0.04	13.59±0.06	13.20±0.06
Other woven	0.00514±0.00003	10.80±0.06	20.65±0.13	21.90±0.15

Source: own production

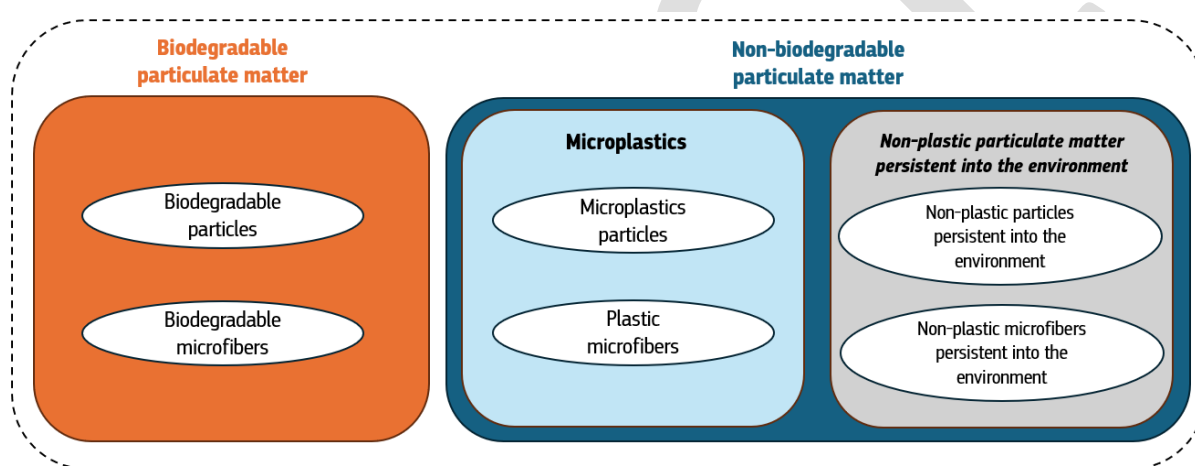
10.2 Fragmented fibre release into the environment

The analysis reported in this section is based on an ongoing JRC study reporting more details on the topics investigated. The study will be available to the public in the following months. The study will be referenced in the preparatory study as JRC (in prep.).

10.2.1 The products of textile apparel fragmentation

The fragmentation of textile apparel generates particulate matter as described in **Figure 48**. The definitions reported in this section, while in line with REACH ⁽²¹⁰⁾ and a previous JRC study (DG JRC, 2024a), are proposed for the first time to the technical community, because this is the first time that the fragmentation of the textile apparel is addressed from the perspective of the ecodesign and the lifecycle of the product. This novelty is driven by the lack of consensus of terminology among studies addressing the same topic. This difference is most probably caused by the diverse aim and scope of the over 200 literature studies consulted (JRC (in prep.)).

Figure 48. Description of type of particulate matter generated from the fragmentation of textile products



Source: own production

The particulate matter is described based on its *shape*, *size* and *composition*. Two types of particulate matter can be distinguished when shape and size are considered: particles and microfibres.

A particle is a particulate matter whose shape might resemble spheres or any irregular form, such as fragments or flakes. The maximum length-to-width ratio of a particle is less than or equal to 3, and all dimensions measure less than 5 mm.

A microfibre is an elongated particulate matter whose length-to-diameter ratio is greater than 3, and the length is less than 15 mm.

Based on its composition, regardless its shape and size, the particulate matter can be:

- *Biodegradable*, when it is broken down by living organisms (e.g. bacteria, fungi, and moulds) into harmless substances under specific environmental conditions within a defined timeframe.
- *Non-biodegradable*, when it does not meet the biodegradability requirements, and therefore it persists into the environment.

Based on this description, further classification is adopted in this report:

- **Biodegradable particulate matter** is a particulate matter that is biodegradable.

²¹⁰ Regulation (EC) No 1907/2006 of the European Parliament and of the Council of 18 December 2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH), establishing a European Chemicals Agency. Available at [this link](#).

— **Microplastics** (MP) are non-biodegradable particulate matter containing synthetic polymers and/or chemically modified natural polymers.

- **Microplastics particles** are MP with the shape and the size of particles.

- **Plastic microfibres** are MP with the shape and the size of microfibres.

— **Non-plastic particulate matter persistent into the environment** (nPM) are non-biodegradable particulate matter, which contain material different from synthetic polymers and/or chemically modified natural polymers.

- **Non-plastic particles persistent into the environment** are nPM with the shape and the size of particles.

- **Non-plastic microfibres persistent into the environment** are nPM with the shape and the size of microfibres.

These definitions consider three main aspects, which are crucial when assessing textile fragmentation:

(a) the complex fibre composition of textile products, which are often made of textile fibre blends (section 5.4.1),

(b) the numerous chemical treatments textile fibres usually undergo during the manufacturing processes,

(c) the persistence of the particulate matter in the environment.

Aspects a) and b) lead to an analysis that looks at the textile fibre types alongside with any other substances that could be attached to the textile product. Additionally, aspect c) leads to investigate the accepted persistency of the particulate matter into the environment.

However, the definitions reported above are based on the generic concept of **biodegradability**, which is the *capability of a defined amount of material to be broken down by living organisms (e.g. bacteria, fungi, and moulds) into harmless substances under specific environmental conditions within a defined timeframe*. Therefore, there are many factors affecting the biodegradability of the material under investigation:

1. The amount of material, which translates the accepted level of material concentration in the investigated environmental compartment (e.g. sea, air, etc.). The amount can be expressed as mass, volume, number of particulates, considering the peculiarity of the material and the environmental compartment.

2. The presence of the living organisms capable to biodegrade the material.

3. The specific environmental conditions in terms of:

- (a) Definition of the environmental compartment where the material is expected to be degraded: such as, marine water, fresh water, atmosphere, soil, etc.

- (b) Presence of oxygen (aerobic or anaerobic conditions), which largely affect the type of living organisms that can perform the degradation.

- (c) pH and other chemical conditions.

- (d) Temperature and other physical conditions.

- (e) Presence of substances that could accelerate or obstacle the degradation process.

- (f) Any other environmental factor specific for the material investigated.

4. The timeframe within the degradation should occur.

These factors highlight that **a material cannot be defined as biodegradable in general**, but a specific amount of material must be tested considering conditions that include specific values of the abovementioned factors.

To the authors knowledge, the literature does not report a definition of biodegradability of fragmented textiles which contains the specification of factors reported above. This lack of definition affected also the types of particulate matter investigated by the literature. Since plastic materials are commonly considered non-biodegradable, most of the literature consulted focused on MP (JRC (in prep)).

10.2.2 Impacts on the environment and human health

Regardless from their origin, MP were found in the most remote locations on Earth — ranging from the highest mountains to the deepest ocean trenches, and from the polar regions to human bloodstreams (Amobonye, Bhagwat, Singh, et al., 2021; Leslie et al., 2022; Lim, 2021). The proliferation of these materials across the globe and in many environmental compartments poses directly significant risks to ecosystems and humans (Danopoulos et al., 2022).

MP were largely studied for their capacity to carry other pollutants in several environmental compartments (Picó and Barceló, 2019; Wang et al., 2019; Landrigan et al., 2020; A Kumar et al., 2023; Barrick et al., 2025), however it is unknown if they have a different effect from any other carrier naturally present in very large quantities in these environments, such as soil dust and particulate matter generated by plants. In this context, the real problem to be tackled should be the harmful substances rather than the carriers. More research is needed to shed light on this aspect.

On **soil**, MP pollution causes:

- The impairment of microbial activity, as well as negatively affects the diversity and the reproduction of soil organisms (Amobonye, Bhagwat, Singh, et al., 2021; Bhardwaj et al., 2024).
- The reduction of seed germination, the stunting of plants, the compromised reduction of earthworm biomass, and alterations to soil pH and stability, which are potential disruptions of the ecosystem's functionality (Boots et al., 2019).
- Obstruction of water and nutrient uptake by plants, when they adhere to the root surface, and oxidative damage to crops, release cytotoxic and genotoxic substances (Wang et al., 2019; Zhao et al., 2022; Iqbal et al., 2023).
- Exacerbation of toxic effects when interacts with heavy metals (Iqbal et al., 2023).
- Disruption of essential ecological processes, e.g. nitrogen cycle, negatively affecting microbial diversity, enzyme functioning, and oxygen fluxes (Dissanayake et al., 2022; Shen et al., 2022; Jia et al., 2023; A Kumar et al., 2023).

In **aquatic environments**, MP pollution causes:

- Blockages of digestive systems of marine life, which may lead to reduced feeding and starvation (Romeo et al., 2015; Li et al., 2022; Mkuye et al., 2022; Tursi et al., 2022; Pradit et al., 2023).
- Leaching of harmful substances due to their degradation (OECD, 2019; Barrick et al., 2025).
- Shifting nutrient dynamics and affecting microbial communities in water sediments (Barrick et al., 2025).
- Biodiversity threatening due to interference with reproduction of aquatic organisms (Barrick et al., 2025).
- Disruption of the endocrine systems of marine animals (Jung et al., 2020).
- Alteration of oceans' efficiency in capturing and storing carbon dioxide from the atmosphere (Sharma et al., 2023).

In **atmospheric ecosystems**, MP pollution acts as any other cloud condensation nuclei or ice-nucleating particles, which determine the cloud reflectivity, rainfall patterns and duration (Aeschlimann et al., 2022). However, further research is ongoing regarding the real impact of MP that could differ from the function of any other particles naturally present in nature.

Human health

Humans are largely exposed to the MP because can potentially breathe air with MP, drink water with MP and eat food with MP (Yee et al., 2021; Ebrahimi et al., 2022; Bošković et al., 2023; O'Brien et al., 2023).

Upon entering the body, MPs can trig inflammatory responses, potentially leading to a spectrum of health issues, including but not limited to cancer, reproductive difficulties, and developmental anomalies (Jaishankar et al., 2014; Qiao et al., 2019; Hirt and Body-Malapel, 2020; Landrigan et al., 2020; Kannan and Vimalkumar, 2021; Caputi et al., 2022; Palaniappan et al., 2022; Roursgaard et al., 2022; Sinha et al., 2022; A Kumar et al.,

2023; Niu et al., 2023; Osman et al., 2023). Additionally, MPs may exacerbate antibiotic resistance by acting as breeding grounds for bacteria (Chatterjee and Sharma, 2019; Shi et al., 2021).

Respiratory irritation and the possibility of contributing to chronic conditions like chronic-obstructive pulmonary disease and asthma have been linked to microplastic inhalation (Amato-Lourenço et al., 2020; Lu et al., 2022). However, more potential health impacts may not be entirely understood yet, due to missing data on very small particulates (DG JRC, 2024b).

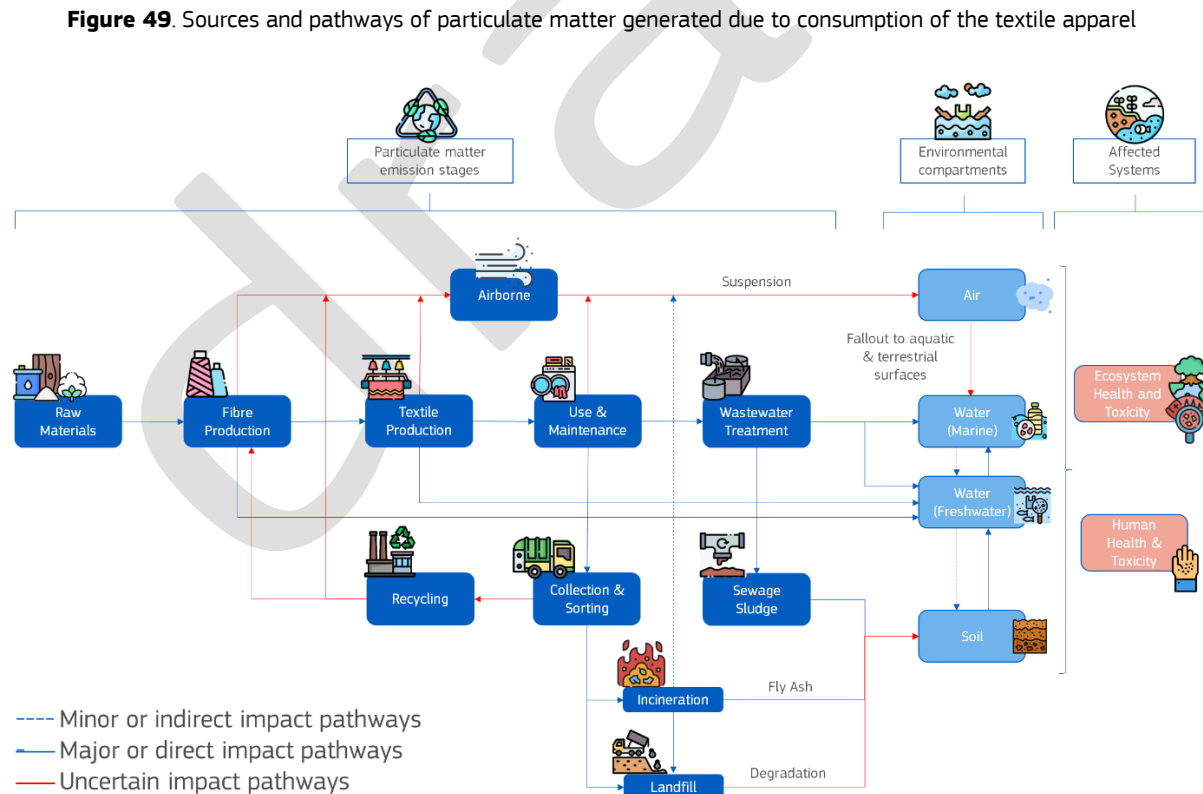
Dermal exposure has been associated with skin irritation, inflammation, and the potential for endocrine disruption and reproductive toxicity (Campanale et al., 2020; Dubey et al., 2022).

Lastly, studies have indicated that MPs can disrupt circadian rhythms and melatonin production, potentially leading to sleep disorders (Moralia et al., 2022; Sinha et al., 2022; Meléndez-Fernández et al., 2023). They may also interfere with hormone regulation and metabolism, contributing to obesity and an increased risk of diabetes, as oxidative stress and inflammation (Ghosh et al., 2023).

The environmental and health impacts of **non-plastic particulate matter persistent into the environment** are strictly related to the substances that compose this particulate matter and those attached to it. Therefore, their assessment overlaps with the investigation performed on the substances of concern carried out in section 12.

10.2.3 Sources and pathways due to textile apparel consumption

Figure 49 shows a simplification of the complex network describing sources and pathways of particulate matter generated due to consumption of the textile apparel. Particulate matter is emitted at all main stages of the product's life cycle and emitted into atmosphere, water and indirectly to soil. Once emitted, the particulate matter can travel across several environmental compartments and affect ecosystems and human health.



Source: adapted from (Henry et al., 2019). Icons from www.flaticon.com

Manufacturing stages

5032 In all manufacturing stages, the particulate matter is emitted into the air and via water discharges (Dris et al.,
5033 2016; Brahney et al., 2020; Deng et al., 2020; Zhou et al., 2020; O'Brien et al., 2023). This is most probably
5034 because particulate matter escapes the treatments that the materials undergo in the several processes without
5035 use of pollution control systems. At these stages, especially emissions into the air directly affect workers, who
5036 stay for many hours in enclosed spaces.

5037 **Transportation, use and maintenance**

5038 Particulate matter is emitted into the air and water discharges during transportation, use and maintenance.
5039 This occurs due to physical and chemical stresses on the product, such as friction and detergent interaction
5040 (Browne et al., 2011; Napper and Thompson, 2016; Carr, 2017; Dreillard et al., 2022; Tiffin et al., 2022;
5041 Hazlehurst et al., 2023; O'Brien et al., 2023; Yang et al., 2023; Tedesco et al., 2024). Exposure to sunlight,
5042 particularly ultraviolet (UV) radiation, can also contribute to the breakdown of textile fibres (Sørensen et al.,
5043 2021).

5044 Many studies were carried out to investigate the **numerous and synergic factors** affecting the particulate
5045 matter released during washing operations, where three different types of products interact with each other's:
5046 the textile product, the washing machine and the detergent. Additionally, two external factors play a role: water
5047 hardness and user behaviour (JRC (in prep.)).

5048 At level of the textile product, the many interacting factors refer to (Hearle and Morton, 2008; Hearle, 2009;
5049 Lawrence, 2015; Cai et al., 2020; Özkan and Gündoğdu, 2021; Zambrano et al., 2021; Periyasamy and Tehrani-
5050 Bagha, 2022; Allen et al., 2024):

- 5051 — the characteristics of the fibres (e.g. length, hairiness and resistance),
- 5052 — the way fibres are spun into yarns,
- 5053 — the fabric construction, and
- 5054 — the finishing processes applied.

5055 The literature does not report which should be the intrinsic characteristics of the textile apparel to decrease
5056 the emission of particulate matter.

5057 At level of the washing machine, the many interacting factors refer to (Hartline et al., 2016; Balasaraswathi
5058 and Rathinamoorthy, 2022; Saskia Manshoven et al., 2022):

- 5059 — the characteristics of the mechanical actions on the textile products, and
- 5060 — the washing temperature.

5061 At level of the detergent and softener, the many interacting factors refer to the chemical composition and their
5062 status (liquid or powder) (Hartline et al., 2016; Balasaraswathi and Rathinamoorthy, 2022; Saskia Manshoven
5063 et al., 2022).

5064 On top of the interaction among these factors belonging to three different types of products, there are two
5065 external factors that play a role, but cannot be controlled by product designers: the hardness of water used
5066 and the user behaviour for washing choices. The higher hardness of water was found to increase the release
5067 of particulate matter from textile products (Balasaraswathi and Rathinamoorthy, 2022); whereas the user
5068 behaviour during washing choices is affected by social factors that vary all over EU (section 6.3.1).

5069 The literature reports that a significant reduction of particulate matter release is achieved after about 5 washes
5070 (De Falco et al., 2019; S Choi et al., 2022; Hazlehurst et al., 2023). The amount of particulate matter released
5071 can increase again after further washing cycles if the quality of a garment deteriorates significantly (Carney
5072 Almroth et al., 2018).

5073 **Wastewater treatment**

5074 The treatment of discharged water is a crucial node of the particulate matter release into the environment.
5075 Although these facilities can remove more than 90% of MP from the wastewater, these pollutants are still
5076 present in the effluent water and the sewage sludge (Iyare et al., 2020; Sol et al., 2021). The effluent water is
5077 considered an important source of particulate matter discharged into natural water basins (Estahbanati and
5078 Fahrenfeld, 2016; Talvitie et al., 2017; Hidayaturrahman and Lee, 2019; Kazour et al., 2019; Lv et al., 2019).

5079 According to Directive 86/278/EEC ⁽²¹¹⁾, in EU, sewage sludge can be applied on agricultural land after specific
5080 stabilization processes, which do not target the removal of particulate matter. Via this pathway, the particulate
5081 matter reaches soil for agricultural purposes (Weithmann et al., 2018; Corradini et al., 2019; Dreillard et al.,
5082 2022; Tagg et al., 2022; Weber et al., 2022).

5083 **Treatment and management of textile waste**

5084 Textile waste is disposed of in landfills, incinerated and recycled. All these processes imply further pathways
5085 for the particulate matter to be generated and reach the environment and humans.

5086 Once disposed of in a landfill, the waste degrade and generates new particulate matter that can escape the
5087 landfill especially via leachate, but also via rainwater run-off and wind transport (Rillig et al., 2017; Henry et
5088 al., 2019; Wang et al., 2019; Palacios-Mateo et al., 2021). If properly collected, the leachate is further treated
5089 in wastewater treatment plants.

5090 The incineration of textile products generates fly and bottom ashes that are usually landfilled, as well as
5091 emissions into the atmosphere treated with air pollution abatement technologies (DG JRC, 2024a; Quashie-
5092 Idun, 2022).

5093 Recycling of textiles may involve mechanical processes, such as shredding or grinding, which generate
5094 particulate matter that is released into the air and water discharge similarly to what happens in the
5095 manufacturing stages (Mušič, 2023).

5096 **10.2.4 Analysis of pollution hotspots**

5097 The analysis of the literature showed that the identification of hotspots, where particulate matter is mostly
5098 released into the environment, **is not currently possible** (JRC (in prep.)). This is mainly driven by:

- 5099 — the absence of a standardised definition on particulate matter biodegradability, and
5100 — the lack of standardised measurement methods for the sampling, characterization and
5101 quantification of these pollutants in all release points of the value chain.

5102 Without a clear and common definition of biodegradability for these particulate matter into specific
5103 environmental compartment, the problem cannot be addressed because its scope is not defined. Additionally,
5104 without standardised measurement methods, the few current emission rates reported by the literature cannot
5105 provide a reliable identification of the main emitting sources upon which to potentially focus the optimised
5106 mitigation strategies.

²¹¹ Council Directive 86/278/EEC of 12 June 1986 on the protection of the environment, and in particular of the soil, when sewage sludge is used in agriculture. Available at [this link](#).

11 Assessment of the design options

A variety of design options are defined with the aim of addressing the identified hotspots and potentially improve the environmental performance of the textile products under scope without causing disproportionate costs. This was done based on the knowledge gathered and developed on:

- The environmental and economic analysis of the base cases (section 10)
- The product technologies (section 9.2),
- The user behaviour (section 6), and
- The market analysis (section 5).

For each product category, the design options are based on the identified Best Available Technologies for each product category and product aspect, as described in section 9. The goal of this section is to estimate the potential environmental benefit of implementing different design options first individually and, when in synergy, then in combination.

This assessment corresponds to the Task 6 of the MEERP, which means that the design options should be understood as potential ecodesign requirements under study and not as final proposals. In a following step, Task 7 will perform a scenario analysis to extrapolate the results to the total EU stock and project the most important impacts. The latter will propose policy options that will serve as a basis for the Impact Assessment that the Commission will undertake after the preparatory study. The final ecodesign requirements will be assessed and defined only then.

As in the analysis of the base cases in section 10.1, the assessment of the design options uses the Life Cycle Thinking approach as described in ISO 14040⁽²¹²⁾ and ISO 14044⁽²¹³⁾, using to the extent possible the Product Environmental Footprint (PEF) framework as defined in the Commission Recommendation (EU) 2021/2279⁽²¹⁴⁾. This analysis was complemented by an assessment of the requirements for substances in section 12.3.

11.1 Definition of design options

Based on the environmental and economic assessment of the base cases, different design options and potential ecodesign requirements have been defined for knitted, denim and other woven products, in order to address the hotspots identified in section 10.1.2 (**Table 70**).

Table 70. Objectives, design options and potential ecodesign requirements for knitted, denim and other woven products

Generic objectives	Specific objectives	Design Options (DOs)	(Potential) Ecodesign requirement
Decrease the environmental impact of products by diluting one-off effects	Extend products lifetime	DO1 – Product with an increased robustness	Information requirement on the robustness of the product (robustness score)
Increase the availability of secondary materials	Increase the share of the waste going to recycling	DO2 – Product with a higher share of weight going to recycling	Information requirement on the recyclability of the product (recyclability score)
Decrease the environmental impact of raw materials	Increase the use of secondary materials	DO3 – Product with higher recycled content	3.1 Information requirement on recycled content 3.2 Performance requirement on the recycled content
Decrease the environmental impact of manufacturing	Decrease the environmental footprint of the manufacturing	DO4.1 – Product with a decreased environmental footprint	Information requirement on the environmental footprint during manufacturing

²¹² ISO 14040:2006 Environmental management — Life cycle assessment — Principles and framework. Available at this link. Last accessed on 4 September 2025.

²¹³ ISO 14044:2006 Environmental management — Life cycle assessment — Requirements and guidelines. Available at this link. Last accessed on 4 September 2025.

²¹⁴ Commission Recommendation (EU) 2021/2279 of 15 December 2021 on the use of the Environmental Footprint methods to measure and communicate the life cycle environmental performance of products and organisations. Available at this link.

Generic objectives	Specific objectives	Design Options (DOs)	(Potential) Ecodesign requirement
	Decrease the carbon footprint of the manufacturing	DO4.2 – Product with a decreased carbon footprint	Information requirement on the carbon footprint during manufacturing

One-off effect refers to a unique and singular occurrence that happens only once and is not repeated (i.e. raw materials production, manufacturing, end of life).

Source: own elaboration.

Details and reasoning for the design options (DO1 – information requirement on **robustness**, DO2 – information requirement on **recyclability**, DO3 – information and performance requirements on the **recycled content** and DO4 – information requirement on the **carbon/environmental footprint during manufacturing**) are provided in sections from 11.1.1 to 11.1.4.

Design options on other specific aspects (i.e. reparability, sustainable material sourcing and release of particulate matter from textile products) were also assessed, with the limitations reported below.

First, although enhanced **reparability** could potentially increase the service lifetime of the product (as DO1), the concept of a design option focused on it presents several challenges. While theoretically, any item of textile apparel can be repaired, the complexity of reparability cannot be objectively quantified. The factors influencing it often conflict with the creative nature of the textile and fashion industry, hence hindering its regulation (see section 9.2.3).

Additionally, mandating the provision of repair instructions appears unwarranted, as the analysis of the users' behaviour showed that these services are normally performed by professionals who do not require such guidance, and thereby, in the view of the authors do not justify the administrative burden for manufacturers to produce them across the board, as a general requirement. Moreover, the analysis in section 6.3.3 does not identify the lack of information as a barrier to repair versus discard or replacement decisions. Instead, factors like emotional attachment, fashion trends, and repair costs are highlighted, none of which are addressed by the considered requirements.

Consequently, only information requirements on available repair services could be considered on voluntary basis. Informing about independent repair services is deemed unsuitable, as these are generally local, small businesses, making it impractical for brands to identify relevant location-specific details without knowing where the garment will be used. Instead, information requirements could rather focus on repair services directly offered by the brand (if any), including their details and contact information, via the Digital Product Passport. This requirement only applies when the service is provided by the retailer, ensuring minimal effort and burden on manufacturers while still offering a benefit to the citizen. The verification of the presence of the repair service should be based on a declaration by the economic operator. Given this is not a mandatory requirement that will affect all economic operators, its impacts were not modelled in the following sections.

Moreover, the offering by the Administrations of financial incentives to subsidise repair services could be a viable approach. Its usefulness could be explored under a legislative framework different from the ESPR.

Secondly, a design option about the **sustainability of fibre production** would be suitable to be explored as the hotspot analysis showed that this stage is responsible for 60–63% of the environmental impacts over the lifecycle (section 10.1.2.1).

However, setting a preference towards the selection of specific types of fibres would overlook the fact that each type of fibre has specific characteristics to meet the demand of the supply chain and the users of the textile product. Therefore, this would not appear to be a suitable way to address the problem. Each type of fibre could be produced in numerous ways, which could have different environmental impacts. Identifying the most sustainable way of producing a specific type of fibre is possible, but it is not a straightforward undertaking because specific sustainability parameters should be selected. These parameters should in theory include at least:

- the trade-offs among the several EU strategies and action plans (e.g. Biodiversity strategy for 2030 ⁽²¹⁵⁾ and Zero Pollution Action Plan ⁽²¹⁶⁾) and their objectives and ambitions, i.e. identifying political priorities when addressing environmental and economic problems, and
- the systemic competition with other industrial sectors that use the same resources. This should avoid distortion of other markets.

This complex analysis cannot be performed within the development of this preparatory study and could potentially be addressed by a different study, which could take as a starting point the background analysis offered by the numerous private schemes certifying different sustainable fibre claims.

In this sense, given the Organic Regulation (EU) 2018/848 includes specific ambitious environmental criteria for the production of cotton, wool and silk, it could offer relevant information to consumers that value them. Thus, the voluntary reporting of fibre content compliant with this EU regulation would enable consumers to choose apparel containing organically produced fibres over others.

Since this requirement is proposed not to be mandatory, its impacts were not modelled in the following sections. Nevertheless, verification of reported voluntary information should rely on chain of custody systems supporting the specific declaration made by the economic operator placing the product on the market. This is the case because the Organic Regulation (EU) 2018/848 sets criteria on the production of the fibres, but it does not put in place mechanisms to certify the quantity of these fibres that was used in the textile apparel. Moreover, this preparatory study addresses the environmental impacts generated by raw material production via other design options (D01, D02 and D03).

Finally, while the topic of the pollution generated by the **fragmentation of textile apparel** is analysed, a design option addressing it is not defined at this stage as some preliminary technical gaps should be first addressed (section 10.2):

1. The definition of particulate matter biodegradability should be standardised. This definition should specify:
 - (a) the amount of addressed material,
 - (b) the presence of living organisms capable to biodegrade the material,
 - (c) the specific environmental conditions, and
 - (d) the timeframe required for the degradation.

This standardisation should consider the numerous environmental compartments which receive this type of pollution.
2. Standardised measurement methods should be agreed for sampling, characterization, and quantification of these pollutants in all release points of the value chain.
3. Standardised measurements should be performed along the whole value chain to identify the main emitting sources of this pollutant.
4. Potential mitigation strategies should be based on points from 1 to 3 and consider the driving factors of the main emitting sources.

11.1.1 D01 - Product with an increased robustness

The service lifespan of products is a key aspect that can influence significantly their environmental performance. Due to the primary contributions of the raw materials and manufacturing stages (81-90% of total impacts - **Figure 33**), a longer use-time would seem to have great potential for improving the environmental impact by spreading one-off effects throughout an extended lifetime, thus diluting them.

However, in examining a potential increase of the service lifetime of apparel, it was found that ecodesign has limited influence, as disposal is not solely determined by physical durability or product-specific attributes like robustness and other design features. According to the analysis of the consumer behaviour (section 6.4), only one-third of disposal reasons relate to intrinsic quality. Factors such as the perceived value, fit, and fashion

²¹⁵ Biodiversity strategy for 2030. Available at [this link](#). Last accessed on 21 October 2025.

²¹⁶ Zero Pollution Action Plan. Available at [this link](#). Last accessed on 21 October 2025.

trends significantly influence disposal decisions, indicating a complex interplay of factors which, in addition, are not addressable under the framework of ESPR. A scenario analysis in section 10.1.4.1 assesses the effects of user behaviour on the premature disposal of the products.

Nonetheless, to address the share of the disposal related to the physical durability and potentially encourage reusability, the impact of a potential information requirement is assessed. The objective of this would be twofold. On the one hand, it would help consumers to make informed purchases of more durable products. On the other hand, it would encourage manufacturers to produce more durable products than those of competitors.

Because measuring the physical durability is challenging due to the lack of a suitable method to simulate the aging process of textile apparel, a potential information requirement is proposed in terms of the robustness of the product:

However, the robustness framework has two main limitations (section 9.2.1.4):

1. Since it does not consider product aging, it is not capable to describe the product performances over a defined number of uses, which are a common unit to express service lifespan of a product (see section 5.8). In other words, the robustness framework is not capable to describe how many times a person can use an item before the probable appearance of a physical failure due to its normal use ⁽²¹⁷⁾. Therefore, the robustness framework is **not capable to describe the foreseen service lifetime related to intrinsic product properties**. The robustness framework is capable to describe only the resistance to external stresses, which do not correspond to the stresses occurring during product aging.
2. A known increase of performance level of key parameters corresponds to an unknown increase of physical durability. This implies that the consequent **increase of the service lifespan can only be based on expert judgements, which will not be verifiable**. Textile experts cannot base their judgement on direct experience, because nobody performs such measurements due to lack of aging tests.

Table 45 reports a parallel description of the durability and robustness framework. The implementation of the durability framework fully backed by scientific evidence would require the development of standards that reliably simulate the aging process of textile apparel.

A performance requirement on 'robustness' is not considered at this stage precisely because of the reported uncertainties surrounding its applicability as a reliable measure of textile durability. **The setting of performance requirements could be considered following the analysis of potential additional stakeholder input and evidence supporting this aspect.**

11.1.1.1 Potential information requirement on robustness

A scoring system has been developed to award products a robustness value in a scale from 0 to 10. It consists in a cumulative points-based evaluation, designed to assess products across selected key parameters: spirality, dimensional change and colour, fabric, seams and non-textile parts as sub-parameters identified by visual inspection (as described in section 9.2.1.4).

Each parameter is assigned a score corresponding to different performance levels based on test results, according to the description of the base cases and best available technologies for knitted and woven products reported in **Table 44**. **Table 71** provides a summary of how the points should be awarded for each of the parameters.

For spirality, the analysis performed in section 9.2.1.4 concluded that test results following the ISO standard 16322-3 of the base cases would range from 5% to 6% for woven products and from 5% to 7% for knitted products. Results above those would not be awarded, as they would correspond to the lowest performing range. Products performing at the lower end of the range indicated for the base cases would be awarded with 1 pt. Products performing at the higher end of the range indicated for the base cases would be awarded with 2 pt. Finally, test results below the indicated ranges would be awarded with 3 pt, corresponding to the best available technologies in terms of this parameter.

²¹⁷ This concept is strictly related to the reliability of the product as reported in the ESPR (see **Table 2**). It is still suitable to report it together with the concept of durability because these two product aspects were gathered in the same group as reported in section 9.1.3.

For dimensional change, the test results of the base cases following the ISO standard 3759 would range between $\pm 3\%$ and $\pm 4\%$ for woven products and $\pm 5\%$ and $\pm 6\%$ for knitted products. Again, results above those would not be awarded, as they would correspond to the lowest performing range. Products performing at the lower end of the range indicated for the base cases would be awarded with 1 pt. Products performing at the higher end of the range indicated for the base cases would be awarded with 2 pt. Finally, test results below the indicated ranges would be awarded with 3 pt, corresponding to the best available technologies in terms of this parameter.

Finally, for visual inspection, applying the ISO standard 15487 to the base cases of both knitted and woven products would result in “3. Moderate change in appearance”. Results below those would not be awarded, as they would correspond to the lowest performing products. Products performing as the base cases would be awarded with 2 pt. Finally, products performing better would be awarded with 4 pt. As compared to the spirality and dimensional change, the results of the visual inspection are more heavily weighted, contributing more to the total score because of the number of sub-parameters considered and the insights gathered from the analysis of failure modes (section 9.2.1.3).

Thus, this system enables a comprehensive evaluation by considering both the overall performance and the relative importance of each parameter in the final score, with 0 indicating performance below the Base Case and 10 indicating the BAT level performance across all parameters.

Table 71. Points to be awarded for the robustness score

Parameter	ISO standard	Fabric construction	Test result	Points
Spirality	16322-3	Woven	Spirality > 6%	0 pt
			$5.5\% < \text{spirality} \leq 6\%$	1 pt
			$5\% < \text{spirality} \leq 5.5\%$	2 pt
			Spirality $\leq 5\%$	3 pt
		Knitted	Spirality > 7%	0 pt
			$6\% < \text{spirality} \leq 7\%$	1 pt
			$5\% < \text{spirality} \leq 6\%$	2 pt
			Spirality $\leq 5\%$	3 pt
Dimensional change	3759	Woven	Change > $\pm 4\%$	0 pt
			$\pm 3.5\% < \text{change} \leq \pm 4\%$	1 pt
			$\pm 3\% < \text{change} \leq \pm 3.5\%$	2 pt
			Change $\leq \pm 3\%$	3 pt
		Knitted	Change > $\pm 6\%$	0 pt
			$\pm 5.5\% < \text{change} \leq \pm 6\%$	1 pt
			$\pm 5\% < \text{change} \leq \pm 5.5\%$	2 pt
			Change $\leq \pm 5\%$	3 pt
Colour, fabric, seams and non-textile parts via visual inspection	15487	All	< 3. Moderate change in appearance	0 pt
			3. Moderate change in appearance	2 pt
			4. Negligible change in appearance	4 pt

The performance level of the visual inspection is given by the least performing sub-parameter.

Source: own elaboration

The option of defining classes of performances has not been further pursued in this study due to insufficient data on the distribution of products in the market according to this framework. This matter could be reassessed in the future if more detailed information about the distribution of characteristics of products on the market becomes available.

11.1.1.2 Estimates for the assessment

The assessment of the impact of an information requirement on the robustness of the products is based on the following assumptions:

- The average **service lifetime** will increase by 5%. This is estimated based on the experience of the Energy Labelling framework, which has showed that products for which information

5296 requirements have been defined have improved their performance by 6 – 14% over a period of 5
5297 years, as compared to the Business-as-Usual scenario ⁽²¹⁸⁾.

5298 While acknowledging the differences between energy efficiency and robustness, this assumption
5299 is made in the absence of a more accurate benchmark. This estimate is based on the assumption
5300 that, regardless of the product aspect reported, consumers would respond similarly to information
5301 presented on a label, adjusting their purchasing decisions towards better-performing products.
5302 This consumer behaviour is expected to lead to comparable improvements, irrespective of the
5303 dimension (or product aspect) being considered. Analogously, the manufacturers are expected to
5304 also behave in the same way to avoid showing performance below those of their competitors.

5305 A higher impact of the requirement on the average service lifetime is not assumed since available
5306 information indicated that the criteria mostly applied by consumers when purchasing apparel
5307 products are price and perceived quality, which do not necessarily imply longer service lifespan
5308 (section 6.2).

5309 This estimate could be revised only based on specific surveys analysing how consumers and
5310 manufacturers would react to an information label on the robustness of textile apparel products.

5311 — The robustness increase would come with improved performances of key parameters (i.e. spirality,
5312 dimensional change and colour, fabric, seams and non-textile parts as sub-parameters identified
5313 by visual inspection). A consultation with textile experts revealed that the performances of the key
5314 parameters could be improved increasing the energy and water used during the manufacturing,
5315 and incrementing the quantity of the fibres and chemicals. The estimates are shown in **Table 72**,
5316 based on expert judgement of the most common processes needed. As reported in section 9.2.1.4,
5317 these expert judgements are not based on experience with measurements on specific products,
5318 but they are just **unverifiable hypothesis made by experts from the textile industry**. To
5319 the authors' knowledge, nobody has ever measured the correlation between the increase of key
5320 parameters and the increase of the service lifetime.

5321 The increase in the **energy** and **water** used relates to processes such as extra stabilisation in
5322 knitted products, mercerisation, curing, double heat-setting or sanforisation in denim products and
5323 sanforisation in other woven products. In the case of the water use, it is slightly higher in the case
5324 of denim because of the high-water demand of mercerisation of cotton ⁽²¹⁹⁾.

5325 As for the changes in the **fibres**, higher robustness is normally achieved with denser constructions
5326 of the fabrics as well as reinforced seams in knitted products, heavier pocketing in denim and
5327 better interlinings in other woven products. This is modelled by an increase in the mass of the
5328 products.

5329 Finally, regarding the **chemicals**, higher robustness is normally achieved in knitted products
5330 through enzymatic treatments and mild surface stabilisers, in denim through alkanile treatment
5331 agents and stabilising finishes, and in other woven products through light stabilising finishes and
5332 mild softeners.

5333 **Table 72.** Change in parameters related to the robustness (for a 5% service lifetime increase).

Product category	Energy	Water	Fibres	Chemicals
Knitted	2% ↑	1% ↑	0.6% ↑	2% ↑
Denim	2.5% ↑	2.5% ↑	0.5% ↑	1.7% ↑

²¹⁸ The figures for the Energy Labelling have been calculated based on the data published in Leo Wierda, (2024), for selected product groups for which the effect of the energy labelling could be decoupled from the introduction of ecodesign implementing measures to remove the effect of any performance requirement.

²¹⁹ **Extra Stabilisation** involves techniques like additional stitching or application of stabilising agents to improve the structure and strength of knitted fabrics. **Mercerisation** is a chemical treatment used primarily on cotton, to strengthen fibres, increase their luster, and improve dye absorption. **Curing** typically involves heat treatment to set or stabilise finishes and dyes. **Double Heat-Setting**, used to stabilise synthetic fibres, involves subjecting the fabric to two cycles of heat treatment to lock in the desired shape and dimensions. **Sanforisation** is a pre-shrinking process that ensures that the garment maintains its size and shape.

Product category	Energy	Water	Fibres	Chemicals
Other woven	2% ↑	1.4% ↑	0.44% ↑	1.25% ↑

Knitted: knitted products; Denim: products with denim fabric; Other woven: woven products without denim fabric.

Source: own elaboration based on textile expert consultation.

- The additional **manufacturing costs** for achieving a higher robustness are calculated based on the additional energy, water, mass and chemicals used.
- An additional **cost of testing** is considered and estimated to be of 0.05 €/unit. This is based on practices of the testing labs who reported that the standard price for a set of relevant robustness tests is around 500€, excluding potential discounts related to testing volume, membership status, or other business practices. Considering that brands may test one item per model or the same item in each colour variant, it is assumed that such price could cover around 10 000 units according to the testing labs consulted.
- An additional **cost of labelling** is also considered and estimated to be of 0.02 €/unit. This is based on the analysis conducted by Marschinski, R. et al., (2025). The smallest value from their analysis was chosen since apparel products already possess labels at the point of sale in the form of hang tags and, therefore, it would not be necessary to create a new one; the information can simply be included in the existing format.

11.1.1.3 Results – DO1

The results of the environmental and economic assessment of the DO1 are shown in **Table 73**, **Table 74**, **Table 75** for knitted, denim and other woven products, respectively. The results indicate that the information requirement on robustness could lead to a reduction of the environmental impact by approximately 2% across all representative products as compared to the base cases: 2.2% for knitted products, 2% for denim and other woven products. This decrease is achieved by spreading the impacts of all lifecycle stages (except the use stage) over a longer product lifetime. This occurs despite increased consumption of water, energy, and chemicals, and the higher usage of fibres affecting lifecycle stages 1 and 2, which leads to less pronounced improvement compared to the benefits seen in the distribution and end of life stages.

The changes also result in cost savings, even when accounting for the additional costs of testing and labelling. Internal costs, as determined by conventional LCC, decrease for all representative products: 0.8% for knitted products, 1.3% for denim, and 1.1% for other woven products. Furthermore, when environmental impacts are internalised in the cost assessment using the societal LCC, improvements of 1.2% in knitted products and 1.4% in both denim and other woven products are observed compared to the base cases.

Table 73. Results of the design option 1 (robustness) for knitted products

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score (environmental points)	1.54E-03	5.69E-04	1.38E-04	2.02E-04	9.57E-06	2.46E-03
Single score compared to the base case	-2.5%	-1.5%	-4.8%	0.0%	-4.8%	-2.2%
Internal costs (LCC)	1.97 €	4.09 €	0.28 €	1.22 €	0.05 €	7.61 €
LCC compared to the base case	-2.7%	0.4%	-4.8%	-0.3%	-5.0%	-0.8%
Societal costs (sLCC = internal costs + environmental externalities)	4.00 €	5.31 €	0.55 €	1.56 €	0.09 €	11.51 €

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
sLCC compared to the base case	-2.6%	0.0%	-4.8%	-0.2%	-4.9%	-1.2%

LCS: Life Cycle Stage; LCC: Life Cycle Costing; sLCC: societal Life Cycle Costing

Source: own elaboration.

Table 74. Results of the design option 1 (robustness) for denim products

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score (environmental points)	2.99E-03	1.08E-03	1.70E-04	7.92E-04	1.35E-05	5.04E-03
Single score compared to the base case	-2.9%	-0.6%	-4.8%	0.0%	-4.8%	-2.0%
Internal costs (LCC)	2.88 €	6.23 €	0.22 €	4.63 €	0.05 €	14.02 €
LCC compared to the base case	-2.9%	-0.9%	-4.8%	-0.4%	-5.1%	-1.3%
Societal costs (sLCC = internal costs + environmental externalities)	6.23 €	8.77 €	0.54 €	5.94 €	0.11 €	21.60 €
sLCC compared to the base case	-2.9%	-0.8%	-4.8%	-0.3%	-4.9%	-1.4%

LCS: Life Cycle Stage; LCC: Life Cycle Costing; sLCC: societal Life Cycle Costing

Source: own elaboration.

Table 75. Results of the design option 1 (robustness) for other woven products

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score (environmental points)	1.64E-03	8.01E-04	1.36E-04	1.43E-04	9.59E-06	2.73E-03
Single score compared to the base case	-2.5%	-0.9%	-4.8%	0.0%	-4.8%	-2.0%
Internal costs (LCC)	2.30 €	4.85 €	0.26 €	0.85 €	0.05 €	8.30 €
LCC compared to the base case	-2.6%	-0.2%	-4.8%	-0.3%	-5.1%	-1.1%
Societal costs (sLCC = internal costs + environmental externalities)	4.69 €	6.64 €	0.52 €	1.08 €	0.09 €	13.01 €
sLCC compared to the base case	-2.6%	-0.4%	-4.8%	-0.3%	-4.9%	-1.4%

LCS: Life Cycle Stage; LCC: Life Cycle Costing; sLCC: societal Life Cycle Costing

Source: own elaboration.

Section 13.11.10.5 reports the results of the environmental assessment of the D01, for each impact category and expressed in their respective units.

11.1.2 D02 - Product with a higher share of weight going to recycling

Looking at the results of the hotspot analysis in the previous section 10.1.2, several design options are defined to address the major contribution of the raw materials to the total lifecycle impacts (60-63%). Although this is more directly tackled with a third design option on recycled content, it would go hand-in-hand with the need to increase the availability of secondary materials. For this reason, this design option analyses the potential impact of an information requirement that would improve the recyclability of the products. Moreover, despite the minor contribution of the end of life to the lifecycle impacts, it is expected to also slightly improve the environmental performance of the products given that recycling is the second-least impacting waste treatment pathway for most environmental impact categories after preparing for reuse, as reported in Solis et al. (2024).

However, the amount of textile waste that goes to recycling is not only influenced by the recyclability of the products by design. In fact, while Huygens et al. (2023) indicate that only 10% of the textile waste goes to recycling, Bakowska, O. et al., (2025) showed that around 20% of garments on their sample were readily available for recycling even without any further processing. This underscores the findings from the analysis of the ecosystem in section 9.2.5., that highlighted the low collection rates, the readiness of certain recycling technologies and the recycling capacities as limiting factors for textile recyclability beyond design.

Despite the limitations, this information requirement is expected to have an impact by guiding consumer choices towards more recyclable products and encouraging manufacturers to produce more recyclable products than those of competitors.

The concept of a performance requirement that prohibits the entry on the market of products that cannot be recycled is discarded for the following reasons. While the analysis of the recycling ecosystem indicates that certain disruptors currently hinder product recyclability, these can be effectively removed or managed during the pre-treatment stages. The only component found to significantly impede recycling is elastane, when present above a specific threshold. Despite this limitation, a prohibition on elastane use is not considered proportionate nor feasible, as the material imparts essential functional properties to textiles that cannot be eliminated without compromising the performance of certain products (further analysis of elastane can be found in section 12.3.8). Instead, its use should be minimised and limited to applications where it is most necessary. The established information requirement will indirectly discourage excessive use, as products containing elastane above the defined limit will be labelled as “Non-recyclable”, as described in the following section.

11.1.2.1 Potential information requirement on recyclability

A scoring system has been developed to award products regarding their recyclability in a scale from 0 to 10. It consists in a cumulative points-based evaluation, designed to assess products according to the main limitations found in the analysis in section 9.2.5.

The scoring considers all ecosystem-related parameters that allow, facilitate or hinder the recycling of textile products, from the sorting to the recycling techniques. It leaves aspects related to the collection out of the scope as these are already addressed in the WFD, requiring Member States to set up separate collection for textiles by 1 January 2025 (Article 11).

With regards to the sorting, the scoring considers that the main obstacle relates to the identification of the composition of the fabric. Thus, it awards the avoidance of layered fabrics whose outer fibres are different from the inner ones since they limit the performance of near-infrared spectroscopy sorting systems. Other limiting aspects such as materials in compositions below 5% cannot be included in the framework despite hindering the sorting due to the issues for its verification related to the accuracy of the tests. In fact, the TLR also acknowledges this technical limitation, and permits the use of tolerances of up to 5% regarding both manufacturing (2%) and testing (3%) processes, cumulatively applicable for all textile content. Since the compliance with the requirement would not be reliably tested, no enforceable threshold is set to cover this aspect in the framework at this stage.

As for the pre-treatment, the scoring criteria cover the presence/avoidance of certain materials that are considered disruptors as they can hinder the recycling process. It is the case of elastane, which does not allow recycling when it is in compositions higher than 15% for high-quality mechanical recycling and 5% for chemical recycling (20% in the case of nylon PA6 chemical recycling). Moreover, the score awards the avoidance of non-

textile difficult to remove parts, such as sequins, or design aspects that affect the quality of the materials recovered, such as coatings and printings.

Finally, the scoring addresses the availability of recycling techniques operational at scale. The mechanical recycling is the most common technique. However, it often results in shorter fibres and have lower mechanical properties than virgin fibres, limiting their application. Moreover, while mechanical recycling can handle all types of fibres and blends, the characteristics of the feedstock are directly transferred to the recycled fibres, resulting in a mix of fibres with different properties that have more difficult application in textile products, if the feedstock is not homogenous. Consequently, single-fibre compositions or most common blends such as polycotton are more likely to be recycled.

There are other recycling techniques that are operational at scale, which in addition provide high-quality outputs to be used back in the textiles sector, thus are further awarded. These are chemical recycling of pure cotton for the production of MMCFs, thermochemical treatments to recover cotton from cotton-rich blends, chemical recycling of PA6-rich blends and thermomechanical recycling of acrylics and polyester-rich blends.

Considering all above, a product would be non-recyclable and thus awarded a score of 0 pt if contains elastane in proportions higher than 15% (20% for PA6-rich (> 80%) blends). If it contains less, the product would be considered recyclable, starting with 1 pt and summing points up to 10 pt for the reasons described in **Table 76**. The highest scores would be awarded to those products for which textile-to-textile recycling is feasible in line with the objectives set in the EU textile strategy.

Table 76. Points to be awarded for the recyclability score

Area	Criteria	Points
Because of being recyclable	If it contains elastane in proportions lower than 15% (20% for PA6-rich (> 80%) blends)	1 pt
Because of facilitating sorting	If it has same inner and outer composition	+1 pt
Because of facilitating pre-treatment	If it is free of printings	+1 pt
	If it is free of coatings	+1 pt
	If it is free of sequins	+1 pt
	If it is free of dyes	+1 pt
	If it is mono-material	+1 pt
Because of the recycling techniques at operational scale	If it can be mechanically recycled (*)	+1 pt
	If it is pure cotton, since it can be chemically recycled	+2 pt
	If it is composed of cotton-rich blends (> 60%), since they can be thermo-chemically recovered to cotton	+2 pt
	If it is composed of PA6-rich blends (> 80%), since they can be chemically recycled	+2 pt
	If it is composed of acrylics or polyester-rich blends (> 80%), since they can be thermo-mechanically recycled	+2 pt

(*) Note that all garments containing elastane in proportions lower than 15% can be mechanically recycled, thus scoring at least 2 pt.

Source: own elaboration.

11.1.2.2 Estimates for the assessment

The assessment of the impact of an information requirement on the recyclability of the products will be based on the following assumptions:

- The average **share going to recycling** will increase by 5%. As in the case of robustness, this is estimated based on the experience obtained from the Energy Labelling framework, which has shown that products for which information requirements have been defined have improved their performance by 6 – 14% over a period of 5 years, as compared to the Business-as-Usual scenario ⁽²²⁰⁾. While acknowledging the differences between energy efficiency and recyclability, this assumption is made in the absence of a more accurate benchmark. It is based on the assumption that, regardless of the product aspect reported, consumers would respond similarly to information presented on a label, adjusting their

²²⁰ These figures have been calculated based on the data published in Leo Wierda, (2024), for selected product groups for which the effect of the energy labelling could be decoupled from the introduction of ecodesign implementing measures to remove the effect of any performance requirement.

purchasing decisions towards better-performing products. This consumer behaviour is expected to lead to comparable improvements, irrespective of the dimension (or product aspect) being considered. Analogously, manufacturers are expected to also behave in the same way to avoid showing performance below that of their competitors. A higher impact of the requirement on the average share going to recycling is not assumed since the criteria mostly applied by consumers when purchasing apparel products are price and perceived quality, which are not connected to the recyclability (section 6.2).

For the revision of this estimate, specific surveys would need to be performed to analyse how consumers and manufacturers would react to an information label on the recyclability of textile apparel.

- An additional **cost of labelling** is also considered and estimated to be of 0.02 €/unit. This is based on the analysis conducted by Marschinski, R. et al., (2025). The smallest value from their analysis was chosen since apparel products already possess labels and, therefore, it would not be necessary to create a new one; the information can simply be included in the existing format.

Additional processes or costs during the manufacturing are not foreseen under this scenario. Moreover, no additional cost for testing is considered as the criteria can be assessed by the manufacturer based on the product's known composition. Similarly, the products can be verified by the Market Surveillance Authorities.

11.1.2.3 Results – D02

The results of the environmental and economic assessment of the D02 are shown in **Table 77**, **Table 78** and **Table 79** for knitted, denim and other woven products, respectively. The results indicate that the information requirement on recyclability could achieve a slight reduction of the environmental impact at the end of life: 0.2% for denim and 0.1% for knitted and other woven products as compared to the base cases. However, this improvement is not noticeable in the overall results due to the relatively small contribution of this lifecycle stage to the entire lifecycle. Nevertheless, as previously indicated, this design option is expected to have additional benefits by increasing the availability of secondary materials that can be used as recycled content, which is addressed separately in the following design option.

Moreover, when implemented alone, the information requirement on recyclability leads to increased costs both during manufacturing—due to the additional labelling costs—and during the end-of-life stage—being recycling more expensive than the waste management in landfills commonly used in the base cases. This results in an increase in costs according to the conventional LCC: 0.4% for knitted products, 0.2% for denim, and 0.3% for other woven products. When environmental benefits are internalised, these cost increases are slightly offset, yet societal costs remaining higher than the base cases by 0.3% for knitted products, 0.1% for denim, and 0.2% for other woven products.

Table 77. Results of the design option 2 (recyclability) for knitted products

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score (environmental points)	1.58E-03	5.78E-04	1.45E-04	2.02E-04	1.00E-05	2.51E-03
Single score compared to the base case	0.0%	0.0%	0.0%	0.0%	-0.1%	0.0%
Internal costs (LCC)	2.02 €	4.10 €	0.30 €	1.22 €	0.05 €	7.70 €
LCC compared to the base case	0.0%	0.8%	0.0%	0.0%	0.2%	0.4%
Societal costs (sLCC = internal costs + environmental externalities)	4.11 €	5.35 €	0.58 €	1.56 €	0.09 €	11.69 €

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
sLCC compared to the base case	0.0%	0.6%	0.0%	0.0%	-0.1%	0.3%

LCS: Life Cycle Stage; LCC: Life Cycle Costing; sLCC: societal Life Cycle Costing

Source: own elaboration.

Table 78. Results of the design option 2 (recyclability) for denim products

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score (environmental points)	3.07E-03	1.08E-03	1.78E-04	7.92E-04	1.41E-05	5.14E-03
Single score compared to the base case	0.0%	0.0%	0.0%	0.0%	-0.2%	0.0%
Internal costs (LCC)	2.97 €	6.32 €	0.23 €	4.65 €	0.06 €	14.22 €
LCC compared to the base case	0.0%	0.4%	0.0%	0.0%	0.3%	0.2%
Societal costs (sLCC = internal costs + environmental externalities)	6.42 €	8.86 €	0.57 €	5.96 €	0.11 €	21.93 €
sLCC compared to the base case	0.0%	0.3%	0.0%	0.0%	-0.2%	0.1%

LCS: Life Cycle Stage; LCC: Life Cycle Costing; sLCC: societal Life Cycle Costing

Source: own elaboration.

Table 79. Results of the design option 2 (recyclability) for other woven products

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score (environmental points)	1.68E-03	8.08E-04	1.43E-04	1.43E-04	1.01E-05	2.79E-03
Single score compared to the base case	0.0%	0.0%	0.0%	0.0%	-0.1%	0.0%
Internal costs (LCC)	2.36 €	4.89 €	0.27 €	0.85 €	0.05 €	8.42 €
LCC compared to the base case	0.0%	0.5%	0.0%	0.0%	0.2%	0.3%
Societal costs (sLCC = internal costs + environmental externalities)	4.81 €	6.69 €	0.55 €	1.09 €	0.09 €	13.22 €

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
sLCC compared to the base case	0.0%	0.4%	0.0%	0.0%	-0.1%	0.2%

LCS: Life Cycle Stage; LCC: Life Cycle Costing; sLCC: societal Life Cycle Costing

Source: own elaboration.

Section 13.11.10.5 reports the results of the environmental assessment of the D02, for each impact category and expressed in their respective units.

11.1.3 D03 - Product with higher recycled content

Section 10.1.2 reports that 60-63% of environmental impacts are generated by raw material production. The main measure proposed for reducing these environmental impacts relates to the use of secondary materials by increasing the recycled content in apparel products through an information and/or a performance requirement.

On the one hand, an information requirement is expected to encourage consumers and manufacturers towards products with higher recycled content. On the other hand, a performance requirement would establish minimum thresholds.

11.1.3.1 Potential information and performance requirements on recycled content

In contrast to the scoring systems defined for the other product aspects, a potential information requirement on the recycled content would report the fraction (in weight) of the secondary material used in the product. Further details and rules for its calculation should be defined in a separate study supporting the policy making process.

In addition, including information on the type of waste from which the material originates could further encourage the preference for post-consumer waste recycling over post-industrial, or for fibre-to-fibre recycling over open-loop, such as that derived from PET bottles. This additional level of granularity in the information would, foreseeably, be useful to articulate other policy instruments such as Green Public Procurement and EPR modulation fees.

As far as it concerns a potential performance requirement for recycled content, setting the thresholds requires an optimisation exercise that examines the environmental and economic impacts of different performance levels that could be proposed for each of the representative products, as well as the analysis of the availability of secondary materials (more details in section 13.12). Table 80 describes the different design options assessed for the recycled content of each of the product categories (knitted, denim and other woven products).

Table 80. Description of the design options assessed for the recycled content in knitted, denim and other woven products

Scenario	Denim products	Knitted and other woven products
BC	0% recycled content	0% recycled content
D03.2.1	10% recycled cotton	5% recycled nylon
D03.2.2	15% recycled cotton	5% recycled nylon + 5% recycled wool
D03.2.3	20% recycled cotton	5% recycled nylon + 10% recycled wool
D03.2.4	20% recycled cotton + 5% recycled polyester	5% recycled nylon + 10% recycled wool + 5% recycled polyester
D03.2.5	20% recycled cotton + 10% recycled polyester	5% recycled nylon + 10% recycled wool + 10% recycled polyester
D03.2.6	20% recycled cotton + 15% recycled polyester	5% recycled nylon + 10% recycled wool + 15% recycled polyester

Knitted: knitted products; Denim: products with denim fabric; Other woven: woven products without denim fabric; BC: base case; DO: design option.

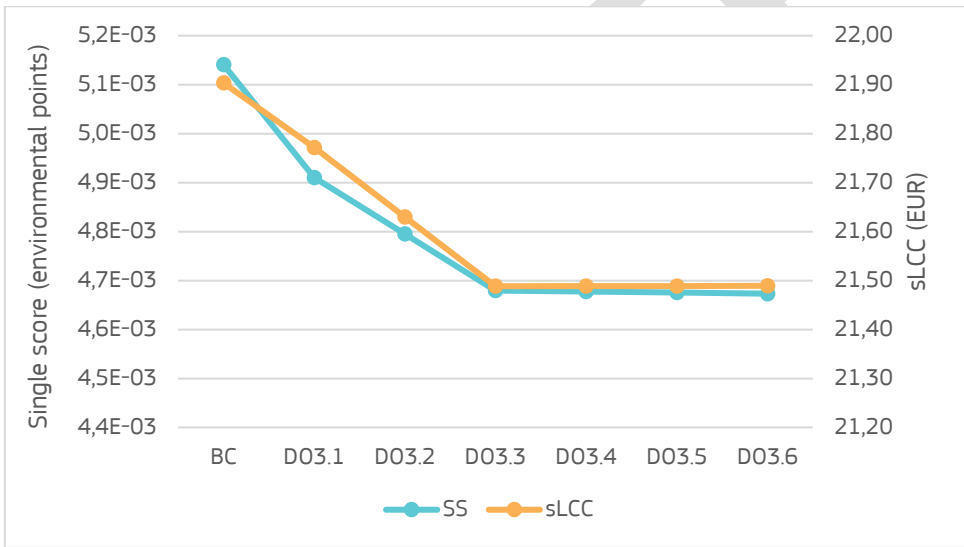
Source: own elaboration.

For denim products, the possibility to incorporate recycled cotton and polyester is considered for being the primary fibres in this product type (Figure 50). The assessment indicates that increasing the recycled cotton content to 20% enhances the environmental performance by up to 8.9% compared to the base case. In contrast, using recycled polyester offers minimal additional benefits, with only a 9.1% improvement even at a 15%

inclusion rate. Furthermore, the economic assessment shows that societal costs decrease significantly by up to 1.9% with recycled cotton, whereas the monetisation of the environmental benefits does not improve the results further in the case of the recycled polyester and increases internal costs due to the high price of secondary materials.

As a result, a threshold of 20% recycled cotton is proposed for denim products. This threshold is deemed technically feasible by Maud Kuppen (2024), which asserts that adding up to 30% pre-consumer recycled textiles generally does not affect product quality or processability. Moreover, an assessment of secondary material availability confirms there is sufficient recycled cotton to meet this requirement. According to the Textile Exchange (2024), the global market currently holds 0.32 million tons of recycled cotton. Estimating that 18% of the global market corresponds to the EU (UniformMarket, 2025), approximately 0.06 million tons would be available for use in the EU market. This accounts for 28% of the cotton used in denim, based on apparent consumption data (PRODCOM dataset³²) and composition data from Bakowska, O. et al., (2025).

Figure 50. Results of the environmental and economic assessment of including different fractions of recycled content in denim products



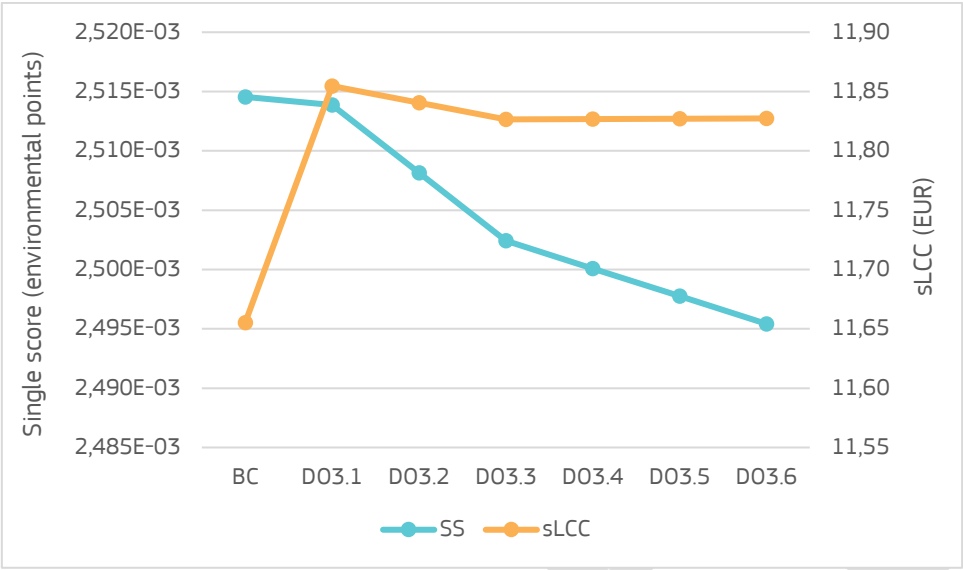
SS: single score; sLCC: societal Life cycle costing

Source: Own elaboration

This would leave the consideration of thresholds on recycled nylon, wool and polyester in knitted and other woven products. The results in **Figure 51** and **Figure 52** show that increasing the fractions of secondary materials for any of these fibres would allow a consistent decrease of the environmental impact of the products. Furthermore, the assessment indicates that although there is an initial rise in costs due to one-off expenses on certification, increasing the recycled content ultimately lowers societal costs, reflecting a stable trend when the recycled polyester shares are considered. Overall, the analysis shows a 0.8% improvement of the environmental impact of knitted product that would translate into a 1.5% increase of the societal costs for the configuration of the following recycled thresholds: 5% for nylon, 10% for wool and 15% for polyester. For other woven products, the same thresholds would allow the improvement of the environmental impact by 0.9%, with a rise in the societal costs of 1.0%.

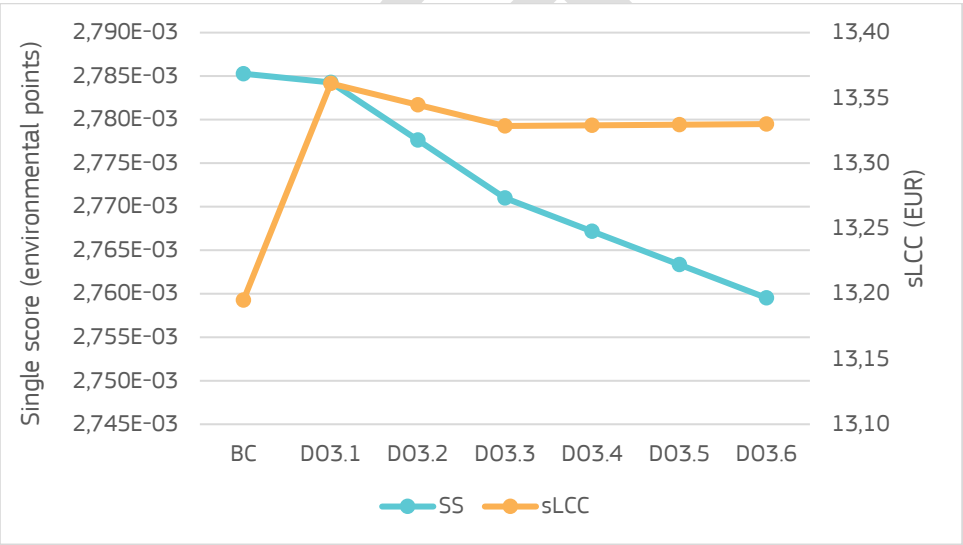
In parallel, the analysis confirms the availability of secondary materials to meet the potential requirements. According to the Textile Exchange (2024), the global market contains 0.1 million tons of recycled nylon. When adjusted for the EU market size, this equates to 0.02 million tons available for use, accounting for 9% of the nylon used in knitted and other woven products, based on apparent consumption data (PRODCOM dataset) and composition data from Bakowska, O. et al., (2025). A similar assessment indicates that the supply of recycled polyester is sufficient to meet the entire demand for this fibre in EU knitted and woven products, with 8.9 million tons available globally and 1.6 million tons allocated for the EU. Lastly, data from the recycling industry in the Prato district (Italy) reveals that 0.2 million tons of high-quality wool are recovered annually, allowing for up to 17% recycled content in knitted and other woven products, according to apparent consumption (PRODCOM dataset) and composition data from Bakowska, O. et al., (2025).

Figure 51. Results of the environmental and economic assessment of including different fractions of recycled content in knitted products



SS: single score; sLCC: societal Life cycle costing
Source: Own elaboration

Figure 52. Results of the environmental and economic assessment of including different fractions of recycled content in other woven products



SS: single score; sLCC: societal Life cycle costing
Source: Own elaboration

11.1.3.2 Estimates for the assessment

The assessment of the impact of information and performance requirement on the recycled content of the products is based on the following assumptions:

- When only information is considered, the effect of the requirement would be an **increase in the share of secondary materials** for certain fibres. For denim products, a 5% recycled content is estimated for cotton and a 2% for polyester. For knitted and other woven products, a 2% is estimated for polyester and 0.5% for nylon, wool and cotton. The estimates are based on expert judgement reflecting the

product's primary composition, the availability of secondary materials and the potential effect of a label on consumer and manufacturer choices.

When the performance requirements are considered, the impacts are determined by the thresholds set in the previous section 11.1.3.1. These are: 20% of the cotton in denim; 5% of the nylon, 10% of the wool and 15% of the polyester in knitted and other woven products.

If coupled together with an information requirement, an additional increase of 5% over those fractions is estimated, in line with the assumptions on improvements due to information requirements on the different product aspects in the previous sections 11.1.1.2 and 11.1.2.2.

For the revision of this estimate, specific surveys would need to be performed to analyse how consumers and manufacturers would react to an information label on the recycled content of textile apparel products.

- An additional **cost of labelling** is considered and estimated to be of 0.02 €/unit. This is based on the analysis conducted by Marschinski, R. et al., (2025). The smallest value from their analysis was chosen since apparel products already possess labels and, therefore, it would not be necessary to create a new one; the information can simply be included in the existing format.

- Moreover, the additional **cost of certification** is considered and estimated to be of 0.13€/ unit. This assumption is based on the price of certification declared by existing schemes that report ranges around €2 600–€4 333. When trying to allocate this cost to one item, the prices would range between €0.26 – €0.43 for small annual facility outputs of 10 000 units; €0.052 – €0.087 for medium annual facility outputs of 50 000 units; €0.026 – €0.043 for large annual facility outputs of 100 000 units; and <€0.0087 and <€0.0043 for even larger annual facility outputs for 500 000 and 1 000 000 units. For the purpose of this report, a value between the small and medium ranges is used.

- Finally, the **prices of the secondary materials** need to be considered, as different from the virgin ones. The following **Table 81** reports the prices used for the assessment.

Table 81. Prices of recycled fibres used in the assessment

Location	Fibre type	Price (€/kg)	Sources
GLO	Recycled PET	1.840	Link
CN	Recycled PET	1.068	Elaborated from link , link
EU	Recycled PET	2.430	Link
Non-EU	Polyamide Recycled	3.375	Link
EU	Polyamide Recycled	3.375	Elaborated from link and link
Non-EU	Cotton Recycled	2.648	Elaborated from link and Khan et al., (2025)
EU	Cotton Recycled	2.693	Elaborated from link
Non-EU	Wool Recycled	3.560	Link
EU	Wool Recycled	3.560	Link

GLO: global; CN: China; EU: European Union; PET: polyethylene terephthalate.

Source: own elaboration.

11.1.3.3 Results – D03

The results of the environmental and economic assessment of an information requirement on the recycled content (D03.1) are shown in **Table 82**, **Table 83** and **Table 84** for knitted, denim and other woven products, respectively. The results indicate that this design option could achieve a reduction of the environmental impact of the raw materials: 3.8% in denim products, 0.4% in knitted and 0.3% for other woven products, as compared to the base cases. Consequently, the environmental impacts would decrease by 2.3% for denim, 0.2% for knitted products and other woven products. As expected, the greatest improvement is shown for denim products due to their primary composition of cotton and polyester, where incorporating fractions of secondary materials more significantly increases the overall recycled content in the product.

Cost analysis reveals varying outcomes across product categories due to differences in composition and fluctuating prices of secondary materials by fibre type. While internal costs consistently rise at the manufacturing stage—due to additional costs of labelling and certification—showing increases of 5.7% for knitted, 2.8% for denim, and 4.0% for other woven products, the costs of the raw materials differ. Specifically,

their change is not noticeable in knitted and other woven products while decrease by 0.4% for denim, owing to the lower price of recycled cotton compared to virgin cotton. In total, this results in internal cost increases of 3.0% for knitted, 1.2% for denim, and 2.3% for other woven products relative to the base cases.

When considering the societal LCC, the savings in the external costs due to the environmental benefits partially compensate the rises in the internal cost. Overall, the increase in costs is limited to 1.9% for knitted products and 1.4% for woven, while denim achieves an almost complete costs offset (increase by 0.2%).

Table 82. Results of the design option 3.1 (recycled content, information only) for knitted products

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score (environmental points)	1.57E-03	5.78E-04	1.45E-04	2.02E-04	1.00E-05	2.51E-03
Single score compared to the base case	-0.4%	0.0%	0.0%	0.0%	0.0%	-0.2%
Internal costs (LCC)	2.02 €	4.30 €	0.30 €	1.22 €	0.05 €	7.90 €
LCC compared to the base case	0.0%	5.7%	0.0%	0.0%	0.0%	3.0%
Societal costs (sLCC = internal costs + environmental externalities)	4.10 €	5.55 €	0.58 €	1.56 €	0.09 €	11.88 €
sLCC compared to the base case	-0.2%	4.4%	0.0%	0.0%	0.0%	1.9%

LCS: Life Cycle Stage; LCC: Life Cycle Costing; sLCC: societal Life Cycle Costing

Source: own elaboration.

Table 83. Results of the design option 3.1 (recycled content, information only) for denim products

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score (environmental points)	2.96E-03	1.08E-03	1.78E-04	7.92E-04	1.41E-05	5.02E-03
Single score compared to the base case	-3.8%	0.0%	0.0%	0.0%	0.0%	-2.3%
Internal costs (LCC)	2.96 €	6.47 €	0.23 €	4.65 €	0.06 €	14.36 €
LCC compared to the base case	-0.4%	2.8%	0.0%	0.0%	0.0%	1.2%
Societal costs (sLCC = internal costs + environmental externalities)	6.28 €	9.02 €	0.57 €	5.96 €	0.11 €	21.94 €
sLCC compared to the base case	-2.2%	2.0%	0.0%	0.0%	0.0%	0.2%

LCS: Life Cycle Stage; LCC: Life Cycle Costing; sLCC: societal Life Cycle Costing

Source: own elaboration.

Table 84. Results of the design option 3.1 (recycled content, information only) for other woven products

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score (environmental points)	1.68E-03	8.08E-04	1.43E-04	1.43E-04	1.01E-05	2.78E-03
Single score compared to the base case	-0.3%	0.0%	0.0%	0.0%	0.0%	-0.2%
Internal costs (LCC)	2.36 €	5.06 €	0.27 €	0.85 €	0.05 €	8.59 €
LCC compared to the base case	0.0%	4.0%	0.0%	0.0%	0.0%	2.3%
Societal costs (sLCC = internal costs + environmental externalities)	4.80 €	6.86 €	0.55 €	1.09 €	0.09 €	13.39 €
sLCC compared to the base case	-0.1%	2.9%	0.0%	0.0%	0.0%	1.4%

LCS: Life Cycle Stage; LCC: Life Cycle Costing; sLCC: societal Life Cycle Costing

Source: own elaboration.

When the information requirement on the recycled content is assessed in combination with a performance requirement (**Table 85, Table 86, Table 87**), the environmental benefits increase for the same additional costs of labelling and certification. Specifically, an improvement in the impact of the raw materials of up to 15.8% is achieved for denim products, and some 1.5% for knitted and other woven products. This translates in 9.4% improvement of the overall environmental profile of the denim products, 1.0% for other woven and 0.8% for knitted. Consequently, there is a further offset of the internal costs. The societal costs of denim products decreased by 1.9% respect to the base case and their increase is limited to 1.7% and 1.2% in the case of knitted and other woven, respectively.

Table 85. Results of the design options 3.1 and 3.2 (recycled content, information and performance requirements) for knitted products

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score (environmental points)	1.56E-03	5.78E-04	1.45E-04	2.02E-04	1.00E-05	2.49E-03
Single score compared to the base case	-1.3%	0.0%	0.0%	0.0%	0.0%	-0.8%
Internal costs (LCC)	2.02 €	4.30 €	0.30 €	1.22 €	0.05 €	7.90 €
LCC compared to the base case	0.1%	5.7%	0.0%	0.0%	0.0%	3.1%
Societal costs (sLCC = internal costs + environmental externalities)	4.07 €	5.55 €	0.58 €	1.56 €	0.09 €	11.86 €
sLCC compared to the base case	-0.8%	4.4%	0.0%	0.0%	0.0%	1.7%

LCS: Life Cycle Stage; LCC: Life Cycle Costing; sLCC: societal Life Cycle Costing

Source: own elaboration.

Table 86. Results of the design options 3.1 and 3.2 (recycled content, information and performance requirements) for denim products

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score (environmental points)	2.59E-03	1.08E-03	1.78E-04	7.92E-04	1.41E-05	4.66E-03
Single score compared to the base case	-15.8%	0.0%	0.0%	0.0%	0.0%	-9.4%
Internal costs (LCC)	2.92 €	6.47 €	0.23 €	4.65 €	0.06 €	14.32 €
LCC compared to the base case	-1.7%	2.8%	0.0%	0.0%	0.0%	0.9%
Societal costs (sLCC = internal costs + environmental externalities)	5.83 €	9.02 €	0.57 €	5.96 €	0.11 €	21.48 €
sLCC compared to the base case	-9.3%	2.0%	0.0%	0.0%	0.0%	-1.9%

LCS: Life Cycle Stage; LCC: Life Cycle Costing; sLCC: societal Life Cycle Costing

Source: own elaboration.

Table 87. Results of the design options 3.1 and 3.2 (recycled content, information and performance requirements) for other woven products

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score (environmental points)	1.65E-03	8.08E-04	1.43E-04	1.43E-04	1.01E-05	2.76E-03
Single score compared to the base case	-1.6%	0.0%	0.0%	0.0%	0.0%	-1.0%
Internal costs (LCC)	2.36 €	5.06 €	0.27 €	0.85 €	0.05 €	8.60 €
LCC compared to the base case	0.3%	4.0%	0.0%	0.0%	0.0%	2.4%
Societal costs (sLCC = internal costs + environmental externalities)	4.77 €	6.86 €	0.55 €	1.09 €	0.09 €	13.35 €
sLCC compared to the base case	-0.8%	2.9%	0.0%	0.0%	0.0%	1.2%

LCS: Life Cycle Stage; LCC: Life Cycle Costing; sLCC: societal Life Cycle Costing

Source: own elaboration.

Section 13.11.10.5 reports the results of the environmental assessment of the D03, for each impact category and expressed in their respective units.

11.1.4 D04 - Product with a decreased environmental or carbon footprint

The results of the hotspot analysis in the previous section 10.1.2 also show the significant impact of the manufacturing stage (over 20% of the whole lifecycle), ranking second following the impact of raw materials. Within this impact, the climate change category contributes the most as compared to the rest of the impact categories (29%-33% of the single score of the LCS2 and 6% of the whole lifecycle).

With the aim of addressing these impacts, an information requirement on the environmental footprint (i.e., in terms of the single score) (DO4.1) or on the carbon footprint (the most contributing impact category) (DO4.2) of the manufacturing stage is proposed. While the former is expected to have a greater effect in terms of environmental improvement because of the broader scope that includes all the impact categories, the latter could represent less effort for reporting and verification of fewer data elements and parameters.

Despite the high impact of the raw materials and thus the interest in capturing it in a footprint indicator to provide a more comprehensive picture to the consumer, this lifecycle stage is left out of the scope of the proposed indicators. The reason for its exclusion is the inequivalent choice of system boundaries used for the datasets available for the assessment of the several types of fibres⁽²²¹⁾, which prevent their fair comparison. The design options on robustness, recyclability and recycled content (detailed in the sections from 11.1.1 to 11.1.3) were defined to specifically address the hotspot of LCS1 through alternative requirements.

On the one hand, the information requirement would help consumers to make informed purchases on products manufactured through processes that are less harmful to the environment. On the other hand, it would provide a harmonised framework for manufacturers to assess and present their performance, encouraging them to improve their processes and invest in better technologies and techniques to produce products with lower environmental impacts than their competitors.

As a limitation of these design options, it is important to highlight the high administrative burdens associated to the calculation of such indicators because of the complexity of the apparel supply chain, which is exceptionally long and geographically dispersed with numerous tiers. This presents significant challenges for manufacturers in obtaining information, as well as in terms of difficulty of verification.

The concept of setting a performance requirement in terms of on the environmental or carbon footprints is not further taken forward in this report due to the lack of data, that prevented to appropriately benchmark the products currently on the market.

11.1.4.1 Potential information requirement on the environmental or carbon footprint

For the calculation of the environmental or carbon footprint indicator, it is proposed that the rules defined in the PEFCR for apparel and footwear are followed (as applicable to the lifecycle stage 2 only – manufacturing stage). This allows the use of secondary datasets and default values that facilitate the calculations in absence of primary data.

Since displaying the absolute value of the environmental / carbon footprint as a standalone indicator might be challenging for consumers to interpret (mainly in the absence of a benchmark), an information requirement is proposed that would simply indicate whether a product's performance is better than the average (and to which extent, in %). The underlying concept is that manufacturers who achieve high performance would be motivated to calculate the environmental / carbon footprint to showcase their 'excellence'. Specifically, a system could be defined whereby a label would be awarded to products where the manufacturer demonstrates that their manufacturing environmental / carbon footprint is superior to the benchmark set by the PEFCR for the representative product in that product category, indicating also by how much (in %).

The proposed approach would allow the voluntary reporting of the footprint indicator. Those not providing the calculations would simply not be eligible for the 'excellence' labelling. This would alleviate administrative burdens for manufacturers compared to requiring everyone to report absolute values.

The manufacturers reporting the footprint indicator would be required to also indicate whether primary or secondary data was used. A different label or label elements would be used for the former, thereby encouraging the use of primary data.

This approach provides manufacturers with an incentive to assess their performance accurately, while allowing consumers to easily identify products that exceed average performance standards.

²²¹ For natural materials, a dataset based on the closest animal or vegetal species shall be chosen. For synthetic materials, the closest material based on the synthesis route of the polymer shall be selected (Quantis, 2025a).

11.1.4.2 Estimates for the assessment

The assessment of the impact of an information requirement on the environmental and carbon footprint of the manufacturing of the textiles apparel products will be based on the following assumptions:

- For the environmental footprint (DO4.1), the average **impact of the manufacturing processes** will decrease by 3%. This results from assuming that all the impact categories improve homogeneously (Quantis, 2025a). Compared to the impacts of the robustness and recyclability scores, the effect of this information requirement is estimated to be lower for two reasons. On the one hand, the manufacturers may opt not to calculate their footprints, as the current definition of the requirement allows that only those wishing to showcase above-average performance do so. This voluntary nature could reduce the effect of the requirement. On the other hand, allowing the use of secondary data may not accurately reflect real performance. Manufacturers might exploit this by presenting their performance better than it actually is, in the cases where their primary data reflects worse performance than that calculated using the secondary data, thereby limiting the potential effectiveness of this requirement.

As regards the carbon footprint (DO4.2), the average **impact of the manufacturing processes under the climate change category** is assumed to decrease by 3%. Compared to the impacts of the robustness and recyclability scores, the effect of this information requirement is also estimated to be lower as the rationale described for the DO4.1. also applies in this case.

A higher impact from these requirements is not assumed since the criteria mostly applied by consumers when purchasing apparel products are price and perceived quality, which are not connected to the environmental impacts of the product (section 6.2).

For an improved appraisal of these estimates, specific surveys would need to be performed to analyse how consumers and manufacturers would react to an information label on the carbon and environmental footprint addressing about 6% and 20%, respectively, of the lifecycle environmental impacts of textile apparel products.

- An additional **cost of labelling** is considered and estimated to be of 0.02 €/unit for both design options. This is based on the analysis conducted by Marschinski, R. et al., (2025). The smallest value from their analysis was chosen since apparel products already possess labels and, therefore, it would not be necessary to create a new one; the information can simply be included in the existing format.

- Additional **administrative costs** for the implementation of these design options are considered and estimated based on the impact assessment carried out in support of Regulation (EU) 2023/1542 concerning carbon footprint thresholds for industrial and EV batteries, as these measures would require similar calculation and verification of indicators. It is however acknowledged that the differences between the value chains of batteries and textile apparel could largely affect the uncertainties of these estimates.

In the case referred, the impact assessment estimated one-off costs per battery type in the range of €100–5 000, depending on the availability of the company-specific data needed and consultancy fees. Additional verification costs would be €2 000–7 000 per battery type with additional small follow-up costs.

To adapt cost assessments for the apparel sector, an average cost scenario is assumed, with an initial cost of €2 550 and a verification cost of €4 500 per garment model (representing the mean values of the figures provided in the batteries report). However, attributing these costs to individual products is challenging due to variations in production volumes influenced by manufacturer size, market demand, and product type.

Knitted fabrics are very common in garments, which generally implies higher production volumes (with some 100 000 items per model). Woven fabrics are common in products like shirts and pants and production volumes may be lower than for knitted fabrics due to less frequent style rotations (with some 80 000 items per model). Finally, denim products are typically higher-cost items, with less dynamic production cycles focused on quality and timeless designs, resulting in lower production volumes (with some 50 000 items per model).

Thus, estimates for the medium cost and medium scenario for each product type would result in €7 050 / 100 000 = €0.07 per item for knitted products, €7 050 / 50 000 = €0.14 per item for denim, and €7 050 / 80 000 = €0.09 per item for other woven products.

Since the approach proposed allows manufacturers to omit reporting their footprint indicator when they do not want to showcase a superior performance, they would not need to incur in the administrative cost of their calculation and verification. Thus, when assessing the average additional costs associated to these design options, it is important to consider their proportionality in relation to the percentage of manufacturers who would adopt them. Assuming a 50% adoption rate, the costs per item would be halved to €0.035 for knitted products, €0.07 for denim, and €0.045 for other woven products.

Nevertheless, it is important to note that costs can vary significantly based on production volumes and levels of spending on implementation and verification. In low-cost, high-production scenarios, the cost per item can be minimal. Conversely, with high implementation costs and lower production volumes, the cost per item increases substantially.

The costs estimates are assumed to be the same for both design options (environmental and carbon footprint) in absence of any reference that would allow fine tuning of these estimates, at the time of writing. However, it is acknowledged that collecting primary data and verifying it could be more costly for the former, due to the greater scope and thereby the additional flows to be considered.

Finally, the authors highlight that estimation of the administrative costs should include the verification of two types of information:

- (a) The input data (primary or secondary data) from specific industrial sites, which are representative of the supply chain of the specific product model described. The representativeness of input data should be documented and verified.
- (b) The correct application of the PEFCR framework.

As a limitation of the study, it is important to highlight that additional costs due to the use of more costly technologies or techniques for achieving a better performance are not considered because of the lack of sufficient data to provide an accurate estimate.

11.1.4.3 Results – DO4

The results of the environmental and economic assessment of the DO4.1 (environmental footprint during manufacturing) are shown in **Table 88**, **Table 89**, **Table 90** for knitted, denim and other woven products, respectively. This information requirement could reduce the environmental impact of that lifecycle stage by some 3.0%, meaning 0.7%, 0.6% and 0.9% reduction for knitted, denim and other woven products respectively over the whole lifecycle.

Implementing this design option would result in a rise in the internal costs of 1.9% for knitted, 1.3% for denim and 1.7% for other woven products, mainly because of the additional administrative costs for the calculation and verification of this indicator. Overall, while the reduction of the environmental impact results in lower external costs as compared to the base cases, the total societal costs are still higher across all product categories: by 1% for knitted products, 0.7% for denim and 0.8% for other woven.

Table 88. Results of the design option 4.1 (environmental footprint) for knitted products

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score (environmental points)	1.58E-03	5.60E-04	1.45E-04	2.02E-04	1.00E-05	2.50E-03
Single score compared to the base case	0.0%	-3.0%	0.0%	0.0%	0.0%	-0.7%
Internal costs (LCC)	2.02 €	4.16 €	0.30 €	1.22 €	0.05 €	7.75 €

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
LCC compared to the base case	0.0%	2.1%	0.0%	0.0%	0.0%	1.1%
Societal costs (sLCC = internal costs + environmental externalities)	4.11 €	5.36 €	0.58 €	1.56 €	0.09 €	11.70 €
sLCC compared to the base case	0.0%	0.9%	0.0%	0.0%	0.0%	0.4%

LCS: Life Cycle Stage; LCC: Life Cycle Costing; sLCC: societal Life Cycle Costing

Source: own elaboration.

Table 89. Results of the design option 4.1 (environmental footprint) for denim products

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score (environmental points)	3.07E-03	1.05E-03	1.78E-04	7.92E-04	1.41E-05	5.11E-03
Single score compared to the base case	0.0%	-3.0%	0.0%	0.0%	0.0%	-0.6%
Internal costs (LCC)	2.97 €	6.40 €	0.23 €	4.65 €	0.06 €	14.31 €
LCC compared to the base case	0.0%	1.7%	0.0%	0.0%	0.0%	0.8%
Societal costs (sLCC = internal costs + environmental externalities)	6.42 €	8.87 €	0.57 €	5.96 €	0.11 €	21.93 €
sLCC compared to the base case	0.0%	0.3%	0.0%	0.0%	0.0%	0.1%

LCS: Life Cycle Stage; LCC: Life Cycle Costing; sLCC: societal Life Cycle Costing

Source: own elaboration.

Table 90. Results of the design option 4.1 (environmental footprint) for other woven products

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score (environmental points)	1.68E-03	7.84E-04	1.43E-04	1.43E-04	1.01E-05	2.76E-03
Single score compared to the base case	0.0%	-3.0%	0.0%	0.0%	0.0%	-0.9%
Internal costs (LCC)	2.36 €	4.95 €	0.27 €	0.85 €	0.05 €	8.48 €
LCC compared to the base case	0.0%	1.7%	0.0%	0.0%	0.0%	1.0%
Societal costs (sLCC = internal costs + environmental externalities)	4.81 €	6.69 €	0.55 €	1.09 €	0.09 €	13.23 €

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
sLCC compared to the base case	0.0%	0.5%	0.0%	0.0%	0.0%	0.2%

LCS: Life Cycle Stage; LCC: Life Cycle Costing; sLCC: societal Life Cycle Costing

Source: own elaboration.

When considering only the carbon footprint instead of all the impact categories through the single score indicator, lower environmental improvements are achieved. The results of the environmental and economic assessment of the D04.2 are shown in **Table 91**, **Table 92** and **Table 93** for knitted, denim and other woven products, respectively. The results indicate that the information requirement on the carbon footprint of the manufacturing could reduce the environmental impact of that lifecycle stage by some 1%, meaning some 0.2% reduction over the whole lifecycle for all the representative products.

Implementing this design option would result in a rise in the internal costs of 1.1% for knitted, 0.8% for denim and 1.0% for other woven products, mainly because of the additional administrative costs for the calculation and verification of this indicator. Overall, while the reduction of the environmental impact results in lower external costs as compared to the base cases, the total societal costs are still higher across all product categories: by 0.6% for knitted products, 0.3% for denim and 0.4% for other woven.

Table 91. Results of the design option 4.2 (carbon footprint) for knitted products

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score (environmental points)	1.58E-03	5.73E-04	1.45E-04	2.02E-04	1.00E-05	2.51E-03
Single score compared to the base case	0.0%	-0.9%	0.0%	0.0%	0.0%	-0.2%
Internal costs (LCC)	2.02 €	4.16 €	0.30 €	1.22 €	0.05 €	7.75 €
LCC compared to the base case	0.0%	2.1%	0.0%	0.0%	0.0%	1.1%
Societal costs (sLCC = internal costs + environmental externalities)	4.11 €	5.38 €	0.58 €	1.56 €	0.09 €	11.72 €
sLCC compared to the base case	0.0%	1.2%	0.0%	0.0%	0.0%	0.6%

LCS: Life Cycle Stage; LCC: Life Cycle Costing; sLCC: societal Life Cycle Costing

Source: own elaboration.

Table 92. Results of the design option 4.2 (carbon footprint) for denim products

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score (environmental points)	3.07E-03	1.07E-03	1.78E-04	7.92E-04	1.41E-05	5.13E-03

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score compared to the base case	0.0%	-1.0%	0.0%	0.0%	0.0%	-0.2%
Internal costs (LCC)	2.97 €	6.40 €	0.23 €	4.65 €	0.06 €	14.31 €
LCC compared to the base case	0.0%	1.7%	0.0%	0.0%	0.0%	0.8%
Societal costs (sLCC = internal costs + environmental externalities)	6.42 €	8.90 €	0.57 €	5.96 €	0.11 €	21.96 €
sLCC compared to the base case	0.0%	0.7%	0.0%	0.0%	0.0%	0.3%

LCS: Life Cycle Stage; LCC: Life Cycle Costing; sLCC: societal Life Cycle Costing

Source: own elaboration.

Table 93. Results of the design option 4.2 (carbon footprint) for other woven products

Indicator	LCS1 Raw Materials	LCS2 Manufacturing	LCS3 Distribution	LCS4 Use	LCS5 End of life	Total
Single score (environmental points)	1.68E-03	8.01E-04	1.43E-04	1.43E-04	1.01E-05	2.78E-03
Single score compared to the base case	0.0%	-0.9%	0.0%	0.0%	0.0%	-0.3%
Internal costs (LCC)	2.36 €	4.95 €	0.27 €	0.85 €	0.05 €	8.48 €
LCC compared to the base case	0.0%	1.7%	0.0%	0.0%	0.0%	1.0%
Societal costs (sLCC = internal costs + environmental externalities)	4.81 €	6.71 €	0.55 €	1.09 €	0.09 €	13.25 €
sLCC compared to the base case	0.0%	0.8%	0.0%	0.0%	0.0%	0.4%

LCS: Life Cycle Stage; LCC: Life Cycle Costing; sLCC: societal Life Cycle Costing

Source: own elaboration.

Section 13.11.10.5 reports the results of the environmental assessment of the DO4.1 and DO4.2, for each impact category and expressed in their respective units.

11.2 Combination of design options: paths

In this section, the design options are studied in combination, clustering in paths those that run in synergy and evaluating the possible trade-offs. To validate the cluster, the DOs are assessed in a cumulative way to ensure

that the environmental benefits of their combination are greater than those achieved by applying a single option alone and that the costs are not disproportionate. In the cumulative assessment, one DO is added at a time to evaluate different configurations within each path and identify the one providing the highest benefit.

In a **first path (P1)**, the following design options are clustered together: the information requirements on robustness, recyclability, recycled content and environmental footprint, as well as the performance requirement on recycled content.

As an alternative **second path (P2)**, the information requirement on the carbon footprint would replace the environmental footprint. These are understood as mutually exclusive, as the carbon footprint is a subset of the broader environmental footprint. Presenting both metrics simultaneously might confuse consumers, as they essentially convey similar insights into the environmental consequences of the production process, so they are studied under different paths to avoid potential misinformation. Thus, P2 clusters the information requirements on robustness, recyclability, recycled content and carbon footprint, as well as the performance requirement on recycled content.

In a **third path (P3)**, the focus is narrowed to only consider the requirements on the recyclability and recycled content. This cluster is explored separately as a less ambitious scenario, where the performance requirement is maintained, paired with the information requirements of only the strictly related parameters. The design options on the robustness and environmental/carbon footprint would thus not be included, relieving manufacturers from the additional efforts and costs of their implementation. Limiting the number of information requirements also reduces the uncertainty of the assessment, as the impact of the path relies less on the consumer choices (which cannot be as predictable as the effect of the thresholds imposed by the performance requirements).

The exclusion of the design options on robustness and environmental or carbon footprint avoids the challenges associated to their limitations described in sections 11.1.1 and 11.1.4.

The appropriateness and proportionality of pursuing one path or another could be reassessed in subsequent stages, following additional stakeholder input and further information that may be obtained during impact assessment.

11.2.1 Estimates for the assessment

Although optimising product aspects simultaneously might require design configurations that may not be compatible, most of the proposed measures are limited to information requirements meaning no inherent contradiction between the potential ecodesign requirements. For instance, increasing recycled content could potentially clash with improving robustness. Yet, providing information on both attributes does not as such pose any product design issues. In fact, their synergy could be explored as it may lead to cost optimisation—where at the same cost, a single label accommodates two items of information.

Nonetheless, the information could sometimes be contradictory since the same product could score high in one of the product aspects and low in the rest. Consequently, when assessing design options collectively, it is assumed that the impact of different information requirements will not be simply additive. Instead, a reduction in the label's effect related to the selective choices made by consumers is considered according to the following scenarios:

- Full Synergy Scenario: it is assumed that 50% of the time the information interacts synergistically, achieving the full improvement effect of 5% for each product aspect.
- Partial Adjacency Scenario: it is assumed that 35% of the time the information is not completely synergistic, leaving on the consumer the choice of which product aspects to prioritise. This would lead to a reduced improvement as compared to the full 5% improvement estimate for each product aspect.
- Independent Choice Scenario: it is assumed that 15% of the time, the consumer is predominantly influenced by one indicator that conflicts with the rest, achieving a 5% improvement only in one of the product aspects.

With these assumptions and their respective probabilities, the effect of the requirements is calculated as a weighted average for each of the configurations in the defined path.

1. When one design option is analysed alone, a 5% improvement is assumed.
2. When two design options are assessed together, 50% of the time both product aspects would improve by 5%. In 35% of cases, the effectiveness of the information would be reduced due to lack of full synergy.

Thus, for instance, while one product aspect would maintain a 5% improvement, the other would be reduced by half to 2.5%. Distributing this equally between parameters results in an effect of 3.75% for each product aspect. Lastly, in the scenario of independent choice, only one of the two pieces of information is considered, resulting in a 5% improvement which, when distributed between both product aspects, provides an improvement of 2.5% for each. Therefore, a 4.2% ($0.5 \cdot 5\% + 0.35 \cdot 3.75\% + 0.15 \cdot 2.5\%$) improvement is considered.

3. When a third design option is added, 50% of the time all product aspects would improve by 5%. In 35% of cases, the effectiveness of the label would decrease and, while one product aspect would maintain a 5% improvement, the others would be reduced by half to 2.5%. Equally distributing this results in an effect of 3.33% for each product aspect. Lastly, in the scenario of independent choice, only one of the pieces of information is considered, resulting in a 5% improvement, which when distributed between the three product aspects provides an improvement of 1.67% for each. Therefore, a 3.9% ($0.5 \cdot 5\% + 0.35 \cdot 3.33\% + 0.15 \cdot 1.67\%$) improvement is considered.

4. Finally, adding the fourth design option would decrease the improvement to some 3.7% ($0.5 \cdot 5\% + 0.35 \cdot 3\% + 0.15 \cdot 1.25\%$). This would result from the combination of the full synergy scenario with the partial adjacency in which a reduced improvement of 3% is estimated for each product aspect and with the independent choice scenario in which the 5% improvement is distributed between the four parameters.

The case of the effect of the information requirements on the environmental and carbon footprints is particular, since a lower impact on the improvement of these product aspects is expected because of its voluntary nature also when studied in combination with other design options. While the impact was estimated to be of 3% improvement when analysed in isolation, it is assumed to be of 2.5% when assessed within the cluster.

Other than that, the estimates for the rest of the parameters are kept as described in the sections 11.1.1.2, 11.1.2.2, 11.1.3.2 and 11.1.4.2 above.

11.2.2 Results

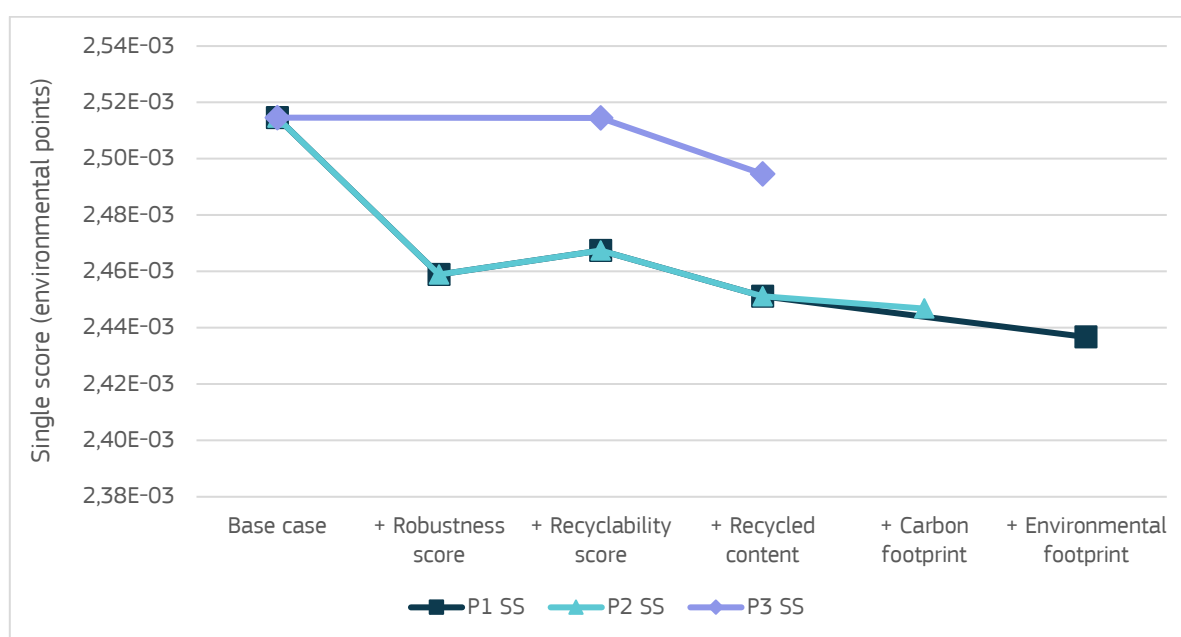
For each of the product categories, each path is assessed in a cumulative way, adding one DO at a time, showing how the different combinations of the design options would work together from an environmental and economic point of view. The cumulative assessment allows the identification of the configuration that provides the highest benefit within each of the paths. Then, the paths can be ranked by comparing their most convenient configurations.

Starting with knitted products (**Figure 53**), the results of the first path (P1) show that combining design options consistently improves the environmental profile of the product except for the cluster of the robustness and recyclability scores. This happens because the label's impact diminishes with additional information and the reduction of the effect of the robustness score is not compensated in this case by the improvement due to the increased recyclability. Any of the other combinations would offer environmental benefits as compared to the base case, with the complete path providing the greatest advantage (-3.1%).

Regarding the second path (P2), the cumulative clustering of design options behaves as described for P1. However, an information requirement on the carbon footprint would have a lower impact than that on the environmental footprint because of the consideration of additional impact categories. Consequently, the overall environmental benefit of this path would be reduced to 2.7% as compared to the base case.

In the third path (P3), the assessment is more conservative in the estimates of the impacts of the requirements, discarding the design options on the information requirements on robustness and environmental/carbon footprint and considering only those related to the recyclability and the recycled content. Also in this case, the environmental impact of the product would decrease (by 0.8%) as compared to the base case, offering a higher improvement than when applying one of the design options alone.

Figure 53. Results of the environmental assessment of the paths (cumulative LCA) for knitted products



SS: single score

Source: Own elaboration

Disaggregating the results of the environmental assessment into the most impactful impact categories (**Table 94**) allows the following conclusions:

- From the assessment in section 10.1.2.1, the impact categories contributing the most to the single score are water use (35%), climate change (16%), resource use – fossil (8.5%), particulate matters (6%), resource use – minerals and metals (5.5%) and acidification (5%).
- The complete path 1 shows improvements by 2% for water use (-2.4 m³ water eq of deprived water), 3.5% for climate change (-0.5 kg CO₂ eq), 3% for resource use fossil (-5.04 MJ), 4% for particulate matters (-3.9E-8 disease incidences), -5.5% for resource use – minerals and metals (-6.4E-6 kg Sb eq/person), -4.3% for acidification (-4.8E-03 mol H⁺ eq.) as compared to the base case.
- The complete path 2 shows improvements by 2% for water use (-2.3 m³ water eq of deprived water), 3.5% for climate change (-0.5 kg CO₂ eq), 2% for resource use fossil (-3.4 MJ), 3.3% for particulate matters (-3.2E-8 disease incidences), -5.1% for resource use – minerals and metals (-5.9E-6 kg Sb eq/person), -3.8% for acidification (-4.2E-03 mol H⁺ eq.) as compared to the base case.
- The complete path 3 shows improvements by 0% for water use (-0.05 m³ water eq of deprived water), 0.8% for climate change (-0.11 kg CO₂ eq), 0.5% for resource use fossil (-0.8 MJ), 2% for particulate matters (-1.6E-8 disease incidences), -3.6% for resource use – minerals and metals (-4.2E-6 kg Sb eq/person), -2% for acidification (-2.2E-03 mol H⁺ eq.) as compared to the base case.

Table 94. Results of environmental assessment for the impact categories with >5% contribution to the single score, for knitted products

Impact category	Unit	P1	P2	P3	P1	P2	P3
					As compared to the base case		
Water use	m ³ water eq of deprived water	-2.40	-2.34	-0.05	-2.0%	-2.0%	0.0%
Climate change	kg CO ₂ eq.	-0.50	-0.50	-0.11	-3.5%	-3.5%	-0.8%
Resource use fossils	MJ	-5.04	-3.37	-0.82	-3.0%	-2.0%	-0.5%
Particulate matter	disease incidences	-3.90E-08	-3.19E-08	-1.58E-08	-4.1%	-3.3%	-1.7%

Impact category	Unit	P1	P2	P3	P1	P2	P3
					As compared to the base case		
Resource use minerals and metals	kg Sb eq./person	-6.40E-06	-5.94E-06	-4.21E-06	-5.5%	-5.1%	-3.6%
Acidification	mol H+ eq.	-4.80E-03	-4.15E-03	-2.22E-03	-4.3%	-3.8%	-2.0%

P1: Path 1 (robustness score, recyclability score, recycled content information and performance requirement, environmental footprint), P2: Path 2 (robustness score, recyclability score, recycled content information and performance requirement, carbon footprint), P3: Path 3 (recyclability score, recycled content information and performance requirement)

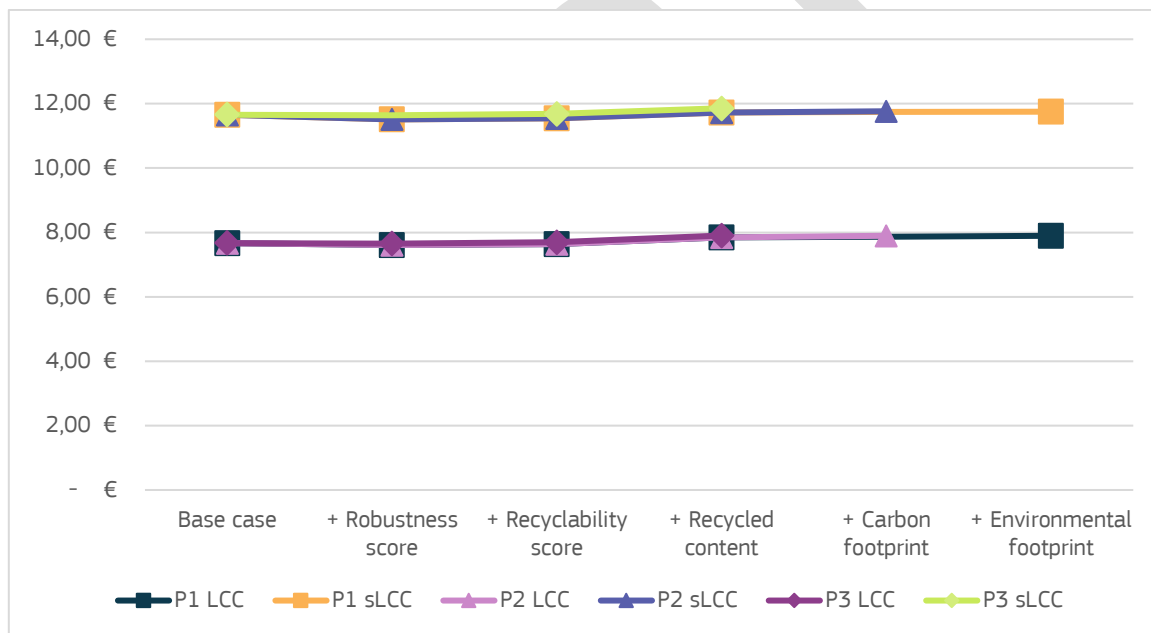
Source: own elaboration.

When looking at the economic assessment (**Figure 54**), the internal costs (LCC) in the first path decrease when implementing the robustness score but increase as we add other design options. For the complete path, the analysis shows that the 3.1% environmental improvement can be achieved with an increase in the internal costs of 3.0%. For the societal LCC, the internalisation of the environmental improvement limits the increase in costs to 0.8%.

The second path behaves similarly, with the same increase in internal costs. Since the environmental externalities are higher in this case than in the previous path, the societal costs increase slightly more (by 0.9% as compared to the base case).

As for the third path, all the options in the cluster lead to increases in costs. The 0.8% environmental improvement would be achieved with a rise in the internal costs of 3.1% (LCC), which is reduced to 1.7% when the environmental externalities are considered (sLCC).

Figure 54. Results of the economic assessment of the paths (cumulative LCC and sLCC) for knitted products

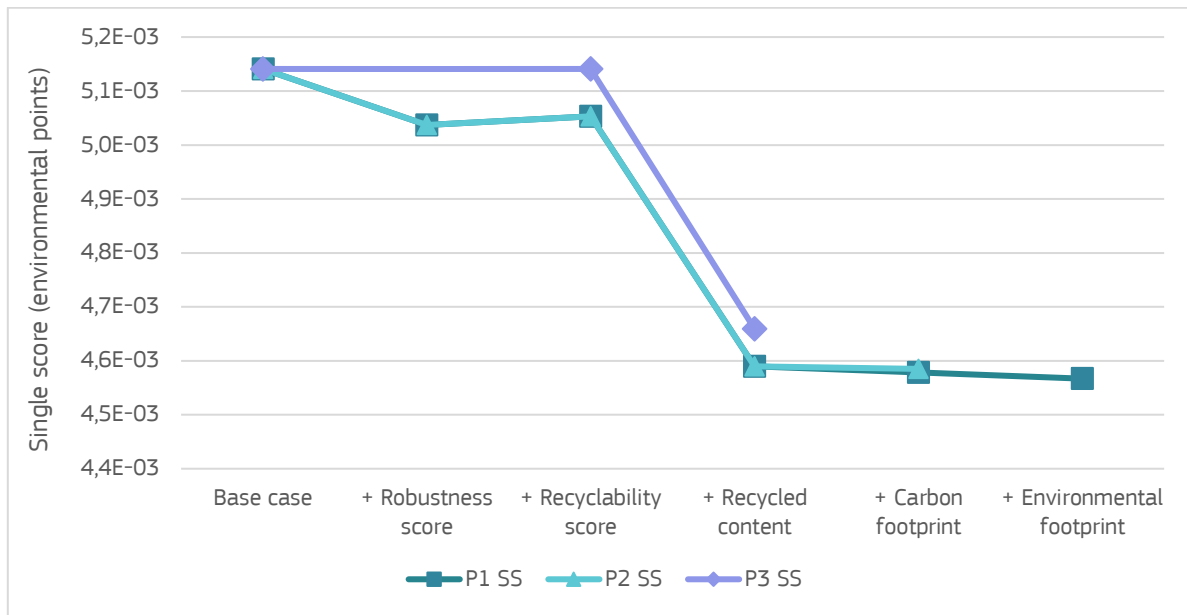


LCC: Life cycle costing; sLCC: societal Life cycle costing

Source: Own elaboration

Following with denim products (**Figure 55**), the results of the first path show again that combining design options consistently improves the environmental profile of the product except for the cluster of the robustness and recyclability scores. Any of the other combinations would offer environmental benefits as compared to the base case, with the complete path providing the greatest advantage (-11.2%). The second path behaves similarly, achieving slightly lower environmental benefit (-10.8%). Regarding the third path, the environmental impact of the product would also decrease as compared to the base case (by 9.3%), offering a higher improvement than when applying one of the design options alone.

Figure 55. Results of the environmental assessment of the paths (cumulative LCA) for denim products



SS: single score

Source: Own elaboration

Disaggregating the results of the environmental assessment into the most impactful impact categories (**Table 95**) allows the following conclusions:

- From the assessment in section 10.1.2.1, the impact categories contributing the most to the single score are water use (42%), climate change (15%), resource use - fossil (9%), eutrophication of freshwater (5%).
- The complete path 1 shows improvements by 20% for water use (-57.3 m³ water eq of deprived water), 4.3% for climate change (-1.15 kg CO₂ eq) and 3% for resource use fossil (-10 MJ) as compared to the base case. However, it also shows an increase in the impact on the eutrophication of freshwater of 6.4% (9.9E-04 kg P equivalents).
- The complete path 2 shows improvements by 20% for water use (-57.2 m³ water eq of deprived water), 4.3% for climate change (-1.15 kg CO₂ eq) and 2% for resource use fossil (-6.7 MJ) as compared to the base case. However, it also shows an increase in the impact on the eutrophication of freshwater of 7.1% (1.1E-03 kg P equivalents).
- The complete path 3 shows improvements by 18% for water use (-52.5 m³ water eq of deprived water), 2.3% for climate change (-0.6 kg CO₂ eq) and 1.3% for resource use fossil (-4.4 MJ) as compared to the base case. However, it also shows an increase in the impact on the eutrophication of freshwater of 8.3% (1.3E-03 kg P equivalents).

Table 95. Results of environmental assessment for the impact categories with >5% contribution to the single score, for denim products

Impact category	Unit	P1	P2	P3	P1	P2	P3
					As compared to the base case		
Water use	m ³ water eq of deprived water	-57.31	-57.24	-52.51	-19.8%	-19.8%	-18.2%
Climate change	kg CO ₂ eq.	-1.15	-1.15	-0.61	-4.3%	-4.3%	-2.3%
Resource use fossils	MJ	-10.00	-6.70	-4.39	-2.9%	-1.9%	-1.3%
Eutrophication freshwater	kg P equivalents	9.91E-04	1.09E-03	1.29E-03	6.4%	7.1%	8.3%

5993 P1: Path 1 (robustness score, recyclability score, recycled content information and performance requirement, environmental footprint), P2:
5994 Path 2 (robustness score, recyclability score, recycled content information and performance requirement, carbon footprint), P3: Path
5995 3 (recyclability score, recycled content information and performance requirement)

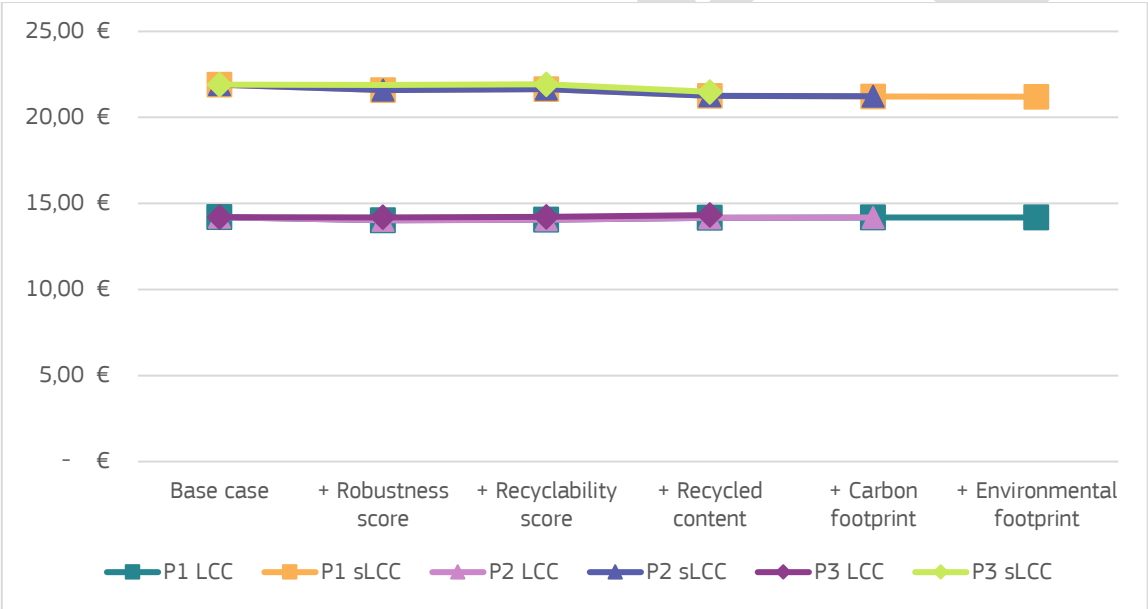
5996 Source: own elaboration.

5997 When looking at the economic assessment (**Figure 56**), the internal costs (LCC) in the first path decrease when
5998 implementing the robustness score but increase as we add other design options. For the complete path, the
5999 analysis shows that the 11.2% environmental improvement could be achieved with a slight increase in the
6000 internal costs of 0.1%. For the sLCC, the monetisation of the environmental improvement compensates the
6001 internal costs, decreasing the societal costs by 3.2% as compared to the base case.

6002 Similar analysis applies to the second path. The 11.8% environmental improvement of the complete path could
6003 be achieved with a slight increase in the internal costs of 0.1%. For the sLCC, the monetisation of the
6004 environmental improvement compensates the internal costs, decreasing the societal costs by 3.1% as
6005 compared to the base case.

6006 As for the third path, the 9.4% environmental improvement for the whole path would be also achieved with
6007 slight internal costs (0.9%) (LCC). When the environmental externalities are considered, the societal cost are
6008 reduced by 1.9% as compared to the base case (sLCC).

6009 **Figure 56.** Results of the economic assessment of the paths (cumulative LCC and sLCC) for denim products

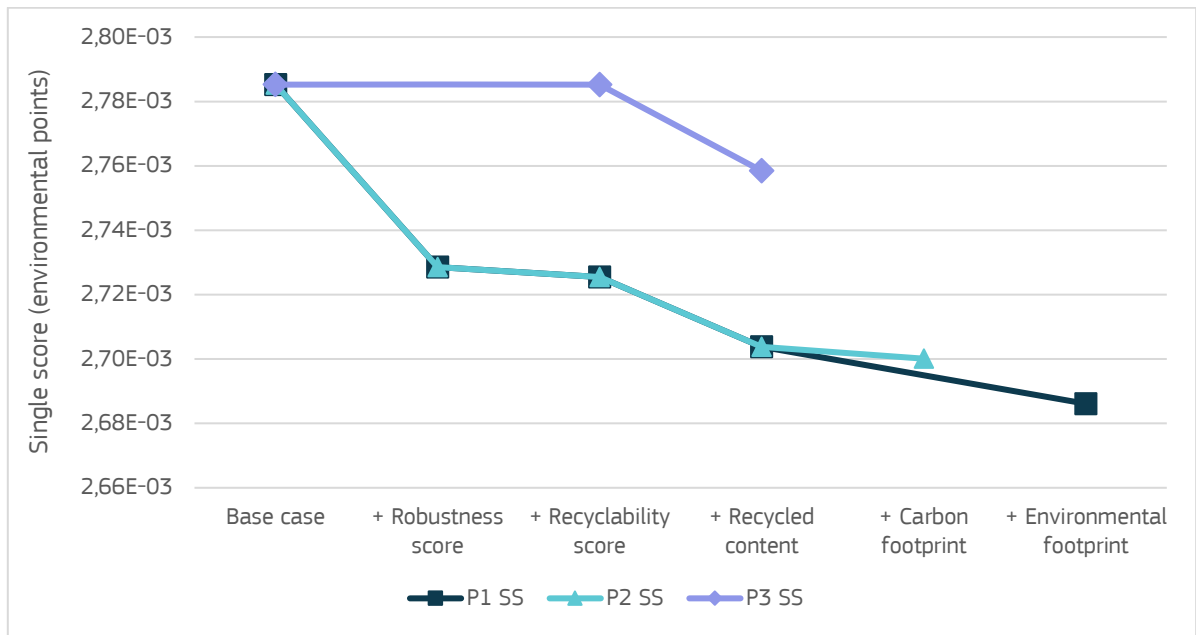


LCC: Life cycle costing; sLCC: societal Life cycle costing

6012 Source: Own elaboration

6013 Finally, for other woven products (**Figure 57**), the results of the first path show that combining design options
6014 consistently improves the environmental profile of the product, even for the cluster of the robustness and
6015 recyclability scores. The complete path would still provide the greatest advantage, achieving a reduction of the
6016 environmental impact of 3.6% as compared to the base case. The second path behaves similarly, achieving
6017 improvements of up to 3.1% for the complete cluster as compared to the base case. Regarding the third path,
6018 the environmental impact of the product would also decrease as compared to the base case (by 1.0%), offering
6019 a higher improvement than when applying each of the design options alone.

Figure 57. Results of the environmental assessment of the paths (cumulative LCA) for other woven products



SS: single score

Source: Own elaboration

Disaggregating the results of the environmental assessment into the most impactful impact categories (**Table 94**) allows the following conclusions:

- From the assessment in section 10.1.2.1, the impact categories contributing the most to the single score are water use (27%), climate change (18%), resource use – fossil (9%), particulate matters (7%), resource use – minerals and metals (7%) and acidification (6%).
- The complete path 1 shows improvements by 2.4% for water use (-2.5 m³ water eq of deprived water), 3.7% for climate change (-0.65 kg CO₂ eq), 3.4% for resource use fossil (-6.8 MJ), 4% for particulate matters (-5.4E-8 disease incidences), -6.5% for resource use – minerals and metals (-1.03E-5 kg Sb eq./person), -4.4% for acidification (-6.2E-03 mol H⁺ eq.) as compared to the base case.
- The complete path 2 shows improvements by 2.3% for water use (-2.4 m³ water eq of deprived water), 3.7% for climate change (-0.65 kg CO₂ eq), 2.2% for resource use fossil (-4.4 MJ), 3.2% for particulate matters (-4.3E-8 disease incidences), -6.2% for resource use – minerals and metals (-9.8E-6 kg Sb eq./person), -3.7% for acidification (-5.3E-03 mol H⁺ eq.) as compared to the base case.
- The complete path 3 shows improvements by 0.1% for water use (-0.06 m³ water eq of deprived water), 0.9% for climate change (-0.15 kg CO₂ eq), 0.7% for resource use fossil (-1.4 MJ), 1.4% for particulate matters (-1.9E-8 disease incidences), -4.3% for resource use – minerals and metals (-6.8E-6 kg Sb eq./person), -1.9% for acidification (-2.7E-03 mol H⁺ eq.) as compared to the base case.

Table 96. Results of environmental assessment for the impact categories with >5% contribution to the single score, for other woven products

Impact category	Unit	P1	P2	P3	P1	P2	P3
					As compared to the base case		
Water use	m3 water eq of deprived water	-2.47	-2.39	-0.06	-2.4%	-2.3%	-0.1%
Climate change	kg CO2 eq.	-0.65	-0.65	-0.15	-3.7%	-3.7%	-0.9%
Resource use fossils	MJ	-6.79	-4.40	-1.43	-3.4%	-2.2%	-0.7%
Particulate matter	disease incidences	-5.43E-08	-4.34E-08	-1.87E-08	-4.0%	-3.2%	-1.4%

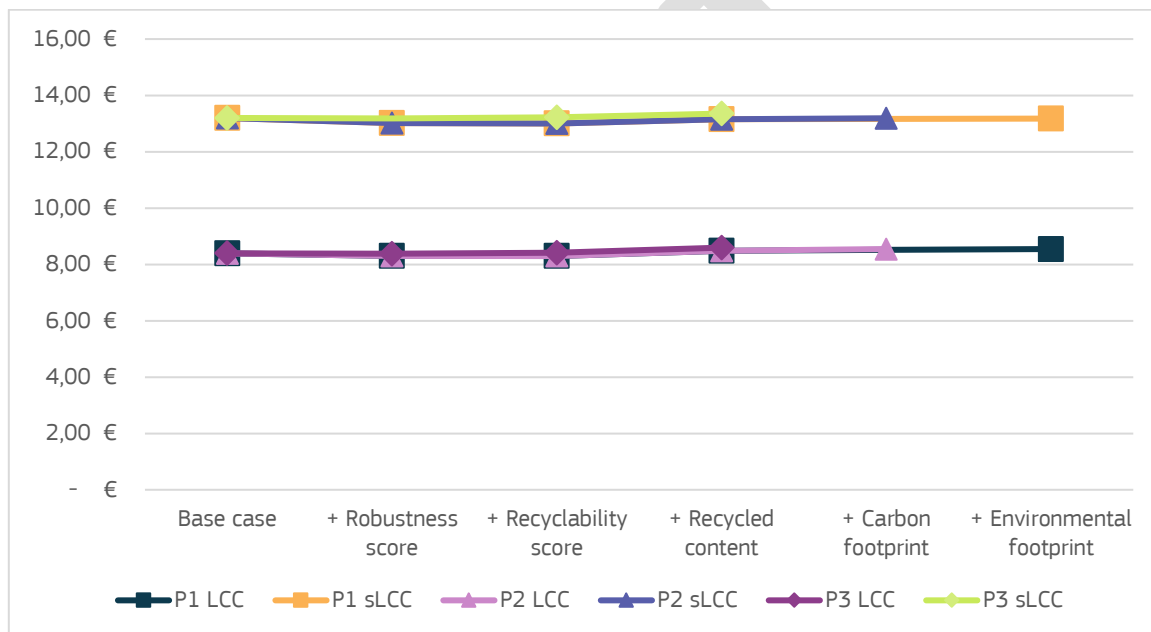
Impact category	Unit	P1	P2	P3	P1	P2	P3
					As compared to the base case		
Resource use minerals and metals	kg Sb eq./person	-1.03E-05	-9.75E-06	-6.82E-06	-6.5%	-6.2%	-4.3%
Acidification	mol H+ eq.	-6.22E-03	-5.25E-03	-2.66E-03	-4.4%	-3.7%	-1.9%

P1: Path 1 (robustness score, recyclability score, recycled content information and performance requirement, environmental footprint), P2: Path 2 (robustness score, recyclability score, recycled content information and performance requirement, carbon footprint), P3: Path 3 (recyclability score, recycled content information and performance requirement)

Source: own elaboration.

Regarding the economic assessment (**Figure 58**), the internal costs (LCC) in the first path decrease when implementing the robustness and the recyclability scores but increase as we add the rest of the design options. For the complete path, the analysis shows that the 3.6% environmental improvement could be achieved with an increase in the internal costs of 1.8%. For the sLCC, the monetisation of the environmental improvement compensate the internal costs, achieving a reduction in the societal costs of 0.2% as compared to the base case. As for the second path, the 3.1% environmental improvement translates into an increase in the internal costs of 1.8%, which are again offset in terms of the societal LCC. For the third path, the 1.0% environmental improvement would be achieved with a rise in the internal costs of 2.4% (LCC), which is reduced to 1.2% when the environmental externalities are considered (sLCC).

Figure 58. Results of the economic assessment of the paths (cumulative LCC and sLCC) for other woven products



LCC: Life cycle costing; sLCC: societal Life cycle costing

Source: Own elaboration

11.2.3 Propagation of uncertainties

In this section, the results of the Monte Carlo analysis for the propagation of the uncertainties in each of the paths are reported. Section 13.11.11 describes the assessment performed.

From the 500 iterations, the standard deviation and assuming a normal distribution, it was possible to estimate a 95% confidence interval for the results of each one of the three product categories, as shown in **Table 97**, **Table 98** and **Table 99** for the cost from the producer's, consumer's and society's perspective.

Table 97. Results with 95% Confidence interval for knitted products – Path 1, 2 and 3

Value	Path 1			Path 2			Path 3		
	Cost for producer	Cost for consumer	Societal cost	Cost for producer	Cost for consumer	Societal cost	Cost for producer	Cost for consumer	Societal cost
Max	6.66	10.50	9.90	6.63	10.51	9.72	6.59	10.30	9.65
Min	8.79	14.34	13.37	8.77	14.19	13.39	8.94	14.28	13.26
Central	7.68	12.16	11.31	7.56	12.53	11.76	7.67	12.14	11.42

P1: Path 1 (robustness score, recyclability score, recycled content information and performance requirement, environmental footprint), P2: Path 2 (robustness score, recyclability score, recycled content information and performance requirement, carbon footprint), P3: Path 3 (recyclability score, recycled content information and performance requirement)

Source: Own elaboration

Table 98. Results with 95% Confidence interval for denim products – Path 1, 2 and 3

Value	Path 1			Path 2			Path 3		
	Cost for producer	Cost for consumer	Societal cost	Cost for producer	Cost for consumer	Societal cost	Cost for producer	Cost for consumer	Societal cost
Max	9.60	15.58	14.97	9.75	15.31	14.82	9.75	16.25	15.49
Min	13.13	24.18	24.22	13.34	24.40	24.75	13.59	25.00	25.77
Central	11.28	19.17	19.00	11.30	19.10	18.92	11.39	19.31	19.20

P1: Path 1 (robustness score, recyclability score, recycled content information and performance requirement, environmental footprint), P2: Path 2 (robustness score, recyclability score, recycled content information and performance requirement, carbon footprint), P3: Path 3 (recyclability score, recycled content information and performance requirement)

Source: Own elaboration

Table 99. Results with 95% Confidence interval for other woven products – Path 1, 2 and 3

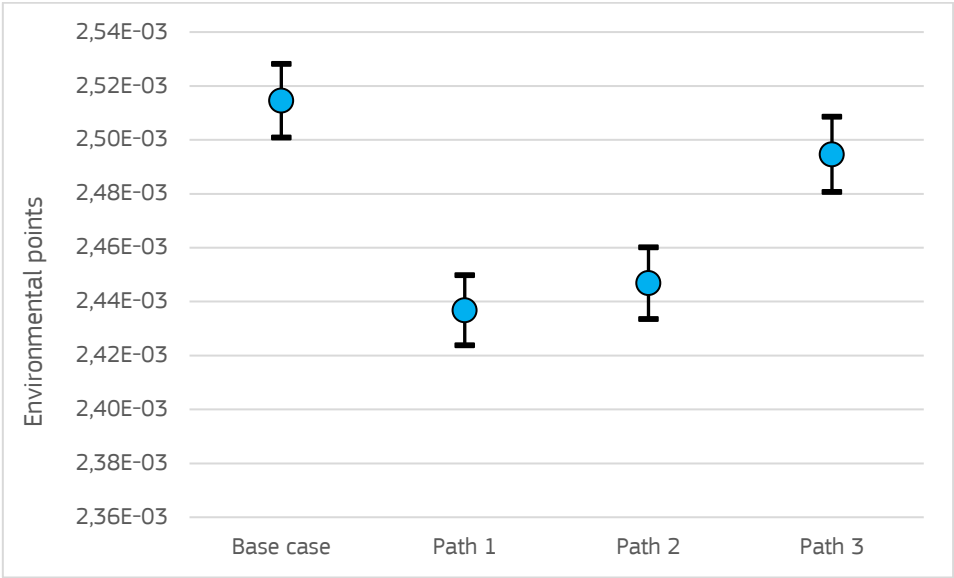
Value	Path 1			Path 2			Path 3		
	Cost for producer	Cost for consumer	Societal cost	Cost for producer	Cost for consumer	Societal cost	Cost for producer	Cost for consumer	Societal cost
Max	7.75	11.76	11.02	7.74	11.67	10.99	7.65	11.73	11.21
Min	10.53	16.32	15.10	10.32	15.91	14.67	10.34	16.19	15.55
Central	8.93	13.77	12.93	8.96	13.82	13.00	8.99	13.88	13.15

P1: Path 1 (robustness score, recyclability score, recycled content information and performance requirement, environmental footprint), P2: Path 2 (robustness score, recyclability score, recycled content information and performance requirement, carbon footprint), P3: Path 3 (recyclability score, recycled content information and performance requirement)

Source: Own elaboration

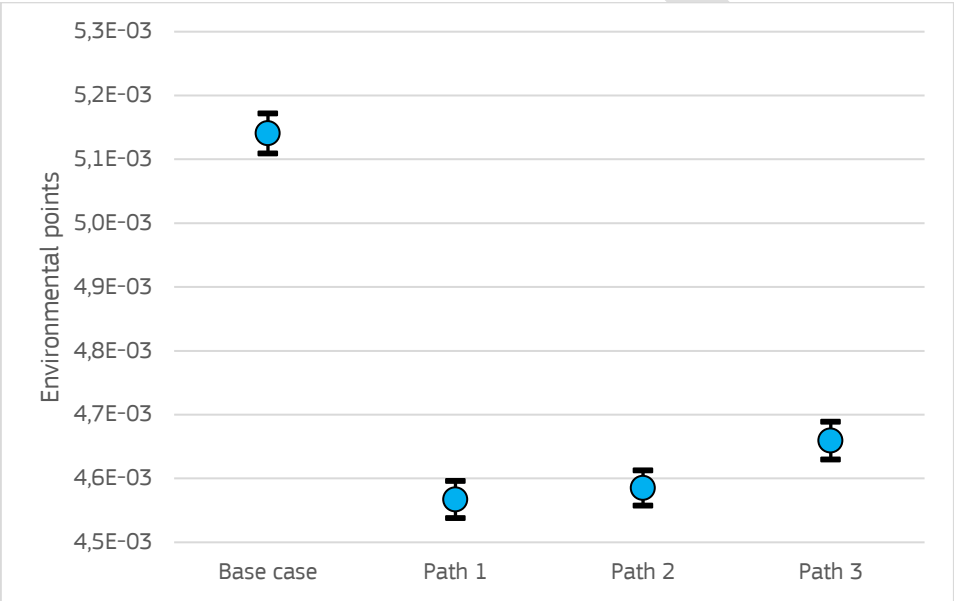
As compared to the results of the base cases, the combinations of design options in the Paths 1 and 2 achieve improvements of the environmental impacts without any overlap in the uncertainties calculated. As shown in **Figure 59**, **Figure 60** and **Figure 61**, the environmental impact of the base cases is always above the ranges reported for the design options, meaning improvements regardless the uncertainties with a 95% confidence. In the case of the Path 3, the same can be concluded for denim products. However, the lower thresholds proposed for the recycled content in knitted and other woven products do not deliver sufficient environmental benefits to avoid the results to overlap with the outcomes of the base cases once uncertainties are considered.

6088 **Figure 59.** Results of the uncertainty propagation of the environmental impact (single score) for knitted products



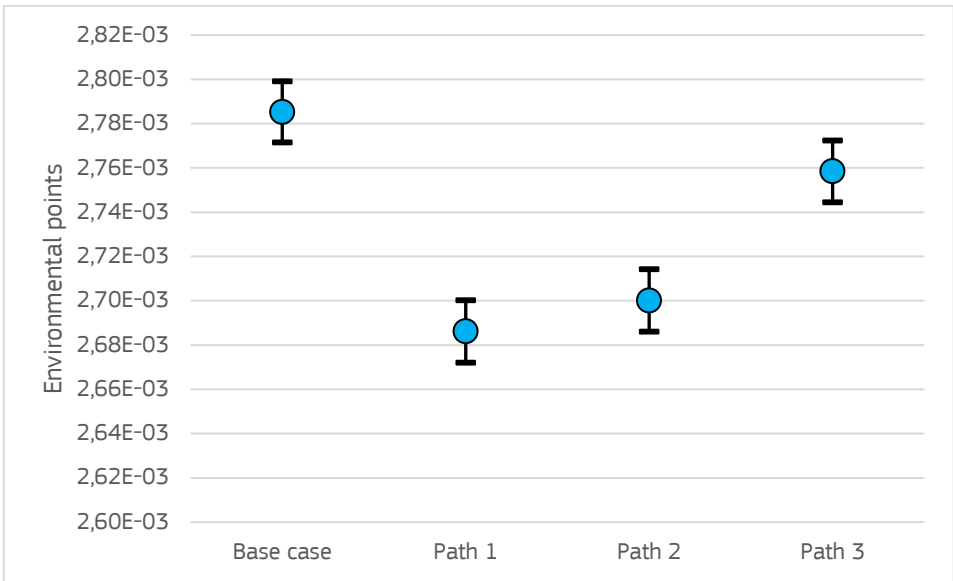
6089
6090 *Source: Own elaboration*

6091 **Figure 60.** Results of the uncertainty propagation of the environmental impact (single score) for denim products



6092
6093 *Source: Own elaboration*

Figure 61. Results of the uncertainty propagation of the environmental impact (single score) for other woven products



Source: Own elaboration

11.3 Main conclusions of the assessment of the design options

The following bullet points aim to summarise the main conclusions from the assessment in this chapter:

- The results are highly dependent on the assumptions underpinning the expected impact of each of the potential ecodesign requirements, introducing very considerable uncertainty, which should not be confused with the model uncertainty assessed and propagated in the model results (section 11.2.3). The fact that most of the design option proposed take the form of information requirements further complicates the analysis, as their effectiveness relies heavily on consumer behaviour and choices, which cannot be accurately predicted in this study.
- The results indicate that, in all cases, combining design options into paths yields greater environmental benefits than analysing them individually.
- Path 1 is the one that achieves the greatest environmental impact reductions across all product categories (knitted, denim and other woven products) followed by Paths 2 and 3.
- The environmental improvements compensate for the increase in internal costs associated with implementing the paths. However, these improvements are sufficient to achieve lower societal costs than the base cases only in the case of denim products, which experienced minimal increases in internal costs. For other woven products, the internal costs are only totally offset in the case of the first path. For the rest of the configurations and for the knitted products, the societal costs increase by 1-2%.
- The internal costs of implementing Path 3 are consistently lower than those of Paths 1 and 2. Despite offering more modest environmental improvements, Path 3 represents a more conservative approach, as the results are subject to less uncertainty in estimates related to the impact of the requirements.
- Paths 1 and 2 show improvements of the environmental impacts of the products even once the uncertainties are considered, with a 95% confidence. For Path 3, the results after the propagation of uncertainties overlap with those of the base cases for knitted and other woven products.

With all, this assessment corresponds to the Task 6 of the MEErP, which means that the design options should be understood as potential ecodesign requirements under study and not as final proposals. In a following step, Task 7 will perform a scenario analysis to extrapolate the results from one representative product to the total EU stock and project the most important impacts. Based on the latter, policy options will be proposed to feed into and inform the envisaged Impact Assessment that the Commission will undertake after the preparatory study. There, the policy options to define the final ecodesign requirements will be reassessed and finally proposed towards inclusion in a forthcoming delegated act on textile apparel.

12 Chemical substances in textile apparel

12.1 ESPR and chemical substances

The use and presence of chemical substances in products in the scope of the Ecodesign for Sustainable Products Regulation (²²²), in particular of Substances of Concern (SoCs), as defined in Article 2(27) of the Regulation, is an important product aspect under the new ecodesign regulation, representing a significant development compared to the depth with which chemicals were addressed under its predecessor, the Ecodesign Directive.

In particular, the ESPR sets the obligation to set information requirements on Substances of Concern and possible performance requirements on substances. The new ecodesign framework expands the scope of the previous Ecodesign Directive, to cover, in principle, all products on the European market (²²³) and expands the focus to cover the full environmental sustainability of products throughout their life cycle, with special attention to the circularity aspects of products.

The focus on chemical substances in products is, under ESPR, placed on Substances of Concern as those substances may have a negative effect upon human health and the environment because of their adverse effects and may negatively affect the recycling and reuse of materials in the product in which SoCs are present. Substances of Concern, which are defined in its **Article 2(27)** include:

- SoCs (a) – Substances of Very High Concern, which meet the criteria laid down in Article 57 of Regulation (EC) No 1907/2006 and are identified in accordance with Article 59(1) of that Regulation, i.e. those listed in the ‘Candidate List’ of substances published on the website of the European Chemicals Agency (ECHA). The list can be consulted here: [Candidate List of substances of very high concern for Authorisation - ECHA](#)
- SoCs (b) – Substances with a harmonised classification listed in Part 3 of Annex VI of the CLP Regulation in the following hazard classes or categories:
 - (i) carcinogenicity categories 1 and 2;
 - (ii) germ cell mutagenicity categories 1 and 2;
 - (iii) reproductive toxicity categories 1 and 2;
 - (iv) endocrine disruption for human health categories 1 and 2;
 - (v) endocrine disruption for the environment categories 1 and 2;
 - (vi) persistent, mobile and toxic or very persistent, very mobile properties;
 - (vii) persistent, bioaccumulative and toxic or very persistent, very bioaccumulative properties;
 - (viii) respiratory sensitisation category 1;
 - (ix) skin sensitisation category 1;
 - (x) hazardous to the aquatic environment - categories chronic 1 to 4;
 - (xi) hazardous to the ozone layer;
 - (xii) specific target organ toxicity – repeated exposure categories 1 and 2;
 - (xiii) specific target organ toxicity – single exposure categories 1 and 2;

²²² Regulation (EU) 2024/1781 of the European Parliament and of the Council of 13 June 2024 establishing a framework for the setting of ecodesign requirements for sustainable products, amending Directive (EU) 2020/1828 and Regulation (EU) 2023/1542 and repealing Directive 2009/125/EC. Available at: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32024R1781&qid=1719580391746>

²²³ Except food and feed, living plants, animals and microorganisms, products of human origin, veterinary and medicinal products, products of plants and animals relating directly to their future reproduction and aspects related to vehicles covered by other sector-specific legislation.

6163 In addition to the consolidated version of the CLP Regulation (direct link: [https://eur-](https://eur-lex.europa.eu/eli/reg/2008/1272/oj/eng)
6164 [lex.europa.eu/eli/reg/2008/1272/oj/eng](https://eur-lex.europa.eu/eli/reg/2008/1272/oj/eng)), harmonised classifications of substances can be consulted as part of
6165 ECHA's online chemical database (ECHA CHEM) ([CLP Annex VI – Harmonised classification](#))

6166 — SoCs (c) – Persistent Organic Pollutants regulated under Regulation (EU) 2019/1021 (the POPs
6167 Regulation). A list of substances in scope can be checked in Annexes I – IV of the Regulation. Direct
6168 link: <http://data.europa.eu/eli/reg/2019/1021/oj>

6169 — SoCs (d) – Substances that negatively affect the reuse and recycling of materials in the product
6170 in which they are present.

6171 Substances known to be in a product can be cross-checked, by means of their substance identifiers (e.g. CAS,
6172 EC number, etc.), against the lists of SVHCs, substances with a harmonised classification in CLP and POP
6173 substance lists that can be found, among other sources, in the links provided above. Substances defined as
6174 SoCs (d) have to be determined for each specific product group and have to be listed in product-specific
6175 delegated acts developed under the ESPR. Elements that can lead a substance to be defined as a SoC (d) are
6176 listed under **Article 5(14)** of the ESPR. These include substances that:

6177 (a) based on standard technologies, make the reuse, or recycling process more complicated, costly,
6178 environmentally impactful, or energy- or resource-demanding;

6179 (b) impair the technical properties or functionalities, the usefulness or the value of the recycled
6180 material coming from the product or products manufactured from that recycled material;

6181 (c) negatively impact aesthetic or olfactory properties of the recycled material.

6182 When mapping SoCs in a particular product group, these can usually be divided into:

6183 a) substances which due to existing regulatory, or client imposed (contractual) limitations, impede
6184 material recycled from the product to be placed on the market or used (usually related to the chemical
6185 safety, for instance restrictions to the placing on the market imposed under REACH or the POPs Regulation
6186 for products containing certain substances) or;

6187 b) substances in the product that, for reasons other than primarily their chemical safety considerations,
6188 interfere with sorting or other parts of the recycling process, are responsible for a substantial increase in the
6189 associated costs or environmental impacts, or have a negative effect on the technical properties or
6190 functionality of the recycled material. Pursuant to the limitations established under Article 6(3) of the ESPR,
6191 this latter group of substances could be potential candidates for consideration towards setting substance-
6192 specific performance requirements.

6193 According to **Article 7(5)** of ESPR, and subject to certain exceptions, information on substances of concern
6194 constitutes a default information requirement, to be, generally, provided in the Digital Product Passport (DPP).
6195 This information should include at least the following items:

6196 a) 'the name or numerical code of the substances of concern present in the product, as follows:

6197 (i) name in the International Union of Pure and Applied Chemistry (IUPAC) nomenclature, or
6198 another international name when IUPAC name is not available;

6199 (ii) other names, including usual name, trade name, abbreviation;

6200 (iii) European Community (EC) number, as indicated in the European Inventory of Existing
6201 Commercial Chemical Substances (EINECS), the European List of Notified Chemical Substances
6202 (ELINCS) or the No Longer Polymer (NLP) list or the number assigned by the European Chemicals
6203 Agency (ECHA), if available and appropriate;

6204 (iv) the Chemical Abstract Service (CAS) name and number, if available;

6205 b) the location of the substances of concern within the product;

6206 c) the concentration, maximum concentration or concentration range of the substances of concern, at
6207 the level of the product, its relevant components, or spare parts;

6208 d) relevant instructions for the safe use of the product;

6209 e) information relevant for disassembly, preparation for reuse, reuse, recycling and the environmentally
6210 sound management of the product at end-of-life.'

6211 It should be noted that the definitions above are not mutually exclusive and, therefore, a given substance
6212 may meet the criteria to be identified simultaneously under several of the SoC categories above.

6213 The purpose of providing this information is to make it possible to track the substances of concern,
6214 throughout the life cycle of the products concerned. Pursuant to **Article 7(6)**, duly justified **exemptions** for
6215 information requirements for substances of concern, or some of their information elements, can be set in the
6216 delegated acts to be proposed by the Commission. Specifically, these can be based on 'the technical
6217 feasibility or relevance of tracking substances of concern, the existence of analytical methods to detect and
6218 quantify them, the need to protect confidential business information or in other duly justified cases'. Article
6219 7(6) further clarifies that 'substances of concern within the meaning of Article 2(27), point (a), shall not be
6220 exempted if they are present in products, their relevant components or spare parts in a concentration above
6221 0,1 % weight by weight'.

6222 In addition to setting information requirements for substances of concern in products, the ESPR envisages the
6223 possibility that specific performance requirements may be imposed on substances in products. **Article 6(2)**
6224 of ESPR states that performance requirements, as appropriate, may include either or both of the following:
6225 '(a) minimum or maximum levels in relation to a specific product parameter or a combination thereof and; (b)
6226 non-quantitative requirements that aim to improve performance in relation to one or more of such product
6227 parameters.' When defining a possible performance requirement for a substance in a product, this can most
6228 often be envisaged to take the form of a maximum (or minimum) content level, similar in a certain way to
6229 what is frequently done for substances in restrictions listed in Annex XVII of the REACH regulation.

6230 To this regard it is important to note that although **recital (26)** of the ESPR acknowledges that chemical
6231 safety is a recognised element of product sustainability, it also states that the Regulation 'should not provide
6232 for the restriction of substances based primarily on chemical safety', given this is already done under other
6233 Union law, such as REACH and the POPs Regulation. This is subsequently defined in the enacting terms of the
6234 Regulation, under its Article 6(3) which states that:

6235 'Performance requirements based on the product parameter referred to in Annex I, point (f), shall not restrict,
6236 for reasons relating primarily to chemical safety, the presence of substances in products. However, the
6237 setting of performance requirements shall also, where appropriate, reduce significant risks to human health
6238 or the environment.'

6239 Consequently, the setting of performance requirements on substances, be it for substances of concern or
6240 otherwise, by means of provisions laid out in product-specific ESPR delegated acts is possible, but in principle
6241 limited to cases where such substances would have negative impacts on sustainability that are not primarily
6242 related to chemical safety. It should be noted that, as stated in Article 6(3) such measures could potentially
6243 also reduce risks to human health and the environment (to the extent this is possible).

6244 In addition, **Article 5(1)** of the ESPR lists the product aspects that should be addressed by ecodesign
6245 requirements:

- 6246 (c) durability;
- 6247 (d) reliability;
- 6248 (e) reusability;
- 6249 (f) upgradability;
- 6250 (g) repairability;
- 6251 (h) the possibility of maintenance and refurbishment;
- 6252 (i) the presence of substances of concern;
- 6253 (j) energy use and energy efficiency;
- 6254 (k) water use and water efficiency;
- 6255 (l) resource use and resource efficiency;
- 6256 (m) recycled content;
- 6257 (n) the possibility of remanufacturing;
- 6258 (o) recyclability;

- 6259 (p) the possibility of recovery of materials;
- 6260 (q) environmental impacts, including carbon and environmental footprint;
- 6261 (r) expected generation of waste.

6262 **12.2 Methodology: substances and substances of concern**

6263 The approach used to scope substances relevant to textile apparel products and subsequently propose
6264 possible information and performance requirements, builds upon the existing Methodology for **Ecodesign of**
6265 **Energy-related Products (MEErP)** ⁽²²⁴⁾, which has been used successfully in the framework of the
6266 Ecodesign Directive. The method, as further developed by the JRC ⁽²²⁵⁾, comprises the following main steps:

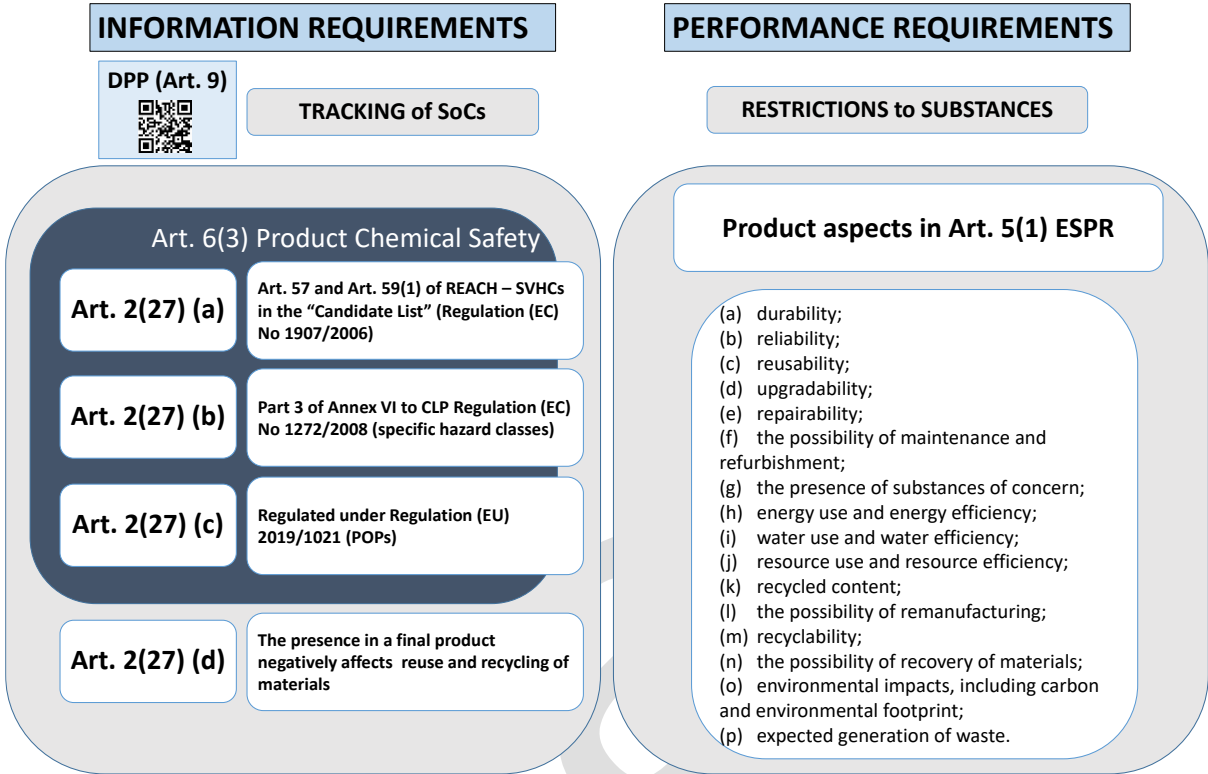
- 6267 — **Preliminary data gathering** resulting in an inventory of substances (carried out to the extent
6268 feasible), to enable the chemical composition of the product Base Case(s) to be defined.
- 6269 — **Identification of substances of concern**, focusing on substances that remain in the textile
6270 apparel products. Initially, substances meeting the criteria to be identified as SoCs (a), (b) and (c) are
6271 determined, via cross-checking the inventory against the relevant substance lists. In addition, based
6272 on available information and following criteria defined under the **ESPR Article 5(14)**, as further
6273 developed for their practical implementation and interpreted by the JRC Preparatory Study team, an
6274 indicative list of SoCs is identified.
- 6275 — The appropriateness of applying **specific thresholds, exceptions and potentially, delayed**
6276 **entry into application timelines** for information requirements on SoCs in the product group are
6277 proposed and justified based on available product information.
- 6278 — Considerations on the appropriateness of developing **performance requirements** on substances
6279 occurring in the product that negatively affect product aspects in **Article 5(1)**, in particular as
6280 regards to the potential for product aspects such as physical durability, repairability, waste
6281 generation or recycled content and recyclability, are analysed and justified (to the extent such
6282 relevant substances, including SoCs, could be identified).

6283 **Figure 62** illustrates the concept of Substances of Concern subject to information requirements and
6284 performance requirements for substances under the ESPR in a schematic way.

²²⁴ European Commission, Joint Research Centre, Gama Caldas, M., Eynard, U., Spiliotopoulos, C., Blengini, G., Alfieri, F., Mancini, L., Mathieux, F. and Ardente, F., *Review of the MEErP - Methodology for Ecodesign of Energy-related Products*, Publications Office of the European Union, Luxembourg, 2024, <https://data.europa.eu/doi/10.2760/24524>, JRC136406.
<https://publications.jrc.ec.europa.eu/repository/handle/JRC136406>.

²²⁵ Perez Camacho, M.N., Cappucci, G.M., Faraca, G., Bracalente, G., Bennett, M.J. et al., *Method for the identification and tracking of substances of concern in products and for the preparation of restriction measures on the use of substances in products*, Publications Office of the European Union, Luxembourg, 2025, <https://data.europa.eu/doi/10.2760/9706783>, JRC143683.

Figure 62. Method concerning the presence and use of substances under the ESPR.



6286

6287

Source: JRC’s own elaboration based on the ESPR.

6288 **12.3 Substances in textile apparel**

6289 The textile industry uses a **very large number of chemical substances** with some studies showing that
6290 during the whole manufacturing process, between 1 kg and 5 kg of chemicals are used per kg of textiles
6291 produced (Millet, 2022) ⁽²²⁶⁾. Often, manufacturers cannot know at which stage and what chemicals
6292 substances were used in the production of textile apparel due not only the large variety of chemicals used
6293 but also because of the long value chains involved in the production of these products which usually take
6294 place outside Europe (more information to this regard is provided in **Section 9.2.7 Presence of substances**
6295 **of concern, of this report**). Derived from these challenges, **several initiatives** have been developed in
6296 the last years in order to shed some light on the use of chemical substances in products and in particular in
6297 the textile sector. Several examples for textiles include industry initiatives compiled in good practice guides
6298 such as those from AFIRM ⁽²²⁷⁾ and ZDHC ⁽²²⁸⁾ as well as labels like OEKO-TEX® ⁽²²⁹⁾, GOTS ⁽²³⁰⁾, BlueSign ⁽²³¹⁾
6299 or the Ecolabels ⁽²³²⁾ ⁽²³³⁾ ⁽²³⁴⁾.

6300 The REACH4Textiles ⁽²³⁵⁾ project highlights significant challenges in enforcing EU chemical regulations for
6301 textiles, particularly for imported goods. This project acknowledges that ~28 billion garments circulate in
6302 Europe annually, with 80% imported from outside the EU. Ensuring that non-compliant products are kept
6303 outside the European market could present major challenges for surveillance authorities. This fact is

226 Millet, A., 2022. Report How toxic are the textiles we consume? And how can the EU trade tools tackle it? The Greens/EFA in the European Parliament (presented by Belgian MEP Saskia Bricmont on 30 January 2023). Available at: https://issuu.com/saskiabricmont/docs/20231701_hazardous_chemicals_textile_report_def

227 <https://afirm-group.com/>

228 <https://www.zdhc.org/>

229 <https://www.oeko-tex.com/en/>

230 <https://global-standard.org/>

231 <https://www.bluesign.com/en/>

232 <https://www.nordic-swan-ecolabel.org/criteria/>

233 <https://www.blauer-engel.de/en/products/products-list-a-z>

234 https://environment.ec.europa.eu/topics/circular-economy/eu-ecolabel/product-groups-and-criteria_en

235 <https://euratex.eu/reach4textiles/>

supported by 2 testing campaigns that the REACH4Textiles project carried out and where 2 374 REACH-compliance tests were done on 160 articles.. Results identified 26 non-compliant items, and of these, 42% originated from China, 23% were of unknown origin, 1.5% from Bangladesh, and the remaining 3.9% from Portugal, Pakistan, North Macedonia, Burma, the Netherlands, and India. Despite EU laws like REACH, enforcement is still inconsistent, especially for imported products.

Another project of relevance is the ECHT ⁽²³⁶⁾ (Interreg North-West Europe) which aims to enable Digital Product Passports in relation to the traceability of chemicals in some types of products. In particular, ECHT is developing a traceability strategy with one of the action plans directed to actors of the textile apparel value chain including policymakers at different levels.

However, as claimed by stakeholder comments received after the publication of the 2nd milestone of this Preparatory Study on textile apparel, it is acknowledged that without a clearly defined list of substances of concern, it is challenging to identify which chemicals may influence specific product characteristics in textile apparel. In this line, this report will present how such a list of chemical substances used in the manufacture of textile apparel was built. Nevertheless, it should be noted that such list cannot possibly be considered exhaustive, and some information may still be complemented with further information that may be further provided from stakeholders.

As explained and emphasised in Task 4 of this preparatory study, the number of chemical substances used in textile apparel is large, depends on the type of textile and the process of manufacturing for each type which may vary from one to another. An overview of different schemes providing lists of substances to be controlled in textile apparel is provided below.

The **AFIRM Restricted Substances List** ⁽²³⁷⁾ includes several hundreds of restricted chemical substances such acetophenone and 2-Phenyl-2-Propanol, alkylphenols (APs) and alkylphenols ethoxylates (APEOs), azo-amines and arylamine salts, bisphenols, brominated and organophosphorus substances (formerly flame retardants), chlorinated paraffins, chlorophenols, chlorinated benzenes and toluenes, cyclosiloxanes, dimethylfumarate, dyes (including forbidden and disperse dyes), fluorinated greenhouse gases, formaldehyde, heavy metals (Sb, As, Ba, Cd, Cr, Co, etc.), monomers (styrene and vinyl chloride), N-nitrosamines, organotin compounds, ortho-phenylphenol, ozono-depleting substances, Per- and Polyfluoroalkyl substances (PFAS), pesticides and herbicides, phthalates, polycyclic aromatic hydrocarbons (PAHs), quinoline, solvents and residuals (including Dimethylformamide (DMFa), Formamide, Dimethylacetamide (DMAC) and N-Methyl-2-pyrrolidone (NMP)), UV absorbers/stabilisers and Volatile Organic Compounds (VOCs).

The **ZDHC Manufacturing Restricted Substances List** ⁽²³⁸⁾ includes similar chemical substances as AFIRM. The groups of restricted substances include in this case: alkylphenol (AP) and alkylphenol ethoxylates (APEOs), including all isomers; anti-microbials and biocides (such dimethylfumarate (DMFu) or o-phenylphenol and salts; chlorinated paraffins; chlorobenzenes and chlorotoluenes, chlorophenols; dyes - allergenic disperse dyes; dyes - carcinogenic or equivalent concern; flame retardants; glycols/glycol ethers; halogenated solvents; organic solvents; organotin compounds; perfluorinated and polyfluorinated chemicals (PFAS); phthalates - including all other esters of ortho-phthalic acid; polycyclic aromatic hydrocarbons (PAHs); restricted aromatic amines (cleavable from azo-colourants); heavy metals; UV absorbers; bisphenols; ethoxylated tallow amine; formaldehyde; phenol; potassium permanganate; and other solvents.

The **OEKO-TEX® standards** are labels for textiles tested for substances ⁽²³⁹⁾. While the STANDARD 100 is the most common, textiles with the OEKO-TEX® MADE IN GREEN label also include criteria for a more sustainable production. Other labels from the same organisation are the OEKO-TEX® ORGANIC COTTON which include the testing of cotton products for the use of genetically modified organisms (GMO) and substances, the OEKO-TEX® LEATHER is tailored for the testing of substances in leather products (from materials, semi-finished and finished articles). OEKO-TEX® STeP certifies production facilities that manufacture textiles and leather articles under sustainable conditions, OEKO-TEX® ECO PASSPORT identifies chemical substances used in textile and leather industry which are considered to be not harmful and environmentally friendly.

²³⁶ <https://echt.nweurope.eu/>

²³⁷ https://afirm-group.com/wp-content/uploads/2025/02/2025_AFIRM_RSL_2025_0205b.pdf

²³⁸ <https://mrsl-30.roadmaptozero.com/>

²³⁹ <https://www.oeko-tex.com/en/our-standards/>

6352 **GOTS** is a textile processing standard for organic fibres which includes environmental criteria. Permitted
 6353 fibres within the label GOTS (²⁴⁰) are organic-in-conversion fibres, natural fibres (from plant and animal
 6354 origin), regenerated cellulosic fibres (man-made cellulosic fibres), recycled natural and synthetic fibres,
 6355 biopolymers and some virgin synthetic (polymer) fibres such polyamide, polypropylene, elastomultiester
 6356 (elasterell-p) and polyurethane (elastane). The certified products must comply with limit values for certain
 6357 chemical substances, namely: alkylphenol (ethoxylates), AOX, arylamines, bisphenols, disperse dyes
 6358 (allergenic or carcinogenic), formaldehyde, flame retardants, glyoxal and other aldehydes, chlorophenols,
 6359 pesticides, heavy metals, organotin compounds, PFAS, phthalates, PAHs, benzo[g,h,i]perylene, chlorinated
 6360 paraffins, cyclic siloxanes, chlorinated benzenes and toluenes, styrene, xylene, solvent residues (NMP, DMAc,
 6361 DMF), formamide, quinoline, and other chemical residues (²⁴¹).

6362 **Bluesign** also assesses and provides certification for products and materials in relation to chemicals. In
 6363 addition to criteria, they provide two types of restricted chemical substances lists (²⁴²) for (1) finished
 6364 chemical products, such as auxiliaries or dyes (bluesign System Black Limits (BSBL)) and (2) chemical
 6365 substances in articles and consumer products (bluesign System Substances List (BSSL)). In addition, they
 6366 provide the bluesign RESTRICTED SUBSTANCES LIST (RSL) which is an extract of the BSSL and contains
 6367 consumer safety limits and recommended testing methods for legally restricted substances in textile/leather
 6368 articles and accessories. Chemical substances included in such list for textile apparel (²⁴³) are aldehydes,
 6369 alkylphenolethoxylates (APEOs), alkylphenols (APs), amines, arylamines, biocides, chlorinated benzenes and
 6370 toluenes, chlorinated phenols, colorants, dioxins and furans, flame retardants, glycols, fluorinated greenhouse
 6371 gases, halogenated biphenyls, halogenated terphenyls and halogenated naphthalenes, halogenated
 6372 diarylalkanes, isocyanates, metals, monomers, nitrosamines, ozone depleting substances, pesticides, PFAS,
 6373 plasticizers, polyaromatic hydrocarbons (PAHs), polymers, solvents, tin-organic compounds and UV stabilizers.

6374 According to feedback from stakeholders, the platform developed by Bluesign is used among industries in a
 6375 large extent with private services being provided upon registration in the so-called Cube section (²⁴⁴). In
 6376 addition, there are two platforms, one to search for chemical products approved under the Bluesign criteria
 6377 (Finder²⁴⁵) and another compiling a database of approved articles (Guide²⁴⁶).

6378 In the ecolabels, certain substances are restricted either in the fibres or in the final textile products. Ecolabels
 6379 such as Blue Angel, Nordic Swan and the EU Ecolabel also restrict certain hazard classifications as part of the
 6380 horizontal chemical substance criterion included in all products bearing an ecolabel.

6381 The **Blue Angel** criteria for textiles (²⁴⁷) sets limits or full restrictions to chemical substances such as
 6382 acrylonitrile, organotin compounds, aromatic diisocyanates, metals, cyclohexasiloxanes (D4, D5, D6),
 6383 alkylphenol ethoxylates (APEOs), chlorophenols, VOCs, quaternary ammonium compounds (DTDMAC, DSDMAC
 6384 and DHTDMAC), mineral oil-based defoamers, mordant dyes containing chromium salts, biocides, flame
 6385 retardants, halogenated substances, perfluorinated and polyfluorinated chemicals (PCs), formaldehyde,
 6386 chlorophenols, phthalates and plasticizers, aniline, dimethylformamide (DMF), dimethylacetamide (DMAc)
 6387 and N-methylpyrrolidone (NMP), alkylphenols and alkylphenol ethoxylates, perfluorinated and polyfluorinated
 6388 chemicals (PFCs), and chinoline/quinoline, among others.

6389 The **Nordic Swan** criteria for textiles, hides/skins, and leather (²⁴⁸) set criteria for certain fibres and final
 6390 products including chemical substances restrictions to specific biocides, metal in dyes, detergents, sizing
 6391 agents, bleaching agents, silicones or VOCs to name a few. It also includes prohibitions to flame retardants,
 6392 per- and polyfluoroalkyl substances (PFASs), nanomaterials/-particles, heavy metals, azo dyes, phthalates,
 6393 chlorinated solvents and carriers, including chlorotoluenes, chlorophenols and chlorobenzenes, alkylphenol

²⁴⁰ <https://global-standard.org/resource-library/standard-and-certification>

²⁴¹ <https://global-standard.org/images/resource-library/documents/standard-and-manual/GRTS%20Draft%201.0%20for%20Public%20Consultation-July%202025-Signed.pdf> (this document is under public consultation). Last accessed on 22 September 2025.

²⁴² <https://www.bluesign.com/en/downloads/>

²⁴³ <https://www.bluesign.com/wp-content/uploads/2025/07/RSL-v16.0.pdf>

²⁴⁴ [Log in | CUBE](#)

²⁴⁵ <https://www.bluesignfinder.com/> <https://www.bluesignfinder.com/>

²⁴⁶ <https://www.bluesignguide.com/>

²⁴⁷ Blue Angel – The German Ecolabel: DE-UZ 154 Basic Award Criteria. Edition January 2023, version 3. Available at <https://produktinfo.blauer-engel.de/uploads/criteriafile/en/DE-UZ%20154-202301-en-criteria-V3.pdf>. Last accessed on 23 September 2025.

²⁴⁸ Nordic Swan Ecolabel: Textiles, hides/skins, and leather. Version 5.8. Available at <https://www.nordic-swan-ecolabel.org/criteria/products-of-textiles-hides-skins-and-leather-112/>. Last accessed on 23 September 2025.

ethoxylates (APEO) and other alkylphenol derivatives, organotin compounds, linear alkylbenzene sulphonates (LAS), quaternary ammonium compounds such as DTDMAC, DSDMAC and DHTDMAC, EDTA (ethylene diamine tetra acetic acid) and DTPA (diethylene triamine pentaacetate).

The **EU Ecolabel** for textile products ⁽²⁴⁹⁾ also sets specific restrictions on chemical substances in certain fibres and includes a horizontal requirement to final products in order to fulfil Article 6(6) of the EU Ecolabel Regulation (EU) 66/2010 ⁽²⁵⁰⁾, which establishes restrictions on certain classifications and in particular states: '6. The EU Ecolabel may not be awarded to goods containing substances or preparations/mixtures meeting the criteria for classification as toxic, hazardous to the environment, carcinogenic, mutagenic or toxic for reproduction (CMR), [in accordance with the CLP Regulation...] nor to goods containing substances referred to in Article 57 of Regulation (EC) No 1907/2006 [the REACH Regulation]'. All substances meeting the criteria for hazardous properties classification as specified in REACH Article 57 (i.e. CMR, PBT, vPvB or endocrine disrupting) are therefore restricted by Article 6(6). This is what is referred to as the 'horizontal CLP restrictions' because the restrictions are based exclusively on the CLP hazard of a substance. In this way, the EU Ecolabel for textile products sets restrictions to Substances of Very High Concern (SVHCs), NPEOs, APEOs, chlorophenols (their salts and esters), polychlorinated biphenyls (PCB), organotin compounds, including TBT, TPhT, DBT and DOT, dimethyl fumarate (DMFu), heavy metals, etc.

While the abovementioned initiatives, lists and/or labels show the chemical substances that are restricted by means of positive lists and restrictions and are potentially useful for the establishment of information requirements and performance requirements under the ESPR, the knowledge of **the whole inventory of chemical substances** remaining in final products and/or used along the life cycle of textile apparel is limited.

Chemical substances used in textile apparel differ depending on the fibres included in the garments and on the processes of manufacturing. In the first instance, raw materials determine the chemical substances involved. The textile value chain starts with the production or harvest of raw fibres for which two general categories are established by the textile industry: natural and man-made fibres. This initial classification is shown in **Table 100**. Man-made fibres encompass both purely synthetic materials of petrochemical origin and regenerative cellulosic materials manufactured from wood fibres (the so-called man-made cellulose fibres such as viscose).

Table 100. General classification of fibres.

Natural origin fibres	Animal origin	Raw wool, silk fibre, hair	
	Vegetable origin	Raw cotton fibre, flax, jute	
Chemical fibres (man-made)	Natural polymer fibres /Man-made cellulose fibres (MMCF) (also called regenerated cellulose fibres)	Viscose, cupro, lyocell, acetate, triacetate	
	Synthetic polymer fibres	Organic polymer	Polyester (PES), Polyamide (PA), Acrylic (PAC), Polypropylene (PP), Elastane (EL)

Source: Adapted from Best Available Techniques (BAT) reference document (BREF) for the textiles industry ⁽²⁵¹⁾.

The basic steps in the production of textile apparel once the fibres are manufactured include **yarn production** (spinning process), followed by **fabric production** which mainly involves weaving and knitting processes and finally **garment production** which can include several processes such dyeing, printing, finishing and coating. This is illustrated in **Figure 31** in Chapter 9 of this Preparatory Study.

²⁴⁹ Commission Decision of 5 June 2014 establishing the ecological criteria for the award of the EU Ecolabel for textile products <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A02014D0350-20201201&id=1758617330731>

²⁵⁰ Regulation (EC) No 66/2010 of the European Parliament and of the Council of 25 November 2009 on the EU Ecolabel <https://eur-lex.europa.eu/eli/reg/2010/66/oj/eng>

²⁵¹ Joze Roth, Benoit Zerger, Damien De Geeter, Jorge Gómez Benavides, Serge Roudier, Best Available Techniques (BAT) Reference Document for the Textiles Industry, Publications Office of the European Union, Luxembourg, 2023, doi:10.2760/355887, JRC 131874.

Chemical substances used in the mentioned manufacturing steps can be **process chemicals or auxiliaries** that do not remain in the final textile product and **performance chemicals** that remain in the final textile product. Examples of process chemicals/auxiliaries are the pretreatment chemicals, dye auxiliaries, spin finish and sizing chemicals. The usual performance chemicals that remain in the final textile product are dyes and pigments, some finishing /spin finish chemicals and coating /lamine chemicals. It should be noted that the spin finish chemicals are usually removed from the fibres before dyeing, printing, finishing or coating, because they interfere with these processes. In some cases, spin finishes fulfil some other performance properties, such as for example hydrophilic properties in absorbent products and thus are meant to remain on the fibres.

As reported by KEMI (2014), the most representative functions for the different types of chemicals used in textile production considered have been:

- Auxiliary (process) chemicals, which are involved in some step of textile processing but are not intended to remain in the product. These include organic solvents, surfactants, softeners, salts, acids and bases and biocides (preservatives used in the process or during storage and transport).
- Functional chemicals, which are intended to remain in the final article and are expected to be present at certain concentrations in order to achieve the desirable function. Some examples include dyestuffs and pigments, crease resistant agents, anti-shrinking agents, oil, soil and water repellents, plasticisers, flame retardants, biocides for defined functionalities in articles, e.g. antibacterial agents, stabilisers, stiffening agents and reactive resins for various finishing treatments.
- Unintended chemical substances such as contaminants and degradation products, are chemicals not intended to remain in the finished article, and these have no function in neither the textile production process nor in the finished textiles. These could be formaldehyde released from certain reactive resins, polyaromatic hydrocarbons (PAH) impurities in pyrolysed products e.g. carbon black, arylamines derived from certain azo dyestuffs and pigments, or toxic metals (e.g. heavy metals) due to impurities from the raw material.

A full list of chemical substances per type which are used in the considered manufacturing stages of textile apparel products (as reported in **Figure 31, Chapter 9**), either from natural or synthetic fibres, are summarised in **Table 226, Annex 13.13**.

There are other sources that differentiate two main process types, i.e. **the dry and the wet textile manufacturing processes**. The dry treatments are applied from fibre to garment while the wet processing can be performed to yarn, fabric or garments to provide them with colour and other characteristics.

The **dry processes** generally correspond to tiers 5 to 1, i.e. from raw material production to garment production, including fibre/polymer production, yarn production and fabric production as well.

- **Tier 5 is the raw material production** where chemicals are used to eliminate pests and improve the quality and production of fibres from natural raw materials. Chemical types used to treat synthetic raw materials at this stage (if any) have not been disclosed by manufacturers. However, the main chemicals utilised in the treatment of natural raw materials are sulphates, insecticides, fertilisers and pesticides.
- **Tier 4 corresponds to fibre/polymer production**. Chemicals used for natural fibres are used to eliminate fats, pectin, linens, and impurities from plants and/or animals. In MMCF fibres, chemicals are mainly used to extract cellulose from the substrate chosen as feedstock (wood, waste garments, others). In synthetic fibres, chemicals are used for the production/dissolution of polymers. In all fibre types, the chemicals used are acids, alkalines/bases, antistatic agents, softeners, soaping agents, humectants, and chemical fibre solvents, among others.
- **Tier 3 is the yarn production** (spinning process) where chemical substances are used to confer lubrication, antistatic properties and higher resistance to physical steps to fibres. These chemicals do not remain on the fibres (other than potentially as traces) after the manufacturing process as they are removed during the wet processing of textiles. The amount of chemicals could vary depending on the machinery used and the fibre nature, among other parameters.
- **Tier 2 is the fabric production** which mainly involves weaving and knitting processes where chemicals are also used to confer lubrication, antistatic properties and higher resistance to physical steps to yarns. These chemicals do not remain on the fibres after the manufacturing process as they

- 6480 are removed during the wet processing of textiles as well. The amount of chemicals could also vary
6481 depending on the machinery used and the fibre nature, among other parameters.
- 6482 — **Tier 1 is the garment production** which involves cutting, stitching, ironing, or labelling of the
6483 fabric and where chemical substances are not frequently used unless they are needed for some
6484 specific tasks such small parts attachment (for instance adhesives).
- 6485 Most of chemical substances used in the dry processes, are designed to be only temporally available in the
6486 fibres and are removed in the wet processes.
- 6487 The **wet processes** are applied to yarns, fabrics or garments and involve chemical treatments with chemical
6488 solutions in liquid state. The chemical substances used for wet treatments vary according to many
6489 parameters such the type of facility, the treatment type (exhaust or continuous) or the fibre type (natural
6490 cellulose, man-made cellulose, or synthetic). In this regard, the main differentiation relates to the fibre type:
- 6491 — **Synthetic fibres** are produced through a specific manufacturing process where chemicals may
6492 vary. For example, polyester fibres follow a specific process with certain chemical substances.
6493 Polyamide fibres follow also specific manufacturing processes, similar to animal fibres (such as
6494 wool) processes except for the pretreatment and finishing. Acetate fibres use same chemicals as
6495 polyester, with the exception of some auxiliaries such as: antioligomers, carriers, migration inhibitors
6496 and reductives.
- 6497 — **Cellulose natural fibres** (such as cotton and linen) follow the same processing. While chemical
6498 products used for linen could be considered to be the same as for cotton, it is worth noting that
6499 some chemical substances could be used in different quantities, usually in higher proportions and
6500 mainly in pre-treatment process.
- 6501 — **Animal natural fibres**, such wool uses same chemical products as polyamide, except antiyellowing
6502 agents. Additionally, wool could be treated with an anti-shrink treatment.
- 6503 — **Man-made cellulose fibres** follow same processing than cellulose fibres, differing the
6504 pretreatment steps. For MMCF, the pretreatment is done in a single pre-washing step. The rest of the
6505 processing could be the same.
- 6506 Textile production involves a wide range of chemical substances, each serving different purposes such as
6507 dyeing, finishing, and processing. In particular, the dyes and pigments represent a relevant group of
6508 substances which usually stay in the final product. A brief summary of the main types of **dyes** and their main
6509 function ⁽²⁵²⁾ is explained as follows:
- 6510 — Reactive Dyes → used for dyeing cellulose fibres like cotton.
- 6511 — Direct Dyes → used for dyeing cellulose fibres without the need for a mordant.
- 6512 — Vat Dyes → used for dyeing cellulose fibres to achieve deep and fast colours.
- 6513 — Sulphur Dyes → known for their high colour fastness and used for cotton.
- 6514 — Disperse Dyes → used for dyeing synthetic fibres like polyester.
- 6515 — Acid Dyes → used for dyeing protein fibres like wool and silk.
- 6516 — Basic Dyes → used for dyeing acrylic fibres and for over-dyeing cotton.
- 6517 The most common **finishing agents** also remaining in the final textile apparel garments are:
- 6518 — Softeners → used to improve the handle and drape of fabrics.
- 6519 — Flame Retardants → applied to fabrics to make them resistant to fire.
- 6520 — Water Repellents → used to make fabrics water-resistant.
- 6521 — Anti-Microbial Agents → added to fabrics to prevent the growth of bacteria and fungi.

²⁵² Joze Roth, Benoit Zerger, Damien De Geeter, Jorge Gómez Benavides, Serge Roudier, Best Available Techniques (BAT) Reference Document for the Textiles Industry, Publications Office of the European Union, Luxembourg, 2023, doi:10.2760/355887, JRC 131874.

- 6522 — UV Protectants → used to protect fabrics from ultraviolet radiation.
- 6523 There are **auxiliaries and additives** which do not remain in the final products and are:
- 6524 — Wetting Agents → used to improve the wetting and penetration of liquids into fibres.
- 6525 — Dispersing Agents → used to disperse dyes and pigments evenly in the dye bath.
- 6526 — Levelling Agents → help to achieve even dyeing by preventing uneven absorption.
- 6527 — Fixing Agents → used to enhance the colour fastness of dyes.
- 6528 — Bleaching Agents such as hydrogen peroxide are used to remove natural colours and impurities from
- 6529 fibres.

6530 Whereas the most common **sizing agents** are:

- 6531 — Starch → commonly used to improve the strength and smoothness of yarns.
- 6532 — Polyvinyl Alcohol (PVA) → used for sizing warp yarns in weaving.
- 6533 — Carboxymethyl Cellulose (CMC) → used as a sizing agent for cotton and other cellulose fibres.

6534 The use of formaldehyde in a variety of textile treatments is extended, mainly for creasing resistance, while
 6535 surfactants are used in various stages of textile processing to reduce surface tension and improve the
 6536 effectiveness of other chemicals. In addition, acids and alkalis are commonly used for pH adjustment and in
 6537 various chemical treatments like scouring and bleaching.

6538 12.3.1 Inventory

6539 From literature research and information provided by retail stakeholders, an inventory of substances used in
 6540 textile apparel has been built and classified per **function**, as chemical substances may be needed to provide
 6541 a certain characteristic/property/aspect to the final product or during the life cycle stage as a functional
 6542 process chemical. When possible, the CAS number is also included.

6543 Given the high number of chemical substances used in textile apparel, the inventory mainly includes the
 6544 **substances remaining in the final product** considering the functions that are more representative.
 6545 However, it is worth noting that the list of chemical substances is not exhaustive, and it may be completed if
 6546 further information is shared by stakeholders.

6547 Consequently, to start this inventory on substances, during the 2nd milestone of the PS on textile apparel
 6548 project, the JRC requested information to stakeholders and in particular **a questionnaire on substances**
 6549 was shared to collect such information. To complement, **other sources of information** have been used:

- 6550 — Relevant literature, LCA inventories, general chemical-related policy (REACH, POPs, CLP).
- 6551 — Specific policy instruments on textiles (Best Available Techniques (BAT) reference document (BREF)
- 6552 for the textiles industry ⁽²⁵²⁾ or the already mentioned ecolabels for textiles).
- 6553 — Reference Document on Best Available Techniques in the Production of Polymers (EC, 2007) ⁽²⁵³⁾.
- 6554 — Other initiatives from industry, EU and international projects and information from other labels
- 6555 (substances lists from AFIRM, ZDHC, Bluesign, Ecolabels).

6556 An indication of the **inventory of the chemical composition** of textile apparel products differentiating the
 6557 functional chemicals per component and material type should have ideally been built for each of the
 6558 Representative Products set in this Preparatory Study for textile apparel. Given that such detailed information
 6559 was not possible to obtain, an inventory of all chemical substances identified in textile apparel has been built
 6560 without distinguishing the product type. The inventory shows the manufacturing stage, fibre type and the
 6561 function of the identified chemical substances. This indicative inventory is reported in **Table 227** in **Annex**
 6562 **13.13**.

²⁵³ https://eippcb.jrc.ec.europa.eu/sites/default/files/2019-11/pol_BREF_for_textiles_0807.pdf

12.3.2 Identification of information requirements on Substances of Concern

This section aims to identify chemical substances subject to the mandatory information requirements under the ESPR and in particular for this PS on textile apparel. Using the inventory built in the previous point and the SVHC list, the “Bill of Substances of Concern” is built. Note that SoCs under Article 2(27) (a), (b) and (c) can be unequivocally identified by their definitions; therefore, their identification is not a core task for the PS team. Nevertheless, some information is provided next as well as the existence of overlaps between the three groups of SoCs (a), (b) and (c). It should be noted that the list of substances identified as SoC (a), (b) and (c) below are provided for the purpose of exemplifying relevant substances but should not be understood as an exhaustive list. It remains up to the due diligence efforts of operators along the textile supply chain and, ultimately to those placing textile apparel items on the EU market, to make reasonable efforts to identify and inform, via the DPP, of relevant substances subject to information requirements.

12.3.3 Substances of Concern type (a)

Substances of concern defined under Article 2(27) (a) are those that meet the criteria laid down in Article 57 of Regulation (EC) No 1907/2006 and are identified in accordance with Article 59(1) of that Regulation. The list of Substances of Very High Concern (SVHC) in the **Candidate List for REACH Authorisation** ⁽²⁵⁴⁾ contains 250 entries (1 October 2025) which include more than 490 substances, which have been identified as meeting one of the criteria a-f as referred to in Article 57 of REACH. The **chemical substances identified** as relevant for textile apparel and part of the **SVHC Candidate List** are summarised in **Table 101**, but it may not remain a complete list as the SVHC list is regularly updated. **Table 101** has been built using the current SVHC list and the AFIRM Restricted Substance List (2025) to cross-check SVHC chemical substances used in the textile apparel industry. Manufacturers using these substances along the life cycle of their products would be obliged to report them following the application of the ESPR information requirement mandate. **Table 101** indicates if SoCs (a) are also (b) and (c).

Table 101. Identification of Substances of Concern (a) for textile apparel products according to the ESPR Article 2(27).

Substances	CAS	Reason for inclusion SVHC	Notes
Octamethylcyclotetrasiloxane (D4)	556-67-2	PBT (Article 57d) vPvB (Article 57e)	May be present in silicone pads, as contaminants in formulations that contain silicone, like silicone softeners and those used for prints. D4 is also SoC (b) under the ESPR.
Decamethylcyclopentasiloxane (D5)	541-02-6		
Dodecamethylcyclohexasiloxane (D6)	540-97-6		
Decabromodiphenyl ether (DecaBDE)	1163-19-5	PBT (Article 57d) vPvB (Article 57e)	These substances are brominated and organophosphorus substances, used as flame retardants. These examples of flame retardant substances used historically across the apparel and footwear industry which is not intended to be a complete list. DecaBDE is also SoC (c). Also SoCs (b).
Tetrabromobisphenol A (TBBP A)	79-94-7	Carcinogenic (Article 57a)	
2,2-bis(bromomethyl)-1,3-propanediol (BBMP)	3296-90-0	Carcinogenic (Article 57a)	
Trixylyl phosphate (TXP)	25155-23-1	Toxic for reproduction (Article 57c)	
Tris(2-chloroethyl)phosphate (TCEP)	115-96-8	Toxic for reproduction (Article 57c)	
Triphenyl Phosphate (TPP)	115-86-6	Endocrine disrupting properties (Article 57(f) - environment)	May be used as a flame retardant, an antioxidant for PU materials, or as an alternative plasticizer to orthophthalates.
UV 326	3896-11-5	vPvB (Article 57e)	UV absorber used in finishing steps of the production of textile apparel. Remain in the final product. Also SoC (b).

²⁵⁴ <https://echa.europa.eu/en/candidate-list-table>

Substances	CAS	Reason for inclusion SVHC	Notes
Bisphenol-A (BPA)	80-05-7	Toxic for reproduction (Article 57c) Endocrine disrupting properties (Article 57(f) - environment) Endocrine disrupting properties (Article 57(f) - human health)	BPA may be used in the production of epoxy resins, polycarbonate plastics, flame retardants, and PVC. BPS may be used as a substitute for BPA for some specific uses, including in thermal receipt paper. BPS can be found in polyamide dye-fixing agents. BPA and BPS can be found in recycled polymeric material due to polycarbonate plastic made with bisphenols entering waste streams. Additional restrictions to other bisphenols may be expected. Also SoCs (b).
Bisphenol B (BPB)	77-40-7	Endocrine disrupting properties (Article 57(f) - environment) Endocrine disrupting properties (Article 57(f) - human health)	
Bisphenol S (BPS)	80-09-1	Toxic for reproduction (Article 57c) Endocrine disrupting properties (Article 57(f) - environment) Endocrine disrupting properties (Article 57(f) - human health)	
Di(2-ethylhexyl)-phthalate (DEHP)	117-81-7	Toxic for reproduction (Article 57c) Endocrine disrupting properties (Article 57(f) - environment) Endocrine disrupting properties (Article 57(f) - human health)	Esters of ortho-phthalic acid (Phthalates) are a class of organic compound commonly added to plastics to increase flexibility. They are sometimes used to facilitate the molding of plastic by decreasing its melting temperature. Phthalates can be found in: flexible plastic components (e.g., PVC), print pastes, adhesives, plastic buttons, plastic sleeveings or polymeric coatings. Also SoCs (b).
Butylbenzylphthalate (BBP)	85-68-7	Toxic for reproduction (Article 57c) Endocrine disrupting properties (Article 57(f) - human health)	
Dibutylphthalate (DBP)	84-74-2	Toxic for reproduction (Article 57c) Endocrine disrupting properties (Article 57(f) - environment) Endocrine disrupting properties (Article 57(f) - human health)	
Diisobutylphthalate (DIBP)	84-69-5	Toxic for reproduction (Article 57c) Endocrine disrupting properties (Article 57(f) - human health)	
Di-n-hexylphthalate (DnHP)	84-75-3	Toxic for reproduction (Article 57c)	
Di-n-pentyl phthalate (DPENP)	131-18-0	Toxic for reproduction (Article 57c)	
Dicyclohexyl phthalate (DCHP)	84-61-7	Toxic for reproduction (Article 57c) Endocrine disrupting properties (Article 57(f) - human health)	
1,2-Benzenedicarboxylic acid, di-C6-8-branched alkyl esters, C7-rich	71888-89-6	Toxic for reproduction (Article 57c)	
Bis(2-methoxyethyl) phthalate	117-82-8	Toxic for reproduction (Article 57c)	
Diisopentyl phthalate (DIPP)	605-50-5	Toxic for reproduction (Article 57c)	

Substances	CAS	Reason for inclusion SVHC	Notes
1,2-Benzenedicarboxylic acid, dihexyl ester, branched and linear	68515-50-4	Toxic for reproduction (Article 57c)	
Diisohexyl phthalate (DIHxP)	71850-09-4	Toxic for reproduction (Article 57c)	
1,2-Benzenedicarboxylic acid, di-C7-11-branched and linear alkyl esters (DHNUP)	68515-42-4	Toxic for reproduction (Article 57c)	
1,2-Benzenedicarboxylic acid Dipentyl ester, branched and linear	84777-06-0	Toxic for reproduction (Article 57c)	
1,2-Benzenedicarboxylic acid, mixed decyl and hexyl and octyl diesters	68648-93-1	Toxic for reproduction (Article 57c)	
1,2-Benzenedicarboxylic acid, di-C6-10-alkyl esters	68515-51-5	Toxic for reproduction (Article 57c)	
n-Pentyl-isopentylphthalate (nPIPP)	776297-69-9	Toxic for reproduction (Article 57c)	
Bis(2-ethylhexyl) tetrabromophthalate	26040-51-7	vPvB (Article 57e)	
Nonyl phenol ethoxylates (NPEOs)	9016-45-9	Endocrine disrupting properties (Article 57(f) - environment)	Also SoC (b).
N,N-dimethylacetamide	127-19-5	Toxic for reproduction (Article 57c)	Also SoC (b).
Pentadecafluorooctanoic Acid (PFOA)	335-67-1	Toxic for reproduction (Article 57c)	Also SoC (b) and (c). (Carc.2, Repr. 1B, STOT RE 1) (listed under the Stockholm Convention and listed under the POPs Regulation)
Hexabromocyclododecane (HBCDD)	3194-55-6	PBT (Article 57d)	Brominated and organophosphorus substance, used as flame retardants. It has been used historically across the apparel and footwear industry. Also SoC (c).
Biphenyl-4-ylamine or 4-Aminobiphenyl	92-67-1	Carcinogenic (Article 57a)	Azo dyes and pigments Also SoC (b).
Perfluorohexane Sulfonic acid (PFHxS)	355-46-4	vPvB (Article 57e)	Perfluorohexane sulfonic acid (PFHxS), its salts and PFHxS-related compounds Also SoCs (c)
Perfluorohexane Sulfonic acid, potassium salt (PFHxS-K)	3871-99-6	vPvB (Article 57e)	
Perfluorohexane Sulfonic acid, lithium salt (PFHxS-Li)	55120-77-9	vPvB (Article 57e)	
Perfluorohexane Sulfonic acid, ammonium salt (PFHxS-NH4)	68259-08-5	vPvB (Article 57e)	
Perfluorohexane Sulfonic acid, sodium salt (PFHxS-Na)	82382-12-5	vPvB (Article 57e)	
[4-[[4-anilino-1-naphthyl][4-(dimethylamino)phenyl]methylene]cyclohexa-2,5-dien-1-ylidene] dimethylammonium chloride (C.I. Basic Blue 26)	2580-56-5	Carcinogenic (Article 57a)	This substance is registered under the REACH Regulation but is not currently being manufactured in and / or imported to the European Economic Area.
Disodium 4-amino-3-[[4'-[(2,4-diaminophenyl)azo][1,1'-biphenyl]-4-yl]azo] -5-hydroxy-6-(phenylazo)naphthalene-2,7-disulphonate (C.I. Direct Black 38)	1937-37-7	Carcinogenic (Article 57a)	Some uses of this substance are restricted under Annex XVII of REACH. – also SoC (b) and SoC (d) for regulatory limitations.

Note: List is not exhaustive.

Source: JRC's own elaboration based on the list of Substances of Very High Concern (SVHC) in the [Candidate List for REACH Authorisation](#) and [AFIRM Restricted Substance List \(2025\)](#).

12.3.4 Substances of Concern type (b)

Substances of concern defined under Article 2(27) (b) are those classified in Part 3 of Annex VI to Regulation (EC) No 1272/2008 in one of the following hazard classes or hazard categories: *carcinogenicity categories 1 and 2; germ cell mutagenicity categories 1 and 2; reproductive toxicity categories 1 and 2; endocrine disruption for human health categories 1 and 2; endocrine disruption for the environment categories 1 and 2; persistent, mobile and toxic or very persistent, very mobile properties; persistent, bioaccumulative and toxic or very persistent, very bioaccumulative properties; respiratory sensitisation category 1; skin sensitisation category 1; hazardous to the aquatic environment — categories chronic 1 to 4; hazardous to the ozone layer; specific target organ toxicity – repeated exposure categories 1 and 2; specific target organ toxicity – single exposure categories 1 and 2.*

Substances of Concern (b) for textile apparel can be identified by examining **Annex VI of the CLP Regulation**, which in its Table 3 lists the harmonised classification of hazardous substances [ECHA CHEM | Obligation list](#). The harmonised classification and labelling of substances was also checked consulting the Classification and Labelling Inventory managed by ECHA ⁽²⁵⁵⁾. In addition, ECHA provides excel tables containing all updates to the harmonised classification and labelling of hazardous substances [Table of harmonised entries in Annex VI to CLP – ECHA](#). However, it should be noted that the only official and legally-binding harmonised classification and labelling is available in **Table 3 to Annex VI of CLP** and its subsequent ATPs published in the Official Journal of the European Union [Regulation \(EC\) No 1272/2008 – CLP link](#).

By way of illustration, some chemical substances relevant to textiles identified to meet the definition of Article 2(27)(b) of the ESPR and included in the list of harmonised classification and labelling of hazardous substances of CLP are shown in **Table 102**.

Table 102. Identification of Substances of Concern (b) for textile apparel products according to the ESPR Article 2(27).

Substances	CAS	Hazard class for SoC (b)	Notes
Acrylic acid	79-10-7	Skin Sens. 1	Used in finishing processes such as printing with dyes. Indicated as an environmental hotspot in the LCA (see section on performance requirements).
(benzothiazol-2-ylthio)methyl thiocyanate; TCMTB	21564-17-0	Skin Sens. 1, Aquatic Chronic 1	Biocide
Sodium hypochlorite	7681-52-9	Aquatic Chronic 1	Bleaching agent
Copper	7440-50-8	Aquatic Chronic 2	Copper and its compounds can be found in alloys and pigments, and in textiles as an antimicrobial agent.
Diazinon	333-41-5	Aquatic Chronic 1	Insecticide
Chlorpyrifos	2921-88-2	Aquatic Chronic 1	Pesticide
Carbon disulphide	75-15-0	Repr. 2, STOT RE	Raw material and pre-processing stage in viscose fibres production.
Zinc sulphate	7733-02-0	Aquatic Chronic 1	

²⁵⁵ <https://echa.europa.eu/information-on-chemicals/cl-inventory-database>

Substances	CAS	Hazard class for SoC (b)	Notes
Imidazole	288-32-4	Repr. 1B	Biocide in spinning fibres.
Formaldehyde	50-00-0	Carc. 1B, Muta. 2, Skin Sens. 1	Released from biocides used in finishing stage.
Methanol	67-56-1	STOT SE 1	
Octamethylcyclotetrasiloxane (D4)	556-67-2	Repr. 2 Aquatic Chronic 1 PBT vPvB	Also SoC (a).
Decamethylcyclopentasiloxane (D5)	541-02-6	PBT vPvB	Also SoC (a).
Dodecamethylcyclohexasiloxane (D6)	540-97-6	PBT vPvB	Also SoC (a).
Decabromodiphenyl ether (DecaBDE)	1163-19-5	PBT vPvB	Also SoC (a) and (c).
Triphenyl Phosphate (TPP)	115-86-6	Endocrine disrupting properties	Also SoC (a).
Tetrabromobisphenol A (TBBP A)	79-94-7	Carc. 1B Aquatic Chronic 1	Also SoCs (a).
2,2-bis(bromomethyl)-1,3-propanediol (BBMP)	3296-90-0	Carc. 1B Muta. 1B	
Trixylyl phosphate (TXP)	25155-23-1	Repr. 1B	
Tris(2-chloroethyl)phosphate (TCEP)	115-96-8	Carc. 2 Repr. 1B Aquatic Chronic 2	
Pentadecafluorooctanoic Acid (PFOA)	335-67-1	Carc.2, Repr. 1B, STOT RE 1	Also SoC (a) and (c).
UV 326	3896-11-5	vPvB	Also SoC (a)
Bisphenol-A (BPA)	80-05-7	Repr. 1B Skin Sens. 1 Aquatic Chronic 1	Also SoC (a)
Bisphenol B (BPB)	77-40-7	ED	Also SoC (a)
Bisphenol S (BPS)	80-09-1	Repr. 1B ED	Also SoC (a)
Di(2-ethylhexyl)-phthalate (DEHP)	117-81-7	Repr. 1B ED	Also SoC (a)
Butylbenzylphthalate (BBP)	85-68-7	Repr. 1B Aquatic Chronic 1 ED	Also SoC (a)
Dibutylphthalate (DBP)	84-74-2	Repr. 1B ED	Also SoC (a)

Substances	CAS	Hazard class for SoC (b)	Notes
Diisobutylphthalate (DIBP)	84-69-5	Repr. 1B ED	Also SoC (a)
Di-n-hexylphthalate (DnHP)	84-75-3	Repr. 1B	Also SoC (a)
Di-n-pentyl phthalate (DPENP)	131-18-0	Repr. 1B	Also SoC (a)
Dicyclohexyl phthalate (DCHP)	84-61-7	Repr. 1B Skin Sens. 1 ED	Also SoC (a)
1,2-Benzenedicarboxylic acid, di-C6-8-branched alkyl esters, C7-rich	71888-89-6	Repr. 1B	Also SoC (a)
Bis(2-methoxyethyl) phthalate	117-82-8	Repr. 1B	Also SoC (a)
Diisopentyl phthalate (DIPP)	605-50-5	Repr. 1B	Also SoC (a)
1,2-Benzenedicarboxylic acid, dihexyl ester, branched and linear	68515-50-4	Repr. 1B	Also SoC (a)
Diisohexyl phthalate (DIHxP)	71850-09-4	Repr. 1B	Also SoC (a)
1,2-Benzenedicarboxylic acid, di-C7-11-branched and linear alkyl esters (DHNUP)	68515-42-4	Repr. 1B	Also SoC (a)
1,2-Benzenedicarboxylic acid Dipentyl ester, branched and linear	84777-06-0	Repr. 1B	Also SoC (a)
1,2-Benzenedicarboxylic acid, mixed decyl and hexyl and octyl diesters	68648-93-1	Repr. 1B	Also SoC (a)
1,2-Benzenedicarboxylic acid, di-C6-10-alkyl esters	68515-51-5	Repr. 1B	Also SoC (a)
n-Pentyl-isopentylphthalate (nPIPP)	776297-69-9	Repr. 1B	Also SoC (a)
Bis(2-ethylhexyl) tetrabromophthalate	26040-51-7	vPvB	Also SoC (a)
Nonyl phenol ethoxylates (NPEOs)	9016-45-9	ED	Also SoC (a)
N,N-dimethylacetamide	127-19-5	Repr. 1B	Also SoC (a)
Pentadecafluorooctanoic Acid (PFOA)	335-67-1	Carc. 2 Repr. 1B STOT RE 1	Also SoC (a)
Biphenyl-4-ylamine or 4-Aminobiphenyl	92-67-1	Carc. 1A	Also SoC (a)
Disodium 4-amino-3-[[4'-[(2,4-diaminophenyl)azo][1,1'-biphenyl]-4-yl]azo]-5-hydroxy-6-(phenylazo)naphthalene-2,7-disulphonate (C.I. Direct Black 38)	1937-37-7	Carc. 1B, Repr. 2	Some uses of this substance are restricted under Annex XVII of REACH. – also SoC (d) for regulatory limitations. Also SoC (a).
tetrasodium 3,3'-[[1,1'-biphenyl]-4,4'-diylbis(azo)]bis[5-amino-4-hydroxynaphthalene-2,7-disulphonate]; C.I. Direct Blue 6	2602-46-2	Carc. 1B Repr. 2	Chemical substances used in dyeing of fibres and garments.
disodium 3,3'-[[1,1'-biphenyl]-4,4'-diylbis(azo)]bis[4-aminonaphthalene-1-sulphonate]; C.I. Direct Red 2	573-58-0	Carc. 1B Repr. 2	

Substances	CAS	Hazard class for SoC (b)	Notes
4,4'-(4-iminocyclohexa-2,5-dienylidenemethylene)dianiline hydrochloride; C.I. Basic Red 9	569-61-9	Carc. 1B	
1,4,5,8-tetraaminoanthra quinone; C.I. Disperse Blue 1	2475-45-8	Carc. 1B Skin Sens. 1	
C.I. Disperse Yellow 3; N-[4-[(2-hydroxy-5-methylphenyl)azo]phenyl]acetamide	2832-40-8	Carc. 2 Skin Sens. 1	
2-[N-ethyl-4-[(5-nitrothiazol-2-yl)azo]-m-toluidino]ethyl acetate; C.I. Disperse Blue 124	15141-18-1	Skin Sens. 1A	
1,4,5,8-tetraaminoanthra quinone; C.I. Disperse Blue 1	2475-45-8	Carc. 1B Skin Sens. 1	
C.I. Disperse Yellow 3; N-[4-[(2-hydroxy-5-methylphenyl)azo]phenyl]acetamide	2832-40-8	Carc. 2 Skin Sens. 1	
C.I. Basic Violet 3; 4-[4,4'-bis(dimethylamino)benzhydrylidene]cyclohexa-2,5-dien-1-ylidene]dimethylammonium chloride	548-62-9	Carc. 2 Acute Aquatic Chronic 1	
9-[2-(ethoxycarbonyl)phenyl] - 3,6-bis(ethylamino) -2,7- dimethylxanthylium chloride; Basic Red 1	989-38-8	Skin Sens. 1 Aquatic Chronic 1	

Note: List is not exhaustive.

Source: JRC's own elaboration based on Table 3 to Annex VI of CLP [Regulation \(EC\) No 1272/2008 – CLP link](#).

12.3.5 Substances of Concern type (c)

Substances of concern defined under Article 2(27) (c) are those regulated under Regulation (EU) 2019/1021 on persistent organic pollutants (POPs), the full information on **SoCs (c)** was obtained from the inventory and its comparison with Annexes I to IV to the Persistent Organic Pollutants (POPs) Regulation ⁽²⁵⁶⁾. At the time of writing this report, the Annexes contain 33 entries, although some entries cover multiple substances, thus raising the number of substances to nearly 300. In addition, the AFIRM list was also used to support the cross-checking of substances.

Substances listed in the POPs Regulation have previously been identified internationally as POPs, under the Stockholm Convention ⁽²⁵⁷⁾, at the proposal of a Party to the Convention and following the recommendation of the POPs Review Committee. Besides, the Convention's page lists the "Chemicals proposed for listing under the Convention" ⁽²⁵⁸⁾ which was useful to get an indication of substances likely to be identified as POPs in the coming years. Also, information on POPs can be found at ECHA's dedicated webpage ⁽²⁵⁹⁾.

By way of illustration, some chemical substances identified to meet the definition in Article 2(27)(c) of the ESPR and included in Annexes I to IV to the Persistent Organic Pollutants (POPs) Regulation and subject to be used in textile apparel products are included in **Table 103**. Note that this is not an exhaustive list.

²⁵⁶ <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32019R1021>

²⁵⁷ POPs listed under the Stockholm Convention. <https://www.pops.int/TheConvention/ThePOPs/AllPOPs/tabid/2509/Default.aspx>

²⁵⁸ <https://www.pops.int/TheConvention/ThePOPs/ChemicalsProposedforListing/tabid/2510/Default.aspx>

²⁵⁹ <https://echa.europa.eu/list-of-substances-subject-to-pops-regulation>

6630
6631

Table 103. Identification of Substances of Concern (c) relevant to textile apparel products according to the ESPR Article 2(27).

Substances	CAS	POPs Reg. Annex	Notes
Pentadecafluorooctanoic Acid (PFOA)	335-67-1	Annex I, part A and Annex IV	Also SoC (a) and (b).
Perfluorooctanesulfonic acid (PFOS)	1763-23-1	Annex I, part A and Annex IV	-
Pentabromodiphenyl ether (PentaBDE)	32534-81-9 and others	Annex I, part A and Annex IV	These substances are brominated and organophosphorus substances, used as flame retardants. These examples of flame retardant substances used historically across the apparel and footwear industry which is not intended to be a complete list. HBCDD also SoC (a).
Hexabromocyclododecane (HBCDD)	3194-55-6	Annex I, part A and Annex IV	
Decabromodiphenyl ether (DecaBDE)	1163-19-5	Annex I, part A and Annex IV	Also SoC (a) and (b).
Didecyldimethyl ammonium perfluorooctane sulfonate (PFOS-N(C ₁₀ H ₂₁) ₂ (CH ₃) ₂)	251099-16-8	Annex I, part A and Annex IV	Salts of PFOS
Perfluorooctanesulfonic acid, potassium salt (PFOS-K)	2795-39-3		
Perfluorooctanesulfonic acid, lithium salt (PFOS-Li)	29457-72-5		
Perfluorooctanesulfonic acid, ammonium salt (PFOS-NH ₄)	29081-56-9		
Perfluorooctane sulfonate diethanolamine salt (PFOS-NH(OH) ₂)	70225-14-8		
Perfluorooctanesulfonic acid, tetraethylammonium salt (PFOS-N(C ₂ H ₅) ₄)	56773-42-3		
N-Ethylperfluoro-1-octanesulfonamide (N-Et-FOSA)	4151-50-2	Annex I, part A and Annex IV	PFOS-related compounds
N-Methylperfluoro-1-octanesulfonamide (N-Me-FOSA)	31506-32-8		
2-(N-Ethylperfluoro-1-octanesulfonamido)-ethanol (N-Et-FOSE)	1691-99-2		
2-(N-Methylperfluoro-1-octanesulfonamido)-ethanol (N-Me-FOSE)	24448-09-7		
Perfluoro-1-octanesulfonyl fluoride (POSF)	307-35-7		
Perfluorooctane sulfonamide (PFOSA)	754-91-6		
Sodium perfluorooctanoate (PFOA-Na)	335-95-5	Annex I, part A	Salts of PFOA
Potassium perfluorooctanoate (PFOA-K)	2395-00-8		
Silver perfluorooctanoate (PFOA-Ag)	335-93-3		

Substances	CAS	POPs Reg. Annex	Notes
Perfluorooctanoyl fluoride (PFOA-F)	335-66-0	Annex I, part A	PFOA-related compounds
Ammonium pentadecafluorooctanoate (APFO)	3825-26-1		
1H,1H,2H,2H-Perfluorodecanesulfonic acid (8:2 FTS)	39108-34-4		
Methyl perfluorooctanoate (Me-PFOA)	376-27-2		
Ethyl perfluorooctanoate (Et-PFOA)	3108-24-5		
2-Perfluorooctylethanol (8:2 FTOH)	678-39-7		
1H,1H,2H,2H-Perfluorodecyl acrylate (8:2 FTA)	27905-45-9		
1H,1H,2H,2H-Perfluorodecyl methacrylate (8:2 FTMA)	1996-88-9		
2H,2H-Perfluorodecanoic acid (H2PFDA)	27854-31-5	Annex I, part A and Annex IV	Pesticide
Dicofol	115-32-2		
Perfluorohexane Sulfonic acid (PFHxS)	355-46-4		Perfluorohexane sulfonic acid (PFHxS), its salts and PFHxS-related compounds Also SoCs (a): 355-46-4; 3871-99-6; 55120-77-9; 68259-08-5; 82382-12-5 → vPvB (Article 57e)
Perfluorohexane Sulfonic acid, potassium salt (PFHxS-K)	3871-99-6		
Perfluorohexane Sulfonic acid, lithium salt (PFHxS-Li)	55120-77-9		
Perfluorohexane Sulfonic acid, ammonium salt (PFHxS-NH4)	68259-08-5		
Perfluorohexane Sulfonic acid, sodium salt (PFHxS-Na)	82382-12-5		
N-Methylperfluoro-1-hexanesulfonamide (N-Me-FHxSA)	68259-15-4		
Perfluorohexane sulfonamide (PFHxSA)	41997-13-1		
Methoxychlor	72-43-5	Annex I, part A	Pesticide
Hexachlorobenzene	118-74-1	Annex III, part B and Annex IV	Pesticide
Benzo(a)pyrene	50-32-8	Annex III, part B	Polycyclic aromatic hydrocarbons (PAHs) (For the purpose of emission inventories, the following four compound indicators shall be used: benzo(a)pyrene, benzo(b)fluoranthene, benzo(k)fluoranthene and indeno(1,2,3- cd)pyrene) Listed here only these 4, however, AFIRM lists more (page 36 of AFIRM Restricted Substances List, 2025). 50-32-8: Also (a) Carcinogenic (Article 57a)
Benzo(b) fluoranthene	205-99-2		
Benzo(k)fluoranthene	207-08-9		
Indeno(1,2,3- cd)pyrene	193-39-5		

Substances	CAS	POPs Reg. Annex	Notes
			Mutagenic (Article 57b) Toxic for reproduction (Article 57c) PBT (Article 57d) vPvB (Article 57e) 207-08-9: Also (a) Carcinogenic (Article 57a) PBT (Article 57d) vPvB (Article 57e)
Pentachlorobenzene	608-93-5	Annex III, part B and Annex IV	Chlorinated benzene
Endosulfan	115-29-7	Annex IV	Pesticides
Endosulfan I (alpha)	959-98-8		
Endosulfan II (beta)	33213-65-9		
Polychlorinated naphthalenes	Various	Annex IV	Pesticides Polychlorinated naphthalenes means chemical compounds based on the naphthalene ring system, where one or more hydrogen atoms have been replaced by chlorine atoms
DDT (1,1,1-trichloro-2,2-bis (4-chlorophenyl)ethane)	50-29-3	Annex IV	Pesticides
Chlordane	57-74-9		
Lindane	58-89-9		
Dieldrin	60-57-1		
Endrin	72-20-8		
Heptachlor	76-44-8		
Aldrin	309-00-2		
Mirex	2385-85-5		
Toxaphene	8001-35-2		
Hexabromobiphenyl	36355-01-8		
Pentachlorophenol (PCP)	87-86-5 and various	Annex IV	Chlorophenols. Includes pentachlorophenol, its salts and esters

6632 Note: List is not exhaustive.

6633 Source: JRC's own elaboration based on Persistent Organic Pollutants (POPs) Regulation.

6634 12.3.6 Substances of Concern type (d)

6635 **Substances of Concern under Article 2(27) (d)** are substances that negatively affect the reuse and
6636 recycling of materials in the product in which they are present.

As detailed in **Article 5(14) of ESPR**, 'For each product group concerned by ecodesign requirements, the Commission shall determine, where relevant, which substances fall under the definition in Article 2(27), point (d), taking into account, at least, whether:

(a) based on standard technologies, the substances make the reuse, or recycling process more complicated, costly, environmentally impactful, or energy- or resource-demanding;

(b) the substances impair the technical properties or functionalities, the usefulness or the value of the recycled material coming from the product or products manufactured from that recycled material;

(c) the substances negatively impact aesthetic or olfactory properties of the recycled material'.

Therefore SoCs (d) can be defined considering technical constraints (related to process and quality disturbing SoCs) and regulatory and customer chains limitations. These SoCs (d) could be present in the final product, remain in the waste and hinder its re-use and/or recycling. Substances of concern (d) could therefore be related to:

- **Technical constraints**, interfering with the processes leading to re-use, preparing for reuse or recycling, or with the functionality, usefulness, value and aesthetics of the resulting recycled materials or products for reuse (as specified in Article 5(14)). These could be process-disturbing and quality-disturbing SoCs.
- Existing **regulatory limitations** ⁽²⁶⁰⁾ to the placing on the market or use of substances (in force or in preparation); as well as **customer-driven limitations** ⁽²⁶¹⁾ along the supply chain.

At this stage, information gathered from stakeholders and available literature, is crucial for the identification of SoCs (d) in textile apparel. After the second milestone of this PS, stakeholders emphasised the absence of substances that would comply with Article 2(27) (d) for technical constraints given that in their view, impediments to recycling were exclusively associated to substances restricted in textiles **due to regulatory limitations** contained in chemical legislation. Some of these substances are already considered as SoCs (a), (b) or (c).

The identification of SoCs (d) for **technical constraints** should include the establishment of an **alert concentration (c_{alert})** which can be defined as a concentration indicated by recycling companies and that would lead to disposing of the product via incineration or landfill rather than recycling. This means that products with a certain SoC in a concentration superior to the **c_{alert}** cannot be recycled/reuse and/or the concentration of that **c_{alert}** cannot be lowered by any pretreatments being considered the limit concentration for recycling/reuse.

12.3.7 Technical constraints

Based on the broad narrative criteria defined in Article 5(14), some examples of possible chemical substances that could qualify as substances that hinder recycling and/or reuse due to technical constraints associated with their effect on waste treatment processes or on the quality, performance and use of the resulting recyclate, or when prepared for reuse of the products in which they are contained, are discussed below.

12.3.8 The case of elastane

According to several studies ⁽²⁶²⁾ ⁽²⁶³⁾, **elastane** can hamper the recyclability of the textile products in which it is present. In the first place, however, it is important to clarify in this context whether elastane is to be

²⁶⁰ This type of limitation is in line with the study carried out by Ökopol GmbH, sofia and RPA in 2020, that investigated information flows on substances of concern (but defined differently compared to ESPR) in products from supply chains to waste operators Customer is intended here as the entity purchasing reused or recycled materials (recyclates).

²⁶² European Commission: Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs, Duhoux, T., Maes, E., Hirschnitz-Garbers, M., Peeters, K. et al., Study on the technical, regulatory, economic and environmental effectiveness of textile fibres recycling – Final report, Publications Office, 2021, <https://data.europa.eu/doi/10.2873/828412>.

²⁶³ Kim Phan, Sibel Ügdüler, Lies Harinck, Ruben Denolf, Martijn Roosen, Galahad O'Rourke, Dirk De Vos, Veronique Van Speybroeck, Karen De Clerck, Steven De Meester, Analysing the potential of the selective dissolution of elastane from mixed fiber textile waste, Resources, Conservation and Recycling, Volume 191, 2023, 106903, ISSN 0921-3449, <https://doi.org/10.1016/j.resconrec.2023.106903>.

6677 considered a fibre or a substance, in which case the ESPR definition of SoCs (d) could apply. Elastane is
6678 generally considered a synthetic fibre ⁽⁴²⁾, ⁽⁴³⁾, ⁽²⁶⁴⁾, however it could be argued that elastane is as such a
6679 chemical substance subject to the SoC definition in the ESPR. In all cases, elastane (also known by its trade
6680 names Spandex or Lycra) is formed by polyurethane and urethane ⁽²⁶⁵⁾. In chemical terms, elastane is a
6681 synthetic linear macromolecule with a long chain containing at least 85% of segmented polyurethane along
6682 with the alternating hard and soft segments linked by urethane bonds. When the elastane polymer is spun
6683 into a fibre (constituted essentially by elastane), this fibre is no longer considered a substance under REACH,
6684 but rather an article ⁽²⁶⁶⁾. The 'Textile Labelling Regulation' defines elastane as an 'elastofibre composed of at
6685 least 85% (by mass) of a segmented polyurethane, and which, when stretched to three times its original
6686 length and released, recovers rapidly and substantially to its initial length' ⁽²⁶⁷⁾.

6687 Elastane is used in fabrics to impart stretch properties, providing comfort and enhancing the appearance of
6688 garments and is typically blended in small amounts (1–5%) with other fibres, such as polyester, cotton or
6689 wool, but also in larger amounts (around 20%) with polyamide in sportswear for example ⁽⁴²⁾, ⁽⁴³⁾, ⁽⁴⁴⁾, ⁽⁴⁵⁾. It
6690 is reported that elastane can hinder high-quality mechanical recycling if present in more than 15%, and while
6691 chemical recycling processes can remove it, they require high percentages of other fibres like PET to be cost
6692 effective. Currently some initiatives exist to focus on substituting elastane with recycled, bio-based, or
6693 biodegradable alternatives, but these do not fully address recycling issues. Nevertheless, there may be some
6694 technologies that claim to process elastane-containing textiles, but details are subject to industrial property
6695 ⁽⁴²⁾, ⁽⁴⁴⁾.

6696 Currently, implementing fibre traceability, and advanced sorting technologies can improve recycling efficiency
6697 ⁽⁴²⁾. In this regard, it should be noted that near-infrared (NIR) technology has limitations in detecting elastane
6698 and other fibres in complex blends or dark-coloured textiles ⁽⁴²⁾ to which a solution has not yet been found. In
6699 relation to traceability, the 'Textile Labelling Regulation' (TLR) mandates the marking of the fibre composition
6700 as compulsory to textiles and other products to ensure that correct and uniform information is made
6701 available to consumers and other stakeholders in the value chain. In accordance with Article 9 of the TLR,
6702 textile products must list fibres by name and weight percentage in descending order and fibres accounting
6703 for up to 5 % of individually or up to 15% collectively, can be grouped as "other fibres" with their total
6704 percentage, if details are unknown to manufacturers. Given this, recyclers should already be able to learn
6705 about the content of elastane in the fibres/products to recycle to some extent and adjust their processes
6706 accordingly.

6707 In the ecolabels (Blue Angel, Nordic Swan and EU Ecolabel), elastane is considered as a textile synthetic fibre
6708 to which requirements apply in relation to substances used during the production of that elastane. As an
6709 example, in EU Ecolabel, organotin compounds cannot be used in the manufacturing of elastane and
6710 workplace emissions to air of diphenylmethane-4,4'-diisocyanate (101-68-8), toluene-2,4-diisocyanate (584-
6711 84-9) and N,N-dimethylacetamide (127-19-5) are limited (to 0.005, 0.005 and 10 ppm respectively). More
6712 strict restrictions are applied to N,N-dimethylacetamide (127-19-5) content in the final product containing
6713 elastane.

6714 The analysed evidence in relation to reported difficulties in recycling fibres with more than 15% of elastane
6715 do not qualify for the restriction of such fibre even if these were to be considered as a chemical substance
6716 potentially falling under the ESPR definition for SoCs (d) and the subsequent obligation to tracking. In all
6717 cases, the 'Textile Labelling Regulation' already mandates the labelling of elastane-made textile apparel
6718 products accompanied of the % in weight of elastane in the garments. Consequently, elastane is not
6719 proposed as an SoC(d) to be listed in a possible future delegated act on textile apparel under ESPR, nor as a
6720 substance upon which a potential limitation, in the form of a performance requirement, would be established.

²⁶⁴ Senthilkumar, M., Anbumani, N. and Hayavadana, J. (2011). Elastane fabrics—A tool for stretch applications in sports. *Indian Journal of Fibre and Textile Research*. 36. 300.

²⁶⁵ What Is Elastane Fabric? Sewport. Available here: <https://sewport.com/fabrics-directory/elastane-fabric>

²⁶⁶ See example 17: 'Textile and non-woven processing' in the ECHA Guidance on requirements for substances in articles. ECHA 2017. https://echa.europa.eu/documents/10162/2324906/articles_en.pdf

²⁶⁷ Regulation (EU) No 1007/2011 of the European Parliament and of the Council of 27 September 2011 on textile fibre names and related labelling and marking of the fibre composition of textile products and repealing Council Directive 73/44/EEC and Directives 96/73/EC and 2008/121/EC of the European Parliament and of the Council <https://eur-lex.europa.eu/eli/reg/2011/1007/oj/eng>

12.3.9 Other identified substances

Another chemical substance for which some evidence has been found that it represents some technical constraints to recycling are the **cyclic oligomers** affecting the recycling of PET. Oligomers are unavoidable by-products of the production of poly(ethylene terephthalate) fibres. Oligomers can be linear or cyclic molecules and while linear oligomers do not cause many problems due to their partial water solubility, the cyclic oligomers are insoluble cyclic trimers and cause problems during subsequent processes, as well as resulting aesthetic problems at dark shades. Thus, this substance could be classified as a quality/process disturbing chemical. It has been reported that major problems associated with these oligomer deposits in textile processing are aesthetic effects (cloudy appearance) at dark shades and lower brilliancy of shade due to deposition on the surface of the fibre; efficiency decrease and the need for a more frequent cleaning of the dyeing apparatus due to precipitation of these compounds on the equipment; deposits on the guide rollers of the yarn winding equipment; dusting on air-jet looms; and more frequent cleaning of production units (Eren and Anis, 2009 ⁽²⁶⁸⁾). Other sources explain that these oligomers can adversely affect the processing of recycled PET, such as dye precipitation on fabric and increased yarn friction (Bingnan et al., 2023 ⁽²⁶⁹⁾). Researchers have investigated the effect of the treatment with ozone of the oligomers, especially cyclic trimers. Given that these oligomers would fall under the definition of NIAS ⁽²⁷⁰⁾ which are not intended to remain in the final product and that such substances fall under the exemption from information requirements proposed in the **section 12.3.13 for 'Exemptions'**, there is no proposal for these as SoC (d) subject to information requirements under the ESPR.

12.3.10 Regulatory and customer-driven limitations

Some chemical substances bring about constraints for recycling because of regulatory limitations. Such is the case of for example **NPEOs** and **PFOA** which are already identified as SoCs (a), (b) or (c).

The recent study from Temmink et al. (2025) ⁽²⁷¹⁾ mapped contaminants in polycotton textiles on the Dutch market in the context of the Texpower project which aims to develop a transparent circular polycotton supply chain in the Netherlands. Whether the recycling process is hampered by the presence of certain substances, depends in a major way of the recycling technology used. In general terms, three technologies are widely spread: mechanical, chemical or thermo-mechanical recycling. When **mechanical recycling** is used, textiles are cut in small pieces and later shredded into new fibres, meaning that finishes or dyes present in the textiles remain. Even if the chemical substance may not hinder the recycling process itself, the resulting fibres and later produced textiles, will generally be contaminated with them. It is reported that for a small fraction of the textiles carrying a certain substance, large amounts of materials can be contaminated as for example in the case of finishings as silicones. In **chemical recycling**, materials are separated before, i.e. polyester would be separated from cotton. Textile recycling challenges include material property differences (e.g., polymer chain length, crystallinity) and contaminants such as finishes or dyes. These can release harmful substances, risking worker/environmental safety. Chemical constituents may also be lost via leaching or degradation, depending on the finish and recycling method. **Thermo-mechanical recycling** presents limitations as it can only be applied to synthetic thermoplastic materials and issues are related to the

²⁶⁸ Eren, H. K. and Anis, P., Surface Trimer Removal of Polyester Fibers by Ozone Treatment, Textile Research Journal Vol 79(7): 652–656 DOI: 10.1177/0040517508097521 <https://journals.sagepub.com/doi/epdf/10.1177/0040517508097521>.

²⁶⁹ Bingnan Mu, Yuanyi Shao, Lauren McBride, Hannah Hidalgo, Yiqi Yang, Rapid fiber-to-fiber recycling of poly (ethylene terephthalate) and its dye from waste textiles without damaging their chemical structures, Resources, Conservation and Recycling, Volume 197, 2023, 107102, ISSN 0921-3449, <https://doi.org/10.1016/j.resconrec.2023.107102>.

²⁷⁰ Non-intentionally added substances ('NIAS') in the final product include the unintentional carry-over of auxiliaries that have been used indirectly to give specific functions or characteristics to the product (e.g. dyeing carriers) or not (e.g. pH adjusters), as well as substances present in the product as a result of contamination (e.g. abrasion from machineries, other materials, e.g., cleaning products, etc.). More information here: Food Packaging Forum, 2013, Dossier – Non-intentionally added substances (NIAS). Prepared by Birgit Geueke. Available at: <https://foodpackagingforum.org/resources/background-articles/non-intentionally-added-substances-nias>

²⁷¹ Robin Temmink, Haroutioun Askanian, Richard Groeneveld, Jens Oelerich, 2025. Mapping contaminants in polycotton textiles on the Dutch market: An overview of chemicals causing possible hinder in textile recycling technologies. Available at: [https://urldefense.com/v3/_https://www.government.nl/documents/reports/2025/06/20/mapping-contaminants-in-polycotton-textiles-on-the-dutch-market_!!DOxrgLBm!HonH0sxN5SAGUSKnJsHen9bmy434p7N45bL0t1ewb1tpVVCj97esa_HQLfJN9VPZrMEHJnd_AFTeBBOEUQjW9axG8Mz5Ba0cHyDfsXb\\$.](https://urldefense.com/v3/_https://www.government.nl/documents/reports/2025/06/20/mapping-contaminants-in-polycotton-textiles-on-the-dutch-market_!!DOxrgLBm!HonH0sxN5SAGUSKnJsHen9bmy434p7N45bL0t1ewb1tpVVCj97esa_HQLfJN9VPZrMEHJnd_AFTeBBOEUQjW9axG8Mz5Ba0cHyDfsXb$.)

6758 presence of impurities compromising the extrusion process (material is converted to pellets, then melted,
6759 extruded and drawn into fibres).

6760 The process can be altered by non-thermoplastic constituents or by alterations in the flow due to certain
6761 chemical substances. The study from Temmink et al. found that chemical compounds, particularly **chemical**
6762 **finishes, dyes, pigments, and coatings**, can be major contaminants hindering textile recycling. While
6763 physical contaminants such as fibre blends and trims also pose significant challenges, chemical finishing
6764 stands out due to its dual impact on both environmental safety and the efficiency of recycling processes. The
6765 study also ranks the type of chemical substances and concluded that **flame retardants, anti-pilling,**
6766 **wrinkle resistant chemicals, improving colour fastness chemicals and repellents** have a high
6767 potential to present a risk to recycling (chemical/thermomechanical or chemical). In relation to dyes, their
6768 findings suggest that **reactive and sulphuric dyes and pigments**, represent a potential risk to recycling. In
6769 relation to coatings and prints, the study concluded that polyurethane, polyacrylate, polyvinylchloride,
6770 poly(tetra)fluoro-carbon, silicone and rubber all have a high potential risk for recycling except silicones which
6771 present a medium risk. The reasons to present a risk to recycling can be the release of harmful substances
6772 during the process, to release harmful side products during recycling, derive in problems in subsequent
6773 stages of the process, to interfere with other chemicals during for instance chemical recycling or to
6774 potentially clog the equipment used. In particular, coatings can stick to fibres during mechanical recycling
6775 leading to low quality of recycled fibres or release toxic side-products during chemical recycling. Since no
6776 single recycling technology can effectively manage all types of contaminants, a multifaceted approach is
6777 essential. The study proposes to address contaminations that hinder recycling by developing advanced
6778 analytical tools, conducting real-world testing, and implementing innovative mitigation strategies. In addition,
6779 a transformation of how textiles are designed, finished, and valued at end-of-life may be alternative
6780 strategies to tackle the issues in the recycling of textile apparel products.

6781 Article 5(14) of ESPR mandates to provide the list of SoCs (d) which, as explained, are only identified due to
6782 existing **regulatory** and **customer-driven limitations** along the supply chain.

6783 Substances of concern (d) due **regulatory limitations** would be those included under a number of
6784 restrictions in Annex XVII to REACH (as already mentioned in Task 4 of this PS). The restrictions under **Annex**
6785 **XVII to REACH to consider as those that specifically mention textiles in their scope**. These include
6786 entries: 4 - Tris (2,3 dibromopropyl) phosphate; 7 - Tris(aziridiny)phosphin oxide; 8 - Polybrominated biphenyls;
6787 18 - Mercury compounds; 20 - Organostannic compounds; 43 - Azocolourants and Azodyes; 46/46a -
6788 Nonylphenol and Nonylphenol ethoxylates; 47 - Chromium VI compounds (relevant to leather articles); 68 -
6789 C9-C14 PFCAs and 72 - CMRs in textiles and footwear. Other restrictions, of general application to articles
6790 placed on the market for supply to the general public, or covering all articles placed on the market, may also
6791 apply to textile articles, for instance entries 50 - (certain) polycyclic aromatic hydrocarbons; 51 and 52
6792 (certain phthalates); 61 - Dimethylfumarate; and 63 - Lead and its compounds (all included in **Table 104**).

6793 Substances of concern (d) due to **customer-driven limitations** could relate to specific industry practice.
6794 Such would constitute limitations or pressures exerted by actors in the supply chain, sometimes on behalf of
6795 the final consumer, on the waste chain actors or on the suppliers of the raw materials. These customer-
6796 driven requirements may be voluntary and more stringent compared to those stemming from regulatory
6797 obligations (NPEOs, in particular, were indicated by stakeholders as a one of the current major concerns of
6798 textile recycling companies). In general, these customer-driven pressures/minimum requirements may take
6799 the form of voluntary “negative lists” that identify specific unwanted substances, as well as certification
6800 schemes including specific criteria on the presence of substances. Several examples for textiles are already
6801 exemplified in the good practice guides from AFIRM, ZDHC, BlueSign, OEKO-TEX® or GOTS where SoCs are
6802 included for both reasons, regulatory and customer-driven. However, these lists also include substances
6803 restricted under REACH, POPs and other policy instruments of application worldwide, i.e. outside Europe (just
6804 to cite an example, the restricted substance list from AFIRM includes restrictions for substances under the
6805 ‘California Proposition 65 on Substances’, the ‘Japan Law 112’, etc.). It may also happen that certain textile
6806 companies along the value chain may be contractually obliged to comply with these restricted substances
6807 lists as part of their B2B contractual requirements. Given the number and variability of these lists,
6808 stakeholders are invited to consult them in order to determine if other substances not listed already as part
6809 of SoCs (a), (b), (c) and (d) for regulatory limitations could fall under these ‘customer-driven’ limitations.

6810

6811 **Table 104.** Identification of possible SoCs (d) due to existing regulatory limitations in textile apparel products.

Substance	CAS no.	Existing regulatory limitation/ notes	Other initiatives/lists
Tris (2,3 dibromopropyl) phosphate (TRIS)	126-72-7	Entry 4 of Annex XVII to REACH (it is a brominated and organophosphorus substance/ flame retardant)	Reporting limit: 5 ppm Limit in component material in final product: 10 ppm (AFIRM)
Tris(aziridinyl)phosphin oxide (TEPA)	545-55-1	Entry 7 of Annex XVII to REACH (it is a brominated and organophosphorus substance/ flame retardant)	Reporting limit: 5 ppm Limit in component material in final product: 10 ppm (AFIRM)
Polybrominatedbiphenyls; Polybrominatedbiphenyls (PBB)	59536-65-1	Entry 8 of Annex XVII to REACH (it is a brominated and organophosphorus substance/ flame retardant)	Reporting limit: 5 ppm Limit in component material in final product: 10 ppm (AFIRM)
Mercury compounds	7439-97-6	Entry 18 of Annex XVII to REACH	Reporting limit: Extractable: 0.02 ppm; Total: 0.1 ppm Limit in component material in final product: Extractable: 0.02 ppm; Total: 0.5 ppm (AFIRM)
Organostannic compounds (Diocetyl tin (DOT))	Various	Entry 20 of Annex XVII to REACH (it is an organotin compound)	Reporting limit: 0.1 ppm; Limit in component material in final product: 1 ppm (AFIRM)
Azocolourants and Azodyes Aromatic amines: biphenyl-4-ylamine 4-aminobiphenyl xenylamine benzidine 4-chloro-o-toluidine 2-naphthylamine o-aminoazotoluene 4-amino-2',3-dimethylazobenzene 4-o-tolylazo-o-toluidine 5-nitro-o-toluidine 4-chloroaniline 4-methoxy-m-phenylenediamine 4,4'-methylenedianiline 4,4'-diaminodiphenylmethane 3,3'-dichlorobenzidine 3,3'-dichlorobiphenyl-4,4'-ylenediamine 3,3'-dimethoxybenzidine o-dianisidine 3,3'-dimethylbenzidine 4,4'-bi-o-toluidine 4,4'-methylenedi-o-toluidine 6-methoxy-m-toluidine p-cresidine 4,4'-methylene-bis-(2-chloro-aniline) 2,2'-dichloro-4,4'-methylene-dianiline 4,4'-oxydianiline 4,4'-thiodianiline	Aromatic amines: 92-67-1 92-87-5 95-69-2 91-59-8 97-56-3 99-55-8 106-47-8 615-05-4 101-77-9 91-94-1 119-90-4 119-93-7 838-88-0 120-71-8 101-14-4 101-80-4 139-65-1	Entry 43 of Annex XVII to REACH	Azo-amines and arylamide salts: Reporting limit: 5 ppm each ; Limit in component material in final product: 20 ppm each (AFIRM). Azodyes: Reporting limit: 15 ppm each ; Limit in component material in final product: 30 ppm each (AFIRM).

Substance	CAS no.	Existing regulatory limitation/ notes	Other initiatives/lists
o-toluidine 2-aminotoluene 4-methyl-m-phenylenediamine 2,4,5-trimethylaniline o-anisidine 2-methoxyaniline 4-amino azobenzene Azodyes A mixture of: disodium (6-(4-anisidino)-3-sulfonato-2-(3,5-dinitro-2-oxidophenylazo)-1-naphtholato)(1-(5-chloro-2-oxidophenylazo)-2-naphtholato)chromate(1-); trisodium bis(6-(4-anisidino)-3-sulfonato-2-(3,5-dinitro-2-oxidophenylazo)-1-naphtholato)chromate(1-)	95-53-4 95-80-7 137-17-7 90-04-0 60-09-3 118685-33-9 (C ₃₉ H ₂₃ ClCrN ₇ O ₁₂ S ₂ Na) CAS not allocated Component 2: C ₄₆ H ₃₀ CrN ₁₀ O ₂₀ S ₂ . 3Na		
(a) Nonylphenol C ₆ H ₄ (OH)C ₉ H ₁₉ (b) Nonylphenol ethoxylates (C ₂ H ₄ O) _n C ₁₅ H ₂₄ O	-	Entry 46 of Annex XVII to REACH	0.01 % by weight (corresponding to the regulatory threshold)
Nonylphenol ethoxylates (NPE) (C ₂ H ₄ O) _n C ₁₅ H ₂₄ O	-	Entry 46a of Annex XVII to REACH	0.01 % by weight (corresponding to the regulatory threshold)
Chromium VI compounds (relevant to leather articles);	18540-29-9	Entry 47 of Annex XVII to REACH equal to or greater than 3 mg/kg (0,0003 % by weight)	Textiles: Reporting limit: 0.5 ppm; Limit in component material in final product: 1 ppm and 3 ppm in leather for both (AFIRM).
C9-C14 PFCA	Various	Entry 68 of Annex XVII to REACH < 25 ppb for the sum of C9-C14 PFCA and their salts or 260 ppb for the sum of C9-C14 PFCA-related substances	25 ppb total (AFIRM)
CMRs in textiles and footwear (see Appendix 12)	Various	Entry 72 of Annex XVII to REACH Substances listed in column 1 of the Table in Appendix 12	Various
(certain) polycyclic aromatic hydrocarbons	Various	Entry 50 of Annex XVII to REACH [...]	Various
Certain phthalates: Bis(2-ethylhexyl) phthalate (DEHP) Dibutyl phthalate (DBP) Benzyl butyl phthalate (BBP)	117-81-7 84-74-2 85-68-7	Entry 51 of Annex XVII to REACH (equal to or greater than 0,1 % by weight of the plasticised material, in toys and childcare articles).	Various

Substance	CAS no.	Existing regulatory limitation/ notes	Other initiatives/lists
Diisobutyl phthalate (DIBP)	84-69-5		
Certain phthalates: Di-'isononyl' phthalate (DINP) Di-'isodecyl' phthalate (DIDP) Di-n-octyl phthalate (DNOP)	28553-12-0 and 68515-48-0 6761-40-0 and 68515-49-1 117-84-0	Entry 52 of Annex XVII to REACH (equal to or greater than 0,1 % by weight of the plasticised material, in toys and childcare articles).	Various
Dimethylfumarate	624-49-7	Entry 61 of Annex XVII to REACH (not be used in articles or any parts thereof in concentrations greater than 0,1 mg/kg. Articles or any parts thereof containing DMF in concentrations greater than 0,1 mg/kg shall not be placed on the market).	Reporting limit: 0.05 ppm Limit in component material in final product: 0.1 ppm (AFIRM)
Lead and its compounds	7439-92-1 and various	Entry 63 of Annex XVII to REACH (concentration of lead (expressed as metal) in those articles or accessible parts thereof is equal to or greater than 0,05 % by weight, and those articles or accessible parts thereof may, during normal or reasonably foreseeable conditions of use, be placed in the mouth by children. That limit shall not apply where it can be demonstrated that the rate of lead release from such an article or any such accessible part of an article, whether coated or uncoated, does not exceed 0,05 µg/cm ² per hour (equivalent to 0,05 µg/g/h)).	Reporting limit: Extractable: 0.2 ppm; Total: 10 ppm Limit in component material in final product: Extractable: Adults: 1 ppm; Children and babies: 0.2 ppm; Total 90 ppm (AFIRM)

Note: List is not exhaustive.

Source: JRC's own elaboration using several sources [AFIRM](#), [REACH consolidated 1 Sept.2025](#)

12.3.11 Conclusion - Substances of Concern type (d)

In conclusion, substances considered to **have a negative effect on the recycling or the reuse of the materials in which they are contained are established for textile apparel products as summarised in Table 104 and Table 106**. These substances are proposed as SoCs (d) due to the regulatory and customer-driven limitations which are already imposed upon them (primarily due to chemical safety considerations). No conclusive examples have been found, also as corroborated by stakeholder consultation, of specific substances in textile apparel having a negative effect on reuse and recycling, other than those identified as already regulated, as a consequence of their risks to human and environmental health.

12.3.12 Thresholds for information requirements for substances under Article 2(27)

Thresholds for the identified SoCs subject to information requirements and which trigger their tracking are also to be proposed for each SoC type, as well as the granularity at which the thresholds should be assessed (i.e., definition of what are 'relevant components', if any).

As stated in **Article 7(5)**, *'The Commission may, where appropriate for the product group concerned, set thresholds for when the information requirement regarding substances of concern is to apply'*. When a substance is assigned to more than one SoC category (e.g. a SoC which is both type (a) and (b)), the threshold triggering the information requirement should conservatively be defined considering the lowest value among the criteria identified for the different SoCs. Within this PS, it seems appropriate to suggest the threshold for information requirements for safety-related SoCs (i.e. those under Article 2(27) (a), (b) and (c)), to be inspired by relevant thresholds established in other Union legislation. In particular:

- For **SoCs under Article 2(27) (a) and (c)** present in textile apparel products, it is suggested to set a threshold of **0.1% w/w**. Equally, all POPs are proposed to be treated as if they were listed as SVHCs in the Candidate List of REACH.

Nevertheless, it should be noted that in REACH, the threshold of 0.1% w/w applies in relation to the weight of an article as defined in the REACH Regulation. A 'textile product' under ESPR may not necessarily in all cases correspond with what is considered an 'article' under REACH since the majority of textile products consist of several 'articles'. Thus, the weight of a 'textile product' would in general be much higher than the weight of the 'articles' composing such 'textile products'. Calculating the threshold concentration based on the weight of textile products would therefore lead to fewer triggers for tracking SoCs under Article 2(27) (a) and (c) than if the threshold were calculated based on the weight of an 'article' (i.e. in accordance with Article 33(1) of REACH).

At this stage, the PS team considers that two options for calculation of threshold concentrations can be presented for further assessment and discussion with stakeholders and during the Impact Assessment:

Option 1) Calculation w/w made based on the weight of the final garment. This means that the calculation of the threshold of 0.1% w/w is done taking into account the total weight of the final apparel product. (i.e. including all its parts/components, be it made of textile materials or other materials). This would seem the simplest approach but deviating from the approach currently applied under REACH.

Option 2) Calculation based on two separate groups of materials (i) the textile fibres component of the textile apparel product (aggregated weight of textile materials) and including soft non-fibre components such as leather, and (ii) separate calculation for individual hard components, in particular buttons, sequins, zippers, etc. which would be assessed based on their individual respective weights. The latter approach would allow to apply the relevant thresholds to discrete components that can be easily separated or become detached from the apparel, providing information on SoCs that is more meaningful towards risk management of these components (e.g. in case of ingestion by children during use phase, but also of relevance to waste management of separated hard components, etc).

The two proposed calculation options would be of application to thresholds proposed for both SoCs (a) and SoCs (c). It will have to be evaluated whether choosing a different granularity for SoCs (b) and/or SoCs (d) would make sense. A priori, choosing a different granularity for SoCs

6867 (b) may create workability issues, given the potentially large number of substances involved
6868 and the need to have an efficient data collection and monitoring that will likely be combined
6869 with that for SoCs (a) and SoCs (c). For SoCs (d), which will likely be defined in smaller
6870 numbers, and for very specific reasons, deviation from a common granularity for all other
6871 SoCs may be justified, on a case-by-case basis.

6872

6873 — In the case of **SoCs under Article 2(27) (b)**, and taking into account that over 80% of textile
6874 apparel is produced in third countries, it is proposed to set a generic threshold equal to the cut-off
6875 values as shown in **Table 1.5.1 provided under section 1.5.3. on 'General guidance for**
6876 **compiling safety data sheets' of GHS** ⁽²⁷²⁾. These values indicate the minimum concentration
6877 above which information on hazardous constituents should be available, internationally, in all SDS
6878 produced in compliance with GHS. Note that the hazard code 'hazardous to the ozone layer' is not
6879 listed in Table 1.5.1 of GHS, however this hazard is defined in **Chapter 4.2** 'Hazardous to the ozone
6880 layer' of GHS and therefore it is proposed to use the concentration of 0.1% which is used for the
6881 classification of mixtures in the Montreal Protocol. This 0.1% applied to the hazard class 'hazardous
6882 to the ozone layer' is in line with Table 5.1 of Annex I to CLP and with Table 1.1 of Annex II to REACH
6883 (classification limit for mixture classification). Where a relevant hazard class or category is not listed
6884 in the referred table, but appears in Annex I to the CLP Regulation with a generic concentration limit
6885 (GCL), as applicable for the classification of mixtures, the lowest generic concentration limit value for
6886 the substance in that class/category that leads to classification of the mixture (i.e. in the lower
6887 category of the relevant hazard class) should be used ⁽²⁷³⁾. The lowest of all the applicable
6888 concentration limits and cut-off values (corresponding to the hazard classes applicable to the
6889 substance) should be the threshold for tracking. Precisely the thresholds should be as indicated in
6890 Annex I of CLP for the following hazard classes:

6891 — endocrine disruption for human health, category 1 or 2: Annex I, section 3.11, Table 3.11.2.;

6892 — endocrine disruption for the environment, category 1 or 2: Annex I, section 4.2, Table 4.2.2.;

6893 — persistent, bioaccumulative and toxic: Annex I, section 4.3, point 4.3.3.1.;

6894 — very persistent, very bioaccumulative: Annex I, section 4.3, point 4.3.3.1.;

6895 — persistent, mobile and toxic: Annex I, section 4.4, point 4.4.3.1.;

6896 — very persistent, very mobile: Annex I, section 4.4 point 4.4.3.1.;

6897 As an alternative to this approach, a stricter approach that would refer to Table 1.1 of Annex II to
6898 REACH and, for those hazard classes not listed therein to the generic concentration limits defined in
6899 CLP (as explained above), could be set as threshold concentrations to trigger the information
6900 requirements. This approach would be consistent with the rules applicable to European
6901 manufacturers and may largely be possible to comply with for safety datasheets of chemicals
6902 procured and used in the EU. It is questionable however that safety data sheets used and made
6903 available in third countries would provide information compliant with REACH SDS limits.
6904 Consequently, this approach, subject potentially to further impact assessment, is not proposed by the
6905 PS team of the JRC for this product group.

6906 To guide manufacturers and other stakeholders on the exact thresholds to use, **Table 105** compiles a
6907 summary of thresholds to use for SoCs (b) identified under the ESPR Article 2(27) in textile apparel products.

²⁷² Globally Harmonised System on Classification and Labelling of Chemicals (GHS). Tenth revised edition. United Nations 2023.
<https://unece.org/transport/dangerous-goods/ghs-rev10-2023>

²⁷³ For instance, it could happen that a mixture is classified under Category 1 for a given hazard class when a 1% GCL value of a substance is applicable but that mixture is classified as Category 2 when a GCL of 0.1% is reached. In this case, the lowest value, i.e. 0.1% should be considered for that substance (because the lowest concentration would lead to the classification of the mixture and, therefore, should still be mentioned in the SDS).

6908 **Table 105.** Proposed thresholds for tracking of SoCs (b) identified under the ESPR Article 2(27) in textile apparel products.

ESPR SoC (b) definition	GHS /CLP	
Hazard classes	Hazard class	Proposed concentration limit to trigger tracking of SoCs (b)
carcinogenicity categories 1 and 2;	Carcinogenicity	≥ 0.1 %
germ cell mutagenicity categories 1 and 2;	Germ cell mutagenicity (Category 1)	≥ 0.1 %
	Germ cell mutagenicity (Category 2)	≥ 1.0 %
reproductive toxicity categories 1 and 2;	Reproductive toxicity	≥ 0.1 %
endocrine disruption for human health categories 1 and 2;	endocrine disruption for human health category 1;	≥ 0.1 %
	endocrine disruption for human health category 2;	≥ 1 %
endocrine disruption for the environment categories 1 and 2;	endocrine disruption for the environment category 1;	≥ 0.1 %
	endocrine disruption for the environment category 2;	≥ 1 %
persistent, mobile and toxic or very persistent, very mobile properties;	PMT or vPvM	≥ 0.1 %
persistent, bioaccumulative and toxic or very persistent, very bioaccumulative properties;	PBT or vPvB	≥ 0.1 %
respiratory sensitisation category 1;	Respiratory sensitization	≥ 0.1 %
skin sensitisation category 1;	Skin sensitization	≥ 0.1 %
hazardous to the aquatic environment — categories chronic 1 to 4;	Hazardous to the aquatic environment	≥ 1.0 %
hazardous to the ozone layer;	hazardous to the ozone layer	≥ 0.1 %
specific target organ toxicity — repeated exposure categories 1 and 2;	Specific target organ toxicity (repeated exposure)	≥ 1.0 %
specific target organ toxicity — single exposure categories 1 and 2;	Specific target organ toxicity (single exposure)	≥ 1.0 %

6909 Source: JRC's own elaboration based on Table 1.5.1 [GHS UN](#) and various tables from [CLP](#).

6910

6911 Based on the information summarised in **Table 105**, it is proposed, as example, that some of the identified

6912 SoCs (b) should be tracked from a certain cut-off value as summarised in **Table 106** (these SoCs (b) should

6913 not fall under the categories of SoCs (a) or (c) for which the threshold is 0.1% w/w).

6914

6915 **Table 106.** Proposed thresholds for the tracking of examples of Substances of Concern (b) for textile apparel products.

Substances	CAS	Hazard class for SoC (b)	Tracking from these thresholds/concentration limits (w/w% according to the 2-option calculation considered above)
Acrylic acid	79-10-7	Skin Sens. 1	≥ 0.1 %
(benzothiazol-2-ylthio)methyl thiocyanate; TCMTB	21564-17-0	Skin Sens. 1, Aquatic Chronic 1	≥ 0.1 %
Sodium hypochlorite	7681-52-9	Aquatic Chronic 1	≥ 1.0 %
Copper	7440-50-8	Aquatic Chronic 2	≥ 1.0 %
Diazinon	333-41-5	Aquatic Chronic 1	≥ 1.0 %
Chlorpyrifos	2921-88-2	Aquatic Chronic 1	≥ 1.0 %

Substances	CAS	Hazard class for SoC (b)	Tracking from these thresholds/concentration limits (w/w% according to the 2-option calculation considered above)
Carbon disulphide	75-15-0	Repr. 2, STOT RE	≥ 0.1 %
Zinc sulphate	7733-02-0	Aquatic Chronic 1	≥ 1.0 %
Imidazole	288-32-4	Repr. 1B	≥ 0.1 %
Formaldehyde	50-00-0	Carc. 1B, Muta. 2, Skin Sens. 1	≥ 0.1 %
Methanol	67-56-1	STOT SE 1	≥ 1.0 %

Note: List is not exhaustive.

Source: JRC's own elaboration.

— For the proposed **SoCs (d)**, the thresholds should be established in a case-by-case scenario.

For **SoCs (d) due to technical constraints**, the tracking should start at a certain threshold called **alert concentration** (C_{alert}) as previously defined. This **alert concentration** should have come from discussions with recyclers and other relevant stakeholders.

For **SoCs (d) due to regulatory limitations**, the threshold for tracking substances as information requirements should be (1) equal, or (2) lower than the legal limit, or (3) a generic limit. The proposal in this PS is (1) i.e. equal to the regulatory limitations. In addition, there is the possibility to (a) list those substances included in the final textile product and provide the exact concentration in the product; or (b) indicate that all substances pertinent to information requirements are below the legal limits without the provision of such limit values but provide a positive declaration. Given that possibility (a) could entail a disproportionate effort to stakeholders putting the product in the market, it would be advisable to opt for (b): report that SoCs (d) due to regulatory limitations are below the legal limits.

For **SoCs (d) due to customer-driven limitations**, tracking should be based in case-by-case scenario and if possible, in agreement with stakeholders. Given the absence of such exchange, the thresholds to track these SoCs (d) are proposed to align with those in AFIRM, Bluesign or ZDHC substance lists.

In summary, the two options for the calculation of threshold concentrations presented above for SoCs (a) and (c) and for further assessment and discussion with stakeholders of the PS on textile apparel and during the Impact Assessment, apply also to all SoCs types. Only for SoCs (d), deviation from a common granularity for all other SoCs may be justified, on a case-by-case basis.

12.3.13 Exemptions

Additionally, justified exemptions on reporting from the mandatory information requirements should be considered for this PS on textile apparel. **Article 7(6)(b)** of the ESPR states that 'where the Commission sets out information requirements in a delegated act adopted pursuant to Article 4, it shall, where relevant: [...] (b) provide duly justified exemptions for substances of concern or information elements from such information requirements referred to in the first subparagraph of paragraph 5, based on the technical feasibility or relevance of tracking substances of concern, the existence of analytical methods to detect and quantify them, the need to protect confidential business information or in other duly justified cases; substances of concern within the meaning of Article 2(27), point (a), shall not be exempted if they are present in products, their relevant components or spare parts in a concentration above 0.1 % weight by weight'.

In relation to technical feasibility or relevance of tracking SoCs and analytical methods, specific exemptions are not proposed in this PS given that the list of proposed SoCs is not exhaustive, especially for SoCs (b) and considering that no justified exemption requests, for specific substances, have been put forward by stakeholders during consultation on SoCs done in the context of this Preparatory Study. While the tracking of

SoCs (a) cannot be exempted, it is crucial to address some practical challenges in the tracking of all SoCs, such as the need for harmonised test methods to avoid a proliferation of diverse results for the same samples. In general, a challenge exists when calculating final concentrations of substances in final products as methods may not be suitable and/or not be developed yet. In this line, some industry initiatives already provide guidance on test methods for restricted substances, an example is the [Bluesign Restricted Substances List \(RSL\)](#) or the [Compendium of Analytical Methods recommended by the Forum to check compliance of REACH Annex XVII restrictions](#). Some of the methods listed in such references for some of the SoCs (b) identified in this PS, are:

- Copper (7440-50-8) → DIN EN 16711-2 (2016).
- Formaldehyde (50-00-0) → ISO 14184-1 (2011).
- Tris (2,3 dibromopropyl) phosphate (TRIS) → DIN EN 16377.
- Polybrominated biphenyls (PBB) → DIN EN 16377.
- 4-Aminobiphenyl xenylamine → ISO 14362-2.
- o-Toluidine → EN ISO 14362-1.

A non-exhaustive list of sources also includes:

- List of analytical methods used by European customs laboratories ⁽²⁷⁴⁾;
- ECHA CHEM database ⁽²⁷⁵⁾ – information on analytical methods is often available in REACH registration dossiers;
- OECD eChemPortal ⁽²⁷⁶⁾.

While industry stakeholders have mostly advocated that CBI-based exemptions should be considered, civil society organisations are against such type of exemptions given they consider them not to be justified, referring for instance to provisions in the Aarhus Convention and POPs Regulation. However, following the consultation of the 2nd milestone for the textile apparel PS, precise information on this regard was not shared by any stakeholder so such CBI-based exemptions could not be considered for inclusion.

The concerns highlighted by a majority of industry stakeholders, derived from the need to compile extensive information on chemical substances along the textile supply chain, are acknowledged. This should not be seen as a limitation that exists only for EU suppliers as equivalent tracking requirements apply to non-EU producers, with potentially even greater challenges, with additional burdens to verifying information from non-EU producers. It is acknowledged that operators face challenges in obtaining information along complex, international, supply chains sometimes having in excess of ten tiers. Although a variety of supply chain transparency platforms, some specific to the textile sector, have become available in recent years, the JRC has been unable to determine the level of uptake, nor the costs of these and, therefore, consider it is too early to assume that these incipient tools for traceability of chemical information, are mainstream and fully in place to guarantee smooth information transfer in the sector. In addition, regulatory constraints regarding the content of GHS based safety data sheets, as compared to those done to be REACH compliant in the EU, must be also acknowledged, in particular as regards applicable thresholds for identifying hazardous substances in mixtures and, very importantly, as regards the recognition under GHS of hazard classes that have only recently been incorporated under CLP.

Having the above limitations in mind, which particularly impact SMEs, which are the vast majority of operators in the textile value chain, a number of general exemptions and delayed entry into application of obligations are proposed. These are proposed without prejudice of recognizing the dire need for increased transparency on the use of chemicals in textile supply chains, which to date, is still poorly implemented and regulated in the textile sector. The proposed approach should be understood as a minimum level for compliance, without prejudice for frontrunner companies, or those benefitting from shorter, more tightly controlled supply chains, to provide additional information on a voluntary basis, including up to the level of full material disclosure (subject to this option being possible to accommodate in the DPP).

²⁷⁴ https://taxation-customs.ec.europa.eu/system/files/2022-02/LIADe_List-Analytical-Methods_January2022.pdf

²⁷⁵ <https://chem.echa.europa.eu/>

²⁷⁶ <https://www.echemportal.org/echemportal/substance-search>

7000 According to the above, the following are proposed:

- 7001 • **Exemption 1** - exempting from information requirements those chemical substances that are not
7002 intentionally added to the final products and that are not present in the final product but that are
7003 related to the life cycle of the products. This would mean that only **chemical substances**
7004 **remaining in the final product and intentionally added** should be subject to mandatory
7005 tracking, i.e. to information requirements.

7006 In a first approach to implementing ESPR information requirements to textile apparel and taking due account
7007 to the product' centred wording of Article 7(5), it is justified to exempt substances that are not intentionally
7008 added to a product given their presence is generally unknown and difficult to determine other than by
7009 targeted chemical analysis. These may come from impurities in raw materials, degradation products or cross-
7010 contamination, or due to traces remaining from certain process chemicals, not intended to remain in the final
7011 product. Requiring the provision of information on these substances will place an additional burden on
7012 manufacturers that may not significantly contribute to the overall goal of improving product sustainability (as
7013 compared to focusing on substances that remain in the product).

7014 For substances that are not present in the final product, tracking and reporting requirements may be
7015 impractical and costly. These substances may be used during the manufacturing process but are removed or
7016 transformed before the product reaches the consumer. In such cases, requiring information on these
7017 substances may not provide meaningful insights into the product's environmental impacts.

7018 For process chemicals used in manufacture, but which are not intended to remain in the product, and which
7019 may result in emissions, other regulatory frameworks than ESPR already exist, in particular the Industrial
7020 Emissions Directive, restricting emissions of pollutants. Focusing on these substances within the context of
7021 product information requirements may lead to duplication of efforts and a certain degree of confusion among
7022 stakeholders.

7023 In this way, simplifying the number of information requirements can make them more targeted, practical, and
7024 effective in promoting transparency and accountability throughout the value chain.

7025 It should be noted that if a SoC would be highlighted by stakeholders or literature as being relevant because it
7026 is an **environmental hotspot**, such SoC should still be requested to be tracked as part of the information
7027 requirements set **in the Impact Assessment stage**. In other words, there could still be SoCs relevant
7028 towards information requirements even if related to emissions to air, water and soil along the life cycle
7029 stages. Similarly, substances that are not intentionally added but are present in the product, which are clearly
7030 identified as adversely impacting the recycling and reuse of materials, could also be included for an
7031 information requirement if later on at the IA stage its relevancy is highlighted.

7032 At this stage no exemptions of specific substances, families of substances or types of SoCs can be proposed
7033 due to aspects such as analytical difficulties, protection of confidential business information or considerations
7034 associated to relevance or other practical factors. Consultation with stakeholders carried out in the context of
7035 the Preparatory Study did not provide relevant information or reasoned requests for specific exemptions.
7036 Consequently, the PS team considers it has no solid justification to propose **substance-specific exemptions**
7037 **at this stage of the PS on textile apparel products other than limit the tracking to substances**
7038 **remaining in the final products.**

7039 The option to exempt, for instance until potentially the next review of the delegated act on textile apparel,
7040 certain broad families of SoC, for instance SoC (b) due to the large number of substances comprised therein
7041 and the limited experience of tracking them in the sector (contrary to SVHCs and POPs where there is already
7042 more extensive experience in their tracking both in the EU and internationally) is a possibility that could be
7043 considered as part of the impact assessment, if full credit were to be given to the claims by the sector, of the
7044 high burden, and material impossibility to comply. In the view of the preparatory study team, such difficulties
7045 could however be addressed soundly, without the need of an exemption, via delayed periods of entry into
7046 application. In such a manner the full coverage of information requirements introduced by the co-legislator
7047 would remain in place from the onset, but with the flexibility of delayed application to enable addressing the
7048 clear difficulties in supply chain transparency that today exist in the textile apparel sector.

7049 **12.3.14 Entry into application**

7050 The ESPR legal text, and in particular its Article 7(6), enables some flexibility to be introduced in the entry into
7051 application of information requirements. While Article 4(4) of the ESPR sets the minimum deadline for entry
7052 into application of provisions in ESPR delegated acts at 18 months from the date of entry into force of the

7053 measure (which is generally 20 days following publication in the Official Journal of the European Union
7054 (OJEU)), Article 7(6)(a) requires the Commission to, as relevant, '*lay down dates of application of such*
7055 *information requirements referred to in the first subparagraph of paragraph 5, differentiating between*
7056 *substances of concern where necessary*'. In order to take into account the difficulties in obtaining reliable
7057 information along the supply chain, and aware of the limitations of analytical approaches, the PS team
7058 proposes a **delay in the entry into application of the tracking of SoCs (b)**.

7059 As indicated, gathering all necessary information to comply with the mandatory information requirements
7060 under the ESPR may be difficult in the long supply chain for textile apparel products. Currently, many
7061 transparency platforms are being set up, and the long and international nature of the textile value chain
7062 increases the challenge to comply with the mandatory information requirements, thus the entry into
7063 application of these mandatory requirements may require some delay.

7064 The proposal for **SoCs (a), (c)** and those **SoC (d)** specifically identified in the delegated act is that they are
7065 to be subject to information requirement **18 months** after the future Delegated Act on textile apparel
7066 products is published in the Official Journal of the European Union (OJEU).

7067 For textile-relevant SoC classified with hazard classes currently recognised under GHS, and taking due account
7068 of the high number of additional substances to be potentially tracked, it is proposed that the information
7069 requirement would enter into application **36 months** after publication of the delegated act. In particular this
7070 would apply to SoC (b) classified at least under one of the following hazard classes (included under ESPR and
7071 considered in GHS):

- 7072 — (i) carcinogenicity categories 1 and 2;
- 7073 — (ii) germ cell mutagenicity categories 1 and 2;
- 7074 — (iii) reproductive toxicity categories 1 and 2;
- 7075 — (viii) respiratory sensitisation category 1;
- 7076 — (ix) skin sensitisation category 1;
- 7077 — (x) hazardous to the aquatic environment - categories chronic 1 to 4;
- 7078 — (xi) hazardous to the ozone layer;
- 7079 — (xii) specific target organ toxicity – repeated exposure categories 1 and 2;
- 7080 — (xiii) specific target organ toxicity – single exposure categories 1 and 2.

7081 For those **SoC (b) with a harmonised classification in the EU under CLP and/or for hazard classes**
7082 **currently not recognised under GHS (i.e. ED, PBT, vPvB, PMT, vPvM)** it is proposed that a delayed entry
7083 into application **8 years** after publication of the Delegated Act in the OJEU is set. This would apply to SoC (b)
7084 classified at least under one of the following hazard classes (included under ESPR but not yet considered in
7085 GHS):

- 7086 — (iv) endocrine disruption for human health categories 1 and 2;
- 7087 — (v) endocrine disruption for the environment categories 1 and 2;
- 7088 — (vi) persistent, mobile and toxic or very persistent, very mobile properties;
- 7089 — (vii) persistent, bioaccumulative and toxic or very persistent, very bioaccumulative properties.

7090 The provision of this information, on a **voluntary basis should be possible**, to the extent this information,
7091 based on harmonised classification, is available to operators. It is suggested that well before the expiry of this
7092 delayed period of entry into application, the Commission reassesses the situation of these new hazard
7093 classes, in particular as regards their level of recognition under the UN GHS system, and the extent to which
7094 these may have been incorporated into the chemicals legislation of third countries relevant to textile
7095 manufacture.

7096 Particular attention should be given to the ongoing universal restriction process on **PFAS**, currently being
7097 developed by ECHA ⁽²⁷⁷⁾. Following a possible Commission Regulation issued under REACH defining a broad

²⁷⁷ <https://echa.europa.eu/hot-topics/perfluoroalkyl-chemicals-pfas>

7098 restriction for PFAS, including in textiles, such substances would, most likely, qualify as SoC (d) due to regulatory
7099 limitations. In such a case, it is advised that the Commission would consider whether the transition time that
7100 may be defined in the REACH restriction is sufficient to also allow for the information requirement to apply also
7101 to PFAS, or whether specific provisions would be required via a dedicated amendment of the Delegated Act.

7102 **12.3.15 Identification of performance requirements on substances**

7103 Developing performance requirements for substances requires the investigation of any possible relationship
7104 between the use or the presence of one or several substances, and one or several of the 16 product aspects
7105 listed in Article 5(1) of the ESPR, which could be complemented using the product parameters included in
7106 Annex I of the ESPR. Article 6 of the ESPR 'Performance requirements', set the basis for such performance
7107 requirements:

7108 *"1. Products shall comply with performance requirements related to the product aspects as laid down in the*
7109 *delegated acts adopted pursuant to Article 4.*

7110 *2. The performance requirements shall be based on the relevant product parameters referred to in Annex I*
7111 *and shall, as appropriate, include either or both of the following:*

7112 *(a) minimum or maximum levels in relation to a specific product parameter or a combination thereof;*

7113 *(b) non-quantitative requirements that aim to improve performance in relation to one or more of such*
7114 *product parameters."*

7115 With respect to product parameter (f) of Annex I, Article 6(3) states that *'Performance requirements based on*
7116 *the product parameter referred to in Annex I, point (f), shall not restrict, for reasons relating primarily to*
7117 *chemical safety, the presence of substances in products. However, the setting of performance requirements*
7118 *shall also, where appropriate, reduce significant risks to human health or the environment'.*

7119 Substances with a specific function as outlined in the inventory developed for textile apparel products (see
7120 Annex 13.13, **Table 227**) have been the starting point for the investigation of aspects having a potential
7121 direct relationship with substances.

7122 In particular, product aspects of relevance for textile apparel are physical durability, maintenance,
7123 reparability, waste generation or recycled content and recyclability as analysed in **Section 9** of this PS report.
7124 In addition, product aspect addressing environmental impacts include the assessment of use (and efficiency)
7125 of water, energy and resources for which a Life Cycle Assessment (LCA) was performed in this preparatory
7126 study.

7127 The presence of substances and substances of concern can affect the abovementioned product aspects. While
7128 substances of concern are addressed by information requirements, other substances different from SoCs can
7129 be addressed via performance requirements.

7130 An analysis of the product aspects of relevance for textile apparel products was performed as follows.

7131 Substance-specific feedback has not been received from stakeholders in order to propose performance
7132 requirements on any chemical substance. However, a qualitative assessment of the relevant product aspects
7133 for textiles which possibly could be impacted by substances is carried out supported by the literature and the
7134 results from the Life Cycle Assessment done on the three representative products (RPs) studied in this PS.

7135 The identification of substances detrimental to the product parameters mentioned above is difficult and could
7136 only be carried out to a somewhat limited extent, since such product parameters are mostly related to the
7137 number and combination of materials, rather than to individual substances per se present in them.

7138 It is also worth highlighting the existence of trade-offs when analysing the products aspects relevant for
7139 textiles. For instance, a certain mix of fibres in textile apparel may have been used with the aim of increasing
7140 the durability of the product, but this could later hinder the recycling of these products due to the inherent
7141 nature of the mix of fibres used, where they are difficult to separate. Examples include the use of elastane
7142 (as previously discussed) or the mixing of virgin and recycled wool fibres, which, where the recycled ones have
7143 shorter fibre lengths which can, in turn, decrease the yarn strength²⁷⁸.

²⁷⁸ Wood, E., Overview of Wool Yarn Manufacture, Woolwise, The Australian Wool Education Trust, 2009, <https://www.woolwise.com/wp-content/uploads/2017/07/Wool-482-582-08-T-02.pdf>

12.3.16 Product aspect (a) 'durability' and (p) 'expected generation of waste'

The product aspect 'durability' may be linked to 'expected generation of waste' for different aspects as for instance when consumers discard the products because a loss of the initial properties of the product thus seeing as a loss of the durability or because the product breaks and cannot be repaired. Under the ESPR, durability is intended as 'physical durability'. However, the 'emotional durability' may play a certain role as already explained in previous sections (Section 9 from this PS).

A certain type of chemical substance that relates to durability are UV absorbers and stabilisers (see inventory) which are added to textile apparel products to absorb the high energy of ultraviolet light and to protect the product from degradation due to radiation. This radiation may otherwise affect the technical lifetime of the garment. In this line, there is already an information requirement set for the chemical substance UV 326 (CAS no. 3896-11-5) which is a SVHC under Article 57e of REACH as being very persistent and very bioaccumulative (vPvB). Although this is clearly a case where a chemical substance relates to physical durability of the products, a negative relationship cannot be established (durability is not negatively impacted by the UV absorber) and as such, a performance requirement cannot be considered in this particular case.

Another example of substances potentially limiting maintenance could be related to the presence of dyes that discolour during the use stage of the product, such as dyes in clothes that fade when washed at normal (30-40°C) temperatures, thus impeding the proper maintenance of the product by the (non-specialised) user. No relevant information was identified that would allow to set chemical-related performance requirements for this product aspect either.

Sodium hypochlorite which has been listed in the inventory as a SoC (b), (CAS 7681-52-9), which generally is used in the bleaching process of natural fibres (cotton, wool) or MMCF, is extensively addressed in the BREF document for textiles as problematic for different reasons, including durability:

- For wool treatment, fibres can be discoloured and damaged, thus hydrogen peroxide is used. Dichloroisocyanurate is indicated as a more advantageous alternative to sodium hypochlorite because it has the ability to release chlorine gradually, thereby reducing the risk of fibre damage.
- Bleaching with sodium hypochlorite leads to secondary reactions that form organic halogen compounds, quantified as AOX emission with its use limited in Europe. However, it should be noted that in general chlorine and chlorine-releasing compounds, including sodium dichloroisocyanurate, are capable of reacting with organic compounds to form AOX.

For these reasons, the use of sodium hypochlorite is reported in the BREF for textiles as being limited to a small number of installations. Since agents that are less aggressive on fibres are available (e.g. dichloroisocyanurate), the preliminary course of action could be to assess if limiting the use of sodium hypochlorite and its substitution with dichloroisocyanurate would be warranted. However, given the qualitative nature of available information, further investigations would be needed before considering proposing sodium hypochlorite as a candidate substance for possible performance requirements, to properly understand the conditions of use for the installations that still employ sodium hypochlorite and to then collect data to define a maximum threshold or potential substitution with alternative substances. Considering also that sodium hypochlorite is used as a process chemical in the treatment of natural fibres, and is not intended to remain in the product, potential performance requirements on this substance, although potentially possible, are considered low priority. Therefore, sodium hypochlorite is not currently proposed, based on the data available, as a candidate substance to set a performance requirement on a substance relevant to textile apparel products under the ESPR.

12.3.17 Product aspect (h) 'energy use and energy efficiency', (i) 'water use and water efficiency', (p) 'expected generation of waste' and (j) 'resource use and resource efficiency'

These product aspects are here investigated as a group since, according to the BREF for textiles, they are linked for specific substances, i.e. resource use and efficiency (energy, water) and generation of waste depend on the substances used.

The use of reactive dyes may affect several product aspects. According to the BREF for textiles 'when dyeing with reactive dyes, two rinsing steps at about 80 °C are normally carried out after dyeing, in order to remove the unfixed dye. This leads to higher consumption of water (approximately 30% more [270, CRAB 2001]) and energy.' (BREF for textiles, page 693).

7195 Due to the poor fixation rate of this family of dyes, significant water discharges arise, compared to other
7196 families of dyes. Indeed, for reactive dyes, the percentage of non-fixed dye that may be discharged in the
7197 effluent may be up to 60% (BREF for textiles, Table 2.16).

7198 Alternative dyes may be considered for wool, cashmere and acrylic fibres, for example acid dyes, whose upper
7199 limit for percentage of non-fixed dye is 20%. This value could be considered for being proposed as the
7200 maximum value of the product parameter (r) *amounts of waste generated, including plastic waste and*
7201 *packaging waste and their ease of reuse, and amounts of hazardous waste generated* (Annex I, ESPR). The
7202 generation of waste considered in this section refers to the manufacturing stage of the product.

7203 Acid dyes show lower ranges of unfixed dye (5-20%), thus reducing the generation of waste (i.e. non-fixed dye
7204 in wastewater). Moreover, acid dyes are compatible with wool fibres. Thus, a preliminary suggestion could be to
7205 consider the ban of (certain) reactive dyes and promoting their substitution with acid dyes, due to their impact
7206 on the product aspect (p) 'expected generation of waste'. Elements of proportionality and whether the main
7207 motivation for considering a restriction would be primarily chemical safety (for instance linked to the sensitising
7208 potential of some reactive dyes), would have to be taken into consideration before formulating any proposal.
7209 Further investigations with input from stakeholders would be needed to properly estimate the corresponding
7210 savings in water, energy and resource consumption achievable with different families of dyes, while ensuring
7211 compatibility with the considered type of fibres. Therefore, reactive dyes are currently not proposed as a possible
7212 performance requirement for textile apparel products under the ESPR.

- 7213 ➤ Substances having an impact on physical durability, repairability, waste generation or recycled
7214 content and recyclability, have not been identified with the sufficient relevance at this stage, thus the
7215 PS is not proposing performance requirements on those.
- 7216 ➤ Product aspect '(o) environmental impacts, including carbon footprint and environmental footprint'
7217 can still be explored in the context of substances.

7218 **12.3.18 Product aspect (o) 'environmental impacts, including carbon footprint and environmental** 7219 **footprint'**

7220 The **Life Cycle Assessments** (LCAs) performed on the RPs considered for textile apparel, may identify
7221 substances as environmental hotspots and/or at least flag them as being of some relevance.

7222 To analyse the product aspect (o) '*environmental impacts, including carbon footprint and environmental*
7223 *footprint*' the LCA performed on the RPs considered for textile apparel was analysed in detail. The PS team
7224 has scrutinised the substances included in the Life Cycle Inventory (LCI) and identified those that resulted in
7225 environmental hotspots or provided some indication of a hotspot. This means this PS analysed (i) the LCI and (ii)
7226 the life cycle impact assessment (LCIA). This was done to find any possible **relationship between**
7227 **substances, their concentrations (or amounts) and the impacts on the environmental footprint of**
7228 **the RPs**, for each individual impact category.

7229 **12.3.19 Chemical substances in the life cycle assessment**

7230 Task 5 provides the **life cycle assessments** (LCAs) of the three representative products in this PS, one for
7231 each product category: (1) knitted products; (2) products made of denim fabric; and (3) woven products made
7232 without denim fabric (**Section 10** of this PS).

7233 According to the LCA conducted on the RPs, the use of chemicals is the third main contributor to the raw
7234 material production stage (LCS1), with values between 7 and 12% of the overall impact contribution. Whereas
7235 the use of chemicals is the second main contributor to the environmental impacts in the manufacturing stage
7236 (LCS2) with a contribution of 25-41%, which correspond to 5-10% of the total environmental impacts.

7237 The production of raw materials (LCS1) and the manufacturing processes (LCS2) are the two stages to which
7238 chemical substances can contribute to a major extent. Chemical substances involved in the distribution stage
7239 (LCS3), the use of the textile apparel products (LCS4) and End-of-life (LCS5) are negligible according to the
7240 life cycle assessment. It should be noted that the number of chemical substances included in the LCA has
7241 been limited to those having characterisation factors in the databases used. Sometimes these were
7242 aggregated datasets.

7243 In particular, the nine chemicals with higher contribution to the environmental profile are found to be the
7244 same for the three RPs (1) knitted, (2) denim, (3) other woven. Some changes are found with a slightly

7245 relevant single score contribution to the assessment generally for knitted products. The summary of the
7246 results of the analysis of the environmental hotspots performed is summarized as follows:

- 7247 — **Acrylic acid** (used in the manufacturing process, LCS2): according to the LCA it contributes to a
7248 single score of 2% for knitted and other woven products and 1% for denim products. In all 3 RPs,
7249 the highest impact is in the environmental impact of human toxicity non-cancer (40.3%), followed by
7250 resource use of fossils (3.8%), photochemical ozone formation (human health) of 3.2% and climate
7251 change (2.2%).
- 7252 — **Antimony** (from raw material production, LCS1): single score of 2% for knitted and other woven and
7253 1% for denim products. In all 3 RPs, the largest impact is due to resource use of minerals and
7254 metals (39.8%).
- 7255 — **Detergent/surfactant** (from LCS1 and LCS4, as detergents and surfactants are used in the
7256 production of fibres/garments in the pretreatment and in dyeing processes and also in the consumer
7257 use phase): single score of 2% for knitted products, and 1% for denim and other woven and largest
7258 impacts due to land use (5.2%), ecotoxicity-fresh water (4.3%) and human toxicity cancer (4.1%).
- 7259 — **Disperse dyes** (from LCS2, dyeing of fibres and/or garments): single score of 1% in knitted
7260 products only (and 0% in the other two RPs). The largest impacts in the LCA are due to resource use
7261 minerals and metals (2.4%).
- 7262 — **Polydimethylsiloxane** (from LCS2, as they are used as lubricant in spinning fibres, and softener in
7263 finishing processes): score of 2% for knitted products and 1% for denim and other woven and
7264 largest impacts due to ozone depletion (45.3%) and human toxicity cancer (8.6%).
- 7265 — **Purified terephthalic acid** (from LCS1): score of 1% in knitted and other woven products and 0%
7266 for denim products. The largest impacts are due to ozone depletion (6.2%), photochemical ozone
7267 formation (human health) of 2.6% and climate change (2.1%).
- 7268 — **Sodium hydroxide**, without water, in 50% solution state (from LCS1 and 2 as this chemical
7269 substance is used for bleaching and solvents in all fibre types). The impact is associated to a single
7270 score of 1% in all 3 RPs' LCAs. The largest impacts are due to human toxicity cancer (6%), ionising
7271 radiation (3.3%) and photochemical ozone formation (human health) and eutrophication of
7272 freshwater with 2.2% of impacts in both environmental impact categories.
- 7273 — **Tetrafluoroethylene** (from LCS2). According to the LCA, it contributes to a single score of 2% for
7274 knitted products and 1% for other woven and denim products. The largest impacts are due to ozone
7275 depletion (47.8%) and climate change (4.3%).
- 7276 — **Urea** (LCS1) score of 1% in knitted and other woven products. The largest impacts are due to
7277 human toxicity cancer (3.8%) and resource use fossils (1.7%).

7278 **In section 13.13, Table 228** the indicative inventory of the LCA is compiled. This LCI only contains the
7279 substances with a characterization factor in the databases used.

7280 Further discussion on these environmental hotspots:

7281 **Acrylic acid** is a substance used in the manufacturing process and considered in the LCS2 of the LCA. It is
7282 usually used in finishing processes such as printing with dyes, thus remaining in the final product. However, the
7283 PS team has not received any further information from stakeholders either from industry or NGOs for its
7284 relevance as a possible performance requirement. It is acknowledged that workers exposed to this substance
7285 should receive proper handling procedures and protective equipment to minimize the possible risks associated
7286 with its handling which falls under safety consideration for which the ESPR should not be the right regulatory
7287 tool (in fact the main environmental impact of this substance was human toxicity non-cancer).

7288 **Antimony** has been highlighted in the LCA results to arise from the raw material production in the LCS1. As a
7289 proxy, antimony trioxide (CAS no. 1309-64-4) can be considered because it is a chemical that, during dyeing,
7290 can arise from the matrix to the surface of fibres (becoming a breakdown product) due to raw material

7291 content on antimony ⁽²⁷⁹⁾ ⁽²⁸⁰⁾ but it could be the result of a non-intentional addition. Antimony trioxide is also
7292 used as a catalyst in the production of PET from terephthalic acid and this could be the main reason for
7293 appearing in the LCA. This substance would, in any event be relevant primarily only in terms of chemical
7294 safety so it can only be flagged as a hotspot for possible further consideration, e.g. under REACH (currently,
7295 Antimony is restricted under REACH Annex XVII – Appendix 13 – but does not include textile products).
7296 However, there is a lack of sufficient background information to set a performance requirement.

7297 **Detergent/surfactants** are highlighted in the LCA in LCS1 and LCS4. These chemical products are used in
7298 the production of fibres, i.e. in the pretreatment and in the final manufacturing of the garments in the dyeing
7299 processes but also in the consumer use phase. Detergents/surfactants are not meant to remain in the final
7300 products. In addition, the LCA did not provide sufficient data or information on the specific chemical substance
7301 used as detergent/surfactant. Given that, they are not proposed for the establishment of performance
7302 requirements.

7303 At this stage it can be highlighted that NPEOs, which are SoCs (a) and fall also under the categories of SoCs (b),
7304 and (d), can be derived from detergents. ZDHC & Quantis, 2025 ⁽²⁸¹⁾ have examined alternative products to
7305 NPEO-based detergents which are based on fatty alcohol ethoxylates (FAEs). These FAEs are a widely adopted
7306 class of non-ionic surfactants known for their effectiveness in low dosages and lower ecotoxicity profiles
7307 compared to NPEOs. They carried out a comparative LCA, incorporating production phases of both NPEOs and
7308 FAEs. The results demonstrated clear environmental benefits when substituting NPEO with tridecyl alcohol
7309 ethoxylate across both the production and usage phases. Although a 51% reduction in freshwater ecotoxicity
7310 was observed (note that in the LCA in the PS one of the largest impacts of detergent/surfactants was
7311 ecotoxicity-fresh water). This is notably important as surfactants mainly enter wastewater systems during
7312 textile wet processing. However, this freshwater ecotoxicity credit is supplemented with a 21% increase in
7313 climate change impact, driven mainly by the greater production footprint of tridecyl alcohol ethoxylate and the
7314 higher usage dosage. This serves to highlight the existence of trade-offs in the substitution of chemicals.

7315 **Disperse dyes** are highlighted in LCS2 of the LCA. These substances are used for the dyeing of fibres and/or
7316 garments. In general, disperse dyes are a type of dyes commonly used in the textile industry, particularly for
7317 dyeing synthetic fibres. While disperse dyes are widely used due to their excellent colorfastness and brightness,
7318 there are several concerns regarding potential water pollution as well as sensitisation. These dyes are an
7319 environmental hotspot in the LCA due to minerals and metals resource use. Information from stakeholders was
7320 not received other than from NGOs advocating for their inclusion as information requirement. Further evidence
7321 for the setting of performance requirements would be needed as reasons other than chemical safety are to be
7322 found.

7323 **Polydimethylsiloxane** has been highlighted in the LCA results as to originate in LCS2. These substances can
7324 be used as lubricants in spinning fibres, and softeners in finishing processes but should not remain in the final
7325 products. Further evidence would need to be gathered in order to justify the setting of a performance
7326 requirement on this substance.

7327 **Purified terephthalic acid** has been highlighted in the LCA results as to come from LCS1 but only in knitted
7328 and woven products. Further evidence would need to be gathered in order to set performance requirements.

7329 **Sodium hydroxide** has been highlighted in the LCA results as to come from LCS1 and 2. This chemical
7330 substance is used for bleaching and as a solvent in all fibre types. The impact is mainly associated to human
7331 toxicity cancer which does not allow to the establishment of performance requirements. This chemical
7332 substance is not meant to remain in the final products. In addition, other policy instruments may already set
7333 requirements for bleaching processes in the production of fibres.

7334 **Tetrafluoroethylene** (from LCS2). This substance is used primarily in the industrial preparation of
7335 fluoropolymers, i.e. the polymerization of tetrafluoroethylene produces polytetrafluoroethylene (PTFE) polymers

²⁷⁹ Rujido-Santos I, Herbelo-Hermelo P, Barciela-Alonso MC, Bermejo-Barrera P, Moreda-Piñeiro A. Metal Content in Textile and (Nano)Textile Products. International Journal of Environmental Research and Public Health. 2022; 19(2):944. <https://doi.org/10.3390/ijerph19020944>

²⁸⁰ Brigden, K., Santillo, D., Hetherington, M., Wang, S., Johnston, P., Greenpeace, Hazardous chemicals in branded luxury textile products on sale during 2013, Greenpeace Research Laboratories Technical Report 01/2012, <https://www.greenpeace.to/greenpeace/wp-content/uploads/2014/02/Technical-Report-01-2014.pdf>

²⁸¹ ZDHC and Quantis, 2025. A Decade of ZDHC. Assessing how ZDHC's Manufacturing Restricted Substances List, Wastewater Guidelines, and Air Emissions Guidelines reduce impact on nature. June 2025. Available here: <https://www.roadmap20zero.com/post/new-report-zdhcs-chemical-management-tools-drive-up-to-96-reduction-in-nature-impacts>

7336 such as Teflon and Fluon. PTFE is included in the PFAS category. According to the LCA, it contributes to ozone
7337 depletion (47.8%) and climate change (4.3%). In the LCA, this substance is used as a proxy of the polymer PTFE
7338 which is used for waterproofing finishes in some textiles. TFE is the monomer from which PTFE is made. These
7339 impacts would not allow to the establishment of performance requirements.

7340 **Urea** (LCS1) largest impacts are due to human toxicity cancer (3.8%) and resource use fossils (1.7%). This
7341 hotspot could be related to the production of polyurethane, the main component of elastane fibres. However,
7342 there is not sufficient granularity in the database used to trace back the origin of this impact. In all cases, as
7343 the main contribution is to human toxicity, the establishment of a performance requirement cannot be
7344 sufficiently justified.

7345 From the LCA perspective, performance requirements are not proposed for any specific substance given that
7346 there are not sufficiently relevant individual hotspots for substances as extracted from the LCA. In some
7347 cases, the environmental hotspot in the LCA comes from an impact on toxicity, which cannot generally be
7348 taken under the ESPR unless enough evidence is found. In addition, and not only for toxicity, the LCA showed
7349 an environmental hotspot for environmental impacts for aspects which are not either addressed under the
7350 ESPR for textile apparel products.

7351 **12.3.20 Conclusion – performance requirements**

7352 Given the analysis provided above, the PS study team considers that there is not sufficient scientific evidence
7353 to the setting of performance requirements for specific chemical substances identified in the inventory
7354 associated to textile apparel products.

References

- Aakko M, Niinimäki K. 2022. Quality matters: reviewing the connections between perceived quality and clothing use time. *J Fash Mark Manag Int J* **26**(1): 107–125. doi: 10.1108/JFMM-09-2020-0192
- Aakko M, Niinimäki K. 2021. Quality matters: reviewing the connections between perceived quality and clothing use time. *J Fash Mark Manag Int J*, in press.
- AB-REOC and BV-OECO. 2019. De Belgische consument en kleding.
- Accenture. 2022. Drugie życie ubrań. Available at <https://www.accenture.com/content/dam/accenture/final/a-com-migration/manual/r3/pdf/pdf-174/Accenture-Drugie-Zycie-Ubran.pdf>.
- Adam Barone. 2022 Nov 28. Quota. Investopedia. Available at <https://www.investopedia.com/terms/q/quota.asp>. Accessed 2023 Aug 17.
- Adigüzel F, Linkowski C, Olson E. 2020. Do Sustainability Labels Make Us More Negligent? Rebound and Moral Licensing Effects in the Clothing Industry. In: Muthu SS, Gardetti MA, editors. *Sustainability in the Textile and Apparel Industries*. Cham: Springer International Publishing. p. 1–18. doi: 10.1007/978-3-030-38532-3_1
- Aeschlimann M, Li G, Kanji ZA, Mitrano DM. 2022. Potential impacts of atmospheric microplastics and nanoplastics on cloud formation processes. *Nat Geosci* **15**(12): 967–975. Nature Publishing Group. doi: 10.1038/s41561-022-01051-9
- Ahlström F, Ferning C, Cheniere MK, Sorooshian S. 2020. Performance Indicators of Textile Reverse Logistics. *IOP Conf Ser Earth Environ Sci* **442**(1): 012012. doi: 10.1088/1755-1315/442/1/012012
- A.I.S.E. 2017a. Pan-European consumer habits survey 2017. Perceptions of cleanliness and hygiene. Available at https://www.aise.eu/documents/document/20180528165059-aise_consumershabetssurvey2017_summary_final.pdf.
- A.I.S.E. 2017b. Low temperature washing. Available at https://www.aise.eu/documents/document/20171215161844-aise_exp017_laundry-b2-low_temp_def.pdf.
- A.I.S.E. 2017c. Pan-European consumer habits survey 2017. Perceptions of cleanliness and hygiene. Available at https://www.aise.eu/documents/document/20180528165059-aise_consumershabetssurvey2017_summary_final.pdf.
- A.I.S.E. 2020. Pan-European consumer habits survey 2020. Available at <https://www.aise.eu/our-activities/information-to-end-users/consumer-research.aspx>.
- AK Wienn, Greenpeace. 2023a. FASHION CONSUMPTION IN AUSTRIA: a high level of awareness, but potential to do more. Available at <https://www.akeuropa.eu/en/sustainable-fashion-consumption-austria-high-level-awareness-potential-do-more>.
- AK Wienn, Greenpeace. 2023b. Fashion Consumption in Austria: a high level of awareness, but potential to do more. Available at <https://www.akeuropa.eu/en/sustainable-fashion-consumption-austria-high-level-awareness-potential-do-more>.
- Aldalbahi A, El-Naggar M, El-Newehy M, Rahaman M, Hatshan M, Khattab T. 2021. Effects of Technical Textiles and Synthetic Nanofibers on Environmental Pollution. *Polymers* **13**(1): 155. doi: 10.3390/polym13010155

- 7395 Allen E, Henninger CE, Garforth A, Asuquo E. 2024 Feb 21. Microfiber Pollution: A Systematic Literature Review
7396 to Overcome the Complexities in Knit Design to Create Solutions for Knit Fabrics. *Environ Sci Technol*:
7397 acs.est.3c05955. doi: 10.1021/acs.est.3c05955
- 7398 Alliance of Commerce, Deloitte. 2022. Les Cahiers de la mode durable. Available at
7399 [https://www.alliancecommerce.org/cahier-de-la-mode-durable-n3-prolonger-la-vie-des-produits-un-](https://www.alliancecommerce.org/cahier-de-la-mode-durable-n3-prolonger-la-vie-des-produits-un-defi-pour-les-marques/)
7400 [defi-pour-les-marques/](https://www.alliancecommerce.org/cahier-de-la-mode-durable-n3-prolonger-la-vie-des-produits-un-defi-pour-les-marques/).
- 7401 Allwood JM, Childs THC, Clare AT, De Silva AKM, Dhokia V, Hutchings IM, Leach RK, Leal-Ayala DR, Lowth S,
7402 Majewski CE, et al. 2016. Manufacturing at double the speed. *J Mater Process Technol* **229**: 729–757.
7403 doi: 10.1016/j.jmatprotec.2015.10.028
- 7404 Altroconsumo, IPSOS. 2024. Italiani a tu per tu con la moda sostenibile. Available at
7405 [https://www.impegnatiacambiare.org/altrovestire/news/italiani-moda-](https://www.impegnatiacambiare.org/altrovestire/news/italiani-moda-sostenibile?utm_content=Box1.0-italiani-moda-sostenibile&utm_term=&utm_source=db-int_nmb&utm_campaign=fus-newsletter-apr-4-b-purple&utm_medium=nl_nmb&utm_content=242634&utm_id=2a765871-0be5-42de-885b-766ad9ffdd86&sfmc_activityid=376cb9df-47f2-4363-b138-e4459e2bfe94&utm_medium=nl_nmb)
7406 [sostenibile?utm_content=Box1.0-italiani-moda-sostenibile&utm_term=&utm_source=db-](https://www.impegnatiacambiare.org/altrovestire/news/italiani-moda-sostenibile?utm_content=Box1.0-italiani-moda-sostenibile&utm_term=&utm_source=db-int_nmb&utm_campaign=fus-newsletter-apr-4-b-purple&utm_medium=nl_nmb&utm_content=242634&utm_id=2a765871-0be5-42de-885b-766ad9ffdd86&sfmc_activityid=376cb9df-47f2-4363-b138-e4459e2bfe94&utm_medium=nl_nmb)
7407 [int_nmb&utm_campaign=fus-newsletter-apr-4-b-](https://www.impegnatiacambiare.org/altrovestire/news/italiani-moda-sostenibile?utm_content=Box1.0-italiani-moda-sostenibile&utm_term=&utm_source=db-int_nmb&utm_campaign=fus-newsletter-apr-4-b-purple&utm_medium=nl_nmb&utm_content=242634&utm_id=2a765871-0be5-42de-885b-766ad9ffdd86&sfmc_activityid=376cb9df-47f2-4363-b138-e4459e2bfe94&utm_medium=nl_nmb)
7408 [purple&utm_medium=nl_nmb&utm_content=242634&utm_id=2a765871-0be5-42de-885b-](https://www.impegnatiacambiare.org/altrovestire/news/italiani-moda-sostenibile?utm_content=Box1.0-italiani-moda-sostenibile&utm_term=&utm_source=db-int_nmb&utm_campaign=fus-newsletter-apr-4-b-purple&utm_medium=nl_nmb&utm_content=242634&utm_id=2a765871-0be5-42de-885b-766ad9ffdd86&sfmc_activityid=376cb9df-47f2-4363-b138-e4459e2bfe94&utm_medium=nl_nmb)
7409 [766ad9ffdd86&sfmc_activityid=376cb9df-47f2-4363-b138-e4459e2bfe94&utm_medium=nl_nmb.](https://www.impegnatiacambiare.org/altrovestire/news/italiani-moda-sostenibile?utm_content=Box1.0-italiani-moda-sostenibile&utm_term=&utm_source=db-int_nmb&utm_campaign=fus-newsletter-apr-4-b-purple&utm_medium=nl_nmb&utm_content=242634&utm_id=2a765871-0be5-42de-885b-766ad9ffdd86&sfmc_activityid=376cb9df-47f2-4363-b138-e4459e2bfe94&utm_medium=nl_nmb)
7410 Accessed 2024 May 21.
- 7411 Alyssa Hardy. 2020 Jan 9. Let's Stop Pretending We Need New Clothes Every Season. InStyle. Available at
7412 <https://www.instyle.com/fashion/future-of-fashion-is-seasonless>. Accessed 2023 Sept 15.
- 7413 Amadei A, De Laurentiis V, Sala S. 2021. A review of monetary valuation in life cycle assessment: state of the
7414 art and future needs. *JOURNAL OF CLEANER PRODUCTION*. USA: ELSEVIER SCI LTD. doi:
7415 10.1016/j.jclepro.2021.129668%20(online)
- 7416 Amasawa E, Kimita K. 2023. Deducing environmental implication of clothing rental from consumer voices and
7417 behaviors: a social experiment in Japan. 5th PLATE Conference. Available at
7418 <https://aaltdoc.aalto.fi/handle/123456789/122687>.
- 7419 Amato-Lourenço LF, dos Santos Galvão L, de Weger LA, Hiemstra PS, Vijver MG, Mauad T. 2020. An emerging
7420 class of air pollutants: Potential effects of microplastics to respiratory human health? *Sci Total Environ*
7421 **749**: 141676. doi: 10.1016/j.scitotenv.2020.141676
- 7422 Amicarelli V, Bux C, Spinelli MP, Lagioia G. 2022. Life cycle assessment to tackle the take-make-waste paradigm
7423 in the textiles production. *Waste Manag* **151**: 10–27. doi: 10.1016/j.wasman.2022.07.032
- 7424 Amini R, Ghaffarianhoseini Amirhosein, Ghaffarianhoseini Ali, Berardi U. 2021. Numerical investigation of indoor
7425 thermal comfort and air quality for a multi-purpose hall with various shading and glazing ratios. *Therm*
7426 *Sci Eng Prog* **22**: 100812. doi: 10.1016/j.tsep.2020.100812
- 7427 Amobonye A, Bhagwat P, Raveendran S, Singh S, Pillai S. 2021. Environmental Impacts of Microplastics and
7428 Nanoplastics: A Current Overview. *Front Microbiol* **12**: 768297. doi: 10.3389/fmicb.2021.768297
- 7429 Amobonye A, Bhagwat P, Singh S, Pillai S. 2021. Plastic biodegradation: Frontline microbes and their enzymes.
7430 *Sci Total Environ* **759**: 143536. doi: 10.1016/j.scitotenv.2020.143536
- 7431 Anisha, Kalaivani. 2016. Factors influencing buyers behaviour while purchasing. *Shanlax International Journal*
7432 *of Commerce* **4**(3). Available at https://www.shanlaxjournals.in/pdf/COM/V4N3/COM_V4_N3_020.pdf.
- 7433 Anthesis. 2015. Clothing Durability Report. Banbury: WRAP. Available at [https://www.teraz-](https://www.teraz-srodowisko.pl/media/pdf/aktualnosci/1694-Clothing-Durability.pdf)
7434 [srodowisko.pl/media/pdf/aktualnosci/1694-Clothing-Durability.pdf](https://www.teraz-srodowisko.pl/media/pdf/aktualnosci/1694-Clothing-Durability.pdf). Accessed 2023 Dec 16.
- 7435 Asdecker B, Sucky E. 2019. Der Weg aus der Retourenfalle. *Best in procurement: BIP; das Magazin für Manager*
7436 *in Einkauf und Logistik* **10**(6): 46–48.

- 7437 Aslan B, Stevenson M, Hendry LC. 2015. The applicability and impact of Enterprise Resource Planning (ERP)
7438 systems: Results from a mixed method study on Make-To-Order (MTO) companies. *Comput Ind* **70**:
7439 127–143. doi: 10.1016/j.compind.2014.10.003
- 7440 Asociación para la Gestión del Residuo Textil y el Calzado. 2024. Usos, costumbres y nuevos desafíos en la
7441 industria textil. Available at [https://gestionresiduotextil.org/wp-content/uploads/2024/04/estudio-](https://gestionresiduotextil.org/wp-content/uploads/2024/04/estudio-gestion-residuo-textil.pdf)
7442 [gestion-residuo-textil.pdf](https://gestionresiduotextil.org/wp-content/uploads/2024/04/estudio-gestion-residuo-textil.pdf).
- 7443 Athey SN, Erdle LM. 2021. Are We Underestimating Anthropogenic Microfiber Pollution? A Critical Review of
7444 Occurrence, Methods, and Reporting. *Environ Toxicol Chem* **41**(4): 822–837. doi: 10.1002/etc.5173
- 7445 Aus R, Moora H, Vihma M, Unt R, Kiisa M, Kapur S. 2021. Designing for circular fashion: integrating upcycling
7446 into conventional garment manufacturing processes. *Fash Text* **8**(1): 34. doi: 10.1186/s40691-021-
7447 00262-9
- 7448 Bachmann TM, Van Der Kamp J, Bianchi M, Pihkola H, Saavedra Del Oso M. 2024. Life cycle costing as part of
7449 a life cycle sustainability assessment of products: methodology and case studies. *Int J Life Cycle Assess*
7450 **29**(10): 1863–1879. Springer Science and Business Media LLC. doi: 10.1007/s11367-024-02347-1
- 7451 Badgett JO. 2017. An Evaluation of the Quality of Mens 100% Cotton Jersey Knit T-Shirts Representing Three
7452 Retail Categories. , in press. University of Kentucky Libraries. doi: 10.13023/ETD.2017.289
- 7453 Bakowska, O., Mora, I., Walsh, S., Van Duijn, H., Novak, M., Cherubini, G., Joshi, R., Morbiato, A., Visileanu, E., Vesel,
7454 A., et al. 2025. *Fate and Composition of Textile Waste from Italy, the Czech Republic and Romania*. LU:
7455 Publications Office. Available at <https://data.europa.eu/doi/10.2760/3332076>. Accessed 2025 Oct 7.
- 7456 Bakowska O, Mora I, Walsh W. 2025. Fate and Composition of Textile Waste from Italy, the Czech Republic and
7457 Romania. Luxembourg: European Union. Available at
7458 <https://publications.jrc.ec.europa.eu/repository/handle/JRC141441>. Accessed 2025 Aug 11.
- 7459 Balasaraswathi SR, Rathinamoorthy R. 2022. Synthetic Textile and Microplastic Pollution: An Analysis on
7460 Environmental and Health Impact. In: Muthu SS, editor. *Sustainable Approaches in Textiles and Fashion: Circular Economy and Microplastic Pollution*. Singapore: Springer. p. 1–20. doi: 10.1007/978-981-19-
7461 0530-8_1
7462
- 7463 Balevičiūtė J. 2021. The Impact of Durability and Sustainability Cues of Eco- Labels on Consumer Purchase
7464 Intention Towards Sustainable Apparel Products. , in press.
- 7465 Banale AK. 2017. Investigation of Properties of Silk Fiber Produced in Ethiopia. *J Mater* **2017**: 1–5. doi:
7466 10.1155/2017/7691797
- 7467 Barrick A, Boardwine AJ, Nguyen NHA, Sevcu A, Novotna J, Hoang TC. 2025. Acute toxicity of natural and
7468 synthetic clothing fibers towards *Daphnia magna*: Influence of fiber type and morphology. *Sci Total*
7469 *Environ* **967**: 178751. doi: 10.1016/j.scitotenv.2025.178751
- 7470 Batra SK, Pourdeyhimi B. 2012. *Introduction to Nonwovens Technology*. DEStech Publications, Inc.
- 7471 Bauer B, Watson D, Gylling A, Remmen A, Lysemose MH, Hohenthal Chatarina, Jönbrink A-K. 2018. Potential
7472 Ecodesign Requirements for Textiles and Furniture. Nordic Council of Ministers. Report No.: TemaNord
7473 2018:535. Available at [http://norden.diva-](http://norden.diva-portal.org/smash/record.jsf?pid=diva2%3A1221509&dswid=6042)
7474 [portal.org/smash/record.jsf?pid=diva2%3A1221509&dswid=6042](http://norden.diva-portal.org/smash/record.jsf?pid=diva2%3A1221509&dswid=6042).
- 7475 Bauer B, Watson D, Gylling A, Remmen A, Lysemose MH, Hohenthal Catharina, Jönbrink A-K. 2018. *Potential*
7476 *Ecodesign Requirements for Textiles and Furniture*. 2018:535. Copenhagen: Nordic Council of Ministers.
7477 (TemaNord). doi: 10.6027/TN2018-535
- 7478 Bedez Ute T, Celik P, Bunyamin Uzumcu M. 2019. Utilization of Cotton Spinning Mill Wastes in Yarn Production.
7479 In: Körlü A, editor. *Textile Industry and Environment*. IntechOpen. doi: 10.5772/intechopen.85127

- 7480 Bernon M, Rossi S, Cullen J. 2011. Retail reverse logistics: a call and grounding framework for research. *Int J*
7481 *Phys Distrib Logist Manag* **41**(5): 484–510. doi: 10.1108/09600031111138835
- 7482 Beton A, Dias D, Farrant L, Gibon T, Le GY, Desaxce M, Perwuelz A, Boufateh I. 2014 Jan 28. Environmental
7483 Improvement Potential of textiles (IMPRO Textiles). *JRC Publ Repos*, in press. doi: 10.2791/52624
- 7484 BEUC. 2023. FASHION MAKEOVER: MAKING SUSTAINABLE TEXTILES THE CONSUMER NORM. Available at
7485 [https://www.beuc.eu/sites/default/files/publications/BEUC-X-2023-](https://www.beuc.eu/sites/default/files/publications/BEUC-X-2023-099_Fashion_makeover_making_sustainable_textiles_fit_consumers.pdf)
7486 [099_Fashion_makeover_making_sustainable_textiles_fit_consumers.pdf](https://www.beuc.eu/sites/default/files/publications/BEUC-X-2023-099_Fashion_makeover_making_sustainable_textiles_fit_consumers.pdf). Accessed 2024 Aug 13.
- 7487 Bhardwaj LK, Rath P, Yadav P, Gupta U. 2024. Microplastic contamination, an emerging threat to the freshwater
7488 environment: a systematic review. *Environ Syst Res* **13**(1): 8. doi: 10.1186/s40068-024-00338-7
- 7489 Boots B, Russell CW, Green DS. 2019. Effects of Microplastics in Soil Ecosystems: Above and Below Ground.
7490 *Environ Sci Technol* **53**(19): 11496–11506. American Chemical Society. doi: 10.1021/acs.est.9b03304
- 7491 Boschmeier E, Ipsmiller W, Bartl A. 2024. Market assessment to improve fibre recycling within the EU textile
7492 sector. *Waste Manag Res J Sustain Circ Econ* **42**(2): 135–145. doi: 10.1177/0734242X231178222
- 7493 Bošković N, Joksimović D, Bajt O. 2023. Microplastics in mussels from the Boka Kotorska Bay (Adriatic Sea) and
7494 impact on human health. *Food Chem Toxicol* **173**: 113641. doi: 10.1016/j.fct.2023.113641
- 7495 Botta V. 2021. Durable, repairable and mainstream: How ecodesign can make our textiles circular. ECOS.
7496 Available at [https://ecostandard.org/section_documents/report-durable-repairable-and-mainstream-](https://ecostandard.org/section_documents/report-durable-repairable-and-mainstream-how-ecodesign-can-make-our-textiles-circular/)
7497 [how-ecodesign-can-make-our-textiles-circular/](https://ecostandard.org/section_documents/report-durable-repairable-and-mainstream-how-ecodesign-can-make-our-textiles-circular/).
- 7498 Boucher J, Friot D. 2017. *Primary Microplastics in the Oceans: A Global Evaluation of Sources*. IUCN International
7499 Union for Conservation of Nature. doi: 10.2305/IUCN.CH.2017.01.en
- 7500 Boulay A-M, Bare J, Benini L, Berger M, Lathuillière MJ, Manzardo A, Margni M, Motoshita M, Núñez M, Pastor
7501 AV, et al. 2018. The WULCA consensus characterization model for water scarcity footprints: assessing
7502 impacts of water consumption based on available water remaining (AWARE). *Int J Life Cycle Assess*
7503 **23**(2): 368–378. doi: 10.1007/s11367-017-1333-8
- 7504 Boulay A-M, Lenoir L. 2020. Sub-national regionalisation of the AWARE indicator for water scarcity footprint
7505 calculations. *Ecol Indic* **111**: 106017. doi: 10.1016/j.ecolind.2019.106017
- 7506 Brahney J, Hallerud M, Heim E, Hahnenberger M, Sukumaran S. 2020. Plastic rain in protected areas of the
7507 United States. *Science* **368**(6496): 1257–1260. American Association for the Advancement of Science.
7508 doi: 10.1126/science.aaz5819
- 7509 Brice JE, Thorpe H. 2021. Toward More-Than-Human Understandings of Sport and the Environment: A New
7510 Materialist Analysis of Everyday Fitness Practices. *Front Sports Act Living* **3**: 660935. doi:
7511 10.3389/fspor.2021.660935
- 7512 Britannica. 2019. Spinning | Synthetic Fibers, Textile Machinery & Spinning Processes |. Britannica. Available at
7513 <https://www.britannica.com/technology/spinning-yarn-manufacturing>. Accessed 2023 July 31.
- 7514 British Fashion Council's Institute of Positive, Roland Berger. 2023. Solving fashion's product returns-How to
7515 keep value in a closed-loop system.
- 7516 Brondino G. 2022. Global value chain analysis of the automotive and garment sectors: a study of Germany,
7517 Spain, Romania, Indonesia and Mexico for 2000 2014. , in press. doi: doi:10.2760/966324
- 7518 Browne MA, Crump P, Niven SJ, Teuten E, Tonkin A, Galloway T, Thompson R. 2011. Accumulation of Microplastic
7519 on Shorelines Worldwide: Sources and Sinks. *Environ Sci Technol* **45**(21): 9175–9179. American
7520 Chemical Society. doi: 10.1021/es201811s

- 7521 Brun P. 2025. It takes many - Mobilising collective action to enable more sustainable consumer choices in
7522 fashion. Available at [https://corporate.zalando.com/en/people-planet/sustainability/sustainability-](https://corporate.zalando.com/en/people-planet/sustainability/sustainability-reports/zalando-publishes-updated-sustainability)
7523 [reports/zalando-publishes-updated-sustainability](https://corporate.zalando.com/en/people-planet/sustainability/sustainability-reports/zalando-publishes-updated-sustainability).
- 7524 Bye E, McKinney E. 2007. Sizing up the Wardrobe—Why We Keep Clothes That Do Not Fit. *Fash Theory* **11**(4):
7525 483–498. doi: 10.2752/175174107X250262
- 7526 Cai Y, Yang T, Mitrano DM, Heuberger M, Hufenus R, Nowack B. 2020. Systematic Study of Microplastic Fiber
7527 Release from 12 Different Polyester Textiles during Washing. *Environ Sci Technol* **54**(8): 4847–4855.
7528 American Chemical Society. doi: 10.1021/acs.est.9b07395
- 7529 Cai Y-J, Choi T-M. 2020. A United Nations' Sustainable Development Goals perspective for sustainable textile
7530 and apparel supply chain management. *Transp Res Part E Logist Transp Rev* **141**: 102010. doi:
7531 10.1016/j.tre.2020.102010
- 7532 Campanale C, Massarelli C, Savino I, Locaputo V, Uricchio VF. 2020. A Detailed Review Study on Potential Effects
7533 of Microplastics and Additives of Concern on Human Health. *Int J Environ Res Public Health* **17**(4):
7534 1212. Multidisciplinary Digital Publishing Institute. doi: 10.3390/ijerph17041212
- 7535 Caputi S, Diomede F, Lanuti P, Marconi GD, Di Carlo P, Sinjari B, Trubiani O. 2022. Microplastics Affect the
7536 Inflammation Pathway in Human Gingival Fibroblasts: A Study in the Adriatic Sea. *Int J Environ Res*
7537 *Public Health* **19**(13): 7782. Multidisciplinary Digital Publishing Institute. doi: 10.3390/ijerph19137782
- 7538 Carney Almroth BM, Åström L, Roslund S, Petersson H, Johansson M, Persson N-K. 2018. Quantifying shedding
7539 of synthetic fibers from textiles; a source of microplastics released into the environment. *Environ Sci*
7540 *Pollut Res* **25**(2): 1191–1199. doi: 10.1007/s11356-017-0528-7
- 7541 Carr SA. 2017. Sources and dispersive modes of micro-fibers in the environment. *Integr Environ Assess Manag*
7542 **13**(3): 466–469. doi: 10.1002/ieam.1916
- 7543 CECU. 2023. Estudio sobre conductas de sobreconsumo textil. Available at [https://cecu.es/wp-](https://cecu.es/wp-content/uploads/2023/12/Informe-sobreconsumo-textil_cecu2023.pdf)
7544 [content/uploads/2023/12/Informe-sobreconsumo-textil_cecu2023.pdf](https://cecu.es/wp-content/uploads/2023/12/Informe-sobreconsumo-textil_cecu2023.pdf). Accessed 2024 May 21.
- 7545 CECU, Amigos de la Tierra. 2022. INFORME SOBRE CONDUCTAS DE CONSUMO Y REPARABILIDAD. Universidad
7546 de Salamanca.
- 7547 Centobelli P, Abbate S, Nadeem SP, Garza-Reyes JA. 2022. Slowing the fast fashion industry: An all-round
7548 perspective. *Green Sustain Chem* **38**.
- 7549 Chang Y, Wang X. 2023. Sweat and odor in sportswear – A review. *iScience* **26**(7): 107067. doi:
7550 10.1016/j.isci.2023.107067
- 7551 Chapagain AK, Hoekstra AY, Savenije HHG, Gautam R. 2006. The water footprint of cotton consumption: An
7552 assessment of the impact of worldwide consumption of cotton products on the water resources in the
7553 cotton producing countries. *Ecol Econ* **60**(1): 186–203. doi: 10.1016/j.ecolecon.2005.11.027
- 7554 Chapter 3. Fibre fineness and transverse dimensions. 2008. Woodhead Publishing Limited,. Available at
7555 [http://182.160.97.198:8080/xmlui/bitstream/handle/123456789/1269/Chapter%203-](http://182.160.97.198:8080/xmlui/bitstream/handle/123456789/1269/Chapter%203-4.pdf?sequence=3)
7556 [4.pdf?sequence=3](http://182.160.97.198:8080/xmlui/bitstream/handle/123456789/1269/Chapter%203-4.pdf?sequence=3). Accessed 2024 June 11.
- 7557 Chatterjee S, Sharma S. 2019. Microplastics in our oceans and marine health. *Field Actions Sci Rep J Field*
7558 *Actions*(Special Issue 19): 54–61. Institut Veolia.
- 7559 Choi S, Kim J, Kwon M. 2022. The Effect of the Physical and Chemical Properties of Synthetic Fabrics on the
7560 Release of Microplastics during Washing and Drying. *Polymers* **14**(16): 3384. doi:
7561 10.3390/polym14163384

- 7562 Choi SY, Jo J, Lee Y, Ha J, Lee J. 2022. A cross-cultural study of the proximity of clothing to self between
7563 millennial women in South Korea and Mongolia. *Fash Text* **9**(1): 19. doi: 10.1186/s40691-022-00291-
7564 y
- 7565 Choudhary AK, Bansal P. 2017. 8 - Drape measurement technique using manikins with the help of image
7566 analysis. In: Nayak R, Padhye R, editors. *Manikins for Textile Evaluation*. Woodhead Publishing. p. 173–
7567 195. doi: 10.1016/B978-0-08-100909-3.00008-X
- 7568 Christensen TH, editor. 2010. *Solid Waste Technology & Management: Christensen/Solid Waste Technology &*
7569 *Management*. Chichester, UK: John Wiley & Sons, Ltd. doi: 10.1002/9780470666883
- 7570 Circle Economy. 2020. #WHATSINMYCLOTHES: THE TRUTH BEHIND THE LABEL. Available at [https://www.circle-](https://www.circle-economy.com/blogs/whatsinmyclothes-the-truth-behind-the-label)
7571 [economy.com/blogs/whatsinmyclothes-the-truth-behind-the-label](https://www.circle-economy.com/blogs/whatsinmyclothes-the-truth-behind-the-label).
- 7572 Clark M. 2011. *Handbook of Textile and Industrial Dyeing: Principles, Processes and Types of Dyes*. Elsevier.
- 7573 Clavreul J, Guyonnet D, Christensen TH. 2012. Quantifying uncertainty in LCA-modelling of waste management
7574 systems. *Waste Manag* **32**(12): 2482–2495. Pergamon. doi: 10.1016/j.wasman.2012.07.008
- 7575 Claxton S, Kent A. 2020a. The management of sustainable fashion design strategies: An analysis of the
7576 designer's role. *J Clean Prod* **268**: 122112. doi: 10.1016/j.jclepro.2020.122112
- 7577 Claxton S, Kent A. 2020b. The management of sustainable fashion design strategies: An analysis of the
7578 designer's role. *J Clean Prod* **268**.
- 7579 Cluver BG. 2008a. Consumer Clothing Inventory Management. Oregon State University.
- 7580 Cluver BG. 2008b. Consumer Clothing Inventory Management. [Oregon]: Oregon State University. Available at
7581 https://www.academia.edu/4969181/Consumer_Clothing_Inventory_Management.
- 7582 COFREET. 2023. Baromètre COFREET 2022 : Les Français et l'étiquette d'entretien textile. Available at
7583 [https://www.lavermonlinge.com/article/FR/cofreet/quelques_chiffres/resultats-du-5eme-barometre-](https://www.lavermonlinge.com/article/FR/cofreet/quelques_chiffres/resultats-du-5eme-barometre-cofreet-ipsos)
7584 [cofreet-ipsos](https://www.lavermonlinge.com/article/FR/cofreet/quelques_chiffres/resultats-du-5eme-barometre-cofreet-ipsos).
- 7585 Conrady T, Kruschwitz A, Stamminger R. 2014. Influencing the sustainability of washing behavior by using
7586 motivational interviewing. *Energy Effic* **7**(2): 163–178. doi: 10.1007/s12053-013-9215-9
- 7587 Consumers, Health, Agriculture and Food Executive Agency., LE Europe., VVA Europe., IPSOS., ConPolicy.,
7588 Trinomics. 2018. *Behavioural Study on Consumers' Engagement in the Circular Economy: Final Report*.
7589 LU: Publications Office. Available at <https://data.europa.eu/doi/10.2818/956512>. Accessed 2024 Feb 8.
- 7590 Cook SC, Yurchisin J. 2017. Fast fashion environments: consumer's heaven or retailer's nightmare? *Int J Retail*
7591 *Distrib Manag* **45**(2): 143–157. doi: 10.1108/IJRDM-03-2016-0027
- 7592 Cooper T, Claxton S. 2022a. Garment failure causes and solutions: Slowing the cycles for circular fashion. *J*
7593 *Clean Prod* **351**: 131394. doi: 10.1016/j.jclepro.2022.131394
- 7594 Cooper T, Claxton S. 2022b. Garment failure causes and solutions: Slowing the cycles for circular fashion. *J*
7595 *Clean Prod* **351**: 131394. doi: 10.1016/j.jclepro.2022.131394
- 7596 Cooper T, Claxton S. 2022c. Garment failure causes and solutions: Slowing the cycles for circular fashion. *J*
7597 *Clean Prod* **351**: 131394. doi: 10.1016/j.jclepro.2022.131394
- 7598 Cordella M, Alfieri F, Sanfelix Forner JV. 2019. Analysis and development of a scoring system for repair and
7599 upgrade of products. Luxembourg: Publications Office of the European Union. Report No.: EUR 29711
7600 EN. doi: 10.2760/725068,

7601 Cordella M, Sanfeliix J, Alfieri F. 2019. Analysis and development of a scoring system for repair and upgrade of
7602 products: final report. LU: Publications Office. Available at <https://data.europa.eu/doi/10.2760/725068>.
7603 Accessed 2024 Aug 14.

7604 Corradini F, Meza P, Eguiluz R, Casado F, Huerta-Lwanga E, Geissen V. 2019. Evidence of microplastic
7605 accumulation in agricultural soils from sewage sludge disposal. *Sci Total Environ* **671**: 411–420. doi:
7606 10.1016/j.scitotenv.2019.03.368

7607 Cotton Incorporated. 2014. Winning in the U.S. Activewear Market. Available at
7608 <https://lifestylemonitor.cottoninc.com/winning-in-the-u-s-activewear-market/>.

7609 Cotton L, Hayward AS, Lant NJ, Blackburn RS. 2020. Improved garment longevity and reduced microfibre release
7610 are important sustainability benefits of laundering in colder and quicker washing machine cycles. *Dyes*
7611 *Pigments* **177**: 108120. doi: 10.1016/j.dyepig.2019.108120

7612 Crenna E, Secchi M, Benini L, Sala S. 2019. Global environmental impacts: data sources and methodological
7613 choices for calculating normalization factors for LCA. *Int J Life Cycle Assess* **24**(10): 1851–1877. doi:
7614 10.1007/s11367-019-01604-y

7615 Cura K, Rintala N, Kamppuri T, Saarimäki E, Heikkilä P. 2021. Textile Recognition and Sorting for Recycling at an
7616 Automated Line Using Near Infrared Spectroscopy. *Recycling* **6**(1): 11. doi: 10.3390/recycling6010011

7617 Dahlbom M, Martvall A. 2025. *Maturity of Fibre-to-Fibre Recycling in Europe: Assessment of Recycling*
7618 *Companies in Europe*.

7619 Damhorst M. 2019 Nov 6. Stories We Wear: Promoting Sustainability Practices with the Case of Patagonia. *Fam*
7620 *Consum Sci Res J*, in press. Available at
7621 [https://www.academia.edu/110827371/Stories_We_Wear_Promoting_Sustainability_Practices_with_t](https://www.academia.edu/110827371/Stories_We_Wear_Promoting_Sustainability_Practices_with_the_Case_of_Patagonia)
7622 [he_Case_of_Patagonia](https://www.academia.edu/110827371/Stories_We_Wear_Promoting_Sustainability_Practices_with_the_Case_of_Patagonia). Accessed 2024 Oct 10.

7623 Dan MC, Ostergaard T. 2021. Circular Fashion: The New Roles of Designers in Organizations Transitioning to a
7624 Circular Economy. *Des J* **24**:6.

7625 Danopoulos E, Twiddy M, West R, Rotchell JM. 2022. A rapid review and meta-regression analyses of the
7626 toxicological impacts of microplastic exposure in human cells. *J Hazard Mater* **427**: 127861. doi:
7627 10.1016/j.jhazmat.2021.127861

7628 D&B. 2020a. Gedragsonderzoek kleding. Available at [https://open.overheid.nl/documenten/ronl-6e59a9f4-](https://open.overheid.nl/documenten/ronl-6e59a9f4-1e95-4ffd-9316-a49affe5b271/pdf)
7629 [1e95-4ffd-9316-a49affe5b271/pdf](https://open.overheid.nl/documenten/ronl-6e59a9f4-1e95-4ffd-9316-a49affe5b271/pdf).

7630 D&B. 2020b. Gedragsonderzoek kleding. Available at [https://open.overheid.nl/documenten/ronl-6e59a9f4-](https://open.overheid.nl/documenten/ronl-6e59a9f4-1e95-4ffd-9316-a49affe5b271/pdf)
7631 [1e95-4ffd-9316-a49affe5b271/pdf](https://open.overheid.nl/documenten/ronl-6e59a9f4-1e95-4ffd-9316-a49affe5b271/pdf).

7632 De Almeida A, Fonseca P, Schlomann B, Feilberg N. 2011. Characterization of the household electricity
7633 consumption in the EU, potential energy savings and specific policy recommendations. *Energy Build*
7634 **43**(8): 1884–1894. doi: 10.1016/j.enbuild.2011.03.027

7635 De Falco F, Di Pace E, Cocca M, Avella M. 2019. The contribution of washing processes of synthetic clothes to
7636 microplastic pollution. *Sci Rep* **9**(1): 6633. Nature Publishing Group. doi: 10.1038/s41598-019-43023-
7637 x

7638 De Klerk HM, Lubbe S. 2008. Female consumers' evaluation of apparel quality: exploring the importance of
7639 aesthetics. *J Fash Mark Manag Int J* **12**(1): 36–50. doi: 10.1108/13612020810857934

7640 De Laurentiis V, Secchi M, Bos U, Horn R, Laurent A, Sala S. 2019. Soil quality index: Exploring options for a
7641 comprehensive assessment of land use impacts in LCA. *J Clean Prod* **215**: 63–74. doi:
7642 10.1016/j.jclepro.2018.12.238

7643 Deng H, Wei R, Luo W, Hu L, Li B, Di Y, Shi H. 2020. Microplastic pollution in water and sediment in a textile
7644 industrial area. *Environ Pollut* **258**: 113658. doi: 10.1016/j.envpol.2019.113658

7645 DG ENV. 2023. Combatting microplastic pollution in the European Union (Part 3/3). European Commission,
7646 Directorate-General for Environment. SWD(2023)332 Report No.: part 3/3. Available at
7647 [https://ec.europa.eu/transparency/documents-](https://ec.europa.eu/transparency/documents-register/detail?ref=SWD(2023)332&lang=en&lang=en)
7648 [register/detail?ref=SWD\(2023\)332&lang=en&lang=en](https://ec.europa.eu/transparency/documents-register/detail?ref=SWD(2023)332&lang=en&lang=en).

7649 DG GROW. 2021a. Data on the EU textile ecosystem and its competitiveness – Final report. Luxembourg: European
7650 Commission, Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs. Available
7651 at [https://op.europa.eu/en/publication-detail/-/publication/574c0bfe-6142-11ec-9c6c-](https://op.europa.eu/en/publication-detail/-/publication/574c0bfe-6142-11ec-9c6c-01aa75ed71a1/language-en/format-PDF/source-298907666)
7652 [01aa75ed71a1/language-en/format-PDF/source-298907666](https://op.europa.eu/en/publication-detail/-/publication/574c0bfe-6142-11ec-9c6c-01aa75ed71a1/language-en/format-PDF/source-298907666). Accessed 2023 Apr 12.

7653 DG GROW. 2021b. Study on the technical, regulatory, economic and environmental effectiveness of textile fibres
7654 recycling: final report. Luxembourg: European Commission, Directorate-General for Internal Market,
7655 Industry, Entrepreneurship and SMEs. Available at <https://data.europa.eu/doi/10.2873/828412>.
7656 Accessed 2023 July 4.

7657 DG JRC. 2024a. Analytical methods to measure microplastics in drinking water: review and evaluation of
7658 methods. LU: Publications Office. Available at <https://data.europa.eu/doi/10.2760/109944>. Accessed
7659 2024 May 13.

7660 DG JRC. 2024b. Micro- and nanoplastics in the air: sources, transport & analytical methods for their
7661 determination. LU: Publications Office. Available at <https://data.europa.eu/doi/10.2760/028991>.
7662 Accessed 2024 May 13.

7663 Dissanayake PD, Kim S, Sarkar B, Oleszczuk P, Sang MK, Haque MN, Ahn JH, Bank MS, Ok YS. 2022. Effects of
7664 microplastics on the terrestrial environment: A critical review. *Environ Res* **209**: 112734. doi:
7665 10.1016/j.envres.2022.112734

7666 Djafarova E, Bowes T. 2021. 'Instagram made Me buy it': Generation Z impulse purchases in fashion industry. *J*
7667 *Retail Consum Serv* **59**: 102345. doi: 10.1016/j.jretconser.2020.102345

7668 Dodd N, Gama CM. 2017 June. Revision of the EU Green Public Procurement (GPP) Criteria for Textile Products
7669 and Services: Technical Report with final criteria. *Jt Res Cent JRC*: 80. doi: 10.2760/906174

7670 Donatello S, Danneck J, Löw C, Watson D, Konstantas A, Trzepacz S, Liu R, Köhler A, Faraca G. 2021. Circular
7671 economy perspectives in the EU textile sector: final report. LU: European Commission. Directorate
7672 General Joint Research Centre. Available at <https://data.europa.eu/doi/10.2760/858144>. Accessed
7673 2023 Apr 17.

7674 Dreicer M, Tort V, Manen P, European Commission R Directorate-General XII, Science, Development
7675 (Luxembourg). 2025. ExternE: Externalities of energy Vol. 5. Nuclear.

7676 Dreillard M, Barros CDF, Rouchon V, Emonnot C, Lefebvre V, Moreaud M, Guillaume D, Rimbault F, Pagerey F.
7677 2022. Quantification and morphological characterization of microfibers emitted from textile washing.
7678 *Sci Total Environ* **832**: 154973. doi: 10.1016/j.scitotenv.2022.154973

7679 Dris R, Gasperi J, Saad M, Mirande C, Tassin B. 2016. Synthetic fibers in atmospheric fallout: A source of
7680 microplastics in the environment? *Mar Pollut Bull* **104**(1): 290–293. doi:
7681 10.1016/j.marpolbul.2016.01.006

7682 Drycleaning Institute. 2015. International Fair claims guide for consumer textiles products. Alexandria - New
7683 South Wales: Drycleaning Institute of Australia Ltd. Available at [https://irp-](https://irp-cdn.multiscreensite.com/670b4da8678441db9ca09d24ced7de7c/files/uploaded/Fair%20Claims%20Guide.pdf)
7684 [cdn.multiscreensite.com/670b4da8678441db9ca09d24ced7de7c/files/uploaded/Fair%20Claims%20](https://irp-cdn.multiscreensite.com/670b4da8678441db9ca09d24ced7de7c/files/uploaded/Fair%20Claims%20Guide.pdf)
7685 [Guide.pdf](https://irp-cdn.multiscreensite.com/670b4da8678441db9ca09d24ced7de7c/files/uploaded/Fair%20Claims%20Guide.pdf). Accessed 2024 Feb 5.

- 7686 Dubey I, Khan S, Kushwaha S. 2022. Developmental and reproductive toxic effects of exposure to microplastics:
7687 A review of associated signaling pathways. *Front Toxicol* **4**. Available at
7688 <https://www.frontiersin.org/articles/10.3389/ftox.2022.901798>. Accessed 2024 Jan 23.
- 7689 Duhoux T, Lingås D, Mortensen LF. 2024. Volumes and destruction of returned and unsold textiles in Europe's
7690 circular economy. European Topic Centre on Circular economy and resource use. Report No.: ETC CE
7691 Report 2024/4. Available at [https://www.eionet.europa.eu/etcs/etc-ce/products/etc-ce-report-2024-4-](https://www.eionet.europa.eu/etcs/etc-ce/products/etc-ce-report-2024-4-volumes-and-destruction-of-returned-and-unsold-textiles-in-europes-circular-economy)
7692 [volumes-and-destruction-of-returned-and-unsold-textiles-in-europes-circular-economy](https://www.eionet.europa.eu/etcs/etc-ce/products/etc-ce-report-2024-4-volumes-and-destruction-of-returned-and-unsold-textiles-in-europes-circular-economy). Accessed
7693 2024 Aug 2.
- 7694 Duhoux T, Maes E, Hirschnitz-Garbers M, Peeters K, Asscherickx L, Christis M, Stubbe B, Colignon P, Hinzmann,
7695 M, Sachdeva A. 2021. Study on the technical, regulatory, economic and environmental effectiveness of
7696 textile fibres recycling: final report. LU: European Commission. Directorate General for Internal Market,
7697 Industry, Entrepreneurship and SMEs. Available at <https://data.europa.eu/doi/10.2873/828412>.
7698 Accessed 2023 Sept 19.
- 7699 Ebrahimi P, Abbasi S, Pashaei R, Bogusz A, Oleszczuk P. 2022. Investigating impact of physicochemical
7700 properties of microplastics on human health: A short bibliometric analysis and review. *Chemosphere*
7701 **289**: 133146. doi: 10.1016/j.chemosphere.2021.133146
- 7702 ECOS, Botta V, Cabral I. 2021. Durable, repairable and mainstream - How ecodesign can make our textiles
7703 circular. ECOS - Environmental Coalition on Standards. Available at [https://ecostandard.org/wp-](https://ecostandard.org/wp-content/uploads/2021/04/ECOS-REPORT-HOW-ECODESIGN-CAN-MAKE-OUR-TEXTILES-CIRCULAR.pdf)
7704 [content/uploads/2021/04/ECOS-REPORT-HOW-ECODESIGN-CAN-MAKE-OUR-TEXTILES-CIRCULAR.pdf](https://ecostandard.org/wp-content/uploads/2021/04/ECOS-REPORT-HOW-ECODESIGN-CAN-MAKE-OUR-TEXTILES-CIRCULAR.pdf).
7705 Accessed 2023 Feb 11.
- 7706 EEA. 2022a. An overview of Europe's repair sector. Available at
7707 <https://circulareconomy.europa.eu/platform/sites/default/files/2022-12/Repair%20sector.pdf>.
- 7708 EEA. 2022b. Textiles and the Environment - The role of design in Europe's circular economy. European
7709 Environment Agency. Report No.: ETC/CE 2022/2. Available at [https://www.eionet.europa.eu/etcs/etc-](https://www.eionet.europa.eu/etcs/etc-ce/products/etc-ce-products/etc-ce-report-2-2022-textiles-and-the-environment-the-role-of-design-in-europes-circular-economy)
7710 [ce/products/etc-ce-products/etc-ce-report-2-2022-textiles-and-the-environment-the-role-of-design-](https://www.eionet.europa.eu/etcs/etc-ce/products/etc-ce-products/etc-ce-report-2-2022-textiles-and-the-environment-the-role-of-design-in-europes-circular-economy)
7711 [in-europes-circular-economy](https://www.eionet.europa.eu/etcs/etc-ce/products/etc-ce-report-2-2022-textiles-and-the-environment-the-role-of-design-in-europes-circular-economy). Accessed 2024 Feb 5.
- 7712 EEA. 2024. ETC CE Report 2024/4 Volumes and destruction of returned and unsold textiles in Europe's circular
7713 economy. Available at [https://www.eionet.europa.eu/etcs/etc-ce/products/etc-ce-report-2024-4-](https://www.eionet.europa.eu/etcs/etc-ce/products/etc-ce-report-2024-4-volumes-and-destruction-of-returned-and-unsold-textiles-in-europes-circular-economy)
7714 [volumes-and-destruction-of-returned-and-unsold-textiles-in-europes-circular-economy](https://www.eionet.europa.eu/etcs/etc-ce/products/etc-ce-report-2024-4-volumes-and-destruction-of-returned-and-unsold-textiles-in-europes-circular-economy).
- 7715 EEA, Manoochchri S, Schluep M, Dams Y, Mehlhart G, Lingås DB, Marin G, Nicolau M, Colgan S. 2022. An overview
7716 of Europe's repair sector. European Environment Agency. Report No.: 2022/6. Available at
7717 <https://www.eionet.europa.eu/etcs/etc-ce/products>.
- 7718 Elhawary IA. 2015a. Fibre to Yarn: Staple-Yarn Spinning. In: *Textiles and Fashion*. p. 191–212.
- 7719 Elhawary IA. 2015b. Chapter 9 - Fibre to Yarn: Staple-Yarn Spinning. In: Sinclair R, editor. *Textiles and Fashion*.
7720 Woodhead Publishing. p. 191–212. doi: 10.1016/B978-1-84569-931-4.00009-X
- 7721 Ellen MacArthur Foundation. 2017. A new textiles economy: Redesigning fashion's future. Ellen MacArthur
7722 Foundation. Available at <http://www.ellenmacarthurfoundation.org/publications>.
- 7723 Ellen MacArthur Foundation. 2021. Circular business models: redefining growth for a thriving fashion industry.
- 7724 Elmogahzy Y. 2019. 1 - Structure and mechanics of yarns. In: Schwartz P, editor. *Structure and Mechanics of*
7725 *Textile Fibre Assemblies (Second Edition)*. Woodhead Publishing. p. 1–25. doi: 10.1016/B978-0-08-
7726 102619-9.00001-8
- 7727 Elmogahzy YE. 2020. Performance characteristics of traditional textiles: Denim and sportswear products. In:
7728 *Engineering Textiles*. Elsevier. p. 319–346. doi: 10.1016/B978-0-08-102488-1.00013-7

- 7729 Engelberg J, Brassell E. 2019. Differences in Fuel Usage in the United States Housing Stock. U.S. Department of
7730 Housing and Urban Development and U.S. Census Bureau. Report No.: H150-19. Available at
7731 <https://www.census.gov/programs-surveys/ahs/research/publications/h150-19.html>.
- 7732 EPA Circular Economy Programme, B&A. 2021. TEXTILES: ATTITUDES & BEHAVIOURS NATIONAL SURVEY 2021.
7733 Available at [https://www.epa.ie/our-services/monitoring--assessment/circular-](https://www.epa.ie/our-services/monitoring--assessment/circular-economy/textiles/national-attitudes-and-behaviours-survey-2021/)
7734 [economy/textiles/national-attitudes-and-behaviours-survey-2021/](https://www.epa.ie/our-services/monitoring--assessment/circular-economy/textiles/national-attitudes-and-behaviours-survey-2021/).
- 7735 Eppinger E, Slomkowski A, Behrendt T, Rotzler S, Marwede M. 2022. Design for recycling of e-textiles: current
7736 issues of recycling of products combining electronics and textiles and implications for a circular design
7737 approach. In: *Recycling Strategy and Challenges Associated with Waste Management Towards*
7738 *Sustaining the World*.
- 7739 Estahbanati S, Fahrenfeld NL. 2016. Influence of wastewater treatment plant discharges on microplastic
7740 concentrations in surface water. *Chemosphere* **162**: 277–284. doi:
7741 10.1016/j.chemosphere.2016.07.083
- 7742 Eur Eau. 2020. The governance of water services in Europe. Belgium: The European Federation of National
7743 Associations of Water Services. Available at [https://www.eureau.org/resources/publications/eureau-](https://www.eureau.org/resources/publications/eureau-publications/5219-the-governance-of-water-services-in-europe-2020-edition/file)
7744 [publications/5219-the-governance-of-water-services-in-europe-2020-edition/file](https://www.eureau.org/resources/publications/eureau-publications/5219-the-governance-of-water-services-in-europe-2020-edition/file). Accessed 2023
7745 May 8.
- 7746 EURATEX. 2020. Facts & Key Figures 2020 of the European Textile and Clothing Industry. EURATEX, Economic
7747 and Statistics. Available at [https://euratex.eu/wp-content/uploads/EURATEX-Facts-Key-Figures-2020-](https://euratex.eu/wp-content/uploads/EURATEX-Facts-Key-Figures-2020-LQ.pdf)
7748 [LQ.pdf](https://euratex.eu/wp-content/uploads/EURATEX-Facts-Key-Figures-2020-LQ.pdf).
- 7749 EURATEX. 2022a. Facts & Key Figures 2022 of the European Textile and Clothing Industry. EURATEX, Economic
7750 and Statistics. Available at <https://euratex.eu/facts-and-key-figures/>.
- 7751 EURATEX. 2022b Dec 16. The EU Textiles industry is highly concerned about the potential loss of
7752 competitiveness, caused by the EU's inaction of the energy crisis, and Chinese and US subsidies to
7753 domestic industry. EURATEX. Available at [https://euratex.eu/news/the-eu-textiles-industry-is-highly-](https://euratex.eu/news/the-eu-textiles-industry-is-highly-concerned-about-the-potential-loss-of-competitiveness-caused-by-the-eus-inaction-of-the-energy-crisis-and-chinese-and-us-subsidies-to-domestic-industry/)
7754 [concerned-about-the-potential-loss-of-competitiveness-caused-by-the-eus-inaction-of-the-energy-](https://euratex.eu/news/the-eu-textiles-industry-is-highly-concerned-about-the-potential-loss-of-competitiveness-caused-by-the-eus-inaction-of-the-energy-crisis-and-chinese-and-us-subsidies-to-domestic-industry/)
7755 [crisis-and-chinese-and-us-subsidies-to-domestic-industry/](https://euratex.eu/news/the-eu-textiles-industry-is-highly-concerned-about-the-potential-loss-of-competitiveness-caused-by-the-eus-inaction-of-the-energy-crisis-and-chinese-and-us-subsidies-to-domestic-industry/). Accessed 2023 Aug 15.
- 7756 EuRIC. 2021. Handling&Sorting Specifications for re-use and recycling of used textiles. The European Recycling
7757 Industries' Confederation. Available at [https://euric.org/images/Position-](https://euric.org/images/Position-papers/EuRIC_Handling_Sorting_Specs_July2021.pdf)
7758 [papers/EuRIC_Handling_Sorting_Specs_July2021.pdf](https://euric.org/images/Position-papers/EuRIC_Handling_Sorting_Specs_July2021.pdf).
- 7759 European Commission. 2009. Special Eurobarometer 314 - Europeans' attitudes toward chemicals in consumer
7760 products: Risk perception of potential health hazards. Chemicals - Chemical Safety. Available at
7761 <https://europa.eu/eurobarometer/surveys/detail/752>.
- 7762 European Commission. 2010. Special Eurobarometer 357 - International Trade report. Report No.:
7763 Eurobarometer 74.1. Available at <https://europa.eu/eurobarometer/surveys/detail/905>.
- 7764 European Commission. 2019. Special Eurobarometer 501 - Attitudes of European citizens towards the
7765 environment. Environment. doi: 10.2779/902489
- 7766 European Commission. 2021. Commission Recommendation (EU) 2021/2279 of 15 December 2021 on the use
7767 of the Environmental Footprint methods to measure and communicate the life cycle environmental
7768 performance of products and organisations C/2021/9332 ELI. Available at
7769 <http://data.europa.eu/eli/reco/2021/2279/oj>.
- 7770 European Commission. 2022. Environmental Footprint reference packages. European Platform on LCA | EPLCA.
7771 Available at <https://eplca.jrc.ec.europa.eu/LCDN/developerEF.html>.
- 7772 European Commission. 2023. Retail. Internal Market, Industry, Entrepreneurship and SMEs. Available at
7773 https://single-market-economy.ec.europa.eu/single-market/services/retail_en. Accessed 2023 Sept 19.

- 7774 European Commission. Directorate General for Environment. 2014. Flash Eurobarometer 388 - Attitudes of
7775 Europeans towards waste management and resource efficiency. Publications Office. Available at
7776 <https://data.europa.eu/doi/10.2779/14825>.
- 7777 European Commission. Directorate General for Environment. 2023. Flash Eurobarometer 535 - The EU ecolabel:
7778 report. LU: Publications Office. Available at <https://data.europa.eu/doi/10.2779/463334>. Accessed 2024
7779 Jan 26.
- 7780 European Commission. Directorate General for Environment., Eunomia. 2023. *Study to Support the Finalisation*
7781 *of the Legal Proposal and the Impact Assessment for the Review of the Packaging and Packaging Waste*
7782 *Directive: Final Report*. LU: Publications Office. Available at <https://data.europa.eu/doi/10.2779/36548>.
7783 Accessed 2025 July 9.
- 7784 European Commission. Directorate General for Internal Market, Industry, Entrepreneurship and SMEs. 2021.
7785 *Study on the Technical, Regulatory, Economic and Environmental Effectiveness of Textile Fibres*
7786 *Recycling: Final Report*. LU: Publications Office. Available at
7787 <https://data.europa.eu/doi/10.2873/828412>. Accessed 2023 Sept 19.
- 7788 European Environment Agency. 2024. Circular economy country profile 2024- France. Report No.: ETC-CE Report
7789 2024/France. Available at [https://www.eea.europa.eu/en/topics/in-depth/circular-economy/country-](https://www.eea.europa.eu/en/topics/in-depth/circular-economy/country-profiles-on-circular-economy/circular-economy-country-profiles-2024/france_2024-ce-coutry-profile_final.pdf)
7790 [profiles-on-circular-economy/circular-economy-country-profiles-2024/france_2024-ce-coutry-](https://www.eea.europa.eu/en/topics/in-depth/circular-economy/circular-economy-country-profiles-2024/france_2024-ce-coutry-profile_final.pdf)
7791 [profile_final.pdf](https://www.eea.europa.eu/en/topics/in-depth/circular-economy/circular-economy-country-profiles-2024/france_2024-ce-coutry-profile_final.pdf).
- 7792 European Parliamentary Research Service. 2019. Environmental impact of the textile and clothing industry -
7793 What consumers need to know. European Parliament. Available at
7794 [https://www.europarl.europa.eu/RegData/etudes/BRIE/2019/633143/EPRS_BRI\(2019\)633143_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/BRIE/2019/633143/EPRS_BRI(2019)633143_EN.pdf).
- 7795 Eurostat. 2023. European business statistics user's manual for PRODCOM: 2023 edition. Publications Office.
7796 Available at <https://data.europa.eu/doi/10.2785/39767>. Accessed 2023 Dec 1.
- 7797 Fantke P, Bijster M, Hauschild MZ, Huijbregts M, Jolliet O, Kounina A, Magaud V, Margni M, McKone TE,
7798 Rosenbaum RK, et al. 2017. USEtox® 2.0 Documentation (Version 1.00). , in press. [object Object]. doi:
7799 10.11581/DTU:00000011
- 7800 Fantke P, Evans J, Hodas N, Apte J, Jantunen M, Jolliet O, McKone TE. 2016. Health impacts of fine particulate
7801 matter. In: *Global Guidance for Life Cycle Impact Assessment Indicators*. Paris: UNEP/SETAC Life Cycle
7802 Initiative.
- 7803 Farrant L, Olsen SI, Wangel A. 2010. Environmental benefits from reusing clothes. *Int J Life Cycle Assess* **15**(7):
7804 726–736. doi: 10.1007/s11367-010-0197-y
- 7805 Fashion Revolution. 2020. A survey of EU consumer attitudes to sustainability and supply chain transparency in
7806 the fashion industry. Available at
7807 https://issuu.com/fashionrevolution/docs/fashrev_consumersurvey_2020_full.
- 7808 Faust M-E. 2014. Apparel size designation and labelling. In: *Anthropometry, Apparel Sizing and Design*. Elsevier.
7809 p. 255–273. doi: 10.1533/9780857096890.2.255
- 7810 Fernandez-Stark K, Bamber P, Couto V. 2022. Analysis of the Textile and Clothing Industry Global Value Chains:
7811 Summary. Inter-American Development Bank. doi: 10.18235/0004663
- 7812 Ferrándiz M, Fages E, Rojas-Lema S, Ivorra J, Gomez-Caturla J, Torres-Giner S. 2021. Development and
7813 Characterization of Weft-Knitted Fabrics of Naturally Occurring Polymer Fibers for Sustainable and
7814 Functional Textiles. *Polymers* **13**: 665. doi: 10.3390/polym13040665
- 7815 Fibre Fineness - an overview | ScienceDirect Topics. n.d. Available at
7816 <https://www.sciencedirect.com/topics/engineering/fibre-fineness>. Accessed 2024 June 11.

- 7817 Finnish Ministry of the Environment. 2023a. Konsumenterna i den cirkulära ekonomist MOT ett ett Samhälle
7818 som främjar cirkulär ekonomist och reparation. Helsinki. Available at [https://urn.fi/URN:ISBN:978-952-](https://urn.fi/URN:ISBN:978-952-361-231-0)
7819 361-231-0.
- 7820 Finnish Ministry of the Environment. 2023b. Konsumenterna i den cirkulära ekonomist MOT ett ett Samhälle
7821 som främjar cirkulär ekonomist och reparation. Helsinki. Available at [https://urn.fi/URN:ISBN:978-952-](https://urn.fi/URN:ISBN:978-952-361-231-0)
7822 361-231-0.
- 7823 Forbrugerrådet Tænk. 2022. Vejen til et tøjforbrug med mindre miljø-og klimaaftryk. Danish Consumer Council.
7824 Available at <https://taenk.dk/det-kaemper-vi-for/sammen-mod-toejspild>.
- 7825 Foresight Factory. 2021. Future of Shopping - Global Report. Available at [http://www.foresightfactory.co/wp-](http://www.foresightfactory.co/wp-content/uploads/2021/07/Snap-Final-Global-Report-.pdf)
7826 content/uploads/2021/07/Snap-Final-Global-Report-.pdf.
- 7827 Franz W-Jl. 2017. Economics of vanity sizing. *J Econ Behav Organ* **134**: 336–355. doi:
7828 10.1016/j.jebo.2017.01.004
- 7829 FRC, Public Eye. 2022. E-commerce “dark patterns” fuel fashion overconsumption. Available at
7830 <https://www.publiceye.ch/en/topics/fashion/e-commerce-dark-patterns-fuel-fashion-overconsumption>.
7831 Accessed 2024 May 21.
- 7832 Frischknecht R, Braunschweig A, Hofstetter P, Suter P. 2000. Human health damages due to ionising radiation
7833 in life cycle impact assessment. *Environ Impact Assess Rev* **20**(2): 159–189. doi: 10.1016/S0195-
7834 9255(99)00042-6
- 7835 Frydendal J, Hansen LE, Bonou A. 2018. Environmental Labels and Declarations. In: Hauschild MZ, Rosenbaum
7836 RK, Olsen SI, editors. *Life Cycle Assessment*. Cham: Springer International Publishing. p. 577–604. doi:
7837 10.1007/978-3-319-56475-3_24
- 7838 Gama Caldas M, Eynard U, Spiliotopoulos C, Blengini G, Alfieri F, Mancini L, Mathieux F, Ardente F. 2024. Review
7839 of the MEeP - Methodology for Ecodesign of Energy-related Products. Luxembourg (Luxembourg):
7840 Publications Office of the European Union. Report No.: KJ-NA-31-866-EN-N (online). doi:
7841 10.2760/24524%20(online)
- 7842 Ghaani Farashahi B, Easter E, Annett-Hitchcock K. 2018a. Price and perceived product quality: a comparison of
7843 denim jeans in three price categories. *J Fash Mark Manag Int J* **22**(3): 369–386. doi: 10.1108/JFMM-
7844 10-2017-0104
- 7845 Ghaani Farashahi B, Easter E, Annett-Hitchcock K. 2018b. Price and perceived product quality: a comparison of
7846 denim jeans in three price categories. *J Fash Mark Manag Int J* **22**(3): 369–386. doi: 10.1108/JFMM-
7847 10-2017-0104
- 7848 Ghosh Shampa, Sinha JK, Ghosh Soumya, Vashisth K, Han S, Bhaskar R. 2023. Microplastics as an Emerging
7849 Threat to the Global Environment and Human Health. *Sustainability* **15**(14): 10821. Multidisciplinary
7850 Digital Publishing Institute. doi: 10.3390/su151410821
- 7851 Giaquinto G, Keuchenius C. 2023. Input circulair textielbeleid Wat stimuleert tot minder kleding kopen? Available
7852 at <https://open.overheid.nl/documenten/dpc-0836e490c8e74293d46f2233016741d9a7c20831/pdf>.
7853 Accessed 2025 May 5.
- 7854 GINETEX. 2017a. TEXTILE CARE SYMBOLS. Available at <https://www.ginetex.net/thematique/GB/textile/a-2>.
- 7855 GINETEX. 2017b. TEXTILE CARE SYMBOLS. Available at <https://www.ginetex.net/thematique/GB/textile/a-2>.
- 7856 GINETEX. 2019 Nov 7. DURABILITY OF CLOTHES: A MAJOR CONCERN FOR EUROPEANS. Available at
7857 [https://www.ginetex.net/thematique/GB/textile/a-barometer-for-textile-care-labelling-in-](https://www.ginetex.net/thematique/GB/textile/a-barometer-for-textile-care-labelling-in-europehttps://www.ginetex.net/userfiles/files/Maarc%20pour%20COFREET_Barometre%20IPSOS_GINETEX.pdf)
7858 europe[https://www.ginetex.net/userfiles/files/Maarc%20pour%20COFREET_Barometre%20IPSOS_GIN](https://www.ginetex.net/userfiles/files/Maarc%20pour%20COFREET_Barometre%20IPSOS_GINETEX.pdf)
7859 ETEX.pdf.

- 7860 GINETEX. 2023. GINGINETEX Newsletter No. 27 – International textile care labelling news. ETEX Newsletter.
- 7861 GINETEX. 2025 Apr 28. GINETEX Barometer 2024: Textile care habits in Europe. GINETEX. Available at
7862 <https://www.ginetex.net/article/ginetex-barometer-2024--textile-care-habits-in-europe>. Accessed
7863 2025 July 16.
- 7864 Gita Gopinath. 2020 Apr 14. The Great Lockdown: Worst Economic Downturn Since the Great Depression. IMF.
7865 Available at [https://www.imf.org/en/Blogs/Articles/2020/04/14/blog-weo-the-great-lockdown-worst-](https://www.imf.org/en/Blogs/Articles/2020/04/14/blog-weo-the-great-lockdown-worst-economic-downturn-since-the-great-depression)
7866 [economic-downturn-since-the-great-depression](https://www.imf.org/en/Blogs/Articles/2020/04/14/blog-weo-the-great-lockdown-worst-economic-downturn-since-the-great-depression). Accessed 2023 Aug 15.
- 7867 Global Fashion Agenda. 2025. Mapping of Global Extended Producer Responsibility (EPR) for Textiles.
- 7868 Gooijer H, Stamminger R. 2016. Water and Energy Consumption in Domestic Laundering Worldwide – A Review.
7869 *Tenside Surfactants Deterg* **53**(5): 402–409. doi: 10.3139/113.110456
- 7870 Gottfridsson M, Zhang Y. 2015. Environmental impacts of shoe consumption – Combining product flow analysis
7871 with an LCA model for Sweden [Master Thesis]. [Gothenburg]: Chalmers University of Technology.
7872 Available at <https://publications.lib.chalmers.se/records/fulltext/218968/218968.pdf>. Accessed 2023
7873 Feb 17.
- 7874 Goudar IV, Kulloli SD. 2022. Flax fibre and blended yarn characteristics. , in press. Available at
7875 <https://www.thepharmajournal.com/archives/2022/vol11issue3/PartZ/11-3-392-331.pdf>.
- 7876 Goworek H, Oxborrow Lynn, Claxton S, Hill H. 2020. Managing sustainability in the fashion business: Challenges
7877 in product development for clothing longevity in the UK. *2020 Journal of Business Research*(117):
7878 629–641.
- 7879 Goworek H, Oxborrow L., Claxton S, McLaren A, Cooper T, Hill H. 2020. Managing sustainability in the fashion
7880 business: Challenges in product development for clothing longevity in the UK. *J Bus Res* **117**: 629–
7881 641. doi: 10.1016/j.jbusres.2018.07.021
- 7882 Gray R, Sabaiduc C, Salvidge C, Doriza A. 2022. Clothing longevity and circular business models receptivity in
7883 the UK. UK: WRAP. Available at [https://wrap.org.uk/resources/report/citizen-insights-clothing-longevity-](https://wrap.org.uk/resources/report/citizen-insights-clothing-longevity-and-circular-business-models-receptivity-uk)
7884 [and-circular-business-models-receptivity-uk](https://wrap.org.uk/resources/report/citizen-insights-clothing-longevity-and-circular-business-models-receptivity-uk).
- 7885 Greenpeace. 2015. Wegwerfware Kleidung Repräsentative Greenpeace-Umfrage zu Kaufverhalten, Tragedauer
7886 und der Entsorgung von Mode. Available at
7887 [https://www.greenpeace.de/sites/default/files/publications/20151123_greenpeace_modekonsum_flye](https://www.greenpeace.de/sites/default/files/publications/20151123_greenpeace_modekonsum_flyer.pdf)
7888 [r.pdf](https://www.greenpeace.de/sites/default/files/publications/20151123_greenpeace_modekonsum_flyer.pdf).
- 7889 Guo Y, Morris KE, Sumner M, Taylor M. 2025. A framework for measuring physical garment durability. *Clean*
7890 *Responsible Consum* **16**: 100245. doi: 10.1016/j.clrc.2024.100245
- 7891 Guy A, Green E, Banim M, editors. 2001. *Through the Wardrobe: Women's Relationships with Their Clothes*. 1st
7892 ed. Oxford International Publishers Ltd. doi: 10.2752/9781847888921
- 7893 Gwilt A. 2021. Caring for clothes: how and why people maintain garments in regular use. **35**(6): 870–882. Taylor
7894 & Francis. doi: 10.1080/10304312.2021.1993572
- 7895 Gwozdz W, Steensen Nielsen K, Müller T. 2017a. An Environmental Perspective on Clothing Consumption:
7896 Consumer Segments and Their Behavioral Patterns. *Sustainability* **9**(5): 762. doi: 10.3390/su9050762
- 7897 Gwozdz W, Steensen Nielsen K, Müller T. 2017b. An Environmental Perspective on Clothing Consumption:
7898 Consumer Segments and Their Behavioral Patterns. *Sustainability* **9**(5): 762. doi: 10.3390/su9050762
- 7899 Harpa R, Piroi C, Buhu A, Ciobanu L. 2024. LEAN SIX SIGMA AND SUSTAINABILITY IN TEXTILE INDUSTRY:
7900 ASYSTEMATIC LITERATURE REVIEW. In: *International Symposium "Technical Textiles - Present and*
7901 *Future". Edition 2023*. Sciendo. p. I–II. doi: 10.2478/9788367405355-fm

- 7902 Harris F, Roby H, Dibb S. 2016. Sustainable clothing: challenges, barriers and interventions for encouraging more
7903 sustainable consumer behaviour: Sustainable clothing. *Int J Consum Stud* **40**(3): 309–318. doi:
7904 10.1111/ijcs.12257
- 7905 Hartline NL, Bruce NJ, Karba SN, Ruff EO, Sonar SU, Holden PA. 2016. Microfiber Masses Recovered from
7906 Conventional Machine Washing of New or Aged Garments. *Environ Sci Technol* **50**(21): 11532–11538.
7907 doi: 10.1021/acs.est.6b03045
- 7908 Hasanbeigi A, Price L. 2012. A review of energy use and energy efficiency technologies for the textile industry.
7909 *Renew Sustain Energy Rev* **16**(6): 3648–3665. doi: 10.1016/j.rser.2012.03.029
- 7910 Hazlehurst A, Tiffin L, Sumner M, Taylor M. 2023. Quantification of microfibre release from textiles during
7911 domestic laundering. *Environ Sci Pollut Res* **30**(15): 43932–43949. doi: 10.1007/s11356-023-25246-
7912 8
- 7913 Hearle JWS. 2009. Fibre structure: its formation and relation to performance. In: *Handbook of Textile Fibre*
7914 *Structure*. Elsevier. p. 3–21. doi: 10.1533/9781845696504.1.3
- 7915 Hearle JWS, Morton WE. 2008. *Physical Properties of Textile Fibres*. Elsevier.
- 7916 Heijungs R, Kleijn R. 2001. Numerical approaches towards life cycle interpretation five examples. *Int J Life Cycle*
7917 *Assess* **6**(3): 141. doi: 10.1007/BF02978732
- 7918 Heiny K, Schneide D. 2021. Attitude-Behavior Gap Report: How the industry and consumers can close the
7919 sustainability Attitude-Behavior Gap in fashion. Zalando. Available at
7920 [https://corporate.zalando.com/en/our-impact/sustainability/sustainability-reports/attitude-behavior-](https://corporate.zalando.com/en/our-impact/sustainability/sustainability-reports/attitude-behavior-gap-report)
7921 [gap-report](https://corporate.zalando.com/en/our-impact/sustainability/sustainability-reports/attitude-behavior-gap-report).
- 7922 Henninger CE, Alevizou PJ, Goworek H, Ryding D, editors. 2017. *Sustainability in Fashion: A Cradle to Upcycle*
7923 *Approach*. Cham: Springer International Publishing. doi: 10.1007/978-3-319-51253-2
- 7924 Henninger CE, Brydges T, Iran S, Vladimirova K. 2021. Collaborative fashion consumption – A synthesis and
7925 future research agenda. *J Clean Prod* **319**: 128648. doi: 10.1016/j.jclepro.2021.128648
- 7926 Henry B, Laitala K, Klepp IG. 2019. Microfibres from apparel and home textiles: Prospects for including
7927 microplastics in environmental sustainability assessment. *Sci Total Environ* **652**: 483–494. doi:
7928 10.1016/j.scitotenv.2018.10.166
- 7929 Hernandez L, Petreca B, Baurley S, Bianchi-Berthouze N, Cho Y. 2024. Examining clothing repair practices, core
7930 competences, techniques, tools and community structures involved in extending the life of garments.
7931 *Int J Sustain Fash Text* **3**(1): 99–121. doi: 10.1386/sft_00041_1
- 7932 Herweyers L, Catarci Carteny C, Scheelen L, Watts R, Du Bois E. 2020. Consumers' Perceptions and Attitudes
7933 toward Products Preventing Microfiber Pollution in Aquatic Environments as a Result of the Domestic
7934 Washing of Synthetic Clothes. *Sustainability* **12**(6): 2244. doi: 10.3390/su12062244
- 7935 Hidayaturrahman H, Lee T-G. 2019. A study on characteristics of microplastic in wastewater of South Korea:
7936 Identification, quantification, and fate of microplastics during treatment process. *Mar Pollut Bull* **146**:
7937 696–702. doi: 10.1016/j.marpolbul.2019.06.071
- 7938 Hildegunn Kyvik Nordås. 2004. *The Global Textile and Clothing Industry Post the Agreement on Textiles and*
7939 *Clothing*. Geneva: World Trade Organization. (WTO discussion papers / World Trade Organization; Vol.
7940 5).
- 7941 Hirt N, Body-Malapel M. 2020. Immunotoxicity and intestinal effects of nano- and microplastics: a review of the
7942 literature. *Part Fibre Toxicol* **17**(1): 57. doi: 10.1186/s12989-020-00387-7

- 7943 Hong Y, Al Mamun A, Yang Q, Masukujjaman M. 2024. Predicting sustainable fashion consumption intentions
7944 and practices. *Sci Rep* **14**(1): 1706. doi: 10.1038/s41598-024-52215-z
- 7945 Horn R, Maier S. 2018. LANCA® - Characterization Factors for Life Cycle Impact Assessment. Version 2.5. , in
7946 press. [object Object]. Available at <http://publica.fraunhofer.de/documents/N-379310.html>. Accessed
7947 2025 Nov 28.
- 7948 Horn S, Mölsä KM, Sorvari J, Tuovila H, Heikkilä P. 2023. Environmental sustainability assessment of a polyester
7949 T-shirt – Comparison of circularity strategies. *Sci Total Environ* **884**: 163821. doi:
7950 10.1016/j.scitotenv.2023.163821
- 7951 Hultén P, Vanyushyn V. 2014. Promotion and shoppers' impulse purchases: the example of clothes. *J Consum*
7952 *Mark* **31**(2): 94–102. doi: 10.1108/JCM-06-2013-0603
- 7953 Huygens D, Foschi J, Caro D, Caldeira C, Faraca G, Foster G, Solis M, Marschinski R, Napolano L, Astrup Fruergaard
7954 T, et al. 2023. Techno-scientific assessment of the management options for used and waste textiles
7955 in the European Union. *JRC134586*, in press. doi: doi:10.2760/6292
- 7956 IEA. 2019. World Energy Balances 2019. Paris: International Energy Agency. Available at
7957 <https://doi.org/10.1787/3a876031-en>.
- 7958 ILO News. 1996 Oct 28. Globalization Changes the Face of Textile, Clothing and Footwear Industries. Available
7959 at http://www.ilo.org/global/about-the-ilo/newsroom/news/WCMS_008075/lang--en/index.htm.
7960 Accessed 2023 Aug 15.
- 7961 INDA. 2023. What are nonwovens. Association of the Nonwoven Fabrics Industry. Available at
7962 <https://www.inda.org/about-nonwovens/>. Accessed 2023 Sept 20.
- 7963 İnovenso Teknoloji Ltd.Şti, Mustafa W. 2023. Breathable waterproof fabrics. , in press. Unpublished. doi:
7964 10.13140/RG.2.2.18928.02565
- 7965 IPCC. 2023. *Climate Change 2021 – The Physical Science Basis: Working Group I Contribution to the Sixth*
7966 *Assessment Report of the Intergovernmental Panel on Climate Change*. 1st ed. Cambridge University
7967 Press. doi: 10.1017/9781009157896
- 7968 Iqbal B, Zhao T, Yin W, Zhao Xin, Xie Q, Khan KY, Zhao Xiaoxun, Nazar M, Li G, Du D. 2023. Impacts of soil
7969 microplastics on crops: A review. *Appl Soil Ecol* **181**: 104680. doi: 10.1016/j.apsoil.2022.104680
- 7970 ITMF. 2021. International production cost comparison. INTERNATIONAL TEXTILE MANUFACTURERS FEDERATION.
7971 Report No.: IPCC 2021. Available at [https://www.itmf.org/publications/statistics-](https://www.itmf.org/publications/statistics-publications/international-production-cost-comparison-ipcc-2021)
7972 [publications/international-production-cost-comparison-ipcc-2021](https://www.itmf.org/publications/statistics-publications/international-production-cost-comparison-ipcc-2021).
- 7973 IWTO. 2022. IWTO Market information - Edition 17. Brussels: International Wool Textile Organization. Available
7974 at <https://www.member.iwto.org/store/viewproduct.aspx?id=22312572>. Accessed 2025 Aug 8.
- 7975 Iyare PU, Ouki SK, Bond T. 2020. Microplastics removal in wastewater treatment plants: a critical review. *Environ*
7976 *Sci Water Res Technol* **6**(10): 2664–2675. The Royal Society of Chemistry. doi: 10.1039/D0EW00397B
- 7977 Jadhav NC, Jadhav AC. 2020. Waste and 3R's in Footwear and Leather Sectors. In: Muthu SS, editor. *Leather*
7978 *and Footwear Sustainability*. Singapore: Springer Singapore. p. 261–293. doi: 10.1007/978-981-15-
7979 6296-9_10
- 7980 Jaffe M, Menczel JD. 2020. Thermal Analysis of Textiles and Fibers, Chapter 1 – Introduction. In: Jaffe M, Menczel
7981 JD, editors. *Thermal Analysis of Textiles and Fibers*. Woodhead Publishing. p. 1–6. doi: 10.1016/B978-
7982 0-08-100572-9.00001-X
- 7983 Jain A. 2021 Mar 31. Influence of consumption behaviours, attitudes and barriers toward clothing repair. , in
7984 press. doi: <https://doi.org/10.7939/r3-bekh-ry58>

- 7985 Jaishankar M, Tseten T, Anbalagan N, Mathew BB, Beeregowda KN. 2014. Toxicity, mechanism and health
7986 effects of some heavy metals. *Interdiscip Toxicol* **7**(2): 60–72.
- 7987 Jana D, Saskia M, Lars FM. 2023. The role of bio-based textile fibres in a circular and sustainable textiles system.
7988 European Environment Agency. Report No.: ETC-CE Report 2023/5. Available at
7989 file:///C:/Users/smoldovan/Downloads/ETC-EEA%20-%20Bio-based%20Textile%20Fibres_FINAL.pdf.
- 7990 Jankoska M, Demboski G. 2017. The influence of the sewing speed and fabric thickness on sewing machine
7991 stitch formation parameters. *Adv Technol* **6**(2): 72–77. doi: 10.5937/savteh1702072J
- 7992 Jia F, Shiyuan Y, Chen L, Chen X. 2020. The circular economy in the textile and apparel industry: A systematic
7993 literature review. *J Clean Prod* **259**.
- 7994 Jia L, Liu L, Zhang Y, Fu W, Liu X, Wang Q, Tanveer M, Huang L. 2023. Microplastic stress in plants: effects on
7995 plant growth and their remediations. *Front Plant Sci* **14**. Available at
7996 <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2023.1226484>. Accessed
7997 2024 Sept 10.
- 7998 Jørgensen P-E, Werner A, Constantinou G. 2022. Research and identification of textile plants globally - focusing
7999 on fiber-to-fiber recycling for the fashion & textile industry. Lifestyle & Design Cluster. Available at
8000 [https://ldcluster.com/wp-content/uploads/sites/4/2022/08/FINAL-Interactive-mapping-of-global-](https://ldcluster.com/wp-content/uploads/sites/4/2022/08/FINAL-Interactive-mapping-of-global-recycling-textile-plants-august-2022-LDC.pdf)
8001 [recycling-textile-plants-august-2022-LDC.pdf](https://ldcluster.com/wp-content/uploads/sites/4/2022/08/FINAL-Interactive-mapping-of-global-recycling-textile-plants-august-2022-LDC.pdf).
- 8002 Joung H-M, Park-Poaps H. 2013. Factors motivating and influencing clothing disposal behaviours: Clothing
8003 disposal behaviours. *Int J Consum Stud* **37**(1): 105–111. doi: 10.1111/j.1470-6431.2011.01048.x
- 8004 Joy A, Sherry JF, Venkatesh A, Wang J, Chan R. 2012a. Fast Fashion, Sustainability, and the Ethical Appeal of
8005 Luxury Brands. *Fash Theory* **16**(3): 273–295. doi: 10.2752/175174112X13340749707123
- 8006 Joy A, Sherry JF, Venkatesh A, Wang J, Chan R. 2012b. Fast Fashion, Sustainability, and the Ethical Appeal of
8007 Luxury Brands. *Fash Theory* **16**(3): 273–295. doi: 10.2752/175174112X13340749707123
- 8008 Jung J-W, Kang J, Choi J, Park J-W. 2020. Chronic toxicity of endocrine disrupting chemicals used in plastic
8009 products in Korean resident species: Implications for aquatic ecological risk assessment. *Ecotoxicol*
8010 *Environ Saf* **192**: 110309. doi: 10.1016/j.ecoenv.2020.110309
- 8011 Kalliala EM, Nousiainen P. 1999. Life Cycle Assessment, environmental profile of cotton and polyester-cotton
8012 fabrics. *Tamp Univ Technol* **1**(1). Available at
8013 [https://proyectaryproducir.com.ar/public_html/Seminarios_Posgrado/Material_de_referencia/LCA%20f](https://proyectaryproducir.com.ar/public_html/Seminarios_Posgrado/Material_de_referencia/LCA%20for%20Cotton%20and%20Polyester-cotton%20fabrics.pdf)
8014 [or%20Cotton%20and%20Polyester-cotton%20fabrics.pdf](https://proyectaryproducir.com.ar/public_html/Seminarios_Posgrado/Material_de_referencia/LCA%20for%20Cotton%20and%20Polyester-cotton%20fabrics.pdf).
- 8015 Kannan K, Vimalkumar K. 2021. A Review of Human Exposure to Microplastics and Insights Into Microplastics
8016 as Obesogens. *Front Endocrinol* **12**. Available at
8017 <https://www.frontiersin.org/articles/10.3389/fendo.2021.724989>. Accessed 2024 Jan 23.
- 8018 Kazour M, Terki S, Rabhi K, Jemaa S, Khalaf G, Amara R. 2019. Sources of microplastics pollution in the marine
8019 environment: Importance of wastewater treatment plant and coastal landfill. *Mar Pollut Bull* **146**: 608–
8020 618. doi: 10.1016/j.marpolbul.2019.06.066
- 8021 KEMI. 2014. Chemicals in textiles – Risks to human health and the environme. Stockholm: Swedish Chemicals
8022 Agency. Report No.: 361 130. Available at
8023 [https://www.kemi.se/download/18.6df1d3df171c243fb23a98f3/1591454110491/rapport-6-14-](https://www.kemi.se/download/18.6df1d3df171c243fb23a98f3/1591454110491/rapport-6-14-chemicals-in-textiles.pdf)
8024 [chemicals-in-textiles.pdf](https://www.kemi.se/download/18.6df1d3df171c243fb23a98f3/1591454110491/rapport-6-14-chemicals-in-textiles.pdf). Accessed 2023 Oct 19.
- 8025 Khairul Akter MMd, Haq UN, Islam MdM, Uddin MA. 2022. Textile-apparel manufacturing and material waste
8026 management in the circular economy: A conceptual model to achieve sustainable development goal
8027 (SDG) 12 for Bangladesh. *Clean Environ Syst* **4**: 100070. doi: 10.1016/j.cesys.2022.100070

- 8028 Khan IA, Malik MH, Ahmad Fraz, Hafsa Ibrahim. 2025. From Waste to Weave: Recycling Industrial and Consumer
8029 Cotton Waste for Sustainable and Cost-Effective Terry Fabric Production. *Tekstilec* **68**(1): 42–56. doi:
8030 10.14502/tekstilec.68.2024103
- 8031 Khanna A. 2021. Factors affecting consumer-buying motivations: An empirical study in the behavioral
8032 economics perspective. *International Journal of Research in Marketing Management and Sales* **3**(2):
8033 19–23.
- 8034 Kim JJ, In Shin K, Ryu HS, Kim EA, Lee M, Wha Oh K. 2004. Luster Properties of Polyester Filament Yarn Woven
8035 Fabrics. *Text Res J* **74**(1): 72–77. doi: 10.1177/004051750407400113
- 8036 Kleinhüchelkotten S, Freudenreich B, Glomb M, Hübner G, Landsbek B, Nebel K, Neitzke H-P, Schaltegger S,
8037 Woznica A. 2019. Slow Fashion: Gestalterische, technische und ökonomische Innovationen für
8038 massenmarktaugliche nachhaltige Angebote im Bedarfsfeld "Bekleidung." Hannover, Hamburg.
- 8039 Kleinhüchelkotten S, Neitzke H-P, Schmidt N. 2018a. Mode, Kleidung und Nachhaltigkeit: Einstellungen und
8040 Verhalten - Ergebnisse der Repräsentativbefragung 2017. Hannover: Innabe. Available at
8041 [https://www.ecolog-institut.de/wp-content/uploads/2021/04/InNaBe_Bericht_5-](https://www.ecolog-institut.de/wp-content/uploads/2021/04/InNaBe_Bericht_5-5_Repr%C3%A4sentativbefragung_2.pdf)
8042 [5_Repr%C3%A4sentativbefragung_2.pdf](https://www.ecolog-institut.de/wp-content/uploads/2021/04/InNaBe_Bericht_5-5_Repr%C3%A4sentativbefragung_2.pdf).
- 8043 Kleinhüchelkotten S, Neitzke H-P, Schmidt N. 2018b. Mode, Kleidung und Nachhaltigkeit: Einstellungen und
8044 Verhalten - Ergebnisse der Repräsentativbefragung 2017. Hannover: Innabe. Available at
8045 [https://www.ecolog-institut.de/wp-content/uploads/2021/04/InNaBe_Bericht_5-](https://www.ecolog-institut.de/wp-content/uploads/2021/04/InNaBe_Bericht_5-5_Repr%C3%A4sentativbefragung_2.pdf)
8046 [5_Repr%C3%A4sentativbefragung_2.pdf](https://www.ecolog-institut.de/wp-content/uploads/2021/04/InNaBe_Bericht_5-5_Repr%C3%A4sentativbefragung_2.pdf).
- 8047 Klepp, Grimstad I. 2001. Hvorfor går klær ut av bruk? Available at <https://hdl.handle.net/20.500.12199/5390>.
- 8048 Klepp IG, Laitala K. 2023a. Washing Clothes. In: *The Routledge History of Fashion and Dress, 1800 to the*
8049 *Present* 1st ed. London: Routledge. p. 491–506. doi: 10.4324/9780429295607-31
- 8050 Klepp IG, Laitala K. 2023b. Washing Clothes. In: *The Routledge History of Fashion and Dress, 1800 to the*
8051 *Present* 1st ed. London: Routledge. p. 491–506. doi: 10.4324/9780429295607-31
- 8052 Klepp IG, Laitala K, Wiedemann S. 2020a. Clothing Lifespans: What Should Be Measured and How. *Sustainability*
8053 **12**(15): 6219. doi: 10.3390/su12156219
- 8054 Klepp IG, Laitala K, Wiedemann S. 2020b. Clothing Lifespans: What Should Be Measured and How. *Sustainability*
8055 **12**(15): 6219. Multidisciplinary Digital Publishing Institute. doi: 10.3390/su12156219
- 8056 Koay KY, Cheah CW, Lom HS. 2022. An integrated model of consumers' intention to buy second-hand clothing.
8057 *Int J Retail Distrib Manag* **50**(11): 1358–1377. doi: 10.1108/IJRDM-10-2021-0470
- 8058 Koszewska M. 2016. Understanding Consumer Behavior in the Sustainable Clothing Market: Model Development
8059 and Verification. In: Muthu SS, Gardetti MA, editors. *Green Fashion*. Singapore: Springer Singapore. p.
8060 43–94. doi: 10.1007/978-981-10-0111-6_3
- 8061 Koukouvinos D. 2012. Psychosocial Factors Influencing Young Consumers' Clothing Disposal Behaviour in
8062 Greece. , in press.
- 8063 Krsteva S, Demboski G. 2011. *DETERMINATION OF GENERATED TEXTILE WASTE IN CLOTHING COMPANIES OF*
8064 *DIFFERENT TECHNOLOGICAL LEVEL*. Zrenjanin: Technical Faculty "Mihajlo Pupin."
- 8065 Kumar A, Mishra S, Pandey R, Yu ZG, Kumar M, Khoo KS, Thakur TK, Show PL. 2023. Microplastics in terrestrial
8066 ecosystems: Un-ignorable impacts on soil characterises, nutrient storage and its cycling. *TrAC Trends*
8067 *Anal Chem* **158**: 116869. doi: 10.1016/j.trac.2022.116869
- 8068 Kumar V, Hernández N, Jensen M, Pal R. 2023. Deep learning based system for garment visual degradation
8069 prediction for longevity. *Comput Ind* **144**: 103779. doi: 10.1016/j.compind.2022.103779

- 8070 Kuppen M. 2024. Technical Report Denim Deal – State of the art for using post-consumer recycled cotton in
8071 denim, in press.
- 8072 Laitala K, Berg LL, I Strandbakken P. 2023. Why won't you complain? Consumer rights and the unmet product
8073 lifespan requirements. 2023. The 5th Product Lifetimes and the Environment (PLATE) Conference. 31-
8074 2 June 2023; Espoo: Aalto University. Available at <http://urn.fi/URN:ISBN:978-952-64-1367-9>.
- 8075 Laitala K, Boks C. 2012. Sustainable clothing design: use matters. *J Des Res* **10**(1/2): 121. doi:
8076 10.1504/JDR.2012.046142
- 8077 Laitala K, Klepp I, Henry B. 2018. Does Use Matter? Comparison of Environmental Impacts of Clothing Based on
8078 Fiber Type. *Sustainability* **10**(7): 2524. doi: 10.3390/su10072524
- 8079 Laitala K, Klepp IG. 2018a. Care and Production of Clothing in Norwegian Homes: Environmental Implications of
8080 Mending and Making Practices. *Sustainability* **10**(8): 2899. doi: 10.3390/su10082899
- 8081 Laitala K, Klepp IG. 2018b. Care and Production of Clothing in Norwegian Homes: Environmental Implications of
8082 Mending and Making Practices. *Sustainability* **10**(8): 2899. doi: 10.3390/su10082899
- 8083 Laitala K, Klepp IG. 2020a. What Affects Garment Lifespans? International Clothing Practices Based on a
8084 Wardrobe Survey in China, Germany, Japan, the UK, and the USA. *Sustainability* **12**(21): 9151.
8085 Multidisciplinary Digital Publishing Institute. doi: 10.3390/su12219151
- 8086 Laitala K, Klepp IG. 2020b. What Affects Garment Lifespans? International Clothing Practices Based on a
8087 Wardrobe Survey in China, Germany, Japan, the UK, and the USA. *Sustainability* **12**(21): 9151. doi:
8088 10.3390/su12219151
- 8089 Laitala K, Klepp IG. 2021a. Clothing Longevity: The Relationship Between The Number of Users, How Long and
8090 How Many Times Garments are Used. 2021. 4th PLATE 2021 Virtual Conference 26-28 May 2021;
8091 Limerick, Ireland. Available at
8092 https://www.researchgate.net/publication/352993692_Clothing_Longevity_The_Relationship_Between_The_Number_of_Users_How_Long_and_How_Many_Times_Garments_are_Used.
8093
- 8094 Laitala K, Klepp IG. 2021b. Clothing Longevity: The Relationship Between The Number of Users, How Long and
8095 How Many Times Garments are Used. 2021. 4th PLATE 2021 Virtual Conference 26-28 May 2021;
8096 Limerick, Ireland. Available at
8097 https://www.researchgate.net/publication/352993692_Clothing_Longevity_The_Relationship_Between_The_Number_of_Users_How_Long_and_How_Many_Times_Garments_are_Used.
8098
- 8099 Laitala K, Klepp IG. 2022. Review of clothing disposal reasons. Available at
8100 <https://clothingresearch.oslomet.no/2022/10/19/review-of-clothing-disposal-reasons/>.
- 8101 Laitala K, Klepp IG, Boks C. 2012. Changing laundry habits in Norway: Changing laundry habits in Norway. *Int J*
8102 *Consum Stud* **36**(2): 228–237. doi: 10.1111/j.1470-6431.2011.01081.x
- 8103 Laitala K, Klepp IG, Haugrønning V, Throne-Holst H, Strandbakken P. 2021a. Increasing repair of household
8104 appliances, mobile phones and clothing: Experiences from consumers and the repair industry. *J Clean*
8105 *Prod* **282**: 125349. doi: 10.1016/j.jclepro.2020.125349
- 8106 Laitala K, Klepp IG, Haugrønning V, Throne-Holst H, Strandbakken P. 2021b. Increasing repair of household
8107 appliances, mobile phones and clothing: Experiences from consumers and the repair industry. *J Clean*
8108 *Prod* **282**: 125349. doi: 10.1016/j.jclepro.2020.125349
- 8109 Laitala K, Klepp IG, Henry B. 2018a. Does Use Matter? Comparison of Environmental Impacts of Clothing Based
8110 on Fiber Type. *Sustainability* **10**(7): 2524. Multidisciplinary Digital Publishing Institute. doi:
8111 10.3390/su10072524

- 8112 Laitala K, Klepp IG, Henry B. 2018b. Use phase of apparel: A literature review for Life Cycle Assessment with
8113 focus on wool. , in press. Unpublished. doi: 10.13140/RG.2.2.25769.90729
- 8114 Laitala K, Klepp IG, Henry BK. 2017. Global laundering practices. Alternatives to machine washing. *H&PC Today*
8115 – *Household and Personal Care Today* **12**(5): 10–16.
- 8116 Laitala K, Klepp IG, Kettlewell R, Wiedemann S. 2020. Laundry Care Regimes: Do the Practices of Keeping Clothes
8117 Clean Have Different Environmental Impacts Based on the Fibre Content? *Sustainability* **12**(18): 7537.
8118 doi: 10.3390/su12187537
- 8119 Lamb P. 2009. Effect of Fibre Properties on Processing Performance: Top to Yarn. *Univ N Engl*, in press.
- 8120 Landrigan PJ, Stegeman JJ, Fleming LE, Allemand D, Anderson DM, Backer LC, Brucker-Davis F, Chevalier N,
8121 Corra L, Czerucka D, et al. 2020. Human Health and Ocean Pollution. **86**(1): 151. Ubiquity Press. doi:
8122 10.5334/aogh.2831
- 8123 Lang C, Armstrong CM, Brannon LA. 2013. Drivers of clothing disposal in the US: An exploration of the role of
8124 personal attributes and behaviours in frequent disposal: Drivers of frequent clothing disposal. *Int J*
8125 *Consum Stud* **37**(6): 706–714. doi: 10.1111/ijcs.12060
- 8126 Lawrence C. 2015. Fibre to Yarn. In: *Textiles and Fashion*. Elsevier. p. 213–253. doi: 10.1016/B978-1-84569-
8127 931-4.00010-6
- 8128 LEITAT. 2017. Implementation of Article 11 under the EU Ecolabel Regulation. Spain: LEITAT-Technological
8129 Center. Report No.: Deliverable 4. FINAL REPORT. Available at
8130 https://ec.europa.eu/environment/ecolabel/documents/180504%20FINAL_REPORT_ARTICLE11_V3%20with%20disclaimer.pdf. Accessed 2024 Jan 19.
- 8131
- 8132 Leo Wierda. 2024. Ecodesign Impact Accounting Status Report 2024.
- 8133 Leslie HA, van Velzen MJM, Brandsma SH, Vethaak AD, Garcia-Vallejo JJ, Lamoree MH. 2022. Discovery and
8134 quantification of plastic particle pollution in human blood. *Environ Int* **163**: 107199. doi:
8135 10.1016/j.envint.2022.107199
- 8136 Lewis T. 2015. Apparel disposal and reuse. In: *Sustainable Apparel*. Elsevier. p. 233–250. doi: 10.1016/B978-1-
8137 78242-339-3.00010-8
- 8138 Li H-X, Shi M, Tian F, Lin L, Liu S, Hou R, Peng J-P, Xu X-R. 2022. Microplastics contamination in bivalves from
8139 the Daya Bay: Species variability and spatio-temporal distribution and human health risks. *Sci Total*
8140 *Environ* **841**: 156749. doi: 10.1016/j.scitotenv.2022.156749
- 8141 Li Z, Zhong P, Tang X, Chen Y, Su S, Zhai T. 2020. A New Method to Evaluate Yarn Appearance Qualities Based
8142 on Machine Vision and Image Processing. *IEEE Access* **8**: 30928–30937. doi:
8143 10.1109/ACCESS.2020.2972967
- 8144 Lim X. 2021 May. Microplastics are everywhere — but are they harmful? *Nature*, in press. doi:
8145 <https://doi.org/10.1038/d41586-021-01143-3>
- 8146 Lingås D, Manshoven S, Mortensen LF, Paulsen F. 2023. EU exports of used textiles in Europe's circular economy.
8147 EEA. Report No.: ETC CE Report 2023/4. Available at <https://www.eionet.europa.eu/etcs/etc-ce/products/etc-ce-report-2023-4-eu-exports-of-used-textiles-in-europe2019s-circular-economy#:~:text=2023>.
- 8148
- 8149
- 8150 Lu K, Zhan D, Fang Y, Li L, Chen G, Chen S, Wang L. 2022. Microplastics, potential threat to patients with lung
8151 diseases. *Front Toxicol* **4**. Available at <https://www.frontiersin.org/articles/10.3389/ftox.2022.958414>.
8152 Accessed 2024 Jan 23.

- 8153 Lu L, Fan W, Meng X, Xue L, Ge S, Wang C, Foong SY, Tan CSY, Sonne C, Aghbashlo M, et al. 2023. Current
8154 recycling strategies and high-value utilization of waste cotton. *Sci Total Environ* **856**: 158798. doi:
8155 10.1016/j.scitotenv.2022.158798
- 8156 Luo Y, Wu X, Ding X. 2023a. Environmental impacts of textiles in the use stage: A systematic review. *Sustain*
8157 *Prod Consum* **36**: 233–245. doi: 10.1016/j.spc.2023.01.006
- 8158 Luo Y, Wu X, Ding X. 2023b. Environmental impacts of textiles in the use stage: A systematic review. *Sustain*
8159 *Prod Consum* **36**: 233–245. doi: 10.1016/j.spc.2023.01.006
- 8160 Luo Y, Wu X, Ding X. 2023c. Environmental impacts of textiles in the use stage: A systematic review. *Sustain*
8161 *Prod Consum* **36**: 233–245. doi: 10.1016/j.spc.2023.01.006
- 8162 Lv X, Dong Q, Zuo Z, Liu Y, Huang X, Wu W-M. 2019. Microplastics in a municipal wastewater treatment plant:
8163 Fate, dynamic distribution, removal efficiencies, and control strategies. *J Clean Prod* **225**: 579–586.
8164 doi: 10.1016/j.jclepro.2019.03.321
- 8165 Machado MAD, Almeida SOD, Bollick LC, Bragagnolo G. 2019. Second-hand fashion market: consumer role in
8166 circular economy. *J Fash Mark Manag Int J* **23**(3): 382–395. doi: 10.1108/JFMM-07-2018-0099
- 8167 Magrini C, Rodriguez Manotas J, Gonzalez Torres M, Senatore V, Gama Caldas M. 2025. Method for the ranking
8168 of potential requirements, based on expected impacts and costs. Luxembourg (Luxembourg):
8169 Publications Office of the European Union. Report No.: JRC143331. doi: 10.2760/1478580
- 8170 Mahmoudi MR. 2010. 4 - Blending and composite yarn spinning. In: Lawrence CA, editor. *Advances in Yarn*
8171 *Spinning Technology*. Woodhead Publishing. p. 102–118. doi: 10.1533/9780857090218.1.102
- 8172 Makov T. 2023. The hidden environmental costs of consumer product returns. 5th PLATE Conference. Available
8173 at <https://aaltodoc.aalto.fi/handle/123456789/122687>.
- 8174 Maldini I, Balkenende. 2017. Reducing clothing production volumes by design: a critical review of sustainable
8175 fashion strategies. 2017. PLATE: Product lifetimes and the environment. Conference proceedings of
8176 PLATE 2017, 8-10 November 2017. Technische Universiteit Delft.; Delft, the Netherlands. Available at
8177 https://pure.hva.nl/ws/portalfiles/portal/3974950/Article_Plate_published.pdf.
- 8178 Maldini I, Stappers PJ, Gimeno-Martinez JC, Daanen HAM. 2019. Assessing the impact of design strategies on
8179 clothing lifetimes, usage and volumes: The case of product personalisation. *J Clean Prod* **210**: 1414–
8180 1424. doi: 10.1016/j.jclepro.2018.11.056
- 8181 Manshoven S, Christis M, Vercalsteren A, Arnold M, Nicolau M, Lafond E, Mortensen, LF, Coscieme L. 2019.
8182 Textiles and the Environment in a circular economy. Belgium: European Topic Centre Waste and
8183 Materials in a Green Economy. Report No.: Eionet Report. Available at
8184 [https://www.eionet.europa.eu/etcs/etc-wmge/products/etc-wmge-reports/textiles-and-the-](https://www.eionet.europa.eu/etcs/etc-wmge/products/etc-wmge-reports/textiles-and-the-environment-in-a-circular-economy)
8185 [environment-in-a-circular-economy](https://www.eionet.europa.eu/etcs/etc-wmge/products/etc-wmge-reports/textiles-and-the-environment-in-a-circular-economy).
- 8186 Mao N. 2016. Methods for characterisation of nonwoven structure, property, and performance. In: Kellie G,
8187 editor. *Advances in Technical Nonwovens*. Woodhead Publishing. p. 155–211. doi: 10.1016/B978-0-08-
8188 100575-0.00006-1
- 8189 Mao N, Wang Y, Qu J. 2016. SMOOTHNESS AND ROUGHNESS: CHARACTERISTICS OF FABRIC-TO-FABRIC SELF-
8190 FRICTION PROPERTIES. , in press.
- 8191 Marsh E, Allen S, Hattam L. 2023. Tackling uncertainty in life cycle assessments for the built environment: A
8192 review. *Build Environ* **231**: 109941. Pergamon. doi: 10.1016/J.BUILDENV.2022.109941
- 8193 de Mattos FB, Esquivel V, Kucera D, Tejani S. 2022. The state of the apparel and footwear industry: Employment,
8194 automation and their gender dimensions. *Backgr Pap N3 GLO/20/82/EUR[ILO_REF]*. Available at

- 8195 [https://www.ilo.org/employment/Whatwedo/Projects/building-partnerships-on-the-future-of-](https://www.ilo.org/employment/Whatwedo/Projects/building-partnerships-on-the-future-of-work/WCMS_835423/lang--en/index.htmç)
8196 [work/WCMS_835423/lang--en/index.htmç](https://www.ilo.org/employment/Whatwedo/Projects/building-partnerships-on-the-future-of-work/WCMS_835423/lang--en/index.htmç).
- 8197 Matusiak Malgorzata, Sikorski Renata. 2011. *Influence_of_the_Structure_of_Woven_Fabrics_on_The.pdf*.
- 8198 Maud Kuppen. 2024. Technical Report Denim Deal – State of the art for using post-consumer recycled cotton
8199 in denim.
- 8200 Maya Forstater. 2010 Jan 6. Sectoral coverage of the global economic crisis. Implications of the global financial
8201 and economic crisis on the textile and clothing sector. *Int Labour Organ*: 32.
- 8202 McKinsey & Company. 2022. Scaling textile recycling in europe turning waste into value. McKinsey Apparel,
8203 Fashion & Luxury Group. Available at [https://www.mckinsey.com/industries/retail/our-insights/scaling-](https://www.mckinsey.com/industries/retail/our-insights/scaling-textile-recycling-in-europe-turning-waste-into-value)
8204 [textile-recycling-in-europe-turning-waste-into-value](https://www.mckinsey.com/industries/retail/our-insights/scaling-textile-recycling-in-europe-turning-waste-into-value).
- 8205 McKinsey & Company. 2024. Reimagining the apparel value chain amid volatility. Available at
8206 [https://www.mckinsey.com/industries/retail/our-insights/reimagining-the-apparel-value-chain-amid-](https://www.mckinsey.com/industries/retail/our-insights/reimagining-the-apparel-value-chain-amid-volatility#/)
8207 [volatility#](https://www.mckinsey.com/industries/retail/our-insights/reimagining-the-apparel-value-chain-amid-volatility#/).
- 8208 McKinsey & Company, BOF. 2016. The State of Fashion 2017. The Business of Fashion and McKinsey &
8209 Company. Available at
8210 [https://www.mckinsey.com/~media/McKinsey/Industries/Retail/Our%20Insights/The%20state%20of%](https://www.mckinsey.com/~media/McKinsey/Industries/Retail/Our%20Insights/The%20state%20of%20fashion/The-state-of-fashion-2017-McK-BoF-report.pdf)
8211 [20fashion/The-state-of-fashion-2017-McK-BoF-report.pdf](https://www.mckinsey.com/~media/McKinsey/Industries/Retail/Our%20Insights/The%20state%20of%20fashion/The-state-of-fashion-2017-McK-BoF-report.pdf). Accessed 2023 Dec 8.
- 8212 McLaren A, Goworek H, Cooper T, Oxborrow L, Hill H. 2015. Clothing longevity perspectives: exploring consumer
8213 expectations, consumption and use. 2015. PLATE conference - Nottingham Trent University, 17/19 June
8214 2015. Available at http://irep.ntu.ac.uk/id/eprint/17978/1/220768_PubSub2644_McLaren.pdf.
8215 Accessed 2023 Oct 30.
- 8216 McLaren A, Goworek H, Cooper T, Oxborrow L, Hill H. 2016. The effect of consumer attitudes on design for
8217 product longevity: The case of the fashion industry. 2016 June 25. Design Research Society Conference
8218 2016. doi: 10.21606/drs.2016.456
- 8219 McNeill L, Moore R. 2015. Sustainable fashion consumption and the fast fashion conundrum: fashionable
8220 consumers and attitudes to sustainability in clothing choice: Sustainable fashion consumption and the
8221 fast fashion conundrum. *Int J Consum Stud* **39**(3): 212–222. doi: 10.1111/ijcs.12169
- 8222 McQueen RH, Jain A, McNeill LS, Kozlowski A. 2023. The role of resources in repair practice: Engagement with
8223 self, paid and unpaid clothing repair by young consumers. *Text Res J* **93**(3–4): 576–591. doi:
8224 10.1177/00405175221123067
- 8225 MEErP. 2011. MEErP: Methodology for Ecodesign of Energy-related Products. European Commission, DG
8226 Enterprise and Industry Unit B1 Sustainable Industrial Policy under. Report No.: Final report. Available
8227 at [https://op.europa.eu/en/publication-detail/-/publication/b7650397-32f1-436c-82c4-](https://op.europa.eu/en/publication-detail/-/publication/b7650397-32f1-436c-82c4-df39aef297a3/language-en/format-PDF/source-295028560)
8228 [df39aef297a3/language-en/format-PDF/source-295028560](https://op.europa.eu/en/publication-detail/-/publication/b7650397-32f1-436c-82c4-df39aef297a3/language-en/format-PDF/source-295028560).
- 8229 Meléndez-Fernández OH, Liu JA, Nelson RJ. 2023. Circadian Rhythms Disrupted by Light at Night and Mistimed
8230 Food Intake Alter Hormonal Rhythms and Metabolism. *Int J Mol Sci* **24**(4): 3392. Multidisciplinary
8231 Digital Publishing Institute. doi: 10.3390/ijms24043392
- 8232 milieu centraal. 2021. Doelgroepverkenningkoplopers reparerenkoplopers repareren.
- 8233 Ministry of Ecology and Environment, China. 2012. Discharge standards of water pollutants for dyeing and
8234 finishing of textile industry GB 4287-2012. Available at
8235 [https://english.mee.gov.cn/Resources/standards/water_environment/Discharge_standard/201301/t201](https://english.mee.gov.cn/Resources/standards/water_environment/Discharge_standard/201301/t20130107_244749.shtml)
8236 [30107_244749.shtml](https://english.mee.gov.cn/Resources/standards/water_environment/Discharge_standard/201301/t20130107_244749.shtml).

- 8237 Ministry of Ecology and Environment, China. 2021. Guideline on available techniques of pollution prevention
8238 4783 and control for textile industry. Available at
8239 <https://www.mee.gov.cn/ywgz/fgbz/bz/bzwb/kxxjszn/202106/W020210809521494726861.pdf>.
- 8240 Mishra M, Kushwaha R, Gupta N, Sinha A, Dwivedi H. 2023. Survey data to evaluate consumer behaviour and
8241 consumption pattern of sustainable apparel: A study on consumer awareness level. *Data Brief* **49**:
8242 109350. doi: 10.1016/j.dib.2023.109350
- 8243 Mishra SP. 2000. *A Text Book of Fibre Science and Technology*. New Age International.
- 8244 Mkuye R, Gong S, Zhao L, Masanja F, Ndandala C, Bubelwa E, Yang C, Deng Y. 2022. Effects of microplastics on
8245 physiological performance of marine bivalves, potential impacts, and enlightening the future based on
8246 a comparative study. *Sci Total Environ* **838**: 155933. doi: 10.1016/j.scitotenv.2022.155933
- 8247 Mobarak Hossain M. 2016. A Review on Different Factors of Woven Fabrics' Strength Prediction. *Sci Res* **4**(3):
8248 88. doi: 10.11648/j.sr.20160403.13
- 8249 Mohapatra P, Gaonkar O. 2021. An overview of toxic chemicals in textiles. New Delhi: Toxics Links. Available at
8250 <https://toxicslink.org/Publication/ToxicsChemicalinTextileReport>.
- 8251 Monitor, Cotton Incorporated Lifestyle. 2018. Price, Quality and Style: Consumers Seek Shopping Symmetry.
8252 Available at <https://lifestylemonitor.cottoninc.com/price-quality-and-style/>. Accessed 2023 Nov 15.
- 8253 Moralia M-A, Quignon C, Simonneaux M, Simonneaux V. 2022. Environmental disruption of reproductive rhythms.
8254 *Front Neuroendocrinol* **66**: 100990. doi: 10.1016/j.yfrne.2022.100990
- 8255 Morell-Delgado G, Talens Peiró L, Toboso-Chavero S. 2024. Revealing the management of municipal textile
8256 waste and citizen practices: The case of Catalonia. *Sci Total Environ* **907**: 168093. doi:
8257 10.1016/j.scitotenv.2023.168093
- 8258 Morgan L R, Birtwistle G. 2009a. An investigation of young fashion consumers' disposal habits. *Wiley Online Libr*
8259 **33**(2): 190–198. doi: <https://doi.org/10.1111/j.1470-6431.2009.00756.x>
- 8260 Morgan L R, Birtwistle G. 2009b. An investigation of young fashion consumers' disposal habits. **33**: 190–198.
8261 doi: <https://doi.org/10.1111/j.1470-6431.2009.00756.x>
- 8262 Morris KE, Joynes A, Summer M, Scott E, Taylor M. 2025. Measuring Physical Garment Durability: An assessment
8263 of 47 T-shirts. *Proc 6th Prod Lifetimes Environ Conf PLATE2025*(6). doi: 10.54337/plate2025-10277
- 8264 Munasinghe P, Druckman A, Dissanayake DGK. 2021. A systematic review of the life cycle inventory of clothing.
8265 *J Clean Prod* **320**: 128852. doi: 10.1016/j.jclepro.2021.128852
- 8266 Murakami S, Oguchi M, Tasaki T, Daigo I, Hashimoto S. 2010. Lifespan of Commodities, Part I - The Creation of
8267 a Database and Its Review. *Journal of Industrial Ecology* **14**(4): 529–684, i–xi. doi: 0.1111/j.1530-
8268 9290.2010.00250.x
- 8269 Mušič B. 2023. Textile industry as a major source of microplastics in the environment. *Glob J Ecol* **8**(2): 064–
8270 074. doi: 10.17352/gje.000084
- 8271 Mustia DI, Edy S, Nurul A. 2021. Analysis of waste composition as a source of refuse-derived fuel in Cilacap.
8272 *IOP Conf Ser Earth Environ Sci* **896**(1): 012063. doi: 10.1088/1755-1315/896/1/012063
- 8273 Muthu SS. 2015. *Handbook of Life Cycle Assessment (LCA) of Textiles and Clothing*. Elsevier. (Woodhead
8274 Publishing Series in Textiles; Vol. 172). doi: 10.1016/C2014-0-00761-7
- 8275 Muthu SS. 2020. *Assessing the Environmental Impact of Textiles and the Clothing Supply Chain*. Second edition.
8276 Elsevier. doi: 10.1016/B978-0-12-819783-7.00001-6

- 8277 Nam PD, Alvi MAH. 2024. History of Sizing System. In: Khan MQ, Nawab Y, Kim IS, editors. *Garment Sizing and*
8278 *Pattern Making*. Singapore: Springer Nature Singapore. p. 21–33. doi: 10.1007/978-981-97-7683-2_2
- 8279 Napper IE, Thompson RC. 2016. Release of synthetic microplastic plastic fibres from domestic washing
8280 machines: Effects of fabric type and washing conditions. *Mar Pollut Bull* **112**(1): 39–45. doi:
8281 10.1016/j.marpolbul.2016.09.025
- 8282 Nayak R., Padhye R. 2015. 17 - Care labelling of clothing. In: Nayak Rajkishore, Padhye Rajiv, editors. *Garment*
8283 *Manufacturing Technology*. Woodhead Publishing. p. 427–446. doi: 10.1016/B978-1-78242-232-
8284 7.00017-5
- 8285 Nayak R, Padhye R. 2018. Introduction to automation in garment manufacturing. In: Nayak R, Padhye R, editors.
8286 *Automation in Garment Manufacturing*. Woodhead Publishing. p. 1–27. doi: 10.1016/B978-0-08-
8287 101211-6.00001-X
- 8288 Nayak R, Ratnapandian S. 2018. *Care and Maintenance of Textile Products Including Apparel and Protective*
8289 *Clothing*. 1st ed. CRC Press. doi: 10.1201/b22481
- 8290 Neuwahl F, Cusano G, Gómez Benavides J, Holbrook S, Serge R. 2019. Best Available Techniques (BAT) reference
8291 document for waste incineration: Industrial Emissions Directive 2010/75/EU (Integrated Pollution
8292 Prevention and Control). LU: Publications Office. Available at
8293 <https://data.europa.eu/doi/10.2760/761437>. Accessed 2024 Dec 4.
- 8294 Newbold T, Hudson LN, Arnell AP, Contu S, De Palma A, Ferrier S, Hill SLL, Hoskins AJ, Lysenko I, Phillips HRP, et
8295 al. 2016. Has land use pushed terrestrial biodiversity beyond the planetary boundary? A global
8296 assessment. *Science* **353**(6296): 288–291. doi: 10.1126/science.aaf2201
- 8297 Niinimäki K. 2011. *From Disposable to Sustainable: The Complex Interplay between Design and Consumption of*
8298 *Textiles and Clothing*. Helsinki: Aalto University School of Art and Design. Available at
8299 <https://aaltodoc.aalto.fi/handle/123456789/13770>.
- 8300 Niinimäki K, Armstrong C. 2013. From pleasure in use to preservation of meaningful memories: a closer look at
8301 the sustainability of clothing via longevity and attachment. *Int J Fash Des Technol Educ* **6**(3): 190–
8302 199. doi: 10.1080/17543266.2013.825737
- 8303 Niinimäki K, Peters G, Dahlbo H, Perry P, Rissanen T, Gwilt A. 2020. The environmental price of fast fashion. *Nat*
8304 *Rev Earth Environ* **1**(4): 189–200. doi: 10.1038/s43017-020-0039-9
- 8305 Niu H, Liu S, Jiang Y, Hu Y, Li Y, He L, Xing M, Li X, Wu L, Chen Z, et al. 2023. Are Microplastics Toxic? A Review
8306 from Eco-Toxicity to Effects on the Gut Microbiota. *Metabolites* **13**(6): 739. Multidisciplinary Digital
8307 Publishing Institute. doi: 10.3390/metabo13060739
- 8308 Nordic Ecolabelling. 2022. Background document of Nordic Swan Ecolabelled Textiles, hides/skins, and leather
8309 - Version 5.6, 01 March 2022 to 31 December 2026. Available at [https://www.nordic-swan-ecolabel.org/4a7d17/contentassets/2a4b1e9ce1d04ad0a820553f8b716cd9/background-](https://www.nordic-swan-ecolabel.org/4a7d17/contentassets/2a4b1e9ce1d04ad0a820553f8b716cd9/background-document_039_manufacturing-of-textiles-hidesskins-and-leather-039_english2.pdf)
8310 [document_039_manufacturing-of-textiles-hidesskins-and-leather-039_english2.pdf](https://www.nordic-swan-ecolabel.org/4a7d17/contentassets/2a4b1e9ce1d04ad0a820553f8b716cd9/background-document_039_manufacturing-of-textiles-hidesskins-and-leather-039_english2.pdf). Accessed 2024
8311 Nov 27.
8312
- 8313 Nørup N, Pihl K, Damgaard A, Scheutz C. 2019. Replacement rates for second-hand clothing and household
8314 textiles – A survey study from Malawi, Mozambique and Angola. *J Clean Prod* **235**: 1026–1036. doi:
8315 10.1016/j.jclepro.2019.06.177
- 8316 NRF, APPRISS RETAIL. 2023. 2023 Consumer Returns in the Retail Industry. Available at
8317 https://cdn.nrf.com/sites/default/files/2024-01/Customer_Returns_Report_2023_Final.pdf.
- 8318 O'Brien S, Rauert C, Ribeiro F, Okoffo ED, Burrows SD, O'Brien JW, Wang X, Wright SL, Thomas KV. 2023. There's
8319 something in the air: A review of sources, prevalence and behaviour of microplastics in the atmosphere.
8320 *Sci Total Environ* **874**: 162193. doi: 10.1016/j.scitotenv.2023.162193

- 8321 OCU. 2018. Otro consumo para un futuro mejor - Nuevas economías al servicio de las personas y el planeta.
8322 Available at <https://www.ocu.org/consumo-familia/consumo-colaborativo/noticias/otro-consumo-es->
8323 posible.
- 8324 OECD. 2018. Due Diligence Guidance for Responsible Supply Chains in the Garment and Footwear Sector.
8325 Available at <https://doi.org/10.1787/9789264290587-en>.
- 8326 OECD. 2019. *Global Material Resources Outlook to 2060: Economic Drivers and Environmental Consequences*.
8327 Paris: Organisation for Economic Co-operation and Development. Available at [https://www.oecd-](https://www.oecd-ilibrary.org/environment/global-material-resources-outlook-to-2060_9789264307452-en)
8328 [ilibrary.org/environment/global-material-resources-outlook-to-2060_9789264307452-en](https://www.oecd-ilibrary.org/environment/global-material-resources-outlook-to-2060_9789264307452-en). Accessed
8329 2024 Jan 23.
- 8330 OECD. 2022. Best Available Techniques (BAT) for Preventing and Controlling Industrial Pollution. Activity 6: Cross
8331 Country analysis of BAT and BAT-associated emission and environmental performance levels in the
8332 Thermal Power Plants, Cement and Textile industries. *Ser Risk Manag*(71).
- 8333 van Oers L, de Koning A, Guinee JB, Huppes G. 2002. Abiotic Resource Depletion in LCA. Amsterdam, The
8334 Netherlands: Road and Hydraulic Engineering Institute, Ministry of Transport and Water.
- 8335 Ogugbue CJ, Sawidis T. 2011. Bioremediation and Detoxification of Synthetic Wastewater Containing
8336 Triarylmethane Dyes by *Aeromonas hydrophila* Isolated from Industrial Effluent. *Biotechnol Res Int*
8337 **2011**: 1–11. doi: 10.4061/2011/967925
- 8338 Osman AI, Hosny M, Eltaweil AS, Omar S, Elgarahy AM, Farghali M, Yap P-S, Wu Y-S, Nagandran S, Batumalaie
8339 K, et al. 2023. Microplastic sources, formation, toxicity and remediation: a review. *Environ Chem Lett*
8340 **21**(4): 2129–2169. doi: 10.1007/s10311-023-01593-3
- 8341 OVAM. 2021. Ecodesign criteria for consumer textiles. OVAM. Available at
8342 [https://circulareconomy.europa.eu/platform/sites/default/files/ecodesign_criteria_for_consumer_textil](https://circulareconomy.europa.eu/platform/sites/default/files/ecodesign_criteria_for_consumer_textiles.pdf)
8343 [es.pdf](https://circulareconomy.europa.eu/platform/sites/default/files/ecodesign_criteria_for_consumer_textiles.pdf).
- 8344 OVAM. 2022. Ecodesign criteria for consumer textiles. Available at [https://ovam-](https://ovam-english.vlaanderen.be/documents/177280/797580/2021+-+Ecodesign+criteria+for+consumer+textiles.pdf/ed47b6c4-a625-300d-64af-8ef88b44cab2?version=2.1&t=1662551112862&download=true)
8345 [english.vlaanderen.be/documents/177280/797580/2021+-](https://ovam-english.vlaanderen.be/documents/177280/797580/2021+-+Ecodesign+criteria+for+consumer+textiles.pdf/ed47b6c4-a625-300d-64af-8ef88b44cab2?version=2.1&t=1662551112862&download=true)
8346 [+Ecodesign+criteria+for+consumer+textiles.pdf/ed47b6c4-a625-300d-64af-](https://ovam-english.vlaanderen.be/documents/177280/797580/2021+-+Ecodesign+criteria+for+consumer+textiles.pdf/ed47b6c4-a625-300d-64af-8ef88b44cab2?version=2.1&t=1662551112862&download=true)
8347 [8ef88b44cab2?version=2.1&t=1662551112862&download=true](https://ovam-english.vlaanderen.be/documents/177280/797580/2021+-+Ecodesign+criteria+for+consumer+textiles.pdf/ed47b6c4-a625-300d-64af-8ef88b44cab2?version=2.1&t=1662551112862&download=true). Accessed 2023 Dec 2.
- 8348 Oxford dictionary. 2023 Sept 13. retail. Oxford dictionary. Available at
8349 <https://dictionary.cambridge.org/dictionary/english/retail>. Accessed 2023 Sept 19.
- 8350 Özkan İ, Gündoğdu S. 2021. Investigation on the microfiber release under controlled washings from the knitted
8351 fabrics produced by recycled and virgin polyester yarns. *J Text Inst* **112**(2): 264–272. doi:
8352 10.1080/00405000.2020.1741760
- 8353 Page T. 2014. Product attachment and replacement: implications for sustainable design. *Int J Sustain Des* **2**(3):
8354 265. doi: 10.1504/IJSDES.2014.065057
- 8355 Palacios-Mateo C, van der Meer Y, Seide G. 2021. Analysis of the polyester clothing value chain to identify key
8356 intervention points for sustainability. *Environ Sci Eur* **33**(1): 2. doi: 10.1186/s12302-020-00447-x
- 8357 Palaniappan S, Sadacharan CM, Rostama B. 2022. Polystyrene and Polyethylene Microplastics Decrease Cell
8358 Viability and Dysregulate Inflammatory and Oxidative Stress Markers of MDCK and L929 Cells In Vitro.
8359 *Expo Health* **14**(1): 75–85. doi: 10.1007/s12403-021-00419-3
- 8360 Pera R, Ferruli. 2023. Consumers' textile disposal practices and their perceived value in the circular economy: A
8361 platform focused ethnography approach. , in press. doi: <https://doi.org/10.1002/bse.3616>
- 8362 Periyasamy AP, Tehrani-Bagha A. 2022. A review on microplastic emission from textile materials and its
8363 reduction techniques. *Polym Degrad Stab* **199**: 109901. doi: 10.1016/j.polymdegradstab.2022.109901

- 8364 Picó Y, Barceló D. 2019. Analysis and Prevention of Microplastics Pollution in Water: Current Perspectives and
8365 Future Directions. *ACS Omega* **4**(4): 6709–6719. American Chemical Society. doi:
8366 10.1021/acsomega.9b00222
- 8367 Piippo R, Niinimäki K, Aakko M. 2022a. Fit for the Future: Garment Quality and Product Lifetimes in a CE Context.
8368 *Sustainability* **14**(2): 726. doi: 10.3390/su14020726
- 8369 Piippo R, Niinimäki K, Aakko M. 2022b. Fit for the Future: Garment Quality and Product Lifetimes in a CE Context.
8370 *Sustainability* **14**(2): 726. doi: 10.3390/su14020726
- 8371 Pinasseau A, Zerger B, Roth J, Canova M, Roudier S. 2018a. Best available techniques (BAT) reference document
8372 for waste treatment: Industrial Emissions Directive 2010/75/EU (integrated pollution prevention and
8373 control). LU: Publications Office. Available at <https://data.europa.eu/doi/10.2760/407967>. Accessed
8374 2024 Dec 4.
- 8375 Pinasseau A, Zerger B, Roth J, Canova M, Roudier S. 2018b. Best Available Techniques (BAT) Reference Document
8376 for Waste Treatment. *JRC Publ Repos*, in press.
- 8377 Posch M, Seppälä J, Hettelingh J-P, Johansson M, Margni M, Jolliet O. 2008. The role of atmospheric dispersion
8378 models and ecosystem sensitivity in the determination of characterisation factors for acidifying and
8379 eutrophying emissions in LCIA. *Int J Life Cycle Assess* **13**(6): 477–486. doi: 10.1007/s11367-008-
8380 0025-9
- 8381 Pradit S, Noppradit P, Jitkaew P, Sengloyluan K, Yucharoen M, Suwanno P, Tanrattanakul V, Sornplang K,
8382 Nitiratsuwan T. 2023. Microplastic Accumulation in Catfish and Its Effects on Fish Eggs from Songkhla
8383 Lagoon, Thailand. *J Mar Sci Eng* **11**(4): 723. Multidisciplinary Digital Publishing Institute. doi:
8384 10.3390/jmse11040723
- 8385 Pranta AD, Tareque Rahaman Md, Reazuddin Repon Md, Shikder AAR. 2024. Environmentally sustainable apparel
8386 merchandising of recycled cotton-polyester blended garments: Analysis of consumer preferences and
8387 purchasing behaviors. *J Open Innov Technol Mark Complex* **10**(3): 100357. doi:
8388 10.1016/j.joitmc.2024.100357
- 8389 Prasannamedha G, Senthilkumar P. 2021. Chemical compliance and regulations in textiles and fashion. In:
8390 *Chemical Management in Textiles and Fashion*. Elsevier. p. 135–154. doi: 10.1016/B978-0-12-
8391 820494-8.00007-1
- 8392 Qiao R, Sheng C, Lu Y, Zhang Y, Ren H, Lemos B. 2019. Microplastics induce intestinal inflammation, oxidative
8393 stress, and disorders of metabolome and microbiome in zebrafish. *Sci Total Environ* **662**: 246–253.
8394 doi: 10.1016/j.scitotenv.2019.01.245
- 8395 Quantis. 2018. Measuring Fashion, 2018. Environmental impact of the global apparel and footwear industries
8396 study. Full report and methodological considerations. Available at
8397 <https://quantis.com/report/measuring-fashion-report/>.
- 8398 Quantis. 2021. Measuring fashion: Environmental Impact of the Global Apparel and Footwear Industries Study.
8399 Quantis. Available at [https://quantis.com/wp-](https://quantis.com/wp-content/uploads/2018/03/measuringfashion_globalimpactstudy_full-report_quantis_cwf_2018a.pdf)
8400 [content/uploads/2018/03/measuringfashion_globalimpactstudy_full-report_quantis_cwf_2018a.pdf](https://quantis.com/wp-content/uploads/2018/03/measuringfashion_globalimpactstudy_full-report_quantis_cwf_2018a.pdf).
8401 Accessed 2023 Sept 12.
- 8402 Quantis. 2024. Draft Product Environmental Footprint Category Rules: Apparel and Footwear. Report No.: Version
8403 2.0 – Second draft PEFCR.
- 8404 Quantis. 2025a. PRODUCT ENVIRONMENTAL FOOTPRINT CATEGORY RULES (PEFCR) APPAREL AND FOOTWEAR.
8405 Switzerland: Quantis. Report No.: A&FW_PEFCE_v3.1.
- 8406 Quantis. 2025b. Product Environmental Footprint Category Rules: Apparel and Footwear. Report No.: Version
8407 v3.1.

- 8408 Quantis Q. 2022. Draft Product Environmental footprint category 2 rules . Apparel and footwear (PEFCR).
8409 Switzerland. Report No.: Version 1.3. Available at [https://green-business.ec.europa.eu/environmental-](https://green-business.ec.europa.eu/environmental-footprint-methods_en)
8410 [footprint-methods_en](https://green-business.ec.europa.eu/environmental-footprint-methods_en).
- 8411 Quashie-Idun MK Sam. 2022 Aug 8. Garment waste from Nike other leading brands fuel toxic kilns in Cambodia.
8412 Unearthed. Available at [https://unearthed.greenpeace.org/2022/08/08/garment-waste-nike-clarks-](https://unearthed.greenpeace.org/2022/08/08/garment-waste-nike-clarks-cambodia-bonded-workers-toxic/)
8413 [cambodia-bonded-workers-toxic/](https://unearthed.greenpeace.org/2022/08/08/garment-waste-nike-clarks-cambodia-bonded-workers-toxic/). Accessed 2024 Jan 23.
- 8414 R. M, D. T, L. M, M. S, L. N. 2025. *Feasibility Study of Measures to Incentivise Improved Textile Waste Management*
8415 *with a Long Term Perspective*. LU: Publications Office. Available at
8416 <https://data.europa.eu/doi/10.2760/8507408>. Accessed 2025 Oct 7.
- 8417 Rahman Khan MdM, Chen Y, Belsham T, Laguë C, Landry H, Peng Q, Zhong W. 2011. Fineness and tensile
8418 properties of hemp (*Cannabis sativa* L.) fibres. *Biosyst Eng* **108**(1): 9–17. doi:
8419 10.1016/j.biosystemseng.2010.10.004
- 8420 Rahman Md. Mizanur, Mashud M, Rahman Md. Mostafizur, editors. 2023. *Advanced Technology in Textiles: Fibre*
8421 *to Apparel*. Singapore: Springer Nature Singapore. (Textile Science and Clothing Technology). doi:
8422 10.1007/978-981-99-2142-3
- 8423 Raj A, Chowdhury A, Ali SW. 2022. Green chemistry: its opportunities and challenges in colouration and chemical
8424 finishing of textiles. *Sustain Chem Pharm* **27**: 100689. doi: 10.1016/j.scp.2022.100689
- 8425 Ramey HH. 2018. THE MEANING AND ASSESSMENT OF COTTON FIBRE FINENESS. *South Reg Res Cent US Dep*
8426 *Agric*, in press.
- 8427 Ranasinghe L, Jayasooriya VM. 2021. Ecolabelling in textile industry: A review. *Resour Environ Sustain* **6**:
8428 100037. doi: 10.1016/j.resenv.2021.100037
- 8429 Ray S. 2012. Warp knitting machines and knitting elements. p. 144–150. doi: 10.1533/9780857095558.144
- 8430 Refashion. 2023a. Characterisation study of the incoming and outgoing streams from sorting facilities. France:
8431 Refashion. Available at [https://refashion.fr/pro/sites/default/files/rapport-](https://refashion.fr/pro/sites/default/files/rapport-etude/Overview_Characterisation_study_Refashion_2023_EN.pdf)
8432 [etude/Overview_Characterisation_study_Refashion_2023_EN.pdf](https://refashion.fr/pro/sites/default/files/rapport-etude/Overview_Characterisation_study_Refashion_2023_EN.pdf). Accessed 2024 Dec 3.
- 8433 Refashion. 2023b. Characterisation study of the incoming and outgoing streams from sorting facilities. France.
8434 Available at [https://pro.refashion.fr/sites/default/files/rapport-](https://pro.refashion.fr/sites/default/files/rapport-etude/Overview_Characterisation_study_Refashion_2023_EN.pdf)
8435 [etude/Overview_Characterisation_study_Refashion_2023_EN.pdf](https://pro.refashion.fr/sites/default/files/rapport-etude/Overview_Characterisation_study_Refashion_2023_EN.pdf).
- 8436 Ribeiro PR, Batista P, Mendes-Palma F, Pintado M, Oliveira-Silva P. 2023. Consumers' Engagement and
8437 Perspectives on Sustainable Textile Consumption. *Sustainability* **15**(22): 15812. doi:
8438 10.3390/su152215812
- 8439 Rillig MC, Ziersch L, Hempel S. 2017. Microplastic transport in soil by earthworms. *Sci Rep* **7**(1): 1362. Nature
8440 Publishing Group. doi: 10.1038/s41598-017-01594-7
- 8441 Roberts H, Milios L, Mont O, Dalhammar C. 2023. Product destruction: Exploring unsustainable production-
8442 consumption systems and appropriate policy responses. *Sustain Prod Consum* **35**: 300–312. doi:
8443 10.1016/j.spc.2022.11.009
- 8444 Roichman R, Sprecher B, Blass V, Meshulam T, Makov T. 2024. The convenience economy: Product flows and
8445 GHG emissions of returned apparel in the EU. *Resour Conserv Recycl* **210**: 107811. doi:
8446 10.1016/j.resconrec.2024.107811
- 8447 Romeo T, Pietro B, Pedà C, Consoli P, Andaloro F, Fossi MC. 2015. First evidence of presence of plastic debris in
8448 stomach of large pelagic fish in the Mediterranean Sea. *Mar Pollut Bull* **95**(1): 358–361. doi:
8449 10.1016/j.marpolbul.2015.04.048

- 8450 Roos S, Jönsson C, Posner S, Arvidsson R, Svanström M. 2019. An inventory framework for inclusion of textile
8451 chemicals in life cycle assessment. *Int J Life Cycle Assess* **24**(5): 838–847. doi: 10.1007/s11367-018-
8452 1537-6
- 8453 Roos S, Sandin G, Zamani B, Peters G. 2015. Environmental assessment of Swedish fashion consumption. Mistra
8454 Future Fashion. Available at [http://mistrafuturefashion.com/life-cycle-assessment-gives-new-](http://mistrafuturefashion.com/life-cycle-assessment-gives-new-understanding-of-fashions-environmental-impact/)
8455 [understanding-of-fashions-environmental-impact/](http://mistrafuturefashion.com/life-cycle-assessment-gives-new-understanding-of-fashions-environmental-impact/).
- 8456 Roth J, Zerger B, De GD, Gómez BJ, Roudier S. 2023 Jan 16. Best available techniques (BAT) reference document
8457 for the Textiles Industry. *JRC Publ Repos*, in press. doi: 10.2760/355887
- 8458 Roursgaard M, Hezareh Rothmann M, Schulte J, Karadimou I, Marinelli E, Møller P. 2022. Genotoxicity of Particles
8459 From Grinded Plastic Items in Caco-2 and HepG2 Cells. *Front Public Health* **10**. Available at
8460 <https://www.frontiersin.org/articles/10.3389/fpubh.2022.906430>. Accessed 2024 Jan 23.
- 8461 Rowen JW, Gagliardi D. 1947. Properties of water-repellent fabrics. *J Res Natl Bur Stand* **38**(1): 103. doi:
8462 10.6028/jres.038.002
- 8463 Roy Choudhury AK. 2017. 7 - Repellent finishes. In: Roy Choudhury AK, editor. *Principles of Textile Finishing*.
8464 Woodhead Publishing. p. 149–194. doi: 10.1016/B978-0-08-100646-7.00007-2
- 8465 Rulikova M. 2020. “I would never wear those old clodhoppers!”: Age differences and used clothing consumption
8466 in the Czech Republic. *J Consum Cult* **20**(2): 175–193. doi: 10.1177/1469540519891274
- 8467 Sala S, A C, R P. 2017. Development of a weighting approach for the Environmental Footprint. (KJ-NA-28562-
8468 EN-C (print),KJ-NA-28562-EN-N (online)). Luxembourg (Luxembourg): Publications Office of the
8469 European Union. doi: 10.2760/945290%20(online),10.2760/446145%20(print)
- 8470 Saleeshya PG, Raghuram P, Vamsi N. 2012. Lean manufacturing practices in textile industries - a case study.
8471 *Int J Collab Enterp* **3**(1): 18. doi: 10.1504/IJCEN.2012.052367
- 8472 Sandin G, Peters GM. 2018. Environmental impact of textile reuse and recycling – A review. *J Clean Prod* **184**:
8473 353–365. doi: 10.1016/j.jclepro.2018.02.266
- 8474 Sandin G, Roos S, Spak B, Zamani B, Peters G. 2019. Environmental assessment of Swedish clothing consumption
8475 - six garments, sustainable futures. Mistra Future Fashion. Available at
8476 <http://mistrafuturefashion.com/impact-of-swedish-clothing-consumption/>.
- 8477 Sandin G, Roos S, Spak, Björn, Zamani B, Peters GM. 2019. Environmental assessment of Swedish clothing
8478 consumption - six garments, sustainable futures. Mistra Future Fashion Consortium. Available at
8479 [http://mistrafuturefashion.com/wp-content/uploads/2019/08/G.Sandin-Environmental-assessment-of-](http://mistrafuturefashion.com/wp-content/uploads/2019/08/G.Sandin-Environmental-assessment-of-Swedish-clothing-consumption.MistraFutureFashionReport-2019.05.pdf)
8480 [Swedish-clothing-consumption.MistraFutureFashionReport-2019.05.pdf](http://mistrafuturefashion.com/wp-content/uploads/2019/08/G.Sandin-Environmental-assessment-of-Swedish-clothing-consumption.MistraFutureFashionReport-2019.05.pdf).
- 8481 Saouter E, Biganzoli F, Ceriani L, Versteeg D, Crenna E, Zampori L, Sala S, Pant R. 2018. Environmental Footprint:
8482 Update of Life Cycle Impact Assessment Methods – Ecotoxicity freshwater, human toxicity cancer, and
8483 non-cancer. Publications Office of the European Union, Luxembourg.
- 8484 Saskia Manshoven, Christian Malarciuc, Anse Smeets, Anna Tenhunen, Lars Fogh Mortensen. 2022. Microplastic
8485 pollution from textile consumption in Europe. Available at [https://www.eionet.europa.eu/etcs/etc-](https://www.eionet.europa.eu/etcs/etc-ce/products/etc-ce-products/etc-ce-report-1-2022-microplastic-pollution-from-textile-consumption-in-europe)
8486 [ce/products/etc-ce-products/etc-ce-report-1-2022-microplastic-pollution-from-textile-consumption-](https://www.eionet.europa.eu/etcs/etc-ce/products/etc-ce-products/etc-ce-report-1-2022-microplastic-pollution-from-textile-consumption-in-europe)
8487 [in-europe](https://www.eionet.europa.eu/etcs/etc-ce/products/etc-ce-products/etc-ce-report-1-2022-microplastic-pollution-from-textile-consumption-in-europe). Accessed 2024 Jan 23.
- 8488 Schäfer R, Preuschoff PM, Borchers J. 2023. Investigating Visual Countermeasures Against Dark Patterns in
8489 User Interfaces. *Mensch und Computer* 2023: 161–172. ACM. doi: 10.1145/3603555.3603563
- 8490 Schmitz A, Stamminger R. 2014. Usage behaviour and related energy consumption of European consumers for
8491 washing and drying. *Energy Effic* **7**(6): 937–954. doi: 10.1007/s12053-014-9268-4

- 8492 Sekhri S. 2016. *Textbook of Fabric Science Fundamentals to Finishing*. Prentice Hall India Pvt., Limited.
- 8493 Seppälä J, Posch M, Johansson M, Hettelingh J-P. 2006. Country-dependent Characterisation Factors for
8494 Acidification and Terrestrial Eutrophication Based on Accumulated Exceedance as an Impact Category
8495 Indicator (14 pp). *Int J Life Cycle Assess* **11**(6): 403–416. doi: 10.1065/lca2005.06.215
- 8496 Sharma S, Sharma V, Chatterjee S. 2023. Contribution of plastic and microplastic to global climate change and
8497 their conjoining impacts on the environment - A review. *Sci Total Environ* **875**: 162627. doi:
8498 10.1016/j.scitotenv.2023.162627
- 8499 Shen M, Song B, Zhou C, Almatrafi E, Hu T, Zeng G, Zhang Y. 2022. Recent advances in impacts of microplastics
8500 on nitrogen cycling in the environment: A review. *Sci Total Environ* **815**: 152740. doi:
8501 10.1016/j.scitotenv.2021.152740
- 8502 Shi J, Wu D, Su Y, Xie B. 2021. Selective enrichment of antibiotic resistance genes and pathogens on polystyrene
8503 microplastics in landfill leachate. *Sci Total Environ* **765**: 142775. doi: 10.1016/j.scitotenv.2020.142775
- 8504 Shirvanimoghaddam K, Motamed B, Ramakrishna S, Naebe M. 2020. Death by waste: Fashion and textile circular
8505 economy case. *Sci Total Environ*, in press.
- 8506 Shishoo RL. 1994. Environmental Issues Facing the Technical Textile Industry in Europe. *J Coat Fabr* **24**(2): 117–
8507 128. doi: 10.1177/152808379402400204
- 8508 Sigaard AS. 2023. Want Not, Waste Not: Preliminary findings. Available at <https://oda.oslomet.no/oda-xmlui/handle/11250/3064950>.
- 8509
- 8510 Singh HB, Bharati KA. 2014 Jan 1. *Handbook of Natural Dyes and Pigments*. : 1–299.
- 8511 Sinha JK, Vashisth K, Ghosh S. 2022. The importance of sleep studies in improving the health indices of a nation.
8512 *Sleep Med X* **4**: 100049. doi: 10.1016/j.sleepx.2022.100049
- 8513 Sinha SK, Chattopadhyay R. 2007. A STUDY ON SPINNING LIMITS AND YARN PROPERTIES WITH PROGRESSIVE
8514 CHANGE IN YARN COUNT IN FRICTION SPINNING. *AUTEX Res J* **7**(1): 1–8. doi: 10.1515/aut-2007-
8515 070101
- 8516 SMED. 2016. Plockanalyser av textilier i hushållens restavfall. Available at
8517 [https://www.naturvardsverket.se/globalassets/amnen/kretslopp-avlopp-](https://www.naturvardsverket.se/globalassets/amnen/kretslopp-avlopp-avfall/avfallsslag/textil/plockanalyser-av-textilier-i-hushallens-restavfall-smed-rapport-2016-06-17.pdf)
8518 [avfall/avfallsslag/textil/plockanalyser-av-textilier-i-hushallens-restavfall-smed-rapport-2016-06-](https://www.naturvardsverket.se/globalassets/amnen/kretslopp-avlopp-avfall/avfallsslag/textil/plockanalyser-av-textilier-i-hushallens-restavfall-smed-rapport-2016-06-17.pdf)
8519 [17.pdf](https://www.naturvardsverket.se/globalassets/amnen/kretslopp-avlopp-avfall/avfallsslag/textil/plockanalyser-av-textilier-i-hushallens-restavfall-smed-rapport-2016-06-17.pdf).
- 8520 Sohail Y, Sun D. 2019. Life Cycle Assessment of Technical Textile Waste: Pilot Evaluation of Environmental
8521 Impacts. 2019. 2nd International Forum on Textiles for Graduate Students 2018; Tianjin Polytechnic
8522 University, Tianjin, China. Tianjin Polytechnic University. Available at <https://hal.science/hal-02133644>.
8523 Accessed 2023 Feb 17.
- 8524 Sol D, Laca Amanda, Laca Adriana, Díaz M. 2021. Microplastics in Wastewater and Drinking Water Treatment
8525 Plants: Occurrence and Removal of Microfibres. *Appl Sci* **11**(21): 10109. Multidisciplinary Digital
8526 Publishing Institute. doi: 10.3390/app112110109
- 8527 Solis M, Huygens D, Tonini D, Fruergaard Astrup T. 2024. Management of textile waste in Europe: An
8528 environmental and a socio-economic assessment of current and future scenarios. *Resour Conserv*
8529 *Recycl* **207**: 107693. doi: 10.1016/j.resconrec.2024.107693
- 8530 Sørensen L, Groven AS, Hovsbakken IA, Del Puerto O, Krause DF, Sarno A, Booth AM. 2021. UV degradation of
8531 natural and synthetic microfibers causes fragmentation and release of polymer degradation products
8532 and chemical additives. *Sci Total Environ* **755**: 143170. doi: 10.1016/j.scitotenv.2020.143170

- 8533 Spaepen V, J. Van Camp M, Lenaerts. 2021. One Size Does Not Fit All: Consumer Attitudes towards Circular
8534 Business Models in Fashion. , in press. Available at <https://orcid.org/0000-0002-8603-7028>.
- 8535 Stephen MacDonald. 2006 Jan 2. The World Bids Farewell to the Multifiber Arrangement - USDA ERS. Economic
8536 Research Service U.S. Department of Agriculture. Available at [https://www.ers.usda.gov/amber-](https://www.ers.usda.gov/amber-waves/2006/february/the-world-bids-farewell-to-the-multifiber-arrangement/)
8537 [waves/2006/february/the-world-bids-farewell-to-the-multifiber-arrangement/](https://www.ers.usda.gov/amber-waves/2006/february/the-world-bids-farewell-to-the-multifiber-arrangement/). Accessed 2023 Aug
8538 17.
- 8539 Struijs J, Beusen A, van Jaarsveld H, Huijbregts MAJ. 2009. Aquatic Eutrophication. Section 6. In: *ReCiPe 2008 -*
8540 *A Life Cycle Impact Assessment Method Which Comprises Harmonised Category Indicators at the*
8541 *Midpoint and the Endpoint Level*first.
- 8542 Svensson-Hoglund S, Luth Richter J, Maitre-Ekern E, Russell J, Pihlajarinne T, Dalhammar C. 2021. Barriers,
8543 enablers and market governance: A review of the policy landscape for repair of consumer electronics
8544 in the EU and the U.S. *J Clean Prod*, in press.
- 8545 Sveriges Konsumenter. 2020. Ny undersökning: konsumenter oroade för kemikalier. Available at
8546 [https://www.sverigeskonsumenter.se/nyheter-press/nyheter-och-pressmeddelanden/ny-undersokning-](https://www.sverigeskonsumenter.se/nyheter-press/nyheter-och-pressmeddelanden/ny-undersokning-konsumenter-oroad-for-kemikalier/)
8547 [konsumenter-oroad-for-kemikalier/](https://www.sverigeskonsumenter.se/nyheter-press/nyheter-och-pressmeddelanden/ny-undersokning-konsumenter-oroad-for-kemikalier/). Accessed 2024 May 21.
- 8548 Swedish Environmental Protection Agency. 2020. HÅLLBARA TEXTILIER: SLUTMÄTNING Rapport konsument
8549 undersökning 2020. Available at
8550 [https://www.naturvardsverket.se/4a4394/globalassets/amnen/textil/konsumentundersokning-](https://www.naturvardsverket.se/4a4394/globalassets/amnen/textil/konsumentundersokning-2020.pdf)
8551 [2020.pdf](https://www.naturvardsverket.se/4a4394/globalassets/amnen/textil/konsumentundersokning-2020.pdf). Accessed 2025 May 16.
- 8552 Szczepanski M. 2019. A decade on from the crisis: Main responses and remaining challenges. EPRS | European
8553 Parliamentary Research Service. Report No.: PE 642.253. Available at
8554 [https://www.europarl.europa.eu/thinktank/en/document/EPRS_BRI\(2019\)642253](https://www.europarl.europa.eu/thinktank/en/document/EPRS_BRI(2019)642253).
- 8555 Tagg AS, Brandes E, Fischer F, Fischer D, Brandt J, Labrenz M. 2022. Agricultural application of microplastic-rich
8556 sewage sludge leads to further uncontrolled contamination. *Sci Total Environ* **806**: 150611. doi:
8557 10.1016/j.scitotenv.2021.150611
- 8558 Tahvildar A, Ezazshahabi N, Mousazadegan F. 2019. Appearance and comfort properties considering yarn-
8559 spinning system and weave structure in worsted woven fabrics. *J Eng Fibers Fabr* **14**:
8560 155892501984597. doi: 10.1177/1558925019845978
- 8561 Talvitie J, Mikola A, Setälä O, Heinonen M, Koistinen A. 2017. How well is microlitter purified from wastewater?
8562 - A detailed study on the stepwise removal of microlitter in a tertiary level wastewater treatment plant.
8563 *Water Res* **109**: 164–172. doi: 10.1016/j.watres.2016.11.046
- 8564 TAUW. 2023. Performance requirements for textiles Input on EU sustainable design criteria for textiles. Utrecht.
8565 Report No.: R001- 1288851BUM-V03- los- NL. Available at
8566 [https://www.rijksoverheid.nl/documenten/rapporten/2023/09/26/bijlage-eindrapport-onderzoek-](https://www.rijksoverheid.nl/documenten/rapporten/2023/09/26/bijlage-eindrapport-onderzoek-designeisen-textiel)
8567 [designeisen-textiel](https://www.rijksoverheid.nl/documenten/rapporten/2023/09/26/bijlage-eindrapport-onderzoek-designeisen-textiel). Accessed 2024 Feb 20.
- 8568 Tedesco MC, Fisher RM, Stuetz RM. 2024. Emission of fibres from textiles: A critical and systematic review of
8569 mechanisms of release during machine washing. *Sci Total Environ* **955**: 177090. doi:
8570 10.1016/j.scitotenv.2024.177090
- 8571 Terzioğlu N. 2021. Repair motivation and barriers model: Investigating user perspectives related to product
8572 repair towards a circular economy. *J Clean Prod* **289**: 125644. doi: 10.1016/j.jclepro.2020.125644
- 8573 Textile Exchange. 2020. Preferred fiber & Materials Market Report 2020. Textile Exchange. Available at
8574 [https://textileexchange.org/app/uploads/2021/04/Textile-Exchange_PREFERRED-Fiber-Material-Market-](https://textileexchange.org/app/uploads/2021/04/Textile-Exchange_PREFERRED-Fiber-Material-Market-Report_2020.pdf)
8575 [Report_2020.pdf](https://textileexchange.org/app/uploads/2021/04/Textile-Exchange_PREFERRED-Fiber-Material-Market-Report_2020.pdf).
- 8576 Textile Exchange. 2021. Preferred Fiber and Materials Market Report 2021. Textile Exchange.

- 8577 Textile Exchange. 2022a. Preferred Fiber & Materials Market Report 2022. Textile Exchange.
- 8578 Textile Exchange. 2022b. Material Change Insights 2021. The state of fiber and materials sourcing. Textile
8579 Exchange. Available at [https://textileexchange.org/app/uploads/2022/06/Textile-Exchange_Material-
8580 Change-Insights_Report_2021.pdf](https://textileexchange.org/app/uploads/2022/06/Textile-Exchange_Material-Change-Insights_Report_2021.pdf). Accessed 2023 July 6.
- 8581 Textile Exchange. 2023. Materials Market Report 2023. Textile Exchange. Available at
8582 <https://textileexchange.org/knowledge-center/documents/materials-market-report-2023/>. Accessed
8583 2023 Dec 19.
- 8584 Textile Exchange. 2024a. Materials Market Report 2024. Textile Exchange. Available at
8585 <https://2d73cea0.delivery.rocketcdn.me/app/uploads/2024/09/Materials-Market-Report-2024.pdf>.
8586 Accessed 2025 Aug 8.
- 8587 Textile Exchange. 2024b. Materials Market Report.
- 8588 Textile exchange - Fiber Conversion Methodology. 2019.
- 8589 Textile exchange - Preferred Fiber & Materials Report 2019. 2019.
- 8590 Thredup. 2023. thredUP's annual Resale Report. Available at [https://cf-assets-
8591 tup.thredup.com/resale_report/2023/thredUP_2023_Resale_Report_FINAL.pdf](https://cf-assets-tup.thredup.com/resale_report/2023/thredUP_2023_Resale_Report_FINAL.pdf).
- 8592 Tiffin L, Hazlehurst A, Sumner M, Taylor M. 2022. Reliable quantification of microplastic release from the
8593 domestic laundry of textile fabrics. *J Text Inst* **113**(4): 558–566. Taylor & Francis. doi:
8594 10.1080/00405000.2021.1892305
- 8595 Trzepacz S, Bekkevold Lingås D, Asscherickx L, Peeters K, van Duijn H, Akerboom M. 2023. LCA-based
8596 assessment of the management of European used textiles. NORION CONSULT and EuRIC TEXTILES.
8597 Available at [https://circulareconomy.europa.eu/platform/sites/default/files/2023-02/LCA-
8598 based%20assessment%20of%20the%20management%20of%20European%20used%20textiles_co
8599 rrected.pdf](https://circulareconomy.europa.eu/platform/sites/default/files/2023-02/LCA-based%20assessment%20of%20the%20management%20of%20European%20used%20textiles_corrected.pdf).
- 8600 Tuovila H. 2024. Consumer views on circular textiles. Available at
8601 [https://telaketju.turkuamk.fi/uploads/01bbc5fb-telavalue_consumer-views-on-circular-textiles_2024-
8602 1.pdf](https://telaketju.turkuamk.fi/uploads/01bbc5fb-telavalue_consumer-views-on-circular-textiles_2024-1.pdf).
- 8603 Tursi A, Baratta M, Easton T, Chatzisyneon E, Chidichimo F, De Biase M, De Filpo G. 2022. Microplastics in
8604 aquatic systems, a comprehensive review: origination, accumulation, impact, and removal technologies.
8605 *RSC Adv* **12**(44): 28318–28340. doi: 10.1039/d2ra04713f
- 8606 Turunen LLM, Gossen M. 2024. From Preloved to Reloved: How Second-Hand Clothing Companies Facilitate the
8607 Transaction of Used Garments. *J Sustain Res* **6**(1). doi: 10.20900/jsr20240002
- 8608 Tyagi GK. 2010. 5 - Yarn structure and properties from different spinning techniques. In: Lawrence CA, editor.
8609 *Advances in Yarn Spinning Technology*. Woodhead Publishing. p. 119–154. doi:
8610 10.1533/9780857090218.1.119
- 8611 UNECE. 2022. Enhancing traceability and transparency of sustainable value chains in the garment and footwear
8612 sector. Recommendation n°46. Geneva: United Nations Centre for Trade facilitation and Electronic
8613 Business. Report No.: ECE/TRADE/463.
- 8614 UNECE, ECLAC. 2024. Reversing direction in the used clothing crisis: Global, European and Chilean perspectives.
8615 Geneva. Report No.: ECE/TRADE/484. Available at [https://unece.org/trade/publications/reversing-
8616 direction-used-clothing-crisis-global-european-and-chilean](https://unece.org/trade/publications/reversing-direction-used-clothing-crisis-global-european-and-chilean). Accessed 2024 Dec 7.
- 8617 UNECE, UNTRADE. 2020. Accelerating action for a sustainable and circular garment and footwear industry:
8618 which role for transparency and traceability of value chains? Geneva: United Nations Economic

- 8619 Commission for Europe. Report No.: ECE/TRADE/449. Available at
8620 https://unece.org/DAM/trade/Publications/ECE_TRADE_449-AcceleratingTransparencyTraceabilityTextile.pdf.
8621 Accessed 2023 Dec 7.
- 8622 UNEP. 2016. *Global Guidance for Life Cycle Impact Assessment Indicators*. (Vol. 1). Available at
8623 [https://www.lifecycleinitiative.org/activities/life-cycle-assessment-data-and-methods/global-](https://www.lifecycleinitiative.org/activities/life-cycle-assessment-data-and-methods/global-guidance-for-life-cycle-impact-assessment-indicators-and-methods-glam/lcia-cf/)
8624 [guidance-for-life-cycle-impact-assessment-indicators-and-methods-glam/lcia-cf/](https://www.lifecycleinitiative.org/activities/life-cycle-assessment-data-and-methods/glam/lcia-cf/).
- 8625 UNEP. 2020. Sustainability and Circularity in the Textile Value Chain - Global Stocktaking. Nairobi, Kenya.: United
8626 Nation Environment Programme. Available at
8627 [https://www.unep.org/resources/publication/sustainability-and-circularity-textile-value-chain-global-](https://www.unep.org/resources/publication/sustainability-and-circularity-textile-value-chain-global-stocktaking)
8628 [stocktaking](https://www.unep.org/resources/publication/sustainability-and-circularity-textile-value-chain-global-stocktaking). Accessed 2023 June 12.
- 8629 UNEP. 2023a. Sustainability and Circularity in the Textile Value Chain - A Global Roadmap. Paris: United Nations
8630 Environment Programme (UNEP). Available at
8631 [https://www.unep.org/resources/publication/sustainability-and-circularity-textile-value-chain-global-](https://www.unep.org/resources/publication/sustainability-and-circularity-textile-value-chain-global-roadmap)
8632 [roadmap](https://www.unep.org/resources/publication/sustainability-and-circularity-textile-value-chain-global-roadmap).
- 8633 UNEP. 2023b. Environmental labels in Ukraine. Information for Producers in the EU's Eastern Neighbourhood.
8634 UNEP, OECD, UNECE, UN, UNIDO, The World Bank. Available at
8635 https://www.eu4environment.org/app/uploads/2023/06/Ecolabelling-PPT_Ukraine_EN.pdf. Accessed
8636 2024 Jan 22.
- 8637 Ungerth L, Carlsson A. 2011. Vad händer sen med våra kläder Enkätundersökning. Available at
8638 [https://docplayer.se/344015-Konsumentforeningen-stockholm-vad-hander-sen-med-vara-klader-](https://docplayer.se/344015-Konsumentforeningen-stockholm-vad-hander-sen-med-vara-klader-enkatundersokning-april-2011.html)
8639 [enkätundersökning-april-2011.html](https://docplayer.se/344015-Konsumentforeningen-stockholm-vad-hander-sen-med-vara-klader-enkatundersokning-april-2011.html).
- 8640 UniformMarket. 2025. Global Apparel Industry Statistics.
- 8641 Van den Bergen J, Parker G. 2023. Textile Processing Guide: Pretreatment, colouration & finishing. Available at
8642 <https://reports.fashionforgood.com/wp-content/uploads/2022/01/Textile-Processing-Guide-3.pdf>.
- 8643 Van Rensburg ML, Nkomo SL, Mkhize NM. 2020. Life cycle and End-of-Life management options in the footwear
8644 industry: A review. *Waste Manag Res* **38**(6): 599–613. doi: 10.1177/0734242X20908938
- 8645 Van Zelm R, Huijbregts MAJ, Den Hollander HA, Van Jaarsveld HA, Sauter FJ, Struijs J, Van Wijnen HJ, Van De
8646 Meent D. 2008. European characterization factors for human health damage of PM10 and ozone in
8647 life cycle impact assessment. *Atmos Environ* **42**(3): 441–453. doi: 10.1016/j.atmosenv.2007.09.072
- 8648 Vanacker H, Lemieux A-A, Bonnier S. 2022. Different dimensions of durability in the luxury fashion industry: An
8649 analysis framework to conduct a literature review. *J Clean Prod* **377**: 134179. doi:
8650 10.1016/j.jclepro.2022.134179
- 8651 Vargas F, Yuleimy Ramírez YR. 2014. Potential of Municipal Solid Waste Calorific Energy as Coal Replacement.
8652 *Tek Rev Científica* **14**(2): 23. doi: 10.25044/25392190.467
- 8653 Vet JMD, Nigohosyan D, Ferrer JN, Gross A-K, Kuehl S, Flickenschild M. 2021 Mar. Impacts of the COVID-19
8654 pandemic on EU industries. , in press.
- 8655 Vinted. 2021. Climate Change Impact Report - Understanding the avoided emissions of second-hand shopping
8656 on Vinted. Available at [https://press-center-](https://press-center-static.vinted.com/Vaayu_x_Vinted_Full_Climate_Impact_Report_2021_045f9e5c4b.pdf)
8657 [static.vinted.com/Vaayu_x_Vinted_Full_Climate_Impact_Report_2021_045f9e5c4b.pdf](https://press-center-static.vinted.com/Vaayu_x_Vinted_Full_Climate_Impact_Report_2021_045f9e5c4b.pdf).
- 8658 Vladimirova D, Boniolo L, Botta V. 2024. ECOS Working paper - Exploring sufficiency approaches to textiles
8659 policy.
- 8660 Vladimirova K, Henninger CE, Joyner-Martinez C, Iran S, Diddi S, Durrani M, Iyer K, Jestratišević I, McCormick H,
8661 Niinimäki K, et al. 2022. Fashion consumption during COVID-19: Comparative analysis of changing

- 8662 acquisition practices across nine countries and implications for sustainability. *Clean Responsible*
8663 *Consum* **5**: 100056. doi: 10.1016/j.clrc.2022.100056
- 8664 VZBV. 2022. Verbraucherbereitschaft zu klimaschonendem Konsum und Einstellungen zu wahren Preisen.
8665 Projekt „Was kostet es wirklich? Internalisierung externer Kosten als elementarer Baustein beim
8666 Klimaschutz. Available at [https://www.vzbv.de/meldungen/umdenken-am-black-friday-bewusster-](https://www.vzbv.de/meldungen/umdenken-am-black-friday-bewusster-konsum-statt-schnaepchenjagd)
8667 [konsum-statt-schnaepchenjagd](https://www.vzbv.de/meldungen/umdenken-am-black-friday-bewusster-konsum-statt-schnaepchenjagd). Accessed 2024 May 21.
- 8668 Wakes S, Dunn L, Penty D, Kitson K, Jowett T. 2020a. Is Price an Indicator of Garment Durability and Longevity?
8669 *Sustainability* **12**(21): 8906. doi: 10.3390/su12218906
- 8670 Wakes S, Dunn L, Penty D, Kitson K, Jowett T. 2020b. Is Price an Indicator of Garment Durability and Longevity?
8671 *Sustainability* **12**(21): 8906. doi: 10.3390/su12218906
- 8672 Wang J, Liu X, Li Y, Powell T, Wang X, Wang G, Zhang P. 2019. Microplastics as contaminants in the soil
8673 environment: A mini-review. *Sci Total Environ* **691**: 848–857. doi: 10.1016/j.scitotenv.2019.07.209
- 8674 Wang X, Disney SM. 2016. The bullwhip effect: Progress, trends and directions. *Eur J Oper Res* **250**(3): 691–
8675 701. doi: 10.1016/j.ejor.2015.07.022
- 8676 Watson D, Trzepacz S, Gravgård Pedersen O. 2018. Mapping of textile flows in Denmark. The Danish
8677 Environmental Protection Agency. Available at [https://www2.mst.dk/Udgiv/publications/2018/08/978-](https://www2.mst.dk/Udgiv/publications/2018/08/978-87-93710-48-1.pdf)
8678 [87-93710-48-1.pdf](https://www2.mst.dk/Udgiv/publications/2018/08/978-87-93710-48-1.pdf).
- 8679 Weber CJ, Santowski A, Chiffard P. 2022. Investigating the dispersal of macro- and microplastics on agricultural
8680 fields 30 years after sewage sludge application. *Sci Rep* **12**(1): 6401. Nature Publishing Group. doi:
8681 10.1038/s41598-022-10294-w
- 8682 Wei W, Wargocki P, Zirngibl J, Bendžalová J, Mandin C. 2020. Review of parameters used to assess the quality
8683 of the indoor environment in Green Building certification schemes for offices and hotels. *Energy Build*
8684 **209**: 109683. doi: 10.1016/j.enbuild.2019.109683
- 8685 Weithmann N, Möller JN, Löder MGJ, Piehl S, Laforsch C, Freitag R. 2018. Organic fertilizer as a vehicle for the
8686 entry of microplastic into the environment. *Sci Adv* **4**(4): eaap8060. American Association for the
8687 Advancement of Science. doi: 10.1126/sciadv.aap8060
- 8688 Wernet G, Bauer C, Steubing B, Reinhard J, Moreno-Ruiz E, Weidema B. 2016. The ecoinvent database version 3
8689 (part I): overview and methodology. *Int J Life Cycle Assess* **21**(9): 1218–1230. doi: 10.1007/s11367-
8690 016-1087-8
- 8691 Williams L, Ackerman J. 2011. Please touch the merchandise. Harvard Business Review. Available at
8692 <https://hbr.org/2011/12/please-touch-the-merchandise>.
- 8693 World Bank. 2024. World development indicators.
- 8694 WRAP. 2017a. Sustainable clothing: A practical guide to enhancing clothing durability and quality. UK. Available
8695 at <https://wrap.org.uk/resources/guide/sustainable-clothing-guide>.
- 8696 WRAP. 2017b. Valuing Our Clothes: the cost of UK fashion. Available at WRAP. (2017). Valuing Our Clothes: the
8697 cost of UK fashion. [http://www.wrap.org.uk/sites/files/wrap/valuing-our-clothes-the-cost-of-uk-](http://www.wrap.org.uk/sites/files/wrap/valuing-our-clothes-the-cost-of-uk-fashion_WRAP.pdf)
8698 [fashion_WRAP.pdf](http://www.wrap.org.uk/sites/files/wrap/valuing-our-clothes-the-cost-of-uk-fashion_WRAP.pdf).
- 8699 WRAP. 2017c. An Economic and Financial Sustainability Assessment of F2F Recycling.
- 8700 WRAP. 2019. Consumer Research for ECAP 2016-2019. Available at [http://www.ecap.eu.com/wp-](http://www.ecap.eu.com/wp-content/uploads/2019/12/Consumer-Research-for-ECAP.pdf)
8701 [content/uploads/2019/12/Consumer-Research-for-ECAP.pdf](http://www.ecap.eu.com/wp-content/uploads/2019/12/Consumer-Research-for-ECAP.pdf).

- 8702 WRAP. 2022. Citizen Insights: Clothing Longevity and Circular Business Models Receptivity in the UK. Available
8703 at [https://wrap.org.uk/resources/report/citizen-insights-clothing-longevity-and-circular-business-](https://wrap.org.uk/resources/report/citizen-insights-clothing-longevity-and-circular-business-modelsreceptivity-uk)
8704 [modelsreceptivity- uk](https://wrap.org.uk/resources/report/citizen-insights-clothing-longevity-and-circular-business-modelsreceptivity-uk).
- 8705 WRAP. 2025. Displacement Rates Untangled: A Standardised Methodology. Available at
8706 [https://www.wrap.ngo/sites/default/files/2025-02/WRAP-Textiles-2030-Displacement-rate-report-](https://www.wrap.ngo/sites/default/files/2025-02/WRAP-Textiles-2030-Displacement-rate-report-REV1.pdf)
8707 [REV1.pdf](https://www.wrap.ngo/sites/default/files/2025-02/WRAP-Textiles-2030-Displacement-rate-report-REV1.pdf).
- 8708 WRAP Cymru. 2022. Composition of Textiles in Wales: Waste composition analysis of kerbside and HWRC
8709 collected textiles. Report No.: COL202- 016. Available at
8710 [https://wrapcymru.org.uk/sites/default/files/2022-12/COL202-](https://wrapcymru.org.uk/sites/default/files/2022-12/COL202-016%20Textiles%20Composition%20Summary%20Report%20FINAL_0.pdf)
8711 [016%20Textiles%20Composition%20Summary%20Report%20FINAL_0.pdf](https://wrapcymru.org.uk/sites/default/files/2022-12/COL202-016%20Textiles%20Composition%20Summary%20Report%20FINAL_0.pdf).
- 8712 WTO. 2020. World Trade Statistical Review 2020. Geneva: World Trade Organization. Available at
8713 https://www.wto.org/english/res_e/statis_e/wts2020_e/wts2020_e.pdf.
- 8714 Xiang Y, Jiang L, Zhou Y, Luo Z, Zhi D, Yang J, Lam SS. 2022. Microplastics and environmental pollutants: Key
8715 interaction and toxicology in aquatic and soil environments. *J Hazard Mater* **422**: 126843. doi:
8716 10.1016/j.jhazmat.2021.126843
- 8717 Yada Y, Feng J, Matsumoto T, Fukushima N, Kido F, Yamana H. 2022. Dark patterns in e-commerce: a dataset
8718 and its baseline evaluations. 2022 IEEE International Conference on Big Data (Big Data): 3015–3022.
8719 IEEE. doi: 10.1109/BigData55660.2022.10020800
- 8720 Yan R, Yurchisin J, Watchravesringkan K. 2008. Use of care labels: linking need for cognition with consumer
8721 confidence and perceived risk. *J Fash Mark Manag Int J* **12**(4): 532–544. doi:
8722 10.1108/13612020810906173
- 8723 Yang T, Gao M, Nowack B. 2023. Formation of microplastic fibers and fibrils during abrasion of a representative
8724 set of 12 polyester textiles. *Sci Total Environ* **862**: 160758. doi: 10.1016/j.scitotenv.2022.160758
- 8725 Yassen HA. 2017. Study of the Relationship between Sewing and Fabric Parameters and Seam Strength. *Int Des*
8726 *J* **7**(2): 125–129. doi: 10.12816/0046559
- 8727 Yates L, Evans D. 2016. Dirtying Linen: Re-evaluating the sustainability of domestic laundry: Dirtying Linen: Re-
8728 Evaluating the Sustainability of Domestic Laundry. *Environ Policy Gov* **26**(2): 101–115. doi:
8729 10.1002/eet.1704
- 8730 Yee MS-L, Hii L-W, Looi CK, Lim W-M, Wong S-F, Kok Y-Y, Tan B-K, Wong C-Y, Leong C-O. 2021. Impact of
8731 Microplastics and Nanoplastics on Human Health. *Nanomaterials* **11**(2): 496. Multidisciplinary Digital
8732 Publishing Institute. doi: 10.3390/nano11020496
- 8733 YouGov. 2019. Fast fashion: il continuo rinnovo del guardaroba. YouGov. Available at
8734 [https://it.yougov.com/society/articles/23537-fast-fashion-il-rinnovo-del-](https://it.yougov.com/society/articles/23537-fast-fashion-il-rinnovo-del-guardaroba?redirect_from=%2Fnews%2F2019%2F05%2F27%2Ffast-fashion-il-rinnovo-del-guardaroba%2F)
8735 [guardaroba?redirect_from=%2Fnews%2F2019%2F05%2F27%2Ffast-fashion-il-rinnovo-del-](https://it.yougov.com/society/articles/23537-fast-fashion-il-rinnovo-del-guardaroba?redirect_from=%2Fnews%2F2019%2F05%2F27%2Ffast-fashion-il-rinnovo-del-guardaroba%2F)
8736 [guardaroba%2F](https://it.yougov.com/society/articles/23537-fast-fashion-il-rinnovo-del-guardaroba?redirect_from=%2Fnews%2F2019%2F05%2F27%2Ffast-fashion-il-rinnovo-del-guardaroba%2F).
- 8737 YouGov. 2021. European Fashion Report. Available at [https://business.yougov.com/content/39089-european-](https://business.yougov.com/content/39089-european-fashion-report-2021-1)
8738 [fashion-report-2021-1](https://business.yougov.com/content/39089-european-fashion-report-2021-1).
- 8739 Yuille P. 2015. Exploring the Relationship between the Presumed Quality and Durability of Fast-Fashion
8740 Garments by the Generation-Y Consumer. 2015. PLATE Conference, Nottingham, UK, 17–19 June 2015;
8741 Nottingham, UK,. Available at https://www.plateconference.org/pdf/plate_2015_proceedings.pdf.
- 8742 Yun C, Patwary S, LeHew MLA, Kim J. 2017. Sustainable care of textile products and its environmental impact:
8743 Tumble-drying and ironing processes. *Fibers Polym* **18**(3): 590–596. doi: 10.1007/s12221-017-6957-
8744 6

- 8745 Zambrano MC, Pawlak JJ, Daystar J, Ankeny M, Venditti RA. 2021. Impact of dyes and finishes on the microfibers
8746 released on the laundering of cotton knitted fabrics. *Environ Pollut* **272**: 115998. doi:
8747 10.1016/j.envpol.2020.115998
- 8748 Zhang L, Hale J. 2022. Extending the Lifetime of Clothing through Repair and Repurpose: An Investigation of
8749 Barriers and Enablers in UK Citizens. *Sustainability* **14**(17): 10821. Multidisciplinary Digital Publishing
8750 Institute. doi: 10.3390/su141710821
- 8751 Zhang L, Leung MY, Boriskina S, Tao X. 2022. Advancing life cycle sustainability of textiles through technological
8752 innovations. *Nat Sustain* **6**(3): 243–253. doi: 10.1038/s41893-022-01004-5
- 8753 Zhao L, Kim K. 2021. Responding to the COVID-19 Pandemic: Practices and Strategies of the Global Clothing
8754 and Textile Value Chain. *Cloth Text Res J* **39**(2): 157–172. doi: 10.1177/0887302X21994207
- 8755 Zhao S, Zhang Z, Chen L, Cui Q, Cui Y, Song D, Fang L. 2022. Review on migration, transformation and ecological
8756 impacts of microplastics in soil. *Appl Soil Ecol* **176**: 104486. doi: 10.1016/j.apsoil.2022.104486
- 8757 Zhou H, Zhou L, Ma K. 2020. Microfiber from textile dyeing and printing wastewater of a typical industrial park
8758 in China: Occurrence, removal and release. *Sci Total Environ* **739**: 140329. doi:
8759 10.1016/j.scitotenv.2020.140329
- 8760 Zimmermann T, Hauschke F, Memelink R, Reitz A, Pelke N, John R, Eberle U, Ninnemann J. 2021. Die
8761 Ökologisierung des Onlinehandels. Neue Herausforderungen für die umweltpolitische Förderung eines
8762 nachhaltigen Konsums Report No.: 142/2021. Available at
8763 [https://www.umweltbundesamt.de/sites/default/files/medien/479/publikationen/texte_142-](https://www.umweltbundesamt.de/sites/default/files/medien/479/publikationen/texte_142-2021_die_oekologisierung_des_onlinehandels.pdf)
8764 [2021_die_oekologisierung_des_onlinehandels.pdf](https://www.umweltbundesamt.de/sites/default/files/medien/479/publikationen/texte_142-2021_die_oekologisierung_des_onlinehandels.pdf).
- 8765
- 8766

8767 **List of abbreviations**

8768	BAT	Best Available Technologies
8769	BNAT	Best Not Yet Available Technologies
8770	DPP	Digital Product Passport
8771	EC	European Commission
8772	ESPR	Ecodesign for Sustainable Product Regulation
8773	EU	European Union
8774	GPP	Green Public Procurement
8775	MEErP	Methodology for Ecodesign of Energy-related Products
8776	PEFCR A&F	Product Environmental Footprint Category Rules for Apparel and Footwear
8777	PS	Preparatory Study on textile products
8778	REACH	Registration, Evaluation, Authorisation and Restriction of Chemicals
8779	SVHC	Substance(s) of Very High Concern
8780	TLR	Textile Labelling Regulation (Regulation (EU) No 1007/2011)
8781	WFD	Waste Framework Directive

8782	List of figures	
8783	Figure 1. Main textile life-cycle stages.....	27
8784	Figure 3. Geographical breakdown of global apparel production and consumption – representation A.....	48
8785	Figure 4. Geographical breakdown of global apparel production and consumption – representation B.....	48
8786	Figure 5. Top ten exporting countries of second-hand apparel by mass (million tonnes).....	49
8787	Figure 6. Top ten importing countries of second-hand apparel by mass (million tonnes).....	50
8788	Figure 7. EU-27 exports of used textiles.....	51
8789	Figure 8. The EU's main suppliers of textiles and apparel in 2021	52
8790	Figure 9. The EU's main customers of textiles and apparel in 2021	52
8791	Figure 10. Market indicators for textile apparel in the EU-27.....	54
8792	Figure 11. Member States producing textile apparel in 2019	55
8793	Figure 12. Member States exporting textile apparel in 2019.....	56
8794	Figure 15. Current number of textile recycling plants classified by input fibre	61
8795	Figure 16. Scale of current textile recycling plants.....	61
8796	Figure 17. Composition of textile products	62
8797	Figure 18. Composition of textile apparel.....	63
8798	Figure 21. Electricity generation by source	66
8799	Figure 22. Overview of the main factors influencing the physical durability of textile apparel	111
8800	Figure 23. Dimensional change test run in the Durhabi project on 37 T-shirts over washing cycles.....	114
8801	Figure 24. Spirality test run in the Durhabi project on 34 T-shirts over washing cycles	115
8802	Figure 25. Most common textile apparel failure modes collected by the Durhabi project based on consumer surveys.....	117
8804	Figure 26. Example of existing care symbols under ISO 3758:2023.....	122
8805	Figure 27. Care labelling systems in the world	124
8806	Figure 28. Apparent consumption of textile apparel per capita in EU-27	131
8807	Figure 29. Recycling system of textile waste	135
8808	Figure 30. Risks associated to textile apparel.....	150
8809	Figure 31. Processes using process and auxiliary chemicals.....	152
8810	Figure 32. Contribution to the single score of the impact categories to the lifecycle environmental impacts	161
8812	Figure 33. Share of environmental impacts generated in the lifecycle of the three product categories – contribution of lifecycle stages	162
8814	Figure 34. Share of societal costs for the three product categories – contribution of internal costs in all life cycle stages and external costs.....	163
8816	Figure 35. Share of environmental impacts generated by specific contributors in raw material production (LCS1) of the three product categories.....	164
8818	Figure 36. Share of environmental impacts generated in the manufacturing processes (LCS2) of the three product categories.....	165
8820	Figure 37. Share of societal costs for the three product categories – contribution of external costs due to the manufacturing stage (LCS2) and internal costs due to all manufacturing processes.....	166

8822	Figure 38. Share of environmental impacts generated by specific contributors in the manufacturing life cycle stage (LCS2) of the three product categories.....	167
8823		
8824	Figure 39. Share of internal costs for the three product categories – contributors in LCS2 - Manufacturing.....	168
8825	Figure 40. Share of environmental impacts generated by the waste management.....	169
8826	Figure 41. Share of societal costs for the three product categories – contribution of the waste management	
8827		169
8828	Figure 42. Share of costs addressed by the producer.....	170
8829	Figure 43. Share of costs addressed by the consumer	171
8830	Figure 44. Environmental impacts expressed as environmental points for knitted products in scenarios investigating the influence of the location of the raw material production and manufacturing processes – all life cycle stages	173
8831		
8832		
8833	Figure 45. Environmental impacts expressed as environmental points for knitted products in scenarios investigating the influence of the location of the raw material production and manufacturing processes – contributors of the first two life cycles.....	174
8834		
8835		
8836	Figure 46. Societal costs expressed as EUR for knitted products in scenarios investigating the influence of location of raw material production and manufacturing processes – all life cycle stages	174
8837		
8838	Figure 47. Societal costs expressed as EUR for knitted products in scenarios investigating the influence of location of raw material production and manufacturing processes – contributors of the first two life cycles.....	175
8839		
8840	Figure 62. Method concerning the presence and use of substances under the ESPR.....	222
8841	Figure 63. Statistical classifications	308
8842	Figure 64. Market indicators per capita for textile apparel in EU-27	332
8843	Figure 65. Market indicators for T-shirts in EU-27	333
8844	Figure 66. Market indicators for shirts and blouses in EU-27.....	334
8845	Figure 67. Market indicators for sweaters and mid-layers in EU-27.....	335
8846	Figure 68. Market indicators for jackets and coats in EU-27	336
8847	Figure 69. Market indicators for pants and shorts in EU-27.....	337
8848	Figure 70. Market indicators for dresses, skirts and jumps in EU-27	338
8849	Figure 71. Market indicators for leggings, stockings and tights in EU-27.....	339
8850	Figure 72. Market indicators for underwear in EU-27.....	340
8851	Figure 73. Market indicators for swimwear in EU-27	341
8852	Figure 74. Market indicators for accessories in EU-27	342
8853	Figure 76. Production of textile apparel categories in EU-27	345
8854	Figure 77. Import of textile apparel categories in EU-27.....	347
8855	Figure 78. Export of textile apparel categories in EU-27	349
8856	Figure 79. Raw material for fibre production and fibre preparation	350
8857	Figure 80. Yarn manufacturing (spinning process)	352
8858	Figure 82. Fabric/yarn finishing processes	354
8859	Figure 88. Laundry sorting methods	395
8860	Figure 89. Distribution of washing temperatures for different textile products.....	396
8861	Figure 91. Number of successful and failed repairs done by different actors (total 625 repairs)	407

8862	Figure 92. Main reasons for disposal of apparel.....	412
8863	Figure 93. Evolution of licences and products awarded with EU Ecolabel for textile products.....	420
8864	Figure 94. System boundaries, with distinction between European and Rest of the World context.....	509
8865	Figure 95. Flow chart of how the model framework is applied to represent the bleaching, dyeing and printing processes.....	515
8867	Figure 96. Transport legs and pathways (taken from PEFCR Apparel and Footwear v2.0). Notice that ‘D2C’ stands for direct to customer.....	524
8868		
8869	Figure 97. Shares of input parameter according to their type.....	537
8870	Figure 98. LCA results for the knitted RP.....	541
8871	Figure 99. Breakdown for LCS and contributors of LCA results for the knitted RP.....	542
8872	Figure 100. LCC results: societal cost for the knitted RP.....	543
8873	Figure 101. Breakdown for LCS and contributor of societal LCC results for the knitted RP.....	544
8874	Figure 102. LCC results: cost for producer associated with the knitted RP.....	544
8875	Figure 103. LCC results: cost for consumer associated with the knitted RP.....	545
8876	Figure 104. LCA results for the other woven RP.....	545
8877	Figure 105. Breakdown for LCS and contributors of LCA results for the other woven RP.....	546
8878	Figure 106. LCC results: societal cost for the other woven RP.....	547
8879	Figure 107. Breakdown for LCS and contributors of societal LCC results for the other woven RP.....	548
8880	Figure 108. LCC results: cost for producer associated with the other woven RP.....	548
8881	Figure 109. LCC results: cost for consumer associated with the other woven RP.....	549
8882	Figure 110. LCA results for the denim RP.....	549
8883	Figure 111. Breakdown for LCS and contributors of LCA results for the denim RP.....	550
8884	Figure 112. LCC results: societal cost for the denim RP.....	551
8885	Figure 113. Breakdown for LCS and contributors of societal LCC results for the denim RP.....	552
8886	Figure 114. LCC results: cost for producer associated with the denim RP.....	552
8887	Figure 115. LCC results: cost for consumer associated with the denim RP.....	553
8888	Figure 116. Comparisons of LCA results for the knitted reference product assuming different “number of uses” of the product. Note that ‘BC’ stands for base case.....	555
8889		
8890	Figure 117. Comparisons of societal Life Cycle Costing (LCC) results for the knitted reference product assuming different “number of uses” of the product. Note that ‘BC’ stands for base case.....	555
8891		
8892	Figure 118. Comparisons of LCA results for the other woven reference product assuming different “number of uses” of the product. Note that ‘BC’ stands for base case.....	556
8893		
8894	Figure 119. Comparisons of societal Life Cycle Costing (LCC) results for the other woven reference product assuming different “number of uses” of the product. Note that ‘BC’ stands for base case.....	556
8895		
8896	Figure 120. Comparisons of LCA results for the denim reference product assuming different “number of uses” of the product. Note that ‘BC’ stands for base case.....	557
8897		
8898	Figure 121. Comparisons of societal Life Cycle Costing (LCC) results for the denim reference product assuming different “number of uses” of the product. Note that ‘BC’ stands for base case.....	557
8899		
8900	Figure 122. Comparisons of LCA results for the knitted RP modelled according to Base Case (BAU) vs EU vs Non-EU scenario.....	562
8901		

8902	Figure 123. Comparisons of societal LCC results for the knitted RP modelled according to Base Case (BAU) vs	
8903	EU vs Non-EU scenario.....	562
8904	Figure 124. Comparisons of cost for producer results for the knitted RP modelled according to Base Case	
8905	(BAU) vs EU vs Non-EU scenario.....	563
8906	Figure 125. Comparisons of cost for consumer results for the knitted RP modelled according to Base Case	
8907	(BAU) vs EU vs Non-EU scenario.....	563
8908	Figure 126. Comparisons of LCA results for the other woven RP modelled according to Base Case (BAU) vs	
8909	EU vs non-EU scenario.....	564
8910	Figure 127. Comparisons of societal LCC results for the other woven RP modelled according to Base Case	
8911	(BAU) vs EU vs Non-EU scenario.....	564
8912	Figure 128. Comparisons of cost for producer results for the other woven RP modelled according to Base	
8913	Case (BAU) vs EU vs Non-EU scenario.....	565
8914	Figure 129. Comparisons of cost for consumer results for the other woven RP modelled according to Base	
8915	Case (BAU) vs EU vs Non-EU scenario.....	565
8916	Figure 130. Comparisons of LCA results for the denim RP modelled according to Base Case (BAU) vs EU vs	
8917	Non-EU scenario.....	566
8918	Figure 131. Comparisons of societal LCC results for the denim RP modelled according to Base Case (BAU) vs	
8919	EU vs Non-EU scenario.....	566
8920	Figure 132. Comparisons of cost for producer results for the denim RP modelled according to Base Case	
8921	(BAU) vs EU vs Non-EU scenario.....	567
8922	Figure 133. Comparisons of cost for consumer results for the denim RP modelled according to Base Case	
8923	(BAU) vs EU vs Non-EU scenario.....	567
8924		

8925 **List of tables**

8926	Table 1. Development of the preparatory study	12
8927	Table 2. Definition of product aspects	16
8928	Table 3. Function and intended use of subgroups in the textiles and footwear product group	19
8929	Table 4. Economic indicators of textiles and footwear in the EU-27 in 2019	22
8930	Table 5. Literature studies investigating environmental impacts of textile products	23
8931	Table 6. Environmental impacts related to the supply chain of textile products	23
8932	Table 7. Initial product categories of textile apparel included in the scope of the preparatory study	24
8933	Table 8. Life-cycle stages of textile apparel – processes, techniques and resources	27
8934	Table 10. Top five topics addressed by environmental labels for textiles globally in 2021	42
8935	Table 14. Change of apparent consumption between the time intervals 1995-2004 and 2005-2019	55
8936	Table 15. Estimates of global production of textile fibres and group of fibres	57
8937	Table 17. Production of cotton lint in the season 2021-22	58
8938	Table 18. Import of cotton lint in the season 2021-22	59
8939	Table 19. Export of cotton lint in the season 2021-22	59
8940	Table 20. Consumption of cotton lint in the season 2021-22	59
8941	Table 22. Material composition of EU post-consumer textile waste	62
8942	Table 25. Water consumption of fibres	67
8943	Table 30. Distribution and retail models of apparel companies	70
8944	Table 34. Duration in use of textile apparel expressed in days of wear	75
8945	Table 35. Potential application of user behaviour aspects in the next steps of the preparatory study	77
8946	Table 36. Main reasons for apparel disposal	89
8947	Table 38. Scope of the textile Ecolabels used the most in the EU	95
8948	Table 39. Number of Contract Awards procuring apparel in the EU	97
8949	Table 40. Relevance of product aspects for textile apparel	103
8950	Table 41. Product aspects and required characteristics of products	104
8951	Table 42. Interaction between product aspects and product parameters reported in Annex I to ESPR	105
8952	Table 43. Most common textile apparel failure modes investigated by Cooper and Claxton (2022)	116
8953	Table 44. Description of product technologies in the context of robustness	118
8954	Table 45. Description of the two potential frameworks	119
8955	Table 46. Best practices on care labelling implemented by producers and retailers	124
8956	Table 47. Description of product technologies in the context of maintenance of textile apparel applicable to	
8957	the EU	126
8958	Table 48. Description of product technologies from the perspective of repair services	130
8959	Table 49. Amount of textile apparel waste generated in EU in 2019	131
8960	Table 50. Status of recycling techniques for textile waste in 2024	137
8961	Table 51. Description of the textile apparel waste in the context of the recyclability and the recycled content	
8962		140

8963	Table 52. Environmental performance levels for emission into water in specific regions	147
8964	Table 53. Description of investigated impact categories	158
8965	Table 54. Environmental impacts reported for all investigated impact categories	160
8966	Table 55. Environmental impacts expressed as environmental points – Values of all investigated impact	
8967	categories	160
8968	Table 56. Environmental impacts expressed as environmental points for the three product categories –	
8969	contribution of lifecycle stages	162
8970	Table 57. Societal costs expressed as EUR for the three product categories – contribution of internal costs in	
8971	all lifecycle stages and external costs	162
8972	Table 58. Environmental impacts expressed as environmental points for the three product categories –	
8973	contributors in LCS1 – Raw materials	163
8974	Table 59. Environmental impacts expressed as environmental points for the three product categories –	
8975	contribution of manufacturing processes in LCS2	164
8976	Table 60. Societal costs expressed as EUR for the three product categories – contribution of external costs	
8977	due to the manufacturing stage (LCS2) and internal costs due to all manufacturing processes	165
8978	Table 61. Environmental impacts expressed as environmental points for the three product categories –	
8979	contributors in LCS2 - Manufacturing	166
8980	Table 62. Internal costs expressed as EUR for the three product categories – contributors in LCS2 -	
8981	Manufacturing	167
8982	Table 63. Environmental impacts expressed as environmental points for the three product categories –	
8983	contribution of the waste management	168
8984	Table 64. Societal costs expressed as EUR for the three product categories – contribution of the waste	
8985	management	169
8986	Table 65. Producer costs expressed as EUR for the three product categories	170
8987	Table 66. Consumer costs expressed as EUR for the three product categories	170
8988	Table 67. Number of times that the worsening scenario produces higher environmental impacts and societal	
8989	costs compared to the base case	172
8990	Table 68. Decrease of environmental impacts and societal costs of the improving scenario compared to the	
8991	base case expressed as percentage (%)	172
8992	Table 69. Results of the environmental and economic assessment reported as central value and	
8993	corresponding uncertainty expressed as 95% confidence interval	175
8994	Table 70. Objectives, design options and potential ecodesign requirements for knitted, denim and other	
8995	woven products	182
8996	Table 71. Points to be awarded for the robustness score	186
8997	Table 72. Change in parameters related to the robustness (for a 5% service lifetime increase)	187
8998	Table 73. Results of the design option 1 (robustness) for knitted products	188
8999	Table 74. Results of the design option 1 (robustness) for denim products	189
9000	Table 75. Results of the design option 1 (robustness) for other woven products	189
9001	Table 76. Points to be awarded for the recyclability score	191
9002	Table 77. Results of the design option 2 (recyclability) for knitted products	192
9003	Table 78. Results of the design option 2 (recyclability) for denim products	193
9004	Table 79. Results of the design option 2 (recyclability) for other woven products	193

9005	Table 80. Description of the design options assessed for the recycled content in knitted, denim and other	
9006	woven products	194
9007	Table 81. Prices of recycled fibres used in the assessment	197
9008	Table 82. Results of the design option 3.1 (recycled content, information only) for knitted products	198
9009	Table 83. Results of the design option 3.1 (recycled content, information only) for denim products	198
9010	Table 84. Results of the design option 3.1 (recycled content, information only) for other woven products	199
9011	Table 85. Results of the design options 3.1 and 3.2 (recycled content, information and performance	
9012	requirements) for knitted products	199
9013	Table 86. Results of the design options 3.1 and 3.2 (recycled content, information and performance	
9014	requirements) for denim products	200
9015	Table 87. Results of the design options 3.1 and 3.2 (recycled content, information and performance	
9016	requirements) for other woven products	200
9017	Table 88. Results of the design option 4.1 (environmental footprint) for knitted products	203
9018	Table 89. Results of the design option 4.1 (environmental footprint) for denim products	204
9019	Table 90. Results of the design option 4.1 (environmental footprint) for other woven products	204
9020	Table 91. Results of the design option 4.2 (carbon footprint) for knitted products	205
9021	Table 92. Results of the design option 4.2 (carbon footprint) for denim products	205
9022	Table 93. Results of the design option 4.2 (carbon footprint) for other woven products	206
9023	Table 94. Results of environmental assessment for the impact categories with >5% contribution to the single	
9024	score, for knitted products	209
9025	Table 95. Results of environmental assessment for the impact categories with >5% contribution to the single	
9026	score, for denim products	211
9027	Table 96. Results of environmental assessment for the impact categories with >5% contribution to the single	
9028	score, for other woven products	213
9029	Table 97. Results with 95% Confidence interval for knitted products – Path 1, 2 and 3	215
9030	Table 98. Results with 95% Confidence interval for denim products – Path 1, 2 and 3	215
9031	Table 100. General classification of fibres	225
9032	Table 101. Identification of Substances of Concern (a) for textile apparel products according to the ESPR	
9033	Article 2(27)	229
9034	Table 102. Identification of Substances of Concern (b) for textile apparel products according to the ESPR	
9035	Article 2(27)	232
9036	Table 103. Identification of Substances of Concern (c) relevant to textile apparel products according to the	
9037	ESPR Article 2(27)	236
9038	Table 104. Identification of possible SoCs (d) due to existing regulatory limitations in textile apparel products.	
9039		243
9040	Table 105. Proposed thresholds for tracking of SoCs (b) identified under the ESPR Article 2(27) in textile	
9041	apparel products	248
9042	Table 106. Proposed thresholds for the tracking of examples of Substances of Concern (b) for textile apparel	
9043	products	248
9044	Table 107. Stakeholders providing contributions to the initial questionnaire	304
9045	Table 108. Stakeholders providing contributions to the 1 st milestone	305
9046	Table 109. Stakeholders providing contributions to the 2 nd milestone	306

9047	Table 110. NACE codes describing the product group of textiles and footwear	309
9048	Table 111. PRODCOM codes describing the subgroup of textile apparel in 2019	310
9049	Table 112. PRODCOM codes describing the subgroup of home/interior textiles in 2019	314
9050	Table 113. PRODCOM codes describing the subgroup of footwear in 2019	315
9051	Table 114. PRODCOM codes describing the subgroup of technical textiles in 2019.....	317
9052	Table 115. List and description of PRODCOM codes included in the scope.....	318
9053	Table 117. Change of apparent consumption after the removal of EU import quota in 2005	343
9054	Table 118. Change of production after the removal of EU import quota in 2005	345
9055	Table 121. Affinity relationship between dyes and fibres.....	354
9056	Table 122. Working Groups with the involvement of several scientific committees in CEN/ TC 248.....	359
9057	Table 123. Working Groups and subcommittees involved in ISO/TC 38.....	359
9058	Table 124. Standards directly related to intrinsic durability	361
9059	Table 125. Standards used by several frameworks to test parameters related to durability of textile products	367
9060		367
9061	Table 126. Standards related to the functionality of the textile product	369
9062	Table 128. Standards for the determination of specific substances.....	372
9063	Table 130 Standards and Technical Specifications potentially related to circularity and environmental aspects	375
9064		375
9065	Table 131. Standards related to topics not covered by Table 123 to Table 129	376
9066	Table 132. Environmental labels used in 2021 in Europe: ISO type and addressed topics	378
9067	Table 133. Number of textile recycling plants classified per location	381
9068	Table 134. Number of textile recycling plants classified per input fibres	381
9069	Table 136. Environmental performance levels for emission to water by the Chinese BREF.....	383
9070	Table 141. Aspects that influence the perception of quality in apparel.....	389
9071	Table 143. Average number of uses of apparel categories prior to washing	397
9072	Table 144. Summary of surveys on laundering habits related to apparel.....	401
9073	Table 145. Summary of surveys related to drying of apparel.....	402
9074	Table 146. Summary of surveys on user behaviour interaction with apparel care labels.....	404
9075	Table 147. Summary of surveys indicating the extent to which users associate a care label symbol to its	405
9076	corresponding care instruction.	405
9077	Table 148. Summary of surveys that provide the percentage of respondents who have made repairs to their	407
9078	apparel in the past year.....	407
9079	Table 149. Summary of surveys that showcase how confident users are to mending their apparel depending	408
9080	on the type of repair needed.	408
9081	Table 150. Summary of surveys that showcase overall declared frequency of apparel repairs.	408
9082	Table 152. Summary of apparel disposal reasons in recent consumer studies with high number of	412
9083	participants.	412
9084	Table 153. Surveys responses on main types of alternative apparel disposal.....	413
9085	Table 156. Figures of EU Ecolabel for textile products in September 2023	419

9086	Table 157. Type of respondents that contributed to the section on EU Ecolabel during the initial questionnaire	420
9087		
9088	Table 158. Suggestions received by respondents to the initial questionnaire on EU ecolabel criteria 1-9, which	
9089	focus on fibres	420
9090	Table 159. Suggestions received by respondents to the initial questionnaire on EU Ecolabel criteria 10-12,	
9091	which focus on components and accessories	421
9092	Table 160. Suggestions received by respondents to the initial questionnaire on EU ecolabel criteria 13-16	
9093	and appendixes, which focus on chemicals	421
9094	Table 161. Suggestions received by respondents to the initial questionnaire on EU Ecolabel criteria 17-25,	
9095	which focus on fitness for use	422
9096	Table 162. Suggestions received by respondents to the initial questionnaire on EU ecolabel criteria 26-28,	
9097	which focus on Corporate Social Responsibility and supporting information	423
9098	Table 163. Textile fibres - comparison among Ecolabels	424
9099	Table 164. Components and accessories – comparison among Ecolabels	443
9100	Table 165. Fitness for use - comparison among Ecolabels	446
9101	Table 166. Chemicals and processes - comparison among Ecolabels	452
9102	Table 167. Corporate social responsibility - comparison among Ecolabels	457
9103	Table 168. Miscellaneous criteria - comparison among Ecolabels	458
9104	Table 169. Dyes restriction - comparison among Ecolabels	461
9105	Table 170. The Common Procurement Vocabulary (CPV) codes for products in the scope	462
9106	Table 171. Number of Contracts Award procuring apparel in EU Member States in 2015	464
9107	Table 176. Type of respondents that contributed to the section on EU GPP during the initial questionnaire	467
9108	Table 177. Description of the main fibre characteristics	485
9109	Table 178. Definitions of yarn and fabric characteristics	487
9110	Table 179. Definitions of main process parameters	488
9111	Table 180. Influence of the fibre characteristics on yarn and fabric characteristics and parameters of	
9112	manufacturing processes	489
9113	Table 181. Influence of spinning parameters on yarn characteristics	492
9114	Table 182. Influence of weaving parameters on fabric characteristics	493
9115	Table 183. Influence of finishing processes on fabric characteristics	494
9116	Table 184. Description of standardised test methods to assess key parameters of physical durability	495
9117	Table 185. Comparison of international labelling systems	496
9118	Table 186. Comparison between price of professional repair operations and price of new items of textile	
9119	apparel	498
9120	Table 187. Repair shops and dry cleaners consulted for the analysis reported in Table 185	499
9121	Table 188. EF impact categories with respective impact category indicators and characterisation models	502
9122	Table 189. Representative Products (RPs) of textile apparel included in the scope of the preparatory study	
9123	and their main characterizing parameters	505
9124	Table 190. Functional unit (FU) of the LCA-LCC.	506

9125	Table 191. Location of production for specific textile fibres (acronym: Unk = unknown; AU = Australia; CN =	
9126	China; EU = Europe; ID = Indonesia; IN = India; HU = Hungary; NZ = New Zealand; Oth = other countries; PO =	
9127	Poland; RSA = South Africa; TU = Türkiye; TW = Taiwan; USA = United States).	511
9128	Table 192. Detail of modelling and inventory used in LCS 2 manufacturing.	513
9129	Table 193. Losses along the product manufacturing value chain.	516
9130	Table 194. Wages values for each country considered in the model.	518
9131	Table 196. Evolution of wages over the latest years in Poland.	519
9132	Table 197. Gross average monthly salary in the industry in Romania 2022, by industry.	519
9133	Table 198. Geographical breakdown of global apparel production and consumption.	520
9134	Table 199. Global production in tonnes of the textile sector, breakdown for manufacturing phases.	520
9135	Table 200. Detail of each trade route modelled for the transport of materials and semi-finished between	
9136	source Countries and receiving Countries according to mutual availability and requirements obtained by	
9137	applying the transport model (BZ=Brazil, CN=China, EU=Europe, ID=Indonesia, IN=India, PA=Pakistan,	
9138	TU=Turkey, VN=Vietnam, JP =Japan; TZ=Tanzania, TW=Taiwan, SA=South Africa, AU=Australia).	522
9139	Table 201. Shares of products per transport leg	523
9140	Table 202. Distances, utilisation ratio and provenance of transport	525
9141	Table 203. Volume, time, capacity, surface, energy consumption	527
9142	Table 204. Shares of post-consumer wasted garments routes and Shares of primary packaging waste routes	
9143		533
9144	Table 205. Transport Scenarios for apparel End of Life	533
9145	Table 206. Details of energy, water inputs and costs for sorting and (mechanical) recycling processes	
9146	modelled in the waste management and EoL stages.	534
9147	Table 207. Normalization and weighting factors for the 16 impact categories considered in the study.	537
9148	Table 208. List of monetary valuation coefficients (MVCs) used in this study. Reference year:2022	
9149		538
9150	Table 209. List of different scenarios modelled to investigate the influence of the choice made by the	
9151	consumer. Note that 'BC' stands for base case, and 'RP' for reference product.	553
9152	Table 210. Details of the data used to model and analyse the life cycle stage 1 (LCS1 - Raw materials)....	557
9153	Table 211. Details of the data used to model and analyse Life Cycle Stage 2 (LCS2 - Manufacturing).....	558
9154	Table 212. Results of the design options for knitted products.....	566
9155	Table 213. Results of the design options for denim products.....	567
9156	Table 214. Results of the design options for woven products.....	568
9157	Table 215. Results of the design options for knitted products.....	569
9158	Table 216. Results of the design options for denim products.....	569
9159	Table 217. Results of the design options for woven products.....	570
9160	Table 218. List of parameters used for the uncertainty analysis.....	571
9161	Table 219. List of additional parameters used for the uncertainty analysis of design options.....	574
9162	Table 222 Results with 95% Confidence interval for RP1.....	577
9163	Table 223 Results with 95% Confidence interval for RP2.....	577
9164	Table 224 Results with 95% Confidence interval for RP3.....	577
9165	Table 225. Apparel market sizes (2025).....	578

9166	Table 226. Classification of chemical substances used in the production of textile apparel according to the	
9167	process stage and the function/type.	581
9168	Table 227. Inventory of the relevant chemical substances identified in textile apparel products per function	
9169	(non-exhaustive).	583
9170	Table 228. Life cycle inventory of the chemical substances used in the LCA in function on the Life Cycle	
9171	Stage (LCS).S).	591
9172		

13 Annexes

13.1 Consultation with stakeholders

The development of the 3rd milestone of the PS builds on author's analysis and previous consultations with stakeholders:

- **Initial questionnaire**, which addressed definitions, scope, market analysis, user behaviour, product aspects, EU Ecolabel, EU Green Public Procurement.
- **1st milestone**, which addressed scope, market, user behaviour, current EU Ecolabel criteria, current EU Green Public Procurement criteria.
- **2nd milestone**, which addressed product technologies, framework and data gaps of environmental and economic model, and a questionnaire about substances and substances of concern.

The following subsections detail how stakeholders were involved for each step.

13.1.1 Initial questionnaire

In 2023, the involvement of stakeholders with the initial questionnaire included the following steps:

- On 24 March, the JRC communicated to stakeholders the start of the development of the PS and the dates when to contribute to the survey. The topics of the survey were also communicated to stakeholders.
- On 30 March, the JRC made available the survey to stakeholders and communicated its closure on Monday 24 April.
- On 14 April, the JRC communicated the deadline extension to contribute to the survey.
- From 30 March to 8 May, stakeholders were able to contribute to the initial questionnaire, whose copy is available on the project's website ⁽²⁸²⁾.

Table 107 reports some statistics of the 111 stakeholders contributing to the initial questionnaire.

Table 107. Stakeholders providing contributions to the initial questionnaire

Type of stakeholder	Number	Percentage (%)
Certified laboratories	4	4
Governments	22	20
Industries (as single entities)	42	38
Industry associations	13	12
NGOs	9	8
Research institutes including universities	5	5
Others	16	14

Source: own production

13.1.2 1st milestone

In 2024, the 1st milestone of the PS included the following steps:

- On 16 February, the JRC communicated to all registered stakeholders the timeline of the consultation about the 1st milestone.
- On 23 February, the JRC made available to the public the working document about the 1st milestone ⁽²⁸³⁾;

²⁸² Preparatory Study on textiles for product policy instruments – the initial questionnaire. Available at [this link](#). Last accessed on 22 August 2025.

²⁸³ Working document – 1st milestone. Preparatory study on textiles for product policy instruments – 1st milestone. Available at [this link](#). Last accessed on 18 August 2025.

- 9203 — On 8 March, the JRC communicated to all registered stakeholders the agenda of the workshop
- 9204 about the 1st milestone.
- 9205 — On 18 and 19 March, the JRC held the online meeting with registered stakeholders on the 1st
- 9206 milestone. The slides presented at the meeting were made available on the project's website:
- 9207
 - Presentation given by DG Environment on 18 March ⁽²⁸⁴⁾;
- 9208
 - Presentation given by JRC on 18 March ⁽²⁸⁵⁾;
- 9209
 - Presentation given by JRC on 19 March ⁽²⁸⁶⁾;
- 9210 — On 20 March, the JRC provided to all registered stakeholders the directions to contribute to the
- 9211 working document of the 1st milestone. Information was also available on the project's website in
- 9212 the section Project Plan.
- 9213 — From 20 March to 22 April, registered stakeholders provided their comments to the working
- 9214 document via a web form.

9215 **Table 108** reports some statistics of the 76 stakeholders contributing to the 1st milestone.

9216 **Table 108.** Stakeholders providing contributions to the 1st milestone

Type of stakeholder	Number	Percentage (%)
Certified laboratories	1	1
Governments	10	13
Industries (as single entities)	18	24
Industry associations	21	28
NGOs	9	12
Research institutes including universities	5	7
Others	12	16

9217 *Source: own production*

9218 **13.1.3 2nd milestone**

9219 The 2nd milestone of the PS included the following steps:

- 9220 — On 21 October 2024, the JRC communicated to all registered stakeholders the date of the
- 9221 workshop about the 2nd milestone and specified the topics to be addressed.
- 9222 — On 14 November 2024, the JRC shared with all registered stakeholders the working document on
- 9223 Task 4 of the preparatory study ⁽²⁸⁷⁾.
- 9224 — From 14 November to 8 December 2024, registered stakeholders had the opportunity to read the
- 9225 section on product technologies before the online consultation meeting.
- 9226 — On 2 December 2024, the JRC provided to all registered stakeholders:
- 9227
 - The detailed agenda of the workshop about the 2nd milestone.
- 9228
 - The questionnaire on substances and substances of concern ⁽²⁸⁸⁾.
- 9229 — From 2 to 9 December 2024, registered stakeholders had the opportunity to read the questionnaire
- 9230 on substances and substances of concern.
- 9231 — On 9 and 10 December 2024, registered stakeholders and JRC attended the online workshop on
- 9232 the 2nd milestone. The slides presented at the meeting were made available on the project's
- 9233 website:

284 Presentation given by DG Environment on 18 March. Available at [this link](#). Last accessed on 18 August 2025.

285 Presentation given by JRC on 18 March. Available at [this link](#). Last accessed on 18 August 2025.

286 Presentation given by JRC on 19 March. Available at [this link](#). Last accessed on 18 August 2025.

287 Task 4 of the preparatory study on textile products - November 2024. Available at [this link](#). Last accessed on 18 August 2025.

288 SoC Questionnaire Textile Apparel – 2nd milestone. Available at [this link](#). Last accessed on 18 August 2025.

- 9234 • Presentation given on 9 December 2024 ⁽²⁸⁹⁾.
- 9235 • Presentation given on 10 December 2024 on substances and substances of concern ⁽²⁹⁰⁾.
- 9236 • Presentation given on 10 December 2024 on Slides of the workshop – Day 2 on framework
- 9237 of environmental and economic model ⁽²⁹¹⁾.
- 9238 — On 18 December 2024, the JRC communicated to all registered stakeholders how to contribute to
- 9239 the documents related to the 2nd milestone. Information was also available on the project's website
- 9240 in the section Project Plan.
- 9241 — On 27 January 2025, the JRC shared with all registered stakeholders a document with answers to
- 9242 stakeholder's questions on the analysis of technologies in the context of physical durability ⁽²⁹²⁾.
- 9243 — From 18 December 2024 to 3 March 2025, registered stakeholders were able to provide comments
- 9244 in writing to the working document.
- 9245 — From 18 December 2024 to 10 March 2025, registered stakeholders were able to contribute to
- 9246 the questionnaire on substances and substances of concern.
- 9247 — From 18 December 2024 to 17 March 2025, registered stakeholders were able to contribute to
- 9248 improve the inventory and framework of the environmental and economic model.

9249 **Table 108** reports some statistics of the 108 stakeholders contributing to the 2nd milestone.

9250 **Table 109.** Stakeholders providing contributions to the 2nd milestone

Type of stakeholder	Number	Percentage (%)
Certified laboratories	1	1
Governments	12	11
Industries (as single entities)	30	28
Industry associations	42	39
NGOs	13	12
Research institutes including universities	7	6
Others	3	3

9251 *Source: own production*

9252 **13.2 PRODCOM and market analysis**

9253 **13.2.1 General description**

9254 PRODCOM is the classification of goods used for statistics in value and quantity on industrial production in the
 9255 EU. It is abbreviated from the French Production Communautaire and it is regulated by the Commission
 9256 Implementing Regulation (EU) 2020/1197 ⁽²⁹³⁾. **Figure 63** shows the classification of activities, products and
 9257 goods at level of world, EU and Member States. **Figure 63** describes how PRODCOM classification relates to
 9258 the other classifications.

9259 PRODCOM was developed in a close relationship with Combined (tariff and statistical) Nomenclature (CN) for
 9260 external trades, which is strictly related to Harmonised System used in the World Custom Organization. The
 9261 basic building blocks of PRODCOM are the European Classification of Economic Activities (NACE) ⁽²⁹⁴⁾ and the
 9262 European Classification of Products by Activity (CPA) ⁽²⁹⁵⁾. Products are identified in PRODCOM via their eight-

²⁸⁹ Slides of the workshop – Day 1 on Task 4. Available at [this link](#). Last accessed on 18 August 2025.

²⁹⁰ Slides of the workshop – Day 2 on substances and substances of concern. Available at [this link](#). Last accessed on 18 August 2025.

²⁹¹ Slides of the workshop – Day 2 on framework of environmental and economic model. Available at [this link](#). Last accessed on 18 August 2025.

²⁹² Answers to stakeholder's questions on the analysis of technologies in the context of physical durability – 2nd milestone. Available at [this link](#). Last accessed on 18 August 2025.

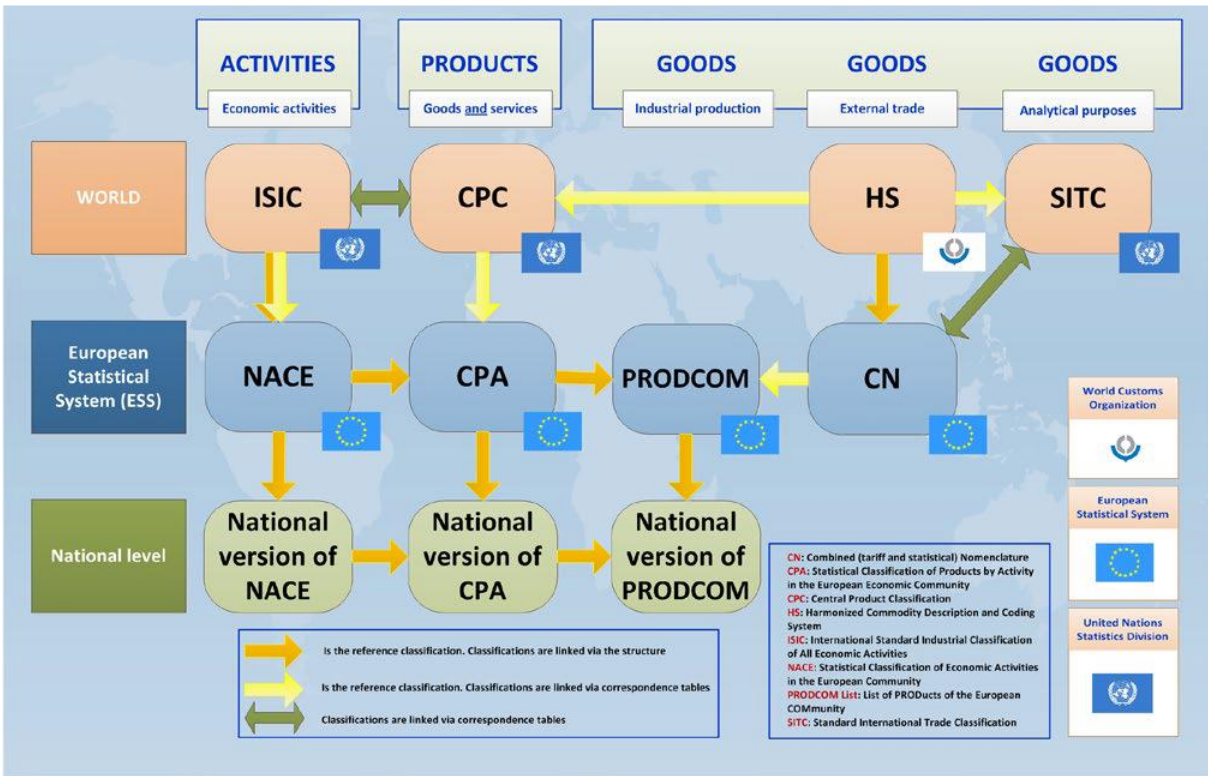
²⁹³ European business statistics. COMMISSION IMPLEMENTING REGULATION (EU) 2020/1197 of 30 July 2020 laying down technical specifications and arrangements pursuant to Regulation (EU) 2019/2152 of the European Parliament and of the Council on European business statistics repealing 10 legal acts in the field of business statistics. Available at [this link](#).

²⁹⁴ The statistical classification of economic activities NACE Revision 2. Regulation (EC) No 1893/2006. Available at [this link](#).

²⁹⁵ Statistical Classification of Products by Activity. COMMISSION REGULATION (EU) No 1209/2014 of 29 October 2014 amending Regulation (EC) No 451/2008 of the European Parliament and of the Council establishing a new statistical classification of products by activity (CPA) and repealing Council Regulation (EEC) No 3696/93. Available at [this link](#).

digit code, the PRODCOM List ⁽²⁹⁶⁾. The first four digits of a PRODCOM code refer to the NACE classification, the first six digits refer to the CPA classification, and the last two digits are created specifically for PRODCOM.

Figure 63. Statistical classifications



Source: (Eurostat, 2023)

NACE is based on the International Standard Industrial Classification of All Economic Activities (ISIC), which is a standard classification of economic activities used by the United Nations ⁽²⁹⁷⁾. Both NACE and ISIC are four-digit codes, which define four levels of description from the first to the last digit: sections, divisions, groups and classes. At the level of division (the first two digits), NACE and ISIC are identical and refer to the same activity. At level of group and classes (the third and the fourth digits), NACE and ISIC refer to different products/activities ⁽²⁹⁸⁾.

The NACE codes were established in 1970. Along with the evolution of the industrial scenery, the NACE codes were subjected to several revisions resulting in changes over time. From 1995 to 2007, the PRODCOM data were based on NACE Rev. 1.1, whereas from 2008 onwards the PRODCOM data were based on NACE Rev. 2. The NACE Rev. 1.1 classification has 21 sections and 88 divisions, whereas the NACE Rev. 2 counts with 17 sections and 62 divisions. The changes between different versions of NACE are supported by conversion tables, which can show sometimes (1) lack of direct correspondence between the old and new codes, and (2) merging of two or more codes into one ⁽²⁹⁸⁾.

Approximately every 2 or 3 years, the PRODCOM Working Group updates the PRODCOM list to reflect changes in the production of goods in the European Union. These changes occur with a frequency set by the technological developments in the industries and as driven by the related nomenclatures (Eurostat, 2023).

13.2.2 Features of PRODCOM data influencing market analysis

PRODCOM dataset DS-056120 includes several indicators ⁽³²⁾. In the PS, the following indicators were used:

²⁹⁶ PRODCOM list 2022. Available at [this link](#). Last accessed on 1 December 2023.

²⁹⁷ The UNSTATS webpage is available at [this link](#). Last accessed on 26 September 2023.

²⁹⁸ The introductory guidelines to NACE Rev. 2. Available at [this link](#). Last accessed on 29 September 2023.

- 9286 — PRODVAL: value of sold production
- 9287 — PRODQNT: quantity of sold production
- 9288 — EXPVAL: value of export
- 9289 — EXPQNT: quantity of export
- 9290 — IMPVAL: value of import
- 9291 — IMPQNT: quantity of import
- 9292 — QNTUNIT: unit used to report quantities

9293 The PRODCOM user's manual reports the following features (Eurostat, 2023).

9294 **PRODQNT**

9295 When a PRODCOM code does not report the unit in QNTUNIT, the figure of PRODQNT is not required to be
 9296 reported. Additionally, figures could be missing due to confidentiality.

9297 **EXPQNT and IMPQNT**

9298 Comext⁽²⁹⁹⁾ is the dataset disseminating the international trades in goods statistics. PRODCOM extracts
 9299 information from Comext. Most of PRODCOM codes have a complete reference to CN, meaning that there is full
 9300 comparability between data from PRODCOM and data from CN. However, data could be missing from PRODCOM
 9301 when:

- 9302 • the corresponding trade data on quantity cannot be provided for PRODCOM codes for which
 9303 unit of measure is not consistent with unit of measure given in CN, even if a PRODCOM code
 9304 has a complete CN reference and/or;
- 9305 • If there is no clear link between PRODCOM and CN.

9306 **Apparent consumption**

9307 The apparent consumption is obtained summing the production to the import and then subtracting the export.
 9308 The use of this economic indicator should consider the following aspects:

- 9309 3. The integration of production and trade data is challenged by the possible heterogeneity between the
 9310 PRODCOM codes and the trade data reported by Comext.
- 9311 4. The thresholds above which businesses are represented in PRODCOM and Comext could differ.
- 9312 5. Data reported in PRODCOM refer to a specific year, but there are temporal delays in production, import
 9313 and exports. However, the delays are reduced when data are considered over several years.
- 9314 6. The value of exports cannot always be compared directly with that of sold production.
 - 9315 (a) Sold production is based on the ex-work selling price⁽³⁰⁰⁾, whereas exports are evaluated at the time
 9316 the goods cross the border.
 - 9317 (b) Imported goods can be exported again with a different value, either without being modified or after
 9318 industrial processing.
- 9319 7. Some figures of quantity of sold production could be missing because of confidentiality or because
 9320 Member States are exempted to report them when quantity units are not specified for the specific
 9321 PRODCOM code.

9322 **13.2.3 Market analysis for the selection of the scope**

9323 The market analysis reported in Section 3.1.2 used the PRODCOM codes reporting about the NACE activities
 9324 specified in **Table 110**.

²⁹⁹ Comext dataset. Available at [this link](#). Last accessed on 2 December 2023.

³⁰⁰ Ex-work selling price does not include any transportation cost, which are addressed by the buyer.

9325 **Table 110.** NACE codes describing the product group of textiles and footwear

Subgroup	NACE codes
Textile apparel	C1412 - Manufacture of workwear C1413 - Manufacture of other outerwear C1414 - Manufacture of underwear C1419 - Manufacture of other wearing apparel and accessories C1431 - Manufacture of knitted and crocheted hosiery C1439 - Manufacture of other knitted and crocheted apparel
Home/interior textiles	C1392 - Manufacture of made-up textile articles, except apparel C1393 - Manufacture of carpets and rugs
Footwear	C1520 - Manufacture of footwear
Technical textiles	C1394 - Manufacture of cordage, rope, twine and netting C1395 - Manufacture of non-wovens and articles made from non-wovens, except apparel C1396 - Manufacture of other technical and industrial textiles C1399 - Manufacture of other textiles n.e.c.

9326

9327 **Table 111, Table 112, Table 113, and Table 114** report all the PRODCOM codes used for the analysis of
 9328 amount of sales and trades in EU. They also report the conversion factors used for the quantification of
 9329 quantities expressed in mass.

9330

Table 111. PRODCOM codes describing the subgroup of textile apparel in 2019

PRODCOM code	Description of the PRODCOM code	QNT UNIT	Type	Conversion factor to kg
14121120	Men's or boys' ensembles, of cotton or man-made fibres, for industrial and occupational wear	p/st	S	0.5
14121130	Men's or boys' jackets and blazers, of cotton or man-made fibres, for industrial and occupational wear	p/st	S	0.95
14121240	Men's or boys' trousers and breeches, of cotton or man-made fibres, for industrial or occupational wear	p/st	S	0.45
14121250	Men's or boys' bib and brace overalls, of cotton or man-made fibres, for industrial or occupational wear	p/st	S	0.45
14122120	Women's or girls' ensembles, of cotton or man-made fibres, for industrial or occupational wear	p/st	S	0.5
14122130	Women's or girls' jackets and blazers, of cotton or man-made fibres, for industrial or occupational wear	p/st	S	0.95
14122240	Women's or girls' trousers and breeches, of cotton or man-made fibres, for industrial or occupational wear	p/st	S	0.45
14122250	Women's or girls' bib and brace overalls, of cotton or man-made fibres, for industrial or occupational wear	p/st	S	0.45
14123013	Men's or boys' other garments, of cotton or man-made fibres, for industrial or occupational wear	p/st	S	0.5
14123023	Women's or girls' other garments, of cotton or man-made fibres, for industrial or occupational wear	p/st	S	0.5
14131110	Men's or boys' overcoats, car-coats, capes, cloaks and similar articles, of knitted or crocheted textiles (excluding jackets and blazers, anoraks, wind-cheaters and wind-jackets)	p/st	S	0.95
14131120	Men's or boys' waistcoats, anoraks, ski-jackets, wind-cheaters, wind-jackets and similar articles, of knitted or crocheted textiles (excluding jackets and blazers)	p/st	S	0.95
14131230	Men's or boys' jackets and blazers, of knitted or crocheted textiles	p/st	S	0.95
14131260	Men's or boys' suits and ensembles, of knitted or crocheted textiles	p/st	S	0.5
14131270	Men's or boys' trousers, breeches, shorts, bib and brace overalls, of knitted or crocheted textiles	p/st	S	0.45
14131310	Women's or girls' overcoats, car-coats, capes, cloaks and similar articles, of knitted or crocheted textiles (excluding jackets and blazers)	p/st	S	0.95
14131320	Women's or girls' waistcoats, anoraks, ski-jackets, wind-cheaters, wind-jackets and similar articles, of knitted or crocheted textiles (excluding jackets and blazers)	p/st	S	0.95
14131430	Women's or girls' jackets and blazers, of knitted or crocheted textiles	p/st	S	0.95
14131460	Women's or girls' suits and ensembles, of knitted or crocheted textiles	p/st	S	0.5
14131470	Women's or girls' dresses, of knitted or crocheted textiles	p/st	S	0.3
14131480	Women's or girls' skirts and divided skirts, of knitted or crocheted textiles	p/st	S	0.25
14131490	Women's or girls' trousers, breeches, shorts, bib and brace overalls, of knitted or crocheted textiles	p/st	S	0.45
14132110	Men's or boys' raincoats	NA	NA	0.5
14132115	Men's or boys' raincoats, overcoats, car-coats, capes, etc.	p/st	NA	0.5
14132116	Men's or boys' overcoats, car coats, capes, cloaks, anoraks (including ski-jackets), wind cheaters, wind-jackets and similar articles (excluding suits, ensembles, jackets, blazers, trousers, bib and brace overalls, breeches and shorts)	NA	S	NA
14132120	Men's or boys' overcoats, car-coats, capes, etc.	NA	NA	0.95
14132130	Men's or boys' waistcoats, anoraks, ski-jackets, wind-jackets and similar articles (excluding jackets and blazers, knitted or crocheted, impregnated, coated, covered, laminated or rubberised)	p/st	NA	0.95
14132200	Men's or boys' suits & ensembles (excluding knitted or crocheted)	p/st	S	0.5
14132210	Men's or boys' suits (excluding knitted or crocheted)	NA	NA	1.4
14132220	Men's or boys' ensembles (excluding knitted or crocheted)	NA	NA	1
14132300	Men's or boys' jackets and blazers (excluding knitted or crocheted)	p/st	S	0.95
14132442	Men's or boys' trousers and breeches, of denim (excluding for industrial or occupational wear)	p/st	S	0.45

PRODCOM code	Description of the PRODCOM code	QNT UNIT	Type	Conversion factor to kg
14132444	Men's or boys' trousers, breeches and shorts, of wool or fine animal hair (excluding knitted or crocheted, for industrial or occupational wear)	p/st	S	0.45
14132445	Men's or boys' trousers and breeches, of man-made fibres (excluding knitted or crocheted, for industrial or occupational wear)	p/st	S	0.45
14132448	Men's or boys' trousers and breeches, of cotton (excluding denim, knitted or crocheted)	p/st	S	0.45
14132449	Men's or boys' trousers, breeches, shorts and bib and brace overalls (excluding of wool, cotton and man-made fibres, knitted or crocheted)	p/st	S	0.45
14132455	Men's or boys' bib and brace overalls (excluding knitted or crocheted, for industrial or occupational wear)	p/st	S	0.45
14132460	Men's or boys' shorts, of cotton or man-made fibres (excluding knitted or crocheted)	p/st	S	0.45
14133110	Woman's or girls' raincoats	NA	NA	0.5
14133115	Woman's or girls' raincoats and overcoats, etc.	p/st	NA	0.5
14133116	Women's or girls' overcoats, car-coats, capes, cloaks, anoraks (including ski jackets), wind-cheaters, wind-jackets and similar articles (excluding suits, ensembles, jackets, blazers, dresses, skirts, divided skirts, trousers, bib and brace overalls, breeches and shorts)	NA	S	NA
14133120	Woman's or girls' overcoats, etc.	NA		0.95
14133130	Women's or girls' waistcoats, anoraks, ski-jackets, wind-jackets and similar articles (excluding jackets and blazers, knitted or crocheted, impregnated, coated, covered, laminated or rubberised)	p/st	NA	0.95
14133200	Women's or girls' suits & ensembles (excluding knitted or crocheted)	p/st	S	0.5
14133210	Women's or girls' suits (excluding knitted or crocheted)	NA	NA	1.4
14133220	Women's or girls' ensembles (excluding knitted or crocheted)	NA	NA	1
14133330	Women's or girls' jackets and blazers (excluding knitted or crocheted)	p/st	S	0.95
14133470	Women's or girls' dresses (excluding knitted or crocheted)	p/st	S	0.3
14133480	Women's or girls' skirts and divided skirts (excluding knitted or crocheted)	p/st	S	0.25
14133542	Women's or girls' trousers and breeches, of denim (excluding for industrial or occupational wear)	p/st	S	0.45
14133548	Women's or girls' trousers and breeches, of cotton (excluding denim, for industrial or occupational wear)	p/st	S	0.45
14133549	Women's or girls' trousers and breeches, of wool or fine animal hair or man-made fibres (excluding knitted or crocheted and for industrial and occupational wear)	p/st	S	0.45
14133551	Women's or girls' bib and brace overalls, of cotton (excluding knitted or crocheted, for industrial or occupational wear)	p/st	S	0.45
14133561	Women's or girls' shorts, of cotton (excluding knitted and crocheted)	p/st	S	0.45
14133563	Women's or girls' bib and brace overalls, of wool or fine animal hair and man-made fibres (excluding cotton, knitted or crocheted, for industrial or occupational wear) and women's or girls' shorts, of wool or fine animal hair (excluding knitted or crocheted)	p/st	S	0.45
14133565	Women's or girls' shorts, of man-made fibres (excluding knitted or crocheted)	p/st	S	0.45
14133569	Women's or girls' trousers, breeches, bib and brace overalls, of textiles (excluding cotton, wool or fine animal hair, man-made fibres, knitted or crocheted)	p/st	S	0.45
14141100	Men's or boys' shirts, knitted or crocheted	p/st	S	0.25
14141220	Men's or boys' underpants and briefs, of knitted or crocheted textiles (including boxer shorts)	p/st	S	0.08
14141230	Men's or boys' nightshirts and pyjamas, of knitted or crocheted textiles	p/st	S	0.15
14141240	Men's or boys' dressing gowns, bathrobes and similar articles, of knitted or crocheted textiles	p/st	S	0.3
14141310	Women's or girls' blouses, shirts and shirt-blouses, of knitted or crocheted textiles	p/st	S	0.25
14141420	Women's or girls' briefs and panties, of knitted or crocheted textiles (including boxer shorts)	p/st	S	0.08
14141430	Women's or girls' nighties and pyjamas, of knitted or crocheted textiles	p/st	S	0.15
14141440	Women's or girls' negligees, bathrobes, dressing gowns and similar articles, of knitted or crocheted textiles	p/st	S	0.3
14141450	Women's or girls' slips and petticoats, of knitted or crocheted textiles	p/st	S	0.5

PRODCOM code	Description of the PRODCOM code	QNT UNIT	Type	Conversion factor to kg
14142100	Men's or boys' shirts (excluding knitted or crocheted)	p/st	S	0.25
14142220	Men's or boys' underpants and briefs (including boxer shorts) (excluding knitted or crocheted)	p/st	S	0.08
14142230	Men's or boys' nightshirts and pyjamas (excluding knitted or crocheted)	p/st	S	0.15
14142240	Men's or boys' singlets, vests, bathrobes, dressing gowns and similar articles (excluding knitted or crocheted)	p/st	S	0.3
14142300	Women's or girls' blouses, shirts and shirt-blouses (excluding knitted or crocheted)	p/st	S	0.25
14142430	Women's or girls' nightdresses and pyjamas (excluding knitted or crocheted)	p/st	S	0.3
14142450	Women's or girls' slips and petticoats (excluding knitted or crocheted)	p/st	S	0.5
14142460	Women's or girls' singlets and other vests, briefs, panties, negligees, bathrobes, dressing gowns, housecoats and similar articles of cotton (excluding knitted or crocheted)	p/st	S	0.08
14142480	Women's or girls' negligees, bathrobes, dressing gowns, singlets, vests, briefs and panties (including boxer shorts), of man-made fibres (excluding knitted or crocheted)	p/st	S	0.08
14142489	Women's or girls' singlets, vests, briefs, panties, negligees, bathrobes, dressing gowns and similar articles, of textiles (excluding cotton, man-made fibres, knitted or crocheted)	p/st	S	0.08
14142530	Brassieres	p/st	S	0.05
14142550	Girdles, panty-girdles and corselettes (including bodies with adjustable straps)	p/st	S	0.5
14142570	Braces, suspenders, garters and similar articles and parts thereof	NA	S	NA
14143000	T-shirts, singlets and vests, knitted or crocheted	p/st	S	0.17
14191100	Babies' garments and clothing accessories, knitted or crocheted including vests, rompers, underpants, stretch-suits, gloves or mittens or mitts, outerwear (for children of height <= 86 cm)	NA	S	NA
14191210	Track-suits, of knitted or crocheted textiles	p/st	S	0.5
14191230	Ski-suits, of knitted or crocheted textiles	p/st	S	0.5
14191240	Men's or boys' swimwear, of knitted or crocheted textiles	p/st	S	0.12
14191250	Women's or girls' swimwear, of knitted or crocheted textiles	p/st	S	0.12
14191290	Other garments, knitted or crocheted (including bodies with a proper sleeve)	kg	S	1
14191300	Gloves, mittens and mitts, of knitted or crocheted textiles	pa	S	0.1
14191930	Shawls, scarves, mufflers, mantillas, veils and the like, of knitted or crocheted textiles	p/st	S	0.1
14191960	Clothing accessories and parts thereof, of knitted or crocheted textiles (excluding gloves, mittens, shawls, scarves, mufflers, mantillas and veils)	NA	S	NA
14192100	Babies' clothing and accessories, of textiles, not knitted or crocheted (for children of height <= 86 cm) including vests, rompers, underpants, stretch-suits, napkins, gloves, mittens and outerwear	NA	NA	NA
14192150	Babies' clothing and accessories, of textiles, not knitted or crocheted (for children of height <= 86 cm) including vests, rompers, underpants, stretch-suits, gloves, mittens and outerwear (excluding sanitary towels and napkins and similar articles)	NA	S	NA
14192210	Other men's or boys' apparel n.e.c., including tracksuits and jogging suits (excluding waistcoats, ski-suits, knitted or crocheted)	p/st	S	0.5
14192220	Other women's or girls' apparel n.e.c., including tracksuits and jogging suits (excluding waistcoats, ski-suits, knitted or crocheted)	p/st	S	0.5
14192230	Ski-suits (excluding of knitted or crocheted textiles)	p/st	S	0.5
14192240	Men's or boys' swimwear (excluding of knitted or crocheted textiles)	p/st	S	0.12
14192250	Women's or girls' swimwear (excluding of knitted or crocheted textiles)	p/st	S	0.12
14192310	Handkerchiefs	p/st	S	0.5
14192333	Shawls, scarves, mufflers, mantillas, veils and the like (excluding articles of silk or silk waste, knitted or crocheted)	p/st	S	0.15
14192338	Shawls, scarves, mufflers, mantillas, veils and the like, of silk or silk waste (excluding knitted or crocheted)	p/st	S	0.15

PRODCOM code	Description of the PRODCOM code	QNT UNIT	Type	Conversion factor to kg
14192353	Ties, bow ties and cravats (excluding articles of silk or silk waste, knitted or crocheted)	p/st	S	0.15
14192358	Ties, bow ties and cravats, of silk or silk waste (excluding knitted or crocheted)	p/st	S	0.15
14192370	Gloves, mittens and mitts (excluding knitted or crocheted)	pa	S	0.1
14192393	Clothing accessories of textiles (excluding shawls, scarves and mufflers, mantillas and veils, ties, bow-ties and cravats, gloves, mittens and mitts, knitted or crocheted)	NA	NA	NA
14192395	Parts of garments or of clothing accessories, of textiles (excluding bras, girdles and corsets, braces, suspenders and garters, knitted or crocheted)	NA	NA	NA
14192396	Clothing accessories, parts of garments or of clothing accessories, of textiles, n.e.c. and parts thereof, (excluding shawls, scarves and mufflers, mantillas and veils, ties, bow-ties and cravats, gloves, mittens and mitts and parts thereof; bras, girdles and corsets, braces, suspenders and garters, knitted or crocheted)	NA	S	NA
14193200	Garments made up of felt or non-wovens, textile fabrics impregnated or coated	p/st	S	0.1
14194130	Hat-forms, hat bodies and hoods, plateaux and manchons of felt (including slit manchons) (excluding those blocked to shape, those with made brims)	p/st	S	0.1
14194150	Hat-shapes, plaited or made by assembling strips of any material (excluding those blocked to shape, those with made brims, those lined or trimmed)	p/st	S	0.1
14194230	Felt hats and other felt headgear, made from hat bodies or hoods and plateaux	p/st	S	0.1
14194250	Hats and other headgear, plaited or made by assembling strips of any material	p/st	S	0.1
14194270	Hats and other headgear, knitted or crocheted or made-up from lace, felt or other textile fabric in the piece (but not in strips); hair-nets of any material	p/st	S	0.1
14194300	Other headgear (except headgear of rubber or of plastics, safety headgear and asbestos headgear); headbands, linings, covers, hat foundations, hat frames, peaks and chinstraps, for headgear	NA	S	NA
14311033	Panty hose and tights, of knitted or crocheted synthetic fibres, measuring per single yarn < 67 decitex	p/st	S	0.07
14311035	Panty hose and tights, of knitted or crocheted synthetic fibres, measuring per single yarn >= 67 decitex	p/st	S	0.07
14311037	Pantyhose and tights of textile materials, knitted or crocheted (excl. graduated compression hosiery, those of synthetic fibres and hosiery for babies)	p/st	S	0.07
14311050	Women's full-length or knee-length knitted or crocheted hosiery, measuring per single yarn < 67 decitex	pa	S	0.01
14311090	Knitted or crocheted hosiery and footwear (including socks; excluding women's full-length/knee-length hosiery, measuring <67decitex, panty-hose and tights, footwear with applied soles)	pa	S	0.07
14391031	Men's or boys' jerseys, pullovers, sweatshirts, waistcoats and cardigans, of wool or fine animal hair (excluding jerseys and pullovers containing >= 50 % of wool and weighing >= 600 g)	p/st	S	0.5
14391032	Women's or girls' jerseys, pullovers, sweatshirts, waistcoats and cardigans, of wool or fine animal hair (excluding jerseys and pullovers containing >= 50 % of wool and weighing >= 600 g)	p/st	S	0.5
14391033	Jerseys and pullovers, containing >= 50 % by weight of wool and weighing >= 600 g per article	p/st	S	0.3
14391053	Lightweight fine knit roll, polo or turtle neck jumpers and pullovers, of cotton	p/st	S	0.5
14391055	Lightweight fine knit roll, polo or turtle neck jumpers and pullovers, of man-made fibres	p/st	S	0.5
14391061	Men's or boys' jerseys, pullovers, sweatshirts, waistcoats and cardigans, of cotton (excluding lightweight fine knit roll, polo or turtle neck jumpers and pullovers)	p/st	S	0.5
14391062	Women's or girls' jerseys, pullovers, sweatshirts, waistcoats and cardigans, of cotton (excluding lightweight fine knit roll, polo or turtle neck jumpers and pullovers)	p/st	S	0.5
14391071	Men's or boys' jerseys, pullovers, sweatshirts, waistcoats and cardigans, of man-made fibres (excluding lightweight fine knit roll, polo or turtle neck jumpers and pullovers)	p/st	S	0.5

PRODCOM code	Description of the PRODCOM code	QNT UNIT	Type	Conversion factor to kg
14391072	Women's or girls' jerseys, pullovers, sweatshirts, waistcoats and cardigans, of man-made fibres (excluding lightweight fine knit roll, polo or turtle neck jumpers and pullovers)	p/st	S	0.5
14391090	Jerseys, pullovers, sweatshirts, waistcoats and cardigans, of textile materials (excluding those of wool or fine animal hair, cotton, man-made fibres)	p/st	S	0.5

9332

9333

9334

9335

9336

QNTUNIT: PRODCOM indicator about unit used to report quantities.

kg = kilogram; p/st = number of items; pa = number of pairs; NA = data not available

Source: own elaboration based on Eurostat data set DS-056120 (data extracted on 15 November 2023). Conversion factors used to convert the unit reported in EUROSTAT to kg based on (Huygens et al., 2023)

Table 112. PRODCOM codes describing the subgroup of home/interior textiles in 2019

PRODCOM code	Description of the PRODCOM code	QNTUNIT	TYPE	Conversion Factor to kg
13921130	Blankets and travelling rugs of wool or fine animal hair (excluding electric blankets)	p/st	S	0.5
13921150	Blankets and travelling rugs of synthetic fibres (excluding electric blankets)	p/st	S	0.5
13921190	Blankets (excluding electric blankets) and travelling rugs of textile materials (excluding of wool or fine animal hair, of synthetic fibres)	p/st	S	0.5
13921230	Bed linen of knitted or crocheted textiles	kg	S	1
13921253	Bed linen of cotton (excluding knitted or crocheted)	kg	S	1
13921255	Bed linen of flax or ramie (excluding knitted or crocheted)	kg	S	1
13921259	Bed linen of woven textiles (excluding of cotton, of flax or ramie)	kg	S	1
13921270	Bed linen of non-woven man-made fibres (excluding knitted or crocheted)	kg	S	1
13921330	Table linen of knitted or crocheted textiles	kg	S	1
13921353	Table linen of cotton (excluding knitted or crocheted)	kg	S	1
13921355	Table linen of flax (excluding knitted or crocheted)	kg	S	1
13921359	Table linen of woven man-made fibres and of other woven or non-woven textiles (excluding of cotton, of flax)	kg	S	1
13921370	Table linen of non-woven man-made fibres	kg	S	1
13921430	Toilet linen and kitchen linen, of terry towelling or similar terry fabrics of cotton	kg	S	1
13921450	Woven toilet linen and kitchen linen, of textiles (excluding terry towelling or similar terry fabrics of cotton)	kg	S	1
13921470	Toilet linen and kitchen linen, of non-woven man-made fibres	kg	S	1
13921530	Curtains and interior blinds, curtain or bed valances, of knitted or crocheted materials	m ²	S	0.25
13921550	Curtains and interior blinds, curtain or bed valances, of woven materials	m ²	S	0.25
13921570	Curtains and interior blinds, curtain or bed valances, of non-woven materials	m ²	S	0.25
13921620	Hand-woven tapestries of the type Gobelins, Flanders, Aubusson, Beauvais, and needle-worked tapestries (including petit point, cross-stitch) whether or not made up	NA	S	NA
13921640	Bedspreads (excluding eiderdowns)	p/st	S	0.5
13921660	Furnishing articles including furniture and cushion covers as well as cushion covers, etc. for car seats (excluding blankets, travelling rugs, bed linen, table linen, toilet linen, kitchen linen, curtains, blinds, valances and bedspreads)	NA	S	NA
13921680	Sets of woven fabrics and yarns for making up into rugs, tapestries, embroidered table cloths, serviettes, or similar textile articles, p.r.s.	NA	S	NA
13922130	Sacks and bags, of cotton, used for packing goods	kg	S	1

PRODCOM code	Description of the PRODCOM code	QNTUNIT	TYPE	Conversion Factor to kg
13922150	Sacks and bags, of knitted or crocheted polyethylene or polypropylene strip, used for packing goods	kg	S	1
13922170	Sacks and bags, of polyethylene or polypropylene strip, used for packing goods (excluding knitted or crocheted)	kg	S	1
13922173	Sacks and bags, of polyethylene or polypropylene strip, weighing <= 120 g/m2, used for packing goods (excluding knitted or crocheted)	NA	NA	NA
13922175	Sacks and bags, of polyethylene or polypropylene strip, weighing > 120 g/m2, used for packing goods (excluding knitted or crocheted)	NA	NA	NA
13922190	Sacks and bags, used for packing goods (excluding of cotton, polyethylene or polypropylene strip)	kg	S	1
13922210	Tarpaulins, awnings and sunblinds (excluding caravan awnings)	kg	S	1
13922230	Tents (including caravan awnings)	kg	S	1
13922250	Sails	kg	S	1
13922270	Pneumatic mattresses and other camping goods (excluding caravan awnings, tents, sleeping bags)	kg	S	1
13922300	Parachutes and rotochutes, parts and accessories (including dirigible parachutes)	kg	S	1
13922430	Sleeping bags	p/st	S	0.5
13922493	Articles of bedding of feathers or down (including quilts and eiderdowns, cushions, pouffes, pillows) (excluding mattresses, sleeping bags)	p/st	S	0.5
13922499	Articles of bedding filled other than with feathers or down (including quilts and eiderdowns, cushions, pouffes, pillows) (excluding mattresses, sleeping bags)	p/st	S	0.5
13922953	Floor-cloths, dish-cloths, dusters and similar cleaning cloths, of non-woven textiles	kg	S	1
13922957	Floor-cloths, dish-cloths, dusters and similar cleaning cloths (excluding knitted or crocheted, articles of non-woven textiles)	kg	S	1
13922990	Floor-cloths, dish-cloths, dusters and similar cleaning cloths, knitted or crocheted; life-jackets, life-belts and other made up articles	NA	NA	NA
13922993	Sanitary towels, tampons and similar article of textile materials (excluding wadding)	NA	NA	NA
13922997	Napkins and napkin liners for babies and similar article of textile materials (excluding wadding)	NA	NA	NA
13922998	Floor-cloths, dish-cloths, dusters and similar cleaning cloths, knitted or crocheted; life-jackets, life-belts and other made up articles (excluding protective face masks, sanitary towels and napkins and similar articles)	NA	S	NA
13922999	Floor-cloths, dish-cloths, dusters and similar cleaning cloths, knitted or crocheted; life-jackets, life-belts and other made up articles (excluding sanitary towels and napkins and similar articles)	kg	NA	1
13931100	Knotted carpets and other knotted textile floor coverings	m ²	S	1.6
13931200	Woven carpets and other woven textile coverings (excluding tufted or flocked)	m ²	S	1.6
13931300	Tufted carpets and other tufted textile floor coverings	m ²	S	1.38
13931930	Needlefelt carpets and other needlefelt textile floor coverings (excluding tufted or flocked)	m ²	S	1.6
13931990	Carpets and other textile floor coverings (excluding knotted, woven, tufted, needlefelt)	m ²	S	1.6

QNTUNIT: PRODCOM indicator about unit used to report quantities.

kg = kilogram; p/st = number of items; NA = data not available, m²=square metres

Source: own elaboration based on Eurostat data set DS-056120 (data extracted on 15 November 2023). Conversion factors used to convert the unit reported in EUROSTAT to kg based on (Huygens et al., 2023)

Table 113. PRODCOM codes describing the subgroup of footwear in 2019

PRODCOM code	Description of the PRODCOM code	QNTUNIT	TYPE	Conversion Factor to kg
15201100	Waterproof footwear, with uppers in rubber or plastics (excluding incorporating a protective metal toecap)	pa	S	0.9
15201210	Sandals with rubber or plastic outer soles and uppers (including thong-type sandals, flip flops)	pa	S	0.5
15201231	Town footwear with rubber or plastic uppers	pa	S	0.9

PRODCOM code	Description of the PRODCOM code	QNTUNIT	TYPE	Conversion Factor to kg
15201237	Slippers and other indoor footwear with rubber or plastic outer soles and plastic uppers (including bedroom and dancing slippers, mules)	pa	S	0.35
15201330	Footwear with a wooden base and leather uppers (including clogs) (excluding with an inner sole or a protective metal toe-cap)	pa	S	0.9
15201351	Men's town footwear with leather uppers (including boots and shoes; excluding waterproof footwear, footwear with a protective metal toe-cap)	pa	S	0.9
15201352	Women's town footwear with leather uppers (including boots and shoes; excluding waterproof footwear, footwear with a protective metal toe-cap)	pa	S	0.9
15201353	Children's town footwear with leather uppers (including boots and shoes; excluding waterproof footwear, footwear with a protective metal toe-cap)	pa	S	0.9
15201361	Men's sandals with leather uppers (including thong type sandals, flip flops)	pa	S	0.5
15201362	Women's sandals with leather uppers (including thong type sandals, flip flops)	pa	S	0.5
15201363	Children's sandals with leather uppers (including thong type sandals, flip flops)	pa	S	0.5
15201370	Slippers and other indoor footwear with rubber, plastic or leather outer soles and leather uppers (including dancing and bedroom slippers, mules)	pa	S	0.35
15201380	Footwear with wood, cork or other outer soles and leather uppers (excluding outer soles of rubber, plastics or leather)	pa	S	0.9
15201444	Slippers and other indoor footwear (including dancing and bedroom slippers, mules) with uppers of textile materials	pa	S	0.35
15201445	Footwear with rubber, plastic or leather outer soles and textile uppers (excluding slippers and other indoor footwear, sports footwear)	pa	S	0.9
15201446	Footwear with textile uppers (excluding slippers and other indoor footwear as well as footwear with outer soles of rubber, plastics, leather or composition leather)	pa	S	0.9
15202100	Sports footwear with rubber or plastic outer soles and textile uppers (including tennis shoes, basketball shoes, gym shoes, training shoes and the like)	pa	S	0.9
15202900	Other sports footwear, except snow-ski footwear and skating boots	pa	S	0.9
15203120	Footwear (including waterproof footwear), incorporating a protective metal toecap, with outer soles and uppers of rubber or of plastics	pa	S	0.9
15203150	Footwear with rubber, plastic or leather outer soles and leather uppers, and with a protective metal toe-cap	pa	S	0.9
15203200	Wooden footwear, miscellaneous special footwear and other footwear n.e.c.	pa	S	0.9
15204020	Leather uppers and parts thereof of footwear (excluding stiffeners)	NA	S	NA
15204050	Uppers and parts thereof of footwear (excluding stiffeners, of leather)	NA	S	NA
15204080	Parts of footwear (excluding uppers) other materials	NA	S	NA

QNTUNIT: PRODCOM indicator about unit used to report quantities.

pa = number of pairs; NA = data not available

Source: own elaboration based on Eurostat data set DS-056120 (data extracted on 15 November 2023). Conversion factors used to convert the unit reported in EUROSTAT to kg based on (Huygens *et al.*, 2023).

Table 114. PRODCOM codes describing the subgroup of technical textiles in 2019

PRODCOM code	Description of the PRODCOM code	QNTUNIT	TYPE	Conversion Factor to kg
13941130	Twine, cordage, rope or cables, of sisal or other textile fibres of 'agave', of jute or other textile bast fibres and hard leaf fibres (excluding binder or baler twine)	kg	S	1
13941133	Twine, cordage, rope or cables, of sisal or other textile fibres of 'agave' measuring >100,000 decitex, of jute or other textile bast fibres and hard leaf fibres (excluding binder or baler twine)	NA	NA	NA
13941135	Twines of sisal measuring <= 100,000 decitex (10 g/m) (excluding binder or baler twine)	NA	NA	NA
13941153	Sisal binder or baler (agricultural) twines	kg	S	1
13941155	Polyethylene or polypropylene binder or baler (agricultural) twines	kg	S	1
13941160	Cordage, ropes or cables of polyethylene, polypropylene, nylon or other polyamides or of polyesters measuring > 50 000 decitex, of other synthetic fibres (excluding binder or baler twine)	kg	S	1
13941170	Twines of polyethylene or polypropylene, of nylon or other polyamides or polyesters measuring <= 50 000 decitex (5 g/m) (excluding binder or baler twine)	kg	S	1
13941190	Twines, cordage, rope and cables of textile materials (excluding jute and other textile bast fibres, sisal, abaca or other hard leaf fibres, synthetic fibres)	kg	S	1
13941233	Made-up fishing nets from twine, cordage or rope of man-made fibres (excluding fish landing nets)	kg	S	1
13941235	Made-up fishing nets from yarn of man-made fibres (excluding fish landing nets)	kg	S	1
13941253	Made-up nets from twine, cable or rope of nylon or other polyamides (excluding netting in the piece produced by crochet, hairnets, sports and fishing nets)	kg	S	1
13941255	Made-up nets of nylon or other polyamides (excluding netting in the piece produced by crochet, hairnets, sports and fishing nets, those made from twine, cable or rope)	kg	S	1
13941259	Knotted netting of textile materials (excluding made-up fishing nets of man-made textiles, other made-up nets of nylon or other polyamides)	kg	S	1
13941280	Articles of twine, cordage, rope or cables	kg	S	1
13951010	Non-wovens of a weight <= 25 g/m ² (including articles made from non-wovens) (excluding articles of apparel, coated or covered)	kg	S	1
13951020	Non-wovens of a weight of > 25 g/m ² but <= 70 g/m ² (including articles made from non-wovens) (excluding articles of apparel, coated or covered)	kg	S	1
13951030	Non-wovens of a weight of > 70 g/m ² but <= 150 g/m ² (including articles made from non-wovens) (excluding articles of apparel, coated or covered)	kg	S	1
13951050	Non-wovens of a weight of > 150 g/m ² (including articles made from non-wovens) (excluding articles of apparel, coated or covered)	kg	S	1
13951070	Non-wovens, coated or covered (including articles made from non-wovens) (excluding articles of apparel)	kg	S	1
13961100	Metallised yarn or metallised gimped yarn	kg	S	1
13961200	Woven fabrics of metal thread and woven fabrics of metallised yarn, used in apparel, as furnishing fabrics or similar purposes	kg	S	1
13961300	Rubber thread and cord, textile covered; textile yarn and strip impregnated, coated, covered or sheathed with rubber or plastics	kg	S	1
13961400	Textile fabrics, impregnated, coated or covered n.e.c.	m ²	S	0.25
13961500	Tyre cord fabrics of high tenacity yarn, of nylon, other polyamides, polyesters or viscose rayon	m ²	S	0.25
13961620	Textile hosepipe and similar textile tubing, whether or not impregnated or coated, with or without lining, armour or accessories of other materials	kg	S	1
13961650	Textile wicks, conveyor belts or belting (including reinforced with metal or other material)	kg	S	1
13961680	Textile fabrics and felts, for paper-making machines or similar machines (including for pulp or asbestos-cement)	kg	S	1
13961730	Narrow woven fabrics other than labels, badges and other similar articles	NA	S	NA
13961750	Labels, badges and similar articles in textile materials (excluding embroidered)	NA	S	NA

9347
9348
9349

PRODCOM code	Description of the PRODCOM code	QNTUNIT	TYPE	Conversion Factor to kg
13961770	Braids in the piece; tassels and pompons, ornamental trimmings (excluding knitted or crocheted)	NA	S	NA
13991130	Tulles and other net fabrics (excluding woven, knitted or crocheted)	NA	S	NA
13991150	Machine-made lace in the piece, in strips or in motifs	NA	S	NA
13991170	Hand-made lace in the piece, in strips or in motifs	NA	S	NA
13991230	Embroidery (without visible ground) in the piece, in strips or in motifs	NA	S	NA
13991250	Cotton embroidery in the piece, in strips or in motifs (excluding embroidery without visible ground)	NA	S	NA
13991270	Embroidery of textiles in the piece, in strips or in motifs (excluding without visible ground, cotton)	NA	S	NA
13991300	Felt, whether or not impregnated, coated, covered or laminated, n.e.c.	kg	S	1
13991400	Textile flock and dust and mill neps	kg	S	1
13991500	Gimped yarn and gimped strip and the like, of man-made textile materials of an apparent width <= 5 mm; chenille yarn; loop wale-yarn	kg	S	1
13991600	Quilted textile products in the piece (excluding embroidery)	m ²	S	0.25
13991900	Powder-puffs and pads for the application of cosmetics or toilet preparations	p/st	S	0.5

QNTUNIT: PRODCOM indicator about unit used to report quantities.

kg = kilogram; p/st = number of items; NA = data not available, m²=square metres

Source: own elaboration based on Eurostat data set DS-056120 (data extracted on 15 November 2023). Conversion factors used to convert the unit reported in EUROSTAT to kg based on (Huygens et al., 2023).

9350

13.2.4 PRODCOM codes included in the scope

9351

Table 115. List and description of PRODCOM codes included in the scope

Category	PRODCOM code	Description of the PRODCOM code	Allocation of the code in the product category (%)	Reported measure unit in PRODCOM	Conversion Factor to kg
1. T-shirts	14143000	T-shirts, singlets and vests, knitted or crocheted	100	p/st	0.17
	14142240	Men's or boys' singlets, vests, bathrobes, dressing gowns and similar articles (excluding knitted or crocheted)	75	p/st	0.3
	14142460	Women's or girls' singlets and other vests, briefs, panties, negligees, bathrobes, dressing gowns, housecoats and similar articles of cotton (excluding knitted or crocheted)	40	p/st	0.08
	14142480	Women's or girls' negligees, bathrobes, dressing gowns, singlets, vests, briefs and panties (including boxer shorts), of man-made fibres (excluding knitted or crocheted)	40	p/st	0.08
	14142489	Women's or girls' singlets, vests, briefs, panties, negligees, bathrobes, dressing gowns and similar articles, of textiles (excluding cotton, man-made fibres, knitted or crocheted)	40	p/st	0.08
	14123013	Men's or boys' other garments, of cotton or man-made fibres, for industrial or occupational wear	25	p/st	0.5
	14123023	Women's or girls' other garments, of cotton or man-made fibres, for industrial or occupational wear	25	p/st	0.5
	14391053	Lightweight fine knit roll, polo or turtle neck jumpers and pullovers, of cotton	20	p/st	0.5
	14142430	Women's or girls' nightdresses and pyjamas (excluding knitted or crocheted)	25	p/st	0.3
	14142230	Men's or boys' nightshirts and pyjamas (excluding knitted or crocheted)	25	p/st	0.15
	14141430	Women's or girls' nighties and pyjamas, of knitted or crocheted textiles	25	p/st	0.15

Category	PRODCOM code	Description of the PRODCOM code	Allocation of the code in the product category (%)	Reported measure unit in PRODCOM	Conversion Factor to kg
	14391055	Lightweight fine knit roll, polo or turtle neck jumpers and pullovers, of man-made fibres	20	p/st	0.5
	14141230	Men's or boys' nightshirts and pyjamas, of knitted or crocheted textiles	25	p/st	0.15
	14191290	Other garments, knitted or crocheted (including bodies with a proper sleeve)	100	kg	1
2. Shirts and blouses	14141100	Men's or boys' shirts, knitted or crocheted	100	p/st	0.25
	14141310	Women's or girls' blouses, shirts and shirt-blouses, of knitted or crocheted textiles	100	p/st	0.25
	14142100	Men's or boys' shirts (excluding knitted or crocheted)	100	p/st	0.25
	14142300	Women's or girls' blouses, shirts and shirt-blouses (excluding knitted or crocheted)	100	p/st	0.25
	14123013	Men's or boys' other garments, of cotton or man-made fibres, for industrial or occupational wear	25	p/st	0.5
	14123023	Women's or girls' other garments, of cotton or man-made fibres, for industrial or occupational wear	25	p/st	0.5
	14141230	Men's or boys' nightshirts and pyjamas, of knitted or crocheted textiles	25	p/st	0.15
	14142430	Women's or girls' nightdresses and pyjamas (excluding knitted or crocheted)	25	p/st	0.3
	14142230	Men's or boys' nightshirts and pyjamas (excluding knitted or crocheted)	25	p/st	0.15
	14141430	Women's or girls' nighties and pyjamas, of knitted or crocheted textiles	25	p/st	0.15
	14391053	Lightweight fine knit roll, polo or turtle neck jumpers and pullovers, of cotton	40	p/st	0.5
	14391055	Lightweight fine knit roll, polo or turtle neck jumpers and pullovers, of man-made fibres	40	p/st	0.5
3. Sweaters and mid-layers	14391031	Men's or boys' jerseys, pullovers, sweatshirts, waistcoats and cardigans, of wool or fine animal hair (excluding jerseys and pullovers containing >= 50 % of wool and weighing >= 600 g)	100	p/st	0.5
	14391032	Women's or girls' jerseys, pullovers, sweatshirts, waistcoats and cardigans, of wool or fine animal hair (excluding jerseys and pullovers containing >= 50 % of wool and weighing >= 600 g)	100	p/st	0.5
	14391033	Jerseys and pullovers, containing >= 50 % by weight of wool and weighing >= 600 g per article	100	p/st	0.3
	14391053	Lightweight fine knit roll, polo or turtle neck jumpers and pullovers, of cotton	40	p/st	0.5
	14391055	Lightweight fine knit roll, polo or turtle neck jumpers and pullovers, of man-made fibres	40	p/st	0.5
	14391061	Men's or boys' jerseys, pullovers, sweatshirts, waistcoats and cardigans, of cotton (excluding lightweight fine knit roll, polo or turtle neck jumpers and pullovers)	100	p/st	0.5
	14391062	Women's or girls' jerseys, pullovers, sweatshirts, waistcoats and cardigans, of cotton (excluding lightweight fine knit roll, polo or turtle neck jumpers and pullovers)	100	p/st	0.5
	14132130	Men's or boys' waistcoats, anoraks, ski-jackets, wind-jackets and similar articles (excluding jackets and blazers, knitted or crocheted, impregnated, coated, covered, laminated or rubberised)	20	p/st	0.95
	14133130	Women's or girls' waistcoats, anoraks, ski-jackets, wind-jackets and similar articles (excluding jackets and blazers, knitted or crocheted, impregnated, coated, covered, laminated or rubberised)	20	p/st	0.95
	14391071	Men's or boys' jerseys, pullovers, sweatshirts, waistcoats and cardigans, of man-made fibres (excluding lightweight fine knit roll, polo or turtle neck jumpers and pullovers)	100	p/st	0.5
	14391072	Women's or girls' jerseys, pullovers, sweatshirts, waistcoats and cardigans, of man-made fibres (excluding lightweight fine knit roll, polo or turtle neck jumpers and pullovers)	100	p/st	0.5

Category	PRODCOM code	Description of the PRODCOM code	Allocation of the code in the product category (%)	Reported measure unit in PRODCOM	Conversion Factor to kg
	14391090	Jerseys, pullovers, sweatshirts, waistcoats and cardigans, of textile materials (excluding those of wool or fine animal hair, cotton, man-made fibres)	100	p/st	0.5
	14131120	Men's or boys' waistcoats, anoraks, ski-jackets, wind-cheaters, wind-jackets and similar articles, of knitted or crocheted textiles (excluding jackets and blazers)	20	p/st	0.95
	14131320	Women's or girls' waistcoats, anoraks, ski-jackets, wind-cheaters, wind-jackets and similar articles, of knitted or crocheted textiles (excluding jackets and blazers)	20	p/st	0.95
4. Jackets and coats	14121130	Men's or boys' jackets and blazers, of cotton or man-made fibres, for industrial and occupational wear	100	p/st	0.95
	14122130	Women's or girls' jackets and blazers, of cotton or man-made fibres, for industrial or occupational wear	100	p/st	0.95
	14131110	Men's or boys' overcoats, car-coats, capes, cloaks and similar articles, of knitted or crocheted textiles (excluding jackets and blazers, anoraks, wind-cheaters and wind-jackets)	100	p/st	0.95
	14131120	Men's or boys' waistcoats, anoraks, ski-jackets, wind-cheaters, wind-jackets and similar articles, of knitted or crocheted textiles (excluding jackets and blazers)	80	p/st	0.95
	14131230	Men's or boys' jackets and blazers, of knitted or crocheted textiles	100	p/st	0.95
	14131310	Women's or girls' overcoats, car-coats, capes, cloaks and similar articles, of knitted or crocheted textiles (excluding jackets and blazers)	100	p/st	0.95
	14131320	Women's or girls' waistcoats, anoraks, ski-jackets, wind-cheaters, wind-jackets and similar articles, of knitted or crocheted textiles (excluding jackets and blazers)	80	p/st	0.95
	14131430	Women's or girls' jackets and blazers, of knitted or crocheted textiles	100	p/st	0.95
	14132115	Men's or boys' raincoats, overcoats, car-coats, capes, etc.	100	p/st	0.5
	14132130	Men's or boys' waistcoats, anoraks, ski-jackets, wind-jackets and similar articles (excluding jackets and blazers, knitted or crocheted, impregnated, coated, covered, laminated or rubberised)	80	p/st	0.95
	14132300	Men's or boys' jackets and blazers (excluding knitted or crocheted)	100	p/st	0.95
	14133110	Woman's or girls' raincoats	100	p/st	0.5
	14133115	Woman's or girls' raincoats and overcoats, etc.	100	p/st	0.5
	14133120	Woman's or girls' overcoats, etc.	100	p/st	0.95
	14133130	Women's or girls' waistcoats, anoraks, ski-jackets, wind-jackets and similar articles (excluding jackets and blazers, knitted or crocheted, impregnated, coated, covered, laminated or rubberised)	80	p/st	0.95
	14133330	Women's or girls' jackets and blazers (excluding knitted or crocheted)	100	p/st	0.95
	14192230	Ski-suits (excluding of knitted or crocheted textiles)	50	p/st	0.5
	14191230	Ski-suits, of knitted or crocheted textiles	50	p/st	0.5
	14191100	Babies' garments and clothing accessories, knitted or crocheted including vests, rompers, underpants, stretch-suits, gloves or mittens or mitts, outerwear (for children of height <= 86 cm)	20	kg	1
	14192150	Babies clothing and accessories, of textiles, not knitted or crocheted (for children of height <= 86 cm) including vests, rompers, underpants, stretch-suits, gloves, mittens and outerwear (excluding sanitary towels and napkins and similar articles)	20	kg	1
	14131260	Men's or boys' suits and ensembles, of knitted or crocheted textiles	40	p/st	0.5

Category	PRODCOM code	Description of the PRODCOM code	Allocation of the code in the product category (%)	Reported measure unit in PRODCOM	Conversion Factor to kg
	14131460	Women's or girls' suits and ensembles, of knitted or crocheted textiles	40	p/st	0.5
	14132200	Men's or boys' suits & ensembles (excluding knitted or crocheted)	40	p/st	0.5
	14133200	Women's or girls' suits & ensembles (excluding knitted or crocheted)	40	p/st	0.5
	14132110	Men's or boys' raincoats	100	p/st	0.5
	14132120	Men's or boys' overcoats, car-coats, capes, etc.	100	p/st	0.95
	14132210	Men's or boys' suits (excluding knitted or crocheted)	50	p/st	1.4
	14132220	Men's or boys' ensembles (excluding knitted or crocheted)	50	p/st	1
	14133210	Women's or girls' suits (excluding knitted or crocheted)	50	p/st	1.4
	14133220	Women's or girls' ensembles (excluding knitted or crocheted)	50	p/st	1
	14192100	Babies' clothing and accessories, of textiles, not knitted or crocheted (for children of height <= 86 cm) including vests, rompers, underpants, stretch-suits, napkins, gloves, mittens and outerwear	20	kg	1
5. Pants and shorts	14131270	Men's or boys' trousers, breeches, shorts, bib and brace overalls, of knitted or crocheted textiles	100	p/st	0.45
	14131490	Women's or girls' trousers, breeches, shorts, bib and brace overalls, of knitted or crocheted textiles	100	p/st	0.45
	14132442	Men's or boys' trousers and breeches, of denim (excluding for industrial or occupational wear)	100	p/st	0.45
	14132444	Men's or boys' trousers, breeches and shorts, of wool or fine animal hair (excluding knitted or crocheted, for industrial or occupational wear)	100	p/st	0.45
	14132445	Men's or boys' trousers and breeches, of man-made fibres (excluding knitted or crocheted, for industrial or occupational wear)	100	p/st	0.45
	14132448	Men's or boys' trousers and breeches, of cotton (excluding denim, knitted or crocheted)	100	p/st	0.45
	14132449	Men's or boys' trousers, breeches, shorts and bib and brace overalls (excluding of wool, cotton and man-made fibres, knitted or crocheted)	100	p/st	0.45
	14141230	Men's or boys' nightshirts and pyjamas, of knitted or crocheted textiles	50	p/st	0.15
	14132455	Men's or boys' bib and brace overalls (excluding knitted or crocheted, for industrial or occupational wear)	100	p/st	0.45
	14132460	Men's or boys' shorts, of cotton or man-made fibres (excluding knitted or crocheted)	100	p/st	0.45
	14133542	Women's or girls' trousers and breeches, of denim (excluding for industrial or occupational wear)	100	p/st	0.45
	14133548	Women's or girls' trousers and breeches, of cotton (excluding denim, for industrial or occupational wear)	100	p/st	0.45
	14133549	Women's or girls' trousers and breeches, of wool or fine animal hair or man-made fibres (excluding knitted or crocheted and for industrial and occupational wear)	100	p/st	0.45
	14133551	Women's or girls' bib and brace overalls, of cotton (excluding knitted or crocheted, for industrial or occupational wear)	100	p/st	0.45
	14142430	Women's or girls' nightdresses and pyjamas (excluding knitted or crocheted)	30	p/st	0.3
	14133561	Women's or girls' shorts, of cotton (excluding knitted and crocheted)	100	p/st	0.45

Category	PRODCOM code	Description of the PRODCOM code	Allocation of the code in the product category (%)	Reported measure unit in PRODCOM	Conversion Factor to kg
	14133563	Women's or girls' bib and brace overalls, of wool or fine animal hair and man-made fibres (excluding cotton, knitted or crocheted, for industrial or occupational wear) and women's or girls' shorts, of wool or fine animal hair (excluding knitted or crocheted)	100	p/st	0.45
	14133565	Women's or girls' shorts, of man-made fibres (excluding knitted or crocheted)	100	p/st	0.45
	14133569	Women's or girls' trousers, breeches, bib and brace overalls, of textiles (excluding cotton, wool or fine animal hair, man-made fibres, knitted or crocheted)	100	p/st	0.45
	14121240	Men's or boys' trousers and breeches, of cotton or man-made fibres, for industrial or occupational wear	100	p/st	0.45
	14121250	Men's or boys' bib and brace overalls, of cotton or man-made fibres, for industrial or occupational wear	100	p/st	0.45
	14122240	Women's or girls' trousers and breeches, of cotton or man-made fibres, for industrial or occupational wear	100	p/st	0.45
	14141430	Women's or girls' nighties and pyjamas, of knitted or crocheted textiles	30	p/st	0.15
	14142230	Men's or boys' nightshirts and pyjamas (excluding knitted or crocheted)	50	p/st	0.15
	14122250	Women's or girls' bib and brace overalls, of cotton or man-made fibres, for industrial or occupational wear	100	p/st	0.45
	14192230	Ski-suits (excluding of knitted or crocheted textiles)	50	p/st	0.5
	14191230	Ski-suits, of knitted or crocheted textiles	50	p/st	0.5
	14131260	Men's or boys' suits and ensembles, of knitted or crocheted textiles	40	p/st	0.5
	14131460	Women's or girls' suits and ensembles, of knitted or crocheted textiles	40	p/st	0.5
	14132200	Men's or boys' suits & ensembles (excluding knitted or crocheted)	40	p/st	0.5
	14133200	Women's or girls' suits & ensembles (excluding knitted or crocheted)	40	p/st	0.5
	14123013	Men's or boys' other garments, of cotton or man-made fibres, for industrial or occupational wear	50	p/st	0.5
	14123023	Women's or girls' other garments, of cotton or man-made fibres, for industrial or occupational wear	50	p/st	0.5
	14132210	Men's or boys' suits (excluding knitted or crocheted)	50	p/st	1
	14132220	Men's or boys' ensembles (excluding knitted or crocheted)	50	p/st	1
	14133210	Women's or girls' suits (excluding knitted or crocheted)	25	p/st	1
	14133220	Women's or girls' ensembles (excluding knitted or crocheted)	25	p/st	1
6. Dresses, Skirts and jumpsuits	14131470	Women's or girls' dresses, of knitted or crocheted textiles	100	p/st	0.3
	14131480	Women's or girls' skirts and divided skirts, of knitted or crocheted textiles	100	p/st	0.25
	14133470	Women's or girls' dresses (excluding knitted or crocheted)	100	p/st	0.3
	14133480	Women's or girls' skirts and divided skirts (excluding knitted or crocheted)	100	p/st	0.25
	14131260	Men's or boys' suits and ensembles, of knitted or crocheted textiles	20	p/st	0.5
	14121120	Men's or boys' ensembles, of cotton or man-made fibres, for industrial and occupational wear	100	p/st	0.5
	14191210	Track-suits, of knitted or crocheted textiles	100	p/st	0.5
	14141430	Women's or girls' nighties and pyjamas, of knitted or crocheted textiles	20	p/st	0.15
	14132200	Men's or boys' suits & ensembles (excluding knitted or crocheted)	20	p/st	0.5

Category	PRODCOM code	Description of the PRODCOM code	Allocation of the code in the product category (%)	Reported measure unit in PRODCOM	Conversion Factor to kg
	14142489	Women's or girls' singlets, vests, briefs, panties, negligees, bathrobes, dressing gowns and similar articles, of textiles (excluding cotton, man-made fibres, knitted or crocheted)	30	p/st	0.08
	14142450	Women's or girls' slips and petticoats (excluding knitted or crocheted)	100	p/st	0.5
	14142480	Women's or girls' negligees, bathrobes, dressing gowns, singlets, vests, briefs and panties (including boxer shorts), of man-made fibres (excluding knitted or crocheted)	30	p/st	0.08
	14142430	Women's or girls' nightdresses and pyjamas (excluding knitted or crocheted)	20	p/st	0.3
	14142240	Men's or boys' singlets, vests, bathrobes, dressing gowns and similar articles (excluding knitted or crocheted)	25	p/st	0.3
	14141440	Women's or girls' negligees, bathrobes, dressing gowns and similar articles, of knitted or crocheted textiles	100	p/st	0.3
	14141450	Women's or girls' slips and petticoats, of knitted or crocheted textiles	100	p/st	0.5
	14141240	Men's or boys' dressing gowns, bathrobes and similar articles, of knitted or crocheted textiles	100	p/st	0.3
	14192210	Other men's or boys' apparel n.e.c., including tracksuits and jogging suits (excluding waistcoats, ski-suits, knitted or crocheted)	100	p/st	0.5
	14192220	Other women's or girls' apparel n.e.c., including tracksuits and jogging suits (excluding waistcoats, ski-suits, knitted or crocheted)	100	p/st	0.5
	14122120	Women's or girls' ensembles, of cotton or man-made fibres, for industrial or occupational wear	100	p/st	0.5
	14131460	Women's or girls' suits and ensembles, of knitted or crocheted textiles	20	p/st	0.5
	14133200	Women's or girls' suits & ensembles (excluding knitted or crocheted)	20	p/st	0.5
	14142460	Women's or girls' singlets and other vests, briefs, panties, negligees, bathrobes, dressing gowns, housecoats and similar articles of cotton (excluding knitted or crocheted)	30	p/st	0.08
	14133210	Women's or girls' suits (excluding knitted or crocheted)	25	p/st	1
	14133220	Women's or girls' ensembles (excluding knitted or crocheted)	25	p/st	1
7. Leggings, Stockings, Tights and socks	14311033	Panty hose and tights, of knitted or crocheted synthetic fibres, measuring per single yarn < 67 decitex	100	p/st	0.07
	14311035	Panty hose and tights, of knitted or crocheted synthetic fibres, measuring per single yarn >= 67 decitex	100	p/st	0.07
	14311037	Pantyhose and tights of textile materials, knitted or crocheted (excl. graduated compression hosiery, those of synthetic fibres and hosiery for babies)	100	p/st	0.07
	14311050	Women's full-length or knee-length knitted or crocheted hosiery, measuring per single yarn < 67 decitex	100	pa	0.01
	14311090	Knitted or crocheted hosiery and footwear (including socks; excluding women's full-length/knee-length hosiery, measuring <67decitex, panty-hose and tights, footwear with applied soles)	95	pa	0.07
8. Underwear	14141220	Men's or boys' underpants and briefs, of knitted or crocheted textiles (including boxer shorts)	100	p/st	0.08
	14141420	Women's or girls' briefs and panties, of knitted or crocheted textiles (including boxer shorts)	100	p/st	0.08
	14142220	Men's or boys' underpants and briefs (including boxer shorts) (excluding knitted or crocheted)	100	p/st	0.08

Category	PRODCOM code	Description of the PRODCOM code	Allocation of the code in the product category (%)	Reported measure unit in PRODCOM	Conversion Factor to kg
	14142530	Brassieres	100	p/st	0.05
	14142550	Girdles, panty-girdles and corselettes (including bodies with adjustable straps)	100	p/st	0.5
	14191100	Babies' garments and clothing accessories, knitted or crocheted including vests, rompers, underpants, stretch-suits, gloves or mittens or mitts, outerwear (for children of height <= 86 cm)	50	kg	1
	14192150	Babies clothing and accessories, of textiles, not knitted or crocheted (for children of height <= 86 cm) including vests, rompers, underpants, stretch-suits, gloves, mittens and outerwear (excluding sanitary towels and napkins and similar articles)	50	kg	1
	14142460	Women's or girls' singlets and other vests, briefs, panties, negligees, bathrobes, dressing gowns, housecoats and similar articles of cotton (excluding knitted or crocheted)	30	p/st	0.08
	14142480	Women's or girls' negligees, bathrobes, dressing gowns, singlets, vests, briefs and panties (including boxer shorts), of man-made fibres (excluding knitted or crocheted)	30	p/st	0.08
	14142489	Women's or girls' singlets, vests, briefs, panties, negligees, bathrobes, dressing gowns and similar articles, of textiles (excluding cotton, man-made fibres, knitted or crocheted)	30	p/st	0.08
	14192100	Babies' clothing and accessories, of textiles, not knitted or crocheted (for children of height <= 86 cm) including vests, rompers, underpants, stretch-suits, napkins, gloves, mittens and outerwear	50	kg	1
9. Swimwear	14191240	Men's or boys' swimwear, of knitted or crocheted textiles	100	p/st	0.12
	14191250	Women's or girls' swimwear, of knitted or crocheted textiles	100	p/st	0.12
	14192240	Men's or boys' swimwear (excluding of knitted or crocheted textiles)	100	p/st	0.12
	14192250	Women's or girls' swimwear (excluding of knitted or crocheted textiles)	100	p/st	0.12
	14191300	Gloves, mittens and mitts, of knitted or crocheted textiles	100	pa	0.1
10. Accessories	14191930	Shawls, scarves, mufflers, mantillas, veils and the like, of knitted or crocheted textiles	100	p/st	0.1
	14191960	Clothing accessories and parts thereof, of knitted or crocheted textiles (excluding gloves, mittens, shawls, scarves, mufflers, mantillas and veils)	100	p/st	1
	14192310	Handkerchiefs	100	p/st	0.5
	14192333	Shawls, scarves, mufflers, mantillas, veils and the like (excluding articles of silk or silk waste, knitted or crocheted)	100	p/st	0.15
	14192338	Shawls, scarves, mufflers, mantillas, veils and the like, of silk or silk waste (excluding knitted or crocheted)	100	p/st	0.15
	14192353	Ties, bow ties and cravats (excluding articles of silk or silk waste, knitted or crocheted)	100	p/st	0.15
	14192358	Ties, bow ties and cravats, of silk or silk waste (excluding knitted or crocheted)	100	p/st	0.15
	14192370	Gloves, mittens and mitts (excluding knitted or crocheted)	100	pa	0.1
	14192393	Clothing accessories of textiles (excluding shawls, scarves and mufflers, mantillas and veils, ties, bow-ties and cravats, gloves, mittens and mitts, knitted or crocheted)	100	NA	1
	14192395	Parts of garments or of clothing accessories, of textiles (excluding bras, girdles and corsets, braces, suspenders and garters, knitted or crocheted)	100	NA	1
	14192396	Clothing accessories, parts of garments or of clothing accessories, of textiles, n.e.c. and parts thereof, (excluding shawls, scarves and mufflers, mantillas and veils, ties, bow-ties and cravats, gloves, mittens and mitts and parts thereof; bras, girdles and corsets, braces, suspenders and garters, knitted or crocheted)	100	NA	1

Category	PRODCOM code	Description of the PRODCOM code	Allocation of the code in the product category (%)	Reported measure unit in PRODCOM	Conversion Factor to kg
	14194130	Hat-forms, hat bodies and hoods, plateaux and manchons of felt (including slit manchons) (excluding those blocked to shape, those with made brims)	100	p/st	0.1
	14194150	Hat-shapes, plaited or made by assembling strips of any material (excluding those blocked to shape, those with made brims, those lined or trimmed)	100	p/st	0.1
	14194230	Felt hats and other felt headgear, made from hat bodies or hoods and plateaux	100	p/st	0.1
	14194250	Hats and other headgear, plaited or made by assembling strips of any material	100	p/st	0.1
	14194270	Hats and other headgear, knitted or crocheted or made-up from lace, felt or other textile fabric in the piece (but not in strips); hair-nets of any material	100	p/st	0.1

9352 kg=kilogram, p/st= Number of items, pa=Number of pairs, NA= Not available

9353 Some codes were assigned to more than one product category with different allocation percentage. This allocation was an estimation of the authors based on the description of the PRODCOM code.

9354 Source: own elaboration based on Eurostat data set DS-056120 (data extracted on 15 November 2023). Conversion factors used to convert the unit reported in EUROSTAT to kg based on (Huygens, Dries et al., 2023).

9355 **Table 116.** Characteristics of data used for the market analysis

PRODCOM codes affected by changes	Subgroup	Product category ID and name (share of the code allocated to the category)	Codes description	Implications of the affected codes (rationale) over the existing codes (merging, newly introduced, complementing, etc.)	Affected indicator	Missing data
14123013	Apparel	1.T-shirts (25%) 2.Shirts and blouses (25%) 5.Pants and shorts (50%)	Men's or boys' other garments, of cotton or man-made fibres, for industrial or occupational wear	The 2 affected codes were added in 2001 to complement the codes about industrial and occupational wear: 14121120 14121130 14121240 14121250 14122120 14122130 14122240 14122250	PRODQNT EXPQNT IMPQNT PRODVAL EXPVAL IMPVAL	NA
14123023	Apparel	1.T-shirts (25%) 2.Shirts and blouses (25%) 5.Pants and shorts (25%)	Women's or girls' other garments, of cotton or man-made fibres, for industrial or occupational wear			NA
14131260	Apparel	4.Jackets and coats (40%) 5.Pants and shorts (40%) 6.Dresses, Skirts and jumpsuits (20%)	Men's or boys' suits and ensembles, of knitted or crocheted textiles	The 4 affected codes were added in 2001 to complement the codes about knitted and crocheted textiles groups: 14131110 14131120 14131230 14131310 14131320	PRODQNT EXPQNT IMPQNT PRODVAL EXPVAL IMPVAL	NA
14131270	Apparel	5.Pants and shorts (100%)	Men's or boys' trousers, breeches, shorts, bib and brace overalls, of knitted or crocheted textiles			NA

PRODCOM codes affected by changes	Subgroup	Product category ID and name (share of the code allocated to the category)	Codes description	Implications of the affected codes (rationale) over the existing codes (merging, newly introduced, complementing, etc.)	Affected indicator	Missing data
14131460	Apparel	4.Jackets and coats (40%) 5.Pants and shorts (40%) 6.Dresses, Skirts and jumpsuits (20%)	Women's or girls' suits and ensembles, of knitted or crocheted textiles	14131430 14131470 14131480)		NA
14131490	Apparel	5.Pants and shorts (100%)	Women's or girls' trousers, breeches, shorts, bib and brace overalls, of knitted or crocheted textiles			NA
14132115	Apparel	4.Jackets and coats (100%)	Men's or boys' raincoats, overcoats, car-coats, capes, etc.	The affected code was added in 2013, merging the following 2 codes: 14132110 14132120	PRODQNT EXPQNT IMPQNT PRODVAL EXPVAL IMPVAL	Data missing for 2022 due to updates of the PRODCOM list.
14132200	Apparel	4.Jackets and coats (40%) 5.Pants and shorts (40%) 6.Dresses, Skirts and jumpsuits (20%)	Men's or boys' suits & ensembles (excluding knitted or crocheted)	The affected code was added in 2013, merging the following 2 codes: 14132210 14132220	PRODQNT EXPQNT IMPQNT PRODVAL EXPVAL IMPVAL	Data missing for 2022.
14132445	Apparel	5.Pants and shorts (100%)	Men's or boys' trousers and breeches, of man-made fibres (excluding knitted or crocheted, for industrial or occupational wear)	The 3 affected codes start in 2001 as to complement the Men's or boys' trousers, breeches, shorts, bib and brace overalls groups divided into specific fibres excluding knitted and crocheted which are complete codes: 14132442 14132444 14132448 14132449 14132460	PRODQNT EXPQNT IMPQNT PRODVAL EXPVAL IMPVAL	Data missing for time interval 1995-2001 most probably due to introduction of new codes.
14132455	Apparel	5.Pants and shorts (100%)	Men's or boys' bib and brace overalls (excluding knitted or crocheted, for industrial or occupational wear)			Data missing for time interval 1995-2001 most probably due to introduction of new codes.
14131270	Apparel	5.Pants and shorts (100%)	Men's or boys' trousers, breeches, shorts, bib and brace overalls, of knitted or crocheted textiles			Data missing for time interval 1995-2001 most probably due to introduction of new codes.

PRODCOM codes affected by changes	Subgroup	Product category ID and name (share of the code allocated to the category)	Codes description	Implications of the affected codes (rationale) over the existing codes (merging, newly introduced, complementing, etc.)	Affected indicator	Missing data
14133115	Apparel	4.Jackets and coats (100%)	Woman's or girls' raincoats and overcoats, etc.	The affected code was added in 2012, merging the 2 following codes: 14133110 14133120	PRODQNT EXPQNT IMPQNT PRODVAL EXPVAL IMPVAL	Data missing for 2022 due to updates of the PRODCOM list
14133130	Apparel	3.Sweaters and mid-layers (20%) 4.Jackets and coats (80%)	Women's or girls' waistcoats, anoraks, ski-jackets, wind-jackets and similar articles (excluding jackets and blazers, knitted or crocheted, impregnated, coated, covered, laminated or rubberised)	NA	PRODQNT EXPQNT IMPQNT PRODVAL EXPVAL IMPVAL	Data missing for 2022 due to updates of the PRODCOM list
14133200	Apparel	4.Jackets and coats (40%) 5.Pants and shorts (40%) 6.Dresses, Skirts and jumpsuits (20%)	Women's or girls' suits & ensembles (excluding knitted or crocheted)	The affected code was added in 2012, merging the following 2 codes: 14133210 14133220	PRODQNT EXPQNT IMPQNT PRODVAL EXPVAL IMPVAL	NA
14133565	Apparel	5.Pants and shorts (100%)	Women's or girls' shorts, of man-made fibres (excluding knitted or crocheted)	The affected code was added in 2001 as to complement the Women's or girls' shorts groups divided into specific fibres (excluding knitted and crocheted which are complete codes): 14133551 14133561 14133563	PRODQNT EXPQNT IMPQNT PRODVAL EXPVAL IMPVAL	Data missing for time interval 1995-2001 most probably due to introduction of new codes.
14192150	Apparel	4.Jackets and coats (20%) 8.Underwear (50%)	Babies clothing and accessories, of textiles, not knitted or crocheted (for children of height <= 86 cm) including vests, rompers, underpants, stretch-suits, gloves, mittens and outerwear (excluding sanitary towels and napkins and similar articles)	The affected code was added in 2012 to complement Babies clothing and accessories: 14192100	PRODVAL EXPVAL IMPVAL	NA

PRODCOM codes affected by changes	Subgroup	Product category ID and name (share of the code allocated to the category)	Codes description	Implications of the affected codes (rationale) over the existing codes (merging, newly introduced, complementing, etc.)	Affected indicator	Missing data
14192100	Apparel	4.Jackets and coats (20%) 8.Underwear (50%)	Babies' clothing and accessories, of textiles, not knitted or crocheted (for children of height <= 86 cm) including vests, rompers, underpants, stretch-suits, napkins, gloves, mittens and outerwear	NA	PRODVAL EXPVAL IMPVAL	Data missing for 2022
14191100	Apparel	4.Jackets and coats (20%) 8.Underwear (50%)	Babies' garments and clothing accessories, knitted or crocheted including vests, rompers, underpants, stretch-suits, gloves or mittens or mitts, outerwear (for children of height <= 86 cm)	NA	NA	PRODQNT EXPQNT
14142240	Apparel	1.T-shirts (75%) 6.Dresses, Skirts and jumpsuits (25%)	Men's or boys' singlets, vests, bathrobes, dressing gowns and similar articles (excluding knitted or crocheted)	NA	NA	EXPQNT IMPQNT
14142460	Apparel	1.T-shirts (40%) 6.Dresses, Skirts and jumpsuits (30%) 8.Underwear (30%)	Women's or girls' singlets and other vests, briefs, panties, negligees, bathrobes, dressing gowns, housecoats and similar articles of cotton (excluding knitted or crocheted)	NA	NA	EXPQNT IMPQNT
14142480	Apparel	1.T-shirts (40%) 6.Dresses, Skirts and jumpsuits (30%) 8.Underwear (30%)	Women's or girls' negligees, bathrobes, dressing gowns, singlets, vests, briefs and panties (including boxer shorts), of man-made fibres (excluding knitted or crocheted)	NA	NA	EXPQNT
14142489	Apparel	1.T-shirts (40%) 6.Dresses, Skirts and jumpsuits (30%) 8.Underwear (30%)	Women's or girls' singlets, vests, briefs, panties, negligees, bathrobes, dressing gowns and similar articles, of textiles (excluding cotton, man-made fibres, knitted or crocheted)	NA	NA	IMPQNT
14191230	Apparel	4.Jackets and coats (50%) 5.Pants and shorts (50%)	Ski-suits, of knitted or crocheted textiles	NA	NA	EXPQNT

PRODCOM codes affected by changes	Subgroup	Product category ID and name (share of the code allocated to the category)	Codes description	Implications of the affected codes (rationale) over the existing codes (merging, newly introduced, complementing, etc.)	Affected indicator	Missing data
14192150	Apparel	4.Jackets and coats (50%) 5.Pants and shorts (50%)	Babies clothing and accessories, of textiles, not knitted or crocheted (for children of height <= 86 cm) including vests, rompers, underpants, stretch-suits, gloves, mittens and outerwear (excluding sanitary towels and napkins and similar articles)	NA	NA	EXPQNT IMPQNT PRODQNT
14192210	Apparel	6.Dresses, Skirts and jumpsuits (100%)	Other men's or boys' apparel n.e.c., including tracksuits and jogging suits (excluding waistcoats, ski-suits, knitted or crocheted)	NA	NA	EXPQNT IMPQNT
14192220	Apparel	6.Dresses, Skirts and jumpsuits (100%)	Other women's or girls' apparel n.e.c., including tracksuits and jogging suits (excluding waistcoats, ski-suits, knitted or crocheted)	NA	NA	EXPQNT
14311037	Apparel	7.Leggings, Stockings, Tights and socks (100%)	Pantyhose and tights of textile materials, knitted or crocheted (excl. graduated compression hosiery, those of synthetic fibres and hosiery for babies)	Code included in 2013 complement Hosiery: 14311090	PRODVAL EXPVAL IMPVAL	NA
14192396	Apparel	10. Accessories	Clothing accessories, parts of garments or of clothing accessories, of textiles, n.e.c. and parts thereof, (excluding shawls, scarves and mufflers, mantillas and veils, ties, bow-ties and cravats, gloves, mittens and mitts and parts thereof; bras, girdles and corsets, braces, suspenders and garters, knitted or crocheted)	Code included in 2015 to complement accessories: 14192393 14192395	PRODVAL EXPVAL IMPVAL	PRODQNT IMPQNT EXPQNT
14192393	Apparel	10. Accessories	Clothing accessories of textiles (excluding shawls, scarves and mufflers, mantillas and veils, ties, bow-ties and cravats, gloves, mittens and mitts, knitted or crocheted)	NA	NA	PRODQNT IMPQNT EXPQNT

PRODCOM codes affected by changes	Subgroup	Product category ID and name (share of the code allocated to the category)	Codes description	Implications of the affected codes (rationale) over the existing codes (merging, newly introduced, complementing, etc.)	Affected indicator	Missing data
14192395	Apparel	10. Accessories	Parts of garments or of clothing accessories, of textiles (excluding bras, girdles and corsets, braces, suspenders and garters, knitted or crocheted)	NA	NA	PRODQNT IMPQNT EXPQNT
14194270	Apparel	10. Accessories	Hats and other headgear, knitted or crocheted or made-up from lace, felt or other textile fabric in the piece (but not in strips); hair-nets of any material	NA	NA	IMPQNT EXPQNT
14191930	Apparel	10. Accessories	Shawls, scarves, mufflers, mantillas, veils and the like, of knitted or crocheted textiles	NA	NA	EXPQNT
14191960	Apparel	10. Accessories	Clothing accessories and parts thereof, of knitted or crocheted textiles (excluding gloves, mittens, shawls, scarves, mufflers, mantillas and veils)	NA	NA	EXPQNT

NA: not applicable

The changes of the affected PRODCOM codes should be understood as merged, changed or newly introduced due to NACE list revision and they are considered complete in the analysis and used.

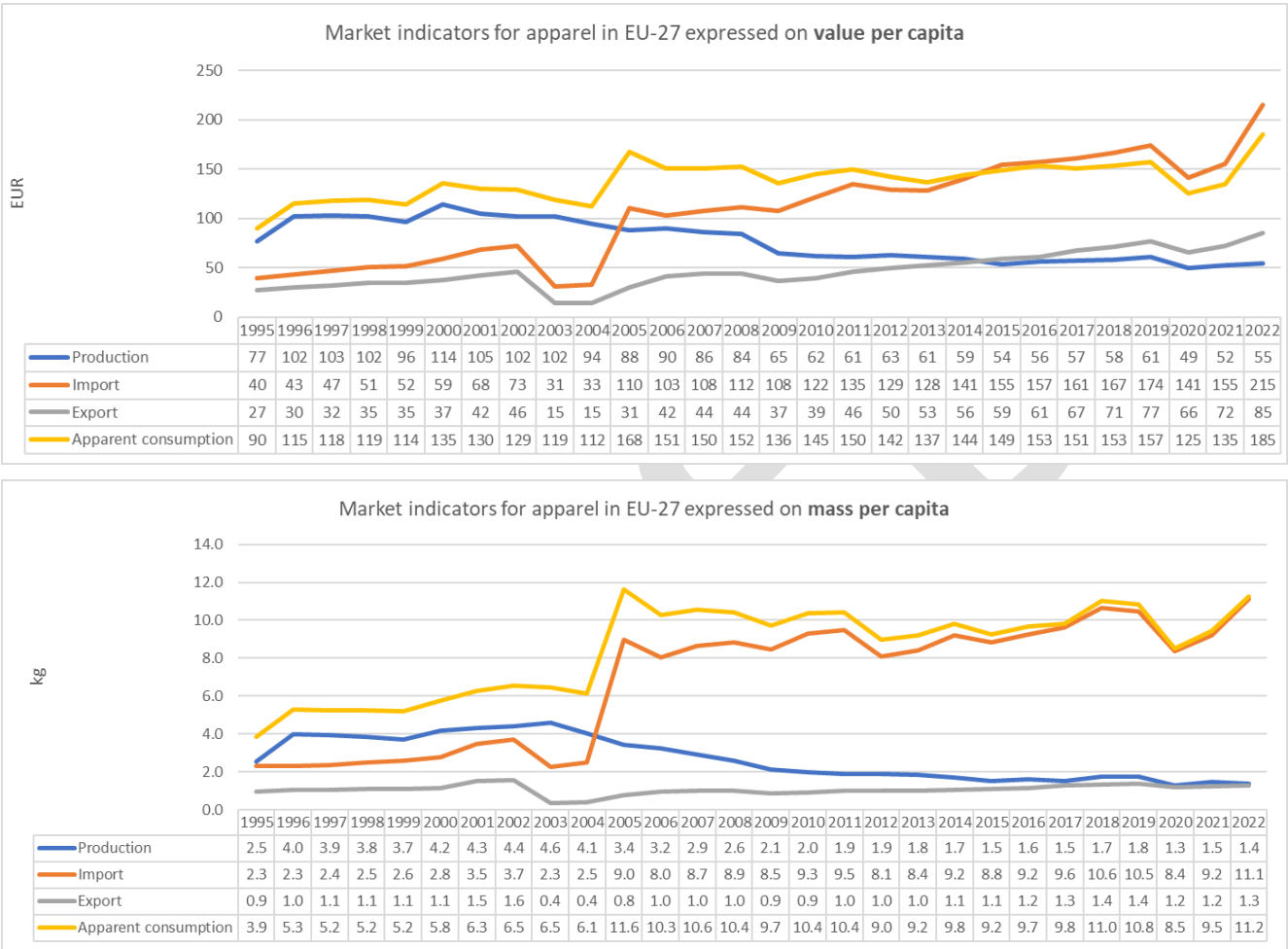
9356

9357

9358

13.2.5 Supplementary information about the EU market

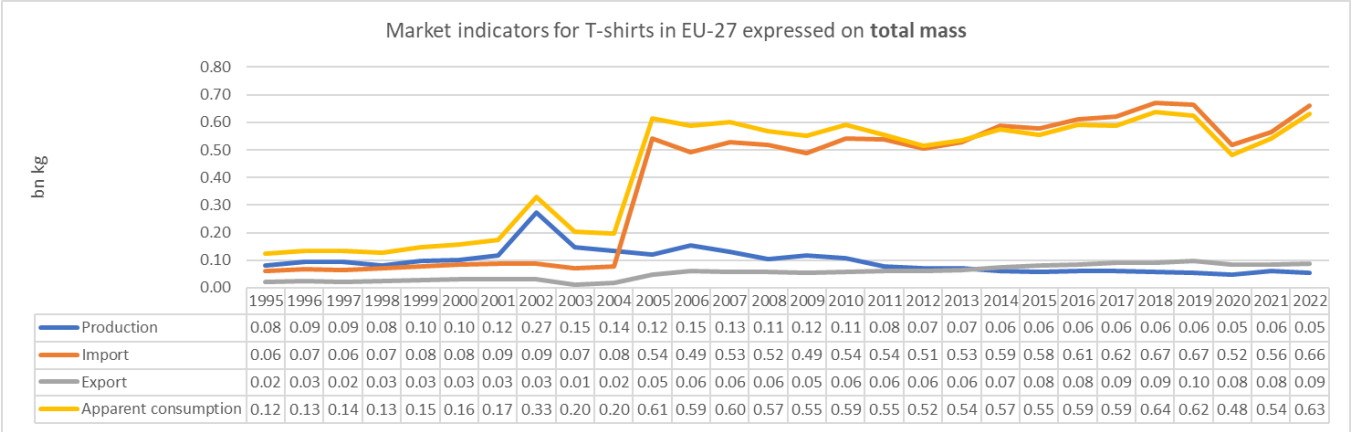
Figure 64. Market indicators per capita for textile apparel in EU-27



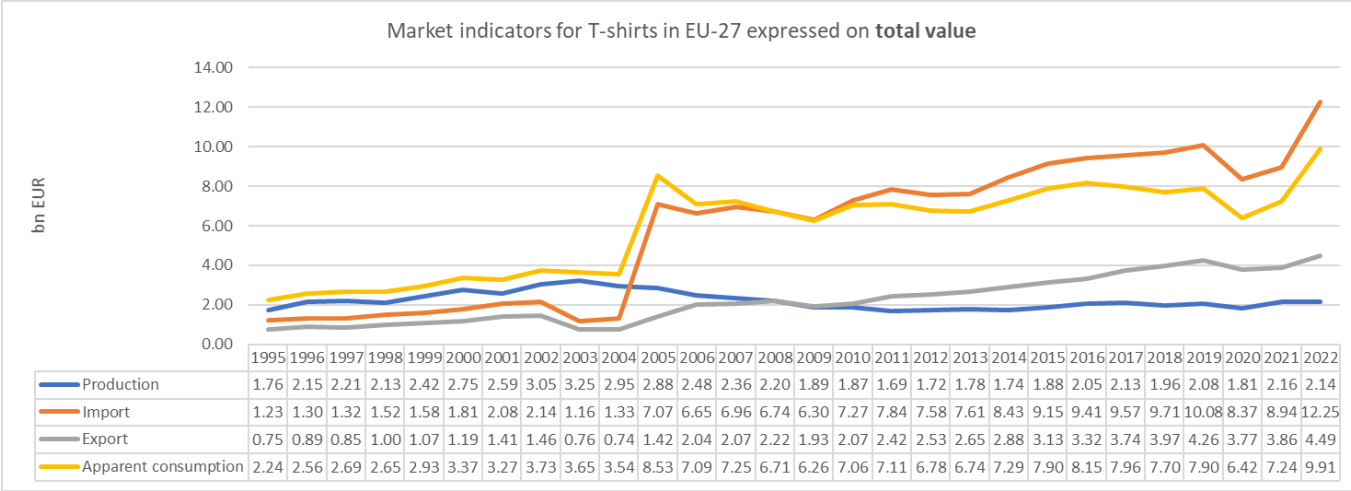
Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

9365

Figure 65. Market indicators for T-shirts in EU-27



9366



9367

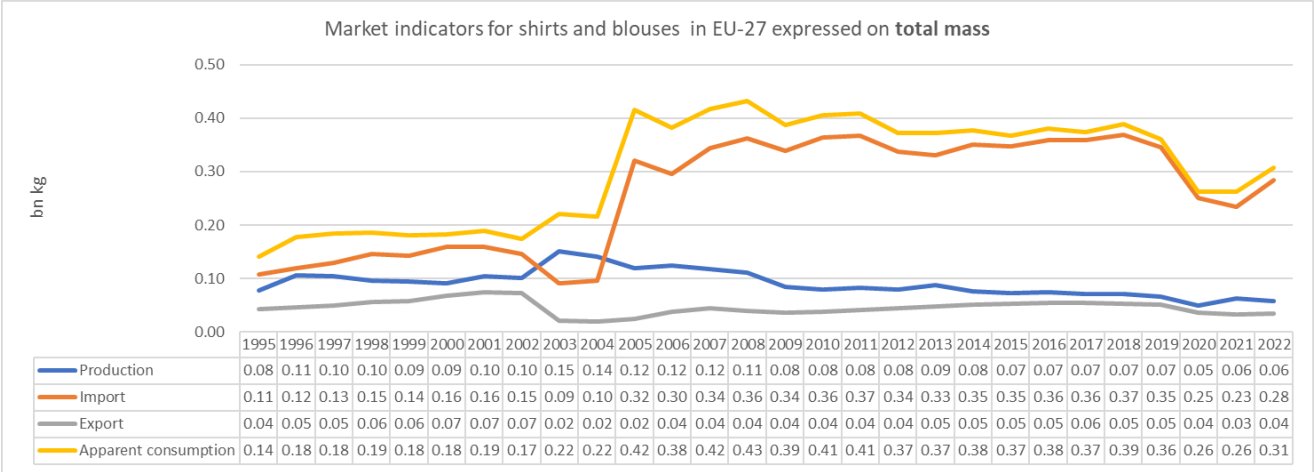
9368

Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

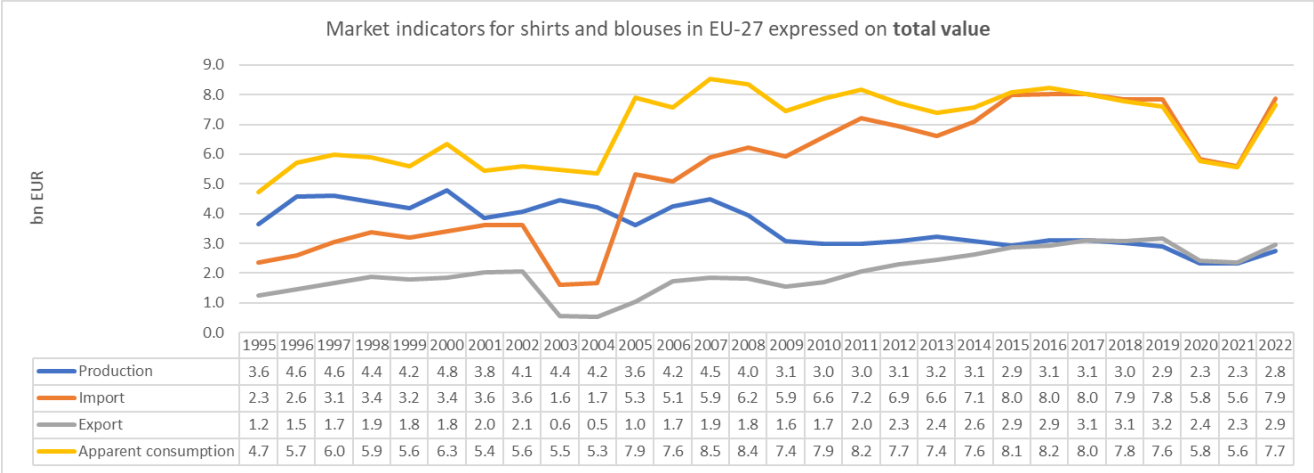
9369

9370

Figure 66. Market indicators for shirts and blouses in EU-27



9371



9372

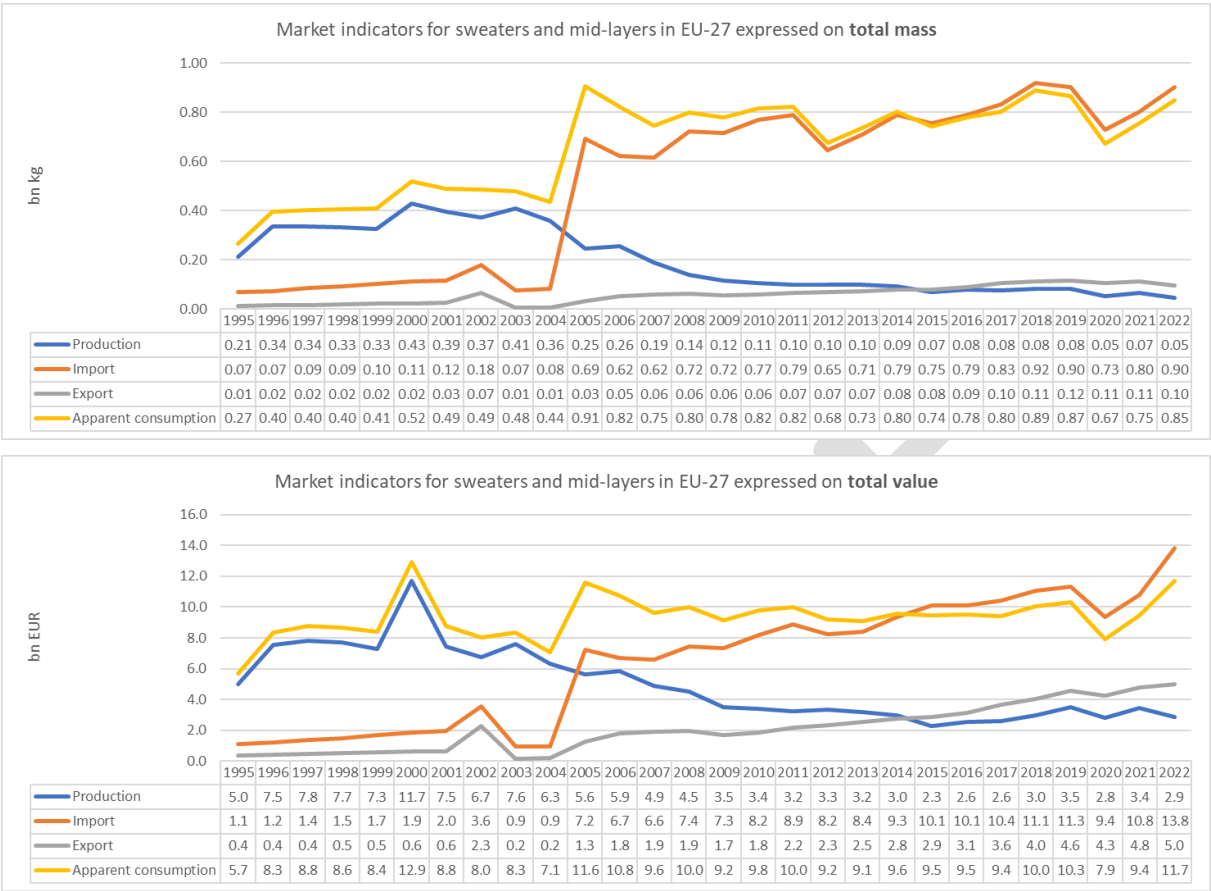
9373

Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

9374

9375

Figure 67. Market indicators for sweaters and mid-layers in EU-27



9376

9377

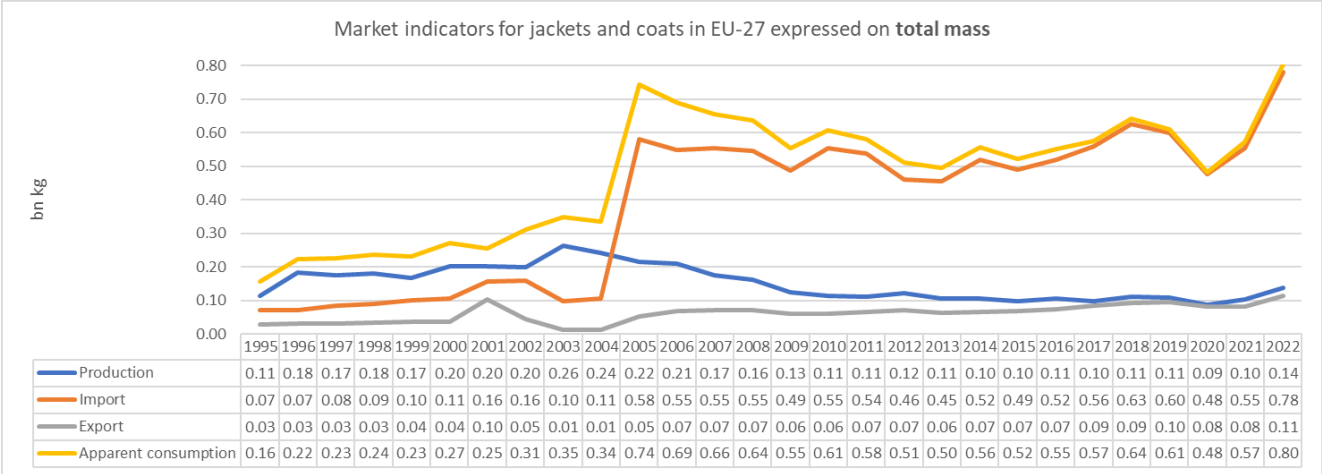
9378

Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

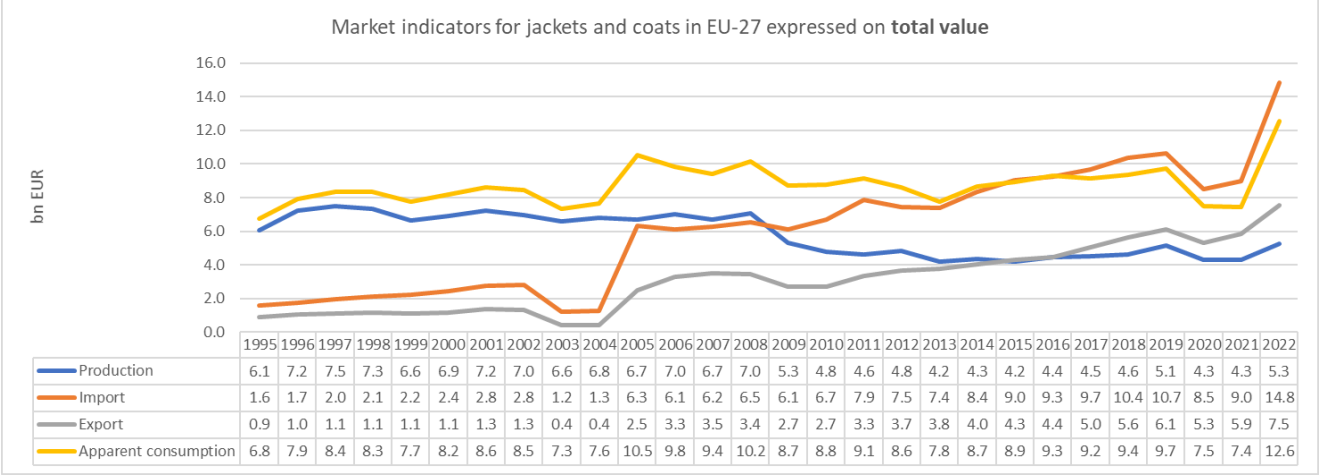
9379

9380

Figure 68. Market indicators for jackets and coats in EU-27



9381



9382

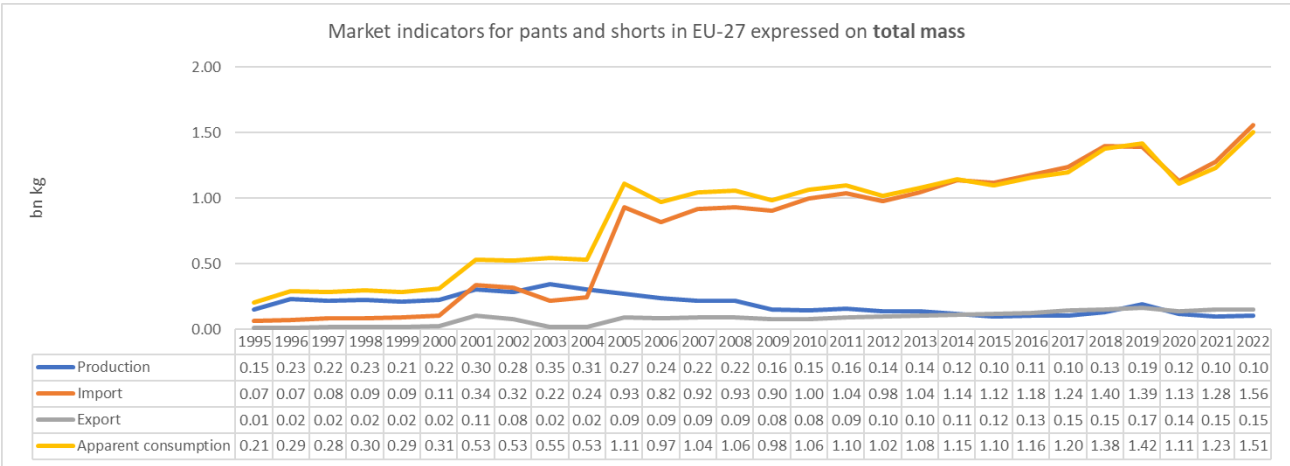
9383

Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

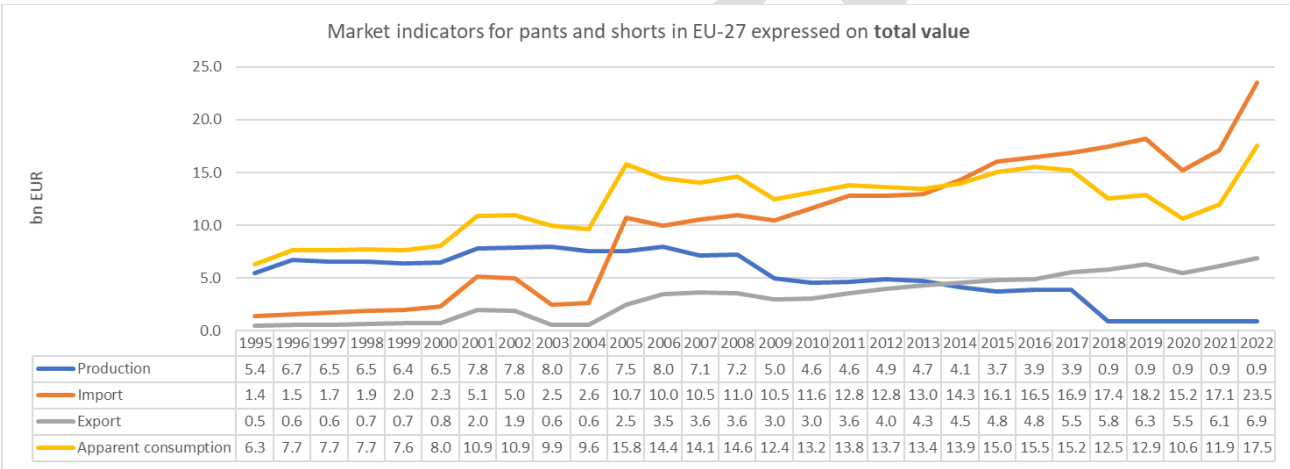
9384

9385

Figure 69. Market indicators for pants and shorts in EU-27



9386



9387

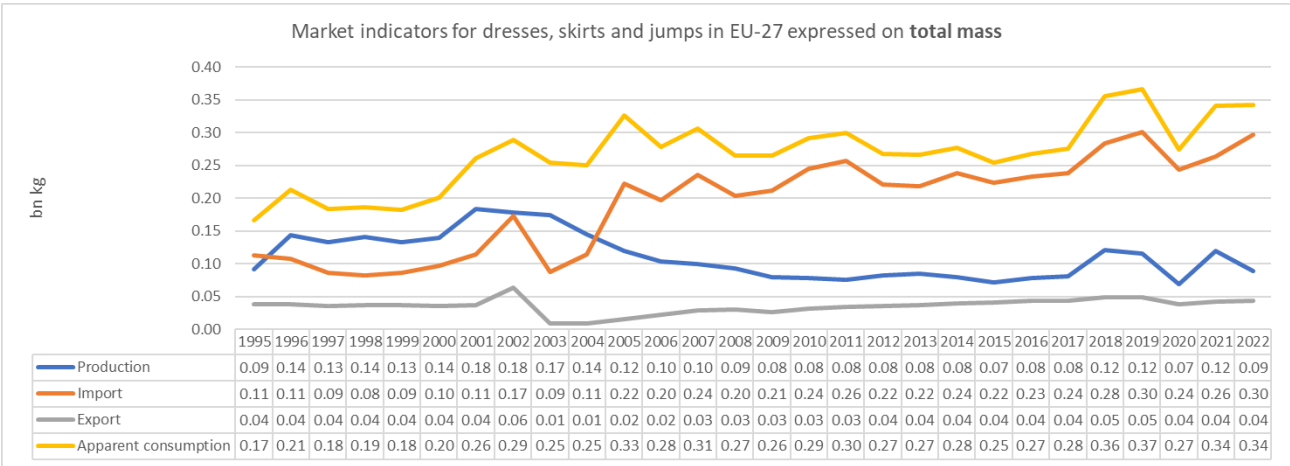
9388

Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

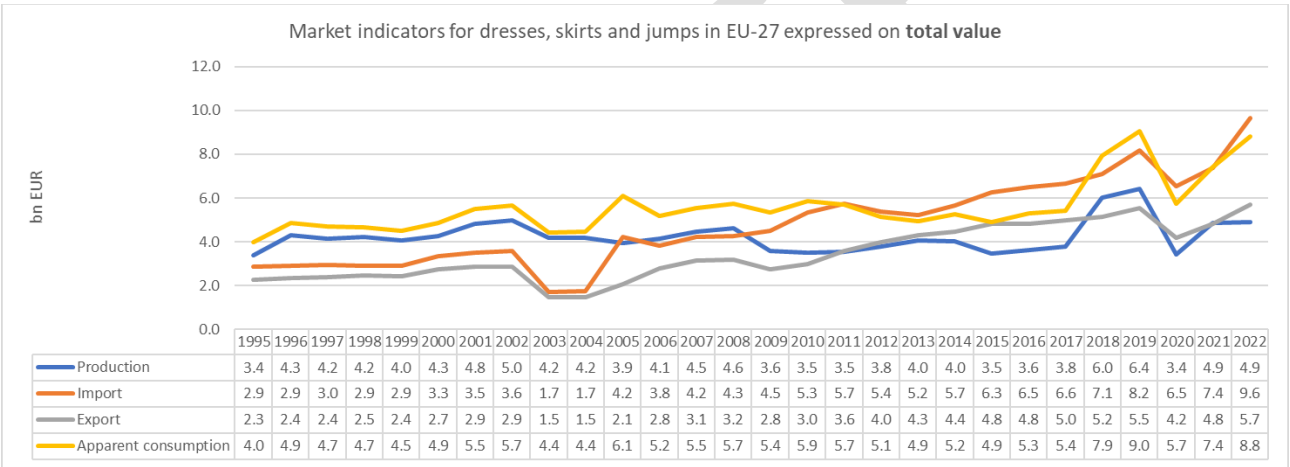
9389

9390

Figure 70. Market indicators for dresses, skirts and jumps in EU-27



9391



9392

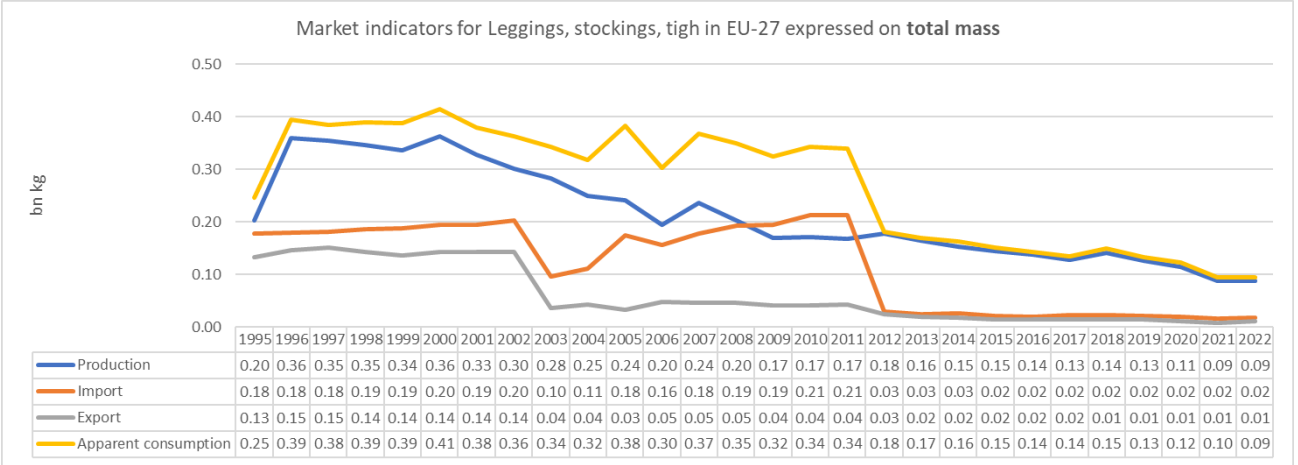
9393

Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

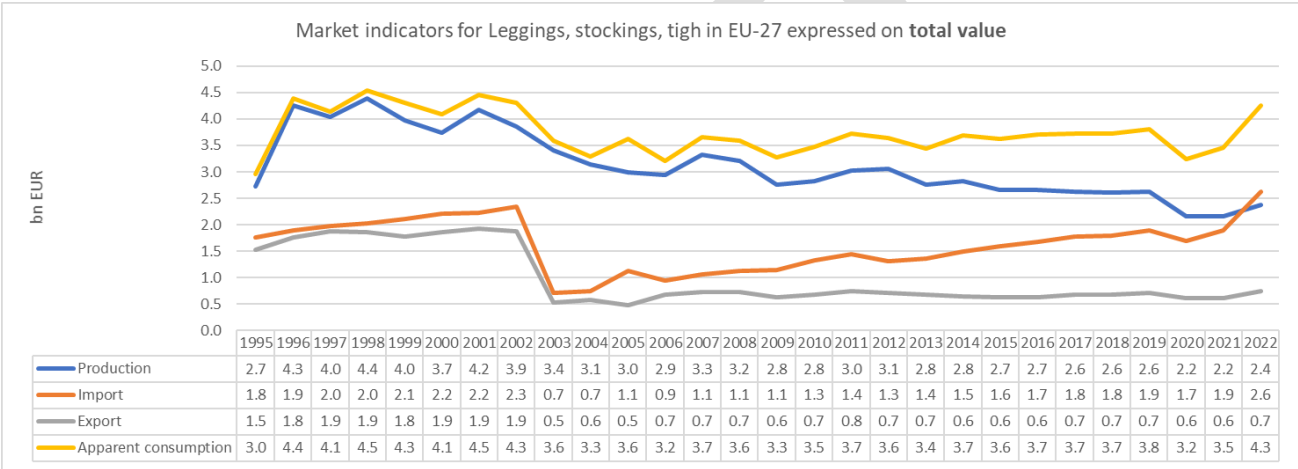
9394

9395

Figure 71. Market indicators for leggings, stockings and tights in EU-27



9396



9397

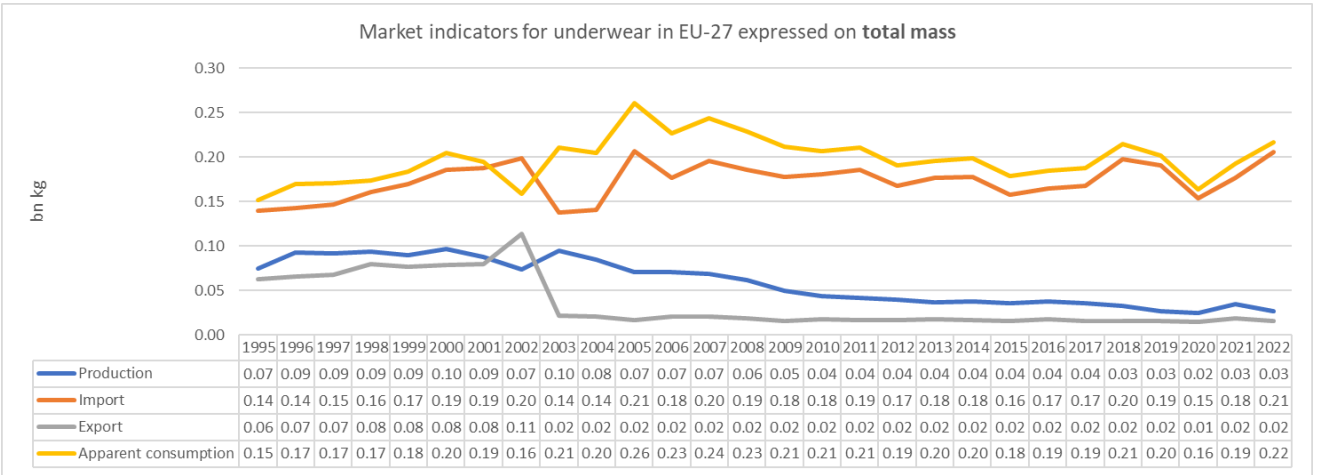
9398

Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

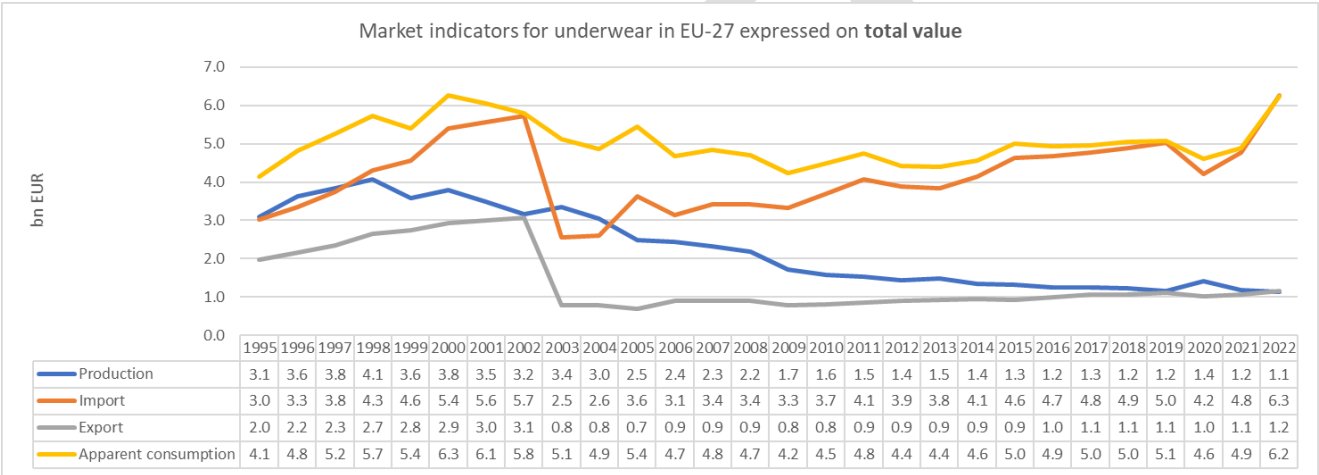
9399

9400

Figure 72. Market indicators for underwear in EU-27



9401



9402

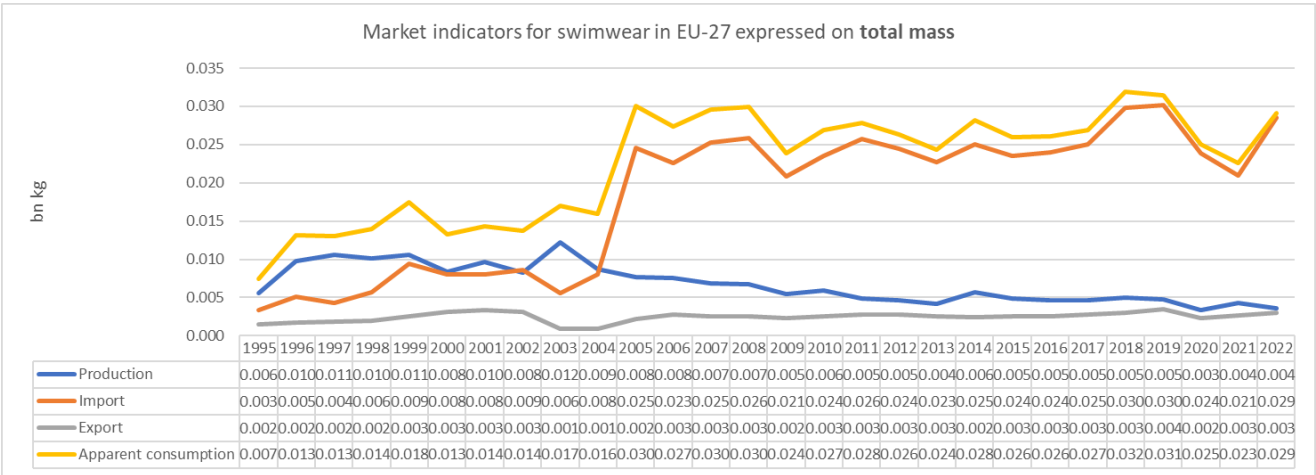
9403

Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

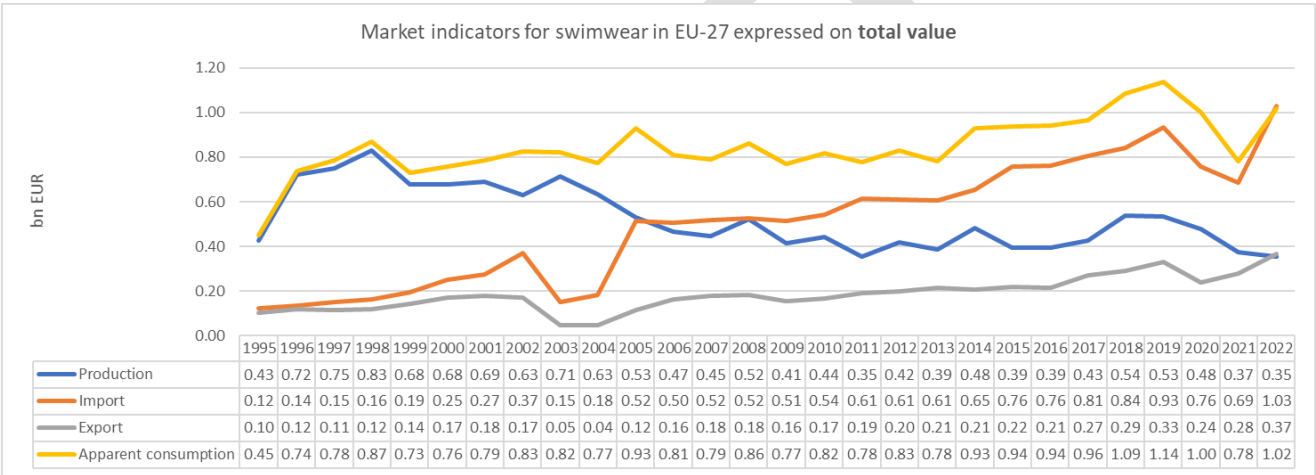
9404

9405

Figure 73. Market indicators for swimwear in EU-27



9406



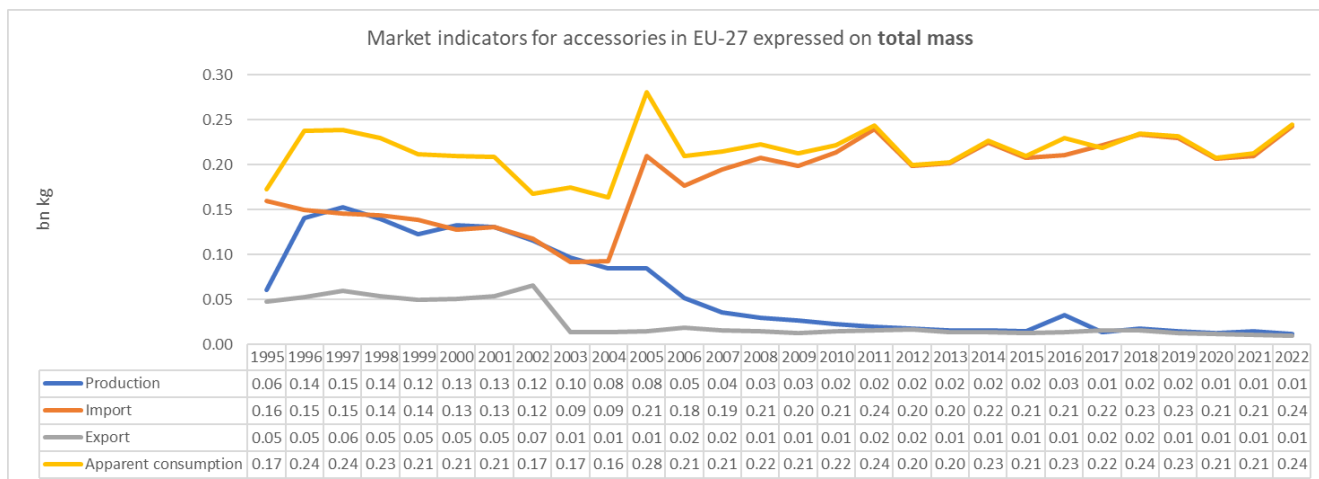
9407

9408

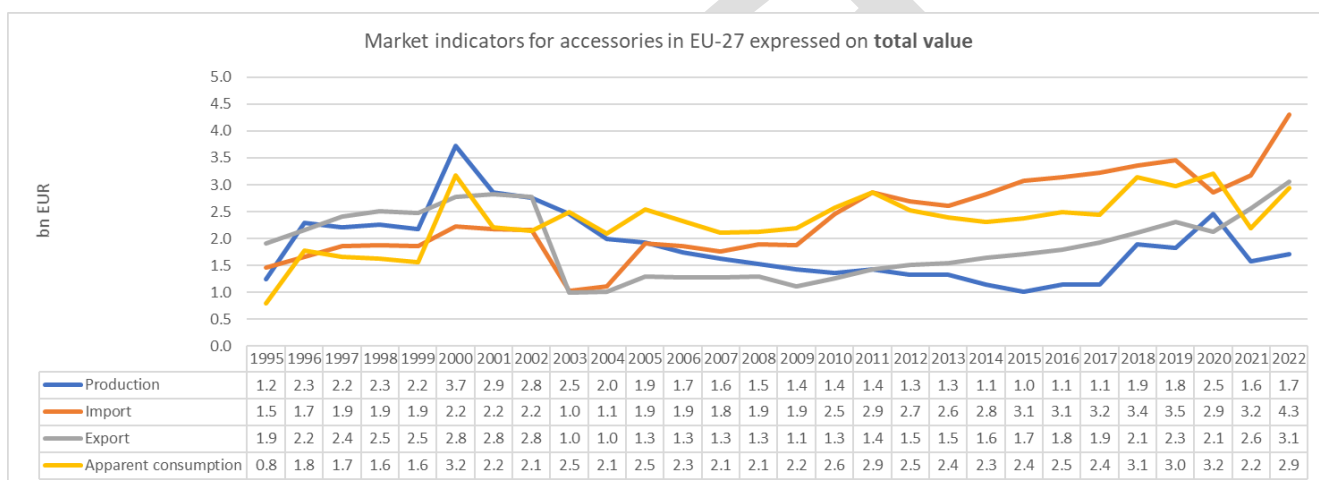
Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

9409

9410

Figure 74. Market indicators for accessories in EU-27

9411



9412

9413

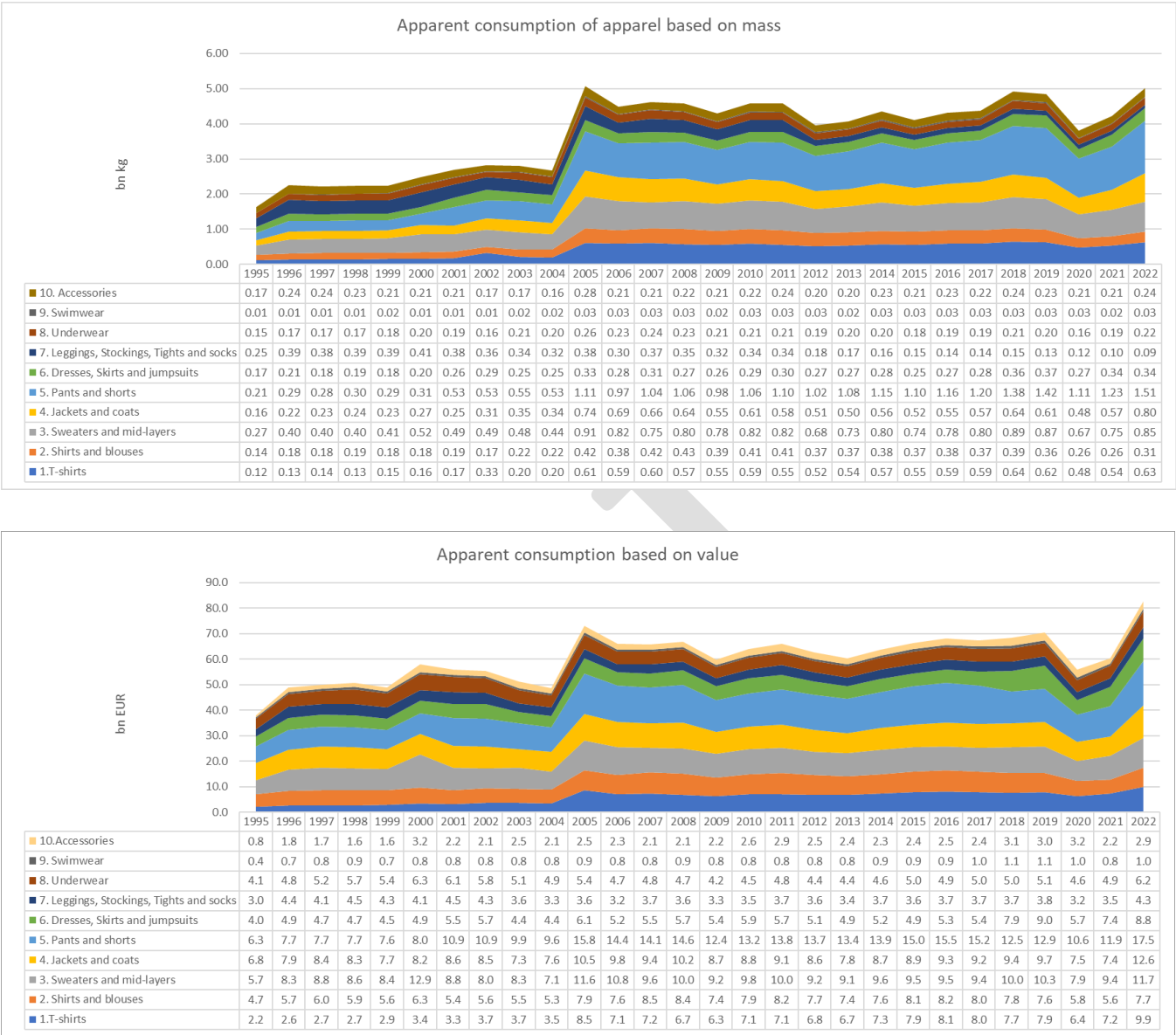
Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

9414

9415

9416

Figure 75. Apparent consumption of textile apparel categories in EU-27



9417

9418

9419

9420

9421

(A) Total mass; (B) Total value

Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

9422 **Table 117.** Change of apparent consumption after the removal of EU import quota in 2005

Textile apparel category	Mass (bn kg)		Change in mass (%)	Value (bn EUR)		Change in value (%)
	Average value in 1995-2004	Average value in 2005-2019		Average value in 1995-2004	Average value in 2005-2019	
1. T-shirts	0.17	0.58	236	3.15	7.36	133
2. Shirts and blouses	0.19	0.39	110	5.71	7.88	38
3. Sweaters and mid-layers	0.43	0.80	86	8.81	9.83	12
4. Jackets and coats	0.26	0.60	129	8.07	9.21	14
5. Pants and shorts	0.38	1.12	194	8.90	14.03	58
6. Dresses, Skirts and jumpsuits	0.22	0.29	33	4.85	5.83	20
7. Leggings, Stockings, Tights and socks	0.36	0.24	-33	4.13	3.59	-13
8. Underwear	0.18	0.21	15	5.48	4.77	-13
9. Swimwear	0.01	0.03	100	0.79	0.89	13
10. Accessories	0.20	0.22	11	2.08	2.49	20

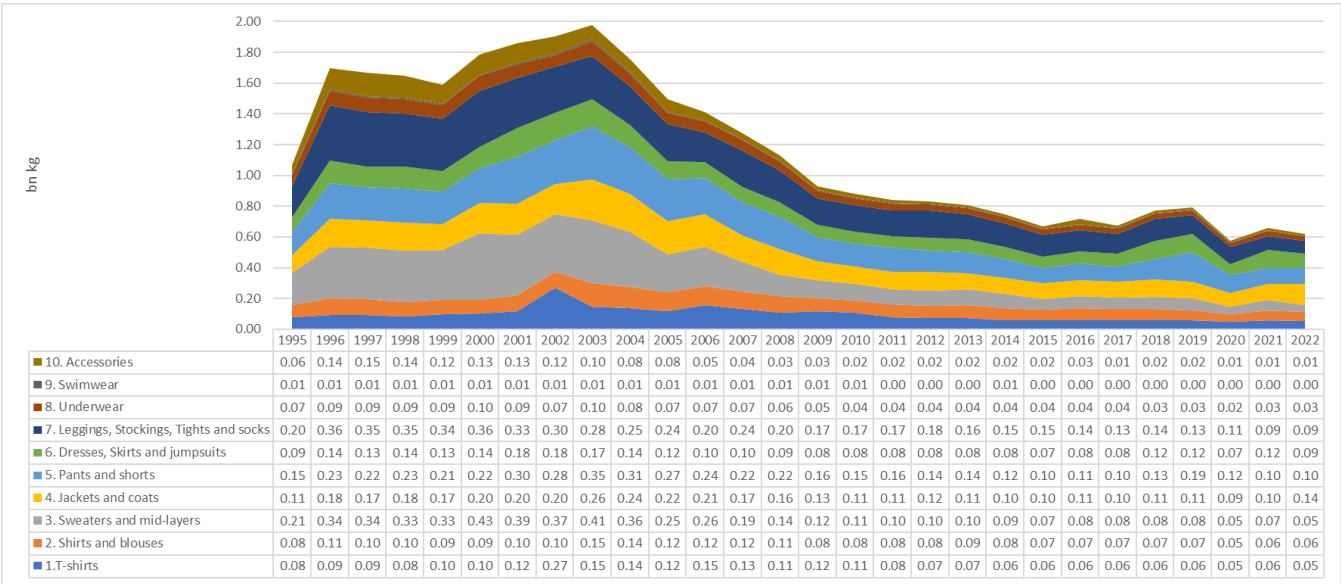
Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

9423

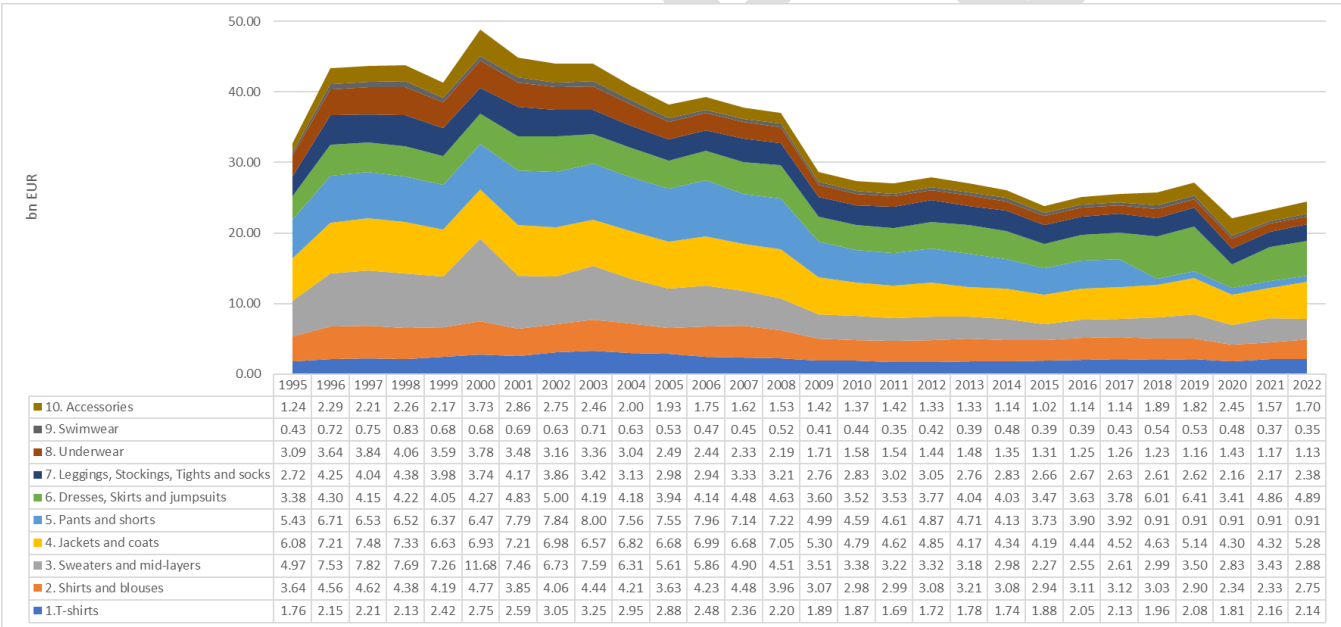
9424

9425

Figure 76. Production of textile apparel categories in EU-27



9426



(A) Total mass; (B) Total value

Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

9427

9428

9429

9430

9431

9432 **Table 118.** Change of production after the removal of EU import quota in 2005

Textile apparel category	Mass (bn kg)		Change in mass (%)	Value (bn EUR)		Change in value (%)
	Average value in 1995-2004	Average value in 2005-2019		Average value in 1995-2004	Average value in 2005-2019	
1. T-shirts	0.12	0.09	-29	2.53	2.05	-19
2. Shirts and blouses	0.11	0.09	-18	4.27	3.32	-22
3. Sweaters and mid-layers	0.35	0.12	-65	7.50	3.63	-52
4. Jackets and coats	0.19	0.13	-32	6.92	5.23	-25
5. Pants and shorts	0.25	0.16	-35	6.92	4.74	-32
6. Dresses, Skirts and jumpsuits	0.15	0.09	-38	4.26	4.20	-1
7. Leggings, Stockings, Tights and socks	0.31	0.17	-45	3.77	2.86	-24
8. Underwear	0.09	0.05	-48	3.50	1.65	-53
9. Swimwear	0.01	0.01	-41	0.68	0.45	-33
10. Accessories	0.12	0.03	-77	2.40	1.46	-39

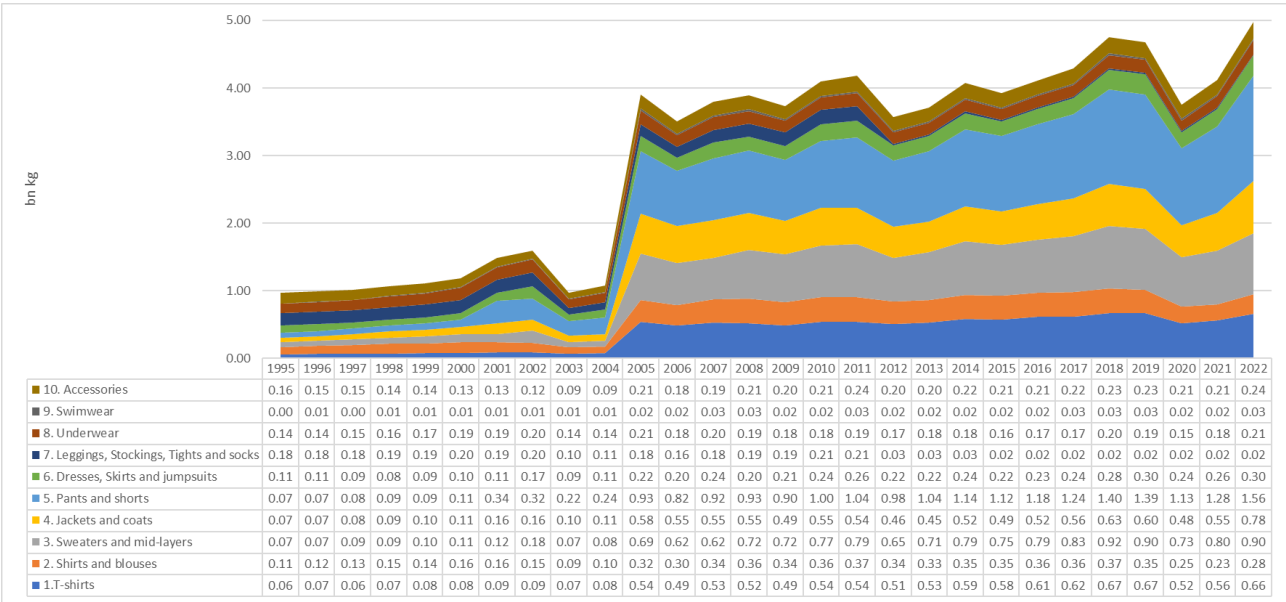
Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

9433

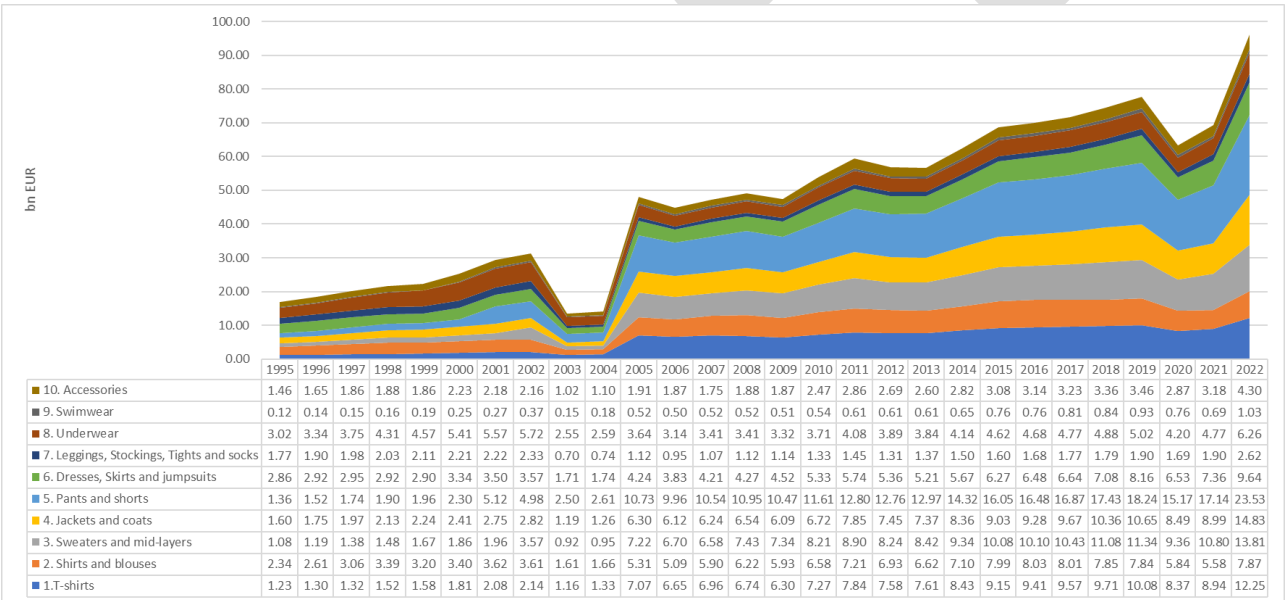
9434

9435

Figure 77. Import of textile apparel categories in EU-27



9436



9437

9438

9439

9440

(A) Total mass; (B) Total value

Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

9441

9442 **Table 119.** Change of import after the removal of EU import quota in 2005

Textile apparel category	Mass (bn kg)		Change in mass (%)	Value (bn EUR)		Change in value (%)
	Average value in 1995-2004	Average value in 2005-2019		Average value in 1995-2004	Average value in 2005-2019	
1.T-shirts	0.07	0.56	650	1.55	8.03	419
2. Shirts and blouses	0.13	0.35	168	2.85	6.84	140
3. Sweaters and mid-layers	0.10	0.75	656	1.60	8.76	446
4. Jackets and coats	0.10	0.54	413	2.01	7.87	291
5. Pants and shorts	0.16	1.07	552	2.60	13.48	419
6. Dresses, Skirts and jumpsuits	0.11	0.24	121	2.84	5.53	95
7. Leggings, Stockings, Tights and socks	0.17	0.10	-41	1.80	1.41	-22
8. Underwear	0.16	0.18	12	4.08	4.04	-1
9. Swimwear	0.01	0.02	274	0.20	0.65	225
10. Accessories	0.13	0.21	63	1.74	2.60	49

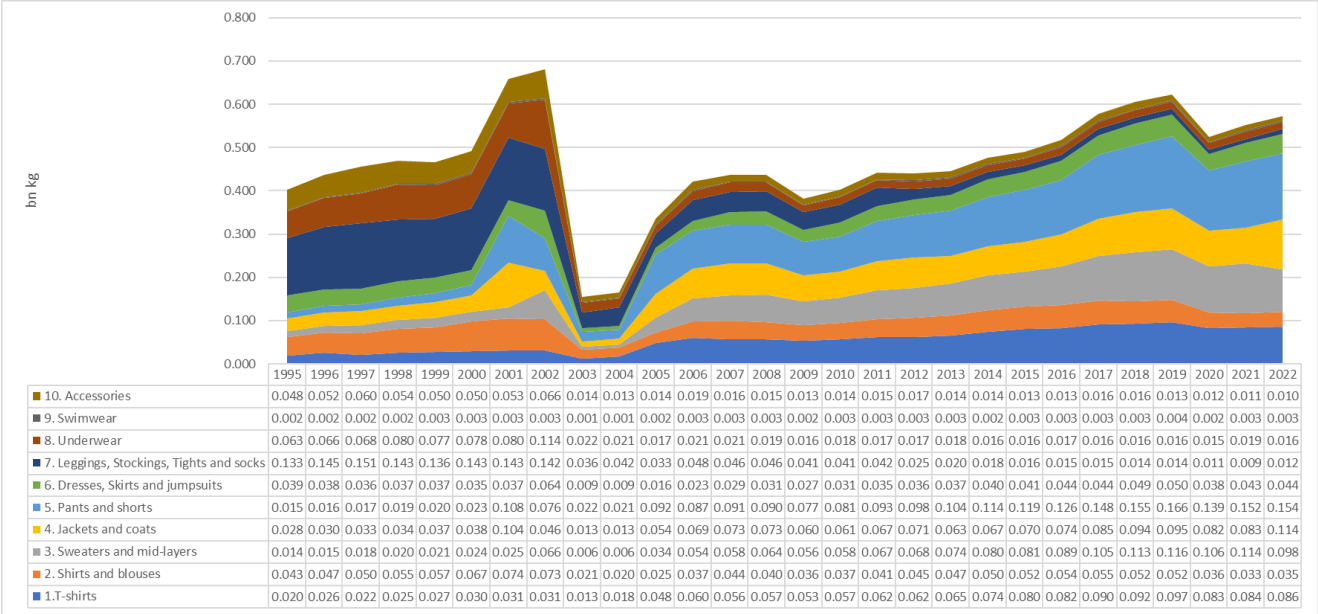
Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

9443

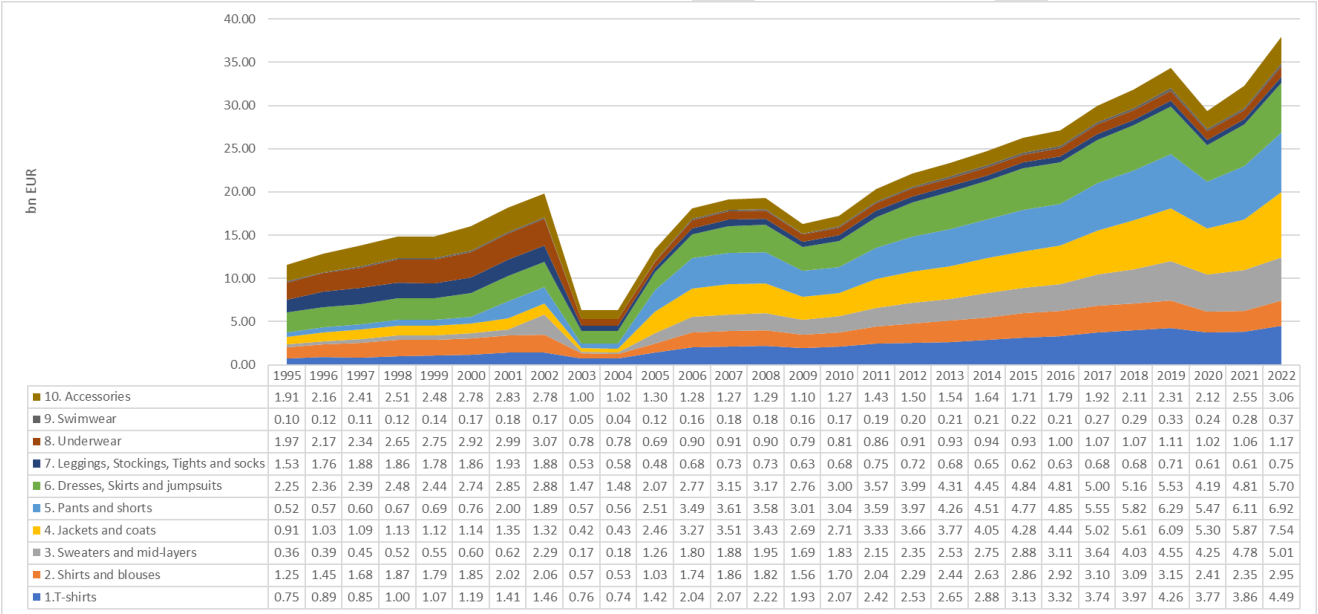
9444

9445

Figure 78. Export of textile apparel categories in EU-27



9446



9447

9448

9449

9450

(A) Total mass; (B) Total value

Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

Table 120. Change of export after the removal of EU import quota in 2005

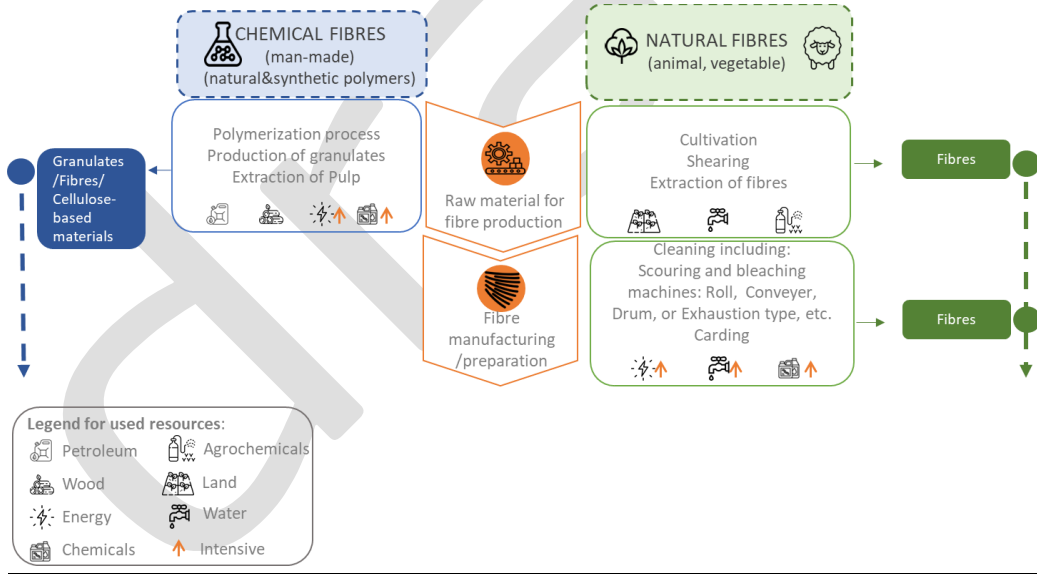
Textile apparel category	Mass (bn kg)		Change in mass (%)	Value (bn EUR)		Change in value (%)
	Average value in 1995-2004	Average value in 2005-2019		Average value in 1995-2004	Average value in 2005-2019	
1.T-shirts	0.02	0.07	186	1.01	2.71	168
2. Shirts and blouses	0.05	0.04	-12	1.51	2.28	51
3. Sweaters and mid-layers	0.02	0.07	246	0.61	2.56	318
4. Jackets and coats	0.04	0.07	91	0.99	3.89	291
5. Pants and shorts	0.03	0.11	226	0.88	4.19	375
6. Dresses, Skirts and jumpsuits	0.03	0.04	4	2.33	3.90	67
7. Leggings, Stockings, Tights and socks	0.12	0.03	-76	1.56	0.67	-57
8. Underwear	0.07	0.02	-74	2.24	0.92	-59
9. Swimwear	0.00	0.00	26	0.12	0.21	70
10. Accessories	0.05	0.01	-68	2.19	1.56	-28

Source: own production based on PRODCOM database (Sold production, exports and imports – DS-056120)

13.3 Life-cycle stages of textile apparel

13.3.1 Raw material and fibre manufacturing

Figure 79. Raw material for fibre production and fibre preparation



Source: production adapted from (McKinsey & Company, 2022), icons from www.flaticon.com

Raw material for fibre production

Natural fibres

The primary raw material for vegetable-based fibres is cellulose, which is the most prevalent natural polymer in nature. Cellulose is sourced from agricultural crops, which often require significant land and water use. The production process typically involves a use of chemicals, like pesticides (Jana et al., 2023) (Roth et al., 2023). After the production process, the result is staple fibres. Following a pre-treatment or preparation phase, these fibres then serve as the input for the yarn production stage.

Animal-based fibres, also known as protein fibres, are sourced from animals through methods such as shearing or collecting silkworm cocoons. This requires farming animals like sheep or silkworms, either on pastures or

9467 through sericulture (Roth et al., 2023) (Jana et al., 2023). The production yields staple fibres, which, after a pre-
9468 treatment phase, are used in yarn production.

9469 *Chemical fibres*

9470 Man-made chemical fibres derived from natural polymers use cellulose, often from sources like wood pulp, as
9471 their primary raw material. This cellulose is then dissolved to produce a solution suitable for the next stage of
9472 fibre production. The key components in this process are the wood feedstock, which come from various sources,
9473 and the chemicals used in the dissolution process (Jana et al., 2023).

9474 Man-made chemical fibres made from synthetic polymers use petroleum as raw material. This is processed
9475 using energy-intensive chemical reactions, which vary depending on the specific synthetic fibre, to produce
9476 synthetic polymers. These polymers are then turned into granules (Roth et al., 2023). These granules are then
9477 prepared for the subsequent fibre production process (Jaffe and Menczel, 2020).

9478

9479 **Fibre manufacturing/preparation**

9480 *Manufacturing of natural and chemical fibres*

9481 Staple fibres of natural origin, man-made chemical staple fibres derived from natural polymers, and synthetic
9482 granules produced in the raw material stage all undergo the manufacturing process. The specific production
9483 steps they undergo vary based on the source of the raw material.

9484 *Preparation of natural fibres*

9485 Natural staple fibres go through similar preparation stages before yarn manufacturing. This typically involves
9486 a scouring or cleaning process before carding. The exact procedures vary based on the fibre type, whether
9487 cellulosic (like cotton) or protein-based (like wool). The primary distinction arises from the type and amount of
9488 impurities present in the natural fibres.

9489 Wool, a protein fibre, undergoes scouring, which is typically a wet cleaning process using detergents to remove
9490 dirt, oils, and other contaminants. After scouring, the wool is usually bleached with hydrogen peroxide, to remove
9491 its natural colour and any remaining impurities, preparing it for subsequent treatments or dyeing. This is then
9492 followed by drying. Overall, this process is intensive in terms of water, energy, and chemicals.

9493 The cotton cleaning process involves dry procedures carried out in a blowing room. This room houses
9494 technologies that move the raw materials through various airflow cleaning systems, integrated within a
9495 mechanical cleaning process.

9496 Furthermore, because natural fibres have inconsistent properties, they may undergo wet chemical processes,
9497 such as bleaching, and other treatments to enhance and modify them.

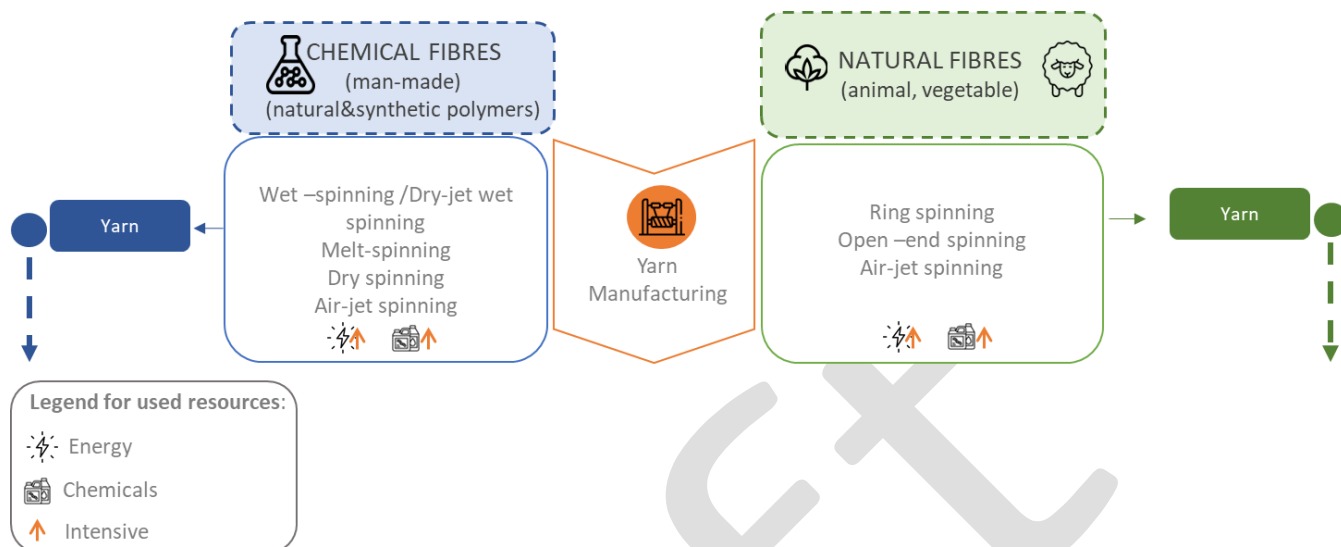
9498 *Preparation of chemical fibres*

9499 Chemical fibres skip the preparation stage, as the staple fibres and granules are used directly for the yarn
9500 production (spinning).

9501

13.3.2 Yarn manufacturing

Figure 80. Yarn manufacturing (spinning process)



Source: own production and adapted from (McKinsey & Company, 2022), icons from www.flaticon.com

Carded natural fibres, cellulose – based material, and polymer granulates serve as inputs for the spinning process, which produces yarns.

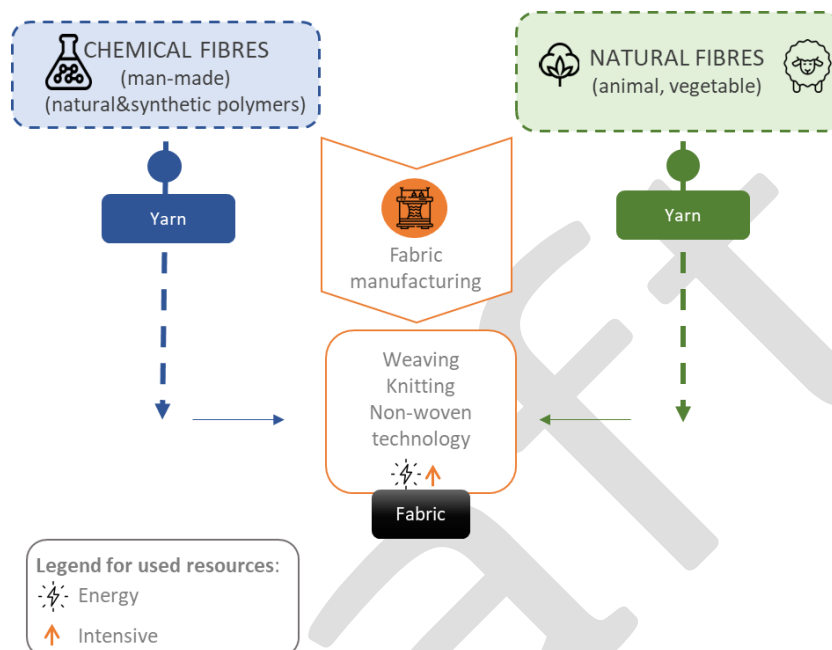
In the textile industry, the definition of spinning varies based on the type of fibre being processed:

- Natural origin fibres: Spinning for these fibres involves passing the staple fibres through a series of rollers to straighten and align them, then twisting them to create a continuous yarn (Britannica, 2019). Each type of natural fibre goes through a unique set of steps for yarn production. For all, the initial step is carding, a mechanical process that further cleans and disentangles the fibres, aligning them. This is followed by other processes such as combing, drawing, and more. Typical spinning machines for natural fibres include ring spinning and open-end spinning (Roth et al., 2023).
- Chemical (man-made) fibres: Spinning refers to the extrusion of a solution, like viscose/cellulose-based materials or dissolved granules, to produce a fibre or yarn. Standard industrial spinning machines for these fibres are:
 - melt spinning: polymer melted in a melt extruder, and is suitable for thermoplastic fibres such as polyester, polyamide, polyolefins, glass fibre and many more;
 - wet spinning: polymer dissolved in a solvent, being suitable for viscose and acrylic fibres dry spinning: polymer dissolved in a solution, applicable for acetate, triacetate and polyacrylonitrile.
 - Man-made cellulosic fibres: Spinning filaments into yarns involves processes where cellulose from wood pulp is dissolved using specific chemical solvents (e.g., sodium hydroxide for viscose, ammonia-copper solution for cupro, NMMO for lyocell, or acetic acid for acetate), and the resulting solution is typically extruded and coagulated using wet spinning techniques. Nevertheless, for staple fibres, ring or open-end spinning is used.

Drawing or air texturizing is performed to give to the chemical yarns a texture similar to natural yarns. Such treatments are carried out with energy demanding equipment (Roth et al., 2023).

13.3.3 Fabric manufacturing

Figure 81. Fabric manufacturing process



Source: own production adopted from (McKinsey & Company, 2022), icons from www.flaticon.com

The central phase of textile production is fabric manufacturing. This involves creating a two-dimensional structure by interlacing yarns. Common methods include weaving, knitting, and producing non-woven fabrics, suitable for all yarn types.

Weaving is the predominant fabric manufacturing technique. It involves interlacing two sets of yarns at right angles on a loom. Most looms are power-driven, termed 'power looms', and are energy intensive. There are also manual looms, which are labour-intensive. Key resources in weaving include chemicals for sizing and desizing textiles and oils for lubricating the loom. Sizing protects the yarn from the loom's abrasive action, while desizing removes these chemicals from the woven fabric after weaving (Roth et al., 2023).

Knitting is the second most used fabric manufacturing method. In this case, yarns are interlooped using needles to form fabric on knitting machines. These machines are broadly categorised as weft³⁰¹ (either flat or circular) and warp³⁰², based on the fabric's formation direction (horizontal or vertical). They're further classified by machine shape and knitting technology. This method is energy intensive. To prevent yarn damage, substances like sizing chemicals and lubricants are used for yarn strengthening and reducing friction (Roth et al., 2023).

Non-woven technology produces textile structures by bonding fibres or filaments together, either mechanically, thermally, or chemically. According to the Association of the Non-woven Fabrics Industry (INDA), they are "flat, porous sheets made directly from separate fibres or molten plastic. They are neither woven nor knitted and don't convert fibres to yarn" (INDA, 2023). Various non-woven processes exist, categorised by web-forming technologies and product consolidation. Common technologies include drylaid⁽³⁰³⁾, wetlaid⁽³⁰⁴⁾, spunmelt⁽³⁰⁵⁾,

³⁰¹ The fabric is formed based on loops made on horizontal way from a single yarn, which is fed and looped in rows by one or more needles at a time (Roth et al., 2023).

³⁰² The yarn is fed into the knitting forming vertical loops, and moves diagonally to knit the next course (yarns generate a zigzag from side to side along the length of the fabric) (Ray, 2012).

³⁰³ The process of fabric web forming with a stream of air, applied on dry fibres by using carding equipment (INDA, 2023).

³⁰⁴ The process of fabric web forming filtering an aqueous suspension of fibre onto a screen conveyor belt or perforated drum. The fibres are retained on the screen, while the water passes through (INDA, 2023).

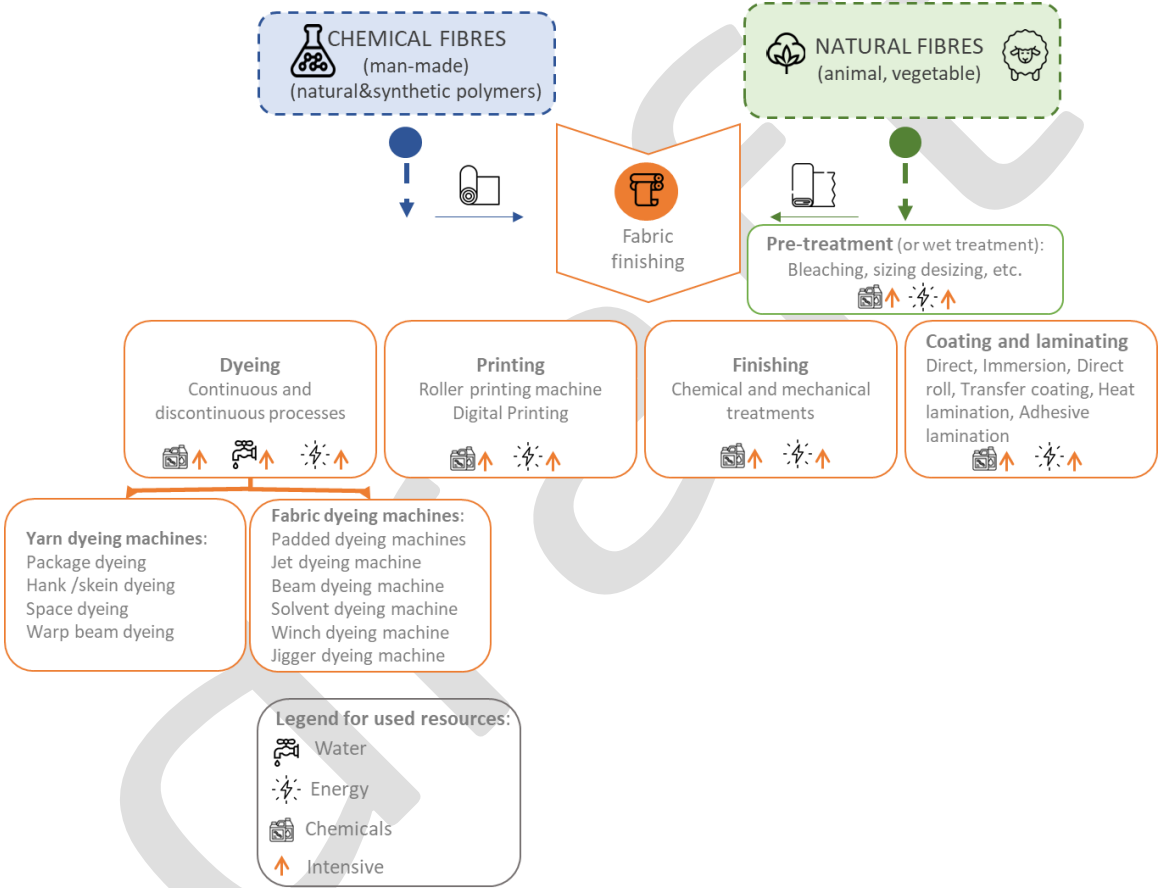
³⁰⁵ The process of non-woven manufacturing through the extrusion of molten polymer spinneret to form fibres, drawing the fibres, and then laying them on a moving screen to form a web. The term "spunlaid" is often used interchangeably with "spunbond" (INDA, 2023).

and needle-punched⁽³⁰⁶⁾. However, many other processes are available⁽³⁰⁷⁾ (Batra and Pourdeyhimi, 2012) (Mao, 2016). Overall, non-woven technology is energy intensive.

In the knitting or weaving process, the type of yarn used, and its count (indicating fabric thickness) influence the durability and strength of the final product, such as its seam or tear resistance (Yassen, 2017) (Jankoska and Demboski, 2017). Non-woven fabrics present recycling challenges due to the fibre composition and the consolidation components used in their production.

13.3.4 Finishing processes

Figure 82. Fabric/yarn finishing processes



Source: own production adapted from (McKinsey & Company, 2022), icons from www.flaticon.com

Finishing processes are an intermediate stage in treating yarn and/or fabrics with colorants and chemicals. This includes various wet treatments aimed at different objectives, such as sizing, desizing, pre-treatment, dyeing, printing, finishing, and more. These treatments are applied based on the specific requirements of the final products, rather than in a sequential manner. Each treatment utilises distinct technologies and chemicals, making this one of the most water and chemically intensive stages

Pre-treatment

Pre-treatment is typically applied to natural fibre fabrics. It encompasses processes like bleaching, scouring, mercerising³⁰⁸, and others, preparing the fabrics for final treatments before dyeing, printing, and finishing. These

³⁰⁶ The mechanical process that uses barbed needles to pull tufts of fibres from the web and insert them vertically into the web, which bonds the fibres together (INDA, 2023).

³⁰⁷ Definitions of the technologies can be found in the Glossary of the Association of the Non-woven Fabrics Industry. Available [here](http://www.associationofnonwovens.com).

³⁰⁸ Mercerising is the process of treatment of cotton fabrics to increase dyeability (generally involving the use of sodium hydroxide) (Roth et al., 2023).

processes heavily rely on chemicals, using inputs like oxidising agents (e.g., chlorine, sodium hypochlorite, calcium hypochlorite, hydrogen peroxide), scouring alkali (e.g., caustic soda), sodium hydroxide, and also consume significant amounts of energy and water.

Dyeing

Dyeing is the process where dye particles diffuse into the textile material. This involves forming chemical bonds between the dye molecules and the textile. This is achieved based on the dye-fibre chemical affinity and the specific dyeing process. Dyes can be natural or man-made (synthetic) and are molecules containing chromophores that interact with light to produce colour (Singh and Bharati, 2014). Textile dyeing uses specific dyes that have an affinity or 'chemical attraction' to particular fibres. **Table 121** reports the fibres and their corresponding dyes, ensuring shades meet performance and fastness standards (Roth et al., 2023).

Table 121. Affinity relationship between dyes and fibres

Fibre type	Fibre name	Dye type
Cellulosic	Cotton	Direct dyes
	Linen	Reactive dyes
	Hemp	Vat dyes
	Jute	Sulphur dyes
	Modal	
	Lyocell	
	Viscose	
Proteic	Wool	Acid dyes
	Silk	Premetallised dyes
	Other fibres of animal hair	Reactive dyes
Man-made	Polyamide	Acid dyes
		Premetallised dyes
		Reactive dyes
		Disperse dyes
	Acrylic	Cationic dyes
		Disperse dyes
	Polyester	Disperse dyes

Source: own elaboration based on (Clark, 2011)

The affinity between the dye and fibre affects the final performance indicators for durability, such as washing and colour fastness. Moreover, various chemicals and auxiliary products are needed to enhance the efficiency and quality of the colouration process.

Dyeing can be conducted in batch, continuous, or semi-continuous modes and is suitable for both yarns and fabrics. The process requires specific temperatures and pressures for set durations. Discontinuous dyeing machines include autoclaves, hank dyeing machines, winches, becks, overflows, jets, jigs, paddles, and drum dyeing machines. Examples of continuous dyeing machines are pad-batch, pad-roll, pad-jig, pad-dry, and thermosol machines.

Overall, the dyeing process consumes significant amounts of water, chemicals, and energy.

Printing

Printing is the colouration process that involves applying dye or pigment to the substrate surface with the aid of specific auxiliaries. This allows for colouring different areas and creating patterns.

Pigments, whether natural or man-made, are insoluble molecules containing chromophores and typically lack affinity to fibres (Singh and Bharati, 2014). The colouring process involves applying a colour paste to the textile using various techniques, followed by fixation. An after-treatment may also be applied, though it can be optional.

- Pigment printing is the most commonly used technique. It does not rely on fibre affinity, allowing for the colouring of all fabric types. The colour's fixing to the fabric surface is determined by the auxiliaries used, such as binders, thickening agents, and others.

- Dye printing requires a more intricate composition for the printing paste. This complexity is influenced by the printing technique, substrate type, application method, and fixation procedures.

For both types of printing, the technology determines how the printing paste is applied. The most common methods include flat-screen printing, rotary-screen printing, roller printing, jet printing, and transfer printing (Roth et al., 2023).

The performance indicators for durability, such as washing and colour fastness, are affected by the composition of the printing pastes and the fixation methods. Since pigments do not bond with fibres, their colour fastness is less efficient as when dyes are used in the colouration process.

Overall, the process requires significant amounts of chemicals and energy.

Finishing

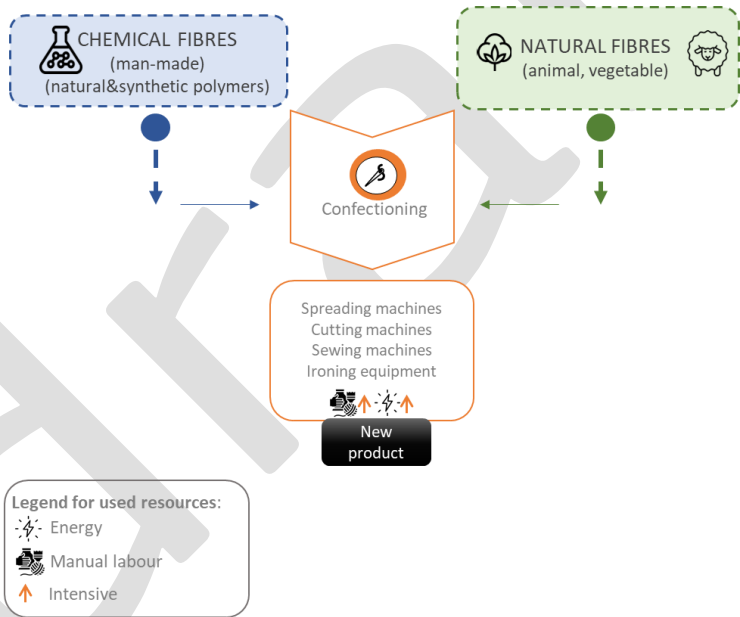
Finishing includes treatments designed to give textile substrates specific end-use properties, like unique visual effects, feel, or functional features such as waterproofing or flame resistance. A variety of chemical and mechanical/physical treatments are associated with this manufacturing stage, and they are typically applied after the colouration process. This stage consumes significant amounts of chemicals and energy (Roth et al., 2023).

Coating and laminating

Coating and laminating involve applying a thin, flexible polymeric film directly to the fabric's surface. The primary technologies employed are roller, spray, and slot die coating. Adding a polymer coating introduces new features to the fabric, such as resistance to dust, liquids, and gases. It can also enhance inherent physical properties, like the fabric's ability to resist abrasion. This stage, too, is chemically and energy-intensive (Roth et al., 2023).

13.3.5 Confectioning

Figure 83. Confectioning process



Source: own production and adapted from (McKinsey & Company, 2022), icons from www.flaticon.com

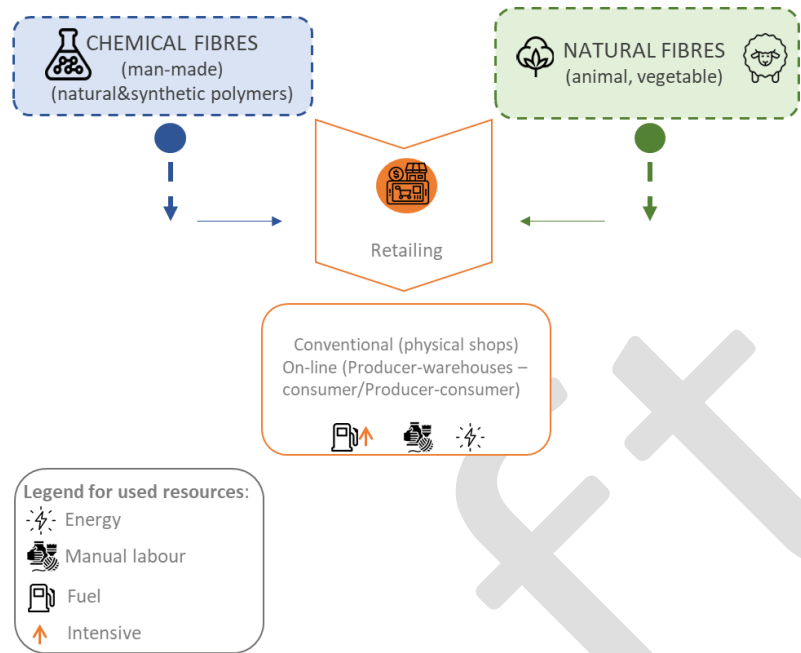
Confectioning is the production stage encompassing product design, fabric cutting, and final product assembly. It is the most manual phase in the production process (Nayak and Padhye, 2018). The fabric spreading and cutting stage is particularly energy-intensive and can be carried out using either mechanised (operator-assisted) or digitalised (automated) machines. The assembly phase is predominantly manual and labour-intensive, involving processes like sewing and culminating in ironing, which is also energy intensive.

In the final product manufacturing phase, the quality of the sewing process, combined with the sewing stitch and yarn count, significantly affects a product's seam strength, a key durability parameter. The type and count of the yarn play a crucial role in determining a textile product's longevity (Yassen, 2017) (Jankoska and Demboski, 2017).

This is the stage that produces post-industrial waste. By automating and digitalising the design and cutting stages, waste quantities can be reduced (de Mattos et al., 2022).

13.3.6 Retailing

Figure 84. Textile products retailing



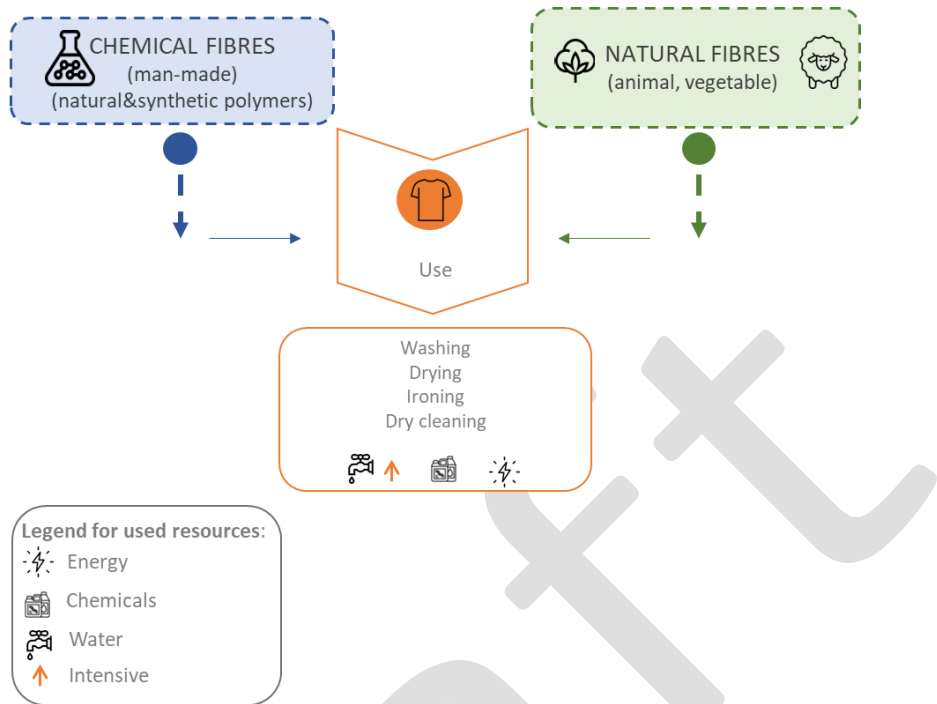
Source: own production adapted from (McKinsey & Company, 2022), icons from www.flaticon.com

Retail, defined by the Oxford dictionary as ‘the activity of selling goods to the public in shops, on the internet, etc.’ (Oxford dictionary, 2023), encompasses the transportation of final products and their sale through various channels (European Commission, 2023). The journey typically involves:

- Moving goods from the production network (manufacturers) to the export network (trade firms), and finally to the marketing networks (retailers) (de Mattos et al., 2022).
- Using various means of transport, including road, sea, rail, air, or a combination of these (multimodal transportation).

The primary resources used in this stage include fuel for transport, which leads to emissions, as well as labour and energy for sales operations. This stage produces pre-consumer waste. To reduce such waste, strategies like production on demand and integrated management systems for efficient inventory management can be employed (Aslan et al., 2015) (de Mattos et al., 2022) (Alyssa Hardy, 2020).

9653 **Figure 85.** Use phase of a textile product



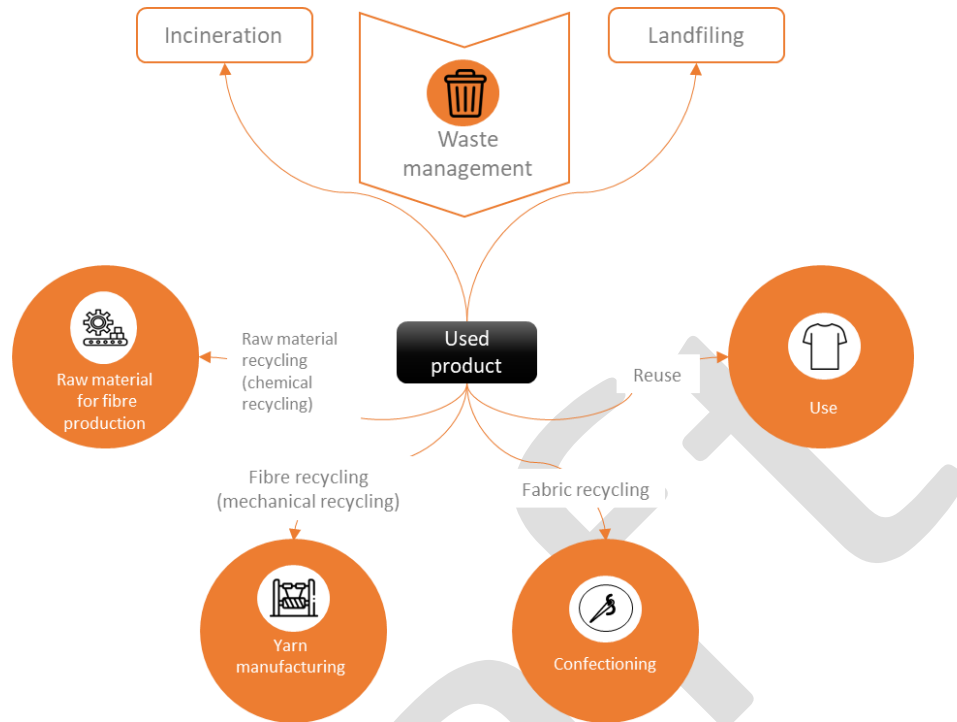
9654
9655 *Source: own production and adapted from (McKinsey & Company, 2022), icons from www.flaticon.com*

9656 The ‘use’ phase of a textile product encompasses activities such as washing, cleaning, drying, ironing, and
9657 steaming (Quantis, 2022). The specific requirements for these activities depend on the product and are detailed
9658 in its care label.

9659 This stage primarily consumes water and chemicals, with energy-intensive processes like ironing and steaming
9660 following closely. The physical durability of a textile product hinges on both the user's adherence to the care
9661 label instructions and the product's inherent characteristics.

13.3.8 Waste management

Figure 86. Textile products waste management



Source: own production and adapted from (McKinsey & Company, 2022), icons from www.flaticon.com

Most disposed textile apparel, at the end of their life, are either incinerated or sent to landfills. This process not only generates waste and emissions into the environment, but it is also energy intensive. Such discarded textiles, primarily consisting of clothing and home textiles, account for about 85% of total textile waste (McKinsey & Company, 2022). Currently, this follows a linear model. However, the energy recovered from incineration can be repurposed for other applications. The calorific value of textile waste is estimated to be between 3 599 and 5 200 MJ/Kg (Vargas and Yuleimy Ramírez, 2014) (Mustia et al., 2021).

Transitioning from this linear model to a circular one involves collecting, sorting, and pre-processing discarded textiles for fibre-to-fibre recycling. Textile waste management avenues include preparation for reuse, fabric recycling, fibre recycling (mechanical recycling), and raw material recycling (chemical recycling). Currently, only 30-35% of textiles are collected, with a significant portion of unsorted waste exported outside Europe (McKinsey & Company, 2022).

Textile recycling technologies fall into three main categories (Jørgensen et al., 2022):

Fibre or Mechanical Recycling: This method uses physical processes and can be applied to all textile waste types. It can also precede other recycling methods like thermo-mechanical, chemical, or biochemical (DG GROW, 2021b).

Raw Material or Chemical Recycling: This involves chemical processes to recycle monomers or polymers, suitable for materials like cotton, PA6, or PET (DG GROW, 2021b).

Thermal Recycling: This method uses heat. Thermo-mechanical recycling melts polymers, mainly for thermoplastic textiles like polyester, while thermo-chemical recycling breaks down polymers into base components, which can then be repurposed (DG GROW, 2021b; McKinsey & Company, 2022).

Additionally, **fabric recycling** involves refurbishing or remanufacturing discarded textiles, resulting in recycled products.

The primary solution to the current textile waste issue is recycling, coupled with waste reduction efforts (DG GROW, 2021b; McKinsey & Company, 2022).

13.4 Supporting information about tests and standards in the textile industry

Table 122. Working Groups with the involvement of several scientific committees in CEN/ TC 248.

Working Group	Topic
CEN/TC248/SC1	Burning behaviour of textiles, textile products and textile containing products
CEN/TC248/SC2	Physical properties
CEN/TC248/SC3	Chemical properties
CEN/TC248/WG 10	Size system of clothing
CEN/TC248/WG 11	Monofilaments
CEN/TC248/WG 12	Physical testing of knitted fabrics and garments
CEN/TC248/WG 13	Determination of resistance of textiles to microbiological attack
CEN/TC248/WG 14	UV protective properties
CEN/TC248/WG 15	Multifilament yarns
CEN/TC248/WG 16	Textiles in the healthcare system
CEN/TC248/WG 17	Hygienic quality of textiles proposed in industrial laundries and used in sectors it is necessary to control biocontamination
CEN/TC248/WG 18	AZO dyestuffs- Detection of certain aromatic amines
CEN/TC248/WG 19	Characterization of fibres
CEN/TC248/WG 20	Safety of children's clothing
CEN/TC248/WG 21	Terry towels
CEN/TC248/WG 22	Elasticity of fabrics
CEN/TC248/WG 23	Burning behaviour of nightwear
CEN/TC248/WG 24	Test methods for the flammability of textiles
CEN/TC248/WG 25	Cosmeto-textiles
CEN/TC248/WG 26	Methods of test for phthalates
CEN/TC248/WG 27	Determination of fibre proof properties of fabrics
CEN/TC248/WG 28	Thermoregulation
CEN/TC248/WG 29	Specifications and test methods for silk articles
CEN/TC248/WG 3	Ropes and agricultural twines and fishing nets
CEN/TC248/WG 30	Quantitative analysis of fibre mixtures
CEN/TC248/WG 31	Smart Textiles
CEN/TC248/WG 32	Organic, green and ecotextiles
CEN/TC248/WG 33	Labelling of superfine wool
CEN/TC248/WG 34	Joint Working Group between CEN/TC248 and CEN/TC 252 Risks in the sleeping environment
CEN/TC248/WG 35	Slide (zip) Fasteners
CEN/TC248/WG 36	Classification of textiles and textile products based on burning behaviour
CEN/TC248/WG 37	Microplastics from textile sources
CEN/TC248/WG 38	Circular Textiles Chain- Requirements and categories
CEN/TC248 WG 39	Circular Economy for textile products and textile chain. This group is fully operative while the norms are under development
CEN/TC248/WG 4	Coated fabrics
CEN/TC248/WG 5	Touch and close fasteners
CEN/TC248/WG 6	Sewing threads
CEN/TC248/WG 7	Cabinet roller towels
CEN/TC248/WG 8	Upholstery fabrics
CEN/TC248/WG 9	Prioritization of research topics
CEN/TC38/WG 35	Environmental aspects
ISO/TC173/SC/WG2	Urinary absorbing aids

Source: CEN official website ⁽³⁰⁹⁾

Table 123. Working Groups and subcommittees involved in ISO/TC 38.

Working Group	Topic
ISO ISO/TC 38/SC 1	Tests for coloured textiles and colorants
ISO/TC 38/SC 2	Cleansing, finishing and water resistance tests
ISO/TC 38/SC 20	Fabric descriptions
ISO/TC 38/SC 23	Fibres and yarns
ISO/TC 38/SC 24	Conditioning atmospheres and physical tests for textile fabrics
ISO/TC 38/CAG	Chair's Advisory Group

³⁰⁹ CEN Technical Committee Working Groups available at [this link](#)

Working Group	Topic
ISO/TC 38/WG 9	Nonwovens
ISO/TC 38/WG 17	Physiological properties of textiles
ISO/TC 38/WG 21	Ropes, cordage, slings and netting
ISO/TC 38/WG 22	Composition and chemical testing
ISO/TC 38/WG 23	Biological properties of textiles
ISO/TC 38/WG 27	Fabric properties relating to moisture
ISO/TC 38/WG 30	Tests for Biodegradability
ISO/TC 38/WG 31	Non-fibrous bio-based material for textiles
ISO/TC 38/WG 32	Smart textiles
ISO/TC 38/WG 33	Animal welfare in the textile supply chain
ISO/TC 38/WG 34	Microplastics from textile sources
ISO/TC 38/WG 35	Environmental aspects
ISO/TC 38/WG 36	Activated carbon fibre

9695

9696 All standards reported in this section include tests on a specific sample of the final product.

9697

Table 124. Standards directly related to intrinsic durability

Specific topic	ID code	Title	Type	Status
Abrasion	ISO 12945-1:2020	Determination of fabric propensity to surface pilling, fuzzing or matting — Part 1: Pilling box method	Standard	Published
Abrasion	ISO 12945-2:2020	Determination of fabric propensity to surface pilling, fuzzing or matting — Part 2: Modified Martindale method	Standard	Published
Pilling and abrasion	ISO 12945-3:2020	Determination of fabric propensity to surface pilling, fuzzing or matting — Part 3: Random tumble pilling method	Standard	Published
Pilling and abrasion	ISO 12945-4:2020	Determination of fabric propensity to surface pilling, fuzzing or matting — Part 4: Assessment of pilling, fuzzing and matting by visual analysis	Standard	Published
Colour fastness	EN ISO 105-A06:1995	Textiles — Tests for colour fastness — Part A06: Instrumental determination of 1/1 standard depth of colour	Standard	Published
Colour fastness	EN ISO 105-B01:2014	Textiles — Tests for colour fastness — Part B01: Colour fastness to light: Daylight	Standard	Published
Colour fastness	EN ISO 105-B02:2014	Textiles — Tests for colour fastness — Part B02: Colour fastness to artificial light: Xenon arc fading lamp test	Standard	Published
Colour fastness	EN ISO 105-B03:2014	Textiles — Tests for colour fastness — Part B03: Colour fastness to weathering: Outdoor exposure	Standard	Published
Colour fastness	ISO 105-C06:2010	Tests for colour fastness — Part C06: Colour fastness to domestic and commercial laundering	Standard	Published
Colour fastness	EN ISO 105-C12:2010	Textiles — Tests for colour fastness — Part C12: Colour fastness to industrial laundering	Standard	Published
Colour fastness	EN ISO 105-D01:2010	Textiles — Tests for colour fastness — Part D01: Colour fastness to drycleaning using perchloroethylene solvent	Standard	Published
Colour fastness	EN ISO 105-E01:2013	Textiles – tests for colour fastness – part E01: colour fastness to water	Standard	Published
Colour fastness	EN ISO 105-E02:2013	Textiles – tests for colour fastness – part E02: colour fastness to sea water	Standard	Published
Colour fastness	EN ISO 105-E03:2010	Textiles – tests for colour fastness – part E03: colour fastness to chlorinated water (swimming-pool water)	Standard	Published
Colour fastness	EN ISO 105-E04:2013	Textiles – tests for colour fastness – part E04: colour fastness to perspiration	Standard	Published
Colour fastness	EN ISO 105-E07:2010	Textiles – tests for colour fastness – part E07: colour fastness to spotting: water	Standard	Published
Colour fastness	EN ISO 105-N01:1993	Textiles – tests for colour fastness – part N01: colour fastness to bleaching: hypochlorite	Standard	Published
Colour fastness	EN ISO 105-N02:1995	Textiles – tests for colour fastness – part N02: colour fastness to bleaching: peroxide	Standard	Published
Colour fastness	EN ISO 105-N03:1995	Textiles – tests for colour fastness – part N03: colour fastness to bleaching: sodium chlorite (mild)	Standard	Published
Colour fastness	EN ISO 105-N04:1995	Textiles – tests for colour fastness – part N04: colour fastness to bleaching: sodium chlorite (severe)	Standard	Published
Colour fastness	EN ISO 105-X11:1994	Textiles — Tests for colour fastness — Part X11: Colour fastness to hot pressing	Standard	Published

Specific topic	ID code	Title	Type	Status
Colour fastness	EN ISO 105-X12:2016	Textiles — Tests for colour fastness — Part X12: Colour fastness to rubbing	Standard	Published
Abrasion resistance on coated fabric	ISO 5470-1:2016	Rubber- or plastics-coated fabrics — Determination of abrasion resistance — Part 1: Taber abrader	Standard	Published
Abrasion resistance on coated fabric	ISO 5470-2:2016	Rubber- or plastics-coated fabrics — Determination of abrasion resistance — Part 2: Martindale abrader	Standard	Published
Abrasion resistance on fabric	ISO 12947-2:2016	Textiles — Determination of the abrasion resistance of fabrics by the Martindale method — Part 2: Determination of specimen breakdown	Standard	Published
Abrasion resistance on fabric	ISO 12947-3:2016	Textiles — Determination of the abrasion resistance of fabrics by the Martindale method — Part 3: Determination of mass loss	Standard	Published
Abrasion resistance on fabric	ISO 12947-4:2016	Textiles — Determination of the abrasion resistance of fabrics by the Martindale method — Part 4: Assessment of appearance change	Standard	Published
Abrasion resistance on fabric	ASTM D3884-22	Standard Guide for Abrasion Resistance of Textile Fabrics (Rotary Platform Abrader Method)	Standard	Published
Abrasion resistance on fabric	ASTM D3885 -07A-19	Standard Test Method for Abrasion Resistance of Textile Fabrics (Flexing and Abrasion Method) (woven or nonwoven fabric)	Standard	Published
Abrasion resistance on fabric	ASTM D3886-22	Standard Test Method for Abrasion Resistance of Textile Fabrics (Inflated Diaphragm Apparatus) (both wet and dry/conditioned samples)	Standard	Published
Abrasion resistance on fabric	ASTM D4966-22	Standard Test Method for Abrasion Resistance of Textile Fabrics (Martindale Abrasion Tester Method)	Standard	Published
Abrasion resistance on fabric	ASTM D4158-08-20	Standard Guide for Abrasion Resistance of Textile Fabrics (Uniform Abrasion)	Standard	Published
Antifungal activity (Functional durability)	ISO 13629-1:2012	Textiles — Determination of antifungal activity of textile products — Part 1: Luminescence method	Standard	Published
Antifungal activity (Functional durability)	ISO 13629-2:2014	Textiles — Determination of antifungal activity of textile products — Part 2: Plate count method	Standard	Published
Antiviral activity (Functional durability)	ISO 18184:2019	Textiles — Determination of antiviral activity of textile products	Standard	Published
Appearance	ISO/TR 16323:2003	Textiles — Three-dimensional measuring apparatus for fabric appearance	Standard	Published
Bursting strength	EN ISO 13938-1:2019	Textiles: Bursting properties of fabrics. Part 1: Hydraulic method for determination of bursting strength and bursting distension	Standard	Published
Bursting strength	ISO 13938-2:2019	Textiles: Bursting properties of fabrics. Part 2: Pneumatic method for determination of bursting strength and bursting distension	Standard	Published
Bursting strength	ISO 9073-3:2023	Nonwovens — Test methods Part 3: Determination of tensile strength and elongation at break using the strip method	Standard	Published
Bursting strength	ISO 9073-4:2021	Nonwovens — Test methods Part 4: Determination of tear resistance by the trapezoid procedure	Standard	Published
Bursting strength	ISO 9073-5:2008	Textiles — Test methods for nonwovens — Part 5: Determination of resistance to mechanical penetration (ball burst procedure)	Standard	
Bursting strength	ASTM D3786/D3786M-18	Standard test method for bursting strength of textile fabrics. Diaphragm bursting strength tester method. Hydraulic or pneumatic diaphragm bursting tester	Standard	Published
Bursting strength	ASTM D3787-16	Standard test method for bursting strength of textile fabrics. The constant rate of traverse (CRT) Ball burst test	Standard	Published

Specific topic	ID code	Title	Type	Status
Bursting strength	ASTM D6797-15	Standard test method for bursting strength of textiles. The constant rate of extension (CRT) Ball burst test	Standard	Published
Colour fastness	BVL B 82.10-1:2011-12	Analysis of commodity goods - Testing of coloured children's toys with respect to their resistance to saliva and perspiration	Standard	Published
Colour fastness	DIN 53160-1:2010-10	Determination of the colourfastness of articles for common use. Part 1 Test with artificial saliva	Standard	Published
Colour fastness	DIN 53160-2:2010-10	Determination of the colourfastness of articles for common use. Part 2 Test with artificial sweat	Standard	Published
Colour fastness	ISO/TR 12116:2008	Textiles — Methods of simulating colour change during actual wear by means of laboratory colour-fastness tests	Standard	Published
Colour fastness	Oeko-Tex Standard 100. Class 1	Colour fastness to saliva and perspiration for children 3 years to younger	Standard	Published
Colour fastness	AATCC8 – 2016e	Test method for colour fastness to crocking	Test method	Published
Colour fastness	AATCC15 –2021e	Test method for colour fastness to perspiration	Standard	Published
Colour fastness	AATCC16 – 2004e	Test method for colour fastness to light	Standard	Published
Crease retention	ISO 7769:2009	Textiles — Test method for assessing the appearance of creases in fabrics after cleansing	Standard	Published
Crease retention	ISO 2313-1:2021	Textiles — Determination of the recovery from creasing of a folded specimen of fabric by measuring the angle of recovery — Part 1: Method of the horizontally folded specimen	Standard	Published
Crease retention	ISO 2313-2:2021	Textiles — Determination of the recovery from creasing of a folded specimen of fabric by measuring the angle of recovery — Part 2: Method of the vertically folded specimen	Standard	Published
Crease retention	AATCC test method 128-2017e	Test Method for wrinkle recovery of fabrics: appearance	Standard	Published. Not approved by ANSI
Crease retention	AATCC test method 66 – 2017e	Test Method for wrinkle recovery of woven fabrics: recovery angle	Standard	Published. Not approved by ANSI
Dimensional stability	ISO 5077:2007	Textiles — Determination of dimensional change in washing and drying	Standard	Published
Dimensional stability	ISO 3005:1978	Textiles — Determination of dimensional change of fabrics induced by free-steam	Standard	Published
Dimensional stability	ISO 7771:1985	Textiles — Determination of dimensional changes of fabrics induced by cold-water immersion	Standard	Published
Dimensional stability	ISO 16322-2:2021	Determination of spirality after laundering- Part 2 Woven and knitted fabrics	Standard	Published
Dimensional stability	ISO 675:2014	Textiles — Woven fabrics — Determination of dimensional change on commercial laundering near the boiling point	Standard	Published
Dimensional stability	ISO 9866-1:1991	Textiles — Effect of dry heat on fabrics under low pressure — Part 1: Procedure for dry-heat treatment of fabrics	Standard	Published
Dimensional stability	ISO 9866-2:1991	Textiles — Effect of dry heat on fabrics under low pressure — Part 2: Determination of dimensional change in fabrics exposed to dry heat	Standard	Published
Dimensional stability	ISO 21765:2020	Textiles — Determination of fabric deformability by forced mechanical distension	Standard	Published

Specific topic	ID code	Title	Type	Status
Dimensional stability	ISO 23231:2008	Textiles — Determination of dimensional change of fabrics — Accelerated machine method	Standard	Published
Dimensional stability (connected standard)	EN ISO 3759:2011	Textiles - Preparation, marking and measuring of fabric specimens and garments in tests for determination of dimensional change	Standard	Published
Dimensional stability (connected standard)	EN ISO 6330:2012	Domestic washing and drying procedures for textile testing	Standard	Published
Dimensional stability (connected standard)	EN ISO 15797:2018	Industrial washing and finishing procedures for testing of workwear	Standard	Published
Dimensional stability (connected standard)	ISO 3175-2:2017	Professional care, drycleaning and wet cleaning of fabrics and garments - Part 2: Procedure for testing performance when cleaning and finishing using tetrachloroethene	Standard	Published
Dimensional stability (connected standard)	ISO 3175-3:2017	Professional care, drycleaning and wet cleaning of fabrics and garments - Part 3: Procedure for testing performance when cleaning and finishing using hydrocarbon solvents	Standard	Published
Dimensional stability (connected standard)	ISO 3175-4:2017	Professional care, drycleaning and wet cleaning of fabrics and garments - Part 4: Procedure for testing performance when cleaning and finishing using simulated wet cleaning	Standard	Published
Easy-Care treatments (functional durability)	ISO 7768:2009	Textiles — Test method for assessing the smoothness appearance of fabrics after cleansing	Standard	Published
Flame retardancy (Functional durability)	ISO 12138:2017	Textiles — Domestic laundering procedures for textile fabrics prior to flammability testing	Standard	Published
Flame retardancy (Functional durability)	BS5651:1978	Method for cleansing procedure for use in the assessment of the effect of cleansing and wetting on the flammability of textile fabrics and fabric assemblies	Standard	Published
Loss of fibre fragments from textiles	AATCC TM212 – 2021e	Fibre fragment release during home laundering	Standard	Published
Loss of fibre fragments from textiles	ISO 4484	Textile and textile products- Microplastics from textile sources. Part1: Determination of material loss from fabrics during washing	Standard	Published
Loss of fibre fragments from textiles	ISO 4484-1:2023	Textile and textile products- Microplastics from textile sources. Part1: Determination of material loss from fabrics during washing	Standard	Published
Loss of fibre fragments from textiles	ISO 4484-2:2023	Textile and textile products- Microplastics from textile sources. Part 2: Qualitative and quantitative analysis of microplastics	Standard	Published
Loss of fibre fragments from textiles	ISO 4484-3	Textile and textile products- Microplastics from textile sources. Part 3: Measurement of collected material mass released from textile end products by domestic washing method	Standard	Published
Matting appearance after cleansing	ISO 16847:2016	Textiles — Test method for assessing the matting appearance of napped fabrics after cleansing	Standard	Published
Antibacterial activity	ISO 20743:2021	Textiles — Determination of antibacterial activity of textile products.	Standard	Published
Oil stain repellency (Functional durability)	ISO 14419:2010	Textiles - Oil repellency - Hydrocarbon resistance test	Standard	Published
Performance requirements	ASTM D5432-12- 19	Standard Performance specification for blanket products for institutional and household use	Standard	Published
Performance requirements	ASTM D411-12 – 19	Standard Performance specification for woven napery and tablecloth fabrics: household and institutional	Standard	Published
Performance requirements	ASTM D5433-12 – 19	Standard Performance specification for towel products for institutional and household use	Standard	Published

Specific topic	ID code	Title	Type	Status
Performance requirements	EN 13569:2001	Cabinet roller towels- Performance requirements and processing	Standard	Published
Performance requirements	EN 14697:2005	Textiles. Terry towels and terry towel fabrics. Specifications and methods for tests sets out requirements based on general applicated test methods	Standard	Published
Physiological comfort (Thermal resistance and Breathability (post-laundering) in PEFCR)	JIS L 1099 – 2021e	Textiles — Physiological effects — Measurement of thermal and water-vapour resistance under steady-state conditions (sweating guarded-hotplate test)	Standard	Published
Physiological comfort (Thermal resistance and Breathability (post-laundering) in PEFCR) (Functional property claims)	ISO 11092:2015	Textiles — Physiological effects — Measurement of thermal and water-vapour resistance under steady-state conditions (sweating guarded-hotplate test)	Standard	Published
Quality of buttons and press fasteners	CEN/TS 17394	Textiles and textile products.	Standard	Published
Quality of buttons and press fasteners	CEN/TS 17394-1:2021	Textiles and textile products – Part 1: Safety of children's clothing – Security of attachment of attached components to infants' clothing – Specification.	Standard	Published
Quality of buttons and press fasteners	CEN/TS 17394-2:2021	Textiles and textile products. Part 2: Safety of Children's clothing security of attachment of buttons. Test method	Standard	Published
Quality of buttons and press fasteners	CEN/TS 17394-3:2021	Textiles and textile products. Part 3: Safety of Children's clothing security of attachment of metal mechanically applied press fasteners. Test method	Standard	Published
Quality of zippers	ASTM D2061-07 (2021)	Standard test methods for strength test for zippers	Standard	Published
Quality of zippers	JIS S3015 – 2019e	Methods for measuring zipper dimensions standard test methods for strength	Standard	Published
Quality of zippers	EN 16732:2016	Slide fasteners (zips) Specifications	Standard	Published
Resistance to chlorinated water	ISO 17608:2015	Textiles — Bare elastane yarns — Determination of resistance to chlorinated water (swimming-pool water)	Standard	Published
Resistance to insect	ISO 3998:1977	Textiles — Determination of resistance to certain insect pests	Standard	Published
Resistance to surface wetting (Functional Durability)	ISO 9865:1991	Textiles - Determination of water repellency of fabrics by the Bundesmann rain-shower test	Standard	Published
Resistance to surface wetting (Thermal resistance and breathability (post-laundering) in PEFCR) (Functional property claims)	ISO 15496:2018	Textiles — Measurement of water vapour permeability of textiles for the purpose of quality control	Standard	Published
Seam slippage (wovens only)	ISO 13936-1:2004	Textiles — Determination of the slippage resistance of yarns at a seam in woven fabrics — Part 1: Fixed seam opening method	Standard	Published
Seam slippage (wovens only)	ISO 13936-2:2004	Textiles — Determination of the slippage resistance of yarns at a seam in woven fabrics — Part 2: Fixed load method	Standard	Published
Seam slippage (wovens only)	ISO 13936-3:2004	Textiles — Determination of the slippage resistance of yarns at a seam in woven fabrics — Part 3: Needle clamp method	Standard	Published
Seam slippage (wovens-only)	ASTM D4035/4034M-19	Standard test method for resistance to yarn slippage at the sewn seam in woven upholstery fabrics	Standard	Published
Seam smoothness	ISO 7770:2009	Textiles — Test method for assessing the smoothness appearance of seams in fabrics after cleansing	Standard	Published
Seam strength	EN-ISO 13935	Textiles — Seam tensile properties of fabrics and made-up textile articles	Standard	Published

Specific topic	ID code	Title	Type	Status
Seam strength	EN-ISO 13935-1:2014	Textiles — Seam tensile properties of fabrics and made-up textile articles — Part 1: Determination of maximum force to seam rupture using the strip method	Standard	Published
Seam strength	EN-ISO 13935-2:2014	Textiles — Seam tensile properties of fabrics and made-up textile articles — Part 2: Determination of maximum force to seam rupture using the grab method	Standard	Published
Seam strength	ASTM D1683-22	Standard test method for failure in sewn seams of woven fabrics	Standard	Published
Seam strength	ASTM D751-19	Standard test method for coated fabrics	Standard	Published
Spirality	ISO 16322-1:2005	Textiles — Determination of spirality after laundering — Part 1: Percentage of wale spirality change in knitted garments	Standard	Published
Spirality	ISO 16322-3:2021	Textiles — Determination of spirality after laundering — Part 3: Woven and knitted garments	Standard	Published
Stain Repellency (Functional durability)	ISO 22958:2021	Textiles — Water resistance — Rain tests: exposure to a horizontal water spray	Standard	Published
Stretch and recovery	ISO 20932-1:2018	Textiles — Determination of the elasticity of fabrics — Part 1: Strip tests	Standard	Published
Stretch and recovery	ISO 20932-2:2018	Textiles — Determination of the elasticity of fabrics — Part 2: Multiaxial tests	Standard	Published
Stretch and recovery	ISO 20932-3:2018	Textiles — Determination of the elasticity of fabrics — Part 3: Narrow fabrics	Standard	Published
Tear Strength	EN ISO 4674-1:2017	Rubber or plastic-coated fabrics- Determination of tear resistance- Part 1: constant rate of tear methods	Standard	Published
Tear Strength	EN-ISO 13937-1:2000	Textiles — Tear properties of fabrics — Part 1: Determination of tear force using ballistic pendulum method (Elmendorf)	Standard	Published
Tear Strength	EN-ISO 13937-2:2000	Textiles — Tear properties of fabrics — Part 2: Determination of tear force of trouser-shaped test specimens (Single tear method)	Standard	Published
Tear Strength	EN-ISO 13937-3:2000	Textiles — Tear properties of fabrics — Part 3: Determination of tear force of wing-shaped test specimens (Single tear method)	Standard	Published
Tear Strength	EN-ISO 13937-4:2000	Textiles — Tear properties of fabrics — Part 4: Determination of tear force of tongue-shaped test specimens (Double tear test)	Standard	Published
Tear Strength	ASTM D1424-21	Standard Test Method for Tearing Strength of Fabrics by falling-pendulum (Elmendorf-Type) apparatus	Standard	Published
Tensile strength and elongation (durability parameter according to OVAM report [3])	ISO 9073-3:1989	Textiles — Test methods for nonwovens — Part 3: Determination of tensile strength and elongation	Standard	Published
Tear Strength	ISO 9073-4:2021	Nonwovens — Test methods — Part 4: Determination of tear resistance by the trapezoid procedure	Standard	Published
Tear Strength	ISO 9073-18:2007	Textiles — Test methods for nonwovens — Part 18: Determination of breaking strength and elongation of nonwoven materials using the grab tensile test	Standard	Published
Tensile strength and elongation (durability parameter according to OVAM report [3])	ISO/FDIS 9073-3	Nonwovens — Test methods — Part 3: Determination of tensile strength and elongation at break using the strip method	DIS	Under development
Tear Strength	ASTM D5587 -15-19	Standard Test Method for Tearing Strength of Fabrics by trapezoid procedure apparatus	Standard	Published

Specific topic	ID code	Title	Type	Status
Tear Strength	ASTM D2261 -13-17	Standard Test method for Tearing Strength of fabric by the tongue (single rip) procedure (constant rate of extension tensile testing machine)	Standard	Published
Tear Resistance	ASTM D2582-21	Standard Test Method for Puncture-Propagation Tear Resistance of Plastic Film and Thin Sheeting	Standard	Published
Tensile strength (durability parameter according to OVAM report [3])	EN-ISO 13934-1:2013	Textiles: Tensile properties of Fabrics Part 1 Determination of maximum force and elongation at maximum force using the strip method	Standard	Published
Tensile strength (durability parameter according to OVAM report [3])	EN-ISO 13934-2:2014	Textiles: Tensile properties of Fabrics. Part 2 Determination of maximum force using the grab method	Standard	Published
Tensile strength and elongation	ASTM D5035-11	Standard test method for breaking force and elongation of textile fabrics (strip method)	Standard	Published
Tensile strength and elongation	ASTM D5034-21	Standard test method for breaking force and elongation of textile fabrics (grab method)	Standard	Published
Tensile strength and elongation	EN ISO 1421 (2016)	Rubber or plastic-coated fabrics - Determination of tensile strength and elongation at break	Standard	Published
Unevenness of textile	ISO 16549:2021	Unevenness of textile strands. Capacitance method	Standard	Published
Visible change after washing	ISO 15487:2018	Textiles — Method for assessing appearance of apparel and other textile end products after domestic washing and drying	standard	Published
Water repellency (Functional Durability)	ISO 4920:2012	Textile fabrics — Determination of resistance to surface wetting (spray test)	Standard	Published
Water resistance after aging	EN 343:2019	Rain Protection	Standard	Published
Wicking (Functional durability)	AATCC 197-2022e	Test method for vertical wicking rate of textiles to specified distances	Standard	NA
Wrinkle resistance	ISO 9867:2022	Textiles — Evaluation of the wrinkle recovery of fabrics — Appearance method	Standard	Published

Standards made of several parts and only the most used are reported in this Table: ¹Made of 4 parts. ²Made of 107 parts. ³Made of 2 parts. ⁴Made of 7 parts. ⁵Made of 2 parts. ⁶Made of 2 parts. ⁷Made of 20 parts. ⁸Made of 3 parts. ⁹Made of 2 parts. ¹⁰Made of 3 parts. ¹¹Made of 2 parts. ¹²Made of 6 parts. ¹³Made of 3 parts. ¹⁴Made of 2 parts. ¹⁵Made of 3 parts. ¹⁶Made of 2 parts. ¹⁷Made of 3 parts. ¹⁸Made of 4 parts. ¹⁹Made of 2 parts. ²⁰Made of 5 parts. ²¹Made of 20 parts. ²²It is made of 2 parts.

Every ID Code indicates the year depending on the Standard reference, as follows:

- ISO/CEN/BVL/BS -> XXX:YEAR (4-DIGIT)
- ASTM ->XXX-YEAR (2-DIGIT)
- AATC/JIS -> XXX-YEARe (4-DIGIT)

Table 125. Standards used by several frameworks to test parameters related to durability of textile products

Parameter	PEFCR A&F	EU Ecolabel	Nordic Ecolabel	Blue Angel
Resistance to pilling and abrasion	ISO 12945-1 (Pilling box method)	ISO 12945-1 Pill box method (Knitted and non-woven products)	ISO 12947-2 Martindale abrasion (Woven fabrics)	ISO 12945-1 Pill box method (Knitted and non-woven products)
	ISO 12947-2 Martindale abrasion (Woven and knitted products) and EN 13770 for knitted footwear garments, ASTM D3939 for snagging resistance and BS 8479 for propensity to snagging	ISO 12945-2 Martindale method (Woven fabrics)		ISO 12945-2 Martindale method (Woven fabrics)

Parameter	PEFCR A&F	EU Ecolabel	Nordic Ecolabel	Blue Angel
Colour fastness to dry rubbing	NA	ISO 105 X12		
Colour fastness to wet rub (Crocking)	NA	ISO 105 X12		
Colour fastness to perspiration	NA	ISO 105 E04 (Acid alkaline)	ISO 105 E04 (Acid alkaline) ISO 105 A06 (saliva)	ISO 105 E04 (acid and alkaline, comparison with multi-fibre fabric)
Colour fastness to light exposure	ISO 105-B02Xenon arc fading lamp test	ISO 105-B02		
Colour fastness to washing	NA	ISO 105 C06 (Domestic washing) ISO 15797 combined with ISO 105 C06 (Commercial washing)	ISO 105 C06 ISO 105 D01 if dry cleaning	ISO 105 C06 (single wash, at temperature marked on the product, with perborate powder)
Colour fastness to water	ISO 105-E02 for sea water	NA		
	ISO 105-E03 for chlorinated water			
Water repellency	Bundesmann ISO 9865 or ISO 4920	NA		ISO 6330 combined with ISO 4920 (Domestic)
				ISO 15797 combined with ISO 4920
				(Industrial washing)
Garment Integrity Test after washing	ISO 6330	ISO 6330 in combination with EN ISO 5077 (Domestic washing) ISO 15797 in combination with EN ISO 5077 (Industrial washing)		ISO 6330 and ISO 5077
Garment Integrity Test after dryclean	ISO 3175-2 to 3175-4	NA		
Oil repellency	NA	ISO 6330 (Domestic) ISO 15797 (Industrial) both in combination with ISO 14419	NA	
Soil or stain Release	NA	ISO 6330 (Domestic) ISO 15797 (Industrial) both in combination with ISO 22958	NA	
Flame retardant functions	NA	ISO 6330 in combination with ISO 12138 (Domestic)	NA	ISO 6330 in combination with ISO 12138 (Domestic)
		ISO 10528 in combination with ISO 12138 (Industrial)		ISO 10528 in combination with ISO 12138 (Industrial)
Easy care function	NA	ISO 7768	NA	ISO 7768
Garment Dimensional Stability Shrinkage & Skew/Twist/Torque	ISO 6330 indications of 5 cycles for machine wash and dry indications.	ISO 6330 combined with ISO 4920 (Domestic)	NA	
	ISO 3175-2 to 3175-4 if dry-clean indications of 3 cycles and ironing after final cycle.	ISO 15797 combined with ISO 4920		
	ISO 5077 and ISO 16322-3 spirality	(Industrial washing)		
Appearance	ISO 15487 with specific conditions for domestic and professional cleaning	NA		
Bursting	ISO 139381 (or-2)	NA		

9708
9709
9710
9711

9712

9713

9714

Parameter	PEFCR A&F	EU Ecolabel	Nordic Ecolabel	Blue Angel
Seam slippage	ISO 13936-2 (Woven fabrics)	NA	ISO 13936-1 o ISO 13936-2 (Woven fabrics)	NA
Fabric Tear Strength	ISO 13937-1	NA	ISO 13937-1 applied on outer fabric not include any inner lining in the product	NA
Fabric Tensile Strength	ISO 13934-2	NA	ISO 13934-2	NA
Seam strength	NA		ISO 13935-2 (Woven fabrics)	NA
Product waterproofness	ISO 811	NA		
Insulation	ISO11092 RCT	NA		
Elasticity of fabrics	ISO 20932-1 and ISO 20932-3	NA		

(a) PEFCR A&F: Product Environmental Footprint Category Rules (PEFCR): apparel and footwear;
(b) EU Ecolabel criteria for textile products. Commission Decision (2014/350/EU).
(c) Blue Angel – The German Ecolabel: DE-UZ 154 Basic Award Criteria. Edition January 2023, version 2.
(d) Nordic Swan Ecolabel: Textiles, hides/skins, and leather. Version 5.4.

Source: own elaboration

Table 126. Standards related to the functionality of the textile product

Specific topic	ID code	Title	Type	Status
Soil or stain release (Functional durability)	AATCC 130-18T, 2018	Washing Procedure selection to match care label	Standard	Published
Waterproofness	ISO 811:2018	Textiles — Determination of resistance to water penetration — Hydrostatic pressure test	standard	Published
Water penetration	ISO 18695:2007	Textiles — Determination of resistance to water penetration — Impact penetration test	Standard	Published
Water adsorption	ISO 18696:2006	Textiles — Determination of resistance to water absorption — Tumble-jar absorption test	Standard	Published
Moisturising effect	ISO 21232:2018	Textiles — Determination of moisturizing effect of textile materials by measurement of microclimate between textiles and simulated human skin using sweating guarded hotplate	Standard	Published
Hygroscopic heat generation	ISO 18782:2015	Textiles — Determination of dynamic hygroscopic heat generation	Standard	Published
Aqueous stain resistance	ISO 23232:2009	Textiles — Aqueous liquid repellency — Water/alcohol solution resistance test	Standard	Published
Thermal resistance	ISO 5085-1:1989	Textiles — Determination of thermal resistance — Part 1: Low thermal resistance	Standard	Published
Thermal resistance	ISO 5085-2:1990	Textiles — Determination of thermal resistance — Part 2: High thermal resistance	Standard	Published
Fire hazard reduction	ISO/TR 9240:1992	Textiles — Design of apparel for reduced fire hazard	Standard	Published
Superhydrophobic characteristics and durability assessment	ISO/CD TS 10818	Nanotechnologies — Textiles containing nanomaterials and nanostructures — Superhydrophobic characteristics and durability assessment	Technical Specifications	Under development
Resistance of cellulose-containing textiles to micro-organisms	ISO 11721-1:2001	Textiles — Determination of resistance of cellulose-containing textiles to micro-organisms — Soil burial test — Part 1: Assessment of rot-retardant finishing	Standard	Published

Specific topic	ID code	Title	Type	Status
Resistance of cellulose-containing textiles to micro-organisms	ISO 11721-2:2003	Textiles — Determination of the resistance of cellulose-containing textiles to micro-organisms — Soil burial test — Part 2: Identification of long-term resistance of a rot retardant finish	Standard	Published
Resistance of cellulose-containing textiles to micro-organisms	ISO 11737-3:2023	Sterilization of health care products — Microbiological methods — Part 3: Bacterial endotoxin testing	Standard	Published
Antifungal activity	ISO 13629-1:2012	Textiles — Determination of antifungal activity of textile products — Part 1: Luminescence method	Standard	Published
Antifungal activity	ISO 13629-2:2014	Textiles — Determination of antifungal activity of textile products — Part 2: Plate count method	Standard	Published
Exothermic and endothermic properties	ISO 16533:2014	Textiles — Measurement of exothermic and endothermic properties of textiles under humidity change	Standard	Published
Deodorant properties	ISO 17299-1:2014	Textiles — Determination of deodorant property — Part 1: General principle	Standard	Published
Deodorant properties	ISO 17299-2:2014	Textiles — Determination of deodorant property — Part 2: Detector tube method	Standard	Published
Deodorant properties	ISO 17299-3:2014	Textiles — Determination of deodorant property — Part 3: Gas chromatography method	Standard	Published
Deodorant properties	ISO 17299-4:2015	Textiles — Determination of deodorant property — Part 4: Condensation sampling analysis	Standard	Published
Deodorant properties	ISO 17299-5:2015	Textiles — Determination of deodorant property — Part 5: Metal-oxide semiconductor sensor method	Standard	Published
Moisture drying rate	ISO 17617:2014	Textiles — Determination of moisture drying rate	Standard	Published
Odour management	ISO 20645:2004	Textile fabrics — Determination of antibacterial activity — Agar diffusion plate test	Standard	Published
Total heat transfer	ISO 20852:2020	Textiles — Determination of the total heat transfer through textiles in simulated environments	Standard	Published
Dust mite resistance	ISO 21326:2019	Textiles — Test methods for determining the efficiency of products against house dust mite	Standard	Published
Burning behaviour	ISO 6940:2004	Textile fabrics — Burning behaviour — Determination of ease of ignition of vertically oriented specimens	Standard	Published
Burning behaviour	ISO 6941:2003	Textile fabrics — Burning behaviour — Measurement of flame spread properties of vertically oriented specimens	Standard	Published
Burning behaviour	ISO 10047:1993	Textiles — Determination of surface burning time of fabrics	Standard	Published
Adsorption	ISO 9073-6:2000	Textiles — Test methods for nonwovens — Part 6: Absorption	Standard	Published
Flexural rigidity	ISO 9073-7:1995	Textiles — Test methods for nonwovens — Part 7: Determination of bending length	Standard	Published
Liquid strike-through time	ISO 9073-8:1995	Textiles — Test methods for nonwovens — Part 8: Determination of liquid strike-through time (simulated urine)	Standard	Published
Drapability	ISO 9073-9:2008	Textiles — Test methods for nonwovens — Part 9: Determination of drapability including drape coefficient	Standard	Published
Lint and particle generation	ISO 9073-10:2003	Textiles — Test methods for nonwovens — Part 10: Lint and other particles generation in the dry state	Standard	Published
Run-off of liquids	ISO 9073-11:2002	Textiles — Test methods for nonwovens — Part 11: Run-off	Standard	Published

Specific topic	ID code	Title	Type	Status
Absorbency of fabric (wettability)	ISO 9073-12:2002	Textiles — Test methods for nonwovens — Part 12: Demand absorbency	Standard	Published
Liquid strike-through time	ISO 9073-13:2006	Textiles — Test methods for nonwovens — Part 13: Repeated liquid strike-through time	Standard	Published
Air permeability	ISO 9073-15:2007	Textiles — Test methods for nonwovens — Part 15: Determination of air permeability	Standard	Published
Resistance to penetration by water	ISO 9073-16:2007	Textiles — Test methods for nonwovens — Part 16: Determination of resistance to penetration by water (hydrostatic pressure)	Standard	Published
Water penetration	ISO 9073-17:2008	Textiles — Test methods for nonwovens — Part 17: Determination of water penetration (spray impact)	Standard	Published
Biodegradability	ISO 21701:2019	Textiles — Test method for accelerated hydrolysis of textile materials and biodegradation under controlled composting conditions of the resulting hydrolysate	Standard	Published
Air permeability	ISO 9237:1995	Textiles — Determination of the permeability of fabrics to air	Standard	Published
Electrostatic property	ISO 18080-1:2015	Textiles — Test methods for evaluating the electrostatic propensity of fabrics — Part 1: Test method using corona charging	Standard	Published
Electrostatic property	ISO 18080-2:2015	Textiles — Test methods for evaluating the electrostatic propensity of fabrics — Part 2: Test method using rotary mechanical friction	Standard	Published
Electrostatic property	ISO 18080-3:2015	Textiles — Test methods for evaluating the electrostatic propensity of fabrics — Part 3: Test method using manual friction	Standard	Published
Electrostatic property	ISO 18080-4:2015	Textiles — Test methods for evaluating the electrostatic propensity of fabrics — Part 4: Test method using horizontal mechanical friction	Standard	Published
Distortion of woven fabric	ISO 13015:2013	Woven fabrics — Distortion — Determination of skew and bow	Standard	Published
Max force and elongation at max force	ISO 24281:2021	Textiles — Biaxial tensile properties of woven fabric — Determination of maximum force and elongation at maximum force using the grab method	Standard	Published
Fire hazard reduction	ISO 12952-1:2010	Textiles — Assessment of the ignitability of bedding items — Part 1: Ignition source: smouldering cigarette	Standard	Published
Fire hazard reduction	ISO 12952-2:2010	Textiles — Assessment of the ignitability of bedding items — Part 2: Ignition source: match-flame equivalent	Standard	Published
Drying rate	ISO 13029:2012	Textiles — Determination of drying rate in dynamic state by the modified sweating-guarded hotplate	Standard	Published
Reduction activity of specific proteins	ISO 4333:2022	Textiles — Determination of reduction activity of specific proteins derived from pollen, mite and other sources on textile products	Standard	Published

Standards made of several parts and only the most used are reported in this Table: ¹ Made of 2 parts. ² Made of 2 parts. ³ Made of 2 parts. ⁴ Made of 5 parts. ⁵ Made of 20 parts. ⁶ Made of 4 parts. ⁷ Made of 2 parts. Every ID Code indicates the year depending on the Standard reference, as follows:

- ISO -> XXX:YEAR (4-DIGIT)
- AATC -> XXX-YEARe (4-DIGIT)

9720 **Table 127.** Standards used for textile characterization

Specific topic	ID code	Title	Type	Status
Mass per unit length and mass per unit area	ISO 3801:1977	Textiles — Woven fabrics — Determination of mass per unit length and mass per unit area	Standard	Published
Mass per unit length and mass per unit area	EN 12127:1998	Textiles. Fabrics. Determination of mass per unit area using small samples.	Standard	Published
Width and length	ISO 22198:2006	Textiles — Fabrics — Determination of width and length	Standard	Published
Colorimetric communication	ISO 10617:2010	Textiles — Standard data format for colorimetric communication — Textiles and related measurements	Standard	Published
Dyestuff identification	ISO 16373-1:2015	Textiles — Dyestuffs — Part 1: General principles of testing coloured textiles for dyestuff identification	Standard	Published
Fibre identification	ISO 18074:2015	Textiles — Identification of some animal fibres by DNA analysis method — Cashmere, wool, yak and their blends	Standard	Published
Fibre identification	ISO 20706-1	Textiles — Qualitative and quantitative analysis of some bast fibres (flax, hemp, ramie) and their blends — Part 1: Fibre identification using microscopy methods	Standard	Published
Thickness	ISO 5084:1996	Textiles — Determination of thickness of textiles and textile products	Standard	Published
Mass per unit area	ISO 9073-1:1989	Textiles — Test methods for nonwovens — Part 1: Determination of mass per unit area	Standard	Published
Thickness	ISO 9073-2:1995	Textiles — Test methods for nonwovens — Part 2: Determination of thickness	Standard	Published
Composition	ISO/TR 11827:2012	Textiles — Composition testing — Identification of fibres	Technical Report	Published

9721 Standards made of several parts and only the most used are reported in this Table: ¹ Made of 3 parts. ² Made of 20 parts.

9722 Every ID Code indicates the year depending on the Standard reference, as follows:

- 9723
- ISO/EN -> XXX:YEAR (4-DIGIT)

9724

9725 **Table 128.** Standards for the determination of specific substances

Specific topic	ID code	Title	Type	Status
Alkylphenol ethoxylates (APEO)	ISO 18254-1:2016	Textiles — Method for the detection and determination of alkylphenol ethoxylates (APEO) — Part 1: Method using HPLC-MS	Standard	Published
Alkylphenol ethoxylates (APEO)	ISO 18254-2:2016	Textiles — Method for the detection and determination of alkylphenol ethoxylates (APEO) — Part 2: Method using NPLC	Standard	Published
Formaldehyde	ISO 141842	Textiles — Determination of formaldehyde	Standard	Published
Formaldehyde	ISO 14184-1:2011	Textiles — Determination of formaldehyde — Part 1: Free and hydrolysed formaldehyde (water extraction method)	Standard	Published
Formaldehyde	ISO 14184-2:2011	Textiles — Determination of formaldehyde — Part 2: Released formaldehyde (vapour absorption method)	Standard	Published
Formaldehyde	ISO/CD 14184	Textiles — Determination of formaldehyde	CD	Under development
Formaldehyde	ISO/CD 14184-3	Textiles — Determination of formaldehyde — Part 3: Free and hydrolysed formaldehyde (extraction method) —Determination by high pressure liquid chromatography	CD	Under development

Specific topic	ID code	Title	Type	Status
Phthalate	ISO 14389:2022	Textiles — Determination of the phthalate content — Tetrahydrofuran method	Standard	Published
Per- and polyfluoroalkyl substances (PFAS)	FprEN 17681-1:2024	Textiles and textile products - Per- and polyfluoroalkyl substances (PFAS) - Part 1: Analysis of an alkaline extract using liquid chromatography and tandem mass spectrometry/	Standard	Draft
Per- and polyfluoroalkyl substances (PFAS)	EN 17681-1:2022	Textiles and textile products - Organic fluorine - Part 1: Determination of non-volatile compounds by extraction method using liquid chromatography	Standard	Published
Phthalate	EN ISO 18856:2004	Water quality- Determination of selected phthalates using gas chromatography/mass spectrometry	Standard	Published
Phthalate	EN 14602:2012	Footwear- Test methods for the assessment of ecological criteria	Standard	Published
Organotin compounds	EN ISO 17353:2007	Water quality- Determination of selected organotin compounds- Gas chromatographic method	Standard	Published
Aromatic amines derived from azo colorants	ISO 14362-1:2017	Textiles — Methods for determination of certain aromatic amines derived from azo colorants — Part 1: Detection of the use of certain azo colorants accessible with and without extracting the fibres	Standard	Published
Organotin compounds	ISO 22744-1:2020	Textiles and textile products — Determination of organotin compounds — Part 1: Derivatisation method using gas chromatography	Standard	Published
Organotin compounds	ISO 22744-2:2020	Textiles and textile products — Determination of organotin compounds — Part 2: Direct method using liquid chromatography	Standard	Published
PAHs	EN 17132:2019	Textiles and textile products — Determination of Polycyclic Aromatic Hydrocarbons (PAH), method using gas chromatography	Standard	Published
PAHs	ISO 16190:2021	Footwear — Critical substances potentially present in footwear and footwear components — Test method to quantitatively determine polycyclic aromatic hydrocarbons (PAHs) in footwear materials	Standard	Published
DMFa	EN 17131:2019	Textiles and textile products - Determination of dimethylformamide (DMF), method using gas chromatography	Standard	Published
DMFa and other aprotic solvents	prEN 17131-1	Textiles and textile products - Determination of certain residual solvents - Part 1: Determination of aprotic solvents, method using gas chromatography	Standard	Draft
Aromatic amines derived from azo colorants	ISO 143625	Textiles — Methods for determination of certain aromatic amines derived from azo colorants	Standard	Published
Aromatic amines derived from azo colorants	ISO 14362-3:2017	Textiles — Methods for determination of certain aromatic amines derived from azo colorants — Part 3: Detection of the use of certain azo colorants, which may release 4-aminoazobenzene	Standard	Published
Detection of disperse dyestuff	DIN 54231:2022	Textiles- Determination of dyes after methanol extraction	Standard	Published
Chlorinated benzenes and toluenes	UNE-EN 17137:2019	Textiles - Determination of the content of compounds based on chlorobenzenes and chlorotoluenes	Standard	Published
Determination of extractable metals	EN ISO 105-E04:2013	Textiles – tests for colour fastness – part E04: colour fastness to perspiration	Standard	Published
Determination of extractable metals	DIN 542337	Testing of Textiles- Determination of metals.	Standard	Published
Determination of extractable metals	UNE-EN 16711-2:2016	Textiles - Determination of metal content - Part 2: Determination of metals extracted by acidic artificial perspiration solution	Standard	Published
Determination of extractable metals	DIN EN 16711-2:2016	Textiles- Determination of metal content- Part 2: Determination of metals extracted by acidic artificial perspiration solution	Standard	Published

Specific topic	ID code	Title	Type	Status
Determination of extractable metals	DIN 38405-24:1987-05	German standard methods for the examination of water, wastewater and sludge: anions (group D; photometric determination of chromium (VI) using 1,5-diphenylcarbonohydrazide (D24)	Standard	Published
Determination of the metallic composition	EN12472-2020	Method for the simulation of the accelerated wear and corrosion for the detection of nickel release from coated items	Standard	Published
Determination of the metallic composition	ISO 1811-1:1998	Copper and copper alloys — Selection and preparation of samples for chemical analysis. Part1: sampling of cast unwrought products	Standard	Published
Hexavalent chromium	ISO 17075-1:2017	Leather — Chemical determination of chromium (VI) content in leather. Part 1: Colorimetric method	Standard	Published
Hexavalent chromium	ISO 17075-1:2017	Leather — Chemical determination of chromium (VI) content in leather. Part 2: Chromatographic method	Standard	Published
Bisphenols	ISO/CD 23377	Footwear — Critical substances potentially present in footwear and footwear components — Test method to quantitatively determine certain bisphenols in footwear materials	Standard	Draft
Bisphenols	ISO 11936:2023	Leather — Determination of total content of certain bisphenols	Standard	Published
Determination of chlorophenols	EN ISO 17070:2015	Leather- Chemical test- Determination of tetra chlorophenol, trichlorophenol content	Standard	Published
Determination of chlorophenols	EN 17134-2:2023	Textiles and textile products - Determination of biocide additives - Part 2: Chlorophenol-based preservatives, method using gas chromatography	Standard	Published
Dyestuff identification	ISO 16373-2:2014	Textiles — Dyestuffs — Part 2: General method for the determination of extractable dyestuffs including allergenic and carcinogenic dyestuffs (method using pyridine-water)	Standard	Published
Dyestuff identification	ISO 16373-3:2014	Textiles — Dyestuffs — Part 3: Method for determination of certain carcinogenic dyestuffs (method using triethylamine/methanol)	Standard	Published
Flame retardance	ISO 17881-1:2016	Textiles — Determination of certain flame retardants — Part 1: Brominated flame retardants	Standard	Published
Flame retardance	ISO 17881-2:2016	Textiles — Determination of certain flame retardants — Part 2: Phosphorus flame retardants	Standard	Published
Flame retardance	ISO/TR 17881-3:2018	Textiles — Determination of certain flame retardants — Part 3: Chlorinated paraffin flame retardants	Standard	Published
Index ingredients	ISO 2219511	Textiles — Determination of index ingredient from coloured textiles	Standard	Published
Index ingredients	ISO 22195-1:2020	Textiles — Determination of index ingredient from coloured textiles — Part 1: Madder	Standard	Published
Index ingredients	ISO 22195-2:2020	Textiles — Determination of index ingredient from coloured textiles — Part 2: Turmeric	Standard	Published
Index ingredients	ISO 22195-3:2023	Textiles — Determination of index ingredient from coloured textile — Part 3: Myrobalan	Standard	Published
Index ingredients	ISO 22195-4:2021	Textiles — Determination of index ingredient from coloured textile — Part 4: Catechu	Standard	Published
Index ingredients	ISO 22195-5:2021	Textiles — Determination of index ingredient from coloured textile — Part 5: Lac	Standard	Published
Index ingredients	ISO 22195-6:2021	Textiles — Determination of index ingredient from coloured textile — Part 6: Punica granatum	Standard	Published
Many substances	CEN/TR 16741:2015	Textiles and textile products - Guidance on health and environmental issues related to chemical content of textile products intended for clothing, interior textiles and upholstery	Technical Report	Published

9726 Standards made of several parts and only the most used are reported in this Table: ¹ Made of 2 parts. ² made of 2 parts. ³ Made of 2 parts. ⁴ Made of 2 parts. ⁵ made of 2 parts. ⁶ Made of 107 parts. ⁷ Made of 2
9727 parts. ⁸ Made of 3 parts. ⁹ Made of 3 parts. ¹⁰ Made of 3 parts. ¹¹ Made of 6 parts.
9728 Every ID Code indicates the year depending on the Standard reference, as follows:
9729 • ISO/DIN/CEN -> XXX:YEAR (4-DIGIT)

9730

9731 **Table 129.** Standards related to loss of fibre fragments from textiles

Specific topic	ID code	Title	Type	Status
Loss of fibre fragments from textiles	ISO 4484-1	Textile and textile products- Microplastics from textile sources. Part1: Determination of material loss from fabrics during washing	Standard	Published
Loss of fibre fragments from textiles	ISO 4484-2	Textile and textile products- Microplastics from textile sources. Part 2: Qualitative and quantitative analysis of microplastics	Standard	Published
Loss of fibre fragments from textiles	ISO 4484-3	Textile and textile products- Microplastics from textile sources. Part 3: Measurement of collected material mass released from textile end products by domestic washing method	Standard	Published
Loss of fibre fragments from textiles	AATCC TM212:2021	Test Method for Fibre Fragment Release During Home Laundering	Standard	Published

9732 Standards made of several parts and only the most used are reported in this Table: ¹ Made of 3 parts.

9733 Every ID Code indicates the year depending on the Standard reference, as follows:

9734 • ISO -> XXX:YEAR (4-DIGIT)

9735

9736 **Table 130** Standards and Technical Specifications potentially related to circularity and environmental aspects

Broader topic	Specific topic	ID code	Title	Type	Status
Definitions	Environmental aspects	ISO 5157:2023 ISO TC 38	Textiles — Environmental aspects — Vocabulary	Standard	Published
Circularity	Principles of circularity	Unknown developed by CEN TC 248 WG 39	Textiles - Circular economy for textile products - General principles and guidance	Technical Specifications	Under development
Circularity	Material use	Unknown developed by CEN TC 248 WG 39	Textiles - Circular economy for textile products – Categorisation of and requirements on non-virgin input materials	Technical Specifications	Under development
Circularity	Design for circularity	Unknown developed by CEN TC 248 WG 39	Textiles - Circular economy for textile products - design for circularity	Technical Specifications	Under development
Environmental claims	Terminology of Environmental claims	CEN/TS 16822:2015	Textiles and textile products - Self-declared environmental claims - Use of the terms	Technical Specifications	Published

9737

9738 **Table 131.** Standards related to topics not covered by **Table 124** to **Table 130**

Broader topic	Specific topic	ID code	Title	Type	Status
Definitions	Definitions of natural fibres	EN ISO 6938:2014	Textiles — Natural fibres — Generic names and definitions	Standard	Published
Definitions	Environmental aspects	ISO 5157:2023 ISO TC 38	Textiles — Environmental aspects — Vocabulary	Standard	Published
Definitions	Stitch types	ISO 4915:1991	Textiles — Stitch types — Classification and terminology	Standard	Published
Definitions	Seam types	ISO 4916:1992	Textiles — Seam types — Classification and terminology	Standard	Published
Definitions	Determination of mass	ISO 6348:1980	Textiles — Determination of mass — Vocabulary	Standard	Published
Definitions	Woven fabric	ISO 2959:2011	Textiles — Woven fabric descriptions	Standard	Published
Definitions	Weaves	ISO 3572:1976	Textiles — Weaves — Definitions of general terms and basic weaves	Standard	Published
Definitions	Knitted fabrics	ISO 8388:1998	Knitted fabrics — Types — Vocabulary	Standard	Published
Definitions	Description of defects	ISO 8498:1990	Woven fabrics — Description of defects — Vocabulary	Standard	Published
Definitions	Description of defects	ISO 8499:2003	Knitted fabrics — Description of defects — Vocabulary	Standard	Published
Definitions	Nonwoven	ISO 9092:2019	Nonwovens — Vocabulary	Standard	Published
Definitions	Nonwoven	ISO 11224:2003	Textiles — Web formation and bonding in nonwovens — Vocabulary	Standard	Published
Definitions	Superfine woven wool	ISO 18103:2015	Superfine woven wool fabric labelling — Requirements for Super S code definition	Standard	Published
Definitions	Man-made fibres	ISO 2076:2021	Textiles — Man-made fibres — Generic names	Standard	Published
Labelling	Labelling	ISO 3758:2012	Textiles — Care labelling code using symbols	Standard	Published
Labelling	Symbols on workwear	ISO 30023:2021	Textiles — Qualification symbols for labelling workwear to be industrially laundered	Standard	Published
Circularity	Principles of circularity	Unknown developed by CEN TC 248 WG 39	Textiles - Circular economy for textile products - General principles and guidance	Standard	Under development
Circularity	Material use	Unknown developed by CEN TC 248 WG 40	Textiles - Circular economy for textile products - Categorisation of and requirements on non-virgin input materials	Standard	Under development
Circularity	Design for circularity	Unknown developed by CEN TC 248 WG 41	Textiles - Circular economy for textile products - design for circularity	Standard	Under development
Environmental claims	Terminology of Environmental claims	CEN/TS 16822:2015	Textiles and textile products - Self-declared environmental claims - Use of the terms	Technical Specifications	Published

9739 Every ID Code indicates the year depending on the Standard reference, as follows:
9740 ISO/CEN -> XXX:YEAR (4-DIGIT)

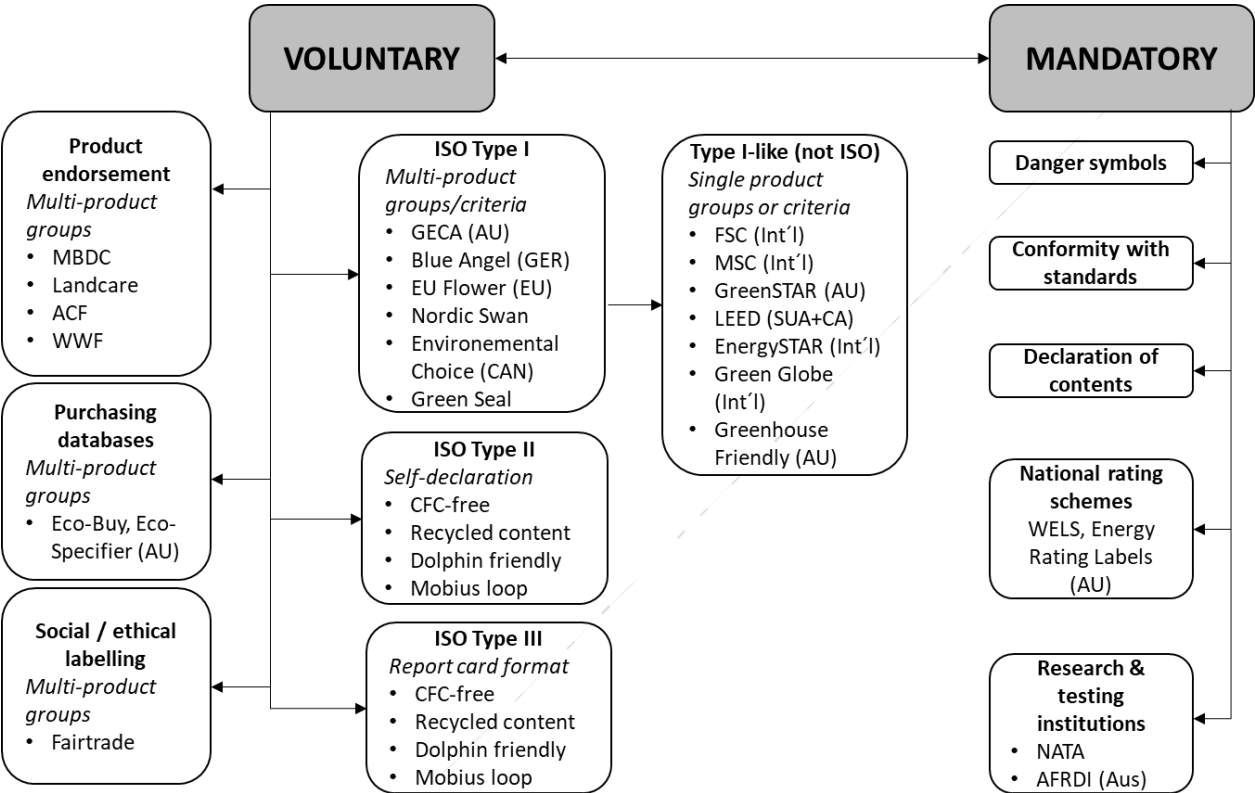
9741

13.5 Supporting information about environmental labels

13.5.1 Types of labels in general

There are many types of labels, addressing single or multiple environmental issues and covering different sectors and regions. **Figure 87** shows the taxonomy of labels, which includes mandatory and voluntary frameworks (Frydendal et al., 2018).

Figure 87. Taxonomy of labels for communication of specific product information



Source: (Frydendal et al., 2018)

The Ecolabel Index⁽³¹⁰⁾ is known to be the largest global directory of labels related to environmental and social aspects. On 12 January 2024, it counted 456 labels in 199 countries, and 25 industry sectors. Due to the high number and types of voluntary environmental labels, the International Organization for Standardization (ISO) established specific guidelines.

ISO Type I environmental labels (ISO 14024:2018)⁽³¹¹⁾, known as **Ecolabels**, are defined as 'voluntary, multi-criteria-based and third party-verified labels that indicate an overall environmental preference in a life cycle perspective of a product or service within a specific product category'. Ecolabels are a subset of the environmental labels that recognise a better environmental performance of a given product or service related to other products in the same product group. An Ecolabel must fulfil the following nine requirements (Frydendal et al., 2018):

1. It is built on the life cycle thinking;

³¹⁰ Ecolabel Index. Website available at [this link](#). Last accessed on 12 January 2024.

³¹¹ ISO 14024:2018. Environmental labels and declarations. Type I environmental labelling. Principles and procedures. Available at [this link](#). Last accessed on 12 January 2024.

- 9761 2. It is based on multiple criteria, therefore it addresses many environmental impact categories, such as
9762 global warming, eutrophication, etc.;
- 9763 3. Criteria are justified by sound scientific and engineering reasons. Their objectivity is ensured by the
9764 involvement of a broad range of stakeholders (industries, consumers, governments, etc...) during the
9765 process of criteria development;
- 9766 4. It comprises requirements related to functional aspects of the product (fitness for use). This
9767 characteristic guarantees a sufficient quality of labelled products and services;
- 9768 5. Criteria are continuously updated according to changes affecting the labelled products. These changes
9769 could be related to technologies, the regulatory context and other aspects;
- 9770 6. All stages of criteria development are transparent;
- 9771 7. It is accessible to all potential applicants;
- 9772 8. It involves third-party certification;
- 9773 9. There is compliance monitoring after the licence is awarded.

9774 **ISO Type II environmental labels** (ISO 14021:2016) ⁽³¹²⁾, known as **Self-declared Environmental Claims**,
9775 are neither third-party verified, nor based on Life Cycle Thinking approach. Many self-declared environmental
9776 claims on the EU market do not necessarily follow the ISO 14021:2016.

9777 **ISO Type III environmental labels** (ISO 14025:2016) ⁽³¹³⁾, known as **Environmental Declarations**, are
9778 labels presenting *quantified environmental information on the life cycle of a product to enable comparisons*
9779 *between products fulfilling the same function*. The Environmental Declarations were mainly envisioned for
9780 business-to-business communication, but their use in business-to-consumer communication is not precluded.
9781 The establishment of Product Category Rules ensures that the life cycle assessment is performed with specific
9782 rules aiming to foster transparency and facilitate comparisons between different Environmental Declarations.
9783 The EC promotes the establishment of commonly recognised Product Environmental Footprint Category Rules
9784 (PEFCRs) ⁽³¹⁴⁾. As any life cycle assessment, the Environmental Declarations are third-party verified.

9785 13.5.2 Voluntary environmental labels used in EU

9786 **Table 132.** Environmental labels used in 2021 in Europe: ISO type and addressed topics

Environmental label	Criteria	ISO TYPE
AnbafaltToxic	other	NA
Better Cotton Initiative	Biodiversity, chemicals, natural resources, pesticides/herbicides/fungicides, soil, wastewater/sewage, water quality, water use, other	NA
Blue Angel	Carbon/GHG emission, Energy use/efficiency, forests, natural resources, recycling, toxics, waste water/sewage, water quality, water use, other	Type I ³¹⁵
bluesign® standard	Carbon/GHG emission, Carbon/GHG offsets, chemicals, energy production/source, energy use/efficiency, material use, natural resources, pesticides/herbicides/fungicides, recycling, toxics, waste, wastewater/sewage, water quality, water use	NA
Carbon Reduction Label	Carbon/GHG emission	NA
Climatop	Animal welfare, biodiversity, Carbon/GHG emission, chemicals, energy production/sources, energy use/efficiency, forests, GMOs, material use, natural resources, pesticides/herbicides/fungicides, recycling, soil, toxics, waste, wastewater/sewage, water quality, water use	NA
Compostability Mark of European Bioplastics	Chemicals, Material use, Natural resources, pesticides/herbicides/fungicides, recycling, toxics, waste, water quality	NA
Coop Naturaline:Switzerland	Biodiversity, chemicals, GMOs, natural resources, pesticides/herbicides/fungicides, soil, toxics, wastewater/sewage	NA

³¹² ISO 14021:2016. Environmental labels and declarations. Self-declared environmental claims (Type II environmental labelling). Available at [this link](#). Last accessed on 12 January 2024.

³¹³ ISO 14025:2006. Environmental labels and declarations. Type III environmental declarations. Principles and procedures. Available at [this link](#). Last accessed on 12 January 2024.

³¹⁴ European platform on LCA | EPLCA. Environmental footprint. Available at [this link](#). Last accessed on 12 January 2024.

³¹⁵ Blue Angel – The German Ecolabel: DE-UZ 154 Basic Award Criteria. Edition January 2023, version 2. Available at [this link](#). Last accessed on 12 January 2024.

Environmental label	Criteria	ISO TYPE
Cradle to Cradle Certified (CM) Products Program	Chemicals, energy production/source, energy use/efficiency, forests, material use, natural resources, recycling, toxics, waste, wastewater/sewage, water quality, water use	NA
Danish Indoor Climate Label	Other (emission to indoor air)	NA
ECOLOGO	Animal welfare, biodiversity, Carbon/GHG emissions, Carbon/GHG offsets, Chemicals, energy production/sources, energy use/efficiency, forests, GMOs, material use, natural resources, pesticides/herbicides/fungicides, recycling, soil, toxics, waste, wastewater/sewage, water quality, water use	Type I ³¹⁶
Ekologicky setrny vyrobek/Environmentally Friendly Product	Carbon/GHG emission, Carbon/GHG offsets, chemicals, energy production/sources, energy use/efficiency, material use, natural resources, pesticides/herbicides/fungicides, recycling, soil, toxics, waste, wastewater/sewage, water quality, water use	Type I **
Environmental product declaration (EPD)	Carbon/GHG emission, Carbon/GHG offsets, chemicals, energy production/source, energy use/efficiency, material use, natural resources, recycling, toxics, waste, water quality, water use	Type III ³¹⁷
EU Ecolabel	Chemicals, energy use/efficiency, forests, material use, natural resources, recycling, toxics, waste, wastewater/sewage, water quality, water use, pesticides, herbicides, fungicides, ectoparasitocides, GMOs, air emissions, fitness for use, and corporate social responsibility	Type I ³¹⁸
Fair for life	Animal welfare, biodiversity, Carbon/GHG emissions, energy use/efficiency, GMOs, natural resources, pesticides/herbicides/fungicides, soil, water use	NA
Fairtrade	Biodiversity, energy use/efficiency, forests, GMOs, natural resources, pesticides/herbicides/fungicides, soil, toxics, waste, water use	Type II***
Global Organic Textile Standard	Chemicals, GMOs, Material use, natural resources, pesticides/herbicides/fungicides, soil, toxics, wastewater/sewage, water quality	NA
Good Environmental choice "Bra Miljoval"	Chemicals, energy production/sources, energy use/efficiency, forests, material use, natural resources, toxics	Type I ³¹⁹
Good Shopping Guide Ethical Award	Animal welfare, Carbon/GHG emissions, Carbon/GHG offsets, chemicals, energy production/sources, energy use/efficiency, GMOs, material use, natural resources, pesticides/herbicides/fungicides, recycling, toxics, waste water quality	NA
Good Weave	Other (workplace environmental conditions)	NA
Green Crane: Ukraine	Biodiversity, Carbon/GHG emissions, chemicals, energy production/sources, energy use/efficiency, forests, material use, natural resources, recycling, toxics, waste, wastewater/sewage, water quality, water use	Type I****
Greenguard	Chemicals, toxics	Type II*
Green Shape	Animal welfare, biodiversity, chemicals, energy use/efficiency, GMOs, material use, natural resources, pesticides/herbicides/fungicides, recycling, toxics, waste, wastewater/sewage, water use	NA
GUT	For floor coverings. Chemicals, toxics, other (indoor air quality)	NA
IMO Certified	Fire testing certification. Chemicals, forests, pesticides/herbicides/fungicides, soil, toxics	NA
Label Step	For carpets. Chemicals, wastewater/sewage, water quality	NA
Milieukeur: the Dutch environmental quality label	Animal welfare, biodiversity, Carbon/GHG emissions, chemicals, energy production/sources, energy use/ efficiency, forests, GMOs, material use, natural resources, pesticides/herbicides/fungicides, recycling, soil, toxics, waste, wastewater/sewage, water quality, water use	Type I**
National Programme of Environmental Assessment and Ecolabelling in the Slovak Republik (NPEHOV)	Carbon/GHG emissions, chemicals, energy production/sources, energy use/efficiency, forests, material use, natural resources, pesticides/herbicides/fungicides, recycling, soil, toxics, waste, wastewater/sewage, water quality, water use	Type I**
Naturland e.V.	Animal welfare, biodiversity, chemicals, forests, GMOs, natural resources, pesticides/herbicides/fungicides, soil, toxics	NA
Nordic Swan Ecolabel	Carbon/GHG emission, chemicals, energy production/sources, energy use/efficiency, material use, natural resources, recycling, soil, toxics, waste,	Type I ³²⁰

³¹⁶ Ecogloballabel website available [here](#)

³¹⁷ EPD website available [here](#)

³¹⁸ Ecolabel website available [here](#)

³¹⁹ Idem

³²⁰ Nordic Swan website available [here](#)

Environmental label	Criteria	ISO TYPE
	wastewater/sewage, water quality, water use, forests, animal welfare, microplastic (loss of fibre fragments)	
Oeko-Tex Standard 100	Toxics prohibited by law, chemicals harmful to health	Other Type I-like voluntary sustainable scheme ³²¹
OK biobased	Carbon/GHG emission, material use, natural resources, waste	NA
ÖkoControl	Forests, natural resources, pesticides/herbicides/fungicides, toxics	NA
Ølabel: Norway	Chemicals, forests, GMOs, pesticides/herbicides/fungicides, soil	NA
Processed Chlorine Free	Biodiversity, Carbon/GHG emissions, Carbon/GHG offsets, chemicals, energy production/sources, energy use/efficiency, forests, material use, natural resources, pesticides/herbicides/fungicides, recycling, toxics, waste, wastewater/sewage, water quality, water use	NA
SEE What You Are Buying Into	Animal welfare, biodiversity, Carbon/GHG emissions, Carbon/GHG offsets, energy production/sources, energy use/efficiency, GMOs, material use, natural resources, recycling, waste, other	NA
Singapore Green Label Scheme (SGLS)	Animal welfare, biodiversity, Carbon/GHG emission, Carbon/GHG offsets, chemicals, energy production/sources, energy use/efficiency, forests, GMOs, material use, natural resources, pesticides/herbicides/fungicides, recycling, soil, toxics, waste, wastewater/sewage, water quality, water use	Type I ³²²
Skal Eko Symbol	Pesticides/herbicides/fungicides	NA
SMArT Consensus Sustainable Product Standards	Animal welfare, biodiversity, Carbon/GHG emissions, Carbon/GHG offsets, Chemicals, energy production/sources, energy use/efficiency, forests, GMOs, material use, natural resources, pesticides/herbicides/fungicides, recycling, soil, toxics, waste, wastewater/sewage, water quality, water use	NA
Soil Association Organic Standard	Animal welfare, biodiversity, chemicals, forests, GMOs, natural resources, pesticides/herbicides/fungicides, soil, toxics	NA
SustentaX	Animal welfare, Carbon/GHG emissions, Carbon/GHG offsets, chemicals, energy production/sources, energy uses/efficiency, forests, material use, natural resources, pesticides/herbicides/fungicides, recycling, soil, toxics, waste, wastewater/sewage, water quality, water use	NA
Terra Cycle	Carbon/GHG offsets, material use, recycling, waste	NA
Totally Chlorine Free	Biodiversity, Carbon/GHG emissions, Carbon/GHG offsets, chemicals, energy production/sources, energy use/efficiency, forests, GMOS, material use, natural resources, pesticides/herbicides/fungicides, recycling, soil, toxics, waste, wastewater/sewage, water quality, water use	NA
WindMade	Carbon/GHG emission, energy production/sources	NA

The topic of 'Toxics' addresses harmful substances that are already prohibited or regulated; whereas the topic of 'Harmful chemicals' address substances that are known to be harmful to health, but are not officially banned. N.B. The authors of the referenced study use the term 'Chemicals' when referring to 'Harmful chemicals'.

The authors of the referenced study do not specify if the topic of 'Waste' addresses any kind of waste generated along the value chain, or waste generated at a specific stage. No further explanation/specification is provided for other topics.

NA: Not available.

Source: own elaboration based on (Ranasinghe and Jayasooriya, 2021) and the Ecolabel Index⁽³²³⁾.
 *(Ranasinghe and Jayasooriya, 2021), ** (LEITAT, 2017), *** (Ziveh and Cinelli, 2023), **** (UNEP, 2023b)

³²¹ Ecogloballabel website available [here](#)

³²² Idem

³²³ Ecolabel Index. Website available at this link. Last accessed on 12 January 2024.

13.6 Supporting information on market analysis

13.6.1 Recycling plants

Table 133. Number of textile recycling plants classified per location

Region	Country	Mechanical				Chemical				Total
		All	Pilot scale	Full scale	Unknown scale	All	Pilot scale	Full scale	Unknown scale	
EU	Austria	0	NA	NA	NA	1	NA	1	NA	1
	Belgium	3	NA	2	1	0	NA	NA	NA	3
	Denmark	3	NA	2	1	1	1	NA	NA	4
	Finland	3	1	2	NA	2	2	NA	NA	5
	France	4	NA	3	1	1	1	NA	NA	5
	Germany	5	NA	5	NA	0	NA	NA	NA	5
	Italy	13	1	9	3	4	1	3	NA	17
	Netherlands	6	2	3	1	1	1	NA	NA	7
	Spain	11	2	9	NA	0	NA	NA	NA	11
	Sweden	0	NA	NA	NA	1	NA	1	NA	1
Europe (Non-EU)	Norway	0	NA	NA	NA	1	NA	1	NA	1
	Switzerland	0	NA	NA	NA	3	1	1	1	3
	UK	1	NA	1	NA	0	NA	NA	NA	1
Middle East	Israel	0	NA	NA	NA	1	NA	1	NA	1
	Türkiye	3	NA	3	NA	0	NA	NA	NA	3
Africa	Mauritius	1	NA	1	NA	0	NA	NA	NA	1
	Morocco	1	NA	1	NA	0	NA	NA	NA	1
Asia	Bangladesh	1	NA	1	NA	0	NA	NA	NA	1
	China	4	1	2	1	5	NA	4	1	9
	India	11	1	7	3	1	NA	NA	1	12
	Indonesia	1	NA	NA	1	1	NA	1	NA	2
	Japan	3	NA	2	1	1	NA	NA	1	4
	Malaysia	1	NA	1	NA	0	NA	NA	NA	1
	Pakistan	3	NA	2	1	0	NA	NA	NA	3
	Singapore	1	NA	NA	1	0	NA	NA	NA	1
	Taiwan	1	NA	1	NA	8	1	6	1	9
America	Thailand	1	NA	1	NA	0	NA	NA	NA	1
	Canada	1	NA	1	NA	1	1	NA	NA	2
	Guatemala	1	NA	1	NA	0	NA	NA	NA	1
	Mexico	1	NA	1	NA	0	NA	NA	NA	1
Oceania	USA	5	NA	5	NA	7	4	1	2	12
	Australia	0	NA	NA	NA	1	NA	1	NA	1

The recycling plants identified were equal to 130. NA: Not available

Source: own elaboration based on Airtable - Sorting for Circularity - Recyclers Database ⁽³²⁴⁾, (Jørgensen et al., 2022; Textile Exchange, 2022a)

Table 134. Number of textile recycling plants classified per input fibres

Input fibre	Mechanical				Chemical				Total
	Tot	Pilot scale	Full scale	Unknown scale	Tot	Pilot scale	Full scale	Unknown scale	
Cotton	53	4	41	8	17	7	8	2	70
Polyester (PES)	51	2	36	13	26	13	8	5	77
Blends of cotton and PES	14	1	12	1	9	3	5	1	23
Man-Made Cellulosic (MMC)	5	NA	5	NA	5	1	3	1	10
Polyamide (PA)	13	1	10	2	6	NA	5	1	19
Wool	40	5	33	2	0	NA	NA	NA	40
Blends of wool and synthetic fibres	21	NA	17	4	0	NA	NA	NA	21
Acrylic (PAC)	13	1	11	1	0	NA	NA	NA	13
Other plant-based	10	NA	10	NA	1	NA	1	NA	11
Other synthetic	8	NA	7	1	0	NA	NA	NA	8
Other blends	6	NA	5	1	1	NA	1	NA	7

³²⁴ Airtable - Sorting for Circularity - Recyclers Database. Available at [this link](#). Last accessed on 31 January 2024.

Input fibre	Mechanical				Chemical				Total
	Tot	Pilot scale	Full scale	Unknown scale	Tot	Pilot scale	Full scale	Unknown scale	
Silk	4	1	3	NA	0	NA	NA	NA	4

Numerous plants could process more than one type of textile fibre.

NA: Not available

Source: own elaboration based on Airtable - Sorting for Circularity - Recyclers Database ⁽³²⁵⁾, (Jørgensen et al., 2022; Textile Exchange, 2022a)

13.6.2 Comparison among BREFs

Table 135. Environmental performance levels for emission to water by BREFs in the World

Country/ organisation Environmental parameter	European Union (EU-BREF)	India (MINAS)	South Korea	United States (US EPA)	World Bank (EHS Guideline)	China	China (GB 4287- 2012)***
AOX (mg/L)	0.1-0.4	NA	NA	NA	1	30-40 (only for a specific wool treatment)	8-12
BOD (mg/L)	No BAT-AEL applies for BOD (**)	30	NA	1.4-35.2 kg/kkg(*) Daily max 0.7-17.6 kg/kkg (*) 30-day avg New 1.4-16.9 kg/kkg (*) Daily max 0.7-8.7 kg/kkg (*) 30-day avg	30	NA	15-50
COD (mg/L)	40-100	100	18-30	21.3-256.8 kg/kkg (*)	160	7-30000	60-200
Chromium (mg/L)	0.01-0.1	2	0.5	Daily: 0.023-0.22 kg/kkg (*) 30-day average: 0.011-0.11 kg/kkg (*)	0.5	NA	Not detectable
Copper (mg/L)	0.03-0.4	NA	1	NA	0.5	NA	NA
Nickel (mg/L)	0.01-0.1	NA	0.25	NA	0.5	NA	NA
Zinc (mg/L)	0.04-0.5	NA	-	NA	2	NA	NA

(*) Kilograms per 1000 Kilograms. The conversion to milligrams per litre (mg/L) is not straightforward. For enhanced accuracy, it is deemed more prudent to retain the unit in its original form in accordance with the United States Best Available Technology (BAT) standards.

(**) As an indication, the yearly average BOD5 level in the effluent from a biological wastewater treatment plant will generally be ≤ 10 mg/L.

(***) The reported data include limits on direct and indirect water pollutant emissions from dyeing and finishing textile enterprises in China, including special limits for certain geographical areas.

Source: (Ministry of Ecology and Environment, China, 2012; OECD, 2022),

³²⁵ Airtable - Sorting for Circularity - Recyclers Database. Available at [this link](#). Last accessed on 31 January 2024.

9816 **Table 136.** Environmental performance levels for emission to water by the Chinese BREF

Process	Fibres	Origin of wastewater	AOX (mg/L)	COD cr Concentration (mg/L)
Processing of raw materials	Hemp	Biological removal	NA	7-3000
		Chemical removal of glue	NA	2000-4000
		Joint removal of glue	NA	1000-3000
	Silk	Boiling cocoon and filament processes	NA	80-400
		Refining: oil removal	NA	20-12000
		Refining: water washing	NA	800-4000
	Wool	Washing	NA	40-9000
		Carbonization (**)	NA	200-400
		Silk-Resistant (***)	30-40	400-600
	Chemical fibres	NA	NA	200-600
Finishing fabrics	Cotton, Hemp & mixed machine fabrics	Slurry removal	NA	30-10000
		Cooking (****)	NA	1000-2000
		Bleaching	NA	200-400
		Mercerization	NA	500-2000
		Dyeing	NA	500-2500
		Printing	NA	1200-2000
		Organizing (*)	NA	2000-10000
	Wool	Dyeing	NA	800-2000
		Organizing (*)	NA	300-1000
	Silk	Pre-Treatment	NA	1500-2500
		Dyeing	NA	500-1500
		Printing	NA	1200-2000
	Chemical Fibres	Refining	NA	10-8000
		Polyester fabric alkali reduction	NA	10000-30000
		Dyeing	NA	500-800
		Printing	NA	1000-2000
		Organizing (*)	NA	2000-5000
	Knitwear	Finishing	NA	500-800
	Yarn	Finishing	NA	1000-2000

(*) Organizing wastewater includes waste finishing fluids and equipment cleaning wastewater.

(**) Carbonization is the process of removing plant impurities before the comb process by chemical means.

(***) The silk photorestriction process is the process of using chlorine as a chemical auxiliary to remove surface scales of wool and apply softeners.

(****) Cooking is the process of further removing impurities such as grease, wax, pectin and other impurities of fibre using thermoaline and surfactants

9824 13.6.3 Service lifespan

9825 **Table 137.** Studies addressing the lifespan of textile apparel

Study	Type of study	Source of data	Comments
(WRAP, 2017a)	Technical report	Online survey	Country: UK Sample: 3 244 people
(Laitala and Klepp, 2020a)	Scientific article	Wardrobe survey and data collected by previous research studies	Wardrobe survey: Countries: China, Germany, Japan, the UK, and the USA Sample: 213–230 respondents per country. Respondents' statistics: 100% 18–64 years old 51.6% Women 48.4% Men Researched studies from:

Study	Type of study	Source of data	Comments
			Australia, Norway, UK (2 studies) Netherlands (2 studies), Denmark, Finland and Norway
(Quantis, 2022)	Technical report	Producer judgement and literature studies	Values proposed by part of the industry within the development of the Product Environmental Footprint Category Rules for apparel and footwear
(Roos et al., 2015)	Technical report	Surveys and expert assumptions supported by national statistics on related topics	The study uses this data to perform an environmental assessment of the Swedish fashion consumption
(Laitala, IG Klepp, et al., 2018a)	Scientific article	Elaboration based on surveys	Average values from surveys carried out in: Norway, Netherlands, Greece, Spain, USA, Sweden, Germany, Poland, Australia, Canada and Finland
(Gray et al., 2022)	Key Findings Report	Surveys	Country: UK Sample: 6 000 interviewed users who purchase clothing at least once a year. Analysis on 44 807 items
(Drycleaning Institute, 2015)	Technical report	Expert judgement	Data are provided by the association of Australian dry cleaners.

9826 Source: Own elaboration

9827 **Table 138.** Parameters related to lifespan of specific types of apparel

Parameter	T-shirts	Shirts	Knitwear	Jeans	Socks
Average wear days per year	25	16	30	75	50
Implied wear days per month	2.1	1.3	2.5	6.2	4.2
Total days of wear for the garment's whole lifetime	112.5	80	150	300	125
Hours of wear during lifetime	1 350	960	1 800	3 600	1 500
Assumed days of wear per wash	2	2	5	10	2
Hours of wear per wash	24	24	60	120	24
Average number of washes for the target lifetime	56	40	30	30	62

9828 Source: (WRAP, 2017a)

9829

9830 **Table 139.** Possession span expressed as years by studies reported in Laitala et al. (2018)

Apparel type	Wardrobe Audit Survey in Seven Countries	Wardrobe Audit Interviews Norway (Textile Waste)	Survey. Norway	Online Survey. UK	Survey. UK	16 Households' Purchases. Netherlands	Survey. Netherlands	Survey (Germany. Poland. Sweden and USA)	Online Survey. Finland
T-shirts	4.6	4.2	NA	4	3.3	6.8	NA	3-4	4.5
Shirts and blouses	4.6	NA	5.6	3.3 (Blouses) 4.3 (shirts)	3.6	7.2	NA	NA	5.7
Sweaters and midlayers	5.8	NA	10.8 (woollen sweaters)	4.5	3.7	7.1	6.17 (Woollen sweaters)	NA	NA
Jackets and coats	5.3 (Jackets) 6.3 (Coats)	4 (Jackets)	6.4 (Coats)	6.5 (Jackets) 6.2 (Coats)	NA	11.5 (Jackets) / 11.6 (Coats)	NA	NA	7.6 (Coats)
Pants and shorts	4.9 (Pants) 3.9 (Jeans)	4.3 (Jeans)	4.4	5.4 (Pants) 3.8 (Jeans)	3.1 (Jeans)	6.2	2.45 (Cotton jeans)	3-4 (Jeans)	5.3
Dresses. skirts and jumpsuits	4.5 (Dresses) 4.8 (Skirts)	4.1 (Skirts)	NA	4.7 (Dresses) 5.2 (Skirts)	NA	15.2 (Skirts)	NA	NA	NA
Leggings, stockings. tights and socks	3.6 (Socks and stockings)	2.9 (Socks)	NA	2.4 (Socks)	1.8 (Socks)	NA	NA	NA	2.3 (Socks)
Underwear	2.5	4.4	NA	2.4	NA	NA	NA	NA	3
Swimwear	NA	NA	NA	NA	NA	NA	NA	NA	NA
Textile apparel accessories	NA	NA	NA	NA	NA	NA	NA	NA	NA

Source: (Laitala, IG Klepp, et al., 2018a)

9831

9832

9833 **Table 140.** Possession span, expressed as years, of types of apparel with specific fibre content

Garment Category	Cotton and blends	Synthetic/Man made	Wool and Blends	Silk
Pants/trousers	4.2	5.1	4.8	NA
Jackets	4.3	4.9	5.7	NA
Coats	5.8	8.4	5.3	NA
Sweaters	5.6	6.5	6.0	NA
Shirts and blouses	3.8	6.2	6.0	8.5
Ties	9.5	12.8	9.3	14.5
Socks and stockings	3.3	4.2	5.5	4.3
Underwear	2.2	3.2	3.9	3.5

Source: (Laitala, IG Klepp, et al., 2018a)

9835

9836 **13.7 Supporting information about user behaviour**

9837 **13.7.1 Pre-purchase aspects**

9838 Approximately, Europeans purchase 14.8 kg of textile products annually (EEA, 2022b; Vladimirova et al., 2024),
9839 which includes 6.0 kg of clothing, or roughly 24 new garments each year. However, these numbers conceal
9840 significant inequalities as purchasing frequency rises with income (WRAP, 2022; Vladimirova et al., 2024).

9841 **13.7.1.1 Reasons for purchasing**

9842 McNeill and Moore (2015) distinguished two specific approaches consumers have towards apparel: 1)
9843 consumers who see apparel as purely functional and 2) those who view apparel as self-representational. This
9844 influences what aspects are the most important for consumers at the point of sale.

9845 A survey among Dutch consumers show that before shopping, almost half of 1 046 users claim to know what
9846 they are going to buy (D&B, 2020a) and they indicate that the main reasons for buying apparel are the following
9847 (listed in order from most to less common):

- 9848 1. Look good or fun;
- 9849 2. Replacing old, worn or broken apparel;
- 9850 3. Buy “essential” apparel, meaning apparel that is basic and for everyday wear, i.e. one-colour t-shirt;
- 9851 4. Want to join the trend;
- 9852 5. Want new apparel;
- 9853 6. Need apparel for a special occasion.

9854 A survey representative of the German population showed that a third of respondents associate fun with buying
9855 apparel. In the 40-70 age group the proportion of those who enjoy buying apparel is significantly higher than
9856 the population average. In the female surveys from this social group, it is more than 60%. Fun shopping for
9857 apparel is also widespread among female respondents from the young group (Kleinhüchelkotten et al., 2018a).

9858 According to a survey by the Swedish Environmental Protection Agency (2020) involving approximately 3 000
9859 respondents, the main reason for purchasing new clothing is that existing garments are worn out. Other
9860 frequently cited reasons include changes in fit and general wear and tear.

9861 Another survey targeting 200 Dutch consumers showcased that consumers may be more likely to buy new
9862 apparel when is outdated, in terms of appearance or when they are afraid the item will break again quickly
9863 after being repaired (milieu centraal, 2021).

9864 It shall be noted that there are a number of factors that may influence consumers buying behaviour: brand
9865 perception and exclusivity, personal factors such as age, gender, culture, etc., marketing campaigns, economic
9866 conditions, etc. (Anisha and Kalaivani, 2016; Khanna, 2021). In particular, malicious interface design strategies
9867 have been investigated to trick users into decisions against their best interests, such as spending money
9868 (Schäfer et al., 2023). These patterns are prevalent in digital services, including online shopping sites (Yada et

al., 2022). An example of dark patterns that may be more prevalent in apparel purchases is the 'low-stock message' which entail a false claim that a product is nearly sold out to create a sense of urgency and encourage quicker purchases (Schäfer et al., 2023). Moreover, the results of an analysis of webshops ran by the Swiss consumer organisation (FRC) and the association Public Eye, showcased that dark patterns frequently tempt consumers to purchase additional items. For instance, 13 shops displayed supposedly matching items as soon as something was added to the online shopping cart, while 10 shops calculated shipping costs to encourage higher spending if the purchase amount is too low (FRC and Public Eye, 2022). Apparel is often bought spontaneously without thinking long beforehand. Only 16% of the respondents clearly disagree with the corresponding statement. In turn, the proportions of spontaneous buyers and those who regularly 'declutter' their wardrobe to make room for new things are above average in the 40-70 age group. More than 60% of female respondents in this population segment are frequent spontaneous buyers. The relatively high proportion of spontaneous buyers as a whole, leads to the conclusion that only a minority inform themselves correctly before buying apparel. The differences between the social groups are not particularly large in this regard (Kleinhückelkotten et al., 2018a). Interestingly, a survey conducted among 1 050 Dutch consumers reveals that while many individuals value apparel for its quality and prefer to creatively combine existing pieces of apparel rather than frequently buy new ones, a significant portion of respondents remain neutral toward overconsumption. For example, 47% of participants preferred styling existing apparel over wearing something new daily, and 45% viewed apparel as valuable and to be treated with care. However, neutrality was highest (up to 37%) when asked about the positivity of buying large quantities of apparel items, indicating that while sustainable attitudes are present, they coexist with ambivalence and ongoing consumption tendencies (Giaquinto and Keuchenius, 2023).

The aspects affecting the purchase decision of apparel among consumers could be influenced by discount offers that stimulate impulsive purchases both in on-line sites and in physical shops (Djafarova and Bowes, 2021). In fact, a phenomenon that affects both type of stores around the globe is the New Year sales which incentivise the purchase motivation (Amasawa and Kimita, 2023). It seems clear that the more impulsive purchases consumers make, the more often they buy apparel per year while also spending more money on apparel each month (D&B, 2020a).

Even though visual aids are used in social media channels to produce a similar purchasing experience to the one in physical stores (Djafarova and Bowes, 2021), the impulse purchases seem to be more associated to the physical stores and affect especially the younger population (Cook and Yurchisin, 2017). However, online retailers have also reported that a significant portion of their customer base comprised individuals who engaged in casual shopping for fashion items without a specific utilitarian purpose often characterized as impulse buyers (Vinted, 2021). When users have events coming up (parties, travelling plans, etc.) then, purchases of apparel become a priority and this may lead to impulsive purchases (Hultén and Vanyushyn, 2014). Impulse purchase tendency may be also triggered more often among consumers who are wealthier and also have more time available for purchasing apparel (Hultén and Vanyushyn, 2014).

Moreover, according to a survey among 2 500 consumers in the United Kingdom, Sweden, Italy, France and Germany, offering discounts can prompt excessive consumption, leading to feelings of regret among buyers. Although the initial thrill of finding a bargain can be exhilarating, it often results in impulsive purchases and a focus on obtaining deals rather than considering actual needs. Zalando survey indicates that 65% of consumers prioritize low prices or discounts when shopping apparel. The allure of a discounted purchase is particularly strong, with 52% of females and 46% of males emphasizing the importance of the excitement associated with buying. This tendency suggests that many consumers may prioritize the thrill of the purchase over making thoughtful, sustainable choices, potentially leading to overconsumption. Interestingly, despite recognizing that impulsive bargain hunting contradicts their sustainability values, consumers frequently experience post-purchase regret. A substantial 82% of respondents admit to feeling some form of regret after shopping, with 28% expressing concerns about environmental impacts and labour conditions during production (Heiny and Schneide, 2021).

13.7.1.2 Criteria used when buying apparel

Studies revealed four main attributes concerning apparel purchase decision: brand, style, place of purchase, and price (Iwanow et al., 2005, as cited in Harris et al., 2016). Price being one of the main drivers in apparel purchase decision (Adıgüzel et al., 2020).

Fashion trends are significantly more important for younger consumers than for older ones (AK Wienn and Greenpeace, 2023a; Spaepen et al., 2021) and they are even seen as a sign of success (AK Wienn and Greenpeace, 2023a). Preference for brand new items and aspects such as comfort when wearing an item of

apparel, its country of origin, type of textile material and even knowledge about the brand are important drivers for older generations of consumers (Generation X and Baby Boomers) (Spaepen et al., 2021).

According to Mishra et al., 2023, perceived value refers to “the subjective evaluation of the worth or benefits that a customer believes they will receive from a product or service”. In this sense, a survey of over 3 000 American adults of over 18 years old indicates that ‘value’ was the first driver influencing purchasing behaviour across apparel followed by quality (Thredup, 2023). Similarly, 1 506 Austrians were surveyed about their apparel consumption and most of them reported functionality (92%) and quality (85%) to be among the most important criteria when purchasing apparel (AK Wienn and Greenpeace, 2023a). The monetary value of the apparel follows the ranking closely with 78% of respondents who claim that a low price is a very important factor when acquiring apparel (AK Wienn and Greenpeace, 2023a). Similarly, among the primary factors influencing Spanish consumers’ apparel purchases are quality and price, with environmental impacts also playing a significant role, being quite or very influential for 44.2% of over 2 000 surveyed consumers (CECU, 2023).

Currently, the EU lacks harmonized labelling regulations that require the disclosure of working conditions where textile products are made. This lack of information on social responsibility is seen as an issue by 65% of respondents to the Textile Labelling Regulation Public Consultation, as it makes it difficult to make informed purchasing decisions (Evaluation report of the Textile Labelling Regulation, under development).

Sustainability and social standards with around 40% of respondents considering them as less important drivers in the purchasing decision (AK Wienn and Greenpeace, 2023a). This is in line with the responses from German consumers who indicate that ecological and social criteria play only a minor role in the purchase of apparel (Kleinhüchelkotten et al., 2018a) and the average of respondents who claim that environmental aspects were unimportant in their last purchase of a coat (Consumers, Health, Agriculture and Food Executive Agency. et al., 2018). Moreover, 35% of 2 500 survey respondents indicate that they often opt for a deal instead of a sustainable item (Heiny and Schneide, 2021). In the young and low-income population segments such criteria are given far less importance according to Kleinhüchelkotten et al., (2018a).

Nonetheless, according to a survey by the Swedish Environmental Protection Agency (2020) (n = 3 002), 66% of respondents consider the global sustainability perspective when purchasing apparel. Awareness is particularly high among women, younger adults (17–29), metropolitan residents, highly educated individuals, low-income earners, and those with an interest in fashion, interior design, or environmental issues. Among respondents who consider sustainability in their purchases (n = 1 973), the most frequently considered aspects are hazardous chemical emissions during production (58%), working conditions (54%), and the consumption of water, energy, and other resources (53%).

Interestingly, the responses of three-quarters of respondents to the survey suggest that a possible pollutant exposure could prevent them from buying certain apparel (Kleinhüchelkotten et al., 2018a). Differently, in a very recent survey from Tuovila, (2024), 1 200 participants (300 per country) in Finland, Sweden, Germany and the Netherlands, were asked to what extent certain factors would influence their purchasing decision when buying apparel, choosing between “very large effect or some effect,” “little or no effect,” or “cannot say.” Among the factors considered, the apparel’s recyclability at the end of its life into new fibre emerged as the most influential. Additionally, the fact that the garment was manufactured more locally within Europe, particularly when part of the material was sourced from recycled textile fibres collected in Europe, was also considered an important factor (Tuovila, 2024).

A survey in Germany explored the social importance given to apparel by gender and by groups with similar socioeconomic positions and cultural values (Kleinhüchelkotten et al., 2018a). In this survey, more than half of the respondents indicated a certain reluctance to buy apparel, with 26% who actually try to get by with few apparel. The proportion of men in the precarious group, who claim to be reluctant to buy apparel is significantly higher at 46.6%. In the focus groups it had already become clear that many of them have inhibitions to throw away apparel. This is in line with the fact that 78% of respondents strongly agree with the statement that apparel should be used for as long as possible and should only be disposed if it is no longer wearable. The intention to use the apparel for as long as possible is significantly lower in the higher age groups (40 to 70 years with higher level of education). Also the statement “I like apparel that I can wear in many occasions” has been repeated in all groups of survey participants. Although in the youth segment, comparatively there is less interest in versatile apparel (Kleinhüchelkotten et al., 2018a).

Additionally, more than half of users below 25 years old (Generation Z) are more likely to shop with a brand that offers second hand alongside new (Thredup, 2023). Moreover, Generation Z values unique apparel and the possibility of ordering and returning items online (Spaepen et al., 2021). Moreover, the Thredup survey (2023)

indicates that resale is increasingly driving the purchasing decisions of Generation Z users, in concrete 82% of them have considered the resale value of apparel before buying it compared to only 58% of the overall consumers.

According to Kleinhückelkotten et al. (2018) purchasing decisions of the German population are not very much influenced by whether the apparel was made nationally or is easy to repair. However, a very large proportion of respondents claim to pay attention to the easy to maintain aspect when buying apparel (Kleinhückelkotten et al., 2018a). In contrast, a survey from Fashion Revolution (2020) revealed that 75% of surveyed German consumers consider important that brands provide repair and care information about an apparel item, the average being 79% of 5 000 consumers across five countries (Fashion Revolution, 2020). The survey participants were between 16 and 75 years old (Fashion Revolution, 2020).

A recent survey from Zalando targeting over 5 000 consumers across France, Germany, Italy, Sweden, and the UK indicates a growing interest in sustainable practices such as purchasing second-hand items, mending clothing, reading product labels, and disposing of apparel responsibly. However, the results also reveal ongoing (and in some areas widening) gaps in consumer engagement, particularly regarding efforts to examine the environmental and social practices of fashion brands. This suggests that although general awareness of sustainability is improving, many consumers still do not routinely factor in sustainability aspects when making apparel purchasing decisions (Brun, 2025).

Price remains paramount for consumers as Tuovila's and Brun's surveys results suggest. When asked about their willingness to pay more for a sustainably manufactured garment, 36.3% of respondents indicated they would be willing to do so. However, the majority either expressed neutrality or stated they were not prepared to pay a higher price for such garments. No significant differences were observed between the surveyed countries, although Finnish respondents showed the highest willingness to pay more compared to respondents from other countries (Tuovila, 2024).

In addition, almost 30% of 5 013 survey participants find it hard to distinguish which products are genuinely more sustainable, around 20% are unsure where to find such options, and over 20% lack clarity on what steps to take. Skepticism also plays a role, as around 20% question the credibility of brands' sustainability claims (Brun, 2025).

It shall be noted that the surveys analysed so far do not specifically address sportswear. On the contrary, a Cotton Incorporated's survey investigated consumer's attitudes and behaviours on sportswear purchases. The survey targeted 1 500 gendered balanced consumers, ages 13 to 70, representative of the United States population found that found that top purchase drivers for sportswear among consumers were: comfort (77%), fit (69%), washes clean (65%), quality (64%) and durability (62%) (Cotton Incorporated, 2014)

13.7.1.3 Quality assessment of apparel

Consumer satisfaction with the quality of an apparel product can be measured in three phases: at the point of purchase, while using the item, and ultimately, when it is disposed. Quality is generally first evaluated "pre-use" and then experienced "during use" (Piippo et al., 2022a).

Fashion is a highly tactile experience in which purchasing decisions are often made based on how the apparel or accessory 'feels' and how the quality is perceived (De Klerk and Lubbe, 2008; Williams and Ackerman, 2011; McLaren et al., 2016; Vladimirova et al., 2022). Yet, for consumers it is very difficult to assess the quality and durability of apparel at the point of purchase (Harris et al., 2016; McLaren et al., 2016). Goworek et al. (2012) reported that consumers usually judge apparel quality by its feel (personal considerations) and the brand. In any case, it shall be noted that the topic of longevity in relation to visual or aesthetic degradation has been minimally explored in existing literature. The lack of information on how expected longevity evolves over time is a significant barrier (V Kumar et al., 2023). This gap not only may prevent consumers from making informed purchasing decisions but also complicates their ability to use the apparel item effectively according to Kumar et al. (2023).

Several researchers have tried to gather the aspects that influence the perception of quality in apparel (Table 141).

Table 141. Aspects that influence the perception of quality in apparel

Intrinsic product attributes	Extrinsic producer factors	Experienced features	Values, convictions
Material	Price	Tactile feeling	Low environmental impact
Manufacturing quality	Brand	Functionality	Local production

Intrinsic product attributes	Extrinsic producer factors	Experienced features	Values, convictions
Fit	Manufacturing location	Durability Emotional value Fit Availability Context Price	Ethical production

Source: Adapted from Niinimäki, (2011); Koszewska, (2016); Henninger et al., (2017)

The aspects reported in **Table 141** are mentioned in different consumer surveys. Overall, it is documented that price is usually associated by consumers as a quality indicator (Keiser et al., 2012 as cited in Wakes et al., 2020). In an American survey, it was documented that 58% of users thought that the apparel with lower price had worse quality compared to higher-priced items, and 78% (ages 13–24 years and 35–70 years) indicated that “you get what you pay for” implying that apparent better quality apparel may be the ones that have higher price (Monitor, Cotton Incorporated Lifestyle, 2018).

Durability and ease of care characteristics seem not to be considered per se, by users when they are buying apparel (De Klerk and Lubbe, 2008; Wakes et al., 2020a). However, this may be because quality and durability are associated by consumers and industry specialists (Yuille, 2015; Wakes et al., 2020a). In other words, apparel with perceived high quality is expected to be durable.

As it happens when assessing the quality of apparel, consumers’ expectations of apparel longevity are mostly related to the price (Forbrugerrådet Tænk, 2022). This is a particular believe for younger generations (Monitor, Cotton Incorporated Lifestyle 2018 as cited in Wakes et al., 2020). Nonetheless, price may not correctly reflect the quality and hence, the durability to laundering, for instance (Ghaani Farashahi et al., 2018b; Wakes et al., 2020a; Badgett, 2017). In fact, some studies have proved that cheaper apparel are not always a synonym of worse quality (Wakes et al., 2020a; Badgett, 2017). Nonetheless, according to Harris et al., (2016) the price is the deal breaker when it comes to decide whether to buy a piece of apparel or not, even among environmentally aware consumers.

Lifetimes expectations for consumers are also linked to where the apparel are purchased, care requirements, material and purpose/context (Forbrugerrådet Tænk, 2022). It is interesting to note that 43% of Austrian survey participants declared their willingness to buy long-lasting apparel despite of the price while 30% of them already buy them (AK Wienn and Greenpeace, 2023a). In addition, 55% out of 1 000 German consumers would be willing to pay more for their apparel if being certain about increased durability (VZBV, 2022). In a Zalando survey, 58% of 2 500 consumers consider long-lasting quality to be significant in apparel, with 52% stating that they frequently consider this aspect while shopping. Similarly, 45% of respondents prioritize value for money (Heiny and Schneide, 2021).

13.7.1.4 Information reported on apparel labels

Apparel labels and tags can have the same meaning. Generally speaking, the word label is used to refer to an attachment that is not meant to be removed from the apparel and provides details about it. The term tag may be referred to any type of attachment, whether made out of paper or cardboard including tags attached at the point of sale intended for removal.

Textile products sold in the EU must have a label and comply with the EU labelling requirements set in the EU Textile Labelling Regulation (TLR) 1007/2011 ⁽¹⁴⁾. According to the mentioned Regulation, the label must be firmly attached to the apparel and must contain information on fibre composition. Moreover, the information on textile composition must be separated from other type of information such as product care.

Information about care of the apparel is voluntary, according to EU law³²⁶ and in general practice, but can also be legally required, notably by Member States and third countries (GINETEX, 2017a). In fact, currently, nine Member States mandate care labelling through their national laws, while others only offer it as a recommended option (Evaluation report of the Textile Labelling Regulation, under development). In most instances, the legislation does not specify how this requirement should be implemented. However, some countries, like Estonia, directly reference ISO standards in their national laws. In Austria, care symbols are provided in an annex to the national legislation (Evaluation report of the Textile Labelling Regulation, under development). When it comes

³²⁶ The ongoing review of the Textile Labelling Regulation is exploring whether to harmonise and even render care labelling mandatory

10071 to care labelling, manufacturers and even retailers can provide the product care instructions using symbols or
10072 offering a written explanation (GINETEX, 2017a).

10073 It is interesting to note that when it comes to making informed consumer decisions, around 75% of respondents
10074 to the TLR Public Consultation consider it a problem the lack of harmonised labelling rules regarding care
10075 information of textile products and textile related products (Evaluation report of the Textile Labelling Regulation,
10076 under development).

10077 Information about user attention to apparel care labels is available in Section 13.7.2.8. However, when it comes
10078 to labels in general, little is known about the extent to which consumers pay attention to them. For instance,
10079 1 056 Portuguese consumers were surveyed about their habit of reading labels before purchasing apparel. The
10080 results were somewhat mixed, with the majority (53%) indicating that they do have this habit, while 39%
10081 reported that they do not typically read labels before making a purchase (Ribeiro et al., 2023).

10082 An attitude-behaviour gap survey from Zalando showed that 58% of 2 500 respondents indicate that they
10083 should understand certain aspects of the apparel item, i.e. the materials used. However, just 38% regularly
10084 check the label for information (Heiny and Schneide, 2021). Similarly, while 60% of the survey participants
10085 express the significance of transparency about apparel aspects, only 20% actively pursue information during
10086 their buying journey (Heiny and Schneide, 2021). In like manner, a significant 74% of a sample of 1 000 Italian
10087 consumers expressed a strong need for clearer and more transparent information regarding the true
10088 sustainability of apparel and the production processes behind them (Altroconsumo and IPSOS, 2024).

10089 Furthermore, a 2010 Eurobarometer survey found that 50% of 26 635 European citizens check the country of
10090 origin when buying textiles and clothing, and over half of those individuals reported that this influences their
10091 decision (28% of all respondents). This indicates that country of origin labelling plays a significant role in
10092 consumers' purchasing choices (European Commission, 2010). Related to this, a survey from the Spanish
10093 Consumer Organization (OCU) targeting 340 Spanish consumers indicated that 42% of respondents admit
10094 looking at the origin of the apparel in order to buy locally or not to buy from countries contributing to labour
10095 exploitation (OCU, 2018). Some of the Spanish consumers who responded to the same survey check the labels
10096 to get information about the material composition as 38% of them have personal preferences towards natural
10097 origin fibres while 25% avoid synthetic fibres due to the microplastic release occurring after washing the
10098 apparel (OCU, 2018).

10099 Overall, textile labels and the information they provide are considered valuable only if they are trusted by
10100 consumers (Circle Economy, 2020 as cited in the Evaluation report of the Textile Labelling Regulation, under
10101 development). This is especially important for voluntary labels that highlight the sustainability features of textile
10102 products (Circle Economy, 2020 as cited in the Evaluation report of the Textile Labelling Regulation, under
10103 development). In this light, users seem to have a growing interest in EU ecolabelled products in the category of
10104 'textiles, clothing/footwear'. In particular, 71% out of 26 635 survey respondents in the EU27 Member States
10105 express a desire to find more of the abovementioned products carrying the EU Ecolabel. This sentiment is
10106 prevalent in all countries, with varying percentages (from 54% in Czechia to 83% in Portugal and Romania)
10107 (European Commission. Directorate General for Environment, 2023). However, a survey from the Swedish
10108 Environmental Protection Agency (2020) indicates that many consumers remain uncertain about the presence
10109 of ecolabels on the apparel they purchase: around 40% did not know whether their garments were ecolabelled,
10110 while only around 30% reported that half or more of their apparel carried such labels. Nonetheless, the survey
10111 highlights that the share of ecolabelled purchases increased slightly between 2018 and 2019, particularly
10112 among women, highly educated individuals, and those interested in fashion or sustainability.

10113 **13.7.1.5 Attitudes towards second-hand apparel purchase**

10114 In recent years, there has been a notable shift in consumer behaviour towards the adoption of second-hand
10115 apparel, reflecting evolving preferences and attitudes. In fact, more than three out of ten (34% of 26 595
10116 respondents) would purchase second-hand textiles (apparel and home textiles included in this category)
10117 (European Commission. Directorate General for Environment, 2014). Another survey indicates that 17% out of
10118 11 483 European consumers purchase their apparel in second-hand shops while 12% buy them on resale apps
10119 (YouGov, 2021). Additionally, Tuovila, (2024) survey reveals divided opinions: on average, 37% of 1 200
10120 participants across Finland, Sweden, Germany and the Netherlands expressed that they like buying second-
10121 hand apparel, while another 38.6% strongly or somewhat disagreed with this statement.

10122 Interestingly, more than 70% of 27 498 respondents across the European Union agree that second-hand
10123 apparel should be promoted more (European Commission, 2019). In fact, 65% of 2 500 Spanish consumers
10124 expressed their desire to give a second life to their apparel items as 87% of them believe that is important

10125 (Asociación para la Gestión del Residuo Textil y el Calzado, 2024). Despite of this, surveys have found that the
10126 attitudes and the consumer behaviours are not always aligned. This is the case for 61% of users who think that
10127 second-hand purchase is important but only 25% of them buy second-hand apparel (Heiny and Schneide, 2021).

10128 There seems to be gender disparities as women exhibit a higher inclination towards buying second-hand apparel
10129 compared to men (European Commission. Directorate General for Environment, 2014; YouGov, 2021). The age
10130 dynamics indicate that younger demographics, particularly 18-25 year-olds, lead in second-hand apparel
10131 adoption, contrasting with relatively lower participation among those aged 26-40 (D&B, 2020a). This is
10132 supported by the findings in AK Wienn and Greenpeace (2023), young consumers (aged 16-29) buy far more
10133 second-hand apparel (both online and offline). Similar results were found in Tuovila (2024) in which among
10134 the survey age groups, the youngest (18-24 years) reported the most positive attitudes towards second hand
10135 purchasing. The primary drivers behind the adoption of second-hand apparel include financial savings,
10136 sustainability/environmental consciousness, and a preference for unique, vintage pieces (D&B, 2020a).

10137 It shall be noted that the second-hand cycle relies on the primary market and consumer behaviour, with
10138 consumers acting as both partners and suppliers by disposing of items in good, reusable condition, thereby
10139 making used apparel available for second-hand use (Machado et al., 2019 as cited in Turunen and Gossen
10140 2024). Consequently, there is a dual role for consumers: those disposing of apparel items with the intention to
10141 receive money from them (sellers) and those interested in purchasing used clothing (buyers) (Turunen and
10142 Gossen, 2024).

10143 Moreover, preliminary observations from the Turunen and Gossen (2024) study indicates that current second-
10144 hand business models also incorporate consumption-promoting marketing strategies. The study also noted an
10145 increase in unworn or slightly used items on second-hand platforms. The use of subscription models to
10146 encourage ongoing buying and selling, may result in shorter garment lifespans. Turunen and Gossen (2024)
10147 also highlight that self-service flea markets, with their lower prices, may attract treasure hunters more for the
10148 shopping experience than the actual purchases, sometimes leading to less responsible buying behaviour of
10149 apparel (Turunen and Gossen, 2024).

10150 The 'replacement or displacing rate' for apparel items is understood as "the degree to which the purchase of
10151 second-hand articles replaces the purchase of similar new items" (Nørup et al., 2019; Trzepacz et al., 2023).
10152 On one hand, there are studies which highlight that second-hand purchases, generally speaking, tend to have a
10153 reduction effect on the purchase of new apparel (Klepp et al., 2020b; Vinted, 2021) thus extending the life of
10154 existing apparel (Klepp et al., 2020a). In particular, some studies operate under the assumption of a one-to-one
10155 replacement ratio, meaning that buying a used apparel item completely offsets the purchase of a new one
10156 (Trzepacz et al., 2023). However, various studies indicate that actual replacement rates differ significantly
10157 (Sandin and Peters, 2018; Trzepacz et al., 2023). For instance, results of a survey based on over 350 000
10158 responses across eight European markets showed that 39 out of 100 people buying a second-hand product on
10159 the retailer's website would have avoided purchasing a new product (Vinted, 2021). Depending on the product
10160 category, significant variation in the avoided purchase rate was observed for instance, coats and jackets were
10161 among the products having high replacement rate (Vinted, 2021). In Farrant et al., (2010), purchasing 100
10162 second-hand apparel items would save between 60 and 85 new ones, depending on the location where they
10163 are reused.

10164 WRAP's (2025) recent research quantifies the environmental benefits of repairing and reselling apparel by
10165 calculating displacement rates, the extent to which these actions replace new purchases. Their findings show
10166 that repairing five items displaces approximately four new purchases (82 % displacement rate), while buying
10167 five resaled items displaces about three new purchases (almost 65% displacement rate).

10168 In any case, more research and other variables should be taken into account when considering estimations of
10169 replacement rates (Trzepacz et al., 2023). As for example, an apparel item with low quality, sold after just one
10170 use, may avoid the purchase of other items. However, due to its low quality, the usability of the item may be
10171 limited. This means there may be cases in which buying brand-new items with higher quality could have better
10172 impact on the environment than buying second-hand apparel items with lower quality.

10173 **13.7.1.6 Attitudes towards the purchase of apparel made without harmful chemicals**

10174 Over 85% of 27 498 respondents of an EU-wide survey expressed concern about how chemicals found in
10175 everyday products may affect their health (European Commission, 2019). In particular, consumers have
10176 reported to be increasingly concerned about the use of hazardous chemicals in apparel (Evaluation report of
10177 the Textile Labelling Regulation, under development).

10178 Based on survey responses from over 2 000 Spanish consumers, 61.2% selected the restriction of hazardous
10179 chemicals in apparel as one of the top three actions authorities should take to promote sustainable products
10180 and reduce apparel consumption. While 23.8% ranked it as their top priority, 22.1% chose it as second, and
10181 15.3% as third indicating strong overall support, though with varying levels of urgency. This measure ranked
10182 just behind ensuring fair working conditions in apparel production (CECU, 2023).

10183 A survey involving 26 718 citizens across the 27 EU Member States found that the chemicals in the apparel
10184 fabrics are generally perceived as minimally risky, with 60% of Europeans viewing them as not posing a risk to
10185 people. Among different fabric types, synthetics are viewed as posing the highest risk, although only 22% of
10186 Europeans categorize them as presenting a significant health risk (European Commission, 2009).

10187 Fabrics with chemical-specific quality labels indicating the absence of harmful chemicals to health and the
10188 environment, are thought to carry lower risks and materials like wool, cotton, or linen are perceived as entirely
10189 risk-free. Europeans hold divergent views on the relevance health and environmental risks when purchasing
10190 apparel – some consider it highly important, while an equivalent proportion does not find it significant (European
10191 Commission, 2009).

10192 Europeans were also surveyed about the significance they attribute to potential health and environmental risks
10193 associated with fabric when purchasing apparel for themselves or their families. Of the respondents, 18%
10194 consider it fairly important, while nearly four in ten respondents find it very important (39%), and a similar
10195 proportion deems it not important (37%). A country-level analysis reveals that approximately two-thirds of
10196 respondents in Germany and Slovakia find environmental and health risks very important when buying apparel.
10197 Conversely, in Denmark and the United Kingdom, more than six in ten respondents consider this aspect not
10198 important (European Commission, 2009).

10199 Additionally, data from the Fashion Revolution (2020) consumer survey shows that an average of 37% of 5 000
10200 consumers in the United Kingdom, France, Germany, Italy and Spain consider important to buy apparel that is
10201 not produced using harmful chemicals. On top of that, a survey from Sveriges Konsumenter (2020) shows that
10202 86% of 1 000 surveyed Swedes would like to access information about the content of harmful chemicals in
10203 apparel at the time of purchase. The Fashion Revolution survey does not provide information about the type of
10204 chemicals. It shall be noted that all age groups, genders and users with different income provided no important
10205 differences in the responses.

10206 **13.7.1.7 Attitudes towards the purchase of apparel made with recycled materials**

10207 The Fashion Revolution (2020) survey indicated that an average of 11% of 5 000 European consumers from
10208 the United Kingdom, France, Germany, Italy and Spain pointed out that it is important that the apparel they buy
10209 contains recycled materials. Italy is considerably above the average with 51% of users considering important
10210 the presence of recycled materials in the apparel they purchase. Spain, France and Germany remain in the
10211 average for the five countries. However, the survey does not offer information about such materials. In this
10212 case, the group of respondents between 55 and 75 years old consider less important whether the apparel they
10213 purchase have recycled materials or not, as only 7% declare so as opposed to the youngest age group (12-24
10214 years old) in which 14% percent of users indicate the importance of buying apparel containing recycled
10215 materials. No notable differences in survey responses exist between different income and gender groups.

10216 Furthermore, three-quarters of 27 498 respondents of an EU-wide survey declared that apparel should only be
10217 made from materials that can be recycled (European Commission, 2019).

10218 Similarly, recent findings show a mixed but overall positive attitude toward apparel made with recycled
10219 materials. According to Tuovila (2024) survey, almost 48% of 1 200 respondents find garments containing
10220 recycled materials interesting (particularly in age group 18 24 and in Finland), and around 45% express interest
10221 in purchasing garments made from recycled textile fibres originating from end-of-life textiles. About 31% have
10222 already bought garments made from recycled fibres. However, concerns remain among consumers: around 43%
10223 perceive these garments as being of lower quality, and about 53% have doubts regarding the cleanliness of
10224 the materials used. A considerable share of respondents remains neutral regarding these aspects, indicating
10225 that while interest is strong, trust and perceptions of quality and hygiene continue to be barriers to wider
10226 acceptance.

10227 **13.7.2 Post-purchase aspects: the user behaviour during use**

10228 The post-purchase aspects of apparel encompass the use and care phases. Studies on laundering procedures
10229 have been investigated from a variety of angles, including their impact on the environment and cleanliness.

Nevertheless, the majority of these research (Gooijer and Stamminger, 2016 as cited in Laitala et al., 2017) solely consider machine washing, paying little attention to other cleaning techniques such as hand washing, dry cleaning, airing, steaming, ironing, and stain removal (Laitala et al., 2017). These maintenance procedures have a different environmental impact than machine cleaning and are widely used specially for woollen and delicate apparel according to Laitala et al., (2017).

Moreover, the underlying drivers and barriers to sustainable consumer use and care are not sufficiently well captured in the literature, especially regarding frequency of laundering apparel, number of uses and product's lifetime (Conrady et al., 2014; Yates and Evans, 2016; Luo et al., 2023b). In fact, retailers claim that the way users take care of apparel determine their physical lifespan (McLaren et al., 2016). This is important as it seems that not many consumers seem to be willing to change their laundering behaviour (Uitdenbogerd 2007 as cited in Luo et al., 2023).

The frequency of laundering, washing temperature, number of days apparel are worn between washes and chemicals used for washing the apparel are among the factors that affect the durability of apparel and also determine the extent of the environmental impacts associated to washing them. All these aspects are heavily dependent on the behaviour of users which vary depending on the culture and climate (Laitala et al., 2017; Klepp and Laitala, 2023b).

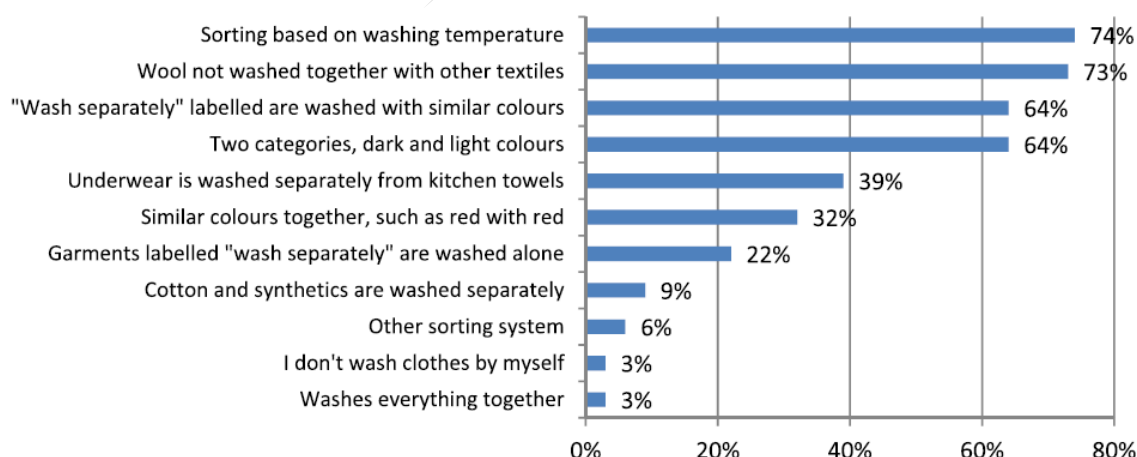
As a result, this chapter brings forward some information about different user laundering practices which encompass the sorting of the apparel before washing, the washing temperature and washing frequency, the drying method, the ironing or steaming process and the storage.

13.7.2.1 Sorting before washing

Textile materials have specific cleaning requirements, hence the importance of sorting before washing in order to preserve the apparel shape and colour (Cooper and Claxton, 2022b; Klepp and Laitala, 2023b) and hence extend the lifetime of the apparel (Cooper and Claxton, 2022b).

Figure 88 gives an overview of different sorting practices and average percentage of users that opt for each of them. The results are based on responses from 545 users in Norway. It seems that sorting based on washing temperature is the most popular sorting method while washing everything together without differentiation between the apparel is not generally chosen by consumers.

Figure 88. Laundry sorting methods



Source: Laitala and Boks, (2012)

The sorting processes in laundry exhibit considerable variation, influenced by factors such as washing temperature, colour, fibre type, usage area, and care labelling. Merely 3% of participants indicated a preference for washing all items together. Younger respondents tend to categorize colours into just two groups (light and dark). Sorting based on washing temperature is more prevalent among female participants and those in higher age brackets. Woollen products are commonly segregated from other fibre types, with 73% of survey respondents affirming that they avoid washing wool alongside other textile materials. Some consumers find it

10267 challenging to amass a sufficient amount of apparel to fill the machine when using multiple sorting categories
10268 (Laitala et al., 2012).

10269 **13.7.2.2** Washing temperature and washing frequency

10270 The average laundry temperatures differ between different countries. The average European washing
10271 temperature was 42.4°C in 2020 (A.I.S.E, 2020), in Scandinavia is 46.2°C and in Southern Europe lower 39.7°C
10272 (A.I.S.E, 2017b). In the USA, the average temperature is 30°C, while in Japan is 20°C (Laitala et al., 2020).

10273 **Figure 89** shows that washing cycles at an average of 40°C have been found as the most commonly selected
10274 washing temperature among consumers. This is also supported by the study from Laitala et al., (2012).

10275 **Table 142.** Preferred washing temperatures for consumers

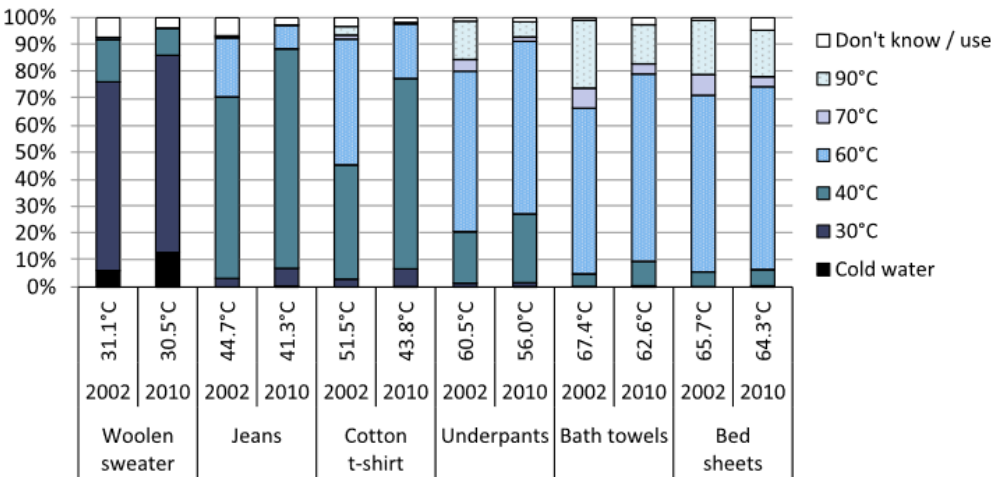
Publication	Sample details and characteristics of the study	Country	Preferred washing temperatures ^(a)			
			20 or cold setting	30	40	60
WRAP (2019)	Two waves of online surveys: Wave 1 was conducted in November 2016 and Wave 2 in June 2019. The sample size in 2016 was 1 000 per country. In 2019 the samples were as follows: Denmark 1 046; Germany 1 113; The Netherlands; 1 117 and Italy 1 226.	Denmark (DK), Germany (DE), The Netherlands (NL) and Italy (IT).	NA	Second most frequently used laundry wash temperature in all five countries.	Most frequently used laundry wash temperature in all five countries	NA

10276 ^(a) There is no information about the correlation between the temperatures indicated and the information reported on the care labels

10277 Source: own production based on references indicated in the first column of the table

10278 Whether the washing temperatures in **Figure 89** are in line with care label recommendations of the apparel in
10279 question is not specified in the study from Laitala et al., (2012). A total of 1 008 users participated in the survey
10280 from Laitala et al., (2012) while the sample in 2010 was a bit more than half the size with 546 respondents.

10281 **Figure 89.** Distribution of washing temperatures for different textile products



10282 Source: Laitala and Boks, (2012)
10283

10284 The frequency of washing is a crucial factor in determining the longevity of apparel because it represents one
10285 of the primary factors that can cause the apparel item to wear out and either extend or shorten its lifespan.

10286 The average washing frequency per household in Europe is estimated in 3.8 washing cycles per week (Schmitz
10287 and Stamminger, 2014; Klepp and Laitala, 2023; Laitala et al., 2018b).

10288 The frequency of washing cycles is closely tied to the household's size. As the number of individuals in the
10289 household rises, so does the overall number of washing cycles. However, the average number of washing cycles
10290 per person decreases, suggesting a more efficient utilization of washing machine capacity (Klepp and Laitala,
10291 2023b).

10292 Other sources indicate that users wash their apparel items on average 10.9 times a month, excluding underwear
10293 and socks (D&B, 2020a). Women (average 11.58 times per month) wash their apparel more often than men
10294 (average 9.93 times per month). Additionally, users living in metropolitan areas tend to wash less often (8.2
10295 times a month) than users whose home is in less urban areas (13.3 times a month in a non-urban area). The
10296 analysis of the survey results from D&B (2020) found no explanation for this behavioural pattern. Age and
10297 educational level, did not offer different results in washing frequency (D&B, 2020a).

10298 The duration of time that apparel is worn before being washed exhibits significant variability among different
10299 types of apparel. Typically, items like underpants and socks are laundered after each use, whereas certain outer
10300 apparel are infrequently washed (Laitala et al., 2018).

10301 **Table 143** displays the difference in number of uses prior to washing depending on the purpose of the category
10302 of apparel.

10303 **Table 143.** Average number of uses of apparel categories prior to washing

Study	T shirts	Shirts and blouses	Sweaters and mid layers	Jackets and coats	Pants and shorts	Dresses skirts and jumpsuits	Leggings stockings tights and socks	Underwear	Swimwear	Textile apparel accessories	Knitwear
Quantis, (2022)	1	2	5	20	3	3	2	1	1	NA	NA
Klepp et al., (2020)	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	4.5	NA	NA
Laitala, IG Klepp, et al., (2018a)	1.5 (Cotton)	2 / 3 (Woollen undershirts)	10 (Wool) / 5 (Cotton)	NA	5.5 (Jeans)	NA	2	1	NA	NA	NA
WRAP, (2017a)	2	2	NA	NA	10 (Jeans)	NA	2	NA	NA	NA	5
Roos et al., (2015)	2	NA	NA	100	9 (Jeans)	3 (Dresses)	NA	NA	NA	NA	NA
Gray et al., (2022)	2.6	2.3	4.7 (Sweatshirt and hoodie)	14.5 (Non-padded) / 16.7 (Padded)	4.2 (Shorts) / 4.5 (Trousers) / 5.5 (Jeans)	3.2 (Skirt) / 2.6 (Dress)	3.3 (Leggings) / 2.4 (Socks and hosiery)	3.5 (Bras)	1.7	NA	4.3

10304 NA: Not available

10305 Knitwear products could belong to more than one category from those considered

10306 Source: own production based on references indicated in the first column of the table

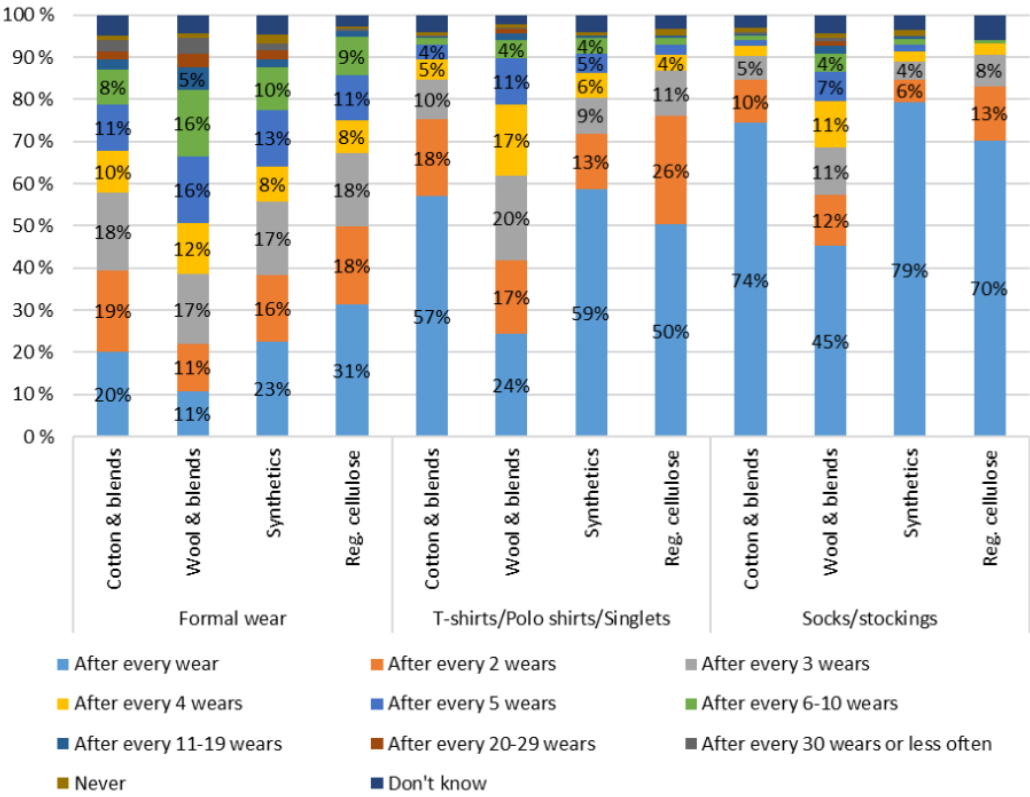
10307

The wardrobe survey conducted by Klepp et al. assessed the cleaning frequency among its respondents. However, there is a big variability among respondents as 39% of them washes the apparel item after every wear, 14% after every 2 wears, 11% after every 3 wears and 29% after every 4 wears or more (Klepp et al., 2020a).

The study conducted by Laitala et al., 2018a presents an overview of studies that provide the average count of wear instances before laundry for distinct types of apparel. It indicates that fibre types are an important factor in determining the number of uses before washing (Laitala et al., 2018a) and overall influence the washing frequency (Laitala et al., 2020)

When it comes to differences in the washing of textile categories and fibres, survey data of 23.392 apparel representing five countries show that socks are washed more frequently compared to t-shirts and formal wear **Figure 90**. In terms of fibres, apparel made out of wool are washed less often compared to other fibres (Laitala et al., 2020).

Figure 90. Frequency of wash or dry clean per fibre content



Source: Laitala et al., (2020).

The type of materials is likely to influence how often apparel is washed, as synthetic fabrics like polyester or polyamide are prone to retaining strong body odours compared to apparel made from natural materials like wool or cotton (Klepp and Laitala, 2023b).

Additionally, even identical types of apparel may undergo washing at distinct frequencies across various countries. A study comparing laundering habits in five countries revealed that, for instance, the Japanese tend to launder their formal wear less frequently (after an average of 8.8 wears) compared to Chinese and American consumers, who wash them after 3.3 and 3.6 wears, respectively (Laitala et al., 2020). In this context, laundry washes include the generic pre-programmed settings the washing machine runs to complete its work (washing, rinsing, and spinning phase). In view of this, Italy has the highest (5.9) number of laundry washes per week, while Germany has the lowest (4.0). However, the number of laundry washes per week have decreased since 2016 in Denmark (from 5.5 to 4.5 per week), Germany (4.4 to 4.0) and The Netherlands (6.0 to 4.8) (WRAP, 2019).

10335 The stronger users feel about the fact that washing apparel less often is not a habit a clean person would
10336 follow, the more often they wash. This association of less washing with poor hygiene is also linked to our image:
10337 we want to appear to others as a “clean” person, because that is the social norm (D&B, 2020a). Additionally,
10338 the more annoying users find the washing of apparel, the less often they do this. Individual beliefs about hygiene
10339 and convenience play a role in how often users wash their apparel. Factors such as time, money and
10340 environmental convictions do not seem to play a role in the frequency of washing apparel (D&B, 2020a).

10341 In general, users do not only throw their apparel in the laundry bin when they are actually dirty, but also ‘after
10342 wearing them several times’. The latter reason indicates customary behaviour also supported by the literature:
10343 the majority of people do not check whether the apparel are dirty before putting them in the laundry bin because
10344 they are used to washing them after a certain number of wears (D&B, 2020a).

10345 It is important also to consider that the cleaning process itself may have environmental and lifespan
10346 implications. Frequent laundering may accelerate the wear and tear on apparel, indirectly reducing their total
10347 lifespan. If laundering fails to effectively remove stains or odours from apparel, it can lead to premature
10348 disposal of apparel (Klepp and Laitala, 2023b). As established in the study based on the survey conducted by
10349 Laitala and Klepp (2020), apparel that are laundered less frequently exhibit longer lifespans. Those washed less
10350 often than every 30 wears have lifespans up to 4.8 years longer than those washed after each wear. On top of
10351 that, apparel washed after each wear are used 94 times less compared to apparel washed less often (washed
10352 every 30 wears) (Laitala and Klepp, 2020a). Nonetheless, it shall be noted that particularities in laundry
10353 frequency behaviour may apply for instance to sportswear. Their type of use and in some cases, distinct fibre
10354 and fabrics, could differ significantly from everyday casual wear apparel (Wei et al., 2020). In particular, certain
10355 types of sportswear can quickly absorb sweat and may also lead to unpleasant odours. This odour can create
10356 social awkwardness and discomfort for both the wearer and those around them (Chang and Wang, 2023).
10357 Therefore, washing sportswear promptly after one single use is generally done by users (Brice and Thorpe,
10358 2021).

10359 As a result, in addition to the frequency of washing, the examination of lifespan can also involve evaluating the
10360 number of cleaning cycles that the specific apparel item can endure in technical assessments or the number of
10361 cleaning cycles they generally experience during consumer studies. This approach may provide more consistent
10362 insights into the durability and potential longevity of apparel.

10363 **13.7.2.3** Choice of softeners and detergents

10364 According to a 2017 A.I.S.E survey, compared to 2008, fewer consumers are reading information on detergent
10365 packaging. Nonetheless, in 2017, 52% claimed to read it before purchasing a product, and 60% mentioned
10366 doing so before using the product (A.I.S.E, 2017c). However, it appears that a majority of consumers continue
10367 to dose detergent arbitrarily, as indicated by A.I.S.E in 2020 hence overdosing is a common mistake, contributing
10368 to increased environmental impact and less effective washing results due to poorer rinsing effects (Paloviita
10369 and Järvi 2008 as cited in Klepp and Laitala, 2023).

10370 Interestingly, when consumers’ attention is compelled, the average time spent on a detergent label is
10371 approximately 20 seconds, irrespective of the label size or content (A.I.S.E, 2017c). When asked if consumers
10372 deliberately check composition information, 37% of European consumers responded affirmatively. They do so
10373 primarily to compare products (59%), avoid specific substances (29%), assess the quantity of a particular
10374 substance (28%), or because they are allergic to a specific substance (24%).

10375 Fabric softeners, also known as fabric conditioners, are chemicals originally developed to reduce static electricity
10376 in synthetic fabrics and are now more commonly used to soften cotton, ease ironing and add fragrance to
10377 laundry (Klepp and Laitala, 2023b). Studies suggest that softeners are utilized in about 55% of washing cycles
10378 in Europe (Stamminger 2016 as cited in Laitala 2023), with significant national variations. In Norway,
10379 approximately 61% of the population use them often, 19% occasionally, and 19% never (Laitala, K., M.
10380 Kjeldsberg, and I.G. Klepp 2012 as cited in Klepp and Laitala, 2023), while in Hungary and Romania, only about
10381 5% use softeners in all wash cycles, and 60% never use (Stamminger 2016 as cited in Laitala, K., M. Kjeldsberg,
10382 and I.G. Klepp 2012 as cited in Klepp and Laitala, 2023).

10383 Further details on the A.I.S.E, 2017a survey results on laundering habits can be found in **Table 144**.

10384 In conclusion, economic factors, such as the price and promotions, play a substantial role in users’ choices of
10385 laundry products. Sustainability is an important criterion for a significant portion of respondents, but less
10386 important than economic aspects. There is still a segment of users that may add more or less detergent than
10387 recommended as it happens with the dosing of softeners.

10388 **13.7.2.4** Consumer awareness on microplastic release during washing cycles

10389 A structured survey among 411 Belgium citizens respondents spanning all age groups, aimed at gauging
10390 consumer awareness regarding microfibre pollution and its influence on washing habits (Herweyers et al., 2020).
10391 Overall, 68% of respondents are aware of the plastic pollution issue. Conversely, only 37% are aware of
10392 microfibre pollution and its association with the problem. Individuals under 25 exhibit the lowest awareness of
10393 synthetic fibre pollution (29%). Those aged 41 to 60 and over 60 are the most knowledgeable about the issue,
10394 yet still, less than half of respondents in these age groups are aware of synthetic fibres' potential harm (43%)
10395 (Herweyers et al., 2020).

10396 Similarly, according to a survey conducted by the Swedish Environmental Protection Agency (2020), microplastic
10397 pollution is commonly associated with apparel care among consumers. Just over half of the around 3000
10398 Swedish respondents (54%) reported that they consider how to wash and care for their apparel in order to
10399 reduce the release of microplastics. However, within this group, 17% indicated that although they are aware of
10400 the issue, they do not know how to effectively reduce microplastic emissions during washing.

10401 **Table 144.** Summary of surveys on laundering habits related to apparel

Publication	Sample details & characteristics of the study	Country	Washing Habits										
			Choice of laundry detergents and/or softeners by users expressed in % of respondents					Laundry detergents dosage habits per user expressed in %					
			Total price of the box	Specials offers and promotions	The detergent form (i.e. liquid/powder)	The fragrance	Sustainability criteria	Aware of the dosing instructions on the package	Find it easy to dose laundry detergents	Usually measure the quantity of detergent to be used	Check the package for dosing instructions	Add more detergent than recommended	Add less detergent than recommended
A.I.S.E (2017)	4 611 participants (aged 18-65) 200 respondents per country. No details on the demographic or social factors of the subjects.	23 European countries	78	71	70	71	Ranked lower than the economic criteria ^(a)	63	74	62	50	24	23

10402 *Source: own production based on references indicated in the first column of the table*

10403

13.7.2.5 Drying

Apparel undergoes diverse drying methods, all of which involve air, causing moisture to evaporate and be carried away. The tools employed by consumers include an array of drying racks and cords utilized both indoors and outdoors, drying cabinets, dryers, and dedicated drying spaces, such as drying ceilings with or without heating. The prevalence of tumble dryers varies significantly even among Western countries, with the USA leading at approximately 80% of households owning one (Engelberg and Brassell, 2019), while the European average stands at around 32% of households (De Almeida et al., 2011).

The data from surveys conducted by Laitala et al. (2020) and GINETEX (2017) provide insights into the prevalence of specific drying methods among participants from different countries and age groups.

A summary of surveys related to drying of apparel is available in **Table 145**.

Table 145. Summary of surveys related to drying of apparel

Publication	Sample details and characteristics of the study	Country	% of users per drying method ^(a)			
			Tumble-drier	Shared tumble-drier	Line dried indoors	Line dried outdoors (natural drying)
Laitala et al. (2020)	1 111 participants Age groups: 18-29 years, 30-49 years and 50-64 years	China, Germany, Japan, United Kingdom and USA	32.5	8	33	27
GINETEX (2017)	6 000 participants (aged 18-65)	Germany, United Kingdom, France, Italy, the Czech Republic and Sweden	NA	NA	NA	62

(a) Whether the results per drying method depend on where the user lives in, is not reported by the studies

Source: own production based on references indicated in the first column of the table

13.7.2.6 Ironing

The final stages of the domestic laundry process involve employing techniques to shape, structure, and refine the appearance of freshly washed fabrics, a step commonly achieved through ironing (Klepp and Laitala, 2023b). In contemporary times, the amount of time dedicated to ironing has notably decreased, particularly over the past few decades (Klepp and Laitala, 2023b). As it seems, ironing is one of the procedures consumers dislike the most (Yun et al., 2017).

Ironing practices exhibit significant variations not only between countries and consumer demographics but also among different types of apparel. In Norway, the statistics indicate diverse ironing habits, with approximately 12% of adults ironing on a weekly basis, 40% opting for monthly ironing, 28% engaging in less frequent ironing, and 20% asserting that they never iron anything (Laitala, IG Klepp, et al., 2018b). Notably, a higher proportion of men and younger respondents tend to forgo ironing, in contrast to women and elderly respondents (Klepp and Laitala, 2023b).

On a different note, the care labels on textiles often contain symbols indicating whether the apparel can be ironed and if so, the most appropriate ironing setting. In some cases, the ironing symbol may be accompanied by dots. The number of dots indicates the temperature ranges for ironing. Ironing at a hot setting can also contribute to the shrinkage of apparel (Chartered Textile Technologist interview as cited in Cooper and Claxon 2022).

13.7.2.7 Storage after washing and drying

The folding and storage of apparel after being washed and dried may have influence on the lifespan of apparel. The way users store their apparel may affect the intrinsic quality they have. Nonetheless, no studies have been found that address the folding and storage of apparel after being washed and dried. Rather some studies on the user behaviour related to the storage of inactive or no-longer-worn-apparel have been identified and are presented in Section 13.7.2.10.

13.7.2.8 Following apparel care label instructions

Care labels may indicate the recommended washing temperature, cycles, detergent to be used, and the way apparel should be hung while still damp, among other things (Cooper and Claxton, 2022b).

Survey evidence (**Table 146**) suggests that a significant percentage of users across various countries tend to follow care instructions on apparel labels, often with reasons including avoiding washing problems and preserving apparel. Both, GINETEX (2017) and GINETEX (2019) surveys report a similar percentage of users (70%) who always follow textile care instructions, suggesting consistency in this behaviour across the two studies. Moreover, the surveys from GINETEX (2017) and AB-REOC and BV-OECO (2019) indicate that around 58% of users follow the care instructions often. This is in line with [Ribeiro et al., \(2023\)](#) study where roughly 53 % of 1 056 Portuguese consumers typically adhere to label instructions for washing and drying their apparel.

Despite consumers recognizing the influence of care and laundering practices on the lifespan of their apparel, adherence to care labels diminished after the initial wash, as indicated by McLaren et al. 2015. In the present survey, 47% of participants acknowledged generally reading apparel care labels. However, when specifically asked about reading care labels for T-shirts, 59% indicated doing so either at the point of purchase (10%) or before cleaning/after use (49%). The percentage of participants who abstained from reading care labels entirely decreased with an increase in T-shirt price. Moreover, survey results from Cotton Incorporated's Lifestyle Monitor revealed a declining trend in the number of consumers who 'always' or 'usually' read care labels before laundering apparel (2003—77%; 2007—64%; 2009—57%), with younger consumers (under 35 years) displaying a reduced inclination to read care instructions.

Furthermore, evidence suggests that there appears to be an imbalance between the essential information consumers need and the volume of details provided on physical care labels in apparel (Evaluation report of the Textile Labelling Regulation, under development). This results in user practices such as cutting labels out reflected in the GINETEX (2017) survey where 62% of the respondents would do so (GINETEX, 2017a). The fact that there are consumers who tend to cut the care labels is also observed in disposed textiles which are collected with missing labels. This affects both, the reselling potential of apparel and their preparation for reuse (WRAP, 2017c; Cura et al., 2021; European Commission. Directorate General for Internal Market, Industry, Entrepreneurship and SMEs, 2021). There are variations in behaviours and opinions, with some users finding care labels uncomfortable (when wearing the apparel item) or too long. GINETEX (2017) report the behaviour of cutting out care labels, suggesting that a portion of users across different countries may engage in this practice. In parallel, 39% of 6 000 consumers in Germany, United Kingdom, France, Italy, the Czech Republic and Sweden would never or rarely buy a piece of apparel without a label, where French consumers may bring up the average with 67% of 1 000 users reporting so (COFREET, 2023).

There is also a desire for more convenient access to care instructions, such as having them directly on the sewn label or accessible through a QR code on a smartphone reported by 65% and 17%, respectively, of 1 000 French survey participants (COFREET, 2023). Similarly, when 1 056 Portuguese consumers were asked how they prefer to access product information in apparel, 49.75% indicated that they favour getting this information directly from the product label. Meanwhile, 33.7% showed a preference for using visual codes, such as QR codes or barcodes, to obtain the information (Ribeiro et al., 2023).

Regarding clarity and transparency of content, approximately half of the respondents in the TLR Public Consultation (117 out of 234) indicated that the information is somewhat clear and transparent. Meanwhile, 27% (64 out of 234) felt that the clarity and transparency are limited, and around 18% (28 out of 234) believed the information is largely clear and transparent (Evaluation report of the Textile Labelling Regulation, under development).

The inclusion of multiple European countries in GINETEX (2017) and GINETEX (2019) contributes to a broader representation. The representativeness may vary based on the scope of coverage. However, these sources offer an overview of common specific behaviours, such as attention to labels, indicate low variability in user practices. The representativeness is influenced by the extent to which these behaviours are widespread.

While the data provides percentages of users following care instructions and engaging in specific behaviours related to care labels, the reasons behind these behaviours are not extensively detailed. Understanding the motivations behind user actions could enhance the representativeness of the findings. Nonetheless, the data provide valuable insights into the way users follow care instructions and engage in specific behaviours related to care labels.

Further details on surveys about user behaviour interaction with apparel care labels can be found in **Table 146**.

10493 **Table 146.** Summary of surveys on user behaviour interaction with apparel care labels

Publication	Sample details and characteristics of the study	Country	% of users who follow the textile care instructions			% of most repeated reasons to follow the care instructions according to users			Other behaviour and opinions related to apparel care labels expressed in % of users					
			Average for the EU population	Follow them 'Always'	Follow them 'Often'	To avoid washing problems such as shrinking	Follow the care label to preserve their apparel and keep them longer	Never or rarely buy a piece of apparel without a label	Find the care labels Useful	Cut the care labels out	Think that care labels itch and irritate the skin	Think that care labels are often too long and uncomfortable	Want to find the maintenance instructions directly on the sewn label on the apparel	Want to consult them on smartphone via a QR code.
GINETEX (2017)	6 000 participants (aged 18-65)	Germany, United Kingdom, France, Italy, the Czech Republic and Sweden	70	13	57	38	31	39	NA	62	74	55	NA	NA
GINETEX (2019)	7 000 participants (aged 18-65)	France, Germany, the Czech Republic, the United Kingdom, Sweden, Italy and Spain	70	NA	NA	NA	NA	NA	82	NA	NA	NA	NA	NA
AB-REOC and BV-OECO (2019)	1 756 Belgian residents (aged 16-80)	Belgium	NA	NA	58	NA	NA	NA	NA	NA	NA	NA	NA	NA
COFREET (2023)	1 000 French residents (aged 18-65)	France	NA	NA	NA	NA	NA	67	NA	NA	NA	NA	65	17

10494 Source: own production based on references indicated in the first column of the table

On a different note, in the investigation conducted by Wakes et al. in 2020, it was observed that the price of the T-shirt had an impact on individuals' attitudes towards care labels: a lower purchase price correlated with a lower likelihood of adhering to care instructions. For T-shirts priced above the lowest tier, over 40% of participants followed care instructions before cleaning or after use. Notably, purchasers of higher-priced T-shirts exhibited a greater tendency to read care labels at the point of sale compared to those who bought lower-priced T-shirts (Wakes et al., 2020a). The Wakes et al., (2020) study was conducted in New Zealand, among females from 18–25 years.

The extent to which users associate a care label symbol to its corresponding care instruction is collected in **Table 147**. Survey evidence indicates that there seems to be a consistent understanding across symbols among participants in both, the GINETEX (2017) and AB-REOC and BV-OECO (2019) surveys. There is a demonstrated high level of understanding for the ironing symbol, with almost all respondents in both surveys correctly associating it with the care instruction. The washing symbols also show a strong association, with 91% in GINETEX (2017) and 89% in AB-REOC and BV-OECO (2019) correctly connecting the symbol with the care instruction.

Table 147. Summary of surveys indicating the extent to which users associate a care label symbol to its corresponding care instruction.

Publication	Sample details and characteristics of the study	Country	% of users who correctly associate the care label symbol with the care instruction				
			Ironing symbol	Washing symbols	Bleaching symbol	Drying symbol	Professional cleaning symbol
GINETEX (2017)	6000 participants (aged 18-65)	Germany, United Kingdom, France, Italy, the Czech Republic and Sweden	97	91	33	32	21
AB-REOC and BV-OECO (2019)	1756 Belgian residents aged between 16 and 80	Belgium	98	89	28	24	15

Source: own production based on references indicated in the first column of the table

However, there is a notable decrease in the correct interpretation and understanding of the bleaching, drying and professional cleaning symbols and their respective care instructions among the participants of the two surveys. This suggests that users may find the bleaching, drying and professional cleaning symbols less intuitive compared to other symbols.

While both surveys provide insights into user associations, the GINETEX (2017) survey encompasses a broader international scope, involving participants from Germany, the United Kingdom, France, Italy, the Czech Republic, and Sweden. In contrast, AB-REOC and BV-OECO (2019) focuses specifically on Belgian residents. Despite this difference, the overall patterns in user understanding of care label symbols are comparable between the two surveys.

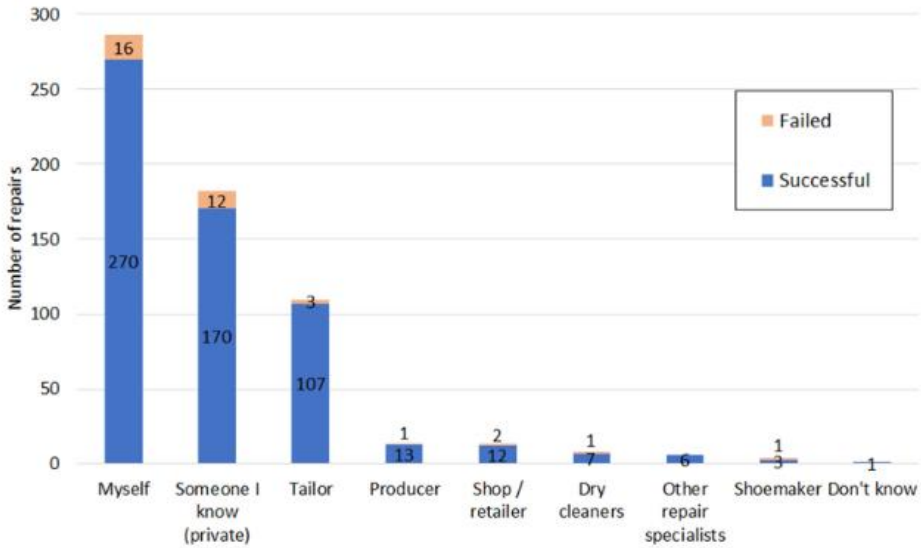
13.7.2.9 Repairing

Consumers nowadays have a quite transitory relationship with their apparel as they are often disposed before they are worn out or broken. Reasons for this may be related to overconsumption and the fact that it is easier and more convenient to replace the unwanted apparel for certain new items rather than repair or modify them (Harris et al., 2016; Pera and Ferruli, 2023). The degree of emotional product-person attachment also influences whether users are willing to engage in repairing certain apparel items (Terzioğlu, 2021; EEA, 2022a). The product-person attachment is described in Section 13.7.3.3.

Consumers can repair their apparel by themselves in private households or at a professional repair shop (EEA, 2022a). The apparel repair industry appears to be predominantly comprised of independent repairers operating as small businesses, often with the owner and a few employees, if any. However, there is a lack of detailed information about the characteristics of this business sector, as European statistical data (Eurostat) on the professional repair of textiles is not available separately (EEA, 2022a).

Figure 91 reflects that most of the apparel repairs are carried by the user followed closely by somebody the user knows, e.g. family member. Bringing the apparel item to repair to a tailor comes as third option. It shall be noted that most repairs are successful in the three repair options users usually opt for.

Figure 91. Number of successful and failed repairs done by different actors (total 625 repairs)



Source: Laitala et al., 2021

A survey in Ireland discovered that 45% of consumers pay to have their apparel repaired while 34% ask family or friends to help them with the repairs. The same survey revealed that around 48% of users would like to repair apparel but do not have the equipment to do it or do not know how to do it. Only 23% of people agree they are not interested in repairing. Interestingly, this behaviour tends to be more common among consumers who purchase apparel often and among young men (EPA Circular Economy Programme and B&A, 2021). Moreover, a survey among 200 Dutch consumers showed that consumers are more likely to help their loved ones with apparel repairs rather than other acquaintances and unknown people (milieu centraal, 2021).

69% of the over 1 000 respondents to a survey in Spain confirm to know where to bring their apparel and shoes for repair (CECU and Amigos de la Tierra, 2022). 67.4% of them add that, if apparel had a label indicating to what extent the product can be repaired, they would use this as a criterion to decide whether to purchase or not the apparel (CECU and Amigos de la Tierra, 2022). Additionally, 52% of 2 000 surveyed consumers declare repairing or bringing their apparel to repair (CECU, 2023).

Furthermore, according to Terzioğlu (2021), individuals typically weigh the costs and benefits before deciding whether to repair an item. For example, how easy is to find a repair solution compared to the price of an apparel item influences the user's decision to engage in repairing activities. As a result, the consumer's trust in commercial repair shops could be promoted by improving the transparency of prices, quality and repair time for example. This would help overcoming the barriers affecting the repair of apparel (EEA, 2022a).

Repair skills for apparel are ceasing to exist. Recent research on sewing skills shows that while different sewing instructions are readily available on the Internet, the main incentive for sewing is trust in one's own skills. A study focused on textile repair in the United Kingdom found that citizen's lack the skills needed to repair compared to previous generations. According to researchers, the decrease in skills is due to the fact that they are no longer taught in schools and the lack of time and equipment in everyday life (Finnish Ministry of the Environment, 2023a).

Laitala and Boks 2012 indicate that users decide to mend their apparel depending on whether they have sewing skills. A relevant number of consumers report to be able to sew a button on (Table 149), and have done so in the past year (Table 148). On the contrary, replacing the zipper is highlighted as a demanding repair which can be seen by the low rate of users who replaced it in the past year (Table 148) and who are confident about doing such work (Table 149). If the zipper breaks, the apparel is disposed unless a strong person-product

attachment exists (Laitala and Boks, 2012). Nonetheless, the users who still may be able to adequately repair apparel may not feel the need to do so anymore (Harris et al., 2016). Moreover, a survey discovered an attitude-behaviour gap among 2 500 users as 58% of them declared to find apparel repair as important while only 23% of them engaged in repair activities (Heiny and Schneide, 2021). This is largely due to the reason indicated above: for some consumers it is more convenient and easier to replace undesired apparel with new items rather than opting for repair or modification (Harris et al., 2016).

Table 148. Summary of surveys that provide the percentage of respondents who have made repairs to their apparel in the past year.

Publication	Sample details and characteristics of the study	Country	% of respondents who have made different repairs to their apparel in the past year						
			Has sewn a button	Fixed seams	Patched apparel	Repaired holes or worn areas in the apparel (darning)	Lengthen or shorten pants (take a hem up or down)	Adjusted size/fit	Changed zipper
Laitala and Boks (2012)	Survey year: 2010 268 participants (aged 15-60+) Survey selection is female dominated (83%)	Norway	73	55	31	27	26	16	10
Laitala and Boks (2012)	Survey year: 2011 1 124 participants (aged 18-60+) Gender of participants equally represented	Norway	64	52	34	NA	NA	NA	NA
Laitala and Klepp (2018)	Survey year: 2011 1 001 participants (aged 18-60+) Gender of participants equally represented	Norway	51	41	NA	25	NA	NA	NA

Source: own production based on references indicated in the first column of the table

The data collected regarding the respondents who have made repairs to their apparel in the past year (**Table 148**) refers to their own willingness to repair and is not always correlated with the user repairing capabilities.

The data collected in **Table 149** regarding how confident users are to repair their apparel is related to their own willingness to repair and is not always correlated with the user repairing capabilities or skills. Additionally, a summary of surveys that showcase how confident users are to mending their apparel depending on the type of repair needed may be found in **Table 150**.

10583 **Table 149.** Summary of surveys that showcase how confident users are to mending their apparel depending on the type
 10584 of repair needed.

Publication	Sample details and characteristics of the study	Country	% of respondents who are confident undertaking apparel repairs					
			Sew a button on	Repair holes or worn areas in the apparel (darning)	Lengthen or shorten pants (take a hem up or down)	Adjust size/fit	Changed zipper	Stain removal
WRAP (2019)	Two waves of online surveys: Wave 1 was conducted in November 2016 and Wave 2 in June 2019. The sample size in 2016 was 1 000 per country. In 2019 the samples were as follows: Denmark 1 046; Germany 1 113; The Netherlands; 1 117 and Italy 1 226	Denmark (DK), Germany (DE), The Netherlands (NL) and Italy (IT)	DK 75 DE 75 NL 70 IT 69	DK 52 DE 46 NL 40 IT 48	DK 46 DE 30 NL 35 IT 32	DK 27 DE 17 NL 25 IT 36	DK 25 DE 25 NL 25 IT 29	DK 72 DE 69 NL 71 IT 61

10585 *Source: own production based on references indicated in the first column of the table*

10586 **Table 150.** Summary of surveys that showcase overall declared frequency of apparel repairs.

Publication	Sample details and characteristics of the study	Country	Overall declared frequency of apparel repairs without specifying the type of repair				
			% of users who have repaired apparel in the past 6 months	% of users who have repaired apparel in the past year	% of users who repair apparel often	% of users who repair apparel sometimes	% of users who never mended apparel
Laitala and Boks (2012)	268 participants (aged 15-60+) Survey selection is female dominated (83%)	Norway	NA	NA	35	51	14
EPA Circular Economy Programme and B&A (2021)	Survey year: 2011 1 000 participants (aged +16) Gender of participants equally represented	Ireland	NA	43	NA	NA	NA
CECU and Amigos de la Tierra (2022)	1 011 participants (+25)	Spain	67.3* (the question included footwear too)	NA	NA	NA	NA

10587 *Source: own production based on references indicated in the first column of the table*

10588 The aptitude to repair is the same or very similar disregarding the country. Denmark and German consistently
 10589 rank higher in repair confidence across various activities, while the Netherlands and Italy generally exhibit lower
 10590 but still varied confidence levels. These variations may stem from cultural, educational, or societal factors
 10591 influencing attitudes toward apparel repairs in each country. However, all countries seem to have high level of
 10592 confidence in sewing a button and repairing holes or worn areas in the apparel.

Finally, there is also research such as [Laitala et al., \(2023\)](#) highlighting the importance on the need for clear guidelines defining what constitutes unacceptable wear and tear versus normal use, and distinguishing between commercial warranties and legal rights. Consumer's complaints on defective or faulty apparel items that fail to meet minimum lifespan requirements, serve as a valuable source of feedback for retailers to understand product performance especially with regards to durability.

Consumers' willingness to repair clothing is strongly influenced by the perceived quality and value of the apparel they own. When apparel is made from low-quality materials, it often leads to a sense of detachment and little emotional connection, making garments less likely to be seen as worth repairing. This is compounded by the convenience and low cost of replacing clothing with new items, which reduces the perceived need to develop or use repair and reuse skills (Pera and Ferruli, 2023). Hence, the importance of a good design that may also contribute to strengthen the emotional product-person attachment with the apparel. Moreover, the study from Pera and Ferruli (2023) highlights that physical proximity to a disposal bin, repair service, or dedicated personnel can serve as a strong enabling factor for engaging in more sustainable practices such as repair. This form of access could help consumers overcome common barriers such as lack of time, or motivation, encouraging behaviours that extend the life of apparel.

13.7.2.10The storage of apparel

The storage of apparel can be done for active or inactive apparel. Storing active apparel would be the action of placing inside wardrobes, for instance, all those apparel items that an individual wears regularly. In other words, the apparel that users move between use and storage frequently (Cluver, 2008a). Inactive apparel items constitute the apparel kept at home without being used for an undetermined time. These are represented by the over 30% of apparel stored in Europeans closets that have not been used for a year or even more time (European Parliamentary Research Service, 2019).

Inactive apparel may comprise apparel that do not fit. In fact, this seems to be generally the case among consumers (Bye and McKinney, 2007). Therefore, why would consumers store items that are not able to use due to their body size? The wishful thinking that the apparel will eventually fit again may trigger the storing of these apparel. In certain cases, this practice may also help some consumers to monitor their weight. This temporary storage of inactive apparel, can be understood as keeping apparel that are not worn for an unknown time period before doing something different with them. Temporary storage may be done while the users are considering the best way to dispose the apparel, when waiting for the apparel to be back in fashion, to repair in the case of wear and tear or just because they want to make sure there was no longer a way to make use of the apparel (Cluver, 2008a). Interestingly enough is the fact that basic apparel that were once very necessary for the user, even though no longer in fashion and/or suffered wear and tear, were stored until new apparel were purchased to substitute them (Cluver, 2008a).

Bye and McKinney, (2007) carried out surveys in which respondents indicated that inactive apparel could be linked to the past personality of the owner who no longer wants to be attached to, hence the reason not to wear them, but still feels like a part of their lives (Guy et al., 2001; Bye and McKinney, 2007). The respondents indicated that they hold onto apparel that were no longer used because they allowed them to recall important people and past experiences they had while wearing them (sentimental value). In other cases, the consumer keeps apparel that were purchased because of a wrong choice just to justify the price paid for them. Finally, some consumers keep certain apparel because a perceived aesthetic value attached to them. On a different note, around 13% of 350 000 Vinted users who are engaged in the resale of apparel items report to store in the closet the items that they do not manage to sell in the online platform Vinted, without the intention of using them (Vinted, 2021). In any case, it is important to note that prolonged inactive storage can eventually cause consumers to discard their apparel due to extended periods of non-use (Accenture, 2022).

All in all, as indicated by [Maldini et al., \(2019\)](#) wardrobes can be seen as a "pull" system, where new apparel items are added only when unsatisfactory ones are replaced. Thus, delaying disposal should delay purchases, decreasing overall demand. However, in practice, wardrobes integrate new apparel items for various reasons, following unpredictable patterns. New items are frequently purchased without considering existing ones, pushing older apparel to the back as more attractive options come in. This behaviour may eventually result in disposing the apparel due to lack of space for new items (Accenture, 2022).

It can be concluded that the storage of apparel items, whether temporary or not, seem to be strongly linked to a person-product attachment. Additionally, it should not be not assumed that longer apparel lifespans and delayed disposal always yield positive environmental impacts given the unpredictability of the users' purchase and storage behaviour, as described above.

13.7.3 User behaviour regarding the disposal of apparel

13.7.3.1 Reasons for the disposal of apparel

Disposal of apparel happens when a user transfers its ownership to another person or entity (Cluver, 2008a).

Henninger et al., (2021) investigated 154 papers of which 32 touched upon the user behaviour related to the disposal of apparel items. Moreover, most of the studies were published after 2015 which showcases that it is a topic that is recently gaining more attention than in the past (Henninger et al., 2021).

The intrinsic or physical durability can be defined as the ability of a product to withstand the tear and wear of time without its functionality and aesthetics being compromised. This is envisaged during the design phase of the product when the selection of materials, modelling, etc. focused on improving the robustness of the article (Alliance of Commerce and Deloitte, 2022).

The intrinsic durability of apparel may be lost due to abrasion, colour changes, broken zipper, soiling that cannot get cleaned, etc. (Laitala and Boks, 2012). However, Laitala and Boks (2012) reported that the examination of disposed apparel showed that there is a great difference in opinions regarding when the apparel is too worn out to be used. Some users did not mind pilling or small holes, whereas for others, these changes indicated the apparel was worn out and were important disposal reasons.

A composition analysis involving 391 composition samples from Swedish municipalities between 2012 and 2014 revealed that 59% of apparel was in a condition suitable for reuse (SMED, 2016).

However, a survey run by Cooper and Claxon in 2022 on 1 476 disposed apparel items unveiled various types of physical issues. The predominant problems were associated with colour fading, particularly in jersey and woven fabrics, and pilling in knitwear and jersey items. Fabric deterioration, characterized by fraying, especially around hems, as well as wear around the crotch of trousers and jeans, and accidental damage such as stains, tears, and rips, were also prevalent. Other forms of failure included the loss of dimensional stability, logo malfunction, discoloration, notably in the collar area of white shirts, holes in seams (including jacket linings), and trim failure. The percentages are calculated based on all apparel in the sample. Notably, 69% of all apparel exhibited either a colour-related or logo problem, while 75% experienced fabric-related issues, indicating that many apparel had multiple problems (Cooper and Claxton, 2022b). The physical failures found in disposed apparel are indicated in **Table 151**.

Table 151. Recorded problems in apparel (% of all apparel items)

Type of Failure	%
Colour fading	53
Fabric breakdown	29
Accidental damage	29
Loss of dimensional stability	20
Logo failure	16
Discolouration	15
Hole(s) in seams	14
Trim failure	8

Source: Cooper and Claxton (2022).

The individual characteristics of the consumers, their habits, demographic context, product traits and quality seem to influence the user's decision to get rid of a product (Cluver, 2008; Goworek et al., 2012, as cited in Harris et al., 2016; Sandin et al., 2019).

The perceived quality of a product is decisive when consumers decide to get rid of a piece of apparel, either because it seems to be worn-out or no longer functional (Aakko and Niinimäki, 2022). In fact, early disposal of apparel may be also accentuated by the loss of the symbolic perceived value of apparel for the consumer (Gwozdz et al., 2017a). Moreover, certain studies document that low quality, may trigger an early disposal of apparel (Piippo et al., 2022a). Additionally, the fact that some consumers often want to own new apparel shortens their time in use too (Klepp and Grimstad, 2001; Piippo et al., 2022a). This is the case for 45% of 982 Italian consumers who indicated that they dispose their apparel items because of they are old (YouGov, 2019). Similarly, 7% of 1 000 Polish consumers indicated to get rid of apparel items as they lacked sufficient space for new ones (Accenture, 2022).

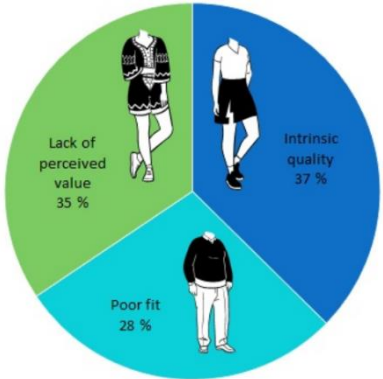
The two most common reasons for the disposal of apparel for both, female and male German participants to a survey were material defects due to wear and tear and an inappropriate size. For German men, the third most common reason to dispose apparel is that it had fallen out of shape. This reason was also mentioned by the

female respondents, but it was more often stated that the apparel simply was not liked anymore. In this regard, the female respondents from the youth group stand out. The percentage of those who disposed apparel because they no longer liked it is significantly larger in this social segment than among female respondents from other social environments and almost twice as much compared to the male respondents from the same social group. Reasons related to taste, fashion or personal style are more commonly mentioned among the young people (Kleinhüchelkotten et al., 2018a).

The decision to dispose apparel, and how frequently it happens, is also influenced by the price of the apparel. There is evidence that consumers see higher value in more pricey apparel items (Morgan and Birtwistle, 2009b; Joy et al., 2012b). This means consumers tend to dispose more frequently the cheapest apparel as they understand that the value proposition in terms of durability and quality is lower (Lewis, 2015).

Across all countries reported in **Table 152**, a significant percentage of participants cite intrinsic quality as a primary reason for the disposal of apparel, indicating a shared concern for durability and material integrity. Additionally, fit problems are a consistent factor influencing apparel disposal, suggesting that comfort and suitability play a universal role in user behaviour. Participants in most countries mention perceived value, indicating that subjective judgments, such as taste-related unsuitability, impact their decision to dispose apparel. These results are in line with the ones presented in the Laitala and Klepp, (2022) review of 17 consumer studies involving around 20 000 participants in which the most common reasons for apparel disposal were identified as intrinsic quality, perceived value and fit issues (**Figure 92**).

Figure 92. Main reasons for disposal of apparel



Source: Laitala and Klepp, (2022).

Also,, changing fashion trends contribute to apparel disposal across various countries, emphasizing the influence of style preferences on user behaviour.

Furthermore, Pera and Ferruli, 2023 identified several internal barriers that hinder sustainable consumer disposal practices. These include a lack of time, energy, or patience; limited skills or know-how and insufficient information leading to skepticism. Easy access to bins, services, or personnel facilitates practices such as repair, donation, and resale by overcoming barriers like idleness and lack of motivation. Additionally, improved communication from companies and local authorities could help consumers better understand the textile disposal system and recognise their role and impact within it (Pera and Ferruli, 2023).

The overall implication of the data reported above is that the reasons for the disposal of apparel exhibit a complex interplay of individual factors. Quality and fit are universal concerns, suggesting a global emphasis on durable and well-fitting apparel. Cultural differences emerge in the importance placed on perceived value, taste-related aspects, situational reasons (e.g lack of space for new apparel), and responsiveness to fashion changes.

While each study contributes valuable insights, the combined dataset offers a more comprehensive understanding of global apparel disposal trends. Variations in sample characteristics and the scope of data collection highlight the importance of considering multiple studies for a holistic perspective.

10727 **Table 152.** Summary of apparel disposal reasons in recent consumer studies with high number of participants.

Publication	Sample details and characteristics of the study	Country	Main reasons for the disposal (expressed in % of respondents)				
			Intrinsic quality (e.g worn-out items)	Fit issues	Perceived value (e.g. taste-related unsuitability)	Fashion changes	Other (e.g. Situational reasons, functional shortcomings)
<u>Greenpeace.</u> (2015)	1 011 participants	Germany	92	72	64	40	NA
<u>Ungerth and Carlsson.</u> (2011)	1 014 participants (aged 16-74)	Sweden	60	9	21	NA	4
<u>Laitala and Boks.</u> (2012))	546 participants (adult population, age groups not defined) 77% of the subjects were women and the 25-39 age group is overrepresented in comparison to the rest age groups	Norway	49	19	11	NA	19
Laitala and Klepp (2020)	1 111 participants (aged 18-65) Women and men are equally represented in all countries	China, Germany, Japan, United Kingdom, and USA	44	13	35	NA	9
WRAP (2017b)	2 058 participants	United Kingdom	18	42	33	NA	NA
Lang, Armstrong, and Brannon (2013)	555 participants	USA	30	31	39	NA	NA
Morell-Delgado et al., (2024)	1 469 participants	Spain	65	44	21.5	NA	2.5
<u>YouGov.</u> (2019)	982 participants	Italy	60	47	16	12	2
<u>Accenture.</u> (2022)	1 000 participants	Poland	38	21	7	6	21

Source: own production based on references indicated in the first column of the table

10728

10729

10730 **13.7.3.2** Disposal channels for apparel

10731 By 1 January 2025, mandatory separate collection of textiles will be set in the EU Member States as required
10732 by Article 11(1) of the Waste Framework Directive. Additionally, the Commission has adopted the establishment
10733 of mandatory extended producer responsibility (EPR) schemes designed to streamline and put in place collection,
10734 sorting, reuse, preparation for reuse, and recycling infrastructure required to address the material collected
10735 once the separate collection obligation takes effect³²⁷. Moreover, the on-going review of the Textile Labelling
10736 Regulation is exploring whether to introduce specifications for physical and digital labelling of textiles, including
10737 end-of-life disposal instructions and other circularity parameters based on requirements under the proposed
10738 Regulation on eco-design for sustainable products and on the implementation of EPR schemes across Member
10739 States.

10740 The person-product type of attachment plays a key role in how apparel may be disposed of. If there are positive
10741 associations with a piece of apparel, this might be gifted to relatives or friends while if there is a negative
10742 association the apparel may end up being donated, swapped or just disposed in the general waste bin (Joung
10743 and Park-Poaps, 2013; Lewis, 2015). Throwing away (usable) apparel is associated with bad conscience and
10744 moral aspects hence they are given to friends, family and charities as the preferred way to dispose of the
10745 apparel (Klepp and Grimstad, 2001). Furthermore, selling apparel is among the alternative methods consumers
10746 are using nowadays when they wish to get rid of apparel they no longer wear. In fact, 21% of 11 483 European
10747 consumers sell their apparel online, and 59% of those do so more than once a year (YouGov, 2021).

10748 The surveys in **Table 153** show the most common disposal trends in different countries. In both surveys,
10749 donating apparel to charity emerges as a popular choice among participants, indicating a shared inclination
10750 towards contributing to charitable causes through apparel. Also, both surveys highlight the significance of giving
10751 apparel to family and friends as a prevalent disposal method, suggesting a common practice of sharing apparel
10752 within one's social network.

10753 By and large, the studies reported in **Table 153** provide valuable insights into apparel disposal practices. While
10754 there are some differences in terms of the represented countries, the similarities in identified disposal methods
10755 across the survey results suggest common trends in how individuals choose to handle apparel they no longer
10756 need or use. The prominence of charitable donations and sharing within social circles appear to be consistent
10757 behaviours, transcending geographical and cultural boundaries to some extent. The survey run by Vinted slightly
10758 differs, as participants were asked what they would do with the apparel items that they did not manage to sell
10759 in the online platform Vinted.

10760 **Table 153.** Surveys responses on main types of alternative apparel disposal

Publication	Sample and characteristics of the study	Country	Main types of alternative apparel disposal (expressed in % of respondents)			
			Donate to charity	Give to family/friends ³²⁸	Used as cleaning cloth ³²⁸	Discarded in the general waste bin at home
Koukouvinos (2012)	201 participants (aged 18-35) 75% of the subjects were women	Greece	51	59	45	NA
Laitala and Boks (2012)	546 participants (adult population, age groups not defined)	Norway	36	12	8	27
Vinted (2021)	350 000 participants	Belgium, Germany, Spain, France, Italy,	19	19	NA	Less than 1

³²⁷ Directive (EU) 2025/1892 of the European Parliament and of the Council of 10 September 2025 amending Directive 2008/98/EC on waste (Text with EEA relevance).. Available here: [Directive - EU - 2025/1892 - EN - EUR-Lex](#)

³²⁸ The quality of these products is unknown.

Publication	Sample and characteristics of the study	Country	Main types of alternative apparel disposal (expressed in % of respondents)			
			Donate to charity	Give to family/friends ³²⁸	Used as cleaning cloth ³²⁸	Discarded in the general waste bin at home
		The Netherlands, Poland and United Kingdom				
YouGov, (2021)	11 483 participants	France, Germany, Austria, Denmark, Finland, Italy, Norway, Spain, Sweden, Switzerland	69	34	NA	13
Morell-Delgado et al. (2024)	1 469 participants (aged 16 to more than 65) 68.8% of the subjects were women	Spain (survey sample focused on Catalonia Autonomous Community)	26	48.9	32.5	7.8

Source: own production based on references indicated in the first column of the table

Finally, it is also important to note that the primary finding of the European Commission's Directorate General Joint Research Centre (2021) study on the quality of clothing found in residual waste is the existence of a correlation between the proportion of separately collected clothing and the average quality and value of clothing discarded in residual waste. According to this study, this correlation is partly due to households making discerning decisions based on the perceived monetary value of apparel, determining which items are suitable for donation or resale for reuse and which have minimal potential for reuse. This idea is supported by a study conducted in Denmark in 2018 (Watson et al., 2018), which strengthens this theory. The study suggests that the 42 000 tonnes of clothing disposed of in Danish residual waste for incineration in 2017 had an estimated value of 12-15 million euros before disposal. In contrast, the 36 000 tonnes of clothing separately collected were sold on reuse markets for an estimated 65 million euros, indicating a value per tonne four to five times higher. As a result, it appears to be commonly understood that textiles with the least potential for re-use are more abundant in the textile portion of mixed municipal waste, but the limited existing research presents inconclusive results according to [Huygens et al., 2023](#). Additionally, an analysis of 38 samples from across five Welsh local authorities emphasises the correlation between the method of textile collection and the reusability of the collected textiles. For instance, in Wales, the reusability rate of textiles collected via container stands at 75.3%, compared to 54.6% for those gathered through door-to-door collection (WRAP Cymru, 2022). Last but not least, findings from [Morell-Delgado et al., \(2024\)](#) indicate that apparel perceived to be in better condition are typically disposed of in street containers for textile waste (78%), with minimal disposal in mixed-fraction containers (3%). Conversely, when apparel are in worse condition, citizens exhibit uncertainty regarding disposal methods, with approximately 30% disposing of them in specialized containers and another 30% opting for mixed-fraction containers. Therefore, the population's decision regarding the disposal destination of unwanted apparel seems to be influenced by the perception of the quality the item may still possess by the consumer (Morell-Delgado et al., 2024).

13.7.3.3 Person-product attachment

The emotional attachment formed between the apparel and the consumer is generally referred to as 'person-product attachment'. This type of connection between a textile item and the user has an influence on how long consumers own a textile and how often they make use of it. The person-product attachment is usually embedded in the term 'emotional durability' which goes beyond just functionality. The Ellen MacArthur Foundation report (Ellen MacArthur Foundation, 2021) defines emotional durability as: 'the product's relevance and desirability to a user, or multiple users, over time'. A product that holds emotional durability tells a compelling narrative, creates history with its users through bonding and aligns with their values.

10793 The ownership of apparel could be 'active' (i.e. use on a daily basis or several times a week), 'seldom' (use
 10794 several times a year) or 'inactive or in storage' (very rarely or never use the apparel) (Niinimäki and Armstrong,
 10795 2013).

10796 Niinimäki and Armstrong (2013) carried out a survey with over 400 participants from United States of America
 10797 (45.1% were men). The sample ranged in age from 18 to 67: 18–24 (7.5%), 25–34 (27.4%), 35–44 (23.5%),
 10798 45– 60 (28.1%), and over 60 (13.5%), most of whom were Caucasian/white (88.0%) and had completed a
 10799 college degree or higher (66.1%). **Table 154** collects the elements that generate person-product attachment
 10800 listed according to the number of times each of them were mentioned by the respondents to the questionnaire.

10801 **Table 154.** Elements that generate person-product attachment (Niinimäki and Armstrong, 2013).

Order of relevance	Element	Description (in order of most referenced by users)
1	Functionality	Comfortable Good fit Multi-function Functional (good for sports and hide body deformation) Easy to match Easy to put on
2	Memory	Memories (youth and childhood) Received from special person Family ties Remind of special person Represent membership to a group (e.g. team/band)
3	Emotional satisfaction	Look/feel good in it Receive compliments wearing the piece of apparel Love the brand
4	Design and style	Good design (e.g. cool looking) In style
5	Fabric and material	Nice colour Pleasant to touch (silky and soft) Fabric aesthetic (not thick, light weight, and sparky) Flexible (not stretched)
6	Personal values	Uniqueness Feel relaxed in it
7	Quality	Durable High quality in manufacturing High quality in material
8	Effort invested	Reward for self Hand made
9	Financial value	Price (good deal and very expensive)

10802 *Source: Niinimäki and Armstrong (2013).*

10803 The findings showed that apparel that were purchased brand-new turned out to have the highest emotional
 10804 attachment for the respondents. This is largely explained by the fact that there is a widespread idea of 'new'
 10805 apparel as something that has a value in itself. New apparel are clean canvas, both in the sense that they are
 10806 physically clean, i.e. free of spots, presentable and hygienic, but it is also 'mental clean' in the sense that it is
 10807 free from stories attached to the apparel in question (Forbrugerrådet Tænk, 2022).

10808 Moreover, it is the frequently used regular T-shirts, dresses and jeans the ones respondents indicated to be
 10809 more meaningful to them. Functionality, a special memory, emotional satisfaction were the most repeated
 10810 attributes by the respondents that help promoting emotional attachment to apparel.

10811 The value found in new apparel is a major barrier to reuse and is also reflected in the fact that apparel are
 10812 used more when they are brand new than when they have several years. In fact, apparel purchased as second
 10813 hand are used 30 per cent less than those purchased as brand new (Forbrugerrådet Tænk, 2022). On top of
 10814 that, it should be noted that reuse may increase the possession span of the apparel, but does not necessarily
 10815 increase the use of the apparel (Forbrugerrådet Tænk 2022; Laitala and Klepp 2021).

10816 The durability and active use of apparel depends on the robustness and longevity of its materials but also on
 10817 the changing trends, needs and wishes of consumers. Emotional durability remains a forward-looking issue,
 10818 which is difficult to measure because it is intrinsically linked to consumer perceptions and brand representations.
 10819 Its understanding is based on social sciences concepts and there is no evaluation methodology that has been
 10820 fully tested (Alliance of Commerce and Deloitte, 2022).

However, apparel may have a certain initial physical or functional lifetime that may be extended if a strong product-user relationship is promoted by retailers. For instance, retailers can maximise the value of the apparel they sold by ensuring that the design meets real long-term needs and that incorporates a particular added value which prevents the user from an early disposal of the textile product (Niinimäki and Armstrong, 2013; Alliance of Commerce and Deloitte, 2022). The aim would be to create articles that can be adapted to the different stages of consumer life, integrating more modularity (multi-seasonality, reversibility, extensibility, etc.) and facilitating alterations, or offering timeless products whose colours and shapes are resistant to changes in trends. To enhance the perceived value of their articles, brands can rely on personalisation levers or marketing methods such as co-creation, tailor-made manufacturing or limited series, which make it possible to create an emotional attachment of the customer to their apparel, while limiting promotions and the continuous inflow of novelties (Alliance of Commerce and Deloitte, 2022).

Bottom line is the role of the quality of the apparel. Laitala et al., (2021) asked a number of consumers what aspects would incentivise them to wear their apparel longer. The most repeated answer both for men and women was if the apparel were of better quality. The same was concluded in *QVAM* (2021). Along the same lines, 92% of 101 visitors to a dress exchange at Greenpeace Hamburg declared that they would wear apparel for more than a year if they were of good quality. 89% of participants to the same survey indicated that if apparel maintained their shape and colour for a longer period would also incentivise them to wear apparel for longer. The third most commented reason for wearing apparel for longer was if there was a bonding with the apparel (product-person attachment) (Kleinhüchelkotten et al., 2019).

All in all, the consumers deal with objects in different ways. This is the main reason why the product-user textile attachment is quite subjective and difficult to draw conclusions from. The designers may play an important role in stimulating the length of ownership and use of apparel when choosing the fabrics, the type of confectionary work and durable design. Research has referred this design as 'emotional durable design' which emphasizes creating enduring experiences over new products. The primary challenge is to develop something that captivates customers, encouraging them to cherish it rather than succumb to the desire for novelty (*Chapman 2015 as cited in Vanacker et al., 2022*). Through each design decision, the long-term customer satisfaction may be increased, resulting in a higher emotional user-product attachment. Nonetheless, there is also research indicating that strategies towards the promotion on person-product attachment showed that their effect had not been empirically validated (Maldini and Balkenende, 2017).

13.7.3.4 Returns of apparel

Convenience seem to be the main reason for buying online as reflected by half of the usual online customers in the survey run by AK Wienn and Greenpeace (2023). However, if the consumer lacks the time to return a parcel to a retailer with a non-flexible returns policy, he or she may be more selective when choosing what to buy to avoid having to keep the purchased items because the time to return the items is up. This may be one of the reasons why longer periods for free returning allow for ordering more apparel as the customer has a more relaxed approach to the potential returns. At the same time, it can happen that for some consumers returning online buy-outs is deemed a barrier because it becomes cumbersome to send it back, despite the type of return policy that the retailer may offer. This means that apparel that do not fit, or apparel that are not up to the customer's preference, end up stored in the cabinet (Forbrugerrådet Tænk, 2022).

A study from the European Environment Agency retrieved that casual dresses and jackets have the highest return rate followed closely by jeans and vests (EEA, 2024). Interestingly, the same EEA study found that expensive products have more probability of being returned. An explanation for this behaviour has not been found.

When it comes to demographics, it is not widely analysed in the scientific literature whether it is females or males who are more prompt to return apparel. A study pinpointed that women return more online purchases than men (AK Wienn and Greenpeace, 2023a). This seems to be supported by the fact that womenswear is the winner category of apparel when it comes to returns (British Fashion Council's Institute of Positive and Roland Berger, 2023). Moreover, young consumers are more likely to order online and return apparel more frequently as they tend to buy more apparel and order several sizes compared to other age groups (AK Wienn and Greenpeace, 2023a).

According to the results of the Foresight Factory survey fit issues is the first reason for returns by consumers (Foresight Factory, 2021; Zimmermann et al., 2021). The second most popular reason to return apparel is taste-related unsuitability of an item and/or its product details for the consumer. In this category, the share of total returns in Zimmermann et al. (2021) corresponds to the dislike of the shape or cut of the apparel (16%), a dislike for the material (8%) and colour or pattern (6%).

Online shopping often leads to mis-buying (Forbrugerrådet Tænk, 2022). The main reasons found in the literature for returning apparel are problems fitting the item and the dislike of the apparel by the consumer. These two reasons are followed almost equally by lack of quality of the apparel and receiving a faulty item. A survey run by Foresight Factory in 2021 with over 20 000 respondents, reports the same main reasons for returning apparel by consumers; 38% of them reported that the items did not fit well while 15% of respondents indicated that the apparel did not suit them. Quality was not enough for 14% of the survey participants and 13% of them received faulty items (Foresight Factory, 2021).

'Bracketing' occurs when consumers purchase several identical or similar items, have them shipped, keep one, and return the others (NRF and APPRISS RETAIL, 2023). It is interesting to see that in the study from Zimmermann 2021, bracketing, is not among the main reasons for returning goods (Zimmermann et al., 2021). A similar behaviour is reported by the survey from Foresight Factory, in which only 6 % of the respondents explicitly indicated that they bought multiple sizes but did not intend to keep all the ordered items. The same percentage of users reported that they purchased multiple styles without the intention to keep everything (Foresight Factory 2021). Another reported behaviour linked returns is what is known as 'wardrobing'. This happens when a consumer purchases an expensive item, wears it, and then returns it (NRF and APPRISS RETAIL, 2023). A survey focusing on retailers, highlighted that 48.8 % of them experienced wardrobing (NRF and APPRISS RETAIL, 2023). No data was found from consumers acknowledging this practice.

The fitting issues that usually come with ordering apparel online are often linked to the inability of consumers to interpret correctly the sizing scales offered in some online sites but there may be also cases of inconsistent and not correct sizing (Vladimirova et al., 2022). Historically, size designation was introduced alongside the development of ready-to-wear clothing as a means to guide consumers in selecting garments that would fit their body dimensions. Size labels, typically expressed through numbers or letters, were originally based on standardized body measurements and proportions. However, over time, the link between size designations and actual body dimensions has weakened significantly. Contemporary research highlights that, particularly in Western markets, consumers — and women in particular — often experience difficulties in finding garments that fit properly when relying solely on the size indicated on the label (Faust, 2014). As a result, the traditional size designation system offers limited support to consumers, frequently leading to incorrect purchases and increased returns as mentioned above. Addressing this gap through improved size information and fit guidance could contribute to more sustainable consumer behaviour and reduced product returns.

Although major corporations tend to concentrate on serving their target markets, many are not interested in adopting a standardised size designation. One reason often cited is that brands and manufacturers use sizing as a marketing tool, commonly referred to as "vanity sizing," to appeal to consumers (Faust, 2014). This is a phenomenon documented in academic research that refers to the practice where garments are labelled with smaller sizes than their true physical measurements, aiming to positively influence consumer perception. This practice is particularly prevalent in women's apparel. Studies have shown that mid-range brands often produce garments with larger actual measurements than discount brands, while brands targeting younger female demographics tend to use smaller sizing standards. The growing inconsistency and expansion of size ranges in women's clothing are closely linked to vanity sizing strategies. By contrast, evidence suggests that vanity sizing is much less common in men's and children's apparel (Franz, 2017). Some studies indicate that the integration of 3D body scanning technologies and adaptive clothing design may hold significant potential to deliver more customised and better-fitting apparel experiences, addressing the needs of a multicultural and diverse global population (Nam and Alvi, 2024). Other studies flag that despite the availability of extensive anthropometric surveys and national 3D body measurement campaigns, few manufacturers have fully adopted these resources in setting their size designations (Faust, 2014).

Although most manufacturers rely on similar key body measurements, the reference to brand-specific size charts leads to considerable inconsistency. As a result, size designations often fail to reliably guide consumers in finding well-fitting garments, particularly given the variations between population data and brand sizing practices. To address these shortcomings, a more comprehensive and standardised approach to size designation is needed. One that balances the needs of both manufacturers and consumers. Building on the original idea of combining size numbers, letters, and symbols, and incorporating pictograms indicating key body measurements, could offer an effective solution. This would allow manufacturers and retailers to continue targeting their specific markets while improving the universality and clarity of size information for consumers (Faust, 2014).

Products with perceived low quality, less durability and the remorse of the consumer due to buying apparel are also highlighted as potential reasons for returning apparel (Bernon et al., 2011).

A lot of consumers have a look at the return policies before purchasing items online (Asdecker and Sucky, 2019; Makov, 2023). In fact, 70% of consumers indicate that whether they repeat a purchase on a site or not, depends

10932 on their return experience (Asdecker and Sucky, 2019). Nonetheless, it seems that most users assume that the
10933 returned items are always put back into sale when this is not necessarily the case due to the complex reverse
10934 logistics of the supply chain (Makov, 2023).

10935 In summary, with the data available, it is not possible to clearly distinguish between the behaviour regarding
10936 apparel returns of men and women. Nonetheless, some studies stress that younger generations return more
10937 items given their tendency to buy apparel more often. The influence different demographic aspects play when
10938 returning products have not been possible to analyse.

10939 Reasons for returning items purchased online tend to have a common root: the difficulties users have choosing
10940 the appropriate size and the right product based on the online descriptions. Furthermore, studies exploring the
10941 reasons for returning products at physical stores have not been found. This may be because online returns are
10942 comparatively much higher than those at physical stores. The impossibility for consumers to try the apparel on
10943 the spot and have a tactile experience contributes to this difference in number of returns between the online
10944 and physical store shopping.

10945 **Table 155.** Reasons for returning apparel to e-commerce

Publication	Sample details and characteristics of the study	Country	Reason for returning (expressed in % of respondents)				Change of mind
			Fit issues	Taste-related unsuitability	Insufficient perceived quality	Faulty items	
Foresight Factory (2021)	20 000 participants	United Kingdom, United States, Canada, Australia, France, Germany, Italy, Netherland, Norway, Japan, Saudi Arabia and UAE	38	15	14	13	8

10946 *Source: own production based on references indicated in the first column of the table*

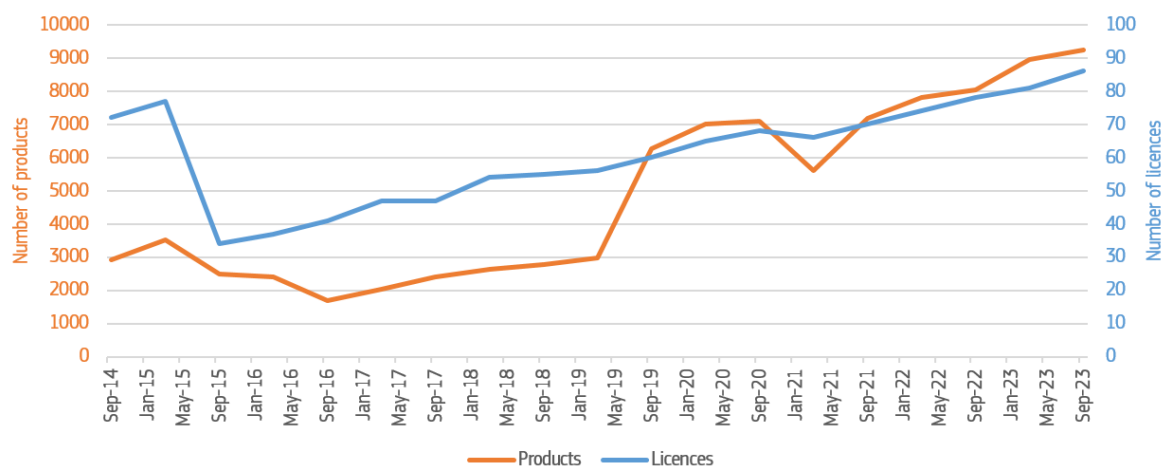
10947

10948

13.8 Supporting information about environmental labels and current EU Ecolabel criteria

13.8.1 Figures of EU Ecolabel for textile products

Figure 93. Evolution of licences and products awarded with EU Ecolabel for textile products



Source: own elaboration based on EU Ecolabel facts and figures ⁽³²⁹⁾

Table 156. Figures of EU Ecolabel for textile products in September 2023

Licences			Products		
Competent body	Number	Percentage compared to the total number (%)	Competent body	Number	Percentage compared to the total number (%)
Denmark	24	28	Portugal	3854	42
Italy	18	21	Italy	3359	36
Norway	11	13	Denmark	1338	14
Austria	6	7	Sweden	172	2
Germany	5	6	Netherlands	164	2
Netherlands	5	6	Norway	161	2
Sweden	5	6	Poland	89	1
Czechia	3	3	Austria	50	1
Bulgaria	2	2	Czechia	31	0
Spain	2	2	Germany	14	0
Poland	2	2	Spain	9	0
Belgium	1	1	Bulgaria	7	0
Portugal	1	1	Belgium	1	0
Romania	1	1	Romania	1	0
Cyprus	0	0	Cyprus	0	0
Estonia	0	0	Estonia	0	0
Greece	0	0	Greece	0	0
Finland	0	0	Finland	0	0
France	0	0	France	0	0
Croatia	0	0	Croatia	0	0
Hungary	0	0	Hungary	0	0
Ireland	0	0	Ireland	0	0
Iceland	0	0	Iceland	0	0
Lithuania	0	0	Lithuania	0	0
Luxemburg	0	0	Luxemburg	0	0
Latvia	0	0	Latvia	0	0

³²⁹ EU Ecolabel facts and figures. Available at [this link](#). Last accessed on 12 January 2024.

Licences			Products		
Competent body	Number	Percentage compared to the total number (%)	Competent body	Number	Percentage compared to the total number (%)
Malta	0	0	Malta	0	0
Slovenia	0	0	Slovenia	0	0
Slovakia	0	0	Slovakia	0	0

The licence is awarded to a company by a competent body. A licence can include the award of the EU Ecolabel to one or more products. The expression 'number of products' does not refer to the number of items related to a specific product.

Source: Own elaboration after consultation with EU Ecolabel Helpdesk

13.8.2 Contributions to the initial questionnaire

Table 157. Type of respondents that contributed to the section on EU Ecolabel during the initial questionnaire

Type of respondent	Number	Percentage compared to the total (%)
University or research institute	2	6
Certified laboratory	0	0
Government (local, regional, or national)	6	18
Non-governmental organisation – Environment	2	6
Non-governmental organisation – Consumers	2	6
Industry – manufacturing	10	29
Industry – manufacturing - association	1	3
Industry – waste collection, sorting and treatment	0	0
Industry – waste collection, sorting and treatment - association	1	3
Distributor/Retailer	2	6
Distributor/Retailer association	1	3
Other	7	21
TOTAL	34	100

Among the respondents of the questionnaire, 7 declared to be licence holders of the EU Ecolabel for textile products, and 4 declared to be competent bodies of the EU Ecolabel.

Source: own elaboration

Table 158. Suggestions received by respondents to the initial questionnaire on EU ecolabel criteria 1-9, which focus on fibres

Criterion	Suggestions
Criterion 1. Cotton and other cellulosic seed fibres	- Make sure that only fibre-to-fibre from post-consumer waste is accepted; - Analyse the real traceability of fibres; - Investigate regenerative farming; - Focus only on organic cotton, disregarding the Integrated Pest Management cotton; - Establish a clear distinction between organic cotton and transitional organic cotton; - Investigate the implications of different minimum percentages of organic cotton; - Extend the analysis of restricted pesticides to recycled cotton; - Consider the expansion of the restricted pesticides following Blue Angel and Oeko-Tex 100.
Criterion 2. Flax and other bast fibres	- Investigate the possible use of the European Flax standards developed by the Alliance for the European Flax-Linen and Hemp; - Revise the COD and TOC limits as absolute values.
Criterion 3. Wool and other keratin fibres	- Assess the possible inclusion of organic fibres; - Consider the inclusion of biological husbandry control; - Specify that also recycled fibres should be checked for ectoparasiticide; - Ban mulesing; - Investigate the possible inclusion of specific certified wool, e.g. recycled, responsible and regenerated wool; - Question the ambition of pH, COD and temperature limit values.
Criterion 4. Acrylic	Limit the residual acrylonitrile in raw fibres to 1.5 mg/kg.
Criterion 5. Elastane	Investigate the possibility to accept 5% of elastane if the current production of elastane does not involve a large use of organotin.
Criterion 6. Polyamide	- Extend the traceability along the entire value chain of any feedstock material: recycled or virgin; - Investigate the effectiveness of verification methods if production occurs outside EU; - Include thresholds of CO₂ emissions.
Criterion 7. Polyester	- Remove the distinction between consumers and commercial or public sector customers; - Allow only fibre-to-fibre recycling; - Analyse the traceability of fibres: virgin and recycled.

Criterion	Suggestions
	- Decrease the limit of antimony and accept its verification via a test. - Investigate the feasibility of verification of VOC emissions.
Criterion 8. Polypropylene	Investigate the relevance of VOC emissions
Criterion 9. Man-made cellulose fibres	- Increase the percentage of fibre coming from sustainable forestry management; - Include the regenerated fibres based on the FAO principles on sustainable forestry management; - Align the ambition of this criterion to the latest EU Ecolabel criteria for absorbent hygiene products; - Investigate how to better support applicants and CBs to check compliance with criterion 9(a); - Consider potential updates of the criterion 9(a) due to latest European Regulation; - Allow only use of organic cotton; - Question the ambition level of AOX emissions; - Investigate the use of the latest learnings about the Roadmap for cleaner viscose by Changing Market Foundation; - Investigate inclusion of other environmental impacts caused by the production of MMCF.

10966 Source: own elaboration

10967 **Table 159.** Suggestions received by respondents to the initial questionnaire on EU Ecolabel criteria 10-12, which focus on
10968 components and accessories

Criterion	Suggestion
Criterion 10. Filling	Include specific sub-criteria addressing: - animal welfare (e.g. prohibition of live plucking), - wastewater production, - hygiene requirements when downs and feathers are used.
Criterion 11. Coating, laminates and membrane	- Include limits about adhesives, organic solvents, and minimum recycled content; - Forbid the use of CFCs as foaming agents; - Include assessment of polyamide in criterion 11(b).
Criterion 12. Accessories	Extend the scope of accessories as done in the criteria of Blue Angel.

10969 Source: own elaboration

10970 **Table 160.** Suggestions received by respondents to the initial questionnaire on EU ecolabel criteria 13-16 and
10971 appendixes, which focus on chemicals

Topic	Suggestion
General points	- Cover the entire supply chain; - Simplify requirements revising their proportionality; - Explicitly report the chemicals that should be analysed; - Improve the comprehension of the criteria.
Criterion 13. Restricted Substances List (RSL)	- Provide directions about the random sampling; - Investigate the use of ZDHC Gateway; - Assess aspects related to Hercosett method on wool; - Transparently report intentionally added chemicals; - Forbid any ingoing SVHC regardless of their concentration; - Use other certification schemes, such as Oeko-Text Standard 100 and Bluesign, as criterion verification.
Criterion 14. Substitution of hazardous substances in dyeing, printing and finishing	- Include restriction to substances that hinder recyclability, and reassess those currently derogated; - Ban substances classified as H400; - Consider the expansion of the following hazard classes: Persistent, bioaccumulative, Toxic (PBTs), very Persistent very Bioaccumulative (vPvBs); Persistent, Mobile and Toxic (PMT), Very Persistent very Mobile (vPvM), endocrine disruption; - Investigate the possibility to completely exclude flame retardants, biocides and optical brighteners.
Criterion 15. Washing, drying and curing energy efficiency	Update the criterion considering the latest evolution of best available technologies.
Criterion 16. Treatment of emissions to air and water	- Update the criterion considering the requirements set by the latest EU-BREF; - Include emission parameters set by the latest version of the Blue Angel criteria; - Investigate verification methods which would facilitate the process when a facility is located outside EU.

Topic	Suggestion
Appendix 1. EU Ecolabel textile restricted substance list	- Analyse the possible ways to verify chemical requirements when the supply chain is outside EU. - Assess the possibility to include in the RSL: - All per and polyfluoroalkyl substances (some exemptions remain for outdoor use) - Flame retardants - SVHC regardless of concentration - Biocides (currently allowed for transportation and storage) - Nanomaterials - Phthalates - Elemental chlorine or hypochlorite - Metal complex dyes - APEOS also in end products - PFAS - Chlorinated solvents - Organotin compounds - Azo dyes - In (f), (ii) the term surfactant is redundant and the 95% is hard to interpret and calculate. - In verification column of (g), (ii) it can be more clear that non-use means no use of easy-care finish. - The term SDS could always be written as "SDS (updated REACH/GHS)" to prevent many outdated incomplete SDS.
Appendix 2. Dye restrictions	- Investigate latest updates adopted by other ISO Type I environmental labels and Oeko-Tex; - Consider inclusion of ZDHC MRSL v.3.0 or higher; - Analyse the possible ways to verify dye requirements when the supply chain is outside EU.
Appendix 3. Best available techniques in the field of washing, drying and curing energy efficiency	- Update the criterion considering the latest evolution of best available technologies; - For washing, assess formulations including surfactants (anionic and nonionic) together with polymers and enzymes with good cleaning performance at low temperatures.

Source: own elaboration

Table 161. Suggestions received by respondents to the initial questionnaire on EU Ecolabel criteria 17-25, which focus on fitness for use

Criterion/topic	Suggestion
Criterion 17. Dimensional changes during washing and drying	- Investigate latest updates adopted by other ISO Type I environmental labels; - Include examples of fabrics with interlock and chunky knit; - Include specific fibre groups that are missing; - Reassess the ambition level of the criterion.
Criterion 18. Colour fastness to washing	Investigate to include restriction to dry cleaning.
Criterion 19. Colour fastness to perspiration (acid, alkaline)	Verify if the depth method is still up to date; Investigate latest updates adopted by other ISO Type I environmental labels.
Criterion 20. Colour fastness to wet rubbing	Investigate latest updates adopted by other ISO Type I environmental labels.
Criterion 21. Colour fastness to dry rubbing	Investigate latest updates adopted by other ISO Type I environmental labels.
Criterion 22. Colour fastness to light	- Verify if the standard depth method is still up to date; - Investigate latest updates adopted by other ISO Type I environmental labels.
Criterion 23. Wash resistance of cleaning products	Investigate latest updates adopted by other ISO Type I environmental labels.
Criterion 24. Fabric resistance to pilling and abrasion	- Extend the possibility to verify the requirement using ISO 12945-3 (RTPT - random tumble); - Include more fibre groups. - Investigate latest updates adopted by other ISO Type I environmental labels.
Criterion 25. Durability of function	- Consider the possibility to include further durability aspects, as reported by latest publications; - Investigate aspects related to oil and stain repellence, which are connected to the use of PFAS; - Assess the exclusion of flame retardants.
Other	Include requirements on tear strength and zipper quality.

Source: own elaboration

10976
10977

Table 162. Suggestions received by respondents to the initial questionnaire on EU ecolabel criteria 26-28, which focus on Corporate Social Responsibility and supporting information

Criterion	Suggestion
Criterion 26. Fundamental principles and rights at work	- Investigate the possibility to use the OECD Guidelines for Multinational Enterprises and the application to the entire value chain; - Assess the latest legislative developments reported in the proposal for the Corporate Sustainability Due Diligence Directive; - Change the headline to better reflects social aspects; - Include the ILO convention 190 and ILO Convention 185, or even going beyond the ILO core criteria; - Investigate the possibility to use other certifications, such as BSCI/Amfori, SMETA, to verify compliancy; - Align with the latest EU Ecolabel requirements for electronic displays and absorbent hygiene products.
Criterion 27. Restriction on the sandblasting of denim	- Investigate the possibility to use the OECD Guidelines for Multinational Enterprises and the application to the entire value chain; - Restrict potassium permanganate.
Criterion 28. Information appearing on the Ecolabel	- Investigate the possibility to include: - a sentence about social aspects; - a specific recycled material content, with its traceability; - Information that sums up the requirements that the product meets; - the specific chain of custody model used for the traceability of virgin and recycled materials; - Consider the latest proposal for Green Claims Directive.

Source: own elaboration

10978
10979

10980 **13.8.3 Comparison among Ecolabels**

10981 **Table 163.** Textile fibres – comparison among Ecolabels

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
Organic cotton	Requirement	Minimum quantity of organic cotton used to manufacture the final product to be labelled: 95%	Minimum quantity of organic cotton used to manufacture the final product to be labelled: 100%	Must be sourced from controlled organic cultivation or fibres from the conversion phase and comply with requirements of regulations and certificates.
		N.A		At all stages of the processing chain, it must be ensured that controlled biological fibres and products are not mixed with conventional fibres and products and controlled biological fibres are not contaminated due to contact with prohibited substances.
	Verification	Common regulation to certify organic cotton: American National Organic Programme (NOP)		
		Council Regulation (EC) No 834/2007 ⁽³³³⁾	Regulation (EU) 2018/848	
		Traceability of organic cotton	N.A	Implied in the requirement text
		Certification annually for each country or origin: Regulation (EC) No 834/2007; The US National Organic Programme (NOP), equivalent legal obligations set by the EU's trade partners. Cotton volume or blend requirements require records, invoices, and documentation from the spinning or fabric production stages.	Fulfilling requirements, emphasising the cultivation of organic cotton in line with the standards. Transaction certificates for suppliers with Global Organic Textile Standard certification are needed.	Compliance verification of requirements. Fibres labelled with the German organic logo (Bio-Siegel), the EU organic logo ("Euro leaf"), or the American National Organic Program (NOP), IFOAM or DIN EN ISO/IEC 17065. For products in the conversion phase, certification of Fibre production is required, and in the case of RAL GmbH, a shipping or transaction certificate is necessary if relevant.

³³⁰ EU Ecolabel criteria for textile products. Commission Decision of 5 June 2014 establishing the ecological criteria for the award of the EU Ecolabel for textile products. Commission Decision (2014/350/EU). Available at [this link](#).

³³¹ Nordic Swan Ecolabel criteria for textiles, hides/skins, and leather. Version 5.6, 01 March 2022 – 31 December 2026. Available at [this link](#). Last accessed on 27 November 2024.

³³² Blue Angel – The German Ecolabel: DE-UZ 154 Basic Award Criteria for textiles. Edition January 2023, version 2. Available at [this link](#). Last accessed on 12 January 2024

³³³ Repealed by Regulation (EU) 2018/848 of the European Parliament and of the Council of 30 May 2018 on organic production and labeling of organic products. Available at [this link](#). Last accessed on 24 April 2025.

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
Other cotton production	Requirement	Minimum content of 20% of cotton grown according to IPM principles, and 60% is the minimum content of cotton grown according to IPM principles for the following products: T-shirts, woman's tops, casual shirts, jeans, pyjamas, and nightwear, underwear, and socks.	Only applicable to clothing (uniforms and workwear) and bed linen, towels, bathrobes, tablecloths, tea towels, cloths, and napkins for, e.g., hotels, hospitals, and other institutions. Cotton cultivated according to different IPM standards (BCI, CmiA, and FairTrade cotton) and not GMO (genetically modified organisms according to EU Directive 2001/18)	Applicable to: the fibres used in the products must not be sourced from genetically modified organisms (GMO). In the case of fine yarns (fineness range > NM100), where the required fibre lengths cannot currently be sourced from the controlled organic cultivation of cotton
		Traceability requirements.		
		List of forbidden pesticides for conventional cotton production and IPM cotton ⁽³³⁴⁾	N.A	N.A
	Verification	N.A	N.A	At all stages of the processing chain, it must be ensured that controlled biological fibres and products are not mixed with conventional fibres and products and controlled biological fibres are not contaminated due to contact with prohibited substances.
		•Need for proper documentation or certification to demonstrate compliance with standards concerning cotton production. •Certified (Government) Integrated Pest Management (IPM) schemes during the cotton cultivation process. •Emphasis on the absence of genetically modified cotton and the traceability of the supply chain.	•Need for proper documentation or certification to demonstrate compliance with standards concerning cotton production. •Certified Integrated Pest Management (IPM) schemes during the cotton cultivation process: BCI, CmiA, and Fair Trade cotton. •Emphasis on the absence of genetically modified cotton and the traceability of the supply chain.	The German Environment Agency can inspect to decide whether the cotton used in the product has been certified by a different certification system (e.g., Cotton Made in Africa (CmiA) and Fairtrade Cotton) can also be approved.
Recycled	Requirement	Pesticide restrictions. Necessity for testing or declarations of non-use from farmers or producer groups. Comprehensive testing methodologies for pesticides such as US EPA 8081 B, 8151 A, 8141 B, and 8270 D for different types of pesticides.	N.A	N.A
		Same as the general recycling requirement	Pre-consumer or post-consumer raw materials, according to ISP 14021 standard Mechanically and chemically recycled fibres are included	The same requirements as the recycled Fibres

⁽³³⁴⁾ As per Criterion 1 (c) , approved IPM schemes (with the exception of organic cotton and cotton from IMP schemes excepted in 1 (b) , must prohibit the pesticides listed under Criterion 1 of the EU Ecolabel criteria for textile products.

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
cotton (fibres or materials)	Verification	Same as the general recycling requirement	Global Recycled Standard certificate 4.0 (or later versions), Recycled Claim Standards (RCS) or other certification approved by Nordic Swan Ecolabelling, AND/OR Documentation indicating the fibre was purchased 100% recycled and indicates the supplier	The same requirements as the recycled Fibres
Flax, linen and other bast fibres	Requirement	N.A	N.A.	Must be sourced from controlled organic cultivation or fibres from the conversion phase and comply with requirements of regulations and certificates. At all stages of the processing chain, it must be ensured that controlled biological fibres and products are not mixed with conventional fibres and products and controlled biological fibres are not contaminated due to contact with prohibited substances.
		Retted under ambient conditions and without thermal energy inputs.	Retting only allowed if the wastewater from the retting ponds is treated to reduce the COD or TOC	If the Fibres are produced using water retting, the water has to be treated to reduce the chemical oxygen demand (COD) or the total organic carbon
		Percentage by which COD or TOC in wastewater from retting should be reduced (at least 75% for hemp and at least 95% for flax and other bast Fibres)		
		N.A	Only cultivated using pesticides permitted according to Regulation (EC) No 1107/2009	N.A
		N.A	N.A	The fibres used in the products must not be sourced from genetically modified organisms (GMOs).
	Verification	For water retting: Test report showing compliance and using the test method: ISO 6060 (COD) Declaration of the retting method used by the farmers and/or other suppliers.	For water retting: Test method for measuring COD and TOC: ISO 6060. Test report from the producer or Proof of a valid EU Ecolabel licence in line with the Commission Decision of July 2014.	For water retting: Test of the COD will be carried out in accordance with ISO 6060 or DIN 38409-41 or DIN 38409-44 or DIN-ISO 15705.
		N.A	For pesticides: Valid certificate from European Flax Standard or equivalent.	Compliance verification Compliance with requirements: Fibres labelled with German organic logo (Bio-Siegel) or Euorganic logo ("Euro leaf") or American National Organic Programme (NOP),-IFOAM or DIN EN ISO/IEC 17065,- For products "in conversion" only if fibre production can be certificated. In the case of RAL GmbH --> a shipping or transaction certificate (if relevant), Regulation (EC) No 834/2007 2018/848 (EC Organic Regulation)

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
				Declaration of compliance with requirements and verification submission for using residues from the agricultural, timber and food industries.
Wool and other keratin fibres	Requirement	The same restrictions on a group of ectoparasiticides with a maximum of 0.5 ppm or a maximum of 2 ppm. The requirement does not apply if specific documentary evidence is presented, together with an independent verification of specific details.		N.A
		Derogation for Wool scourers operating closed-loop water systems that break down the ectoparasiticides above must comply with at least two treatments indicated in EU Ecolabel.	N.A	N.A
		COD values restrictions for wool scouring operations. Limits (same as the Nordic Swan) • 45 g/kg for fine wool • 25 g/kg for coarse wool	COD emissions from wool scouring plants. Limits (same as the EU Ecolabel) • 45 g/kg for fine wool • 25 g/kg for coarse wool Wastewater sent to local/regional treatment works is exempted. Measurement of PCOD, TOC or BOD may also be used if a correlation to COD is evident.	Requirements for wastewater from wool scouring before mixing (indirect discharge) The COD limit is 45 g/kg of greasy wool before mixing with other wastewater (it does not apply to recycled fibres) Requirements for wastewater from wool scouring at the discharge point (direct discharge) The COD limit is 150 mg/l (qualified random sample) or 1.5 mg/l (2-hour mixed sample) of greasy wool.
		N.A	pH value and temperature of wastewater from wool scouring The pH value of the wastewater released to the surface water must be 6-9 (same as Blue Angel), and the temperature must be lower than 40°C (higher temperature than Blue Angel's requirement)	pH value of the wastewater discharged to surface waters must be between 6 and 9 (same as the Nordic Swan) (unless the pH value of the receiving waters is outside this range), and the temperature must be below 35°C (lower temperature than Nordic Swan's requirement) (unless the temperature of the receiving waters is already above this limit). The requirement does not apply to recycled fibres.

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
		Scouring: Measures to recover value from either oxidised grease, fibre, suint or sludge arising from the scouring site used for the ecolabelled wool products •Recovery for sale as a chemical feedstock; •Production of compost or liquid fertiliser; •Manufacturing of products such as building materials; •Treatment and energy recovery by anaerobic digestion or incineration	Scouring agents Must be either readily aerobically biodegradable or inherently aerobically biodegradable by test method: OECD 301 A-F (60% degradability), OECD 310 (60% degradability), OECD 302 A-C (70% degradability) or equivalent.	Scouring: Exclusion of washing agents containing alkylphenol ethoxylates Washing agents containing alkylphenol ethoxylates (APEO) are prohibited. Direct and indirect discharge limits may not exceed 5µg/l APEO (NPEO, OPEO, NP, and OP).
	Verification	Ectoparasiticides: Provide the documentation or the following test methods: IWTO draft test method 59 (same as the Nordic Swan). In case of derogation, applicants must provide evidence confirming the scouring plant configuration and laboratory test reports.	Ectoparasiticides: Tests must be performed in accordance with IWTO Draft Test Method 59 (same as EU Ecolabel). Verify through the wool supplier's declaration that no mulesing has been used. Test report showing that the pesticide requirement has been fulfilled.	N.A
		COD same test for all: Test method ISO 6060 and the following compliance/verification methods		
		COD: Compliance with this criterion shall be based on monthly averages for the six months preceding the application.	COD: Test report or a valid GOTS or EU Ecolabel certificate	COD: Declaration of compliance with the requirements and confirmation from the operator of the wool scouring plant+ method treatment Test of the COD: DIN 38409-41 or, DIN 38409-44 or DIN-ISO 15705. If discharged to an urban wastewater treatment plant, the applicant shall also enclose a notice of approval verifying compliance with Directive (EEC) 91/271.
		Scouring: Provide a report and waste transfer notes confirming the type and proportion of waste recovered and the method used.	Scouring: Declaration from the chemical supplier and safety data sheet for the scouring agents used and/or OECD or ISO test results showing compliance with the requirement.	Scouring: declaration of conformity - Annex 57 of the German Wastewater Ordinance or equivalent international test reports. The following test methods can be used here (based on a qualified random sample or a 2-hour mixed sample): NPEO, OPEO, NP, and OP: ISO 18857-1, ISO 18857-2, ISO 18254-1 or ASTM D7742-17.

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
Organic wool	Requirement	N.A	Certified organic wool: wool fibre certified as organic according to a standard approved in the IFOAM.	Wool, alpaca and cashmere must be sourced from controlled organic cultivation, controlled biological animal husbandry, or fibres from the conversion phase (transition from non-organic to organic farming) and comply with the requirements indicated in "verification". At all stages of the processing chain, it must be ensured that controlled biological fibres and products are not mixed with conventional fibres and products and controlled biological fibres are not contaminated due to contact with prohibited substances.
	Verification	N.A	Certification required: standards approved in the IFOAM, such as Regulation (EU) 2018/848, ISO 17065), USDA National Organic Program (NOP) (same as Blue Angel) APEDA's National Programme for Organic Production (NPOP), China Organic Standard GB/T19630, GOTS certification. Also approved are GOTS and DEMETER and certification as "transitioning to organic cultivation". The certification body must have the accreditation required for the standard, such as IFOAM (same as Blue Angel).	The applicant shall declare compliance with the requirement. Fibres labels accepted: German organic logo (Bio-Siegel), the EU organic logo ("Euro leaf") In the case of alpaca, certification according to the Responsible Alpaca Standard 1.0 © 2021 Textile Exchange. The certification of products "in conversion" is possible. In the case of cashmere, certification according to the Good Cashmere Standard® (GCS) by AbTF can be submitted as an alternative to certification of controlled biological animal husbandry.
Recycled wool	Requirement	Same as the general recycling requirement.	Recycled wool: As per the definition in the ISO 14021 standard. Both mechanically and chemically recycled fibres are included.	Same as the requirements for the recycled fibres.
	Verification	Same as the general recycling requirement.	Fulfilment with either a) or b): a) Global Recycled Standard certificate 4.0, Recycled Claim Standard certificate, or equivalent certification. b) Present documentation demonstrating that the recycled fibre was purchased as recycled and state the supplier.	Same as the requirements for the recycled fibres.

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
Acrylic	Requirement	The emissions to air of acrylonitrile (during polymerisation and up to the solution ready for spinning) shall be less than 1g/kg of fibre produced (same limit as Blue Angel) as an annual average.	<i>If bio-based origin:</i> Must contain at least 90% bio-based raw material. Palm oil, soybean oil, and soy flour must not be used. They must meet either a) or b): a) Waste or residual products defined by (EU) Renewable Energy Directive 2018/2001. b) If sugar cane is used, it must be certified to Bonsucro standard, version 5.1 or later version or certified to a standard that meets the requirements. The producer must have a CoC certification according to the standard by which the raw material is certified. Traceability must at least be ensured by mass balance. Book and claim systems are not accepted. The producer of the bio-based polymer must document its purchase of certified raw materials for polymer production. The raw materials must meet either c) or d): c) Waste or residual products defined by (EU) Renewable Energy Directive 2018/2001. d) Primary raw materials not genetically modified. <i>If fossil origin:</i> 100% recycled material. This must not include recycled plastic from EFSA plants or FDA-approved food contact material or be marketed as compatible with these.	The emissions to air of acrylonitrile (during polymerisation and up to the solution ready for spinning), expressed as an annual average, must be less than 1 g/kg of fibre produced (same limit as EUEL) (this requirement does not apply to recycled fibres unless the recycling process breaks down the materials to the monomer level. If this is the case, this requirement also applies to recycled fibres) The residual acrylonitrile content in raw fibres leaving the fibre production plant must be less than 1.5 mg/kg (this requirement does not apply to recycled fibres unless the recycling process breaks down the materials to the monomer level. If this is the case, this requirement also applies to recycled fibres)
		The workplace emissions to air of N, N-dimethylacetamide (127-19-5) during polymerisation and spinning shall not exceed an Indicative Occupational Exposure Limit Value of 10.0 ppm.		N.A

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
	Verification	Acrylonitrile: Detailed documentation and/or test reports showing compliance with this criterion and a Declaration of Compliance from the fibre manufacturer.	<i>If bio-based origin:</i> -Test according to ISO 16620, ASTM D6866, or equivalent standards showing the content of bio-based raw material. - Declaration by the producer that palm oil, soybean oil and soy flour are not used. - For waste and residual products: Documentation that shows the requirement's definition of waste or residual products and traceability is met. - Sugar cane: Certification system sugar cane is certified for. CoC certificate or a certificate number. Documentation from the producer of the bio-based polymer showing the purchase of bio-based polymer from certified raw material in at least the same annual quantity as is used in the production of the bio-based polymer. Declaration stating that it has not been genetically modified. - For primary raw materials: Declaration stating that they have not been genetically modified. Name and geographical origin of the primary raw materials used. <i>If fossil origin:</i> - Declaration from the producer of the recycled raw material that it is not EFSA or FDA-approved. - Certificate from an independent certifier of the supply chain. - Documentation from the producer showing that the raw material is 100% recycled.	Acrylonitrile Compliance with the requirement and submission of confirmation from the suppliers of the fibres, as well as a test (according to specified methods) report from the suppliers of the fibres verifying compliance with this requirement.
		N, N-dimethylacetamide: Emissions values measured at those process stages in which the substances are used, expressed as an 8-hour average value. Test reports and monitoring data from the fibre manufacturer showing compliance with this criterion.		N.A

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
Elastane	Requirement	Bans organotin compounds. Limits exposure to 3 specific aromatic diisocyanates: (i) diphenylmethane-4,4'-diisocyanate (101-68-8) 0.005 ppm (ii) toluene-2,4-diisocyanate (584-84-9) 0.005 ppm (iii) N, N-dimethylacetamide (127-19-5) 10.0 ppm	<i>If bio-based origin:</i> Must contain at least 90% bio-based raw material. Palm oil, soybean oil, and soy flour must not be used. They must meet either a) or b): a) Waste or residual products defined by (EU) Renewable Energy Directive 2018/2001. b) If sugar cane is used, it must be certified to Bonsucro standard, version 5.1 or later version or certified to a standard that meets the requirements. The producer must have a CoC certification according to the standard by which the raw material is certified. Traceability must at least be ensured by mass balance. Book and claim systems are not accepted. The producer of the bio-based polymer must document its purchase of certified raw materials for polymer production. The raw materials must meet either c) or d) c) Waste or residual products defined by (EU) Renewable Energy Directive 2018/2001. d) Primary raw materials not genetically modified. <i>If fossil origin:</i> 100% recycled material. This must not include recycled plastic from EFSA plants or FDA-approved food contact material or be marketed as compatible with these. Synthetic fibre of fossil origin 100% recycled material in line with ISO 14021, EXCEPTION: • For elastane fibres that are STANDARD 100 by OEKO-TEX certified	Bans organotin compounds. Limits exposure to aromatic diisocyanates a 0.05 mg/m³ value expressed as an 8-hour average. It applies to recycled fibres, only if the recycling process breaks down the materials to the monomer level.

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
	Verification	Declaration of non-use from the Fibre manufacturer and measurement of emissions values, provided as an 8-hour average value. Test reports and monitoring data from the fibre manufacturer(s) demonstrating compliance are necessary.	<i>If bio-based origin:</i> -Test according to ISO 16620, ASTM D6866 or equivalent standards showing the content of bio-based raw material. - Declaration by the producer that palm oil, soybean oil and soy flour are not used. - For waste and residual products: Documentation which shows that the requirement's definition of waste or residual products is met, as well as traceability. - Sugar cane: Certification system sugar cane is certified for. CoC certificate or a certificate number. Documentation from the producer of the bio-based polymer showing the purchase of bio-based polymer from certified raw material in at least the same annual quantity as is used in the production of the bio-based polymer. Declaration stating that it has not been genetically modified. - For primary raw materials: Declaration stating that they have not been genetically modified. Name and geographical origin of the primary raw materials used. <i>If fossil origin:</i> - Declaration from the producer of the recycled raw material that it is not EFSA or FDA-approved. - Certificate from an independent certifier of the supply chain. - Documentation from the producer showing that the raw material is 100% recycled.	Declaration of the non-use of specific compounds and submit confirmation from the fibre suppliers, with the option to use suitable HPLC test methods recognised testing laboratories.

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
Polyester	Requirement	a) The level of antimony present in the polyester fibres shall not exceed 260 ppm (same limit as Blue Angel). Polyester fibres manufactured from recycled PET bottles are derogated (same requirement as Blue Angel). b) Staple fibres shall contain a minimum content of 50 % and filament fibres 20 % of recycled PET (same requirement as Blue Angel). Micro-fibres are derogated from this requirement and shall comply with (c). c) Emissions of VOCs during the production of polyester limit 1.2 g/kg for PET chips (same limit as Blue Angel) and 10.3 g/kg for filament fibre (same limit as Blue Angel).	<i>If bio-based origin:</i> Must contain at least 90% bio-based raw material. Palm oil, soybean oil, and soy flour must not be used. They must meet either a) or b): a) Waste or residual products defined by (EU) Renewable Energy Directive 2018/2001. b) If sugar cane is used, it must be certified to Bonsucro standard, version 5.1 or later version, or certified to a standard that meets the requirements. The producer must have a CoC certification according to the standard by which the raw material is certified. Traceability must at least be ensured by mass balance. Book and claim systems are not accepted. The producer of the bio-based polymer must document its purchase of certified raw materials for polymer production. The raw materials must meet either c) or d): c) Waste or residual products defined by (EU) Renewable Energy Directive 2018/2001. d) Primary raw materials not genetically modified. <i>If fossil origin:</i> 100% recycled material. This must not include recycled plastic from EFSA plants or FDA-approved food contact material or be marketed as compatible with these. Synthetic fibre of fossil origin 100% recycled material in line with ISO 14021, EXCEPTION: • For white polyester (200-220 GG, Tint 0-3 or equivalent according to the CIE Whiteness Index) for professional textiles, an exception is given until June 30th, 2024. When using the exception, it is required that the fibres be STANDARD 100 by OEKO-TEX certified.	a) Limit the amount of antimony present in the polyester fibres 260 ppm (same limit as EUEL) or an elutable amount of 30 mg/kg. This requirement also applies to recycled fibres. b) The use of PET beverage packaging in the production of recycled fibres is permitted for a transitional period of two years (after this period, PET from beverage packaging is not permitted). Staple fibres shall contain a minimum content of 50 % and filament fibres 20 % of recycled PET (same requirement as EUEL) c) VOC emissions must not exceed 1.2 g/kg for PET chips (same limit as EUEL), 10.3 g/kg for filament fibres (same limit as EUEL) or 0.2 g/kg for produced polyester resin. This requirement does not apply to recycled polyester fibres unless the recycling process breaks down the materials to the monomer level. If this is the case, this requirement also applies to recycled polyester fibres.

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
	Verification	a) Declaration of non-use or a test report using the test methods indicated in the criterion. A declaration shall be provided for fibres manufactured from recycled PET bottles. b) Recycled content shall be traceable back to the reprocessing of the feedstock. c) The applicant shall provide monitoring data and/or test reports demonstrating compliance with EN 12619 (same standard as Blue Angel) or standards with an equivalent test method. Monthly averages for the total emissions of organic compounds from production sites for ecolabelled products shall be provided for a minimum of six months preceding the application.	<i>If bio-based origin:</i> -Test according to ISO 16620, ASTM D6866, or equivalent standards showing the content of bio-based raw material. - Declaration by the producer that palm oil, soybean oil and soy flour are not used. - For waste and residual products: Documentation that shows the requirement's definition of waste or residual products and traceability is met. - Sugar cane: Certification system sugar cane is certified for. CoC certificate or a certificate number. Documentation from the producer of the bio-based polymer showing the purchase of bio-based polymer from certified raw material in at least the same annual quantity as is used in the production of the bio-based polymer. Declaration stating that it has not been genetically modified. - For primary raw materials: Declaration stating that they have not been genetically modified. Name and geographical origin of the primary raw materials used. <i>If fossil origin:</i> - Declaration from the producer of the recycled raw material that it is not EFSA or FDA-approved. - Certificate from an independent certifier of the supply chain. - Documentation from the producer showing that the raw material is 100% recycled.	a) Declaration of antimony-free polyester fibres and the corresponding declaration or a test report from the fibre suppliers to verify compliance. Leaching according to DIN EN ISO 105-E04 / determined according to ISO 17294-2 (ICP/MS). The recycling process must be described if recycled fibres are used a) and c) a Declaration of compliance with the requirement and submission of a declaration of conformity from the fibre suppliers, as well as a test report by DIN EN 12619 (same standard as EU Ecolabel). The recycling process must be described if recycled fibres are used.

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
Polyp ropyl ene	Requirement	Bans the use of lead based pigments.	<i>If bio-based origin:</i> Must contain at least 90% bio-based raw material. Palm oil, soybean oil, and soy flour must not be used. They must meet either a) or b): a) Waste or residual products defined by (EU) Renewable Energy Directive 2018/2001. b) If sugar cane is used, it must be certified to Bonsucro standard, version 5.1 or later version or certified to a standard that meets the requirements. The producer must have a CoC certification according to the standard by which the raw material is certified. Traceability must at least be ensured by mass balance. Book and claim systems are not accepted. The producer of the bio-based polymer must document its purchase of certified raw materials for polymer production. The raw materials must meet either c) or d): c) Waste or residual products defined by (EU) Renewable Energy Directive 2018/2001. d) Primary raw materials not genetically modified. <i>If fossil origin:</i> 100% recycled material. This must not include recycled plastic from EFSA plants, FDA-approved as food contact material, or marketed as compatible with these. Synthetic fibre of fossil origin 100% recycled material in line with ISO 14021,	Bans the use of lead based pigments.

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
	Verification	Declaration of non-use.	<i>If bio-based origin:</i> -Test according to ISO 16620, ASTM D6866, or equivalent standards showing the content of bio-based raw material. - Declaration by the producer that palm oil, soybean oil and soy flour are not used. - For waste and residual products: Documentation that shows the requirement's definition of waste or residual products and traceability is met. - Sugar cane: Certification system sugar cane is certified for. CoC certificate or a certificate number. Documentation from the producer of the bio-based polymer showing the purchase of bio-based polymer from certified raw material in at least the same annual quantity as is used in the production of the bio-based polymer. Declaration stating that it has not been genetically modified. - For primary raw materials: Declaration stating that they have not been genetically modified. Name and geographical origin of the primary raw materials used. <i>If fossil origin:</i> - Declaration from the producer of the recycled raw material that it is not EFSA or FDA-approved. - Certificate from an independent certifier of the supply chain. - Documentation from the producer showing that the raw material is 100% recycled.	Declaration of non-use and confirmation from the fibre suppliers. The recycling process must be described if recycled fibres are used.
Man-made cellulose fibres	Requirement	Sustainable Forestry Management: A minimum of 25% of pulp Fibres must be manufactured from wood grown according to sustainable forestry management principles. The remaining proportion of pulp Fibres should be from legal forestry and plantations. Pulp from Cotton Linters: Pulp produced from cotton linters must meet the requirements of either cotton criterion 1a or 1b. Bleaching Process: Requirement (9c): The pulp used for fibre manufacturing must be bleached without elemental chlorine. Total chlorine and organically bound chlorine (OX) in finished Fibres must not exceed 150	<i>If bio-based origin:</i> Must contain at least 90% bio-based raw material. Palm oil, soybean oil, and soy flour must not be used. They must meet either a) or b): a) Waste or residual products defined by (EU) Renewable Energy Directive 2018/2001. b) If sugar cane is used, it must be certified to Bonsucro standard, version 5.1 or later version or certified to a standard that meets the requirements. The producer must have a CoC certification according to the standard by which the raw material is certified. Traceability must at least be ensured by mass balance. Book and claim systems are not accepted. The producer of the bio-based polymer must document its purchase of certified raw materials for polymer production.	N.A

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
		ppm. AOX in wastewater from pulp manufacturing must not exceed 0.170 kg/ADt pulp.	The raw materials must meet either c) or d): c) Waste or residual products defined by (EU) Renewable Energy Directive 2018/2001. d) Primary raw materials not genetically modified.	
		Value Recovery from Spent Process Liquors: Requirement (9d): A minimum of 50% of the pulp used for Fibre manufacturing must be purchased from dissolving pulp mills that recover value from spent process liquors through on-site electricity and steam generation or manufacturing chemical co-products.		
		Sulphur Content in Emissions (for Viscose and Modal Fibres): Requirement (9e): The sulphur content of emissions of sulphur compounds to air from Fibre production processes must not exceed specified performance values.		
	Verification	Sustainable Forestry Management: The applicant needs to obtain valid, independently certified chain of custody certificates from the Fibre manufacturer(s) demonstrating compliance with sustainable forestry management principles. Certifications from FSC, PEFC, or equivalent schemes are accepted. The fibre manufacturer shall demonstrate that due diligence processes have been followed as specified in Regulation (EU) No 995/2010 of the European Parliament and of the Council ⁽³³⁵⁾ in order to ensure that timber has been legally harvested.	If bio-based origin: - Test according to ISO 16620, ASTM D6866, or equivalent standards showing the content of bio-based raw material. - Declaration by the producer that palm oil, soybean oil and soy flour are not used. - For waste and residual products: Documentation that shows the requirement's definition of waste or residual products and traceability is met. - Sugar cane: Certification system sugar cane is certified for. CoC certificate or a certificate number. Documentation from the producer of the bio-based polymer showing the purchase of bio-based polymer from certified raw material in at least the same	N.A

³³⁵ Regulation (EU) No 995/2010 of the European Parliament and of the Council of 20 October 2010 laying down the obligations of operators who place timber and timber products on the market Text with EEA relevance OJ L 295, 12.11.2010, p. 23–34. Available at [this link](#). Last accessed on 24 April 2025.

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
		Pulp from Cotton Linters: Cotton must be certified by an independent control body, complying with regulations and standards. Non-genetically modified varieties of cotton should be verified according to specified regulations. Bleaching Process: The applicant needs to provide a test report showing compliance with OX or AOX requirements using the appropriate test methods (OX: ISO 11480, AOX: ISO 9562). Value Recovery from Spent Process Liquors: The applicant must provide a list of pulp suppliers, documentation, and evidence of energy generation and co-product recovery systems. Sulphur Content in Emissions (for Viscose and Modal Fibres): The applicant needs to provide detailed documentation and/or test reports showing compliance with this criterion, along with a declaration of compliance	annual quantity as is used in the production of the bio-based polymer. Declaration stating that it has not been genetically modified. - For primary raw materials: Declaration stating that they have not been genetically modified. Name and geographical origin of the primary raw materials used.	
Regenerated Cellulose fibres	Requirement	N.A.	Recycled Content: Recycled raw materials for producing new regenerated cellulose fibres must be pre-consumer or post-consumer* cellulosic material. It must be documented that 100% is recycled material.	Recycled Content: Allows the use of recycled materials.
			Chain of Custody Certification: Chain of Custody certification (FSC or PEFC). Manufacturers using only recycled material are exempt from the Chain of Custody requirement	Chain of Custody Certification: Requires certain certifications for cellulose origin.
			Bleaching Restrictions: Prohibits the use of chlorine gas in bleaching.	Bleaching Restrictions: Restricts the use of elementary chlorine. Additional Bleaching Process Restrictions Place restrictions on using hypochlorite and set limits on chlorine dioxide consumption and AOX emissions to wastewater during bleaching.
			Emission Limits: Imposes emission limits for specific substances in the production process.	Emission Limits: Sets limits on emissions to wastewater and air, including COD, nitrogen, phosphorus, sulphur compounds, and dust.
			Tree Species Limitation: Restrictions on using virgin wood Fibres from specific tree species.	N.A
			Origin: Requires traceability of regenerated cellulose fibre or recycled textile fibre	Cellulose Origin: Requires cellulose from wood or bamboo cultivated according to sustainable forestry principles. It also allows using recycled materials and

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
				residues from the agricultural, timber, and food industries.
			N.A	Emission Limits Specific to Fibre Types: Sets specific emission limits for viscose and modal fibres, including sulphur content in emissions.
			N.A	Energy Consumption Limits: specifies limits on specific electrical and heating energy consumption in cellulose production.
			N.A	Halogen Content: Specifies a maximum halogen content for fibres.
			Traceability: Requires traceability of regenerated cellulose fibre or recycled textile fibre	N.A
	Verification	N.A	Recycled Content: Certificate from either Global Recycled Standard or Recycled Claim Standard	Recycled Content: Certificate from either Global Recycled Standard or Recycled Claim Standard or submission of purchasing and procurement receipts for waste material.
			Chain of Custody Certification: Verification with valid FSC/PEFC Chain of Custody certificates.	Chain of Custody Certification: Verification through valid, independently issued certificates from Fibre producers confirming sustainable forestry management for wood or bamboo Fibres.
			Bleaching Restrictions: Declaration from the cellulose mass and regenerated cellulose manufacturers or a valid EU Ecolabel License.	Bleaching Restrictions: Manufacturers to submit test reports and documentation showing adherence to the specified limits
			Emission Limits: Documentation of closed-loop processes and test reports for emission requirements	Emission Limits: compliance declarations, emission values, and test reports for various aspects of production, including emissions to air and water.
			Cellulose Origin: Declaration from the cellulose mass and regenerated cellulose manufacturers or a valid EU Ecolabel License.	Cellulose Origin: For cellulose sourced from bamboo, a transaction certificate must be submitted
			Tree Species Limitation: Documented with a certificate from either the Global Recycled Standard or the Recycled Claim Standard.	N.A
			N.A	Emission Limits Specific to Fibre Types: Compliance with emission limits for specific fibre types
			N.A	Energy Consumption Limits: Manufacturers need to ensure that the specific electrical and heating energy consumption in cellulose production meets the defined limits

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
			N.A	Halogen Content: Manufacturers must ensure that the halogen content of fibres does not exceed the specified maximum limit.
			Traceability: documented with a certificate from either the Global Recycled Standard or the Recycled Claim Standard.	N.A
Polyamide	Requirement	Polyamide products shall comply with at least one of the following: - Fibres shall be manufactured using a minimum content of 20 % recycled from pre and/or post-consumer waste. - The emissions to air of N2O from nylon monomer production, expressed as an annual average, shall not exceed 9.0 g N2O/kg of caprolactam or adipic acid.	<i>If bio-based origin:</i> Must contain at least 90% bio-based raw material. Palm oil, soybean oil, and soy flour must not be used. They must meet either a) or b): a) Waste or residual products defined by (EU) Renewable Energy Directive 2018/2001. b) If sugar cane is used, it must be certified to Bonsucro standard, version 5.1 or later version or certified to a standard that meets the requirements. The producer must have a CoC certification according to the standard by which the raw material is certified. Traceability must at least be ensured by mass balance. Book and claim systems are not accepted. The producer of the bio-based polymer must document its purchase of certified raw materials for polymer production. The raw materials must meet either c) or d): c) Waste or residual products defined by (EU) Renewable Energy Directive 2018/2001. d) Primary raw materials not genetically modified. <i>If fossil origin:</i> 100% recycled material. This must not include recycled plastic from EFSA plants or FDA-approved food contact material or be marketed as compatible with these.	Polyamide products shall comply with at least one of the following: - Fibres shall be manufactured using a minimum content of 20 % recycled from pre and/or post-consumer waste. - The N2O emissions to air during the monomer production, expressed as an annual average, must not exceed 9 g/kg of caprolactam or adipic acid. In addition, reduction technologies must be used to produce caprolactam and adipic acid. It must be ensured that the degree of reduction in the N2O emissions during adipic acid production is at least 95%.

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
	Verification	Recycled content shall be traceable back to the reprocessing of the feedstock. This shall be verified by independent certification of the chain of custody or by documentation provided by suppliers and processors. The applicant shall provide documentation or test reports showing compliance based on monitoring data and a Declaration of Compliance from the fibre manufacturer and their feedstock providers.	<i>If bio-based origin:</i> -Test according to ISO 16620, ASTM D6866 or equivalent standard showing the content of bio-based raw material. - Declaration by the producer that palm oil, soybean oil and soy flour are not used. - For waste and residual products: Documentation that shows the requirement's definition of waste or residual products and traceability is met. - Sugar cane: Certification system sugar cane is certified for. CoC certificate or a certificate number. Documentation from the producer of the bio-based polymer showing the purchase of bio-based polymer from certified raw material in at least the same annual quantity as is used in the production of the bio-based polymer. Declaration stating that it has not been genetically modified. - For primary raw materials: Declaration stating that they have not been genetically modified. Name and geographical origin of the primary raw materials used. <i>If fossil origin:</i> - Declaration from the producer of the recycled raw material that it is not EFSA or FDA-approved. - Certificate from an independent certifier of the supply chain. - Documentation from the producer showing that the raw material is 100% recycled.	Declare compliance with the requirement and submit a DoC from the monomer producer and test reports for the raw and clean gas, verifying that a reduction of at least 95% has been achieved.
Recycled textile fibre	Requirement	Recycled content shall meet the requirements of the Criterion 13 Restricted Substance List.	An extensive list of substances and limits indicated Extractable metals, Organic tin compounds, Chlorophenols, Per- and polyfluorinated compounds, Phthalates, Surfactants, wetting agent residues, Dyes,	Provision of information on the recycled content, the type and composition of the recycled fibre, the origins and composition of the materials fed into the recycling process, and a description of the recycling process.
	Verification	Recycled content shall be traceable back to the reprocessing of the feedstock. This shall be verified by independent third party certification of the chain of custody or by documentation provided by feedstock suppliers and reprocessors. Annually	a) an Oeko-Tex standard 100 class I certificate or b) test report showing that the requirement is complied with. And a description of the procedure confirming an annual test.	Declaration from the manufacturer about the recycled content and origins, accompanied by one of the following certificates: • RCS (Recycled Claim Standard), • GRS (Global Recycled Standard), • International Sustainability and Carbon Certification (ISCC+), • Roundtable on Sustainable Biomaterials (RSB), • RedCert (only in Europe) • or a comparable certification system whose scope and requirement standards is equivalent to one of the named certification systems. An

Topic	Requirement/ Verification	EU Ecolabel ⁽³³⁰⁾	Nordic Swan ⁽³³¹⁾	Blue Angel ⁽³³²⁾
				independent environmental verifier must confirm the equivalence of the certification system.

10982

10983 **Table 164.** Components and accessories – comparison among Ecolabels

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
Fillings	Requirement	Fibre Restrictions: defined fibre restrictions for fillings in alignment with their respective standards		Fibre restrictions: Not explicitly mentioned
		Chemical Substances: emphasis on the chemical substances used in fillings and their laundering processes. The criteria include biocides, formaldehyde emissions, fabric softeners, and other relevant chemicals.	Chemical Substances: restricts chemical substances used in additives and treatments applied to fillings. It does not mention the laundering process.	Chemical Substances: references substance restrictions from the REACH "list of candidates," the CLP Regulation (EC/1272/2008), and limits from the ZDHC MRSL, provided they are stricter than REACH. It does not mention the laundering process.
		Polyurethane and Latex: not specifically address polyurethane and latex fillings.	Polyurethane and Latex: focuses on restrictions related to blowing agents	Polyurethane and Latex: concentration limits and VOC (Volatile Organic Compounds) emissions for polyurethane and latex. The concentration of aromatic diisocyanates and blowing agents are also restricted in PUR.
		Foamed Synthetic Materials: Formaldehyde emissions restrictions resulting from the laundering process.	Foamed Synthetic Materials: Limits on emissions from substances used in foamed synthetic materials, including formaldehyde and others.	Foamed Synthetic Materials: Latex foam and specifies restrictions regarding its concentration and volatile organic compounds (VOC) emissions.
		Polycyclic Aromatic Hydrocarbons (PAHs): Not referenced	Polycyclic Aromatic Hydrocarbons (PAHs): For foamed synthetic materials such as polyurethane, latex, and expanded polystyrene, the content of each PAH stated in the requirement shall be below 0.5 mg/kg.	Polycyclic Aromatic Hydrocarbons (PAHs): Not mentioned

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
	Verification	Fibres: follow their fibre verification methods		Fibres: Not explicitly mentioned
		Emissions from PUR and latex foams: not mentioned	Emissions from PUR and latex foams: same standard for verification, but different verification methods.	
		Polyurethane: not mentioned	Polyurethane: Declaration of Conformity (DoC) regarding blowing agents	Polyurethane: Declaration of Conformity (DoC) and test reports on limit values for additives and emissions (CFCs and VOCs).
		PAHs: not mentioned	PAHs: tested by ISO 18287 or ZEK 01.2-08 (GC/MS)	PAHs: not mentioned
Feathers and downs	Requirement	N.A	Ethical requirements: prohibit using feathers and down plucked from live birds or forced feeding birds	
			Includes recycled down and feather	N.A
			Microbial cleanliness: must document microbial purity and comply with the oxygen index number of max. 10 and fat content	Microbial cleanliness: requirements for down and feathers regarding oxygen index, microbiological state, mesophil aerobic bacteria count, faecal streptococci count, sulphite reducing clostridium count, presence of salmonella, oil and grease content, turbidity, pH
			Labelling: requires labelling feathers and down-filling materials; Blue Angel Ecolabel does not mention it.	Labelling: Does not mention it.
			Wastewater: The given text does not provide specific wastewater discharge standards.	Wastewater: detailed wastewater discharge standards, including limits for various parameters and pH and temperature requirements.
	Verification	N.A	Adherence to the Down Standard or an equivalent.	Submission of audit reports from testing institutions.
			For recycled materials, either a Recycled Global Standard certificate or supplier documentation is required.	N.A
			N.A	For wastewater: compliance with the requirements. A declaration of conformity from the processing plant operator and test reports.

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
Coating, laminates and membranes	Requirement	Focuses on specific criteria for polyurethane and polyester, including workplace emissions, antimony content, and VOC emissions. Addresses restrictions on phthalates in polymers. Allows the use of fluoropolymer membranes and laminates but with restrictions on PFOA. Does not mention adhesives.	It covers requirements for textiles used as substrates, including materials for coatings, laminates, and membranes. Specifies criteria for recycled and biobased raw materials, including palm oil, soybean oil, and soy flour restrictions. Specifically, if a polymer constitutes more than 5% by weight of the finished fabric, it must be either 100% recycled or composed of at least 90% bio-based raw material. Prohibits halogenated polymers and sets requirements for additives (CLP Regulation 1272/2008) in polymers. Lists various prohibited substances and materials, such as siloxanes, flame retardants, PFCs, and heavy metals. Covers adhesives used in the laminating process.	Focuses on requirements for laminates and membranes used in products. Addresses textile and membrane materials. Requires compliance with recycled materials (at least 30%) or avoiding organic solvents for polyurethane membranes. Allows only specific types of adhesives (thermoplastic or reactive hot melt) and prohibits solvent-based adhesives. Using reactive polyurethane-based hot melt adhesives sets limits for aromatic diisocyanates in workplaces. Includes criteria for functional products, with exclusion criteria for certain substances and materials, such as biocides, flame retardants, PFCs, and VOCs.
	Verification	For polyester components, compliance involves meeting standard fibre criteria. There are specific requirements related to antimony content and VOC emissions.	Verification procedures align with specified fibre criteria. A Declaration of Conformity confirms the absence of blowing agents for polyurethane components.	Verifies textiles against fibre criteria. They have specific requirements related to recycled materials and antimony content. Verification relies on the Declaration of Conformity (DoC) process.
Accessories	Requirement	N.A	Zipper, buttons, and non-textile accessories are limited if they do not have a practical purpose.	N.A
		Concentration of metals: Nickel migration 0.5 µg/cm ² /week Lead 90 mg/kg Cadmium -> Intended for children under 3 y -> 50 mg/kg Others 100 mg/kg Chromium 60 mg/kg Mercury 60 mg/kg	Concentration of metals: Lead (Pb): 90 mg/kg. Cadmium (Cd): 40 mg/kg Nickel (Ni): migration limit must be less than 0.5 micrograms/cm ² /week	Concentration of metals: Lead 90mg/kg; Cadmium: 50 mg/kg; Chromium: 60 mg/kg. Nickel: migration value for metal alloys is 0.5 ug/cm ² /week.
		Plastic accessories cannot have phthalates		
		N.A	Restrictions on chlorinated plastics	N.A

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
	Verification	Nickel test migration and testing for other metals.		
		Phthalate Testing	Declaration from the manufacturer of the plastic material that the plastic meets the requirement.	Phthalate Testing
		N.A	Declaration from the Licensee that no details or accessories are used without a practical function.	N.A

10984

10985 **Table 165.** Fitness for use - comparison among Ecolabels

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
Dimensional changes during washing and drying	Requirement	This does not apply to fibres or yarn labelled "dry clean only." or equivalent, and furniture fabrics that are not removable and washable. Knitted products +-4.0% Chunky knit +-6% Interlock +-5% Woven fabrics: cotton, cotton mix +-3%, wool mix and synthetic fibres +-2% Socks and hosiery, bathroom linen +-8% Washable and removable woven upholstery +-2% Woven Mattress ticking +-3% Non-woven mattress ticking +5% All other non-woven fabric +6%	Not apply to fibres or yarn, products labelled "dry clean only", and upholstery fabrics not removable and washable. Knitted products and hosiery +-5.0% 100% Wool knitwear +-10% Curtains and upholstery cover removable and washable +-2% Woven fabrics for duvets and pillows +-5% Woven textiles of wool blend and synthetic fibres +-2% Woven textiles not covered by the categories above +-3% Bedding, tablecloths and napkins +-5% Terry towels and washcloths +-7%	Not apply to fibres or yarn, products labelled "dry clean only", not removable and washable furniture fabrics. Knitted products +-5.0% Chunky knit +-6% House and home textiles +-8% Woven fabrics: cotton, cotton mix, linen, flax and silk +-3%, cotton and cotton mix for bedding and wool mix +-5%, and synthetic fibres +-2% Socks and hosiery +-5% Bathroom linen, including terry towelling and fine rib fabrics +-8% Regenerated and synthetic woven fabrics +-3% Regenerated and synthetic knitted fabrics +-5%
	Verification	For domestic washing, EN ISO 6330 combined with EN ISO 5077. For commercial washings, ISO 15797 combined with EN ISO 5077.	For domestic washing, EN ISO 6330 combined with EN ISO 5077. For commercial washings, ISO 15797 combined with EN ISO 5077. Woven products for duvets and pillows with feathers and down filling tested with EN 13186	DIN EN ISO 6330 combined with EN ISO 5077.

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
Tear Strength	Requirement	N.A	Trousers, shorts, skirts 1.5daN Jackets and coats 1.2daN Sportswear, ski clothing, and other outdoor wear 1.2daN Lingerie, pyjamas, and other nightwear 0.8daN T-shirts, blouses, shirts, and dresses 0.8daN Swimwear 1.0 daN Bed linen and sheets 0.8daN Towels 0.8daN	N.A
	Verification	N.A	ISO 13937	N.A
Tensile Strength	Requirement	N.A	Trousers, shorts, skirts 18daN Jackets and coats 15daN Sportswear, ski clothing, and other outdoor wear 18daN Lingerie, pyjamas, and other nightwear 12daN T-shirts, blouses, shirts, and dresses 12daN Swimwear 15 daN Bed linen and sheets 12daN Towels 12daN	N.A
	Verification	N.A	ISO 13934-2	N.A
Seam Strength	Requirement	N.A	Woven fabrics The seam in the lining 80N Textile with fabric weight <220g/m2 150N Textile with fabric weight >220g/m2 200N Backpacks and bags 200N	N.A
	Verification	N.A	ISO 13935-2	N.A

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
Seam Slippage resistance	Requirement	N.A	<i>If testing with EN-ISO 13936-1</i> Trousers, shorts, skirts 4mm at 14 daN load Jackets and coats 4mm at 14 daN load Sportswear, ski clothing, and other outdoor wear 4mm at 14 daN load Lingerie, pyjamas, and other nightwear 4mm at 10 daN load T-shirts, blouses, shirts, and dresses 4mm at 11 daN load Swimwear 4mm at 14 daN load Bed linen and sheets 4mm at 10 daN load <i>If testing with EN-ISO 13936-2</i> Trousers, shorts, skirts 12 daN load at 3mm Jackets and coats 12 daN load at 4mm Sportswear, ski clothing, and other outdoor wear 12 daN load at 4mm Lingerie, pyjamas, and other nightwear 6 daN load at 3mm T-shirts, blouses, shirts, and dresses 6 daN load Swimwear 4mm at 14 daN load at 3mm Bed linen and sheets 6 daN load at 3mm	N.A
	Verification	N.A	ISO 13936-1 or ISO 13936-2	N.A
Colour fastness to washing	Requirement	At least 3-4 for colour change and staining It does not apply to products labelled "dry clean only", to white products or products that are neither dyed nor printed or to non-washable furniture fabrics.	Level 3-4 for colour change and discolouration It does not apply to white products, products that are neither dyed nor printed or non-washable furniture fabrics.	At least levels 3-4 according to ISO 105-A03 or A04 and A02 or A05. It does not apply to products labelled "dry clean only," indigo-dyed denim, or end products that are neither dyed, printed nor non-washable furniture fabrics.
	Verification	For domestic washing, ISO 105 C06. For industrial washing, ISO 15797 combined with ISO 105 C06.	Test method for wash: ISO 105 C06. Test method for dry cleaning: ISO 105 D01	ISO 105-C06
Colour fastness to perspiration	Requirement	At least levels 3-4 according to ISO 105. It does not apply to white products, products neither printed or dyed, furniture fabrics, curtains, and similar textiles intended for interior decoration	Underwear, sportswear, and t-shirts must meet: For discolouration: level 4 -For staining: level 4	At least levels 3-4 according to ISO 105-A03 or A04 and A02 or A05. It does not apply to printed or dyed products, curtains, and similar textiles intended for interior decoration except cushions.

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
(acid alkaline)			Level 3 is permitted for textiles that are dark in colour and/or made from recycled wool. It does not apply to white products or end products that are neither dyed nor printed.	
	Verification	ISO 105 E04		
Colour fastness to saliva	Requirement	N.A	Baby clothes must meet the following standards: -For discolouration: level 4 -For staining: level 4 Level 3 is permitted for textiles that are dark in colour and/or made from recycled wool. This requirement does not apply to white products or end products that are neither dyed nor printed.	Colour fast level 5 for baby and children products according to DIN 53160-1. This requirement does not apply to end products that are neither dyed nor printed.
	Verification	N.A	ISO 105-A06	§64 of the LFGB, B 82. 92-3 and BVL B 82.92-13 in combination with DIN 53160-1
Colour fastness to wet rubbing	Requirement	At least level 2-3. Level 2 allowed for dark-dyed denim. Level of 1 for all other denim colour shades It does not apply to white products, products neither printed nor dyed.	At least level 3-4 It does not apply to white products, products neither printed nor dyed Dark coloured denim level 1-2 Medium-coloured denim level 2-3	At least levels 2-3. For dark colours level 2, according to DIN EN ISO 105-A03 or A04 and A02 or A05. It does not apply to dyed or printed.
	Verification	ISO105 X12		
Colour fastness to dry rubbing	Requirement	At least level 4. Levels 3-4 are allowed for dark-dyed denim. Level of 2-3 for all other denim colour shades It does not apply to white products, products neither printed nor dyed, curtains, and similar textiles intended for interior decoration.	At least level 4. It does not apply to white products, products neither printed nor dyed, curtains, and similar textiles intended for home furnishing textiles. Dark-coloured denim level 3	At least levels 3-4 for dark colours and level 3 for denim. It does not apply to dyed or printed interior decoration.
	Verification	ISO105 X12		
Colour fastness to light	Requirement	At least level 5 for furniture, curtains, or drapes. At least level 4 for all other products. It does not apply to mattress ticking, mattress protection, or underwear.	At least level 5 for outdoor wear, swimwear, UV-protecting clothing, furniture, curtains, and drapery. The requirement does not apply to white textiles, mattress covers, and mattress protectors.	At least level 5 for furniture, curtains, or drapes. At least level 4 for all other products, This requirement does not apply to mattress ticking, protection, or underwear.

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
	Verification	ISO105 B02		
Wash resistance and absorbency of cleaning products	Requirement	Woven and non-woven products for wet cleaning 80 washes, 40°C Microfibre products for dusting 200 washes, 40°C Products deriving from recycled textile fibres 20 washes, 30°C Mop for washing floors 200 washes, 60°C Cloths for washing floors 5 washes 30°C Absorbency of cleaning products Products deriving from recycled textile fibres, Microfibre products for surface and floor cleaning, woven and non-woven products for wet cleaning, and products for washing floors <= 10 seconds	N.A	N.A
	Verification	EN ISO 6630	N.A	N.A
Resistance to abrasion	Requirement	N.A	Number of rubs/abrasions (Martindale) 50.000 cycles for commercial upholstery 30.000 cycles for workwear for outdoor use and domestic upholstery 20.000 cycles for workwear for indoor use, trousers, shorts and skirts, sportswear, ski clothing and other outdoor wear, swimwear 16.000 cycles for jackets and coats 12.000 cycles for T-shirts, blouses, shirts and dresses 10.000 cycles for lingerie, pyjamas, and other nightwear, bed linen, and sheets 8.000 cycles for knitting	N.A
	Verification	N.A	EN ISO 12947-2	N.A
Resistance to pilling	Requirement	Minimum of 3 for non-woven fabrics and knitted garments, accessories, and blankets made of wool, wool blends, and polyester woven cotton. Minimum of 2 for polyamide tights and leggings.	Minimum of 4 for clothing woven fabric, fleece, and upholstery for professional use. Minimum of 3-4 for upholstery for private use and upholstery of wool or wool blends for professional use. Minimum of 2-3 for clothing woven fabric with raised surface and knitting.	Non-woven fabrics and knitted garments, accessories, and blankets made of wool, wool mixes, and polyester must resist pilling up to a minimum rating of 3. Woven cotton fabrics used for garments must also resist pilling up to a minimum rating of 3. Polyamide tights and leggings must resist pilling up to a minimum rating of 2.
	Verification	-Knitted and non-woven: ISO 12945-1 Pill box method -Woven: ISO 12945-2 Martindale method	ISO 12945-2	-Knitted and non-woven: ISO 12945-1 Pill box method -Woven: ISO 12945-2 Martindale method

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
Durability of function	Requirement	Oil repellents shall retain the functionality of 3.5 out of 4 Stain repellents shall retain functionality of 3.0 out of 5.0 Washable products shall retain functionality after 50 industrial wash and tumble dry cycles at a minimum of 75°C. Non-washable products shall retain their functionality after a soak test. Natural fibres shall achieve an SA-3 fabric smoothness grade. Blended natural and synthetic fibres with an SA-4 fabric smoothness grade after 10 domestic wash and tumble-drying cycles at 40°C.	N.A	Water repellents shall retain the functionality of 80 out of 90 after 5 domestic wash and tumble dry cycles at 40°C or after 5 industrial washing and drying cycles at a minimum of 75°C. Industrial washing temperatures may be reduced to 60°C for garments with taped seams. Care instructions on the reimpregnation of the product must be supplied with the textile. Washable products shall retain functionality after 25 industrial wash and tumble dry cycles by care instructions. Non-washable products shall retain their functionality after a soak test. Natural fibre products shall achieve an SA-3 fabric smoothness grade. Blended natural and synthetic fibres with an SA-4 fabric smoothness grade after 10 domestic wash and tumble-drying cycles at 40°C.
	Verification	Water repellent function Domestic washing ISO 6330 in combination with ISO 4920 Industrial washing ISO15797 in combination with ISO 4920 Oil repellent function Domestic washing ISO 6330 in combination with ISO 14419 Industrial washing ISO 15797 in combination with ISO 14419 Stain repellent function Domestic washing ISO 6330 in combination with ISO 22958 Industrial washing ISO 15797 in combination with ISO 22958 Flame retardant Domestic washing ISO 6330 in combination with ISO 12138 Industrial washing ISO10528 in combination with ISO 12138 Easy care function ISO 7768	N.A	Water repellent function Domestic washing ISO 6330 in combination with ISO 4920 Industrial washing ISO15797 in combination with ISO 4920 Flame retardant Domestic washing ISO 6330 in combination with ISO 12138 Industrial washing ISO10528 in combination with ISO 12138 Easy care function ISO 7768

10986

10987

Table 166. Chemicals and processes - comparison among Ecolabels

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
Substance restrictions or specific chemical requirements	Requirement restricted substances	The 3 labels refer to restrictions imposed by REACH (1907/2006/EC), the Candidate List of substances of very high concern for Authorisation with reference to ECHA, and CLP Regulation (EC/1272/2008) and Restricted Substances List (RSL by REACH), with the following indications: *EUEL and BA approach per process/service; meanwhile, NS focuses on categorising ingoing substances. *at the general level, EUEL lists products included in product categories. Meanwhile, NS and BA highlight the general substance category, except where lists are also indicated. *most substances with requirements indicated in the 3 ecolabels coincide, including most of the testing methods, with some exceptions, and are presented in different manners that could be different naming and quantification (as units of measurement).		
		Halogenated carriers (referred to in Nordic Ecolabel and Blue Angel as chlorinated agents) limitations on their use for bleaching and dyeing (including chlorinated and brominated derivatives)	Chlorinated solvents as carriers agents (referred to as Halogenated carriers in EUEL)	Chlorinated solvents and carriers agents (Halogenated carriers in EUEL) for bleaching. The only label indicating limits for chlorinated benzenes and toluenes.
		Azo dyes (25 azo dyes listed)	Azo dyes (35 azo dyes listed): additional from EUEL and Blue Angel are the 10 following: 2-amino-4-nitrophenol 99-57-0 m-phenylenediamine 108-45-2 2-amino-5-nitrothiazole 121-66-4 2-amino-5-nitrophenol 121-88-0 p-aminophenol 123-30-80 p-phenetidine 156-43-4 2-methyl-phenylenediamine,2,5diaminotoluene 615-50-9 2-methyl-phenylenediamine,2,5diaminotoluene 95-70-5 2-methyl-phenylenediamine, 2,5 diaminotolunene 25376-45-8 6-chloro-2,4-dinitroaniline 3531-19-9	Azo dyes (25 azo dyes listed)
		CMR dyes and Potentially sensitising dyes	Not specifically mentioned	CMR dyes and potentially sensitising dyes
		Chrome mordant dyes are not permitted.	Mention of chromium in metal dyes	Chrome mordant dyes are not permitted
		Metal complex dyes based on copper, chrome, and nickel shall only be permitted for dyeing: wool, polyamide, and blends of wool and/or polyamide with man-made cellulose fibres.	Only metal complex dyes and pigments based on copper that comprise a maximum of 5% by weight may be used to dye wool, polyamide, a blend of wool, and/or polyamide with regenerated cellulose fibre.	No specific mention (but testing measures for direct and indirect discharge of wastewater)
		Plastisol binders banned	mentioned just in Re-used textiles, hides/skins, leather	restriction less than 1,000mg/Kg
		Biocide finishes banned	Biocides and antibacterial banned	Biocides are banned, but in-con preservatives are permitted
		Anti-felting and shrink resistance restricted	not specifically mentioned	Anti-felting and shrink resistance restricted

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
		Water, stain and oil repellent treatments	not specifically mentioned	Water, stain and oil repellent treatments
		Flame retardants (list of banned substances included)	Flame retardants (general term) banned	Flame retardants (general term) banned
		not specifically mentioned	not specifically mentioned	Cerium compounds restricted
		Biodegradability: 95% (additional indication: eliminable in wastewater treatment plants)	Biodegradability: similar requirements 95% indicated	Biodegradability: 90% (additional indication: eliminable in wastewater treatment plants)
		APEO generally indicated		APEO specified in the treatment of wool and water discharges
		EDTA, DTPA, and more are generally prohibited substances	EDTA and DTPA in generally prohibited substances	EDTA and DTPA are mentioned only in bleaching processes in regenerated fibres
		N,N-Dimethylacetamide % limits of 0.001%w/w for babies and children under 3 products, 0.005% w/w for products in direct contact with skin and garments with limited contact and interior textiles	not specifically mentioned	N, N-dimethylacetamide, dimethylformamide, and N-methyl pyrrolidone are restricted to 0.1% by mass.
		Formaldehyde residues measured in ppm (16 ppm), but the same verification tests		Formaldehyde residues are measured in mg/kg (20 mg/kg), but the same verification tests.
		Organotin compounds and Chlorophenols are generally indicated and listed	Organotin compounds and Chlorophenols generally indicated	Organotin compounds and Chlorophenols (limits indicated) extended list compared to EUEL and NS
		Prohibited phthalates listed	Prohibited phthalates generally mentioned	Prohibited phthalates listed
		Extractable metals (mg/kg) apply to products intended for babies and children under 3 years old and accessories; the list does not coincide.	Extractable metals - Referred to as Heavy metals measured in ppm, the list does not coincide.	Extractable metals (mg/kg), the same list as EUEL, are also included in accessories.
		Carcinogenic substances referred to in Article 57 of REACH	Carcinogenic substances indicated in the document	Carcinogenic substances referred to in Article 57 of REACH
		Endocrine disruptors in Article 57 of REACH	Endocrine disruptors referencing the EU ED LISTS	Endocrine disruptors in Chapter 1 of the ZDHC MRSL
		Substances referred to in Article 59 of REACH	not specifically mentioned	Substances referred to in Article 59 of REACH
		not specifically mentioned		refers to ZDHC MRSL
		not specifically mentioned	Nanomaterial generally approached	Nanomaterials with H (hazard) phrases
		not specifically mentioned	Silicones, Linear alkylbenzene sulphonates (LAS)	not specifically mentioned
		not specifically mentioned	Polycyclic aromatic hydrocarbons are not comparable with Blue Angel	Polycyclic aromatic hydrocarbons are not similar with Nordic Swan
		not specifically mentioned		Mineral oil defoamers, enzymatic processes, content of soluble proteins from natural rubber, Free aniline in jeans products,

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
				Perfluorinated and polyfluorinated chemicals (PFCs) in hydrophobised textiles, Chinoline/quinoline
	Requirement hazardous substances	Additional hazards are indicated in comparison to Nordic Swan and Blue Angel. Below are the differentiating elements: Acute toxicity: H304 (R65) and EUH070 (R39/41) -different than Nordic Swan Specific target organ toxicity: H371 (R68/20, R68/21, R68/22) and H373 (R48/20, R48/21, R48/22) -different than Nordic Swan Respiratory and skin sensitisation, carcinogenic, mutagenic or toxic for reproduction (divided in Nordic Swan: Germ cell mutagenicity*, Carcinogenicity*, Reproductive toxicity*), Hazardous to the aquatic environment, (H412 (R52/ 53) and H413 (R53)-different than Nordic Swan)Respiratory and skin sensitisation, carcinogenic, mutagenic or toxic for reproduction (divided in Nordic Swan: Germ cell mutagenicity, Carcinogenicity, Reproductive toxicity), Hazardous to the aquatic environment, (H412 (R52/ 53) and H413 (R53)-different than Nordic Swan) Hazardous to the ozone layer (EUH059 (R59)-different than Nordic Swan).	Less than EUEL and Blue Angel. Below are missing hazard phrases: Acute toxicity: H304 (R65) and EUH070 (R39/41)-different than EUEL and Blue Angel Acute toxicity: H304 (R65) and EUH070 (R39/41)-different than EUEL and Blue Angel Specific target organ toxicity: H371 (R68/20, R68/21, R68/22) and H373 (R48/20, R48/21, R48/22)-different than EUEL and Blue Angel Respiratory and skin sensitisation, carcinogenic, mutagenic or toxic for reproduction (divided in Nordic Swan: Germ cell mutagenicity*, Carcinogenicity*, Reproductive toxicity*) -different than EUEL and Blue Angel Respiratory and skin sensitisation, carcinogenic, mutagenic or toxic for reproduction (divided in Nordic Swan: Germ cell mutagenicity, Carcinogenicity, Reproductive toxicity) -different than EUEL and Blue Angel Hazardous to the aquatic environment: H412 (R52/ 53) and H413 (R53)-different than EUEL and Blue Angel Hazardous to the ozone layer (EUH059 (R59)- different than EU EL but identical with Blue Angel due to H420).	Less than EUEL. Below are missing hazard phrases: Acute toxicity: EUH070 (R39/41) -different than EU EL Specific target organ toxicity: no difference with EU EL Respiratory and skin sensitisation, carcinogenic, mutagenic, or toxic for reproduction (referred to in Blue Angel as Carcinogenic, mutagenic, and reprotoxic substances)-no difference with EU EL Hazardous to the aquatic environment (referred to in Blue Angel as Water-hazardous substances)-no difference with EU EL Hazardous to the ozone layer (EUH059 (R59)-different than EU EL but identical with Nordic Swan due to H420).
	Verification	Mainly, the provision of declarations of compliance with the requirements is supported by evidence such as technical datasheets or test reports showing fulfilment of the requirement. Manufacturing facilities, associated chemical providers, and analytical labs must adhere to the outlined testing procedures. Where required, product analysis tests shall be conducted yearly throughout the licensing term, and the results must be forwarded to the relevant authoritative organisation for confirmation.		
Washing, drying and curing energy efficiency		Similar general BAT themes: 1. general energy management, 2. washing and rinsing, and 3. drying and curing using stretchers		
	Requirement	Similar 15 techniques among the 3 ecolabels	2 additional techniques are almost identical to Blue Angel: 1) Combining multiple wet treatments into one process. 2) Using solar thermal panels, solar photovoltaic panels, or a heat recovery system for hot water, aiming to generate energy amounting to 30% of what the process requires.	Almost identically presented techniques with Nordic Swan
	Verification	The same testing: I SO 50001 or equivalent systems for energy or carbon dioxide emissions		

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
Treatment of emissions to water	Requirement	Emissions to water: <u>TREATED:</u> 20 g COD/kg textiles processed: applied to weaving, dyeing, printing, and finishing processes used to manufacture the product(s). Measured downstream of on-site wastewater treatment plant and/or off-site wastewater treatment plant receiving wastewater from these processing sites. <u>DIRECT DISCHARGE:</u> IF Effluent is treated on-site and discharged directly to surface waters: 1) pH between 6,0 and 9,0 (unless the pH of the receiving water is outside this range) 2) temperature of less than 35C (unless the temperature of the receiving water is above this value) If colour removal is required: 436 nm (yellow sector) 7 m-1, (ii) 525 nm (red sector) 5 m-1, (iii) 620 nm (blue sector) 3 m-1- with BA	Emissions to water: <u>TREATED:</u> COD (chemical oxygen demand) in wastewater from wet processes which is discharged to surface water after treatment shall not exceed 150 mg/L <u>DIRECT DISCHARGE:</u> The pH value of the wastewater released to the surface water shall be between 6 and 9 (unless the pH value in the recipient lies outside this interval). The wastewater released to the surface water shall be lower than 40°C (unless the temperature in the recipient is higher).	Emissions to water: • COD: 100 mg/l (expressed as an average yearly value), • BOD5: 30 mg/l, • Sulphite: 1 mg/l, • Ammonium nitrogen: 10 mg/l, • Total nitrogen: 15 mg/l, • Total phosphorous: 2 mg/l, The dye must comply with the following values: Spectral absorption coefficient at with EUEL *436 nm (yellow spectral region) 7 m-1 *525 nm (red spectral region) 5 m-1 *620 nm (blue spectral region) 3 m-1 • Toxicity to fish eggs GEI: 2 • The pH value of the wastewater discharged to surface waters must be between 6 and 9 (unless the pH value of the receiving waters is outside this range), and the temperature must be below 35 °C (unless the temperature of the receiving waters is already above this limit). Before mixing (direct and indirect discharge) • AOX: 0.5 mg/l, • Sulphide: <=1 mg/l, • Copper: 0.4 mg/l, • Nickel: 0.2 mg/l, • Total chromium: 0.3 mg/l, • Tin: 2 mg/l, • Zinc: 0.8 mg/l, • Antimony: 1.2 mg/l

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
	Verification	Detailed documentation and test reports, using ISO 6060 and ISO 7887 as relevant+ compliance with this criterion based on monthly averages for the six months preceding the application	COD content shall be tested by ISO 6060 or equivalent. test report+ procedure in place for annual testing in line with the requirement and for ensuring compliance with the requirement Report showing average monthly calculations of COD, pH, and temperature for at least three of the past 12 months	Declare compliance, declaration of conformity from the operator of the textile finishing plant and test reports verifying compliance TEST METHODS: COD: ISO 6060 or DIN 38409-41 or DIN 38409-44 or DIN ISO 15705, BOD: DIN EN 1899-2 or ISO 5815-1, Copper and nickel: ISO 8288, Total chromium: ISO 9174 or DIN EN 1233, Sulphide: DIN 38405-27 or ISO 10530, Sulphite: DIN EN ISO 10304-3, Toxicity to fish eggs: DIN EN ISO 15088, AOX (chloride content < 5g/l): DIN EN ISO 9562, Spectral absorption coefficient: DIN 38404-3, Ammonium nitrogen: DIN EN ISO 11732, Total nitrogen: DIN EN ISO 12260, Total phosphorus: DIN EN ISO 11885, Tin: DIN EN ISO 11885, Zinc: DIN EN ISO 11885, Antimony: DIN EN ISO 11885 The wastewater treatment plant must be regularly monitored(at least every six months)
Treatment of emissions to air	Requirement	Emissions to air: Emission of organic compounds (not comparable with the other eco-labels) from thermosetting, thermosoling, coating, and impregnating textiles, including their respective drying (centre) facilities.	Emissions to air: Emissions from printing pastes and foamed synthetic materials indicating different substances (not comparable with other ecolabels)	Emissions to air: Emission of organic compounds to air in the textile finishing process in thermosetting, thermosoling, coating, impregnating, or finishing of textiles (not comparable with the other ecolabels) Additional emission from firing systems in the textile finishing process in thermosetting, thermosoling, coating, impregnating, or finishing of textiles with a strict list of limitations

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
	Verification	Compliance according to EN 12619 or other equivalent standards. Monthly averages for the total emissions of organic compounds from production sites shall be provided for the six months preceding the application.	Declaration from the producer or supplier and emission testing according to the ISO 16000 standard, parts 3, 6, 9, & 11	Compliance by DIN EN 12619 (total gaseous organic carbon), DIN CEN/TS 17638 (formaldehyde) and DIN EN ISO 21877 (ammonia). For firing systems: testing: Depending on the heating capacity, the heating boilers must be tested as follows: - 0.3 MW to 2 MW every 3 years - > 2 MW at least yearly - > 5 MW continuously

10988

10989 **Table 167.** Corporate social responsibility - comparison among Ecolabels

Topic	Requirement / Verification	EU Ecolabel	Nordic Swan	Blue Angel
Fundamental principles and rights at work	Requirement	Worker principles based on ILO Conventions. UN Global Compact and OECD Guidelines for Manufacturing Principles.	Must comply with the relevant national laws and regulations and the ILO Conventions. Some extra requirements were added for product licenses.	Must comply with the social and human rights requirements for the Green Button 2.0
	Verification	Third-party verification, including site visits by auditors. In countries where the ILO Labour Inspection Convention, 1947 (No 81) has been ratified, ILO supervision indicates that the national labour inspection system is effective and verification by labour inspector(s) appointed by a public authority shall be accepted.	Third-party verification (BSCI audit report) or SA8000 certificate for manufacturing. Description of code of conduct, policy and routine for produce licence.	Test report from an auditing body or SA8000 certificate.
Restriction on the sandblasting of denim	Requirement	Manual and mechanical sandblasting is prohibited.	Manual and mechanical sandblasting is prohibited. The use of potassium permanganate is not permitted.	
	Verification	The applicant must provide details of all production sites—evidence of the alternative processes to achieve distressed denim finishes.	Declaration stating the method used and that the requirement is fulfilled.	The applicant must provide details of all production sites. Evidence of the alternative processes used to achieve distressed denim finishes. Confirmation that no mechanical sandblasting or potassium permanganate is used.

10990

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
Information appearing on the Ecolabel	Requirement	The label may optionally contain text like: - More sustainable fibre production or a text defined in ECOLABEL, depending on product content. -Less polluting production processes -Restrictions on hazardous substances -Tested for durability	Mandatory information: - Product Type - Name and the fibre composition -All fabrics, their name, and % by weight -Membranes/coatings, impregnations or laminates -Details/accessories with information on the material type -Fillings and stuffing with information of the material type -Information on all recycled or biobased materials -If reused	Fibres are used by Regulation EC 1007/2011. Care and cleaning information (textile care symbols) by GINETEX or ISO EN DIN3758 and ISO10023 for industrial washing textiles. Information on repairable or replaceable parts of the textile.
	Verification	Label sample + DoC	Schematic overview	Label sample + DoC
Packaging	Requirement	N.A	PVC must not be used -It shall be possible to recycle the main material - Only monomaterials shall be used. Multi-material hangers are allowed if reused in a textile manufacturer's take-back system. - Plastic packaging shall be made from PE, PP, or PET. - Coloured plastic cannot be used for virgin plastic feedstock. Only if at least 50% by weight of the plastic is recycled material is colouring permitted. - Information on how it can be sorted for recycling - Chlorophenols and their salts, PCB, and organotin compounds shall not be used in transport or storage.	Mandatory description. Unnecessary packaging must be avoided. Composite packaging is not permitted and may not contain any dimethyl fumarate. The requirements apply to repackaging and transport, sales, and delivery of packaging directly used by the applicant. Clothes hangers are exempt.
	Verification	N.A	Declaration of Compliance about who is responsible for the product's primary packaging. Declaration of Compliance about PVC from the manufacturer of plastic material. Description of the main material and how it can be recycled. Description of primary packaging documenting compliance. Multi-material hangers: textile manufacturer's procedure describing the take-back system for hangers. Product labels or artwork providing information on recycling. Suppliers must declare that chlorophenols are not used in the yarn, fabric, or end product or have a valid license certificate for the EU Ecolabel.	DoC and a description of the intended packaging solution, designation and composition, raw materials and their origin, and a sample of the product packaging where relevant.
Unsold textiles	Requirement	N.A	Unsold textiles must not be sent for incineration or dumped in landfills. The brand owner must inform Nordic Swan Ecolabelling and state how they deal with unsold products on their website.	N.A

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
			Products with contaminations that are harmful to the environment or health are exempt. Military and police uniforms are exempt.	
	Verification	N.A	Description of how unsold products are dealt with and its procedure.	N.A
Reused textiles	Requirement	N.A	The material shall not come from workwear and other textiles used in the chemical and oil industry -The material must not contain plastisol print - Textiles from the health care sector must be washed, inactivating microorganisms. - Material originally eco-labelled with the Nordic Swan Ecolabel, the EU Ecolabel, GORS, or Bra Miljöval or have Oeko-Tex 100 or only be used for furnishing fabrics such as rugs, tablecloths If re-designed for professional use, the product must fulfil the expected function. If processed with chemical products, the requirements for chemicals must be accomplished.	N.A
	Verification	N.A	Invoice or label on the textile -Documentation showing that the textile used is reused - Declaration that the textile has been washed in an industrial laundry in a microbiological wash - Declaration that The material does not contain PVC	N.A
Reduced washing	Requirement	N.A	The text "Reduce the number of washes and help save energy and reduce climate impact" must be included.	N.A
	Verification	N.A	Photo	N.A
Production Chain	Requirement	N.A	Description of all the production methods/treatment techniques, including production by suppliers. -Name of the fibres, yarns, and fabrics -Information of all the actors in the production chain	N.A
	Verification	N.A	Schematic overview	N.A

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
Materials limitations	Requirement	N.A	The sewing thread is not covered -Embroidery thread applies chemical requirements -Belt buckles of metals must not exceed 25% by weight -Fibre types with less than 5% by weight are exempt. -Small textile elements present between 5% to 10% by weight may be exempted from the requirements of the textile element has an EU-Ecolabel or GOTS or Oeko Tex 100 class I certificate - For elastic bands, up to a total of 25% of the product's weight may be exempted from the requirements if GOTS or Oeko-Tex 100 class I certification - Information printed directly on the textile product must meet the requirements of prohibited substances	N.A
	Verification	N.A	Declaration of Compliance and certificate where applicable. If Oeko Tex 100: Statement regarding fluorinated organic compounds	N.A
Quality and regulatory requirements	Requirement	N.A	Written documentation obtained annually showing that suppliers are familiar with Nordic Swan Ecolabel's requirements. Supplier changes must be approved. An individual responsible for ensuring the fulfilment should be appointed. The licence must guarantee the quality of the labelled product during the validity period. Written notice must be given in case of changes. Nonconformities must be reported. Traceable product.	N.A
	Verification	N.A	Documentation, operational chart and procedures description.	N.A
Traceability of the Nordic Swan Ecolabelled product	Requirement	N.A	The brand owner is responsible for ensuring that a Nordic Swan Ecolabelled product can be traced back to a production licence.	N.A
	Verification	N.A	-The brand owner must submit the information specified in the requirement. - A description of the procedure, which shows how it is ensured that the information held by Nordic Ecolabelling is kept updated.	N.A

10993 **Table 169.** Dyes restriction - comparison among Ecolabels

Topic	Requirement/ Verification	EU Ecolabel	Nordic Swan	Blue Angel
Carcinogenic aromatic amines	Requirement	The same 25 dyes listed, limitation: 30 mg/kg for each amine	Prohibited: the same 25 dyes listed + 10 additional dyes	The same 25 dyes listed limitation: 20 mg/kg for each amine
	Verification	Test results: EN 14362-1 and 3.	Prohibited: Declaration from the chemical manufacturer or chemical supplier that the requirement is fulfilled	Test results: EN 14362-1 and 3.
List of dyes that may cleave to carcinogenic aromatic amines	Requirement	An extended list of prohibited disperse, basic, acid and direct dyes limitation: 30 mg/kg for each amine	There is no specific dyes list, but in the section addressed to recycled fibres, there are limitations for cleavable dyes according to Oeko-Tex 100 Annex 5	Azo dyes that may cleave to one of the aromatic amines must not exceed a limit value of 20 mg/kg.
	Verification	Test results: EN 14362-1 and 3.	Cleavable, classified as carcinogenic in Oeko-Tex Annex 5: Total 20 mg/kg Cleavable aniline as listed in Oeko-Tex Annex 5: Total 100 mg/kg Test reports or Oeko-Tex 100 class I certificate showing fulfilment of the requirement	20 mg/kg for each dye Test results by the test method DIN EN 14362-1 and, DIN EN 14362-3 (for arylamine), and DIN 54231 (for dispersion dyes).
Dyes that are carcinogenic, mutagenic or toxic to reproduction	Requirement	List of prohibited dyes	There is no specific dyes list but criterion 034. Prohibition of CMR substances refers to carcinogenic, mutagenic or toxic to reproduction substances in general.	Extended list of prohibited dyes
	Verification	Prohibited. Declaration of non-use from the chemical supplier supported by SDS.	Declaration from the chemical producer that the requirement has been fulfilled.	Declaration from the chemical producer that the requirement is fulfilled and test results by the test method DIN EN 14362-1 and DIN EN 14362-3 (for arylamine)
Disperse dyes that are potentially sensitising	Requirement	List of prohibited dyes	There is no specific dyes list but criterion 033. Classification of chemical products refers to sensitising on inhalation or skin contact substances.	Extended list of prohibited dyes
	Verification	Prohibited. Declaration of non-use from the chemical supplier supported by SDS.	Declaration from the chemical producer that the requirement has been fulfilled.	Declaration from the chemical producer that the requirement is fulfilled and test results by the test method DIN EN 14362-1 and DIN EN 14362-3 (for arylamine)+

10994

10995

10996

10997 **13.9 Supporting information about public procurement**

10998 **13.9.1 The Common Procurement Vocabulary (CPV) codes for products in the scope**

10999 **Table 170.** The Common Procurement Vocabulary (CPV) codes for products in the scope

Product category	Common Procurement Vocabulary (CPV) code
T-shirts	18000000-9*: Clothing, footwear, luggage articles and accessories 18300000-2*: Garments 18318400-5: Vests 18330000-1*: T-shirts and shirts 18331000-8: T-shirts 18235400-9: Waistcoats 18400000-3*: Special clothing and accessories 18410000-6*: Special clothing 18411000-3*: Baby clothing
Shirts and blouses	18000000-9*: Clothing, footwear, luggage articles and accessories 18300000-2*: Garments 18318000-1*: Nightwear 18318100-2: Nightshirts 18318200-3: Dressing gowns 18318300-4*: Pyjamas 18318500-6: Nightdresses 18330000-1*: T-shirts and shirts 18332000-5: Shirts 18333000-2: Polo shirts 18400000-3*: Special clothing and accessories 18410000-6*: Special clothing 18411000-3*: Baby clothing
Sweaters and mid-layers	18000000-9*: Clothing, footwear, luggage articles and accessories 18235000-5: Pullovers, cardigans and similar articles 18235100-6: Pullovers 18235200-7: Cardigans 18235300-8: Sweatshirts 18300000-2*: Garments 18400000-3*: Special clothing and accessories 18410000-6*: Special clothing 18411000-3*: Baby clothing
Jackets and coats	18000000-9*: Clothing, footwear, luggage articles and accessories 18200000-1: Outerwear 18210000-4: Coats 18211000-1: Capes 18212000-8: Cloaks 18213000-5: Wind jackets 18220000-7: Weatherproof clothing 18221000-4: Waterproof clothing 18221100-5: Waterproof capes 18221200-6: Anoraks 18221300-7: Raincoat 18223000-8: Jackets and blazers 18223100-9: Blazers 18223200-0: Jackets 18224000-5: Clothing made of coated or impregnated textile fabrics 18230000-0: Miscellaneous outerwear 18300000-2*: Garments 18400000-3*: Special clothing and accessories 18410000-6*: Special clothing 18411000-3*: Baby clothing
Pants and shorts	18000000-9*: Clothing, footwear, luggage articles and accessories 18233000-1: Shorts 18234000-8: Trousers 18300000-2*: Garments 18318000-1*: Nightwear 18318300-4*: Pyjamas 18400000-3*: Special clothing and accessories 18410000-6*: Special clothing 18411000-3*: Baby clothing

Product category	Common Procurement Vocabulary (CPV) code
Dresses: Skirts and jumpsuits	18000000-9*: Clothing, footwear, luggage articles and accessories 18222100-2: Suits 18222200-3: Ensembles 18231000-7: Dresses 18232000-4: Skirts 18300000-2*: Garments 18400000-3*: Special clothing and accessories 18410000-6*: Special clothing 18411000-3*: Baby clothing
Leggings: Stockings: Tights and socks	18315000-0: Stockings 18316000-7: Tights 18317000-4: Socks
Underwear	18310000-5: Underwear 18311000-2: Slips 18312000-9: Underpants 18313000-6: Panties 18320000-8: Brassieres, corsets, suspenders and similar articles 18321000-5: Brassieres 18322000-2: Corsets 18323000-9: Suspenders
Swimwear	18412800-8: Swimwear
Accessories	18421000: Handkerchiefs 18422000: Scarves 18423000: Ties 18424000: Gloves 18425000: Belts

* These CPV codes were assigned to more than one product category

Source: own elaboration based on CPV description provided by BIP Solutions ⁽³³⁶⁾

³³⁶ BIP Solutions. CPV codes. Available at [this link](#). Last accessed on 12 January 2024.

11003 **13.9.2 Number of Contract Award procuring apparel in EU**

11004 **Table 171.** Number of Contracts Award procuring apparel in EU Member States in 2015

Country	CPV 181	CPV 182	CPV 183	CPV 184	CPV 351	CPV 358	CPV 374	Total in the country
Austria	2	NA	NA	3	NA	NA	NA	5
Belgium	10	1	9	4	NA	1	NA	25
Bulgaria	31	1	NA	NA	NA	4	NA	36
Croatia	2	4	1	2	NA	7	NA	16
Cyprus	NA	1	NA	NA	NA	1	NA	2
Czechia	5	5	2	11	1	5	1	30
Denmark	4	NA	NA	6	NA	NA	NA	10
Estonia	NA	2	1	3	1	NA	1	8
Finland	3	NA	4	4	NA	NA	NA	11
France	93	6	11	11	3	5	2	131
Germany	17	10	11	19	3	17	1	78
Greece	1	NA	NA	NA	NA	NA	NA	1
Hungary	1	NA	NA	NA	NA	NA	NA	1
Ireland	NA	NA	NA	NA	NA	NA	NA	0
Italy	8	7	2	3	NA	3	NA	23
Latvia	3	NA	NA	2	NA	NA	NA	5
Lithuania	8	1	3	5	2	5	NA	24
Luxembourg	NA	NA	NA	NA	NA	NA	NA	0
Malta	NA	1	NA	NA	NA	NA	NA	1
Netherlands	3	4	1	4	1	1	NA	14
Poland	18	13	9	21	3	11	NA	75
Portugal	7	1	NA	NA	NA	NA	NA	8
Romania		1	1	4	NA	2	NA	8
Slovakia	2	1	NA	5	NA	NA	NA	8
Slovenia	5	2	NA	NA	NA	3	NA	10
Spain	5	NA	NA	1	NA	1	NA	7
Sweden	13	NA	2	2	NA	NA	2	19
Total per CPV code	241	61	57	110	14	66	7	556

11005 CPV 181: CPV 181XXXXX-X Occupational clothing, special workwear and accessories;

11006 CPV 182: CPV 182XXXXX-X Outerwear;

11007 CPV 183: CPV 183XXXXX-X Garments;

11008 CPV 184: CPV 184XXXXX-X Special clothing and accessories;

11009 CPV 351: CPV 351134XX-X Protective and safety clothing;

11010 CPV 358: CPV 3581XXXX-X Individual and support equipment;

11011 CPV 374: CPV 3741XXXX-X Sport goods and equipment.

11012 Source: own elaboration based on Tenders Electronic Daily (TED) (csv subset) – public procurement notices ⁽¹³⁹⁾.

11013

11014 **Table 172.** Number of Contracts Award procuring apparel in EU Member States in 2016

Country	CPV 181	CPV 182	CPV 183	CPV 184	CPV 351	CPV 358	CPV 374	Total in the country
Austria	6	NA	4	1	NA	1	1	13
Belgium	11	4	2	3	NA	3	NA	23
Bulgaria	23	1	NA	1	NA	16	NA	41
Croatia	4	1	1	NA	1	4	NA	11
Cyprus	NA	NA	1	NA	NA	1	NA	2
Czechia	5	9	2	6	2	7	NA	31
Denmark	4	NA	3	1	1	2	1	12
Estonia	2	NA	1	1	NA	1	NA	5
Finland	7	NA	2	1	NA	1	NA	11
France	106	9	11	7	11	10	3	157
Germany	25	10	10	4	6	17	NA	72
Greece	1	NA	NA	NA	NA	NA	NA	1
Hungary	1	NA	NA	1	NA	4	1	7
Ireland	NA	NA	NA	2	NA	NA	NA	2
Italy	5	6	6	1	2	10	NA	30
Latvia	1	NA	1	NA	NA	2	NA	4

Country	CPV 181	CPV 182	CPV 183	CPV 184	CPV 351	CPV 358	CPV 374	Total in the country
Lithuania	3	2	4	9	2	5	1	26
Luxembourg	1	NA	NA	NA	NA	NA	NA	1
Malta	1	NA	NA	NA	NA	NA	NA	1
Netherlands	9	6	1	1	1	1	NA	19
Poland	14	11	4	7	9	12	1	58
Portugal	9	NA	NA	NA	NA	1	NA	10
Romania	2	NA	2	NA	NA	2	NA	6
Slovakia	2	6	2	1	NA	NA	NA	11
Slovenia	11	3	NA	NA	1	7	NA	22
Spain	NA	NA	NA	NA	NA	1	NA	1
Sweden	32	NA	3	NA	1	NA	3	39
Total per CPV code	285	68	60	47	37	108	11	616

CPV 181: CPV 181XXXXX-X Occupational clothing, special workwear and accessories;

CPV 182: CPV 182XXXXX-X Outerwear;

CPV 183: CPV 183XXXXX-X Garments;

CPV 184: CPV 184XXXXX-X Special clothing and accessories;

CPV 351: CPV 351134XX-X Protective and safety clothing;

CPV 358: CPV 3581XXXX-X Individual and support equipment;

CPV 374: CPV 3741XXXX-X Sport goods and equipment.

Source: own elaboration based on Tenders Electronic Daily (TED) (csv subset) – public procurement notices ⁽¹³⁹⁾.

Table 173. Number of Contracts Award procuring apparel in EU Member States in 2017

Country	CPV 181	CPV 182	CPV 183	CPV 184	CPV 351	CPV 358	CPV 374	Total in the country
Austria	5	1	1	2	1	1	1	12
Belgium	13	1	NA	NA	1	3	NA	18
Bulgaria	25	1	2	5	2	22	NA	57
Croatia	5	4	NA	2	NA	5	NA	16
Cyprus	NA	NA	NA	1	NA	NA	NA	1
Czechia	16	32	24	12	2	21	1	108
Denmark	6	1	2	4	NA	1	1	15
Estonia	2	NA	3	2	NA	NA	NA	7
Finland	9	NA	1	NA	NA	1	NA	11
France	126	7	14	12	12	6	2	179
Germany	37	16	8	5	4	16	2	88
Greece	NA	NA	1	NA	NA	1	NA	2
Hungary	5	NA	NA	2	1	4	NA	12
Ireland	1	NA	NA	NA	NA	NA	NA	1
Italy	9	4	2	3	4	8	NA	30
Latvia	3	NA	NA	4	NA	5	NA	12
Lithuania	5	NA	2	8	2	3	NA	20
Luxembourg	NA	NA	NA	NA	NA	NA	NA	0
Malta	1	1	2	NA	NA	1	NA	5
Netherlands	10	4	2	3	2	4	NA	25
Poland	19	16	16	19	13	18	1	102
Portugal	NA	NA	NA	NA	1	NA	NA	1
Romania	1	NA	1	3	NA	2	NA	7
Slovakia	5	3	4	2	1	1	NA	16
Slovenia	23	2	NA	8	2	6	NA	41
Spain	5	NA	NA	NA	NA	NA	NA	5
Sweden	29	NA	3	NA	2	4	NA	38
Total per CPV code	360	93	88	97	50	133	8	829

CPV 181: CPV 181XXXXX-X Occupational clothing, special workwear and accessories;

CPV 182: CPV 182XXXXX-X Outerwear;

CPV 183: CPV 183XXXXX-X Garments;

CPV 184: CPV 184XXXXX-X Special clothing and accessories;

CPV 351: CPV 351134XX-X Protective and safety clothing;

CPV 358: CPV 3581XXXX-X Individual and support equipment;

CPV 374: CPV 3741XXXX-X Sport goods and equipment.

Source: own elaboration based on Tenders Electronic Daily (TED) (csv subset) – public procurement notices ⁽¹³⁹⁾.

11033

11034 **Table 174.** Number of Contracts Award procuring apparel in EU Member States in 2018

Country	CPV 181	CPV 182	CPV 183	CPV 184	CPV 351	CPV 358	CPV 374	Total in the country
Austria	5	5	4	NA	NA	NA	NA	14
Belgium	26	NA	4	2	1	NA	NA	33
Bulgaria	28	5	3	2	4	25	NA	67
Croatia	3	3	2	1	NA	4	NA	13
Cyprus	NA	NA	NA	NA	NA	NA	NA	NA
Czechia	17	31	20	18	10	22	2	120
Denmark	13	1	NA	6	2	2	NA	24
Estonia	4	3	6	4	1	4	NA	22
Finland	3	3	NA	2	NA	1	NA	9
France	108	4	14	7	15	9	3	160
Germany	33	10	10	4	7	6	NA	70
Greece	2	1	NA	NA	NA	1	NA	4
Hungary	10	NA	NA	1	NA	9	NA	20
Ireland	NA	NA	NA	1	NA	NA	NA	1
Italy	22	2	4	4	2	6	NA	40
Latvia	3	NA	4	9	NA	18	NA	34
Lithuania	12	2	4	4	2	6	2	32
Luxembourg	1	NA	NA	NA	NA	NA	NA	1
Malta	1	1	NA	NA	NA	2	NA	4
Netherlands	8	5	1	3	5	1	NA	23
Poland	39	10	15	26	12	22	NA	124
Portugal	NA	NA	NA	NA	1	NA	NA	1
Romania	21	2	4	NA	NA	9	NA	36
Slovakia	5	2	2	4	1	NA	NA	14
Slovenia	21	1	NA	10	5	10	NA	47
Spain	1	NA	NA	NA	1	NA	NA	2
Sweden	45	2	2	3	NA	1	3	56
Total per CPV code	431	93	99	111	69	158	10	971

11035 CPV 181: CPV 181XXXXX-X Occupational clothing, special workwear and accessories;

11036 CPV 182: CPV 182XXXXX-X Outerwear;

11037 CPV 183: CPV 183XXXXX-X Garments;

11038 CPV 184: CPV 184XXXXX-X Special clothing and accessories;

11039 CPV 351: CPV 351134XX-X Protective and safety clothing;

11040 CPV 358: CPV 3581XXXX-X Individual and support equipment;

11041 CPV 374: CPV 3741XXXX-X Sport goods and equipment.

11042 Source: own elaboration based on Tenders Electronic Daily (TED) (csv subset) – public procurement notices ⁽¹³⁹⁾.

11043

11044 **Table 175.** Number of Contracts Award procuring apparel in EU Member States in 2019

Country	CPV 181	CPV 182	CPV 183	CPV 184	CPV 351	CPV 358	CPV 374	Total in the country
Austria	4	NA	6	NA	NA	1	NA	11
Belgium	13	2	NA	1	5	2	1	24
Bulgaria	20	4	2	3	4	38	NA	71
Croatia	2	4	NA	NA	NA	3	NA	9
Cyprus	NA	1	NA	1	NA	1	NA	3
Czechia	26	35	41	22	16	34	NA	174
Denmark	9	1	4	1	1	1	NA	17
Estonia	3	3	5	2	NA	1	NA	14
Finland	6	NA	3	1	NA	2	NA	12
France	156	11	10	14	15	6	8	220
Germany	50	14	18	12	8	18	NA	120
Greece	6	1	2	1	2	2	1	15
Hungary	8	NA	NA	1	2	2	NA	13
Ireland	NA	1	NA	NA	1	NA	NA	2
Italy	15	7	7	5	4	10	NA	48
Latvia	6	1	NA	20	1	3	NA	31
Lithuania	23	2	6	13	2	7	NA	53

Country	CPV 181	CPV 182	CPV 183	CPV 184	CPV 351	CPV 358	CPV 374	Total in the country
Luxembourg	NA	NA	NA	NA	NA	NA	NA	0
Malta	NA	NA	1	NA	NA	NA	NA	1
Netherlands	7	10	1	2	3	3	NA	26
Poland	62	15	16	17	27	28	NA	165
Portugal	1	NA	NA	NA	NA	NA	NA	1
Romania	48	11	6	9	1	36	1	112
Slovakia	2	NA	4	4	NA	NA	NA	10
Slovenia	18	NA	2	8	5	8	NA	41
Spain	3	NA	NA	NA	NA	NA	NA	3
Sweden	45	1	4	5	6	NA	4	65
Total per CPV code	533	124	138	142	103	206	15	1261

CPV 181: CPV 181XXXXX-X Occupational clothing, special workwear and accessories;

CPV 182: CPV 182XXXXX-X Outerwear;

CPV 183: CPV 183XXXXX-X Garments;

CPV 184: CPV 184XXXXX-X Special clothing and accessories;

CPV 351: CPV 351134XX-X Protective and safety clothing;

CPV 358: CPV 3581XXXX-X Individual and support equipment;

CPV 374: CPV 3741XXXX-X Sport goods and equipment.

Source: own elaboration based on Tenders Electronic Daily (TED) (csv subset) – public procurement notices ⁽¹³⁹⁾.

13.9.3 Contributions to the initial questionnaire

Table 176. Type of respondents that contributed to the section on EU GPP during the initial questionnaire

Type of respondent	Number	Percentage compared to the total (%)
University or research institute	1	3
Certified laboratory	1	3
Government (local, regional, or national)	12	35
Non-governmental organisation – Environment	3	9
Non-governmental organisation – Consumers	0	0
Industry – manufacturing	6	18
Industry – manufacturing - association	2	6
Industry – waste collection, sorting and treatment	1	3
Industry – waste collection, sorting and treatment - association	1	3
Distributor/Retailer	0	0
Distributor/Retailer association	1	3
Other	6	18
TOTAL	34	100

Source: own elaboration

13.10 Supporting information on relevant aspects

13.10.1 Qualitative assessment based on technical, socioeconomic and environmental dimensions

In this section, a qualitative assessment is carried out, focusing on three dimensions, with information collected via literature review. Only those aspects that are considered relevant after the assessment based on key guiding questions are evaluated (therefore, every aspect except 'possibility of recovery of materials' is evaluated in this section).

13.10.1.1 Durability

Durability is the ability of a product to maintain over time its function and performance under specified conditions of use, maintenance and repair. In the context of textile apparel, durability can be classified as intrinsic and extrinsic (emotional).

Intrinsic durability refers to physical and measurable characteristics of products, such as fabric strength, abrasion resistance, pilling, wrinkling, colour fastness, dimensional changes, seam slippage, etc. Intrinsic durability is often related with product quality, as described in Piippo et al., (2022).

11072 Quality is a fundamental aspect of the durability and usability of clothing, and thus plays a role on its lifespan
11073 (Aakko and Niinimäki, 2021). Extrinsic durability refers to subjective and perception aspects of products, such
11074 as emotional attachment by users. This section focuses only on the intrinsic durability of textile apparel because
11075 it best corresponds to the durability definition provided by the ESPR.

11076 Characteristics of the products in the scope

11077 The lifetimes of textile apparel are, to a degree, subjective (Cooper and Claxton, 2022c). In fact, there is no
11078 common approach to assess or guarantee the physical durability of textile apparel; nor are there legislative
11079 standards that apply directly to textile apparel, other than the general requirement under consumer legislation
11080 that goods are 'fit for purpose' (Cooper and Claxton, 2022c).

11081 Literature review suggests that there is room for improvement in the average durability of textile apparel. In
11082 the current market, many products are well made and have lifetimes that would generally be considered
11083 satisfactory. However, the lifetime of many other products is unduly short due to inadequate performance, in
11084 part due to pressure to meet predetermined price points (Cooper and Claxton, 2022c). It is possible to influence
11085 the intrinsic durability of products with many factors, which include the use of different fibres: for instance, in
11086 the manufacturing of denim products, the utilization of synthetic fibres, linen and hemp -rather than 100%
11087 cotton- can help to produce high-durability denim products particularly with respect to tensile and abrasion
11088 resistance (Elmogahzy, 2020).

11089 The most common physical failures in textile apparel are related to colour fading (particularly for jersey and
11090 woven fabrics) and pilling of knitwear and jersey items. Fabric breakdown (fraying or thinning around the hems)
11091 and wear round the crotch of trousers and jeans, and accidental damage due to stains, tears and rips, are also
11092 common. Other failures include loss of dimensional stability, logo failure, discolouration, holes in seams and
11093 trim failure (Cooper and Claxton, 2022c). Even though they are not the main cause for physical failures, priority
11094 parts such as zippers, buttons, etc., may also have a relevance in product durability.

11095 Environmental

11096 Extending the lifetime of textile apparel has been highlighted as the most effective method of reducing the
11097 impact of the industry on the environment. For instance, extending the average life of clothes by 3 months'
11098 usage per item could reduce carbon, water and waste footprints by 5-10% (Goworek, Lynn Oxborrow, et al.,
11099 2020). Extending the average life of clothes by nine months of active use per item would typically lead to a
11100 reduction in the carbon, water and waste footprints of 27%, 33% and 22%, respectively (Cooper and Claxton,
11101 2022c).

11102 Moreover, almost all life-cycle stages of textile apparel entail significant energy consumption (as described in
11103 Section 3.3.2). Some processes, such as the extraction of raw materials or production processes -such as
11104 spinning, knitting and weaving- are highly energy demanding. The longer the products are in active use by
11105 consumers (the more durable the products), the more likely it is to offset the manufacturing of new products,
11106 thus reducing the industry's impacts related to energy.

11107 Socioeconomic

11108 Durability in the context of textile apparel collides with one of the prevalent business models, based on fast
11109 trend turnovers and short production times. This business model does not offer space for developments to
11110 improve durability of products, as it focuses on short times between design and product availability, and aims
11111 to low pricing obtained with production of large volumes manufactured in specific locations and conditions (Dan
11112 and Ostergaard, 2021). Therefore, an increase in the durability of products may affect negatively manufacturers
11113 whose strategy is based on such business models, since the prevailing commercial drive is to save costs
11114 (Goworek et al., 2020); whereas it could affect positively manufacturers which are already focusing on the
11115 durability of products.

11116 An increase in the durability of textile apparel could stimulate the establishment of other product-related
11117 businesses. Increasing textile apparel durability could be supported by an increase in the number of companies
11118 providing clothing repair services. Moreover, products with enhanced durability seem a necessary condition for
11119 the existence of businesses based on product renting or sharing. These businesses could extend the use of
11120 products and potentially reduce the purchase of new items if they do not incentivise overconsumption.

11121 It is worth noting where the socioeconomic changes of product durability would be perceived. Whereas any
11122 changes in product manufacturing would mostly take place outside the EU (which is where most manufacturing
11123 takes place today) (Section 5.1), the effects related with repair, renting and sharing services would be
11124 experienced within the EU.

11125 Consumers are increasingly interested in textile products with enhanced durability: around 90% of consumers
 11126 think that apparel should be made to last longer (European Commission, 2019). There are consumers in
 11127 different age groups and markets who consider durability, in connection with quality, when making fashion
 11128 purchases, even if sustainability is not their main motivation (Goworek et al., 2020).

11129 Durability has been highlighted as one of the top three aspects to influence the decision to purchase apparel
 11130 such as coats or jackets (Consumers, Health, Agriculture and Food Executive Agency. et al., 2018). Around 43%
 11131 of consumers in some Member States express a willingness to buy long-lasting apparel despite the price (AK
 11132 Wienn and Greenpeace, 2023b).

11133 It is also worth noting that design for physical durability involves the development and testing of yarns, fabrics
 11134 and textile apparel to meet specified performance standards that can withstand prolonged wear (Claxton and
 11135 Kent, 2020b), which could have an effect on product purchase price.

11136 Qualitative assessment of relevance

11137 Based on the above, durability is considered a relevant product aspect in the context of textile apparel.

11138 **13.10.1.2 Reliability**

11139 Reliability is the probability that a product functions as required under given conditions for a given duration
 11140 without an occurrence which results in a primary or secondary function of the product no longer being
 11141 performed.

11142 In the context of textile apparel, reliability can be understood as the ability of a product to retain the physical
 11143 characteristics that allow its use. Therefore, reliability is directly linked with the intrinsic durability of products.
 11144 Ensuring that textile apparel are durable inherently makes them more reliable. Reliability of textile apparel can
 11145 be interpreted as the ability of a product to avoid or delay damages related with colour fading, discolouring,
 11146 pilling, wear and tear, dimensional stability or seam defects, among others.

11147 Characteristics of the products in the scope

11148 Most of the rationale provided for durability in section 13.10.1.1 is applicable for reliability. There is room for
 11149 improvement in the reliability of textile apparel: many products in the market have lifetimes that would
 11150 generally be considered satisfactory, whereas in other products, lifetime is unduly short due to inadequate
 11151 performance. The choice of fibres and the way they are processed can have an influence on the reliability of
 11152 products: for instance, in the manufacturing of denim products, the utilization of synthetic fibres, linen and
 11153 hemp -rather than 100% cotton- can help to produce denim products with higher tensile and abrasion resistance
 11154 (Elmogahzy, 2020). It must be noted that, on occasions this can affect negatively the recyclability of the product.
 11155 As in the case of durability, even though priority parts are not the main cause for physical failures, they may
 11156 also have a relevance in product reliability.

11157 Environment

11158 Enhancing the reliability of textile apparel can contribute to lifetime extension. Textile lifespan extension is
 11159 essential to avoid virgin natural resources withdrawals, either energy or material technologies (Amicarelli et al.,
 11160 2022). Moreover, it can be concluded that the longer the products are in active use by consumers, the more
 11161 likely it is to offset the manufacturing of new products, thus reducing the industry's impacts related to energy.

11162 Socioeconomic

11163 Reliability in the context of textile apparel is not aligned with business models based on fast trend turnovers
 11164 and short production times. As a textile apparel manufacturer expressed it in (Goworek et al., 2020), "product
 11165 longevity loses sales". Therefore, enhancing the reliability of products may affect negatively those
 11166 manufacturers whose strategy is based on fast trend turnovers and short production times; whereas it could
 11167 affect positively manufacturers which are already focusing on the reliability of products.

11168 An increase in the durability of textile apparel could stimulate the establishment of other product-related
 11169 businesses. For instance, products with enhanced reliability seem one of the necessary conditions for the
 11170 existence of businesses based on product renting or sharing (a more reliable product is more likely to withstand
 11171 the more intensive use within such business models). Quality considered as durability is a crucial aspect of
 11172 textile apparel in the second hand markets, which enable extending the textile apparel use times after the initial
 11173 user (Aakko and Niinimäki, 2021).

11174 As in the case of durability, whereas any changes in product manufacturing to enhance the reliability of products
11175 would mostly take place outside the EU (Section 5.1), the effects related with renting and sharing services would
11176 be experienced within the EU.

11177 Most of the rationale provided for durability in section 13.10.1.1 is applicable for reliability. Around 90% of
11178 consumers think that apparel should be made to last longer (European Commission, 2019), therefore would be
11179 interested in more reliable products.

11180 Qualitative assessment of relevance

11181 Based on the above, reliability is considered a relevant product aspect in the context of textile apparel.

11182 **13.10.1.3 Reusability**

11183 Reusability is the ability of a product or component that is not waste to be used again for the same purpose for
11184 which it was conceived.

11185 In the context of textile apparel, reusability can have different interpretations. For instance, it can refer to
11186 extending the lifetime of products (for instance via repair) so that the same owner can continue using the
11187 product. It can also refer to transferring products to a new owner via donating to charity or second hand shops,
11188 renting, inheriting, trading, swapping and borrowing (Shirvanimoghaddam et al., 2020).

11189 Characteristics of the products in the scope

11190 Reusability and durability are also intertwined (as it happened between reliability and durability). Enhancing the
11191 performance of textile apparel increases durability, and thus enables increased use times (Aakko and Niinimäki,
11192 2021).

11193 There is a great potential to further increase reuse, as clothing items typically are disposed of long before the
11194 end of their technical service life (Sandin and Peters, 2018). Many products in the market have lifetimes that
11195 would generally be considered satisfactory, whereas in other products, lifetime is unduly short due to
11196 inadequate performances. As stated for reliability, the choice of fibres and the way they are processed can have
11197 an influence on the reusability of products: textile apparel with higher tensile and abrasion resistance are more
11198 likely to be reused. As in the case of durability, even though priority parts are not the main cause for physical
11199 failures, they may also have a relevance in product reusability.

11200 An aspect related with textile product reusability is the discrepancy between technical lifetime (how long the
11201 product is designed to last) and average real use time by consumers. In Section 5.8, it has been highlighted that
11202 textile products nowadays remain in use for shorter periods, with an ever decreasing trend (some authors
11203 estimate that t-shirts are used for an average of 22 days in total). It appears that many textile products are
11204 currently underused during their initial lifetime, it can be expected that reusability strategies can counteract the
11205 negative effects of this product underuse.

11206 Environment

11207 Enhancing the reusability of textile apparel can contribute to lifetime extension, which is essential to avoid virgin
11208 natural resources withdrawals (Amicarelli et al., 2022). Moreover, it can be concluded that the longer the
11209 products are in active use by consumers, the more likely it is to offset the manufacturing of new products, thus
11210 reducing the industry's impacts related with energy. If the average life of textile apparel is extended by 3 years
11211 (for instance, via reuse), the carbon and water footprint and waste generation can be reduced by 5-10%
11212 (Shirvanimoghaddam et al., 2020).

11213 Socioeconomic

11214 Reusability in the context of textile apparel is also in opposition with business models based on fast trend
11215 turnovers and short production times. Therefore, enhancing the reusability of products may affect negatively
11216 those manufacturers whose strategy is based on such business models; whereas it could affect positively
11217 manufacturers which are already focusing on the reusability of products (some companies claim that their
11218 products are not designed or made for use by only one user, but many users (Piippo et al., 2022b)).

11219 Textile apparel reuse has been related with collaborative consumption. Increase product reuse could boost
11220 product-service systems, commercial sharing systems and access-based consumption (Shirvanimoghaddam et
11221 al., 2020). These business models can contribute to intensifying their use and reducing the purchase of new
11222 items. In this context, the relationship between performances and durability arises again: high technical
11223 performance enables textile apparel to be used for a longer time and enables further uses through, for example,
11224 renting, leasing and second-hand markets (Piippo et al., 2022b). However, it must be noted that, although reuse

11225 of textile apparel takes place today, supply exceeds demand and many wearable items end up either exported
11226 (Cooper and Claxton, 2022c) or in landfills (Goworek et al., 2020).

11227 Reusability can also be associated with take-back services and second-hand shops. In such schemes, companies
11228 maintain and sell the products again, enabling a new life for the product with a new consumer (Piippo et al.,
11229 2022b).

11230 For some consumers, buying high-quality second-hand clothes (in other words, reusing textile apparel)
11231 represents ethical consumption, since it enables prolonging the lifespan of existing products and avoiding the
11232 traditional fashion production chain (Aakko and Niinimäki, 2021). Moreover, in section 6.2.5 it has been
11233 mentioned that second-hand textile apparel are experiencing fast growth. Around 37% of the population
11234 actively engages in buying second-hand apparel (D&B, 2020b). It must be noted, however, that reusability does
11235 not change overconsumption and overproduction patterns.

11236 Qualitative assessment of relevance

11237 Based on the above, reusability is considered a relevant product aspect in the context of textile apparel.

11238 **13.10.1.4 Upgradability**

11239 Upgradability is the ability of a product to be accessible for implementing actions to enhance its functionality,
11240 performance, capacity, safety or aesthetics of a product.

11241 In the context of textile apparel, upgradability can also be considered a factor that contributes to the overall
11242 durability of a product. An item of textile apparel that is upgradeable will have more chances to extend its
11243 lifetime, and therefore be more durable.

11244 Characteristics of the products in the scope

11245 An upgradable item of textile apparel may be one with modular components –such as detachable or
11246 interchangeable parts- that allow for updates and style changes without replacing the entire item. Although
11247 some upgradable products and designs may exist today, these type of modular designs are innovative, but not
11248 yet part of the mainstream market. Despite this lower relevance in the market today, upgradability is one of
11249 the possible strategies that can contribute to product lifetime extension. Upgradability is mostly associated with
11250 products made of physically durable fabrics.

11251 Environment

11252 Possibly due to the lower relevance in the market today, the authors have not found any study that highlights
11253 the potential environmental benefits of increasing the upgradability of textile products. In any case, if it is
11254 assumed that upgradability can contribute to extend the lifetime of textile apparel, it could also be concluded
11255 that upgrading products may have environmental benefits similar to the ones reported in section 13.10.1.1 on
11256 durability.

11257 Socioeconomic

11258 Although upgradable textile products can still be considered to be a niche in the market today, it can be
11259 concluded that they are also in opposition to business models based on fast trend turnovers and short
11260 production times. Modular products that can be upgraded will last longer and therefore prevent the sales of
11261 new products. Therefore, promoting the upgradability of products may affect negatively those manufacturers
11262 whose strategy is based on such business models; whereas it could affect positively manufacturers which focus
11263 on their upgradability.

11264 There is currently no evidence that suggests that increasing the upgradability of products may stimulate the
11265 creation of alternative business models based on renting or sharing. It could contribute to the creation of jobs
11266 on the clothing repair sector, which could undertake some of the activities related with product upgrade that
11267 cannot be carried out by consumers (replacing older by upgraded modules, for instance).

11268 Possibly due to the lower relevance in the market today, the authors have not found any evidence on the
11269 willingness of consumers to purchase upgradable textile products. The price to upgrade a product, compared
11270 with the price of a new product, may also be a factor for the willingness of consumers to upgrade textile apparel.

11271 Qualitative assessment of relevance

11272 Based on the above, upgradability is considered a relevant product aspect in the context of textile apparel.

11273 **13.10.1.5**Repairability

11274 Repairability is the ability of a defective product or waste object to return to a condition where it fulfils its
11275 intended use. In the context of textile apparel, repairability is directly linked with durability. One of the factors
11276 that makes a product more durable is its potential to be repaired. Repair makes it possible to increase product
11277 lifespans as well as value creation (Laitala and Klepp, 2021b). Either as an activity carried out by the consumer
11278 or as a service provided by a company, repair is another way in which to extend use-time of textile apparel
11279 (Piippo et al., 2022b).

11280 Characteristics of the products in the scope

11281 The repairability of textile apparel can be influenced at the design stage, for instance, considering their ease of
11282 disassembly, ensuring that products can be taken apart to replace worn or damaged components (Cooper and
11283 Claxton, 2022c).

11284 The presence of priority parts has been mentioned in previous sections on durability and reliability. Priority parts
11285 are particularly relevant for repairability, since these are the components with the highest frequency of failure
11286 and functional relevance. In the case of textile apparel, a possible list of priority parts (non-exhaustive)
11287 comprises fabrics, zips, buttons and embellishments such as sequins, gems and beads. It is worth noting that
11288 relatively few pieces of textile apparel with these components have failures, which suggests that the relevance
11289 of repairability is lower than, for instance, the relevance of reliability. Missing buttons and broken zips are the
11290 main problems, followed by missing embellishments (Cooper and Claxton, 2022c).

11291 Environment

11292 Repair of textile apparel has environmental benefits. As highlighted in (Luo et al., 2023c), the frequent use of a
11293 product will cause wear and tear and shorten its service life, while the repair process will extend the service life
11294 of the product and consequently reduce the purchase of new products and excessive waste of resources. It
11295 must be noted that providing spare parts for repairability increases the environmental impact of the product, if
11296 such parts are finally not used.

11297 Socioeconomic

11298 Currently, the professional services used for repair of clothing are companies mostly independent of the
11299 production and sales of clothes (Laitala and Klepp, 2021b). A rise in the availability of more repairable clothing
11300 could stimulate the creation of business related with clothing repair.

11301 Repair could also be stimulated by the same product manufacturers, either providing clothing repair as a service,
11302 or providing information to consumers on how to carry out simple repairs. Some brands offer more service-
11303 oriented solutions, such as repair and alterations, while others are designing modular or more adaptable clothing
11304 (Goworek et al., 2020).

11305 In fact, the number of repairs conducted privately is at least three times more frequent than professional
11306 repairs, since minor mending of textiles often only requires needle and thread, a limited amount of time and
11307 only rather basic skills (Laitala and Klepp, 2021b).

11308 Repair of clothing carried out by consumers is diminishing, due to the current low skills of consumers, influenced
11309 by the scarcity of available time to carry out repairs, the lack of appropriate equipment and price (Finnish
11310 Ministry of the Environment, 2023b). Therefore, only 11% of consumers consider repairability to be important
11311 when purchasing apparel items such as coats and jackets (Consumers, Health, Agriculture and Food Executive
11312 Agency. et al., 2018).

11313 The price of repair is a fundamental factor in the case of textile apparel. The price of repairing clothes depends
11314 a lot on the type of textile apparel and repair needed. A general trend observed is that consumers are not willing
11315 to pay much for the repair of textile apparel, and especially not for low-priced textile apparel. Consumers
11316 compare the price to a new textile apparel, which often is the same or even lower than repair or adjustments
11317 done by the tailor (Laitala and Klepp, 2021b). The authors have not found specific data regarding the willingness
11318 of consumers to pay for the repair of textile apparel. Using as a reference values from other product groups,
11319 the willingness to pay for repairs of small electronics is between 20% and 40% of the replacement cost
11320 (Cordella, Sanfeliix, et al., 2019; Svensson-Hoglund et al., 2021).

11321 In essence, clothes would be worn longer if repair services were cheaper (or products more expensive): 27% of
11322 consumers agree that they would wear their textile apparel longer if they were better at repairing them (Laitala
11323 and Klepp, 2021b). For those few consumers that do carry out repairs, it is worth noting that textile apparel has
11324 a much higher share of successful repairs than other products such as electrical appliances.

11325 Qualitative assessment of relevance

11326 Based on the above, repairability is considered a relevant product aspect in the context of textile apparel.

11327 **13.10.1.6**Possibility of maintenance

11328 The possibility of maintenance is the ability of a product to be kept in a condition where it is able to fulfil its
11329 intended purpose through one or more actions. In the context of textile apparel, it consists of the set of activities
11330 that a consumer carries out, fundamentally during the use phase of the product, in order to maintain it in a
11331 condition that satisfies all their needs associated to the product. This can include activities such as washing,
11332 drying, ironing, folding, storing, wearing in specific environments and conditions, etc.

11333 Characteristics of the products in the scope

11334 Washing is potentially the most relevant maintenance activity for textile apparel, which are prone to getting
11335 dirty or soiled. The ability to remove dirt and stains can be influenced by design, for instance by the choice of
11336 materials. Inherent fibre properties affect the soiling characteristics of textile apparel. Cotton gets dirty easily,
11337 but can be washed efficiently and thus cleaned. In contrast, wool resists staining and develops less odour, but
11338 stains are difficult to remove (Laitala and Klepp, 2018b). Similar trade-offs can be expected for other
11339 maintenance activities such as drying or ironing.

11340 Environment

11341 Textile apparel maintenance activities, because they are repeated often and by all people in the world,
11342 constitutes a significant environmental impact. Washing, drying and wrinkle removal are repeated on a regular
11343 basis. The longer an item of textile apparel is used, the more important this phase will be for the overall
11344 environmental impact of the textile apparel (Laitala and Klepp, 2018b).

11345 Maintenance can contribute to extend the lifetime of products and therefore reduce their environmental impact.
11346 It is important for keeping products in use and for motivating consumers to repair them when needed (Laitala
11347 and Klepp, 2021b). Extending textile apparel lifetimes should also take into account how more durable textile
11348 apparel items can be cared for sustainably by consumers, for example by avoiding dry-clean only fabrics
11349 (Goworek, Lynn Oxborrow, et al., 2020).

11350 Socioeconomic

11351 Manufacturers of textile apparel emphasize that optimal care and maintenance is a significant factor for
11352 extending product life. For them, it is important to determine which products customers need, but also how they
11353 use and care for these products. Since it is often the customers who care for the products, it is important for
11354 companies to share information on correct maintenance (Piippo et al., 2022b).

11355 The possibility of maintenance can also be a factor in take-back services. In such schemes, companies maintain
11356 the products they sold, in order to sell them again, enabling a new life for the product with a new consumer
11357 (Piippo et al., 2022b).

11358 Similarly to previous product aspects, the possibility of maintenance is in opposition with business models based
11359 on fast trend turnovers and short production times. A product which is easier to be maintained will delay the
11360 sale of new products, delaying their replacement. Therefore, promoting the possibility of maintenance of
11361 products may affect negatively those manufacturers whose strategy is based on such business models;
11362 whereas it could affect positively manufacturers which focus on product maintenance.

11363 Maintenance of textile products relies mostly on users. Most of the laundering and storing of textiles is carried
11364 out by the owner of the apparel. Manufacturers already provide maintenance instructions to users on how to
11365 wash and take care of textile products, on a voluntary basis. As pointed out in Section 6.3.2, a significant
11366 percentage of users follow these care instructions, but the adherence diminishes after the initial wash.

11367 Qualitative assessment of relevance

11368 Based on the above, possibility of maintenance is considered a relevant product aspect in the context of textile
11369 apparel.

11370 **13.10.1.7**Possibility of refurbishment

11371 The possibility of refurbishment is the ability of a product or a discarded product to be prepared, cleaned, tested,
11372 serviced and, where necessary repaired to restore its performance or functionality within the intended use and
11373 range of performance originally conceived at the design stage at the time of the placing of the product on the

11374 market. In the context of textile apparel, it can be interpreted as a very similar –or equivalent– aspect to
11375 repairability. In essence, when a product is repaired, it may also follow the steps described in the definition of
11376 possibility of refurbishment (preparing, cleaning, testing, servicing and repairing). As in the case of repair, the
11377 possibility of refurbishing is directly linked with durability.

11378 Characteristics of the products in the scope

11379 As in the case of repairability, the possibility of refurbishing of textile apparel can be influenced at the design
11380 stage, for instance, considering their ease of disassembly, ensuring that products can be taken apart to replace
11381 worn or damaged components (Claxton and Kent, 2020). The presence of priority parts has been mentioned in
11382 previous sections on durability, reliability and repairability. Priority parts are also particularly relevant for the
11383 possibility of refurbishing, since these are the components with the highest frequency of failure and functional
11384 relevance, and therefore the parts which more likely may need to be substituted in a potential refurbishing
11385 activity. Refurbishment is strictly connected to the physical durability of priority parts and the price related to
11386 refurbish a used product or waste rather than manufacturing a new one.

11387 Environment

11388 Possibly due to the similarities with product repair, the authors have not found scientific literature highlighting
11389 the potential environmental benefits of textile apparel refurbishing.

11390 Socioeconomic

11391 Similarly to the case of previous product aspects, the possibility of refurbishment is in contrast with the business
11392 models based on fast trend turnovers and short production times. A product that can be refurbished will reduce
11393 the opportunity of selling new items. Therefore, promoting the possibility of maintenance and refurbishment of
11394 products may affect negatively those manufacturers whose strategy is based on such business models;
11395 whereas it could affect positively manufacturers which focus on product maintenance and refurbishment.

11396 A rise in the availability of products which are easier to be refurbished could stimulate the creation of business
11397 related with textile apparel repair.

11398 Qualitative assessment of relevance

11399 Based on the above, possibility of refurbishment is considered a relevant product aspect in the context of textile
11400 apparel.

11401 **13.10.1.8** Presence of substances of concern

11402 The presence of substances of concern should be understood as the existence of chemicals in products that for
11403 reasons of their intrinsic hazards pose risks to human health or the environment, or that otherwise negatively
11404 affects the reuse and recycling of materials in the product in which they are present. This may bring about the
11405 need to consider regulatory action, improved management and/or substitution with safer alternatives, whenever
11406 feasible. ESPR specifically requires to track the presence of these substances in products.

11407 In ESPR, a substance of concern means a substance that:

- 11408 (a) meets the criteria laid down in Article 57 of Regulation (EC) No 1907/2006 and is identified in
11409 accordance with Article 59(1) of that Regulation;
- 11410 (b) is classified in Part 3 of Annex VI to Regulation (EC) No 1272/2008 in one of the following hazard
11411 classes or hazard categories:
 - 11412 i. carcinogenicity categories 1 and 2;
 - 11413 ii. germ cell mutagenicity categories 1 and 2;
 - 11414 iii. reproductive toxicity categories 1 and 2;
 - 11415 iv. endocrine disruption for human health categories 1 and 2;
 - 11416 v. endocrine disruption for the environment categories 1 and 2;
 - 11417 vi. persistent, mobile and toxic or very persistent, very mobile properties;
 - 11418 vii. persistent, bioaccumulative and toxic or very persistent, very bioaccumulative
11419 properties;
 - 11420 viii. respiratory sensitisation category 1;

- 11421 ix. skin sensitisation category 1;
- 11422 x. hazardous to the aquatic environment — categories chronic 1 to 4;
- 11423 xi. hazardous to the ozone layer;
- 11424 xii. specific target organ toxicity — repeated exposure categories 1 and 2;
- 11425 xiii. specific target organ toxicity — single exposure categories 1 and 2;
- 11426 (c) is regulated under Regulation (EU) 2019/1021; or
- 11427 (d) negatively affects the reuse and recycling of materials in the product in which it is present;

11428 Characteristics of the products in the scope

11429 As already pointed out in Section 3.3.2, the raw materials extraction and manufacturing of textile apparel involve
 11430 the use of a wide variety of chemical substances and mixtures, which includes pesticides, solvents, surfactants,
 11431 dyes, pigments, stain repellents, flame retardants or biocides, among others, as well as water. The demand of
 11432 chemical compounds by the textile industry is estimated to use around 25% of global chemical production (Raj
 11433 et al., 2022). Many of the chemicals used during textile manufacturing are associated with spinning and weaving
 11434 and wet processing. A single European textile-finishing company uses over 466 g of chemicals per 1 kg of
 11435 textile (Niinimäki et al., 2020).

11436 Chemicals are used in the manufacturing of textile apparel with a specific purpose, such as providing a certain
 11437 level of resistance to abrasion, adding colour, waterproofing, etc. The presence of substances of concern can be
 11438 influenced at the design stage, via substitutions by safer alternatives (often with trade-offs) (Zhang and Hale,
 11439 2022).

11440 Environment

11441 Chemicals have an influence in almost all lifecycle stages of a textile product. Chemicals used during
 11442 manufacturing are often discharged or results in releases and get in direct contact with soil and water bodies.
 11443 Consumers are directly exposed to chemicals in the textile during the use phase. Chemicals may also have an
 11444 influence at end-of-life, either hindering processes such as recycling or being released to the environment if
 11445 apparel is landfilled or incinerated.

11446 Eliminating or reducing the presence of substances of concern has been highlighted as one of the key areas
 11447 with potential of improvement regarding ecodesign (Ellen MacArthur Foundation, 2017; Bauer, Watson, Gylling,
 11448 Remmen, Lysemose, Catharina Hohenthal, et al., 2018; Niinimäki et al., 2020). For instance, natural cotton fabric
 11449 dyeing using dyes of plant origin (white onion) can reduce the impacts on human health and ecosystems
 11450 (Amicarelli et al., 2022).

11451 Socioeconomic

11452 Companies in the textile apparel sector aim to reduce production costs through manufacturing in locations with
 11453 lower production costs and where often environmental regulation is laxer or pollution-mitigating technologies
 11454 are not legally required. This approach to manufacturing often leads not only to high environmental impacts
 11455 from use of chemicals but increased health risks for factory workers, cotton farmers and fashion consumers
 11456 (Niinimäki et al., 2020).

11457 Despite being in direct contact with chemicals when wearing apparel, 60% of respondents to a survey in the EU
 11458 perceived chemicals in apparel fabrics as representing a minimal risk (European Commission, 2019).

11459 Qualitative assessment of relevance

11460 Based on the above, presence of substances of concern is considered a relevant product aspect in the context
 11461 of textile apparel.

11462 **13.10.1.9**Energy use and energy efficiency

11463 Energy use can be defined as the total use of energy in all lifecycle stages of a product. Energy efficiency is
 11464 the ratio of output of performance, service, goods or energy to input of energy.

11465 Characteristics of the products in the scope

11466 Almost all life-cycle stages of apparel have significant energy consumption, particularly the extraction of fibres
 11467 such as silk; production processes such as spinning, knitting and weaving; or thermal treatments during
 11468 production (see Section 3.3.1). Although manufacturing covers about 70-80% of total lifecycle energy

11469 consumption (Sandin, Roos, Spak, et al., 2019; Quantis, 2021), energy is also consumed in the use phase for
11470 laundering, ironing and drying.

11471 There is a wide range of materials available for manufacturers, with different performance in terms of energy
11472 use and energy efficiency. Considering the manufacturing stage, some fibres are less energy intensive than
11473 others. However, reductions in energy use may come with trade-offs in other properties of the fibre (Niinimäki
11474 et al., 2020).

11475 Considering the use phase, the electricity consumed during domestic washing amounts to 2% of household
11476 usage, while tumble drying accounts for 4.5% (Zhang et al., 2022). There are also examples of energy efficiency
11477 improvement potential areas in the use phase, described in (Zhang et al., 2022). For instance, products made
11478 from hygroscopic fibres require higher energy in tumble drying than their hydrophobic fast-drying synthetic
11479 counterparts (Zhang et al., 2022). Some fibres require lower temperatures for laundering than others (less
11480 energy use). Some materials require ironing to achieve a minimum aesthetic performance, whereas others do
11481 not need to be ironed. However, different fibres will provide different performance and aesthetics. Design
11482 strategies tend to be aimed at reducing rather than eliminating the negative impacts of the use phase, typically
11483 in the design of clothes for less frequent washing and use of lower laundering temperatures to prolong textile
11484 apparel life (Claxton and Kent, 2020b)

11485 There is room for improvement in the energy efficiency of textiles manufacturing as well. In (Zhang et al., 2022),
11486 several examples are provided: low-twist spinning technology introduces false twist between the front roller
11487 and the yarn guide, and can achieve a twist reduction of 20-40%; significant energy saving in air humidification
11488 can be achieved by using energy-efficient nozzles and variable-frequency drives based on the real-time
11489 humidity conditions in the spinning and weaving process; or high-volume, low-pressure nozzles can save up to
11490 26% of energy thanks to their optimized nozzle geometry (Zhang et al., 2022). It is worth noting that, in the
11491 context of the Industrial Emissions Directive, the Reference Document for the Textiles Industry (Roth et al.,
11492 2023) addresses energy use for installations located in the EU and points out the Best Available Techniques.

11493 Environmental

11494 As pointed out in Section 5.4.3, considering the manufacturing stage, the energy consumption in the global
11495 textile industry was estimated to be 2% of the global energy consumption. Section 3.3.1 reports that almost all
11496 stages of the textile apparel value chain are energy demanding. Indirectly related with energy, the apparel
11497 industry is reported to be responsible for about 6.5% of global GHG emissions (Niinimäki et al., 2020).

11498 The substantial consumption of electricity in the use stage of textiles was also revealed to have a significant
11499 impact on the environment due to repeated use and care operations, in some cases even exceeding the
11500 contribution of the production stage (Luo et al., 2023c).

11501 Socioeconomic

11502 As mentioned in Section 5.4.3, the cost of energy plays an important role in the textile industry. In fact, the
11503 increase of the cost of energy in the EU in 2022 negatively affected the EU textile production. Moreover, the
11504 national energy strategies influence the establishment of textile industry focussing on specific stages of the
11505 value chain.

11506 Textile apparel are generally not perceived by consumers as energy-related products. Possibly because of that,
11507 the authors have not found any evidence on the willingness of consumers to purchase energy efficient textile
11508 products.

11509 Users can have an influence on the energy consumed during the lifecycle of apparel, mostly in the laundering,
11510 ironing and drying activities. Following the instructions available in labelling in terms of water temperature can
11511 help to reduce energy consumption. Reducing the frequency of ironing may also save energy. Substituting the
11512 use of tumble driers by air-drying (when possible) can also have a positive effect.

11513 Qualitative assessment of relevance

11514 Based on the above, energy use and energy efficiency is considered a relevant product aspect in the context of
11515 textile apparel.

11516 **13.10.1.10** Water use and water efficiency

11517 Water use is the use of water in all lifecycle stages of a product. Water efficiency is the ratio of output of
11518 performance, service, goods to input of water.

11519 Characteristics of the products in the scope

11520 Apparel manufacturing requires enormous volumes of water in fabric production (Jia et al., 2020). Typically,
 11521 70-250 l of water are used for every kilogram of finished textiles (Zhang et al., 2022). Water is also consumed
 11522 during the use phase in laundering activities.

11523 There is a wide range of materials available for manufacturers, with different performance in terms of water
 11524 use and water efficiency. Cotton and hemp are the most water-demanding among textile fibres, whereas
 11525 polyester and polypropylene need the least amount of water. However, reductions in water use may come with
 11526 trade-offs in other properties of the fibre (Niinimäki et al., 2020).

11527 There is room for improvement in the water efficiency of textiles manufacturing. In (Zhang et al., 2022), several
 11528 examples are provided: a foam-laying technique has been studied as a replacement for the wet-laying non-
 11529 woven process, using 20% of the water and saving energy simultaneously; for synthetic fibres, colourless
 11530 polymers can be structurally coloured with nano-sized pigment inclusions via spin dyeing, achieving 50% water
 11531 savings; recycling wastewater generated from spent dyeing and rinsing baths through catalytic ozonation with
 11532 carbon aerogel is a method to minimize water consumption.

11533 There are also examples of water efficiency improvement potential areas in the use phase (for laundering),
 11534 described in (Zhang et al., 2022). For instance, products made from hygroscopic fibres require more water than
 11535 their hydrophobic fast-drying synthetic counterparts. It is worth noting that, in the context of the Industrial
 11536 Emissions Directive, the Reference Document for the Textiles Industry (Roth et al., 2023) addresses water use
 11537 for installations located in the EU and points out the Best Available Techniques.

11538 Environmental

11539 Considering the initial stages of textile apparel production, water consumption for materials extraction and
 11540 manufacturing is estimated to account for 4% of global freshwater extraction (Ellen MacArthur Foundation,
 11541 2017). Just cotton is estimated to account for 2.5% of water consumed globally every year (Amicarelli et al.,
 11542 2022). In the EU, the production of textile apparel, footwear and household textiles purchased in 2020 was
 11543 around 4 000 million m³ of blue water⁽³³⁷⁾.

11544 Considering the use stage of textile apparel, the substantial consumption of water in the use stage of textiles
 11545 was revealed to have a significant impact on the environment due to repeated use and care operations, in some
 11546 cases even exceeding the contribution of the production stage (Luo et al., 2023c).

11547 Socioeconomic

11548 As described in Section 5.7, the global value chain of textile apparel causes most of its environmental impacts
 11549 in the production stages, and most of these stages occur outside of the EU. In particular, in countries that allow
 11550 production at lower costs (due to poor labour conditions and less stringent measures on environmental
 11551 protection). This is particularly relevant for water consumption, an aspect which is directly related with the
 11552 specific location. For instance, cotton has the highest water footprint of any fashion fibre (Niinimäki et al., 2020).
 11553 For making a cotton t-shirt, 2 700 litres of water are used (Shirvanimoghaddam et al., 2020).

11554 Textile apparel are generally not perceived by consumers as water-related products. Possibly because of that,
 11555 the authors have not found any evidence on the willingness of consumers to purchase water efficient textile
 11556 products. However, water is also consumed during the use phase in laundering activities, which are mostly
 11557 influenced by the users. As highlighted in previous sections, different washing habits are observed across
 11558 countries, with factors such as age and societal norms also having an influence. User perceptions about hygiene
 11559 and convenience may impact washing frequency, influencing the consumption of water during the use phase.
 11560 There is also a relevant interaction with laundry detergents (some of them might require more water than
 11561 others to achieve the same level of cleanliness).

11562 Qualitative assessment of relevance

11563 Based on the above, water use and water efficiency is considered a relevant product aspect in the context of
 11564 textile apparel.

11565 **13.10.1.11** Resource use and resource efficiency

11566 Resource use is the use of raw materials, mainly abiotic (minerals, metals, fossil fuels), in all lifecycle stages.
 11567 It can also include biotic resources such as land, air, ecosystems. The use of natural resources can be accounted

³³⁷ Blue water refers to fresh surface and groundwater

11568 for as the volumes of resources consumed (materials) or used (land, air, ecosystems), or the impacts derived
11569 from the use of resources. Water and energy are not considered within resources under the scope of ESPR given
11570 these are addressed as separate product aspects. Resource efficiency is the ratio of output of performance,
11571 service, goods to input of resources, raw materials, air, land, soil and ecosystem services.

11572 Characteristics of the products in the scope

11573 The textile apparel industry is resource-intensive (Piippo et al., 2022b). Most of these resources are consumed
11574 in the raw material extraction and manufacturing stages. This product aspect is therefore related with product
11575 parameters, reported in Annex I to ESPR, such as 'use or content of sustainable renewable materials' and
11576 'reduction of materials consumption', among others.

11577 Considering textile manufacturing, there are a wide variety of techniques, each of them with a different level
11578 of resource efficiency. Taking cotton as an example, it may be produced following organic farming, conventional
11579 farming or regenerative agriculture and using different quantities of land, water, chemicals and energy
11580 according to the climate and the ecosystem where it is cultivated. Each specific cultivation practice in a given
11581 location has a specific crop yield (or efficiency in the use of resources). Similarly, manmade cellulosic fibres can
11582 be produced using cellulose from recycled wood, agricultural waste, virgin wood coming from sustainably
11583 managed forests, virgin wood coming from deforestation-free forests or from unknown sources, etc. All these
11584 options use different resources and affect different ecosystems.

11585 Resource efficiency can also be linked with the process of reducing the weight of products to improve their
11586 environmental performance. This practice, aimed at decreasing material use throughout the product lifecycle,
11587 it is also known as lightweighting. Denim fabrics, for instance, can be made in many weights. Light denim will
11588 be suitable for dresses or shirts where drape, softness and flexibility are required. Heavy denims are typically
11589 used for blue jeans trousers and skirts (Elmogahzy, 2020). Although light weights use less material, they also
11590 provide different properties to textile apparel. Heavier weight fabrics are generally more durable and less
11591 susceptible to abrasion and wear, while fabrics with a tighter, more compact construction are also more
11592 resistant to damage (Cooper and Claxton, 2022c). Therefore, the authors consider that lightweighting is not a
11593 relevant product parameter for textile apparel.

11594 Environmental

11595 The textile industry is the fifth industrial sector for primary use of materials, mainly related to the extensive
11596 production of natural origin fibres and man-made cellulosic fibres. The estimates of textile fibre production
11597 were around 116-124 million tonnes in 2022 (Textile Exchange, 2022b).

11598 Socioeconomic

11599 Similarly to energy efficiency and water efficiency, resource efficiency does not appear to be a relevant aspect
11600 influencing the behaviour of consumers when purchasing textile apparel.

11601 Qualitative assessment of relevance

11602 Based on the above, resource use and resource efficiency is considered a relevant product aspect in the context
11603 of textile apparel.

11604 **13.10.1.12** Recycled content

11605 Recycled content is the proportion, by mass, of recycled material in a product or packaging. In the case of textile
11606 apparel, it is generally expressed with reference to the amount of fibres, which are at least 80% of the weight
11607 of the product.

11608 Characteristics of the products in the scope

11609 The possibility to use recycled material in textile apparel has been mentioned in some studies as a relevant
11610 product aspect to consider (Ellen MacArthur Foundation, 2017; Bauer, Watson, Gylling, Remmen, Lysemose,
11611 Catharina Hohenthal, et al., 2018; Niinimäki et al., 2020).

11612 When considering the possibility of having recycled content in a product, a key aspect is the availability of
11613 recycled material in the market. In the case of textile apparel, as pointed out in Section 3.3.2, the EU generated
11614 about 12.6 Mt of textile waste in 2019, which could be recycled and reintroduced in the manufacturing process
11615 of textile apparel. In Section 5.4.1, it has also been mentioned that a 30% increase in textile waste is expected
11616 for 2030-2035, so the supply of feedstock for recycled material seems guaranteed for the coming years,
11617 provided that the corresponding collection and sorting system is well-established. In addition, other products

11618 such as PET bottles can also be recycled into fibres. In 2023, the industry recycled 8.9 million tons of polyester,
11619 with 98% coming from bottles (Textile Exchange, 2024b).

11620 However, when designing a product with recycled content it must be taken into account that, depending on the
11621 recycling technology, recycled fibres tend to be of lower performance than their virgin equivalents and are
11622 normally limited to single fibre content fabrics (Claxton and Kent, 2020b). For instance, a maximum of 20%
11623 post-consumer mechanically recovered cotton fibres can be blended with virgin cotton before strength is
11624 compromised (Niinimäki et al., 2020). There is thus room for improvement to increase the recycled content in
11625 textile apparel.

11626 Chemical recycling works by breaking down fibres through a chemical process to the level of a polymer. The
11627 process preserves fibres better than the mechanical recycling and is, therefore, anticipated to enable textile
11628 apparel to be produced with higher percentages of recycled fibres (Niinimäki et al., 2020).

11629 Environmental

11630 Some studies highlight that the use of recycled materials in textile apparel reduces their environmental impact.
11631 For instance, by using recycled cotton for spinning it is possible to decrease 60% of the emitted CO₂eq, reduce
11632 the consumption of oil equivalent by 11% and water by almost 80%. Similarly, the use of recycled cotton fibre
11633 has a huge potential to reduce environmental impacts, since it is possible to avoid the production of virgin
11634 cotton, with a reduction of up to 98% in water use. Substituting primary by secondary materials in a polyester
11635 jacket shows a reduction in impacts between 50%-80% in the categories of photochemical smog formation,
11636 human toxicity and water scarcity (Amicarelli et al., 2022).

11637 Socioeconomic

11638 Increasing the content of recycled material in textile apparel could contribute to job creation in different areas.
11639 For instance, in the collection and sorting of textiles, as well as in transport and recycling operations. The
11640 increase in recycled content might also incentivise the recycling of end-of-life waste material.

11641 In contrast with other aspects such as durability or repairability, consumers give little importance to the use of
11642 recycled materials in apparel (11%, as highlighted in Section 6.2.7).

11643 Qualitative assessment of relevance

11644 Based on the above, recycled content is considered a relevant product aspect in the context of textile apparel.

11645 **13.10.1.13** Possibility of remanufacturing

11646 The possibility of remanufacturing is the possibility of producing through actions a new product from objects
11647 that are waste, products or components and through which at least one change is made that may affect the
11648 performance, purpose or type of the product.

11649 In the context of textile apparel, it must be noted that product attributes that make it fit for repair (such as
11650 ease of disassembly) also make it fit for remanufacturing.

11651 Characteristics of the products in the scope

11652 As in the case of repairability, the possibility of remanufacturing of textile apparel can be influenced in the
11653 design stage, for instance, considering their ease of disassembly, ensuring that products can be taken apart to
11654 replace worn or damaged components (Claxton and Kent, 2020b).

11655 The presence of priority parts has been mentioned in previous sections on durability, reliability, repairability and
11656 possibility of refurbishing. Priority parts are also particularly relevant for the possibility of remanufacturing,
11657 since these are the components with the highest frequency of failure and functional relevance, and therefore
11658 the parts which more likely may need to be substituted in a potential remanufacturing activity.

11659 Environment

11660 Possibly due to the similarities with product repair, the authors have not found scientific literature highlighting
11661 the potential environmental benefits of textile apparel remanufacturing.

11662 Socioeconomic

11663 Similarly with previous product aspects, the possibility of remanufacturing is in opposition with business models
11664 based on fast trend turnovers and short production times. A product which is easier to be remanufactured will
11665 reduce the opportunity of selling new items. Therefore, promoting the possibility of remanufacturing of products

11666 may affect negatively those manufacturers whose strategy is based on such business models; whereas it could
11667 affect positively manufacturers which focus on product remanufacturing.

11668 A rise in the availability of products which are easier to be remanufactured could stimulate the creation of
11669 business related with textile apparel repair. The cost of remanufacturing may also be a significant factor for
11670 the expansion of this activity.

11671 Qualitative assessment of relevance

11672 Based on the above, possibility of remanufacturing is considered a relevant product aspect in the context of
11673 textile apparel.

11674 **13.10.1.14** Recyclability

11675 Recyclability is the ability of waste materials originating from products to be reprocessed into products,
11676 materials or substances whether for the original or other purposes. It includes the reprocessing of organic
11677 material but does not include energy recovery and the reprocessing into materials that are to be used as fuels
11678 or for backfilling operations.

11679 Characteristics of the products in the scope

11680 It is possible to recycle textile apparel, either via mechanical, chemical or enzymatic technologies. However, due
11681 to different fibre material combinations, auxiliaries such as buttons and zippers, and various thermal
11682 treatments, fibre to fibre recycling is a complex and hardly practiced treatment (Eppinger et al., 2022).
11683 Mechanical recycling is the most common recycling technology nowadays.

11684 Recycling may be closed loop (when materials are recycled back into the same or similar products) or open loop
11685 (when materials are converted into different types of products). In the case of textile apparel, most of the
11686 recycling processes today are open loop. As the length of the fibres and the constituent molecules are reduced
11687 by wear and laundry, fabric and fibre recycling typically yields materials of lower performances than materials
11688 made from virgin fibres (Sandin and Peters, 2018). For instance, materials recycled via mechanical recycling
11689 can be used to produce insulating materials for cars and buildings, fibre-reinforced composites and disposable
11690 non-woven products such as sanitary wipes, napkins and diapers (Zhang et al., 2022). Moreover, there are many
11691 opportunities to use cotton and textile waste for various applications including composites, microbial fuel cell,
11692 potassium-ion exchange, biochar applications, sound absorbents, thermal insulation, EMI shielding, etc.
11693 (Shirvanimoghaddam et al., 2020).

11694 Mechanically recycled fibres generally show weakened properties (Zhang et al., 2022). This process gives the
11695 best results when the material still has some level of performance. If very poor (high wear and tear) material
11696 is mechanically recycled, up to 30-50% of virgin material needs to be added so that the performance of the
11697 material becomes sufficient (Piippo et al., 2022b). This can also be observed in the case of denim products. The
11698 common approach to recycle denim products is through fabric shredding into fibres, which can be used in many
11699 applications. The success will primarily depend on the characteristics of fibres obtained from the recycling
11700 process (Elmogahzy, 2020).

11701 There is room for improvement in the possibilities of recycling textile apparel and this may be influenced at the
11702 design stage. For instance, most apparel products currently in use are fibre blends of multiple polymers.
11703 Recycling is complicated by textile apparel being constructed of fibre blends, which require separation (Niinimäki
11704 et al., 2020). Mono-materiality in textiles has long been recognized for its great recycling efficiency without
11705 need for disassembly, making it one of the most effective design strategies for proactive material recovery and
11706 perhaps the basis for a new textile paradigm (Zhang et al., 2022).

11707 Environmental

11708 As reported in Solis et al. (2024), recycling is the second-most preferred pathway for most environmental impact
11709 categories after preparing for reuse. It has been proved to be more beneficial than landfilling and incineration.
11710 Some studies highlight that the use of recycled materials in textile products decreases their environmental
11711 impact (Cai and Choi, 2020; Niinimäki et al., 2020; Amicarelli et al., 2022). The environmental analysis of section
11712 13.10.1.12 on recycled content provides some insight on the benefits of recycling textile apparel. Upcycling
11713 textile waste into new products or even recycling textile waste in another sector (e.g. as part of composite
11714 material in the automotive industry) is an important approach in a circular economy (Piippo et al., 2022b).
11715 Socioeconomic

11716 In Section 5.4.1, a summary was provided in terms of availability of textile recycling plants. Europe, Asia and
11717 North America host 50%, 33% and 11% of global textile recycling plants, respectively. Most of these plants are

11718 able to process textile products made of many textile fibres. Increasing the recyclability in textile products could
11719 contribute to job creation in different areas: textile collection, sorting, transport and recycling.

11720 However, at global level, only 12% of textile materials are currently recycled (Zhang et al., 2022). Only 15% of
11721 post-consumer textile waste was collected separately for recycling purposes in 2015, and less than 1% of total
11722 production was recycled in closed loop. Most of the recycled textiles were recycled into other, lower-value
11723 applications, such as insulation material, wiping cloths and mattress stuffing (Niinimäki et al., 2020).

11724 In Solis et al (2024), the authors investigated the socioeconomic impacts of different waste management
11725 scenarios for textiles. They concluded that mechanical recycling of cotton, polyester, polycotton and polyamide
11726 achieved lower costs than incineration. However, if other recycling pathways (such as mechanical recycling of
11727 viscose, open-loop recycling, chemical recycling of cotton, polyester and polycotton) are to be competitive from
11728 the socio-economic perspective relative to incineration, technology costs need to decrease significantly.

11729 In terms of employment, preparing for reuse and recycling create significantly more jobs than incineration and
11730 landfilling (Solis et al., 2024).

11731 Recyclability does not seem a relevant aspect for consumers when purchasing textile apparel.

11732 It has been mentioned above that fibre blends make the process of recycling more complex. To overcome this,
11733 mono-material apparel and footwear engineered for mechanical recycling may be manufactured from
11734 thermoplastic materials via a combination of existing machinery and processes. However, the huge variation in
11735 consumer preference is not easily satisfied by a single material (Zhang et al., 2022).

11736 The recyclability of textile apparel can also be influenced by increasing their collection at end of life, an aspect
11737 where consumers also have an influence. Moreover, the Waste Framework Directive requires Member States to
11738 set up separate collection for textiles by 1 January 2025. It also promotes the establishment of Extended
11739 Producer Responsibility Schemes for textile products in several Member States.

11740 Qualitative assessment of relevance

11741 Based on the above, recyclability is considered a relevant product aspect in the context of textile apparel.

11742 **13.10.1.15** Environmental impacts

11743 Environmental impacts refer to any change to the environment, whether adverse or beneficial, wholly or partially
11744 resulting from a product during its life cycle.

11745 Characteristics of the products in the scope

11746 Textile apparel include a wide variety of products, including t-shirts, shirts and blouses, sweaters, jackets, pants,
11747 dresses and accessories, among others. These products have different applications and therefore a very diverse
11748 range of performance requirements. Related with the variety of products and applications, a wide variety of
11749 possible materials and manufacturing processes arise, each one providing different characteristics to textile
11750 apparel.

11751 This wide variety of possibilities in terms of products, materials and manufacturing processes causes that the
11752 nature of the environmental impacts is also wide. Some materials and processes might have a larger impact
11753 due to energy use and greenhouse gas emissions, whereas other have a larger impact due to water use, resource
11754 use, release of chemicals or generation of waste.

11755 Environmental aspects

11756 As highlighted in Section 3.3.2, textile apparel has relevant effects on the environment throughout its lifecycle,
11757 on different areas. For instance:

- 11758 — The textile industry's water consumption is estimated to account for 4% of global freshwater
11759 extraction (Ellen MacArthur Foundation, 2017).
- 11760 — In 2004, the energy consumption in the global textile industry was estimated to be equal to 2%
11761 of the global energy consumption. Additionally, the production of 1 kg of generic textile product
11762 was estimated to require about 126 MJ (about 35 kWh) of energy (Muthu, 2015).
- 11763 — Numerous life-cycle stages of textile apparel involve the use of chemical substances and mixtures.
11764 Some of them are pesticides, solvents, surfactants, dyes and pigments, water and stain repellents,
11765 flame retardants, biocides and many more (Ellen MacArthur Foundation, 2017).

- 11766 — In 2019, the EU generated about 12.6 Mt of textile waste, including post-industrial, pre-consumer
11767 and post-consumer waste, representing 11%, 3% and 86% of the total, respectively.
- 11768 — The European Environment Agency estimated that textile consumption in the EU in 2020 emitted
11769 121 million tonnes of greenhouse gases. Around 75% of the emissions occurred outside Europe,
11770 specifically in Asian countries (EEA et al., 2022).
- 11771 — Textile fragmentation is also a relevant aspect for the environmental impacts of textile apparel. It
11772 has been reported that one of the leading sources of microplastics pollution is the fragmentation
11773 of synthetic textiles (Boucher and Friot, 2017). Current patterns indicate that emissions of
11774 microplastics from textiles are projected to rise by approximately 22% by the year 2030 (DG ENV,
11775 2023).
- 11776 — The BREF on textiles (Roth et al., 2023) identifies and addresses the main pollutants during the
11777 manufacturing process of textile apparel, such as COD, SOx, NOx, energy consumption and water
11778 consumption.

11779 Socioeconomic

11780 Many of the impacts mentioned above occur outside the EU. Third countries producing textiles often have less
11781 stringent environmental and labour requirements. This allows manufacturers to reduce costs and offer final
11782 products at a lower purchase price than EU production, at the expense of environmental degradation and health
11783 issues of the workers, farmers and the communities working around the manufacturing plants. Promoting
11784 products with a lower environmental impact would contribute to the reduction of impacts across the whole
11785 value chain.

11786 As pointed out in Section 6.2.2, most of EU consumers argue that environmental impact of products is 'very' or
11787 'rather important' in their purchasing decisions (European Commission. Directorate General for Environment.,
11788 2023). Information is a key aspect to convey messages on the environmental performance of products: around
11789 82% of consumers believe there is insufficient information available on environmental aspects associated with
11790 apparel (European Commission, 2019).

11791 Qualitative assessment of relevance

11792 Based on the above, environmental impact is considered a relevant product aspect in the context of textile
11793 apparel.

11794 **13.10.1.16** Expected generation of waste

11795 The expected generation of waste is the generation of any substance, or object that the holder discards or
11796 intends or is required to discard.

11797 Characteristics of the products in the scope

11798 Waste is generated in different lifecycle stages of textile apparel. Depending on that, waste can be classified
11799 as (Huygens et al., 2023):

- 11800 • Post-industrial waste: waste generated during the manufacturing of textile products and their
11801 precursors.
- 11802 • Pre-consumer waste: waste generated at retail stages (e.g. discarded unsold textiles).
- 11803 • Post-consumer waste: textiles that have been disposed of after consumption and use by the
11804 citizen or end-users of commercial and industrial activities (hotel, healthcare, etc.), commonly
11805 referred to as household and commercial post-consumer textile waste, respectively.

11806 Therefore, waste is relevant in different stages of the textile product value chain. There seems to be room for
11807 improvement in the expected generation of waste. In terms of post-industrial waste, methods have been
11808 developed to design textile apparel that minimises cutting waste and puts nearly all offcuts into production
11809 (Niinimäki et al., 2020). Most of the leftover material from textile apparel manufacturing (post-industrial waste)
11810 ends up dumped or burned (Aus et al., 2021). It is technically feasible to increase the separate collection of
11811 textile products, especially those with highly valuable materials. Directive (EU) 2025/1892 requires Member
11812 States to ensure the separate collection of textiles by 2025. Moreover, a proposal for a targeted revision of this
11813 Directive is currently in co-decision and is expected to introduce more detailed collection and sorting
11814 requirements and the establishment of Extended Producer Responsibility (EPR) for certain textile and footwear
11815 products in Member States.

11816 Environmental aspects

11817 The impacts from the fashion industry include over 92 million tonnes of waste produced per year, accounting
11818 for up to 22% of mixed waste worldwide (Niinimäki et al., 2020). As mentioned in Section 3.3.2, the EU
11819 generated in 2019 around 12.6 Mt of textile waste. Around 86% of that waste is post-consumer waste. The
11820 current management of textile waste is considered suboptimal, as landfilling and incineration are the
11821 predominant treatment methods (Solis et al., 2024), releasing harmful substances and contributing to global
11822 warming, among other environmental impacts.

11823 In Europe, on average 37% of textiles waste are separately collected (Amicarelli et al., 2022). The majority of
11824 post-consumer waste that is separately collected is sent outside the EU as used products intended for re-use.
11825 However due to the saturation of the global market for such used products many items are instead discarded
11826 in third countries causing negative environmental and social impacts (Huygens et al., 2023).

11827 Strategies aiming at reducing the amount of waste generated, or at increasing the amount of textile waste
11828 collected and sorted at end of life, would have a significant benefit on the environment.

11829 Socioeconomic

11830 As pointed out in Section 5.6, the exports of used textiles from the EU reached 1 700 000 tonnes in 2019.
11831 Usually, these products are exported to Africa for a first screening in local markets, then down-cycled into
11832 industrial rags or filling, or re-exported to Asia for recycling. A significant amount of this waste ends up in Asian
11833 and South American landfills.

11834 Reducing the amount of waste that is sent to landfill could also stimulate the creation of jobs, related to the
11835 collection, sorting and recycling of textiles.

11836 The expected generation of waste is directly related with consumers. The apparent consumption of textile
11837 apparel has doubled after the opening of the EU market to products coming from China, as explained in section
11838 5.2. Section 6.4.1 already highlighted the main factors influencing disposal decisions (perceived quality being
11839 one of the key aspects). Material defects, inappropriate size, loss of shape, or not liking the item, are also
11840 important factors (Kleinhückelkotten et al., 2018b). Sharing with social circles, donating or swapping are
11841 common ways to dispose of apparel no longer being used. Throwing away usable apparel is generally perceived
11842 as a socially reproachable behaviour (despite this, large amounts of textile waste are still generated nowadays,
11843 as pointed out above).

11844 Qualitative assessment of relevance

11845 Based on the above, expected generation of waste is considered a relevant product aspect in the context of
11846 textile apparel.

11847

11848 **13.10.2 Supporting information on parameters affecting the physical durability**

11849 **Table 177** reports the main fibre characteristics playing a role in the physical durability of textile apparel.
11850 Additionally, **Table 178** and **Table 179** report definitions of yarn and fabric characteristics and main process
11851 parameters, respectively. This information supports the description of interlinks among several factors in the
11852 textile apparel production.

11853 **Table 180** reports the influence of fibre characteristics on yarn and fabric characteristics. **Fibre length**
11854 influences the strength, smoothness, and uniformity of the fabric, with longer fibres enhancing the physical
11855 performance of yarns and fabrics. **Fibre strength** is critical to the physical durability of the yarn and has a
11856 direct impact on the ability of the textile apparel to withstand wear and tear. **Fibre fineness** is important for
11857 producing finer yarns, which are essential for creating smoother, softer and more uniform products, enhancing
11858 the texture, comfort and appearance of textile apparel. **Fibre elasticity**, which is the ability of fibres to return
11859 to their original shape after stretching, is vital for the resilience and shape retention of the fabric. Additionally,
11860 the type of fibre, whether natural or chemical, plays a significant role. More details are reported in **Table 180**.

11861 Fibre characteristics are largely influenced by various external factors. **Environmental conditions** such as
11862 climate and soil characteristics significantly affect natural fibres, while the diet and health of animals impact
11863 the characteristics of fibres like wool and silk. **Chemical exposures**, including pesticides and pollutants, can
11864 degrade fibre characteristics. Additionally, the mechanical and chemical processing techniques used during fibre
11865 production can alter their strength, elasticity, and overall integrity. Finally, **moisture and temperature during**

11866 **the storage** of fibres plays a crucial role in preventing damage like mould growth or material degradation,
11867 ensuring fibres maintain their desired properties (Hearle and Morton, 2008) (Mishra, 2000) (Rahman et al.,
11868 2023).

11869 Understanding the influence of fibre characteristics on yarn, processing and fabric is essential for optimizing
11870 manufacturing processes and achieving desired textile performance and aesthetics.

11871

11872

Table 177. Description of the main fibre characteristics

Fibre characteristic	Definition	Description of natural fibres	Description of chemical fibres	Source
Fibre length	The length of the individual staple fibre ³³⁸	Natural fibre lengths are generally categorised into: Short (S) – Less than 25mm; Medium (M) – 25-30mm; Long (L) – 30-37mm; and Extra Long Staple (ELS) – Over 37mm. Specific natural fibre lengths: Silk (continuous): 50 000 to 1 500 000 cm Hemp: average 182.88 cm Flax: up to 90 cm Wool: - Fine wool: 3.8 to 12.7 cm - Medium wool 6.35 to 15.24 cm - Courser or long 12.7 to 38.1 cm Cotton: - Very short-staple cotton: ≤2.1 cm - Short staple cotton: >2.2 cm and <2.5 cm - Medium Staple cotton: >2.6 cm and < 2.8 cm - Ordinary long-staple cotton: >2.9 cm and < 3.4 cm - Extra-long staple cotton: ≥3.4 cm	Chemical fibre lengths can be considered continuous	https://textileexchange.org/glossary/fibre-length/ https://www.textilecoach.net/post/cotton-fibre#:~:text=Short%20staple%20cotton%3A%20%3E22mm%20and,staple%20cotton%3A%20%E2%89%A534%20mm https://www.fao.org/natural-fibres-2009/about/15-natural-fibres/en/
Fibre strength	The strength of an individual fibre	It is generally measured by tenacity, defined as the breaking force of the fibre divided by the weight of 1000 m of fibre (e.g. cN/tex³³⁹). Minimum fibre strength =6 cN/tex is technically acceptable for spinning processes.	Significant breaking strengths of most used chemical fibres for textile apparel: Polyamide: 65-82 cN/tex Polyester: 35-60 cN/tex Acrylic: 19.6-29.4 cN/tex Viscose: 19.6 cN/tex Acetate: 12.7 cN/tex	(Lamb, 2009) https://textilelearner.net/properties-of-silk-fibre/#:~:text=Tenacity%3A%20Silk%20loses%20strength%20on%20wetting https://omnexus.specialchem.com/polymer-property/strength-at-break-tensile (Rahman Khan et al., 2011) https://cmdgroup.in/physical-properties-of-fibres.php https://oldswicofil.com/products/223polyamideimide.html https://www.tp-industrial.com/files/specs-polyamide-1.pdf

³³⁸ Considering that man-made fibres are assigned infinite length.

³³⁹ cN/tex= centi Newton / Tex is a unit allowing to qualify the elastic capacity of a textile or a yarn

Fibre characteristic	Definition	Description of natural fibres	Description of chemical fibres	Source
Fibre cleanliness	Refers to the degree of fibre contamination with impurities ³⁴⁰	Impurities are classified into ranges and according to the impurities grade and are mainly common in natural fibres, but they can appear in chemical fibres due to the treatment agents such as lubricants and oils in chemical fibres production. Range of impurities: • ≥7 %- Very dirty • 4-7 %- Dirty • 2-4%- Medium • 1.2-2- Clean • ≤1.2- Very clean Impurities grade (AFIS³⁴¹): • ≤15 µm- Breathable dust • 15-50 µm- Micro dust • 50-500 µm- Dust • ≥500 µm-Trash		http://www.definetextile.com/2013/04/fibre-cleanliness.html
Fibre fineness	Determines the number of fibres that can made up the cross-section of a yarn with a specific thickness.	Measures units of mass per unit of length to assess linear density (e.g. dtex³⁴²) According to the classification of Sekhri (Sekhri, 2016), fibre fineness can be classified as it follows: • Thick fibres >7 dtex • Semi-fine fibres: 7-2.4 dtex • Fine fibres: 2.4-1 dtex • Microfibres: 1-0.3 dtex • Super microfibres <0.3 dtex Hemp: 48.1–186.9 dtex Wool: 3.5 dtex Flax: 1.39- 1.7 dtex Cotton: 1.6 dtex Silk: 0.95-1.6 dtex	The minimum fineness that the raw material and technology permits obtaining for the conventional applications is presented below, considering that the fineness can be increased depending on the application: Polyester: 0.8-3 dtex (ultrafine or microfibre 0.55 dtex) Polyamide: minimum 1 dtex Acrylic: minimum 0.9 dtex Viscose: minimum 1.2 dtex Acetate: minimum 1.5 dtex	http://www.definetextile.com/2013/04/fibre-fineness.html (Ramey, 2018) (Fibre Fineness - an overview ScienceDirect Topics, n.d.) (Rahman Khan et al., 2011) (Banale, 2017) (Chapter 3. Fibre fineness and transverse dimensions, 2008) (Goudar and Kulloli, 2022)

³⁴⁰ The impurities may be natural from the cultivation process, but also generated by processing. Generally, impurities must be eliminated via a cleaning process.

³⁴¹ AFIS (Advanced Fibre Information System)

³⁴² dtex (deci-tex): Grams per 10.000 metres of yarn

Fibre characteristic	Definition	Description of natural fibres	Description of chemical fibres	Source
Fibre colour	The colour of the fibre affects the ability of a yarn or fabric to be dyed or bleached.	The performance of yarn during dyeing or bleaching is influenced by the natural colour of the fibre. This characteristic is mainly relevant to natural fibres. The colour of man-made fibres can be controlled during the production process.	This does not apply to chemical fibres.	https://www.fao.org/natural-fibres-2009/about/15-natural-fibres/en/
Fibre stiffness	Fibre stiffness refers to the ability of a fibre to resist deformation or bending when subjected to a load	Commonly measured using parameters like Young's modulus or bending modulus, which characterise the material resistance to deformation. It depends on the fibre substance and the relationship between fibre length and fibre fineness: - The length of the fibre is proportional to the bending modulus. The following relationship can be indicated: - Thick fibres are generally stiff, firm and wrinkle-resistant. Fine fibres are soft, flexible and generate good drape.		(Ferrándiz et al., 2021)

11874 Source: own knowledge and sources indicated in the table

11875 **Table 178.** Definitions of yarn and fabric characteristics

Intermediate textile product	Characteristic	Definition	Source
Yarn	Evenness	Yarn evenness refers to the uniformity of the yarn in terms of thickness and weight along its length.	(Elmogahzy, 2019)
	Hairiness	Yarn hairiness refers to the presence and extent of fibre ends and loops protruding from the main body of the yarn.	(Elmogahzy, 2019)
	Strength	Yarn strength refers to the ability of a yarn to withstand tensile forces without breaking.	(Elmogahzy, 2019)
	Appearance	Yarn appearance refers to the visual and physical characteristics of yarn, including diameter, hairiness, and defects, which affect its performance and the physical durability of the final fabric.	(Tahvildar et al., 2019; Li et al., 2020)
Fabric	Handle	Fabric handle refers to the tactile qualities of a fabric, encompassing its softness, smoothness, flexibility, and overall sensation when touched, which contribute to its comfort and user experience	(Tahvildar et al., 2019; Li et al., 2020)
	Lustre	The visual aspect of a fabric or yarn that reflects light in a way that creates a shiny or glossy appearance.	(Kim et al., 2004)
	Strength	Fabric strength can be defined as the ability of a fabric to withstand various forces and stresses without breaking or deforming	(Mobarak Hossain, 2016)
	Appearance	Fabric appearance encompasses the overall look and visual appeal of the fabric, which is influenced by factors such as weave structure, yarn compactness, and finishing processes.	(Tahvildar et al., 2019)
	Drape	Fabric drape refers to how the fabric hangs in three dimensions, retaining its shape from when it is laid flat or adapting to new contours.	(Choudhary and Bansal, 2017)
	Smoothness	Fabric smoothness refers to the fabric surface resistance to a sliding tangential force applied.	(Mao et al., 2016)
	Softness	Fabric softness refers to the tactile sensation or comfort level experienced when touching or wearing a fabric.	(Ferrándiz et al., 2021)
	Breathability	Breathability in fabrics is the ability of water vapours to successfully permeate through that fabric via diffusion and, henceforth, facilitate cooling via evaporation.	(Inovenso Teknoloji Ltd.Şti and Mustafa, 2023)
	Insulation	Fabric insulation refers to a fabric's ability to retain heat and provide warmth by reducing the transfer of heat from the body to the surrounding environment.	(Matusiak Malgorzata and Sikorski Renata, 2011)
	Anti-wrinkle	Anti-wrinkle is a finishing method for textiles that prevents creases and wrinkles, enhancing the appearance of the articles.	(Van den Bergen and Parker, 2023)

Intermediate textile product	Characteristic	Definition	Source
	Resistance to stains and water	Resistance to stains refers to the ability of a fabric to repel or prevent the absorption of various staining substances, thereby maintaining its appearance and cleanliness over time. Resistance to water, also known as water repellency, describes the capacity of a fabric to prevent water from penetrating its surface, thus keeping the material dry and maintaining its structural integrity.	(Rowen and Gagliardi, 1947) (Roy Choudhury, 2017)

11876 *Source: AITEX's knowledge and references reported in the last column*

11877 **Table 179.** Definitions of main process parameters

Process parameter	Definition	Sources
Spinning method	The spinning method encompasses various techniques used to transform raw fibres into yarn. These methods involve processes that draw out and twist the fibres, creating a continuous strand of yarn. For detailed descriptions of different spinners, refer to section 13.3.2.	(Elmogahzy, 2019) (Sinha and Chattopadhyay, 2007) (Mahmoudi, 2010) (Tyagi, 2010) (Mobarak Hossain, 2016) (Rahman et al, 2023)
Spinning limit	The spinning limit defines the range of yarn counts achievable from a given fibre using a specific spinning system. It depends on the number and thickness of individual fibres (for natural fibres) or filaments (for chemical fibres) twisted together.	
Spinning Productivity	Spinning productivity can be defined as the production of the finest yarn count from a given fibre ensuring acceptable performance and minimising end breakage rates.	
Twist level	Twist level is defined by the number of turns per unit length (per inch or per meter) in a spun yarn. It describes yarn density.	
Twists number	The twist number quantifies the number of turns about its axis per unit length of a yarn or textile strand. It is expressed as turns per inch (TPI), turns per meter (TPM), or turns per centimetre (TPCM).	
Spinning tension	Spinning tension represents the resistance or force applied to the yarn during the spinning process.	
Speed of spinning	The speed of spinning defines the rate at which fibres are converted into yarn during spinning.	
Rolling, revolving, and twisting movements	These fundamental movements in spinning: Rolling: Rotation of fibres to align them, Revolving: Circular motion of the spindle, Twisting: Imparting twists to the fibres, resulting in yarn formation	
Yarn manufacturing process development	Encompasses the life cycle stage for converting raw fibres into yarn.	
Conditioning (steaming)	It involves moistening the yarn using saturated steam or wet steam.	
Spinning twist limit	The spinning twist limit defines the upper range of yarn count achievable for a given fibre and spinning system, determining the maximum fineness of yarn produced.	
Temperature and humidity control	Maintaining optimal temperature and humidity levels is crucial. High humidity improves yarn properties, while excessive moisture can lead to issues.	
Dyeing process	Dyeing involves applying colour to yarn or fabric.	
Characteristics of components and machinery	The characteristics of components and machinery directly affects yarn production. Well-maintained, precise equipment ensures consistent yarn properties.	
Defects control	Defect control focuses on minimising flaws in yarn, such as knots or irregularities, enhancing yarn desired performances, and reducing waste.	
Thread count	Thread count refers to the number of yarns (threads) per unit length in woven fabrics, and it affects fabric density and texture.	
Weight of fabric	The weight of fabric is determined by the total mass of yarn used per unit area, influencing fabric thickness and durability.	

11878

11879

Process parameter	Definition	Sources
Weave pattern	The weave pattern describes how warp and weft yarns interlace to form fabric	

Source: AITEX's knowledge and references reported in the last column

Table 180 Influence of the fibre characteristics on yarn and fabric characteristics and parameters of manufacturing processes

Characteristic or parameter		Fibre length ⁽¹⁾	Fibre strength	Fibre cleanliness	Fibre fineness	Fibre colour	Fibre stiffness ⁽²⁾
Yarn characteristics	Evenness	Yarn evenness increases during drafting ³⁴³ proportionally with the fibre length.	Yarn evenness increases proportionally with the increase of fibre strength (smaller extent).		Yarn evenness increases proportionally with the increase of fibre fineness. ⁽⁵⁾		
	Hairiness ⁽³⁾	Yarn hairiness decreases with the increase of fibre length.	Yarn hairiness decreases with the increase of fibre strength.		Yarn hairiness increases with the increase of fibre fineness. ⁽⁵⁾		
	Strength	Yarn strength increases proportionally with fibre length increase.					
	Appearance			Yarn appearance improves with the increase of fibre cleanliness. ⁽⁴⁾			
Yarn manufacturing (spinning) parameters	Twist limit				Twist limit decreases with the increase of fibre fineness, after the optimal working range.		
	Spinning limit	Spinning limit increases proportionally with the increase of fibre length.	Spinning limit increases proportionally with the increase of fibre strength.		Spinning limit increases proportionally with the increase of fibre fineness.		
	Spinning productivity	Spinning productivity increases with the increase of fibre length.	Spinning productivity increases with the increase of fibre strength.		Spinning productivity increases with the increase of fibre fineness.		

³⁴³ Drafting is the process of reducing the thickness of fibres to achieve specific fineness and strength for yarns (Rahman et al., 2023).

Characteristic or parameter		Fibre length ⁽¹⁾	Fibre strength	Fibre cleanliness	Fibre fineness	Fibre colour	Fibre stiffness ⁽²⁾
	Rolling, revolving, and twisting movements						
	Spinning process development			Spinning process improves with the increase of fibre cleanliness. ⁽⁴⁾			Spinning process worsens with the increase of fibre stiffness.
Finishing processes parameters	Dyeing					Dyeing process efficiency decreases with the increase of fibre natural colour. ⁽⁶⁾	
Fabric characteristics	Handle	Fabric softness increases with the increase of fibre length.			Fabric softness increases with the increase of fibre fineness.		
	Luster	Fabric shininess increases with the increase of fibre length.			Fabric shininess increases with the increase of fibre fineness.		
	Strength		Fabric resistance to tear strength increases with the increase of fibre strength.				
	Appearance			Fabric appearance improves with the increase of fibre cleanliness. ⁽⁴⁾			
	Drape				Fabric drape improves with the increase of fibre fineness.		

11880 ⁽¹⁾ Most commonly related to natural fibres and mechanically recycled synthetic or other man-made fibres. It defines the spinnability of the fibre.

11881 ⁽²⁾ A fibre that is too stiff has difficulty in adapting to process movements. The lower the springiness generates more creation of neps (entangled and knotted fibres generated by increased stiffness).

11882 ⁽³⁾ Unwanted parameter

11883 ⁽⁴⁾ Impurities generate a bad yarn drying, with broken ends increased, generating a worse appearance. Additionally, impurities interfere with the process development affecting the strength of the material to be processed. Finally, impacts the appearance and physical characteristics of the fabric and finished product.

11884 ⁽⁵⁾ A finer the fibre allows a multitude of fibres/filaments in the cross-section, allowing a better distribution in the yarn. Fibre fineness decreases after the optimal working range and yarn hairiness will increase.

11885 ⁽⁶⁾ The intrinsic colour of the fibre interferes with the dyeing process.

11887 Source: AITEX's knowledge and **Figure 22**

11888 **Yarn manufacturing**, the spinning process combining rolling, revolving and twisting movements, is defined by parameters that influence the final yarn characteristics.

11889 **Table 181** reports the effect of main process parameters on the characteristics of the yarn. The selection of the spinning method influences the evenness, softness and
11890 strength of the yarns. Nevertheless, the twist level and number impact the strength and softness of the yarns. Controlling optimal spinning tension and speed is important
11891 due to the impact on the strength of the yarn. Proper yarn conditioning has the potential to enhance characteristics such as flexibility, strength and evenness. Manufacturing
11892 process parameters such as temperature and humidity control and the characteristics of components and machinery, completed with performance control, influence overall
11893 the mechanical-related strength of the yarns.

11894 Understanding and controlling these processing parameters enables manufacturers to customise yarn characteristics for precise end applications, like textiles for fashion
11895 such as underwear, jackets, or accessories. Adjustments in these parameters can result in notable differences in yarn physical performance and durability, demonstrating
11896 the importance of precise control and monitoring during the spinning process.

11897

11898 **Table 181.** Influence of spinning parameters on yarn characteristics

Yarn characteristics	Spinning method	Twist level	Twists number	Spinning tension ⁽²⁾	Speed of spinning	Conditioning (steaming)	Temperature and humidity control	Characteristics of components and machinery	Defects control
Yarn evenness	Air-Jet Spinning → highly uniform yarns			Constant tension: uniform yarn		Enhances evenness	Proper humidity prevents static electricity, material processing issues, equipment damage, and fabric shrinkage.	High quality of the machinery produces consistent yarns. Ensures improved efficiency, safety, productivity, and product performance	Critical for maintaining high-performance standards and preventing imperfections in yarn and fabric
Yarn elasticity						Enhances flexibility			
Yarn softness	Air-Jet Spinning → highly smooth yarns	Low Twist: softer yarns							
Yarn strength	Ring Spinning → very high-strength and finer yarns Rotor (Open-End) Spinning → bulkier yarns with reduced strength Air-Jet Spinning → less durable yarns	High Twist: stronger yarns	Yarn strength increases with twist to a peak ⁽¹⁾ , then decreases.	Uneven Tension: can cause breakages during subsequent processing	Higher Speeds: can cause breakages	Enhances strength			

11899 ⁽¹⁾ Optimal twist depends on fibre length, fineness, strength, and friction.

11900 ⁽²⁾ Spinning tension: Essential for evaluating ring spinning machine performance.

11901 Source: AITEX's knowledge and **Figure 22**

11902 In the **fabric manufacturing**, the industry opts for either weaving or knitting techniques, depending on the intended use of the final products. The parameters involved in
11903 weaving significantly impact the characteristics of the fabric. These are the yarn count, weight of fabric, and weave pattern, among others. **Table 182** reports the influence
11904 of the main weaving parameters on the fabric characteristics. A higher thread count defines smoother, softer, and more breathable fabrics. Meanwhile, the weight of the
11905 fabric improves the resistance to tear but decreases breathability and drape. Finally, the selection of the weave pattern impacts the resistance to tear and is done according
11906 to the desired functionality, such as smoothness, anti-wrinkle, and resistance to stains and water.

11907 **Table 182.** Influence of weaving parameters on fabric characteristics

Parameter	Type	Fabric characteristic							
		Smoothness	Softness	Breathability	Tear strength	Insulation	Anti-wrinkle	Resistance to stains and water	Drape
Thread count ⁽¹⁾	Higher the number of threads	Increases fabric smoothness	Increases fabric softness	Decreases fabric breathability					
Weight of fabric ⁽²⁾	Heavier fabric			Decreases fabric breathability	Increases resistance to tear	Increases fabric insulation properties			Decreases fabric drape
Weave pattern ⁽³⁾	Plain (eg. Chiffon, organza)	Increases fabric smoothness		Increases fabric breathability					
	Diagonal (eg. twill)				Increases resistance to tear		Increases fabric anti-wrinkle capacity	Increases fabric resistance to stains and water	
	Satin	Increases fabric smoothness			Decreases fabric resistance to tear				

11908 ⁽¹⁾ Generally, more important for fabrics intended to be used in contact with the skin (e.g. beddings, textile apparel). The optimal thread count can vary depending on the desired fabric use and feel, and sometimes
11909 higher thread counts are achieved by using multiple-ply yarns, which can affect the fabric differently.
11910 ⁽²⁾ The weight is influenced by the yarn's thickness and the tightness of the weave or knit. Generally more important for fabrics such as upholstery or drapery.
11911 The fabric construction parameters, including weave pattern, yarn count, fabric thickness, and compaction, influence tear strength. Fabrics with denser weaves, higher yarn counts, and thicker constructions tend to
11912 exhibit higher tear resistance.
11913 The weight or mass of the fabric per unit area is indicative of fibre concentration and arrangement. Heavier fabrics with higher yarn density typically offer better tear strength. The denser packing of fibres in the
11914 fabric structure increases resistance to tearing forces.
11915 ⁽³⁾ Each weaving pattern offers unique characteristics that can be leveraged for specific applications, impacting not only the aesthetics of the fabric but also its performance in various uses

11916 Source: stakeholder's consultation and **Figure 22**

11917

Table 183 reports the influence of finishing processes on fabric characteristics. **Fabric finishing** is a crucial stage in manufacturing that imparts specific aesthetic and technical functionalities to the fabric. **Dyeing** plays a significant role, as the choice of dye and dyeing technique can greatly influence colour fastness⁽³⁴⁴⁾, hand feel, colour depth, and evenness. **Table 121** reports information on the relationship between fibre type and dye affinity. **Printing** affects the vibrancy of colours and the handle of the fabric. **Chemical finishing** imparts various properties to the fabric, such as improved handle and repellency to water, oil, and stains, thereby enhancing its physical durability. Mechanical finishing alters the fabric's physical properties, contributing to smoothness or improving the handle, such as adding fluffiness. **Coating and laminating** processes are used to enhance fabric performance by adding waterproofing capabilities and increasing breathability or resistance to abrasion (Rahman et al., 2023).

Table 183. Influence of finishing processes on fabric characteristics

Finishing process type	Primary influence on fabric characteristics
Dyeing	Colour depth, evenness, and fastness
Printing	Pattern precision, vibrancy, and hand feel
Chemical Finishing	Functional properties like softness, repellence
Mechanical Finishing	Physical properties like smoothness, plushness
Coating and Laminating	Protective and performance-enhancing characteristics such as waterproofing, breathability

Source: AITEX's knowledge and **Figure 22**

Three main components affect the physical durability during **confectioning** of the textile apparel: the sewing physical characteristics, precision of cuts, and seam strength. **Sewing physical characteristics** include the straightness of seams, consistency of stitch length, and the absence of skipped stitches. These affect both the appearance and physical durability of the textile apparel. The **strength and type of seams** used in fabric assembly determine the physical durability of the textile apparel and its ability to endure wear and stress. Inadequately constructed seams may result in rapid deterioration. **Precise cutting** following patterns is essential for ensuring that the textile apparel fits properly. Mistakes in cutting can result in poorly fitting textile apparel, directly affecting consumer satisfaction (Rahman et al., 2023) (Elhawary, 2015b).

³⁴⁴ Colour fastness refers to the resistance of a fabric's colour to fading or running when exposed to various conditions, such as washing, sunlight, rubbing, and perspiration

11939 **13.10.3 Supporting information on test methods to describe the physical durability**

11940 **Table 184.** Description of standardised test methods to assess key parameters of physical durability

Parameter	Testing method	Short description of the scope of the test	Type of result obtained
Dimensional change	ISO 3759:2011 ³⁴⁵ Textiles — Preparation, marking and measuring of fabric specimens and textile apparel in tests for determination of dimensional change	Provides a method for preparing, marking, and measuring textile fabrics, textile apparel, and fabric assemblies to test how they change in size after treatments like washing, dry cleaning, soaking, or steaming. These treatments must follow procedures from specific ISOs. It applies to woven and knitted fabrics and finished textile articles but does not apply to certain upholstery coverings.	Percentage (%) This percentage reflects the dimensional change, such as shrinkage or stretching of the fabric after washing or dry cleaning. The percentage change is calculated based on the difference between the original and the post-treatment dimensions.
Visual inspection for: (1) Colour change (2) Pilling (3) Trimmings aspect (4) Self-staining	ISO 15487:2018 ³⁴⁶ Textiles — Method for assessing appearance of apparel and other textile end products after domestic washing and drying	It specifies a method for assessing the appearance of apparel and textile products after one or more cleaning cycles. The evaluation includes factors such as colour change, pilling, fuzzing, matting, fabric smoothness, seam appearance, retention of pressed-increases, and damage to components like buttons or fasteners. It applies to any washable textile product, regardless of fabric construction, as supplied by the manufacturer.	Grading scale The grading scale usually ranges from 1 to 5, where 1 indicates severe distortion or damage, and 5 indicates no visible distortion or damage. Grade 1: Severe distortion or damage, significant loss of fabric integrity. Grade 2: Pronounced distortion or damage, noticeable but not severe. Grade 3: Moderate distortion or damage, visible but not significantly affecting the fabric's usability. Grade 4: Slight distortion or damage, minimal and only noticeable upon close inspection. Grade 5: No visible distortion or damage, fabric maintains its original appearance.

11941 *Source: cited standards*

³⁴⁵ ISO 3759:2011 Preparation, marking and measuring of fabric specimens and textile apparel in tests for determination of dimensional change. Available at this [link](#).

³⁴⁶ ISO 15487:2018 Textiles — Method for assessing appearance of apparel and other textile end products after domestic washing and drying. Available at this [link](#).

11942 **13.10.4 Supporting information on maintenance**

11943 **Table 185.** Comparison of international labelling systems

Characteristic	International Care Labelling System (Ginetex) ³⁴⁷	ASTM Care Labelling System ³⁴⁸	Canadian Care Labelling System ³⁴⁹	British Care Labelling System ³⁵⁰	Japanese Care Labelling System ³⁵¹	Australian Care Labelling System ³⁵²	China Care Labelling System ³⁵³	ISO Care Labelling System ³⁵⁴
Governing Body	GINETEX	ASTM International	Canadian General Standards Board (CGSB)	British Standards Institution (BSI)	Japanese Industrial Standards (JIS)	Standards Australia and ACCC	China National Textile and Apparel Council (CNTAC)	International Organization for Standardization (ISO)
Region	Global	United States	Canada	United Kingdom	Japan	Australia	China	Global (especially for export procedures)
Primary Symbols	Washing, Bleaching, Drying, Ironing, Professional Care	Washing, Bleaching, Drying, Ironing, Professional Care	Washing, Bleaching, Drying, Ironing, Professional Care	Washing, Bleaching, Drying, Ironing, Professional Care	Washing, Bleaching, Drying, Ironing, Professional Care	Washing, Bleaching, Drying, Ironing, Professional Care	Washing, Bleaching, Drying, Ironing, Professional Care	Washing, Bleaching, Drying, Ironing, Professional Care
Temperature Units	Celsius	Fahrenheit (with some symbols using Celsius)	Celsius and Fahrenheit	Celsius	Celsius	Celsius	Celsius	Celsius (with some flexibility for export)
Text Requirement	No text required	Text required alongside symbols	Text required alongside symbols (bilingual: English and French)	No text required	Text is often included for clarity	Text required alongside symbols	Text required alongside symbols (Chinese characters)	No text required, but can include text for clarity
Drying Symbols	Includes natural and tumble drying symbols	Tumble drying symbols with specific temperature settings	Tumble and natural drying symbols	Natural and tumble drying symbols	Unique symbols for indoor/outdoor and natural drying	Tumble and line drying symbols	Symbols for tumble drying and line drying, with specific instructions for sunlight exposure	Similar to GINETEX; focuses on export needs

³⁴⁷ Complete information about International Care Labelling System (Ginetex) available [here](#).

³⁴⁸ Complete information about ASTM Care Labelling System available [here](#).

³⁴⁹ Complete information about Canadian Care Labelling System available [here](#).

³⁵⁰ Complete information about British Care Labelling System available [here](#).

³⁵¹ Complete information about Japanese Care Labelling System available [here](#).

³⁵² Complete information about Australian Care Labelling System available [here](#).

³⁵³ Complete information about Chinese Care Labelling System available [here](#).

³⁵⁴ Complete information about ISO Care Labelling System available [here](#).

Characteristic	International Care Labelling System (Ginetex) ³⁴⁷	ASTM Care Labelling System ³⁴⁸	Canadian Care Labelling System ³⁴⁹	British Care Labelling System ³⁵⁰	Japanese Care Labelling System ³⁵¹	Australian Care Labelling System ³⁵²	China Care Labelling System ³⁵³	ISO Care Labelling System ³⁵⁴
Bleaching Symbols	Basic symbols for chlorine and non-chlorine bleach	Detailed symbols for bleach types	Specific symbols for non-chlorine bleach	Similar to GINETEX	Specific symbols for bleach types, including traditional care	Similar to GINETEX	Specific symbols for chlorine and non-chlorine bleach, with detailed instructions	Basic symbols for chlorine and non-chlorine bleach
Ironing Symbols	Dots indicating heat level	Dots indicating heat level	Dots indicating heat level	Dots indicating heat level	Dots with additional instructions	Dots indicating heat level	Dots indicating heat level, with additional instructions on steam use	Dots indicating heat level
Professional Care Symbols	Basic dry cleaning and wet cleaning symbols	Detailed symbols for professional care	Basic symbols for dry and wet cleaning	Similar to GINETEX	Detailed symbols including specific letters for different cleaning methods	Similar to GINETEX	Detailed symbols including specific letters for different cleaning methods	Basic dry cleaning and wet cleaning symbols
Language and Accessibility	Universal symbols for easy understanding	Text and symbols for detailed instructions	Symbols with bilingual text (English/French)	Universal symbols to minimize language barriers	Uses both symbols and text, considering Japanese consumers	Uses both symbols and text	Symbols with accompanying text in Chinese for clarity	Universal symbols for global understanding
Legal Requirements	Voluntary but widely adopted	Mandatory compliance with FTC regulations	Mandatory under the Canadian Textile Labelling Act	Voluntary but widely followed in Europe	Mandatory for domestic manufacturers	Mandatory compliance with Australian Standards	Mandatory compliance under Chinese standards	Voluntary for global standardization

Source: own production

11944

11945

11946

13.10.5 Supporting information on repairability

This section reports a simple analysis of professional repair prices against the prices of the new items of textile apparel. **Table 186** reports the interval of the relative repair price in EU, which takes into account the price intervals of some repair operations carried out by professional repairers and the price intervals of some new items. The analysis shows that repairing an item of textile apparel could be many times more expensive than buying a new item.

Table 186. Comparison between price of professional repair operations and price of new items of textile apparel

Priority part	Failure type	Repair price (EUR)		Items of textile apparel	New item price (EUR)		Relative repair price (%)	
		Min	Max		Min	Max	Min	Max
Fabric	Fabric breakdown	3	9	T-Shirts	3	22	14	300
	Pilling	8	24	Apparel accessories	2	41	20	1 200
				Dresses, Skirts & Jumpsuits	6	90	9	400
				Jackets & Coats	7	589	1	343
				Leggins, stockings, tights and socks	1	19	42	2 400
				Pants & Shorts	3	66	12	800
				Shirts & Blouses	3	90	9	800
				Sweaters and midlayers	3	85	9	800
				Swimsuits	8	74	11	300
				T-Shirts	3	22	36	800
	Stains	3	51	Apparel accessories	2	41	7	2 550
				Dresses, Skirts & Jumpsuits	6	90	3	850
				Jackets & Coats	7	589	1	729
				Leggins, stockings, tights and socks	1	19	16	5 100
				Pants & Shorts	3	66	5	1 700
				Shirts & Blouses	3	90	3	1 700
				Sweaters and midlayers	3	85	4	1 700
				Swimsuits	8	74	4	638
				T-Shirts	3	22	14	1 700
Seam	Hole(s) in seams	3	32	Apparel accessories	2	41	7	1 600
				Dresses, Skirts & Jumpsuits	6	90	3	533
				Jackets & Coats	7	589	1	457
				Leggins, stockings, tights and socks	1	19	16	3 200
				Pants & Shorts	3	66	5	1 067
				Shirts & Blouses	3	90	3	1 067
				Sweaters and midlayers	3	85	4	1 067
				Swimsuits	8	74	4	400
				T-Shirts	3	22	14	1067
Trim	Buttons	1	8	Apparel accessories	2	41	2	400
				Dresses, Skirts & Jumpsuits	6	90	1	133
				Jackets & Coats	7	589	0	114
				Leggins, stockings, tights and socks	1	19	5	800
				Pants & Shorts	3	66	2	267

Priority part	Failure type	Repair price (EUR)		Items of textile apparel	New item price (EUR)		Relative repair price (%)	
		Min	Max		Min	Max	Min	Max
				Shirts & Blouses	3	90	1	267
				Sweaters and midlayers	3	85	1	267
				Swimsuits	8	74	1	100
				T-Shirts	3	22	5	267
	Zips	6	32	Apparel accessories	2	41	15	1 600
				Dresses, Skirts & Jumpsuits	6	90	7	533
				Jackets & Coats	7	589	1	457
				Leggins, stockings, tights and socks	1	19	32	3 200
				Pants & Shorts	3	66	9	1 067
				Shirts & Blouses	3	90	7	1 067
				Sweaters and midlayers	3	85	7	1 067
				Swimsuits	8	74	8	400
				T-Shirts	3	22	27	1 067

11954 Prices are reported without VAT. Repair price were sourced from websites reported in **Table 187**.

11955
$$\text{Relative repair price (\%)} = \frac{\text{price of repair}}{\text{price of new apparel textile}} * 100$$

11956 Source: own elaboration based on (Cooper and Claxton, 2022c) and Statista ⁽³⁵⁵⁾.

11957 **Table 187.** Repair shops and dry cleaners consulted for the analysis reported in **Table 186**

Type	Country	Link
Repair shop	Spain	https://arreglosderopabarcelona.es/tarifas-precios-arreglos-ropa/ https://www.pepevera.com/lista_precios
	Romania	https://www.croitoriepentrutoti.ro/preturi/ https://www.urbanwash.ro/images/lista-croitorie-arad.pdf https://atelierdecroitorie.ro/servicii-si-preturi https://www.atelier-croitorie.com/lista-preturi-croitorie-retus
	Ireland	https://paulhenrytailoring.ie/alterations-dublin
	Sweden	https://www.masterarnes.se/skr%C3%A4dderi.html https://www.jeansverket.se/en/jeans/repairs/ https://secondsunrise.se/pages/repairs
	Italy	https://endelea.it/pages/wear-again-textile-apparel-restoration-hub
	Spain	https://tintoreriasronsel.es/tienda/precios-tarifas-y-promociones-de-tintoreria/
Dry cleaners	Romania	https://ecoclean.ro/preturi/ https://www.elisse.ro/preturi-spalatorie-haine.html https://www.extraclean.ro/preturi-curatorie-spalatorie-calcatorie-haine-bucuresti/

11958 Source: reported websites

11959

³⁵⁵ Statista. Available at [this link](#). Last accessed 8 August 2024.

13.11 Supporting information on the environmental and economic assessment – the Life Cycle Thinking model

The purpose of this section is to deliver a comprehensive and transparent description of the Life Cycle Assessment (LCA) and Life Cycle Cost (LCC) textile model, underlying assumptions, and inherent limitations.

This section is following the approach outlined in Task 5 - Environment and Economic analysis (Base case LCA & LCC) of the Methodology for Ecodesign of Energy-related Products (MEErP) framework. It builds upon the extensive data and insights collected from Tasks 1 - Scope (definitions, standards and legislation) through Task 4 - Technologies (product supply side, includes both BAT and BNAT) of the MEErP.

The integration of information from the initial Tasks ensures a comprehensive understanding of the textile sector's current landscape, encompassing market analysis, user behaviours and technical and economic considerations. By systematically utilizing this information, Task 5 focuses on identifying and evaluating potential environmental and economic impacts, with the aim to prepare the ground for Task 6, which focuses on assessing the impacts of design options, aiming to enhance the sustainability of textile products throughout their life cycle.

The LCA-LCC textile model has been developed following the guidelines and standards mentioned below:

- ISO 14040³⁵⁶ that provides the “principles and framework for LCA, including the goal and scope definition, life cycle inventory analysis, life cycle impact assessment, and life cycle interpretation, limitations, etc.”
- ISO 14044³⁵⁷ that provides the requirements and guidelines for LCA including relevant terms and definitions.
- Product Environmental Footprint (PEF) (Commission Recommendation (EU) 2021/2279)³⁵⁸
- Product Environmental Footprint Category Rules (PEFCR)

An attributional LCA approach was used to estimate the environmental and economic impacts related to textile apparel placed on EU market. If necessary, the multifunctionality of the system was addressed using mass or economic allocation.

13.11.1 Intended application of the results

The intended application of the results from this study is to support and inform decision-making processes within the European Union (EU), in the context of textile product policy development.

By quantifying the economic and environmental costs and benefits of textile apparel available in the EU market, the results are designed to provide stakeholders—such as EU policymakers, industry leaders, and Member States—with a comprehensive understanding of the current impacts of textile products.

The findings are intended to guide the formulation and implementation of ecodesign requirements, ensuring that these interventions effectively minimize environmental impacts without causing disproportionate economic burdens. Ultimately, the results aim to underpin policy decisions that promote sustainable textile practices, stimulate positive changes in the industry, and enhance the overall environmental and economic performance of textile products consumed within the EU.

The analyses conducted allowed to obtain the results, presented in Section 13.11.9.7, relating to a model developed to represent textile apparel placed on the EU market. The results are presented in relation to representative products with a reference base case, used to describe a broad category of textile apparel.

³⁵⁶ ISO 14040:2006 Environmental management — Life cycle assessment — Principles and framework. Available at this [link](#). Last accessed on 4 September 2025.

³⁵⁷ ISO 14044:2006 Environmental management — Life cycle assessment — Requirements and guidelines. Available at this [link](#). Last accessed on 4 September 2025.

³⁵⁸ Commission Recommendation (EU) 2021/2279 of 15 December 2021 on the use of the Environmental Footprint methods to measure and communicate the life cycle environmental performance of products and organisations [link](#)

In addition, scenario analyses were conducted, investigating worsening and improving conditions to test the influence of the consumer choice when disposing of a product regardless of its physical performances. An additional scenario sought to test the influence of location of raw material production and manufacturing processes.

For the LCC analysis, the results obtained were examined from three different perspectives: (i) society, (ii) producer, and (iii) consumer.

13.11.2 Limitations due to methodological choices

In conducting a comprehensive assessment of the environmental and economic performances of textile apparel placed in the EU market, methodological limitations have been identified. These limitations are primarily rooted in the complexities involved in representing a wide array of textile products (section 9). Understanding these limitations is crucial to interpreting the results of the study in the appropriate context and to ensuring that they are used wisely in decision-making processes.

Some of the limitations identified are:

- **Abstract representation of product categories:** A significant methodological choice in this study involves the use of Representative Products (RPs) to describe a broad category of textile apparel. RPs are virtual products, which do not represent necessarily a product that can be found in the EU market, and their compositions is based on information outlined in previous sections of the preparatory study (Section 9.3). Given the diversity and variation in apparel products available on the market, creating a model that incorporates each specific product was impractical. Instead, an abstract representation was employed to capture the fibres composition, quantitative and qualitative characteristics (e.g. mass, colour, fabric structure, etc.) and performance metrics (e.g. resistance to pilling, colour fasteners, etc.) of numerous products within a single category. While this approach facilitates a more manageable analysis, it introduces a limitation in that the RP may not perfectly reflect the nuances and specificities of individual market products. For instance, in the case of the RP Knitted, it encompasses garments produced by knitting processes, designed to cover specific parts of the body, incorporating a variety of items such as t-shirt and polo, sweaters and mid-layers, trousers and shorts, dresses, skirts and jumpsuit, jackets and coats, socks and stoking tights, underwear, swimwear, and accessories. The characteristics of this RP are derived from the aggregate features and compositions of these items, reflecting an average rather than specific product details.
- **Life cycle assessment and life cycle costing methodology:** Any model represents an imperfect representation of the reality. LCA and LCC attempt to simulate complex systems and interactions within the textile lifecycle, from resource extraction to end-of-life disposal. Despite advanced modelling techniques, LCAs and LCCs cannot entirely replicate the multifaceted real-world processes they aim to represent. As a result, there are certain approximations and simplifications embedded within the assessment that could lead to potential discrepancies between modelled outcomes and actual impacts.
- **Life cycle impact assessment:** The assessment framework adopted in this study incorporates all 16 impact categories as outlined by the Product Environmental Footprint (PEF) methodology. This comprehensive approach allows for a holistic evaluation of textile lifecycle impacts within the assumptions of the LCIA methods adopted for each impact category. **Table 188** provides a default list of EF impact categories and related assessment methods. The impact assessment methods do not include impact categories focusing on impacts related to textile fragmentation and biodiversity loss. Established impact methods to quantify such impacts are lacking and, therefore, they were not included for the analysed products in the quantitative assessment. Instead, these impacts were addressed qualitatively in section 10.2. This methodological choice recognizes practical constraints while ensuring that significant environmental concerns are nonetheless acknowledged within the study.
- **Monetisation of environmental impacts:** for two impact categories (i.e., ionising radiation, terrestrial eutrophication) an agreement on the monetisation factor to be used was not reached

because of a lack of scientific consensus in literature. Therefore, these categories are not part of the results of the societal LCC. For more information on the monetisation factors, the report by Magrini et al. (2025) and Gama Caldas et al. (2024) can be consulted (Gama Caldas et al., 2024; Magrini et al., 2025).

Table 188. EF impact categories with respective impact category indicators and characterisation models

EF impact category	Impact category indicator	Unit	Characterisation model
Climate change, total	Global warming potential (GWP100)	kg CO ₂ eq	Bern model - Global warming potentials (GWP) over a 100-year time horizon (based on IPCC 2013)
Ozone depletion	Ozone depletion potential (ODP)	kg CFC-11 eq	EDIP model based on the ODPs of the World Meteorological Organisation (WMO) over an infinite time horizon (WMO 2014 + integrations)
Human toxicity, cancer	Comparative toxic unit for humans (CTUh)	CTUh	based on USEtox2.1 model (Fantke et al., 2017), adapted as in (Saouter et al., 2018)
Human toxicity, noncancer	Comparative toxic unit for humans (CTUh)	CTUh	based on USEtox2.1 model (Fantke et al., 2017), adapted as in (Saouter et al., 2018)
Particulate matter	Impact on human health	Disease incidence	PM model (Fantke et al., 2016; UNEP, 2016)
Ionising radiation, human health	Human exposure efficiency relative to U235	kBq U ₂₃₅ eq	Human health effect model as developed by (Dreier et al., 2025) (Frischknecht et al., 2000)
Photochemical ozone formation, human health	Tropospheric ozone concentration increase	kg NMVOC eq	LOTOS-EUROS model (Van Zelm et al., 2008) as applied in ReCiPe 2008
Acidification	Accumulated exceedance (AE)	mol H ⁺ eq	Accumulated exceedance (Seppälä et al., 2006; Posch et al., 2008)
Eutrophication, terrestrial	Accumulated exceedance (AE)	mol N eq	Accumulated exceedance (Seppälä et al., 2006; Posch et al., 2008)
Eutrophication, freshwater	Fraction of nutrients reaching freshwater end compartment (P)	kg P eq	EUTREND model (Struijs et al., 2009) as applied in ReCiPe
Eutrophication, marine	Fraction of nutrients reaching marine end compartment (N)	kg N eq	EUTREND model (Struijs et al., 2009) as applied in ReCiPe
Ecotoxicity, freshwater	Comparative toxic unit for ecosystems (CTUe)	CTUe	based on USEtox2.1 model (Fantke et al., 2017), adapted as in (Saouter et al., 2018)
Land use	Soil quality index	Dimensionless (pt)	Soil quality index based on LANCA model (De Laurentiis et al., 2019) and on the LANCA CF version 2.5 (Horn and Maier, 2018)
Water use	User deprivation potential (deprivationweighted water consumption)	m ³ water eq of deprived water	Available WATER REmaining (AWARE) model (UNEP, 2016; Boulay et al., 2018)
Resource use, minerals and metals	Abiotic resource depletion (ADP ultimate reserves)	kg Sb eq	(van Oers et al., 2002) as in CML 2002 method, v.4.8
Resource use, fossils	Abiotic resource depletion – fossil fuels (ADP-fossil)18	MJ	(van Oers et al., 2002) as in CML 2002 method, v.4.8

13.11.3 Limitations due to LCI information availability

Along with the methodological limitations described in the previous subsection, limitations in the inventory have also been identified and some of them are displayed in the following points:

- **Use of assumptions in the absence of inventory data:** In instances where detailed inventory data is unavailable, the study relies on assumptions grounded in literature review and expert judgment. These assumptions serve to fill data gaps and maintain the completeness of the assessment process, with the exception of elastane, modal and acetate materials for which a suitable dataset or proxy could not be found and used. However, relying on assumptions introduces unknown variability and uncertainty into the assessment.
- **Use of proxies:** To address areas where direct inventory data does not correspond with available datasets, proxies were employed, particularly utilizing datasets from Ecoinvent 3.10³⁵⁹ (Wernet et al., 2016). While proxies are a pragmatic solution in such scenarios, they involve an element of substitution that may not entirely reflect the specific dynamics of the original data. The use of proxies requires careful consideration and introduces an additional layer of uncertainty. An example is given by the modelling of the spinning process (one of the often relevant sub-stage in the production phases); of the four main processes used in the textile sector (ring, melt, dry and wet spinning) only ring and melt spinning were modelled and used to carry out the analysis; of which melt spinning was used as a proxy to represent the wet and dry processes as it was not possible to obtain and find any primary data and information that would allow modelling these two processes. For further details on the data used to support the analysis, see section 13.11.9.
- **Geographical, temporal and technological representativeness:** The textile sector's value chain is highly fragmented and internationally dispersed. This structure reflects cost-cutting and regional specialization. It is therefore difficult to provide a complete and comprehensive representation of this characteristic territorial fragmentation. The study was conducted by identifying and modelling the system by considering the countries around the world that play a leading role in the various processes comprising the entire value chain. However, when specific supporting data were not found, primary and/or secondary data relating to a global average context were used. The reference year for the study is 2022. Many data supporting the analysis (e.g. unit costs) are specific to that year. For other types of information, sources, and data collected, the reference time horizon dates to years prior to the one used as a base point (e.g. some Ecoinvent datasets are built with supporting information dating back to the early 2000s or even earlier). Similarly, from the technological point of view, production and transformation processes globally involve different technologies and machinery. The analysis was conducted with data primarily from the Ecoinvent database. The datasets are built using multiple sources of information (e.g., publications, literature, and surveys conducted by companies); however, in some cases the data source is more limited (including information from a single company used as a proxy to represent the production process of an entire nation or even the global context). For further details on the data used to support the analysis, see section 13.11.9. These methodological approaches highlight practical constraints, whilst ensuring that process and material environmental burdens and costs are still recognized and accounted for within the study.
- The difficulty in retrieving country-specific information on taxes and tariffs applied to the disposal or other treatments of post-industrial waste (i.e. waste generated during the production phases) has led to modelling the system without quantifying these cost items for the LCC analysis. The waste treatments were quantified including only the direct process costs (which should in any case represent the most relevant fraction).
- **Chemical substances:** Chemical substances are involved in many parts of the lifecycle, especially in the dry and the wet textile manufacturing processes, as finishing agents (softeners, water repellents, etc) that can stay in the final products, as auxiliaries and additives (wetting agents, dispersing agents, bleaching agents, etc) which do not remain in the final products and sizing agents like starch and

³⁵⁹ CUT-OFF system model

Polyvinyl Alcohol (PVA). Although this study tried to include as many as possible, limitations relative to lack of information on the specific chemicals have to be acknowledged. In some cases, the information acquired was not granular enough to allow for a full mapping of the chemicals used in a specific process. This could be since the contribution to the total mass could be lower than a specific threshold applied in a cut-off criterion (e.g. inputs contributing with less than 1% could be excluded). Of course, a chemical substance can have a small contribution to a process input, but their impacts could be significant and not negligible (e.g. contribution more than 5% to a toxicity related impact category). Nevertheless, the adverse effects of using some of the chemicals might not be apparent when adding the impact of their production that can be found in a dataset, because it is their discharge to the environment that could be of greater concern. Along with the lack of information there could also be a methodological limitation of missing LCIA method for translating emissions into impacts. Finally, even if the chemicals can be identified, a representative dataset might not be available, and a proxy might have been used. For example, acrylic acid was used as a proxy for pigment paste thickener (polyacrylate component).

By clarifying these methodological and inventory limitations, this section aims to provide transparency regarding the constraints faced in the study. Acknowledging these constraints also highlights areas for future research and data collection, which could lead to more precise and comprehensive evaluations of textile products in future assessments. More details on the nature of the assumptions made and the proxies used are provided in section 13.11.9.

13.11.4 Target audience

The development of this study actively involves a diverse range of stakeholders, including policymakers, EU Member States, industry representatives, researchers, and non-governmental organizations.

13.11.5 Comparative study to be disclosed to the public

The aim 2 of the study (see section 13.11.1) requests a comparative analysis of the current performances of textile apparel with alternative design options. This approach assesses how potential ecodesign interventions could enhance environmental performance of products placed on the EU market without causing disproportionate costs.

13.11.6 Commissioner of the study and influential actors

As described in the foreword of this document, the commissioners of the study are the Directorate-General for Environment (DG ENV) and the Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs (DG GROW). The influential actors are the stakeholders mentioned in Section 13.11.4.

13.11.7 Definitions

- **Unit process:** elementary physical and/or chemical process that can be considered independently within the sequence of transformations targeted to generate a final product. The unit process implies the transformation of raw materials and/or semi-finished products by manual and/or automatic operations to obtain value added output, such as intermediate- or final- product. The unit process also represents the smallest element considered in the life cycle inventory analysis for which input and output data may be consistently quantified.
- **Production system:** the ensemble of sequential interconnected unit process to obtain products belonging to the same product category.
- **Inventory input:** inflow of raw materials, energy, other physical resources from the environment and semi-finished products or products to the analysed unit process.
- **Inventory output:** ensemble of outflows of waste, by-products, and emissions to air, soil and water that are generated by the analysed unit process.
- **Product category:** a group of products displaying common features, as reported in section 9.3.

- 12155 — **Life Cycle Stage (LCS):** one of the specific phases within the product life cycle that span from
12156 “cradle” (i.e., raw materials sourcing) to “grave” (i.e., product end of life).
- 12157 — **System Boundaries:** the logical and physical boundaries that define which processes and phases of
12158 a product’s life cycle are considered in the assessment and which are not all while moving within the
12159 boundaries indicated by the Technosphere and the Ecosphere.
- 12160 — **Technosphere:** It includes the processes and intermediate products generated by human activities. It
12161 is the set of all structures, systems, and objects made by humans through technology that modify or
12162 influence the natural environment.
- 12163 — **Ecosphere:** It is the set of all ecosystems on Earth: it includes the physical environment (air, water,
12164 soil) and all living things (biosphere). It can be considered as an interconnected system in equilibrium.
12165 It includes the interactions between living organisms and non-living factors, such as climate, geology,
12166 and natural cycles (water, carbon, nitrogen, etc.).
- 12167 — **CAPEX:** Capital Expenditures (CAPEX) refer to the investments made by a company to purchase,
12168 improve, or maintain durable assets such as buildings, plants, equipment, machinery, or
12169 infrastructure.
- 12170 — **OPEX:** Operating Expenditures (OPEX) are the costs incurred by a company to manage its day-to-day
12171 operations. These are recurring expenses, necessary to keep the business running, but which do not
12172 generate a durable asset over time.

12173 **13.11.8 Goal and scope**

- 12174 This model has mainly two goals:
- 12175 1. Investigating the environmental and economic performances of textile apparel included in the
12176 scope of this preparatory study;
 - 12177 2. Assessing the environmental and economic effects of design options that could be considered to
12178 develop future potential ecodesign requirements on the same products.
- 12179 Goal 1 supports the analysis to be performed in Task 5 of the MEeRP (section 10.1), whereas goal 2 supports
12180 the analysis to be performed in Task 6 of the MEeRP (section 11).

12181 **13.11.8.1 Products included in the scope**

12182 The scope of the LCA-LCC study covers all products defined in the PS (Section 9.3). **Table 189** presents the
12183 representative products (RPs) considered in the study, along with details on their mass, service lifespan, number
12184 of uses, and the average number of uses prior to washing. More information about the driver and rationale
12185 based on which the categorisation and selection of the Representative products was undertaken can be found
12186 in Section 9.3.

12187 The numerical values of the characterizing parameters of each RP were obtained from several research studies,
12188 namely Bakowska et al. (2025), Drycleaning Institute (2015), Gray et al. (2022),; Laitala et al. (2018a, 2018b),
12189 Quantis (2024), Roos et al. (2015), WRAP (2017). The values were determined by averaging what is reported
12190 in the literature.

12191 **Table 189.** Representative Products (RPs) of textile apparel included in the scope of the preparatory study and their main
12192 characterizing parameters.

Representative product	Parameter full name	Final mass RPs	Number of uses (expressed in days of wear)	Average number of uses prior to washing (expressed in days of wear)	Fixed timeframe*	Service lifespan of 1 product
	Unit	kg	day	day	years	years
RP1	Knitted	0.23	72	5.27	6	3.83
RP2	Woven w/o denim fabric	0.28	78	7.70	6	4.59

Representative product	Parameter full name	Final mass RPs	Number of uses (expressed in days of wear)	Average number of uses prior to washing (expressed in days of wear)	Fixed timeframe*	Service lifespan of 1 product
	Unit	kg	day	day	years	years
RP3	Denim	0.47	237	7.23	6	5.09

*The fixed timeframe is equal to the longest service lifespan within the products considered in the RP.

Source: own elaboration, based on Bakowska et al. (2025), Drycleaning Institute (2015), Gray et al. (2022), Laitala et al. (2018a, 2018b), Quantis (2024), Roos et al. (2015), and WRAP (2017).

The entire list of parameters characterizing each RP used to conduct the LCA and LCC analyses are reported in the tab "Base Case - Input data" of spreadsheet "Textile-Prep-Study_3rd-Milestone_Model" ⁽³⁶⁰⁾.

13.11.8.2 Functional unit

The functional unit (FU) of the LCA and LCC is defined as *the coverage of the body using a specified item that fulfils the consumer's needs over 6 years within the European Union (EU)*. **Table 190** defines the main aspects of the FU.

Table 190. Functional unit (FU) of the LCA-LCC.

Aspect of the FU	Definition
What	Coverage of the body
How much	With an item that satisfies the need of the consumer during a certain timeframe
Where	In EU
For how long	Timeframe (T) expressed in years, 6 years in this study
How well	Keeping conditions that satisfy the owners until disposal

Source: own elaboration.

The timeframe for the FU is determined based on the service lifespan of investigated products (section 3.2.1). The product with the longest service lifespan serves as the benchmark for setting this fixed timeframe. For example, if the product jackets and coats is identified as having the longest service lifespan at 6 years, then this 6-year period becomes the standard fixed timeframe (T expressed in years) for evaluating all other RPs in the study.

To each RP is assigned a specific service lifespan, which is closely linked to the number of uses and care cycles it can withstand. For example, the RP knitted products might have a duration of service of 3.83 years, during which it is expected to endure around 72 uses. These parameters help define the product's lifespan and performance under typical consumer behaviour based on the literature.

Translating the number of uses and care cycles into a parameter, such as "average number of uses prior to washing", provides a valuable metric for understanding the interaction between how often an item is worn and how frequently it is cleaned. For the example of the RP knitted products with a service lifespan of 3.83 years, 72 uses, and 13.66 care cycles, this can be expressed as 5.27 uses per care cycle. This metric is important for assessing the wear-and-care patterns associated with a specific apparel item. Understanding the balance between uses and care cycles helps in evaluating the environmental impacts of washing, such as water and energy consumption, as well as the wear and tear on the garment. It also provides insights into consumer behaviour and the potential for extending the lifespan of apparel through optimized washing practices.

For example, the main function of the RP knitted products is designed to provide adequate body coverage for a consumer in the European Union over a 6-year period. The study assumes that a single RP knitted product has a service lifespan of approximately 3.83 years, accommodating around 72 uses and 13.66 care cycles. A consumer would require a total of 1.57 RP knitted product over the 6-year timeframe. By considering the entire service lifespan of these 1.57 RP T-shirts, the analysis can effectively assess the environmental impacts associated with exploitation of the consumer needs in 6 years, considering their production, distribution, usage, and end-of-life phases, offering valuable insights into resource use, waste generation, cost and potential sustainability enhancements within the specified geographic and temporal context.

³⁶⁰ Textile-Prep-Study_3rd-Milestone_Model is available on the project website I the section "Documents" at [this link](#).

12229 “Keeping conditions that satisfy the owners until disposal” means that the product does not present any of the
12230 defects that would trigger the consumer to discard it.

12231 **13.11.8.3 Reference flow**

12232 The reference flow (RF) is defined as the total number of items required to satisfy the consumer needs over a
12233 specified timeframe as reported in the FU.

12234 To express the reference flow in terms of mass, the mass of the single item should be determined. Suppose
12235 the weight of one item is denoted as W expressed in kg. If the item's service lifespan is L expressed in years,
12236 and the fixed timeframe is T expressed in years, then the consumer would need T/L items for the entire period.
12237 The total weight of these items is calculated by multiplying the number of items by the weight of a single item.

12238 For example, where a consumer needs 1.57 RPs to ensure adequate body coverage over 6 years, the reference
12239 flow is 1.57 RPs. To express this reference flow in terms of mass, we consider the weight of a single RP, which
12240 for knitted products is 0.230 kg. Therefore, the total weight of the 1.57 RPs needed over the 6-year period is
12241 calculated by multiplying the number of RPs by their individual weight. Thus, the reference flow in mass is
12242 0.361 kg, representing the total material requirement to satisfy the consumer's apparel needs over the
12243 specified fixed timeframe.

12244 **13.11.8.4 System boundaries and completeness requirements**

12245 The analysis is based on a model developed in accordance with information reported in the PS and allows
12246 assessing representative products' impacts and cost.

12247 The analysed RPs are defined by the specification of features, properties and attributes reported in terms of
12248 descriptors and quantifiers, which include:

- 12249 — Type and amount of fibre.
- 12250 — Kind and number of trims composing the products.
- 12251 — Reference conditions of use (number of uses and care cycles).
- 12252 — Quantitative information of the amount of knitted and/or woven yarns.
- 12253 — Proportions of white, mono-colour, multicolour textile apparel.
- 12254 — Identifying confectioning options.

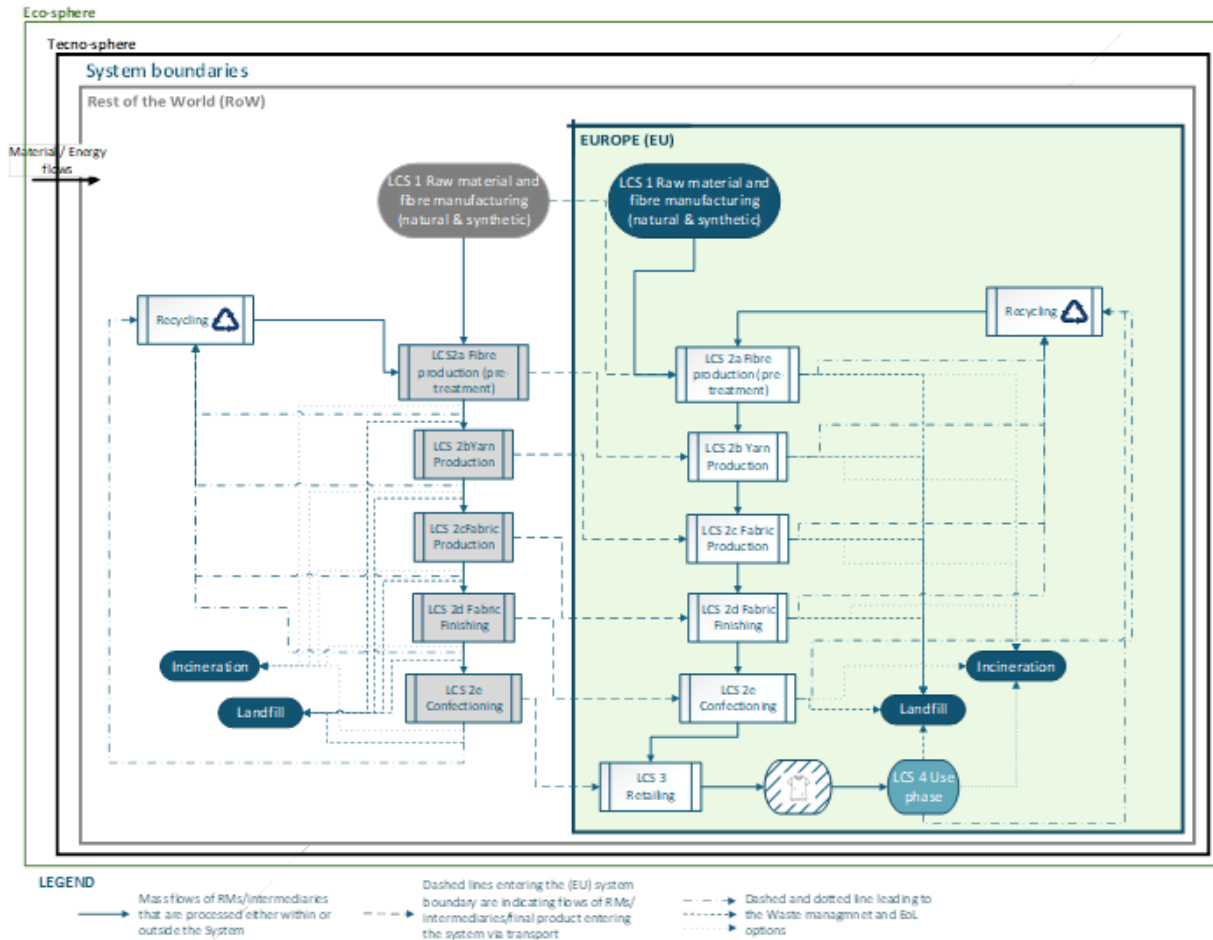
12255 The model supports a “cradle-to-grave” analysis approach considering the entire life cycle of each RP, which is
12256 described by accounting for the corresponding raw material sourcing, the specified production system and
12257 distribution model, the products conditions of use, as well as its end-of-life options. Specifically, the following
12258 Life Cycle Stages (LCSs) are addressed in the LCA and LCC analyses:

- 12259 — LCS 1 RAW MATERIAL AND FIBRE MANUFACTURING
- 12260 — LCS 2 PRODUCTION PROCESS:
 - 12261 ○ LCS 2a Fibre pre-treatment
 - 12262 ○ LCS 2b Yarn production
 - 12263 ○ LCS 2c Fabric production
 - 12264 ○ LCS 2d Fabric finishing processes
 - 12265 ○ LCS 2e Confectioning
 - 12266 ○ LCS 2f Waste management of post-industrial
- 12267 — LCS 3 RETAILING AND TRANSPORT
 - 12268 ○ LCS 3a Retailing
 - 12269 ○ LCS 3b Transport
 - 12270 ○ LCS 3c Waste management of pre-consumer waste.
- 12271 — LCS 4 USE

— LCS 5 END OF LIFE (Waste management of post-consumer waste)

The flowchart presented in **Figure 94** represents the analysed system by graphically identifying the system boundaries, flows and life cycle stages of a textile apparel product. The global geographical context that the analysis is able to examine is also represented, with production and life stages taking place in Europe and the Rest of the World (non-EU). The model assumes that the initial stages of textile apparel life cycle occur globally (i.e. partially in Europe and mainly outside Europe), then converge during the distribution phase within Europe. A more detailed description of the geographical context where fibre production and the manufacturing and transportation stages of semi-finished and finished products take place is presented in section 13.11.9.

Figure 94. System boundaries, with distinction between European and Rest of the World context.



Source: own production.

The boundaries of the system are set within the techno-sphere. The material-energy flows entering the system originate both from the eco-sphere and from the techno-sphere.

The upstream boundaries entry point into the system is represented by the type of raw materials and components, mainly represented by fibres and trims. The model enables quantifying environmental and cost impacts of all the energy and materials flows through each LCSs and within LCS2 through each unit process of the production system.

The system considers waste management (post-industrial, pre-consumer) and the end-of-life of the RP itself (post-consumer). In both cases, incineration, landfill and recycling processes have been modelled.

For each LCS the analysis allows a regionalised assessment with impact drivers derived as a weighted average of energy and material flows related to each specific region considered on a global scale. In fact, it can be seen

12293 from the flow chart how there is a strong relationship and exchange of materials/intermediates throughout all
 12294 stages of the life cycle (up to the production of the finished product) between the EU and the rest of the world.

12295 Coherently, impact drivers associated to transport from source nodes (e.g. yarn production company) to
 12296 destination nodes (e.g. fabric production company) of raw materials, intermediaries and finished textile apparel
 12297 products are modelled accordingly by computing the weighted average based on routing and shipment
 12298 quantities.

12299 The environmental and economic analyses adopt the same system boundaries with limited exceptions. One of
 12300 these is the quantification of environmental impacts and costs associated with capital goods (i.e., the production
 12301 and use of equipment). The environmental impacts related to the capital goods are excluded from the
 12302 assessment, as processes are carried out in machines that are used to produce such a large number of products
 12303 that their specific impact per unit mass of product can be considered negligible. Nevertheless, the capital
 12304 expenditures relative to acquiring the machinery are accounted for in the economic analysis. Finally, with
 12305 respect to infrastructure, the impacts and costs related to it are not considered in neither the environmental
 12306 nor the economic assessments.

12307 Another small exception is the model of LCS1: indeed, the economic analysis is directly based on the market
 12308 prices of fibres, while in the environmental assessment a detailed analysis of all the processes was performed
 12309 (see section 13.11.9.1).

12310 For the confectioning sub-stage, all consumables that could be used during this phase were neglected (and
 12311 therefore excluded from the system boundaries). Indeed, an initial assessment of the main materials
 12312 (consumables) related to this process was conducted and revealed that their environmental impacts are
 12313 negligible compared to the ones of the ancillary material and direct energy requirements of the other modelled
 12314 unit processes.

12315 **13.11.9 Life Cycle Inventory analysis**

12316 The model supports a “cradle-to-grave” analysis approach, and considers the entire life cycle of each product,
 12317 which is described by considering the corresponding raw material sourcing, the specified production system
 12318 and distribution model, the products conditions of use, as well as its end-of-life options. Specifically, the
 12319 following LCSs are addressed in the Life Cycle Assessment and Life Cycle Costing analyses:

- 12320 — LCS 1 RAW MATERIAL AND FIBRE MANUFACTURING.
- 12321 — LCS 2 PRODUCTION PROCESS:
- 12322 — LCS 3 RETAILING AND TRANSPORT
- 12323 — LCS 4 USE
- 12324 — LCS 5 END OF LIFE (Waste management of post-consumer waste).

12325 For each LCS the model allows to perform a regionalised assessment with impact factors coming from
 12326 Ecoinvent. One of the main data sources used to build the inventory was the Ecoinvent³⁶¹ (Wernet et al., 2016)
 12327 database. The spreadsheet contains details of the specific datasets from which the information used to model
 12328 the analysed system was extracted. Additional information for the datasets used (i.e. detailed product or
 12329 process description, geographical, temporal, and technological references) can be accessed through the
 12330 [ecoquery webpage](#)³⁶² provided by Ecoinvent (in the search bar provided on the page, it is possible to search for
 12331 the specific dataset of interest using the name provided in the spreadsheet).

12332 **13.11.9.1 Raw Material and Fibre Manufacturing (LCS 1)**

12333 **Description of the stage**

12334 The first phase of the life cycle is linked to the production and supply of raw materials used in the textile sector.
 12335 The raw materials include not only the fibres but also the trims and packaging materials.

³⁶¹ CUT-OFF system model

³⁶² Ecoquery webpage link: <https://ecoquery.ecoinvent.org/3.11/cutoff> [Accessed 10/10/2025]

LCA and LCC modelling choices and shared inventory

This is the only stage of the life cycle where there is a differentiation of inventory and model data as regards the LCA and LCC analysis. In fact, for the quantification of environmental impacts, it was preferred to model the production of the main fibres used in the textile sector with a "cradle to gate" approach. Instead, the economic analysis is directly based on the market prices of a fibre. The choice to use market prices to quantify the costs of LCS1 is twofold, firstly, the focus of the study is on the life cycle of textile products and not specifically on the fibres themselves used in their production, thus removing the need to identify the contributions to the cost items for the production of the fibres themselves (the situation is different for the environmental dimension where there is a strong need to determine the most significant contributions to the environmental burden associated with the production of the fibres themselves). The second justification is given by the fact that in the final quantification of the cost generated throughout the entire life cycle of a textile apparel (for producer, consumer, and society) the fibres themselves are always assumed to be purchased on the market (therefore, it is not necessary to quantify the direct costs of their production). This means that, in the case of an analysis conducted from the perspective of the producer of a textile product, it was assumed that the producer itself plays no active role in the production of the fibres themselves. The market unit prices used to quantify the costs of the fibres themselves are reported in the spreadsheet "Textile-Prep-Study_3rd-Milestone_Model" (³⁶³) tab containing all the parameters characterizing the 3RPs, while the inventory in the same spreadsheet only reports the input/output flows used to model the fibres themselves and subsequently conduct the environmental analysis. Specific input/output data were identified and quantified for the production of cotton, organic cotton, wool, polyester, recycled polyester and viscose fibres. The model considers as a starting point for the quantification of the environmental and economic impacts the data extracted from internationally recognized databases (e.g. Ecoinvent 3.10³⁶⁴ Wernet et al., 2016) and market prices (several sources including: United States department of agriculture report on Cotton and Product Update³⁶⁵; Emerging Textiles reports³⁶⁶; United States department of agriculture Economic Research Service and reports³⁶⁷; Textile Value Chain³⁶⁸; ICAC - International Cotton Advisory Committee³⁶⁹; UN Comtrade data³⁷⁰; Zaubas website³⁷¹; ChemAnalyst³⁷² website; Renewcell annual report³⁷³), respectively as a starting point for the quantification of the environmental and economic impacts. Fibre market prices for 2022 were used.

The mass of the fibres used for the subsequent production phases of the RPs were determined by the composition and the final product mass from **Table 189**. This quantity is identical for both LCA and LCC analyses and considers the losses of the production phase and the distribution phase.

When identifying the data to support the analysis, attention was paid to the geographical context from which the fibres originate. It should be noted that most of the fibres produced globally are of unknown origin due to the difficulties in tracking information. However, the available information (**Table 191**) shows the global dimension of fibre production, with China and India producing the majority of fibres and many other countries in all continents producing specific fibres. The main source of information is reported in **Figure 3** adapted from Textile Exchange, 2022, integrated with additional data for wool with data from International Wool Textile Organisation³⁷⁴, for Polyester and Polyester recycled from KBV³⁷⁵.

When it comes to recycled polyester, the recycling of PET bottles was considered, with the possibility of utilising impact data from polyester textile recycling as well.

³⁶³ Textile-Prep-Study_3rd-Milestone_Model is available on the project website in the section "Documents" at [this link](#).

³⁶⁴ CUT-OFF system model

³⁶⁵ <https://www.usda.gov/>

³⁶⁶ <https://www.emergingtextiles.com/sample-reports/>

³⁶⁷ <https://www.usda.gov/>

³⁶⁸ <https://textilevaluechain.in/>

³⁶⁹ <https://icac.org/>

³⁷⁰ <https://comtradeplus.un.org/>

³⁷¹ <https://www.zaubas.com/>

³⁷² <https://www.chemanalyst.com/>

³⁷³ <https://www.circulose.com/en/>

³⁷⁴ WTO, "World Sheep Numbers & Wool Production" report. Link: <https://iwto.org/> [Accessed July 2025]

³⁷⁵ KBV, Link: [Europe Polyester Fiber Market Size & Growth Forecast by 2030](#) report [Accessed July 2025]

Table 191. Location of production for specific textile fibres (acronym: Unk = unknown; AU = Australia; CN = China; EU = Europe; ID = Indonesia; IN = India; HU = Hungary; NZ = New Zealand; Oth = other countries; PO = Poland; RSA = South Africa; TU = Türkiye; TW = Taiwan; USA = United States).

Fibre	Breakdown by country					
Polyester (PES)	Unk	CN	Oth	TU	EU	
	57.70%	10.70%	8.20%	5.80%	17.60%	
Cotton	Unk	IN	Oth	CN	USA	EU
	33.80%	20.70%	6.20%	22.70%	15.00%	1.40%
Man-made cellulosic fibres (MMCF)	Unk	CN	Oth	IN	ID	
	58%	18%	16%	5%	3%	
Polyamide (PA)	Unk	CN	TW	USA		
	81%	15%	3%	1%		
Wool	Unk	AU	RSA	NZ	Oth	EU
	44.20%	22%	2.90%	9.40%	17.30%	4.10%
Other animal fibres	CN	Unk	HU	PO	EU	
	88%	9%	2%	1%	3%	
Polyester recycled	Unk	CN	Oth	TU	EU	
	57.70%	10.70%	8.20%	5.80%	17.60%	
Organic Cotton	Unk	IN	Oth	CN	USA	EU
	33.80%	20.70%	6.20%	22.70%	15.10%	1.40%

Source: own elaboration based on **Figure 3** adapted from Textile Exchange, 2022, integrated with additional data for wool with data from International Wool Textile Organisation³⁷⁶, for Polyester and Polyester recycled from KBV³⁷⁷.

For fibres in the textile sector for which primary data could not be acquired (as their use is of limited relevance on a global scale), the inventories were based on Ecoinvent datasets (specific and/or proxies) and market prices were used, as done similarly for the other fibres.

For trims, based on the information provided in the PEFCR v3.1 (Quantis, 2025b), it was assumed that they are made of Polyester (50%), Polyethylene terephthalate (33%) and brass (17%). For packaging, based on PEFCR v3.1 and own estimation, the assumption made is that per kg of product the required materials will be: 60 grams of carton board, 60 grams of paper bag, 0 grams of polyethylene for a thin bag and 20 grams of polyethylene for a thick bag.

13.11.9.2 Production Process (LCS 2)

The model of the production stage of textile apparel has been divided into seven sub-stages. Five stages represent actual production phases. Two stages consider the waste management of post-industrial waste and the transport that occurs along the entire supply chain.

As mentioned previously these are the sub-stage considered:

- LCS 2a Fibre pre-treatment
- LCS 2b Yarn production
- LCS 2c Fabric production
- LCS 2d Fabric finishing processes
- LCS 2e Confectioning
- LCS 2f Waste management (post-industrial)
- Transport LCS 1 & LCS 2

³⁷⁶ IWT, "World Sheep Numbers & Wool Production" report. Link: <https://iwt.org/> [Accessed July 2025]

³⁷⁷ KBV, Link: [Europe Polyester Fiber Market Size & Growth Forecast by 2030](#) report [Accessed July 2025]

Description of the stage

Fibre pre-treatment (LCS 2a) applies only to natural fibres, which go through similar preparation steps before yarn production. Fibre pre-treatment process involves “scouring” and is applied only to wool fibres (Section 13.310.1.2.4). Cotton fibres are usually cleaner than wool fibres and their pre-treatment process is usually performed in a blowing room (this step is considered in the following production phase that is the ring spinning). Although other natural fibres often require specific cleaning treatments, these are not included in the model. The choice is supported by the analysis conducted in the work Refashion 2023b), in which it is observed that the use of natural fibres, other than cotton and wool, is less than 1% by weight compared to 95% of the total non-reusable textile raw materials.

One of the key processes in textile apparel manufacturing is yarn production (LCS 2b). Spinning is the process that converts fibres into yarns, and it is applied to all types of fibres (i.e. natural and chemical). The yarn production phase has been developed considering as possible main spinning processes ring spinning, dry spinning, melt spinning and wet spinning. Only two of these, however, have been modelled and implemented to conduct the analysis, i.e. ring and melt spinning. The reason that led to making this choice was the complete lack of information that would have allowed modelling the dry and wet spinning processes (both from literature and from requests for primary data from stakeholders).

Based on a technological point of view, it has been assumed that the ring spinning process can be applied for the assessment of environmental burdens and costs to all natural fibres; instead, acrylic, acetate and elastane fibres could be spun using a dry spinning process; melt spinning is applied for the treatment of polyamide, polyester and PTFE fibres, while man-made cellulosic fibres could be spun using a wet spinning process. The melt spinning process has been used as a proxy to replace the dry and wet spinning processes.

Fabric production (LCS 2c) is the second central phase of textile production and involves the creation of a two-dimensional structure through the interlacing of yarns. Common methods include weaving and knitting that are suitable for all types of yarn. These methods are energy intensive and to prevent yarn damage, substances like sizing chemicals and lubricants are used for yarn strengthening and reducing friction.

LCS 2d Fabric finishing processes - includes a variety of processes that are an intermediate step in the treatment of yarns and/or fabrics with dyes and chemicals. These treatments are applied based on the specific fibre requirements and on the features of the products analysed, which include the shares of uncoloured (white) multi-coloured and single-coloured items for the products in the considered RP category. This step includes various wet treatments aimed at different objectives, such as bleaching, dyeing, printing and wet finishing (as reported in section 13.3). These treatments, especially the dyeing process, can be performed at different stages of textile production, i.e. at fibres, yarns and fabrics stages depending on the technical requirements and practice. The finishing stage is used to give textile substrates specific properties for the end use, such as unique visual effects, tactile sensations or specific functional characteristics and this applied to all type of fibres.

After the finishing treatments, the fabric is sent to a confectioning process (LCS 2e). The activities carried out concern pattern design, fabric spreading, cutting, sewing and finishing processes (such as pressing, ironing, blocking and setting) of the final product. This is the stage where the larger portion of post-industrial waste is generated (according to the information reported in section 9.2.4).

Each of the production stages generates a quantity of textile waste defined as post-industrial (Huygens et al., 2023). The analysis modelled an ad-hoc process for their disposal (described in section 13.11.9.6).

LCA and LCC modelling choices and shared inventory

The following table lists the model and inventory details for each LCS 2 sub-stage.

The original data were collected and quantified with reference to the mass of 1 kg of fibre/semi-finished product entering the analysed production process. These values were then recalculated and presented in the inventory file “Textile-Prep-Study_3rd-Milestone_Model”⁽³⁷⁸⁾. The values reported in the file were allocated based on the actual quantity of fibre and/or semi-finished product of each RP entering each individual process. More detailed information (e.g. list of all inputs/outputs, their numerical values and data source) is provided in the inventory contained in the spreadsheet.

³⁷⁸ Textile-Prep-Study_3rd-Milestone_Model is available on the project website in the section “Documents” at [this link](#).

12448 **Table 192.** Detail of modelling and inventory used in LCS 2 manufacturing.

Life Cycle sub-stage	Modelling and inventory details
LCS 2a Fibre pre-treatment	The modelling of scouring process is based on Roth et al. (2023). The input and output data are to be understood as applied for 1 kg of fibre entering the process. The input flows used to model the process refer to contributions of energy, chemical, and water consumption; these are used to quantify and evaluate both the environmental impacts and costs associated with this process. The direct emission flows into air, water or soil compartments have not been modelled and quantified because the supporting data come from the work of Roth et al. (2023), where no mention is made of these inputs/outputs. The lack of quantification of possible direct emissions, however, should not represent a significant issue in terms of completeness of the results, since the fact that these are not indicated in the work of Roth et al. (2023) (that is strongly focused on the identification of the main pollution sources and key environmental indicators), allows to assume that the process under consideration does not generate (or generates only negligibly) this type of flows. The economic analysis, in addition to the input flows indicated above, is modelled by considering further cost sources/drivers characterizing the specific process analyzed: (i) Machine economic value; (ii) Machine average lifetime (useful life); (iii) Average process duration; (iv) Employment. All these data, relating to 1 kg of fibre mass entering the process, were provided by industrial sector experts and own elaboration based on data reported in the (ITMF, 2021).
LCS 2b Yarn production	It has been assumed that the ring spinning process can be applied for the assessment of environmental burdens and costs to all natural fibres; whereas melt spinning is applied for the treatment of all synthetic and man-made cellulosic fibres: polyamide, polyester, PTFE, acrylic, elastane, viscose, lyocell and modal. The choice was dictated by the lack of qualitative and quantitative information to support modelling of the wet and dry spinning processes. In case the input/output data were obtained from the Ecoinvent 3.10 database they were processed to consider the incidence of production in different countries of the world. Indeed, the value reported in a specific dataset for a country was weighted by the percentage of production that actually occurs in that country compared to the global context (data reported in Table 198). In addition to the consumption of energy, water, materials and chemicals, defined as process inputs, the inventory for spinning processes is completed by the quantification of direct emissions and process waste (i.e. elementary emission flows into water, air and soil that occur directly during the process). The economic analysis, in addition to the input flows indicated above, is modelled by considering further cost sources/drivers characterizing the specific process analyzed: (i) Machine economic value; (ii) Machine average lifetime (useful life); (iii) Average process duration; (iv) Employment. All these data, relating to 1 kg of fibre mass entering the process, were provided by industrial sector experts and own elaboration based on data reported in the (ITMF, 2021). For melt spinning data related to (iii) and (iv) are estimation obtained from the web³⁷⁹ because no other sources are available.
LCS 2c Fabric production	The choice of the specific process for fabric production (weaving or knitting) is associated with the representative product analysed. In case the input/output data were obtained from the Ecoinvent 3.10 database they were processed to consider the incidence of production in different countries of the world. Indeed, the value reported in a specific dataset for a country was weighted by the percentage of production that actually occurs in that country compared to the global context (data reported in Table 198). The environmental assessment quantifies the impacts associated with specific direct emission/waste flows that occur in the air, water and soil compartments and that are commonly associated with the weaving and knitting processes.

³⁷⁹ <https://www.spic-emmen.com/> and <https://www.ri.se/en/expertise-areas/expertises/melt-spinning>

	For the economic analysis the models of weaving and knitting processes have been enriched with the following inputs: (i) Machine economic value; (ii) Machine average lifetime (useful life); (iii) Average process duration; (iv) Employment. All these data, relating to 1 kg of mass of yarn entering the process, were provided by industrial sector experts, data received after the first consultation from the stakeholder's through the questionnaire and own elaboration based on data reported in the (ITMF, 2021).
LCS 2d Fabric finishing processes	The model at first approximation level simplifies the inherent stage complexity providing a more general representation of the LCS 2d and assumes that the bleaching and colouring treatments are applied only at fabric stage. — The modelled processes are bleaching, dyeing, printing and finishing. Bleaching, dyeing and printing are applied to the unit mass of fabric in different proportions, depending on the specific RP considered and the type of fibre it is made of. In this case the model is structured so that each LCS 2d process can be applied to the specific mass of fabric input at this stage; — Each RP is distinguished by a percentage³⁸⁰ of products classified as: white, mono-colour, multi-coloured; — The bleaching process is applied only to natural fibres based on the specific share of white product for the analyzed RP; — For the fraction of white product, made from chemical (synthetic) fibres, it was assumed to use the dyeing process; — Mono-coloured products whether made of natural or synthetic fibres, are created through the dyeing process; — It was also assumed to equally divide the production of multi-coloured product as generated through the dyeing and printing process (equal share); — The wet finishing process is assumed to be always used in the production phase of each product and applied to the mass of fabric exit bleaching/dyeing/printing processes. Figure 95 shows the division of the flow of fabric during the LCS 2d; the fabric quantity coming out of the previous process is hypothetically split between natural and chemical fibres and each of these is subsequently divided between the bleaching, dyeing and printing processes. This division is based on the percentages of final product composed of white, mono and multi-coloured products. Figure 95. Flow chart of how the model framework is applied to represent the bleaching, dyeing and printing processes <pre> graph LR Fabric[Fabric] --> Natural[Natural fibers] Fabric --> Chemical[Chemical fibers] Natural -- White --> Bleaching[Bleaching] Natural -- Multi-coloured --> Printing1[Printing] Natural -- Mono-coloured & multi-coloured --> Dyeing1[Dyeing] Chemical -- Multi-coloured --> Printing2[Printing] Chemical -- White & mono & multi-coloured --> Dyeing2[Dyeing] </pre>

³⁸⁰ Percentage defined based on data published on the work of (Bakowska et al., 2025)).

	<i>Source: own production</i> The input/output flows that allow the modelling of the processes have been obtained from different data sources: — When input/output data were obtained from the Ecoinvent 3.10 database, these were processed to consider the incidence of production in different countries of the world. The value reported in a specific dataset for a country was weighted by the percentage of production that actually occurs in that country compared to the global context (data reported in Table 198); — More information than that obtained from the Ecoinvent 3.10 database, in particular on the type of chemicals (i.e. process support chemicals, dyes and pigment pastes) was obtained from the data reported in the work of Roth et al. (2023); — For data obtained from other sources, the value or indication found in the literature was simply reported in the inventory. Given the technological variety of dyeing processes, inventory data report averages representative data from batch and continuous processes. Given the simplification to the inherent complexity of the different existing dyeing processes, the unit process modelled may be regarded as a meta-model providing a representation to account for the associated impacts of many real dyeing processes. It was assumed that the dyeing process would be applied to cover the percentages of final products classified as mono-coloured and the 50% of final products classified as multi-coloured (in the latter case, in fact, all products with a colour or graphic, that can be created with a dyeing or printing process are classified with the multi-coloured label). In addition, the dyeing process is also applied to the product portion consisting of white garments, produced with synthetic fibres (these in fact do not undergo a bleaching process, but are assumed to be coloured in this phase). Printing is the colouration process that involves applying dye or pigment to the substrate surface with the aid of specific auxiliaries. This allows for colouring different areas and creating patterns. The colouring process involves applying a colour paste to the textile using various techniques, followed by fixation (see section 13.3). It was assumed to model the system in such a way that the printing process is applied to 50% of the fraction of products classified as multi-coloured. The actual wet finishing phase involves a variety of chemical and mechanical-physical treatments, which are generally applied after the coloring process (as reported in section 13.3). To represent and model the possible processes that make up this last phase, a general "wet finishing" process was used, extrapolating the supporting data from Ecoinvent. The environmental assessment, in addition to evaluating the environmental burdens due to process inputs, quantifies the impacts associated with specific direct emission/waste flows that occur in the air, water and soil compartments and that are commonly associated with each of these activities. Similarly to the previous sub-stages, the following inputs were also quantified for these processes: (i) Machine economic value; (ii) Machine average lifetime (useful life); (iii) Average process duration; (iv) Employment. All these data, relating to 1 kg of mass of fabric entering the process, were provided by industrial sector experts, data received after the first consultation from the stakeholder's through the questionnaire and own elaboration based on data reported in the (ITMF, 2021).
LCS 2e Confectioning	This sub-stage was modelled considering the processes of: • Pattern design • Spreading • Cutting • Sewing • Finishing Processes (pressing, ironing, blocking and setting) A further division was applied in the model by subdividing each of these processes for woven or knitted products.

	An initial assessment of what could be the main material/energy flows related to each process was performed, concluding that the material and energy flows involved in these processes could be negligible compared to the material flows and direct energy requirements of the other unit processes modelled. However, data on the average energy consumption for each process were collected and used in the model, while eventual materials inputs to the processes were neglected. If the environmental dimension is not very relevant in this sub-stage, the same cannot be said for the economic one, there is in fact a strong component of use of machinery and manpower. For the processes of spreading, cutting, sewing and finishing, inventory data were collected in relation to economic aspects such as: (i) Machine economic value; (ii) Machine average lifetime (useful life); (iii) Average process duration; (iv) Employment. Only the average process duration and employment were considered for the pattern design processes. All these data, relating to 1 kg of fabric mass entering the process, were provided by industrial sector experts and data received after the first consultation from the stakeholders through the questionnaire.
LCS 2f Waste management (post-industrial)	Textile material losses occurring during LCS 2 (from LCS 2a a to LCS 2e sub-stages) are accounted for in this dedicated section of the model. The model for analysing environmental burdens and costs associated with end-of-life management of textile waste from the production stage will be described in more detail in section 13.11.9.6 In general, the developed model encompasses the contributions of waste transport, recycling, incineration and landfilling processes. The model has been implemented to represent both the European and extra Europe (named non-EU) context. The mass quantity of textile waste generated along the LCS 2 was determined by summing up the losses generated at each stage. As in LCS 5 the economic data for the quantification of costs related to waste management processes have been taken from Solis et al. 2024) for the European context. To represent the non-EU scenario, costs have been recalculated by normalizing the OPEX inputs for electricity, water and labor costs with respect to the average cost values of non-EU countries. The sorting phase was not considered for the extra-European waste management model. The model developed for the LCC analysis of waste management in LCS2 does not consider any waste treatment/disposal tax or tariff as a cost item. This is because, given the global context of the production phases of a textile product, it has been difficult to identify and allocate specific national taxes and tariffs to the modelled waste treatment processes. The waste management in LCS2 only quantifies the direct costs due to the use of materials, energy, and human resources in the recycling, incineration, and landfill processes themselves.

Source: own elaboration

General aspects

The LCS 2 analytical model allows quantifying the **losses** at each unit process within the production described in LCS 2. Losses rates are estimated based on the average between two different sources: Ecoinvent 3.10³⁸¹ (Wernet et al., 2016) and EF3.1 databases (European Commission, 2022).

The overall losses in the material manufacturing processes and the finishing of textile apparel is presented in **Table 193**.

Table 193. Losses along the product manufacturing value chain.

Manufacturing unit process	Losses
Ring spinning	16.7%
Melt spinning	1.5%
Weaving	4.8%
Knitting	2.2%

³⁸¹ CUT-OFF system model

Manufacturing unit process	Losses
Bleaching	0.0%
Dyeing	5.2%
Printing	1.2%
Wet finishing	0.5%
Confectioning	19.8%

Source: own elaboration based on Ecoinvent 3.10³⁸² (Wernet et al., 2016) and EF3.1 databases (European Commission, 2022).

For each unit process of LCS 2 the fraction of lost material is identified in percentage, calculated according to the following equation:

Equation 3. Losses (%)

$$\text{Losses (\%)} = 100 \times \frac{\text{Amount of initial material}_i - \text{Amount of output material}_i}{\text{Amount of initial material}_i}$$

Where "i" indicates the unit process that is being modelled (the "i" varies from 1 to 5, thus covering from LSC 2a to LCS 2e).

For all the processes in LCS 2 the impact and cost drivers of **operating infrastructure** (e.g. HVAC: Heating, Ventilating, and Air Conditioning) are represented by the energy flows only, as they may not be neglected with respect to the ones associated to direct production (Ozturk, 2005). According to a conservative modelling approach and based on the data reported in literature (Ozturk, 2005), infrastructure energy flows are set as a fixed share equal to 30% of the direct consumption for each process analysed (e.g. infrastructure energy consumption in spinning process with respect to electricity and fuel consumption for spinning). This assumption is assumed for all the countries considered during the production unit processes. The increase in energy consumption was considered as an input to be accounted for in both the environmental and economic analysis.

Additional details for LCC analysis

As introduced in **Table 192** data for conducting environmental analyses have been enriched with additional input items describing additional economic drivers. All production processes, encompassed in LCS 2, were quantified including unit cost drivers such as:

- CAPEX
- Machine average lifetime
- Process duration
- Human labor

LCS 2 is the only life cycle stage for which the cost quantification of machinery has been considered as a cost factor in the analysis. The allocation of CAPEX per unit of mass, to each analysed unit process (e.g. ring spinning process), has been performed by applying the formula reported in Equation 4.

Equation 4. CAPEX

$$\text{CAPEX} \left[\frac{\text{€}}{\text{kg}} \right] = \frac{A \left[\frac{\text{€}}{\text{year}} \right]}{W \left[\frac{\text{hr}}{\text{year}} \right]} / D \left[\frac{\text{kg}}{\text{hr}} \right]$$

Where: A= annual initial cost allocation (defined as in Eq. 3); W= total working hour per year (assuming 2 shift per day, thus 4080 hr/year); D= process duration

Equation 5. Initial cost allocation

³⁸² CUT-OFF system model

12488

$$A = \frac{P}{\left[\frac{(1 + ir)^n - 1}{ir(1 + ir)^n} \right]}$$

12489 Where: P= machinery present value; ir= interest rates; n= machine average lifetime.

12490 The interest rate was assumed to be equal to 9%.

12491 Another analysis item included in the model was the “employment”, quantified as Full Time Equivalent (FTE)
 12492 for each process main output. The employment value was used to estimate human labour associated to each
 12493 unit process of LCS 2. The basic assumption is that 1 FTE is equal to 40 hr/week.

12494 The table below shows the wages values used in the model and the source of information. The values are
 12495 reported for each country considered in the model as a possible location of the unit processes that make up
 12496 the textile apparel value chain.

12497 **Table 194.** Wages values for each country considered in the model.

Countries in the model	Euro/month	Source
Bangladesh	129.36	Chart: The Low Wages of Textile Workers Statista
Brazil	517.710	ILOSTAT Data Explorer
China	763.224	Chart: The Low Wages of Textile Workers Statista
EU (28)	5294.407	Home - Eurostat
India	180.18	Chart: The Low Wages of Textile Workers Statista
Indonesia	179.256	Chart: The Low Wages of Textile Workers Statista
Pakistan	124.74	Chart: The Low Wages of Textile Workers Statista
Russia	915.426	ILOSTAT Data Explorer
Turkey	506.352	Chart: The Low Wages of Textile Workers Statista
USA	7650.048	ILOSTAT Data Explorer
Vietnam	303.072	Chart: The Low Wages of Textile Workers Statista
Other	1091.426	Average of the values determined for all nations except Europe

12498 Source: own elaboration the specific sources are reported in the table.

12499 The evolution of wages over the latest years was also investigated. The values in dollars were converted to
 12500 EUR using the rates reported by IRS³⁸³ for the relevant year.

12501 **Table 195.** The evolution of wages over the latest years in several countries.

Evolution of wages	Average monthly manufacturing wage of textile workers	Monthly basic wage of garment workers in selected Asian countries in 2020-2021	Minimum monthly wage in the textile industry in selected countries, 2020
	Source: Statista (2023 ³⁸⁴)	Source: Statista (2021 ³⁸⁵)	Source: Statista (2020 ³⁸⁶)
Country	euro/month	euro/month	euro/month
Bangladesh	129.36	80.37	83
China	763.22	n.a.	194
India	180.18	122.67	127
Indonesia	179.26	205.58	n.a.
Pakistan	124.74	n.a.	91

³⁸³ [Yearly average currency exchange rates | Internal Revenue Service](#)

³⁸⁴ <https://www.statista.com/chart/34510/wages-in-textile-producing-countries/>

³⁸⁵ Clean Clothes Campaign. (July 16, 2021). Average monthly basic wage of garment workers in selected Asian countries between March 2020 to March 2021 (in U.S. dollars) [Graph]. In Statista. Retrieved June 13, 2025, from <https://www.statista.com/statistics/1281264/average-monthly-wage-garment-workers-asia/>

³⁸⁶ <https://www.statista.com/chart/17903/monthly-minimum-wage-in-the-global-garment-industry/>

Evolution of wages	Average monthly manufacturing wage of textile workers	Monthly basic wage of garment workers in selected Asian countries in 2020-2021	Minimum monthly wage in the textile industry in selected countries, 2020
	Source: Statista (2023) ³⁸⁴	Source: Statista (2021) ³⁸⁵	Source: Statista (2020) ³⁸⁶
Country	euro/month	euro/month	euro/month
Turkey	506.35	n.a.	314
Vietnam	303.07	n.a.	n.a.

Source: own elaboration the specific sources are reported in the table.

Moreover, the proposed values for EU were validated by checking available data for Poland³⁸⁷ and Romania³⁸⁸.

Table 196. Evolution of wages over the latest years in Poland.

Evolution of wages over the latest years in Poland	2020	2023	2024
Manufacture of textiles (euro/month)	977.64	1186.55	1242.01
Manufacture of wearing apparel (euro/month)	780.85	1005.08	1078.49

Source: own elaboration based on data for Poland³⁸⁹.

Table 197. Gross average monthly salary in the industry in Romania 2022, by industry.

Gross average monthly salary in the industry in Romania	2022
Manufacture of textiles (euro/month)	945.22
Manufacture of leather and related products (euro/month)	901.16
Manufacture of wearing apparel (euro/month)	749.38

Source: own elaboration based on data for Romania³⁹⁰.

In determining the **cost to the producer**, the import tariffs for semi-finished and finished products in Europe and the Value Added Tax (VAT) were included in the economic evaluation. These values, expressed as percentages, are specific to each RP and were obtained from the TARIC database³⁹¹. The VAT was assumed equal to 21.80%.

The **cost for the consumer** related to the LCS 1 and LCS 2 stages has been defined as a price, increasing the costs determined for the producer by a percentage that represents the possible marginality and in addition also the VAT paid at the time of purchase.

³⁸⁷ Source: Central Statistical Office of Poland. (April 28, 2025). Average monthly gross wages and salaries in FMCG industry in Poland from 2020 to 2024, by sector (in zloty) [Graph]. In Statista. Retrieved June 13, 2025, from <https://www.statista.com/statistics/1197708/poland-average-wages-and-salaries-in-fmcg-by-sector/>. Exchange rate from: <https://www.exchange-rates.org/exchange-rate-history/pln-eur-2020> (1st January of every year)

³⁸⁸ Source: National Institute of Statistics Romania. (December 21, 2022). Gross average monthly salary in the industry in Romania in 2022, by industry (in Romanian lei) [Graph]. In Statista. Retrieved June 13, 2025, from <https://www.statista.com/statistics/1394149/romania-gross-average-monthly-salary-in-the-industry-by-activities/>. Exchange rate from: <https://www.exchange-rates.org/exchange-rate-history/ron-eur-2022> (3rd January)

³⁸⁹ Source: Central Statistical Office of Poland. (April 28, 2025). Average monthly gross wages and salaries in FMCG industry in Poland from 2020 to 2024, by sector (in zloty) [Graph]. In Statista. Retrieved June 13, 2025, from <https://www.statista.com/statistics/1197708/poland-average-wages-and-salaries-in-fmcg-by-sector/>. Exchange rate from: <https://www.exchange-rates.org/exchange-rate-history/pln-eur-2020> (1st January of every year)

³⁹⁰ Source: National Institute of Statistics Romania. (December 21, 2022). Gross average monthly salary in the industry in Romania in 2022, by industry (in Romanian lei) [Graph]. In Statista. Retrieved June 13, 2025, from <https://www.statista.com/statistics/1394149/romania-gross-average-monthly-salary-in-the-industry-by-activities/>. Exchange rate from: <https://www.exchange-rates.org/exchange-rate-history/ron-eur-2022> (3rd January)

³⁹¹ TARIC https://taxation-customs.ec.europa.eu/customs/calculation-customs-duties/customs-tariff/eu-customs-tariff-taric_en

13.11.9.3 Supply Chain Transport

In this section the assumptions made for **transport** occurring between LCS 1 and LCS 2 and between the sub-stages of LCS2 are described. To quantify the costs and environmental burdens associated with the transportation of raw materials and intermediaries a transportation model has been developed.

The model implemented to represent the transport that occurs along the value chain includes: the transport of raw materials and intermediates (i.e. semi-finished products) on a global scale. The transport that allows the connection between the various production stages, up to the production gate of the final textile apparel product to be marketed (i.e. from LCS 1 to LCS 2e), are then modelled.

The transportation model has been developed considering the global scale and considering routes combining intercontinental and regional freights with the corresponding means of transport. This model quantifies in detail the distances, means of transportation and costs for the transportation of fibres, yarn, fabric, and dyed and finished products.

The transport model is based on the availability of raw material and intermediaries in different geographical locations. The data on production shares referred to each unit process and to the corresponding geographical location are derived from the **Figure 4** and are reported in **Table 198**. The data refers to the year 2018. The production shares of fibres, yarns, fabrics and related processing are reported for different countries; the percentages are to be understood as calculated with respect to the relative global production. The data reported in **Table 198** has been associated with the data reported in **Table 199** to be able to estimate the production at country and fibre level for 2018.

Table 198. Geographical breakdown of global apparel production and consumption.

	Bangladesh	Brazil	China	EU (27+1)	India	Indonesia	Pakistan	Russia	Turkey	USA	Vietnam	Other
Fibre production	N.A.	1%	57%	7%	13%	2%	N.A.	1%	N.A.	4%	N.A.	15%
Yarn production	3%	N.A.	64%	1%	9%	N.A.	4%	N.A.	N.A.	N.A.	N.A.	19%
Fabric production	3%	1%	60%	N.A.	12%	N.A.	2%	N.A.	5%	N.A.	N.A.	17%
Dyeing & Finishing	28%	N.A.	44%	11%	N.A.	N.A.	N.A.	N.A.	17%	N.A.	N.A.	0.0%
Assembly	7%	N.A.	35%	11%	7%	N.A.	N.A.	N.A.	N.A.	N.A.	6%	34%

Source: own elaboration based on **Figure 4**

Table 199. Global production in tonnes of the textile sector, breakdown for manufacturing phases.

PROCESS	Global production 2018 (tonnes)	Source
Fibre production	107 000 000	(Textile exchange - Preferred Fibre & Materials Report 2019, 2019)
Yarn production	97 833 092	Textile exchange - Fibre Conversion Methodology, 2019
Fabric production	86 693 816	Textile exchange - Fibre Conversion Methodology, 2019
Dyeing & Finishing	79 394 197	Calculation based on the amount reported in the previous step integrating data on process losses from EF3.1 database (losses = 8.42%)
Assembly	63 658 267	Calculation based on the amount reported in the previous step integrating data on process losses from EF3.1 database (losses = 19.82%)

Source: own elaboration based on the specific sources stated in the table.

The transport model has been developed considering the materials' mass flows that need to be exchanged between countries to supply the unit processes (i.e., fibre production, yarn production, fabric production, dyeing and finishing processes, final assembly and confectioning) for obtaining the available unit process' output in each country. As showed in **Table 198** it is possible to identify a limited number of countries within which the various production processes are carried out without the need to import raw materials and/or intermediaries.

The quantification of transport in terms of distances, means of transport, loads and costs was conducted through quotations made by logistics service providers³⁹².

The transport network for the exchange of materials between the various production stages and across countries has been implemented in the model based on absolute values (in mass) determined for each production stage and for each country (obtained by multiplying the values of **Table 199** by the percentages of **Table 198**.

The transport model makes use of the following definitions:

- In the context of the production system, the cardinal number "n" has been assigned to one of the unit processes, such as that pertaining to fabric production.
- The preceding unit process is labelled with the cardinal number "n-1"
- The subsequent unit processes is labelled with the cardinal number "n+1".
- The unit process located in the specific country "C", is labelled as "nC", for the preceding and subsequent unit processes in the same country the labels are (n-1)C and (n+1)C.

Further, the transport model establishes that:

- If the amount of raw materials or intermediaries (semi-finished product) available in one Country from the preceding unit process transformation step (n-1)C are balanced, i.e. meet the internal demand to feed the local transformation unit process nC within the same Country, then no import of raw materials or intermediaries is required from other Countries to comply with the amount needed to feed unit process nC;
- If the amount of raw materials or intermediaries (semi-finished product) available in one Country from the preceding unit process transformation step (n-1)C are not enough to feed the local transformation unit process nC of the production system within the same Country, then import of the needed quantity of raw material or semi-finished products is required from other Countries to comply with the amount needed to feed unit process nC;
- If the output amount by the unit process nC is balanced with the production requirement of the subsequent unit process (n+1)C in the same Country, then no imports are considered;
- If the output amount by the unit process nC is exceeding the production capacity of the subsequent unit process (n+1)C in the same Country, then surplus of the output intermediary is rendered available for export to be processed in other countries;
- Raw material and intermediary exchanges among countries are set according to the "proximity" rule, favouring the shortest transport route, i.e. the exporting Country providing intermediary surplus will send to the nearest importer Country (C') needing input for its subsequent unit process (n+1)C';
- The implemented model takes into account the material losses that occur at each production step;
- The transport of the various materials has been estimated considering the main packaging methods (e.g. bales, boxes and cartons) used to store the goods inside standard size containers (unit of transport);
- Full load (load factor=1) has been assumed, in compliance with the weight and volume limitations;
- The distances of intercontinental routes were modelled according to the routing associated to location of specific source and receiving nodes;
- The distance assumed for internal transport of each country is 1000 km, of which 50% is achieved by road freight and 50% by rail freight.

To better understand how the transport model works, an example is herein provided. If the quantity (in metric tonnes) of fibre needed to produce yarn in Bangladesh is greater than the quantity of available fibre generated in the same Country, it is assumed that the missing amount must be imported from another Country which produces an excess of fibre compared to its internal consumption (e.g. India).

Table 200 reports for each stage of the production process the details of each trade route modelled for the transport of materials (e.g. transport of fibres from Europe to China => EU - CN); the table reports the countries of origin and destination, the distance is expressed in km and the main means of transport used is specified.

³⁹² SeaRates website: <https://www.searates.com/logistics-explorer/> [Accessed September 2024]

Table 200. Detail of each trade route modelled for the transport of materials and semi-finished between source Countries and receiving Countries according to mutual availability and requirements obtained by applying the transport model (BZ=Brazil, CN=China, EU=Europe, ID=Indonesia, IN=India, PA=Pakistan, TU=Turkey, VN=Vietnam, JP =Japan; TZ=Tanzania, TW=Taiwan, SA=South Africa, AU=Australia).

Reference production phase	Origin - Destination	Distance (km)	Main means of transport for the route
FIBRE TRANSPORTATION	EU - CN	15 911	Ship freight
	RU - PA	5 971	Road freight
	USA - Other (JP)	8 390	Ship freight
	ID - PA	5 832	Ship freight
	IN - CN	6 841	Ship freight
	IN - BD	314	Road freight
	Other (BZ) - PA	14 672	Ship freight
	Average country inside	1 000	50% Road freight; 50% Rail freight
YARN TRANSPORTATION	CN - IN	6 841	Ship freight
	EU - TU	2 000	Rail freight
	EU - TU	2 000	Road freight
	PA - TU	6 726	Ship freight
	Other (TZ) - BZ	10 451	Ship freight
	CN - TU	14 590	Ship freight
	Other (TZ) - TU	7 277	Ship freight
	Average country inside	1 000	50% Road freight; 50% Rail freight
FABRIC TRANSPORTATION	CN - BD	4 297	Road freight
	Other (TZ) - EU	8 611	Ship freight
	PA - TU	6 726	Ship freight
	IN - BD	314	Road freight
	Other (TZ) - TU	7 277	Ship freight
	IN - TU	7 026	Ship freight
	BZ - TU	11 049	Ship freight
	Average country inside	1 000	50% Road freight; 50% Rail freight
DYEING and FINISHED PRODUCTS TRANSPORTATION	BD - IN	317	Road freight
	BD - VN	4 013	Ship freight
	TU - Other (TW)	13 730	Ship freight
	BD - Other (SA)	8 860	Ship freight
	CN - Other (AU)	7 493	Ship freight
	Average country inside	1 000	50% Road freight; 50% Rail freight

Source: own elaboration.

13.11.9.4 Retailing (LCS 3)

Description of the stage

The 'Retailing' LCS 3 includes the distribution of the products, the storage and the management of the waste resulting during these and the unsold products.

1. **Distribution:** This is the process of getting the product from the manufacturer to the retail store. This includes logistics, transportation, and handling.

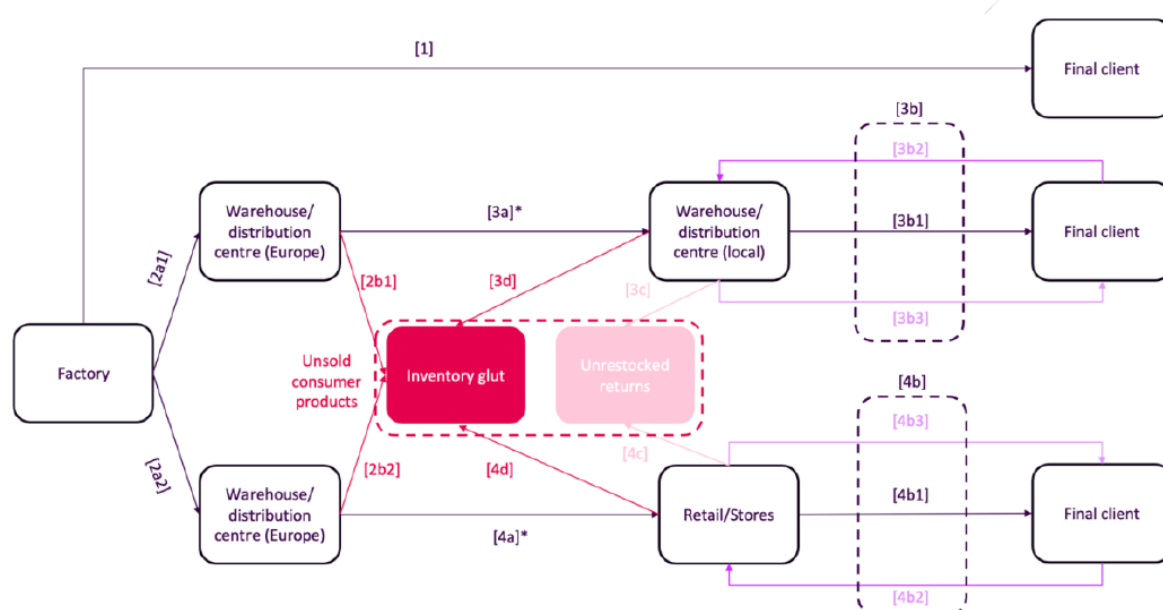
2. **Storage:** Once the products reach the retail location, they need to be stored properly until they are ready to be displayed for sale. Storage should ensure that products are kept in good condition.
3. **Waste Management:** This involves the proper disposal or recycling of packaging materials, damaged goods, and by-products of the retail process.

LCA and LCC modelling choices and shared inventory

Distribution

The activities considered in this life cycle stage are related to the transport of the finished product to the final client and the storage of the products at Distribution centres and retail stores. The final client is defined as the individual purchaser of the apparel product. The model is based on three default transport pathways as described in PEFCR Apparel and Footwear v2.0, which are visualised in **Figure 96**.

Figure 96. Transport legs and pathways (taken from PEFCR Apparel and Footwear v2.0). Notice that 'D2C' stands for direct to customer.



* Distribution losses apply up to point of sale (= to [3a] and [4a]). For D2C, factories are considered as the point of sale.

Source: PEFCR Apparel and Footwear v2.0.

The shares of products per transport leg that were used are shown in **Table 201** that follows.

Table 201. Shares of products per transport leg

Transport leg	Default values (%)	Default values with distribution losses (%) Share[X]	Source/Assumption
[1c]	0.6%	0.6% (no distribution losses considered)	PEFCR Apparel & Footwear v2.0
[2a1]	42.8%	43.2%	
[2a2]	69.1%	69.8%	
[2b1]	3.0%	3.0% (no distribution loss considered)	
[2b2]	4.8%	4.8% (no distribution loss considered)	
[3a]	39.8%	40.2%	
[3b]	34.2%	34.2% (no distribution loss considered)	
[3b1]	39.8%	39.8% (no distribution loss considered)	
[3b2]	11.1%	11.1% (no distribution loss considered)	
[3b3]	5.6%	5.6% (no distribution loss considered)	

[3c]	5.6%	5.6% (no distribution loss considered)	
[4a]	64.2%	64.9%	
[4b]	62.0%	62.0% (no distribution loss considered)	
[4b1]	64.2%	64.2% (no distribution loss considered)	
[4b2]	4.5%	4.5% (no distribution loss considered)	
[4b3]	2.2%	2.2% (no distribution loss considered)	
[4c]	2.2%	2.2% (no distribution loss considered)	
[3d]	0%	0%	
[4d]	0%	0%	

Source: own elaboration based on the source(s) reported in the table.

Direct to customer (D2C)

In the direct to customer (D2C) pathway, the finished products are transported to the final client directly, which is represented by transport leg [1] in **Figure 96**. The assumption is that the final client purchases the product directly from the manufacturer. The product is shipped to final client without requiring any storage in between. The share of D2C assumed in the model is 3.8% (Quantis, 2024).

Classic e-commerce

Classic e-commerce includes four transport legs. First, the product is transported from the factory to the warehouse/distribution centre (DC) located in Europe ([2a1] in **Figure 96**), and then it is transported from the same warehouse/distribution centre located in Europe to a local warehouse/distribution centre ([3a] in **Figure 96**). Then, from the mentioned local warehouse/distribution centre the product is transported to the final client ([3b1] and [3b3] in **Figure 96**) and, finally, the product can be transported from the final client to a local warehouse/distribution centre for (possible) returns (28% according to Quantis, 2024) ([3b2] in **Figure 96**). It is assumed that 34.2% of the products are transported according to the e-commerce pathway and that a default loss rate of 1% was used for the distribution stages until the point of sale (Quantis, 2024).

Retail

According to Quantis (2024), 62% of the products are transported following the retail pathway. This includes first the transport of the product from the factory to the warehouse/distribution centre (DC) located in Europe ([2a2] in **Figure 96**), followed by its transport from the same warehouse/distribution centre located in Europe to retail/stores ([4a] in **Figure 96**). Then, the product is transported from the mentioned retail/ stores to the final client ([4b1] and [4b3] in **Figure 96**), and, finally, the product can be transported from the final client to the retail/ stores for (possible) returns (7% according to Quantis, 2024) ([4b2] in **Figure 96**). Finally, the rate of restocking (i.e., the rate at which returns are restocked to retail stores to be sold again) is assumed to be 50%, while the loss rate of the distribution stage until the point of sale (POS) is assumed to be 1% according to (Quantis, 2024).

A common problem to the distribution model is the inventory glut, which is intended as surplus, excessive inventory, overstock and deadstock, which can either be retained at the point of sale, or at the European warehouse/distribution centre. Yet, the share of inventory glut retained either at European warehouse/distribution centre or at local warehouse/distribution centre is unknown. The PEFCR recommends to assume that the inventory gut, both for the classic e-commerce and the retail pathways, is retained at the European warehouse/distribution centre (i.e., $\text{InventoryGlut}_{\text{POS}} = 0$ and, therefore, $[3d] = [4d] = 0$ in Figure 3).

The total amount of unsold products can be calculated as the sum of inventory glut and unrestocked items ($[2b1] + [2b2] + [3c] + [4c] + [3d] + [4d]$) and a value of 15.6% was obtained.

Transport requirements

The main factors affecting the distribution impacts are the mass/volume being transported (including packaging), the distance travelled, and the mode of transport.

The distances of each leg, the mode of transport and the share of local, intracontinental and international supply chains (Provenance) in each transport leg are given in **Table 202** based on the information reported in (Quantis, 2024).

12659 **Table 202.** Distances, utilisation ratio and provenance of transport

Transport Leg [X]	Transport Leg description	Supply chain location	Distance	Utilisation Ratio	Provenance (% of total transport)	Transport Mode [Y]	Source/Assumption
1	Factory to final client (direct to consumer)	Local	1200	64%	5%	Truck (>32t, EURO 4)	Commission Recommendation (EU) 2021/2279
		Intracontinental	3500	64%	15%	Truck (>32t, EURO 4)	Commission Recommendation (EU) 2021/2279
		International (Ship)	1000	64%	0%	Truck (>32t, EURO 4)	Commission Recommendation (EU) 2021/2279
			18000	N/A		Ship (Transoceanic container)	Commission Recommendation (EU) 2021/2279
		International (Plane)	1000	64%	80%	Truck (>32t, EURO 4)	Commission Recommendation (EU) 2021/2279
			10000	N/A		Cargo Plane	Commission Recommendation (EU) 2021/2279
2a1 2a2	Factory to warehouse/DC located in Europe	Local	1200	64%	5%	Truck (>32t, EURO 4)	Commission Recommendation (EU) 2021/2279
		Intracontinental (Barge)	2500	64%	3%	Truck (>32t, EURO 4)	Based on PEFCR v1.2
			800	N/A		Ship (Barge)	Based on PEFCR v1.2
		Intracontinental (Train)	800	64%	14%	Truck (>32t, EURO 4)	Based on PEFCR v1.2
			2500	N/A		Train (Average freight train)	Based on PEFCR v1.2
		International (Ship)	1000	64%	66%	Truck (>32t, EURO 4)	Commission Recommendation (EU) 2021/2279
			18000	N/A		Ship (Transoceanic container)	Commission Recommendation (EU) 2021/2279
		International (Plane)	1000	64%	10%	Truck (>32t, EURO 4)	Commission Recommendation (EU) 2021/2279
			10000	N/A		Cargo Plane	Commission Recommendation (EU) 2021/2279
		International (Train)	1000	64%	100%	Truck (>32t, EURO 4)	Commission Recommendation (EU) 2021/2279
			10000	N/A		Train (Average freight train)	Commission Recommendation (EU) 2021/2279
3a	From warehouse/DC located in Europe to local warehouse/DC	Intracontinental (Plane)	500	64%	100%	Truck (>32t, EURO 4)	Commission Recommendation (EU) 2021/2279
			10000	N/A		Cargo Plane	Commission Recommendation (EU) 2021/2279
3b1 3b3	From local warehouse/DC to final client	Local	250 (round trip)	20%*	100%	Van (Lorry <7.5t, EURO 3)	Commission Recommendation (EU) 2021/2279
							Commission Recommendation (EU) 2021/2279

3b2	From final client to local warehouse/ DC (returns)	Local	250 (round trip)	20%*	100%	Van (Lorry <7.5t, EURO 3)	Commission Recommendation (EU) 2021/2279
4a	From warehouse/ DC located in Europe	Local	1200	64%	100%	Truck (>32t, EURO 4)	Commission Recommendation (EU) 2021/2279
4b1 4b3	From retail/ stores to final client (consumer travel)	Local	5	N/A	62%	Passenger car (Average)	Commission Recommendation (EU) 2021/2279
			5	20%	5%	Van (Lorry <7.5t, EURO 3)	Commission Recommendation (EU) 2021/2279
			N/A	N/A	33%	No impact modelled (walking, etc.)	Commission Recommendation (EU) 2021/2279
4b2	Final client to retail. Stores (customer travel, returns)	Local	5	N/A	62%	Passenger car (Average)	Commission Recommendation (EU) 2021/2279
			5	20%	5%	Van (Lorry <7.5t, EURO 3)	Commission Recommendation (EU) 2021/2279
			N/A	N/A	33%	No impact modelled (walking, etc.)	Commission Recommendation (EU) 2021/2279

Source: own elaboration based on the source(s) reported in the table

The transport requirements were calculated considering as unit of weight 1kg and unit of volume 1m³. For weight-limited transport methods, the weight of primary packaging was also taken into consideration. In the present assessment only primary packaging is considered for the sake of simplicity and this is represented by a polybag (0.02 kg). A default utilization ratio (mass of the real load divided by the mass of the payload) of 64% (PEF section 4.4.2.7) was used for trucks to take into account empty returns and default utilization ratio of 50% should be used for van transport (PEF section 4.4.3.2) (European Commission, 2021). The goods can also be transported via a Lorry (<7.5 t) for which a 20% utilization ratio is assumed according to PEF.

The transport requirements are calculated for unit weight/unit volume of final product using Equation 6 and Equation 7.

Equation 6. Weight-limited transport

$$\text{Activity } W_{XY} = \text{Share}[X] * \text{Weight}[X] * \text{Distance}[XY] * \text{Provenance}[XY]$$

Where;

Share [X] (%): Share of product per transport leg X (values reported in **Table 202**)

Weight [X] (kg/product): Weight transported per transport leg X. Unit weight was used in this model.

Distance [XY] (km): Transport distance per transport leg X and transport mode Y (values reported in **Table 160Table 202**)

Provenance [XY] (%): Provenance share per transport leg X and transport mode Y (values reported in **Table 202**)

For volume-limited transport, the transport requirement in km for the distribution impact calculation was calculated using Equation 7:

Equation 7. Volume-limited transport

12685
$$\text{Activity } V_{XY} = \text{Share}[X] * \frac{\text{Volume}[X]}{\text{VolumeMax}[Y]} * \text{Distance}[XY] * \text{Provenance}[XY]$$

12686 Where;

12687 Share [X] (%): Share of product per transport leg X (values reported in **Table 202**)

12688 VolumeMax [Y] (m³): Maximum volume transported by transport mode Y. For passenger cars, this is 0.2 m³

12689 (European Commission, 2021).

12690 Volume [X] (m³/product): Volume transported per transport leg X. Unit volume was used in this model.

12691 Distance [XY] (km): Transport distance per transport leg X and transport mode Y (values reported in **Table 202**)

12692 Provenance [XY] (%): Provenance share per transport leg X and transport mode Y (values reported in **Table 202**)

12693 **Table 202**

12694 **Storage**

12695 Storage was modelled for pathways including warehouse/distribution centres and retail/stores, as defined in

12696 **Figure 96**, hence excluding the D2C pathway. The data used to model the storage characteristics required for

12697 each transport leg are based on the information of the PEFCR Apparel and Footwear v2.0. The emissions and

12698 resources used by the storage systems were allocated to the products stored. The allocation is based on the

12699 space and time occupied by the product; default storage durations are reported in **Table 203**.

12700 **Table 203.** Volume, time, capacity, surface, energy consumption

Trans port leg	Volume stored (m ³ /prod uct)	Def ault stor age time (wee ks)	Storage capacity (m ³ *week s/year)	Storage surface (m ²)	Storage energy consumption (kWh/m ² /year or MJ/m ² /year)	Other (kWh/product)	Source/Ass umption
[1a1] [1a3]	N/A	N/A	N/A	N/A	N/A	DigitalInfrastr	PEFCR Apparel & Footwear v2.0
[1a2]	$\text{Volume}_{\text{prod}} * \text{Volume}_{\text{Fd}}$	3.5 week s	$\text{Volume}_{\text{ware}} * 52 \text{ weeks} / \text{year}$	$\frac{\text{Volume}_{\text{warehouse}} * \text{ShelfHeight}}{\text{ShelfHeight}}$	$\text{Consumption}_{\text{Electric}} + \text{Consumption}_{\text{GasW}}$	N/A	PEFCR Apparel & Footwear v2.0
[1c]	N/A	N/A	N/A	N/A	N/A	N/A	PEFCR Apparel & Footwear v2.0
[2a1] [2a2]	$\text{Volume}_{\text{prod}} * \text{Volume}_{\text{Fd}}$	3.5 week s	$\text{Volume}_{\text{ware}} * 52 \text{ weeks} / \text{year}$	$\frac{\text{Volume}_{\text{warehouse}} * \text{ShelfHeight}}{\text{ShelfHeight}}$	$\text{Consumption}_{\text{Electric}} + \text{Consumption}_{\text{GasW}}$	N/A	PEFCR Apparel & Footwear v2.0
[2b1] [2b2]	$\text{Volume}_{\text{prod}} * \text{Volume}_{\text{Fd}}$	3 year s	$\text{Volume}_{\text{ware}} * 52 \text{ weeks} / \text{year}$	$\frac{\text{Volume}_{\text{warehouse}} * \text{ShelfHeight}}{\text{ShelfHeight}}$	$\text{Consumption}_{\text{Electric}} + \text{Consumption}_{\text{GasW}}$	N/A	PEFCR Apparel & Footwear v2.0
[3a]	$\text{Volume}_{\text{prod}} * \text{Volume}_{\text{Fd}}$	3.5 week s	$\text{Volume}_{\text{ware}} * 52 \text{ weeks} / \text{year}$	$\frac{\text{Volume}_{\text{warehouse}} * \text{ShelfHeight}}{\text{ShelfHeight}}$	$\text{Consumption}_{\text{Electric}} + \text{Consumption}_{\text{GasW}}$	N/A	PEFCR Apparel & Footwear v2.0
[3b]	N/A	N/A	N/A	N/A	N/A	N/A	PEFCR Apparel & Footwear v2.0
[3c]	N/A	N/A	N/A	N/A	N/A	N/A	PEFCR Apparel & Footwear v2.0

Transport leg	Volume stored (m ³ /product)	Default storage time (weeks)	Storage capacity (m ³ *weeks/year)	Storage surface (m ²)	Storage energy consumption (kWh/m ² /year or MJ/m ² /year)	Other (kWh/product)	Source/Assumption
[3d]	N/A	N/A	N/A	N/A	N/A	N/A	PEFCR Apparel & Footwear v2.0
[4a]	$Volume_{product} * VolumeFactor$	3.5 weeks	$Volume_{Retail} * 52 \text{ weeks/year}$	$\frac{Volume_{Retail} * ShelfCoverage}{ShelfHeight}$	$Consumption_{Electricity}$	N/A	PEFCR Apparel & Footwear v2.0
[4b]	N/A	N/A	N/A	N/A	N/A	DigitalInfrastructure	PEFCR Apparel & Footwear v2.0
[4c]	N/A	N/A	N/A	N/A	N/A	N/A	PEFCR Apparel & Footwear v2.0
[4d]	N/A	N/A	N/A	N/A	N/A	N/A	PEFCR Apparel & Footwear v2.0

Source: own elaboration based on the source(s) reported in the table

Space-based allocation is based on the volume of the storage location, which is either a warehouse (Volume_{Warehouse}) or a retail store (Volume_{Retail}), as well as the average storage shelf height (ShelfHeight) and the shelf coverage of the storage location surface (ShelfCoverage). The default values of 48 000 m³ for Volume_{Warehouse}, 2 000 m³ for Volume_{Retail}, 2 m for ShelfHeight and 50% for ShelfCoverage were used (European Commission, 2021). The product volume stored is defined as the volume of the product including primary packaging (Volume_{Product}) and the default values in **Table 203** were used. Additionally, a storage volume factor (VolumeFactor) of 4 is used for ambient storage to account for the additional space the product takes in the storage facility, meaning the product volume will be multiplied by 4 (European Commission, 2021).

As for the energy consumption, it is assumed that warehouses require 30 kWh/ m²*year (Consumption_{ElectricityWarehouse}) and 360 MJ/m²/year of gas (Consumption_{GasWarehouse}), while 150 kWh/m²/year are required by retail stores Consumption_{ElectricityRetail} according to the PEF method (European Commission, 2021).

The energy consumption also takes into account the impacts generated by the digital infrastructure (DigitalInfrastructure) used for online shopping in the classic e-commerce and D2C pathways. For each product ordered, 0.030 kWh of electricity (European grid mix) are needed based on expert judgement as noted in PEFCR Apparel and Footwear v2.0(Quantis, 2024).

The impact linked to energy consumption for storage and distribution for each transport leg [X] is calculated according to Equation 8:

Equation 8. Energy consumption

$$Energy\ Consumption\ [X] = \frac{Share[X] * Volume\ stored[X] * StorageTime[X] * StorageSurface[X]}{StorageCapacity[X]} * (Consumption_{Energy}[X])$$

Where;

Share [X] (%): Share of product per transport leg X (values reported in **Table 203**)

Volume stored [X] (m³): Volume of product stored in transport leg X (Values reported in **Table 202Table 203**)

12727	Storage Time [X] (weeks): The default storage time of a product in transport leg X (Values reported in Table
12728	203)
12729	Storage Surface [X] (m ²): Surface area available for storage in transport leg X (Values reported in Table 203)
12730	Storage Capacity[X] (m ³ *weeks/year): Obtained as the product of Volume of storage facility and number of
12731	weeks in a year (52 weeks/year)
12732	Consumption _{Energy} [X]: Energy consumption, both electricity (kWh/m ² /year) and Gas (MJ/m ² /year), of the storage
12733	facility in transport leg X (Consumption _{ElectricityWarehouse} , Consumption _{GasWarehouse} , & Consumption _{ElectricityRetail})
12734	Waste management of pre-consumer waste
12735	Pre-consumer textile waste is generated at the retailing stage (LCS3) and includes the products lost during the
12736	distribution and storage as well as the following unsold products:
12737	- Finished products that the manufacturers do not send to their customers due to order changes or
12738	cancellation;
12739	- Products that are placed on the market but that were not chosen by consumers;
12740	- Products returned to the retailer after being purchased;
12741	- Products that the retailer decides not to place on the market due to marketing strategies.
12742	This waste is assumed to be treated in the same way as post-consumer waste products (described in section
12743	1.2.8); yet, the quantities of pre-consumer waste refer to the amount of the distribution losses plus the
12744	unsold items , which, at this stage, they are considered equal to 1% and 15.6% respectively based on values
12745	reported in the PEFCR Apparel and Footwear v2.0 (Quantis, 2024).
12746	13.11.9.5 Use (LCS 4)
12747	Description of the stage
12748	The activities considered in the Use stage (LCS 4) are washing, drying and ironing. Neither storage and nor
12749	detergent packaging waste treatment were considered because they are outside of the system boundaries and
12750	of negligible contribution respectively.
12751	1. Washing: The process of cleaning clothes typically requires water, energy (to heat the water and run
12752	the washing machine), and detergents, which can have various environmental impacts.
12753	2. Drying: Drying clothes can be done naturally by air or through the use of a tumble dryer. Air drying is
12754	more environmentally friendly, but not always practical. Tumble dryers, on the other hand, are
12755	significant energy consumers.
12756	3. Ironing: Ironing consumes electricity to heat up the iron, and while it uses less energy compared to
12757	washing and drying, it still contributes to the overall energy footprint of clothing maintenance.
12758	LCA and LCC modelling choices and shared inventory
12759	Important factors to consider for this stage are the weight of the product as well as the washing frequency,
12760	which is expressed by the combination of number of uses and care cycles. Supporting data are provided in
12761	Table 189.
12762	Washing
12763	The electricity, water and detergent needs per wash were estimated based on the information reported in
12764	PEFCR Heavy Duty Liquid Laundry Detergents (HDLLD) (A.I.S.E, 2019) for machine wash and in the subsection
12765	6.3 Based on these, the full load wash weight, detergent, and water required per wash of full load can be
12766	assumed to be 4.5 kg, 76.5 ml and 50 litres, respectively. Based on (Bakowska et al., 2025), an average of
12767	38.1°C has been found to be the most commonly selected washing temperature among consumers; a value
12768	also supported by the study from Laitala et al (2012). Based on these values, as well as the equations found
12769	in PEFCR HDLLD, it is estimated that for the wash of 1 kg of product 0.51024 MJ of electricity, 0.01111 m3 of
12770	water and 0.017 kg of detergent are required. For the wastewater, it is assumed that the weight can be equal

12771 to the sum of the water used for the wash plus the detergent, which makes 11.12811 kg per kg of product
12772 washed.

12773 **Drying**

12774 For the drying, the only input considered was the electricity used. Based on the subsection 6.3 the prevalence
12775 of tumble dryers varies significantly even among Western countries, while the European average stands at
12776 around 27% of households (De Almeida et al., 2011 and Bakowska et al., 2025 and own elaboration). Based
12777 on the Directorate-General for Energy 'Energy Efficient Products' website referring to the 'Tumble dryers',
12778 assuming 370kWh per year and 160 cycles, it can be estimated that the tumble dryer consumption per cycle
12779 is 2.3125kWh and, assuming a full load dry weight of 5 kg,(Commission et al., 2017) we can estimate that the
12780 electricity required per kg of product is 0.452 MJ.

12781 **Ironing**

12782 Electricity consumption is the only input considered to model ironing. It is necessary to know the share of
12783 garments ironed and the time spent per garment as well as the iron device wattage. It can be assumed that
12784 an iron can use 800 to 2 000 watts, with an average iron using 1 100 watts when heated on high and in our
12785 case a 1.1kW iron was selected. There is data gap for the average wattage of irons in the EU market so this
12786 value was chosen arbitrarily. The share of garments ironed and the time spent per garment depends on the
12787 specific garment and, at present, the data that could be considered in the ironing of garments are presented in
12788 PEFCR Apparel and Footwear v2.0 Table 35 (Laitala, 2018a; Daystar, 2019; Sandin, 2019) for the specific
12789 products. Based on these, as well as the database from the Bakowska et al., 2025 and own elaboration, for the
12790 example of a knitted garment which is assumed to be ironed 27% and for 2.5 minutes, an electricity
12791 requirement of 0.045 MJ can be estimated.

12792 Based on the above the life cycle inventory for the LCS4 for the 3 RPs was calculated and reported in the file
12793 "Textile-Prep-Study_3rd-Milestone_Model" (³⁹³).

12794

12795 **Additional details for LCC analysis**

12796 The LCC analysis was conducted using the model, parameters and input/output data defined for the
12797 environmental analysis.

12798 The costs associated with the use phase can be classified as Operational Expenses - OPEX (along the life cycle
12799 of the RP), as they are borne by the user throughout the entire time interval for which the RP is used. In this
12800 case, a correction of the costs of this phase was applied by incorporating a discount factor, through the
12801 evaluation of the Present Worth Factor - PWF.

12802 Equation 9. LCS 4 cost

12803
$$LCS\ 4\ cost = OPEX * PWF$$

12804 Where:

12805 LCS 4 cost is the cost to end-users in € associated only with the use phase of the RP,

12806 OPEX is the annual operating expense ³⁹⁴ in €,

12807 PWF is the Present Worth Factor calculated as

12808 Equation 10. Present Worth factor

³⁹³ Textile-Prep-Study_3rd-Milestone_Model is available on the project website I the section "Documents" at [this link](#).

³⁹⁴ According to Annex II of the Ecodesign Directive 2009/125/EC, "the operating expenses cover primarily energy consumption and additional expenses in other resources (such as water or detergent)".

12809

$$PWF = \frac{1 - \frac{1}{(1+d)^N}}{d} \quad (d \neq 0)$$

12810

Where:

12811

N is the product's life in years,

12812

d is the discount rate in %,

12813

in case d=0 the value of PWF=N.

12814

The discount rate was assumed equal to 3%.

12815

13.11.9.6 End Of Life (LCS 5)

12816

Description of the stage

12817

The End of Life (EoL) stage refers to the fate of textile apparel at the end of its life (therefore after the product's use stage). This section describes how the model also considers textile waste utilising the circular footprint formula (CFF). Post-consumer textile waste is a textile product that has been disposed of after consumption and used by the citizen or end-users of commercial and industrial activities (hotel, care, automotive, etc.). For this reason, it is commonly referred to as 'household post-consumer textile waste' and 'commercial post-consumer textile waste', respectively (Huygens et al., 2023).

12823

LCS5 is composed of three substages, namely: i) LCS5a End-of-life of 1 product, ii) LCS5b End-of-life of packaging and iii) LCS5c Transport for EoL of 1 product and packaging. The next paragraphs describe the end-of-life processes for the product and the packing based on the application of the Circular Footprint Formula (CFF) and details on the transport for the EoL are given in the next subsection 'LCA and LCC modelling choices and shared inventory'.

12828

Detailed modelling of the fate of textile waste generated in the above-mentioned phases is particularly complex because several variables come into play:

12830

- Different properties and characteristics of waste based on the phase in which it is generated;

12831

- Treatment processes specific to the geographical context in which the waste is produced;

12832

- Lack of adequate qualitative and quantitative information that allows the identification of flows and representative solutions for the treatments.

12834

The modelled possible treatment solutions include recycling, landfill and incineration (with energy recovery).

12835

According to the information provided by Huygens et al. (2023), it can be appreciated that only a small percentage of waste, both post-industrial, pre-consumer and post-consumer, is destined for recycling. Such recycling can be achieved with various techniques (mechanical, chemical, open-loop, thermo-mechanical, etc.); however, the real applications of these processes seem to find little response. The model considers the possibility that the fraction of textile products destined for recycling could, through mechanical processes, generate wipers and insulation materials (as indicated by Huygens et al., 2023).

12841

For the End of Life, which refers to the post-consumer waste, the assessments should follow the CFF, based on the PEF method (European Commission, 2021) as shown in the following equations and described below.

12843

Equation 11. MATERIAL

12844

$$(1 - R_1)E_v + R_1 * \left(A * E_{Recycled} + (1 - A)E_v * \frac{Q_{Sin}}{Q_p} \right) + (1 - A)R_2 * \left(E_{RecyclingEoL} - E_v^* * \frac{Q_{Sout}}{Q_p} \right)$$

12845

12846

Equation 12. ENERGY

12847

$$(1 - B)R_3 * (E_{ER} - LHV * X_{ER,heat} * E_{SE,heat} - LHV * X_{ER,elec} * E_{SE,elec})$$

12848

12849

Equation 13. DISPOSAL

$$(1 - R_2 - R_3)E_D$$

12851

12852 Where:

12853 A: allocation factor of burdens and credits between supplier and user of recycled materials.

12854 B: allocation factor of energy recovery processes. It applies both to burdens and credits.

12855 Q_{Sin}: quality of the ingoing secondary material, i.e., the quality of the recycled material at the point of
12856 substitution.

12857 Q_{Sout}: quality of the outgoing secondary material, i.e., the quality of the recyclable material at the point of
12858 substitution.

12859 Q_p: quality of the primary material, i.e., quality of the virgin material.

12860 R₁: the proportion of material in the input to the production that has been recycled from a previous system.

12861 R₂: the proportion of the material in the product that will be recycled (or reused) in a subsequent system.
12862 Therefore, R₂ shall take into account the inefficiencies in the collection and recycling (or reuse) processes. R₂
12863 shall be measured at the output of the recycling plant.

12864 R₃: the proportion of the material in the product that is used for energy recovery at EoL.

12865 E_{recycled}: specific emissions and resources consumed (per functional unit) arising from the recycling process
12866 of the recycled (reused) material, including collection, sorting and transportation process.

12867 E_{recyclingEoL}: specific emissions and resources consumed (per functional unit) arising from the recycling
12868 process at EoL, including the collection, sorting and transportation processes.

12869 E_V: specific emissions and resources consumed (per functional unit) arising from the acquisition and pre-
12870 processing of virgin material.

12871 E_{V*}: specific emissions and resources consumed (per functional unit) arising from the acquisition and pre-
12872 processing of virgin material assumed to be substituted by recyclable materials.

12873 E_{ER}: specific emissions and resources consumed (per functional unit) arising from the energy recovery process
12874 (e.g., incineration with energy recovery, landfill with energy recovery, etc.).

12875 E_{SE,heat} and E_{SE,elec}: specific emissions and resources consumed (per functional unit) that would have arisen
12876 from the specific substituted energy source, heat and electricity respectively.

12877 E_D: specific emissions and resources consumed (per functional unit) arising from the disposal of waste material
12878 at the analysed product's EoL, without energy recovery.

12879 X_{ER,heat} and X_{ER,elec}: the efficiency of the energy recovery process for both heat and electricity.

12880 LHV: lower heating value of the material in the product used for energy recovery

12881 These parameters are defined for each one of the waste materials (i.e textile, paper and plastic packaging).

12882 The model considers the lifespan of the product disregarding the number of users it has. Transport to potential
12883 second-hand retailer is outside the system boundaries. Therefore, all garments are considered to be collected
12884 as municipal waste. The model has been structured in such a way as to divide the flow of textile waste treated
12885 in the EU and those sent outside the EU (mainly towards Africa and Asia). The value of 50% was used to divide
12886 the flow remaining in the EU from that sent externally. According to Annex C of the PEF method (European
12887 Commission, 2021), in the European geographic context, the share of municipal waste collected that is
12888 incinerated R₃ equals to 39%, the calculated fraction recycled is around 17% and the rest (1-R₂-R₃=44%) is
12889 landfilled (see next section for all values). As default, the B allocation factor of energy recovery process is
12890 considered zero and no energy recovery process for heat and electricity are considered. For non-EU countries
12891 the same default values have been used.

12892 The end of life of the primary packaging is considered in LCS 5, and the burdens and credits of the end of life
12893 of are allocated using the CFF, using the default parameters available in the latest version of Annex C of the
12894 PEF method (European Commission, 2021). In this case it is assumed that the entire mass of waste is treated
12895 in EU. The share of waste landfilled, incinerated and recycled is reported in the next section.

The approach modelled for the End-of-Life management of post-consumer textile apparel and their packaging has been adopted as such in the accounting of impacts and costs for waste originating along the entire value chain: post-industrial and pre-consumer (accounted for in LCS 2 and LCS 3 - waste management sections). Waste management in LCS 3 includes the possible shipment of final waste outside the EU (as in LCS 5). Here the same share of waste is considered as sent to non-EU countries for final treatment, i.e. 50%.

LCA and LCC modelling choices and shared inventory

This section shows the numerical value of the share of waste sent to landfill, incineration and recycling, both for the EU and non-EU geographical context. Data on packaging waste management are also reported.

As mentioned, the percentage of post-consumer textile waste treated in the EU is 50%, while that sent to countries outside Europe is 50%.

Table 204. Shares of post-consumer wasted garments routes and Shares of primary packaging waste routes

Geographical context	Shares of textile apparel post-consumer waste routes	
EU	Landfill	44%
EU	Incineration	39%
EU	Recycling - total	17%
Non-EU	Landfill	44%
Non-EU	Incineration	39%
Non-EU	Recycling - total	17%
Geographical context	Shares of primary packaging waste routes	
EU	Landfill for paper	12.7%
EU	Incineration for paper	12.7%
EU	Recycling for paper	75%
EU	Landfill for plastics (LDPE)	35.6%
EU	Incineration for plastics (LDPE)	35.6%
EU	Recycling for plastics (LDPE)	29%

Source: own elaboration, based on Annex C of the PEF method (European Commission, 2021).

For the transportation at the end of life in substage LCS5c, the packaging weight (0.068 kg) - weighted average based on scenario route shares was used for all modes of transport except the car. For the car, the transport requirements depend on the distance only and not the mass. However, assuming that it is very unusual for a consumer to travel to the specific sorting bin for only 1 garment, we estimated the number of items that could fit in a common 20lt bag would be 5 and hence we divided the distance by 5.

Table 205 shows the values used to model the transport of waste from user to final treatment (distances and means of transport shares). These values apply to both pre-consumer (generated in LCS 3), post-consumer (LCS 5) and packaging waste.

The table report the estimated distance for shipping waste outside the EU for final disposal.

Table 205. Transport Scenarios for apparel End of Life.

Transport Scenarios for textile apparel End of Life	Distance (km)	Share of means of transport
Final client to landfill (km)	30	80.5%
Final client to incineration (km)	30	80.5%
Final client to collection point	1	19.5%
Collection point to sorting point – Truck	130	19.5%
Collection point to sorting point – Train	240	19.5%
Collection point to sorting point – Ship	270	19.5%
Sorting point to recycling	100	16.9%
Sorting point to incineration	30	2.6%

Transport Scenarios for textile apparel End of Life	Distance (km)	Share of means of transport
International shipping outside EU - with truck	1000	100%
International shipping outside EU - with container ship	18000	100%

Source: own elaboration, based on Annex C of the PEF method (European Commission, 2021).

Part of the entries of **Table 205** for the estimated distance for shipping waste outside the EU for final disposal are also used for the waste management of LCS2 (post industrial waste). The two last entries referring to the international shipping outside EU are excluded. However, in the LCS2f substage, there is also differentiation between the EU participation (7%) and non, while in LCS3 and LCS5 all of the waste are supposed to be located in EU, except the two last entries referring to the international shipping outside EU.

It should be noted that the inventory data reported in the file "Textile-Prep-Study_3rd-Milestone_Model" ⁽³⁹⁵⁾ indicates the mass of waste destined for the recycling and/or sorting process. This approach was used for the waste management of LCS2 and LCS3 as well as the LCS5 end-of-life. In accordance with the reports of (Solis et al., 2024), the sorting and recycling processes (of textile materials) are modelled using energy and water inputs and costs as indicated in **Table 206**.

Table 206. Details of energy, water inputs and costs for sorting and (mechanical) recycling processes modelled in the waste management and EoL stages.

■	Input	Value	Unit of measurement
Sorting	Electricity	0.017	kWh/kg
	Heat	0.03	MJ/kg
	Water	0.00011	l/kg
	Unit cost	0.4	€/kg
Recycling (mechanical)	Electricity	0.5	kWh/kg
	Water	0.02	l/kg
	Unit cost	0.63	€/kg

Source: own elaboration, based on data of Solis et al., 2024.

Additional details for LCC analysis

The data reported in the previous section allow to evaluate the costs incurred during waste management according to a value flow logic. This logic evaluates the actual cost resulting from all cost factors and is used to quantify the costs incurred by society (in the sLCC).

The unit costs reported in the (Solis et al., 2024) research (shown here in **Table 206**) were quantified with respect to the European geographical context. To obtain an estimate of cost that happened in countries located in geographical areas outside Europe, the possible costs were recalculated by normalizing the OPEX inputs for electricity, water, and labor costs compared to global average values. The CAPEX value was instead kept the same as the European context. It should be noted that the extra-European waste management model was created by excluding the sorting phase.

The quantification of the costs incurred by consumers in LCS 5 was instead carried out by referring only to the taxes paid by the citizen and not following the value flow logic. The tax values were obtained from the work of (Solis et al., 2024) and represent a European average for taxes related to incineration and landfill processes (in fact, no taxes are allocated to waste destined for recycling). The values used are: tax for landfill 0.043€/kg and tax for incineration 0.021€/kg.

The economic analysis of waste management at the expense of the producer (therefore waste labelled post-industrial and pre-consumer, generated in LCS 2 and LCS 3) was conducted with the logic of value flow. This is because given the fragmentation of production over various locations and the complicit absence of information to identify taxation in each geographical context, it was preferred to adopt an approach of evaluation of the actual costs of each waste management process.

³⁹⁵ Textile-Prep-Study_3rd-Milestone_Model is available on the project website I the section "Documents" at [this link](#).

Similarly to what was applied for the quantification of the discounted costs in LCS4 (see Equation 9 and Equation 10), the results of LCS5 were also discounted by applying the PWF to the obtained value.

Equation 14. LCS 5 cost

$$LCS\ 5\ cost = EoL * PWF$$

Where:

LCS 5 cost is the cost in € associated with the EoL of the RP when its useful life end,

EoL is the total cost obtain from the activity of disposal, recycling and incineration as modelled,

PWF is the Present Worth Factor calculated as reported in Equation 10.

13.11.9.7 Model parameterisation

The LCI described in the previous paragraphs was elaborated based on the parameterisation feature of the model. It is worth noting that the model is built in Microsoft office Excel (Microsoft Corporation, 2024) and utilises values to calculate the LCA/LCC impacts of interest for our study for the type of garment we define. Some of inputs values remain constant for all types of garments we would like to model and some of them are allowed to be changed by the user to make the specific garment model more representative. The overall model is allowed to be modified by the developers to update constant values and improve it but for simplicity we could distinguish two main types of input values:

- The input values that are not allowed to be changed by the user and which **remain the same for all type of garments** to be assessed and these are:
 - Values for Impacts per kg/unit of reference, for example the value for the CO2-eq for the production of 1 kg of cotton calculated for the Climate Change EF impact category
 - Values for LCC impacts per unit (i.e. costs)
 - Values for the normalisation and weighting factors for the 16 EF impact categories

The input values that are not allowed to be changed by the user and which remain the same for all type of garments amount up to thousands and the investigation of all these within an uncertainty analysis would not be feasible within the remits of this work. Therefore, the parameters that are used which allow the configuration of a specific type of product according to the needs of the analysis are chosen from the set of input values described as follows.

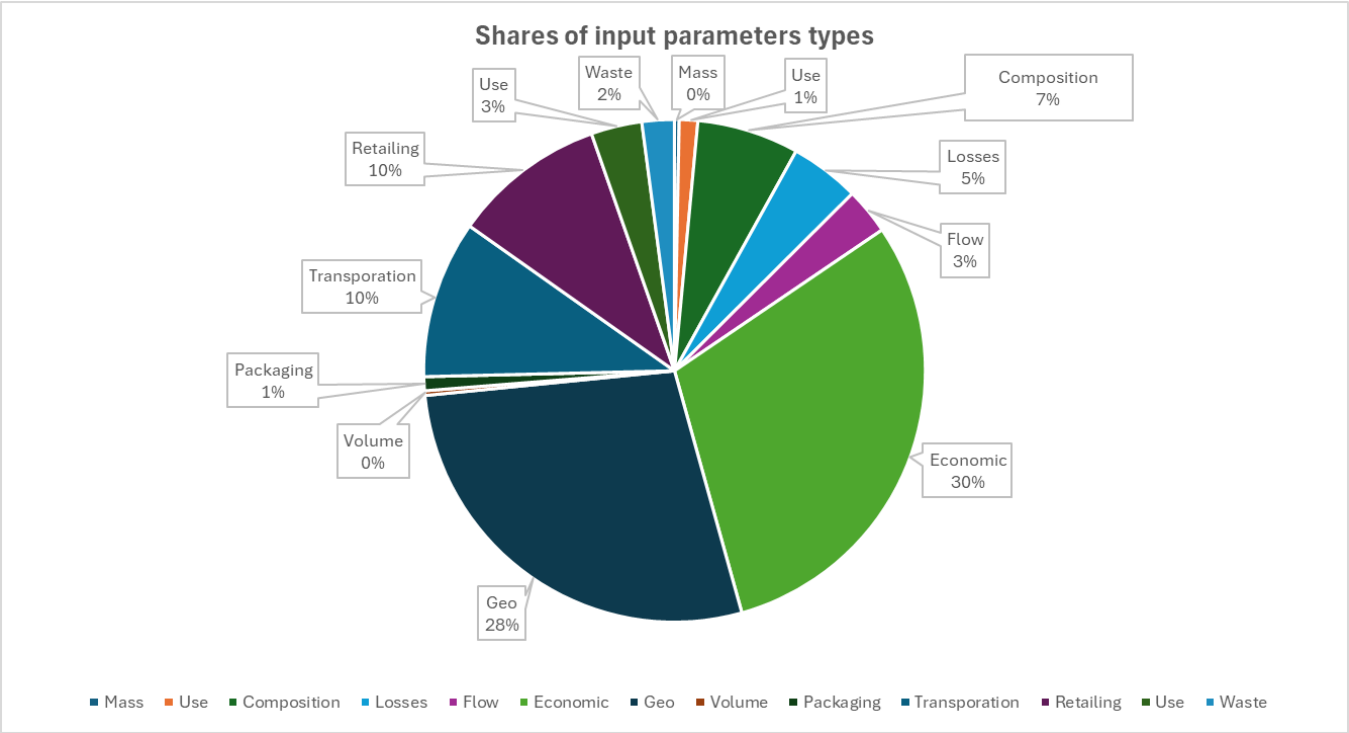
The input values that are allowed to be changed by the user and need **to be defined separately for each type of garment** to be assessed are values related to:

- **Composition** of the representative product (i.e. shares of types of fibres in the total makeup of the RP)
- **Geographic** region/country of origin of the specific type of fibre (e.g. percentage of polyester fibres coming from China) or the location of the specific type of manufacturing process (e.g. share of assembly taking place in Vietnam)
- **Economics**, including prices for various fibres, unit costs for electricity, monetisation factors etc
- **Flow** of material within the various manufacturing processes (e.g. percentage of fabric entering the cutting process)
- **Losses** during the various production and distribution processes (e.g. losses during knitting process)
- **Mass**, referring to the final mass of the representative product
- **Packaging**, referring to the mass of the polybag, paper bag and cardboard used for the packing of the product
- **Retailing**, referring to the routes and the way the final products are transported from the place of production to the customer including their storage. These parameters include distances between the factory and the clients and the warehouses in between, the percentages of the products transported between specific legs of these routes, means of transportation, energy consumption during storage etc
- **Transportation** between the various locations of production (e.g. share of the route EU-Turkiye in the yarn transportation)

- **Use**, referring to the number of uses, duration of service, percentages of the garments that are washed, dried and ironed as well as technical characteristics based on which energy and water use are estimated
- **Volume**, referring to the volume of the representative product
- **Waste**, referring to the percentages of the garments collected and following a specific route at the end of life (e.g. percentage of garments that go to municipal waste after being collected and sorted)

The total number of these parameters is 345 and the share of their contribution can be found in **Figure 97**.

Figure 97. Shares of input parameter according to their type



Source: own production

A set of these values are used for the sensitivity and uncertainty analysis after aa refinement process described in the subsection 13.11.11.1.1.

13.11.10 Assessment of Life Cycle Environmental Impacts and Costs

This section presents the main results of the environmental impact and cost analysis carried out using the LCA and LCC methodologies.

The first two paragraphs describe key methodological aspects of both types of analysis. The subsequent paragraphs present the results, starting with the base case, which includes a detailed assessment of the 3 RPs.

In addition, two alternative scenarios were modelled to compare different conditions:

- The influence of consumer choices, considering both worsening and improving behaviours.
- The effect of the location of raw material production and manufacturing processes.

13.11.10.1 Assessment Method: Environmental LCA

Based on the LCA guidelines and standards mentioned in section 13.11, the LCI can be used to model the product under study and together with the appropriate databases and impact assessment methods they can be combined (usually in a specialised software) to assess the potential environmental impacts.. The LCA and LCC models used in this study were created in Microsoft Office Excel (Microsoft Corporation, 2024), where the

Life Cycle Inventory was compiled following the way described in section 13.11.9 and combined with a database of impacts as well as normalisation and weighting factors. This model was developed in a parameterised manner so that various parameters can be changed to accommodate a large number of product variations and scenario analyses.

The database of impacts contains the results per unit (kg, kWh, etc.) that is used to describe each dataset used as input (e.g. low voltage electricity for country 'X'). These impacts have been derived using OpenLCA v.2.4 and combining information from Ecoinvent 3.10 database and other sources. For the impact assessment, the EF3.1 (adapted) method was used which contains some regionalised methods for regionalised calculations and more characterization factors and elementary flows than the ecoinvent LCIA method. This allowed to calculate results for the 16 impact categories of the Environmental Footprint method. The 16 impact categories, the normalisation and weighting factors based on Sala et al. (2017) are shown in **Table 207**. Applying normalisation and weighting factors on midpoint results allows to present results as a single score.

Table 207. Normalization and weighting factors for the 16 impact categories considered in the study.

Impact categories	Unit	Normalisation Factor	Weighting Factor [%]
Acidification	mol H ⁺ eq./person	5.56E+01	6.20%
Climate change	kg CO ₂ eq./person	7.55E+03	21.06%
Ecotoxicity, freshwater	CTUe/person	5.67E+04	1.92%
EF-particulate matter	disease incidences/person	5.95E-04	8.96%
Eutrophication, freshwater	kg P eq./person	1.61E+00	2.80%
Eutrophication, marine	kg N eq./person	1.95E+01	2.96%
Eutrophication, terrestrial	mol N eq./person	1.77E+02	3.71%
Human toxicity, cancer	CTUh/person	1.73E-05	2.13%
Human toxicity, non-cancer	CTUh/person	1.29E-04	1.84%
Ionising radiation	kBq U-235 eq./person	4.22E+03	5.01%
Land use*	pt/person	8.19E+05	7.94%
Ozone depletion	kg CFC-11 eq./person	5.23E-02	6.31%
Photochemical ozone formation	kg NMVOC eq./person	4.09E+01	4.78%
Resource depletion, fossils	MJ/person	6.50E+04	8.32%
Resource depletion, minerals and metals	kg Sb eq./person	6.36E-02	7.55%
Water use*	m ³ water eq of deprived water/person	1.15E+04	8.51%

* The calculation of the normalisation factor of the impact categories "land use" and "water use", are based on literature (Crenna et al., 2019).

Source: own elaboration based on (Sala et al., 2017).

The adoption of the EF 3.1 was supported by the following factors:

- High-Level Consensus Building. The EF method was developed through a rigorous consensus-building process among numerous European bodies representing the Technical Advisory Body (TAB). This ensures a broad, validated basis for its application.
- Member States. The framework incorporates essential input and scrutiny from the Steering Group (representing the Member States). This integration provides crucial political acceptance and guarantees alignment with EU-wide policy objectives.
- Policy Relevance. The EF is already extensively employed in numerous other politically relevant cases across the EU. Utilising it ensures methodological consistency and interoperability within the wider European context.
- Scientific Integrity. The entire process is firmly anchored by the principle of scientific correctness, which serves as the starting point for every aspect of the methodology.

13.11.10.2 Assessment method: Life Cycle Costing

Life Cycle Costing is a methodology for evaluating the economic performance of a product or service throughout its entire life cycle. In its broadest sense, this methodology allows the estimation of the total cost of a product or process by evaluating all direct and indirect cost sources.

The methodological approach of the LCC is inspired by the Life Cycle Assessment (LCA methodology). The LCA methodology is defined by internationally recognised standards, namely ISO 14040 and ISO 14044, while the one of LCC is not standardised. Nevertheless, the LCC was conducted with reference to the LCA standards, to be able to combine the approach, phases of analysis and results with those of the environmental analysis. The LCC shares the same goal and scope, FU, and system boundaries of the LCA, with small exceptions as reported in the sections 13.11.8 and 13.11.9.

As indicated by (Bachmann et al., 2024) it is more properly defined Conventional LCC – cLCC, the economic evaluation conducted in the most coherent way possible with the environmental assessment, therefore with the same goal and scope, functional unit and system boundary. cLCC is based on the quantification of internal costs (i.e. capital expenditures, operational expenditures, EoL cost, etc.) applying discounting. The discount rate is applied to all costs that will be incurred in a future period of time with respect to the time of the analysis; therefore, the costs related to the use phase and the management of waste relating to the end-of-life phase of the analysed product are involved (see paragraph “Additional details for LCC analysis” of the sections Use (LCS 4) and End Of Life (LCS 5)).

In general, the cost evaluation includes internal costs (budget costs and transfers); budget costs are the costs borne by the different actors involved in the value chain, while transfers refer to the money redistributed among the stakeholders (taxes and value added tax - VAT and tariffs).

The cLCC analysis was carried out following a “value flow” logic, considering the cost flows relating to the system analysed along each phase of the life cycle. More specifically, the cost flows are quantified concerning “cost drivers” evaluated per unit process (defined by LCA as the smallest element of the model created to evaluate the analysed system, for which the input and output data are quantified). The “cost drivers” represent all the “cost items” making up internal costs that are part of the analysis as, for example, materials, energy, labour costs, machinery, tariff etc.

To have a more extensive economic evaluation that provides a general framework for systematic assertions, including the LCA results, the Societal Life Cycle Costing - sLCC analysis was conducted. This type of economic analysis considers internal costs (from which the costs of environmental taxes, covered by the cLCC, have been subtracted) and external costs, trying to consider in perspective the set of costs borne by society. These external costs are not currently covered by the market price of goods, i.e. they are not internalized in the current price of products paid by consumers. For the evaluation of environmental externalities, the monetization factors of environmental impacts provided reported in **Table 208** were used. All the cost values refer to 2022 taking the Harmonised Indices of Consumer Prices (HICP)³⁹⁶.

Table 208. List of monetary valuation coefficients (MVCs) used in this study. Reference year:2022

Impact category	Unit of measure	Value	Method
Climate change	[€/kg CO ₂]	1.19E-01	Avoidance cost approach
Ozone depletion	[€/kg CFC-11 eq.]	6.28E+01	Damage cost
Human toxicity, cancer	[€/CTUh]	1.04 E+06	Damage cost
Human toxicity, non-cancer	[€/CTUh]	1.88E+05	\
Particulate matter	[€/disease incidence]	8.66E+05	Damage cost
Ionizing radiations	NA	NA	NA

³⁹⁶ https://doi.org/10.2908/PRC_HICP_AIND

Impact category	Unit of measure	Value	Method
Photochemical ozone formation	[€/kg NMVOC eq.]	1.43E+00	Average damage cost in EU28.
Acidification	[€/ mol H+ eq.]	3.96 E-01	Damage and avoidance (restoration)
Eutrophication, freshwater	[€/kg P eq.]	2.21E+00	Mix of damage and abatement cost approaches
Eutrophication, marine	[€/kg N eq.]	3.70E+00	Mix of damage and abatement cost approaches
Ecotoxicity, freshwater	[€/CTUe]	4.40E-05	Damage cost
Land use	[€/pt]	2.01E-04	Abatement cost
Water use	[€/m ³ water eq. of deprived water]	5.75E-03	Resource depletion costs
Resource use, minerals and metals	[€/kg Sb eq.]	1.89E+00	Market price
Resource use, Fossils	[€/MJ]	1.49E-03	Market price

NA: Not Available

Source: own elaboration based on Magrini et al. (2025) and Gama Caldas et al. (2024).

The two types of LCC analyses are then reflected in different points of view for the representation of the results. In fact, the values attributed to each cost driver, for each unit process, flow into the final cost and three different final cost perspectives are reported:

- **costs borne by the producer** (cLCC): the amount of costs sustained by the producer up until the moment in which the product enters into the possession of the customer. Cradle to gate approach, where the final gate encompasses the distribution phase to the customer and/or to the shop from which the customer himself will buy the product. VAT and tariffs (for the import of semi-finished and finished products into the EU) are included;
- **costs borne by the consumer** (cLCC): costs that the consumer must bear to own, use and dispose of the product. Here the costs incurred by the producer, increased by the margin obtained from the sale, plus the costs of the use phase of the product and its final disposal (quantified as direct process costs and taxes) are included;
- **costs for the society** (societal LCC): it is determined by following the cLCC approach, which quantifies the costs that originate from each unit process and for each stage of the product's life cycle (from cradle to grave), to which the costs of externalities deriving from environmental impacts are added; neither tariffs nor taxes are included in this calculation.

13.11.10.3 Results: Base Case

This section presents the main results obtained for the three RPs analyzed on the basis of the system boundary, functional unit and inventory described in the previous sections, defined as the Base Case. More detailed information and results are reported in the spreadsheet: "Textile-Prep-Study_3rd-Milestone_Model" ⁽³⁹⁷⁾.

The spreadsheet shows the LCA and LCC results of the three RP (i.e. knitted, other woven and denim) divided for:

- Full life cycle results (for LCC differentiation for society, producer and consumer);
- Full life cycle results: breakdown by life cycle stages and contributors*;
- LCS 2 – Manufacturing (focus on LCS2 that encompass all manufacturing processes);
- Waste management (results related to the waste management for post-industrial, pre-consumer and post-consumer textile waste);
- An additional section in the lower part of each tab where is possible to find more detail for the results:
 - LCS2 manufacturing breakdown of impacts/costs for sub-stage, processes and contributors;
 - Waste management (that occurs in LCS2, LCS3 and LCS5), breakdown of impacts/costs for sub-stage, country where the impact/cost occurs (based on the distribution of waste treated in the EU and non-EU) and processes.

³⁹⁷ Textile-Prep-Study_3rd-Milestone_Model is available on the project website I the section "Documents" at [this link](#).

**Contributors: Labels created to group together the type of inputs/outputs of the system that generate an environmental impact and cost on the product. In detail, the inputs/outputs (and therefore the related impacts and costs) have been classified according to the following groupings:*

- Chemicals: any chemical substance that takes part in a unit process;
- Emission: elementary emission flows into air, water and soil (effect only on environmental analysis);
- Energy: any energy input (e.g. electricity, natural gas, etc.);
- Land: inputs associated with land use (effect only on environmental analysis)
- Other: inputs that do not fall into one of the other categories, mainly non-textile waste, packaging and cost items (i.e. inputs like human resources (HR) and CAPEX, which only influence the economic analysis);
- Transport: includes all items associated with the transport of raw materials, semi-finished products, products and waste;
- Water: water consumption and wastewater treatment.

Additional there are:

- Aggregated datasets: used only for LCA, it includes the input/output impacts that could not be broken down into previous contributions (e.g., in the case of fibres, for which Ecoinvent datasets could be used directly);
- Fibre: used only for LCC, it includes the costs of all fibres (in fact, these were evaluated using market prices).

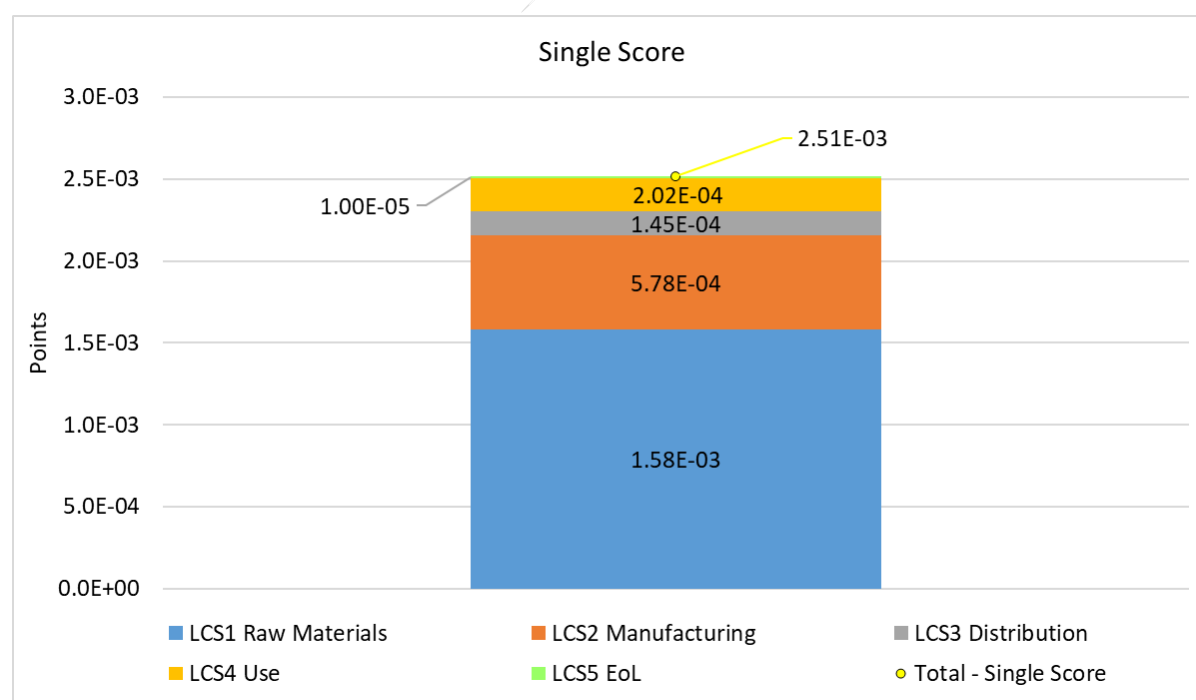
The tab called **"Base Case - Input data"** shows the list of parameters characterizing the 3 RPs.

Below are reported the results related to the full life cycle of the three RPs.

13.11.10.3.1 Full life cycle results of knitted product:

Below are the LCA results of knitted RP, calculated as single score. The graph shows the breakdown of environmental impact by life cycle stage and the total in relation to the FU defined in section 13.11.8.2.

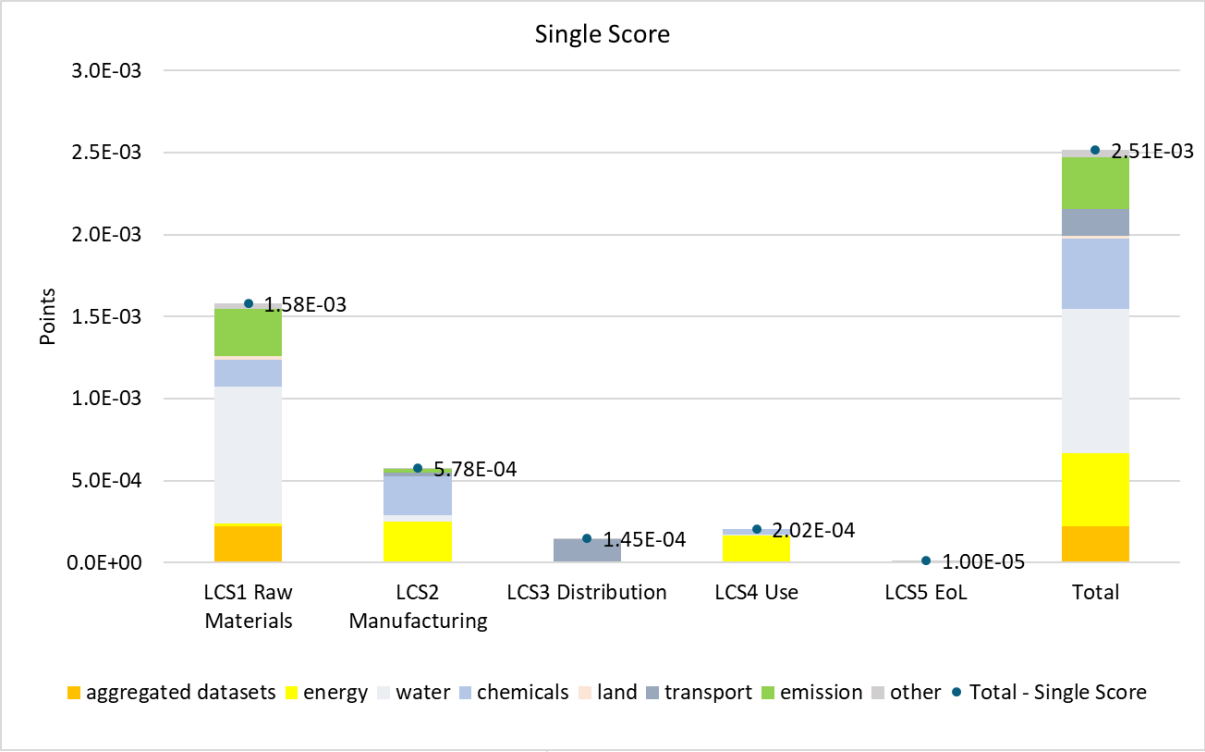
Figure 98. LCA results for the knitted RP.



Source: own production

13154 The following graph shows the breakdown of LCA results per life cycle stages and per contributors (i.e. labels
13155 that group environmental impacts by input/output categories used in the inventory, e.g. energy, water,
13156 chemicals, etc.).
13157 The last column on the right side of the graph “Total”, shows the sum of each stage of the life cycle.

13158 **Figure 99.** Breakdown for LCS and contributors of LCA results for the knitted RP.

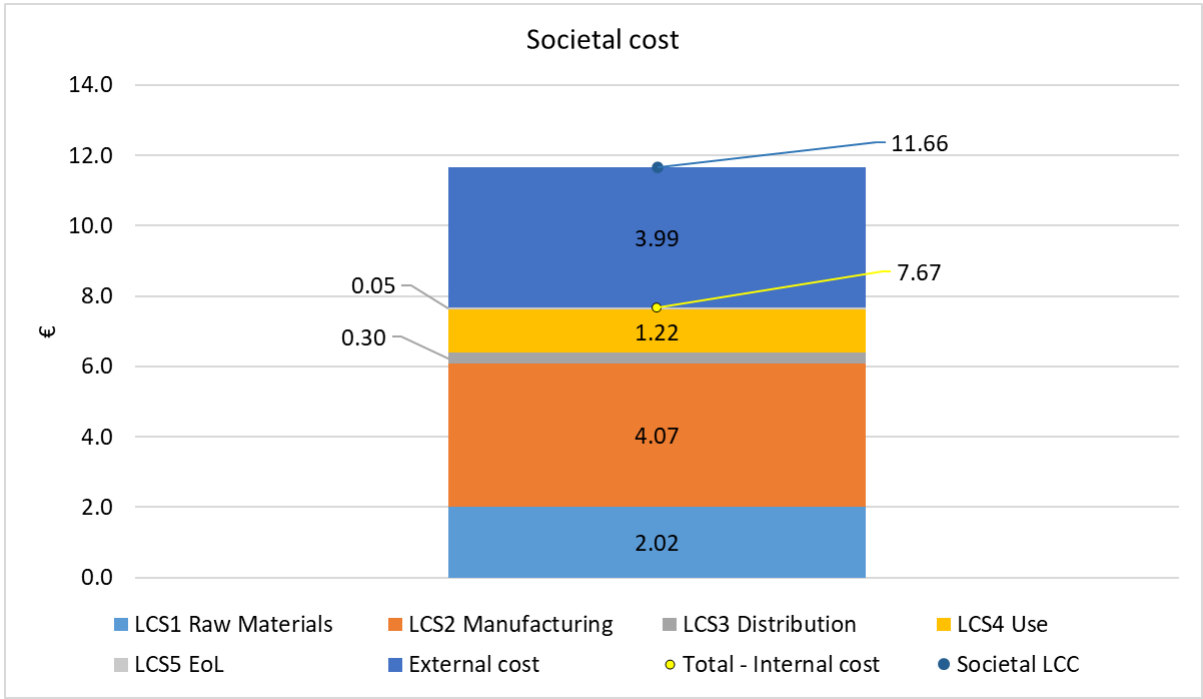


Source: own production

13161 The results of the LCC analysis are shown in the following three figures - societal cost, cost for producer and
13162 cost for consumer.

13163

Figure 100. LCC results: societal cost for the knitted RP.



Source: own production

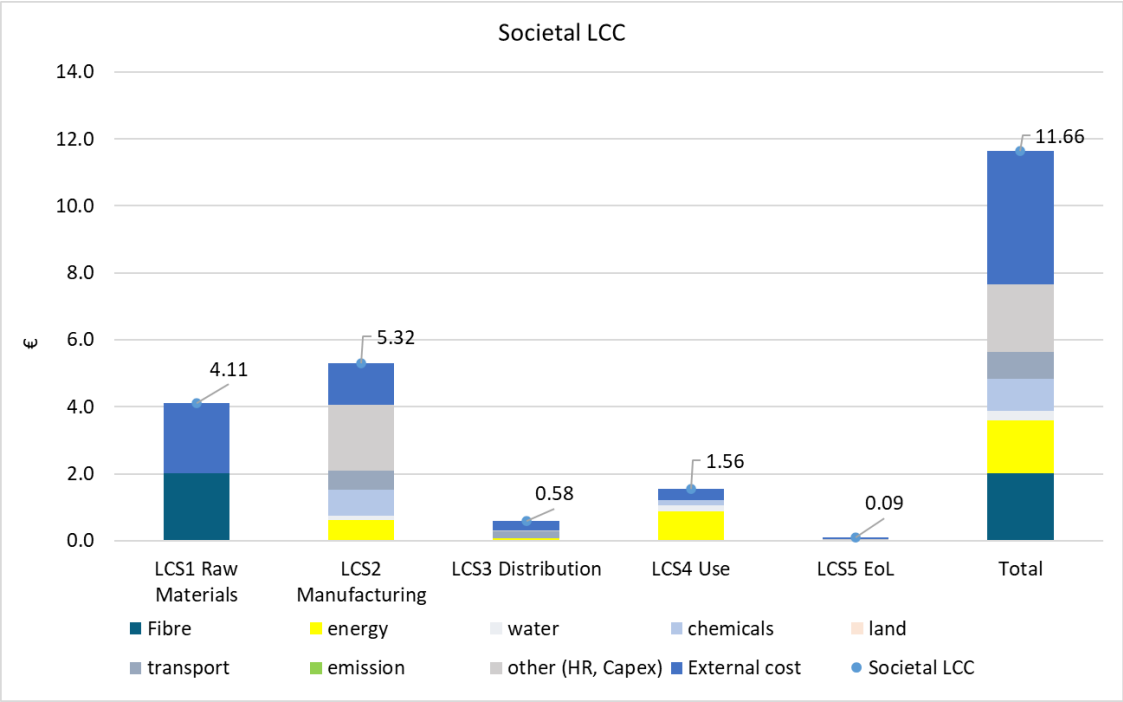
13164
13165

13166 In the previous graph “External costs” are due to the monetization of environmental impacts obtained through
13167 the LCA analysis. “Internal costs”, on the other hand, are the costs determined for all life cycle stages due to
13168 the use of resources, materials, energy, etc. “Societal LCC” is the total cost given by the sum of internal and
13169 external costs.

13170 Similar to what is shown for the LCA, the breakdown of costs related to the societal cost of the RP analysed by
13171 life cycle stages and contributors constituting the cost items is shown. The “Other” label includes costs due to
13172 HR and CAPEX related to machinery used in the processes. The last column on the right side of the graph “Total”,
13173 shows the sum of each stage of the life cycle.

13174

Figure 101. Breakdown for LCS and contributor of societal LCC results for the knitted RP.



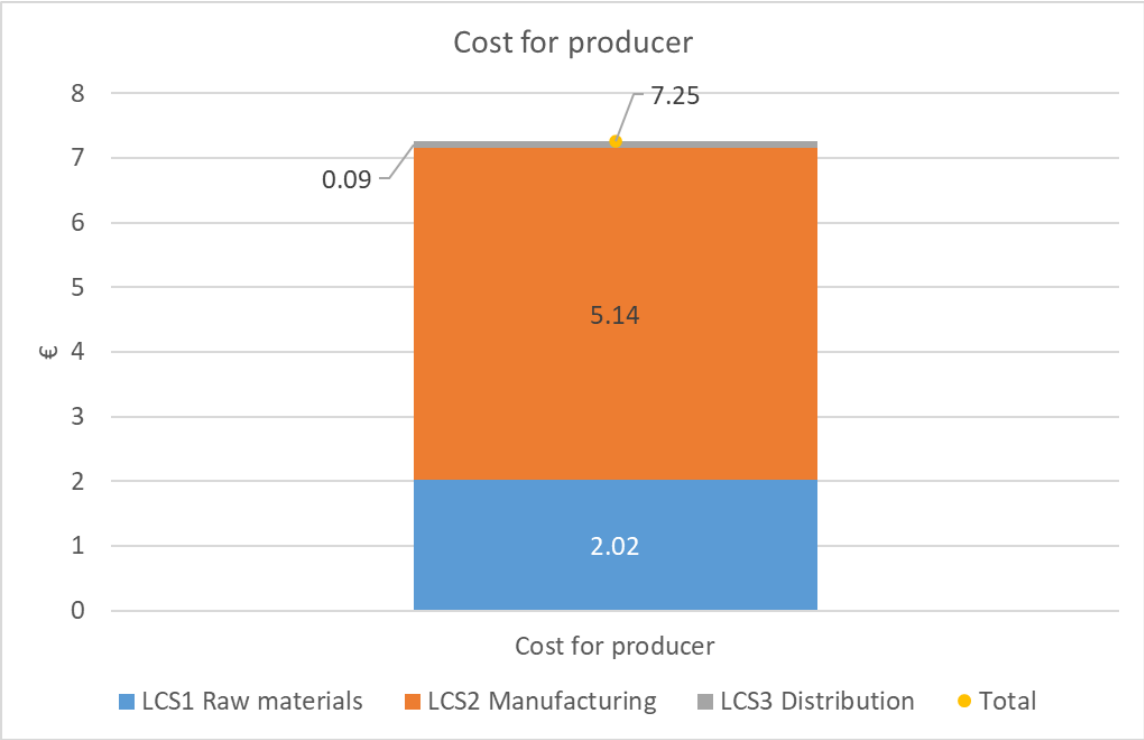
Source: own production

13175

13176

13177

Figure 102. LCC results: cost for producer associated with the knitted RP.



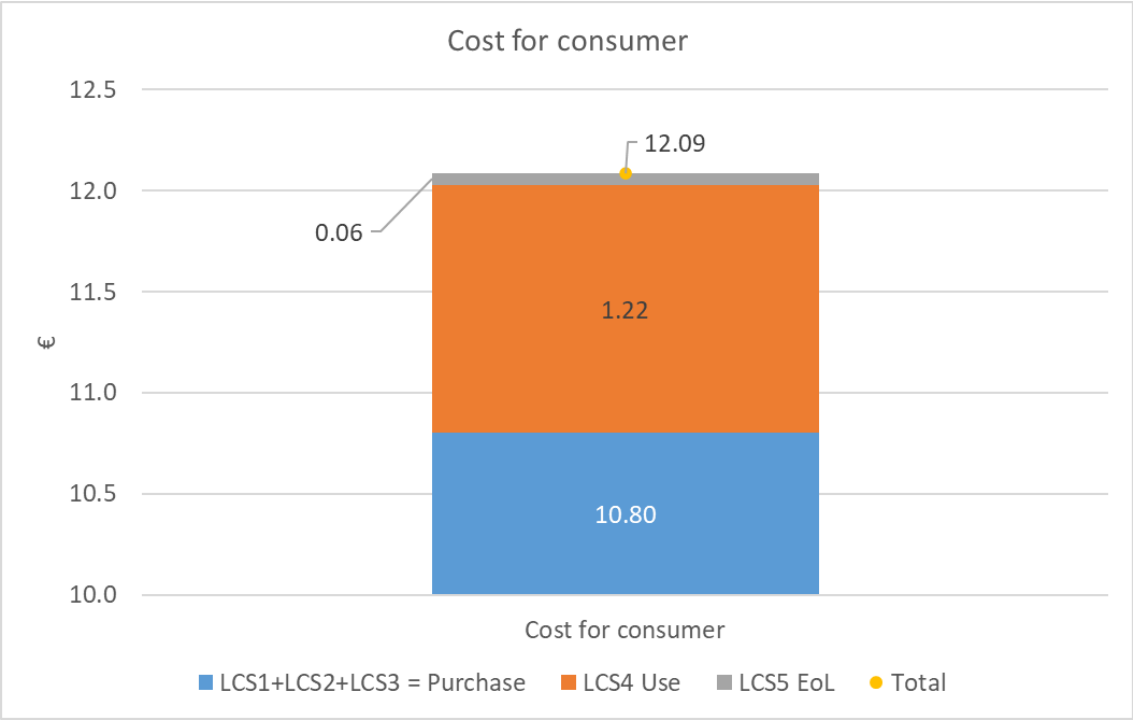
Source: own production

13178

13179

13180

Figure 103. LCC results: cost for consumer associated with the knitted RP.



Source: own production

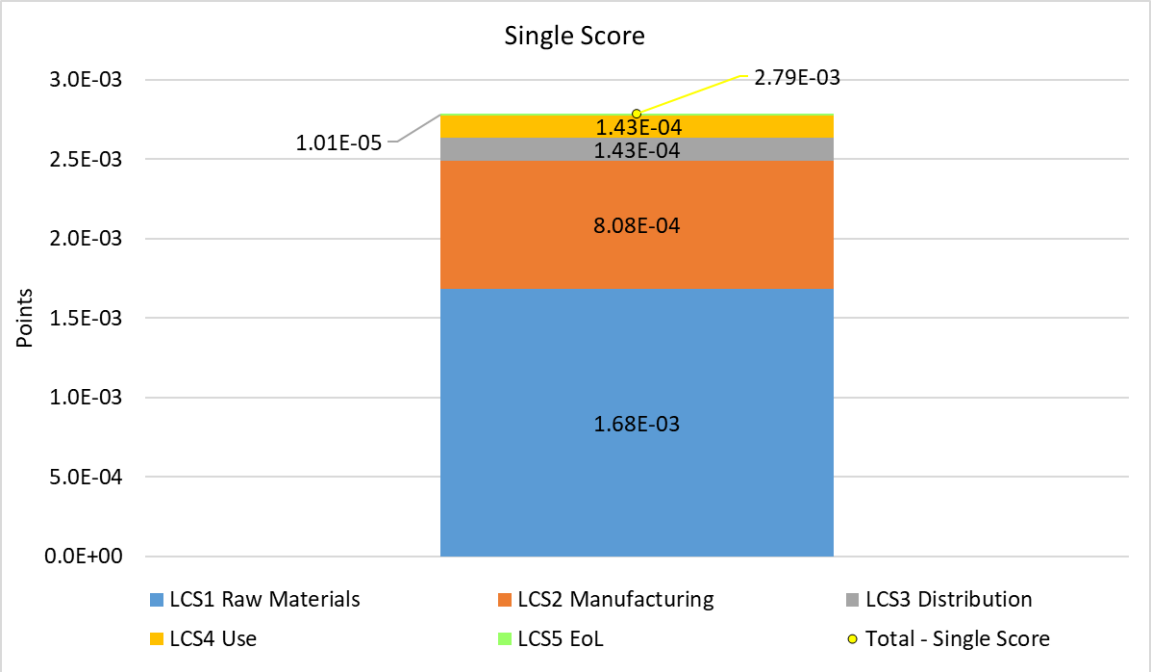
13181
13182

13183 13.11.10.3.2 Full life cycle results of other woven product:

13184 Below are the LCA results of other woven RP, calculated as single score. The graph shows the breakdown of
13185 environmental impact by life cycle stage and the total in relation to the FU defined in section 13.11.8.2.

13186

Figure 104. LCA results for the other woven RP.

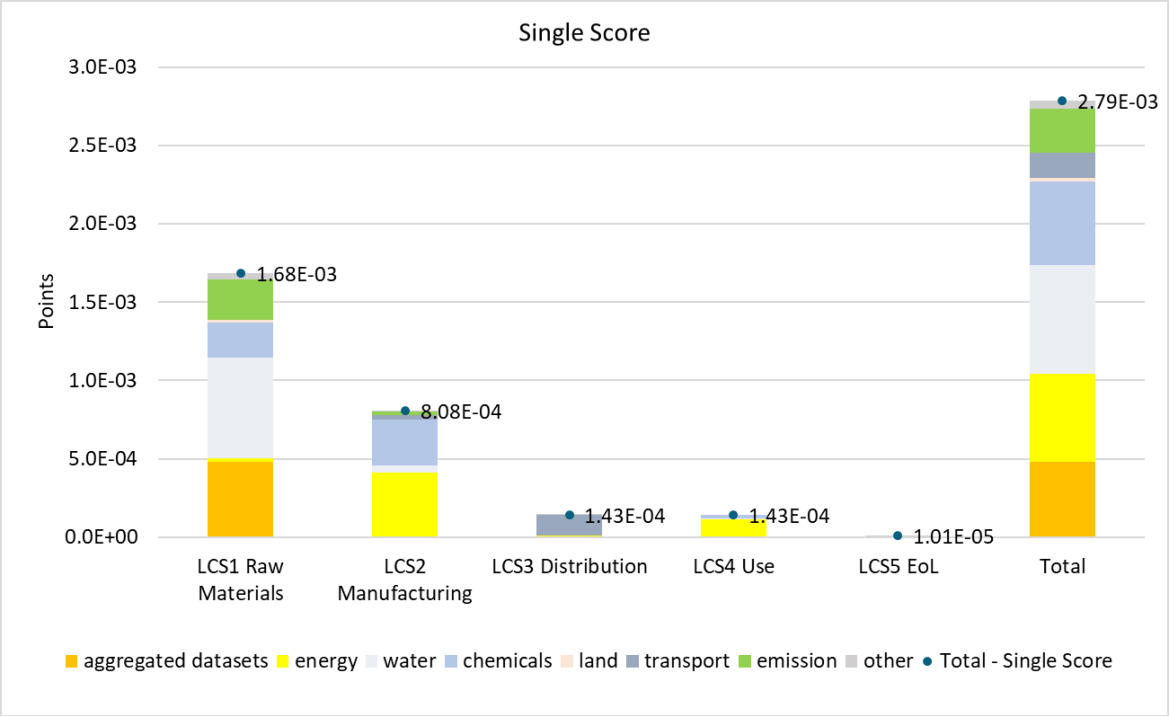


Source: own production

13187
13188

13189 The following graph shows the breakdown of LCA results per life cycle stages and per contributor (i.e. labels
13190 that group environmental impacts by input/output categories used in the inventory, e.g. energy, water,
13191 chemicals, etc.).
13192 The last column on the right side of the graph “Total”, shows the sum of each stage of the life cycle.

13193 **Figure 105.** Breakdown for LCS and contributors of LCA results for the other woven RP.

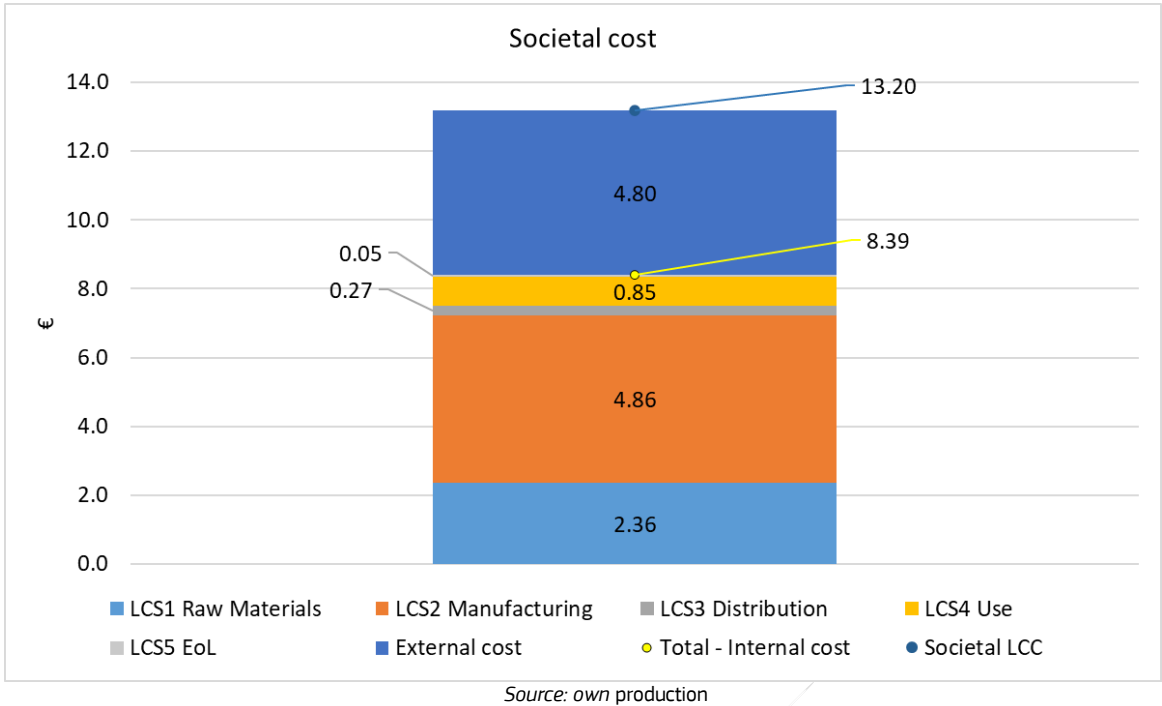


Source: own production

13196 The results of the LCC analysis are shown in the following three figures - societal cost, cost for producer and
13197 cost for consumer.

13198

Figure 106. LCC results: societal cost for the other woven RP.



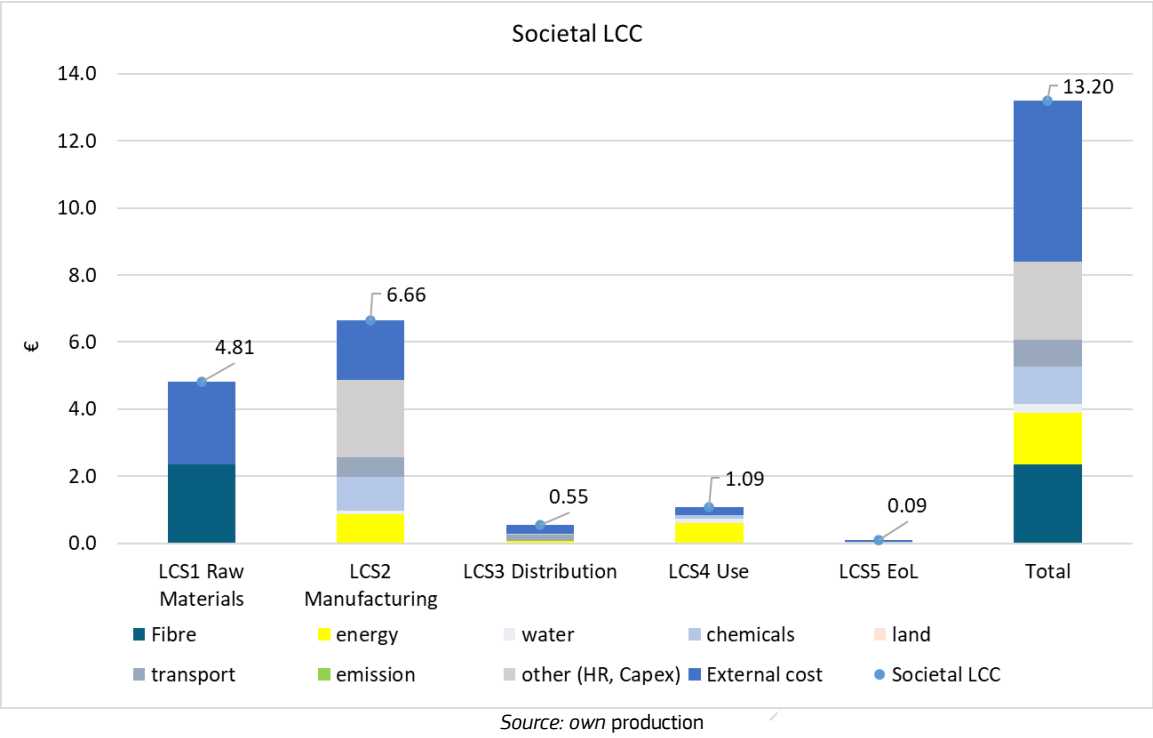
13199
13200

13201 In the previous graph “External costs” are due to the monetization of environmental impacts obtained through
13202 LCA analysis. “Internal costs”, on the other hand, are the costs determined for all life cycle stages due to the
13203 use of resources, materials, energy, etc. “Societal LCC” is the total cost given by the sum of internal and external
13204 costs.

13205 Similar to what is shown for the LCA, the breakdown of costs related to the societal cost of the RP analysed by
13206 life cycle stages and contributors constituting the cost items is shown. The "Other" label includes costs due to
13207 HR and CAPEX related to machinery used in the processes. The last column on the right side of the graph “Total”,
13208 shows the sum of each stage of the life cycle.

13209

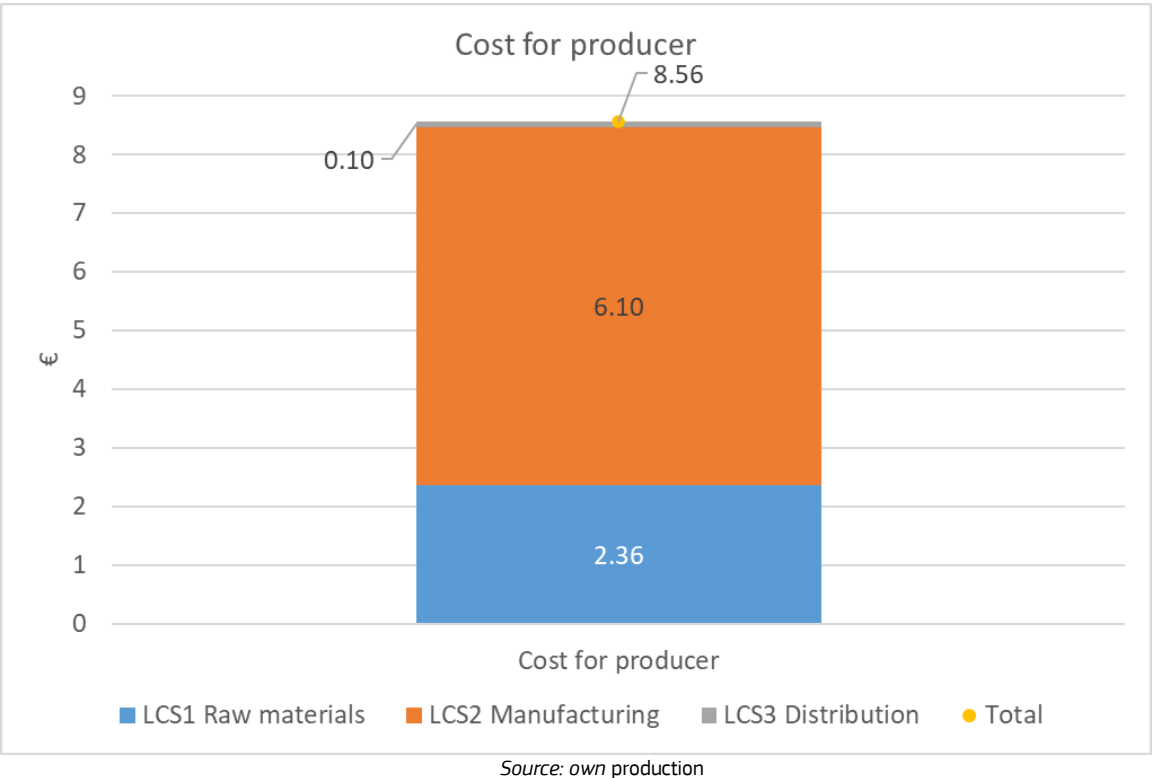
Figure 107. Breakdown for LCS and contributors of societal LCC results for the other woven RP.



13210
13211

13212

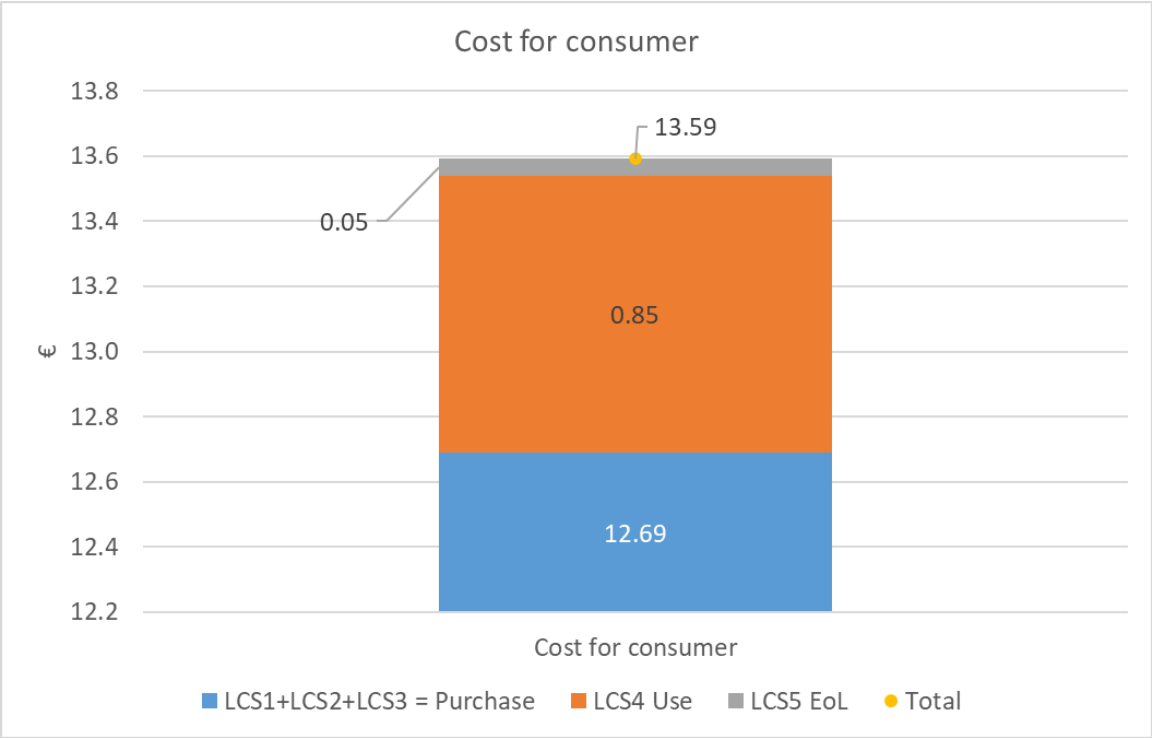
Figure 108. LCC results: cost for producer associated with the other woven RP.



13213
13214

13215

Figure 109. LCC results: cost for consumer associated with the other woven RP.



Source: own production

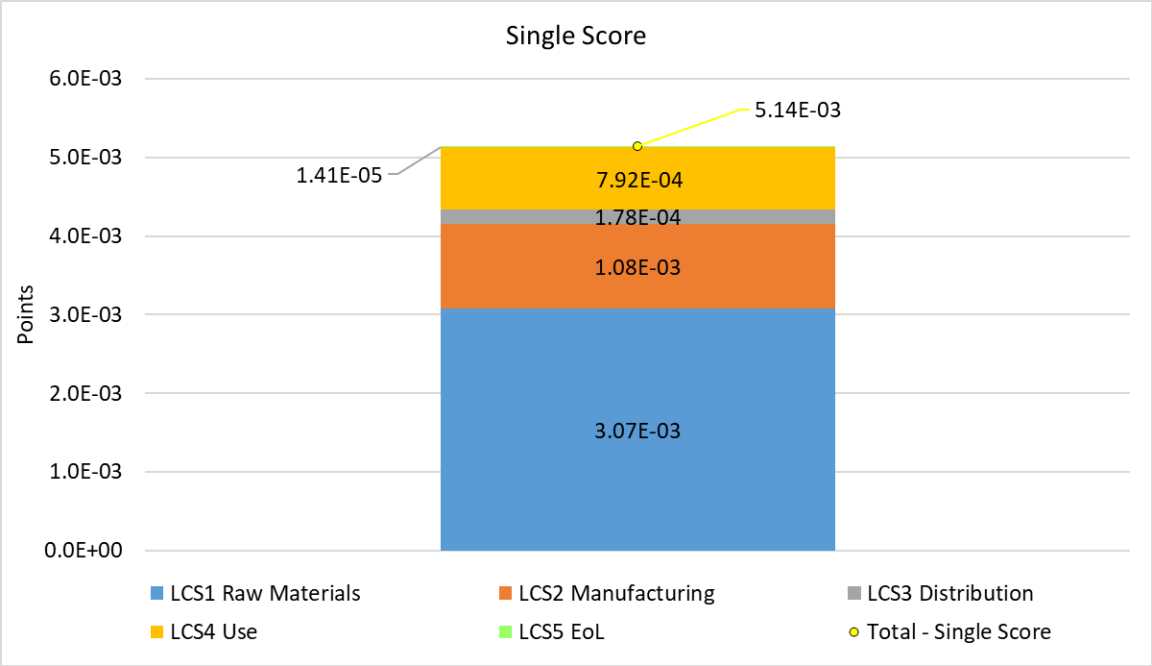
13216
13217

13218 13.11.10.3.3 Full life cycle results of denim:

13219 Below are the LCA results of denim RP, calculated as single score. The graph shows the breakdown of
13220 environmental impact by life cycle stage and the total in relation to the FU defined in section 13.11.8.2.

13221

Figure 110. LCA results for the denim RP.

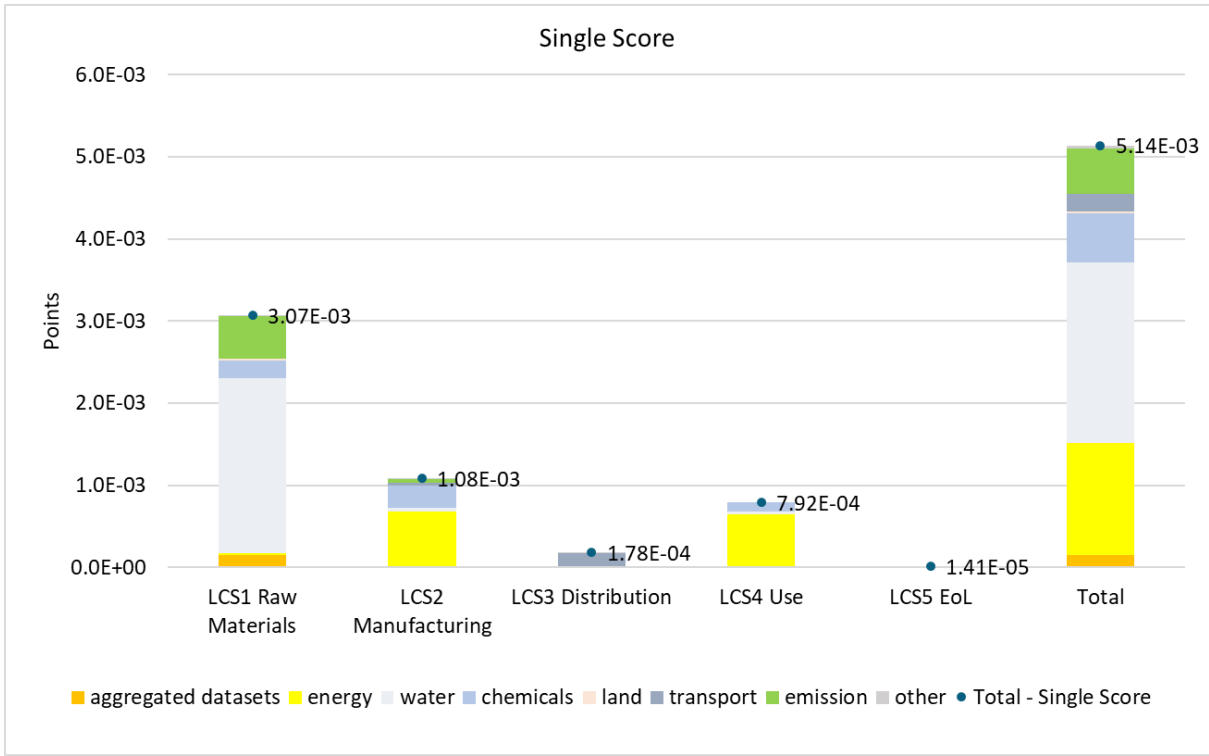


Source: own production

13222
13223

13224 The following graph shows the breakdown of LCA results per life cycle stages and per contributors (i.e. labels
13225 that group environmental impacts by input/output categories used in the inventory, e.g. energy, water,
13226 chemicals, etc.).
13227 The last column on the right side of the graph “Total”, shows the sum of each stage of the life cycle.

13228 **Figure 111.** Breakdown for LCS and contributors of LCA results for the denim RP.

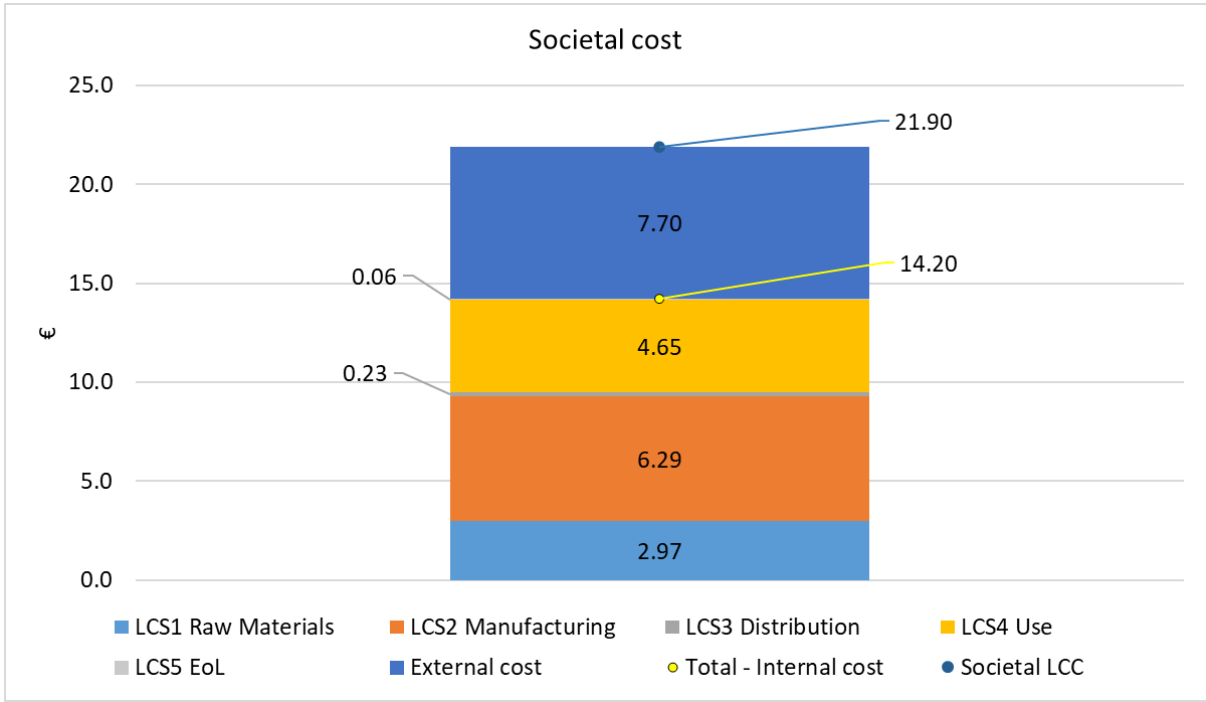


Source: own production

13231 The results of the LCC analysis are shown in the following three figures - societal cost, cost for producer and
13232 cost for consumer.

13233

Figure 112. LCC results: societal cost for the denim RP.



Source: own production

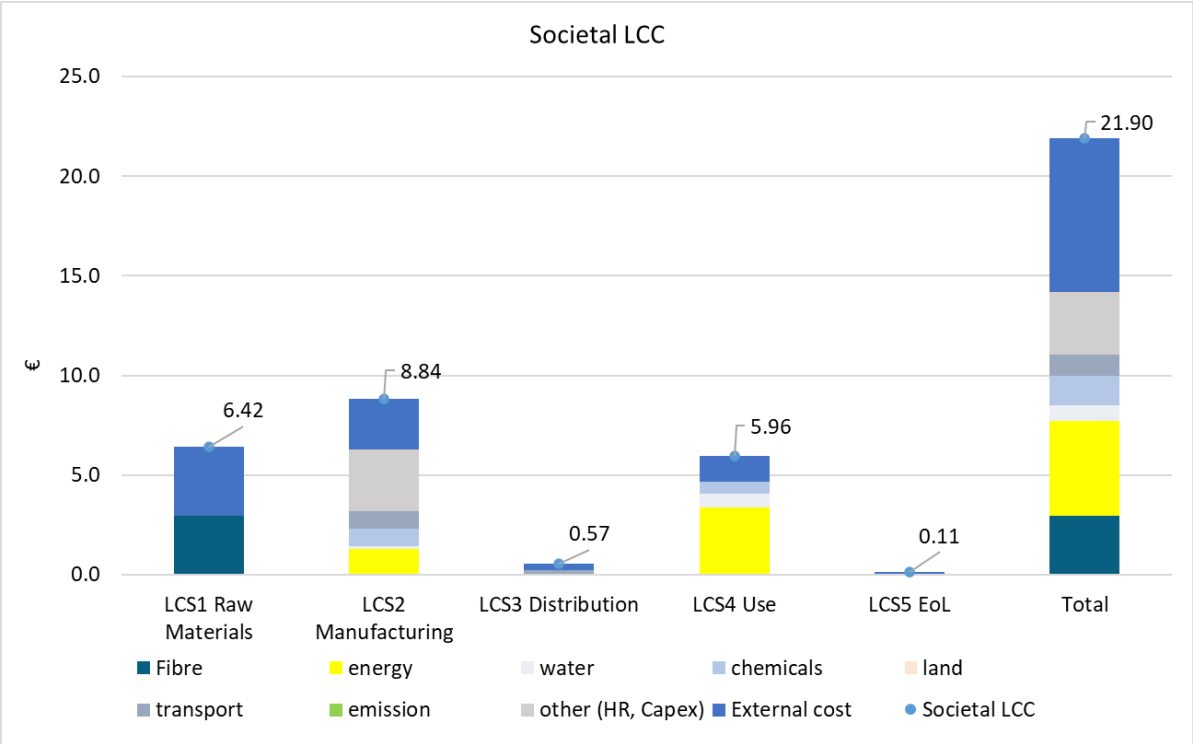
13234
13235

13236 In the previous graph “External costs” are due to the monetization of environmental impacts obtained through
13237 the LCA analysis. “Internal costs”, on the other hand, are the costs determined for all life cycle stages due to
13238 the use of resources, materials, energy, etc. “Societal LCC” is the total cost given by the sum of internal and
13239 external costs.

13240 Similar to what is shown for the LCA, the breakdown of costs related to the societal cost of the RP analyzed by
13241 life cycle stages and contributors constituting the cost items is shown. The “Other” label includes costs due to
13242 HR and CAPEX related to machinery used in the processes. The last column on the right side of the graph “Total”,
13243 shows the sum of each stage of the life cycle.

13244

Figure 113. Breakdown for LCS and contributors of societal LCC results for the denim RP.

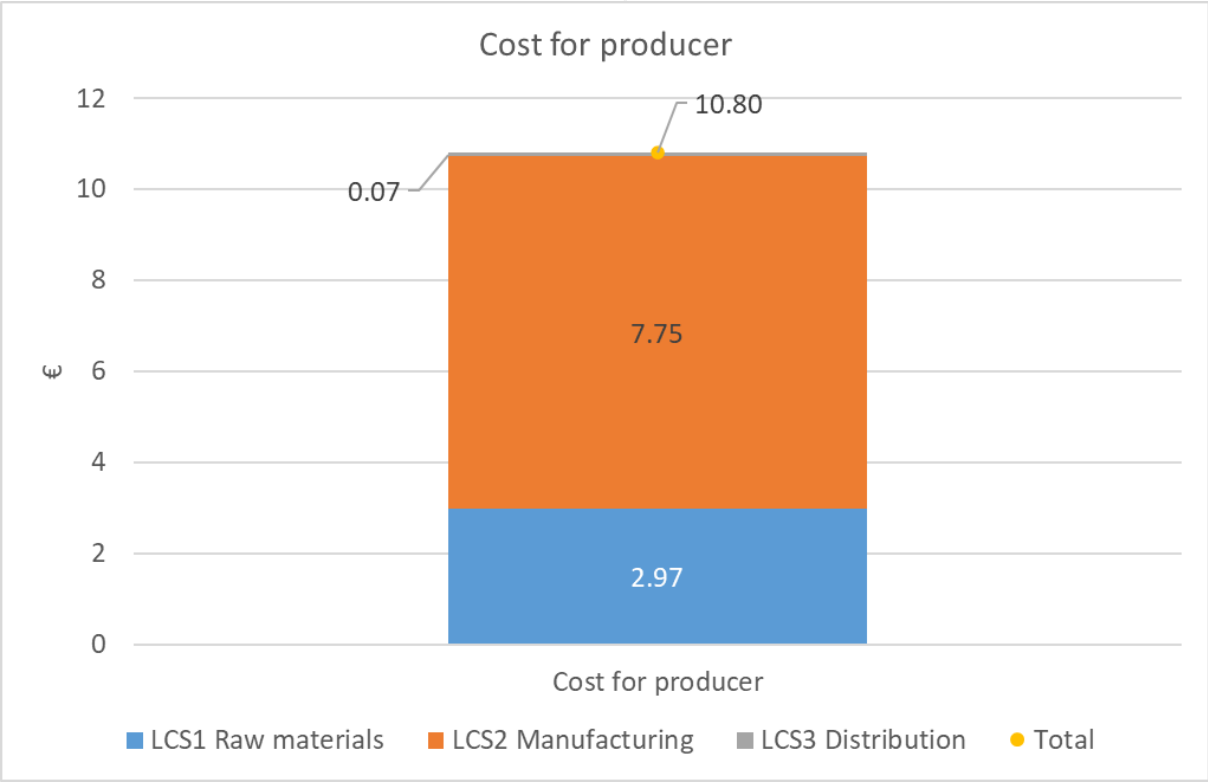


13245

13246

13247

Figure 114. LCC results: cost for producer associated with the denim RP

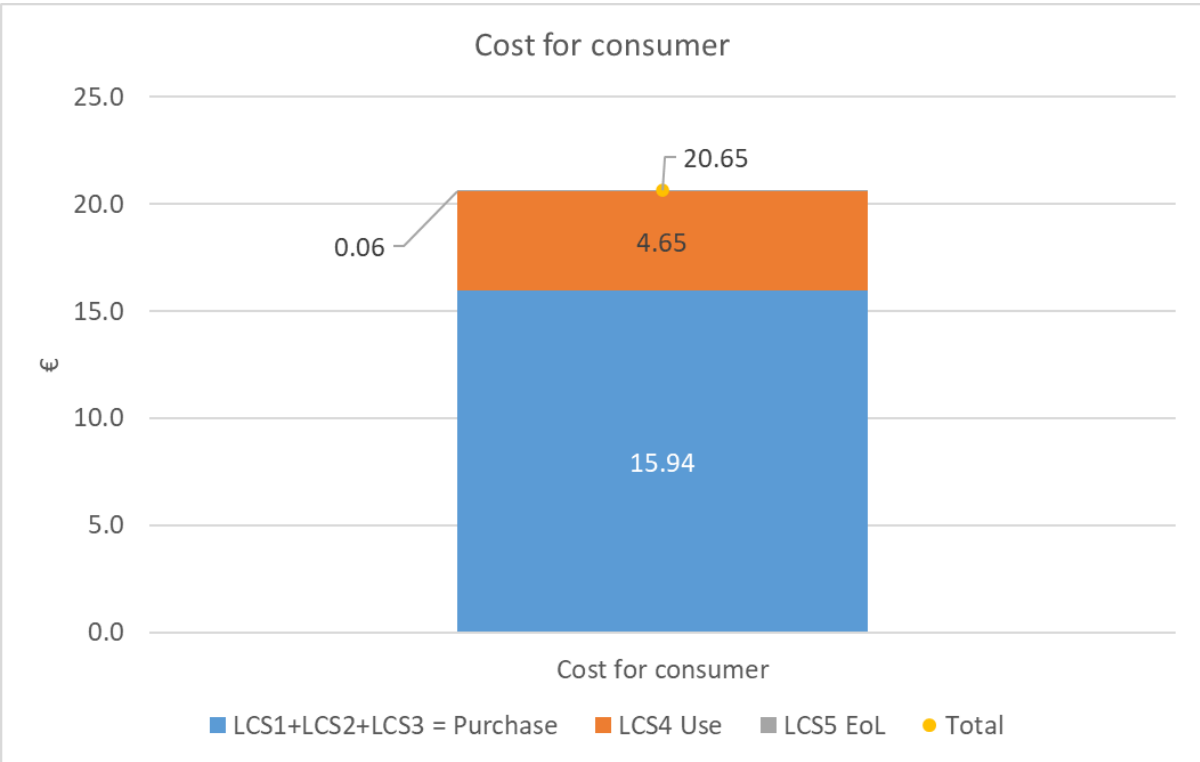


13248

13249

13250

Figure 115. LCC results: cost for consumer associated with the denim RP.



Source: own production

13251
13252

13253 **13.11.10.4** Scenario Analysis

13254 This section shows the results of two scenarios evaluated in the preparatory study:

- 13255 — Influence of the choice made by the consumer: worsening and improving scenarios - variations in the
- 13256 parameters of the number of uses.
- 13257 — Location of raw material production and manufacturing processes: products manufactured according
- 13258 to the base case (BC) model compared with products entirely made in Europe (EU), compared with
- 13259 products manufactured outside Europe (non-EU);

13260 A section is dedicated to each of them with further details and explanations.

13261 13.11.10.4.1 Influence of the choice made by the consumer

13262 This scenario was modelled to allow comparisons to test the influence of consumer choice when disposing a
13263 product regardless of its physical performance. The parameter identified to evaluate this aspect is the number
13264 of products uses. Therefore, the parameter "Number of Uses" was modified from what is reported in the base
13265 case of each RP. At the same time the parameter "Actual service duration" was made dependent on the number
13266 of uses, assuming as a starting point the ratio of the two parameters determined in the base case for each RP.

13267 More details and the complete set of results can be found in this spreadsheet: "Textile-Prep-Study_3rd-
13268 Milestone_Model" ⁽³⁹⁸⁾.

13269 **Table 209** lists the modelled scenarios and briefly describes the changes applied in each case. These scenario
13270 analyses were carried out for all RPs.

³⁹⁸ Textile-Prep-Study_3rd-Milestone_Model is available on the project website I the section "Documents" at [this link](#).

Table 209. List of different scenarios modelled to investigate the influence of the choice made by the consumer. Note that 'BC' stands for base case, and 'RP' for reference product.

Case reference name	Explanation
Base case	<i>Parameter values "Number of Uses" and "Actual duration of service" as indicated in the BC for each RP.</i>
Worst (1 use)	<i>Scenario corresponding to a single use of the garment = disposable garment.</i>
BC plus 10% uses	<i>Base case with 10% increase in the number of uses.</i>
BC plus 30% uses	<i>Base case with 30% increase in the number of uses.</i>
BC plus 50% uses	<i>Base case with 50% increase in the number of uses.</i>
BC weighted MIN uses	<i>The minimum values of "number of uses" for each RP were selected from the literature, then the weighted average was made for the share of each item that makes up one of the 3 final RP categories.</i>
BC weighted MAX uses	<i>The maximum values of "number of uses" for each RP were selected from the literature, then the weighted average was made for the share of each item that makes up one of the 3 final RP categories.</i>
1 year	<i>A "Number of uses" has been set which makes the "Effective length of service" of the RP as close as possible to 1 year, i.e. use of the item for only one season</i>

Source: own elaboration.

The detail of the parameters varied in each tested case is reported in the reference file "Textile-Prep-Study_3rd-Milestone_Model" ⁽³⁹⁹⁾, as well as the numerical and graphical results for each RP are reported for:

- LCA - Single Score
- Cost per producer
- Cost per consumer
- Societal cost

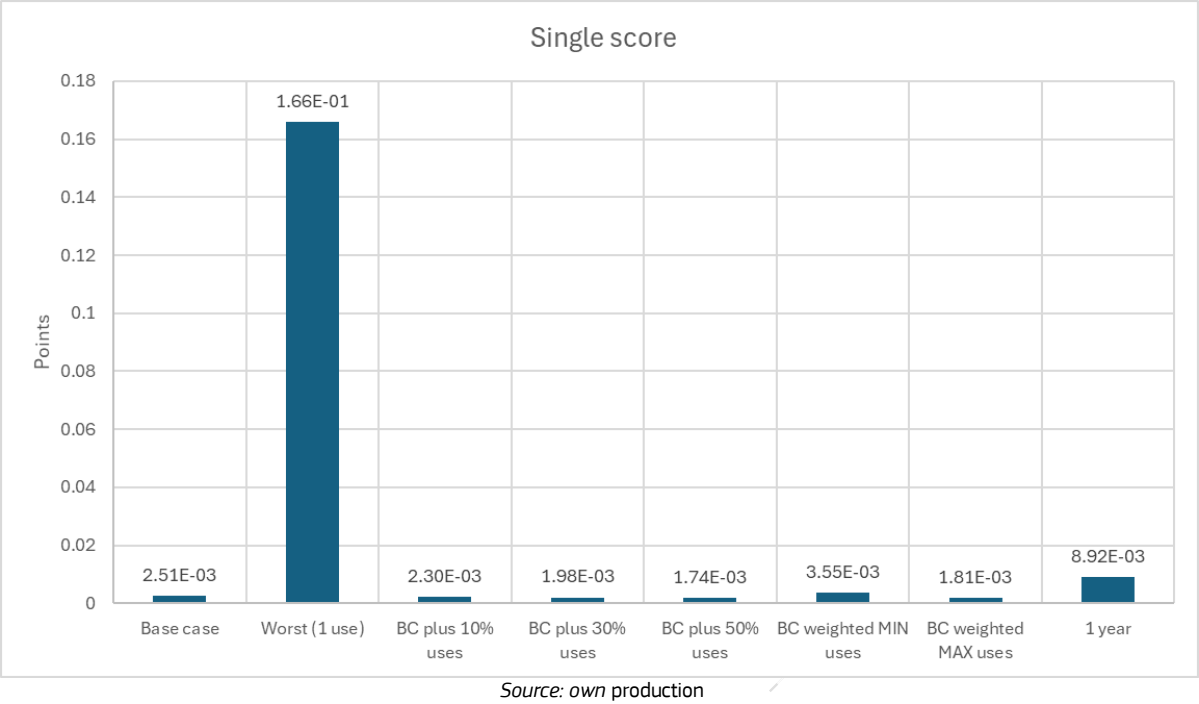
The reference flow is always the one indicated by the original functional unit, relating to the entire life cycle of the RPs.

Results for knitted RP

Comparison of LCA results for the different tested cases. The reference benchmark is the base case, that corresponds to what is shown as the reference in the file "Textile-Prep-Study_3rd-Milestone_Model".

³⁹⁹ Textile-Prep-Study_3rd-Milestone_Model is available on the project website I the section "Documents" at [this link](#).

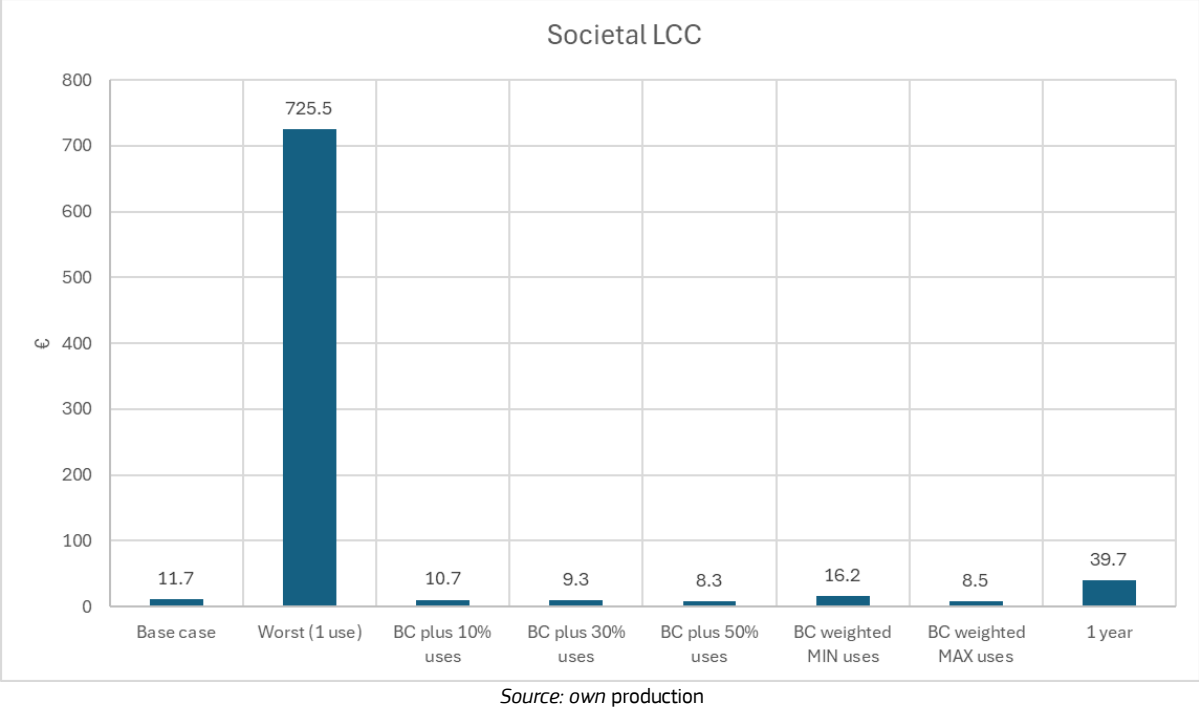
13285 **Figure 116.** Comparisons of LCA results for the knitted reference product assuming different "number of uses" of the
13286 product. Note that 'BC' stands for base case.



13287
13288

13289 Below the equivalent societal LCC results are displayed.

13290 **Figure 117.** Comparisons of societal Life Cycle Costing (LCC) results for the knitted reference product assuming different
13291 "number of uses" of the product. Note that 'BC' stands for base case.

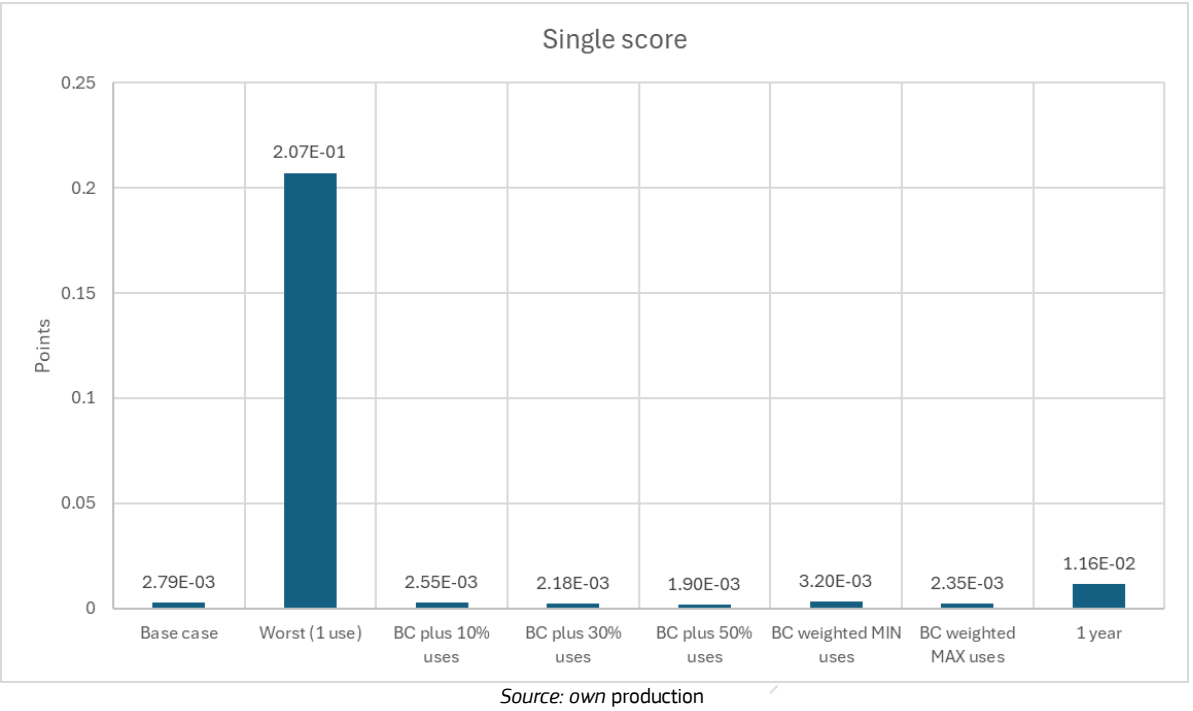


13292
13293

13294 **Results for other woven RP**

13295 Comparison of LCA results for the different tested cases. The reference benchmark is the base case, which
13296 corresponds to what is shown as the reference in the file "Textile-Prep-Study_3rd-Milestone_Model".

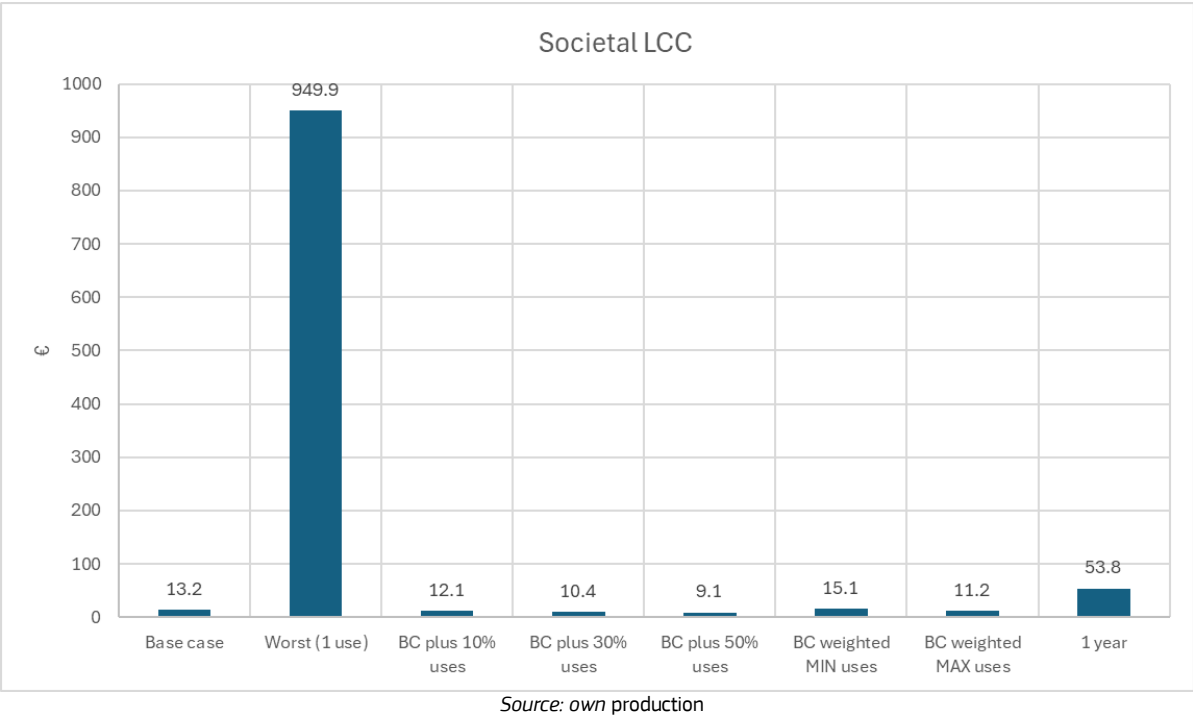
13297 **Figure 118.** Comparisons of LCA results for the other woven reference product assuming different "number of uses" of
13298 the product. Note that 'BC' stands for base case.



13299
13300

13301 Below the equivalent societal LCC results are displayed.

13302 **Figure 119.** Comparisons of societal Life Cycle Costing (LCC) results for the other woven reference product assuming
13303 different "number of uses" of the product. Note that 'BC' stands for base case.



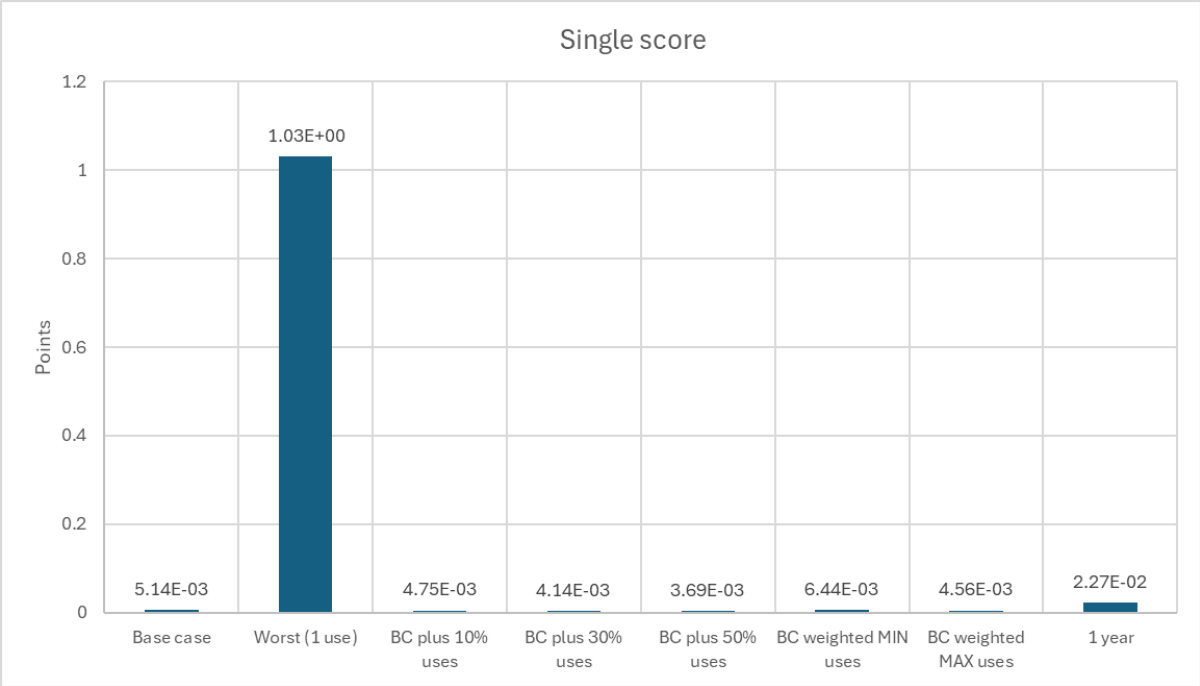
13304
13305

13306 **Results for denim RP**

13307 Comparison of LCA results for the different tested cases. The reference benchmark is the base case, that
13308 corresponds to what is shown as the reference in section Results: Base Case.

13309
13310

Figure 120. Comparisons of LCA results for the denim reference product assuming different "number of uses" of the product. Note that 'BC' stands for base case.

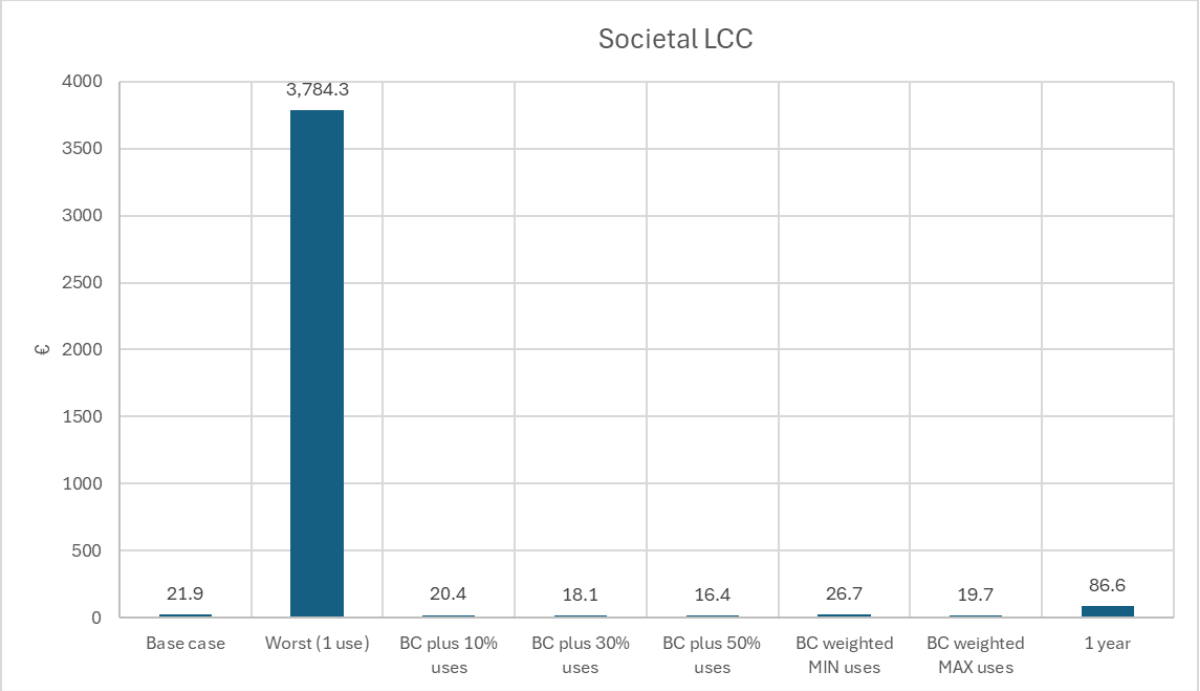


Source: own production

13311
13312

Below the equivalent societal LCC results are displayed.

Figure 121. Comparisons of societal Life Cycle Costing (LCC) results for the denim reference product assuming different "number of uses" of the product. Note that 'BC' stands for base case.



Source: own production

13316
13317

13318 13.11.10.4.2 Location of raw material production and manufacturing processes

13319 This scenario was modelled to allow for the comparison of the influence of location of raw material production
13320 and manufacturing processes. Each RP was evaluated considering different geographical contexts regarding
13321 the supply of raw materials and the implementation of the production phases. The RPs have been modelled
13322 and analysed in different situation:

- 13323 — A Base Case (BC) raw materials and manufacturing - this is the equivalent situation to the base case,
13324 with fibre production and manufacturing processes taking place in a global context;
- 13325 — Raw materials and manufacturing stages entirely made in Europe (EU) - all the stages from the
13326 production and supply of fibres to the marketing of the product, thus including all manufacturing
13327 processes, modelled as carried out in Europe;
- 13328 — A manufacturing of the product totally realised in a geographical context outside Europe (Non-EU) -
13329 the product life cycle was modelled on a European basis only from the distribution phase onwards
13330 (therefore LCS 4 and LCS5), all previous stages were modelled for a non-European situation.

13331 The life cycle stages affected by the change of geographical location in which the products were produced
13332 concern LCS1 and LCS2; the detail description of how the fibres (in LCS1) and the production processes (in
13333 LCS2) were modelled and analysed are reported in the following tables.

13334 **Table 210.** Details of the data used to model and analyse the life cycle stage 1 (LCS1 - Raw materials)

Process	Description Non-EU	Description EU
Viscose fibre	The following data were used as supporting information: China - Ecoinvent dataset for viscose fibre production (Global) India - Ecoinvent viscose fibre production dataset (Global dataset) Indonesia - Ecoinvent dataset for viscose fibre production (Global) Others - Ecoinvent dataset for viscose fibre production (Global) Unknown - Ecoinvent dataset for viscose fibre production (Global)	Ecoinvent dataset for viscose fibre production (Global) was used to model EU scenario by replacing the providers for energy related inputs to providers from EU
Cotton fibre	The following data were used as supporting information: India - Ecoinvent dataset for cotton production and ginning (India) USA - Ecoinvent dataset for cotton production and ginning (Rest-of-World) China - Ecoinvent dataset for cotton production and ginning (Rest-of-World) Unknown - Ecoinvent dataset for cotton production and ginning (Rest-of-World) Others - Ecoinvent dataset for cotton production and ginning (Rest-of-World)	Ecoinvent dataset for cotton production and ginning (Rest-of-World) were used to model the inventory and the provider for energy inputs were changed to corresponding EU provider.
Organic cotton fibre	The following data were used as supporting information: India - Ecoinvent dataset for organic cotton production and ginning (India) China - Ecoinvent dataset for organic cotton production and ginning (Rest-of-World) USA - Ecoinvent dataset for organic cotton production and ginning (Rest-of-World) Others - Ecoinvent dataset for organic cotton production and ginning (Rest-of-World) Unknown - Ecoinvent dataset for organic cotton production and ginning (Rest-of-World))	Ecoinvent dataset for organic cotton production and ginning (Rest-of-World) were used to model the inventory and the provider for energy inputs were changed to corresponding EU provider.
Wool	The following data were used as supporting information: Australia - Ecoinvent dataset for sheep fleece production (Rest-of-World) South Africa - Ecoinvent dataset for sheep fleece production (Rest-of-World) New Zealand - Ecoinvent dataset for sheep fleece production (Rest-of-World)	Ecoinvent dataset for sheep fleece production (Rest-of-World). There were no energy related inputs within the process to change within the disaggregated dataset.

Process	Description Non-EU	Description EU
	Others – Ecoinvent dataset for sheep fleece production (Rest-of-World) Unknown – Ecoinvent dataset for sheep fleece production (Rest-of-World)	
PET	The following data were used as supporting information: China – Ecoinvent dataset for bottle grade PET granulate production (Rest-of-World) Turkiye – Ecoinvent dataset for bottle grade PET granulate production (Rest-of-World) Others – Ecoinvent dataset for bottle grade PET granulate production (Rest-of-World) Unknown – Ecoinvent dataset for bottle grade PET granulate production (Rest-of-World)	Ecoinvent dataset for bottle grade PET granulate production for Europe (RER) was used.
Acrylic	Ecoinvent dataset for acrylonitrile production (Rest -of-world) has been used to model Non-EU scenario	Ecoinvent dataset for acrylonitrile production for Europe (RER) has been used to model EU scenario
Silk	Ecoinvent dataset for Silk textile production (Rest -of-world) has been used to model Non-EU scenario	The same dataset was used as for the Non-EU scenario, i.e., the dataset relating to the Rest-of-World context. This choice was made because the percentage relevance of this fibre's use in the three RPs is very limited (therefore, any potential variation in impact between the EU and non-EU contexts is considered negligible).

Source: Own elaboration

Table 211. Details of the data used to model and analyse Life Cycle Stage 2 (LCS2 – Manufacturing).

Process	Description Non-EU scenario		Description EU scenario
	Input data to model the system ^(*)	Dataset details for impact assessment ^(*)	
Scouring	The data used to model the inventory were taken from Roth et al. (2023), which describes the EU techniques. These are identical for each country (i.e. Brazil, China, India, Indonesia, Russia, USA and Rest-of-World excluding EU).	Each country's impact was assessed using its own energy datasets (e.g. for Brazil specific electricity impact produced in Brazil was used to evaluate the energy consumption of scouring process in this country). Chemical inputs were assessed for all countries using datasets for the Rest-of-World or the Global region.	The process carried out in Europe was modelled using the inventory data from Roth et al., 2023 (the same used for all countries in the non-EU scenario), where datasets specific to the EU geographical context (i.e. EU provider) were used to evaluate both energy and chemical inputs.
Ring spinning	The data used to model the inventory were taken For Bangladesh: Ecoinvent ring spinning dataset (relating to the geographical area of Bangladesh). For India: Ecoinvent ring spinning dataset (relating to the geographical area of India). For all the other countries (China, Pakistan, Russia, - Turkiye, USA, Vietnam and Rest-of-World excluding EU): average of the two available Ecoinvent ring spinning dataset (relating to the geographical area Global)	Energy inputs (electricity and heat) were assessed using the environmental burdens generated by the specific countries they were intended to represent (e.g. if the Global scale spinning process was used for China, the specific impact for the Chinese territorial context was used as a reference for quantifying the environmental impacts of electricity). Chemical inputs were assessed for all countries using datasets for the Rest-of-World or the Global region.	The process carried out in Europe was modelled using the Ecoinvent ring spinning dataset (relating to the geographical area Global), where datasets specific to the EU geographical context (i.e. EU provider) were used to evaluate both energy and chemical inputs.
Melt spinning	The data used to model the inventory were taken from the Ecoinvent dataset for "polyester fibre production". Ecoinvent has two datasets available, one for India and the other for Rest-	As before, specific datasets from each represented country were used to assess the impact of electricity and chemicals (here, in the absence	The process carried out in Europe was modelled using the inventory data obtained from the Ecoinvent dataset relating

Process	Description Non-EU scenario		Description EU scenario
	Input data to model the system ^(*)	Dataset details for impact assessment ^(*)	
	of-World, but the reported inventory quantities are identical in both. The data were processed by eliminating the polyester input from the Ecoinvent inventory, to represent only the melt spinning process. The inventory data are identical for each country (i.e. Bangladesh, China, India, Pakistan and Rest-of-World excluding EU).	of specific datasets, Rest-of-World and Global were used). For example, the melt spinning process in China has been evaluated by modifying the electricity and chemical providers in Rest of the World dataset to data representing electricity consumption and chemical production in China.	to the India/Rest-of World (the same used for all countries in the non-EU scenario), where datasets specific to the EU geographical context (i.e. EU provider) were used to evaluate both energy and chemical inputs.
Weaving	The data used to model the inventory were taken For Bangladesh: Ecoinvent weaving dataset (relating to the geographical area of Bangladesh). For all the other countries (Brazil, China, India, Pakistan, - Turkiye, Rest-of-Word): Ecoinvent weaving dataset (relating to the geographical area of Rest-of-World excluding EU).	Except for Bangladesh, where the inventory data were already country-specific, for the other countries specific datasets from each represented country were used to assess the impact of electricity and chemicals (here, in the absence of specific datasets, Rest-of-World and Global were used).	The Rest-of-Word dataset was used to represent the model inventory for Europe, but to represent the specific environmental burdens generated in Europe, the Rest-of-World inputs were replaced with those of EU providers for both electricity and chemicals.
Knitting	The data used to model the inventory were taken For Bangladesh: Ecoinvent knitting dataset (relating to the geographical area of Bangladesh). For India: Ecoinvent knitting dataset (relating to the geographical area of India). For all the other countries (Brazil, China, Pakistan, - Turkiye, Rest-of-Word excluding EU): Ecoinvent knitting dataset (relating to the geographical area of Rest-of-World).	For Bangladesh and India, country specific inventory data were used as it was available . For the other countries, Rest of the World, or Global datasets of the process was used after changing electricity and chemical datasets to country specific data.	The Rest-of-Word dataset was used to represent the model inventory for Europe, but to represent the specific environmental burdens generated in Europe, the Rest-of-World inputs were replaced with those of EU providers for both electricity and chemicals.
Bleaching	The data used to model the inventory were taken For Bangladesh: Ecoinvent bleaching dataset (relating to the geographical area of Bangladesh). For all the other countries (China and - Turkiye): Ecoinvent bleaching dataset (relating to the geographical area of Rest-of-World excluding EU).	For Bangladesh, country specific inventory data were used as it was available. For the rest of the countries, country specific data were used to assess the impact of electricity and chemicals (In the absence of country specific datasets for chemicals and energy, Rest-of-World or Global datasets were used).	The Rest-of-Word dataset was used to represent the model inventory for Europe, but to represent the specific environmental burdens generated in Europe, the Rest-of-World inputs were replaced with those of EU providers for both electricity and chemicals.
Dyeing	The data used to model the inventory were taken For Bangladesh: Ecoinvent batch dyeing dataset (relating to the geographical area of Bangladesh). For all the other countries (China and - Turkiye): Ecoinvent continuous dyeing dataset (relating to the geographical area of Global).	For Bangladesh and India, country specific inventory data were used as it was available. For the other countries, Rest of the World, or Global datasets of the process was used after changing electricity and chemical datasets to country specific data.	The Global dataset was used to represent the model inventory for Europe, but to represent the specific environmental burdens generated in Europe, the Global inputs were replaced with those of EU providers for both electricity and chemicals.
Printing	As for Scouring the data used to model the inventory were taken from Roth et al., 2023. These are identical for each country (i.e. Bangladesh, China and - Turkiye).	Each country's impact was assessed using its own energy datasets (e.g. for Bangladesh specific electricity impact produced in Bangladesh was used to evaluate the energy consumption of printing process in this country).	The process carried out in Europe was modelled using the inventory data from Roth et al., 2023 (the same used for all countries in the non-EU scenario), where datasets specific to the EU geographical context (i.e.

Process	Description Non-EU scenario		Description EU scenario
	Input data to model the system ^(*)	Dataset details for impact assessment ^(*)	
		Chemical inputs were assessed for all countries using datasets for the Rest-of-World or the Global region.	EU provider) were used to evaluate both energy and chemical inputs.
Finishing (wet finishing process)	All countries (i.e. Bangladesh, China and Turkiye) were modelled on the same data derived from the average of two Ecoinvent datasets, both relating to wet finishing processes on a Global scale and integrated with data obtained from the literature Roth et al., (2023), which describes EU techniques.	Each country's impact was assessed using its own energy datasets (e.g. for Bangladesh specific electricity impact produced in Bangladesh was used to evaluate the energy consumption of wet finishing process in this country). Chemical inputs were assessed for all countries using datasets for the Rest-of-World or the Global region.	The Global datasets with data from literature were used to represent Europe (the same used for all countries in the non-EU scenario), where datasets specific to the EU geographical context (i.e. EU provider) were used to evaluate both energy and chemical inputs.
All processes belonging to the sub-stage: Confectioning	The only inventory input used for the LCA analysis is electricity, with values obtained from different sources as reported in the inventory file. The amount of electricity inputs are identical for each nation of each process constituting the sub-stage (i.e. Bangladesh, China, India, Vietnam and Rest-of-World excluding EU).	Each country's impact was assessed using its own energy datasets (e.g. for Bangladesh specific electricity impact produced in Bangladesh was used to evaluate the energy consumption of the process in this country).	The process carried out in Europe was modelled using the inventory data defined for all the countries in the non-EU scenario, where datasets specific to the EU geographical context (i.e. EU provider) were used to evaluate both energy and chemical inputs.

^(*) From an operational methodological point of view, it is first necessary to define an inventory with input and output values, as reported in column "Input data to model the system" and subsequently these input/output values are converted into environmental impact through the use of specific datasets that include characterization factors for each flow making up the inventory, details reported in "Dataset details for impact assessment".

Source: Own elaboration

More details and the complete set of results can be found in this spreadsheet: "Textile-Prep-Study_3rd-Milestone_Model" ⁽⁴⁰⁰⁾.

The file also contains a tab (named "Location sc - Input data") where the input parameters varied to model each scenario are reported.

The file "Textile-Prep-Study_3rd-Milestone_Model" ⁽⁴⁰¹⁾ contains the results for the 3 RP: knitted, other woven and denim, reported in different tabs.

Each tab shows a LCA section with:

- The detail of the full life cycle results for each scenario;
- Full life cycle results: breakdown by life cycle stages and contributors;
- Comparison of LCA results BAU vs EU non-EU.

And a LCC section with:

- The detail of the full life cycle results for each scenario, from the point of view of society, producer and consumer;
- Full life cycle results: for societal LCC the breakdown by life cycle stages and contributors;
- Comparison of LCC results BAU vs EU Non-EU (for producer, consumer and society).

In this document the LCA and LCC comparisons of results for the 3 RPs are reported, for additional details refer to the reference file "Textile-Prep-Study_3rd-Milestone_Model" ⁽⁴⁰²⁾.

⁴⁰⁰ Textile-Prep-Study_3rd-Milestone_Model is available on the project website I the section "Documents" at [this link](#).

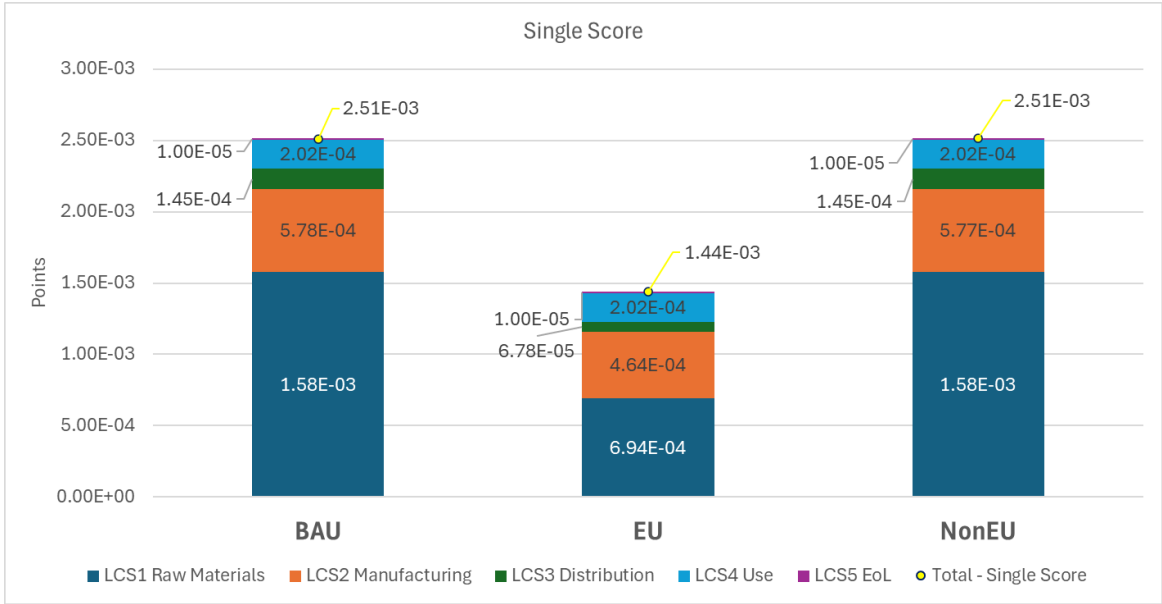
⁴⁰¹ Textile-Prep-Study_3rd-Milestone_Model is available on the project website I the section "Documents" at [this link](#).

⁴⁰² Textile-Prep-Study_3rd-Milestone_Model is available on the project website I the section "Documents" at [this link](#).

13361 **Results Knitted RP**

13362 Comparisons of LCA results are shown in the following graph.

13363 **Figure 122.** Comparisons of LCA results for the knitted RP modelled according to Base Case (BAU) vs EU vs Non-EU
13364 scenario.



13367 Below the results of LCC (for society, producer and consumer) are displayed.

13368 **Figure 123.** Comparisons of societal LCC results for the knitted RP modelled according to Base Case (BAU) vs EU vs Non-
13369 EU scenario.

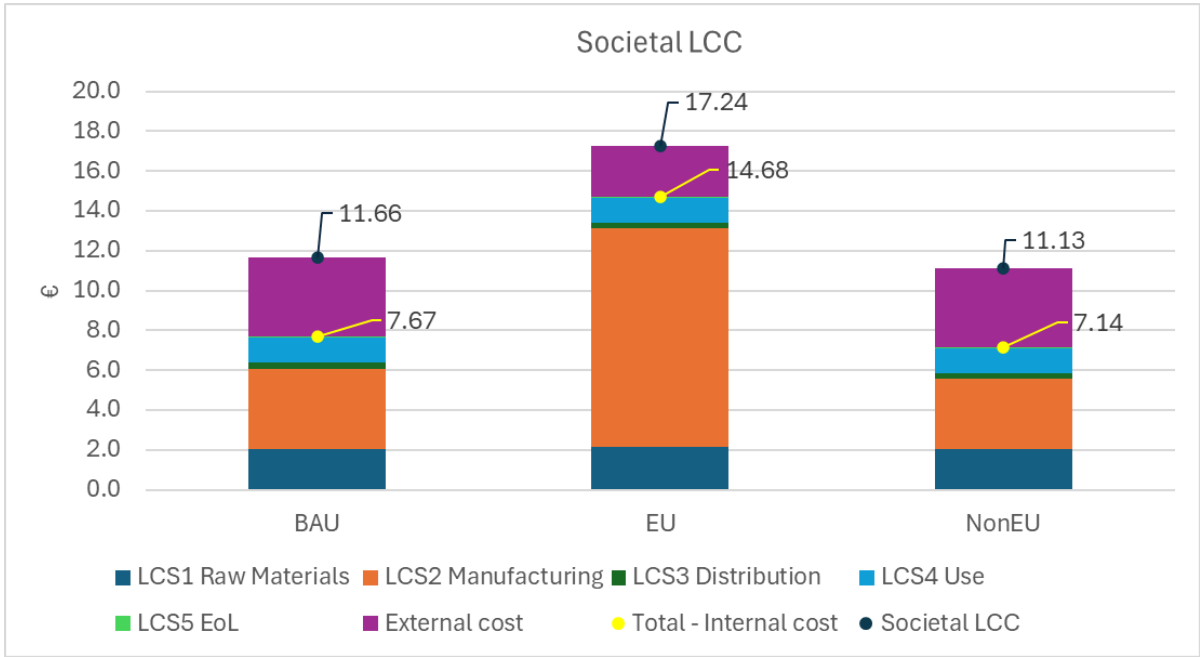


Figure 124. Comparisons of cost for producer results for the knitted RP modelled according to Base Case (BAU) vs EU vs Non-EU scenario

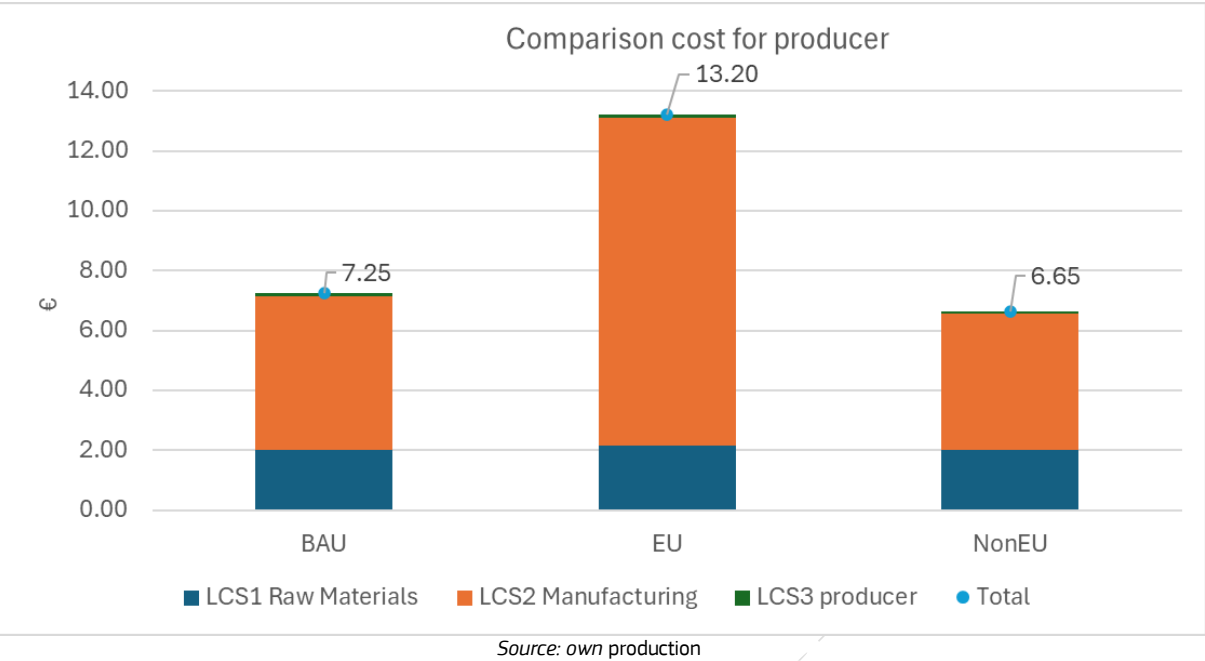
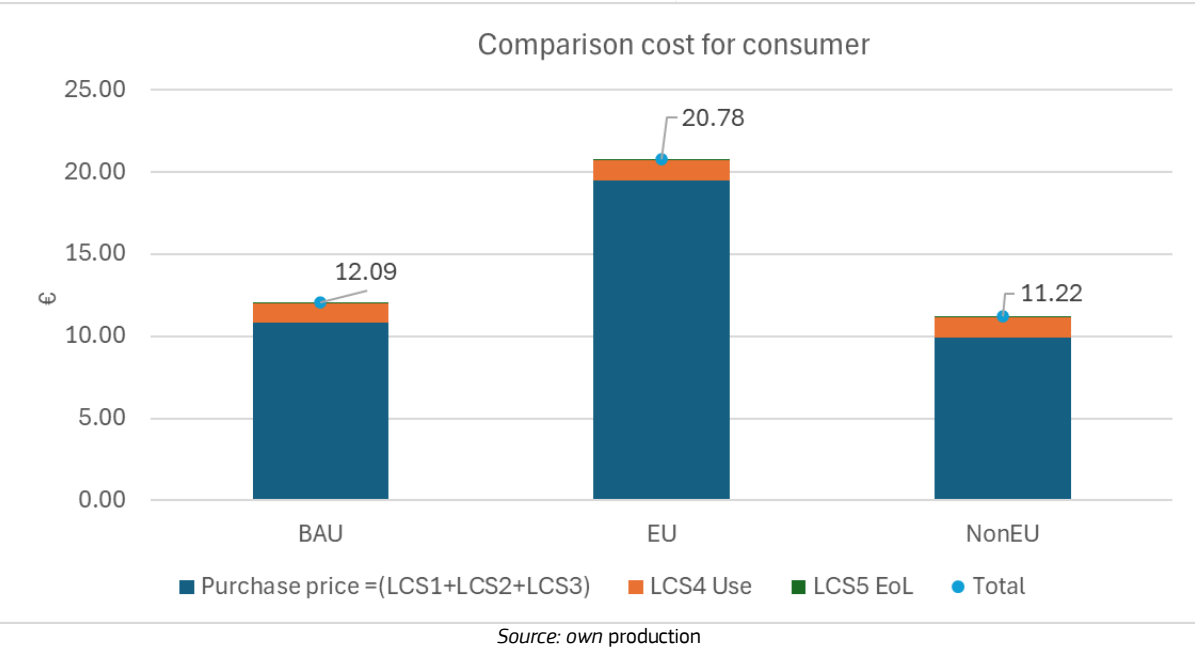


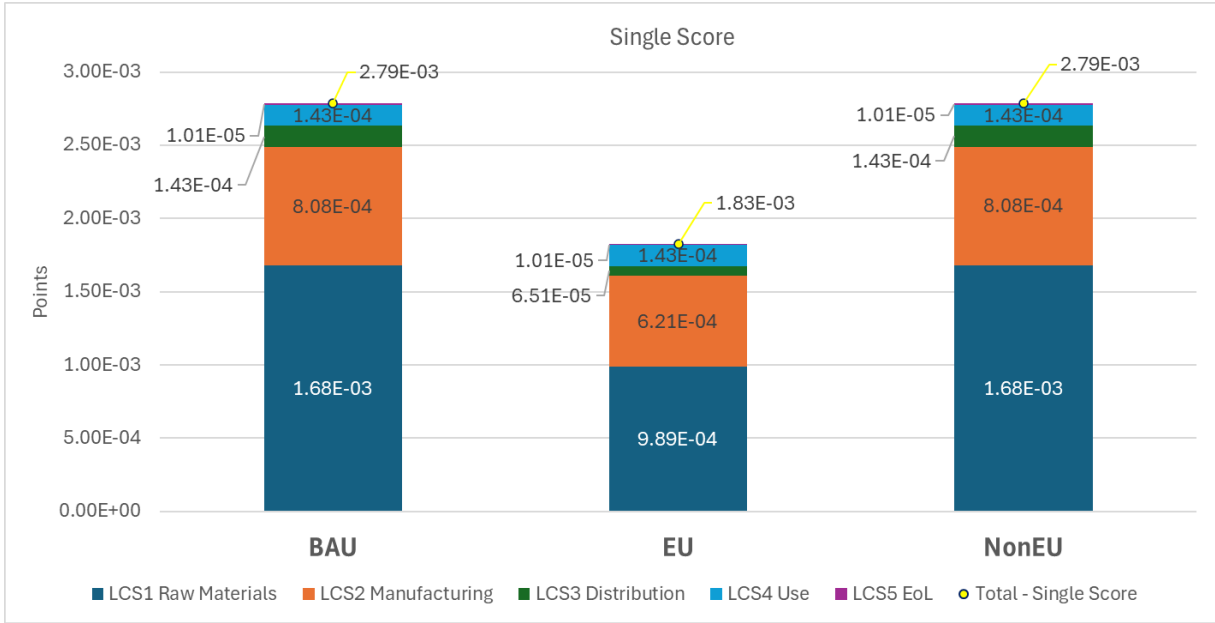
Figure 125. Comparisons of cost for consumer results for the knitted RP modelled according to Base Case (BAU) vs EU vs Non-EU scenario



Results Other woven RP

Comparisons of LCA results are shown in the following graph.

Figure 126. Comparisons of LCA results for the other woven RP modelled according to Base Case (BAU) vs EU vs non-EU scenario



Below the results of LCC (for society, producer and consumer) are displayed.

Figure 127. Comparisons of societal LCC results for the other woven RP modelled according to Base Case (BAU) vs EU vs Non-EU scenario.

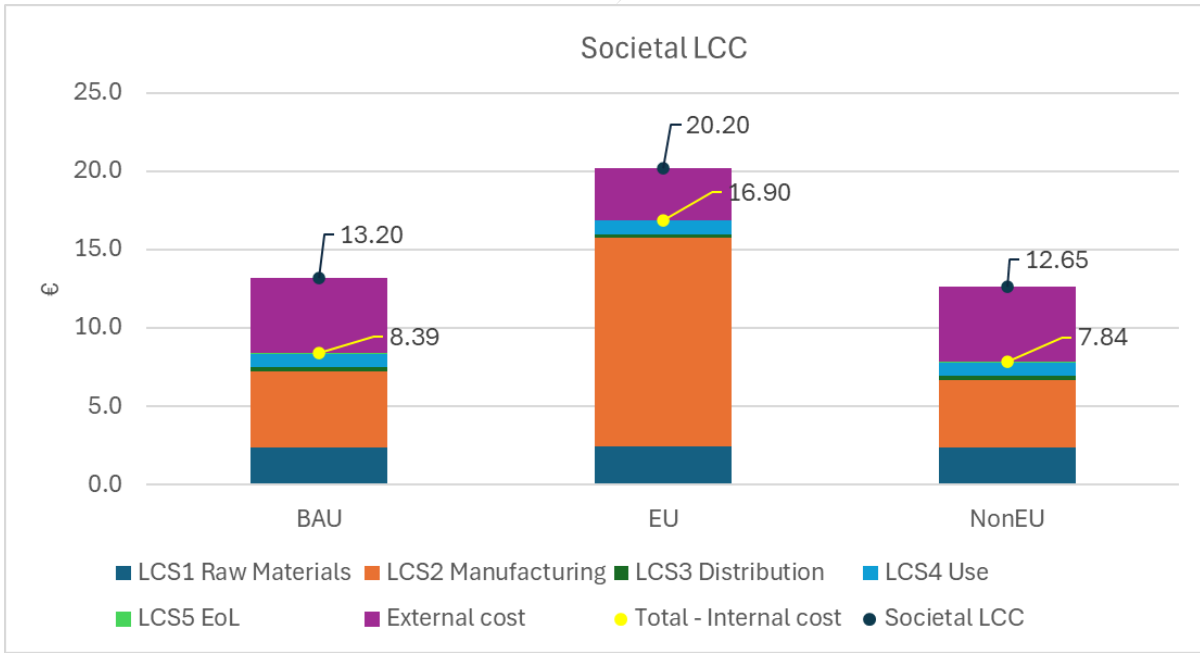
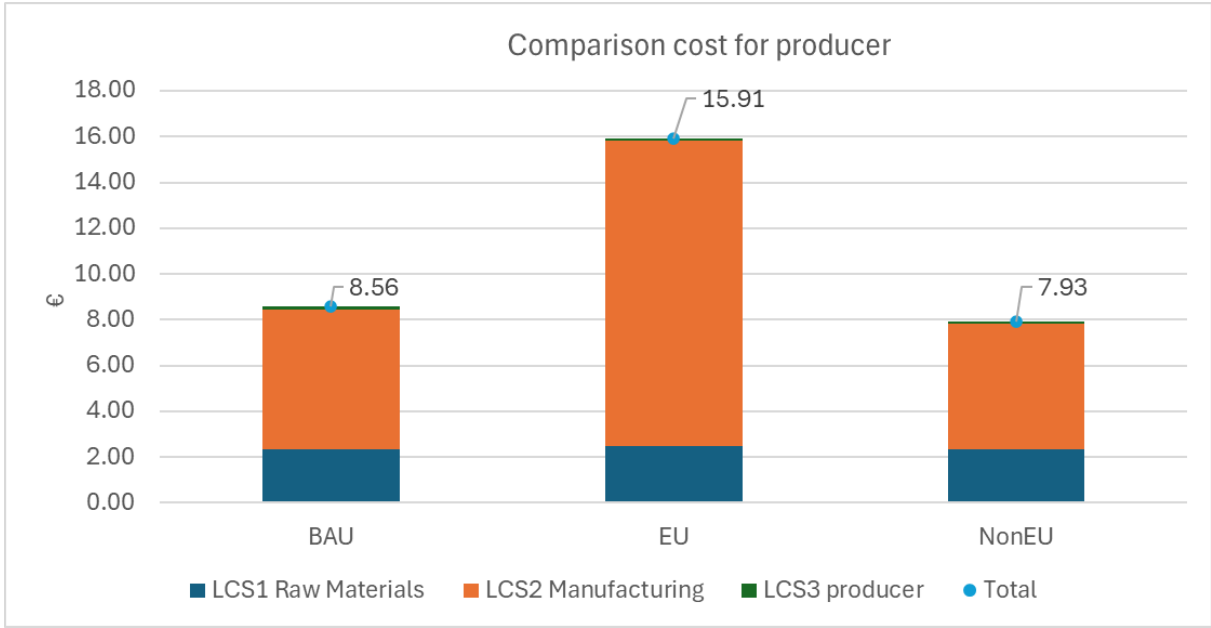
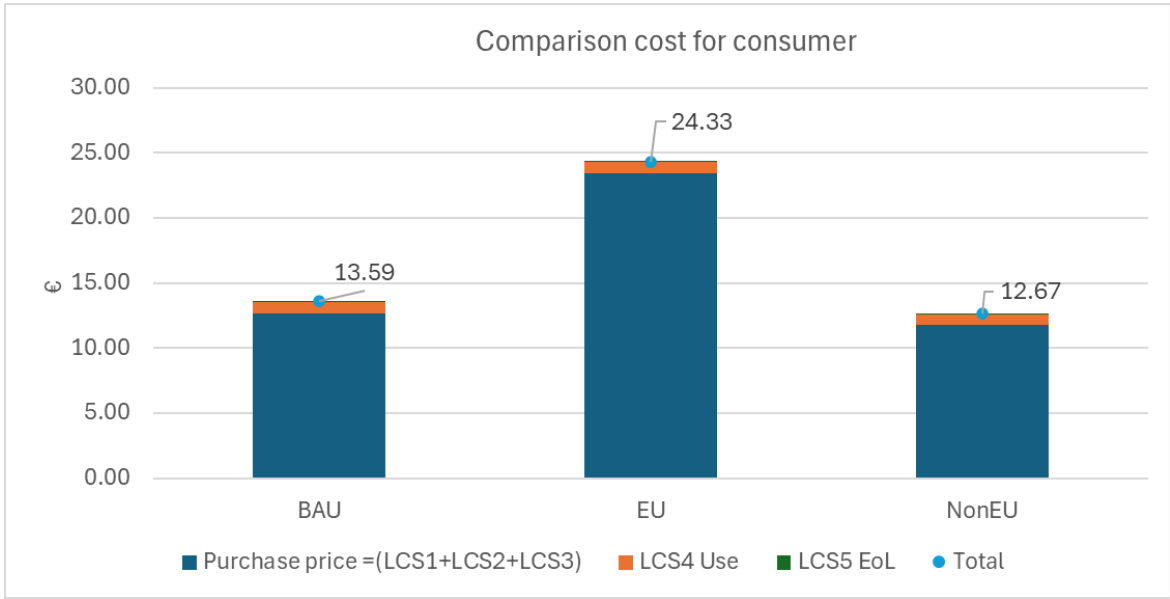


Figure 128. Comparisons of cost for producer results for the other woven RP modelled according to Base Case (BAU) vs EU vs Non-EU scenario.



Source: own production

Figure 129. Comparisons of cost for consumer results for the other woven RP modelled according to Base Case (BAU) vs EU vs Non-EU scenario.



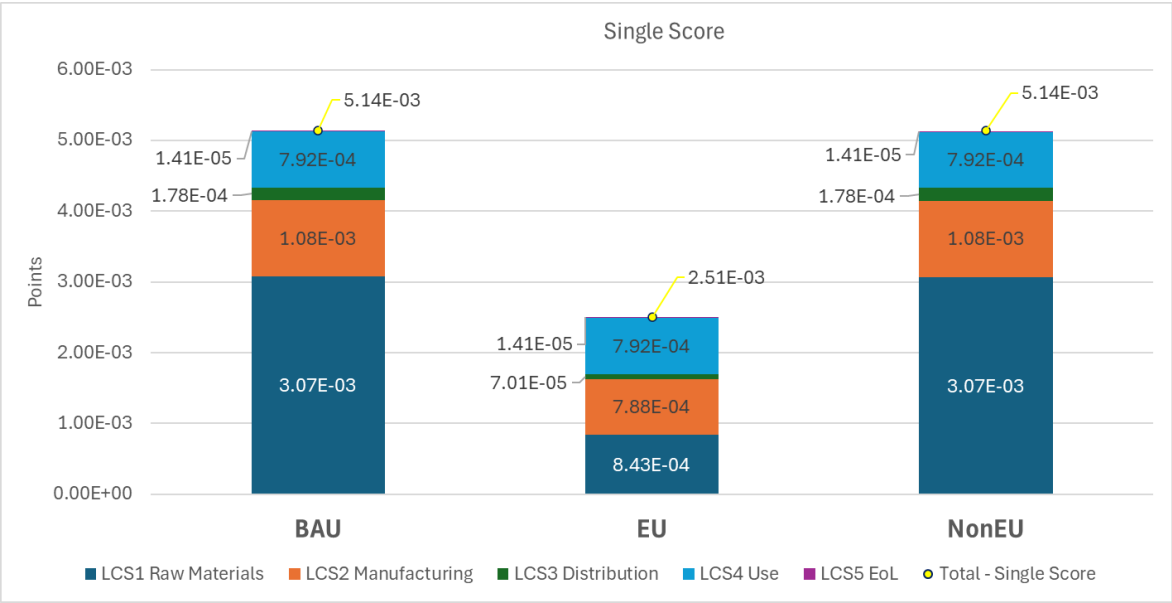
Source: own production

Results Denim RP

Comparisons of LCA results are shown in the following graph.

13402
13403

Figure 130. Comparisons of LCA results for the denim RP modelled according to Base Case (BAU) vs EU vs Non-EU scenario

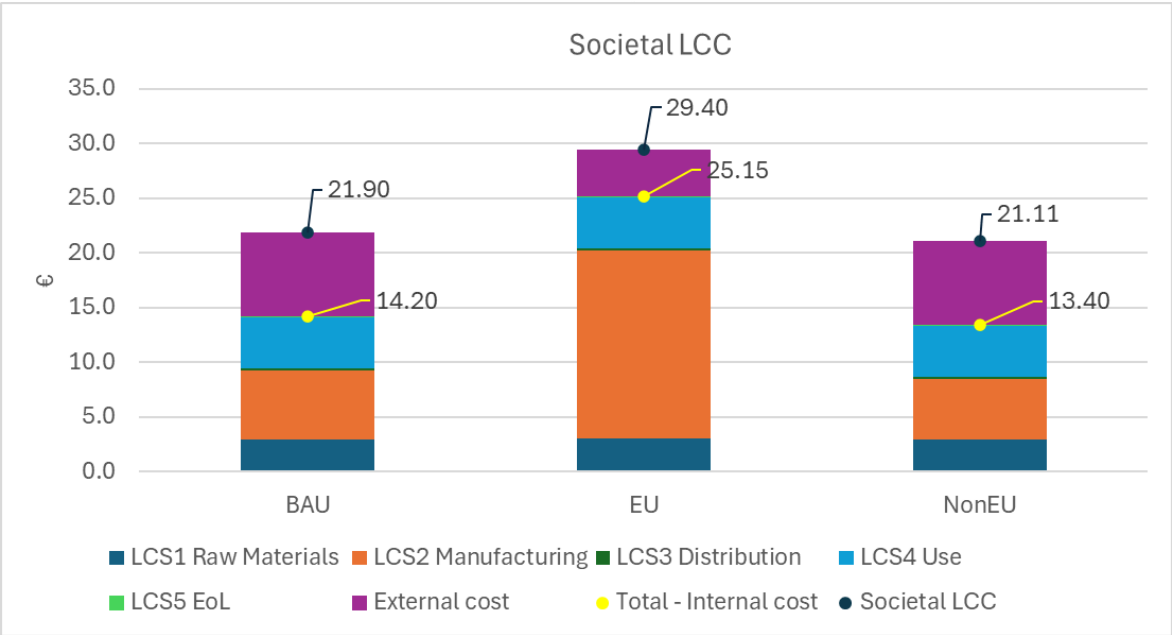


13404
13405

13406 Below the results of LCC (for society, producer and consumer) are displayed.

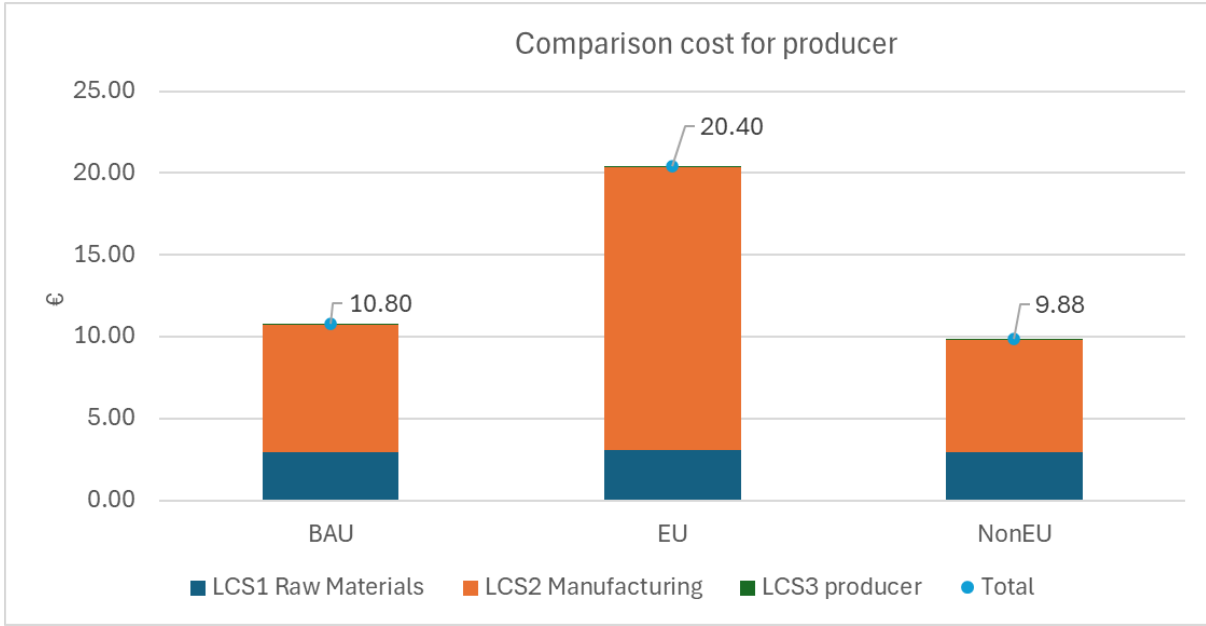
13407
13408

Figure 131. Comparisons of societal LCC results for the denim RP modelled according to Base Case (BAU) vs EU vs Non-EU scenario.



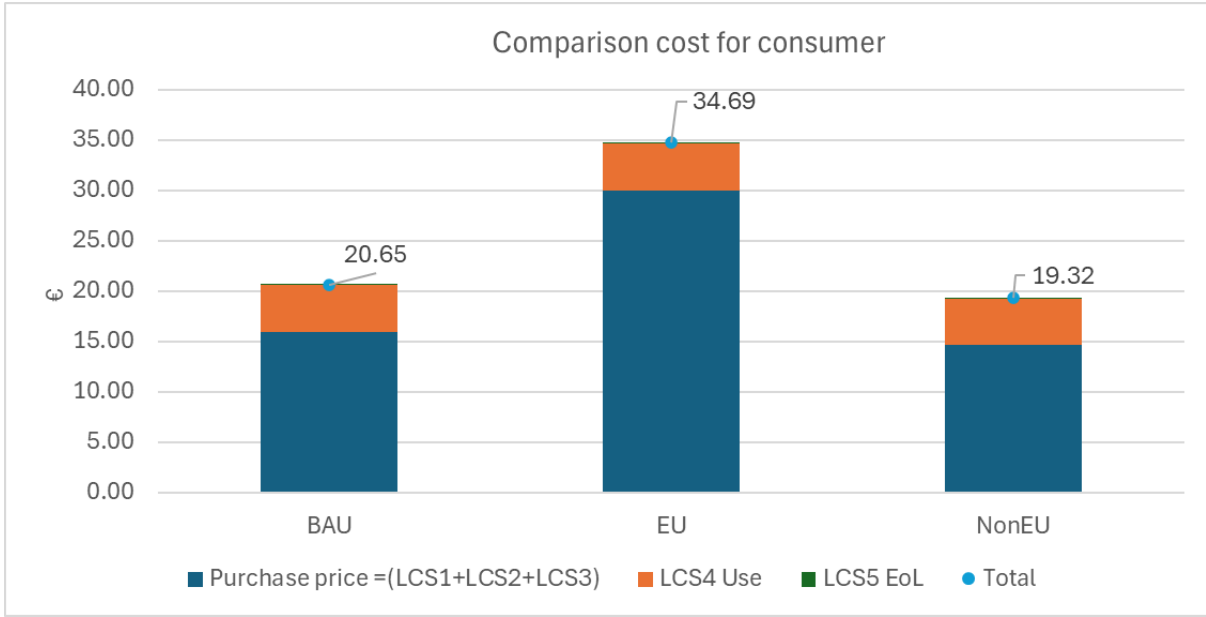
13409
13410

Figure 132. Comparisons of cost for producer results for the denim RP modelled according to Base Case (BAU) vs EU vs Non-EU scenario.



Source: own production

Figure 133. Comparisons of cost for consumer results for the denim RP modelled according to Base Case (BAU) vs EU vs Non-EU scenario.



Source: own production

13.11.10.5 Results: Design options

To complement the figures provided in section 11, the following tables report the results for each of the impact categories for the assessment of the design options when studied individually. The results of the analogous analysis performed for the clusters of design options in paths are reported directly in section 11.2 .

Table 212. Results of the design options for knitted products

Impact category	Unit	D01	D02	D03.1	D03.1+D03.2	D04.1	D04.2
Climate change	kg CO2 eq/person	13.76	14.05	14.04	13.93	13.87	13.87

Impact category	Unit	D01	D02	D03.1	D03.1+D03.2	D04.1	D04.2
Ozone depletion	kg CFC-11 eq./person	3.27E-05	3.32E-05	3.32E-05	3.30E-05	3.23E-05	3.32E-05
Human toxicity cancer	CTUh/person	4.42E-08	4.51E-08	4.50E-08	4.49E-08	4.44E-08	4.51E-08
Human toxicity non-cancer	CTUh/person	4.22E-07	4.30E-07	4.30E-07	4.30E-07	4.20E-07	4.30E-07
Particulate matter	disease incidences/person	9.36E-07	9.56E-07	9.55E-07	9.40E-07	9.48E-07	9.56E-07
Ionising radiation (human health)	kBq U-235 eq./person	1.33	1.34	1.34	1.34	1.33	1.34
Photochemical ozone formation (human health)	kg NMVOC eq./person	5.06E-02	5.18E-02	5.18E-02	5.16E-02	5.12E-02	5.18E-02
Acidification	mol H+ eq./person	0.11	0.11	0.11	0.11	0.11	0.11
Eutrophication terrestrial	mol N eq./person	0.37	0.37	0.37	0.37	0.37	0.37
Eutrophication freshwater	kg P eq./person	6.79E-03	6.95E-03	6.96E-03	6.92E-03	6.90E-03	6.95E-03
Eutrophication marine	kg N eq./person	5.81E-02	5.95E-02	5.93E-02	5.91E-02	5.94E-02	5.95E-02
Ecotoxicity: freshwater	CTUe/person	199.00	203.32	202.88	201.78	201.93	203.32
Land use	pt/person	317.53	325.34	324.45	315.19	324.71	325.33
Water use	m ³ water eq of deprived water/person	115.44	118.35	117.85	118.30	118.28	118.35
Resource use minerals and metals	kg Sb eq./person	1.15E-04	1.17E-04	1.17E-04	1.13E-04	1.17E-04	1.17E-04
Resource use fossils	MJ/person	164.16	167.30	167.14	166.36	165.26	167.28

D01 - Robustness score, D2 - Recyclability score, D03.1 - Information on recycled content, D03.1 + D03.2 - Information and performance requirement on recycled content, D04.1 - Environmental footprint, D04.2 - Carbon footprint

Source: own elaboration

Table 213. Results of the design options for denim products

Impact category	Unit	D01	D02	D03.1	D03.1+D03.2	D04.1	D04.2
Climate change	kg CO2 eq./person	26.43	26.74	26.60	26.14	26.37	26.37
Ozone depletion	kg CFC-11 eq./person	4.02E-05	4.07E-05	4.07E-05	4.07E-05	3.95E-05	4.07E-05
Human toxicity cancer	CTUh/person	7.07E-08	7.18E-08	7.16E-08	7.10E-08	7.08E-08	7.17E-08
Human toxicity non-cancer	CTUh/person	5.26E-07	5.37E-07	5.34E-07	5.26E-07	5.26E-07	5.37E-07
Particulate matter	disease incidences/person	1.50E-06	1.52E-06	1.50E-06	1.44E-06	1.50E-06	1.52E-06
Ionising radiation (human health)	kBq U-235 eq./person	4.34	4.35	4.34	4.33	4.33	4.35
Photochemical ozone formation (human health)	kg NMVOC eq./person	9.26E-02	9.41E-02	9.35E-02	9.18E-02	9.28E-02	9.41E-02
Acidification	mol H+ eq./person	0.18	0.18	0.18	0.17	0.18	0.18
Eutrophication terrestrial	mol N eq./person	0.53	0.55	0.53	0.49	0.54	0.55
Eutrophication freshwater	kg P eq./person	1.52E-02	1.54E-02	1.58E-02	1.67E-02	1.53E-02	1.54E-02
Eutrophication marine	kg N eq./person	1.30E-01	1.33E-01	1.28E-01	1.14E-01	1.32E-01	1.33E-01
Ecotoxicity: freshwater	CTUe/person	396.23	404.58	395.53	366.61	402.28	404.57

Impact category	Unit	D01	D02	D03.1	D03.1+D03.2	D04.1	D04.2
Land use	pt/person	438.60	448.11	437.85	405.06	446.78	448.10
Water use	m ³ water eq of deprived water/person	280.98	289.11	276.51	236.20	289.03	289.11
Resource use minerals and metals	kg Sb eq/person	1.91E-04	1.94E-04	1.92E-04	1.90E-04	1.93E-04	1.94E-04
Resource use fossils	MJ/person	345.74	348.97	347.83	344.51	344.97	348.95

D01 - Robustness score, D2 - Recyclability score, D03.1 - Information on recycled content, D03.1 + D03.2 - Information and performance requirement on recycled content, D04.1 - Environmental footprint, D04.2 - Carbon footprint

Source: own elaboration

Table 214. Results of the design options for woven products

Impact category	Unit	D01	D02	D03.1	D03.1+D03.2	D04.1	D04.2
Climate change	kg CO ₂ eq/person	17.23	17.53	17.52	17.38	17.27	17.27
Ozone depletion	kg CFC-11 eq./person	3.82E-05	3.88E-05	3.88E-05	3.86E-05	3.78E-05	3.88E-05
Human toxicity cancer	CTUh/person	5.89E-08	6.01E-08	6.01E-08	5.99E-08	5.93E-08	6.01E-08
Human toxicity non-cancer	CTUh/person	5.28E-07	5.37E-07	5.37E-07	5.37E-07	5.24E-07	5.37E-07
Particulate matter	disease incidences/person	1.35E-06	1.37E-06	1.37E-06	1.35E-06	1.36E-06	1.37E-06
Ionising radiation (human health)	kBq U-235 eq./person	1.23	1.24	1.24	1.24	1.23	1.24
Photochemical ozone formation (human health)	kg NMVOC eq./person	6.51E-02	6.65E-02	6.64E-02	6.62E-02	6.56E-02	6.65E-02
Acidification	mol H ⁺ eq./person	0.14	0.14	0.14	0.14	0.14	0.14
Eutrophication terrestrial	mol N eq./person	0.47	0.48	0.48	0.47	0.48	0.48
Eutrophication freshwater	kg P eq./person	7.65E-03	7.80E-03	7.81E-03	7.77E-03	7.72E-03	7.80E-03
Eutrophication marine	kg N eq./person	5.66E-02	5.79E-02	5.77E-02	5.74E-02	5.77E-02	5.79E-02
Ecotoxicity: freshwater	CTUe/person	219.62	224.27	223.90	222.44	222.47	224.27
Land use	pt/person	386.73	396.14	395.26	384.36	395.17	396.13
Water use	m ³ water eq of deprived water/person	100.28	102.78	102.40	102.72	102.69	102.78
Resource use minerals and metals	kg Sb eq./person	1.54E-04	1.58E-04	1.57E-04	1.51E-04	1.57E-04	1.58E-04
Resource use fossils	MJ/person	195.81	199.01	198.79	197.54	196.10	199.00

D01 - Robustness score, D2 - Recyclability score, D03.1 - Information on recycled content, D03.1 + D03.2 - Information and performance requirement on recycled content, D04.1 - Environmental footprint, D04.2 - Carbon footprint

Source: own elaboration

To complement the figures above, the following tables show how the individual design options compare to the base cases in terms of each of the impact categories.

13437 **Table 215.** Results of the design options for knitted products

Impact category	D01	D02	D03.1	D03.1+D03.2	D04.1	D04.2
Climate change	-2.1%	0.0%	-0.1%	-0.8%	-1.3%	-1.3%
Ozone depletion	-1.6%	0.0%	-0.1%	-0.5%	-2.8%	0.0%
Human toxicity cancer	-2.0%	0.0%	0.0%	-0.3%	-1.4%	0.0%
Human toxicity non-cancer	-1.9%	0.0%	0.0%	-0.1%	-2.5%	0.0%
Particulate matter	-2.2%	0.0%	-0.2%	-1.7%	-0.9%	0.0%
Ionising radiation (human health)	-0.7%	0.0%	0.0%	-0.1%	-0.6%	0.0%
Photochemical ozone formation (human health)	-2.4%	0.0%	-0.1%	-0.4%	-1.2%	0.0%
Acidification	-2.2%	0.0%	-0.2%	-2.1%	-0.7%	0.0%
Eutrophication terrestrial	-2.4%	0.0%	-0.3%	-2.6%	-0.5%	0.0%
Eutrophication freshwater	-2.2%	0.0%	0.1%	-0.4%	-0.7%	0.0%
Eutrophication marine	-2.4%	0.0%	-0.3%	-0.6%	-0.3%	0.0%
Ecotoxicity: freshwater	-2.1%	0.0%	-0.2%	-0.8%	-0.7%	0.0%
Land use	-2.4%	0.0%	-0.3%	-3.1%	-0.2%	0.0%
Water use	-2.5%	0.0%	-0.4%	0.0%	-0.1%	0.0%
Resource use minerals and metals	-2.0%	0.0%	-0.5%	-3.6%	-0.5%	0.0%
Resource use fossils	-1.9%	0.0%	-0.1%	-0.6%	-1.2%	0.0%

13438 D01 - Robustness score, D2 - Recyclability score, D03.1 - Information on recycled content, D03.1 + D03.2 - Information and performance
13439 requirement on recycled content, D04.1 - Environmental footprint, D04.2 - Carbon footprint

13440 *Source: own elaboration*

13441 **Table 216.** Results of the design options for denim products

Impact category	D01	D02	D03.1	D03.1+D03.2	D04.1	D04.2
Climate change	-1.2%	0.0%	-0.6%	-2.3%	-1.4%	-1.4%
Ozone depletion	-1.3%	0.0%	-0.1%	-0.1%	-2.8%	0.0%
Human toxicity cancer	-1.4%	0.0%	-0.3%	-1.1%	-1.4%	0.0%
Human toxicity non-cancer	-2.0%	0.0%	-0.5%	-2.0%	-2.1%	0.0%
Particulate matter	-1.5%	0.0%	-1.3%	-5.5%	-1.3%	0.0%
Ionising radiation (human health)	-0.2%	0.0%	-0.1%	-0.3%	-0.4%	0.0%
Photochemical ozone formation (human health)	-1.6%	0.0%	-0.6%	-2.4%	-1.3%	0.0%
Acidification	-1.7%	0.0%	-1.6%	-6.8%	-1.0%	0.0%

Impact category	D01	D02	D03.1	D03.1+D03.2	D04.1	D04.2
Eutrophication terrestrial	-2.2%	0.0%	-2.3%	-9.6%	-0.7%	0.0%
Eutrophication freshwater	-1.4%	0.0%	2.0%	8.4%	-0.8%	0.0%
Eutrophication marine	-2.5%	0.0%	-3.4%	-14.1%	-0.3%	0.0%
Ecotoxicity: freshwater	-2.1%	0.0%	-2.2%	-9.4%	-0.6%	0.0%
Land use	-2.1%	0.0%	-2.3%	-9.6%	-0.3%	0.0%
Water use	-2.8%	0.0%	-4.4%	-18.3%	0.0%	0.0%
Resource use minerals and metals	-1.4%	0.0%	-0.6%	-1.8%	-0.5%	0.0%
Resource use fossils	-0.9%	0.0%	-0.3%	-1.3%	-1.1%	0.0%

D01 - Robustness score, D2 - Recyclability score, D03.1 - Information on recycled content, D03.1 + D03.2 - Information and performance requirement on recycled content, D04.1 - Environmental footprint, D04.2 - Carbon footprint

Source: own elaboration

Table 217. Results of the design options for woven products

Impact category	D01	D02	D03.1	D03.1+D03.2	D04.1	D04.2
Climate change	-1.7%	0.0%	-0.1%	-0.9%	-1.5%	-1.5%
Ozone depletion	-1.6%	0.0%	-0.1%	-0.7%	-2.7%	0.0%
Human toxicity cancer	-2.0%	0.0%	0.0%	-0.3%	-1.3%	0.0%
Human toxicity non-cancer	-1.8%	0.0%	0.0%	-0.1%	-2.6%	0.0%
Particulate matter	-1.9%	0.0%	-0.1%	-1.4%	-1.0%	0.0%
Ionising radiation (human health)	-0.7%	0.0%	0.0%	-0.2%	-0.9%	0.0%
Photochemical ozone formation (human health)	-2.0%	0.0%	-0.1%	-0.5%	-1.4%	0.0%
Acidification	-2.0%	0.0%	-0.2%	-1.9%	-0.8%	0.0%
Eutrophication terrestrial	-2.3%	0.0%	-0.2%	-2.4%	-0.5%	0.0%
Eutrophication freshwater	-2.0%	0.0%	0.1%	-0.4%	-1.0%	0.0%
Eutrophication marine	-2.2%	0.0%	-0.3%	-0.8%	-0.4%	0.0%
Ecotoxicity: freshwater	-2.1%	0.0%	-0.2%	-0.8%	-0.8%	0.0%
Land use	-2.4%	0.0%	-0.2%	-3.0%	-0.2%	0.0%
Water use	-2.4%	0.0%	-0.4%	-0.1%	-0.1%	0.0%
Resource use minerals and metals	-2.1%	0.0%	-0.6%	-4.4%	-0.4%	0.0%
Resource use fossils	-1.6%	0.0%	-0.1%	-0.7%	-1.5%	0.0%

D01 - Robustness score, D2 - Recyclability score, D03.1 - Information on recycled content, D03.1 + D03.2 - Information and performance requirement on recycled content, D04.1 - Environmental footprint, D04.2 - Carbon footprint

Source: own elaboration

13.11.11 Sensitivity and uncertainty Analysis

Uncertainty is inherent in modelling and, when this refers to the input values, a sensitivity analysis is necessary. The data uncertainty could be related to the choices that are made, and which refer to more fundamental and qualitative choices of the model such as where the products are manufactured or variations in the parameters of the number of uses to test possible improvements and worsening. Such issues are dealt with separate scenario analyses presented in the relevant subsections. For the uncertainty which is related to the values acquired for specific inputs there are two methods that can be used to analyse how sensitive or robust the model is: a Perturbation analysis and a Monte Carlo analysis. The application of these two analyses on the Representative Products are described in the subsections that follow the description of the parameters used.

13.11.11.1 Sensitivity and uncertainty analysis parameters

Both analyses mentioned above utilise a selected number of parameters based on a choice of the input variables that were used for the model which is described in subsection 13.11.9.7. The refinement process that allowed to choose the most appropriate parameters is described below.

13.11.11.1.1 Refinement of chosen parameters

From the set 345 input values that are used as parameters, 98 of them were allowed to have their values changed in the uncertainty analysis. There are three main reasons why specific parameters were excluded as explained in the points below:

- Parameters that are not independent because the increase of one of them would mean that the others would have to decrease, and this cannot be guaranteed when running a Monte Carlo using a random number generator. Monte Carlo methods rely on independent random variables (inputs) and when these are not independent, the accuracy of the simulation can be compromised and lead to incorrect conclusions. These parameters are the ones relevant to: the geographic origin of the production sites, the transportation share of the route between the various production sites, the waste, the specific flow distribution (colouring type and percentage of yarn entering weaving/knitting).
- Parameters whose contribution to the final scores within the same product category was negligible such as the ones relating to: the retailing, the waste, the losses during the various manufacturing processes and distribution, the packaging as well as the interest rate and VAT which are expected to show minimal variation.
- Parameters that are not relevant to the comparison between the BC and the Design Options based on the fact that the second goal of the study is to compare BC with DOs for the same category. If these parameters are not important for the DOs, they were excluded. This refers to parameters related to the retailing as well as parameters related to the functional unit definition such as the: number of uses (expressed in days of wear), average number of uses prior to washing (expressed in days of wear), final mass of the RPs, desired duration of service and actual duration of service of 1 product.

The full list of the parameters used for the uncertainty analysis is provided in the table that follows.

Table 218. List of parameters used for the uncertainty analysis.

Reference category	Description
Losses	Losses in ring spinning
Losses	Losses in weaving
Losses	Losses in dyeing
Losses	Losses in confectioning
Losses	Manufacturing Leftover Losses fraction
Economic	Full time equivalent
Economic	Working hours per year (considering 2 shift/day)
Economic	Discount rate
Economic	Tariff for textile intermediate product
Economic	Tariff for textile final product
Volume	Product volume

Reference category	Description
Use	Share of garments machine washed
Use	Temperature of average EU Washing
Use	Full load wash weight
Use	Detergent required per wash of full load
Use	Water required per wash of full load
Use	Share of garments tumble dried
Use	Tumble dryer electricity consumption per cycle
Use	Full load dry weight
Use	Share of garments ironed
Use	Iron device wattage
Use	Time spent for ironing of 1 garment
Economic	Price for LCS1j – Acrylic (Rest of the world)
Economic	Price for LCS1j – Acrylic (EU)
Economic	Price for LCS1k – Elastane (Global)
Economic	Price for LCS1g – Polyamide (Unknown)
Economic	Price for LCS1g – Polyamide (China)
Economic	Price for LCS1g – Polyamide (Taiwan)
Economic	Price for LCS1g – Polyamide (USA)
Economic	Price for LCS1g – Polyamide (EU)
Economic	Price for LCS1e – PET (Unknown)
Economic	Price for LCS1e – PET (China)
Economic	Price for LCS1e – PET (Others)
Economic	Price for LCS1e – PET (Turkey)
Economic	Price for LCS1e – PET (EU)
Economic	Price for LCS1a – viscose fibre (Unknown)
Economic	Price for LCS1a – viscose fibre (China)
Economic	Price for LCS1a – viscose fibre (Others)
Economic	Price for LCS1a – viscose fibre (India)
Economic	Price for LCS1a – viscose fibre (Indonesia)
Economic	Price for LCS1a – viscose fibre (EU)
Economic	Price for LCS1x – Acetate (Global)
Economic	Price for LCS1b – cotton fibre (Unknown)
Economic	Price for LCS1b – cotton fibre (India)
Economic	Price for LCS1b – cotton fibre (Others)
Economic	Price for LCS1b – cotton fibre (China)
Economic	Price for LCS1b – cotton fibre (USA)
Economic	Price for LCS1b – cotton fibre (EU)
Economic	Price for LCS1d – Wool (Unknown)
Economic	Price for LCS1d – Wool (Australia)
Economic	Price for LCS1d – Wool (South Africa)
Economic	Price for LCS1d – Wool (New Zealand)
Economic	Price for LCS1d – Wool (Others)
Economic	Price for LCS1d – Wool (EU)
Economic	Price for silk (Non-EU)

Reference category	Description
Economic	Price for silk (EU)
Economic	Unit cost for Electricity MV (Bangladesh)
Economic	Unit cost for Electricity MV (Pakistan)
Economic	Unit cost for Electricity MV (Turkey)
Economic	Unit cost for Electricity MV (Vietnam)
Economic	Unit cost for Electricity MV (China)
Economic	Unit cost for Electricity MV (Global)
Economic	Unit cost for Electricity MV (India)
Economic	Unit cost for Electricity MV (EU)
Economic	Unit cost for Electricity MV (USA)
Economic	Unit cost for Electricity MV (Rest of the World)
Economic	Unit cost for Electricity LV (EU)
Economic	Unit cost for Electricity LV (Global)
Economic	Unit cost for heat, district or industrial, natural gas (Rest of the world)
Economic	Unit cost for heat, district or industrial, natural gas (Global)
Economic	Unit cost for heat, district or industrial, natural gas (Bangladesh)
Economic	Unit cost for heat, district or industrial, natural gas (China)
Economic	Unit cost for heat, district or industrial, natural gas (India)
Economic	Unit cost for heat, district or industrial, natural gas (Turkey)
Economic	Unit cost for heat, district or industrial, natural gas (USA)
Economic	Unit cost for heat, district or industrial, natural gas (EU)
Economic	Unit cost for heat, central or small-scale, other than natural gas (Bangladesh)
Economic	Unit cost for heat, central or small-scale, other than natural gas (China)
Economic	Unit cost for heat, central or small-scale, other than natural gas (EU)
Economic	Unit cost for heat, central or small-scale, other than natural gas (India)
Economic	Unit cost for heat, central or small-scale, other than natural gas (Pakistan)
Economic	Unit cost for heat, central or small-scale, other than natural gas (Turkey)
Economic	Unit cost for heat, central or small-scale, other than natural gas (USA)
Economic	Unit cost for heat, central or small-scale, other than natural gas (Vietnam)
Economic	Unit cost for heat, central or small-scale, other than natural gas (Rest of the world)
Economic	Unit cost for heat, central or small-scale, other than natural gas (Global)
Economic	Unit cost for heat, district or industrial, other than natural gas (Rest of the world)
Economic	Unit cost for heat, district or industrial, other than natural gas (Global)
Economic	Unit cost for heat, district or industrial, other than natural gas (EU)
Economic	Unit cost for HR [h/kg] (Bangladesh)
Economic	Unit cost for HR [h/kg] (China)
Economic	Unit cost for HR [h/kg] (EU)
Economic	Unit cost for HR [h/kg] (India)
Economic	Unit cost for HR [h/kg] (Pakistan)
Economic	Unit cost for HR [h/kg] (Turkey)
Economic	Unit cost for HR [h/kg] (USA)
Economic	Unit cost for HR [h/kg] (Vietnam)
Economic	Unit cost for HR [h/kg] (Rest of the world)

13485

Source: own elaboration.

13486 There is an additional set of indicators which is required for the investigation of the design options only.
 13487 These parameters are listed in the table that follows and they have used only for the Monte Carlo
 13488 investigation.

13489 **Table 219.** List of additional parameters used for the uncertainty analysis of design options.

Reference category	Description
Durability – Robustness	Change in Electricity because of the length of fibres
Durability – Robustness	Change in Mass because of the thickness of yarn
Durability – Robustness	Change in Electricity because of the thickness of yarn
Durability – Robustness	Change in Mass because of the number of yarn in fabrics
Durability – Robustness	Change in Electricity because of the number of yarn in fabrics
Durability – Robustness	Change in Mass because of the threads counts in fabrics
Durability – Robustness	Change in Electricity because of the threads counts in fabrics
Durability – Robustness	Change in Mass because of the type of patterns in fabrics
Durability – Robustness	Change in Electricity because of the type of patterns in fabrics
Durability – Robustness	Change in Chemical because of the dimensional stability in fabrics
Durability – Robustness	Change in Water in dyeing
Durability – Robustness	Change in Electricity in dyeing
Durability – Robustness	Change in Chemical in dyeing
Durability – Robustness	Change in Electricity in printing
Durability – Robustness	Change in Chemical in printing
Durability – Robustness	Change in Electricity in coating
Durability – Robustness	Change in Chemical in coating
Economic	Price for LCS1f – Recycled PET (Global)
Economic	Price for LCS1f – Recycled PET (China)
Economic	Price for LCS1f – Recycled PET (EU)
Economic	Price for LCS1u – Recycled Polyamide (Non-EU)
Economic	Price for LCS1u – Recycled Polyamide (EU)
Economic	Price for LCS1v – Recycled Cotton (Non-EU)
Economic	Price for LCS1v – Recycled Cotton (EU)
Economic	Price for LCS1w – Recycled Wool (Non-EU)
Economic	Price for LCS1w – Recycled Wool (EU)

13490 *Source: own elaboration.*

13491 **13.11.11.2** Sensitivity - Perturbation analysis

13492 The core concept of perturbation analysis is that minor changes in input parameters can translate into either
 13493 small or large variations in the output (Heijungs and Kleijn, 2001). Identifying which parameters cause

significant deviations and which have only minor effects can provide valuable insights. The identified parameters are changed individually by a small amount and the resulting changes in the results are observed to investigate the Sensitivity ratio, which is the ratio between relative changes (Clavreul et al., 2012).

Whereas, the Sensitivity Ratio, $SR = \frac{\frac{\Delta \text{result}}{\text{initial_result}}}{\frac{\Delta \text{parameter}}{\text{initial_parameter}}}$

When a 10% perturbation was applied on each of the selected parameters in the model across all grouped RPs, none of the results across the single score, based on 16 EF impact categories, as well as the 3 cost categories showed a sensitivity ratio over 1. i.e., none of the results showed a variation of over 10% when the parameter values were increased by 10%. The parameters that showed maximum variation for all the grouped RPs is given in the table below.

Table 220. Parameters with maximum variations in perturbation analysis

RP	Impact category	Sensitivity Ratio (S.R)			
		Positive variation		Negative variation	
		Parameter	Value	Parameter	Value
Knitted	Cost for producer	full time equivalent	0.311	working hours per year (Considering 2 shifts/day)	-0.027
	Cost for consumer	full time equivalent	0.273	Full load wash weight	-0.054
	Societal cost	Losses in confectioning	0.200	Full load wash weight	-0.071
	Single score	Losses in confectioning	0.216	Full load wash weight	-0.037
Other woven	Cost for producer	full time equivalent	0.309	working hours per year (Considering 2 shifts/day)	-0.023
	Cost for consumer	full time equivalent	0.285	Full load wash weight	-0.035
	Societal cost	Losses in confectioning	0.217	Full load wash weight	-0.046
	Single score	Losses in confectioning	0.226	Full load wash weight	-0.025
Denim	Cost for producer	full time equivalent	0.222	working hours per year (Considering 2 shifts/day)	-0.123
	Cost for consumer	full time equivalent	0.170	Full load wash weight	-0.190
	Societal cost	Losses in confectioning	0.136	Full load wash weight	-0.178
	Single score	Losses in confectioning	0.204	Full load wash weight	-0.072

Source: own elaboration.

For 'Knitted' fabrics, the single score showed maximum positive variation of 2.16% (S.R = 0.216) for the parameter 'Losses in confectioning' (Lc) and 'Full load wash weight (wei_was) showed the most negative variation of 0.37% (S.R = -0.037). 'Cost for producer' showed maximum positive variation of 3.11% (S.R = 0.311) for the parameter 'full time equivalent' (fte) and 'working hours per year' (wor_h) showed the most negative variation of 0.27% (S.R = -0.027). 'Cost for consumer' showed maximum positive variation of 2.73% (S.R = 0.273) for the parameter 'full time equivalent' (fte) and 'Full load wash weight (wei_was) showed the most negative variation of 0.54% (S.R = -0.054). 'Societal cost' showed maximum positive variation of 2.00% (S.R = 0.200) for the parameter 'Losses in confectioning' (Lc) and 'Full load wash weight (wei_was) showed the most negative variation of 0.71% (S.R = -0.071).

For 'Other woven' fabrics, the single score showed maximum positive variation of 2.26% (S.R = 0.226) for the parameter 'Losses in confectioning' (Lc) and 'Full load wash weight (wei_was) showed the most negative variation of 0.25% (S.R = -0.025). Cost for producer showed maximum positive variation of 3.09% (S.R = 0.309) for the parameter 'full time equivalent' (fte) and 'working hours per year' (wor_h) showed the most negative variation of 0.23% (S.R = -0.023). Cost for consumer showed maximum positive variation of 2.85% (S.R = 0.285) for the parameter 'full time equivalent' (fte) and 'Full load wash weight (wei_was) showed the most

negative variation of 0.35% (S.R = -0.035). Societal cost showed maximum positive variation of 2.17% (S.R = 0.217) for the parameter 'Losses in confectioning' (Lc) and 'Full load wash weight (wei_was) showed the most negative variation of 0.46% (S.R = -0.046).

For 'Denim', the single score showed maximum positive variation of 2.04% (S.R = 0.204) for the parameter 'Losses in confectioning' (Lc) and 'Full load wash weight (wei_was) showed the most negative variation of 0.72% (S.R = -0.072). Cost for producer showed maximum positive variation of 2.22% (S.R = 0.222) for the parameter 'full time equivalent' (fte) and 'working hours per year' (wor_h) showed the most negative variation of 1.23% (S.R = -0.123). 'Cost for consumer' showed maximum positive variation of 1.70% (S.R = 0.170) for the parameter 'full time equivalent' (fte) and 'Full load wash weight (wei_was) showed the most negative variation of 1.90% (S.R = -0.190). 'Societal cost' showed maximum positive variation of 1.36% (S.R = 0.136) for the parameter 'Losses in confectioning' (Lc) and 'Full load wash weight (wei_was) showed the most negative variation of 1.78% (S.R = -0.178).

Table below shows the ranges in which Sensitivity Ratios of each of the 98 parameters across all the 4 impact categories considered falls under after the perturbation analysis.

Table 221. Number of parameters across different S.R ranges

RP	Impact category	Sensitivity Ratio (S.R)				
		1>S.R>0.1	0.1>S.R>0	0	0>S.R>-0.1	-0.1>S.R>-1
Knitted	Cost for producer	3	61	33	1	0
	Cost for consumer	2	72	20	4	0
	Societal cost	2	70	22	4	0
	Single score	2	17	77	2	0
Other woven	Cost for producer	3	68	26	1	0
	Cost for consumer	2	79	13	4	0
	Societal cost	2	77	15	4	0
	Single score	2	13	81	2	0
Denim	Cost for producer	2	2	1	92	1
	Cost for consumer	2	7	1	86	1
	Societal cost	3	7	1	85	2
	Single score	2	13	81	2	0

Source: own elaboration.

This classification shows that most of the parameters have Sensitivity Ratios between -0.1 and 0.1. The main parameters which showed a positive variation with a S.R >0.1 were the 'full time equivalent', 'Losses in confectioning', 'Losses in ring spinning' and 'share of garments machine washed' depending on the RP. The parameters which showed a negative variation with a S.R < -0.1 were only present for Denim and these were 'Full load wash weight', 'Full load dry weight' and 'Working hours per year'.

13.11.11.3 Monte Carlo uncertainty analysis

The common method used for the simulation of uncertainty propagation in the model results are sampling methods which samples model inputs from input distributions. Monte Carlo (MC) is the most widely used sampling method for uncertainty propagation, allowing the combined impact of individual input uncertainties to be evaluated (Marsh et al., 2023). This process requires a sample of randomly generated input values for each model iteration. Repeated simulations based on this sampling approach can be used to derive the propagated uncertainty of the results. In our case, the chosen parameters and the values were randomly generated based on the ranges defined by minimum and maximum values elaborated based on the expertise of the modelling team. Then the model was run for 500 iterations using the fixed values that were not chosen as parameters as well the chosen parameters which were allowed to receive values randomly in the range defined by the minimum and maximum values defined. The ranges used for the parameters referring to losses, use phase, volume and economics were acquired by the same sources mentioned in section 13.11.9 and own elaboration. For the parameter values referring to the monetisation factors, the minimum and maximum values

were gathered from the Trinomics report⁴⁰³ for Human toxicity, cancer, Human toxicity, non-cancer, Acidification, Eutrophication, freshwater, Eutrophication, marine, Ecotoxicity, freshwater, Land use, Water use, Resource use, minerals and metals, Resource use, fossils. For Ozone depletion, the minimum and maximum values were taken from (Amadei et al., 2021). Climate change range reported in DG MOVE report⁴⁰⁴ was used (Short and medium run, up to 2030). For Photochemical ozone formation the values for each Country reported in DG MOVE report were analysed: the lowest and the highest values were considered as representative of minimum and maximum values. For Particulate matter, the procedure reported in Annex IV of the report by (Gama Caldas et al., 2024) was followed, starting from the minimum and maximum values of the parameters PM2.5 transport metropole, PM2.5 transport city, PM2.5 transport rural, PM10 average reported in Table 14 of DG MOVE report. All the values were updated to the reference year (2022) using the Harmonised Index of Consumer Prices (HICP) reported by Eurostat⁴⁰⁵.

Based on the results, acquired from the 500 iterations, it was possible to calculate the standard deviation and assuming that these iterations are normally distributed it was possible to estimate a 95% confidence interval for the results of each one of the three RPs, as also shown in **Table 222**, **Table 223**, **Table 224**.

Table 222 Results with 95% Confidence interval for RP1

RP1	Single score	Cost for producer	Cost for consumer	Societal cost
Max	0.00253	7.29	12.15	11.71
Min	0.00250	7.21	12.03	11.60
Central value	0.00251	7.25	12.09	11.66

Table 223 Results with 95% Confidence interval for RP2

RP2	Single score	Cost for producer	Cost for consumer	Societal cost
Max	0.00280	8.61	13.66	13.25
Min	0.00277	8.52	13.53	13.14
Central value	0.00279	8.56	13.59	13.20

Table 224 Results with 95% Confidence interval for RP3

RP3	Single score	Cost for producer	Cost for consumer	Societal cost
Max	0.00517	10.86	20.78	22.05
Min	0.00511	10.74	20.53	21.76
Central value	0.00514	10.80	20.65	21.90

⁴⁰³ European Commission, Directorate-General for Energy, Smith, M., Moerenhout, J., Thuring, M., et al, External costs: energy costs, taxes and the impact of government interventions on investments: final report, Publications Office, 2020.

⁴⁰⁴ European Commission, Directorate-General for Mobility and Transport, Essen, H., El Beyrouty, K., Bieler, C., et al, Handbook on the external costs of transport: version 2019, Publications Office, 2019.

⁴⁰⁵ Eurostat, Harmonised Index of Consumer Prices (HICP) - annual data (average index).

13.12 Supporting information on the assessment of the availability of secondary materials

The following sections from 13.12.1 to 13.12.4 describe the calculations, assumptions and sources for the assessment of the availability of secondary materials.

13.12.1 Detailed assessment of recycled cotton availability

Global and EU Market Allocation

- Input Data:**

- Global Recycled Cotton: **0.32 Mton**
 - Source: Textile Exchange, (2024).

- Assumption:**

- Share of EU in Global Market: 18%
- Source for Market Size Data: UniformMarket, (2025) (**Table 225**)
- Source for Gross domestic product (GDP): World Bank, (2024)

Table 225. Apparel market sizes (2025)

Country	Apparel Market Size (billion \$)
Germany	72.57
Italy	57.86
France	40.02
Spain	24.96
EU*	330
Global	1840

* The EU figure is estimated based on the shares of the EU countries for which data is available to the total EU GDP.
Source: *Elaborated from UniformMarket, (2025) and World Bank, (2024)*

- Calculation:**

- Recycled cotton available for EU market: $0.32 \text{ Mton} \times 18\% = 0.06 \text{ Mton}$

Denim Cotton Usage in the EU

- Apparent Consumption:**

- Denim trousers (men and women): $0.11 \text{ Mton} + 0.09 \text{ Mton} = 0.21 \text{ Mton}$
 - Source: PRODCOM dataset
- Denim jacket and coat: 0.51 Mton
 - Source: PRODCOM dataset assuming that the share of denim to the total jacket and coat category is the same as in trousers.

- Composition Data:**

- Cotton in denim trousers: 71%
- Cotton in denim jackets: 57%
 - Source: Bakowska, O. et al., (2025)

- Cotton in Denim Calculation:**

- Cotton in denim trousers: 0.15 Mton
- Cotton in denim jackets: 0.05 Mton
- Estimated Cotton in Denim: 0.2 Mton
 - Source: estimated from the apparent consumption and composition data.

Maximum Threshold for Recycled Cotton in EU Denim

- **Calculation:**
 - *Maximum Possible Threshold:* $\frac{0.06 \text{ Mton (available recycled cotton)}}{0.2 \text{ Mton (cotton in denim)}} = 28\%$

13613 **13.12.2 Detailed assessment of recycled nylon availability**

13614 **Global and EU Market Allocation**

- **Input Data:**
 - Global Recycled Nylon: **0.1 Mton**
 - *Source:* Textile Exchange (2024)
- **Assumption:**
 - Share of EU in Global Market: 18%
 - *Source for Market Size Data:* UniformMarket (2025) (Table 225)
 - *Source for Gross domestic product (GDP):* World Bank (2024)
- **Calculation:**
 - Recycled nylon available for EU market: 0.1 Mton×18% = 0.02 Mton

13624 **Nylon Usage in Knitted and Other Woven Products in the EU**

- **Apparent Consumption:**
 - Apparel excluding denim = 4.34 Mton
 - *Source:* PRODCOM dataset
- **Composition Data:**
 - Nylon in knitted and other woven products: 5%
 - *Source:* Bakowska, O. et al. (2025)
- **Nylon consumption Calculation:**
 - Nylon in knitted and other woven products: 0.21 Mton
 - *Source:* estimated from the apparent consumption and composition data.

13634 **Maximum Threshold for Recycled Nylon in EU Knitted & other woven products**

- **Calculation:**
 - *Maximum Possible Threshold:* $\frac{0.02 \text{ Mton (available recycled nylon)}}{0.21 \text{ Mton (nylon in knitted and other woven products)}} = 9\%$

13637 **13.12.3 Detailed assessment of recycled polyester availability**

13638 **Global and EU Market Allocation**

- **Input Data:**
 - Global Recycled Polyester: **8.9 Mton**
 - *Source:* Textile Exchange (2024)
- **Assumption:**
 - Share of EU in Global Market: 18%
 - *Source for Market Size Data:* UniformMarket (2025) (Table 225)
 - *Source for Gross domestic product (GDP):* World Bank (2024)
- **Calculation:**
 - Recycled polyester available for EU market: 8.9 Mton×18% = 1.6 Mton

Polyester Usage in Knitted and Other Woven Products in the EU

- **Apparent Consumption:**

- Apparel excluding denim = 4.34 Mton

- *Source:* PRODCOM dataset

- **Composition Data:**

- Polyester in knitted and other woven products: 29%

- *Source:* Bakowska, O. et al., (2025)

- **Polyester consumption Calculation:**

- Polyester in knitted and other woven products: 1.27 Mton

- *Source:* estimated from the apparent consumption and composition data.

Maximum Threshold for Recycled Polyester in EU Knitted & other woven products

- **Calculation:**

- *Maximum Possible Threshold:* $\frac{1.6 \text{ Mton (available recycled polyester)}}{1.27 \text{ Mton (polyester in knitted and other woven products)}} = 125\%$

13.12.4 Detailed assessment of recycled wool availability

Recycled Wool Production

- **Input Data:**

- Recovered Wool from Prato District: 0.02 Mton

- *Source:* *Prato District Data*

Wool Usage in Knitted and Other Woven Products in the EU

- **Apparent Consumption:**

- Apparel excluding denim = 4.34 Mton

- *Source:* PRODCOM dataset

- **Composition Data:**

- Wool in knitted and other woven products: 3%

- *Source:* Bakowska, O. et al., (2025)

- **Wool Consumption Calculation:**

- Estimated Wool in Knitted & Other Woven: 0.13 Mton

Maximum Threshold for Recycled Wool in EU Knitted & Woven Products

- **Calculation:**

- *Maximum Possible Threshold:* $\frac{0.2 \text{ Mton (available recycled wool)}}{0.13 \text{ Mton (wool in knitted and other woven products)}} = 17\%$

13.13 Supporting information for the identification of substances and substances of concern

Table 226. Classification of chemical substances used in the production of textile apparel according to the process stage and the function/type.

Process stage	Function /type of chemical substance used
Pretreatment (process chemicals, which are not intended to remain in the final textile product)	Fibre protecting agents in pretreatment Scouring auxiliaries Bleaching auxiliaries Mercerising and causticizing auxiliaries Carbonizing assistants Desizing agents Hydrophilizing agents Enzymes Sequestering /chelating agents Stabilizers Detergents Others (e.g. acids, bases, salts, solvents, etc.)
Spin finishing and sizing chemicals (remain in the final textile product, usually till fabric treatment)	Spinning solution additives Spinning additives Spinning bath additives Lubricants Coning oils, warping and twisting oils Conditioning and stabilizing agents Anti-electrostatic agents Sizing agents (natural / synthetic) Sizing additives Hydrophilising agent Hydrophobising agent
Dyeing (dyes remain in the final textile product)	Dyestuff solubilizing and hydrotropic agents Dispersing agents and protective colloids Wetting-, dearation-, anti-foaming agents Soaping agents Levelling agents Dye accelerators Crease-preventing agents Dyestuffs protecting agents Padding Auxiliaries Anti-migration agents Anti-frosting auxiliaries Anti-back staining agents Products increasing wet-pick-up Fix accelerator for continuous dyeing and printing Others (e.g. detergents, salts, solvents, acids, bases)
Printing (colours remain in the final textile product)	Agents to remove printing thickeners Printing and edge adhesives Oxidizing agents Reducing agents Discharging agents and discharging assistants Resistant agents Mordants Brightening and stripping agents pH-regulators, acid and alkali dispensers Agents to reduce or deepen shade
Finishing (remain in the final textile product at least until the first washing)	Crease and shrink resistance agents (easy-care, easy-ironing) Additives for non-creasing + non-shrinking finishes Catalysts for non-creasing + non-shrinking finishes Handle-imparting agents Weighting agents Filling and stiffening agents Softening agents Anti-electrostatic agents Water, stain and oil repellents Crosslinking agent for water repellents and coatings Soil release agents Felting Agents Anti-felting agents

Process stage	Function /type of chemical substance used
	Antisnagging Tear/seam and abrasion resistance Rub fastness, abrasion resistance Conditioning agents Anti-microbiotics (finish, incl. insects repellents) Anti-microbiotics (storage stability) Odour absorbers Agents and additives to promote bonding of fibres and threads UV protection Fragrance finishes (incl. microencapsulated) Anti-pilling agents Flame retardants Resins
Common chemicals to provide colours in dyeing /printing /pretreatment processes	Reactive dyes / inks (CO, PA, WO) Mordant dyes Direct dyes (CO) Disperse dyes / inks (mainly PES) Basic dyes (cationic dye) Acid dyes / inks Metal complex dyes / inks (WO, PA) Vat dyes / inks Sulphur dyes Pigments Solvent dyes Natural dyes Optical brighteners (pretreatment)
Common chemicals in dyeing /printing processes	Aftertreatment agents for dye fastness improvement Bonding agent (binders) for pigment printing Printing thickeners (native / synthetic) Printing pastes (water based, solvent based, multicomponent ready-to-use, foam building, ...) Softening agents / plasticisers for printing Emulsifiers Adhesives Agents to reduce or deepen shade Anti-foaming agents for printing and coating Printing Oils
Coating chemicals (remain in the final textile product)	Delustering agents Non-slip, ladder-proof and anti-snap agents Flame retardants Anti-microbiotics (coating, storage, insects) Anti-microbiotics (storage stability) Coating agents and additives Laminating agents and according additives UV protection Resins Polymer coating dispersions (water-/solvent based) Crosslinking agents for coatings Accelerators for solvent coating Fragrance finishes (incl. microencapsulated) Anti-pilling agents

Note: List is not exhaustive.

Source: Adapted from EUCTL – European Chemistry for Textile and Leather, the European Association representing the companies that operate in Europe, producing and putting on the market chemicals used in the textiles and leather value chains.

13687 **Table 227.** Inventory of the relevant chemical substances identified in textile apparel products per function (non-exhaustive).

Process step	Fibre	Function	Chemical substance	CAS number	Notes/Reference
Raw material production - agricultural phase	cotton, wool including cashmere	Pesticide	Diazinon (wool, cashmere)	333-41-5	The most common ectoparasiticides used for treating sheep is diazinon, an organophosphorous insecticide. Other chemicals with same properties are: propetamphos, cypermethrin and cyromazine (Source: BREF Textiles).
			Cypermethrin (wool, cashmere, cotton)	52315-07-8	
			Cyromazine (wool, cashmere)	66215-27-8	
			Propetamphos (wool, cashmere)	31218-83-4	
			Chlorpyrifos (cotton)	2921-88-2	Possible pesticides (residues) detected in the cotton products include chlorpyrifos, malathion, profenofos and cypermethrin (Source: BREF Textiles)
			Profenofos (cotton)	41198-08-7	
			Malathion (cotton)	121-75-5	
Fibre production	viscose	Solvent	Carbon disulphide	75-15-0	(Source: BREF Polymers as MMCF are addressed in that BREF and not in the BREF Textiles)
		Acid agent	Zinc sulphate	7733-02-0	
		Solvent	Sodium hydroxide	1310-73-2	
		Solvent	Sulphuric acid	7664-93-9	
Spinning of fibres	cotton, wool, cashmere, viscose	Lubricant (also used in knitting)	White mineral oil	8042-47-5	(Source: BREF Textiles, EUCTL)
			Silicones (e.g. copolymers of silicone-like polydimethylsiloxane (PDMS) with glycols)	9006-65-9	(Source: BREF Textiles, EUCTL) May be present in silicone pads, as contaminants in formulations that contain silicone, like silicone softeners and those used for prints (AFIRM).
		Surfactant (as emulsifier) (also used in knitting, washing as pretreatmen)	NPEOs (non-ionic)	9016-45-9	Toxic to aquatic life (Source: BREF Textiles)
			Ethylene oxide (non-ionic)	75-21-8	(Source: BREF Textiles)
			Propylene oxide (non-ionic)	75-56-9	

Process step	Fibre	Function	Chemical substance	CAS number	Notes/Reference
			Fatty amine ethoxylates (non ionic) (no example chemicals in BREF): Tallow amine ethoxylate Coco amine ethoxylate Stearyl amine ethoxylate Oleyl amine ethoxylate Lauryl amine ethoxylate	61791-26-2 61791-14-8 26635-93-8 26635-94-9 31017-83-1	
			Lignin sulphonates (anionic) (no example chemicals in the BREF) e.g. lignosulfonic acid, sodium salt	8061-51-6	
			Quaternary ammonium compounds (cationic) e.g. distearyl dimethyl ammonium chloride	107-64-2	(Source: BREF Textiles)
		Biocide	Formaldehyde-containing compounds (e.g. Formaldehyde)	50-00-0	(Source: BREF Textiles)
			Heterocyclic compounds (imidazol and isothiazolinone derivatives) Isothiazolinone	2682-20-4	(Source: BREF Textiles)
			Heterocyclic compounds (imidazol and isothiazolinone derivatives) Imidazole	288-32-4	(Source: BREF Textiles)
Washing (pre-treatment)	All fibre types	Complexing agent	see surfactant (as emulsifier) above		
			Ethylenediaminetetraacetic acid (EDTA) (polycarboxylic acids)	60-00-4	(Source: BREF Textiles)
			Diethylenetriaminepentaacetic acid (DTPA) (polycarboxylic acids)	67-43-6	

Process step	Fibre	Function	Chemical substance	CAS number	Notes/Reference
Bleaching	viscose, cotton, wool, cashmere, polyester, polyamide, acetate		Tripolyphosphate (polyphosphates)	7758-29-4	
			Diethylenetriaminepenta(methylenephosphonic acid) (DTPMP) (phosphonates)	15827-60-8	
			Ethylenediaminetetra(methylenephosphonic acid) (EDTMP) (phosphonates)	1429-50-1	
		Bleaching agent	Sodium hypochlorite	7681-52-9	AOX emissions (BREF textiles and polymers)
			Hydrogen peroxide	7722-84-1	(BREF textiles and polymers)
		Sequestering/stabilizer	see complexing agent above		
		Reductive	sodium dithionite	7775-14-6	(BREF textiles and polymers)
		Alkaline	Sodium hydroxide	1310-73-2	(BREF textiles and polymers)
			sodium carbonate	497-19-8	(BREF textiles and polymers)
		Surfactant	see surfactant above		
		Optical Brightening Agent	Stilbene Pyrazoline Triazines Benzoxazoline	-	(Information provided by retail industry)
Finishing (easy care/cross linking/no crease)	cotton, viscose	Easy care/cross linking agents	Hydroxymethyl urea (formaldehyde-rich) and other similar species	1000-82-4	Emissions of formaldehyde, methanol and other volatile organic substances in the exhaust air in the application of these agents (BREF textiles and polymers)

Process step	Fibre	Function	Chemical substance	CAS number	Notes/Reference
		Catalyst	Ammonium sulphate	7783-20-2	Ammonium salts (e.g. chloride, sulphate and nitrate) commonly used in the past, particularly with cross-linkers based on urea and formaldehyde or melamine, are in decline today, following the corresponding decline in importance of these cross-linking agents (BREF textiles and polymers)
			Ammonium nitrate	6484-52-2	
			Ammonium chloride	12125-02-9	
		Softeners	[dispersion of] polysiloxanes	161755-53-9; 9006-65-9 (dimethyl polysiloxanes)	Sources of emissions of VOCs in the exhaust air from curing and drying operations. The poor biodegradability of the above-mentioned polymer dispersions has also to be taken into account when considering water pollution (BREF textiles and polymers)
			[dispersion of] polyethylene waxes (partially oxidised polyethylenes)	68441-17-8	
			[dispersion of] polyacrylates (e.g. as in Polymer dispersions (aqueous formulations, 50% water) for Coating (BREF page 869): - poly(meth)acrylate (butyl, ethyl, methyl etc.); - polyacrylic acid; - polyacrylonitrile; - polyacryloamide)	9011-14-7 (Poly(methyl methacrylate)) 9003-63-8 (Poly(butyl methacrylate)) 9003-42-3 (Poly(ethyl methacrylate)) 9003-01-4 (polyacrylic acid) 9009-54-5 (polyurethane) 9003-05-8 (polyacryloamide)	

Process step	Fibre	Function	Chemical substance	CAS number	Notes/Reference
Dyeing	Natural cellulosic fibres, MMCF	Reactive Dyestuff	Monochlorotriazine, Vinylsulphone, Monofluorotriazine, Dichloroquinoxaline, Difluorochlorpyrimidine, Dichlorotriazine, Metal azo-complex, Triphenodioxazine, phthalocyanine	-	(Information provided by retail industry)
	Natural cellulosic fibres, MMCF	Direct Dyestuff	Azoic based dyestuff	-	
	Natural cellulosic fibres, MMCF	Vat Dyestuff	Anthraquinone based dyestuff	-	
	Natural cellulosic fibres, MMCF	Sulphur Dyestuff	Azoic based dyestuff, Anthraquinone based dyestuff	-	
	Natural cellulosic fibres, MMCF	Mordants	Quaternary ammonium polymers, Polyethylene polyamine Aliphatic polyamine Diciandiamide polymer	-	
	Natural cellulosic fibres, MMCF	Fixers	Quaternary ammonium polymers Polyethylene polyamine Aliphatic polyamine Diciandiamide polymer	-	
	Polyester, acetate	Disperse Dyestuff	Anthraquinone based dyestuffs Azoic based dyestuffs Thiphen based dyestuffs (Azoic based dyestuff) Others: Methyne, Styryl, Acrylene benzimidazol, Quinonaphthalon, Amino naphthyl amide, Napthoquinone imine and Nitro Disperse dyes	-	(Information provided by retail industry)
Dyeing	Polyamide, wool	Acid dyestuff	Azoic based dyestuff Anthraquinone based dyestuff	-	(Information provided by retail industry)
	Polyamide, wool	Fixing agent	Anionic-hidroxifenisulfone Cationic Poliamine	-	(Information provided by retail industry)
Printing (with pigments)	Natural cellulosic fibres, polyester, MMCF, acetate	Binder	Self-crosslinker polyacrylate Non-self-crosslinker polyacrilate Aliphatic polyurethane Aromatic polyurethane (Polyvinil chloride) PVC	-	(Information provided by retail industry)
	Natural cellulosic fibres,	Thickeners	Polyacrylates Other polymers without formaldehyde Polyacrilates	-	(Information provided by retail industry)

Process step	Fibre	Function	Chemical substance	CAS number	Notes/Reference
	polyester, MMCF, acetate		Acrylic acids		
	Natural cellulosic fibres, polyester, MMCF	Fixers/ catalysts (crosslinkers)	Azoic based pigment Naphthol derivative based pigments Phthalocyanine based pigment Carbon black pigment	-	(Information provided by retail industry)
	Natural cellulosic fibres, polyester, MMCF, acetate	Pigments	Azoic based pigment Naphthol derivatives based pigments Phthalocyanine based pigment Carbon black pigment	-	(Information provided by retail industry)
Printing (with Soluble Dyestuffs)	Natural cellulosic fibres, polyester, MMCF, acetate	Thickeners	Natural thickeners (Sodium alginate, Carboxymethylcellulose / Hydroxyethylcellulose, Starch/Carboxymethylstarch, Locust bean/ british / arabic gum, Guar carboxymethylate /ydroxipropilenated /depolymerized, Tamarind, Bentonite)	-	(Information provided by retail industry)
Positional Printing or Finishing	Natural cellulosic fibres, MMCF	Resins Paste Plasticizer Lacquer Adhesives	Compounds that contain aromatic surfactants Compounds that contain PVC Compounds that contain formaldehyde free polymers Compounds that contain aromatic polyurethane	-	(Information provided by retail industry)
	Natural cellulosic fibres, MMCF	Metallic Effects (i.e. glitter, purpurine, pearl)	Aromatic polyurethane Self- crosslinker polyacrylate DMHEU polymer/Glyoxal monomer Melamine-formaldehyde polymer	-	According to Bluesign BSSL : Melamine 108-78-1 is used in all types of fibres and it has a usage ban (<200 mg/kg)
Finishing	Natural cellulosic fibres, MMCF	Resinate	Aliphatic polyurethane	-	(Information provided by retail industry)
	Natural cellulosic fibres, MMCF	3D effects	Self-crosslinker polyacrilate Non self-crosslinker polyacrilate	-	(Information provided by retail industry)
	Natural cellulosic fibres, MMCF	No creases effect	METIOL UREA (e.G. GLYOXAL) Melamine-formaldehyde polymer	-	(Information provided by retail industry)

Process step	Fibre	Function	Chemical substance	CAS number	Notes/Reference
Finishing	Natural cellulosic fibres, MMCF	Catalyzers (Crosslinkers)	Polyvinyl acetate Etilvinyl acetate	-	(Information provided by retail industry)
	Natural cellulosic fibres, MMCF	Extenders	LATEX (Estire butandiene resins) PVC Dispersion	-	(Information provided by retail industry)
	Natural cellulosic fibres, MMCF	Adhesives	EPOXI Resins	-	(Information provided by retail industry)
	Natural cellulosic fibres	Thickeners	Natural or synthetic thickeners	-	(Information provided by retail industry)
	Natural cellulosic fibres	Foaming Agents	Ammonium stearate Oxid amine Sulphates	-	(Information provided by retail industry)
	Natural cellulosic fibres, polyester, MMCF, acetate	Flame Retardant	Halogenated Non halogenated Phosphorus derivates Phosphor free Compound without halogenates and phormol Compound with halogenates or phormol Metal oxide	-	(Information provided by retail industry)
	Natural cellulosic fibres, MMCF	Antislip	Silica	-	(Information provided by retail industry)
Finishing	Polyester, Polyamide, Wool, acetate	Water repellent	Silicone Polymer Paraffin Fat modified melamine Natural oils Fluorocarbons	-	(Information provided by retail industry)
	Polyester, acetate	UV filters	Hydroxyphenone derivatives	-	
	Polyester, acetate	Increase Hygroscopicity	Polyamide polymer	-	
	Polyester, acetate	Antistatics	Polyester resin polymer	-	
	Polyamida	Antiyellowing	Derived from ethylpyrolidine	-	
	Polyamida	Increase Hygroscopicity	Polyamide resin	-	
	Polyamida	Antistatics	Polyester resin	-	

Process step	Fibre	Function	Chemical substance	CAS number	Notes/Reference
	Wool	Unshrinkage	Acid permonosulphuric salt Various polymers Dichloroisocyanuric	-	
Finishing	All type of fibres	Biocides	Pyrethione sodium	3811-73-2	Bluesign: Limitation 50 / 200 / 200 mg/kg
			2-Chloroacetamide	79-07-2	Bluesign (usage ban: 1 mg/kg)
			Dichlorophen	97-23-4	Bluesign (usage ban: 1 mg/kg)
			Dimethylfumarate	624-49-7	Bluesign (usage ban: 0.1 mg/kg)
			N-Methylol-chloroacetamide	2832-19-1	Bluesign (usage ban: 1 mg/kg)
			Permethrin	52645-53-1	Bluesign (usage ban: 1 mg/kg)
			Pyrethione zinc	13463-41-7	Bluesign (usage ban: 10 mg/kg)
			Triclosan	3380-34-5	Bluesign (usage ban: 1 mg/kg)
			All isothiazolinones: CIT, MIT, Mixture (3:1) of CIT and MIT, MBIT, OIT, BIT, DCOIT,	26172-55-4 2682-20-4 55965-84-9 2527-66-4 26530-20-1 2634-33-5 64359-81-5	Bluesign (usage ban: 1, 10, 1, 10, 10, 25, 10 mg/kg)
			(benzothiazol-2-ylthio)methyl thiocyanate; TCMTB	21564-17-0	(Avagyan et al., 2015) ⁽⁴⁰⁶⁾ The mentioned source is about benzothiazole derivative. Here, a derivative approved for use as active substance under BPR ⁽⁴⁰⁷⁾ is considered.
Washing Auxiliaries	Polyamida	Fixing agent	Arylsulphonate in aqueous solution	-	(Information provided by retail industry)

Note 1: In **bold**, chemical substances that remain in the final textile product.

Note 2: List is not exhaustive.

Source: JRC's own elaboration using literature and stakeholder feedback.

⁴⁰⁶ Avagyan, R., Luongo, G., Thorsén, G. et al. Benzothiazole, benzotriazole, and their derivatives in clothing textiles—a potential source of environmental pollutants and human exposure. Environ Sci Pollut Res 22, 5842–5849 (2015). <https://doi.org/10.1007/s11356-014-3691-0>

⁴⁰⁷ Regulation (EU) No 528/2012 of the European Parliament and of the Council of 22 May 2012 concerning the making available on the market and use of biocidal products (BPR).

13692 **Table 228.** Life cycle inventory of the chemical substances used in the LCA in function on the Life Cycle Stage (LCS).

LCS	Substances
LCS1	[sulfonyl]urea-compound
LCS1	[thio]carbamate-compound
LCS1	acetamide-anillide-compound, unspecified
LCS1	ammonium nitrate
LCS1	antimony
LCS1	benzimidazole-compound
LCS1	benzoic-compound
LCS1	Carbon dioxide, in air (Elementary flows/Resource/in air)
LCS1	carbon disulfide
LCS1	chemical, inorganic
LCS1	chemical, organic
LCS1	cyclic N-compound
LCS1	diethylene glycol
LCS1	dinitroaniline-compound
LCS1	diphenylether-compound
LCS1	ethylene glycol
LCS1	glyphosate
LCS1	inorganic nitrogen fertiliser, as N
LCS1	inorganic phosphorus fertiliser, as P2O5
LCS1	natural gas, high pressure
LCS1	nitrile-compound
LCS1	nitrogen, liquid
LCS1	organophosphorus-compound, unspecified
LCS1	oxygen, liquid
LCS1	pendimethalin
LCS1	pesticide, unspecified
LCS1	phosphate rock, beneficiated
LCS1	phosphoric acid, industrial grade, without water, in 85% solution state
LCS1	potassium chloride
LCS1	purified terephthalic acid
LCS1	pyrethroid-compound
LCS1	single superphosphate
LCS1	soap
LCS1	sodium chloride, powder
LCS1	sodium hydroxide, without water, in 50% solution state
LCS1	sodium hypochlorite, without water, in 15% solution state
LCS1	sulfate pulp, bleached
LCS1	sulfur dioxide, liquid
LCS1	sulfuric acid
LCS1	triazine-compound
LCS1	triple superphosphate
LCS1	urea
LCS1	urea ammonium nitrate mix

LCS	Substances
LCS1	zinc monosulfate
LCS1	zinc oxide
LCS2	Acetic acid
LCS2	Acrylic acid
LCS2	Alkylphenol ethoxylates (APEOs)/ Nonylphenol ethoxylates (NPEOs)
LCS2	Ammonia
LCS2	Ammonium sulfate
LCS2	Average dye
LCS2	Benzo[thia]diazole-compound
LCS2	Chelating agent/stabilizer
LCS2	Detergent/surfactant
LCS2	Diammonium phosphate
LCS2	Diesel (thickener for pigment)
LCS2	Disperse dyes
LCS2	Hydrogen peroxide
LCS2	Lubricating oil
LCS2	Melamine formaldehyde resin
LCS2	Other chemicals
LCS2	Other chemicals - pigment
LCS2	Paraffin
LCS2	Polydimethylsiloxane
LCS2	Polymethyl methacrylate, beads
LCS2	Rosin size
LCS2	Soda ash
LCS2	Sodium chloride
LCS2	Sodium hydrosulfide
LCS2	sodium hydroxide, without water, in 50% solution state
LCS2	Sodium silicate
LCS2	Sodium sulphate
LCS2	Solvent
LCS2	Starch
LCS2	Tetrafluoroethylene
LCS2	(blank)
LCS2	Acrylic acid
LCS2	Detergent/surfactant
LCS2	Melamine formaldehyde resin
LCS2	Polydimethylsiloxane
LCS4	Detergent/surfactant

13693 Source: JRC's own elaboration from LCA model.

13694

draft